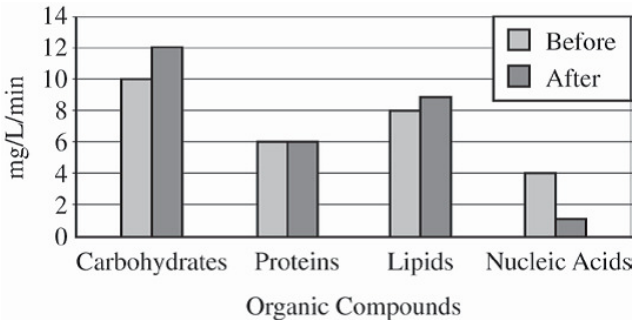
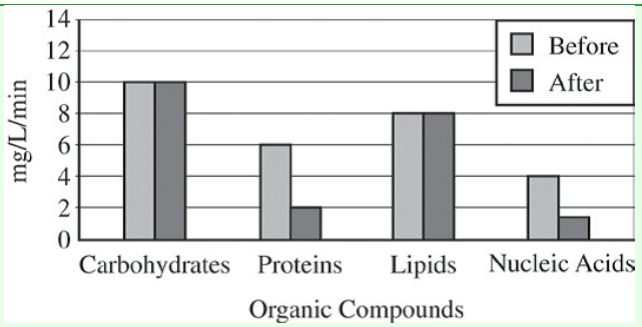
1 Chemistry of Life MCQ

1. A culture of *Spirogyra* (an autotrophic alga) is maintained in a water solution containing dissolved carbon dioxide and a source of phosphates but lacking nitrogen compounds. A researcher determines the rates of synthesis of several organic compounds found in the *Spirogyra* before and after several weeks in the water solution. Which of the following graphs best illustrates a likely result of the experiment?

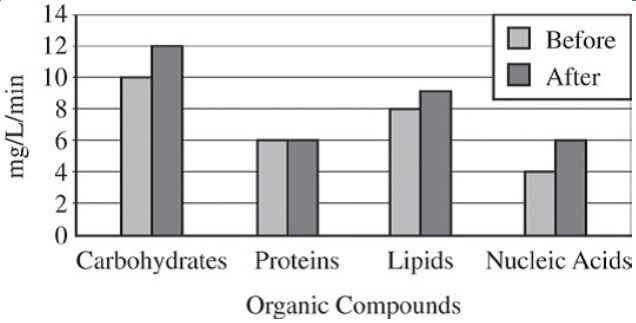
(A)



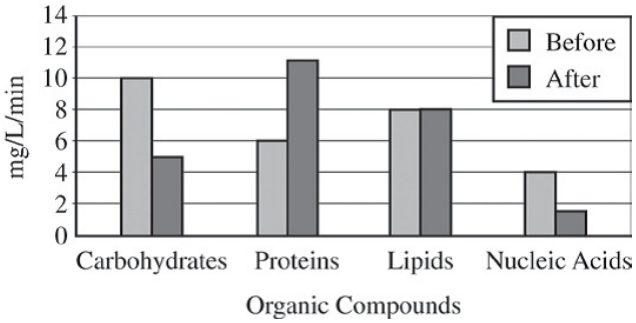
(B) ✓



(C)



(D)



1 Chemistry of Life MCQ

Answer B

This option is correct. Without a source of nitrogen, the *Spirogyra* cannot synthesize proteins or nucleic acids, both of which contain nitrogen. This answer reflects that the student is able to represent graphically the exchange of molecules between an organism and its environment and the use of these molecules in synthesis.

2. A feature of organic compounds NOT found in inorganic compounds is the presence of

(A) ionizing chemical groups

(B) electrons

(C) carbon atoms covalently bonded to each other ✓

(D) oxygen

(E) hydrogen bonds

3. tRNA molecules deliver amino acids to ribosomes where the amino acid next

(A) forms an ionic bond with another amino acid

(B) is deaminated and catabolized

(C) becomes an anticodon

(D) is covalently bonded to a growing polypeptide ✓

(E) forms hydrogen bonds with esters

4. The CFTR protein is made up of 1,480 amino acids linked together in a chain. Some humans produce a version of the CFTR protein in which phenylalanine (an amino acid) has been deleted from position 508 of the amino acid chain.

Which of the following best predicts how the amino acid deletion will affect the structure of the CFTR protein?

(A) It will have no observable effect on the structure of the CFTR protein.

(B) It will affect the primary structure of the CFTR protein, but the other levels of protein structure will not be affected.

(C) It will affect the secondary and tertiary structures of the CFTR protein, but the primary structure will not be affected.

(D) It will affect the primary, secondary, and tertiary structures of the CFTR protein. ✓

Answer D

Correct. A change in the primary structure of the CFTR protein will lead to changes in the secondary

and tertiary structures. As a result, the protein will not fold properly and will not function normally.

5. Which of the following describes a key difference among the 20 amino acids that are used to make proteins?

(A) Only some amino acids have an R-group.

(B) Only some amino acids have a carboxyl group (COOH).

(C) Some amino acids are hydrophobic. ✓

(D) Some amino acids contain the element phosphorus.

Answer C

Correct. Due to having nonpolar R-groups, 10 of the 20 amino acids are hydrophobic. Interactions between hydrophobic amino acids play an important role in determining protein structure and function.

6. Scientists examined the folded structure of a purified protein resuspended in water and found that amino acids with nonpolar R groups were primarily buried in the middle of the protein, whereas amino acids with polar R groups were primarily on the surface of the protein. Which of the following best explains the location of the amino acids in the folded protein?

(A) Polar R groups on the surface of the protein can form ionic bonds with the charged ends of the water molecules.

(B) Polar R groups are too bulky to fit in the middle of the protein and are pushed toward the protein's surface.

(C) Nonpolar R groups that cannot form hydrogen bonds with water are pushed into the middle of the protein. ✓

(D) Nonpolar R groups from different parts of the protein form covalent bonds with each other to maintain the protein's structure.

1 Chemistry of Life MCQ

7. Figure 1 shows three amino acids that are part of a polypeptide chain. Figure 2 shows the same section of the chain after a mutation has occurred.

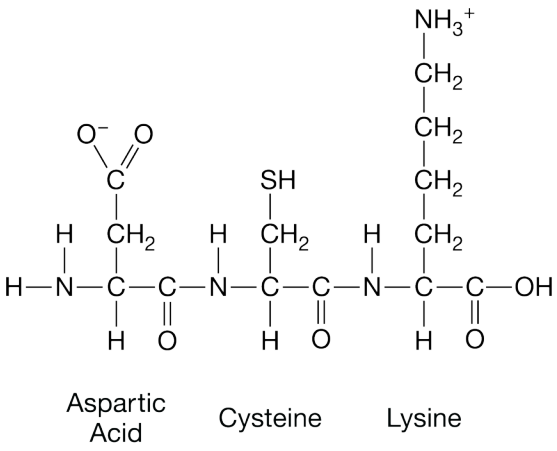


Figure 1: Original Amino Acid Chain

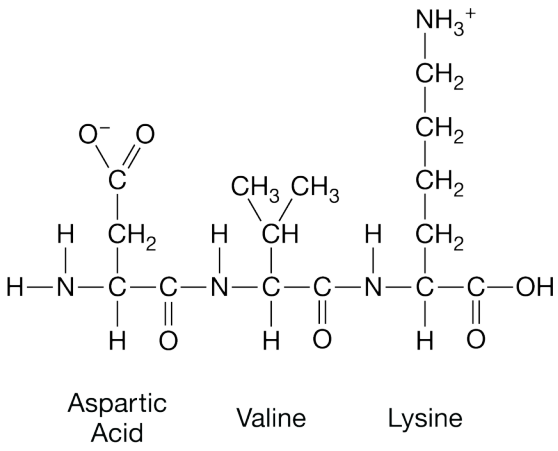


Figure 2: Mutated Amino Acid Chain

How might this change affect the structure and function of the protein?

(A) The R-group of the new amino acid, valine, has different chemical properties than the R-group of cysteine. This will cause the protein to misfold and not function properly in the cell. ✓

(B) The new amino acid, valine, has replaced cysteine in the new protein. Since the number of amino acids has remained the same, there will be no change in the three-dimensional folding, or function, of the protein.

(C) Since this is a linear section, it does not influence protein folding. Thus, there will be no change in protein structure or function.

(D) Since the new amino acid is bounded on one side by an amino acid with a negatively charged R-group and by an amino acid on the other side with a positively charged R-group, the charges will balance and the protein will fold as usual.

Answer A

Correct. Three-dimensional folding of a protein is due to interactions among the R groups of the amino acids. Cysteine has a sulfhydryl group, which may form a disulfide bridge with another part of the polypeptide chain. Valine has no sulfhydryl group and is nonpolar, which will affect how the polypeptide will fold.

1 Chemistry of Life MCQ

Students investigated the effect of light on the carbon cycle in aquatic ecosystems by performing the controlled experiment summarized below. The students placed equal amounts of water (pH 7.0) from a large aquarium in glass beakers. The students transferred aquatic plants from the aquarium to several of the beakers, and then they placed equal numbers of the beakers in the light or the dark (Figure 1: groups I and II). Similarly, the students transferred goldfish from the same aquarium to other beakers, and then they placed equal numbers of those beakers in the light or dark (Figure 1: groups III and IV). Finally, the students placed an equal number of beakers containing water only in the light or dark (Figure 1: groups V and VI).

After exposing the samples to light or dark for one hour, the students recorded the pH of the water in each beaker. Carbon dioxide dissolved in water will lower the pH of an aqueous solution. In the experiment, the students used changes in pH to monitor changes in the amount of carbon dioxide in the water. For each treatment group, the students calculated the mean pH and standard error, as documented in the table below.

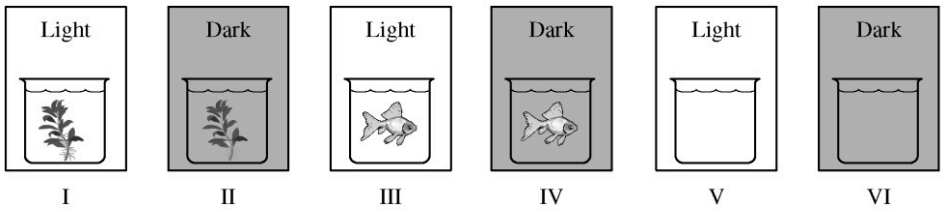


Figure 1. Treatment groups

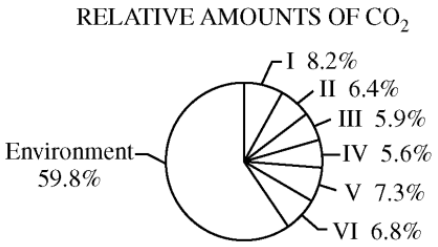
MEAN pH OF TREATMENT GROUPS AFTER 1 HOUR

Treatment group (n = 10)	I	II	III	IV	V	VI
Mean pH	8.2	6.4	5.9	5.6	7.3	6.8
Standard error of the mean	0.3	0.3	0.2	0.3	0.5	0.4

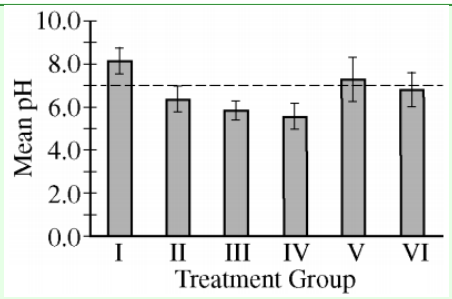
8. Which of the following graphs is the most appropriate representation of the experimental results documented in the table?

1 Chemistry of Life MCQ

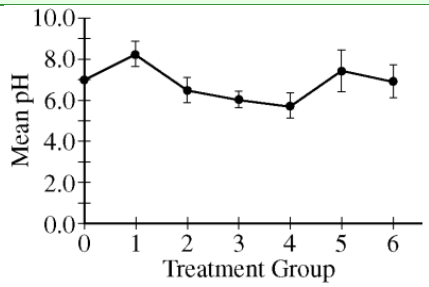
(A)



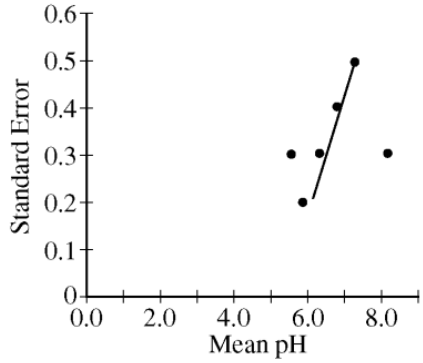
(B)



(C)



(D)



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9. Bacteriophages are viruses that infect bacteria. In an experiment, bacteriophages were labeled with either radioactive phosphorus or radioactive sulfur. The labeled bacteriophages were incubated with bacteria for a brief amount of time and then removed. The infected bacteria cells were found to contain significant amounts of radioactive phosphorus but not radioactive sulfur.

Based on the results of the experiment, which of the following types of molecules did the bacteriophages most likely inject into the bacteria cells?

1 Chemistry of Life MCQ

(A) Simple carbohydrate

(B) Amino acid

(C) DNA ✓

(D) Polypeptide

10.

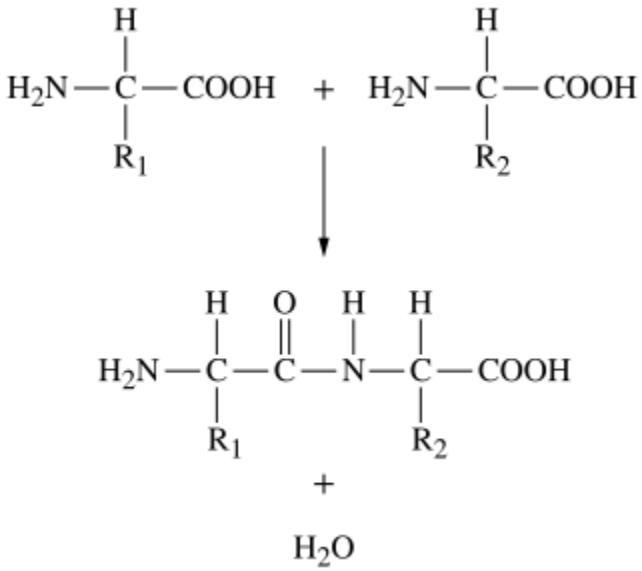


Figure 1. A representation of a reaction involved in the formation of a biological molecule

Which of the following best explains the reaction represented in Figure 1 ?

(A) It is a dehydration synthesis reaction involved in the formation of a lipid.

(B) The reaction is the hydrolysis of a nucleic acid and results in the formation of a nucleotide.

(C) The reaction is the hydrolysis of a nucleic acid and results in the formation of a phospholipid.

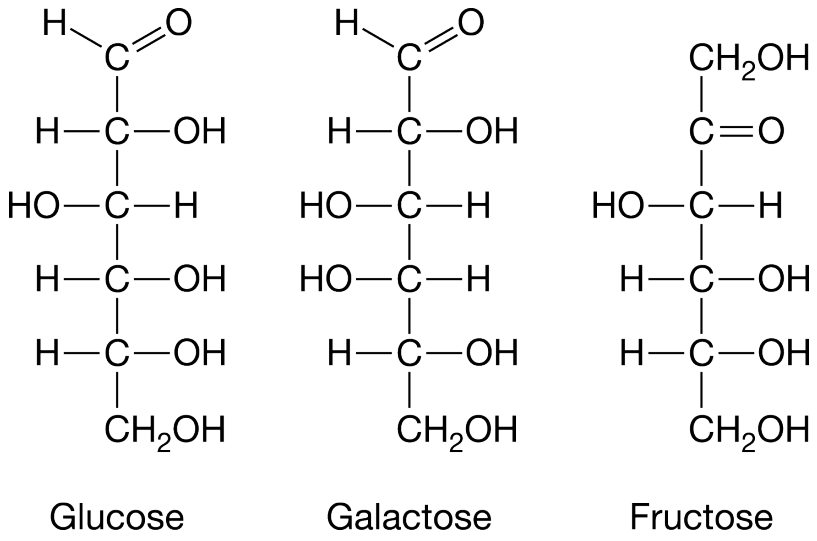
(D) It is a dehydration synthesis reaction involved in the formation of a polypeptide. ✓

Answer D

Correct. Two amino acids undergo a dehydration synthesis reaction, resulting in the formation of a peptide bond between them, the process involved in the growth of a polypeptide.

1 Chemistry of Life MCQ

11. The carbohydrates glucose, galactose, and fructose have the same chemical formula $(C_6H_{12}O_6)$ but different structural formulas, as represented in the figure.



Which of the following statements about glucose, galactose, and fructose is most likely true?

(A) The carbohydrates have the same properties because they have the same number of carbon, hydrogen, and oxygen atoms.

(B) The carbohydrates have the same properties because they each have a single carbon-oxygen double bond.

(C) The carbohydrates have different properties because they have different arrangements of carbon, hydrogen, and oxygen atoms. ✓

(D) The carbohydrates have different properties because they have different numbers of carbon-carbon bonds.

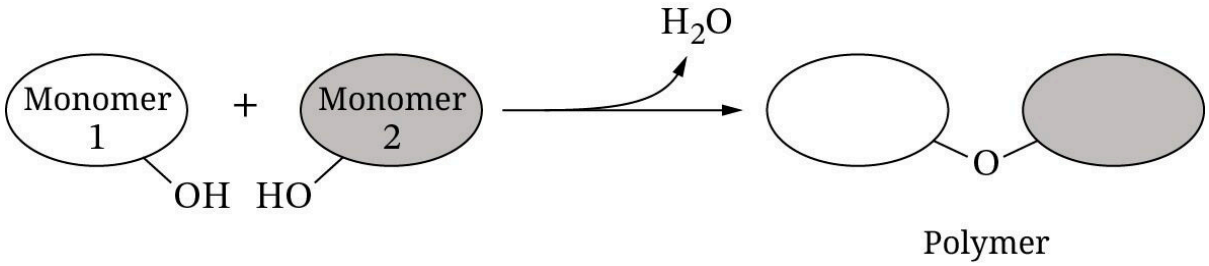
Answer C

Correct. Compounds that have the same chemical formula but different structural formulas usually have different properties.

1 Chemistry of Life MCQ

12. Figure 1 shows the formation of a carbohydrate polymer.

Figure 1. Formation of a carbohydrate polymer



Which of the following best describes how the carbohydrate polymer is being built in Figure 1?

(A) A water molecule is added to form a hydrogen bond between two monomers.

(B) A water molecule is added to form a covalent bond between two monomers

(C) A water molecule is removed to form a hydrogen bond between two monomers.

(D) A water molecule is removed to form a covalent bond between two monomers. ✓

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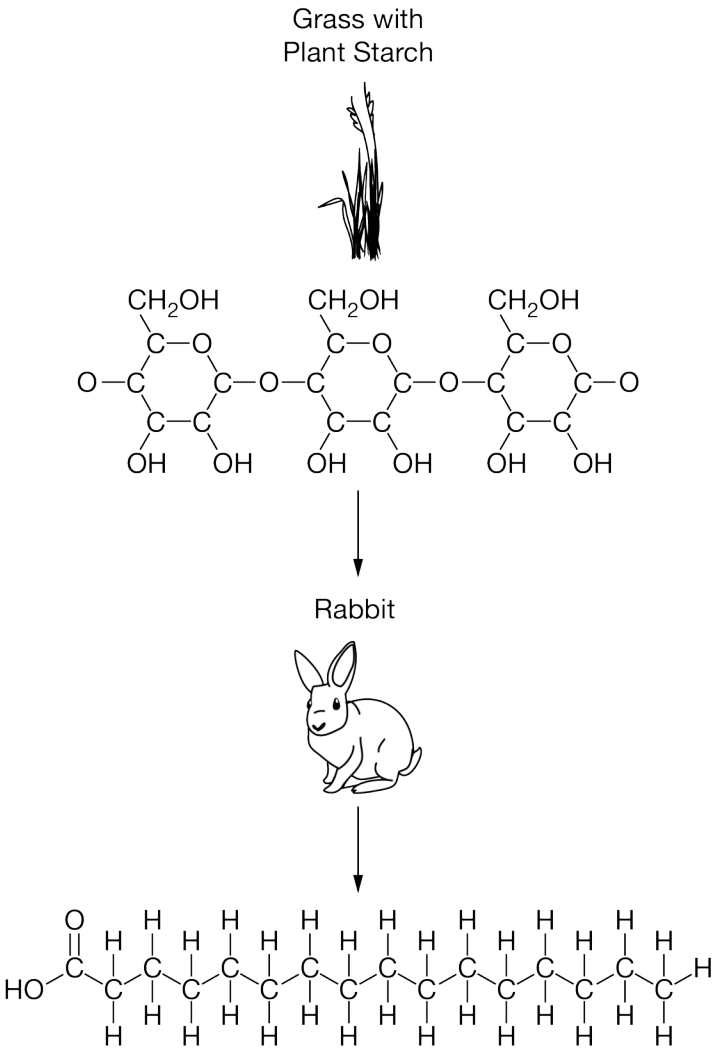
13.



Which of the following statements best describes how organisms such as rabbits obtain the carbon necessary for building biological molecules?

(A) Rabbits eat plants and use energy absorbed from the plants to make carbon atoms from electrons, protons, and neutrons in the air.

(B) Rabbits eat plants and break down plant molecules to obtain carbon and other atoms that they rearrange into new carbon-containing molecules. ✓

(C) Rabbits eat plants and use water absorbed from the plants to hydrolyze CO_2, which the rabbits breathe in from the air and use as a carbon source.

(D) Rabbits eat plants and make carbon-containing molecules by using carbon atoms that the plants absorbed from the soil and stored in the cells of their leaves.

1 Chemistry of Life MCQ

Answer B

Correct. Rabbits break down the plant molecules to obtain the individual atoms or small molecules that can be built into larger, complex biological molecules.

Different polysaccharides are used by plants for energy storage and structural support. The molecular structures for two common polysaccharides are shown in Figure 1. Starch is used by plants for energy storage, and cellulose provides structural support for cell walls. The monomer used to construct both molecules is glucose.

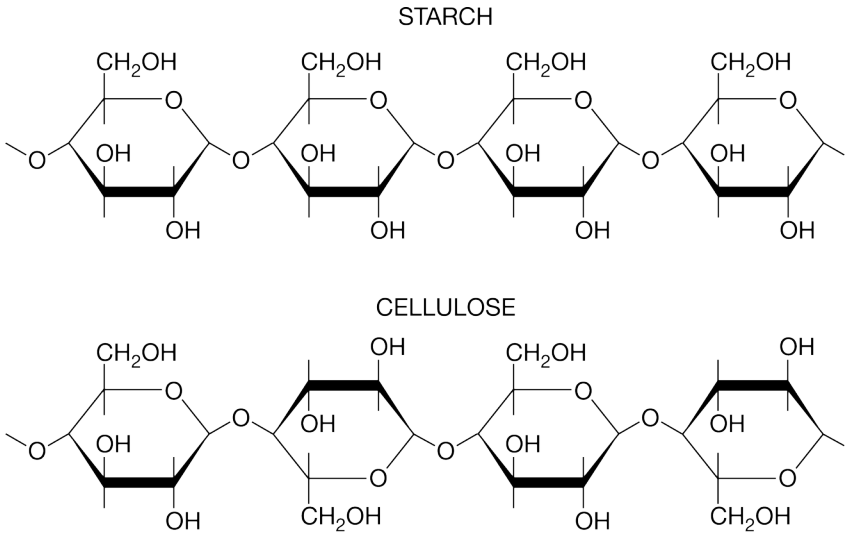


Figure 1. Comparison of segments of starch and cellulose

A study determined the effect of two different digestive enzymes, A and B, on these two polysaccharides. Table 1 presents the data from the study.

Table 1. Effect of Enzymes A and B on Cellulose and Starch

Test Tube	Polysaccharide Added	Enzyme Added	Glucose Detected after 5 Minutes at 37°C
1	Cellulose	A	No
2	Cellulose	B	Yes
3	Starch	A	Yes
4	Starch	B	No

Mammals do not produce digestive enzyme B. However, sheep and cattle are two types of mammals that contain microorganisms in their digestive tract that produce enzyme B.

1 Chemistry of Life MCQ

14. Which of the following would most likely occur if cattle lost the ability to maintain a colony of microorganisms in their digestive tract?

(A) Cattle would no longer be able to synthesize cellulose.

(B) Cattle would have to convert cellulose to starch before digesting it.

(C) Cattle would have to start producing enzyme B without the help of the bacteria.

(D) Cattle would no longer be able to use cellulose as a primary source of glucose. ✓

Answer D

Correct. Without the enzyme B produced by microorganisms in their digestive tract, cellulose would pass through the digestive tract without being digested.

15. Which of the following best describes the process that adds a monosaccharide to an existing polysaccharide?

(A) The monosaccharide is completely broken down by a specific enzyme and then the atoms are reorganized and made into a polysaccharide.

(B) Ionic bonds are formed between adjacent carbon atoms of the monosaccharide and the polysaccharide by adding water $(\mathrm{H_2O})$ and a specific enzyme.

(C) A specific enzyme removes the hydrogen (H) from the monosaccharide and the hydroxide (OH) from the polysaccharide, creating a bond between the two and creating a water $(\mathrm{H_2O})$ molecule. ✓

(D) A specific enzyme removes two hydroxides (OH), one from the monosaccharide, and one from the polysaccharide, creating a bond between the two monosaccharides and creating a hydrogen peroxide ($\mathrm{H_2O_2}$) molecule.

Answer C

Correct. This is a description of dehydration synthesis, which joins multiple monosaccharides to create a polysaccharide and produces water $(\mathrm{H_2O})$ molecules.

16. Based on Figure 1, which of the following best compares the atomic structures of starch and cellulose?

(A) Starch is composed of carbon, hydrogen, and oxygen, while cellulose also contains nitrogen.

(B) Starch and cellulose are composed of repeating glucose monomers; however, in cellulose every other glucose monomer is rotated 180 degrees. ✓

(C) Starch is composed of monomers that each have a $\mathrm{CH_2OH}$ group, while cellulose only has a $\mathrm{CH_2OH}$ group on every other monomer.

(D) Starch and cellulose are composed of identical monomers and therefore have identical structures.

1 Chemistry of Life MCQ

Answer B

Correct. Both starch and cellulose are both composed of repeating glucose molecules; however, the orientation of every other glucose in cellulose is upside down compared with the ones on either side.

17. Which of the following statements best describes the different functions of starch and cellulose in plants?

(A) The differences in the assembly and organization of the monomers of these two polymers result in different chemical properties. ✓

(B) Since starch and cellulose are composed of identical monomers, the cellular environment where they are located controls their function.

(C) The monomers of cellulose are connected by covalent bonds, making it idea for structural support.

(D) The monomers of starch are connected by ionic bonds, making it ideal for energy storage for plants.

Answer A

Correct. The identical orientations of the glucose monomers in starch create a polysaccharide with alpha bonds that is easy to break down into glucose for energy use. The alternating orientations of the glucose monomers in cellulose create beta bonds that produce a rigid polymer that is difficult to digest for energy use.

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18. Cholinesterase is an enzyme that catalyzes the hydrolysis of the signaling molecule acetylcholine. Which of the following best explains how cholinesterase carries out its function?

(A) Cholinesterase breaks covalent bonds within acetylcholine by the addition of a water molecule. ✓

(B) Cholinesterase breaks covalent bonds within acetylcholine by removing a water molecule.

(C) Cholinesterase forms covalent bonds between acetylcholine and another molecule by the addition of a water molecule.

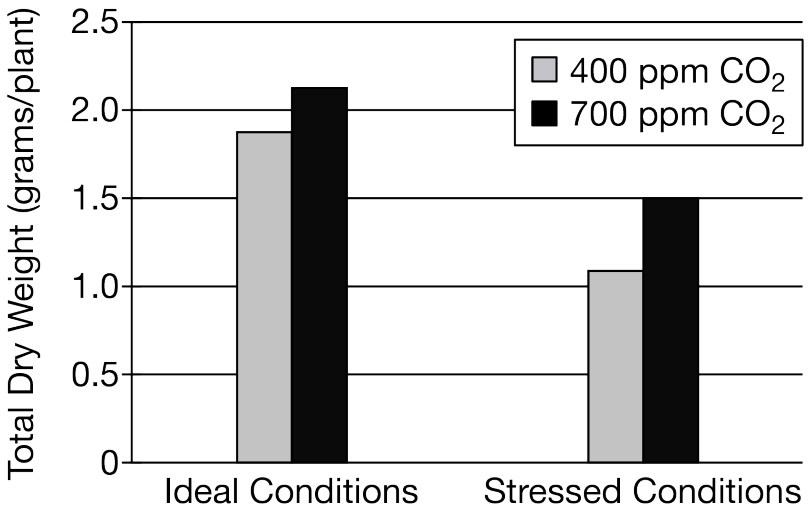
(D) Cholinesterase forms covalent bonds between acetylcholine and another molecule by removing a water molecule.

Answer A

Correct. Hydrolase enzymes catalyze the cleavage of covalent bonds by adding a water molecule to the bond.

1 Chemistry of Life MCQ

19. The figure shows the results of an experiment to investigate the effects of an enriched CO_2 environment on plant growth. Identical plants were separated into different groups and grown either in a standard CO_2 environment (400 ppm CO_2) or in an enriched CO_2 environment (700 ppm CO_2). Of the plants in each environment, half were grown under ideal conditions and half were grown under stressed conditions.



Based on the figure, which statement best describes the observed relationship between atmospheric CO_2 enrichment and plant growth under ideal and stressed conditions?

(A) The increase in atmospheric CO_2 had no observable effect on plant growth under either ideal or stressed conditions.

(B) The increase in atmospheric CO_2 resulted in a greater increase in plant growth under ideal conditions than under stressed conditions.

(C) The increase in atmospheric CO_2 resulted in a greater increase in plant growth under stressed conditions than under ideal conditions. ✓

(D) The increase in atmospheric CO_2 resulted in an inhibition of plant growth under both ideal and stressed conditions.

Answer C

Correct. Based on the figure, the increase in atmospheric CO_2 resulted in a greater increase in plant growth under stressed conditions than under ideal conditions.

20. Which of the following is responsible for the cohesive property of water?

1 Chemistry of Life MCQ

(A) Hydrogen bonds between the oxygen atoms of two adjacent water molecules

(B) Covalent bonds between the hydrogen atoms of two adjacent water molecules

(C) Hydrogen bonds between the oxygen atom of one water molecule and a hydrogen atom of another water molecule ✓

(D) Covalent bonds between the oxygen atom of one water molecule and a hydrogen atom of another water molecule

(E) Hydrogen bonds between water molecules and other types of molecules

21. Which of the following communities is likely to be most stable?

(A) One where a keystone predator has been removed

(B) One with uniformly spaced vegetation

(C) One with high species diversity ✓

(D) One that lacks decomposers

22. Researchers compared similar proteins from related organisms in different habitats. They found that the proteins from organisms living in harsh environments had a greater number of cysteine amino acids than did proteins from organisms not living in harsh environments. The structure of cysteine is shown. Bonds can form between the sulfur atom of different cysteine amino acids (S-S bonds).

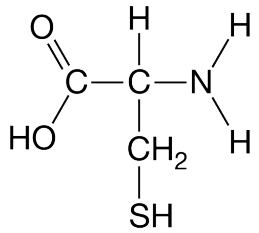


Figure 1. Chemical structure of cysteine

Which of the following best describes the effect of a greater number of cysteine amino acids on the stability of the proteins?

(A) The change has no effect on the stability of the protein because only one type of amino acid is involved.

(B) The change leads to increased protein stability because of an increased number of S-S bonds in the tertiary structure of the proteins. ✓

(C) The change leads to decreased protein stability because of an increased number of S-S bonds in the tertiary structure of the proteins.

(D) The change leads to increased protein stability only when the added cysteine amino acids are next to other cysteine amino acids in the primary structure.

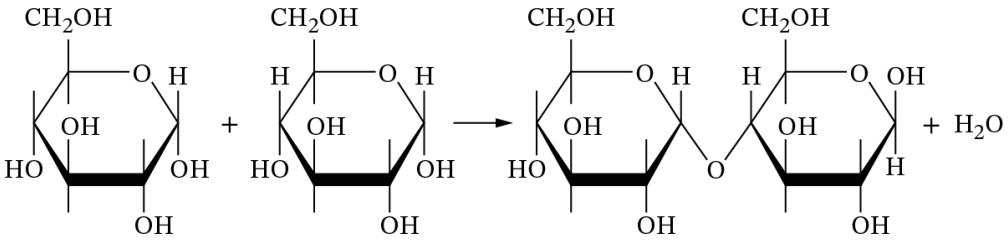
1 Chemistry of Life MCQ

Answer B

Correct. An increased number of S-S bonds are possible with the addition of more cysteine in the proteins. The S-S covalent bonds should add more structural stability to the proteins.

23. Figure 1 shows the formation of maltose (a disaccharide) from two glucose molecules.

Figure 1. Diagram of the formation of maltose



Which of the following statements best describes the molecular changes that occur during the reaction shown in Figure 1?

(A) Two hydroxyl groups are removed from each glucose molecule, and one water molecule is formed.

(B) A water molecule is added to the hydroxyl groups to form the covalent bond between the two glucose molecules.

(C) A hydroxyl group from one glucose molecule and a hydrogen atom from the other glucose molecule are removed, and one water molecule is formed. ✓

(D) A hydrogen atom in one glucose molecule and a hydroxyl group in the other glucose molecule form a hydrogen bond.

1 Chemistry of Life MCQ

24. A polypeptide is polymer of amino acids held together by peptide bonds. The process of dehydration synthesis creates these peptide bonds, as shown in Figure 1.

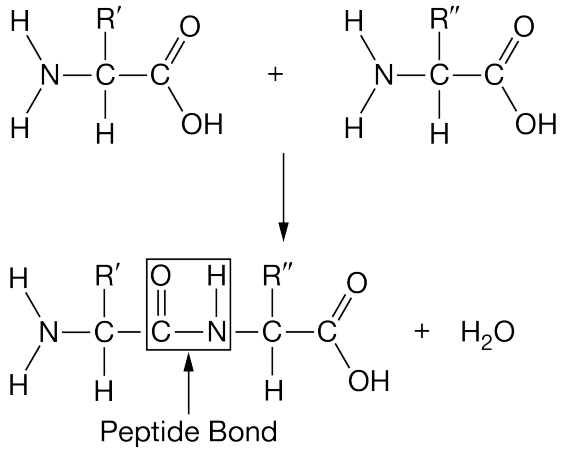


Figure 1. Amino acids are linked through the formation of peptide bonds.

As shown in Figure 1, an amino acid must have which of the following properties in order to be incorporated into a polypeptide?

(A) The ability to remain stable in the presence of water molecules

(B) An R-group that is compatible with the R-group of the last amino acid incorporated

(C) A central carbon atom that reacts with a nitrogen atom to form the peptide bond

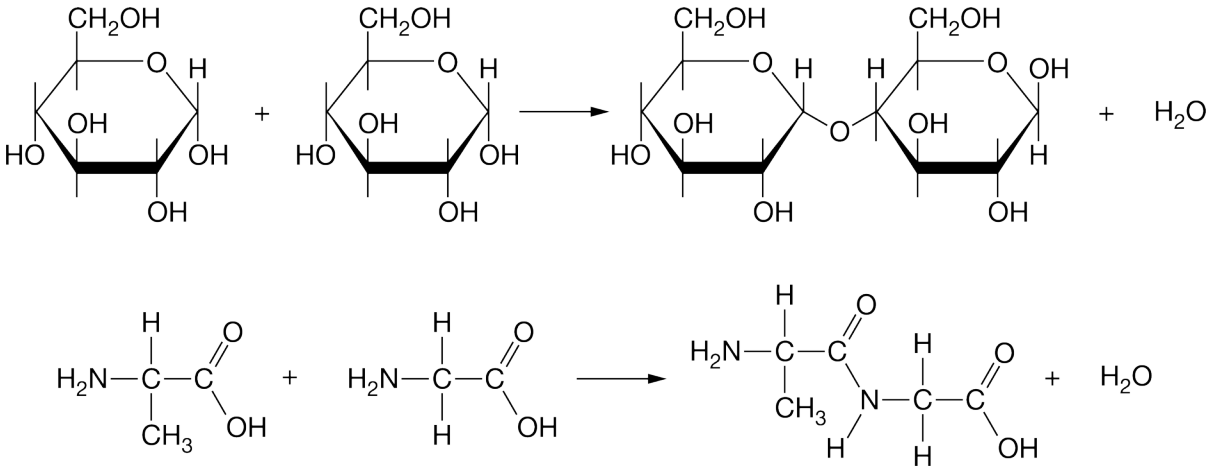
(D) The ability to form a covalent bond with both its NH_2 group and its COOH group ✓

Answer D

Correct. An amino acid needs to participate in the formation of two peptide bonds in order to be part of a polypeptide. The first bond may occur when the NH_2 group interacts with the COOH group of another amino acid. The second bond occurs when the COOH group interacts with the NH_2 group of a third amino acid.

1 Chemistry of Life MCQ

25.



Which of the following is common feature of the illustrated reactions showing the linking of monomers to form macromolecules?

(A) Two identical monomers are joined by a covalent bond.

(B) Two different monomers are joined by a covalent bond.

(C) Monomers are joined by a covalent bond, and a water molecule is produced. ✓

(D) Monomers are joined by ionic bonds, and a water molecule is produced.

Answer C

Correct. The monomers of the two reactions illustrated are joined by covalent bonds with the production of a water molecule.

26. Which of the following correctly illustrates a dipeptide and an amino acid in the optimal position to form a tripeptide?

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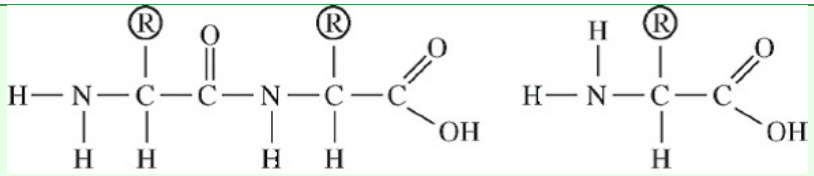
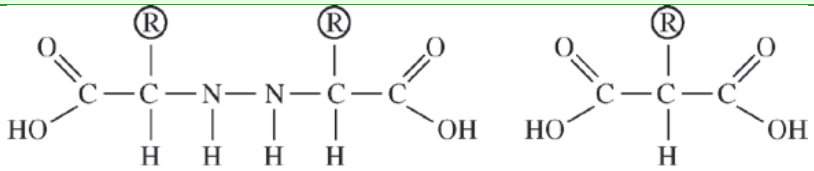
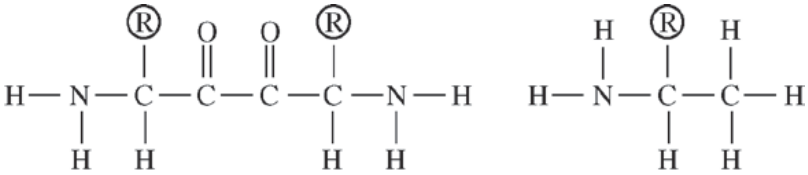
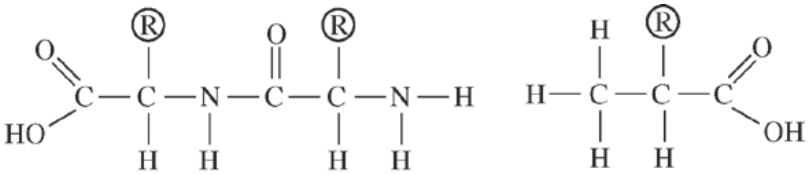
(A) 

(B) 

(C) 

(D) 

Answer A

This option is correct. It demonstrates an ability to refine and/or interpret a representation to explain the synthesis of a biological polymer. Peptides or proteins are polymers of amino acid monomers arranged in a unique linear sequence. Each of the 20 amino acids consists of a carbon atom surrounded by an amine group (NH_2), a carboxyl group (COOH), a hydrogen, and an R (variable) group. When two amino acids are positioned so that the carboxyl group of one amino acid backbone is adjacent to the amine group of another amino acid backbone, they can join by a dehydration reaction.

1 Chemistry of Life MCQ

27.

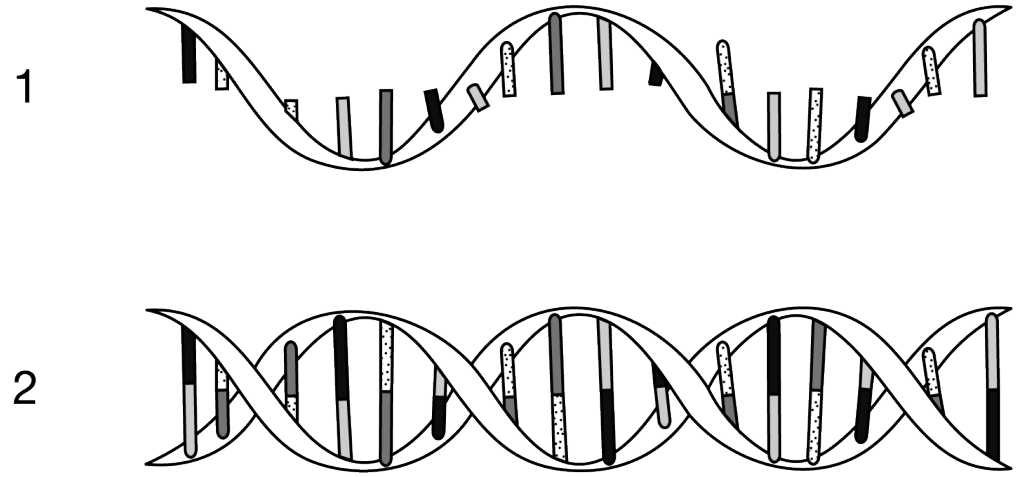


Figure 1. Molecule 1 represents RNA, and molecule 2 represents DNA.

Which of the following best describes a structural similarity between the two molecules shown in Figure 1 that is relevant to their function?

(A) Both molecules are composed of the same four nucleotides, which allows each molecule to be produced from the same pool of available nucleotides.

(B) Both molecules are composed of the same type of five-carbon sugar, which allows each molecule to act as a building block for the production of polysaccharides.

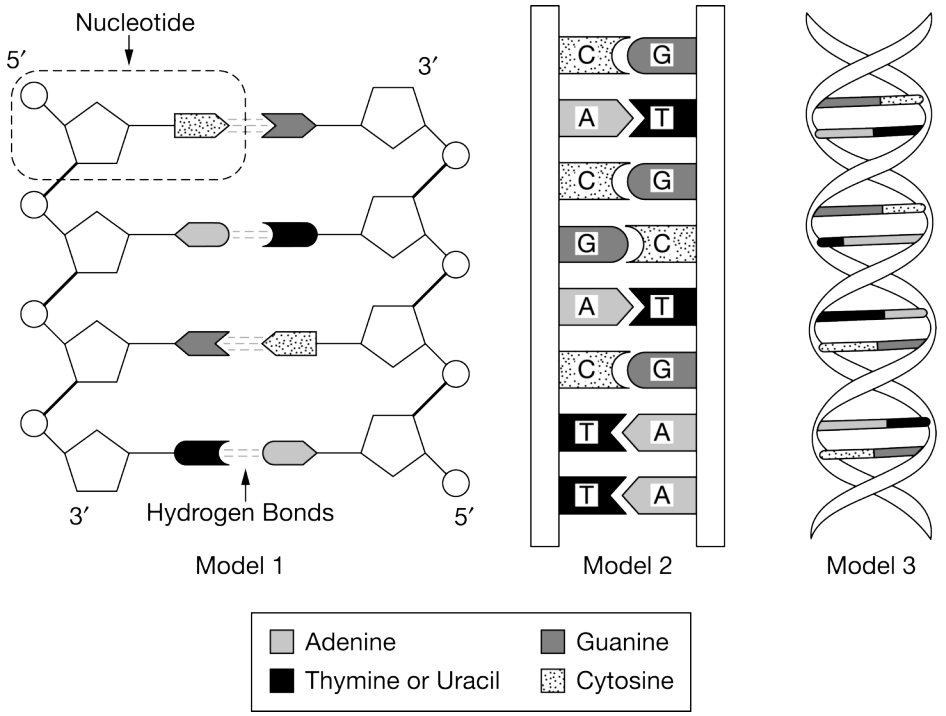
(C) Both molecules contain nucleotides that form base pairs with other nucleotides, which allows each molecule to act as a template in the synthesis of other nucleic acid molecules. ✓

(D) Both molecules contain nitrogenous bases and phosphate groups, which allows each molecule to be used as a monomer in the synthesis of proteins and lipids.

Answer C

Correct. Nucleotides form base pairs with other nucleotides. The base pairing allows a strand of RNA or DNA to act as a template in the synthesis of other nucleic acid molecules. Examples include the cellular processes of DNA replication (DNA is used as a template to make DNA), transcription (DNA is used as a template to make RNA), and reverse transcription (RNA is used as a template to make DNA).

1 Chemistry of Life MCQ



28.

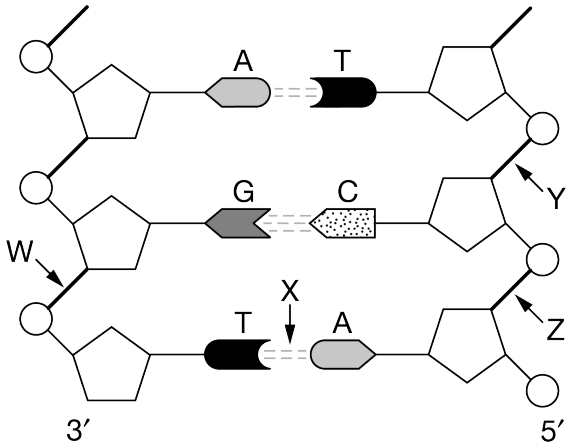


Figure 1. Four different bonds (W, X, Y, and Z) in a DNA molecule

Figure 1 represents a segment of DNA. Radiation can damage the nucleotides in a DNA molecule. To repair some types of damage, a single nucleotide can be removed from a DNA molecule and replaced with an undamaged nucleotide. Which of the four labeled bonds in Figure 1 could be broken to remove and replace the cytosine nucleotide without affecting the biological information coded in the DNA molecule?

(A) Bond X only

(B) Bond W only

(C) Bonds Y and Z at the same time ✓

(D) Bonds W and Z at the same time

1 Chemistry of Life MCQ

Answer C

Correct. By breaking bonds Y and Z at the same time, the cytosine nucleotide could be removed from the DNA molecule and replaced with an undamaged cytosine without changing the biological information stored in the DNA.

29. A student wants to modify model 1 so that it represents an RNA double helix instead of a DNA double helix. Of the following possible changes, which would be most effective in making model 1 look more like RNA than DNA ?

(A) Changing the sequence of the base pairs

(B) Changing the deoxyriboses to riboses by adding $-OH$ groups ✓

(C) Changing the shapes of the nitrogenous bases to match those shown in model 2

(D) Changing the sugar-phosphate backbone to a ribbon, as shown in model 3

Answer B

Correct. RNA contains ribose, whereas DNA contains deoxyribose. A ribose sugar has an $-OH$ group linked to the $2'$ carbon that a deoxyribose sugar does not have.

30. Which feature of model 1 best illustrates how biological information is coded in a DNA molecule?

(A) The $5'$ and $3'$ labels at the ends of each strand

(B) The labeling of the hydrogen bonds between base pairs

(C) The lines connecting sugars and phosphate groups that represent covalent bonds

(D) The linear sequence of the base pairs ✓

Answer D

Correct. The sequence of base pairs in a DNA molecule plays a central role in the coding of biological information.

1 Chemistry of Life MCQ

31. Figure 1 shows a short segment of a double-stranded nucleic acid molecule.

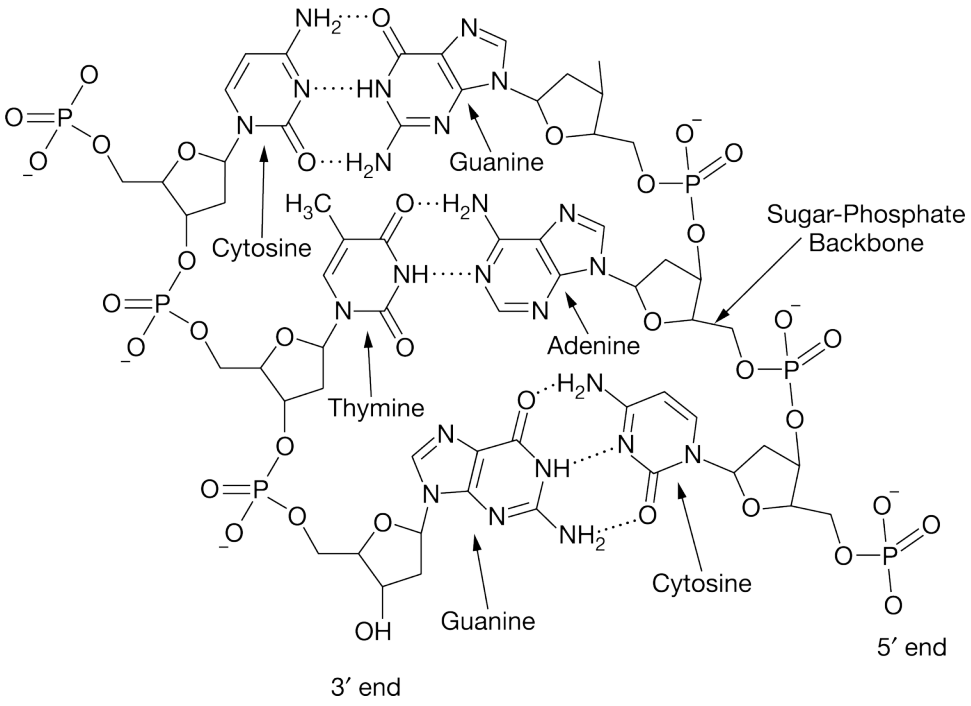


Figure 1. A short segment of a double-stranded nucleic acid molecule

Which of the following statements is correct about the molecule shown in Figure 1 ?

(A) It is RNA because of the relative direction of the two strands.

(B) It is RNA because of the number of different nucleotides found in the molecule.

(C) It is DNA because of the nature of the hydrogen bonds between guanine and cytosine.

(D) It is DNA because of the nucleotides present. ✓

Answer D

Correct. The figure indicates that one of the nucleotides is thymine. Thymine is found in DNA and not in RNA.

1 Chemistry of Life MCQ

32. Figure 1 represents a nucleic acid fragment that is made up of four nucleotides linked together in a chain.

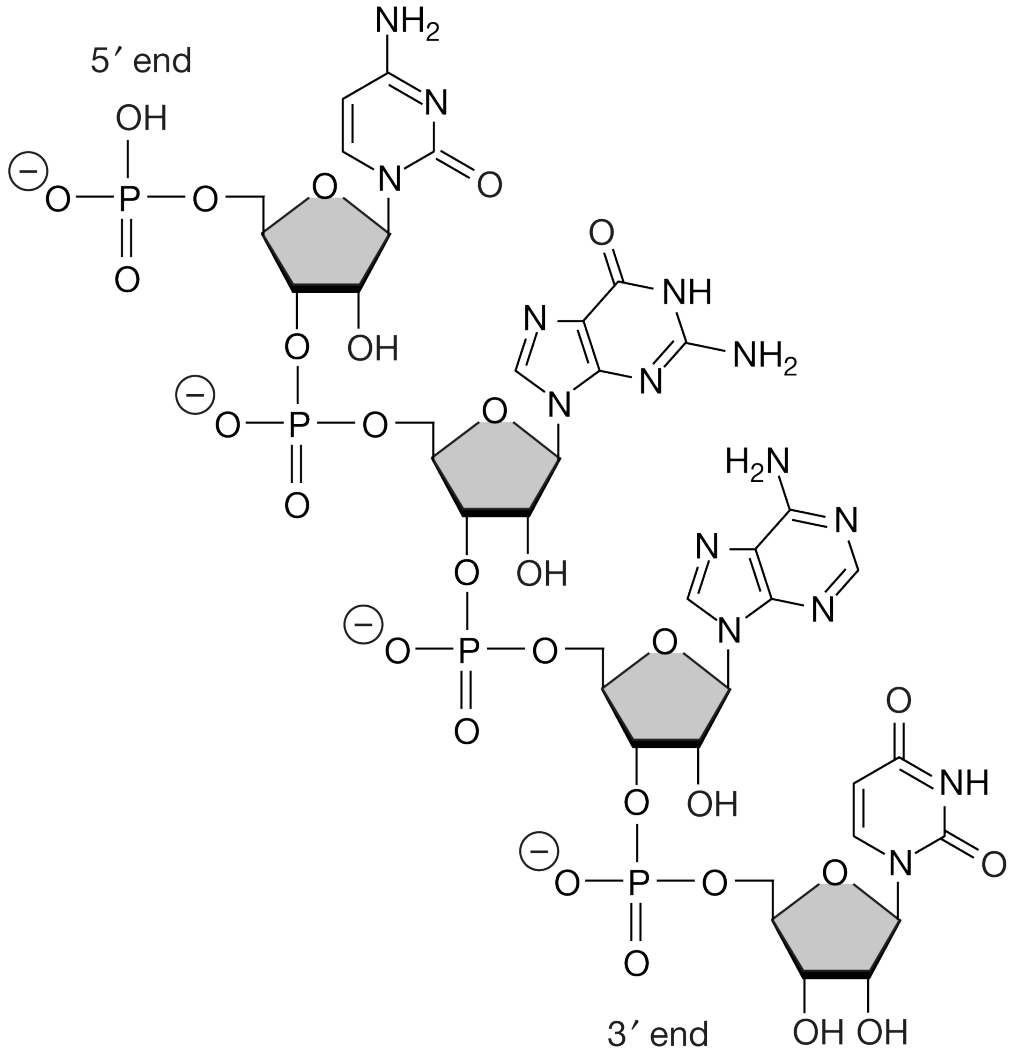


Figure 1. Nucleic acid fragment

Which of the following characteristics of Figure 1 best shows that the fragment is RNA and not DNA?

(A) The 5′ to 3′ orientation of the nucleotide chain

(B) The identity of each nitrogenous base ✓

(C) The charges on the phosphate groups

(D) The type of bond linking the nucleotides together

1 Chemistry of Life MCQ

Answer B

Correct. RNA contains uracil, whereas DNA contains thymine.

33.

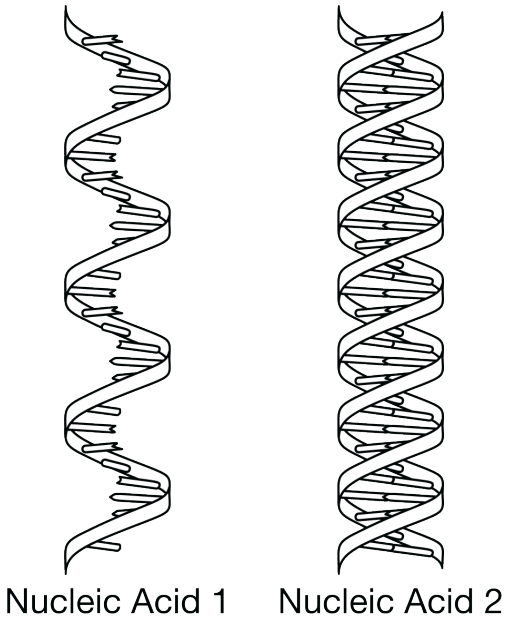


Figure 1. Nucleic acid segments

DNA and RNA are nucleic acids that can store biological information based on the sequence of their nucleotide monomers. Figure 1 shows a short segment of each of the two types of nucleic acids.

Which of the following best describes a structural difference between DNA and RNA?

(A) DNA contains four types of nitrogenous bases, whereas RNA contains only two types of nitrogenous bases.

(B) The backbone of DNA contains deoxyribose, whereas the backbone of RNA contains ribose. ✓

(C) A DNA molecule is composed of two parallel strands with the same $5'$ to $3'$ directionality, whereas an RNA molecule is composed of only one $5'$ to $3'$ strand.

(D) Phosphate groups provide rigidity to DNA, but RNA is flexible and contains no phosphate groups.

Answer B

Correct. The backbones of DNA and RNA are composed of an alternating sugar and base. In DNA, the sugar is deoxyribose, whereas the sugar in RNA is ribose.

1 Chemistry of Life MCQ

Figure 1. Domain structure of two proteins

Investigators studied the interactions between two different proteins, protein I and protein II. Each protein has two structural domains (Figure 1) that represent different functional parts of the protein. Different combinations of the protein domains were tested for their ability to bind to a known DNA sequence, interact with each other, and activate transcription. The results are shown in Table 1.

TABLE 1. INTERACTION AND FUNCTION OF COMBINATIONS OF STRUCTURAL VARIANTS OF PROTEINS I AND II

Test	Domain(s) of Protein I Used	Domain(s) of Protein II Used	DNA Binding	Protein-Protein Interaction	Transcription Activation
1	FG	LM	+	+	+
2	FG	None	+	−	−
3	None	LM	−	−	−
4	F	LM	+	−	−
5	G	LM	−	+	−
6	FG	L	+	−	−
7	FG	M	+	+	+

34. Which of the following claims is best supported by the data from test 5 in Table 1 ?

(A) Domain L is required for transcription activation.

(B) Domain F is not required for protein-protein interaction. ✓

(C) Protein-protein interaction is necessary for DNA binding.

(D) Protein-protein interaction is sufficient for transcription activation.

1 Chemistry of Life MCQ

Answer B

Correct. Based on the data for test 5 in which domain F is omitted but protein-protein interaction occurs, the G domain but not the F domain of protein I interacts with protein II. This is also confirmed in test 4 in which the F domain but not the G domain of protein I is present and in which protein-protein interaction does not occur.

35. Based on the data, which structural domain is most likely required for DNA binding?

(A) Domain F ✓
(B) Domain G
(C) Domain L
(D) Domain M

Answer A

Correct. Based on the data in Table 1, DNA binding occurs anytime both domains of protein I are present. The data for test 4 indicate that DNA binding also occurs when only domain F of protein I is present. The data for test 5 confirm that the presence of domain G is insufficient to allow DNA binding. Therefore, domain F is responsible for DNA binding by protein I.

36. Which of the following claims is best supported by the data in Table 1 ?

(A) Protein-protein interaction always results in transcription activation.
(B) DNA binding is required for protein-protein interaction.
(C) Protein-protein interaction is necessary for, but not adequate for, transcriptional activation. ✓
(D) DNA binding always results in transcription activation.

Answer C

Correct. Based on the data for tests 1 and 7, transcription activation occurs when both DNA binding and protein-protein interaction occur. The data for test 5 demonstrate that protein-protein interaction alone is insufficient for transcription activation to occur.

1 Chemistry of Life MCQ

37. Which of the following is the most likely result of testing a combination of domains G and M?

(A) No functionality

(B) DNA binding only

(C) Transcriptional activation only

(D) Protein-protein interaction only ✓

Answer D

Correct. Based on the data from tests 4 and 5, domain G (and not domain F) enables protein-protein interaction. Based on the data from tests 7 and 8, domain M (and not domain L) enables protein-protein interaction. Therefore, protein-protein interaction is expected to occur when domains G and M are combined.

---

38. A chemical binds to a protein composed of a single polypeptide chain and prevents the formation of an alpha helix that is typically formed in the absence of the chemical. Which of the following best describes the effect the chemical has on the structure of the protein?

(A) The primary structure held together by covalent bonds is affected.

(B) The secondary structure held together by hydrogen bonds is affected. ✓

(C) The secondary, tertiary, and quaternary structures are affected.

(D) All levels of protein structure are affected.

Answer B

Correct. Alpha helices are a form of secondary protein structure and are held together by hydrogen bonds. This chemical prevents alpha helices from forming, and therefore, it is most likely that the chemical affects hydrogen bonding within the local region of amino acids.

1 Chemistry of Life MCQ

39.

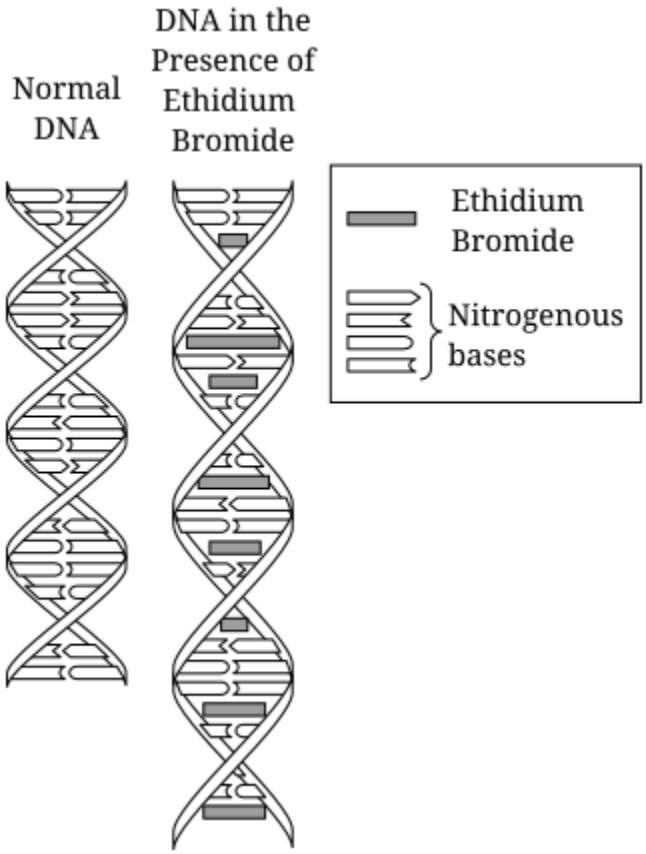


Figure 1. Model of DNA structure in the absence and presence of ethidium bromide

Ethidium bromide is a chemical that disrupts the replication of DNA. Based on the model shown in Figure 1, which of the following best describes the effect of ethidium bromide on DNA structure?

(A) It is inserted between complementary nitrogenous bases, causing the two strands of DNA to denature.

(B) It is inserted between complementary nitrogenous bases, causing hydrogen bonds to form between noncomplementary nitrogenous bases.

(C) It is inserted between sequential nitrogenous base pairs, increasing the length of the DNA molecule. ✓

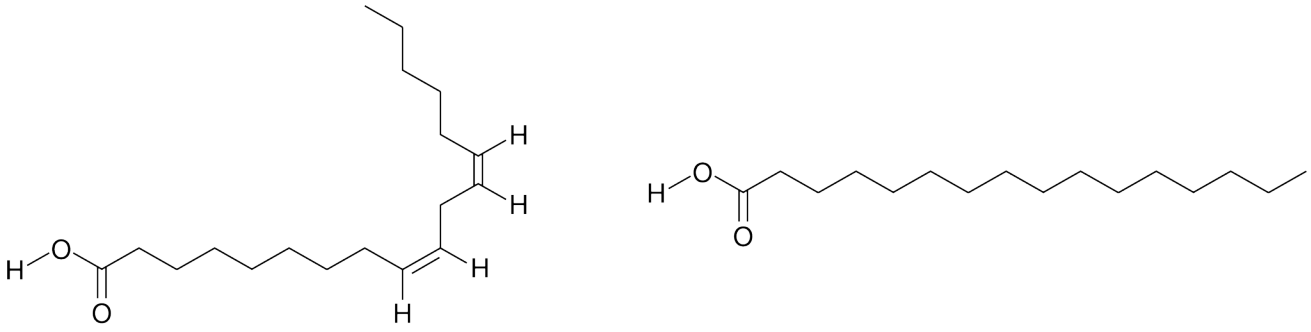
(D) It is inserted in place of certain nitrogenous base pairs, decreasing the length of the DNA molecule.

Answer C

Correct. Ethidium bromide is inserted between sequential base pairs, resulting in a longer DNA molecule.

1 Chemistry of Life MCQ

40. The molecular structures of linoleic acid and palmitic acid, two naturally occurring substances, are shown in the figure.



Linoleic Acid

Palmitic Acid

Based on the molecular structures shown in the figure, which molecule is likely to be solid at room temperature?

(A) Linoleic acid, because the absence of carbon-carbon double bonds allows the molecules to pack closely together.

(B) Linoleic acid, because the presence of carbon-carbon double bonds prevents the molecules from packing closely together.

(C) Palmitic acid, because the absence of carbon-carbon double bonds allows the molecules to pack closely together. ✓

(D) Palmitic acid, because the presence of carbon-carbon double bonds prevents the molecules from packing closely together.

Answer C

Correct. Palmitic acid is a saturated fatty acid. The absence of carbon-carbon double bonds in palmitic acid allows the molecules to pack closely together and form a solid at room temperature.

41. The fluid mosaic model of cell membranes describes proteins and steroids associated with a phospholipid bilayer. Which of the following membrane components affect membrane stability?

(A) Embedded proteins give plasma membranes less stability.

(B) Steroids, such as cholesterol, give membranes more stability. ✓

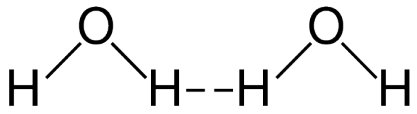
(C) Proteins in the hydrophilic region of the plasma membrane create more stability.

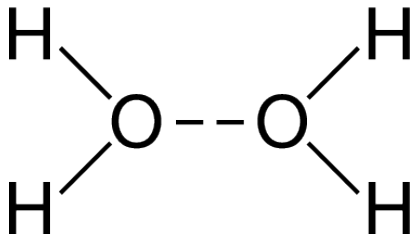
(D) Polar heads of phospholipids give plasma membranes less stability.

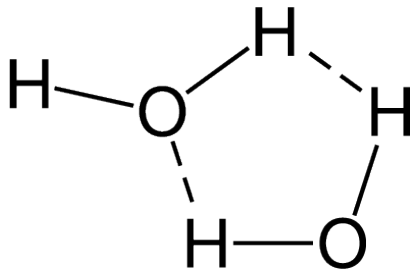
1 Chemistry of Life MCQ

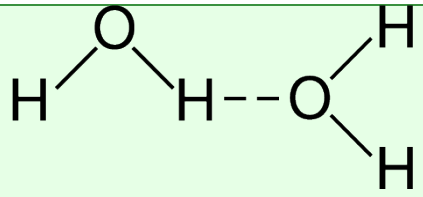
42. Water molecules are polar covalent molecules. There is a partial negative charge near the oxygen atom and partial positive charges near the hydrogen atoms due to the uneven distribution of electrons between the atoms, which results in the formation of hydrogen bonds between water molecules. The polarity of water molecules contributes to many properties of water that are important for biological processes.

Which of the following models best demonstrates the arrangement of hydrogen bonds between adjacent water molecules?

(A) 

(B) 

(C) 

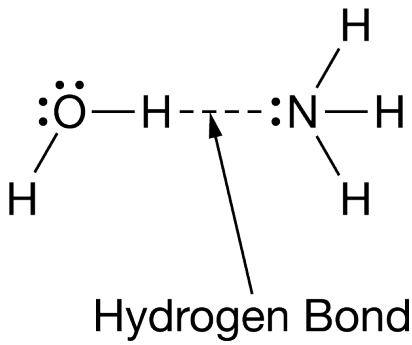
(D) 

Answer D

Correct. The hydrogen bonds between these water molecules correctly show the attractive force between the hydrogen atom of one water molecule and the oxygen atom of the adjacent water molecule.

1 Chemistry of Life MCQ

43. Water and ammonia interact to form hydrogen bonds, as represented in the figure.



Which statement best helps explain the formation of the hydrogen bond represented in the figure?

(A) The oxygen has a partial positive charge, and the nitrogen has a partial negative charge.

(B) The nitrogen has a partial negative charge, and the hydrogen attached to the oxygen has a partial positive charge. ✓

(C) The hydrogen attached to the oxygen has a partial negative charge, and the nitrogen also has a partial negative charge.

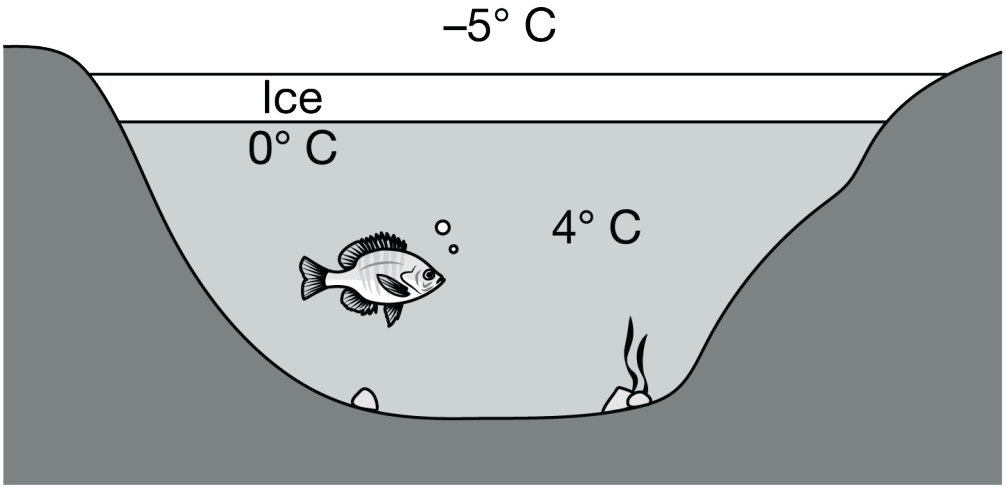
(D) The nitrogen has a partial positive charge, and the hydrogen attached to the oxygen also has a partial positive charge.

Answer B

Correct. The greater electronegativity of oxygen and nitrogen compared with hydrogen makes for unequal sharing of electrons, which results in partial negative charges associated with the oxygen and nitrogen atoms and partial positive charges associated with the hydrogen atoms in both molecules. The attraction between opposite charges results in the formation of the hydrogen bond represented in the figure.

1 Chemistry of Life MCQ

44.



As shown in the diagram, when environmental temperatures drop below freezing, a layer of ice typically forms on the surface of bodies of freshwater such as lakes and rivers.

Which of the following best describes how the structure of ice benefits the organisms that live in the water below?

(A) The water molecules in ice are closer together than those in liquid water, so the ice prevents the passage of air to the water, maintaining a constant gas mixture in the water.

(B) The water molecules in ice are closer together than those in liquid water, so the ice forms a barrier that protects the organisms in the water from the freezing air temperatures.

(C) The water molecules in ice are farther apart than those in liquid water, so the ice floats, maintaining the warmer, denser water at the lake bottom. ✓

(D) The water molecules in ice are farther apart than those in liquid water, so the ice floats, preventing the escape of gases from the liquid water.

Answer C

Correct. The water molecules in ice are farther from each other than are water molecules in liquid water, so ice is less dense than liquid water and floats on its surface, while the denser water at 4 degrees Celsius sinks to the bottom, maintaining a steady temperature all winter.

1 Chemistry of Life MCQ

45. The process of hydrogenation is used in the food industry to convert lipids that are liquids at room temperature into solids at room temperature. The process involves the chemical addition of hydrogen atoms to the fatty acid tails of lipids, breaking some or all of the carbon-carbon double bonds in the fatty acid tails.

Which of the following best explains why hydrogenation converts liquid lipids to solid lipids?

(A) Hydrogenation makes the fatty acids more saturated, enabling them to be more densely packed together. ✓

(B) Hydrogenation makes the fatty acids less saturated, enabling them to be more densely packed together.

(C) Hydrogenation makes the fatty acids more saturated, resulting in fewer lipid molecules fitting within a given space.

(D) Hydrogenation makes the fatty acids less saturated, resulting in fewer lipid molecules fitting within a given space.

Answer A

Correct. Adding hydrogen atoms results in fatty acid tails that are more saturated. The hydrocarbon chains of saturated fatty acids are straighter than those of unsaturated molecules and can therefore pack more closely together. More intermolecular forces result in higher melting points, converting the lipids to solids at room temperature.

46. One form of the genetic disorder lipodystrophy involves the lack of an enzyme that is required for the synthesis of triglycerides, a type of lipid. Triglycerides serve as an energy reserve in fat cells.

Which of the following best describes the most likely effect of lipodystrophy on cell function?

(A) There will be a decrease in phospholipids in cell membranes.

(B) There will be a decrease in cholesterol in cell membranes.

(C) There will be a decrease in fat storage in cells. ✓

(D) There will be a decrease in fatty acid synthesis in cells.

1 Chemistry of Life MCQ

47.

TABLE 1. PERCENT CHEMICAL COMPOSITION OF FOUR MACROMOLECULES

Macromolecule Sample	C (%)	O (%)	H (%)	N (%)	P (%)	S (%)
A	25	25	50	0	0	0
B	15	20	40	20	0	5
C	40	4	45	0	1	0
D	25	20	35	15	5	0

A researcher analyzed four different samples of macromolecules, where all macromolecules in each sample are of the same type. The researcher measured the percent of carbon, oxygen, hydrogen, nitrogen, phosphorus, and sulfur atoms in each sample. The results are shown in Table 1.

Which of the following claims is best supported by the data in Table 1 ?

(A) Sample A contains nucleic acids.

(B) Sample B contains protein. ✓

(C) Sample C contains nucleic acids.

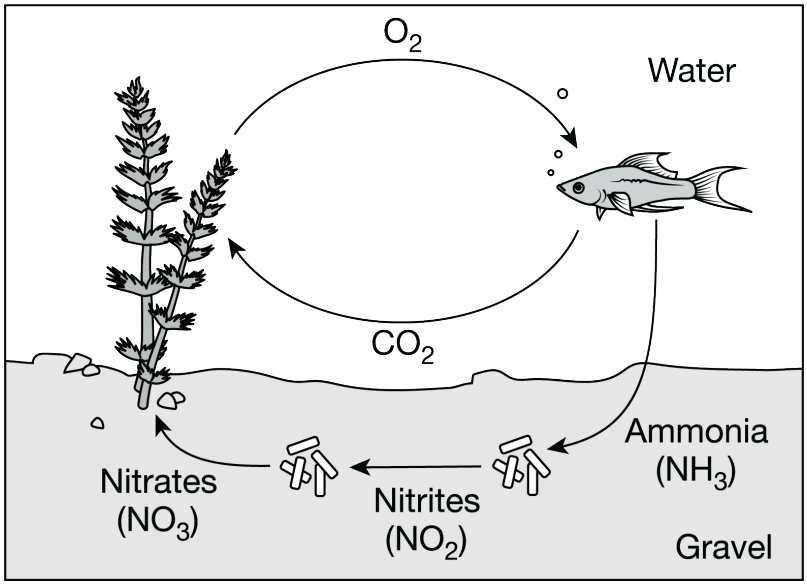
(D) Sample D contains protein.

Answer B

Correct. This sample is consistent with the atoms found in the largest quantities in proteins (C, H, O, N). A small amount of sulfur is also typically found in proteins. This sample does not have phosphorus, which would be indicative of a nucleic acid and not a protein.

1 Chemistry of Life MCQ

48. The figure shows a model of the exchange of matter between the organisms that live together in an aquarium. The model includes matter exchange between plants, fish, and bacteria. The bacteria are represented as rod-shaped organisms living in the gravel at the bottom of the aquarium.



Which of the following statements best describes how molecules released by the fish become nutrients for the plants?

(A) The carbon dioxide molecules released by the fish are converted by the bacteria to oxygen atoms, which are used by the plants to make water molecules.

(B) The oxygen molecules released by the fish are converted by the bacteria to ammonia molecules, which are used by the plants to make lipids and fatty acids.

(C) The nitrites released by the fish are converted by the bacteria to carbon dioxide molecules, which are used by the plants to make carbohydrates.

(D) The ammonia molecules released by the fish are converted by the bacteria to nitrates, which are used by the plants to make proteins and nucleic acids. ✓

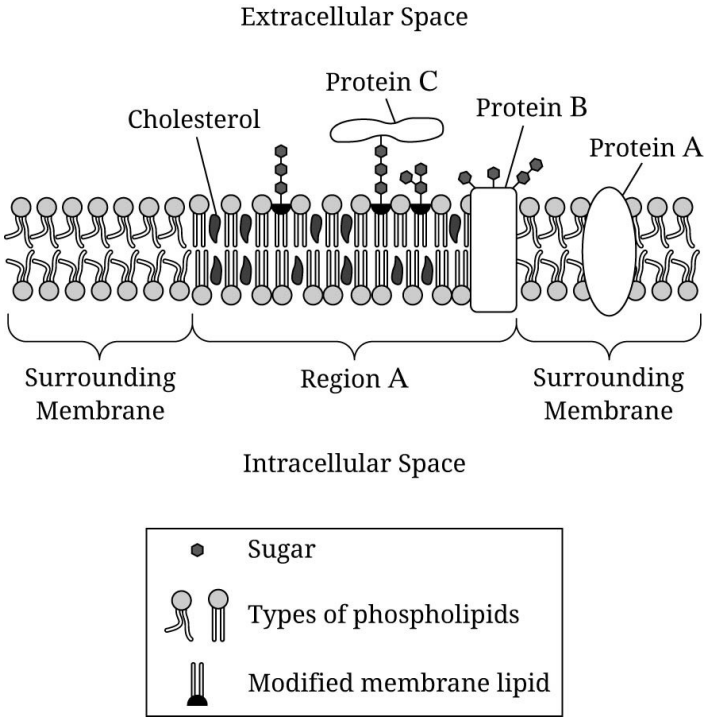
Answer D

Correct. As shown in the figure, the ammonia molecules released by the fish are converted by the bacteria to nitrites and nitrates. The plants absorb the nitrates, which are nitrogen-containing nutrients that the plants use to make the amino acids and nucleotides needed for building proteins and nucleic acids.

1 Chemistry of Life MCQ

49. Researchers have identified regions, including region A in Figure 1, of the cell membrane that have a different density and composition than the surrounding membrane.

Figure 1. Structure of a portion of the cell membrane



Which of the following best describes the chemical composition of the different types of macromolecules that make up the membrane represented in Figure 1?

(A) All the molecules contain carbon and oxygen, but only some contain nitrogen. ✓

(B) All the molecules contain carbon and hydrogen, but only some contain oxygen.

(C) All the molecules contain carbon and oxygen, but only some contain hydrogen.

(D) All the molecules contain carbon and nitrogen, but only some contain oxygen.

1 Chemistry of Life MCQ

50.

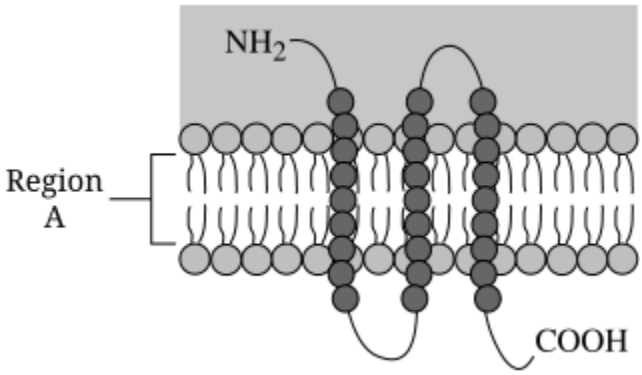


Figure 1. Structure of a membrane protein

Based on Figure 1, the amino acids in region A are most likely to have which of the following characteristics?

(A) Most amino acids will be hydrophobic because they interact most favorably with the phospholipid tails. ✓

(B) Most amino acids will be hydrophilic because they interact most favorably with the phospholipid heads.

(C) Most amino acids will be ionic amino acids because they interact most favorably with the phospholipid tails.

(D) Most amino acids will be polar amino acids because they interact most favorably with the phospholipid heads.

Answer A

Correct. Region A brackets the hydrophobic tails of the membrane phospholipids. In order for a protein to be embedded in the membrane among the phospholipid tails, the amino acids of the portion of the protein in region A must also be hydrophobic.

51. Which of the following statements best describes the structure of complex carbohydrates?

(A) Several polysaccharide chains are linked via hydrogen bonds.

(B) Monosaccharides are connected by covalent bonds. ✓

(C) Several polysaccharide chains are linked via ionic bonds.

(D) Monosaccharides are joined by peptide bonds.

1 Chemistry of Life MCQ

52. A small protein is composed of 110 amino acids linked together in a chain. As shown in Figure 1, the first and last five amino acids in the chain are hydrophobic (have nonpolar and uncharged R-groups), whereas the remaining 100 amino acids are hydrophilic (have charged or polar R-groups). The nature of the R-group determines if the amino acid is hydrophobic or hydrophilic.

A mutation results in the production of a version of the small protein that is only 105 amino acids long, as shown in Figure 2. Five of the hydrophobic amino acids are missing from one end of the chain.

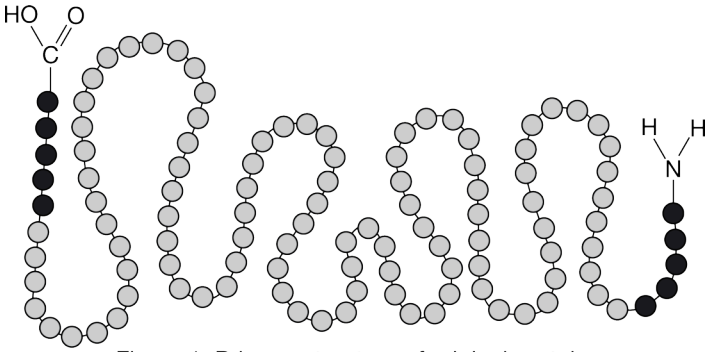
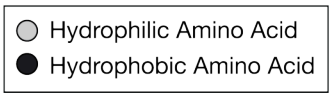


Figure 1. Primary structure of original protein



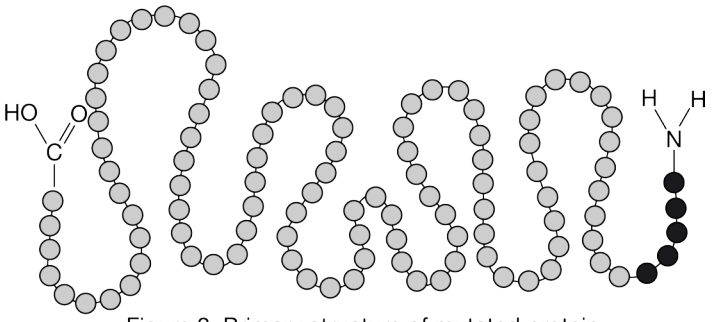
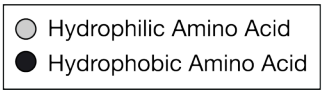
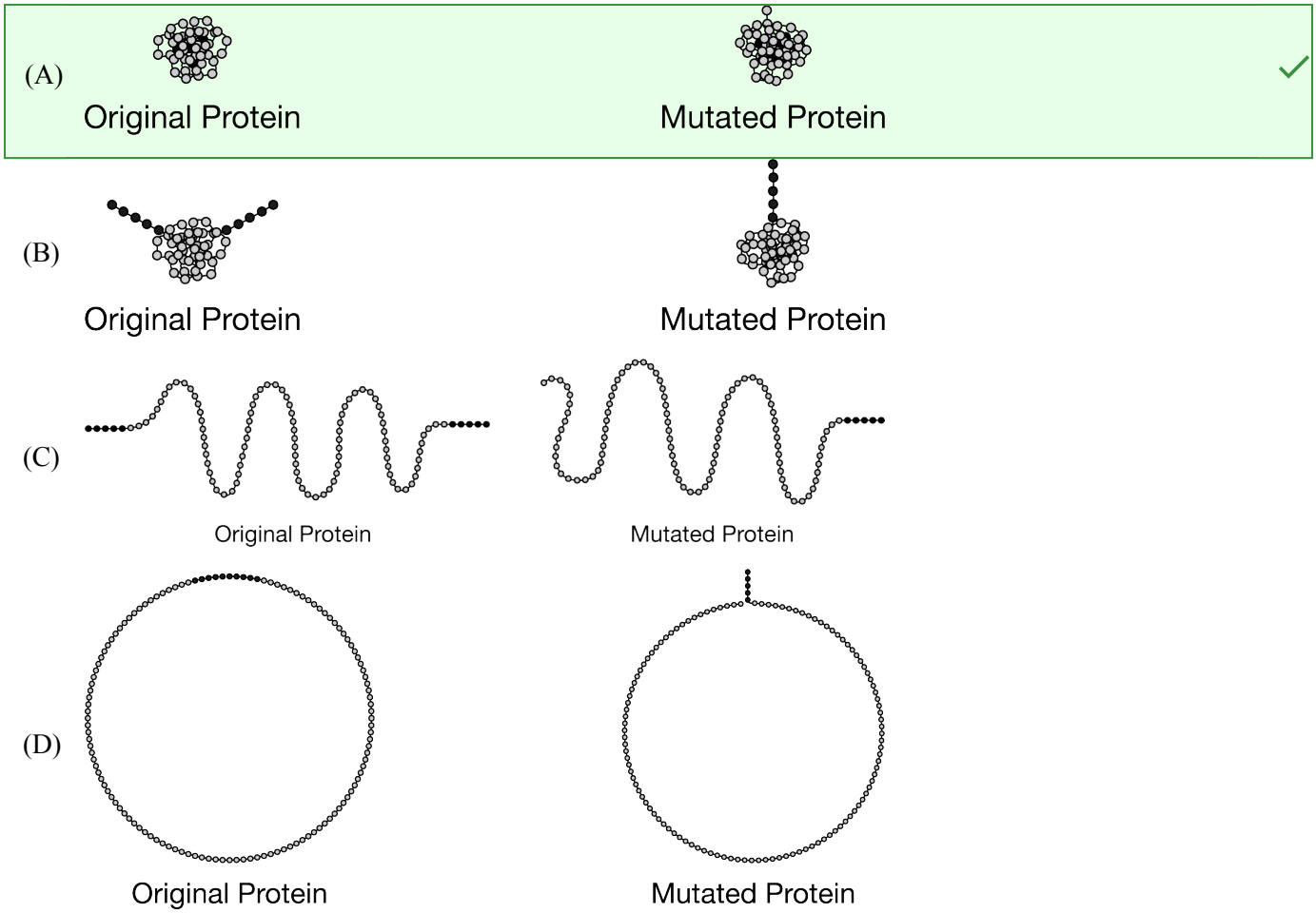


Figure 2. Primary structure of mutated protein



Which of the following best depicts the tertiary structures of the two proteins in water? The diagrams in the options are not drawn to the same scale as those in Figure 1 and Figure 2.

1 Chemistry of Life MCQ

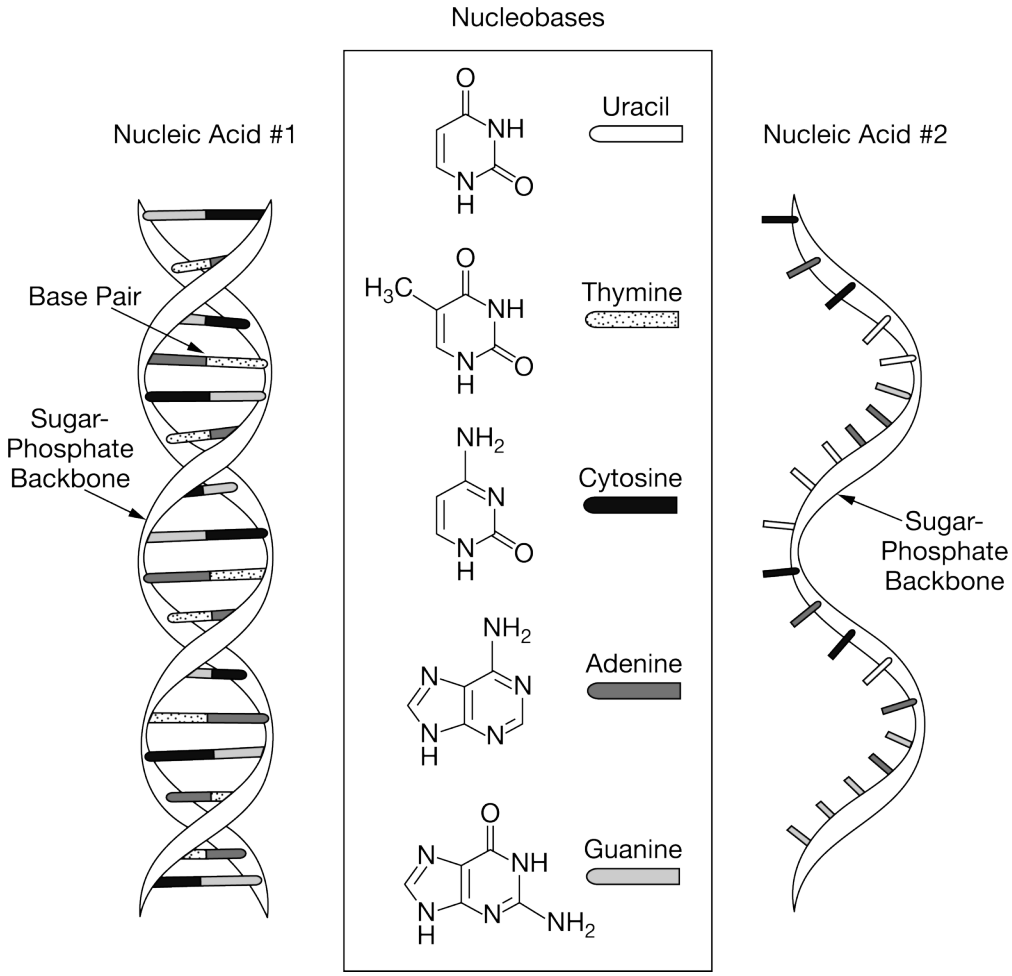


Answer A

Correct. The hydrophilic middle of the protein chain would be attracted to the water molecules, and the hydrophobic ends would be repelled. Hydrophobic portions of proteins in water tend to be found near the center of the tertiary structure, as far from the water as possible.

1 Chemistry of Life MCQ

53.



Which of the following conclusions is most clearly supported by the representations of nucleic acid #1 and nucleic acid #2 ?

(A) Nucleic acid #1 contains only purines, whereas nucleic acid #2 contains only pyrimidines.

(B) Nucleic acid #1 contains the sugar ribose, whereas nucleic acid #2 contains the sugar deoxyribose.

(C) Nucleic acid #1 contains positively charged phosphate groups, whereas nucleic acid #2 does not.

(D) Nucleic acid #1 contains adenine-thymine base pairs, whereas nucleic acid #2 does not. ✓

Answer D

Correct. The representation of nucleic acid #1 shows adenine-thymine base pairs, whereas the representation of nucleic acid #2 shows only uracil in the nucleotide chain and does not show any base pairing.

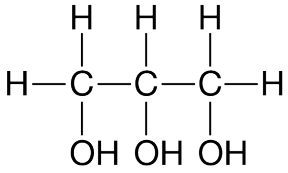
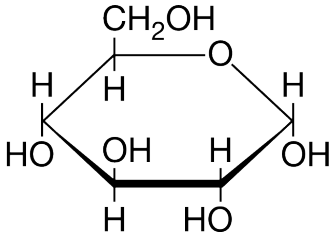
1 Chemistry of Life MCQ

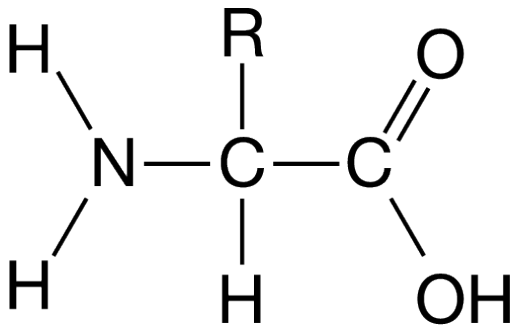
54. Students conducted a controlled experiment to investigate whether sawdust provides enough nutrients to support plant growth. The students separated ten nearly identical sunflower seedlings into two groups. They grew the seedlings in the first group in potting soil and the seedlings in the second group in sawdust composed mostly of cellulose. After twenty days, the students recorded observations about the seedlings in each group. The students' observations are presented in the table.

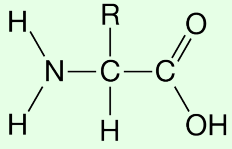
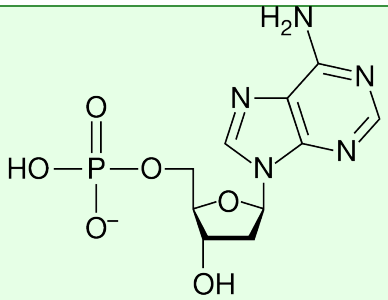
Treatment Group	Mean Seedling Height	Observations
Seedlings in potting soil	18.5 cm	The leaves have a dark green color and are normal in size.
Seedlings in sawdust	4.8 cm	The leaves have a grayish color and are small in size.

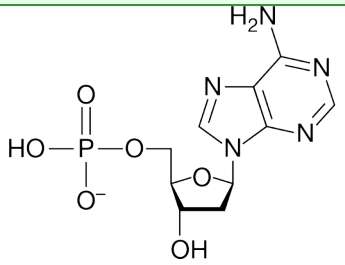
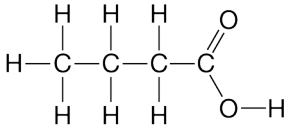
The observed differences between the groups most likely resulted from differences in the ability of the seedlings to produce which of the following monomers?

1 Chemistry of Life MCQ

(A)  and 

(B)  only

(C)  and 

(D)  and 

Answer C

Correct. Sawdust will not provide sufficient amounts of nitrogen and phosphorus, resulting in the seedlings having a reduced ability to produce both of the monomers shown (amino acids and nucleotides).

1 Chemistry of Life MCQ

55. Figure 1 represents a common process that occurs in organisms.

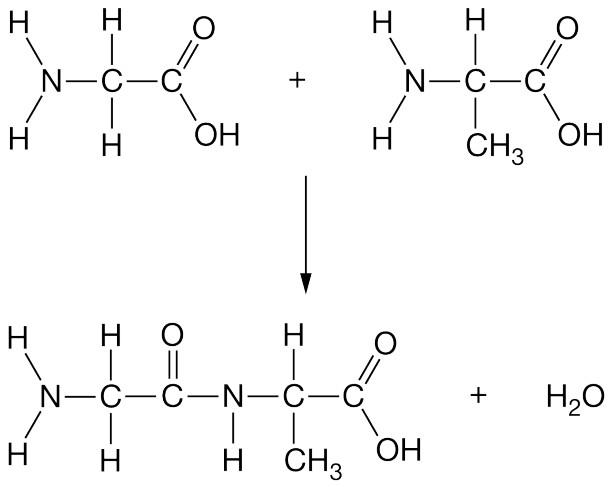


Figure 1. Structural formula for a common biological reaction

Which of the following is an accurate description of the process shown in Figure 1 ?

(A) The linking of amino acids with an ionic bond as an initial step in the protein synthesis process

(B) The formation of a more complex carbohydrate with the covalent bonding of two simple sugars

(C) The hydrolysis of amino acids with the breaking of covalent bonds with the release of water

(D) The formation of a covalent peptide bond in a dehydration synthesis reaction ✓

Answer D

Correct. This reaction combines two smaller amino acids to form a dipeptide (a step in the formation of a protein) with the release of water. The presence of nitrogen is consistent with both amino acids and proteins. The combining of smaller molecules to form a larger molecule with the release of water is a dehydration synthesis reaction.

1 Chemistry of Life MCQ

56.

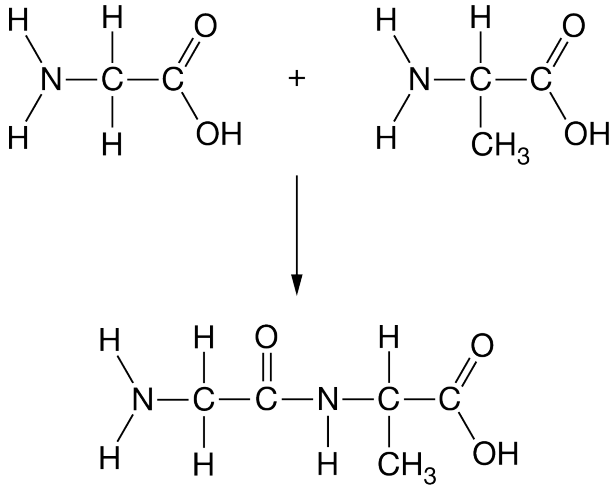


Figure 1. Formation of a peptide bond

Which of the following best describes the formation of the bond shown in Figure 1 ?

(A) An ionic bond is formed between a carbon atom of one amino acid and the nitrogen atom of the other amino acid.

(B) An ionic bond is formed when the negative charge of an OH group is balanced by the positive charge of a hydrogen ion.

(C) A covalent bond is formed between a carbon atom and a nitrogen atom along with the formation of H_2O . ✓

(D) A covalent bond is formed that replaces the hydrogen bond between the OH group and the H atom.

Answer C

Correct. Even though the water molecule that is produced is not shown in Figure 1, it shows the formation of the peptide bond and the missing H and OH.

57. Which of the following best describes the role of phospholipids?

(A) Phospholipids provide structural stability to animal cell membranes.

(B) Phospholipids support physiological functions, including growth.

(C) Phospholipids provide energy storage and support cell function.

(D) Phospholipids form the lipid bilayer found in cell membranes. ✓

1 Chemistry of Life MCQ

58. Phosphorous (P) is an important nutrient for plant growth. Figure 1 shows *Arabidopsis thaliana* plants grown under phosphorus-sufficient (left) and phosphorus-starved (right) conditions for six weeks.

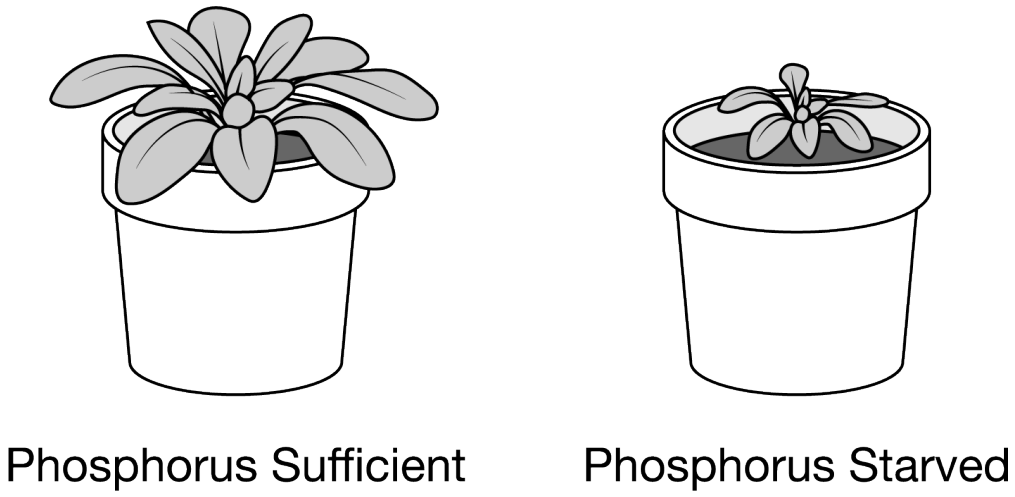


Figure 1. *Arabidopsis thaliana* plants grown for six weeks.

Which of the following is the most likely reason for the difference in leaf growth?

(A) The phosphorus-starved plant was unable to synthesize both the required proteins and lipids, limiting growth.

(B) The phosphorus-starved plant was unable to synthesize both the required proteins and carbohydrates, limiting growth.

(C) The phosphorus-starved plant was unable to synthesize both the required nucleic acids and lipids, limiting growth. ✓

(D) The phosphorus-starved plant was unable to synthesize both the required carbohydrates and nucleic acids, limiting growth.

Answer C

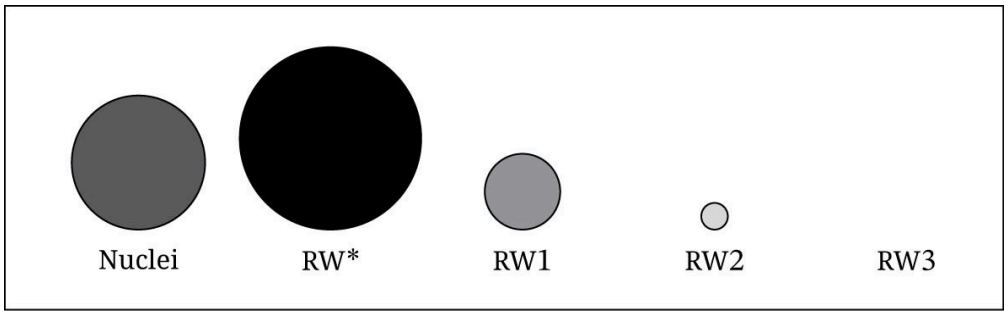
Correct. Phosphorus is used to make nucleic acids and certain lipids. Without phosphorus atoms, nucleic acids and lipids cannot be made for the plant to use for growth.

1 Chemistry of Life MCQ

59. A researcher conducted an experiment in a brightly lit aquarium containing a large population of single-celled green algae. Water containing radioactive phosphorus was added to the aquarium. After 7 days, the researcher took a water sample containing the algae and separated the algae from the radioactive water (RW*).

The researcher added deionized water to rinse the algae and saved the rinse water and then repeated that process twice. The rinse water from each of the three rinses was saved as RW1, RW2, and RW3, respectively. The researcher also removed and kept the nuclei of the algal cells. The nuclei were mixed with deionized water, and a drop of the mixture was placed on a piece of filter paper. A drop of the radioactive water and a drop of each of the rinse water samples were also placed on the same paper. These samples were exposed to X-ray to determine their radioactivity. The results after 24 hours are shown in Figure 1.

Figure 1. Radioactivity of saved algal water samples. Darker color and larger circle size indicate a higher level of radioactivity.



Based on Figure 1, which of the following statements best explains why the nuclei sample had a higher level of radioactivity than any of the rinse water samples?

(A) Phosphorus is used in the building of nucleic acids in the nucleus. ✓

(B) The nuclei sample was contaminated with radioactive water.

(C) Radioactive phosphorus is dissolved from rocks and becomes part of the sugar molecules created during photosynthesis.

(D) Radioactive phosphorus is dissolved from rocks and is used to create cholesterol stored in the nucleus.

60. Which of the following statements best describes the formation of complex carbohydrates?

(A) Monosaccharides are converted to simple sugars.

(B) Monomers of polysaccharides are linked together through peptide bonds.

(C) Nitrogen-containing bases are used to build complex carbohydrates.

(D) Monosaccharides are joined together through covalent bonds. ✓

61. Which of the following best describes the structures of carbohydrates?

1 Chemistry of Life MCQ

(A) They only occur as disaccharides.

(B) They occur as monomers, chains of monomers, and branched structures. ✓

(C) They only occur as long and branched structures.

(D) They occur as chains of monomers that hydrogen bond with complementary chains of monomers.

Answer B

Correct. Carbohydrates occur in many forms. Glucose is an example of a monomer, sucrose is a chain of two carbohydrate monomers, and starch is an example of a carbohydrate with a branched structure.

62. The synthesis of protein or carbohydrate polymers always produces which of the following as a byproduct?

(A) ATP

(B) Oxygen

(C) Carbon dioxide

(D) Urea

(E) Water ✓

63. Which of the following statements best describes a difference between proteins and nucleic acids?

(A) Proteins contain sulfur, while nucleic acids do not. ✓

(B) Proteins contain phosphorus, while nucleic acids contain sulfur.

(C) Both proteins and nucleic acids contain sulfur as a key element.

(D) Both proteins and nucleic acids contain nitrogen and sulfur.

1 Chemistry of Life MCQ

64. Polypeptides are continuously being formed and degraded. One of these processes is shown.

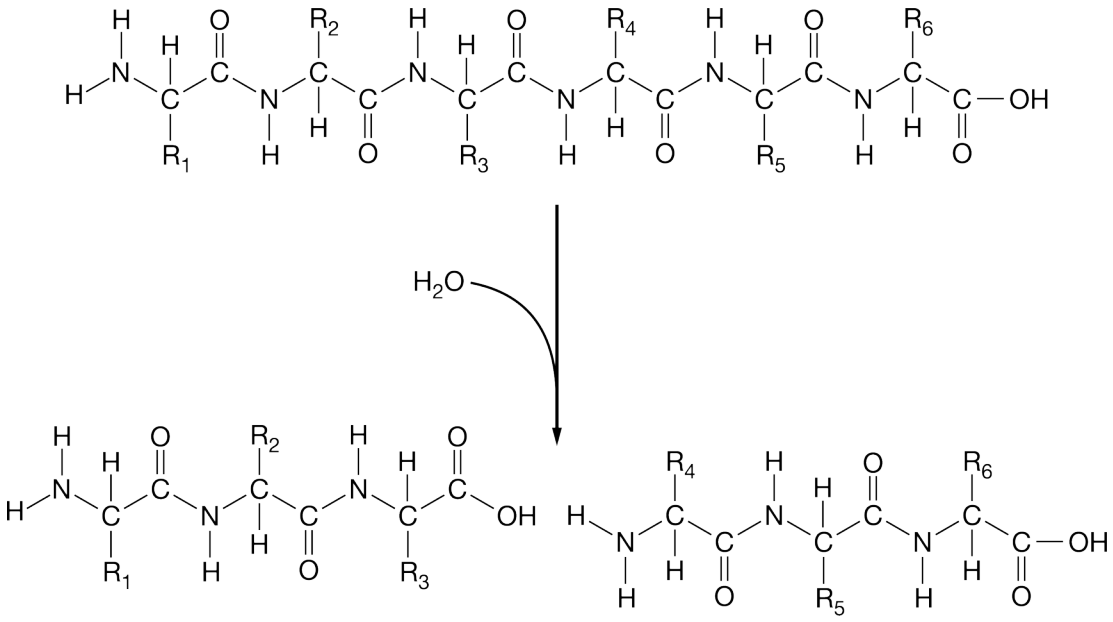


Figure 1. Polypeptide reaction

Which statement is the most accurate description of the reaction shown in Figure 1?

(A) It represents monomers linked by dehydration synthesis.

(B) It represents a polypeptide chain that folds to form the tertiary structure.

(C) It represents a polypeptide chain that is denatured into the primary structure.

(D) It represents a polypeptide chain that is broken down through a hydrolysis reaction. ✓

Answer D

Correct. The reaction shows the polypeptide being split into two smaller units with the addition of a hydroxide ion and a hydrogen ion from water. This facilitates the breaking of the bond holding the polypeptide together.

1 Chemistry of Life MCQ

For following group of questions first study the description of the situation or data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

In the first step of an experiment, rat liver cells were exposed for 5 minutes to amino acids labeled with a radioactive isotope. The cells were then washed to stop any further incorporation of radioactive amino acids. The cells were sampled periodically thereafter, and the radioactivity of a certain protein (protein X) was measured in various cell components, as shown below.

RADIOACTIVE COUNTS (PROTEIN X)

Sampling Time (minutes)	Cell Component						
	Endoplasmic Reticulum	Golgi Apparatus	Cytoplasm	Nucleus	Mitochondria	Lysosomes	Extracellular Space
5	980	20	50	4	2	6	2
20	135	760	20	3	5	120	9
45	50	195	4	6	2	800	3
120	15	30	8	6	8	975	4

65. What is the most likely identity of protein X ?

(A) An enzyme that is involved in transcription

(B) An electron transport protein that participates in respiration reactions

(C) A protein secreted from liver cells into the blood stream that permits proper capillary function

(D) An enzyme that participates in the degradation and recycling of cell components ✓

(E) A building block of the microtubules that are part of the cytoskeleton

66. A researcher is studying a specific type of cellular defect in animal cells. She notices that cells with this defect have reduced energy storage and decreased cell membrane stability.

Which of the following predicts the type of cellular defect that is the most likely cause of the researcher's observation?

(A) A defect in fatty acid synthesis, reducing the formation of fats and phospholipids ✓

(B) A defect in cholesterol synthesis, leading to reduced cholesterol in cell membranes

(C) A defect in steroid hormone production, leading to a depletion of energy reserves

(D) A defect in unsaturated fatty acid production, affecting membrane fluidity

67. Which of the following best describes a characteristic of saturated fatty acids?

1 Chemistry of Life MCQ

(A) They contain at least one double bond between carbon atoms.

(B) They have a kinked carbon chain structure.

(C) They contain only single bonds between carbon atoms. ✓

(D) They are more liquid at room temperature than unsaturated fatty acids

Answer C

Correct. Saturated fatty acids contain only single bonds between carbon atoms. This characteristic allows the carbon chains to pack tightly together, giving saturated fats their solid nature at room temperature.

68.

Segment 1:	5′ - ATATGAGTAGT - 3′ 3′ - TATACTCATCA - 5′	Segment 2:	5′ - GCGCAGACGAC - 3′ 3′ - CGCGTCTGCTG - 5′

The sequences for two short fragments of DNA are shown above. Which of the following is one way in which these two segments would differ?

(A) Segment 1 would not code for mRNA because both strands have T, a base not found in RNA.

(B) Segment 1 would be more soluble in water than segment 2 because it has more phosphate groups.

(C) Segment 1 would become denatured at a lower temperature than would segment 2 because A-T base pairs have two hydrogen bonds whereas G-C base pairs have three. ✓

(D) Segment 1 must be from a prokaryote because it has predominantly A-T base pairs.

Answer C

This option is correct. It demonstrates an understanding of the relationship between structure and function at the molecular level and the ability to make predictions about how change(s) in structure affect functionality. Because G-C base pairs have three hydrogen bonds, they are more stable structurally. Thus, because DNA Segment 2 has more G-C base pairs than DNA Segment 1, Segment 2 is more stable and thus would denature at higher temperatures than Segment 1.

69. Which of the following best explains why surface tension is created on the surface of bodies of water?

1 Chemistry of Life MCQ

(A) Water molecules are attracted to each other because the hydrogen atoms donate electrons to oxygen atoms.

(B) Water molecules are attracted to each other because the sharing of electrons between hydrogen and oxygen is unequal. ✓

(C) Water molecules are attracted to substances dissolved in the water because most dissolved substances are positively charged.

(D) Water molecules are attracted to substances dissolved in the water because most dissolved substances are negatively charged.

Answer B

Correct. The unequal sharing of electrons between hydrogen and oxygen in a water molecule results in a partial positive charge and a partial negative charge on the hydrogen and oxygen atoms, respectively. These partial charges attract other water molecules, resulting in hydrogen bond formation and cohesive forces between the water molecules, especially at the surface of the water.

70. Figure 1 is a diagram of water molecules at the air-water interface at the surface of a pond.

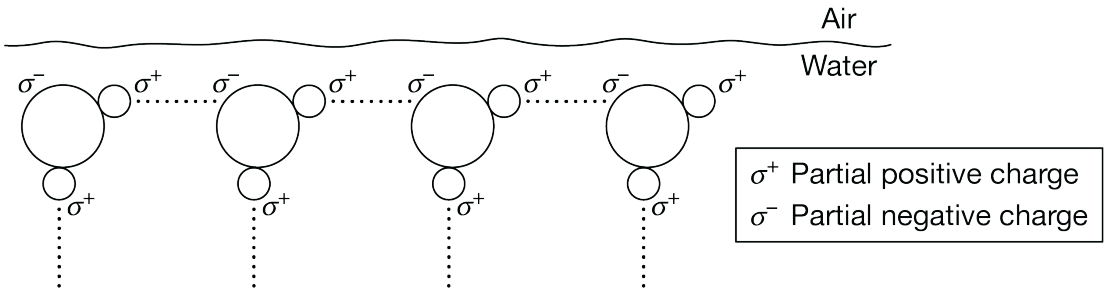


Figure 1. Alignment of water molecules at air-water interface

Based on Figure 1, which of the following best describes how the properties of water at an air-water interface enable an insect to walk on the water's surface?

(A) Covalent bonds between water molecules and the air above provide cohesion, which causes tiny bubbles to form under the feet of the insect.

(B) Ionic bonds between molecules at the surface of the water provide an electric charge, which attracts the feet of the insect, keeping it on the surface.

(C) Polar covalent bonds between molecules at the surface of the water provide adhesion, which supports the weight of the insect.

(D) Hydrogen bonds between molecules at the surface of the water provide surface tension, which allows the water surface to deform but not break under the insect. ✓

1 Chemistry of Life MCQ

Answer D

Correct. Hydrogen bonds form between water molecules due to the partial charges produced as a result of the polarity of the water molecule. The strength of the attraction between the water molecules allows organisms to walk across the surface without breaking the hydrogen bonds.

71. Which of the following best describes how amino acids affect the tertiary structure of a protein?

(A) The number of amino acids determines the tertiary structure of the protein.

(B) The interactions of the different R-groups with other R-groups and with their environment determine the tertiary structure of the protein. ✓

(C) The R-group of the last amino acid that is added to a growing polypeptide chain determines the next amino acid that is added to the chain.

(D) The sequence of the amino acids in the polypeptide chain determines the protein's primary structure but has no effect on its tertiary structure.

Answer B

Correct. The tertiary structure is determined by the attractions, repulsions, and bonding between different amino acids. Changing the sequence of even two amino acids may affect the tertiary shape of the protein.

72. Researchers analyzed the composition of cell membranes of bacteria living in different environments. Thermophiles are bacteria that live at extremely high temperatures, typically at or above $70^{\circ}C$.

Which of the following best predicts the effect of a greater proportion of saturated fatty acids, as compared with unsaturated fatty acids, on the stability of the cell membrane in thermophiles?

(A) A greater proportion of saturated fatty acids has no effect on cell membrane stability because both saturated and unsaturated fatty acids are hydrophobic.

(B) A greater proportion of saturated fatty acids leads to increased membrane stability at lower temperatures because saturated fatty acids increase the stability of the membrane.

(C) A greater proportion of saturated fatty acids leads to decreased membrane stability because saturated fatty acids decrease the fluidity of the membrane.

(D) A greater proportion of saturated fatty acids leads to increased membrane stability at higher temperatures because saturated fatty acids pack tightly together. ✓

Answer D

Correct. Saturated fatty acids pack tightly, which reduces membrane fluidity. This is critical for maintaining membrane stability in thermophiles exposed to extremely high temperatures.

1 Chemistry of Life MCQ

73. In vascular plants, water flows from root to leaf via specialized cells called xylem. Xylem cells are hollow cells stacked together like a straw. A student explains that evaporation of water from the leaf pulls water up from the roots through the xylem, as shown in Figure 1.

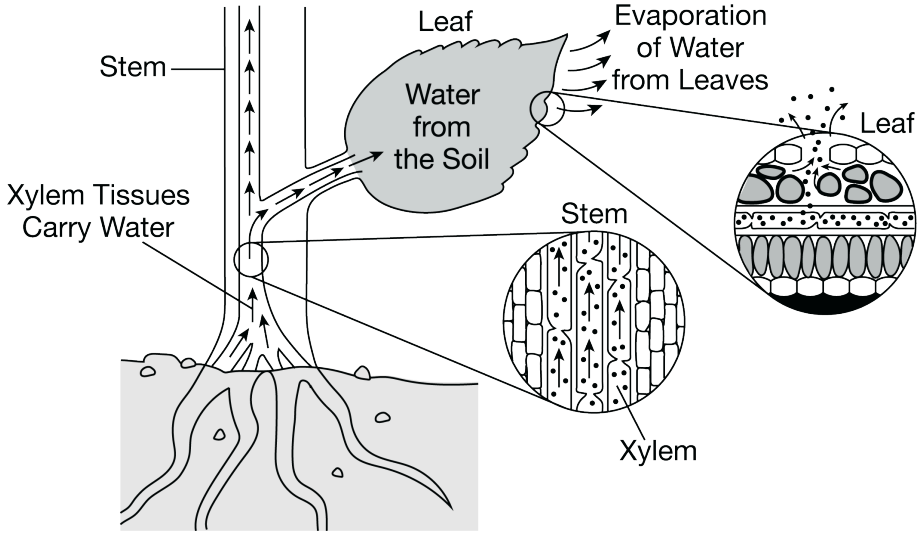


Figure 1. Model of water movement through the xylem, with magnified models of water movement in the stem and leaf.

Which statement describes how water is pulled up through the xylem to the leaves of the plant?

(A) As water exits the leaf, hydrogen bonding between water molecules pulls more water up from below. ✓

(B) As water exits the leaf, signals are sent to the roots to pump more water up to the leaves through the xylem by adhesion.

(C) Evaporation from the leaf decreases the hydrogen bonds that form between the water molecules in the xylem, which helps the water molecules to be pulled up the xylem.

(D) Evaporation of water from the leaf increases the hydrogen bonds that form between water molecules in the air, providing the energy for transport.

Answer A

Correct. As water exits the leaf, hydrogen bonding pulls more water molecules up through the leaf and xylem by cohesion.

74. Which of the following is most directly responsible for water's unique properties?

1 Chemistry of Life MCQ

(A) It contains oxygen atoms.

(B) It contains hydrogen atoms.

(C) It is an ionic compound.

(D) It forms hydrogen bonds. ✓

(E) It is nonpolar.

2 Cell MCQ

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it and fill in the corresponding circle on the answer sheet.

Dialysis tubing is permeable to water molecules but not to sucrose. Four dialysis tubes are half filled with 5 percent, 10 percent, 20 percent, and 40 percent sucrose solutions, respectively, and two dialysis tubes are half filled with distilled water. The dialysis tubes are all sealed at both ends, and the initial masses are determined. Five dialysis tubes are placed into beakers containing distilled water, and the sixth dialysis tube, containing distilled water, is placed into a 40 percent sucrose solution. The masses of the dialysis tubes are recorded at 30-minute intervals for 90 minutes, as shown in the table below.

Dialysis Tube Number	Dialysis Tube Contents	Beaker Contents	Initial Mass	Mass (grams)		
				30 Min	60 Min	90 Min
1	5% Sucrose	Distilled water	12.8	13.3	13.4	13.8
2	10% Sucrose	Distilled water	15.6	16.7	17.0	17.3
3	20% Sucrose	Distilled water	13.7	15.4	16.2	16.5
4	40% Sucrose	Distilled water	13.4	16.3	18.1	19.6
5	Distilled water	Distilled water	11.2	11.2	11.2	11.2
6	Distilled water	40% Sucrose	13.8	10.1	8.3	6.3

1. The contents of which dialysis tube are initially isotonic to the distilled water in the beaker?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5 ✓

2. A net movement of water into the beaker occurs in which of the following dialysis tubes?

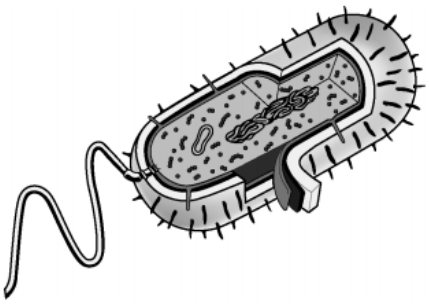
2 Cell MCQ

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6 ✓

3. To model a plant cell, a permeable, nonflexible case is placed around each piece of dialysis tubing. The greatest pressure potential will develop within which dialysis tube number?

(A) 2
(B) 3
(C) 4 ✓
(D) 5
(E) 6

4.

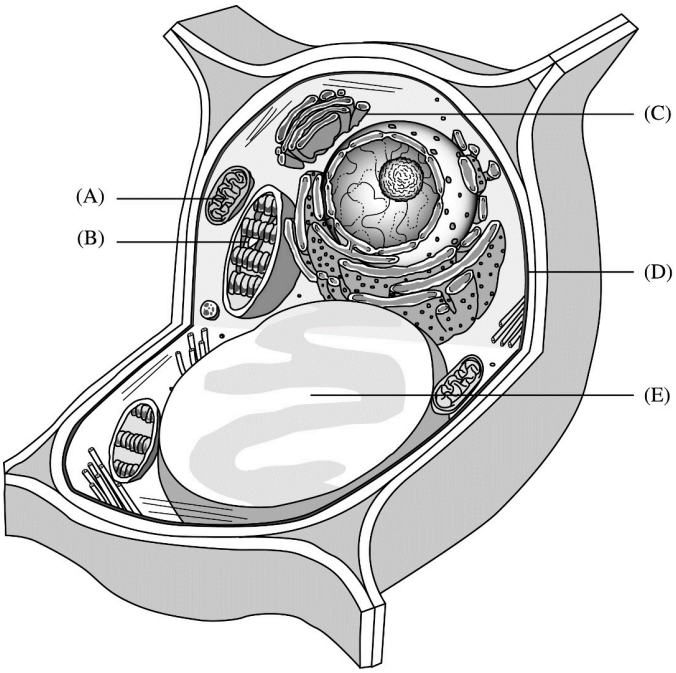


Which of the following is true about the structure of the prokaryotic cell represented in the figure?

(A) It possesses cilia and lysosomes.
(B) It has a diploid genome and contains linear DNA.
(C) It has a cell wall exterior to the plasma membrane and contains ribosomes. ✓
(D) It possesses membrane-bound organelles and a flagellum.

2 Cell MCQ

The following questions refer to the following diagram. For each phrase or sentence, select the labeled part to which it is most closely related. Each option may be used once, more than once, or not at all for each group.



5. Evolved from a photoautotrophic prokaryote

(A) A
(B) B ✓
(C) C
(D) D
(E) E

6. Site of modification and packaging of proteins and lipids prior to export from the cell

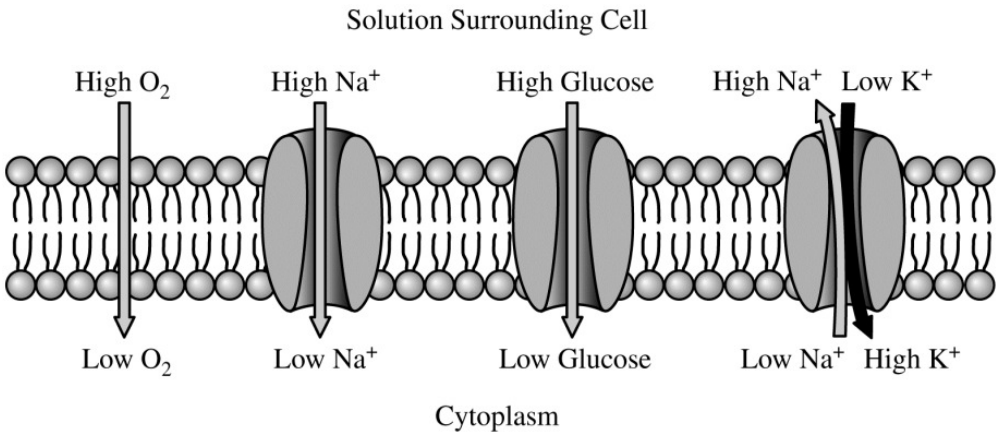
(A) A
(B) B
(C) C ✓
(D) D
(E) E

7. Site of transport of materials into and out of the cell

2 Cell MCQ

(A) A
(B) B
(C) C
(D) D ✓
(E) E

8.



The manner in which several different ions and molecules move through a cell membrane is shown in the diagram above. For each ion or molecule, the relative concentration on each side of the membrane is indicated. Which of the following accurately describes one of the movements taking place?

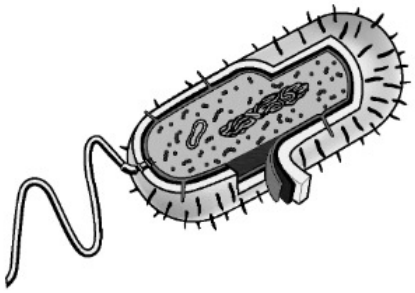
(A) Glucose is transported into the cell by active transport.
(B) Na^+ is transported into the cell by active transport.
(C) The movement of glucose through the membrane requires ATP hydrolysis.
(D) Na^+ transport out of the cell requires ATP hydrolysis. ✓

9. Intravenous (IV) solutions administered to patients are normally isotonic. Which of the following is most likely if an IV of distilled water is administered to a patient?

(A) The cells that are exposed to hypotonic solutions will shrink as a result of salt moving into the blood.
(B) The liver will secrete additional bile salts into the blood to raise the tonicity of the administered fluid.
(C) The cells that are exposed to hypotonic solutions will expand as water moves osmotically into the cells from the blood. ✓
(D) The patient's respiration rate will slow to compensate for the higher levels of circulating blood.

2 Cell MCQ

10.



The diagram above represents a typical rod-shaped bacterium. Which of the following best describes a feature shown in the diagram that is unique to archaea and bacteria?

(A) The organism is surrounded by a cell wall.

(B) The organism contains ribosomes.

(C) The organism does not have a nuclear membrane surrounding its genetic material. ✓

(D) The organism is not capable of making or providing itself with ATP.

11. Water crosses the plasma membranes of certain eye cells at a higher rate than it does in other cell types. Which of the following best explains why the rate of water transport is higher in the eye cells?

(A) The eye cells move water against the concentration gradient, which is faster than moving water with the concentration gradient.

(B) The eye cells have protein channels that allow water to pass through in greater quantities than the phospholipid bilayer allows. ✓

(C) The eye cells have more ATP than other cells do, which allows them to use active transport to move water across the membrane more rapidly.

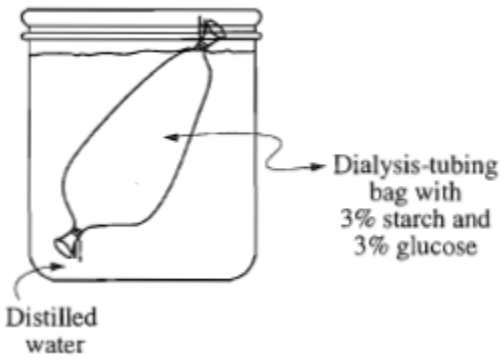
(D) The eye cells modify water molecules so that they are positively charged, allowing them to cross the phospholipid bilayer more easily than unmodified water molecules.

Answer B

Correct. Water is able to cross the cell membrane when moving with its concentration gradient. The presence of protein channels such as aquaporins allows water to move faster than when it simply passes between the phospholipids of the bilayer.

2 Cell MCQ

The following questions refer to an experiment in which a dialysis-tubing bag is filled with a mixture of 3% starch and 3% glucose and placed in a beaker of distilled water, as shown below. After 3 hours, glucose can be detected in the water outside the dialysis-tubing bag, but starch cannot.



12. From the initial conditions and results described, which of the following is a logical conclusion?

(A) The initial concentration of glucose in the bag is higher than the initial concentration of starch in the bag.

(B) The pores of the bag are larger than the glucose molecules but smaller than the starch molecules. ✓

(C) The bag is not selectively permeable.

(D) A net movement of water into the beaker has occurred.

(E) The molarity of the solution in the bag and the molarity of the solution in the surrounding beaker are the same.

13. Which of the following best predicts the condition expected after 24 hours?

(A) The bag will contain more water than it did in the original condition. ✓

(B) The contents of the bag will have the same osmotic concentration as the surrounding solution.

(C) Water potential in the bag will be greater than water potential in the surrounding solution.

(D) Starch molecules will continue to pass through the bag.

(E) A glucose test on the solution in the bag will be negative.

14. If, instead of the bag, a potato slice were placed in the beaker of distilled water, which of the following would be true of the potato slice?

(A) It would gain mass. ✓

(B) It would neither gain nor lose mass.

(C) It would absorb solutes from the surrounding liquid.

(D) It would lose water until water potential inside the cells is equal to zero.

(E) The cells of the potato would increase their metabolic activity.

2 Cell MCQ

15. A student used a microscope to observe a wet-mount slide of red onion epidermal cells that were suspended in a 1% NaCl solution. The student then added a 15% NaCl solution to the slide and observed the changes that occurred. The student's observations are represented in Figure 1.

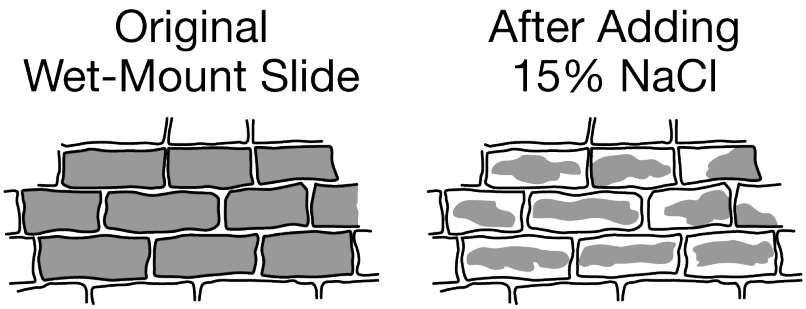


Figure 1. Student's observations of onion cells

Which of the following most directly explains the changes in the cells?

(A) The degradation of DNA in the nuclei of the cells

(B) The lysis of chloroplasts in the cells

(C) The movement of water from the central vacuoles of the cells into the solution ✓

(D) The movement of NaCl from the solution into the cytoplasm of the cells

16. Which of the following best supports the statement that mitochondria are descendants of endosymbiotic bacteria-like cells?

(A) Mitochondria and bacteria possess similar ribosomes and DNA. ✓

(B) Mitochondria and bacteria possess similar nuclei.

(C) Glycolysis occurs in both mitochondria and bacteria.

(D) Both mitochondria and bacteria have microtubules.

(E) Neither mitochondria nor bacteria possess chloroplasts.

17. Which of the following components of the cell membrane is responsible for active transport?

(A) Phospholipid

(B) Protein ✓

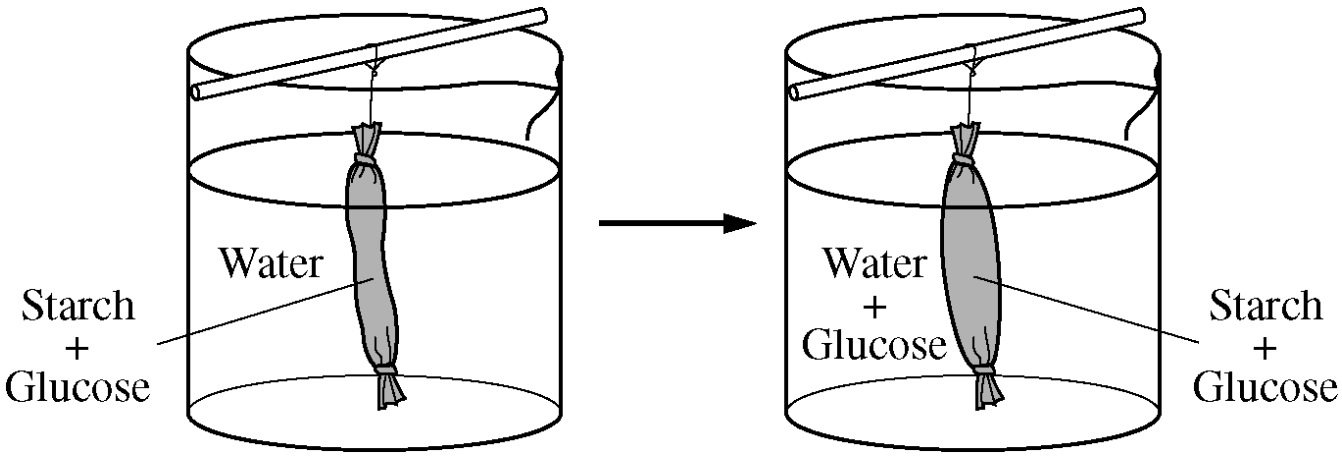
(C) Lipid

(D) Phosphate

(E) Cholesterol

2 Cell MCQ

18. A common laboratory investigation involves putting a solution of starch and glucose into a dialysis bag and suspending the bag in a beaker of water, as shown in the figure below.

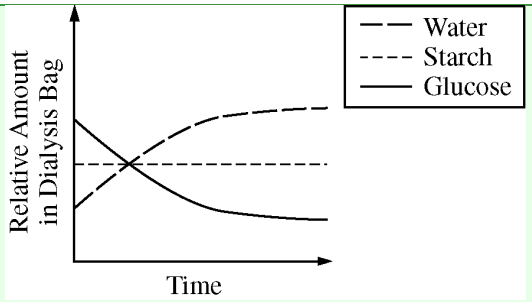


The investigation is aimed at understanding how molecular size affects movement through a membrane.

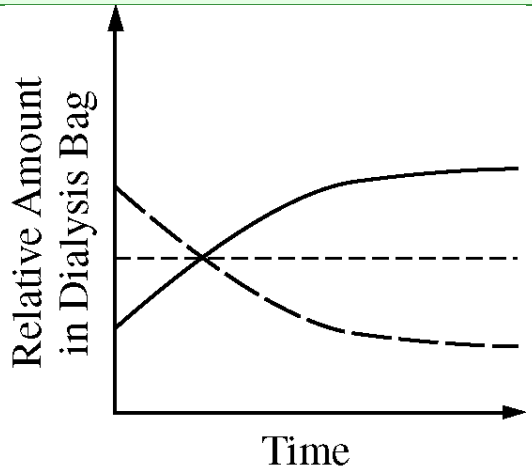
Which of the following best represents the amount of starch, water, and glucose in the dialysis bag over the course of the investigation?

2 Cell MCQ

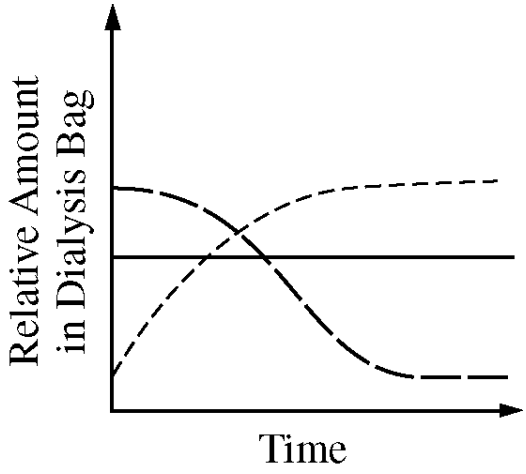
(A)



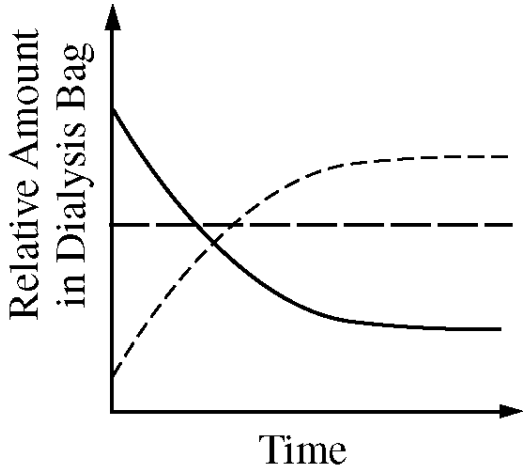
(B)



(C)

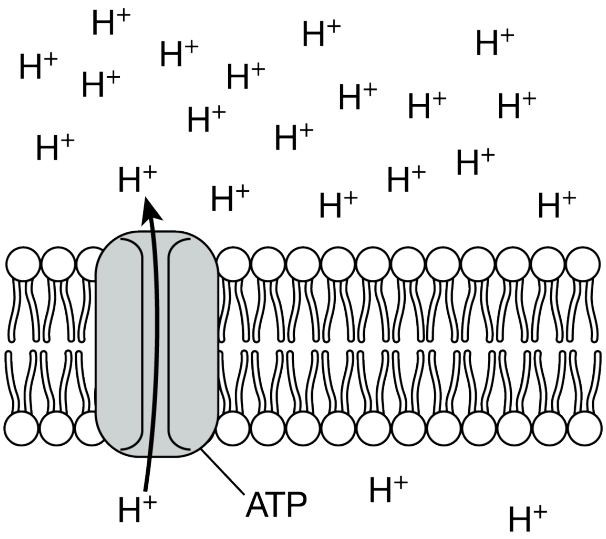


(D)



2 Cell MCQ

19. The illustration shows the active transport of hydrogen ions through a membrane protein.



Which of the following best predicts the effect of not having ATP available to supply energy to this process?

(A) H^+ ions will stop moving through the protein. ✓

(B) H^+ ions will move in the other direction through the protein.

(C) H^+ ions will continue to move through the protein in the original direction but at a slower rate.

(D) H^+ ions will begin to move through the phospholipid portion of the membrane in the original direction.

Answer A

Correct. Metabolic energy such as ATP is required for active transport of ions against a concentration gradient.

20. Which of the following transport mechanisms will be affected most directly by a temporary shortage of ATP molecules inside the cell?

(A) The movement of water molecules through aquaporins

(B) The diffusion of oxygen molecules across the plasma membrane

(C) The transport of glucose molecules against a concentration gradient ✓

(D) The facilitated diffusion of Ca^{2+} ions into the cell

2 Cell MCQ

Answer C

Correct. The active transport of glucose molecules against a concentration gradient requires an input of energy. Biological processes that require an input of energy typically incorporate the hydrolysis of ATP, an energy-rich molecule.

21. Some membrane proteins help maintain the concentrations of ions inside a cell by transporting the ions across the cell's plasma membrane. Other membrane proteins form pores in the plasma membrane through which the ions can diffuse. A model showing the influence of membrane proteins on the movement of sodium (Na^+) and potassium (K^+) ions across a plasma membrane is presented in Figure 1.

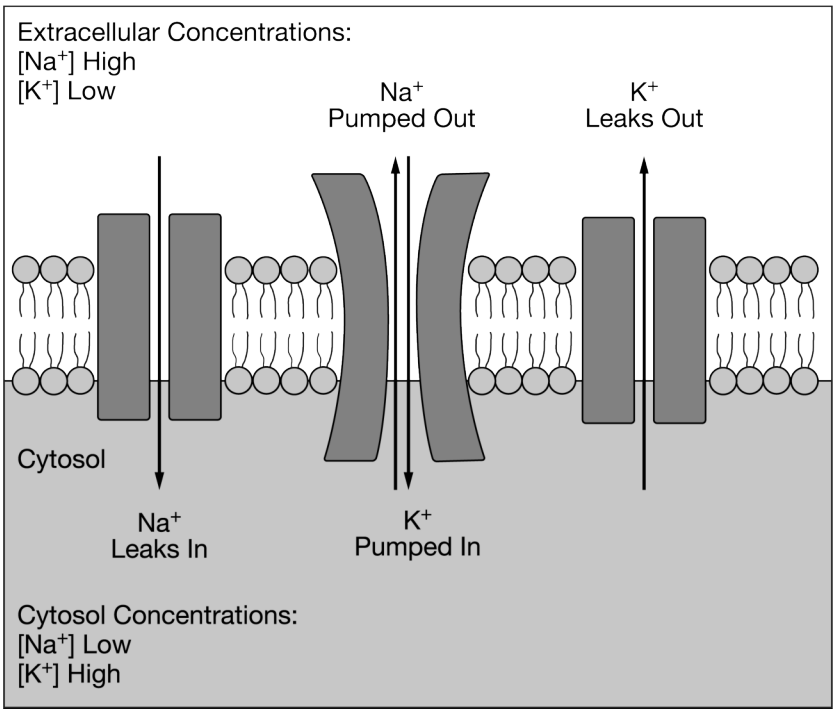


Figure 1. Section of a cell's plasma membrane, showing ion concentrations and membrane proteins

Based on the model presented in Figure 1, which of the following changes will most likely result from a depletion of available ATP stores inside the cell?

(A) The Na^+ concentration outside the cell will increase.

(B) The Na^+ concentration inside the cell will increase. ✓

(C) The K^+ concentration inside the cell will increase.

(D) The K^+ concentration outside the cell will decrease.

2 Cell MCQ

Answer B

Correct. Without ATP, the Na^+ ions will continue to leak into the cell but will not be pumped out of the cell, resulting in a concentration increase inside the cell.

22. The active transport pump used to move sodium ions across the membranes of gill cells in a freshwater fish has each of the following characteristics EXCEPT:

(A) It uses osmosis to carry sodium ions into the cells. ✓

(B) It requires energy.

(C) It has a specific binding site for ATP.

(D) It has a specific binding site for sodium ions.

(E) It is a protein.

2 Cell MCQ

23. Figure 1 shows a model of how a channel protein influences the movement of a particle across a cell's plasma membrane.

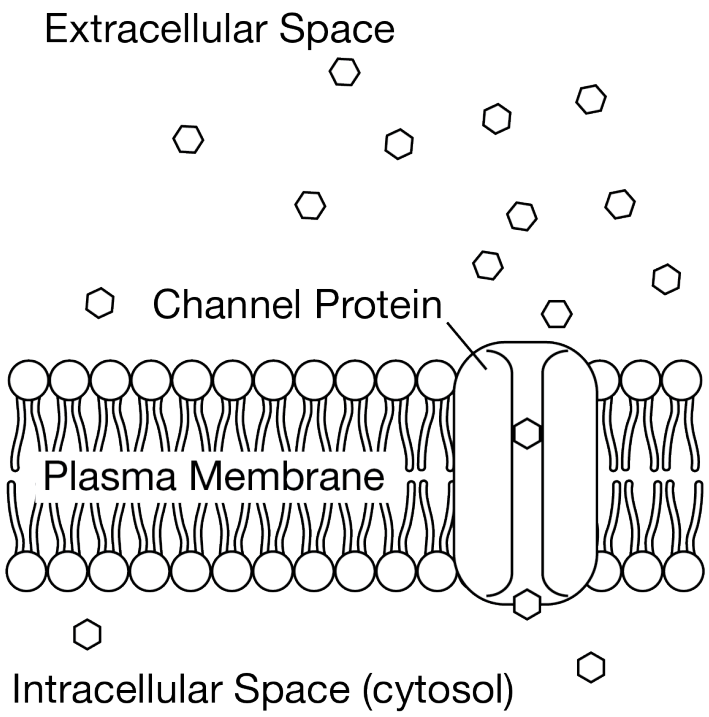


Figure 1. A section of a cell's plasma membrane, showing a channel protein and a concentration gradient across the membrane

An investigator wants to understand whether a newly found membrane protein is involved in membrane transport of a certain particle. Which investigation will help determine whether the new membrane protein is a channel protein involved in membrane transport?

(A) Add small nonpolar molecules to the extracellular space and measure the direction of particle movement of the molecules.

(B) Measure the rate of extracellular fluid movement into the intracellular space.

(C) Add more of the proteins to the plasma membrane and measure the rate of the particle movement. ✓

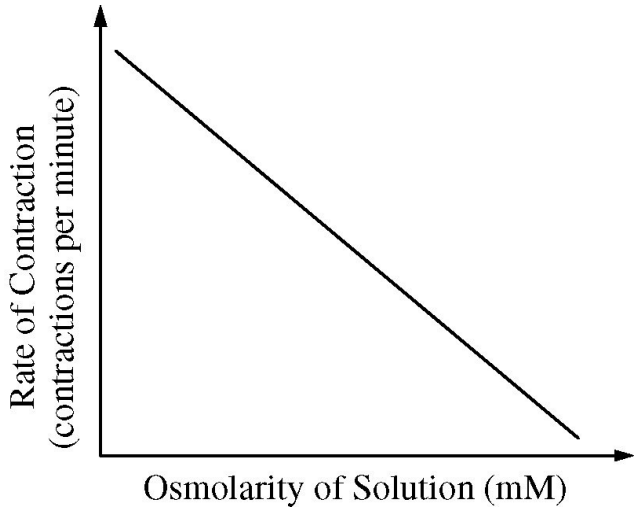
(D) Remove ATP from the intracellular space and measure the rate of the particle movement into the intracellular space.

Answer C

Correct. If the movement of particles is mediated by the channel protein, then an increase in proteins in the cell membrane should influence the rate of particle movement across the membrane.

2 Cell MCQ

24.



Paramecia are unicellular protists that have contractile vacuoles to remove excess intracellular water. In an experimental investigation, paramecia were placed in salt solutions of increasing osmolarity. The rate at which the contractile vacuole contracted to pump out excess water was determined and plotted against osmolarity of the solutions, as shown in the graph. Which of the following is the correct explanation for the data?

(A) At higher osmolarity, lower rates of contraction are required because more salt diffuses into the paramecia.

(B) The contraction rate increases as the osmolarity decreases because the amount of water entering the paramecia by osmosis increases. ✓

(C) The contractile vacuole is less efficient in solutions of high osmolarity because of the reduced amount of ATP produced from cellular respiration.

(D) In an isosmotic salt solution, there is no diffusion of water into or out of the paramecia, so the contraction rate is zero.

25. A human kidney filters about 200 liters of blood each day. Approximately two liters of liquid and nutrient waste are excreted as urine. The remaining fluid and dissolved substances are reabsorbed and continue to circulate throughout the body. Antidiuretic hormone (ADH) is secreted in response to reduced plasma volume. ADH targets the collecting ducts in the kidney, stimulating the insertion of aquaporins into their plasma membranes and an increased reabsorption of water.

If ADH secretion is inhibited, which of the following would initially result?

(A) The number of aquaporins would increase in response to the inhibition of ADH.

(B) The person would decrease oral water intake to compensate for the inhibition of ADH.

(C) Blood filtration would increase to compensate for the lack of aquaporins.

(D) The person would produce greater amounts of dilute urine. ✓

2 Cell MCQ

26. A human kidney filters about 200 liters of blood each day. Approximately two liters of liquid and nutrient waste are excreted as urine. The remaining fluid and dissolved substances are reabsorbed and continue to circulate throughout the body. Antidiuretic hormone (ADH) is secreted in response to reduced plasma volume. ADH targets the collecting ducts in the kidney, stimulating the insertion of aquaporins into their plasma membranes and an increased reabsorption of water.

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(C) Blood filtration would increase to compensate for the lack of aquaporins.

(D) The person would produce greater amounts of dilute urine. ✓

27. All of the following cell components are found in prokaryotic cells EXCEPT

(A) DNA

(B) ribosomes

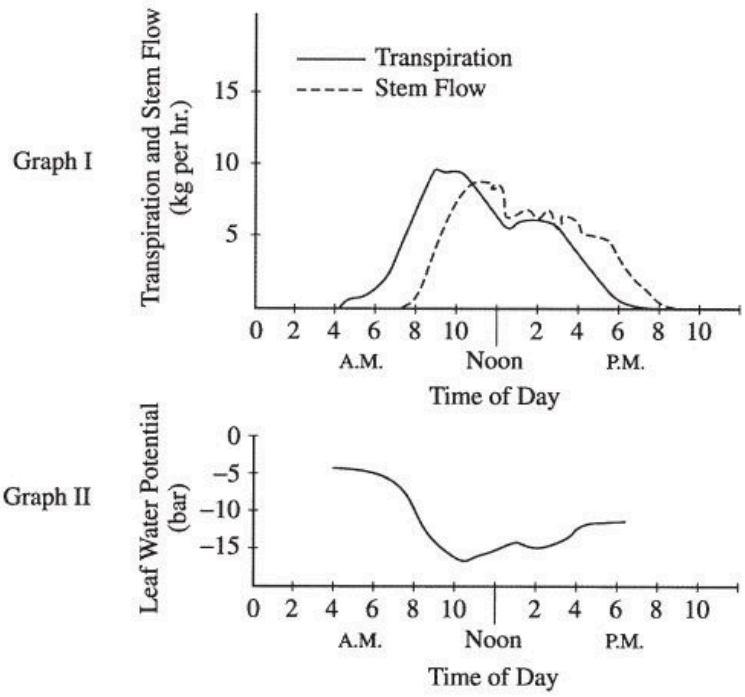
(C) cell membrane

(D) nuclear envelope ✓

(E) enzymes

2 Cell MCQ

28. The rate of transpiration, the flow of water through the stem, and leaf water potential are measured in a tree during a 24-hour period under normal environmental conditions. The results from these measurements are shown in the graphs below.



All of the following changes would be likely to decrease the rate of transpiration at 8 A.M. EXCEPT

(A) causing the stomata to close

(B) increasing the humidity of the atmosphere

(C) increasing the water potential of the atmosphere

(D) increasing the water potential of the soil ✓

(E) placing the plant in total darkness

29. Researchers claimed that a particular organelle originated from a free-living prokaryotic cell that was engulfed by a larger cell, as shown in Figure 1.

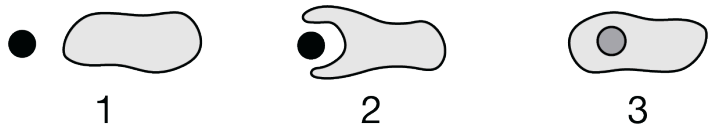


Figure 1. A model showing a cell engulfing a smaller cell

Which of the following provides evidence to best support the researchers' claim?

2 Cell MCQ

(A) The organelle has a phospholipid membrane.
(B) The organelle has protein in the membrane.
(C) The organelle has a double membrane. ✓
(D) The organelle has an internal aqueous environment that is similar to the cytosol of the larger cell.

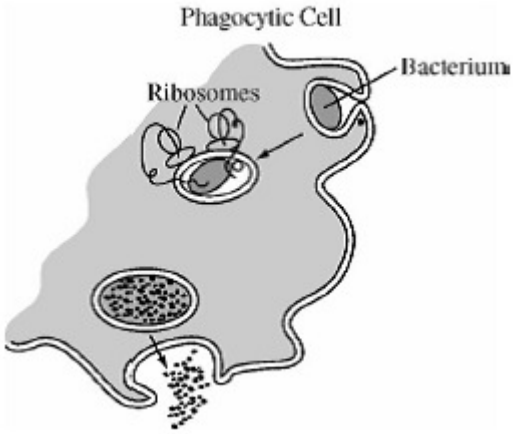
Answer C

Correct. The organelle's double membrane could have resulted from the process of endocytosis. Based on the claim, the organelle's inner membrane originated from the plasma membrane of the smaller cell and the organelle's outer membrane originated from the plasma membrane of the larger cell.

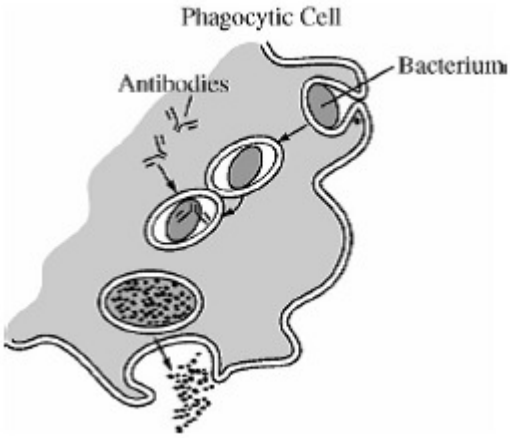
30. A pathogenic bacterium has been engulfed by a phagocytic cell as part of the nonspecific (innate) immune response. Which of the following illustrations best represents the response?

2 Cell MCQ

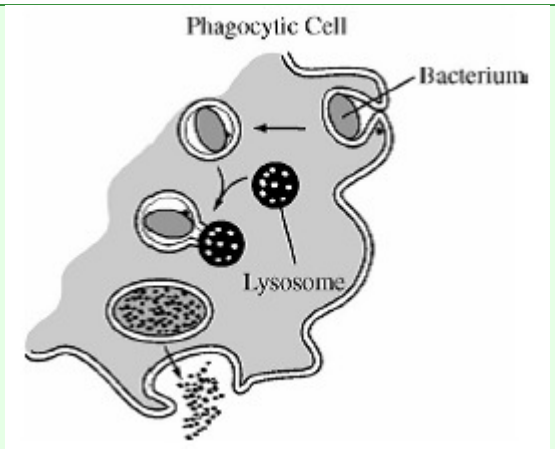
(A)



(B)

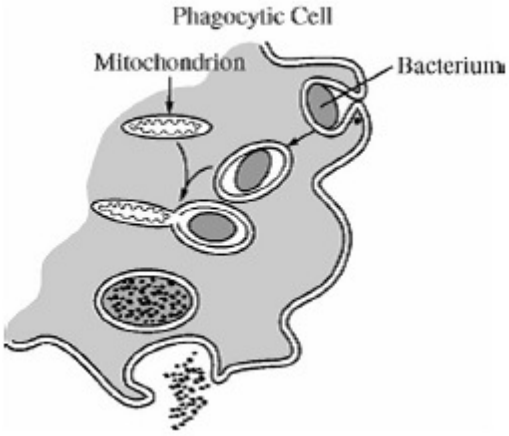


(C)



2 Cell MCQ

(D)



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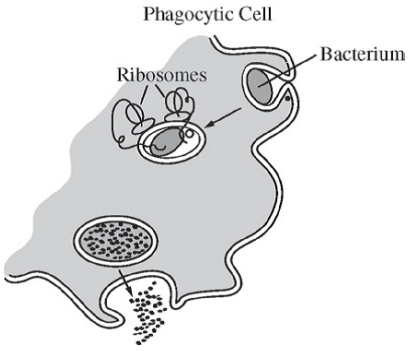
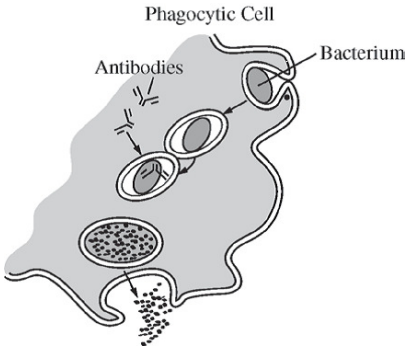
2 Cell MCQ

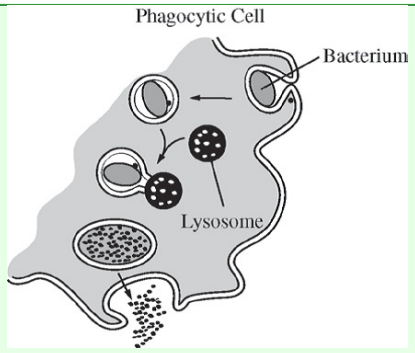
(A)



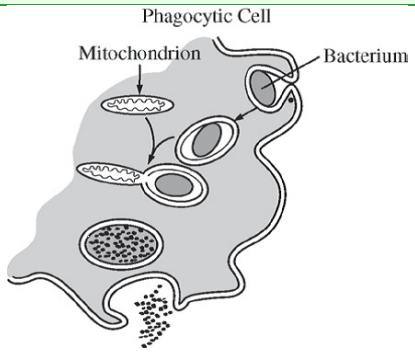
(B)



(C) ✓



(D)



Answer C

This option is correct. It demonstrates an understanding to interpret representations that describe a nonspecific immune response. In phagocytosis, a cell engulfs a bacterium by wrapping pseudopodia

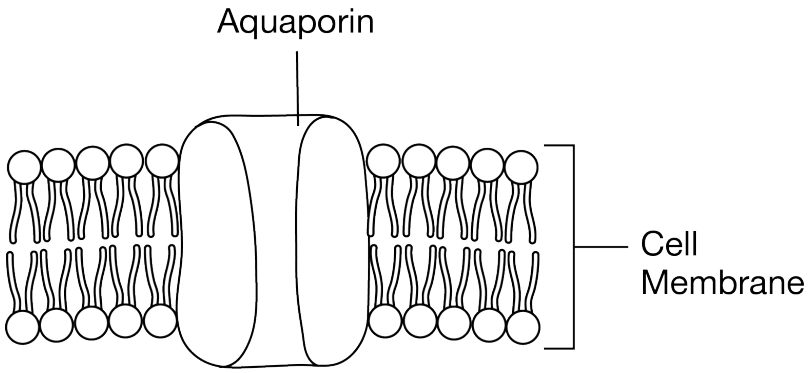
2 Cell MCQ

around it and packaging it within a membrane-enclosed vacuole. The bacterium is digested after the vacuole fuses with a lysosome containing hydrolytic enzymes. Digested particles can be exported out of the cell by exocytosis.

32. A prokaryotic cell has which of the following?

(A) Centrioles
(B) Lysosomes
(C) Plasma membrane ✓
(D) Mitochondria
(E) Endoplasmic reticulum

33.



Which statement best describes the effect on water transport across the cell membrane if the aquaporin in the figure ceases to function?

(A) Water molecules will no longer be able to move across the cell membrane.
(B) Water molecules will still be able to move across the cell membrane but at a slower rate. ✓
(C) Water molecules will only be able to enter the cell by active transport.
(D) Water molecules will move across the cell membrane at a faster rate without the aquaporin regulating their flow.

Answer B

Correct. The aquaporin facilitates water passage, but water can still move across the membrane by osmosis.

2 Cell MCQ

34. Aquaporins are channel proteins that facilitate the transport of water across the cell membrane. One group of researchers hypothesizes that without functional aquaporins, no water will be able to enter the cell. A different group proposes an alternative hypothesis, stating that even with nonfunctional aquaporins, a small amount of water will still cross the cell membrane. An experiment is set up in which plant cells with mutated (nonfunctional) aquaporins and plant cells with normally functioning aquaporins are both placed in distilled water.

Which of the following data would support the alternative hypothesis?

(A) Cells with functional aquaporins exhibit low turgor pressure and are hypertonic.

(B) Cells with functional aquaporins exhibit high turgor pressure and are hypotonic.

(C) Cells with mutated aquaporins exhibit an absence of turgor pressure and are completely plasmolyzed.

(D) Cells with mutated aquaporins exhibit moderate turgor pressure and are hypertonic. ✓

Answer D

Correct. Even with mutated aquaporins, a small amount of water is able to enter and leave the cell. Thus, turgor pressure is not lost and the alternative hypothesis is supported.

35. A study was conducted to understand the factors controlling the rate at which molecules or ions travel across cell membranes. An artificial membrane was created that was composed of a phospholipid bilayer only. The speed at which various substances crossed this membrane was measured.

Some substances can pass through an actual cell membrane much faster than they passed through the artificial membrane in this study. Which of the following statements best explains this finding?

(A) Actual cell membranes have a much thicker phospholipid bilayer than the artificial membrane does.

(B) Actual cell membranes have a variety of proteins embedded in the membrane that are absent in the artificial membrane. ✓

(C) Hydrophobic substances spend more time between the two layers of phospholipid in the artificial membrane than they do between the layers in an actual membrane.

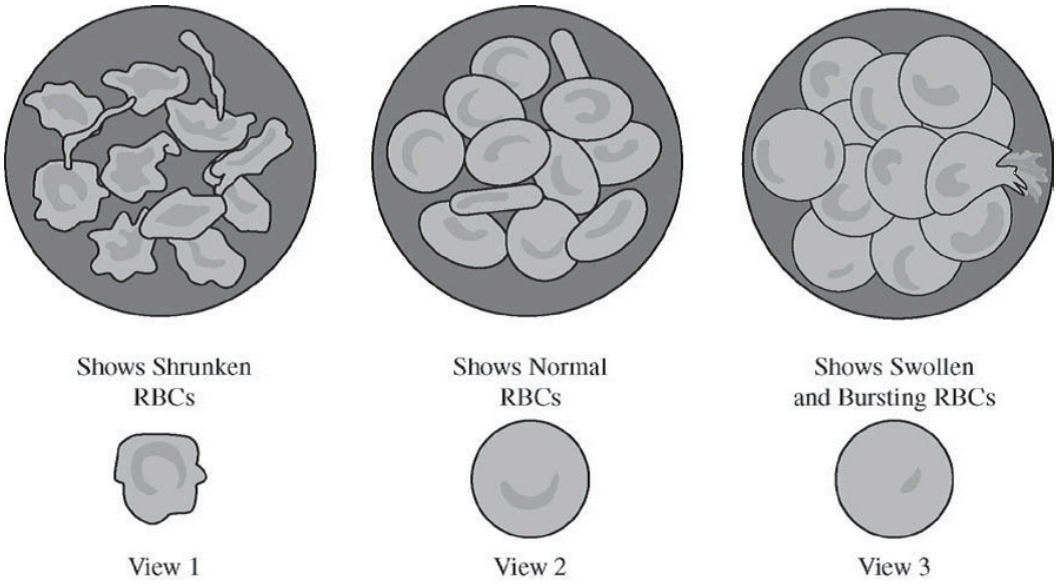
(D) Hydrophilic substances spend more time attached to the polar region of the phospholipids in the artificial membrane than they do attached to the polar region of the phospholipids in an actual membrane.

Answer B

Correct. Actual cell membranes have a variety of embedded proteins that increase the movement of substances across the membrane by both active and passive transport.

2 Cell MCQ

36. A sample of human blood was placed in a test tube containing a physiological saline solution (0.9% sodium chloride). This type of solution is often used intravenously to quickly rehydrate patients. A drop of the blood from the test tube was placed on a slide and red blood cells (RBCs) were observed under a microscope. Three possible outcomes are diagrammed below.



Which of the following best predicts which diagrammed microscope view the laboratory worker would see and best explains why?

(A) View 1 because RBC membranes are freely permeable to water

(B) View 2 because the RBCs use energy to allow sodium entry and to pump water out

(C) View 2 because the rate of water movement into the RBCs equals the rate of water movement out of the cells ✓

(D) View 3 because the sodium-potassium pumps in the RBC membranes use energy to keep the sodium out but allow water to freely flow into the cells

Answer C

This option is correct. The saline solution is isotonic relative to the RBC's cytoplasm. This will result in a dynamic equilibrium. View 2 shows a RBC in an isotonic environment because it would shrink in a hypertonic solution and swell in a hypotonic solution.

37. ATPase requires the use of ATP to move ions from one side of a cell membrane to another. The amino acid cysteine is important to the function of ATPase. A change in the cysteine prevents the ATPase from functioning.

Which of the following best describes what will happen to the movement of ions if there is a change in the cysteine of ATPase?

2 Cell MCQ

(A) Ions will move directly through the cell membrane.
(B) The electrochemical gradient will remain the same.
(C) The electrochemical gradient will change. ✓
(D) Ions will be unable to move through channel proteins.

38. If ATP breakdown (hydrolysis) is inhibited, which of the following types of movement across cell membranes is also inhibited?

(A) Movement of oxygen into a cell
(B) Movement of water through aquaporins
(C) Passage of a solute against its concentration gradient ✓
(D) Facilitated diffusion of a permeable substance

39. The antibiotic blasticidin S is a large molecule that can inhibit protein synthesis in mammalian cells. A researcher is investigating how blasticidin S enters mammalian cells. The researcher identified two proteins that might be associated with blasticidin S transport into the cell.

Which of the following claims most likely explains how the two proteins are directly involved in the antibiotic's ability to enter the cell?

(A) They form a receptor complex that selectively binds to blasticidin S in the cell cytoplasm.
(B) They are regulatory proteins that control the expression of other proteins involved in antibiotic transport.
(C) They are transport proteins that facilitate the movement of blasticidin S across the cell membrane. ✓
(D) They are enzymes that chemically modify blasticidin S to make it more soluble inside the cell.

40. Researchers have proposed a model of chloroplast evolution. According to the model, chloroplasts evolved from a small prokaryotic organism that was engulfed by an ancestral eukaryote. The engulfed prokaryote then formed an endosymbiotic relationship with the eukaryotic host.

Which of the following observations best supports the model?

(A) Chloroplasts are separated from other subcellular compartments by semipermeable membranes.
(B) Prokaryotic and eukaryotic organisms both acquire nutrients from the surrounding environment.
(C) Eukaryotes evolved after prokaryotes and have more complex structures.
(D) Chloroplasts and some prokaryotes share similar photosynthetic reactions. ✓

Answer D

Correct. Photosynthesis is a feature of only a limited number of organisms, including cyanobacteria, algae, and plants. The fact that this specialized process is similar in prokaryotes, such as cyanobacteria,

and in eukaryotes, such as algae and plants, provides evidence in support of the model.

41. Cholesterol is an important component of animal cell membranes. Cholesterol molecules are often delivered to body cells by the blood, which transports the molecules in the form of cholesterol-protein complexes. The complexes must be moved into the body cells before the cholesterol molecules can be incorporated into the phospholipid bilayers of cell membranes.

Based on the information presented, which of the following is the most likely explanation for a buildup of cholesterol molecules in the blood of an animal?

(A) The animal's body cells are defective in exocytosis.

(B) The animal's body cells are defective in endocytosis. ✓

(C) The animal's body cells are defective in cholesterol synthesis.

(D) The animal's body cells are defective in phospholipid synthesis.

Answer B

Correct. The cholesterol-protein complexes are most likely moved into the body cells by endocytosis. A defect in endocytosis is likely to result in a buildup of cholesterol in the blood.

42. When a substance moves across the plasma membrane along a concentration gradient at a rate faster than would be expected by simple diffusion alone but without the expenditure of metabolic energy, the process is best described as

(A) facilitated diffusion ✓

(B) osmosis

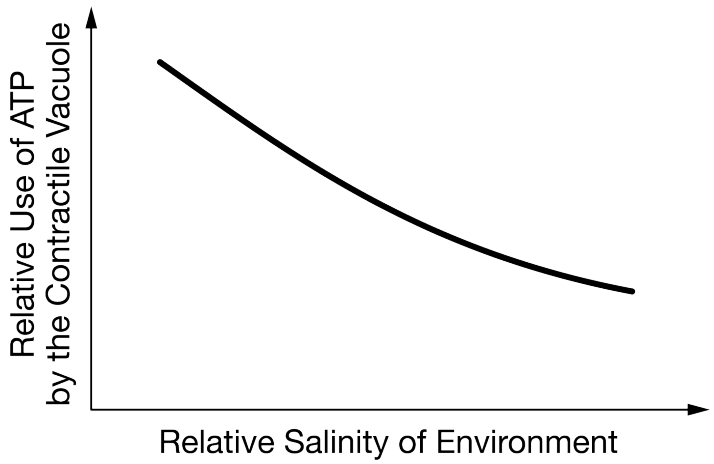
(C) active transport

(D) pinocytosis

(E) the work of an ion pump

2 Cell MCQ

43. Water is constantly diffusing into the cytosol of freshwater single-celled organisms. In order to maintain the proper solute concentrations in the cytosol, contractile vacuoles pump out the excess water. An experimenter placed single-celled organisms into various saline concentrations and recorded the ATP used by the contractile vacuole. The data are shown in the graph.



Of the following, which additional investigation can be used to determine when the cells are in an isotonic solution?

(A) Decreasing the salinity of the environment a little at a time until the ATP usage reaches a maximum

(B) Decreasing the salinity of the environment a little at a time until ATP usage reaches a minimum

(C) Increasing the salinity of the environment a little at a time until ATP usage reaches a maximum

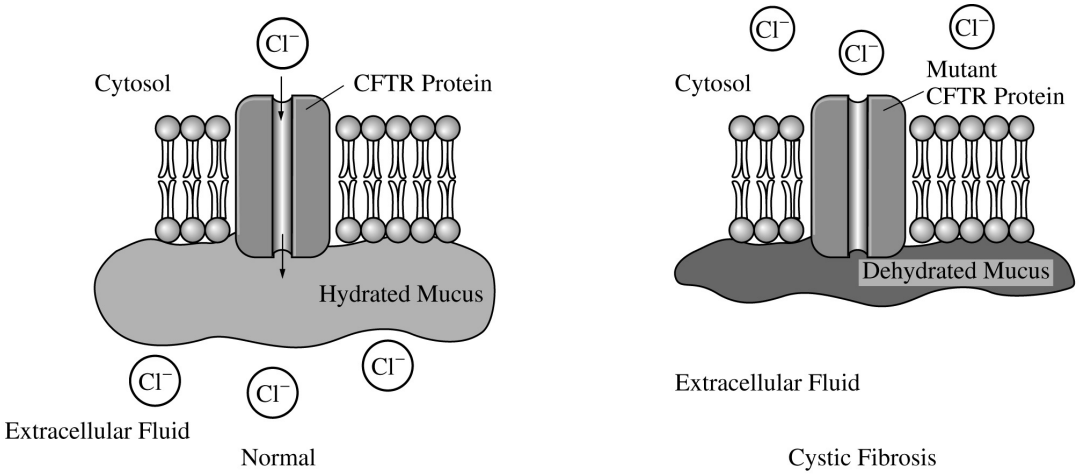
(D) Increasing the salinity of the environment a little at a time until the ATP usage reaches a minimum ✓

Answer D

Correct. If the environment is isotonic to the cell, the net flow of water due to diffusion is zero, and ATP is not used to pump water out of the cell.

2 Cell MCQ

44.



Cystic fibrosis is a recessively inherited disorder that results from a mutation in the gene encoding CFTR chloride ion channels located on the surface of many epithelial cells. As shown in the figure, the mutation prevents the normal movement of chloride ions from the cytosol of the cell to the extracellular fluid. As a consequence of the mutation, the mucus layer that is normally present on the surface of the cells becomes exceptionally dehydrated and viscous.

An answer to which of the following questions would provide the most information about the association between the CFTR mutation and the viscous mucus?

(A) Is the mucus also secreted from the cells through the CFTR proteins?

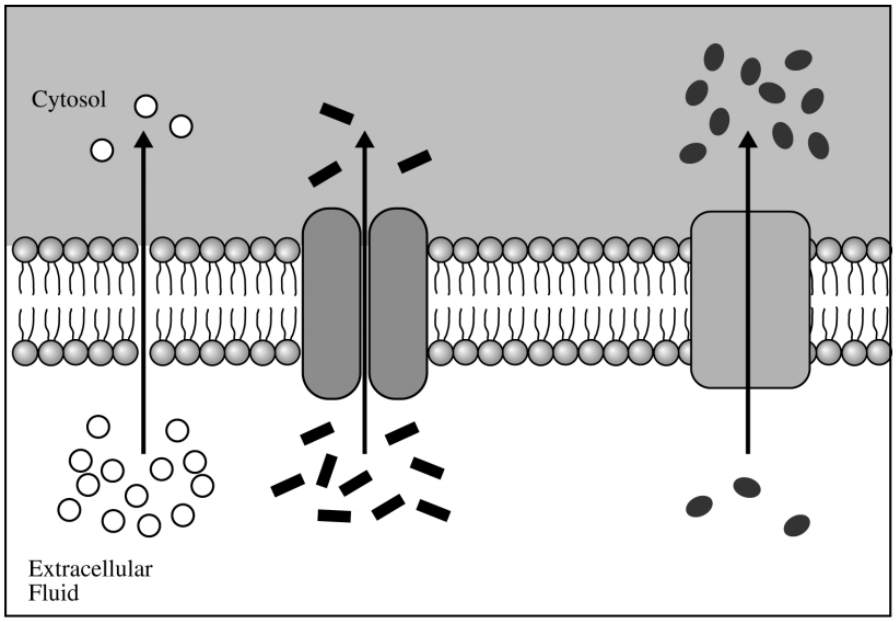
(B) How does the disrupted chloride movement affect the movement of sodium ions and water by the cell? ✓

(C) How does the mutation alter the structure of the CFTR proteins?

(D) What is the change in nucleotide sequence that results in the CFTR mutation?

2 Cell MCQ

45.



Which of the following scientific questions is most relevant to the model represented in the figure above?

(A) Is ATP required for the transportation of sugars across the outer mitochondrial membrane?

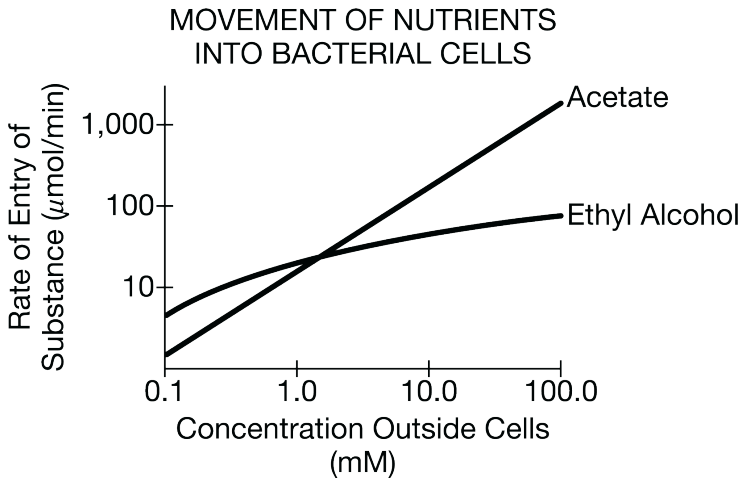
(B) Do the types of phospholipids in a membrane affect the rate at which molecules enter a cell by passive diffusion?

(C) Which molecular substance is actively transported across the plasma membrane? ✓

(D) How does temperature affect the movement of molecules into lysosomes?

2 Cell MCQ

46. Certain bacteria can use both ethyl alcohol and acetate as sources of nutrients. In an experiment where both nutrients are available to a bacterial population, the following results were obtained and graphed.



What additional procedure would best help determine whether these movements are due to active transport or to passive transport?

(A) Repeat the original experiment, but at three different temperatures. Compare the transport rates among the three temperatures.

(B) Repeat the original experiment, but add a substance known to block movement of molecules across aquaporins. Compare the rates on the two graphs.

(C) Use two additional treatments, one containing only ethyl alcohol and one containing only acetate. Compare the graphs of these two treatments with the original graph.

(D) Use two additional treatments, one containing only ethyl alcohol and one containing only acetate. Include a substance known to block ATP use by the plasma membrane. Compare the graphs of these two treatments to the original graph. ✓

Answer D

Correct. Active transport requires the availability of ATP. If the nutrients are actively transported, blocking ATP use should reduce the rate of transport along the whole concentration range. If these two graphs are the same as the original graph, the transport is passive. Separation of the nutrients eliminates the possibility that the two transport processes could interfere with each other.

2 Cell MCQ

47.

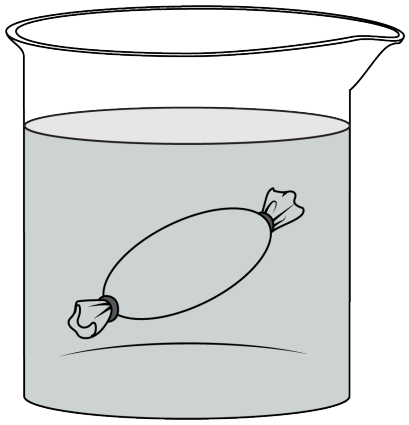


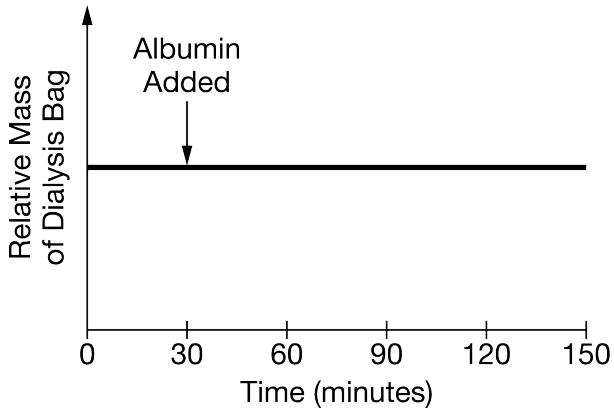
Figure 1

A student is using dialysis bags to model the effects of changing solute concentrations on cells. The student places one dialysis bag that contains $25\ \mathrm{mL}$ of distilled water into each of two beakers that are filled with $200\ \mathrm{mL}$ of distilled water. (Figure 1). The membrane of each dialysis bag membrane contains pores that allow small solutes such as monoatomic ions to pass through but are too small for anything larger to pass. After 30 minutes, $5\ \mathrm{mL}$ of a concentrated solution of albumin (a medium-sized, water-soluble protein) is added to one of the two beakers. Nothing is added to the other beaker. After two more hours at room temperature, the mass of each bag is determined. There is no change in the mass of the dialysis bag in the beaker to which no albumin was added.

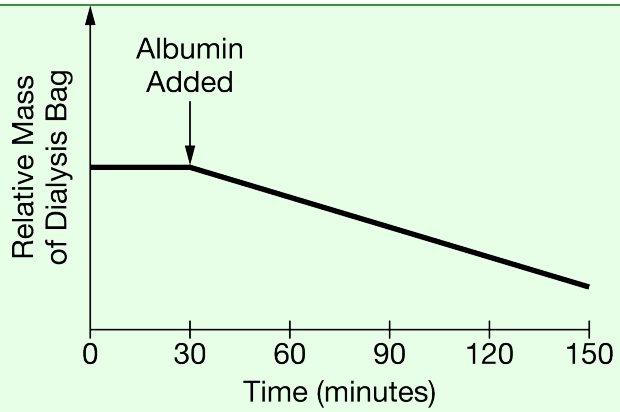
Which of the graphs below best represents the predicted change in mass over time of the dialysis bag in the beaker to which albumin was added?

2 Cell MCQ

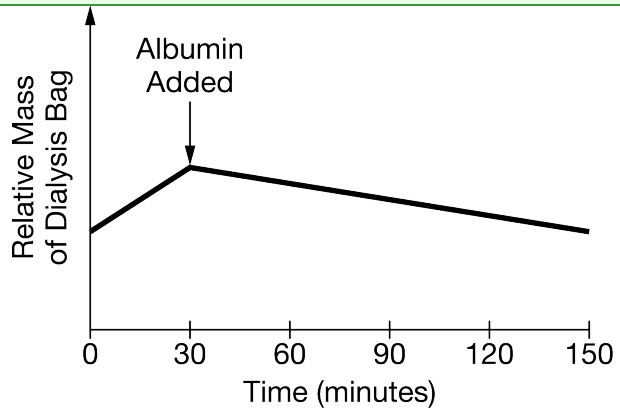
(A)



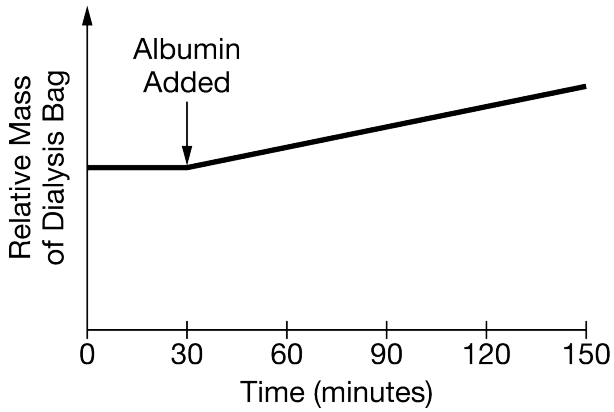
(B) ✓



(C)



(D)



2 Cell MCQ

48. Which of the following statements best supports the claim that certain organelles within eukaryotic cells evolved from free-living prokaryotic cells?

(A) The cytoplasm of both eukaryotes and prokaryotes is surrounded by a plasma membrane.

(B) Eukaryotes and prokaryotes both contain ribosomes, but the ribosomes of eukaryotes are more complex in structure than those of prokaryotes.

(C) Eukaryotes exchange segments of internal membranes between the endoplasmic reticulum and Golgi apparatus, but prokaryotes have no such internal membranes.

(D) Some organelles contain their own DNA that is more similar to prokaryotic DNA in structure and function than to the eukaryotic DNA found in the cell's nucleus. ✓

Answer D

Correct. The mitochondria and chloroplasts contain DNA more similar to prokaryote DNA in structure and function than to eukaryotic DNA. Eukaryotic DNA in the nucleus has additional structural proteins and a more complex process of protein synthesis. The most reasonable explanation involves the incorporation of one free-living prokaryotes that eventually integrated with the structures and processes of their eukaryotic hosts.

49. Which of the following is a characteristic of mitochondria and chloroplasts that supports the endosymbiotic theory?

(A) Both have bacteria-like polysaccharide cell walls.

(B) Both can reproduce on their own outside of the cell.

(C) Both contain DNA molecules. ✓

(D) Both contain endoplasmic reticulum and Golgi bodies.

(E) Both contain ribosomes that are identical to ribosomes of the eukaryotic cytoplasm.

2 Cell MCQ

50. Euglenids are single-cell eukaryotes that live in aquatic environments. The chloroplasts found inside euglenids are enveloped by three membranes, as represented in Figure 1. The inner membrane of euglenid chloroplasts resembles the thylakoid membrane.

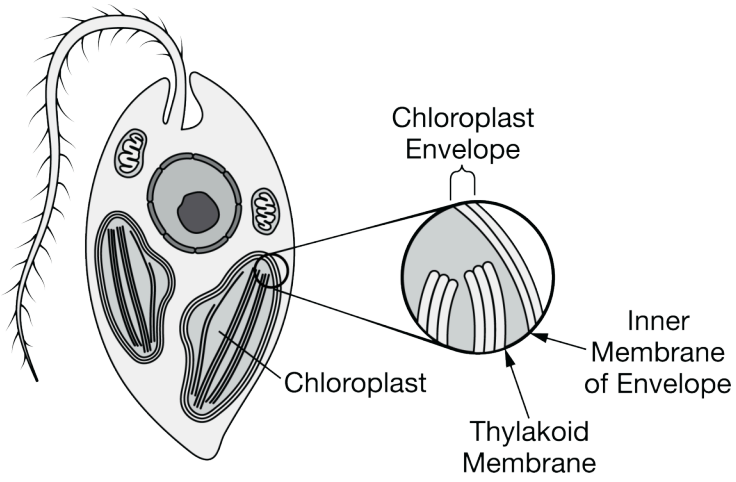


Figure 1. Simplified diagram of a euglenid, showing the structure of the chloroplast envelope

Which of the following claims about the origin of the euglenid chloroplast is best supported by the three-membrane structure of the envelope?

(A) It originated from the spontaneous assembly of organic molecules into a lipid bilayer inside a free-living prokaryote.

(B) It originated from the fusion of the plasma membranes of two different free-living photosynthetic prokaryotes.

(C) It originated from the incorporation of a photosynthetic prokaryote into a eukaryotic cell by a single endosymbiotic event.

(D) It originated from the incorporation of a photosynthetic prokaryote into a eukaryotic cell by two endosymbiotic events. ✓

Answer D

Correct. The three-membrane structure of the chloroplast envelope best supports the claim presented in this answer choice. The three-membrane structure most likely resulted from two endosymbiotic events, with the middle membrane originating from the first event and the outer membrane originating from the second event.

2 Cell MCQ

51. Gaucher disease is an inherited disorder in which cells of the body are unable to break down a particular type of lipid, resulting in a buildup of the lipid in some tissues and organs.

Based on the information provided, Gaucher disease results most directly from a defect in the function of which of the following organelles?

(A) The smooth endoplasmic reticulum

(B) The nucleus

(C) The lysosome ✓

(D) The mitochondrion

Answer C

Correct. The lysosome contains specific enzymes used to break down a variety of molecules and cellular waste products. A defect in the function of the lysosomal enzymes that are needed to break down lipids is the most direct cause of Gaucher disease.

52. Lysosomes digest food particles brought into a cell by endocytosis. After a vesicle containing food particles fuses with a lysosome, H^+ ions are transported into the lysosome from the cytosol. This significantly lowers the pH of the lysosome relative to the cytosol and activates the enzymes that digest the particles.

Which of the following best predicts what will happen to the lysosomal enzymes if the proteins that transport H^+ ions from the cytosol into the lysosome are damaged?

(A) The lysosomal enzymes will not become active, since there will be no active transport of H^+ ions. ✓

(B) The lysosomal enzymes will not become active, since H^+ ions will diffuse out of the lysosome.

(C) The lysosomal enzymes will become active, since facilitated diffusion will move H^+ ions into the lysosome.

(D) The lysosomal enzymes will become active, since passive diffusion will move H^+ ions into the lysosome.

Answer A

Correct. Since the pH of the lysosome has to become lower than that of the cytosol, active transport is needed to move H^+ ions into the lysosome, against the H^+ ion concentration gradient, to activate the enzymes.

2 Cell MCQ

53. Changing the shape or morphology of the mitochondrial inner membrane can change the efficiency of mitochondrial function.

Which of the following outcomes will most likely result from a change in the shape of the mitochondrial inner membrane from a highly folded surface to a smooth, flat surface?

(A) Mitochondria will become more efficient because the inner mitochondrial membrane will become more permeable to ions.

(B) Mitochondria will become more efficient because the total volume of the mitochondria will increase.

(C) Mitochondria will become less efficient because the inner mitochondrial membrane will become less permeable to large molecules.

(D) Mitochondria will become less efficient because the surface area of the inner mitochondrial membranes will decrease. ✓

Answer D

Correct. The change to the inner mitochondrial membrane will result in a decrease in the membrane's surface area, which will reduce the efficiency of mitochondrial function because there will be less surface area for reactions to occur.

2 Cell MCQ

Stickleback fish are found in both marine and freshwater habitats. The marine fish have no scales but have hardened, armorlike plates along their sides. The plates are thought to protect sticklebacks from certain predators.

In the late 1980s, sticklebacks from a marine population colonized Loberg Lake, a freshwater lake in Alaska. Starting in 1990, researchers sampled fish from the lake every four years and recorded the armor-plate phenotypes of the male sticklebacks in each sample. The armor-plate phenotypes were categorized as either complete (plates extending from head to tail), partial (plates extending from head to abdomen), or low (a few plates near the head only). The results are shown in the table below.

ARMOR-PLATE VARIATION IN THE STICKLEBACK POPULATION OF LOBERG LAKE

Percent of Males in the Sample with Each Armor-Plate Phenotype

Year	Low	Partial	Complete
1990	1%	2%	97%
1994	45%	14%	41%
1998	58%	16%	26%
2002	76%	15%	9%
2006	90%	6%	4%

54. Intact cells of two unknown cell types were placed into solutions with different concentrations of NaCl. Type I cells swelled and burst in the solution with the lowest concentration of NaCl. Type II cells swelled but did not burst in the solution with the lowest concentration of NaCl.

Which of the following descriptions of cell type I and cell type II are most consistent with the data?

2 Cell MCQ

(A)

Cell Type I	Cell Type II
Animal cell surrounded by a plasma membrane only	Plant cell surrounded by a plasma membrane and a cell wall

(B)

Cell Type I	Cell Type II
Plant cell surrounded by a plasma membrane only	Bacterial cell surrounded by a cell wall only

(C)

Cell Type I	Cell Type II
Plant cell surrounded by a plasma membrane only	Animal cell surrounded by a plasma membrane and a cell wall

(D)

Cell Type I	Cell Type II
Animal cell surrounded by a cell wall only	Bacterial cell surrounded by a plasma membrane only

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55. Which of the following statements accurately describes the facilitated diffusion of materials across a cell membrane?

(A) Energy is required to move charged substances across the membrane.

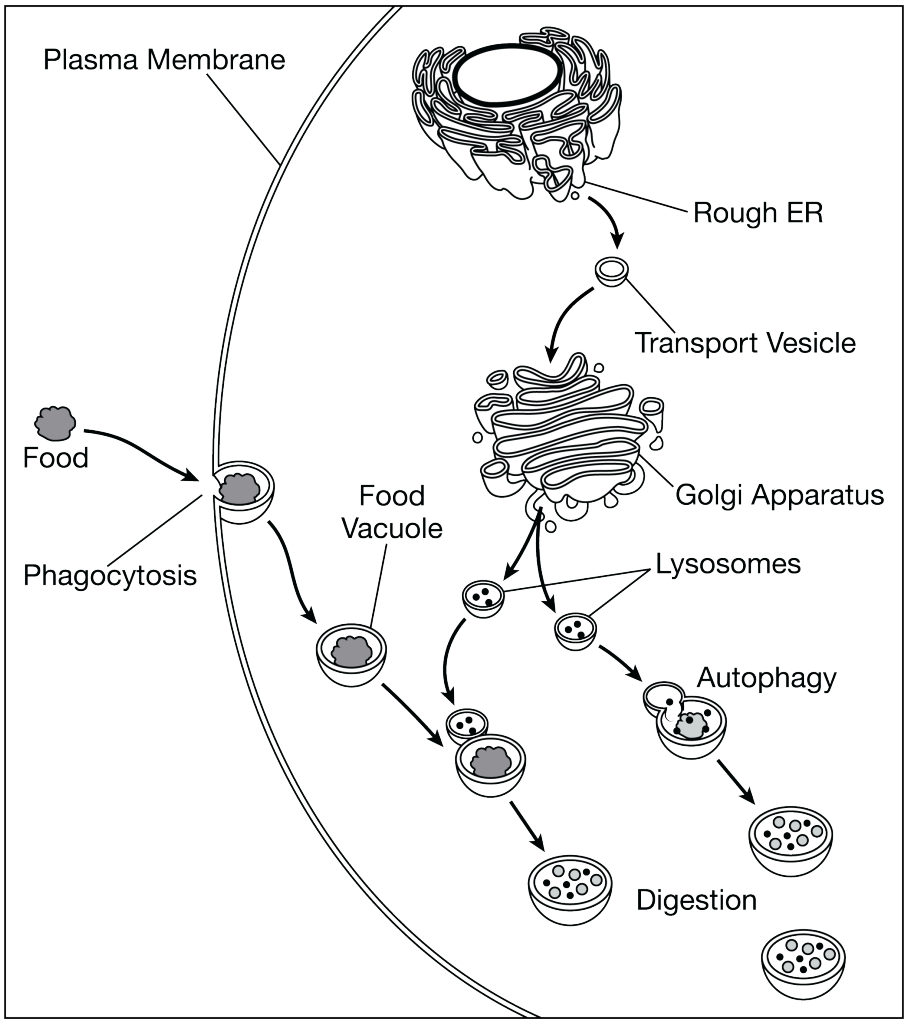
(B) Small, polar molecules require transport proteins to cross the membrane.

(C) Oxygen molecules move directly across the membrane.

(D) Ions move down their concentration gradient across the membrane.

2 Cell MCQ

56. The figure shows a process by which a cell might absorb food from its surrounding environment and break it down for use as a source of energy and matter. The process involves lysosomes, which are membrane-bound organelles that contain hydrolytic enzymes. Activation of the hydrolytic enzymes requires an acidic pH, and lysosomes maintain an internal acidic pH by using ion pumps.



Which of the following outcomes will most likely result from a loss of ion pump function in the cell's lysosomes?

(A) The internal pH of the lysosomes will decrease, which will prevent the activation of hydrolytic enzymes and interfere with the intracellular digestion of food.

(B) The internal pH of the lysosomes will increase, which will prevent the activation of hydrolytic enzymes and interfere with the intracellular digestion of food. ✓

(C) The internal pH of the lysosomes will decrease, which will activate hydrolytic enzymes and enhance the intracellular digestion of food.

(D) The internal pH of the lysosomes will increase, which will activate hydrolytic enzymes and enhance the intracellular digestion of food.

2 Cell MCQ

Answer B

Correct. The lysosomes rely on the ion pumps to maintain an internal acidic pH. A loss of ion pump function will most likely result in an increase in the internal pH of lysosomes from acidic to neutral, which will prevent the activation of hydrolytic enzymes and interfere with intracellular digestion of food.

57. Some viral infections can lead to the rupture of the lysosome membrane. Which prediction of the effect of this disruption of cellular compartmentalization is most likely correct?

(A) Enzymes will be released that will specifically target the virus.

(B) Cellular osmotic concentrations will change, preventing viral entry into the cell.

(C) Hydrolytic enzymes will be released, which will cause cell death. ✓

(D) Intracellular digestion of organic materials will increase, which will increase the energy available to the cell for fighting the virus.

Answer C

Correct. Hydrolytic enzymes will be released, resulting in cell death and preventing further viral reproduction.

2 Cell MCQ

58. A cell's membrane potential is maintained by the movement of ions into and out of the cell. A model showing the influence of membrane proteins on the movement of sodium (Na^+) and potassium (K^+) ions across the plasma membrane is presented in Figure 1.

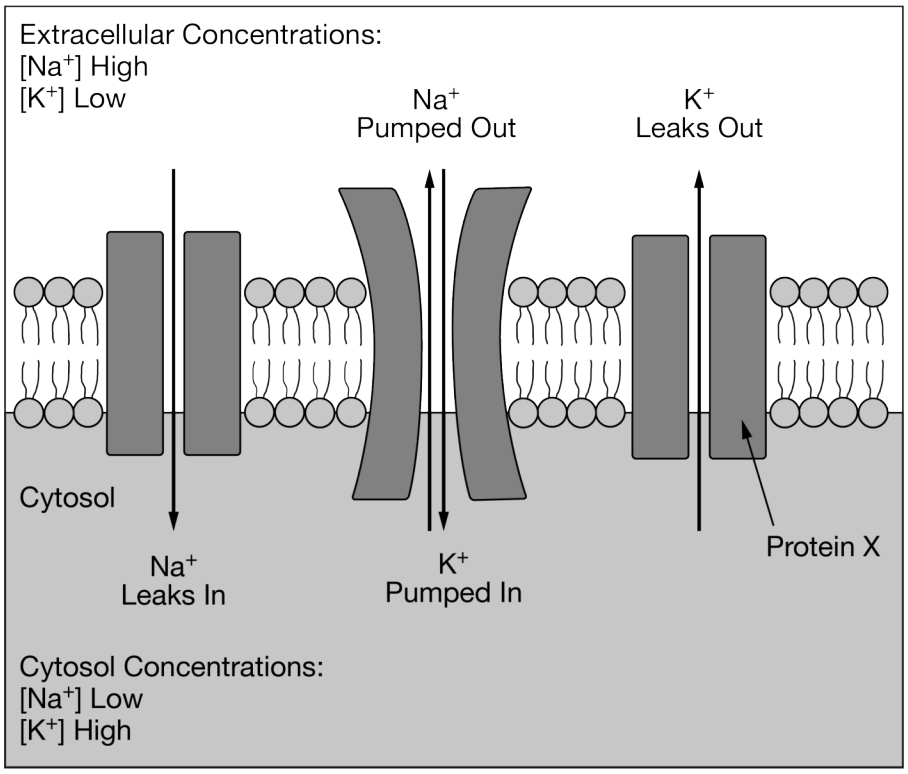


Figure 1. Section of a cell's plasma membrane, showing ion concentrations and membrane proteins

Based on the model presented in Figure 1, which of the following outcomes will most likely result from a loss of protein X function?

(A) The membrane potential will be disrupted by an increase in Na^+ concentration inside the cell.

(B) The membrane potential will be disrupted by an increase in K^+ concentration inside the cell. ✓

(C) The membrane potential will be maintained by the $Na^+ - K^+$ pump moving more K^+ ions into the cell.

(D) The membrane potential will be maintained by the diffusion of Na^+ ions into the cell.

Answer B

Correct. The intracellular K^+ concentration will increase because the $Na^+ - K^+$ pump will continue to

transport K^+ into the cell without any diffusing out of the cell. As a result, the cell's membrane potential will be disrupted.

59. Researchers investigate the transport of a certain protein into cells by endocytosis. In an experiment, the researchers incubate the cells in the presence of the protein and measure the amount of the protein that is absorbed into the cells over a five-minute period.

Based on their observations, what should the researchers do to further clarify how the availability of the protein outside the cells affects the rate of endocytosis of the protein?

(A) Incubate the cells in the absence of the protein.

(B) Incubate the cells in the presence of several different proteins.

(C) Incubate the cells in the presence of several different concentrations of the protein. ✓

(D) Incubate the cells in the presence of the protein for several different lengths of time.

Answer C

Correct. Changing the concentration of the protein will change the availability of the protein outside the cells.

60. Membrane-bound organelles have been an important component in the evolution of complex, multicellular organisms. Which of the following best summarizes an advantage of eukaryotic cells having internal membranes?

(A) Eukaryotic cells are able to reproduce faster because of the presence of organelles.

(B) Some organelles, such as mitochondria and chloroplasts, are similar to prokaryotic cells in structure.

(C) Organelles isolate specific reactions, increasing metabolic efficiency. ✓

(D) Compartmentalization leads to a higher mutation rate in DNA, which leads to more new species.

Answer C

This option is correct. It demonstrates understanding and provides an explanation of how internal membrane-bound organelles contribute to cell function by providing a favorable local environment for metabolic reactions, and by protecting the cell from potentially damaging metabolic reactions.

61. Mitochondria are found in most eukaryotic cells and contain their own DNA and ribosomes that are similar to those typical of many prokaryotic cells. Which of the following statements is justified by these observations?

2 Cell MCQ

(A) The mitochondrion is the only location in which eukaryotic cells can synthesize ATP.

(B) An ancestral cell most likely engulfed an aerobic prokaryote in a relationship that proved beneficial for both cells. ✓

(C) Mitochondrial membranes provide abundant surface area for reactions because of the infoldings called cristae.

(D) The mitochondrion plays a role in respiration, but it also stores the cell's extra mRNA.

62. The endosymbiont theory proposes a model for the evolution of mitochondria. According to the model, an ancestral eukaryote engulfed a small, free-living prokaryotic organism. The engulfed prokaryote then formed an endosymbiotic relationship with the eukaryotic host.

Which of the following observations best supports the model?

(A) Prokaryotes and eukaryotes acquire nutrients from the surrounding environment.

(B) Organelles such as mitochondria and the endoplasmic reticulum have membranes composed of phospholipids.

(C) Mitochondria and some prokaryotes share similar metabolic reactions that produce ATP. ✓

(D) Eukaryotes evolved after prokaryotes and have more complex structures.

Answer C

Correct. The observation that mitochondria and some prokaryotes share similar metabolic reactions that produce ATP supports the model that mitochondria are endosymbionts that originated from a free-living prokaryote.

63. Which of the following observations best supports the claim that mitochondria evolved from once-free-living prokaryotic cells by the process of endocytosis?

(A) Mitochondria produce ATP.

(B) Mitochondria contain proteins.

(C) Mitochondria exchange substances with the cytosol.

(D) Mitochondria are surrounded by a double membrane. ✓

Answer D

Correct. The double membranes of mitochondria provide evidence that an ancestor of mitochondria, which was most likely a type of free-living aerobic bacterium, was ingested via endocystosis by a primitive eukaryotic cell.

2 Cell MCQ

64. Which of the following statements best predicts the effect of increasing the permeability of the mitochondrial membranes to large molecules?

(A) ATP production will increase because of an increase in the rate at which proteins diffuse out of mitochondria.

(B) ATP production will increase because of an increase in the mixing of mitochondrial and cytosolic substances.

(C) ATP production will decrease because of an increase in the occurrence of uncontrolled chemical reactions. ✓

(D) ATP production will decrease because of an increase in the surface area of the mitochondrial membranes.

Answer C

Correct. ATP production will most likely decrease. Increasing the permeability of the mitochondrial membranes to large molecules will likely result in a variety of uncontrolled substrate-enzyme interactions that will disrupt normal mitochondrial function.

65. Which of the following accurately describes a model of active transport?

(A) A membrane channel protein facilitates the movement of molecules from an area of high concentration on the outside of the cell to an area of lower concentration on the inside.

(B) A cell in a hypotonic solution swells as water moves into it.

(C) A membrane protein binds a specific molecule outside of the cell and moves the molecule into the cell where there is already a high concentration of the molecule compared with the outside. ✓

(D) A membrane protein binds a specific molecule on the inside of the cell and moves the molecule to an area of lower concentration on the outside of the cell.

Answer C

Correct. Active transport uses a membrane protein to move a specific molecule up the concentration gradient from an area of low concentration to an area of high concentration.

66. The Na^+/K^+ pump requires ATP to move the ions from one side of a membrane to the other. Which of the following explains the role ATP plays in the pump's activity?

2 Cell MCQ

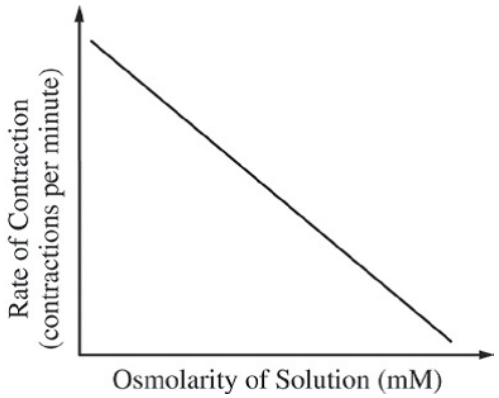
(A) ATP provides energy for the ions to move down the concentration gradient.

(B) ATP adds a phosphate to the K^+ ion so that it is not charged and can be moved by the pump.

(C) ATP provides energy for the ions to move against the concentration gradient. ✓

(D) ATP adds a phosphate to the Na^+ ion so that it is not charged and can be moved by the pump.

67.



Paramecia are unicellular protists that have contractile vacuoles to remove excess intracellular water. In an experimental investigation, paramecia were placed in salt solutions of increasing osmolarity. The rate at which the contractile vacuole contracted to pump out excess water was determined and plotted against osmolarity of the solutions, as shown in the graph. Which of the following is the correct explanation for the data?

(A) At higher osmolarity, lower rates of contraction are required because more salt diffuses into the paramecia.

(B) The contraction rate increases as the osmolarity decreases because the amount of water entering the paramecia by osmosis increases. ✓

(C) The contractile vacuole is less efficient in solutions of high osmolarity because of the reduced amount of ATP produced from cellular respiration.

(D) In an isosmotic salt solution, there is no diffusion of water into or out of the paramecia, so the contraction rate is zero.

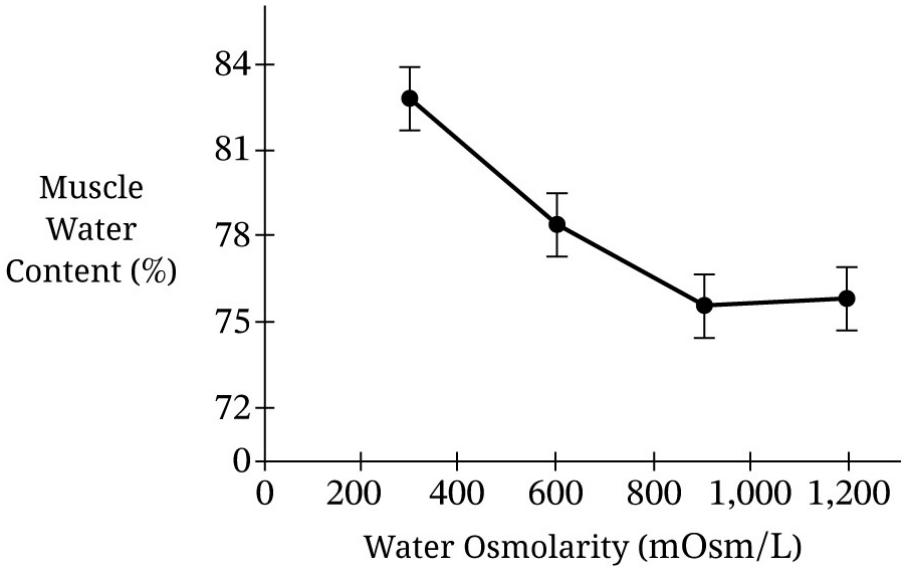
Answer B

This option is correct. The lower the concentration of solute, the larger the gradient between the paramecium and the solution. When the paramecium is hypertonic to the low concentration solution, water will enter the paramecium by osmosis. If the paramecium does not pump excess water back out, it will burst. As the solution gradient decreases, less water will enter, and, therefore, there will be a decreased need to pump excess water out.

2 Cell MCQ

68. Researchers studying osmoregulation in a specific crab species measured the change in water content within the crabs' muscles when the crabs were exposed to water with different salt concentrations (Figure 1).

Figure 1. Muscle water content in crabs at different saltwater osmolarities. Error bars represent $\pm 2SE_{\bar{x}}$



Which of the following claims about the crab's muscle water content is most likely supported by the data shown in Figure 1?

(A) The water content remains relatively constant across different levels of water osmolarity.

(B) The water content increases and then decreases as water osmolarity increases.

(C) The water content decreases as water osmolarity increases. ✓

(D) The water content increases as water osmolarity increases.

2 Cell MCQ

69. **TABLE 1. INTERNAL AND EXTERNAL WATER POTENTIALS OF FOUR DIFFERENT PARAMECIA**

Paramecium	Internal Ψ	External Ψ
1	-1.44 MPa	-0.08 MPa
2	-1.56 MPa	-1.01 MPa
3	-1.33 MPa	-1.76 MPa
4	-0.37 MPa	-1.73 MPa

A paramecium is a single-celled organism that contains one or more organelles called contractile vacuoles. In its freshwater environment, water may accumulate inside the paramecium and the contractile vacuoles. Contraction of the vacuoles removes excess water from the organism, thus preventing the paramecium from bursting. Table 1 provides internal and external water potentials (Ψ) for four different paramecia living in four different environments.

In which paramecium in Table 1 will the contractile vacuoles most likely have the highest contraction rate?

(A) Paramecium 1 ✓

(B) Paramecium 2

(C) Paramecium 3

(D) Paramecium 4

Answer A

Correct. Paramecium 1 has the largest difference of the four paramecia between the internal and external water potential, and the internal water potential is more negative than the external is. This will cause a net movement of water into the paramecium and will require the contractile vacuole to contract more in order to keep the paramecium from bursting.

70. Which of the following statements best explains the processes of passive and active transport?

2 Cell MCQ

(A) Passive transport is the net movement of substances down a concentration gradient that requires metabolic energy. Active transport is the movement of substances up a concentration gradient that does not require energy.

(B) Passive transport is the net movement of substances down a concentration gradient that does not require metabolic energy. Active transport is the movement of substances up a concentration gradient that requires energy. ✓

(C) Passive transport is the net movement of substances up a concentration gradient that requires metabolic energy. Active transport is the movement of substances down a concentration gradient that does not require metabolic energy.

(D) Passive transport is the net movement of substances up a concentration gradient that does not require metabolic energy. Active transport is the movement of substances down a concentration gradient that requires energy.

Answer B

Correct. Passive transport is the net movement of substances down (with) a concentration gradient that does not require metabolic energy and active transport is the movement of substances up (against) a concentration gradient that requires energy.

71. In an experiment, cells were isolated from an aquatic plant and suspended in pond water, a sucrose sugar solution, or distilled water. All of the cells were then viewed under a microscope. Compared with the cell in the pond water, the cell in the sugar solution appeared shriveled, and the cell in the distilled water appeared inflated. The results of the experiment are represented in Figure 1.

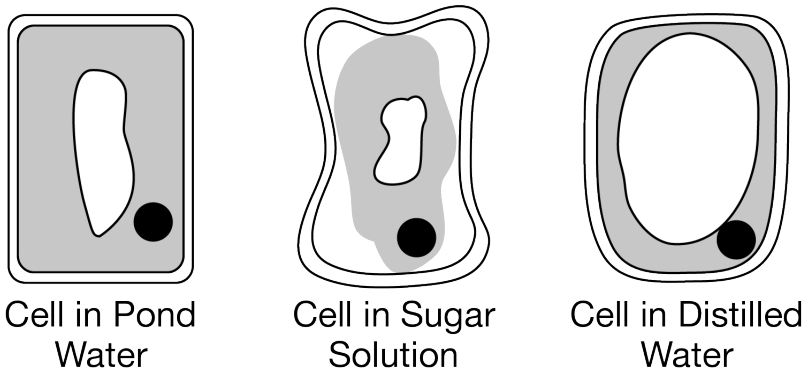


Figure 1. The results of an experiment using aquatic plant cells

Which of the following statements best explains the observations represented in Figure 1 ?

2 Cell MCQ

(A) There was a net movement of sucrose out of the cell suspended in the sugar solution and a net movement of sucrose into the cell suspended in the distilled water.

(B) There was a net movement of sucrose into the cell suspended in the sugar solution and a net movement of sucrose out of the cell suspended in the distilled water.

(C) There was a net movement of water out of the cell suspended in the sugar solution and a net movement of water into the cell suspended in the distilled water. ✓

(D) There was a net movement of water into the cell suspended in the sugar solution and a net movement of water out of the cell suspended in the distilled water.

Answer C

Correct. The plasma membrane that surrounds a living cell is typically permeable to water but not sucrose. In the experiment represented in Figure 1, there was a net movement of water out of the cell suspended in the sugar solution, causing the cell to appear shriveled, and a net movement of water into the cell suspended in the distilled water, causing the cell to appear inflated.

72. Which of the following are characteristic of both prokaryotic and eukaryotic cells?

(A) Cytoplasm and a well-defined nucleus surrounded by a membrane

(B) Membranous sites of ATP synthesis, Golgi complex, and ribosomes

(C) Mitochondria, nucleus, and ribosomes

(D) Cell wall, several chromosomes, and cytoplasm

(E) Cell membrane, ribosomes, DNA, and RNA ✓

73. Which of the following groups of cellular components are found in eukaryotic cells but not prokaryotic cells?

(A) Ribosomes, a nucleus, and chloroplasts

(B) Circular chromosomes, mitochondria, and an endoplasmic reticulum

(C) A nucleus, ribosomes, and cell walls

(D) An endoplasmic reticulum, mitochondria, and a nucleus ✓

2 Cell MCQ

74.

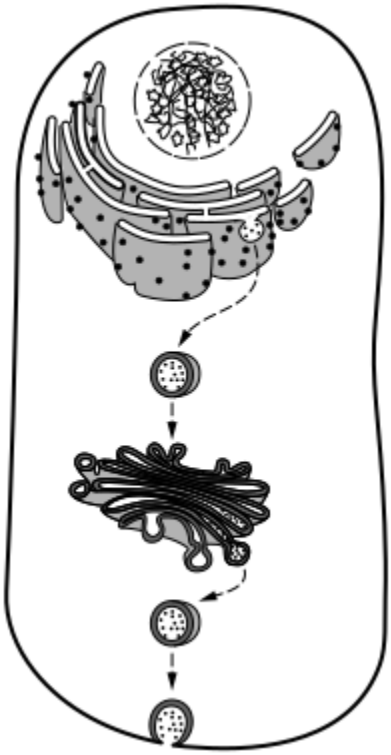


Figure 1. The path of insulin export from a cell

Insulin is a protein hormone that helps to regulate blood sugar levels. Figure 1 shows the path of insulin from its production in a cell to its export out of the cell.

Based on Figure 1, which of the following best describes how insulin leaves the cell?

(A) A transport protein carries insulin from the cytosol through the plasma membrane by facilitated diffusion.

(B) A transport protein in the plasma membrane moves insulin by active transport from the Golgi complex through the plasma membrane.

(C) A vesicle containing insulin moves through the plasma membrane by simple diffusion.

(D) A vesicle containing insulin fuses with the plasma membrane, enabling exocytosis of insulin. ✓

Answer D

Correct. Because insulin is secreted from the cell, it moves progressively through the components of the endomembrane system (rough endoplasmic reticulum and Golgi complex). This involves packaging insulin in a Golgi-derived transport vesicle, which then fuses with the plasma membrane, releasing

insulin by exocytosis.

75. The transport of a substance across a plasma membrane of a specific organelle requires energy. The rate at which the transport takes place also depends on temperature. A scientist isolated the specific organelle and then used the following treatments to determine the conditions that will result in the maximal transport. All treatments contained the extracted organelle and were maintained at $25^\circ\mathrm{C}$.

The data from this experiment indicate that maximal rate of transport of protein X at $25^\circ\mathrm{C}$ occurs at an ATP concentration of $1.0\ \mu\mathrm{m/mL}$.

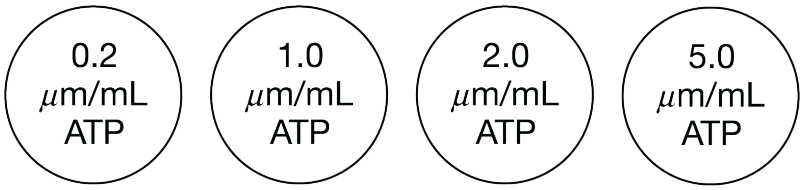


Figure 1. The four ATP concentrations used in the experiment

Which procedure should be done next to gather data needed to meet the scientist's objective?

(A) Incubate samples with the same four ATP concentrations at $30^\circ\mathrm{C}$.

(B) Incubate samples containing $5.0\ \mu\mathrm{m/mL}$ of ATP at four temperatures other than $25^\circ\mathrm{C}$.

(C) Incubate samples containing $1.0\ \mu\mathrm{m/mL}$ of ATP at four temperatures other than $25^\circ\mathrm{C}$. ✓

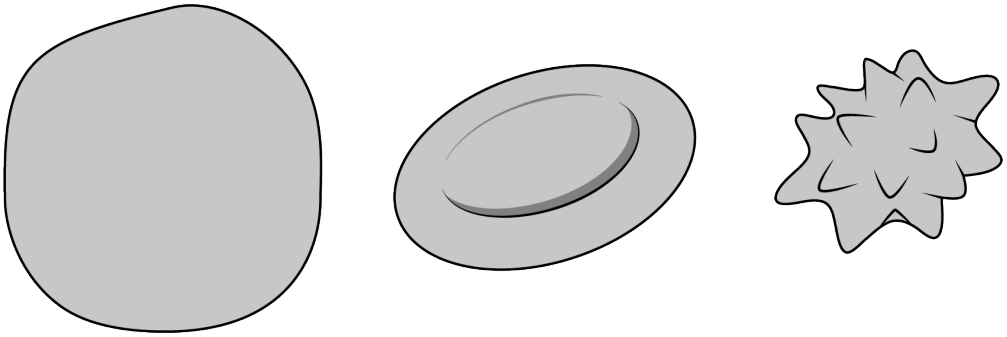
(D) Incubate samples containing $1.0\ \mu\mathrm{m/mL}$ of ATP at $25^\circ\mathrm{C}$ and determine the rate of transport for four other proteins.

Answer C

Correct. This procedure combines the optimal ATP concentration for transport with data for transport rates at four other temperatures.

2 Cell MCQ

76. Intravenous (IV) therapy is used for fluid replacement in instances of dehydration in humans and other animals. One type of IV fluid is essentially a saltwater solution. To determine the best concentration for therapy in people, a team of students is researching the effects of solutions of different salt concentrations on red blood cells. The following observations were made from three different red blood cell samples viewed under a microscope.



0.3% Saline (Cells swell) 0.9% Saline (Cells unchanged) 1.5% Saline (Cells shrink)

The team wants to extend the research project. What should the team of students do next to obtain data that are more conclusive?

(A) Repeat the process with other salt concentrations. ✓

(B) Develop a model to explain why the cells react differently to different salt concentrations.

(C) Repeat the process using red blood cells from other animals.

(D) Develop an experimental procedure that uses a stain that makes the organelles of red blood cells more visible.

Answer A

Correct. The figures illustrate the extremes of hypotonic (swollen cells) and hypertonic (shriveled cells) environments, so an extension would be to investigate the small range of acceptable salt concentrations in IV fluid solutions.

77. The salinity of a small inland lake has recently started to increase. Researchers are planning to study the lake over several decades to investigate how freshwater organisms survive significant changes in their natural habitat. Which of the following physiological mechanisms will the researchers most likely observe among the surviving organisms in the lake?

2 Cell MCQ

(A) Prokaryotic organisms will use various mechanisms to counteract swelling of cells as a result of increased water uptake.

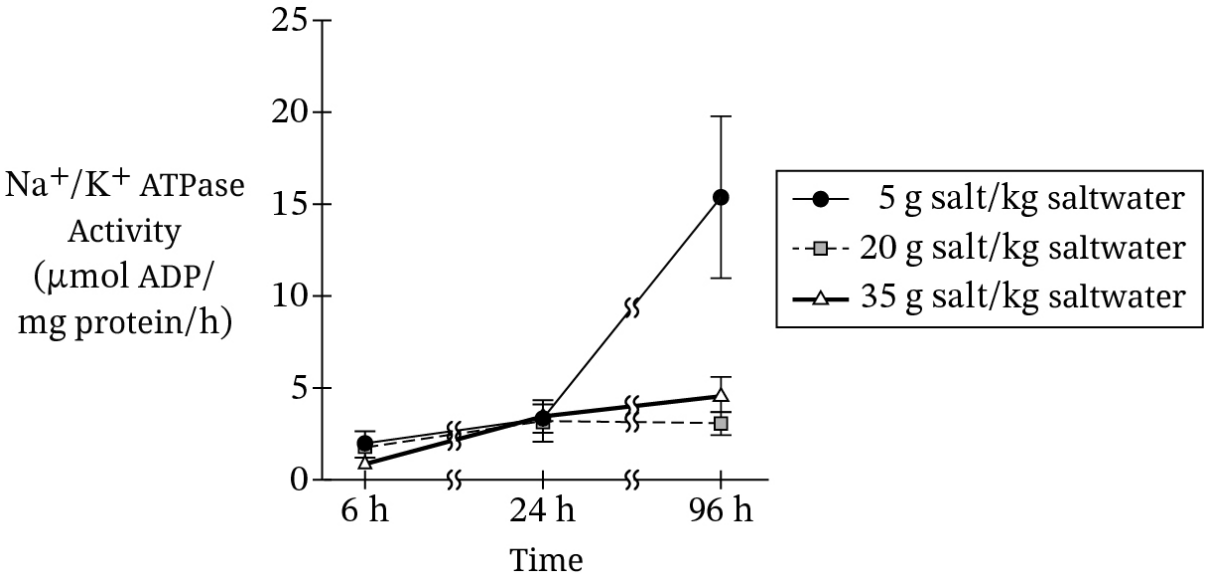
(B) Single-celled organisms will use various mechanisms to counteract the increased flow of water from cells to the environment. ✓

(C) Eukaryotic organisms will use various mechanisms to counteract the diffusion of positively charged ions across the cell membrane.

(D) Multicellular organisms will use various mechanisms to counteract the loss of cell adhesion as a result of calcium deficiencies.

78. A saltwater fish has specialized cells that contain Na^+/K^+ATPase to help with osmoregulation. Scientists studying this fish investigated the effect of the different concentrations of salt in the saltwater on the activity of Na^+/K^+ATPase between six and 96 hours (Figure 1).

Figure 1. Activity of Na^+/K^+ATPase at different saltwater concentrations over time. Error bars represent $\pm SE_{\overline{x}}$



Which of the following statements accurately supports the data shown in Figure 1?

(A) \(Na^+/K^+\)ATPase activity increases steadily as saltwater concentration and time increase.

(B) \(Na^+/K^+\)ATPase activity increases steadily as saltwater concentration decreases and time increases.

(C) \(Na^+/K^+\)ATPase activity stays approximately the same throughout the experiment and at all saltwater concentrations.

(D) \(Na^+/K^+\)ATPase activity increases only after 24 hours at the lowest saltwater concentration. ✓

Answer D

Correct. The data show that a significant increase in ATPase activity occurs only in the \(5\ \text{g}\ salt\

2 Cell MCQ

/\text{kg}\ water\) (lowest) concentration condition and only after the 24-hour mark. The other concentration treatments (20 and \(35\ \text{g}\ salt\ /\text{kg}\ water\)) show minimal changes over time.

79. Simple diffusion and facilitated diffusion are related in that both

(A) require protein carriers

(B) depend on a concentration gradient ✓

(C) occur via contractions of cytoskeletal elements attached to membrane proteins

(D) are endergonic processes and thus require the hydrolysis of ATP

(E) occur in eukaryotic cells but not in prokaryotic cells

80. Which of the following best supports the endosymbiotic theory?

(A) Mitochondria and chloroplasts have their own genomes. ✓

(B) Enzymes used in photosynthesis differ from those in cellular respiration in all eukaryotes.

(C) Bacteria and plants both have cell walls.

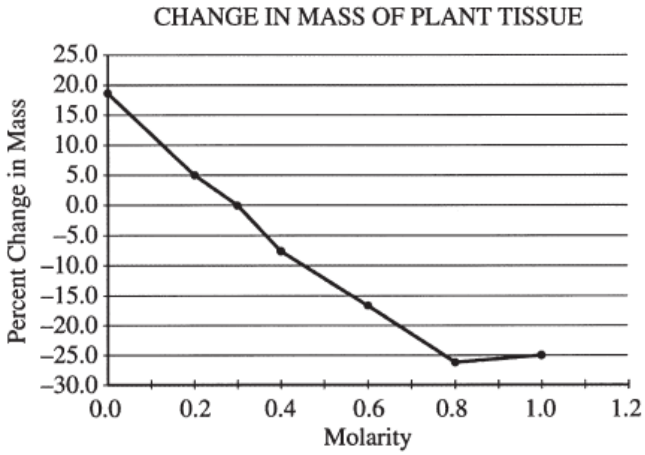
(D) Bacterial flagella are structurally different from eukaryotic flagella.

Answer A

Correct. The endosymbiotic theory claims that mitochondria and chloroplasts originated from early prokaryotic cells being engulfed and incorporated by another cell. These engulfed cells would have a double membrane due to endocytosis and would retain some of their original genomes. Mitochondria and chloroplasts possess both of these characteristics.

2 Cell MCQ

81. The following question refer to the graph below, which illustrates the percent change in the mass of pieces of plant tissue placed in solutions of different sucrose molarities.



The approximate molarity of the solution in which the mass of the plant pieces would not change is

(A) 0.0 *M*

(B) 0.1 *M*

(C) 0.3 *M* ✓

(D) 0.5 *M*

(E) 0.7 *M*

82. A magnesium sulfate solution taken orally can cause a net movement of water into the large intestine, which results from water molecules diffusing through aquaporins embedded in the cells of the intestinal lining.

By which of the following mechanisms do the water molecules most likely move into the large intestine?

(A) By passive transport from an area of low osmolarity to an area of high osmolarity ✓

(B) By passive transport from an area of high osmolarity to an area of low osmolarity

(C) By active transport from an area of low osmolarity to an area of high osmolarity

(D) By active transport from an area of high osmolarity to an area of low osmolarity

Answer A

Correct. Based on the information presented, the water molecules move through aquaporins by diffusing from an area of low osmolarity to an area of high osmolarity, which is an example of passive transport.

2 Cell MCQ

83.

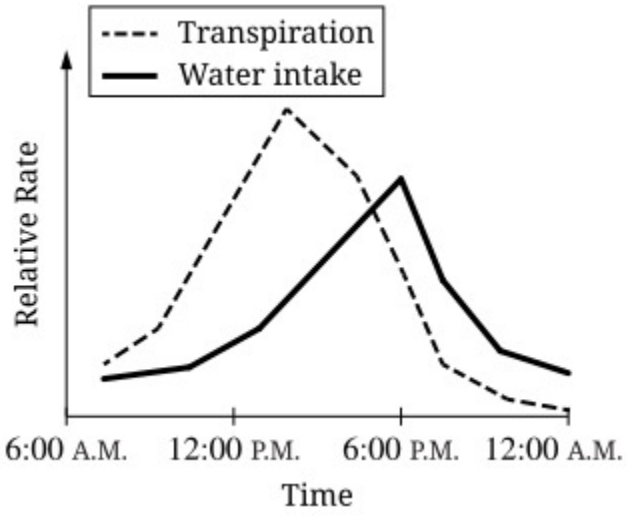


Figure 1. Relative rates of transpiration and water intake for a plant over 18 hours

Figure 1 shows the relative rates of transpiration (water loss) and water intake for a plant over an 18-hour period. Which of the following describes the most likely relationship between transpiration and water intake in the plant?

(A) Transpiration from the leaves of the plant stimulates subsequent water intake by the roots of the plant. ✓

(B) Transpiration and water intake peak at the same time of day.

(C) Water intake by the roots of the plant drives transpiration at the leaves of the plant.

(D) Transpiration and water intake have an inverse relationship.

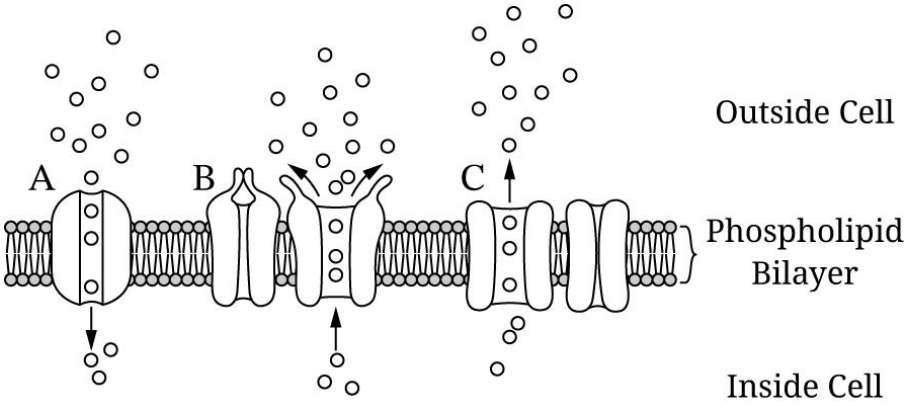
Answer A

Correct. In order for transpiration to be a stimulus for water intake, the rate of transpiration must peak before there is a peak in the rate of water intake. This relationship is supported by the data in Figure 1.

2 Cell MCQ

84. A researcher is studying three transport proteins (A, B, and C) in the membrane and has observed that protein A exhibits passive transport. Figure 1 shows the movement of solute particles in the different transport proteins. In the figure, two different conformations of the same protein are shown for protein B and protein C.

Figure 1. Transport proteins A, B, and C in the membrane



Based on Figure 1, which of the following experiments would best confirm that protein A utilizes passive transport, while proteins B and C do not?

(A) Measuring the rate of solute transport through each protein at various temperatures.

(B) Comparing the ATP consumption of cells expressing each protein type during solute transport. ✓

(C) Analyzing the amino acid sequences of all three proteins to identify structural similarities.

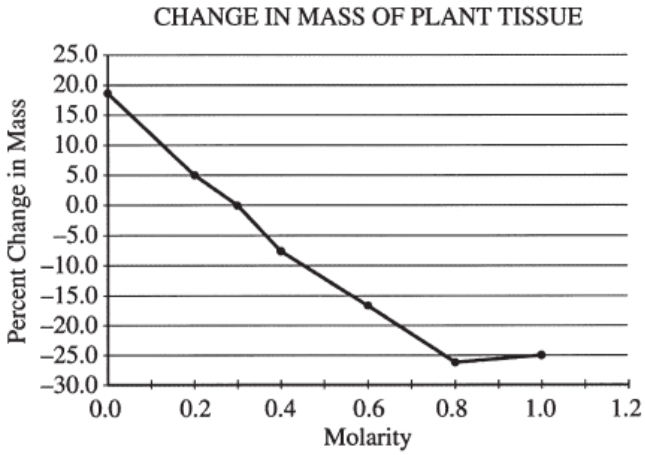
(D) Determining the direction and rate of solute movement through each protein.

Answer B

Correct. Since passive transport (protein A) does not require energy input while active transport (proteins B and C) requires ATP, measuring ATP consumption would clearly differentiate between these mechanisms. Cells expressing protein A would show no additional ATP consumption during transport.

2 Cell MCQ

85. The following question refer to the graph below, which illustrates the percent change in the mass of pieces of plant tissue placed in solutions of different sucrose molarities.



Water enters and leaves the plant cells primarily by

(A) endocytosis

(B) phagocytosis

(C) osmosis ✓

(D) active transport

(E) facilitated diffusion

86. Water is transported from the soil into the roots of a plant, and then up the stem to the leaves. A student determines that the water potential of root cells in the plant is -0.2 MPa. Which of the following claims is best supported by this measurement?

(A) The water potential of the soil will be higher than -0.2 MPa, whereas the water potential of the leaf will be lower than -0.2 MPa. ✓

(B) The water potential of the soil will be lower than -0.2 MPa, whereas the water potential of the leaf will be higher than -0.2 MPa.

(C) The water potential of both the soil and the leaf will be higher than -0.2 MPa.

(D) The water potential of both the soil and the leaf will be equal to -0.2 MPa.

Answer A

Correct. Water will move toward a region with lower water potential. If the water potential of the soil is greater than that of the roots and the water potential of the leaves is less than that of the roots, then water will move from the soil into the roots, up the stem, and to the leaves.

2 Cell MCQ

87. Which of the following provides the weakest evidence that mitochondria were once free-living prokaryotes?

(A) Mitochondrial ribosomes resemble those of prokaryotes.

(B) Mitochondria have DNA that is circular and does not have associated protein.

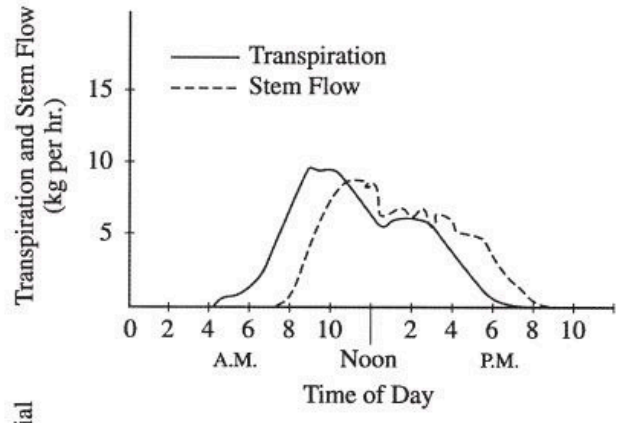
(C) Enzyme pathways on mitochondrial membranes resemble those found on modern prokaryote membranes.

(D) Mitochondria reproduce by a process similar to binary fission.

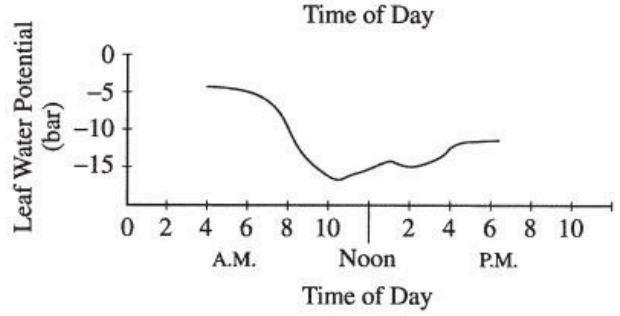
(E) Mitochondria and prokaryotes both are found in a variety of sizes. ✓

88. The rate of transpiration, the flow of water through the stem, and leaf water potential are measured in a tree during a 24-hour period under normal environmental conditions. The results from these measurements are shown in the graphs below.

Graph I



Graph II



What can be deduced from graph I?

(A) The rate of transpiration is constant through-out this 24-hour period.

(B) There is no correlation between the rate of transpiration and the flow of water through the stem.

(C) The maximal flow of water through the stem lags behind the maximal rate of transpiration. ✓

(D) The rate of transpiration never reaches zero.

(E) At noon, the rate of transpiration exceeds the rate of water flow through a stem.

89. Which of the following best describes an advantage that eukaryote organisms have over prokaryote organisms?

2 Cell MCQ

(A) Prokaryotes lack a cell membrane and therefore are unable to control what enters or exits the cell.

(B) Eukaryotes have a nuclear envelope separating their DNA from the rest of the cell, which increases the likelihood of advantageous mutations.

(C) Eukaryotes have mitochondria and chloroplasts that contain their own genome, which allows the cells to reproduce more rapidly.

(D) Eukaryotes have organelles that allow for compartmentalization of cellular processes, which increases the efficiency of those processes. ✓

For following group of questions first study the description of the situation or data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

In the first step of an experiment, rat liver cells were exposed for 5 minutes to amino acids labeled with a radioactive isotope. The cells were then washed to stop any further incorporation of radioactive amino acids. The cells were sampled periodically thereafter, and the radioactivity of a certain protein (protein X) was measured in various cell components, as shown below.

RADIOACTIVE COUNTS (PROTEIN X)

Sampling Time (minutes)	Cell Component						
	Endoplasmic Reticulum	Golgi Apparatus	Cytoplasm	Nucleus	Mitochondria	Lysosomes	Extracellular Space
5	980	20	50	4	2	6	2
20	135	760	20	3	5	120	9
45	50	195	4	6	2	800	3
120	15	30	8	6	8	975	4

90. What is the most likely identity of protein X ?

(A) An enzyme that is involved in transcription

(B) An electron transport protein that participates in respiration reactions

(C) A protein secreted from liver cells into the blood stream that permits proper capillary function

(D) An enzyme that participates in the degradation and recycling of cell components ✓

(E) A building block of the microtubules that are part of the cytoskeleton

91. The same procedure was repeated with a second protein (protein Y). If the function of Y is protein processing and packaging, the most radioactivity would be found in the

(A) Golgi apparatus ✓

(B) Nucleus

(C) Mitochondria

(D) Lysosomes

(E) Cytoplasm

92. Which of the following correctly shows the order in which protein X moves through the cell?

2 Cell MCQ

(A) Endoplasmic reticulum→Golgi apparatus→lysosomes ✓

(B) Endoplasmic reticulum→Golgi apparatus→nucleus→cytoplasm→lysosomes

(C) Extracellular space→mitochondria→cytoplasm

(D) Golgi apparatus→cytoplasm→lysosomes

(E) Cytoplasm→mitochondria→extracellular space

93. In graphing the data from this experiment, the sampling time would be shown on the *x*-axis because

(A) cell components were shown on the *y*-axis

(B) radioactivity level shows the most variation

(C) time values are always located on the horizontal axis

(D) all quantitative values are represented on the *x*-axis

(E) sampling time is the independent variable ✓

---

94. The fluid mosaic model of cell membranes describes proteins and steroids associated with a phospholipid bilayer. Which of the following membrane components affect membrane stability?

(A) Embedded proteins give plasma membranes less stability.

(B) Steroids, such as cholesterol, give membranes more stability. ✓

(C) Proteins in the hydrophilic region of the plasma membrane create more stability.

(D) Polar heads of phospholipids give plasma membranes less stability.

95. Which of the following processes is most likely to occur as a result of an animal cell receiving a signal to initiate apoptosis?

(A) Ribosomes will translate mRNA to produce proteins.

(B) Vesicles will release extracellular growth factors via exocytosis.

(C) Lysosomes will release digestive enzymes into the cytosol. ✓

(D) Vacuoles will fuse with the cellular membrane.

96. Which of the following organelles modifies and packages for secretion the materials produced by the ribosomes?

(A) The chloroplast

(B) The Golgi apparatus ✓

(C) The nucleus

(D) The nucleolus

(E) The mitochondrion

2 Cell MCQ

97. All eukaryotic cells contain at least one Golgi complex, typically located in the cytoplasm and near the endoplasmic reticulum.

Which of the following best describes a process that occurs within the Golgi complex?

(A) Enzymatic modification of newly synthesized integral membrane proteins ✓

(B) Synthesis of cytosolic proteins based on the nucleotide sequences of mRNAs

(C) Degradation of proteins by hydrolytic enzymes contained within the complex

(D) Synthesis of various types of lipids

Answer A

Correct. The Golgi complex is a membrane-bound structure to which some proteins are moved from the rough endoplasmic reticulum, where they are enzymatically modified before being sent to a cellular membrane.

98. Organelles such as mitochondria and the endoplasmic reticulum have membranes that compartmentalize reactions and other metabolic processes. To function properly, the organelles must move substances across their membranes.

Which of the following statements describes a feature shared by mitochondria and the endoplasmic reticulum that increases the efficiency of their basic functions?

(A) They have rigid, nonfluid membranes.

(B) They have highly folded membranes. ✓

(C) They have membranes composed of many carbohydrates.

(D) They have double membranes, with one membrane enclosed within the other.

Answer B

Correct. Compared with a smooth membrane, a highly folded membrane has a greater surface area that increases the rate of material transport across the membrane and provides more support for membrane-associated processes.

2 Cell MCQ

99. A cell is treated with a drug that prevents the formation of new lysosomes. The cell continues to transcribe the genes that code for the hydrolytic enzymes that are normally found in lysosomes and continues to translate the mRNAs for those proteins on membrane-bound ribosomes.

The hydrolytic enzymes are most likely to accumulate in which of the following cellular structures?

(A) Nucleus
(B) Mitochondrion
(C) Smooth endoplasmic reticulum
(D) Golgi complex ✓

100. Which of the following best describes a cell that is likely to have an extensive Golgi complex?

(A) A muscle cell that uses the energy stored in glucose molecules
(B) An intestinal cell that requires a large surface area
(C) A plant cell that uses chemiosmosis
(D) A stomach cell that produces and releases digestive enzymes ✓

Answer D

Correct. Stomach cells that produce and secrete digestive enzymes require an extensive Golgi complex because this organelle is responsible for processing, packaging, and preparing proteins, such as digestive enzymes, for secretion from the cell.

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the following groups of biological compounds.

(A) Proteins
(B) Carbohydrates
(C) Nucleic acids
(D) Lipids
(E) Steroids

101. Synthesized at the ribosome

2 Cell MCQ

(A) Proteins ✓
(B) Carbohydrates
(C) Nucleic acids
(D) Lipids
(E) Steroids

---

102. Contains hydrolytic enzymes associated with the intracellular digestion of macromolecules

(A) Centriole
(B) Lysosome ✓
(C) Nucleolus
(D) Peroxisome
(E) Ribosome

103. Liver cells manufacture glycoproteins, while adipose cells store fat. Which of the following subcellular structures is likely to be more prominent in liver cells than in adipose cells?

(A) Nucleus
(B) Golgi apparatus ✓
(C) Cytoskeleton
(D) Plasma membrane

104. Which of the following best explains why hydrolytic enzymes are typically transported to the lysosome after they are synthesized by ribosomes?

(A) These enzymes are essential for breaking down organic molecules. ✓
(B) These enzymes function most efficiently at low concentrations.
(C) These enzymes require a continuous input of ATP to function.
(D) These enzymes require post-translational modifications to be bound to the cell membrane.

105. Prokaryotic and eukaryotic cells generally have which of the following features in common?

(A) A membrane-bound nucleus
(B) A cell wall made of cellulose
(C) Ribosomes ✓
(D) Flagella or cilia that contain microtubules
(E) Linear chromosomes made of DNA and protein

2 Cell MCQ

106. Muscle cells have high ATP demands. Which of the following is a scientific claim about how the structure of the mitochondria in muscle cells should be different than it is in other cells because of the high energy demands of mitochondria?

(A) The inner membrane of the mitochondria in muscle cells should have more folds to increase the surface area, allowing more ATP to be synthesized. ✓

(B) The inner membrane of the mitochondria in muscle cells should be more permeable to large enzymes, allowing the same reactions to occur in both compartments of the mitochondria.

(C) The outer membrane of the mitochondria in muscle cells should be thicker, allowing more rapid diffusion of molecules into the mitochondria.

(D) The outer membrane of the mitochondria of muscle cells should have more folds, increasing the surface area for faster diffusion of molecules from the cytoplasm.

Answer A

Correct. The amount of ATP produced is related to the number of ATP synthase molecules located on the inner mitochondria membrane. Increasing the folds increases the number of ATP generation sites and results in more ATP production.

107. Which of the following cellular deficiencies would most likely be related to mutations in mitochondrial proteins?

(A) The cell is unable to synthesize most proteins required for normal cell functions.

(B) The cell is unable to break down toxic materials and would accumulate large volumes of these materials.

(C) The cell is able to synthesize proteins, but the proteins would not fold properly and would not contain the correct molecular tags for export from the cell.

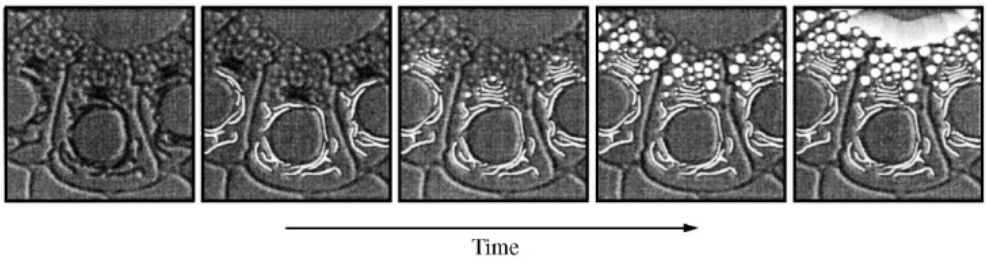
(D) The cell is unable to complete reactions related to electron transport and ATP production. ✓

Answer D

Correct. The electron transport chain and ATP production are associated with proteins in the inner membrane of the mitochondrion. Nonfunctional proteins in the mitochondrion are likely to result in reduced ATP production.

2 Cell MCQ

108.



The figure above shows a series of microscope images taken over time of ovalbumin production in chick oviduct cells in response to stimulation with estrogen. The ovalbumin protein was detected using a fluorescent dye as shown by the white areas in the time-lapse sequence.

The microscope images indicate that ovalbumin

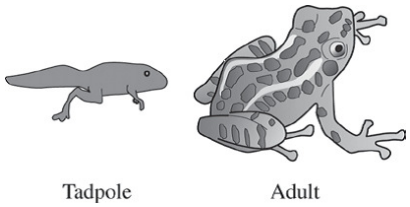
(A) is synthesized on soluble ribosomes and packaged into lysosomes, from which it is secreted into the oviduct

(B) is synthesized on ribosomes bound to the rough endoplasmic reticulum, and then secreted directly into the oviduct

(C) crosses the endoplasmic reticulum membrane as it is translated, then moves to the Golgi apparatus, then to vesicles from which it is secreted ✓

(D) crosses the Golgi apparatus membrane post-translationally, is taken up by the Golgi apparatus and endoplasmic reticulum, then is packaged into secretory vesicles

109.



The last part of the metamorphosis of a tadpole to an adult frog results in the disappearance of the tail. This stage of development most likely occurs by

(A) cells of the tail dying and the nutrients being absorbed and reused by the body ✓

(B) shedding of the tail so energy is not spent on maintenance of an unneeded part

(C) bilateral division of the tail and fusion with the developing hind limbs

(D) individual cells of the tail migrating to the developing gonads

Answer A

This option is correct. It demonstrates understanding of dynamic homeostasis in the reuse of nutrients by the body in development.

2 Cell MCQ

110. The organelle that is a major producer of ATP and is found in both heterotrophs and autotrophs is the

(A) chloroplast

(B) nucleus

(C) ribosome

(D) Golgi apparatus

(E) mitochondrion ✓

111. Which of the following claims is scientifically accurate and consistent with an observation that a decrease in lysosome production within a cell leads to a decline in mitochondrial activity?

(A) A lack of lysosomes will cause a decrease in the synthesis of enzymes necessary for cellular respiration.

(B) Fewer lysosomes will be available to break down macromolecules to provide the necessary nutrients for cellular respiration. ✓

(C) Fewer lysosomes will be available to store materials required for the functioning of the mitochondria.

(D) Lysosomes will not be available to modify proteins so that they are targeted to the mitochondria.

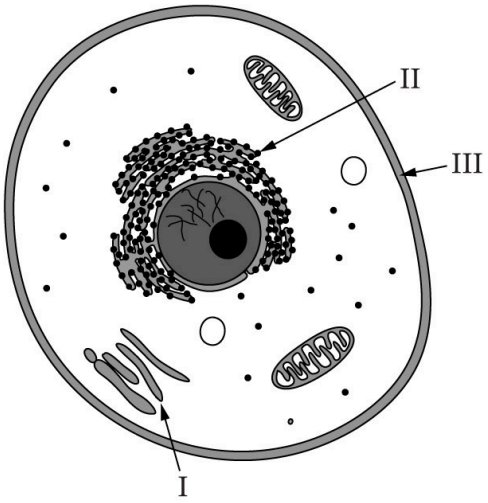
Answer B

Correct. A function of the lysosomes is the breakdown and recycling of macronutrients, such as carbohydrates. Products of carbohydrate breakdown such as glucose are utilized by the mitochondria in the process of cell respiration.

2 Cell MCQ

112. Figure 1 shows three labeled organelles (I, II, III) in a eukaryotic cell.

Figure 1. A eukaryotic cell



Which of the following is a claim that could be made about the relationship between organelles I, II, and III in Figure 1?

(A) All three organelles play a direct role in the production, modification, and transport of proteins that are released from the cell. ✓

(B) All three organelles play a direct role in the breakdown of materials that enter the cell.

(C) Organelles I and II synthesize macromolecules that are then transported by organelle III to the electron transport chain to become part of cellular respiration.

(D) Organelles II and III synthesize and transport proteins that are then digested into amino acids by organelle I.

113. Cells contain smaller components called organelles that are necessary for a cell's survival. Organelle functions have often been compared to components of larger systems.

Which of the following functional differences between the rough and smooth endoplasmic reticulum (ER) is explained by the structural differences between them?

(A) Rough ER breaks down toxic substances, and smooth ER only transports them out of the cell.

(B) Rough ER can synthesize and package lipids for export, and smooth ER cannot.

(C) Rough ER can produce ATP, and smooth ER cannot.

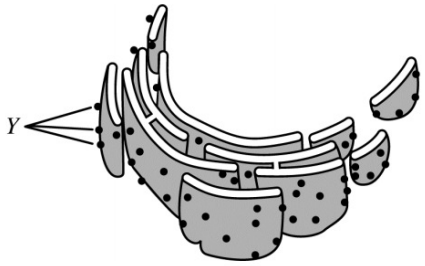
(D) Rough ER can synthesize and package proteins for export, and smooth ER cannot. ✓

2 Cell MCQ

Answer D

Correct. The ribosomes associated with rough ER synthesize proteins that are then exported from the cell.

114.



The figure above represents a rough endoplasmic reticulum. Which of the following best describes the role of the structure labeled *Y*?

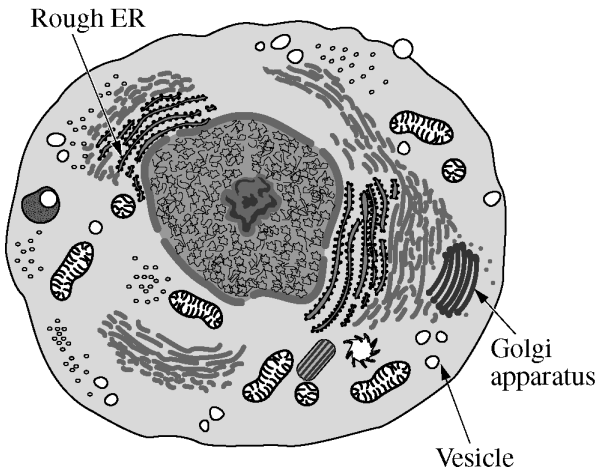
(A) Structure *Y* contributes the raw materials required for the synthesis of proteins.

(B) Structure *Y* packages proteins for export from the cell.

(C) Structure *Y* is the location where proteins are synthesized. ✓

(D) Structure *Y* contains enzymes that cut and activate proteins.

115. The figure below illustrates a eukaryotic cell. Which of the following best describes how the three structures indicated by the arrows work together?



(A) To synthesize lipids and modify toxic substances in order to render them harmless

(B) To synthesize and isolate proteins for secretion or for use in the cell ✓

(C) To catabolize nutrients and produce ATP for intracellular energy storage

(D) To synthesize all ribosomal proteins

2 Cell MCQ

116. A certain type of specialized cell contains an unusually large amount of rough endoplasmic reticulum (ER).

Which of the following functions is this cell type most likely specialized to perform?

(A) The production and secretion of steroids

(B) The destruction of toxic materials produced in other cells of the organism

(C) The synthesis of polysaccharides for energy storage

(D) The production and secretion of proteins ✓

Answer D

Correct. The cytosolic surface of the rough ER is covered by ribosomes that synthesize proteins that are then transported into the rough ER, then to the Golgi complex, and finally out of the cell.

117. A scientist is studying the various prokaryotic and eukaryotic species found floating in a sample of water taken from a marine ecosystem.

Which cellular component will be found in the widest range of organisms in the sample?

(A) The chloroplast, since all organisms need a source of energy.

(B) The ribosome, since all organisms need to synthesize proteins. ✓

(C) The mitochondrion, since all organisms need to break down glucose.

(D) The cell wall, since all marine organisms need them for support.

Answer B

Correct. Ribosomes are found in all forms of life, allowing for comparison of the widest possible range of plankton species.

2 Cell MCQ

118. In an experiment, researchers provided a radiolabeled amino acid to living plant cells. After one hour, the researchers determined the amount of the radiolabeled amino acid that was in each of several subcellular compartments. The results of the experiment are represented in the table.

RELATIVE AMOUNTS OF RADIOLABELED AMINO ACID

Nucleus	Mitochondria	Endoplasmic Reticulum	Cytosol
2.1	2.7	1.9	1

Which of the following conclusions about the radiolabeled amino acid is best supported by the results of the experiment?

(A) It was mostly incorporated into nucleic acids that store the biological information.

(B) It was mostly incorporated into proteins that regulate and manage metabolic reactions. ✓

(C) It was mostly incorporated into lipids that help separate cells from their surrounding environment.

(D) It was mostly incorporated into carbohydrates that form protective structures outside the cells.

Answer B

Correct. Amino acids are the building block of proteins, and the data indicate that most of the radiolabeled amino acid was in the mitochondria. Therefore, the data best support the conclusion that most of the amino acid molecules were incorporated into proteins that regulate and manage the metabolic reactions that occur in mitochondria.

2 Cell MCQ

119. Researchers have proposed a model of the process by which a newly synthesized protein is transported to the plasma membrane and secreted into the extracellular space. The model is represented in Figure 1.

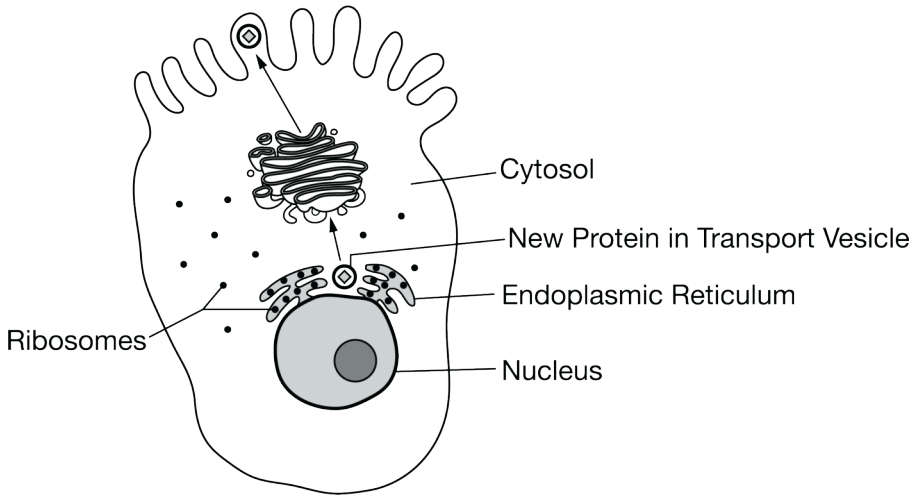


Figure 1. A model of the intracellular transport of a newly synthesized secreted protein

Based on the model, the newly synthesized protein is transported directly from the endoplasmic reticulum to which of the following?

(A) The nucleus

(B) The plasma membrane

(C) The Golgi complex ✓

(D) The extracellular space

Answer C

Correct. Membrane proteins and secreted proteins are synthesized by ribosomes bound to the surface of the endoplasmic reticulum. As shown in the figure, a newly synthesized secreted protein is transported by vesicles from the endoplasmic reticulum directly to the Golgi complex.

120. Macrophages and B lymphocytes are cells that help fight infections. Macrophages engulf and destroy bacteria. B lymphocytes produce proteins that bind to bacteria in the blood.

Based on the functions of macrophages and B lymphocytes, which of the following claims is most likely true of the organelles needed in both types of cells?

2 Cell MCQ

(A) They need organelles that contain enzymes for digesting materials.

(B) They need organelles that store macromolecules and water.

(C) They need organelles that process and transport proteins. ✓

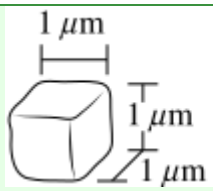
(D) They need organelles that produce lipids for the plasma membrane.

Answer C

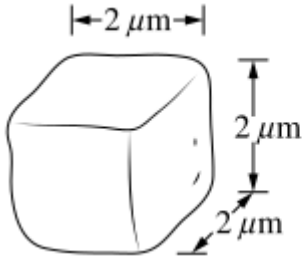
Correct. Both cell types require an extensive endoplasmic reticulum and Golgi complex to process and transport proteins. Macrophages need these organelles for producing digestive enzymes, while B lymphocytes need them for protein (antibody) production and secretion.

121. Which one of the following cells is likely to be the most efficient at eliminating waste by diffusion?

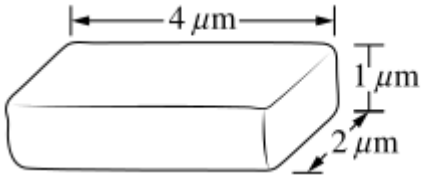
(A) ✓



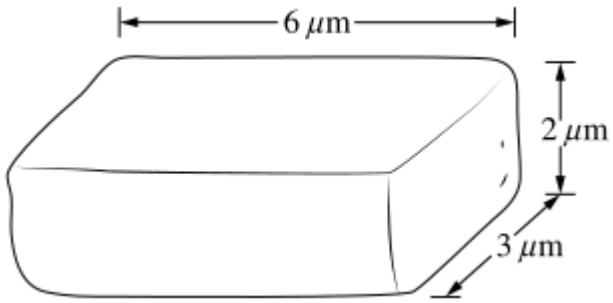
(B)



(C)



(D)



2 Cell MCQ

Answer A

Correct. In cells, the elimination of wastes by diffusion is limited by the ratio of surface area to volume. The greater the ratio of surface area to volume is, the more efficient the diffusion of materials from the cell is. This cell has the greatest surface-area-to-volume ratio ($6:1$), making it the most efficient at eliminating waste.

122. Which of the following best describes the relationship between the size and shape of a cell and the ability of the cell to move substances across the membrane?

(A) Larger and more round-shaped cells are more efficient at moving substances across the membrane than small round-shaped cells.

(B) Smaller and more oval-shaped cells release waste more slowly than large oval-shaped cells.

(C) Larger cube-shaped cells obtain nutrients more quickly than small cube-shaped cells.

(D) Smaller irregularly shaped cells are more efficient at dissipating heat than large irregularly shaped cells. ✓

123. **TABLE 1. DIMENSIONS OF SHAPES USED TO MODEL EFFICIENT MATERIAL EXCHANGE**

Cell Shape	Width at Widest Point (μm)	Height (μm)	Surface Area (μm^2)	Volume (μm^3)
Right circular cylinder	30	90	9,896	63,617
Triangular prism	30	90	8,879	35,074
Rectangular prism	30	90	12,600	81,000
Square-based pyramid	30	90	6,374	27,000

A scientist is developing a mathematical model of cells of different shapes. To construct the model, the scientist has specified that the width of each cell at its widest point must be $30\ \mu m$ and the height of each cell must be $90\ \mu m$. Table 1 shows the three-dimensional shapes that the scientist is considering for the model cells. Which of the proposed shapes for the model cells will allow the most efficient exchange of materials with the surrounding environment?

(A) Right circular cylinder

(B) Triangular prism ✓

(C) Rectangular prism

(D) Square-based pyramid

2 Cell MCQ

Answer B

Correct. This cell shape has the greatest surface area-to-volume ratio (0.253), allowing for the most efficient exchange of materials with the surrounding environment.

124.

Cellular Shape	Ratio of Surface Area to Volume
Cube	1.7
Regular Tetrahedron	0.5
Cylinder	0.9
Sphere	0.6

In an experiment, the efficiency of oxygen exchange across the plasma membrane is being assessed in four artificial red blood cells. The table above lists some properties of those artificial cells. Other conditions being equal, which artificial cell is predicted to be the most efficient in exchanging oxygen with the environment by diffusion?

(A) The cuboidal cell ✓

(B) The tetrahedral cell

(C) The cylindrical cell

(D) The spherical cell

125. Some cells, such as intestinal cells, exchange a lot of material with their surroundings. The surface-to-volume ratio of these cells affects the efficiency of material exchange.

The table provides measurements of four different eukaryotic cells.

Cell	1	2	3	4
Total surface area (μm^2)	40	60	80	100
Total volume (μm^3)	20	10	30	20

Based on the data, which cell is likely to be most effective in the exchange of materials?

(A) Cell 1

(B) Cell 2 ✓

(C) Cell 3

(D) Cell 4

2 Cell MCQ

Answer B

Correct. The surface area to volume calculation is $\frac{60}{10} = 6$. Of the four cells, this one has the highest ratio of surface area to volume and is likely to be most effective in the exchange of materials.

126.

TABLE 1. AVERAGE DIMENSIONS OF FOUR CELL TYPES

	Surface Area (mm^2)	**Volume** (mm^3)
Cell type 1	96	64
Cell type 2	60	10
Cell type 3	54	27
Cell type 4	24	8

Which of the cell types represented in Table 1 is most efficient at acquiring nutrients by diffusion?

(A) Cell type 1

(B) Cell type 2 ✓

(C) Cell type 3

(D) Cell type 4

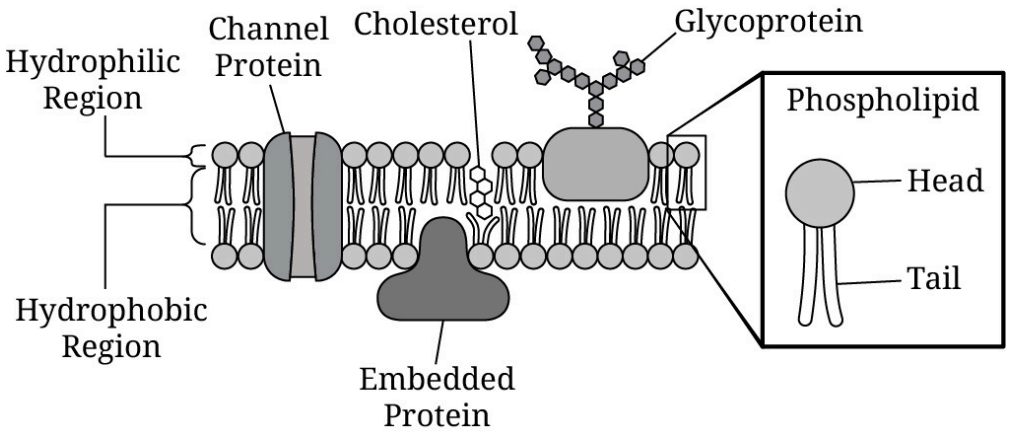
Answer B

Correct. Nutrient acquisition by diffusion relies on surface area to volume ratio. $\frac{60}{10} = 6$. Cell type 2 has the largest SA /V ratio and would therefore be the most efficient at acquiring nutrients by diffusion.

2 Cell MCQ

127. Figure 1 shows a cell membrane that is composed of a phospholipid bilayer and several other components.

Figure 1. Components of a cell membrane



Based on Figure 1, which of the following membrane components are most likely nonpolar?

(A) The phospholipid tails and cholesterol ✓

(B) The phospholipid heads and the channel protein

(C) The phospholipid heads and the glycoprotein

(D) The phospholipid tails and the embedded protein

128. Which of the following best describes the position of the phospholipids within the phospholipid bilayer of the plasma membrane?

(A) The hydrophilic heads face the inner region of the membrane, and the hydrophobic tails point outward.

(B) The hydrophilic tails face the inner region of the membrane, and the hydrophobic heads point outward.

(C) The hydrophobic heads face the inner region of the membrane, and the hydrophilic tails point outward.

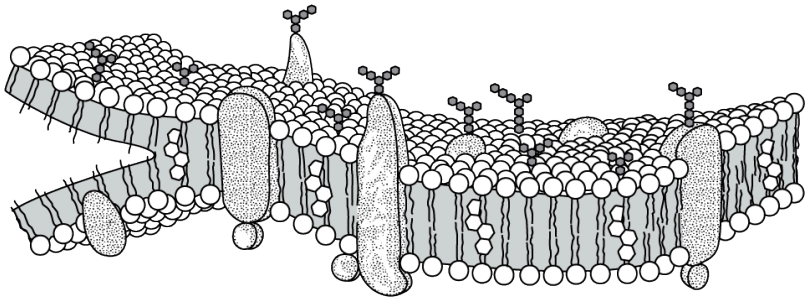
(D) The hydrophobic tails face the inner region of the membrane, and the hydrophilic heads point outward. ✓

Answer D

Correct. Because both the extracellular fluid and the cytosol are aqueous, the hydrophilic heads of the phospholipids face these two fluids, while the hydrophobic tails face away from them toward the interior of the membrane.

2 Cell MCQ

129. The model below shows the structure of a portion of a plasma membrane in an animal cell.



Which statement best explains the orientation of the phospholipid molecules in this model?

(A) The nonpolar portions of the phospholipid molecules are attracted to the internal and external environments.

(B) The hydrophilic phosphate groups of the phospholipid molecules are attracted to the aqueous internal and external environments. ✓

(C) The embedded proteins attract the fatty acid tails of the phospholipids, so the tails point away from the internal and external aqueous environments.

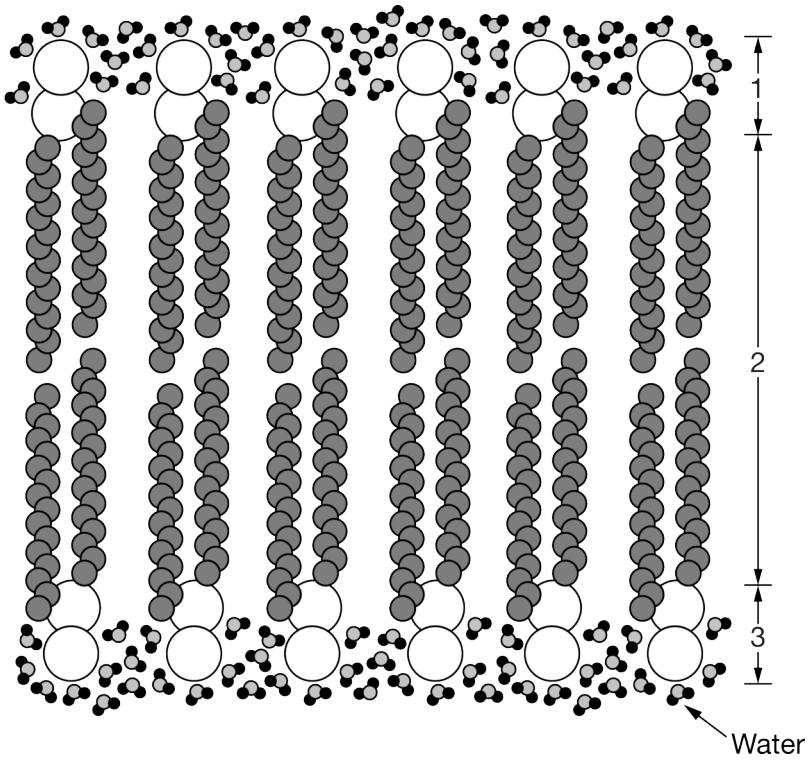
(D) The fatty acid tails of the phospholipid molecules are hydrophilic and are repelled by the internal and external aqueous environments.

Answer B

Correct. The hydrophilic polar phosphate groups of the phospholipid molecules orient toward the polar internal and external aqueous environments, forming a bilayer that keeps the hydrophobic portions of the phospholipids away from the polar molecules.

2 Cell MCQ

130.



Which of the following best describes the numbered areas?

(A) Areas 1 and 3 are polar, since the membrane molecules are aligned with water molecules. ✓

(B) Area 2 is polar, since water has been excluded from this area of the membrane.

(C) Areas 1 and 3 are hydrophilic, since membrane molecules formed covalent bonds with water.

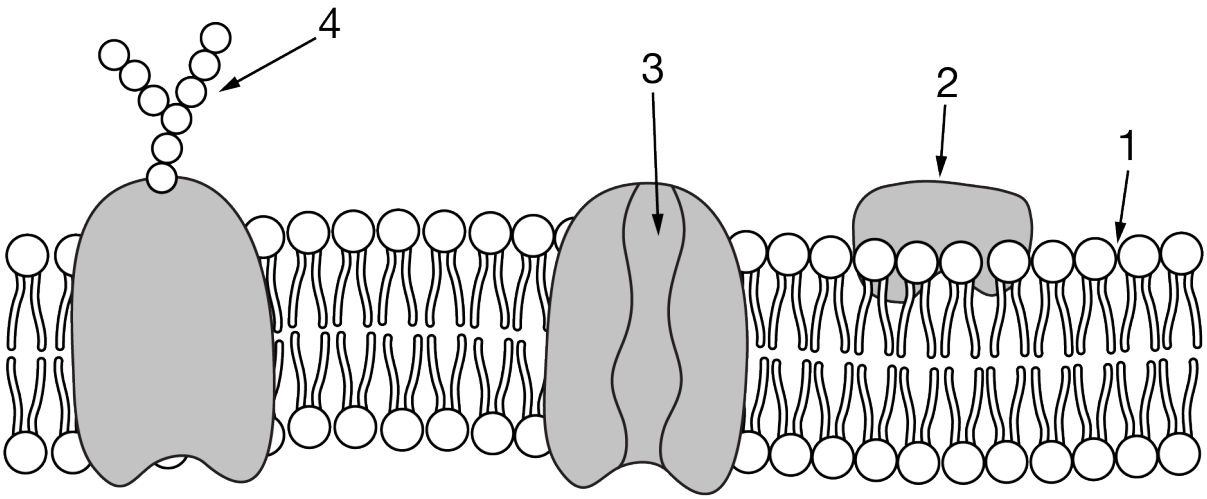
(D) Area 2 is nonpolar, since hydrogen bonds between the adjacent lipids hold the membrane together.

Answer A

Correct. As indicated in the diagram, the heads of the phospholipids in areas 1 and 3 are adjacent to water molecules, which shows that the phospholipid heads are hydrophilic.

2 Cell MCQ

131. Figure 1. Testosterone movement across the cellular membrane



Testosterone is a small steroid hormone that is important in cell signaling. Which of the following describes where testosterone enters a cell and why it is able to cross at that point?

(A) 1, testosterone is nonpolar and can diffuse through the membrane. ✓

(B) 2, testosterone covalently binds to a surface protein and transports into the cell.

(C) 3, testosterone dissolves in water and flows through the channel.

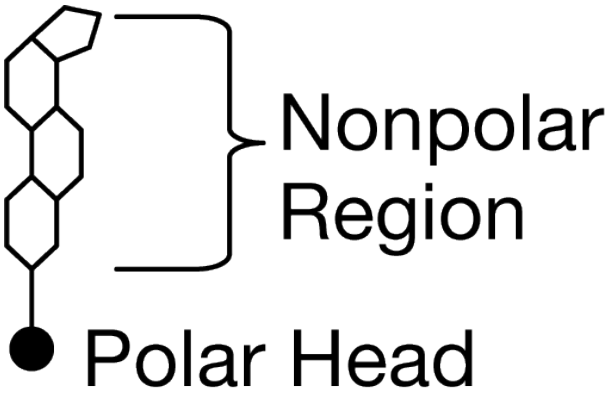
(D) 4, testosterone is filtered out of the extracellular fluid and taken into the cell by endocytosis.

Answer A

Correct. Steroids such as testosterone are hydrophobic lipids. Therefore, testosterone can cross the hydrophobic inner region of the phospholipid bilayer.

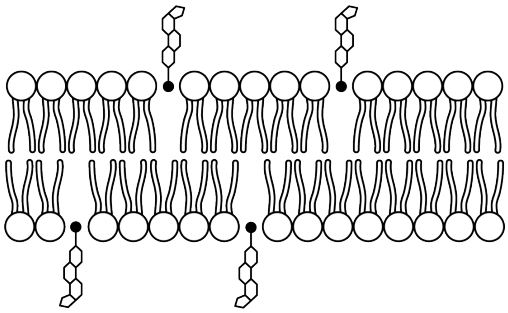
2 Cell MCQ

132. Cholesterol is a naturally occurring substance that helps regulate the fluidity of a cell's plasma membrane. A cholesterol molecule can be represented as having a polar head and a nonpolar region, as shown in the figure.

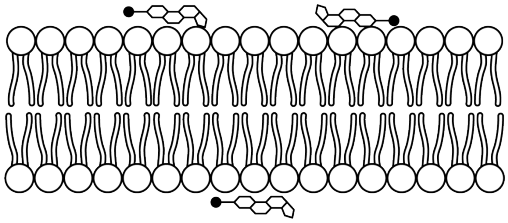


Which of the following models shows how cholesterol molecules most likely interact with the phospholipid bilayer of a cell's plasma membrane?

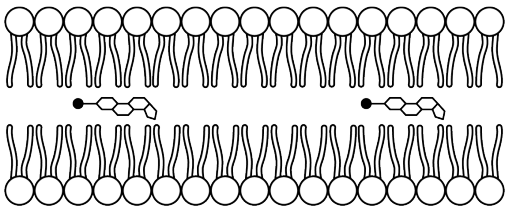
(A)



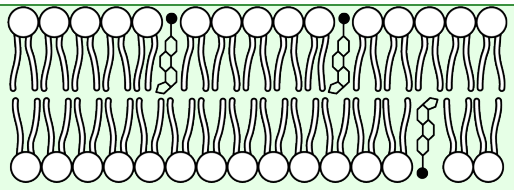
(B)



(C)



(D) ✓



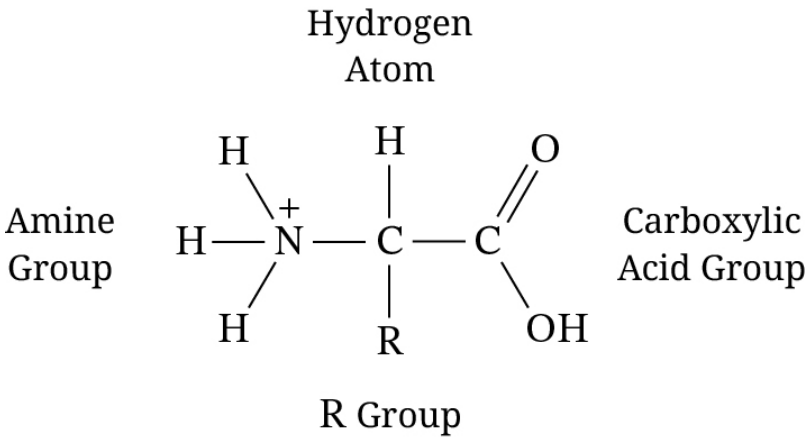
2 Cell MCQ

Answer D

Correct. The model correctly shows the polar heads of the cholesterol molecules interacting with the polar heads of the phospholipids. Also, the model correctly shows the nonpolar regions of the cholesterol molecules interacting with the hydrophobic interior of the phospholipid bilayer.

133. Figure 1 shows the general structure of an amino acid.

Figure 1. Structure of an amino acid



Which of the following amino acid components shown in Figure 1 best determines the interaction between an embedded protein and the hydrophobic regions of the cell membrane?

(A) The hydrogen atom bonded to the central carbon

(B) The carboxyl group bonded to the central carbon

(C) The R group bonded to the central carbon ✓

(D) The amine group bonded to the central carbon

134. A spherical bacterial cell has a radius of $3\ \mu\text{m}$. The human egg cell has a radius of $100\ \mu\text{m}$.

Which statement correctly indicates the cell that is able to more efficiently exchange materials with the external environment and provides a correct explanation?

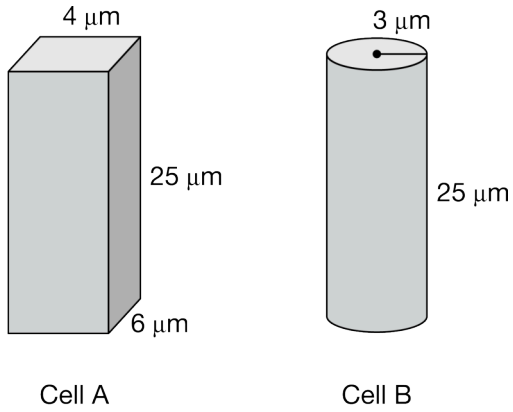
2 Cell MCQ

(A) The egg cell, because it has the smallest surface-to-volume ratio.
(B) The egg cell, because it has the largest surface-to-volume ratio.
(C) The bacterial cell, because it has the smallest surface-to-volume ratio.
(D) The bacterial cell, because it has the largest surface-to-volume ratio. ✓

Answer D

Correct. The bacterial cell is more efficient in the exchange of materials with the external environment. While the egg cell is much larger than the bacterial cell, the egg cell has a surface-to-volume ratio that is 33 times smaller than the bacterial cell's surface-to-volume ratio.

135. Two different models of a living cell are represented in the figure.



Note: Figures not drawn to scale.

Of the two cells represented in the figure, which would likely be more efficient at exchanging substances with the surrounding environment?

(A) Cell A, because it has the larger surface-area-to-volume ratio. ✓
(B) Cell A, because it has the smaller surface-area-to-volume ratio.
(C) Cell B, because it has the larger surface-area-to-volume ratio.
(D) Cell B, because it has the smaller surface-area-to-volume ratio.

Answer A

Correct. Of the two cells represented in the figure, Cell A has the larger surface-area-to-volume ratio of

2 Cell MCQ

0.91 compared to a surface-area-to-volume ratio of 0.75 for Cell B. Cell A would likely exchange substances with the surrounding environment at a greater rate.

136. Stomata are pores on the surfaces of the leaves and stems of plants that regulate gas exchange between the plants and the atmosphere.

Researchers found that the stomata density on the leaves of a species of plant change as the concentration of CO_2 in the atmosphere changes. When grown at 350 $\text{ppm } CO_2$ the plant has an average density of 300 stomata per mm^2, but when grown at 400 $\text{ppm } CO_2$ the plant has an average density of 250 stomata per mm^2.

Which of the following best describes how the ratio of the density of stomata (stomata per mm^2) per CO_2 concentration ($\text{ppm } CO_2$) changes as the CO_2 concentration increases?

(A) The ratio decreases from 0.86 to 0.63, because fewer stomata are needed at higher CO_2 concentrations. ✓

(B) The ratio decreases from 1.6 to 1.2, because fewer stomata are needed at higher CO_2 concentrations.

(C) The ratio increases from 0.63 to 0.86, because more stomata are needed at higher CO_2 concentrations.

(D) The ratio increases from 1.2 to 1.6, because more stomata are needed at higher CO_2 concentrations.

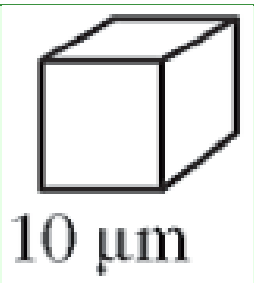
Answer A

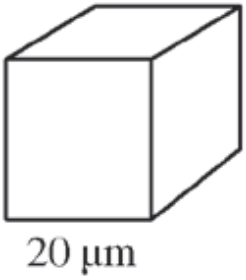
Correct. The ratio of 300 stomata per mm^2 to 350 $\text{ppm } CO_2$ is 0.86, and the ratio of 250 stomata per mm^2 to 400 $\text{ppm } CO_2$ is 0.63. This reflects that fewer stomata are needed as the concentration of CO_2 increases.

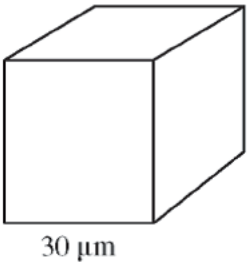
137. Simple cuboidal epithelial cells line the ducts of certain human exocrine glands. Various materials are transported into or out of the cells by diffusion. (The formula for the surface area of a cube is 6 X S^2, and the formula for the volume of a cube is S^3, where S = the length of a side of the cube.)

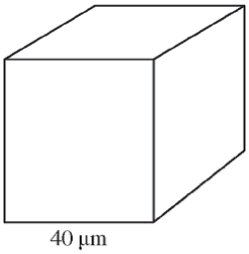
Which of the following cube-shaped cells would be most efficient in removing waste by diffusion?

2 Cell MCQ

(A) 

(B) 

(C) 

(D) 

Answer A

This option is correct. It demonstrates the ability to calculate surface area-to- volume ratios and predict their effects on a cell's ability to exchange materials with its environment. The cell shown in A has the greatest surface area-to- volume ratio at 0.6:1.

2 Cell MCQ

138. Plant cell walls are composed of cellulose, while fungal cell walls are composed of chitin. A group of scientists hypothesize that this difference means the cell wall has largely different functions in plant cells and fungal cells. Alternatively, another group of scientists hypothesize that despite their biochemical differences, plant and fungal cell walls serve similar functions.

Which of the following observations would best support the alternative hypothesis described above?

(A) Plant cell walls are found just outside the plasma membrane, while fungal cell walls are found just beneath the plasma membrane.

(B) In both plant cells and fungal cells, the cell wall surrounds the outside of the cell membrane. ✓

(C) Some plant cells have secondary cell walls that confer additional rigidity, while fungal cells do not.

(D) Photosynthesis occurs in plant cells, but it does not occur in fungal cells.

Answer B

Correct. In both plants and fungi, a major function of the cell wall is to confer structural rigidity to the cell. Thus, in both cell types, the cell wall's location is similar. This is evidence supporting the alternative hypothesis proposed above.

139. Which of the following best describes the primary mechanism of transport whereby carbon dioxide moves from inside a cell to outside the cell?

(A) Carbon dioxide is released when vesicles containing carbon dioxide fuse with the plasma membrane.

(B) Carbon dioxide is actively pumped into the membrane between the halves of the bilayer and then passively diffuses out.

(C) Carbon dioxide passively diffuses between the membrane lipids. ✓

(D) Carbon dioxide passively moves through membrane channel proteins.

Answer C

Correct. Because CO_2 is a nonpolar gas, it moves directly between the cell membrane's hydrophobic lipid molecules, from an area of high concentration to an area of lower concentration.

2 Cell MCQ

140. Beetroot cells contain a family of dark red pigments called betalains. The selectively permeable nature of the beetroot cells keeps the internal environment of the cell separate from the external environment of the cell. Researchers are interested in determining whether the selective permeability of beetroot cells is due to the cell membrane or if it is due to the cell wall.

Exposure to cellulase is known to damage the structure of the cell wall. An experiment is set up in which beetroot cells are placed in an aqueous solution with cellulase and in one without cellulase.

Which of the following results best refutes the alternative hypothesis that selective permeability is a consequence of the cell wall?

(A) When beetroot cells are placed in a solution with cellulase, the solution turns dark red.

(B) When beetroot cells are placed in a solution with cellulase, the solution remains clear. ✓

(C) When beetroot cells are placed in a solution, it turns dark red with or without cellulase present.

(D) Since plant cells contain cell membranes, not cell walls, the alternate hypothesis cannot be tested.

Answer B

Correct. Cellulase digests cellulose and damages the structure of the cell wall, not the cell membrane. The lack of color change in the solution indicates that the betalain is not leaking out of the beetroot cells even though the cell wall has been damaged. This refutes the alternative hypothesis.

2 Cell MCQ

141. A model of the plasma membrane showing several biological molecules, including a transmembrane protein, is shown in Figure 1.

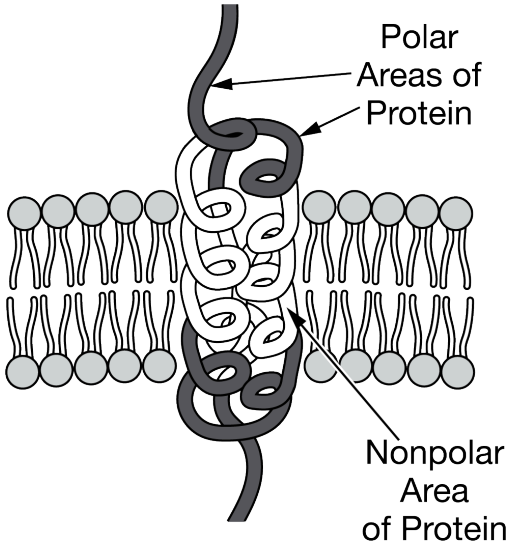


Figure 1. Phospholipid bilayer with transmembrane protein

Which statement best explains why correct protein folding is critical in the transmembrane protein shown above?

(A) Interactions of the hydrophobic and hydrophilic amino acids help to anchor the protein in the membrane. ✓

(B) Interactions of the peptide bonds of the protein with the membrane will affect the rate at which substances can cross the membrane.

(C) Interactions of the protein and phospholipids increase membrane fluidity.

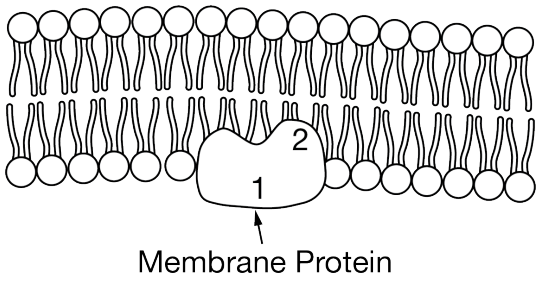
(D) Interactions of the quaternary structure of the protein will increase hydrogen bonding in the membrane and make the membrane less fluid.

Answer A

Correct. Tertiary structure of the protein resulting from hydrophobic and hydrophilic interactions plays a key role in how the protein interacts with the hydrophobic tails and hydrophilic head of the membrane phospholipids.

2 Cell MCQ

142. The figure shows a representation of a protein embedded in a cell membrane. The numbers indicate different structural regions of the protein.



Based on the figure, which of the following statements best describes the relationship between regions 1 and 2 of the protein?

(A) Region 1 is hydrophilic because it interacts with the interior of the membrane, whereas region 2 is hydrophobic because it interacts with an aqueous environment.

(B) Region 1 is hydrophilic because it interacts with an aqueous environment, whereas region 2 is hydrophobic because it interacts with the interior of the membrane. ✓

(C) Region 1 is hydrophobic because it interacts with the interior of the membrane, whereas region 2 is hydrophilic because it interacts with an aqueous environment.

(D) Region 1 is hydrophobic because it interacts with an aqueous environment, whereas region 2 is hydrophilic because it interacts with the interior of the membrane.

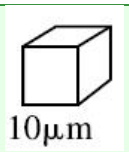
Answer B

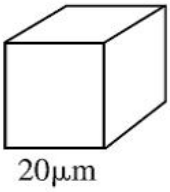
Correct. A cell membrane is a phospholipid bilayer that separates one aqueous environment from another. The interior of a phospholipid bilayer is a hydrophobic environment. Because region 1 interacts with the aqueous environment on one side of the phospholipid bilayer, it is most likely hydrophilic. Because region 2 interacts with the interior of the phospholipid bilayer, it is most likely hydrophobic.

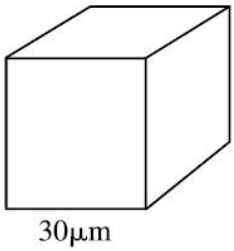
143. Simple cuboidal epithelial cells line the ducts of certain human exocrine glands. Various materials are transported into or out of the cells by diffusion. (The formula for the surface area of a cube is 6 X S^2, and the formula for the volume of a cube is S^3, where S = the length of a side of the cube.)

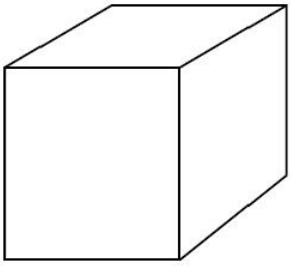
Which of the following cube-shaped cells would be most efficient in removing waste by diffusion?

2 Cell MCQ

(A) 

(B) 

(C) 

(D) 

144. Researchers propose a model to explain variation in phytoplankton cell sizes in a marine environment. They base their model on the idea that smaller cells absorb nutrients more efficiently. The researchers predict that the mean diameter of phytoplankton cells will change by 50 micrometers for every 5-kilometer increase in distance from the shore because of a gradual decrease in nutrient availability. To test their model, the researchers determine that the phytoplankton cells found closest to shore have a mean diameter of 900 micrometers.

Based on the model, what will be the mean diameter of the phytoplankton cells that are found 25 kilometers from shore?

(A) 650 micrometers

(B) 875 micrometers

(C) 925 micrometers

(D) 1150 micrometers

2 Cell MCQ

Answer A

Correct. The model predicts that phytoplankton cell size will decrease with increasing distance from the shore, resulting in increased efficiency of nutrient absorption. The correct value was arrived at by subtracting 250 from 900.

145. The table below provides a comparison of nitrogenous waste production in selected organisms.

Nitrogenous Waste	Solubility in Water	Amount of Water Required for Excretion	Organisms that Primarily Produce the Nitrogenous Waste
Ammonia	High	High	Freshwater fish, aquatic invertebrates
Urea	Medium	Medium	Mammals, sharks
Uric acid	Very low	Very low	Birds, reptiles, and most terrestrial insects and arthropods

Which of the following statements is most consistent with the data in the table?

(A) In response to the hypotonic environment in which freshwater fish live, they excrete ammonia in concentrated urine or across their gills.

(B) The kidneys of reptiles and birds are highly efficient because little water is needed to excrete uric acid. ✓

(C) Birds excrete ammonia in addition to uric acid, and the ratio of the two substances is independent of whether the birds are primarily terrestrial or aquatic species.

(D) The similar regulation of extracellular fluid volume and composition in all the organisms suggests conservation of kidney structure throughout evolution.

146. Hereditary spherocytosis (HS) is a disorder of red blood cells that causes the cells to be smaller and spherical instead of having the usual flattened, biconcave shape. The average diameter of normal red blood cells is $7.2\ \mu\mathrm{m}$, and the average diameter of red blood cells in a person with HS was found to be $6.7\ \mu\mathrm{m}$. The normal red blood cell has an average surface area of $136\ \mu\mathrm{m}^2$ and an average volume of $91\ \mu\mathrm{m}^3$.

Which of the following provides an accurate calculation of the surface area to volume ratio of an HS red blood cell, as well as a prediction of its effect on the efficient transferring of oxygen compared to a normal red blood cell?

(A) The ratio is 0.45, and the cells are more efficient at transferring oxygen.

(B) The ratio is 1.12, and the cells are less efficient at transferring oxygen.

(C) The ratio is 0.89, and the cells are less efficient at transferring oxygen. ✓

(D) The ratio is 141, and the cells are more efficient at transferring oxygen.

2 Cell MCQ

Answer C

Correct. The correct calculation of the surface area to volume ratio of the HS cell is 0.89. This ratio is less than the ratio found in a normal red blood cell, 1.49, so the cell would be less efficient at transferring oxygen.

EFFECT OF WATER-SOLUBLE POLLUTANTS ON MEMBRANE PERMEABILITY

Treatment Group	Treatment Solution*	Mean Absorbance of 460 nm Light ($n = 5$)	2×Standard Error of the Mean (2×SEM)
I	70% isopropanol; 30% water	0.164	0.032
II	90% isopropanol; 10% water	0.125	0.100
III	50% acetone; 50% water	0.215	0.034
IV	70% acetone; 30% water	0.274	0.018
V	100% water	0.095	0.004
*Concentrations are reported as percent by volume.			

A student formulated a hypothesis that water-soluble pollutants damage living organisms by increasing the permeability of cellular membranes. To test the hypothesis, the student investigated the effect of isopropanol and acetone on beet root cells. The vacuoles of beet root cells contain large amounts of betacyanin, a water-soluble pigment that is released into the extracellular environment as a result of increased membrane permeability.

The student prepared identical samples of beet root tissue and incubated each sample for 15 minutes in the specific solution for that group. At the end of the incubation period, the student measured the absorbance of 460 nm light for each sample. A greater concentration of betacyanin in the solution surrounding the beet root cells results in a greater absorbance of 460 nm light. The results of the experiment are shown in the table above.

147. Based on the data from the investigation, which of the following is the best scientific question about organisms living in water that is polluted with organic solvents?

(A) Do organisms without pigments have a selective advantage in polluted environments?

(B) Will organisms living in polluted environments exhibit detrimental effects from an increased permeability of their cellular membranes? ✓

(C) Will organic solvents from the environment become incorporated into the cell membranes of organisms living in polluted environments?

(D) Should governments place more stringent restrictions on the release of organic solvents into aquatic ecosystems?

148. The student analyzed the data from the investigation and concluded that the estimate of the mean of one treatment group was unreliable. Which of the following identifies the treatment group most likely to have provided an unreliable estimate of the mean, and correctly explains why the estimate appears unreliable?

2 Cell MCQ

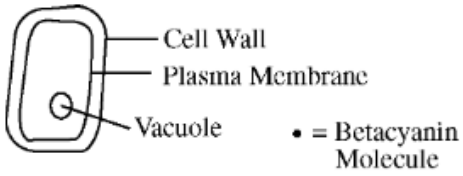
(A) Treatment group II; it has a lower than expected mean absorbance and the largest standard error of the mean. ✓

(B) Treatment group III; it has a higher than expected mean absorbance and the largest standard error of the mean.

(C) Treatment group IV; it has a higher than expected mean absorbance and the smallest standard error of the mean.

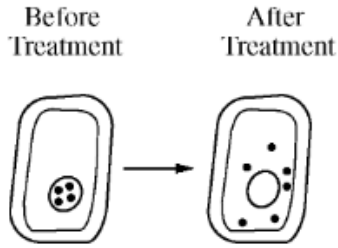
(D) Treatment group V; it has a lower than expected mean absorbance and the smallest standard error of the mean.

149.

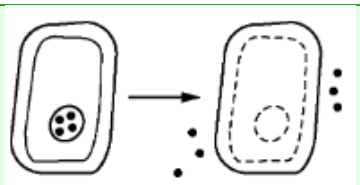


The illustration above is a model of a typical beet root cell. Based on the experimental results, which of the following best represents the effect of acetone on the permeability of cellular membranes?

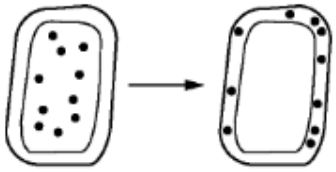
(A)



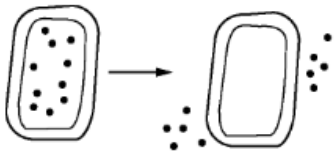
(B) ✓



(C)



(D)



150. Which of the following is the dependent variable in the experiment?

2 Cell MCQ

(A) The percent by volume of isopropanol in the treatment solutions

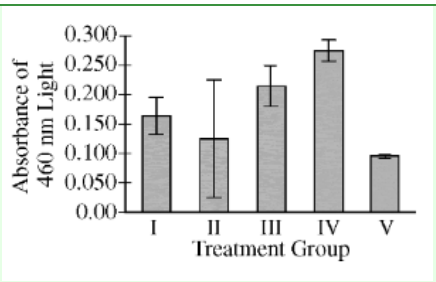
(B) The percent by volume of water in the treatment solutions

(C) The length of time each sample was incubated in the treatment solutions

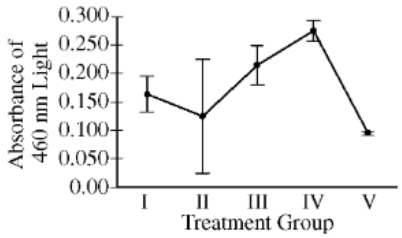
(D) The absorbance of 460 nm light by the treatment solutions ✓

151. Which of the following graphs is the most appropriate representation of the experimental data?

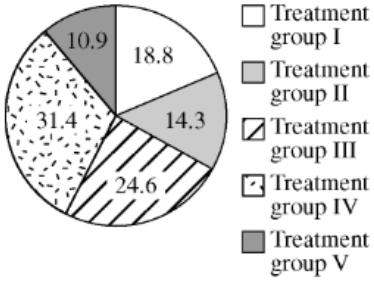
(A) ✓



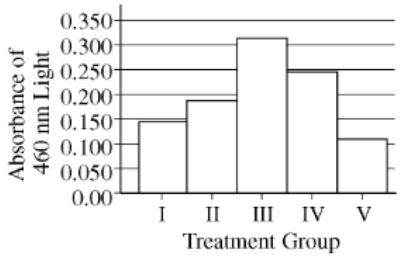
(B)



(C)



(D)



2 Cell MCQ

152. Aldosterone (a steroid hormone) is a small, nonpolar, hydrophobic molecule that enters a target cell by moving across the plasma membrane, down a concentration gradient.

Based on the information presented, how does aldosterone most likely enter target cells?

(A) By simple diffusion because aldersterone can enter a cell by moving across the plasma membrane and moving down the concentration gradient. ✓

(B) By facilitated diffusion because aldersterone would need the assistance of a protein channel to move across the plasma membrane.

(C) By active transport because aldersterone would need energy to move across the plasma membrane.

(D) By endocytosis because aldersterone would need to be transported by vesicle across the plasma membrane

Answer A

Correct. Small, nonpolar, hydrophobic molecules can enter a cell by moving across the plasma membrane and down a concentration gradient by simple diffusion.

153. Two competing hypotheses exist regarding the cell membrane structure. One hypothesis states that membrane structure is static and membrane components throughout the bilayer are rigidly bound. Alternatively, the other hypothesis states that cell membranes are a fluid mosaic in which membrane components may drift within the bilayer around the surface of the cell. An experiment is set up in which membrane proteins of two different cells are fluorescently labeled with two different colors and then fused as shown in Figure 1.

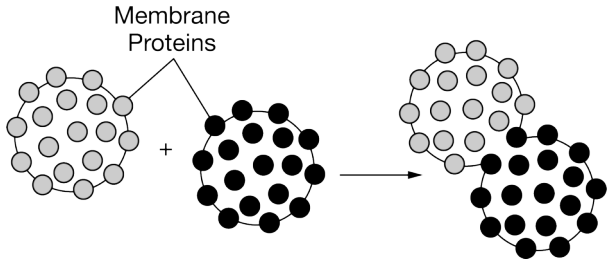
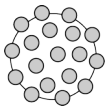
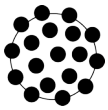


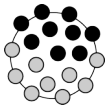
Figure 1. Model of initiation of cell fusion experiment

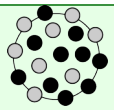
Which of the following results, one hour after membrane fusion, best supports the alternative hypothesis that the cell membrane is a fluid mosaic?

2 Cell MCQ

(A) 

(B) 

(C) 

(D) 

Answer D

Correct. The membrane proteins from each cell have mixed and drifted within the bilayer, which supports the fluid mosaic model of cell membranes. Cell membranes consist of a structural framework of phospholipid molecules that is embedded with proteins and steroids that may flow around the surface of the cell within the bilayer.

154. Researchers exposed identical cells to solutions containing different solutes. Each solution had a higher concentration of the solute outside the cells than inside the cells. The solutes varied in size, charge, and polarity. After a set time, the researchers measured the concentration of each solute inside the cells to determine the permeability of the cell membranes to each solute. The results are shown in Table 1.

Table 1. Permeability of Cell Membranes to Different Solutes

Solute	Size	Charge	Polarity	Initial Concentration outside the cell	Concentration Inside the Cell
A	Small	Neutral	Nonpolar	High	High
B	Large	Positive	Nonpolar	High	Low
C	Small	Negative	Polar	High	Medium

Based on the data in Table 1, which of the following statements best supports the hypothesis that only certain types of molecules can move freely across the cell membranes?

2 Cell MCQ

(A) Large, nonpolar solutes showed an increase in the concentration inside the cell, while small, nonpolar solutes showed a slow increase in the concentration inside the cell.

(B) Ionic solutes showed equal movement of solutes inside and outside the cell internal and external concentrations.

(C) Small, nonpolar solutes showed the greatest increase in internal concentration, while large, nonpolar solutes showed little to no increase. ✓

(D) None of the solutes showed any change in internal concentration, regardless of their size, polarity, and charge.

155. Which of the following statements is true regarding the movement of substances across cell membranes?

(A) Ions are unable to move through the phospholipid bilayer because the nonpolar tail regions of the phospholipids are hydrophobic. ✓

(B) Ions are able to move through the phospholipid bilayer because the polar head regions of the phospholipids are charged.

(C) Water is able to move through the phospholipid bilayer because the nonpolar tail regions of the phospholipids are charged.

(D) Water is unable to move through the phospholipid bilayer because the polar head regions of the phospholipids are charged.

156. A team of biologists develop a new drug, and one team member hypothesizes that the drug is incapable of freely passing across the plasma membrane and requires the help of membrane proteins to enter cells. Alternatively, another biologist on the team hypothesizes that the drug can diffuse passively across the plasma membrane like O_2 and CO_2 can.

Which of the following, if true about the drug, best supports the alternative hypothesis that the new drug will exhibit simple diffusion across plasma membranes?

(A) The drug is a small nonpolar molecule. ✓

(B) The drug is a small charged molecule.

(C) The drug is a large polar molecule.

(D) The drug is a large charged molecule.

Answer A

Correct. Small nonpolar molecules freely pass across the plasma membrane. If the drug is a small nonpolar molecule, this supports the alternative hypothesis.

2 Cell MCQ

157. The cell membrane is selectively permeable due to its structure. Thus, the internal environment of the cell is distinct from the external environment of the cell. One biologist hypothesizes that small nonpolar molecules readily pass through the membrane. Another biologist alternatively hypothesizes that these types of molecules require channel and transport proteins that are embedded in the membrane in order to move across the membrane.

Which of the following data would best refute this alternative hypothesis?

(A) Ethanol is found in the cytosol of cells when they are briefly exposed to a ten percent ethanol solution.

(B) Cells become oxygen deficient when membrane protein activity is blocked.

(C) CO_2 and N_2 movement in and out of cells is unaffected when membrane protein activity is blocked. ✓

(D) Sodium ions cannot move across the cell membrane when membrane protein activity is blocked.

Answer C

Correct. Both CO_2 and N_2 are small nonpolar molecules. Their ability to enter and leave cells normally when membrane protein activity is blocked would refute the alternative hypothesis.

3 Cellular Energetics MCQ

1. A researcher claims that the incorporation of carbon dioxide into organic molecules during photosynthesis does not violate the second law of thermodynamics.

Which of the following statements best helps justify the researcher's claim?

(A) Organisms contain enzymes that lower the activation energies of specific chemical reactions.

(B) An ecosystem is formed by the interaction of a community of organisms with their surrounding environment.

(C) Photosynthetic organisms use the organic molecules produced during photosynthesis for growth and repair.

(D) The total system that includes photosynthetic organisms and the Sun becomes less ordered over time. ✓

Answer D

Correct. Because the total system that includes photosynthetic organisms and the Sun becomes less ordered over time, the incorporation of carbon dioxide molecules into organic molecules that occurs during photosynthesis does not violate the second law of thermodynamics.

2. Which two cellular organelles in eukaryotes have both electron transport systems and chemiosmotic mechanisms?

(A) Ribosomes and endoplasmic reticulum

(B) Chloroplasts and endoplasmic reticulum

(C) Chloroplasts and mitochondria ✓

(D) Mitochondria and nuclei

(E) Nuclei and Golgi apparatus

3. According to the chemiosmotic theory (chemiosmotic coupling), the energy required to move protons from the mitochondrial matrix to the intermembrane space against a concentration gradient comes most directly from

(A) photons of red or blue light

(B) the hydrolysis of ATP

(C) the breakdown of high-energy fatty acids in the mitochondrial matrix

(D) electrons flowing along the electron transport chain ✓

(E) substrate-level phosphorylation

3 Cellular Energetics MCQ

An experiment to measure the rate of respiration in crickets and mice at 10^{o} C and 25^{o} C was performed using a respirometer, an apparatus that measures changes in gas volume. Respiration was measured in mL of O_2 consumed per gram of organism over several five-minute trials, and the following data were obtained.

Organism	Temperature (°C)	Average respiration (mL O_2/g/min)
Mouse	10	0.0518
Mouse	25	0.0321
Cricket	10	0.0013
Cricket	25	0.0038

4. According to the data, the mice at 10^{o} C demonstrated greater oxygen consumption per gram of tissue than did the mice at 25^{o} C. This is most likely explained by which of the following statements?

(A) The mice at 10^{o} C had a higher rate of ATP production than the mice at 25^{o} C. ✓

(B) The mice at 10^{o} C had a lower metabolic rate than the mice at 25^{o} C.

(C) The mice at 25^{o} C weighed less than the mice at 10^{o} C.

(D) The mice at 25^{o} C were more active than the mice at 10^{o} C.

5. During aerobic cellular respiration, oxygen gas is consumed at the same rate as carbon dioxide gas is produced. In order to provide accurate volumetric measurements of oxygen gas consumption, the experimental setup should include which of the following?

(A) A substance that removes carbon dioxide gas ✓

(B) A plant to produce oxygen

(C) A glucose reserve

(D) A valve to release excess water

3 Cellular Energetics MCQ

An experiment to measure the rate of respiration in crickets and mice at 10°C and 25°C was performed using a respirometer, an apparatus that measures changes in gas volume. Respiration was measured in mL of O_2 consumed per gram of organism over several five-minute trials and the following data were obtained.

Organism	Temperature (°C)	Average respiration (mL O_2/g/min)
Mouse	10	0.0518
Mouse	25	0.0321
Cricket	10	0.0013
Cricket	25	0.0038

6. According to the data, the mice at 10°C demonstrated greater oxygen consumption per gram of tissue than did the mice at 25°C. This is most likely explained by which of the following statements?

(A) The mice at 10°C had a higher rate of ATP production than the mice at 25°C. ✓

(B) The mice at 10°C had a lower metabolic rate than the mice at 25°C.

(C) The mice at 25°C weighed less than the mice at 10°C.

(D) The mice at 25°C were more active than the mice at 10°C.

Answer A

This option is correct. Mice are endotherms and at cold temperatures will increase their rate of ATP production in order to shiver. This increase in metabolism will produce heat according to the second law of thermodynamics. This heat production will help the mouse maintain a constant internal environment.

7. During aerobic cellular respiration, oxygen gas is consumed at the same rate as carbon dioxide gas is produced. In order to provide accurate volumetric measurements of oxygen gas consumption, the experimental setup should include which of the following?

(A) A substance that removes carbon dioxide gas ✓

(B) A plant to produce oxygen

(C) A glucose reserve

(D) A valve to release excess water

Answer A

This option is correct. There is no change in gas volume measured by the respirometer unless carbon

3 Cellular Energetics MCQ

dioxide is removed, because oxygen is consumed at the same rate that carbon dioxide is produced during cellular respiration.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

8. According to the results of this experiment, germination of tobacco seeds during the first week is

(A) increased by exposure to light ✓

(B) unaffected by light intensity

(C) prevented by paper towels

(D) accelerated in green-leaved seedlings

Answer A

This option is correct. Dish B, which was uncovered, had 20 seeds germinate, while the covered dish A had only 12 seeds germinate in the first week.

9. Additional observations were made on day 21, and no yellow-leaved seedlings were found alive in either dish. This is most likely because

(A) yellow-leaved seedlings were unable to absorb water from the paper towels

(B) taller green-leaved seedlings blocked the light and prevented photosynthesis

(C) yellow-leaved seedlings were unable to convert light energy to chemical energy ✓

(D) a higher rate of respiration in yellow-leaved seedlings depleted their stored nutrients

3 Cellular Energetics MCQ

Answer C

This option is correct. The only difference between the two plants after day 14 was the leaf color. Without chlorophyll, the yellow seedlings could not absorb enough light energy to convert to chemical energy to sustain growth. Therefore, no yellow seedlings were found alive in either dish after day 21.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
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(D) accelerated in green-leaved seedlings

11. Additional observations were made on day 21, and no yellow-leaved seedlings were found alive in either dish. This is most likely because

(A) yellow-leaved seedlings were unable to absorb water from the paper towels

(B) taller green-leaved seedlings blocked the light and prevented photosynthesis

(C) yellow-leaved seedlings were unable to convert light energy to chemical energy ✓

(D) a higher rate of respiration in yellow-leaved seedlings depleted their stored nutrients

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3 Cellular Energetics MCQ

12. Some cells release active signaling proteins when membrane-bound precursor proteins are cleaved by proteolytic enzymes. The signaling proteins can then bind to receptors on the surface of a target cell, thereby activating an intracellular signaling pathway and eliciting a response from the target cell.

This mechanism of activating receptor-binding signaling proteins has been observed in a variety of organisms from bacteria to humans. Many of the enzymes responsible for proteolysis of membrane-bound precursor proteins have been isolated and characterized.

Which of the following questions would be most appropriate to investigate whether the proteolytic enzymes are evolutionarily conserved among species?

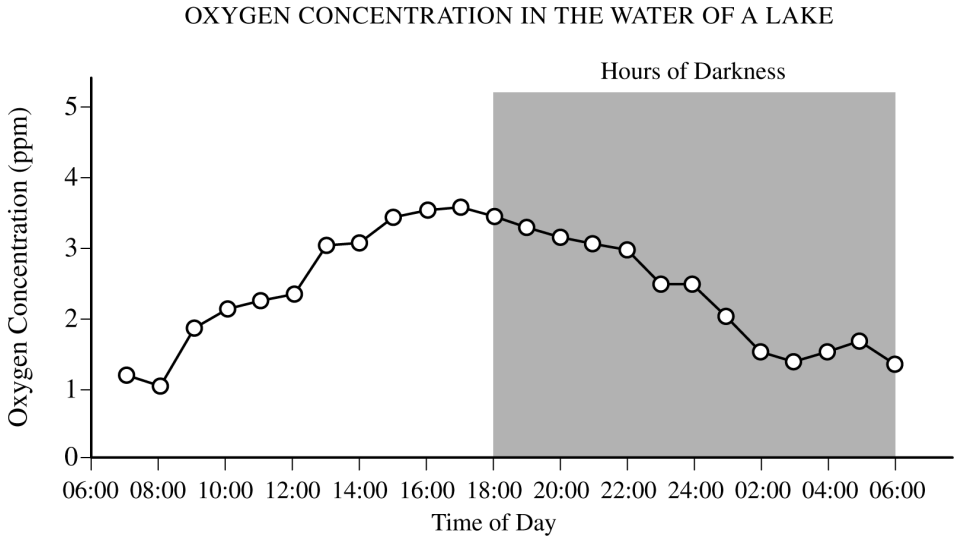
(A) Are the genes encoding the proteolytic enzymes expressed in the same cell types in all species?

(B) Once the precursor proteins of different species are cleaved, do the active signaling proteins bind to the same receptors on different target cells?

(C) If a proteolytic enzyme from one species is incubated with a precursor protein from another species, does correct cleavage occur? ✓

(D) Are the proteolytic enzymes synthesized in the rough endoplasmic reticulum of all species?

13.



What most likely causes the trends in oxygen concentration shown in the graph above?

(A) The water becomes colder at night and thus holds more oxygen.

(B) Respiration in most organisms increases at night.

(C) More organisms are respiring at night than during the day.

(D) Photosynthesis produces more oxygen than is consumed by respiration during the day. ✓

3 Cellular Energetics MCQ

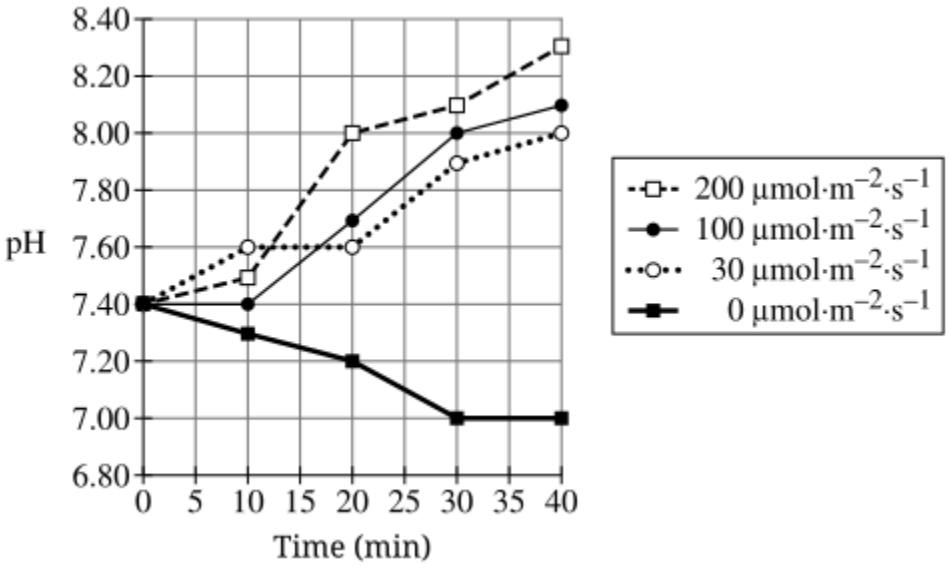


Figure 1. Effect of light intensity ($\mu mol \cdot m^{-2} \cdot s^{-1}$) on the pH of a solution containing algae

Students investigated the effect of light intensity on the rates of photosynthesis and cellular respiration in algae. They placed equal numbers of algal cells into test tubes containing a solution at $pH\ 7.40$. The test tubes were divided into four treatment groups, and each treatment group was exposed to a different light intensity. The pH in the tubes was measured every 10 minutes and used as an indicator of CO_2 concentration because CO_2 produces an acidic solution when dissolved in water (Figure 1).

14. Which of the following claims is best supported by the data?

(A) Algae only carry out photosynthesis when exposed to $200\ \mu mol \cdot m^{-2} \cdot s^{-1}$.

(B) Algae double the rate of photosynthesis when the light intensity doubles.

(C) Algae that are exposed to $30\ \mu mol \cdot m^{-2} \cdot s^{-1}$ of light perform photosynthesis instead of cellular respiration.

(D) Algae that are exposed to $100\ \mu mol \cdot m^{-2} \cdot s^{-1}$ of light have a greater rate of photosynthesis than of cellular respiration over time. ✓

Answer D

Correct. As indicated by the increase in pH, the rate of photosynthesis at $100\ \mu mol \cdot m^{-2} \cdot s^{-1}$ increased over time as the CO_2 produced by respiration was consumed in photosynthesis. This contrasts with the pH of the treatment group exposed to no light ($0\ \mu mol \cdot m^{-2} \cdot s^{-1}$) in which the pH decreased over time as CO_2 accumulated, because respiration but not photosynthesis was occurring.

3 Cellular Energetics MCQ

15. Scientists claim that changes in the rate of photosynthesis will be reflected by changes in pH.

Which of the following best supports their claim?

(A) Algae performing photosynthesis remove $\mathrm{CO_2}$ from the solution, increasing pH. ✓

(B) Algae performing photosynthesis release $\mathrm{CO_2}$ into the solution, decreasing pH.

(C) Algae performing photosynthesis remove $\mathrm{O_2}$ from the solution, decreasing pH.

(D) Algae performing photosynthesis release $\mathrm{O_2}$ into the solution, increasing pH.

Answer A

Correct. As compared with the "no light" control, the pH of each treatment group increased over time as the $\mathrm{CO_2}$ produced by respiration was consumed in photosynthesis. As the rate of photosynthesis increases, so does the pH.

16. Based on Figure 1, which of the following environmental changes will result in the greatest increase in the net photosynthetic productivity by the algae?

(A) A change in light intensity from $0\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $30\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ ✓

(B) A change in light intensity from $30\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $100\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$

(C) A change in light intensity from $100\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $200\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$

(D) A change in light intensity from $200\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $30\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$

Answer A

Correct. At the end of the experiment, the greatest increase in net photosynthetic productivity, as measured by a change in pH, was between $0\ \mu\mathrm{mol} \cdot \mathrm{m^{-2}} \cdot \mathrm{s^{-1}}$ and $30\ \mu\mathrm{mol} \cdot \mathrm{m^{-2}} \cdot \mathrm{s^{-1}}$. There was a 10-fold increase in pH between these two treatments, from 7.0 to 8.0.

17. Based on Figure 1, which of the following was used as a control in the experiment?

(A) A test tube placed directly in front of a light source

(B) The $0\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ light-intensity treatment ✓

(C) The $200\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ light-intensity treatment

(D) A test tube containing no algal cells

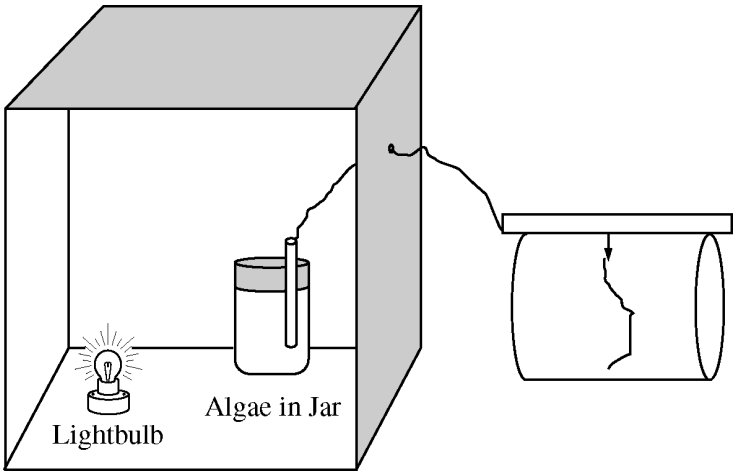
3 Cellular Energetics MCQ

Answer B

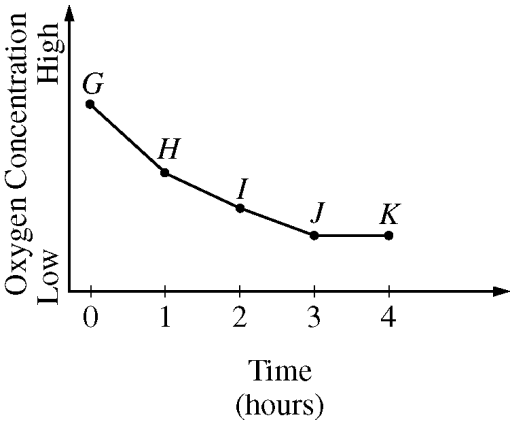
Correct. No light served as a control when measuring the effect of various light intensities.

For following group of questions first study the description of the situation and diagrams and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

A student studied the effects of light intensity on oxygen production in green algae. The algae were suspended in water inside a sealed glass jar, and the jar was placed into a constant-temperature, lightproof box containing a light source. A probe was inserted into the jar to record the concentration of oxygen. The probe was connected to a recording device. The setup is shown below.



The student decreased the intensity of the light hourly and recorded the corresponding changes in oxygen concentration. The graph below shows the results from the recording device.



3 Cellular Energetics MCQ

18. An increase in the rate of oxygen production by algae would be accompanied by a comparable increase in the rate of production of which of the following substances?

(A) $C_6H_{12}O_6$ ✓

(B) CO_2

(C) CH_4

(D) NH_3

(E) H_2O

19. Based on the data shown, changes in the light intensity resulted in changes in the rate of which of the following processes?

(A) Excretion

(B) Photosynthesis ✓

(C) Respiration

(D) Translation

(E) Transcription

20. The rate of oxygen production equaled the rate of oxygen consumption during which of the following time periods?

(A) *G* to *H*

(B) *H* to *I*

(C) *I* to *J*

(D) *J* to *K* ✓

(E) *G* to *K*

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to molecules of the following substances.

(A) Cytochrome
(B) $FADH_2$
(C) NAD^+
(D) $NADP^+$
(E) Oxygen (O_2)

21. An intermediate electron acceptor for oxidations that occur in both glycolysis and in Krebs cycle reactions

3 Cellular Energetics MCQ

(A) Cytochrome
(B) $FADH_2$
(C) NAD^+ ✓
(D) $NADP^+$
(E) Oxygen (O_2)

22. Coenzyme that transfers electrons from the Krebs cycle to the mitochondrial electron-transport chain at a lower energy level than that of electrons entering at the beginning of the chain

(A) Cytochrome
(B) $FADH_2$ ✓
(C) NAD^+
(D) $NADP^+$
(E) Oxygen (O_2)

---

23. A researcher claims that increased atmospheric carbon dioxide levels cause increased growth rates in plants.

Which of the following statements best supports the researcher's claim?

(A) Atmospheric carbon dioxide is produced by the burning of fossil fuels, which are formed from the remains of living organisms such as plants.
(B) Atmospheric carbon dioxide is a byproduct of cellular respiration, which is a metabolic process that occurs in plants and other living organisms.
(C) Atmospheric carbon dioxide typically enters plant leaves through stomata, which plants rely on for regulating gas exchange with the atmosphere.
(D) Atmospheric carbon dioxide is the raw material for photosynthesis, which plants rely on for producing sugars and other organic compounds. ✓

Answer D

Correct. Atmospheric carbon dioxide is the raw material for photosynthesis, which is the metabolic process by which green plants use energy in sunlight to produce sugars and other organic compounds. The plants use the sugars and organic compounds to support growth, repair, and reproduction.

24. A researcher claims that the synthesis of ATP from ADP and inorganic phosphate $(\mathrm{P_i})$ is essential to cellular function.

Which of the following statements best helps justify the researcher's claim?

3 Cellular Energetics MCQ

(A) ADP is a small molecule that some cells release into their environment as a way of communicating with other cells.

(B) ATP hydrolysis is an energy-releasing reaction that is often coupled with reactions that require an input of energy. ✓

(C) Inorganic phosphate $(\mathrm{P_i})$ is a substance that cells typically acquire from their environment.

(D) ATP synthase is a mitochondrial enzyme that catalyzes the conversion of ADP and $\mathrm{P_i}$ to ATP.

Answer B

Correct. Many cellular processes necessary for maintaining the internal organization of the cell are coupled with the hydrolysis of ATP. Without a supply of ATP, these activities will stop and the cells will die.

25. During respiration, most ATP is formed as a direct result of the net movement of

(A) potassium against a concentration gradient

(B) protons down a concentration gradient ✓

(C) electrons against a concentration gradient

(D) electrons through a channel

(E) sodium ions into the cell

26. A scientist performed an experiment to test the ability of chloroplasts to produce ATP in the dark. The scientist maintained chloroplasts under dark conditions and placed them first in a pH 4 solution for 15 seconds and then in a pH 8 solution containing ADP and P_i (inorganic phosphate). Under these conditions, the pH of the stroma increased to 8, the pH of the thylakoid space remained at 4, and the chloroplasts produced ATP.

Which of the following best explains how the difference in pH between the stroma and the thylakoid space enabled the chloroplasts to produce ATP in the dark?

(A) The difference in pH created a proton gradient that enabled protons to pass through ATP synthase, activating it to produce ATP. ✓

(B) The difference in pH enabled complexes of protons and ADP in the thylakoid space to move through ATP synthase to the stroma, producing ATP.

(C) The difference in pH stimulated the electron flow needed to pump protons into the stroma and induce P_i and ADP to bond and form ATP.

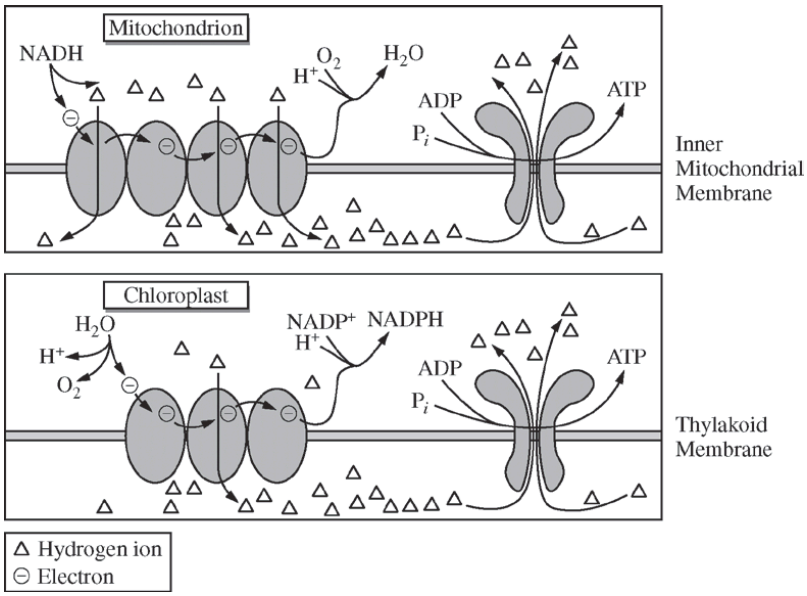
(D) The difference in pH caused the photosystems to release high energy electrons that provided enough energy for ADP and P_i to bond and form ATP.

3 Cellular Energetics MCQ

Answer A

Correct. Normal chloroplast function produces ATP by using photosystems to capture light, which generates high-energy electrons that are passed down an electron transport chain (ETC). Members of the ETC pump protons, creating a proton gradient between the thylakoid space and the stroma. Protons then move down their gradient through ATP synthase. The passage of protons provides energy to ATP synthase to form ATP from ADP and P_i. The scientist achieved this same effect by using low, then high, pH solutions in the absence of light.

27. The figures below illustrate the similarities between ATP synthesis in mitochondria and chloroplasts.



The figures can best assist in answering which of the following questions?

(A) Do electron transport chains create a gradient so that ATP synthase can generate ATP molecules? ✓

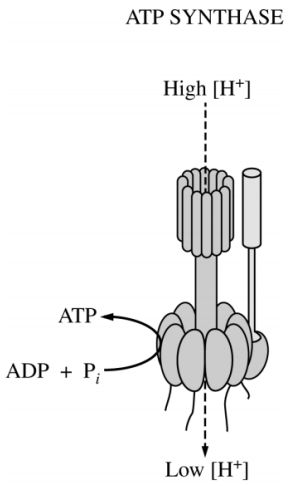
(B) What are the sources of energy that drive mitochondrial and chloroplast electron transport systems?

(C) What is the optimal temperature at which ATP synthase chemically converts ADP and a phosphate group into one molecule of ATP?

(D) What is the evolutionary relationship between the ATP synthase in mitochondria and the ATP synthase in chloroplasts?

3 Cellular Energetics MCQ

28.



Which of the following questions will best direct an investigation of the mechanism of ATP synthase?

(A) What is the source of the inorganic phosphate that is used to generate ATP from ADP?

(B) Is the phosphorylation of ADP by ATP synthase dependent on the formation of a proton gradient? ✓

(C) Can ATP synthase use the energy released by phosphorylation of ADP to pump protons against a concentration gradient?

(D) Can oxidative phosphorylation be uncoupled from the electron transport chain?

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29. Brown fat is a type of fat tissue found in hibernating mammals. Inside the mitochondria of these fat tissue cells, these mammals have an uncoupling protein embedded in the inner mitochondrial membrane. This uncoupling protein allows hydrogen ions to leak from the intermembrane space back into the mitochondrial matrix. Figure 1 shows details of the processes in the inner mitochondrial membrane.

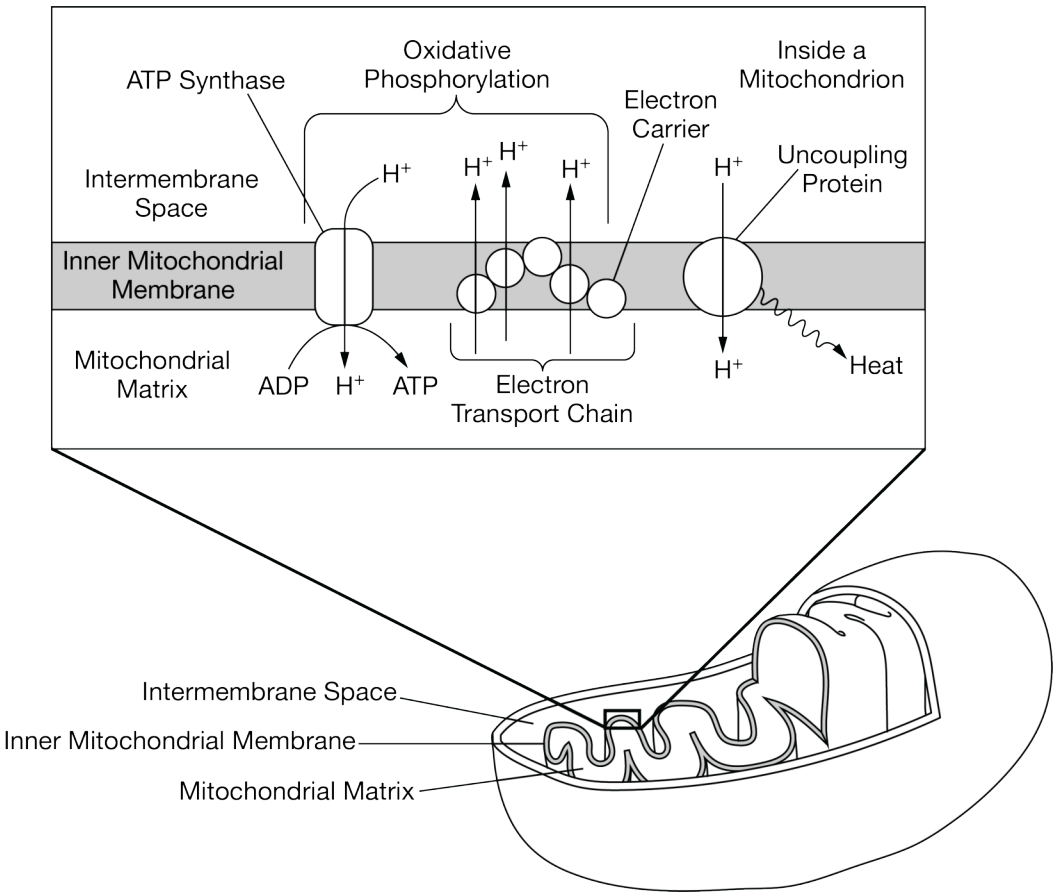


Figure 1. Processes in the inner mitochondrial membrane

Which of the following statements provides reasoning that supports the claim that brown fatty tissue keeps an animal warm?

(A) The uncoupling protein in this tissue increases the production of ATP and causes more body heat to be produced to warm the animal.

(B) The uncoupling protein in this tissue reduces the proton gradient across the membrane and thus produces heat to warm the animal without ATP production. ✓

(C) The uncoupling protein in this tissue causes an increase in the proton gradient, which causes more ATP to be produced that helps to warm the animal.

(D) The uncoupling protein in this tissue reduces the production of ATP and creates an increase in the proton gradient that allows more heat energy to be produced to warm the animal.

3 Cellular Energetics MCQ

Answer B

Correct. The uncoupling protein causes a reduction in the proton gradient across the inner mitochondrial membrane because protons constantly leak into the matrix. Heat energy is released as the protons move, but less ATP is made because of the decreased concentration of hydrogen ions available to move across ATP synthase.

30. The energy required to run the Calvin cycle reactions of photosynthesis comes from which two substances produced during the light-dependent reactions?

(A) ATP and NADPH ✓
(B) ADP and PO_4
(C) H^+ and PO_2
(D) O_2 and CO_2
(E) H_2O and CO_2

31. A researcher claims that only a portion of the light energy captured by green plants is available for growth and repair.

Which of the following observations best helps justify the researcher's claim?

(A) Light-capturing pigment molecules in green plants absorb red, blue, and violet light but reflect green light.
(B) The energy of a photon of light is proportional to its frequency and inversely proportional to its wavelength.
(C) As light energy is converted to chemical energy by metabolic processes, some of the energy is lost as heat. ✓
(D) Captured energy is stored in the molecular bonds of organic molecules, including simple sugars and starch.

Answer C

Correct. During photosynthesis, some of the captured light energy is lost to the environment in the form of heat energy. As a result, only some of the captured light energy is converted to chemical energy and stored for later use in growth and repair.

32. Carbohydrate-synthesizing reactions of photosynthesis directly require

3 Cellular Energetics MCQ

(A) light

(B) products of the light reactions ✓

(C) darkness

(D) O_2 and H_2O

(E) chlorophyll and CO_2

33. The carbon that makes up organic molecules in plants is derived directly from

(A) combustion of fuels

(B) carbon fixed in photosynthesis ✓

(C) carbon dioxide produced in respiration

(D) carbon in the lithosphere

(E) coal mines

34. Which of the following statements about mitochondrial chemiosmosis is NOT true?

(A) A proton gradient is established across the inner membrane of the mitochondrion.

(B) The potential energy released from the mitochondrial proton gradient is used to produce ATP.

(C) The mitochondrial proton gradient provides energy for muscle contraction.

(D) Proteins embedded in the inner mitochondrial membrane play an important role in ATP synthesis.

(E) Heat energy is required to establish the electron transport chain. ✓

35. Different photosynthetic organisms have different types of chlorophyll molecules. The distribution of chlorophylls in several different groups of organisms is shown in Table 1. A plus sign (+) in the table indicates the presence of a chlorophyll, while a minus sign (−) indicates its absence.

Table 1. The distribution of chlorophylls in several groups of organisms

	Chlorophyll *a*	**Chlorophyll *b***	**Chlorophyll *c***	**Chlorophyll *d***
Flowering plants	+	+	−	−
Green algae	+	+	−	−
Brown algae	+	−	+	−
Red algae	+	−	−	+
Cyanobacteria	+	−	−	−

Based on the data, which of the following most likely describes the evolutionary relationship among the organisms?

3 Cellular Energetics MCQ

(A) Because brown algae, red algae, and cyanobacteria lack chlorophyll *b*, they evolved before green algae and flowering plants did.

(B) Because green algae and flowering plants contain chloroplasts, they evolved more recently than brown algae, red algae, and cyanobacteria did.

(C) Because increasingly complex forms of chlorophyll are found in red algae, brown algae, green algae, and flowering plants, respectively, this reflects the order of their appearance.

(D) Because all of the organisms contain chlorophyll *a*, the organisms share a common ancestor. ✓

36. Which of the following best describes the function of the coenzymes NAD^+ and FAD in eukaryotic cellular respiration?

(A) They participate in hydrolysis reactions by accepting protons from water molecules.

(B) They participate directly in the phosphorylation of ADP to ATP.

(C) They serve as final electron acceptors in the electron transport chain.

(D) They aid vitamins such as niacin in the breakdown of glucose.

(E) They accept electrons during oxidation-reduction reactions. ✓

37. Within the cell, many chemical reactions that, by themselves, require energy input (have a positive free-energy change) can occur because the reactions

(A) may be coupled to the hydrolysis of ATP ✓

(B) take place very slowly

(C) take place when the cells are at unusually high temperatures

(D) are catalyzed by enzymes

(E) are aided by various metal ions that act as catalysts

3 Cellular Energetics MCQ

38. A researcher claims that the initial rise of oxygen in Earth's early atmosphere, which occurred approximately 2.3 billion years ago, resulted from the metabolic activity of prokaryotic organisms. The claim is based on an interpretation of the geochemical and fossil evidence represented in Figure 1.

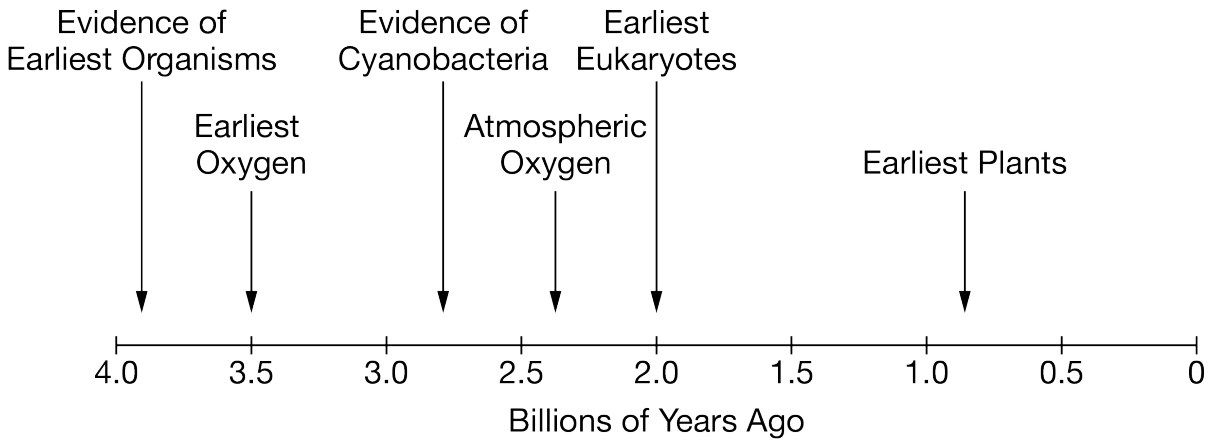


Figure 1. Selected events in geologic time based on geochemical and fossil evidence

Which of the following types of evidence will best support the researcher's claim?

(A) Evidence that some of the earliest eukaryotes used oxygen to produce ATP by cellular respiration

(B) Evidence that the earliest plants produced oxygen as a by-product of photosynthesis

(C) Evidence that some of the earliest organisms carried out photosynthesis without producing oxygen

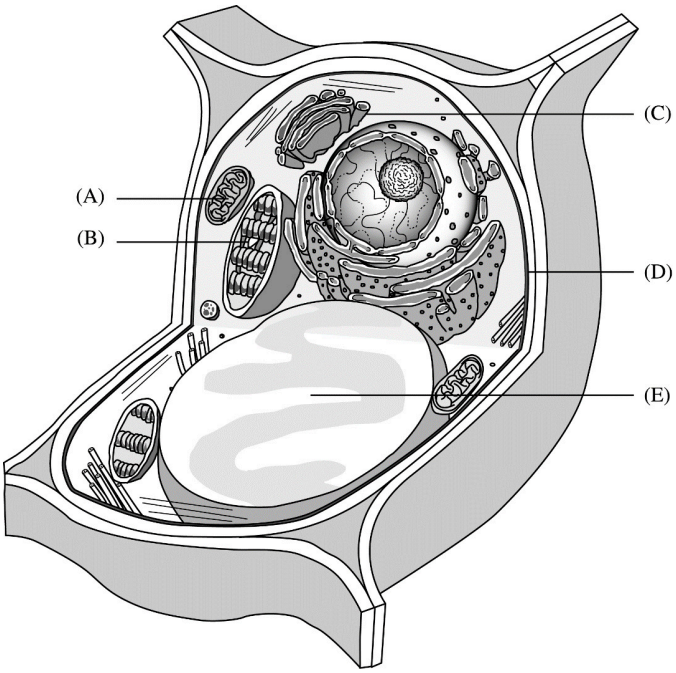
(D) Evidence that the cyanobacteria produced oxygen as a by-product of photosynthesis ✓

Answer D

Correct. Based on Figure 1, cyanobacteria appear in the fossil record before the rise of oxygen in Earth's early atmosphere. The researcher's claim will be supported by evidence that the cyanobacteria that were on Earth more 2.3 billion years ago produced oxygen as a by-product of photosynthesis. Also, cyanobacteria are prokaryotes, which is consistent with the researcher's claim.

3 Cellular Energetics MCQ

The following questions refer to the following diagram. For each phrase or sentence, select the labeled part to which it is most closely related. Each option may be used once, more than once, or not at all for each group.



39. Site of glucose synthesis

(A) A

(B) B ✓

(C) C

(D) D

(E) E

40. Site of conversion of chemical energy of glucose to ATP

(A) A ✓

(B) B

(C) C

(D) D

(E) E

3 Cellular Energetics MCQ

41.

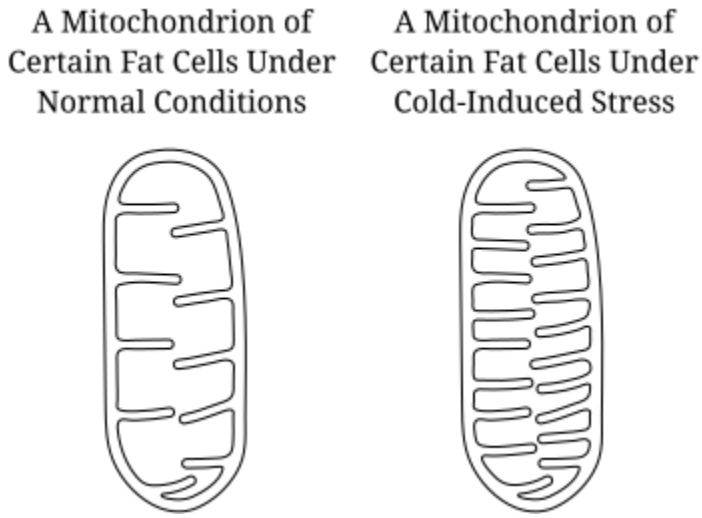


Figure 1. Model of a mitochondrion in cells under normal conditions and in cells experiencing cold-induced stress

In a certain type of fat cells, cold temperatures trigger a series of events that include changes to the structure of the mitochondria. A model of a mitochondrion from these fat cells under normal environmental conditions and when experiencing cold-induced stress is shown in Figure 1.

Which of the following best explains how the changes in mitochondrial structure observed when the cell experiences cold-induced stress affect cellular respiration?

(A) The volume of the mitochondrial intermembrane space decreases, preventing the reactions of the Krebs cycle from occurring.

(B) The surface area of the inner membrane decreases, preventing the transport of oxygen molecules into the mitochondria.

(C) The volume within the mitochondria increases, providing more space for the reactions of glycolysis to take place.

(D) The surface area of the inner membrane increases, providing more space for the components of the electron transport chain. ✓

Answer D

Correct. The increased folding of the inner membrane in cold-induced mitochondria results in an increase in surface area. This allows for more components of the electron transport chain (ETC) that is localized to the inner mitochondrial membrane. This in turn enables an increased rate of cellular respiration that can help maintain the body temperature of an endotherm exposed to cold temperatures.

3 Cellular Energetics MCQ

42.

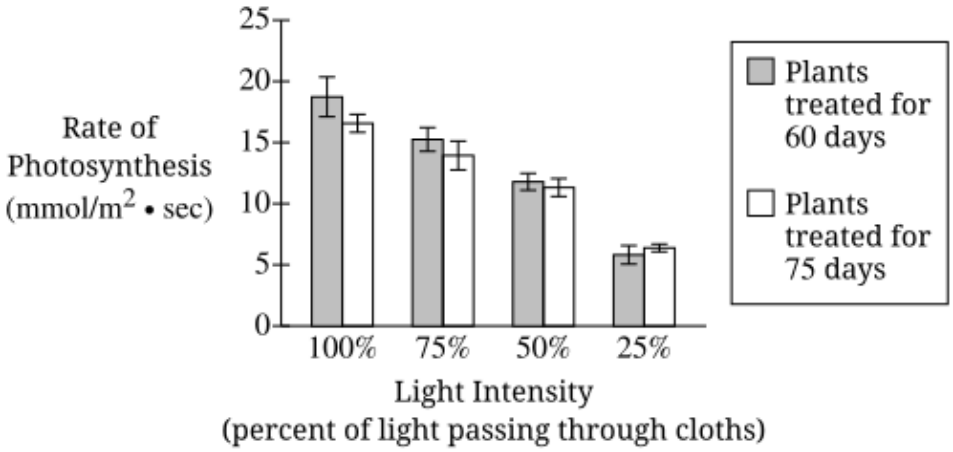


Figure 1. Effect of light intensity and growth period on the rate of photosynthesis in sage plants. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

Using light-filtering cloths, scientists exposed sage plants to four different light intensities. Half of the sage plants were exposed to light for 60 days, while the other half were exposed to light for 75 days. The rate of photosynthesis in each plant was then determined (Figure 1).

Which of the following statements best describes the data?

(A) At 100% light intensity, the rate of photosynthesis was higher in plants that were treated for 60 days than in plants that were treated for 75 days.

(B) At 75% light intensity, the rate of photosynthesis was unaffected by an increase in treatment time. ✓

(C) At 50% light intensity, the rate of photosynthesis was the same as the rate of photosynthesis at 75% light intensity, regardless of the length of treatment time.

(D) At 25% light intensity, the rate of photosynthesis was higher in plants treated for 75 days than it was in plants treated for 60 days.

Answer B

Correct. Because the 60 days and 75 days error bars overlap, the rate of photosynthesis is the same for both treatments.

3 Cellular Energetics MCQ

43. Researchers conducted an experiment to investigate the effects of a valinomycin treatment on skeletal muscle cells. Valinomycin is a naturally occurring substance that can be used as a drug. The results of the experiment are presented in the table.

	Relative Rates of ATP Production	
Time after Treatment	Untreated Cells	Valinomycin-Treated Cells
5 minutes	1.0	0.3
10 minutes	7.7	2.7

Which of the following claims about the effects of the valinomycin treatment is best supported by the data presented in the table?

(A) The valinomycin treatment caused an increase in the activity of the rough endoplasmic reticulum.

(B) The valinomycin treatment caused an increase in the activity of the Golgi complex.

(C) The valinomycin treatment caused a decrease in the activity of the lysosome.

(D) The valinomycin treatment caused a decrease in the activity of the mitochondria. ✓

Answer D

Correct. The data indicate that the valinomycin treatment caused a decrease in the relative rate of ATP production, which likely resulted from impaired mitochondrial function.

44. Students in a class measured the mass of various living organisms. They then kept the organisms in the dark for 24 hours before remeasuring them. None of the organisms were provided with nutrients during the 24-hour period. The data are as follows.

Organism	Starting Mass (g)	Final Mass (g)
Elodea (submerged aquatic plant)	15.10	14.01
Goldfish	10.10	9.84
Sea anemone	25.60	24.98

Which of the following is the best explanation for the pattern of change in mass of the organisms over time?

3 Cellular Energetics MCQ

(A) Water loss due to evaporation

(B) Cellular respiration ✓

(C) The law of conservation of matter

(D) Growth and reproduction

45. The electron transport chain (ETC) is made up of several carrier molecules that transport electrons. These carrier molecules are found within membranes.

Which of the following statements best explains why these carrier molecules are typically found within membranes?

(A) The electrons that move through the ETC are directly used to drive the processes of endocytosis and exocytosis.

(B) The electrons that move through the ETC are coupled with the pumping of ATP out of the intermembrane space.

(C) The energy derived from the ETC creates a difference in the concentration of protons on either side of the membrane. ✓

(D) The energy derived from the ETC is used to fully oxidize an ATP molecule, releasing stored energy outside the membrane.

46.

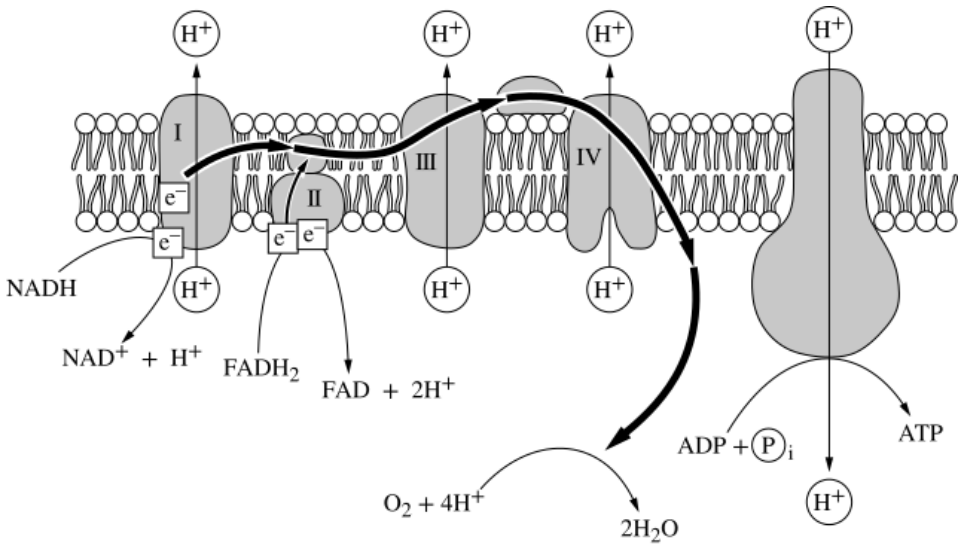


Figure 1. The electron transport chain of cellular respiration. The bold arrows passing through the complexes in the membrane represent the path of electron flow in mitochondria.

Compound X binds to complex IV of the mitochondrial electron transport chain and prevents complex IV from accepting electrons.

Based on Figure 1, which of the following best explains why the cells of an animal exposed to compound X have an increased ratio of $NADH$ to NAD^+?

3 Cellular Energetics MCQ

(A) $NADH$ increases because compound X directly prevents ATP synthase from producing the ATP necessary for the oxidation of $NADH$ to NAD^+.

(B) $NADH$ increases because the binding of compound X to complex IV also prevents the transport of NAD^+ out of the mitochondria.

(C) $NADH$ cannot be oxidized to NAD^+ because complexes I, II, and III cannot accept electrons if electrons cannot be passed to complex IV. ✓

(D) $NADH$ cannot be oxidized to NAD^+ because H_2O cannot bind to the active site of complex IV if this complex cannot accept electrons.

Answer C

Correct. If complex IV is blocked, complexes I, II, and III will remain reduced and unable to accept any more electrons. $NADH$ will be unable to transfer electrons to complex I and be oxidized to NAD^+, resulting in an increase in the ratio of $NADH$ to NAD^+ in the cells.

47. Experimental evidence shows that the process of glycolysis is present and virtually identical in organisms from all three domains, Archaea, Bacteria, and Eukarya. Which of the following hypotheses could be best supported by this evidence?

(A) All organisms carry out glycolysis in mitochondria.

(B) Glycolysis is a universal energy-releasing process and therefore suggests a common ancestor for all forms of life. ✓

(C) Across the three domains, all organisms depend solely on the process of anaerobic respiration for ATP production.

(D) The presence of glycolysis as an energy-releasing process in all organisms suggests that convergent evolution occurred.

48. Experimental evidence shows that the process of glycolysis is present and virtually identical in organisms from all three domains, Archaea, Bacteria, and Eukarya. Which of the following hypotheses could be best supported by this evidence?

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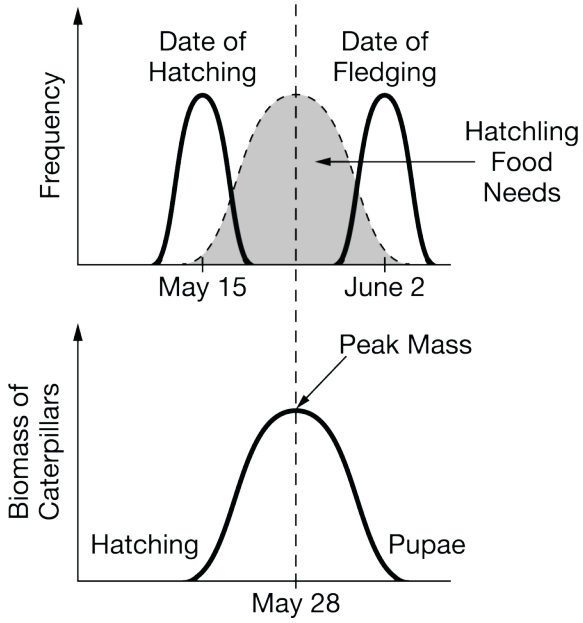
3 Cellular Energetics MCQ

49. European flycatchers feed caterpillars to their hatchlings. Graph 1 shows the average dates of hatching and fledging (leaving the nest), and the biomass of the caterpillars between early May (when flycatcher young hatch) and June (when fledging of young occurs).



Graph 1. Comparison of European flycatcher hatching and fledging dates and caterpillar biomass

Based on the data, scientists claim that the reproductive behavior of European flycatchers is influenced by the availability of energy sources. Which of the following statements best justifies this claim?

(A) Young European flycatchers hatch from eggs when caterpillar biomass is available for the young birds to consume and convert into energy for growth. ✓

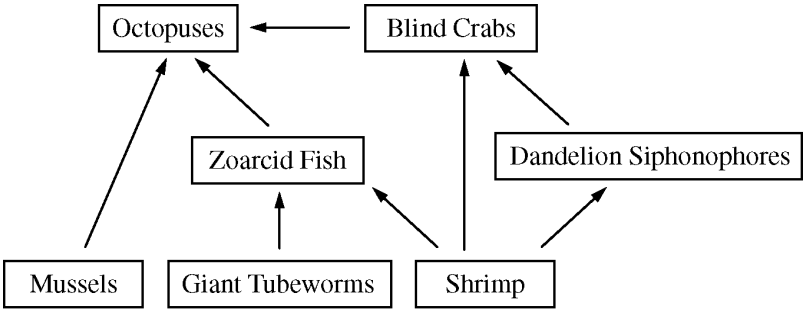
(B) European flycatcher hatchlings begin to need energy to leave the nest only after the caterpillars have turned into pupae.

(C) Female European flycatchers require energy to lay eggs, so they lay their eggs when the caterpillar biomass is maximal.

(D) The energy requirements for hatching European flycatchers and caterpillars are proportional to each other.

3 Cellular Energetics MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



The food web above represents feeding relationships in a biological community near a deep-sea hydrothermal vent. Hydrothermal vents are geysers on the seafloor that gush super-heated, mineral-rich water. The seawater surrounding hydrothermal vents typically contains carbon dioxide (CO_2), molecular hydrogen (H_2), hydrogen sulfide (H_2S), and methane (CH_4). Sunlight, however, fails to reach the seafloor where deep-sea hydrothermal vents are located.

As part of an investigation, researchers collected living specimens from an area near a deep-sea hydrothermal vent. Mussels in the collection were found to be dependent on molecular hydrogen in seawater. Also, the researchers discovered multiple species of bacteria living in the gills of the mussels. Mussels use gills for filter-feeding and gas exchange with the surrounding seawater. On the basis of their experimental results, the researchers hypothesized that some bacteria living in the gills of the mussels are capable of chemosynthesis.

50. Which of the following best explains how biological communities near deep-sea hydrothermal vents can exist in a habitat lacking sunlight?

(A) Environmental conditions on some distant planets resemble those experienced by organisms living near hydrothermal vents.

(B) Heterotrophs metabolize carbon-containing compounds produced by the photosynthetic organisms that live on the seafloor.

(C) Some organisms rely on energy captured from inorganic compounds to drive basic biological processes. ✓

(D) Some organisms that can tolerate high temperatures are single celled, whereas others are multicellular.

3 Cellular Energetics MCQ

51. A researcher claims that a certain herbicide suppresses plant growth by inhibiting chloroplast function. To test the claim, the researcher treats isolated chloroplasts with increasing concentrations of the herbicide. The data from the experiment are presented in Table 1.

Table 1. The effect of an herbicide on the function of isolated chloroplasts

Herbicide Concentration ($\mu g/L$)	pH Difference across Thylakoid Membrane	ATP Production
0	3.5	Very high
2	2.5	Moderate
8	0.5	Very low
32	0	Very low

Which of the following statements best clarifies how the data support the researcher's claim?

(A) The thylakoid membrane is more permeable to carbon dioxide than to polar molecules.

(B) ATP synthase activity depends on a proton gradient across the thylakoid membrane. ✓

(C) Some enzymes embedded in the thylakoid membrane catalyze the hydrolysis of ATP.

(D) Carbon fixation in the Calvin-Benson cycle takes place in the stroma of chloroplasts.

Answer B

Correct. The results of the experiment indicate that ATP production in the isolated chloroplasts decreases with increasing concentrations of the herbicide. The data also show that the decrease in ATP production is associated with a decrease in the pH difference across the thylakoid membrane. The pH difference is created by a proton gradient, which ATP synthase relies on to produce ATP.

52. Glycolysis is a metabolic pathway that converts glucose into pyruvate and is observed in each of the three domains. The hexokinase family of enzymes is required during glycolysis to phosphorylate six-carbon sugars. Researchers designed a general hexokinase inhibitor that is effective in the neurons of rats.

Which of the following best predicts the effect of adding this inhibitor to a culture of plant cells?

(A) Plant cells will be unaffected by the inhibitor as they do not perform glycolysis.

(B) Plant cells will be unable to perform glycolysis due to the inhibitor and will die. ✓

(C) Plant cells will be unable to perform photosynthesis due to the inhibitor and will die.

(D) Plant cells will still be able to perform glycolysis since plant hexokinase is not structurally similar to animal hexokinase.

3 Cellular Energetics MCQ

Answer B

Correct. The inhibitor is a general hexokinase inhibitor, which will prevent any form of hexokinase from functioning.

53. In a mesophyll cell of a leaf, the synthesis of ATP occurs in which of the following?

I. Ribosomes
II. Mitochondria
III. Chloroplasts

(A) I only
(B) II only
(C) III only
(D) II and III only ✓
(E) I, II, and III

54. In an experiment, a scientist isolates mitochondria from living cells and suspends them in two different buffered solutions. One solution is maintained at pH 4, while the other solution is maintained at pH 9. The scientist finds that mitochondria in the solution at pH 4 continue to produce ATP but those in the pH 9 solution do not.

The results of the experiment can be used as evidence in support of which of the following scientific claims about mitochondrial activity?

(A) Mitochondria in a cell-free environment are unable to convert thermal energy into ATP.
(B) The electron transport chain pumps electrons from the cytosol to the mitochondrial matrix.
(C) ATP production in mitochondria requires a hydrogen ion gradient that favors movement of protons into the mitochondrial matrix. ✓
(D) ATP synthase molecules change their orientation in relation to the proton gradient across the mitochondrial membrane.

55. In most freshwater fish, nitrogenous waste is primarily excreted as ammonia, which is highly soluble in water and is toxic at low concentrations. In terrestrial mammals, ammonia is converted to urea before it is excreted. Urea is also highly soluble in water but is less toxic than ammonia at low concentrations. Which of the following best explains why freshwater fish do <u>not</u> convert ammonia to urea for excretion?

(A) The metabolic pathways of fish do not normally involve nitrogen consumption.
(B) The dilution of ammonia by direct excretion into freshwater conserves energy. ✓
(C) Ammonia is concentrated in tissues, where it is stored prior to excretion.
(D) The nitrogen in ammonia is recycled for use in protein and nucleotide synthesis.

3 Cellular Energetics MCQ

Answer B

This option is correct. There is no need to convert ammonia to urea because it can be excreted often in dilute, nontoxic concentrations. This process requires less energy than the additional step of urea conversion. Dehydration due to water loss is not a problem for freshwater fish because water is plentiful.

56. Which of the following statements is true about the Krebs (citric acid) cycle and the Calvin (light-independent) cycle?

(A) They both result in a net production of ATP and NADH.
(B) They both require a net input of ATP.
(C) They both result in a release of oxygen.
(D) They both take place within the cytoplasmic matrix.
(E) They both are carried out by enzymes located within an organelle matrix. ✓

57. Which of the following best explains why the inner mitochondrial membrane has more embedded proteins of the electron transport chain than the outer mitochondrial membrane?

(A) The inner membrane is more efficient at capturing light than the outer membrane.
(B) The inner membrane has a smaller surface area than the outer membrane.
(C) The Krebs cycle occurs in the mitochondrial matrix and not the intermembrane space. ✓
(D) The outer membrane has a greater surface area for electron transfer.

Answer C

Correct. Since the Krebs cycle occurs in the mitochondrial matrix, having the electron transport chain in the inner membrane is more efficient as it allows direct access to the NADH and \(FADH_2\) produced by the Krebs cycle.

58. In chloroplasts, ATP is synthesized from ADP plus inorganic phosphate $(\mathrm{P_i})$ in a reaction catalyzed by ATP synthase molecules that are embedded in the thylakoid membrane.

Which of the following statements provides evidence to support the claim that no ATP will be synthesized in the absence of a proton gradient across the thylakoid membrane?

3 Cellular Energetics MCQ

(A) Blocking electron flow from one carrier to the next in the electron transport chains blocks formation of a proton gradient in the thylakoid.

(B) Increasing the proton concentration difference across the thylakoid membrane is not associated with a parallel increase in the rate of ATP synthesis.

(C) No ATP is synthesized when channel proteins that allow the free passage of protons are inserted into the thylakoid membrane. ✓

(D) No ATP is synthesized while the Calvin cycle is synthesizing carbohydrates and using ATP and NADPH at a high rate.

Answer C

Correct. Channel proteins that allow the passage of protons will dissipate the proton concentration difference, and ATP will not form. Under normal circumstances, the only way for protons to cross the thylakoid membrane is by passing through the ATP synthase molecules.

59. Which of the following is an important difference between light-dependent and light-independent reactions of photosynthesis?

(A) The light-dependent reactions occur only during the day; the light-independent reactions occur only during the night.

(B) The light-dependent reactions occur in the cytoplasm; the light-independent reactions occur in chloroplasts.

(C) The light-dependent reactions utilize CO_2 and H_2O; the light-independent reactions produce CO_2 and H_2O.

(D) The light-dependent reactions depend on the presence of both photosystems I and II; the light-independent reactions require only photosystem I.

(E) The light-dependent reactions produce ATP and NADPH; the light-independent reactions use energy stored in ATP and NADPH. ✓

60. Which of the following describes a metabolic consequence of a shortage of oxygen in muscle cells?

(A) An increase in blood pH due to the accumulation of lactic acid

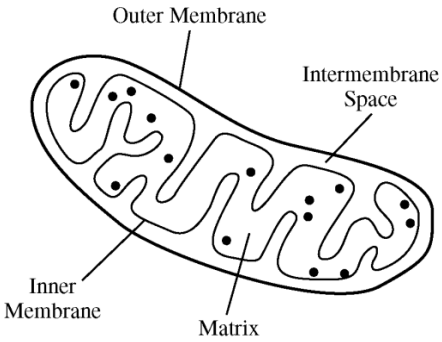
(B) No ATP production due to the absence of substrate-level phosphorylation

(C) A buildup of lactic acid in the muscle tissue due to fermentation ✓

(D) A decrease in the oxidation of fatty acids due to a shortage of ATP

3 Cellular Energetics MCQ

61.



The figure above shows an organelle typically found in eukaryotic cells. Which of the following best describes the function of the double membrane system of this organelle?

(A) The outer membrane allows the transport of all molecules into the intermembrane space, while the inner membrane serves as the regulatory boundary.

(B) The inner membrane has specialized proteins that create a hydrogen ion concentration gradient between the intermembrane space and the matrix. ✓

(C) The outer membrane contains transport proteins that establish a sodium ion concentration gradient used for ATP production, while the inner membrane contains transport proteins that establish a hydrogen ion concentration gradient used for glucose production.

(D) The toxins and wastes entering a cell cross the outer membrane and are detoxified by digestive enzymes stored within the intermembrane space.

62. In an experiment, researchers compared the growth of two different plants, plant X and plant Y. The researchers maintained the plants under nearly identical conditions and observed that plant X grew faster than plant Y. The researchers also observed that the inner mitochondrial membranes of plant X had more folds than did those of plant Y.

Which of the following conclusions about increasing the number of folds in the inner mitochondrial membrane is best supported by the results of the experiment?

(A) It increases the efficiency of photosynthesis, which results in faster cell growth.

(B) It increases the surface area available for ATP production, which results in faster cell growth. ✓

(C) It increases the amount of space available for storing cellular wastes, which results in faster cell growth.

(D) It increases the rate of protein transport to the plasma membrane, which results in faster cell growth.

Answer B

Correct. The increased surface area of the folds will contain more ATP synthase, allowing for more efficient use of the chemiosmotic gradient and more efficient production of ATP. The observation that plant X grew faster than plant Y supports this conclusion.

3 Cellular Energetics MCQ

63. More than 90 percent of the nitrogenous waste that is processed and excreted by humans is derived from the breakdown of proteins. Most of the remaining nitrogenous waste material is derived from the breakdown of

(A) fats

(B) plant cell walls

(C) steroids

(D) nucleic acids ✓

(E) glycogen

64. Muscle contraction depends on ATP hydrolysis. During periods of intense exercise, muscle cells rely on the ATP supplied by three metabolic pathways: glycolysis, mitochondrial respiration, and the phosphagen system. Figure 1 shows the rates at which the three metabolic pathways produce ATP following the start of an intense period of exercise.

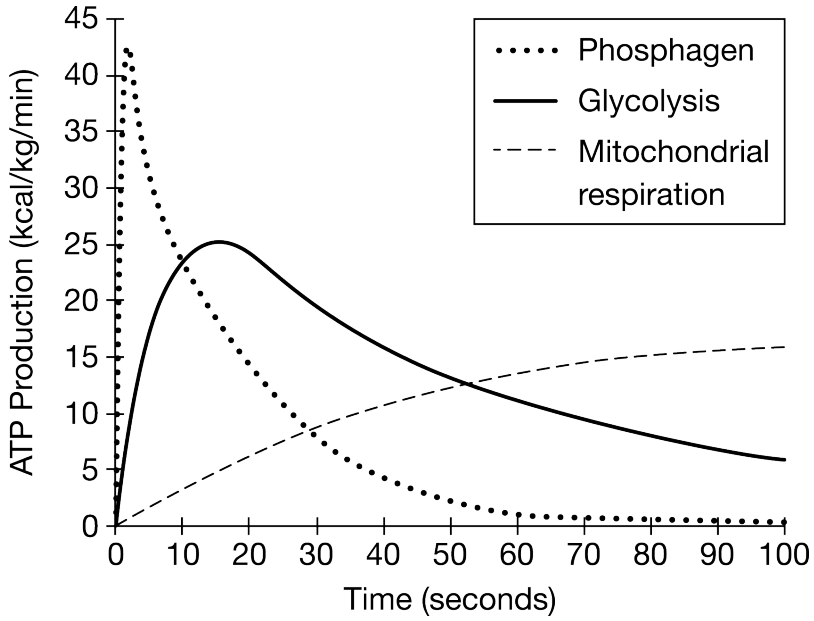


Figure 1. ATP production by three metabolic pathways following the start of an intense period of exercise

Which of the following correctly uses the data to justify the claim that the phosphagen system is an immediate, short-term source of ATP for muscle cells?

(A) ATP production by the phosphagen system increases and decreases rapidly following the start of the exercise period. ✓

(B) ATP production by the phosphagen system increases gradually and continuously throughout the entire exercise period.

(C) The ATP produced by the phosphagen system contains more energy per molecule than does the ATP produced by the other pathways.

(D) ATP hydrolysis in muscle cells occurs immediately after the start of the exercise period but stops before the end of the exercise period.

3 Cellular Energetics MCQ

Answer A

Correct. Based on Figure 1, the phosphagen system reaches its maximum rate of ATP production approximately 5 seconds after the start of the exercise period. Because ATP production by the phosphagen system increases and decreases rapidly following the start of the exercise period, the phosphagen system is an immediate, short-term source of ATP.

65. Two nutrient solutions are maintained at the same pH. Actively respiring mitochondria are isolated and placed into each of the two solutions. Oxygen gas is bubbled into one solution. The other solution is depleted of available oxygen. Which of the following best explains why ATP production is greater in the tube with oxygen than in the tube without oxygen?

(A) The rate of proton pumping across the inner mitochondrial membrane is lower in the sample without oxygen. ✓

(B) Electron transport is reduced in the absence of a plasma membrane.

(C) In the absence of oxygen, oxidative phosphorylation produces more ATP than does fermentation.

(D) In the presence of oxygen, glycolysis produces more ATP than in the absence of oxygen.

66. The function of which of the following organelles directly requires oxygen?

(A) Ribosome

(B) Mitochondrion ✓

(C) Nucleus

(D) Centriole

(E) Golgi apparatus

67. Which of the following best supports the claim that organisms of different domains share a common ancestor?

(A) Photosynthesis occurs in chloroplasts, and the citric acid cycle occurs in mitochondria.

(B) Glycolysis occurs in both prokaryotic cells and eukaryotic cells. ✓

(C) Introns are present in eukaryotic DNA but not in prokaryotic DNA.

(D) Errors in DNA synthesis provide some of the genetic variation in a population.

68. Which of the following statements best supports the claim that organisms share fundamental processes as a result of evolution?

3 Cellular Energetics MCQ

(A) All organisms that are introduced into new environments have the capacity to fill vacant ecological roles.

(B) All organisms have the ability to utilize oxygen to harness energy from the chemical breakdown of organic compounds.

(C) All organisms share a genetic code organized into triplet codons, making it possible for one organism to express a gene from another organism. ✓

(D) All organisms possess structures such as chloroplasts and mitochondria within their cells that reflect past symbiotic relationships between prokaryotic precursors.

Answer C

This option is correct. It demonstrates an understanding that all organisms, both extant and extinct, share a universal genetic code with the justification that the hereditary information encoded in DNA directs the development of biochemical, anatomical, and physiological traits by the fundamental processes of replication, transcription, and translation. Through the process of transformation, genetic information from one organism can change the genotype and phenotype of another organism, thus providing additional evidence for the universality of the genetic code.

69. Oxygen consumption can be used as a measure of metabolic rate because oxygen is

(A) necessary for ATP synthesis by oxidative phosphorylation ✓

(B) necessary to replenish glycogen levels

(C) necessary for fermentation to take place

(D) required by all living organisms

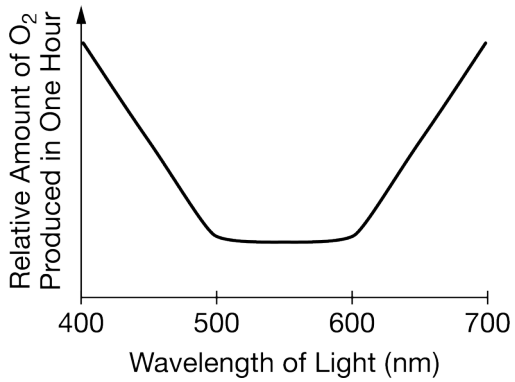
(E) required to break down the ethanol that is produced in muscles

70. A researcher claims that spinach leaves capture the most energy from light waves in the range of 500 nm to 600 nm. To test the claim, the researcher will place spinach leaves in separate chambers and expose the leaves to different wavelengths of light. For each chamber, the researcher will measure the amount of oxygen gas (O_2) that is produced in one hour.

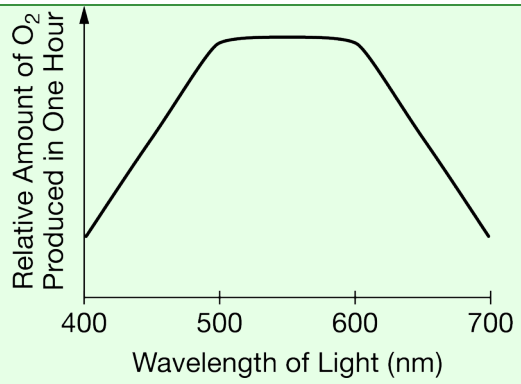
Which of the following graphs best represents data from the experiment that will support the researcher's claim?

3 Cellular Energetics MCQ

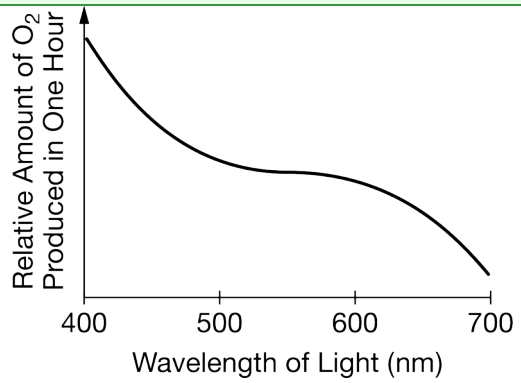
(A)



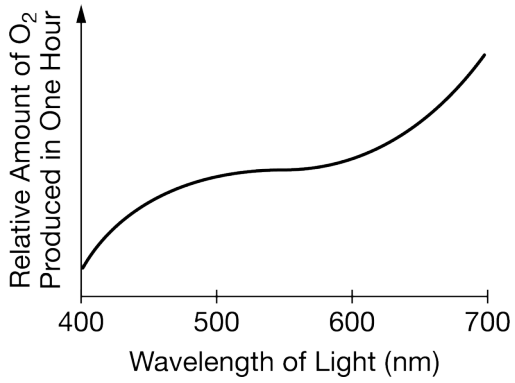
(B) ✓



(C)



(D)



3 Cellular Energetics MCQ

Answer B

Correct. The high levels of O_2 produced at wavelengths between 500 nm and 600 nm directly support the researcher's claim. Because oxygen is produced as a by-product of photosynthesis, a greater level of oxygen production indicates a greater level of energy-capture from light.

71. It is estimated that oxygen production first evolved in photosynthetic prokaryotes approximately 2.7 billion years ago. The first photosynthetic prokaryotes are presumed to be similar to today's cyanobacteria.

Which of the following best supports the claim that photosynthetic prokaryotes were responsible for the oxygen in Earth's atmosphere?

(A) The light reactions of photosynthesis split carbon dioxide into carbon and oxygen.

(B) The light reactions of photosynthesis split water into hydrogen ions and oxygen. ✓

(C) The Calvin cycle splits glucose into carbon, hydrogen, and oxygen.

(D) The Calvin cycle splits water into hydrogen ions and oxygen.

72. A scientist claims that *Elysia chlorotica,* a species of sea slug, is capable of photosynthesis.

Which of the following observations provides the best evidence to support the claim?

(A) *Elysia chlorotica* will die if not exposed to light.

(B) *Elysia chlorotica* grows when exposed to light in the absence of other food sources. ✓

(C) *Elysia chlorotica* grows faster when exposed to light than when placed in the dark.

(D) *Elysia chlorotica* grows in the dark when food sources are available.

Answer B

Correct. Growth requires an energy source. If the only energy source available is light, then the growth must be the result of photosynthesis capturing light energy for use by the sea slug.

73. To test the hypothesis that a particular plant synthesizes storage lipids by using glyceraldehyde 3-phosphate (G3P) from photosynthesis, a researcher plans to use radiolabeled precursors to track the molecules through the biosynthetic pathway. Which of the following radiolabeled precursors is most appropriate for the researcher to use?

3 Cellular Energetics MCQ

(A) ^{15}N-labeled N_2, because atmospheric nitrogen is fixed to amino acids by photosynthesis

(B) ^{14}C-labeled CO_2, because atmospheric carbon is fixed to carbohydrates by photosynthesis ✓

(C) ^{35}S-labeled methionine, because amino acids are incorporated into lipids during photosynthesis

(D) ^{32}P-labeled phosphate, because lipids are stored in plants as phospholipids

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- **Directions:** This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

(A) Glysolysis
(B) Krebs cycle (citric acid cycle)
(C) Calvin cycle (light-independent reactions of photosynthesis)
(D) Light-dependent reactions of photosynthesis
(E) Chemiosmosis

74. Process found in both photosynthesis and cellular respiration

(A) A
(B) B
(C) C
(D) D
(E) E ✓

75. Process in which carbon from CO_2 is incorporated into organic molecules

(A) A
(B) B
(C) C ✓
(D) D
(E) E

76. Process in which CO_2, is released as a by-product of oxidation-reduction reactions

(A) A
(B) B ✓
(C) C
(D) D
(E) E

77. Process in which O_2 is released as a by-product of oxidation-reduction reactions

3 Cellular Energetics MCQ

(A) A
(B) B
(C) C
(D) D ✓
(E) E

78. Process in which sugar is oxidized to pyruvic acid

(A) A ✓
(B) B
(C) C
(D) D
(E) E

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79. Some organisms use a weaker terminal electron acceptor than oxygen during cellular respiration. Which of the following best explains why the cells of these organisms might generate less ATP per molecule of glucose than cells that use oxygen as the terminal electron acceptor?

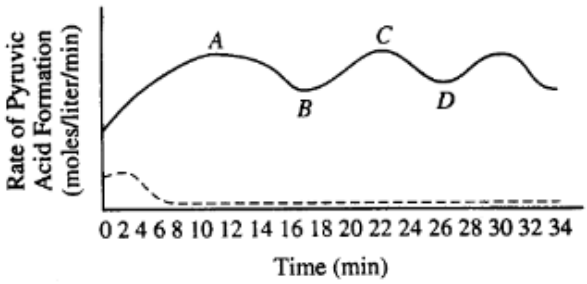
(A) The cells of these organisms remove fewer electrons from glucose than cells that use oxygen.
(B) The cells of these organisms create a smaller proton gradient than cells that use oxygen. ✓
(C) The cells of these organisms convert pyruvate to lactate more often than cells that use oxygen.
(D) The cells of these organisms generate more metabolic heat than cells that use oxygen.

80. When hydrogen ions are pumped out of the mitochondrial matrix, across the inner mitochondrial membrane, and into the space between the inner and outer membranes, the result is

(A) damage to the mitochondrion
(B) the reduction of NAD
(C) the restoration of the Na-K balance across the membrane
(D) the creation of a proton gradient ✓
(E) the lowering of pH in the mitochondrial matrix

3 Cellular Energetics MCQ

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.



A tissue culture of vertebrate muscle was provided with a constant excess supply of glucose under anaerobic conditions starting at time zero and the amounts of pyruvic acid and ATP produced were measured. The solid line in the graph above represents the pyruvic acid produced in moles per liter per minute. ATP levels were also found to be highest at points *A* and *C*, lowest at *B* and *D*. A second culture was set up under the same conditions, except that substance **X** was added, and the results are indicated by the dotted line.

81. Which of the following is most likely to result if oxygen is added to the tissue culture?

(A) Lactic acid formation will increase.

(B) For each glucose molecule consumed, more ATP will be formed. ✓

(C) The levels of ATP produced will decrease.

(D) Ethyl alcohol will be produced.

(E) No change in the production of pyruvic acid will be observed.

82. It is most reasonable to hypothesize that, in the breakdown of glucose, substance **X** is

(A) an activator

(B) an inhibitor ✓

(C) a substrate

(D) a coenzyme

(E) a cofactor

83. Which of the following best accounts for the shape of the solid line between points *A* and *D*?

(A) After ten minutes the cellular enzymes became ineffective.

(B) Respiration became uncontrolled.

(C) ATP acted as an allosteric inhibitor on one or more of the enzymes. ✓

(D) The measurements of pyruvic acid were unreliable.

(E) The cells required more glucose than was being provided.

3 Cellular Energetics MCQ

84. Which of the following questions is most relevant to understanding the Calvin cycle?

(A) How does chlorophyll capture light?

(B) How is ATP used in the formation of 3-carbon carbohydrates? ✓

(C) How is $NADP^+$ reduced to NADPH?

(D) How is ATP produced in chemiosmosis?

85. The chemical reaction for photosynthesis is

$6\ CO_2 + 12\ H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6\ O_2 + 6\ H_2O$

If the input water is labeled with a radioactive isotope of oxygen, ^{13}O, then the oxygen gas released as the reaction proceeds is also labeled with ^{13}O. Which of the following is the most likely explanation?

(A) During the light reactions of photosynthesis, water is split, the hydrogen atoms combine with the CO_2, and oxygen gas is released.

(B) During the light reactions of photosynthesis, water is split, removing electrons and protons, and oxygen gas is released. ✓

(C) During the Calvin cycle, water is split, regenerating NADPH from $NADP^+$, and oxygen gas is released.

(D) During the Calvin cycle, water is split, the hydrogen atoms are added to intermediates of sugar synthesis, and oxygen gas is released.

Answer B

This option is correct. This choice correctly describes the events of the light- dependent reactions and indicates that the student can justify the selection of data regarding the types of molecules that an organism will take up as necessary building blocks.

86. The chemical reaction for photosynthesis is

$6CO_2 + 12H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$

If the input water is labeled with a radioactive isotope of oxygen, ^{18}O, then the oxygen gas released as the reaction proceeds is also labeled with ^{18}O. Which of the following is the most likely explanation?

(A) During the light reactions of photosynthesis, water is split, the hydrogen atoms combine with the CO_2, and oxygen gas is released.

(B) During the light reactions of photosynthesis, water is split, removing electrons and protons, and oxygen gas is released. ✓

(C) During the Calvin cycle, water is split, regenerating NADPH from $NADP^+$, and oxygen gas is released.

(D) During the Calvin cycle, water is split, the hydrogen atoms are added to intermediates of sugar synthesis, and oxygen gas is released.

3 Cellular Energetics MCQ

87. The reactions of glycolysis occur in the

(A) cytosol ✓

(B) nucleus

(C) matrix of the mitochondrion

(D) membranes of the mitochondrion

(E) stroma of the chloroplast

88. Which metabolic process is common to both aerobic cellular respiration and alcoholic fermentation?

(A) Krebs cycle

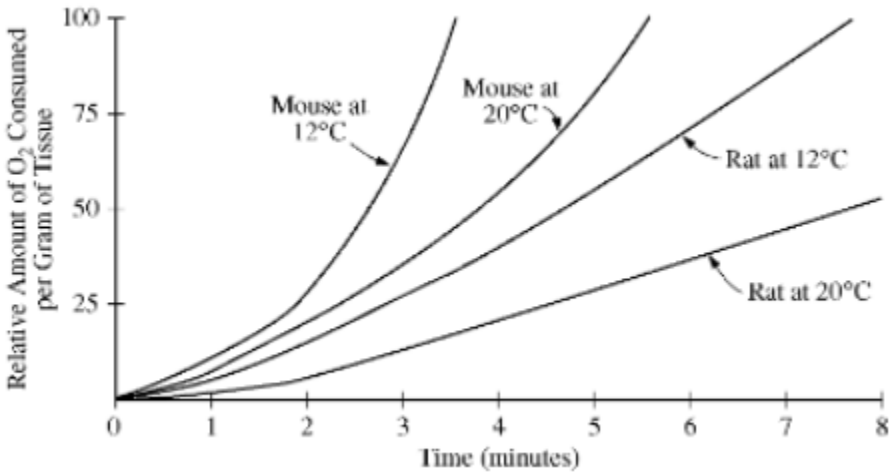
(B) Glycolysis ✓

(C) Electron transport chain

(D) Conversion of pyruvic acid to acetyl CoA

(E) Production of a proton gradient

A respirometer is a container used to measure the amount of oxygen consumed by an organism. A respirometer was used to determine how environmental temperature affects the uptake of oxygen in one 300-gram rat and one 50-gram mouse. The results of this experiment are shown on the graph below.



89. Which of the following hypotheses is best supported by the results of this experiment?

(A) Metabolic rate per gram of tissue is higher in smaller mammals. ✓

(B) Metabolic rate per gram of tissue is not related to body mass.

(C) Mice produce less CO_2 per gram of tissue than do rats.

(D) Rats consume more food per gram of tissue than do mice.

(E) Rats lose more heat per gram of tissue than do mice.

3 Cellular Energetics MCQ

90. Which of the following would most adversely affect the validity of this experiment?

(A) Large variation in metabolic rates between individuals of a species ✓

(B) Repeating the experiment several times

(C) Using a brown rat and a white mouse

(D) Using a thermometer that consistently read 2°C too low

(E) Using a scale that consistently indicated 2 grams too much

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91. Which of the following best describes the role of ATP and NADPH in the process of photosynthesis?

(A) ATP is required for the chemiosmotic production of NADPH during photosynthesis.

(B) ATP and NADPH are necessary to initiate the Krebs cycle in photosynthesis.

(C) ATP and NADPH are carriers of chemical energy that is used in the production of carbohydrates. ✓

(D) ATP and NADPH are required to boost electrons to a higher energy state in photosystems I and II.

Answer C

Correct. The energy captured from light during the photosynthetic light-dependent reactions is stored in ATP and NADPH for use in the Calvin cycle when carbon dioxide is fixed to enable the production of carbohydrates.

92. The FtsZ protein is present in prokaryotes and in chloroplasts. The protein is structurally and functionally similar to tubulin proteins of eukaryotic cells. Which of the following is a likely conclusion to draw from this information?

(A) FtsZ and tubulin proteins were both present in a common ancestor.

(B) Microtubules are involved in the mechanics of photosynthesis.

(C) Tubulin genes are evolutionarily derived from the gene that codes for the FtsZ protein. ✓

(D) The sequences of the genes encoding the FtsZ and tubulin proteins are identical.

Answer C

This option is correct. The evidence states that the FtsZ gene is present in prokaryotes and in chloroplasts. Chloroplasts evolved from endosymbiotic prokaryotes. The tubulin protein is structurally and functionally similar to FtsZ and is found in eukaryotes, inferring evolutionary descent.

3 Cellular Energetics MCQ

93. The O_2 released during photosynthesis comes from

(A) CO_2

(B) H_2O ✓

(C) NADPH

(D) RuBP (RuDP)

(E) $C_6H_{12}O_6$

94.

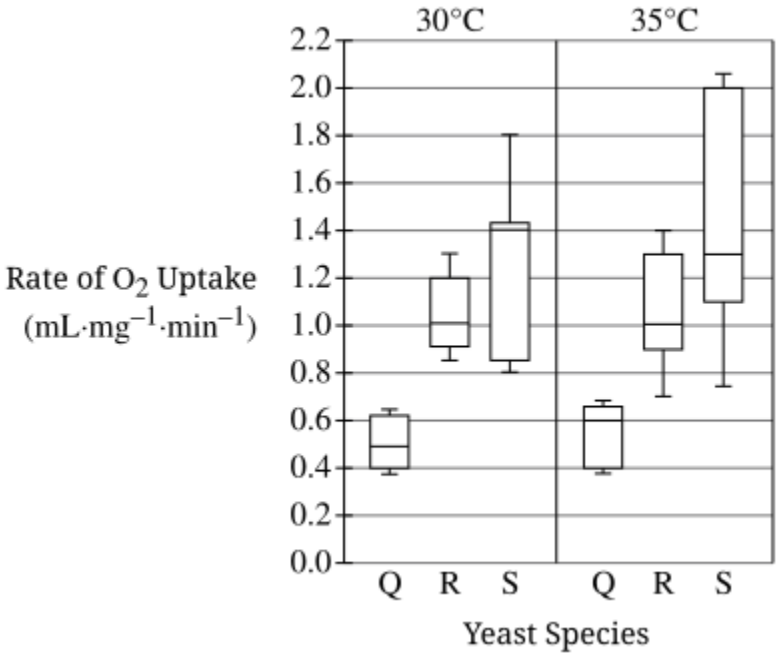


Figure 1. The rate of O_2 uptake for three species of *Saccharomyces* yeast at $30^\circ C$ and $35^\circ C$

Researchers studied the effect of increased temperature on the rate of respiration of three species of *Saccharomyces* yeast (Q, R, and S) by measuring the rate of oxygen uptake in each species at $30^\circ C$ and at $35^\circ C$ (Figure 1).

Based on Figure 1, which of the following statements about the effect of the temperature increase on the respiratory rate of one of the species, Q, R, or S, is most likely true?

(A) The median O_2 uptake rate for species S at $35^\circ C$ is $0.1\ \mu L \cdot mg^{-1} \cdot min^{-1}$ less than it is at $30^\circ C$. ✓

(B) The range from minimum to maximum O_2 uptake rate for species R at $35^\circ C$ is $0.1\ \mu L \cdot mg^{-1} \cdot min^{-1}$ more than it is at $30^\circ C$.

(C) The median O_2 uptake rate for species Q at $35^\circ C$ is the same as it is at $30^\circ C$.

(D) The range of the O_2 uptake rate for species S at $35^\circ C$ is the same as it is at $30^\circ C$.

3 Cellular Energetics MCQ

Answer A

Correct. In a box-and-whisker plot, the line inside the box is the median value. For species S at $30\,^{\circ}C$, the median O_2 uptake rate is 1.4, while at $35\,^{\circ}C$, the median O_2 uptake rate is 1.3.

95. Acetylcholinesterase $(AChE)$ is a protein that catalyzes the conversion of acetylcholine to acetate and choline. When the concentration of $AChE$ in an aqueous solution is held constant, the rate of the reaction catalyzed by $AChE$ increases with increasing concentrations of substrate. At low concentrations of acetylcholine, a small increase in the substrate concentration results in a large increase in the reaction rate. At high concentrations of acetylcholine, however, a large increase in the substrate concentration results in only a small increase in the reaction rate.

Which of the following statements correctly explains the observed effect of the acetylcholine concentration on the rate of the enzyme-catalyzed reaction?

(A) The active site of $AChE$ is specific for acetylcholine, and only one substrate molecule can occupy the active site at a time. ✓

(B) $AChE$ begins converting product into substrate as the acetylcholine concentration changes from low to high.

(C) The $AChE$ protein becomes denatured as the acetylcholine concentration changes from low to high.

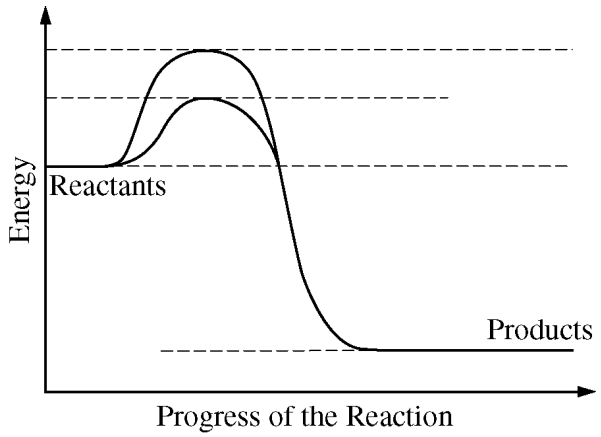
(D) The substrate specificity of $AChE$ changes as the acetylcholine concentration changes from low to high.

Answer A

Correct. Acetylcholine is the substrate for $AChE$, and since the shape and charge of the active site is specific for acetylcholine, only one substrate molecule can occupy the active site at a time. Therefore, at high concentrations of acetylcholine, $AChE$ becomes saturated with substrate. When an enzyme is saturated with substrate, adding more substrate to the solution will not lead to an observable increase in the reaction rate.

3 Cellular Energetics MCQ

96. The diagram below shows energy changes in a specific chemical reaction with and without the addition of an enzyme to the reaction.



Which of the following questions can best be answered by the diagram?

(A) Does the addition of an enzyme reduce the activation energy required for a reaction? ✓

(B) Does the addition of an enzyme result in the formation of covalent bonds?

(C) Does the addition of an enzyme produce a greater amount of products?

(D) Does the addition of an enzyme change the pathway for the reaction?

97. A researcher proposes a model of an enzyme-catalyzed reaction in which a reactant is converted to a product. The model is based on the idea that the reactant passes through a transition state within the enzyme-substrate complex before the reactant is converted to the product.

Which of the following statements best helps explain how the enzyme speeds up the reaction?

(A) The enzyme's active site binds to and stabilizes the reactant, which decreases the free-energy change of the reaction.

(B) The enzyme's active site binds to and stabilizes the transition state, which decreases the activation energy of the reaction. ✓

(C) The enzyme's active site binds to and stabilizes the product, which increases the amount of energy released by the reaction.

(D) The enzyme's active site binds to and stabilizes both the reactant and the product at the same time, which increases the reaction's equilibrium constant.

Answer B

Correct. An enzyme typically speeds up a reaction by stabilizing a transition state in the reaction, which decreases the reaction's activation energy. The enzyme's active site binds more tightly to the transition state than to either the substrate or the product, which speeds up the reaction and allows the enzyme to

catalyze the same reaction repeatedly.

98. Amylase is a protein that catalyzes the conversion of starch to simple sugars. Amylase activity in an aqueous solution can be measured by using iodine as a starch indicator. A solution containing iodine and starch will have a dark-blue color, whereas a solution containing iodine but no starch will have a light-brown color. The color change of an iodine solution from dark blue to light brown can be used to measure the rate at which starch is converted to simple sugars.

A student designs an experiment to investigate the effect of environmental pH on amylase function. The design of the experiment is presented in Table 1.

Table 1. An experiment for investigating the effect of pH on amylase function

Test Tube	Environmental pH	Starch Added	Amylase Added	Iodine Added	Pretreated by Boiling
I	5	Yes	Yes	Yes	No
II	6	Yes	Yes	Yes	No
III	7	No	Yes	Yes	No
IV	7	Yes	No	Yes	No
V	7	Yes	Yes	Yes	Yes
VI	7	Yes	Yes	Yes	No
VII	8	Yes	Yes	Yes	No
VIII	9	Yes	Yes	Yes	No

Which of the following statements best justifies the inclusion of test tube V as a control in the experiment?

(A) It will provide a measurement of amylase activity at an acidic pH.

(B) It will provide a measurement of amylase activity at a basic pH.

(C) It will show the color change that occurs in the absence of enzyme activity. ✓

(D) It will show the color change that occurs in the absence of the amylase protein.

Answer C

Correct. Test tube V will contain all the components of the reaction mixture, but the amylase protein will be inactive because it will have been denatured during the pretreatment. Test tube V is an appropriate control in the experiment because it will show the color change that occurs in the absence of amylase activity.

3 Cellular Energetics MCQ

99. A researcher claims that some bacteria contain factors that influence the function of a particular enzyme but other bacteria do not. To test the claim experimentally, the researcher will grow two different bacterial strains in separate liquid cultures and isolate the contents of the cells in each culture. The researcher will add different combinations of cellular contents, substrate, and enzyme to test tubes containing a buffer solution adjusted to the optimal pH of the enzyme and then measure the rate of product formation. The design of the researcher's experiment is presented in Table 1.

Table 1. An experiment for investigating the influence of bacterial factors on enzyme function

Test Tube	Source of Cellular Contents	Cellular Contents Added	Substrate Added	Enzyme Added
1	Strain 1	No	Yes	Yes
2	Strain 1	Yes	No	Yes
3	Strain 1	Yes	Yes	No
4	Strain 1	Yes	Yes	Yes
5	Strain 2	No	Yes	Yes
6	Strain 2	Yes	No	Yes
7	Strain 2	Yes	Yes	No
8	Strain 2	Yes	Yes	Yes

Which of the following statements best justifies the inclusion of test tubes 3 and 7 in the experiment?

(A) They will show whether the isolated cellular contents have enzymatic activity. ✓

(B) They will show whether environmental pH affects the function of the enzyme.

(C) They will show the rate of product formation in the absence of bacterial factors.

(D) They will show the rate of product formation in the absence of the substrate.

Answer A

Correct. The researcher will add cellular contents and the substrate but not the enzyme to test tubes 3 and 7. As such, test tubes 3 and 7 will show whether the cellular contents have enzymatic activity.

100. A researcher designs an experiment to investigate the effect of environmental temperature on the function of an enzyme. For each trial included in the experiment, the researcher will add the enzyme and its substrate to an aqueous buffer solution and then measure the amount of product formed over 20 minutes.

Which of the following must remain the same for all trials of this experiment?

3 Cellular Energetics MCQ

(A) The initial concentration of the substrate ✓

(B) The final concentration of the product

(C) The three-dimensional structure of the enzyme

(D) The temperature of the aqueous buffer solution

Answer A

Correct. Keeping the initial substrate concentration constant allows the researcher to compare the reaction rates under different environmental conditions. The reaction rates will provide the researcher with information about the enzyme's function in different environments.

101. Catalase is an enzyme found in yeast cells that facilitates the chemical breakdown of hydrogen peroxide to water and oxygen gas. An experiment was conducted to determine the effect of pH on catalase function. Five buffer solutions of varying pH (2, 4, 6, 8, and 10) were prepared and added to separate test tubes. Hydrogen peroxide (H_2O_2) was added to each test tube. Yeast was added, and the reactions were timed. After 1 minute the amount of oxygen gas released was determined by measuring the foam layer produced in each test tube. Figure 1 illustrates the experimental setup.

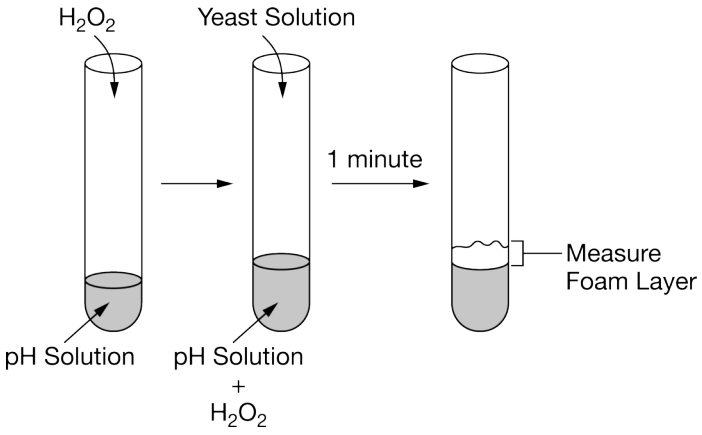


Figure 1. Illustration of experimental procedure

A set of five additional test tubes were prepared and used as controls. Which of the following best describes the contents expected to be contained in one of the five control test tubes?

(A) pH 4 buffer solution and hydrogen peroxide only ✓

(B) All five pH buffer solutions combined and hydrogen peroxide only

(C) Water, hydrogen peroxide, and yeast

(D) Water and yeast only

3 Cellular Energetics MCQ

Answer A

Correct. Since the experiment is studying the effect of pH on catalase in yeast, the control set should determine what would happen without yeast present. Each test tube in the control set should contain hydrogen peroxide and one of the pH solutions.

For following group of questions first study the description of the data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

To study the actions of the enzyme catalase on hydrogen peroxide, students performed the following experiment. Catalase was extracted from potatoes by blending raw potatoes in a blender with cold distilled water. The filtrate was stored on ice. The following hydrogen peroxide solutions were made: 1 percent, 5 percent, 10 percent, and 15 percent. Filter paper disks were soaked in the catalase filtrate and dropped into beakers containing the various solutions. The activity of the enzyme was measured by the amount of time it took for the disks to float to the surface of the solution on the bubbles produced by the reaction. The following data were obtained.

Hydrogen Peroxide Solution	Average Time, in Seconds, for Disks to Float
1%	30 sec
5%	25 sec
10%	20 sec
15%	10 sec

102. Which of the following experimental designs should the students use as a control for the experiment?

(A) Place a catalase-soaked disk in a beaker of water. ✓

(B) Drop the disks from different heights into the solution.

(C) Poke the floating disks back down to the bottom of the beaker and retime the rise of the disks.

(D) Shake the beakers during the time the disks are rising.

(E) Dip the disks in the 5% solution before putting them in any other solution.

103. Which of the following best describes why ice was used during this experiment?

(A) To increase the activity of the enzyme

(B) To retard the breakdown of the catalase ✓

(C) To purge the solution of excess air trapped during blending

(D) To slow the catalase molecules to increase the probability of contact with the hydrogen peroxide molecules

(E) To increase the size of the active site on the enzyme

3 Cellular Energetics MCQ

104. Which of the following best explains why the disks rose to the surface faster in the more concentrated hydrogen peroxide solutions?

(A) There was more enzyme present in the more concentrated solutions.

(B) A greater amount of heat was generated in the more concentrated solutions.

(C) The more concentrated solutions lowered the activation energy of the reaction.

(D) The higher substrate concentrations in the more concentrated solutions speeded the reaction. ✓

(E) The density of the water was lower in the more concentrated solutions.

105. If the potato solution was boiled for 10 minutes and cooled for 10 minutes before being tested, the average time for the disks to float to the surface of the hydrogen peroxide solution would be

(A) less than 1 second

(B) 5 seconds

(C) 10 seconds

(D) 30 seconds

(E) more than 30 seconds ✓

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106. Which of the following statements best helps explain the reaction specificity of an enzyme?

(A) The free energy of the reactants is greater than the free energy of the products.

(B) The equilibrium constant of the reaction is much greater than 1.

(C) The shape and charge of the substrates are compatible with the active site of the enzyme. ✓

(D) The concentration of the enzyme inside living cells is greater than the concentration of substrate.

Answer C

Correct. The reaction specificity of an enzyme is determined by the compatibility of the substrate with the enzyme's active site. A substrate's shape and charge affect how it fits into an enzyme's active site and whether it forms favorable interactions with the enzyme.

3 Cellular Energetics MCQ

107. Gelatin is a protein that is derived from collagen which is found in the bones, skin, and connective tissue of animals. To investigate the ability of various enzymes to digest gelatin, a group of students set up an assay involving camera film. Camera film contains gelatin and appears black when exposed to light but turns clear as the gelatin gets broken down. The students incubated pieces of exposed camera film in test tubes, each containing one of three different enzyme solutions (trypsin, lipase, or amylase) as indicated in Figure 1. The students recorded the time it took for the enzymes to digest the gelatin in each test tube, turning the film from black to clear.

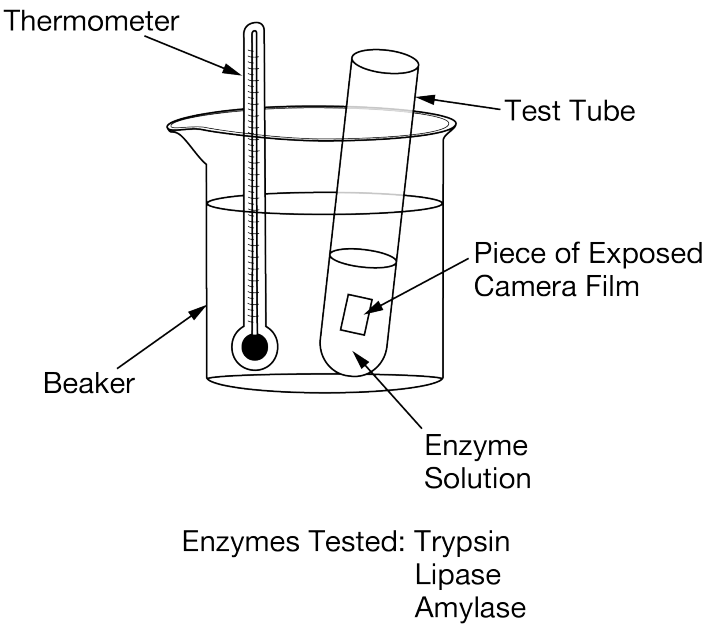


Figure 1. Diagram of experimental setup.

Which of the following would be the most appropriate control for this experiment?

(A) A test tube containing no camera film

(B) A test tube containing only a piece of exposed camera film

(C) A test tube containing a piece of exposed camera film submerged in water ✓

(D) A test tube containing a piece of exposed camera film and all three enzyme solutions

Answer C

Correct. Having one test tube with only water and camera film will show that water is not the substance that causes the film to change color.

3 Cellular Energetics MCQ

108. The enzyme hexokinase catalyzes the conversion of glucose to glucose-6-phosphate, which is an important step in glycolysis. The reaction involves the transfer of a phosphate group from ATP to glucose.

Either a glucose molecule or a water molecule can fit in the active site of hexokinase. The presence of a water molecule in hexokinase's active site would result in the hydrolysis of ATP to ADP instead of the conversion of glucose to glucose-6-phosphate.

Which of the following statements best helps explain the reaction specificity of hexokinase?

(A) Both glucose and water are polar molecules that form favorable interactions with charged and polar amino acid side chains in hexokinase's active site.

(B) Both glucose and water have oxygen atoms that can form covalent bonds with the phosphorus atoms of phosphate groups.

(C) Glucose is an energy-rich organic molecule that can be broken down by glycolysis to produce ATP, whereas water is an inorganic molecule.

(D) Glucose has the right shape and charge to cause hexokinase to undergo a structural change needed for catalysis, whereas water does not. ✓

Answer D

Correct. Hexokinase catalyzes the conversion of glucose to glucose-6-phosphate but not the hydrolysis of ATP. Hexokinase's reaction specificity is explained by the compatibility of glucose and hexokinase's active site, which results in part from glucose's shape and charge. Because of this compatibility, hexokinase undergoes a structural change when it binds to glucose but not to water, and the structural change is necessary for hexokinase to become catalytically active.

3 Cellular Energetics MCQ

109. A student designs an experiment to investigate the influence of temperature on enzyme function. The student's plan is presented in Table 1.

Table 1. An experiment to investigate the influence of temperature on enzyme function

Test Tube	Substrate Added	Enzyme Added	Temperature (°C)
1	Yes	No	10
2	Yes	Yes	10
3	Yes	No	20
4	Yes	Yes	20
5	Yes	No	30
6	Yes	Yes	30
7	Yes	No	40
8	Yes	Yes	40

Which test tubes are controls in the experiment?

(A) Test tubes 1 and 2 only

(B) Test tubes 5 and 6 only

(C) Test tubes 1, 3, 5, and 7 ✓

(D) Test tubes 2, 4, 6, and 8

Answer C

Correct. The test tubes that do not contain the enzyme are negative controls in the experiment. The negative controls help confirm that the results of the experiment are related to enzyme function and not some other factor.

110. Aminolevulinate dehydratase (ALAD) is an enzyme that relies on zinc as a cofactor. A zinc ion binds to the ALAD active site, where it forms favorable interactions with the side chains of three amino acids. Researchers have found that substituting a lead ion for a zinc ion in the ALAD active site causes inhibition of ALAD.

Which of the following statements best helps explain how the lead ion causes inhibition of ALAD?

3 Cellular Energetics MCQ

(A) It changes the shape and charge of the substrate so that it becomes more compatible with ALAD's active site.

(B) It changes the amino acid sequence of the ALAD protein so that the enzyme catalyzes a different reaction.

(C) It changes the three-dimensional structure of the active site so that ALAD is no longer compatible with its substrate. ✓

(D) It changes the enzyme-substrate complex so that the transition state is more stable and the reaction proceeds at a faster rate.

Answer C

Correct. The substitution of a lead ion for a zinc ion most likely changes the three-dimensional structure of the ALAD active site. The three-dimensional structure of the active site plays an important role in the compatibility between an enzyme and its substrate. Changing the structure of the active site will likely interfere with enzyme function by disrupting the enzyme-substrate interaction.

111. Pectinase is a protein that catalyzes the breakdown of pectic polysaccharides in plant cell walls. A researcher designs an experiment to investigate the effect of salinity on the ability of pectinase to lower the activation energy of the reaction involved. The design of the experiment is presented in Table 1. For each test tube, the researcher will measure the amount of product formed over 20 minutes.

Table 1. An experiment to investigate the effect of salinity on pectinase function

Test Tube	Sodium Chloride Concentration (molar)	Temperature (degrees C)	Substrate Added	Pectinase Added
1	0	23	Yes	No
2	0	23	Yes	Yes
3	0.5	23	Yes	No
4	0.5	23	Yes	Yes
5	1.0	23	Yes	No
6	1.0	23	Yes	Yes
7	1.5	23	Yes	No
8	1.5	23	Yes	Yes

Which of the following statements best helps justify the inclusion of test tube 5 in the experiment?

3 Cellular Energetics MCQ

(A) It will act as a control for test tube 4 by showing the effect of the presence or absence of the substrate.

(B) It will act as a control for test tube 4 by showing the effect of a change in environmental temperature.

(C) It will act as a control for test tube 6 by showing the effect of the presence or absence of the enzyme. ✓

(D) It will act as a control for test tube 6 by showing the effect of a change in sodium chloride concentration.

Answer C

Correct. Test tube 5 will act as a control for test tube 6. The solutions in test tubes 5 and 6 are nearly identical except that the researcher will add pectinase to test tube 6 but not to test tube 5. A comparison of the data for test tubes 5 and 6 will show the effect of the presence or absence of the enzyme.

Researchers investigated the influence of environmental pH on the activity of peroxidase, an enzyme that catalyzes the conversion of hydrogen peroxide to water and oxygen gas. In an experiment, the researchers added a hydrogen peroxide solution containing guaiacol to several identical test tubes and adjusted the solution in each test tube to a different pH. The researchers included the guaiacol because it caused the solutions to change color as the reactions proceeded, which the researchers relied on for measuring reaction rates. Finally, the researchers added the same amount of peroxidase to each test tube and measured the rate of each reaction at $23^{\circ}C$. The results of the experiment are represented in Figure 1.

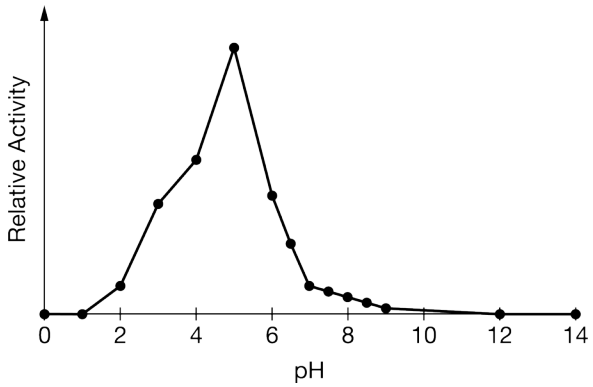


Figure 1. The effect of pH on peroxidase activity

112. One of the researchers proposes using oxygen gas production to measure reaction rates. Which of the following statements best justifies the use of the proposed modification as a way of creating an appropriate control for the investigation?

3 Cellular Energetics MCQ

(A) The experiment can be repeated without hydrogen peroxide, which will help eliminate an uncontrolled variable.

(B) The experiment can be repeated without peroxidase, which will introduce a second independent variable.

(C) The experiment can be repeated without guaiacol, which will reveal the effect of guaiacol on the reaction rates. ✓

(D) The experiment can be repeated without water, which will reveal whether the reaction can occur inside a living cell.

Answer C

Correct. In the original experiment, the researchers used guaiacol as a reaction indicator. By using oxygen gas production to measure reaction rates, the researchers no longer need the guaiacol and can repeat the experiment without it, which will reveal the effect of guaiacol on the reaction rates.

113. Which of the following actions will provide a negative control for the investigation?

(A) Repeating the experiment at $25^\circ C$

(B) Repeating the experiment using twice the amount of hydrogen peroxide

(C) Repeating the experiment using twice the amount of peroxidase

(D) Repeating the experiment using heat-denatured peroxidase ✓

Answer D

Correct. A negative control is a group in which no response is expected. It is the opposite of the positive control, in which a known response is expected. Heat-denatured peroxidase will be inactive. Repeating the experiment using heat-denatured peroxidase will allow the researcher to measure reaction rates in the presence of inactive peroxidase.

114. Based on Figure 1, which of the following statements best predicts the effect that a change from a moderately acidic environment (pH near 6) to a basic environment will have on peroxidase activity?

(A) Peroxidase activity will decrease. ✓

(B) Peroxidase activity will increase.

(C) Peroxidase activity will stay the same.

(D) Peroxidase activity will increase at first and then decrease.

3 Cellular Energetics MCQ

Answer A

Correct. Based on Figure 1, peroxidase's optimal pH is 5. A change from a moderately acidic environment (pH near 6) to a basic environment (pH above 7) will result in a decrease in peroxidase activity.

---

115. A researcher designs an experiment to investigate whether soil bacteria trigger the synthesis of defense enzymes in plant roots. The design of the experiment is presented in Table 1. For each group in the experiment, the researcher will determine the average rate of change in the amount of defense enzymes in the roots of the seedlings.

Table 1. An experiment to investigate the effect of soil bacteria on plant defenses

Group	Number of Seedlings	Type of Soil	Treatment Solution
1	10	Sterile potting soil	Contains actively reproducing soil bacteria
2	10	Sterile potting soil	Contains heat-killed soil bacteria
3	10	Sterile potting soil	Contains no soil bacteria

Which of the following statements best helps justify the inclusion of group 2 as one of the controls in the experiment?

(A) It will show whether the changes observed in group 1 depend on the metabolic activity of soil bacteria. ✓

(B) It will show whether the changes observed in group 1 depend on the type of plants used in the experiment.

(C) It will show the average growth rate of seedlings that are maintained in a nonsterile environment.

(D) It will show the changes that occur in the roots of seedlings following an infection by soil bacteria.

Answer A

Correct. The seedlings in group 1 will be treated with a solution that contains actively reproducing bacteria, which will be metabolically active. In contrast, the seedlings in group 2 will be treated with a solution that contains heat-killed bacteria, which will be metabolically inactive. The researcher will be able to compare the results for group 1 with those for group 2. The comparison will help the researcher determine whether the changes observed in group 1 depend on the metabolic activity of soil bacteria.

3 Cellular Energetics MCQ

116. A researcher claims that increasing the environmental temperature by $10^{\circ}C$ will double the rate of an enzyme-catalyzed reaction. To test the claim, the researcher designs an experiment that uses a particular enzyme isolated from plants. The design of the experiment is presented in Table 1. For each test tube in the experiment, the researcher will measure the rate of product formation.

Test Tube	Temperature ($^{\circ}C$)	Substrate Added	Enzyme Added	Pretreated by Boiling
1	15	Yes	Yes	No
2	25	No	Yes	No
3	25	Yes	No	No
4	25	Yes	Yes	No
5	25	Yes	Yes	Yes
6	35	Yes	Yes	No

Which of the following statements best helps justify the inclusion of test tube 5 as a control in the experiment?

(A) It will provide a measurement of product formation in the absence of the substrate.

(B) It will provide a measurement of product formation in the presence of a denatured enzyme. ✓

(C) It will show the effect of doubling the amount of substrate on the rate of product formation.

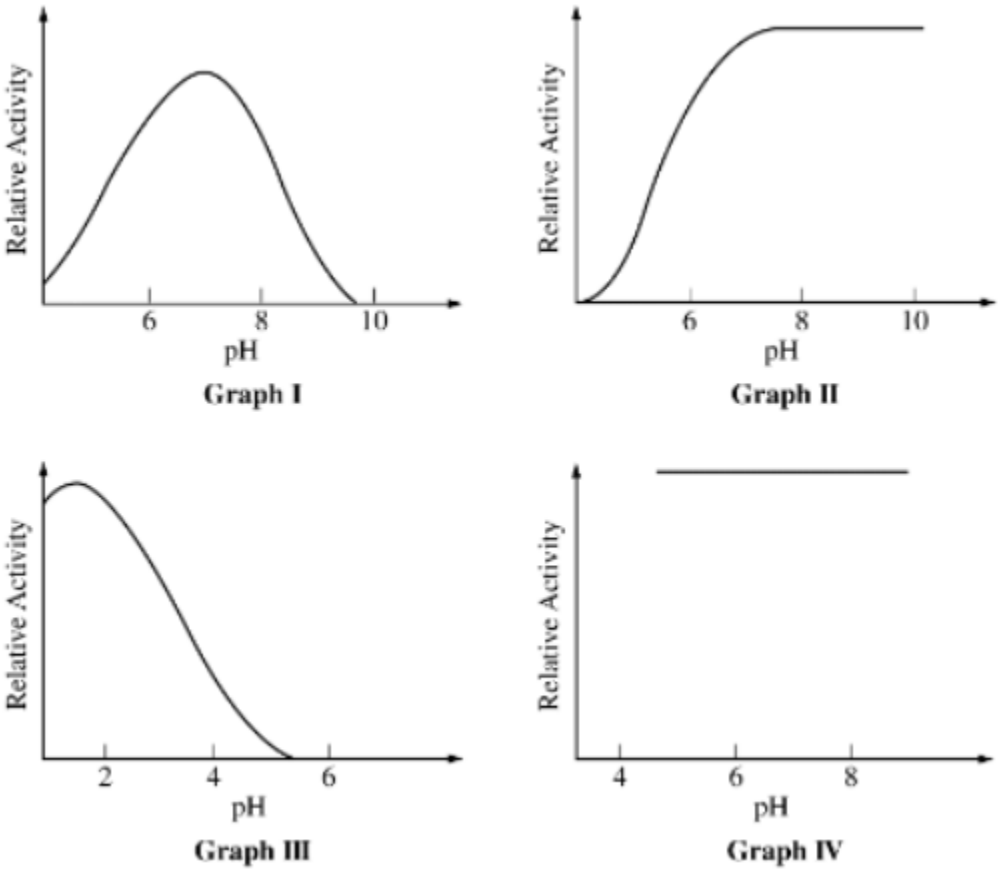
(D) It will show the effect of increased enzyme activity on the rate of product formation.

Answer B

Correct. The contents of test tube 5 will be pretreated by boiling. The pretreatment will denature the plant enzyme used in the experiment. As such, test tube 5 will provide a measurement of product formation in the presence of a denatured enzyme.

3 Cellular Energetics MCQ

For the following questions: Graphs I-IV depict the effect of pH on the activity of four different hydrolytic enzymes.



117. Graphs representing enzymes sensitive to changes in pH include which of the following?

(A) I only
(B) IV only
(C) II and III only
(D) III and IV only
(E) I, II, and III only ✓

118. The most likely explanation for the results shown in Graph I is that

(A) pH affects the shape of the active site of the enzyme ✓
(B) pH affects the temperature of the reaction
(C) the enzyme has a quaternary structure
(D) the enzyme has disulfide bonds
(E) pH affects the primary structure of the enzyme

119. Enzymes with their highest activity at an alkaline (basic) pH are represented by which of the following graphs?

3 Cellular Energetics MCQ

(A) I only

(B) II only ✓

(C) III only

(D) I and III only

(E) I and IV only

---

120. In an experiment, a researcher prepares a reaction mixture by dissolving a substance in a buffered solution. The substance is the substrate of a certain enzyme. The researcher adds a small amount of the enzyme to the reaction mixture and measures the amount of product that is formed over time. The data are represented in Figure 1.

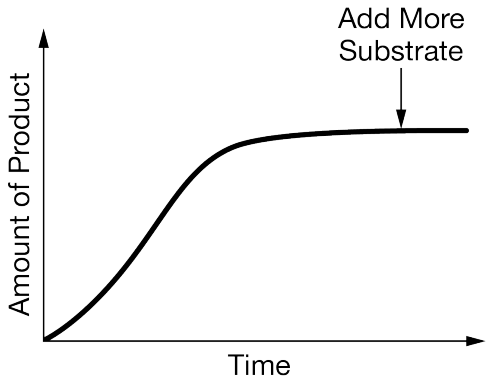


Figure 1. The amount of product formed by an enzyme-catalyzed reaction over time

Which of the following best predicts the immediate result of adding more substrate to the reaction mixture at the point indicated by the arrow in Figure 1?

(A) The amount of product will decrease until the reaction rate goes to zero.

(B) The amount of product will decrease until the reaction reaches its equilibrium point or until the enzyme is been used up by the reaction.

(C) The amount of product will increase until the reaction reaches its equilibrium point or until the substrate is used up by the reaction. ✓

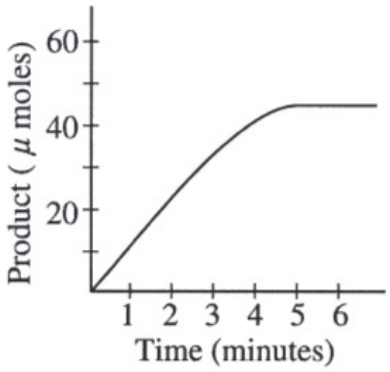
(D) The amount of product will increase without stopping because the enzyme will be unchanged by the reaction.

Answer C

Correct. The graph shown in Figure 1 indicates that the amount of product became constant as a consequence of the reaction reaching its equilibrium point or as a consequence of the substrate being used up. Adding more substrate will likely result in an increase in the amount of product because the reaction catalyzed by the enzyme will convert the added substrate into product.

3 Cellular Energetics MCQ

121. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



During which time interval is the reaction rate lowest?

(A) 0-1 minute
(B) 1-2 minutes
(C) 2-3 minutes
(D) 3-4 minutes
(E) 4-5 minutes ✓

3 Cellular Energetics MCQ

The coral species *Acropora millepora* is a marine animal that has a symbiotic relationship with algae species in the genus *Symbiodinium*. An increase in the concentration of dissolved CO_2 in oceans has a negative impact on this relationship and the health of the coral species. Scientists investigated the effect of CO_2 concentrations on the number of algae cells found in the coral (Figure 1) and on both the gross (total) and net rates of photosynthesis of those algae (Figure 2). The net rate of photosynthesis is equal to the gross rate of photosynthesis minus the rate of respiration by the algae.

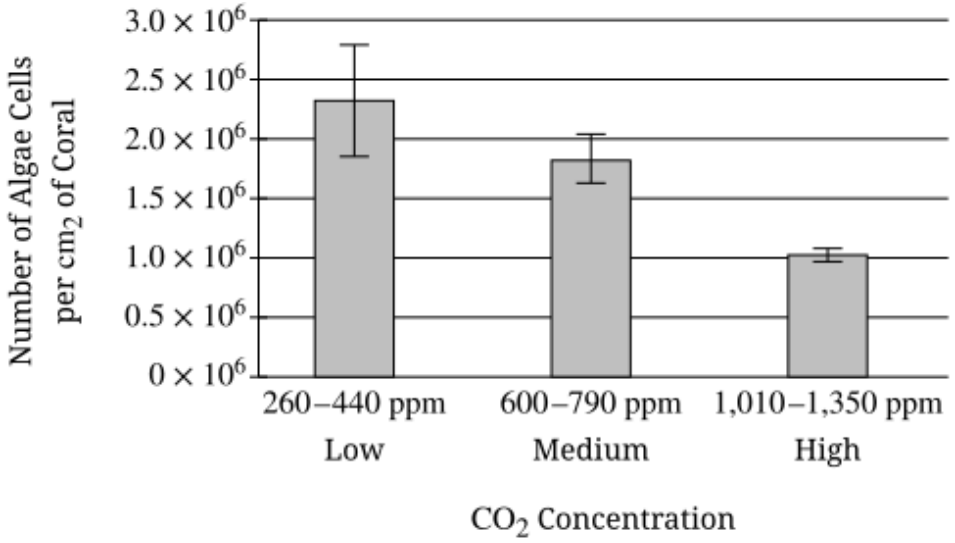


Figure 1. Effect of CO_2 concentration on the number of algae per cm^2 of coral. Error bars represent $\pm 2SE_{\bar{x}}$.

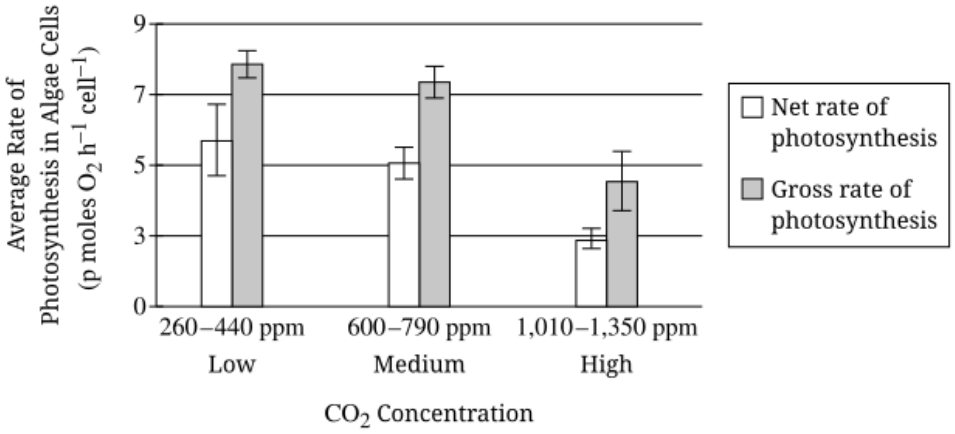


Figure 2. Average gross and net rates of photosynthesis in algae under various concentrations of CO_2. Error bars represent $\pm 2SE_{\bar{x}}$.

3 Cellular Energetics MCQ

122. Researchers observe that the water with the high CO_2 concentration has a lower pH than does the water with the low CO_2 concentration. The researchers claim that lower pH alters the structure of the enzymes of photosynthesis.

Which of the following most likely explains how a lower pH could alter the structure of photosynthetic enzymes?

(A) There will not be enough H^+ ions for the ribosome to correctly form peptide bonds between amino acids during synthesis of the enzymes.

(B) There will not be enough H^+ ions for the necessary hydrogen bonds to form in the secondary, tertiary, and quaternary structures of the enzymes.

(C) There will be additional H^+ ions that can disrupt peptide bonds in the primary structure of the enzymes.

(D) There will be additional H^+ ions that can disrupt hydrogen bonds in the secondary, tertiary and quaternary structures of the enzymes. ✓

Answer D

Correct. Additional H^+ ions compete for hydrogen bonding with amino acid side chains and therefore denature the protein and alter the structure of the photosynthetic enzymes.

123. Based on Figure 2, which of the following best supports the claim that CO_2 concentration and photosynthetic productivity are inversely related?

(A) The net rate of photosynthesis is less than the gross rate of photosynthesis at the high concentration of CO_2.

(B) Both the net and gross rates of photosynthesis at the high concentration of CO_2 are lower than those at the low concentration of CO_2. ✓

(C) The net rate of photosynthesis at the low concentration of CO_2 is lower than the gross rate of photosynthesis at the high concentration of CO_2.

(D) The difference between net and gross rates of photosynthesis is greater at the low concentration of CO_2 than it is at the high concentration of CO_2.

Answer B

Correct. This describes an inverse relationship where one measure increases as the other decreases. Because the rate of photosynthesis decreases as CO_2 concentration increases, this supports the claim that CO_2 concentration is inversely related to photosynthetic rate (productivity).

3 Cellular Energetics MCQ

124. Alcohol dehydrogenase (ADH) is an enzyme that aids in the decomposition of ethyl alcohol ($\mathrm{C_2H_5OH}$) into nontoxic substances. Methyl alcohol acts as a competitive inhibitor of ethyl alcohol by competing for the same active site on ADH. When attached to ADH, methyl alcohol is converted to formaldehyde, which is toxic in the body.

Which of the following statements best predicts the effect of increasing the concentration of substrate (ethyl alcohol), while keeping the concentration of the inhibitor (methyl alcohol) constant?

(A) There will be an increase in formaldehyde because ADH activity increases.

(B) Competitive inhibition will be terminated because ethyl alcohol will bind to methyl alcohol and decrease ADH activity.

(C) The peptide bonds in the active site of the enzyme will be denatured, inhibiting the enzyme.

(D) Competitive inhibition will decrease because the proportion of the active sites occupied by substrate will increase. ✓

Answer D

Correct. The increase in amount of substrate will increase the number of enzyme-ethanol complexes formed, increasing the amount of nontoxic product formed from ethyl alcohol.

125. A researcher proposes a model to explain how enzyme-substrate interactions determine enzyme specificity. The model is based on the idea that substrate molecules form favorable interactions with the amino acid side chains in an enzyme's active site.

Based on the model, which of the following statements best explains an enzyme's specificity for a particular substrate molecule?

(A) A hydrophilic molecule interacts with nonpolar side chains in the enzyme's active site.

(B) A hydrophobic molecule interacts with polar side chains in the enzyme's active site.

(C) A molecule with positive charges interacts with positively charged side chains in the enzyme's active site.

(D) A molecule with negative charges interacts with positively charged side chains in the enzyme's active site. ✓

Answer D

Correct. Because opposite charges attract, a molecule with negative charges will form favorable interactions with positively charged side chains in an enzyme's active site. Based on the model, the favorable interactions helps explain the enzyme's specificity for the molecule as a substrate.

3 Cellular Energetics MCQ

126. A researcher claims that different enzymes exhibit maximal function over different pH ranges. To test the claim, the researcher carries out an experiment that includes three different enzymes: pepsin, salivary amylase, and arginase. The results of the experiment are represented in Figure 1.

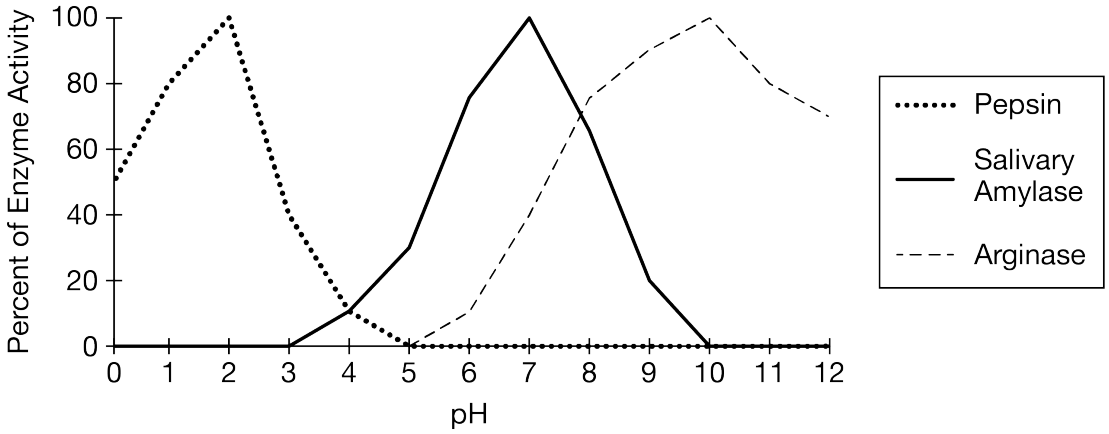


Figure 1. The effect of pH on three different enzymes

Which of the following actions will provide the most appropriate negative control for the experiment?

(A) Repeating the experiment with a fourth enzyme

(B) Repeating the experiment at several different temperatures

(C) Repeating the experiment with denatured enzymes ✓

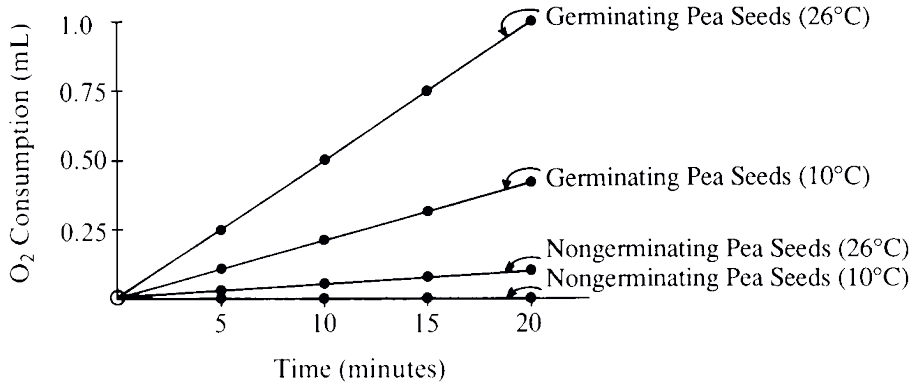
(D) Repeating the experiment using several different methods for measuring pH

Answer C

Correct. Repeating the experiment with denatured enzymes will provide an appropriate negative control for the experiment. The negative control will show that the changes observed in the experiment result from changes in enzyme function.

3 Cellular Energetics MCQ

The following questions refer to an experiment that is set up to determine the relative volume of O_2 consumed by germinating and nongerminating (dry) pea seeds at two different temperatures. The change in volume is detected by using a respirometer over a given period of time. The data are given below.



127. The rate of oxygen consumption in germinating pea seeds at 26ºC is

(A) 0.05 mL / min ✓

(B) 0.25 mL / min

(C) 0.50 mL / min

(D) 0.75 mL / min

(E) 1.00 mL / min

128. Which of the following conclusions is best supported by the data?

(A) Nongerminating pea seeds have a higher rate of respiration than germinating pea seeds do.

(B) Light is required for pea seed germination.

(C) In the nongerminating pea seeds, oxygen consumption is directly proportional to oxygen concentration.

(D) Less carbon dioxide is produced by germinating pea seeds at 26ºC than at 10ºC.

(E) In pea seeds an increase in temperature results in an increase in oxygen consumption. ✓

3 Cellular Energetics MCQ

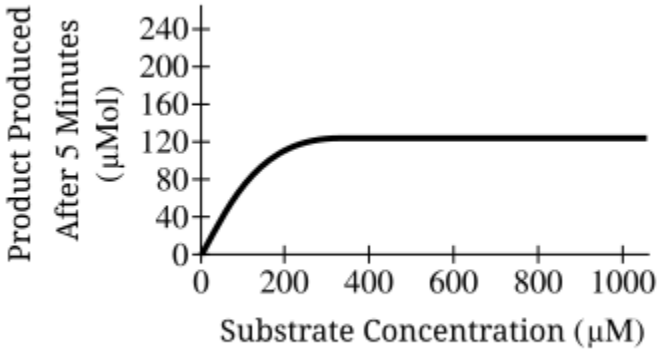


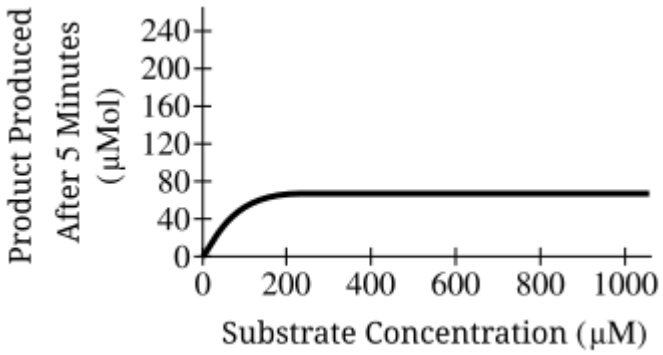
Figure 1. Rate of an enzyme-catalyzed reaction

Researchers investigated the dynamics of a reaction catalyzed by an enzyme. The researchers prepared a series of reactions, each with the same concentration of enzyme but with different concentrations of substrate. The researchers measured the amount of product in each reaction mixture after 5 minutes. The results are shown in Figure 1.

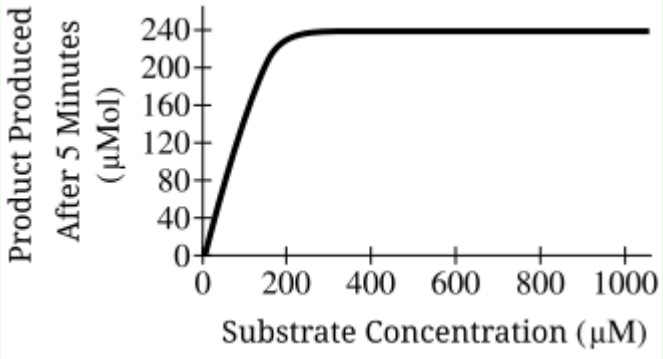
129. Which of the following graphs predicts the most likely rate of the enzyme-catalyzed reaction if the concentration of the enzyme in the mixture were doubled?

3 Cellular Energetics MCQ

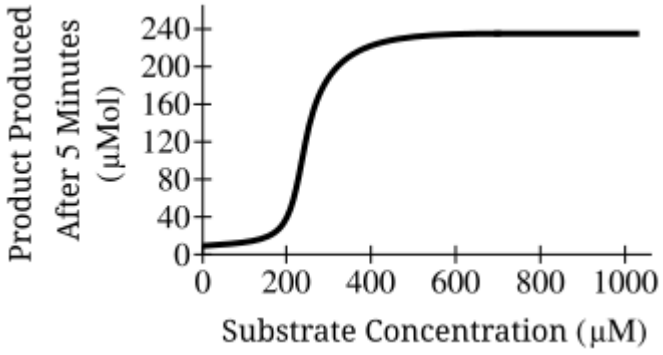
(A)



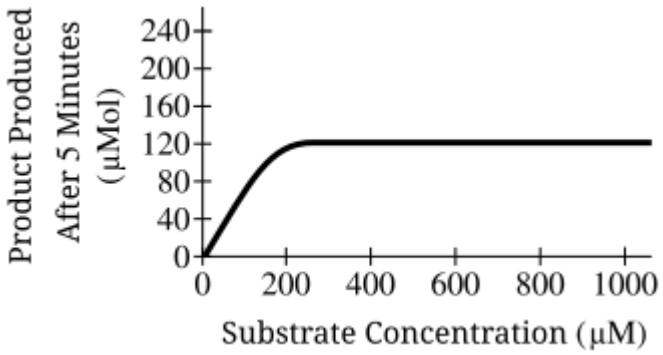
(B)



(C)



(D)



3 Cellular Energetics MCQ

Answer B

Correct. The rate of the reaction is initially proportional to the concentration of enzyme present. When all active sites are occupied, the rate remains constant at a maximum. Thus, doubling the enzyme concentration should double the rate of the reaction at every substrate concentration, and the maximum rate should also double.

130. Which of the following questions best addresses whether a particular inhibitor is competitive or noncompetitive?

(A) Does the inhibitor bind more tightly to the substrate or to the enzyme?

(B) Can the inhibitor be converted to product by the enzyme?

(C) How much inhibitor is necessary to completely inhibit the activity of the enzyme?

(D) Does the inhibitor bind to an allosteric site or the active site of the enzyme? ✓

Answer D

Correct. Competitive enzyme inhibitors affect the rate of reaction by competing with the substrate for binding to the active site of the enzyme. Noncompetitive inhibitors affect the rate of reaction by binding to an allosteric site of the enzyme, altering the shape and thus the function of the enzyme.

131. Which of the following experimental designs would best allow researchers to investigate the effect of pH on the rate of the enzyme-catalyzed reaction?

(A) Measure reaction rates at different pH levels when the enzyme concentration remains the same and the substrate concentration is 100 µM. ✓

(B) Measure the pH of the reaction mixture when the enzyme concentration remains the same and substrate concentrations vary between 0 µM and 1,000 µM.

(C) Measure reaction rates when the enzyme concentration remains the same but the pH of the reaction mixture increases and the substrate concentration increases from 0 µM to 1,000 µM.

(D) Measure reaction rates when the substrate concentration is 100 µM and both the pH of the reaction mixture and the enzyme concentration are increased.

Answer A

Correct. To investigate the role of pH on the reaction rate, the change in pH must be the only independent variable.

3 Cellular Energetics MCQ

132. Which of the following best describes how an increase in the concentration of substrate in the reaction mixture affected the frequency of enzyme-substrate interactions?

(A) The frequency of enzyme-substrate interactions decreased over all substrate concentrations.

(B) The frequency of enzyme-substrate interactions increased exponentially over all substrate concentrations.

(C) The frequency of enzyme-substrate interactions increased until all of the active sites were interacting with substrate. ✓

(D) The frequency of enzyme-substrate interactions increased until all of the enzyme was depleted.

Answer C

Correct. The frequency of enzyme-substrate interactions is determined by the maximum number of enzyme active sites available to bind substrate and convert it to product. The rate of the reaction increases until the active sites of all enzyme molecules are filled, and then the rate of the reaction remains constant.

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133. Which of the following can be used to determine the rate of enzyme-catalyzed reactions?

(A) Rate of disappearance of the enzyme

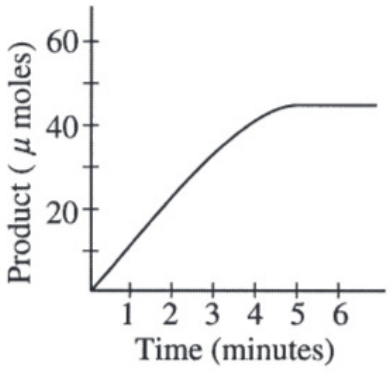
(B) Rate of disappearance of the substrate ✓

(C) Rate of disappearance of the product

(D) Change in volume of the solution

(E) Increase in activation energy

134. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



What is the most likely explanation for the change in the slope of the line between 3 and 5 minutes?

3 Cellular Energetics MCQ

(A) The enzyme had denatured.

(B) The enzyme had achieved its maximum velocity.

(C) A large amount of the substrate had been consumed. ✓

(D) An allosteric inhibitor appeared.

(E) There was a dramatic change in the pH.

135. The enzyme trypsin aids in protein digestion in the small intestine. The relative activity of trypsin at different pH values is shown in Figure 1.

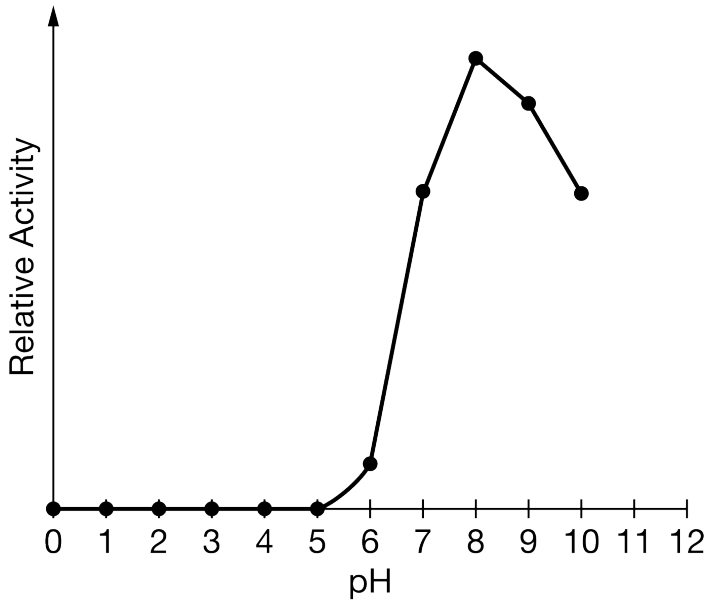


Figure 1. Effect of pH on the activity of trypsin

Which of the following statements best explains the activity levels of trypsin shown in Figure 1 ?

(A) The small intestine releases inhibitor molecules that block the activity of trypsin unless it is at its optimum pH.

(B) The number of effective collisions between trypsin and its substrate increase at higher pH values.

(C) As pH values increase, the substrate concentration decreases, leading to an eventual decline in the rate of the trypsin-catalyzed reaction.

(D) At extremely low pH values, trypsin is denatured and cannot function efficiently. ✓

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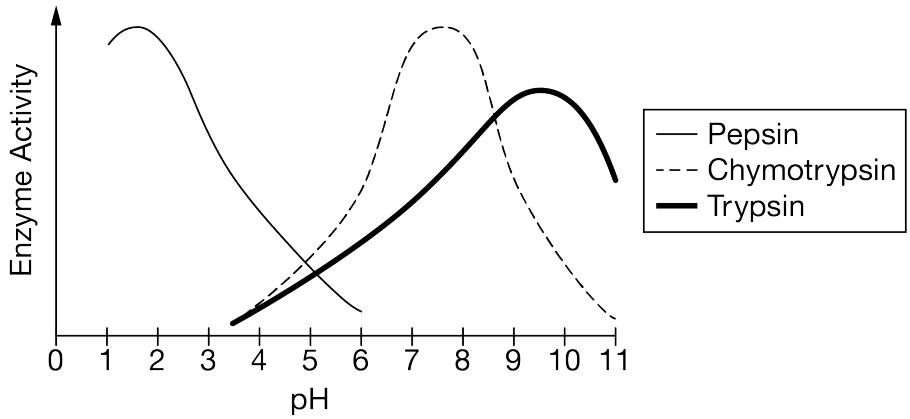
136. Protein digestion in humans is primarily carried out by three enzymes. Pepsin is found in the stomach $(\mathrm{pH}\ 2)$, where it aids in the breakdown of large proteins into smaller peptides, while trypsin and chymotrypsin are found in the small intestine $(\mathrm{pH}\ 8)$, where they aid in the further breakdown of the proteins into amino acids and dipeptides that can be absorbed into the bloodstream. Graph 1 shows the effect of pH on the activity levels of the three enzymes.



Graph 1. Relative activity of pepsin, trypsin, and chymotrypsin at pH 0 through 11

Which of the following best predicts how the structure and function of pepsin will change as it enters the small intestine?

(A) Pepsin will not change in shape and will continue to break down proteins in the small intestine.

(B) Pepsin will not change in shape but may not work due to the basic environment of the small intestine.

(C) Pepsin will change in shape because of the basic environment of the small intestine; therefore, its enzymatic activity will decrease. ✓

(D) Pepsin will change in shape because of the presence of trypsin and chymotrypsin in the small intestine, both of which act as competitive inhibitors.

Answer C

Correct. The significant change in pH will disturb the hydrogen bonding and cause denaturation of pepsin. As a result, pepsin will stop working in the small intestine.

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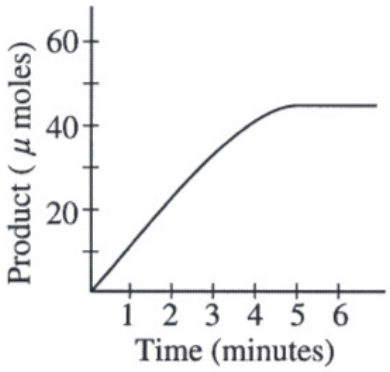
137. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



The rate of the reaction could also be determined by

(A) measuring the change in the amount of enzyme

(B) measuring the change in the amount of substrate ✓

(C) measuring the change in salt concentration

(D) adding more substrate

(E) adding more enzyme

138. Researchers investigated the effect of urea on the three-dimensional structure of a certain enzyme. The researchers dissolved the enzyme in an aqueous buffer solution and added urea to the solution. The enzyme did not appear to have a secondary or tertiary structure. The researchers carefully removed the urea from the solution and determined that the enzyme had the original secondary and tertiary structure again.

Based on the results of the experiment, which of the following statements best predicts the effect of urea on the enzyme's function?

(A) Function will be disrupted by adding the urea and regained by removing the urea. ✓

(B) Function will be disrupted by adding the urea, but it will not be regained by removing the urea.

(C) Function will be gained by adding the urea and disrupted by removing the urea.

(D) Function will be unaffected by the addition and removal of the urea.

Answer A

Correct. Based on the information, adding the urea denatures the enzyme, which will disrupt the enzyme's function by disrupting the active site. In contrast, removing the urea causes the enzyme to refold, which will restore the active site and the enzyme's function.

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139.

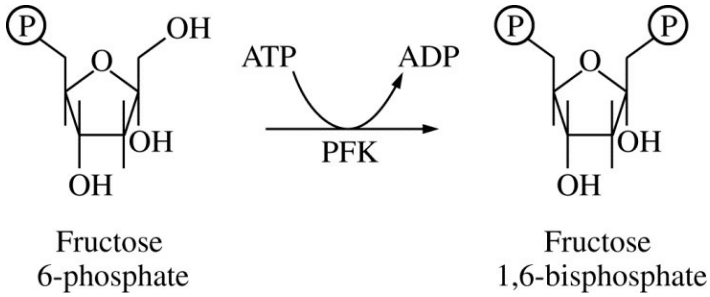


Figure 1. Reaction catalyzed by phosphofructokinase (PFK) during glycolysis

Phosphofructokinase (PFK) is an enzyme that catalyzes the conversion of fructose 6-phosphate to fructose 1,6-bisphosphate during glycolysis, as represented in Figure 1.

PFK can be allosterically inhibited by ATP at high concentrations. Which of the following is the benefit of regulating glycolysis by the concentration of ATP?

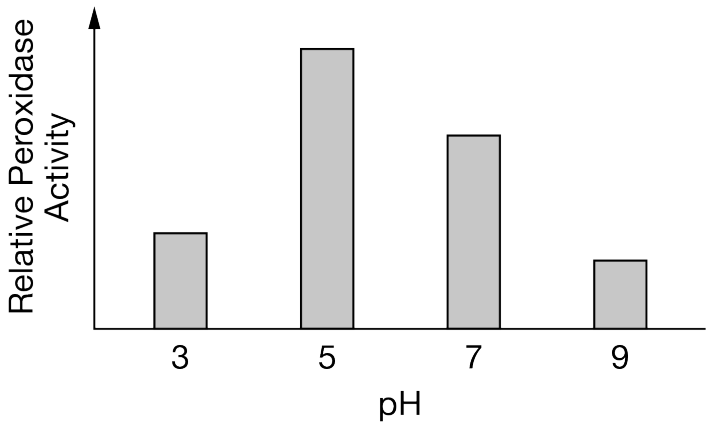
(A) Glycolysis proceeds when the intracellular concentration of ATP is low, which provides ATP to drive cellular reactions. ✓

(B) Glycolysis proceeds when the intracellular concentration of ATP is high and the cell stores ATP for future use.

(C) Glycolysis is inhibited when the intracellular concentration of ATP is low because PFK requires ATP as a substrate for the reaction it catalyzes.

(D) Glycolysis is inhibited when the intracellular concentration of ATP is high because ATP will compete with fructose 1,6-bisphosphate for binding to the active site on the enzyme.

140. The enzyme peroxidase is found in many organisms. It catalyzes the breakdown of hydrogen peroxide into water and oxygen gas. The rate of peroxidase activity at different pH values was assessed by students in the lab. The students' results are shown in graph 1.



Graph 1. Effect of pH on peroxidase activity

If the experiment is repeated at pH 11, the observed activity level of the enzyme will most likely be

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(A) the same as the level at pH 7

(B) lower than the level at pH 9 ✓

(C) greater than the level at pH 9

(D) between the levels observed at pH 5 and pH 7.

Answer B

Correct. Peroxidase activity drops off as the pH increases beyond 5, indicating that the optimal level was closer to pH 5. The enzyme is likely to be further denatured by a pH greater than 9. Since this enzyme occurs in many organisms, it is reasonable that the optimal pH is closer to neutral than to a strongly basic or strongly acidic pH.

141. In plants the enzyme EPSP synthase catalyzes a reaction between two different substrates, S3P and PEP, to form a product used in the synthesis of three amino acids. The structure of glyphosate, the active chemical in many herbicides, is very similar to that of PEP.

Which of the following explains how glyphosate most likely functions as an herbicide?

(A) Glyphosate prevents synthesis of the three amino acids by competing with PEP for binding to the active site of EPSP synthase. ✓

(B) Glyphosate prevents synthesis of the three amino acids by binding to an allosteric site of EPSP synthase, increasing the likelihood that PEP will bind to the active site.

(C) Glyphosate causes too much synthesis of the three amino acids by reducing the activation energy of the reaction catalyzed by EPSP synthase.

(D) Glyphosate alters the structure of the three amino acids by binding to both S3P and PEP in the active site of EPSP synthase.

Answer A

Correct. Because the structure of glyphosate is very similar to that of PEP, glyphosate acts as a competitive inhibitor of the EPSP enzyme by competing with PEP for the active site of the enzyme.

4 Cell Communication and Cell Cycle MCQ

1.

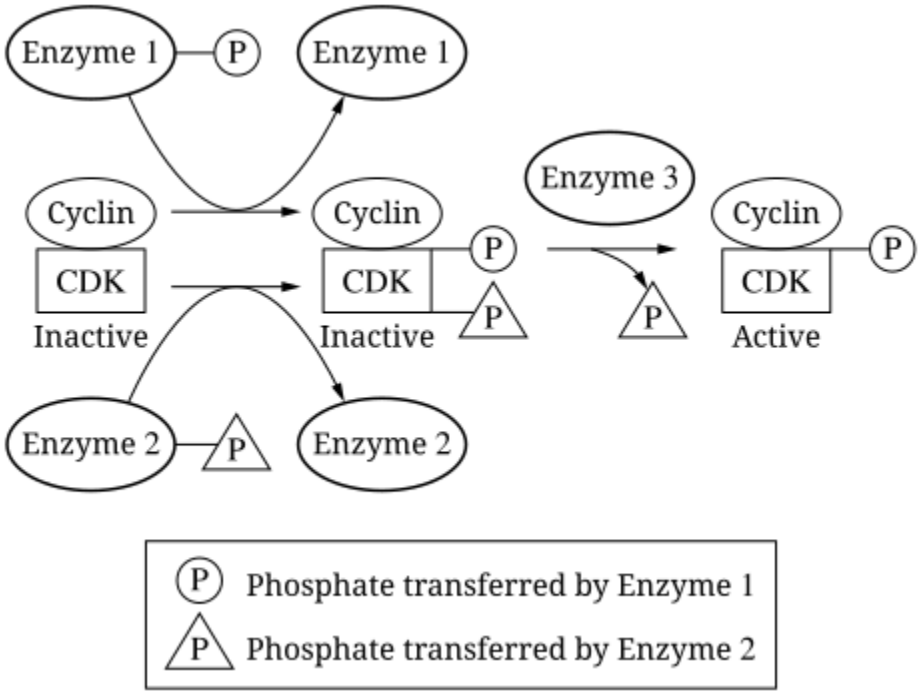


Figure 1. Activation of a cyclin-CDK complex

The regulation of a certain cyclin-CDK complex involves the activity of three enzymes (Figure 1), and the activation of this complex is required for cells to progress through the cell cycle.

Based on Figure 1, which of the following describes a cell in which cell cycle progression continues?

(A) A cell in which enzyme 1 is functional, and enzyme 2 and enzyme 3 are nonfunctional ✓

(B) A cell in which enzyme 1 is nonfunctional, and enzyme 2 and enzyme 3 are functional

(C) A cell in which enzyme 1 and enzyme 2 are functional, and enzyme 3 is nonfunctional

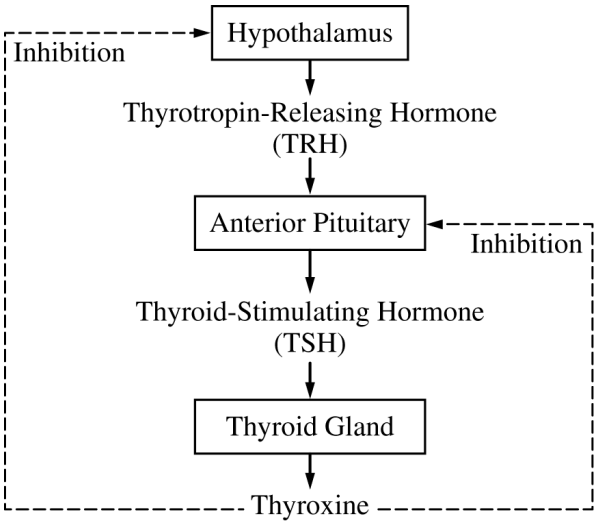
(D) A cell in which enzyme 1 and enzyme 3 are nonfunctional, and enzyme 2 is functional

Answer A

Correct. Enzyme 1 is required to transfer a phosphate group to the cyclin-CDK complex, making it active. Enzyme 2 inactivates the cyclin-CDK complex by transferring a second phosphate group to it, and enzyme 3 removes the phosphate group transferred by enzyme 2 to again activate the cyclin-CDK complex. If only enzyme 1 is functional, then the cyclin-CDK complex is active and can induce cell cycle progression.

4 Cell Communication and Cell Cycle MCQ

2.



The diagram above illustrates feedback control as exerted by the hormone thyroxine. Following surgical removal of the thyroid gland, the level of TSH in the blood will increase. Which of the following best explains this increase?

(A) Residual blood thyroxine, from prior to thyroid gland removal, will bind to cells in the anterior pituitary, signaling more TSH secretion.

(B) Thyroxine will remain bound to thyroxine receptors on various body cells, and these body cells will secrete additional hormones that stimulate the anterior pituitary to secrete TSH.

(C) Thyroxine that was stored in the anterior pituitary prior to thyroid gland removal will signal more TSH secretion.

(D) A decrease in thyroxine levels means a loss of inhibition to the hypothalamus and anterior pituitary, leading to increased TSH secretion. ✓

3. A human cell in prophase of mitosis can be distinguished from a human cell in prophase I of meiosis by the presence of

(A) kinetochores in the mitotic cell

(B) a spindle in the mitotic cell

(C) twice as many chromosomes in the meiotic cell

(D) half as many chromosomes in the meiotic cell

(E) paired homologous chromosomes in the meiotic cell ✓

4 Cell Communication and Cell Cycle MCQ

4. The relative amount of DNA in a cell at various stages of the cell cycle is shown in Figure 1 .

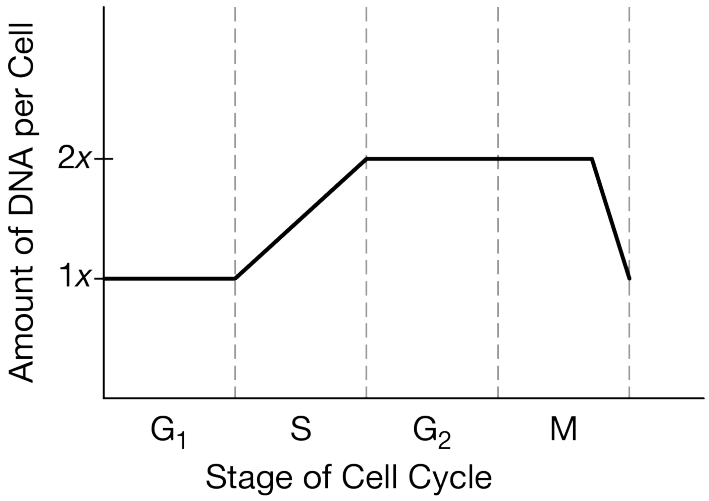


Figure 1. Amount of DNA per cell during different stages of the cell cycle, relative to the beginning of the G_1 stage

Which of the following best describes how the amount of DNA in the cell changes during M phase?

(A) The amount of DNA doubles as the DNA is replicated.

(B) The amount of DNA slightly increases as a result of new organelle synthesis.

(C) The amount of DNA does not change while the cell grows.

(D) The amount of DNA is halved as the cell divides into two daughter cells. ✓

Answer D

Correct. During mitosis, sister chromatids are separated, two nuclei form, and cytokinesis results in the formation of two new cells. Each daughter cell has the same number of chromosomes as the original parent cell.

5. Antidiuretic hormone (ADH) is important in maintaining homeostasis in mammals. ADH is released from the hypothalamus in response to high tissue osmolarity. In response to ADH, the collecting duct and distal tubule in the kidney become more permeable to water, which increases water reabsorption into the capillaries. The amount of hormone released is controlled by a negative feedback loop.

Based on the model presented, which of the following statements expresses the proper relationship between osmolarity, ADH release, and urine production?

4 Cell Communication and Cell Cycle MCQ

(A) As tissue osmolarity rises, more ADH is released, causing less water to be excreted as urine. ✓

(B) As tissue osmolarity rises, less ADH is released, causing less water to be excreted as urine.

(C) As tissue osmolarity rises, more ADH is released, causing more water to be excreted as urine.

(D) As tissue osmolarity rises, less ADH is released, causing more water to be excreted as urine.

6. Researchers have discovered details about apoptosis (programmed cell death) by studying embryologic development of a nematode worm, *Caenorhabditis elegans*. Apoptosis is a normal developmental process in *C. elegans*. They found several genes involved in apoptosis, including $ced-9$ and $ced-3$. The $ced-3$ gene was found to promote cell death, and $ced-9$ to inhibit it. The $ced-9$ gene serves as a regulator that prevents apoptosis in the absence of a signal promoting apoptosis.

Which of the following statements best justifies the claim that changes in the expression of $ced-9$ in *C. elegans* can affect regulation of apoptosis in the cell?

(A) An experiment showed that a mutation in the $ced-9$ gene led to excessive cell death in *C. elegans*. ✓

(B) An experiment showed that the $ced-9$ gene normally produces a protein that promotes excessive cell death in *C. elegans.*

(C) A mutation in $ced-3$ will cause $ced-9$ to be incorrectly transcribed.

(D) Apoptosis is dependent on a signal from the $ced-9$ gene in *C. elegans.*

Answer A

Correct. If $ced-9$ acts as a regulator and normally inhibits apoptosis, then if it loses function, more cell death will occur.

7. A researcher examining a root tip observes a plant cell with condensed sister chromatids, kinetochores with attached microtubules, and individual chromosomes that are aligned at the equatorial plate of the cell. Which of the following best explains what the next process will be in the cell?

(A) Homologous chromosomes (each with two sister chromatids) will move toward opposite poles of the cell.

(B) Paired chromatids will separate, and the new daughter chromosomes will move toward opposite poles of the cell. ✓

(C) The nuclear envelope will break down, and the spindle will begin to form.

(D) The chromatin will decondense, and the daughter cell will enter interphase.

4 Cell Communication and Cell Cycle MCQ

8. Phosphofructokinase (PFK) is a key enzyme in glycolysis. ATP is one of the two substrates for the reaction catalyzed by PFK. ATP is also an allosteric regulator of PFK. Figure 1 shows the enzyme-substrate interactions of PFK.

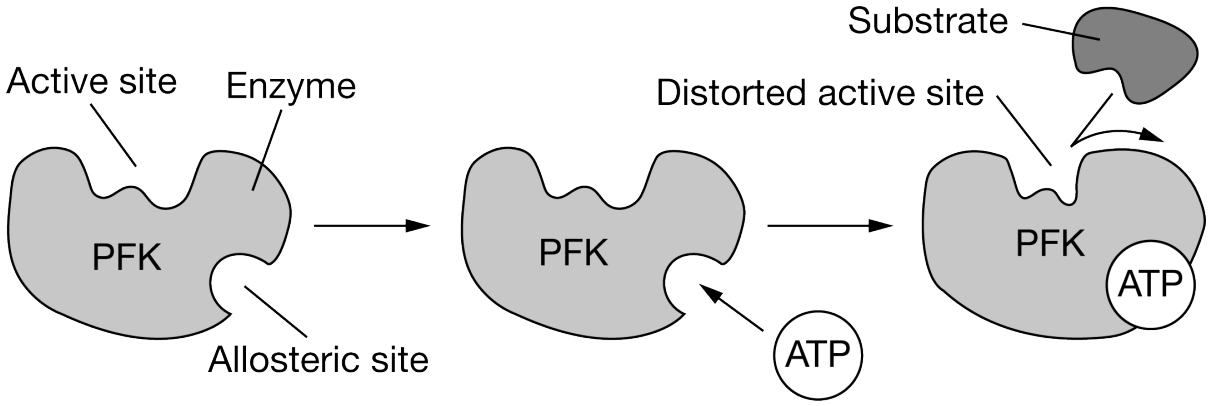


Figure 1. The enzyme-substrate interactions of PFK

A researcher found a mutation that resulted in the PFK enzyme being unable to bind ATP to the allosteric site. Which of the following best predicts the effect of the mutation?

(A) The activity of the enzyme will not be affected because the active site is not involved in substrate binding at the allosteric site.

(B) Negative feedback regulation does not occur, so the enzyme will be active when glycolysis is not needed. ✓

(C) Positive feedback does not occur, and the activity of the enzyme will decrease when glycolysis is needed.

(D) The activity of the enzyme will fluctuate independent of the ATP concentration.

Answer B

Correct. The reaction is normally inhibited when ATP binds to the allosteric site and changes the shape of the active site to the point where it can no longer bind to the substrate. The mutation will not allow this inhibition to occur, and the enzyme will be active even at high ATP concentrations.

4 Cell Communication and Cell Cycle MCQ

In response to elevated blood glucose levels, beta (β) cells in the pancreas release insulin, a regulatory hormone. Insulin signals body cells to take up glucose from the blood, which returns blood glucose levels back to normal. Type 1 diabetes is an autoimmune disorder that destroys β-cells, resulting in elevated blood glucose levels.

Researchers have proposed that diabetes could be treated by implanting human embryonic stem cells (hESCs) that have been induced to develop into β-cells (hESC-β). To test the proposed treatment, the researchers set up two groups of genetically identical mice and implanted the mice from one group with hESC-β cells. Several weeks after the hESC-β implant, both groups of mice were given a drug (STZ) that selectively destroys the naturally occurring mouse β-cells but does not affect the implanted hESC-β cells. Figure 1 shows a comparison of average blood glucose levels in both groups of mice.

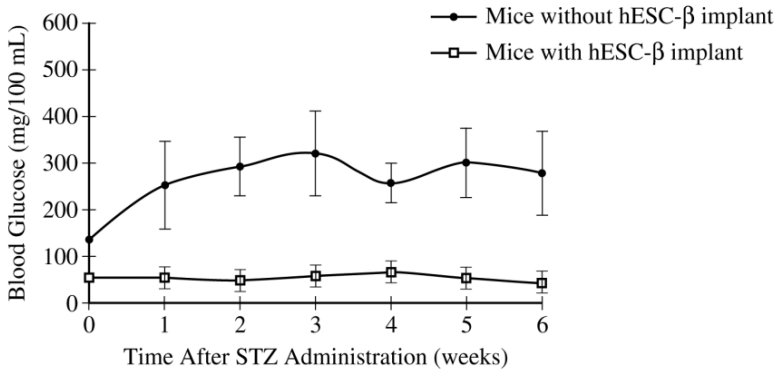


Figure 1. Average blood glucose levels in mice after STZ treatment. Error bars indicate standard deviation.

In a continuation of the experiment, the researchers removed the hESC-β implant from one of the mice 16 weeks after STZ treatment. Figure 2 shows the blood glucose levels in the mouse over the duration of the experiment.

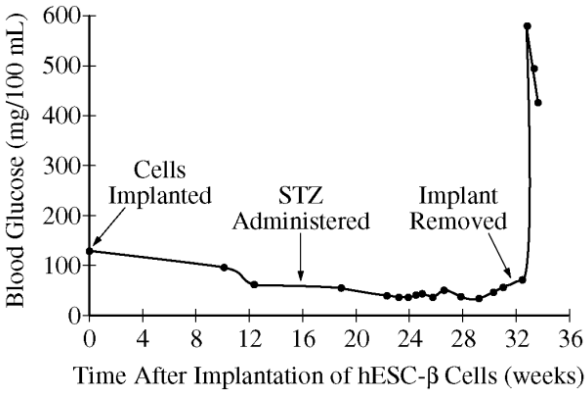


Figure 2. Blood glucose levels for an individual mouse over the duration of the experiment.

4 Cell Communication and Cell Cycle MCQ

9. Based on the data, which of the following best explains how the mice with the implanted hESC-β cells use negative feedback to return blood glucose levels to normal if blood glucose levels increase?

(A) Decreased insulin secretion by body cells → increased glucose uptake by mouse β-cells → increased insulin secretion by body cells

(B) Increased insulin production by mouse β-cells → increased glucose uptake by hESC-β cells → decreased glucose metabolism by body cells

(C) Increased insulin secretion by hESC-β cells → increased glucose uptake by body cells → decreased insulin secretion by hESC-β cells ✓

(D) Increased metabolism of glucose by hESC-β cells → differentiation of hESC-β cell into body cells → increased glucose production by body cells

10. Which of the following modifications to the experimental design would improve the reliability of the results presented in Figure 1 ?

(A) Measuring blood glucose levels in human patients who have type 1 diabetes

(B) Measuring blood glucose levels in a group of mice that were not given STZ ✓

(C) Using a drug that destroys both mouse β-cells and hESC-β cells

(D) Injecting the mice in the experiment with sucrose or fructose

11. Which of the following conclusions is best supported by the data presented in Figure 1 ?

(A) Mouse β-cells are more effective than implanted hESC-β cells at regulating blood glucose levels.

(B) Implanted hESC-β cells are effective at regulating blood glucose levels in the absence of mouse β-cells. ✓

(C) Mouse b-cells do not regulate blood glucose levels in the presence of implanted hESC-β cells.

(D) Implanted hESC-β cells cause type 1 diabetes in laboratory mice.

12. In contrast to type 1 diabetes, where there is no insulin production, type 2 diabetes is characterized by a failure of body cells to respond normally to insulin. Based on the results of the experiment, which of the following best predicts the effectiveness of using hESC-β implantation to treat type 2 diabetes?

(A) The treatment would fail because insulin produced by implanted hESC-β cells will not stimulate glucose uptake by body cells. ✓

(B) The treatment would fail because the implanted hESC-β cells would require more glucose than the body can produce.

(C) The treatment would succeed because the implanted hESC-β cells can express the gene that codes for human insulin.

(D) The treatment would succeed because the body cells would secrete more insulin in the presence of the implanted hESC-β cells.

13. Based on the data in Figure 2, which of the following best justifies the researchers' claim that hESC-β implants can functionally replace the mouse's naturally occurring β-cells?

4 Cell Communication and Cell Cycle MCQ

(A) The mouse with an hESC-β implant demonstrated a decrease in blood glucose levels for the duration of the experiment.

(B) The mouse with an hESC-β implant demonstrated an increase in blood glucose levels before STZ treatment.

(C) The mouse with an hESC-β implant successfully regulated blood glucose levels by regenerating its naturally occurring β-cells.

(D) The mouse with an hESC-β implant successfully regulated blood glucose levels after STZ treatment but not after the implant was removed. ✓

4 Cell Communication and Cell Cycle MCQ

Insulin, a hormone secreted by pancreatic cells, stimulates glucose uptake in skeletal muscle cells by mobilizing glucose transporter proteins (GLUT4) to the plasma membrane. As depicted in Figure 1, binding of insulin to the insulin receptor triggers an intracellular signaling cascade in which certain molecules activate other molecules in a relay of the hormone signal to cell targets. One outcome of the signaling cascade is mobilization of GLUT4 from vesicle storage sites in the cytoplasm to sites at the cell surface, where GLUT4 allows glucose to enter the cell.

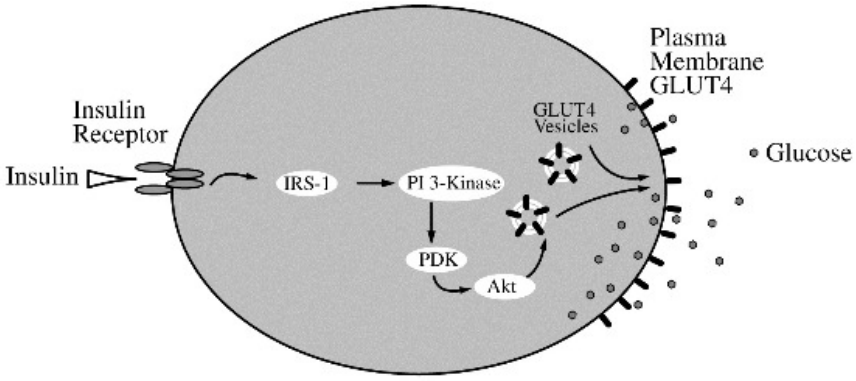


Figure 1. Insulin signaling in muscle cells

In type 2 diabetes, the cellular response to insulin is disrupted, and individuals with type 2 diabetes cannot properly regulate their blood glucose levels. In an investigation of the insulin signaling pathway, samples of skeletal muscle were isolated from individuals who have type 2 diabetes and from individuals who do not. The results of several experiments that were performed on the muscle samples are shown in Figure 2, Figure 3, and Figure 4.

4 Cell Communication and Cell Cycle MCQ

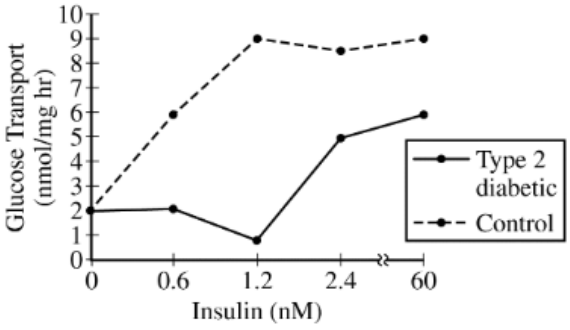


Figure 2. Insulin-stimulated glucose uptake

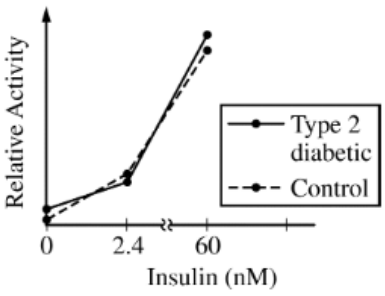


Figure 3. Insulin receptor activation

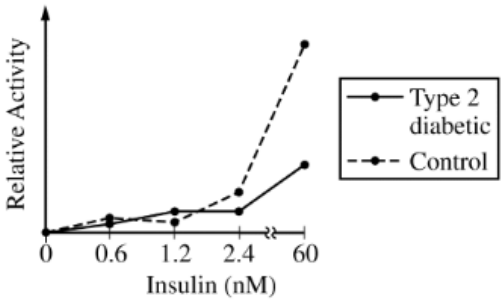


Figure 4. IRS-1 activation

14. Based on the experimental results, which of the following explains the most likely defect in muscle cells of patients with type 2 diabetes?

(A) Insulin receptor proteins do not reach the cell surface.

(B) Insulin does not activate its receptor.

(C) IRS-1 activation is reduced at high insulin concentrations. ✓

(D) GLUT4 blocks glucose from entering cells.

15. Which of the following is a valid interpretation of the experimental results that explains how individuals with type 2 diabetes differ from individuals without diabetes?

(A) The relatively low levels of glucose uptake in individuals with type 2 diabetes indicate that mobilization of GLUT4 to the cell surface is reduced in muscle cells of those individuals. ✓

(B) The relatively low levels of glucose uptake in individuals with type 2 diabetes indicate that no functional GLUT4 protein is produced in the muscle cells of those individuals.

(C) The absence of activated insulin receptors in individuals with type 2 diabetes indicates that no insulin is secreted by the pancreatic cells of those individuals.

(D) The absence of activated IRS-1 in individuals with type 2 diabetes indicates that no functional insulin receptor protein is produced in the muscle cells of those individuals.

4 Cell Communication and Cell Cycle MCQ

16. Based on the information presented, which of the following genetic changes in an individual without diabetes is most likely to result in a disrupted cellular response to insulin signaling similar to that of an individual with type 2 diabetes?

(A) A deletion in the gene encoding the insulin receptor that removes only the cytoplasmic domain of the protein ✓

(B) Duplication of the gene encoding a PI-3 kinase that results in synthesis of a muscle-specific variant of the enzyme in skin cells as well as in muscle cells

(C) A mutation in the gene encoding IRS-1 that causes the protein to be active in muscle cells even in the absence of insulin signaling

(D) Insertion of a small segment of DNA into the promoter of the *Glut4* gene that results in increased synthesis of GLUT4 proteins in muscle cells

4 Cell Communication and Cell Cycle MCQ

Excess intracellular iron is toxic to cells (iron-induced toxicity). Ferritin is an intracellular iron storage protein that binds excess iron. The presence of ferritin can protect cells from iron-induced toxicity.

In an experiment to investigate the effects of dietary iron intake on ferritin synthesis, rats were given food containing different amounts of iron. Subsequently, the levels of ferritin protein in the liver were measured. The results are shown in Figure 1.

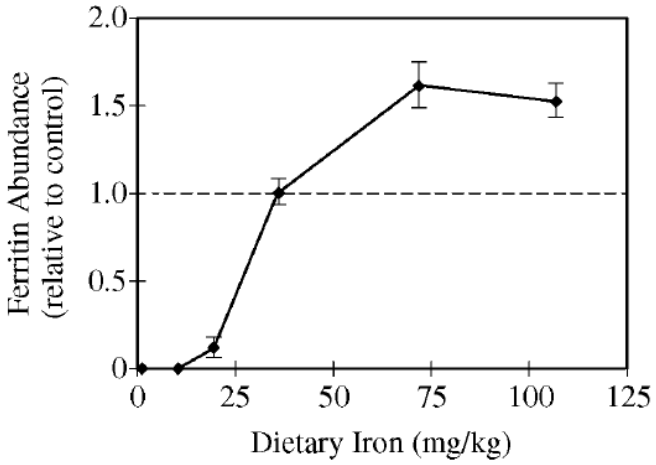
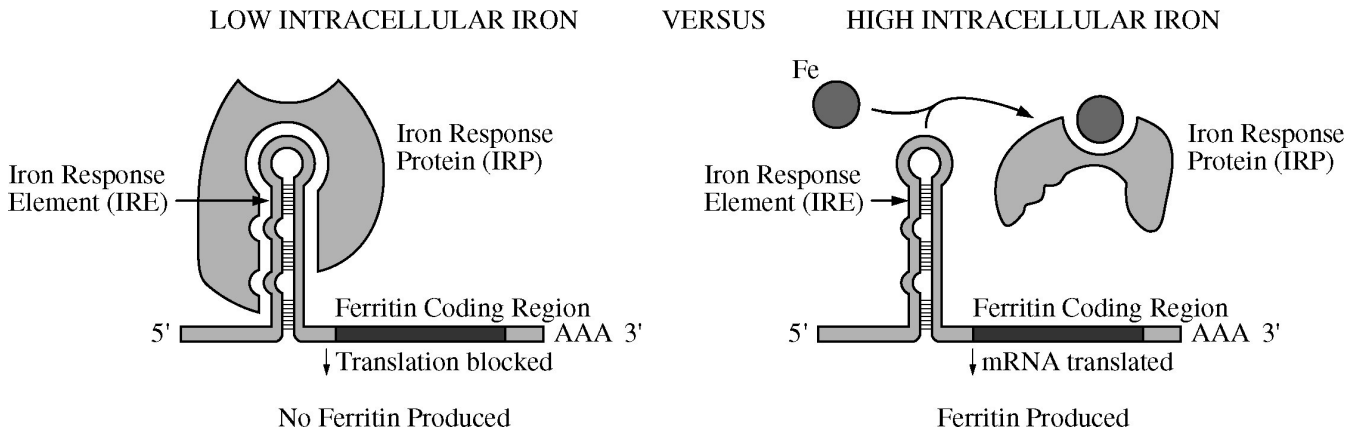


Figure 1. Effects of dietary iron on ferritin levels in rat liver

Based on these and other data, researchers have developed the following model demonstrating how ferritin synthesis is regulated by iron. When iron levels are low, a repressor of translation, iron response protein (IRP), binds to an iron response element (IRE), which is a stem-loop structure near the 5¢ end of ferritin mRNA. When iron levels are high, intracellular iron binds to the IRP, and the iron-IRP complex dissociates from the IRE, permitting ribosomes to proceed with the translation of ferritin mRNA. Figure 2 represents the model of the regulation of ferritin mRNA translation by iron.



4 Cell Communication and Cell Cycle MCQ

Figure 2. Model of regulation of ferritin synthesis by iron

17. Based on the model of ferritin synthesis presented in Figure 2, which of the following describes the role of feedback on the control of intracellular iron levels?

(A) A decrease in iron levels activates the IRP. The IRP in turn activates iron transport proteins in the cell membrane, thereby returning free iron levels to normal.

(B) A decrease in iron levels activates synthesis of ferritin protein. Ferritin protein in turn releases bound iron, thereby returning free iron levels to normal.

(C) An increase in iron levels activates the IRP. The IRP in turn binds iron, thereby decreasing both free iron levels and ferritin synthesis.

(D) An increase in iron levels activates synthesis of ferritin protein. Ferritin protein in turn binds iron, thereby decreasing both free iron levels and ferritin synthesis. ✓

18. Based on the model of ferritin synthesis presented in Figure 2, which of the following best describes the mechanism whereby iron most likely regulates ferritin production?

(A) Translation occurs under low intracellular iron concentration when the IRP recruits ribosomes to the ferritin mRNA.

(B) Translation occurs under low intracellular iron concentration when the IRP stabilizes the stem-loop structure in the ferritin mRNA.

(C) Translation occurs under high intracellular iron concentration when the IRP-iron complex dissociates from ferritin mRNA, permitting ribosomes access to the ferritin coding region. ✓

(D) Translation occurs under high intracellular iron concentration when the IRP-iron complex brings the 5¢ end of the mRNA closer to the ferritin coding region.

4 Cell Communication and Cell Cycle MCQ

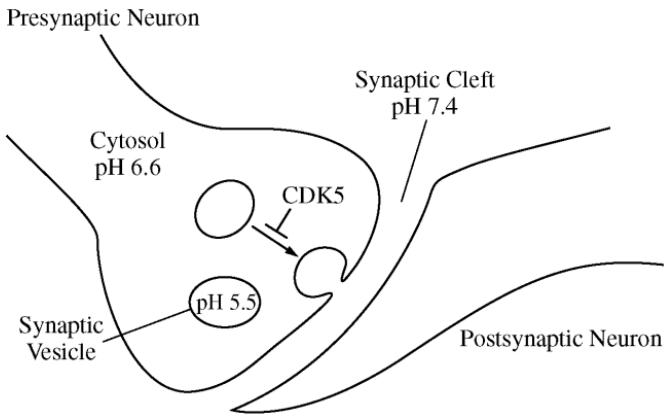


Figure 1. Model of synapse

Researchers investigating the regulation of neurotransmitter release from presynaptic neurons proposed a model (Figure 1) in which CDK5, a protein expressed in axon terminals, inhibits the movement of synaptic vesicles to the presynaptic membrane.

To test their model, the researchers used a modified version of green fluorescent protein (GFP*). In slightly alkaline conditions, GFP* exhibits a bright green fluorescence. In acidic conditions, GFP* exhibits no fluorescence. Using standard techniques, the gene encoding GFP* is easily introduced into living cells. By engineering the expression of GFP* in laboratory-cultured nerve cells, the researchers found that a bright green fluorescence was exhibited only when a presynaptic neuron was given a certain stimulus.

19. Based on the model, which of the following best explains how regulation of neurotransmitter release might increase the range of responses to a stimulus in the nervous system?

(A) In the absence of any stimulus, neurons can still release neurotransmitters.

(B) Different neurons in the same neural network can release different amounts of neurotransmitter. ✓

(C) In the depolarization phase of an action potential, postsynaptic neurons can adjust the amount of neurotransmitter bound to receptors on their surface.

(D) In the recovery phase following a stimulus, enzymes can be mobilized to degrade molecules present in the synaptic vesicles.

20. Previous experiments indicate that CDK5 is active only when attached to a protein called p35. Which of the following best predicts how p35 might play a role in regulating neuron function?

(A) Elevated intracellular levels of p35 result in increased synaptic activity.

(B) Degradation of p35 results in increased synaptic activity. ✓

(C) Reabsorption of p35 from the synaptic cleft results in increased synaptic activity.

(D) Attachment of p35 to synaptic vesicles results in increased synaptic activity.

21. Which of the following observations best supports the hypothesis that CDK5negatively regulates neurotransmitter release?

4 Cell Communication and Cell Cycle MCQ

(A) Introduction of CDK5 protein into neurons results in the movement of synaptic vesicles to the plasma membrane in the absence of any stimulus.

(B) Uptake of a gene encoding CDK5 by neurons results in the movement of synaptic vesicles to the plasma membrane in the absence of any stimulus.

(C) Suppression of CDK5 expression in neurons inhibits the movement of synaptic vesicles to the plasma membrane in response to a specific stimulus.

(D) Inhibition of CDK5 activity in neurons increases the movement of synaptic vesicles to the plasma membrane in response to a specific stimulus. ✓

22. The beta-2 adrenergic receptor is a membrane-bound protein that regulates several cellular processes, including the synthesis and breakdown of glycogen. The receptor binds specifically to the hormone epinephrine. The binding of epinephrine to the beta-2 adrenergic receptor triggers a signal transduction cascade that controls glycogen synthesis and breakdown in the cell. A simplified model of the signal transduction cascade is represented in Figure 1.

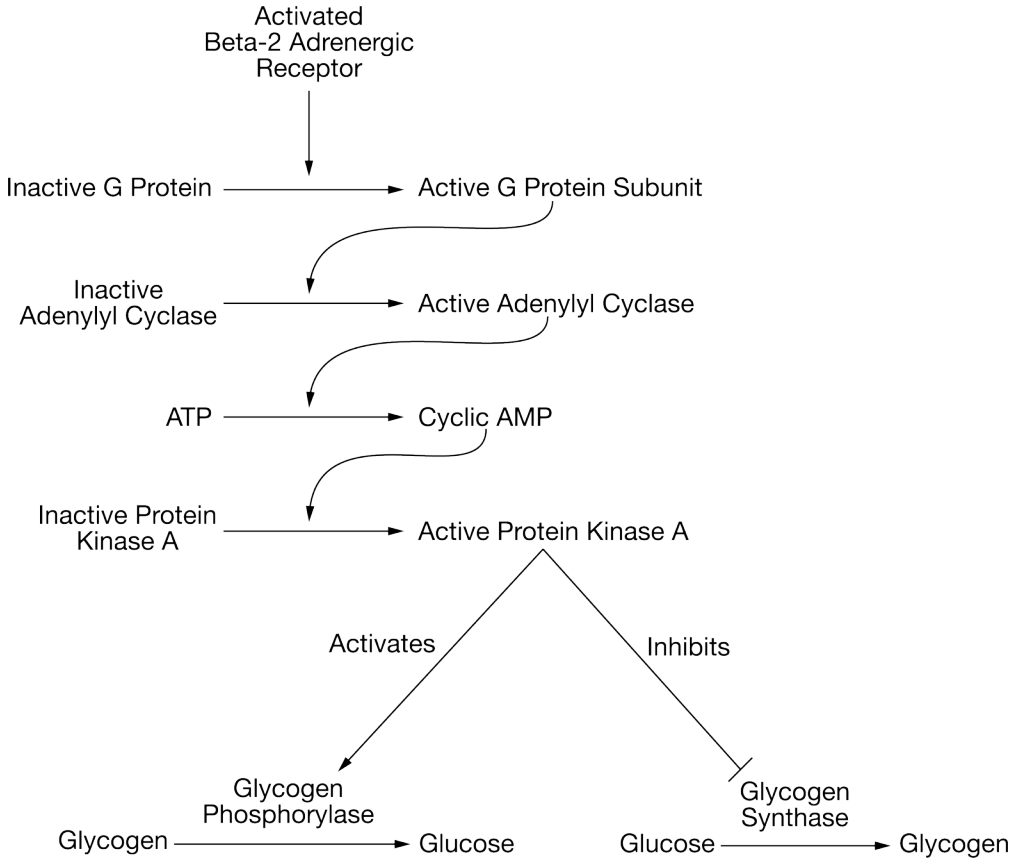


Figure 1. A simplified model of the signal transduction cascade triggered by epinephrine binding to the beta-2 adrenergic receptor

Which of the following outcomes will most likely result from the inactivation of the beta-2 adrenergic receptor?

4 Cell Communication and Cell Cycle MCQ

(A) The cellular concentration of cyclic AMP will increase.
(B) The enzymatic activity of protein kinase A will increase.
(C) The activation of glycogen phosphorylase will increase.
(D) The rate of glycogen synthesis in the cell will increase. ✓

Answer D

Correct. Based on Figure 1, activation of the beta-2 adrenergic receptor results in the stimulation of glycogen breakdown and the inhibition of glycogen synthesis. The immediate effect of inactivating the beta-2 adrenergic receptor will be an increase in the rate of glycogen synthesis in the cell because glycogen synthase will no longer be inhibited. Also, glycogen breakdown in the cell will no longer be stimulated because glycogen phosphorylase will most likely be deactivated.

4 Cell Communication and Cell Cycle MCQ

23. Blood clots are formed by a positive feedback loop. Two pathways exist, the extrinsic and intrinsic pathways, which converge during clot formation. There are many clotting factors involved, most of which are proteins. Vitamin K is required for the formation of the active form of several of the clotting factors, including Factor X. A simplified model of the blood clotting process is shown in Figure 1.

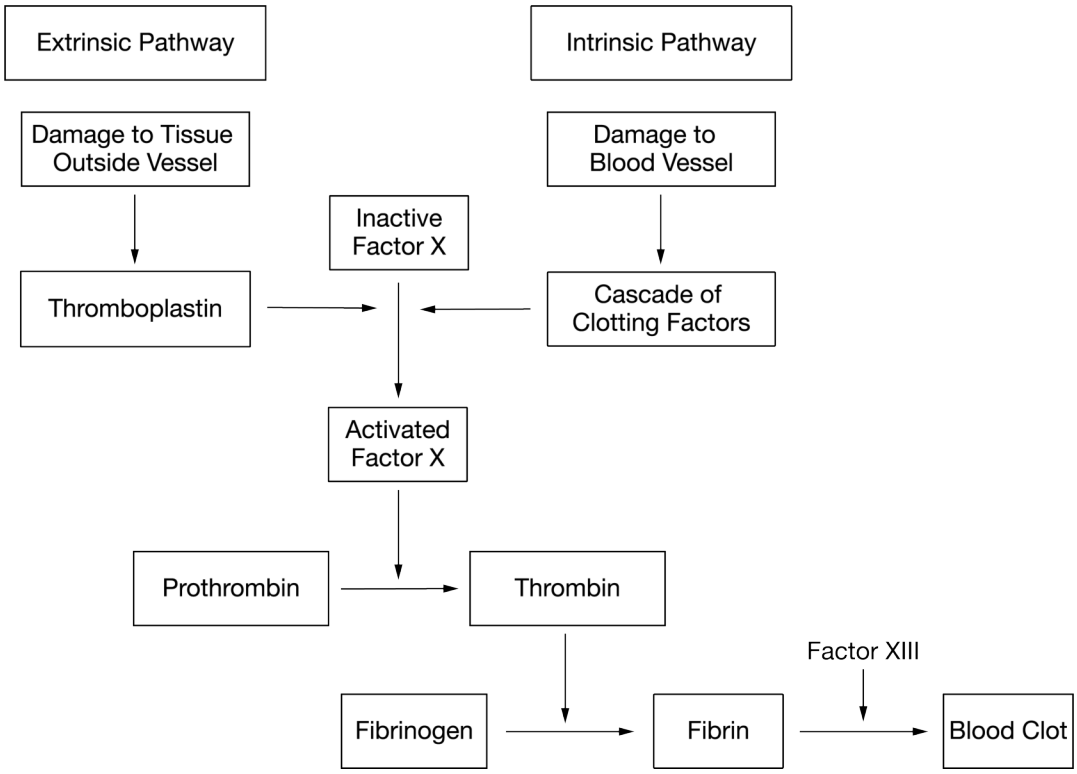


Figure 1. Simplified model of clotting cascade

Warfarin is a drug used to treat certain blood clots. Warfarin blocks the formation of the active form of vitamin K -dependent clotting factors. Based on the model, which of the following best predicts the effects of warfarin on a patient?

(A) Fibrinogen will form fibrin, but the clot will not form because Factor XIII will not be synthesized.

(B) The intrinsic pathway will take over because the clotting factors are part of that pathway.

(C) Thrombin will be converted to prothrombin because Factor X will reverse the reaction.

(D) Factor X will not be activated, which will prevent thrombin from forming. ✓

Answer D

Correct. Vitamin K is required for the synthesis of Factor X and warfarin inhibits that process; thus, clots are prevented from forming.

4 Cell Communication and Cell Cycle MCQ

24. A person's blood glucose level fluctuates during the day, as represented in Figure 1. Two hormones, insulin and glucagon, are directly involved in regulating the blood glucose level to maintain a healthy level. Insulin acts to lower the blood glucose level, and glucagon acts to increase the blood glucose level.

Figure 1. Blood glucose fluctuations of an individual

Which of the following best predicts what will happen to the blood glucose level if the person has another meal at 5 P.M.?

(A) Immediately after the meal, the blood glucose level will decrease because of the increase in glucagon levels.

(B) Immediately after the meal, the blood glucose level will increase, and then insulin will be secreted to counter the increase. ✓

(C) Several hours after the meal, the blood glucose level will increase sharply because of an increase in the amount of glucagon secreted.

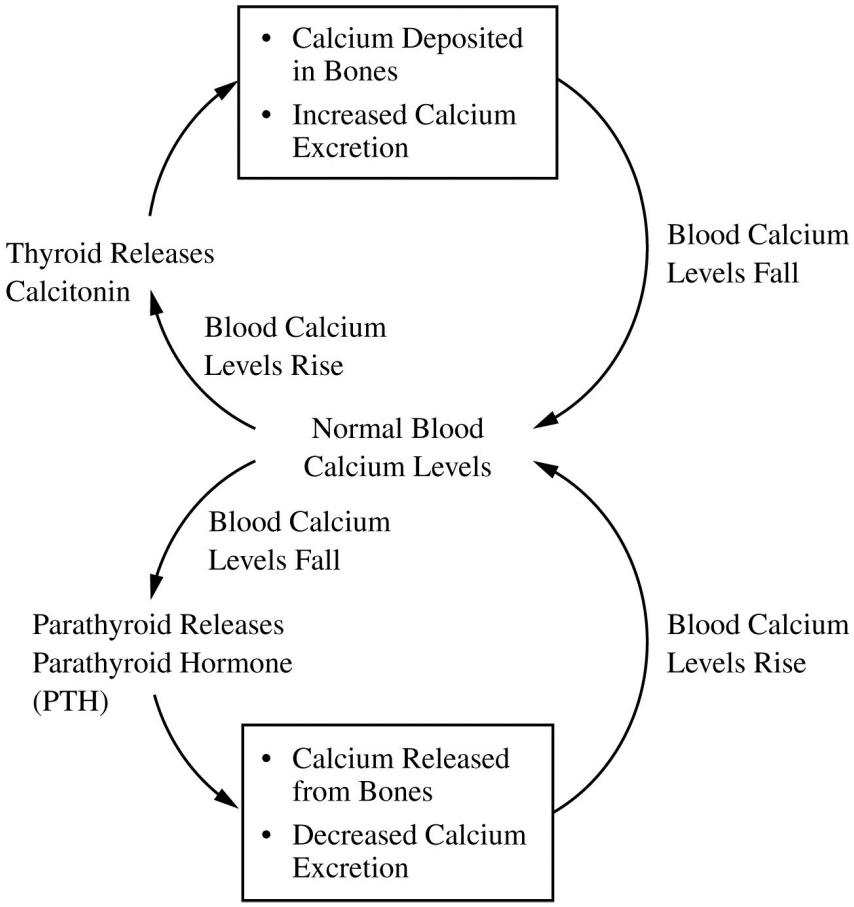
(D) The blood glucose level will not change after the 5 P.M. meal because the person has already consumed two meals and the blood glucose level has been adjusted to a steady-state level.

Answer B

Correct. The blood glucose levels increase due to the nutrients taken in. This increase is a signal for the pancreas to release more insulin that will stimulate cells to take in the excess glucose. As a result, the blood glucose level will then decrease.

4 Cell Communication and Cell Cycle MCQ

25.



The model shown in the figure represents the role of two hormones, calcitonin and parathyroid hormone (PTH), in maintaining normal blood calcium levels in humans. If a dietary change results in an increase in blood calcium concentration above normal levels, which of the following is the most likely effect on calcium homeostasis?

(A) Calcitonin levels will decline, thus stimulating the release of PTH.

(B) Calcitonin levels will rise, thus promoting the deposit of calcium into bones. ✓

(C) PTH levels will decline, thus stimulating the loss of calcium from bones.

(D) PTH levels will increase, thus preventing the release of calcitonin.

26. Cancer can result from a variety of different mutational events. Which of the following is LEAST likely to result in the initiation of a cancerous tumor?

(A) A receptor mutation results in activation of a cell-division pathway in the absence of the appropriate ligand.

(B) A mutation results in the loss of the ability to produce a tumor-suppressor protein.

(C) A defect in a cell-cycle checkpoint prevents a cell from entering the S phase. ✓

(D) At the anaphase checkpoint, separation of chromatids occurs without all centromeres being attached to kinetochore microtubules from both poles.

4 Cell Communication and Cell Cycle MCQ

27. Most cells that have become transformed into cancer cells have which of the following characteristics when compared to normal, healthy cells?

(A) Shorter cell cycle ✓

(B) More carefully regulated rates of cell division

(C) Lower rates of mitosis

(D) Higher rates of protein translation

(E) Identical DNA

28. Cyclin-dependent kinase 7 (CDK 7) binds to cyclin H and a second accessory protein, MAT 1, to form a complex of three proteins. Researchers investigating the role of CDK 7 claim that formation of this complex promotes cell cycle progression.

Which of the following supports the researchers' claim?

(A) Decreased expression of CDK 7 initiates DNA replication.

(B) Decreased expression of CDK 7 results in the formation of spindle fibers.

(C) Increased expression of CDK 7 causes cells to enter G_0.

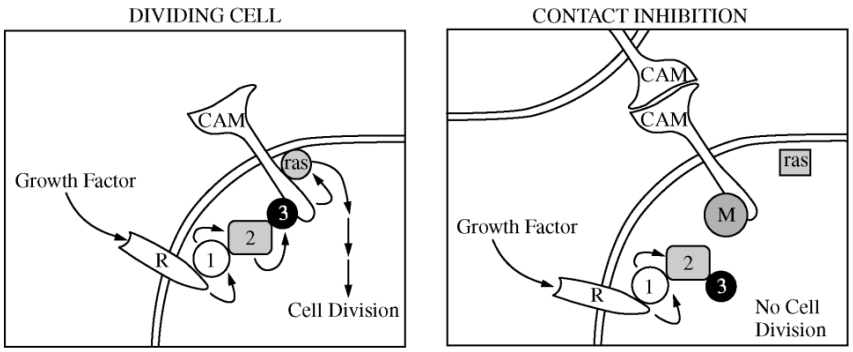
(D) Increased expression of CDK 7 is associated with certain types of cancer. ✓

Answer D

Correct. Cancer is associated with increased cell division. Increased expression of CDK 7 in certain cancers would indicate increased cell cycle progression and would therefore support the researchers' claim.

4 Cell Communication and Cell Cycle MCQ

29. Many human cells can be stimulated to divide by hormonelike growth factors that bind to receptor proteins (R) on the plasma membrane and trigger an internal signal-transduction cascade. In many cases, however, the process of contact inhibition prevents mitosis when cells are in direct contact with one another. Contact inhibition occurs when proteins called cell adhesion molecules (CAMs) interact, causing them to change shape so that the growth-factor signaling proteins that normally associate with CAMs are replaced by another protein, called M. Both pathways are depicted in the figures below.



Which of the following statements accurately uses the information presented to support the hypothesis that interruption of M function in a single body cell can result in cancer?

(A) Protein 3 will be prevented from interacting with CAMs, causing the cell cycle to stop permanently.

(B) The ras protein will remain bound to DNA, blocking expression of genes required for mitosis.

(C) Growth-factor signaling can trigger mitosis in cells that are in direct contact with other cells. ✓

(D) The receptor proteins of body cells will no longer bind to growth-factor proteins.

30. Which of the following best describes a cell that is in the G_1 stage of the cell cycle?

(A) The cell is growing in size and increasing its number of proteins and organelles. ✓

(B) The chromosomes within the cell are checked to make certain that they have been replicated correctly.

(C) The chromosomes are replicating, and the amount of DNA in the cell is doubling.

(D) The cell is in a resting stage and no longer dividing.

Answer A

Correct. The cellular activities described are activities that occur during the G_1 stage of the cell cycle.

4 Cell Communication and Cell Cycle MCQ

31. A cell culture commonly used in research was selected to study the effect of a specific virus on the timing of cell cycle phases. Two separate cultures were started, one untreated and one inoculated with the virus. Both cultures were incubated under identical conditions. After a period of time, 200 cells from each culture were observed and classified as shown in Table 1.

Table 1. Number of normal and infected cells found in three phases of the cell cycle

Phase of Cell Cycle	Untreated Cells ($n = 200$)	Virus-Infected Cells ($n = 200$)
G_0	196	10
Interphase	2	40
Mitosis	2	150

Which of the following most accurately describes an observation and an effect of the viral infection indicated by the data in Table 1?

(A) Normal cells spend 98 percent of their time cycling in and out of interphase. The virus reduces this to 5 percent of the time.

(B) Twenty percent of the virus-infected cells are in interphase. These cells are no longer part of the cell cycle.

(C) Forty percent of the virus-infected cells are in interphase. These cells are preparing for replication of genetic material.

(D) Seventy-five percent of the virus-infected cells are found in mitosis. The virus stimulates frequent cell division. ✓

Answer D

Correct. One hundred fifty out of 200 is 75 percent. The virus activates cells currently in the G_0 phase, leading to rapid cell growth and division.

4 Cell Communication and Cell Cycle MCQ

32. Figure 1 represents the relative time and sequence of the phases of the cell cycle.

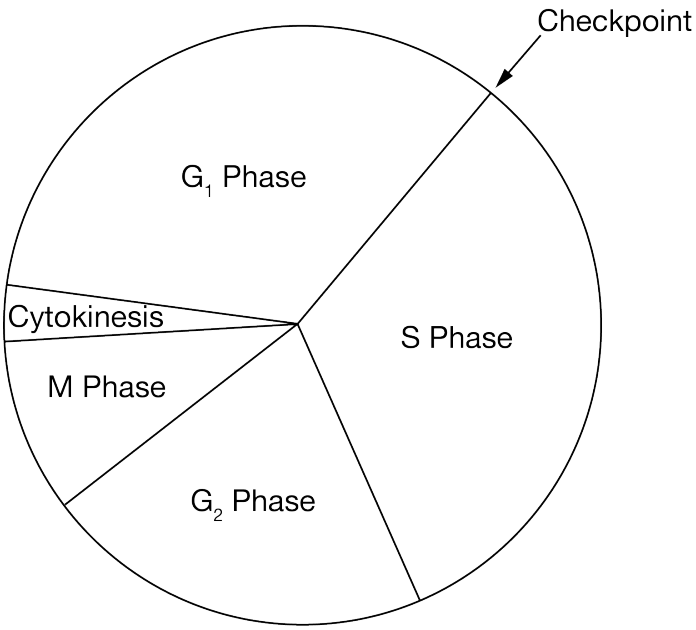


Figure 1. Representation of the cell cycle and identification of the G_1/S checkpoint

Which statement best predicts why a cell's progression through the cell cycle might be halted at the G_1/S checkpoint?

(A) Spindle fibers have not correctly attached to chromosomes.

(B) There are not enough nucleotides available to construct new DNA. ✓

(C) Damage occurred to DNA when it was being copied in G_1.

(D) Proteins necessary for M phase of the cell cycle have not been produced.

Answer B

Correct. DNA replication, which occurs during S phase, requires free nucleotides. If nucleotides are unavailable, the G_1/S checkpoint halts the cell cycle to conserve energy.

4 Cell Communication and Cell Cycle MCQ

33. Researchers performed an experiment to determine the effect of certain genetic mutations on mitosis in tropical fruit fly embryos. They determined the percentage of cells in each of four phases of mitosis as shown in Figure 1.

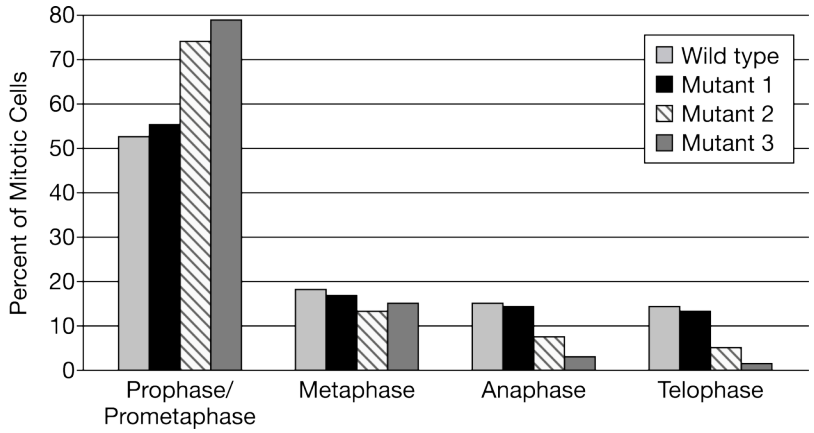


Figure 1. Percent of cells in phases of mitosis

Which of the following patterns is shown by the data?

(A) Mutant 1 cells are more similar to mutant 3 cells than to wild-type cells.

(B) In wild-type cells, the percent of cells in anaphase is twice the amount of those in telophase

(C) In mutant 3 cells, more time is spent in prophase/prometaphase than in the later stages of mitosis. ✓

(D) The percent of mutant 2 cells in anaphase is higher than that of mutant 1 cells.

Answer C

Correct. According the data, mutant 3 had the highest percent of cells in prophase/prometaphase of all cell types and the least time in anaphase and telophase.

34. What is the expected percent change in the DNA content of a typical eukaryotic cell as it progresses through the cell cycle from the start of the G_1 phase to the end of the G_2 phase?

(A) -100%

(B) -50%

(C) $+50\%$

(D) $+100\%$ ✓

4 Cell Communication and Cell Cycle MCQ

Answer D

Correct. The DNA of a typical eukaryotic cell is replicated in the S phase of the cell cycle, which occurs between the G_1 and G_2 phases. DNA replication typically results in a 100% increase in the DNA content of a cell.

35. If chemical signals in the cytoplasm control the progression of a cell to the M phase of the cell cycle, then fusion of a cell in G_1 with a cell in early M phase would most likely result in the

(A) replication of chromosomes only in the G_1 cell

(B) exiting of both cells from the cell cycle and into the G_0 phase

(C) condensation of chromatin in preparation of nuclear division in both cells ✓

(D) transfer of organelles from the G_1 cell to the cell in the M phase

36. Figure 1 shows the number of chromosomes observed in an actively dividing human cell at each stage of cell division.

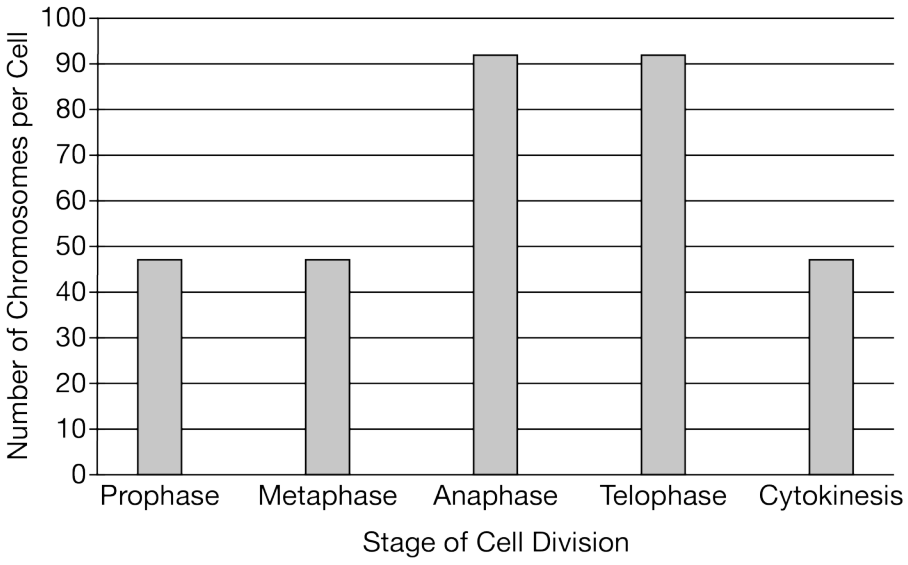


Figure 1. Number of chromosomes in a human cell at different stages of cell division

Which of the following presents a correct interpretation of the changes in chromosome number depicted in Figure 1 ?

4 Cell Communication and Cell Cycle MCQ

(A) DNA replication occurs between metaphase and anaphase, doubling the number of chromosomes. Between telophase and cytokinesis, the cell divides in two, with each cell receiving half of the replicated chromosomes.

(B) New chromosomes formed during prophase are doubled during anaphase and are recombined before cytokinesis.

(C) Chromosomes enter metaphase containing two chromatids attached by a centromere. During anaphase, the chromatids are separated, each becoming a chromosome. Cytokinesis distributes the chromosomes into two separate cells. ✓

(D) At anaphase a cell contains two identical copies of each chromosome, but following telophase, one of the copies is broken down into nucleotides.

Answer C

Correct. This statement accurately describes changes in the chromosome number during the different stages of cell division as represented in the graph. During prophase and metaphase, each chromosome is composed of two chromatids. During anaphase, the two chromatids separate to the opposite ends of the cell, doubling the number of chromosomes in the cell. The number of DNA strands in the cell has not changed, only how they are organized. During cytokinesis, the cell is divided into two daughter cells, each containing half of the chromosomes found in the parent cell during anaphase and telophase.

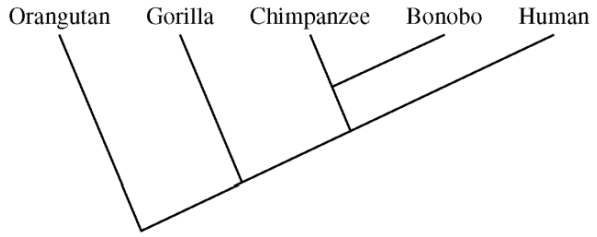
4 Cell Communication and Cell Cycle MCQ

The *TAS2R38* gene encodes a receptor protein that influences the ability to taste bitterness. The gene has two alleles: a dominant, wild-type allele that enables an individual (taster) to taste bitterness and a recessive, mutant allele that interferes with the ability of an individual (nontaster) to taste bitterness. Three single nucleotide mutations in the coding region of the *TAS2R38* gene are associated with the nontaster allele. The nucleotides present at the three positions are shown in the table below.

NUCLEOTIDE VARIATION IN THE *TAS2R38* GENE

	Position in the Nucleotide Sequence		
	145	785	886
Human nontaster	G	T	A
Human taster	C	C	G
Bonobo	C	C	G
Chimpanzee	C	C	G
Gorilla	C	C	G

A cladogram representing the evolutionary relatedness of selected primates is shown below.



37. The TAS2R38 receptor protein has been detected on the surface of cells from individuals who are homozygous for the nontaster allele of the *TAS2R38* gene. Which of the following is the most likely effect of the mutations associated with the nontaster allele on *TAS2R38* gene expression?

(A) The mutations change the primary structure of the encoded receptor protein. ✓

(B) The mutations increase the stability of the *TAS2R38* mRNA.

(C) The mutations prevent transcription of the *TAS2R38* gene.

(D) The mutations prevent translation of the *TAS2R38* mRNA.

4 Cell Communication and Cell Cycle MCQ

38. Researchers grew seedlings of corn, *Zea mays*, in loose and compact sand. The researchers measured the amount of time required for the cells in the growing root tips of the seedlings to double in number. The mean cell doubling times for the two groups of seedlings are shown in Figure 1.

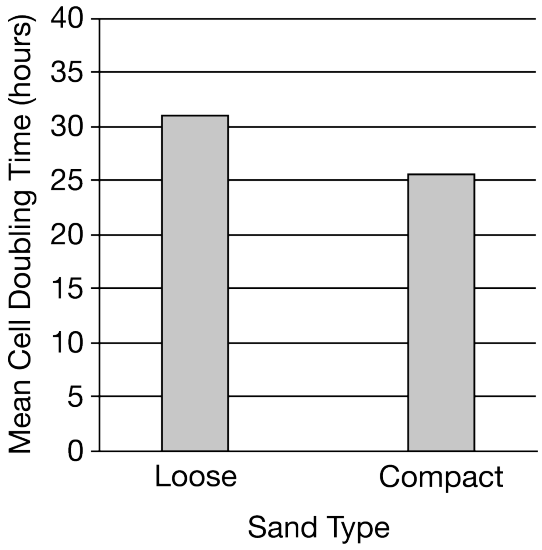


Figure 1. Mean cell doubling times for the growing root tips of *Zea mays* seedlings planted in loose or compact sand

Based on the sample means, which of the following conclusions about the cells in the growing root tips of *Zea mays* seedlings is best supported by the results of the experiment?

(A) The cells of the root tips grow to larger sizes when the seedlings are planted in compact sand than when the seedlings are planted in loose sand.

(B) The average rate of mitotic cell division is greater for the root tips growing in loose sand than for the root tips growing in compact sand.

(C) The average cell cycle time is greater for the root tips growing in compact sand than for the root tips growing in loose sand.

(D) More cells are produced per unit of time in the root tips growing in compact sand than in the root tips growing in loose sand. ✓

Answer D

Correct. According to the figure, the mean cell doubling time of *Zea mays* seedling root tips in compact sand is less than in loose sand. A lower mean cell doubling time indicates an increased rate of mitotic cell division for cells of root tips in compact sand.

4 Cell Communication and Cell Cycle MCQ

39. Cortisol is a hormone produced in response to stress, including starvation, in humans. Which of the following is most likely an immediate effect of a starvation-induced increase in cortisol secretion?

(A) Increased activation of the immune system

(B) Increased urine production by the kidneys

(C) Increased bone and collagen formation

(D) Increased mobilization of fatty acids from fat cells ✓

40. Damaged tissue releases chemicals that activate platelets and stimulate the formation of blood clots. Which of the following statements about the activity of platelets best explains a positive feedback mechanism?

(A) Activated platelets release chemicals that inhibit blood clot formation.

(B) Activated platelets release signaling molecules that inhibit cell division in damaged tissue.

(C) Activated platelets constrict the blood vessels, stopping blood flow.

(D) Activated platelets release chemicals that activate more platelets. ✓

41. Researchers tracked the amount of DNA (measured in picograms) over time beginning with a single cell and continuing through several rounds of cell division. The researchers observed threadlike chromosomes prior to cell division. The threadlike chromosomes disappeared from view shortly after each division. The amount of DNA in picograms per cell over several rounds of cell division is shown in Figure 1.

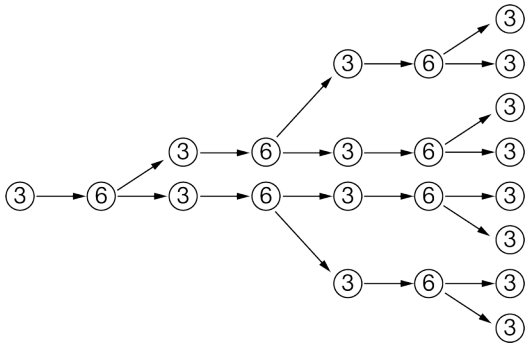


Figure 1. Amount of DNA in picograms per cell over several rounds of cell division

Which of the following statements is consistent with the data in Figure 1?

(A) The cells have a haploid chromosome number of 3.

(B) The cells have a diploid chromosome number of 6.

(C) There is a change from 3 to 6 picograms of DNA because DNA is replicated before each round of cell division. ✓

(D) There is a change from 6 to 3 picograms of DNA after each cell division because the chromosomes lengthen following cell division.

4 Cell Communication and Cell Cycle MCQ

Answer C

Correct. The change from 3 to 6 picograms of DNA prior to replication indicates that the DNA has been replicated. The replicated DNA is condensed into chromosomes during cell division or mitosis. This ensures that the complete genome from a parent cell is passed on to two genetically identical daughter cells.

42. DNA damage can lead to the release of the protein cytochrome *c* from mitochondria. Cytochrome *c* then activates the enzyme caspase-9, resulting in the release of more cytochrome *c*, followed by apoptosis.

Which of the following best explains how the release of cytochrome *c* from mitochondria is regulated by feedback?

(A) It is positive feedback because the release of cytochrome *c* from mitochondria contributes to the production of more mitochondria.

(B) It is negative feedback because the release of cytochrome *c* from mitochondria results in an increase in cell death.

(C) It is positive feedback because the release of cytochrome *c* from mitochondria contributes to the release of more cytochrome. ✓

(D) It is negative feedback because the release of cytochrome *c* from mitochondria is caused by DNA damage.

Answer C

Correct. The release of cytochrome *c* results in the activation of caspase-9, which results in the production of more cytochrome *c*. This is an example of a positive feedback loop.

43. DNA replication occurs

(A) during the S phase of the cell cycle ✓

(B) as the nuclear envelope breaks down in early mitosis

(C) during mitosis but not during meiosis

(D) in animal cells but not in plant cells

(E) only in cells destined to become gametes

4 Cell Communication and Cell Cycle MCQ

44. Human epidermal growth factor (EGF) is a protein that stimulates cell division and growth by binding to its receptor, human epidermal growth factor receptor 2 ($\mathrm{HER\,2}$). Certain types of cancer are caused when the overexpression of EGF activates a cell-signaling pathway, leading to uncontrolled cell division.

Which of the following best explains how a drug can be used to inhibit tumor growth caused by EGF overexpression?

(A) The drug binds to $\mathrm{HER\,2}$, preventing the activation of the signaling pathway. ✓

(B) The drug acts as a positive regulator of transcription of the gene encoding EGF.

(C) The drug activates the receptor in the absence of EGF, leading to the activation of the signaling pathway.

(D) The drug enables the cells to progress through the $\mathrm{S/G_2}$ checkpoint.

Answer A

Correct. If activating the $\mathrm{HER\,2}$ pathway leads to cell division, then an inhibitor that binds to the receptor and blocks its activation, probably by blocking the binding of EGF to $\mathrm{HER\,2}$, will block cell division.

45. During mitosis, which of the following normally occurs?

(A) Homologous chromosomes pair.

(B) Replicated chromosomes line up on the equatorial plate. ✓

(C) Tetrads form.

(D) Unreplicated chromosomes become oriented in the center of the cell.

(E) Maternal and paternal chromatids pair.

4 Cell Communication and Cell Cycle MCQ

46. **TABLE 1. DISTRIBUTION OF CONTROL AND EXPERIMENTAL CELLS IN DIFFERENT STAGES OF THE CELL CYCLE AFTER 24 HOURS**

	G_1	S	G_2	M
Control	96	59	48	81
Experimental	142	142	0	0

Researchers claimed that cells treated with radiation are unable to progress through the cell cycle. They tested their claim by exposing a group of cells to radiation (experimental group). A group of unexposed cells served as the control group. After 24 hours, they determined the number of cells in each stage of the cell cycle and confirmed that all differences were statistically significant.

Based on Table 1, which of the following best supports the researchers' claim?

(A) In the experimental group, no cells were able to enter G_2. ✓

(B) In the control group, the number of cells in G_2 was less than the number of cells in M.

(C) In the control group, the number of cells in each stage decreased as the cell cycle continued.

(D) In the experimental group, the number of cells in G_1 was the same as the number of cells in S.

Answer A

Correct. The lack of cells in the G_2 phase supports the researchers' claim that radiation-treated cells are unable to progress through the cell cycle.

47. The drug 5-fluorouracil inhibits thymine production in eukaryotic cells. Which of the following cell cycle stages will be most directly affected by 5-fluorouracil?

(A) The first growth phase (G_1)

(B) Synthesis of DNA phase (S) ✓

(C) Preparation for mitosis (G_2)

(D) Cytokinesis

Answer B

Correct. DNA synthesis that occurs during S requires thymine−containing nucleotides, so inhibition of thymine production will inhibit DNA synthesis.

4 Cell Communication and Cell Cycle MCQ

48. The epidermal growth factor receptor EGFR is a cell surface receptor. When a growth factor binds to EGFR, the receptor is activated. The activated EGFR triggers a signal transduction pathway, which leads to increased frequency of cell division.

Which of the following best predicts the effect of a mutation that causes EGFR to be active in the absence of a growth factor?

(A) Increased apoptosis will lead to abnormal growth of the tissue.

(B) Increased cell division will lead to the formation of a tumor. ✓

(C) Cells will exit the cell cycle, entering a non-dividing G_0 phase.

(D) Fewer cells will be in any of the stages of mitosis.

Answer B

Correct. Growth factors are important signaling molecules that trigger cells to divide at the correct rate and correct time. An EGFR that is active when a growth factor is absent would lead to increased cell division. Uncontrolled cell growth will likely result in a tumor.

49. The endocrine system incorporates feedback mechanisms that maintain homeostasis. Which of the following demonstrates negative feedback by the endocrine system?

(A) During labor, the fetus exerts pressure on the uterine wall, inducing the production of oxytocin, which stimulates uterine wall contraction. The contractions cause the fetus to further push on the wall, increasing the production of oxytocin.

(B) After a meal, blood glucose levels become elevated, stimulating beta cells of the pancreas to release insulin into the blood. Excess glucose is then converted to glycogen in the liver, reducing blood glucose levels. ✓

(C) At high elevation, atmospheric oxygen is more scarce. In response to signals that oxygen is low, the brain decreases an individual's rate of respiration to compensate for the difference.

(D) A transcription factor binds to the regulatory region of a gene, blocking the binding of another transcription factor required for expression.

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4 Cell Communication and Cell Cycle MCQ

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(C) At high elevation, atmospheric oxygen is more scarce. In response to signals that oxygen is low, the brain decreases an individual's rate of respiration to compensate for the difference.

(D) A transcription factor binds to the regulatory region of a gene, blocking the binding of another transcription factor required for expression.

Answer B

This option is correct. The decrease of blood glucose levels as a result of insulin production is an example of how the endocrine system is involved in negative feedback, allowing organisms to maintain their internal environments.

51. During a fight-or-flight response, epinephrine is released into the body's circulatory system and transported throughout the body. Some cells exhibit a response to the epinephrine while other cells do not.

Which of the following justifies the claim that differences in components of cell signaling pathways explain the different responses to epinephrine?

(A) Cell signaling depends on the ability to detect a signal molecule. Not all cells have receptors for epinephrine. Only cells with such receptors are capable of responding. ✓

(B) Cell signaling depends on the transduction of a received signal by the nervous system. Not all cells are close enough to a synapse to receive the signal and respond.

(C) Cell signaling depends on the signal being able to diffuse through the cell membrane. Epinephrine is incapable of diffusing through some plasma membranes because of the membrane's phospholipid composition.

(D) Cell signaling requires reception, transduction, and response. All cells can receive epinephrine, all cells respond with a pathway, but only select cells have the proper coding in their DNA to respond.

Answer A

Correct. Cell signaling starts with the reception of the signal. Not all cells are equipped to receive all signals. Only those cells with the correct receptor to a signal will be able to respond to a signal such as epinephrine.

4 Cell Communication and Cell Cycle MCQ

The epinephrine signaling pathway plays a role in regulating glucose homeostasis in muscle cells. The signaling pathway is activated by the binding of epinephrine to the beta-2 adrenergic receptor. A simplified model of the epinephrine signaling pathway is represented in Figure 1.

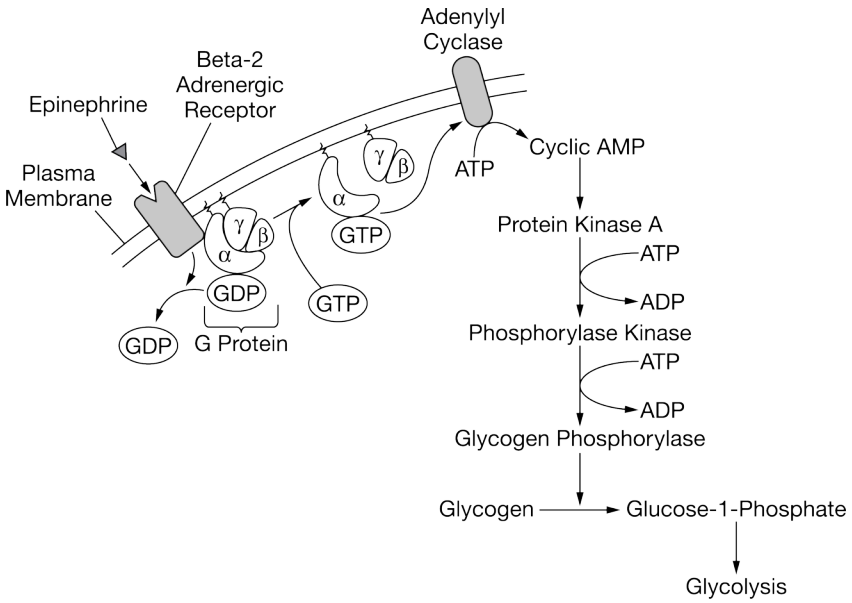


Figure 1. A simplified model of the epinephrine signaling pathway in muscle cells

52. Which of the following outcomes will most likely result from the irreversible binding of GDP to the G protein?

(A) The intracellular concentration of glycogen will increase. ✓

(B) The intracellular concentration of activated protein kinase A will increase.

(C) The intracellular concentration of cyclic AMP will increase.

(D) The intracellular concentration of glucose-1-phosphate will increase.

Answer A

Correct. Activation of the G protein requires the exchange of GDP for GTP. The irreversible binding of GDP to the G protein will block activation of the G protein, which will prevent epinephrine stimulation of glycogen breakdown. Consequently, the intracellular concentration of glycogen will increase.

53. A researcher claims that the epinephrine signaling pathway controls a catabolic process in muscle cells. Which of the following statements best helps justify the researcher's claim?

4 Cell Communication and Cell Cycle MCQ

(A) Epinephrine is a signaling molecule that binds to a transmembrane protein.
(B) The G protein in the epinephrine signaling pathway consists of three different subunits.
(C) Phosphorylase kinase catalyzes the hydrolysis of ATP.
(D) Glycogen phosphorylase catalyzes the conversion of glycogen to glucose-1-phosphate. ✓

Answer D

Correct. Glycogen phosphorylase catalyzes the breakdown of glycogen, which is a catabolic process. Consequently, the statement helps justify the researcher's claim.

54. Cyclic AMP phosphodiesterase is an enzyme that catalyzes the conversion of cyclic AMP to a different molecule. Which of the following best predicts the effect of inhibiting cyclic AMP phosphodiesterase in a muscle cell stimulated by epinephrine?

(A) The concentration of cyclic AMP will decrease because adenylyl cyclase will no longer be activated.
(B) The G protein will diffuse out of the cell because it will no longer bind to the plasma membrane.
(C) Phosphorylase kinase will remain active because protein kinase A will no longer be deactivated. ✓
(D) Glycolysis will stop because epinephrine signaling will no longer stimulate glycogen breakdown.

Answer C

Correct. Phosphorylase kinase is activated by protein kinase A, which is activated by binding to cyclic AMP. Cyclic AMP phosphodiesterase causes the deactivation of protein kinase A by lowering the intracellular levels of cyclic AMP. Inhibiting cyclic AMP phosphodiesterase will result in protein kinase A remaining active, which will result in phosphorylase kinase remaining active.

55. Which of the following statements best describes the role of adenylyl cyclase in the epinephrine signaling pathway?

(A) It converts a polymer to its monomer subunits.
(B) It moves substances across the plasma membrane.
(C) It accelerates the production of a second messenger. ✓
(D) It transfers phosphate groups from ATP to protein substrates.

4 Cell Communication and Cell Cycle MCQ

Answer C

Correct. When activated, adenylyl cyclase accelerates a reaction in which ATP is converted to cyclic AMP, which acts as a second messenger in the epinephrine signaling pathway.

56. Based on Figure 1, which of the following statements best describes the epinephrine signaling pathway?

(A) It involves the opening and closing of ion channels.

(B) In involves enzymes activating other enzymes. ✓

(C) It involves changes in the expression of target genes.

(D) It involves protons moving down a concentration gradient.

Answer B

Correct. Based on Figure 1, the epinephrine signaling pathway involves enzymes activating other enzymes. For example, the pathway includes several protein kinases, enzymes that catalyze the transfer of a phosphate group from ATP to a protein substrate. As represented in Figure 1, protein kinase A catalyzes the transfer of a phosphate group from ATP to phosphorylase kinase, which results in the activation of phosphorylase kinase. The activated phosphorylase kinase activates glycogen phosphorylase in a similar manner.

4 Cell Communication and Cell Cycle MCQ

57. Figure 1 is a proposed model of the feedback system controlling erythrocyte (red blood cell) production.

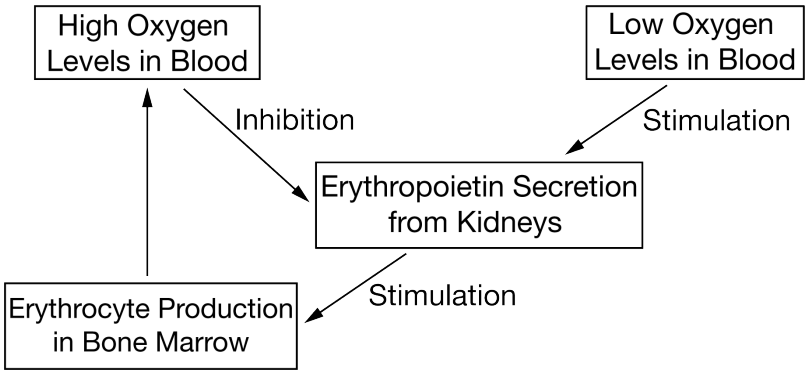


Figure 1. Model of erythrocyte production control

Air is less dense at very high elevations, so less oxygen is available than in the denser air at sea level. Based on the model in Figure 1, if a person travels from sea level to a high elevation location, which of the following correctly predicts the response to the decreased blood oxygen level?

(A) More erythropoietin will be secreted from the kidneys, decreasing production of erythrocytes.

(B) More erythropoietin will be secreted from the kidneys, increasing production of erythrocytes. ✓

(C) Less erythropoietin will be secreted from the kidneys, decreasing production of erythrocytes.

(D) Less erythropoietin will be secreted from the kidneys, increasing production of erythrocytes.

Answer B

Correct. Erythropoietin production is triggered by a negative feedback mechanism. When kidney cells detect low oxygen levels, the kidney secretes more erythropoietin, which triggers the bone marrow to increase erythrocyte production.

58. Ethylene causes fruits to ripen. In a signaling pathway, receptors activate transcription factors, which ultimately leads to ripening.

Which of the following best supports the claim that ethylene initiates the signal transduction pathway that leads to ripening of fruit?

4 Cell Communication and Cell Cycle MCQ

(A) Ethylene is a simple gaseous molecule, which makes it easily detected by receptors.

(B) Fruit will ripen in closed containers without exposure to air.

(C) Ethylene synthesis is under both positive and negative feedback regulation.

(D) Loss-of-function mutations in ethylene receptors result in changes to the ripening process. ✓

Answer D

Correct. If ethylene is not perceived by the receptors, then ripening will not occur. This best justifies the claim that ethylene initiates the signal transduction pathway.

59. Platelets are fragments of larger cells and normally circulate in the blood without adhering to blood vessel walls. When the wall of a blood vessel is damaged, collagen fibers in the wall are exposed to the interior of the blood vessel. The exposed fibers and chemicals released from the endothelial cells that line the blood vessel attract platelets, which start to form a plug and release other chemicals (Figure 1).

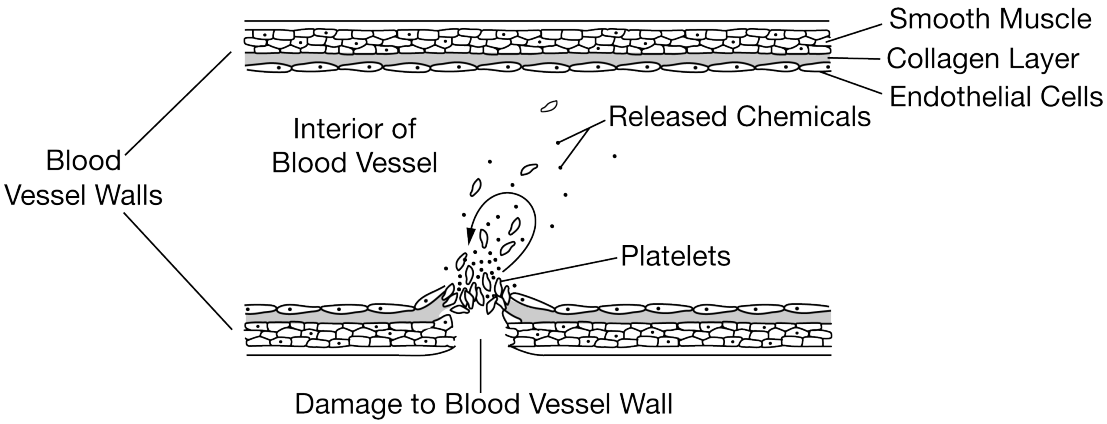


Figure 1. Formation of a platelet plug in a damaged blood vessel wall

Which of the following best explains the feedback mechanism illustrated in Figure 1 ?

(A) This is an example of positive feedback, because the few platelets that initially bind attract more platelets to the damaged area. ✓

(B) This is an example of positive feedback, because it results from the interactions among collagen, endothelial cells, and platelets.

(C) This is an example of negative feedback, because a large clump of platelets can block the blood vessel and prevent blood flow through it.

(D) This is an example of negative feedback, because the accumulation of platelets returns the open blood vessel wall to a closed state.

4 Cell Communication and Cell Cycle MCQ

60. Beta cells, a type of cell found in the pancreas, are responsible for producing, storing, and releasing the hormone (chemical signal) insulin. An increase in blood glucose levels stimulates beta cells to produce insulin, which facilitates the uptake of glucose by body cells and reduces the glucose levels in the blood. As the blood glucose levels decrease, beta cells are no longer stimulated and stop producing insulin.

Which of the following best explains the feedback mechanism that regulates blood glucose levels?

(A) Regulation of blood glucose levels demonstrates positive feedback, because as the blood glucose levels increase, insulin acts to raise blood glucose levels.

(B) Regulation of blood glucose levels demonstrates negative feedback, because as the blood glucose levels increase, insulin acts to lower blood glucose levels. ✓

(C) Regulation of blood glucose levels demonstrates positive feedback, because as the production of insulin by beta cells increases, blood glucose levels increase.

(D) Regulation of blood glucose levels demonstrates negative feedback, because as the production of insulin by beta cells decreases, blood glucose levels decrease.

Answer B

Correct. A negative feedback loop occurs when a signal is detected and causes a decrease in production. As more insulin is released, blood glucose levels decrease, which in turn decreases insulin production.

4 Cell Communication and Cell Cycle MCQ

61. Fibroblast growth factor receptors (FGFRs) are transmembrane proteins that regulate cellular processes such as cell proliferation and differentiation. The extracellular domains of FGFR proteins bind specifically to signaling molecules called fibroblast growth factors. The intracellular domains of FGFR proteins function as protein kinases, enzymes that transfer phosphate groups from ATP to protein substrates.

FGFR activation occurs when binding by fibroblast growth factors causes FGFR proteins in the plasma membrane to become closely associated with each other. The association of two FGFR proteins stimulates protein kinase activity, which triggers the activation of intracellular signaling pathways. A simplified model of FGFR activation is represented in Figure 1.

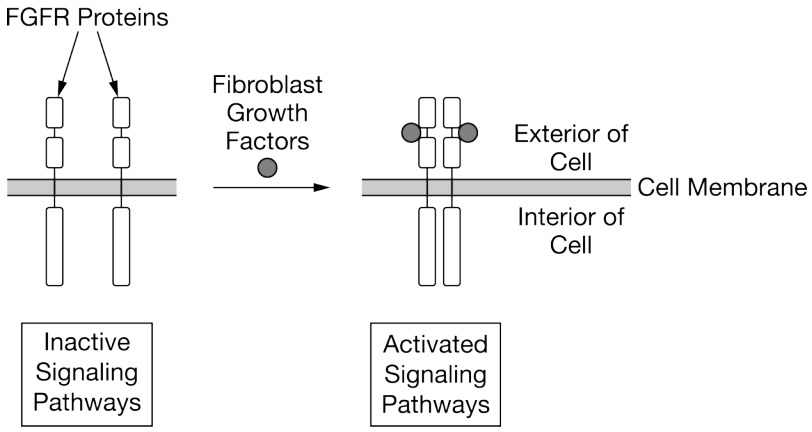


Figure 1. A simplified model of FGFR activation

Which of the following changes in the FGFR signaling pathway is most likely to result in uncontrolled cell proliferation?

(A) The irreversible association of FGFR proteins ✓

(B) The loss of the FGFR protein kinase function

(C) A decrease in the intracellular concentration of ATP

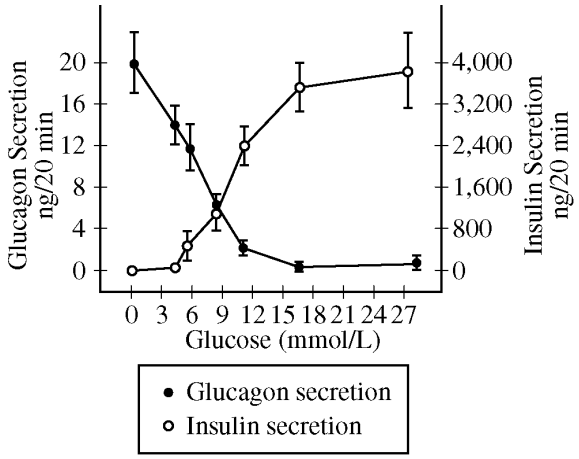
(D) A decrease in the extracellular concentrations of fibroblast growth factors

Answer A

Correct. The irreversible association of FGFR proteins will most likely result in the permanent activation of the FGFR signaling pathway. Because the FGFR signaling pathway regulates cell proliferation, the irreversible association of FGFR proteins will result in uncontrolled cell proliferation.

4 Cell Communication and Cell Cycle MCQ

62.



The graph above shows changes in glucagon and insulin secretions at different concentrations of blood glucose. Which of the following feedback mechanisms is best supported by the data?

(A) A falling glucagon level causes a rise in the insulin level, which maintains equal amounts of both hormones in the blood.

(B) A high glucagon level causes a rise in the insulin level, which maintains high levels of both hormones in the blood.

(C) A low glucose level causes the release of glucagon, which stimulates the release of more glucose from tissues, which in turn lowers the amount of glucagon being released. ✓

(D) A low glucose level causes the release of insulin, which stimulates the release of more glucose from tissues, which in turn increases the amount of insulin being released.

4 Cell Communication and Cell Cycle MCQ

63. Glucocorticoids are steroid hormones that control cellular responses through several different signaling pathways. One of the signaling pathways involves the glucocorticoid receptor, an intracellular protein that is activated by binding to a glucocorticoid molecule. A simplified model of the glucocorticoid receptor signaling pathway is represented in Figure 1.

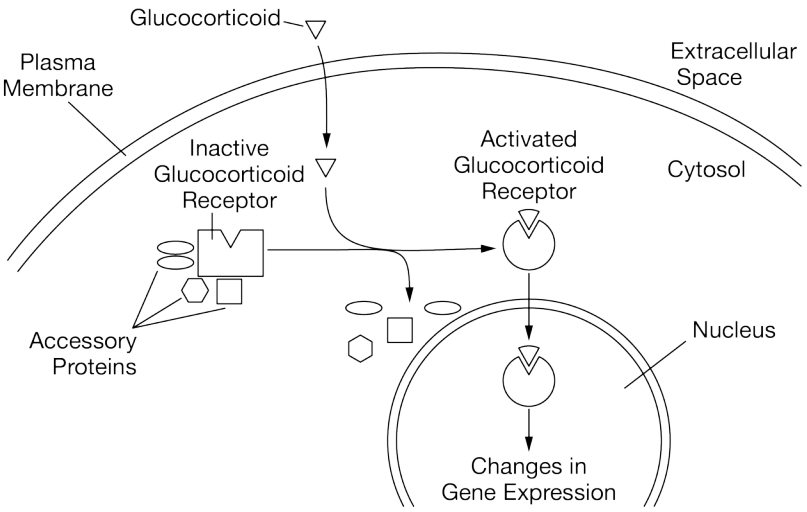


Figure 1. A simplified model of the glucocorticoid receptor signaling pathway

Which of the following statements best predicts the effect of a mutation that results in a loss of the glucocorticoid receptor's ligand binding function?

(A) The transduction of the glucocorticoid signal across the plasma membrane will be blocked.

(B) The glucocorticoid receptor will remain associated with the accessory proteins. ✓

(C) The rate of diffusion of glucocorticoid molecules into the cell will increase.

(D) The concentration of glucocorticoid receptors inside the nucleus will increase.

Answer B

Correct. In the absence of glucocorticoid signaling, the glucocorticoid receptor is maintained in its inactive state in the cytosol by its association with accessory proteins. The binding of a glucocorticoid molecule to the glucocorticoid receptor induces a conformational change in the glucocorticoid receptor that results in its dissociation from the accessory proteins and translocation to the nucleus. A mutation that results in a loss of the glucocorticoid receptor's ligand binding function will result in the glucocorticoid receptor remaining associated with the accessory proteins in a multiprotein complex in the cytosol.

4 Cell Communication and Cell Cycle MCQ

64. Glycogen synthetase kinase 3 beta is a protein kinase that has been implicated in many types of cancer. Depending on the cell type, the gene for glycogen synthetase kinase 3 beta $(GSK3\beta)$ can act either as an oncogene or as a tumor suppressor.

Which of the following best predicts how $GSK3\beta$ mutations can lead to the development of cancer?

(A) Cells with inactive $GSK3\beta$ fail to trigger apoptosis. ✓

(B) Cells with inactive $GSK3\beta$ fail to proceed past the $\mathrm{G_2/M}$ checkpoint.

(C) Cells with overactive $GSK3\beta$ are more likely to repair DNA damage.

(D) Cells with overactive $GSK3\beta$ have longer cell cycles.

Answer A

Correct. If apoptosis cannot be initiated in cells with $GSK3\beta$ mutations, those cells are more likely to proliferate, which could lead to cancer.

65. In animals, the *hox* genes encode a family of transcription factors that are important for proper development of embryonic segments and are widely conserved in organisms. The figure below shows the embryonic segments in which one such gene, *Hoxc6*, is expressed in the embryo of a mouse, a chick, and a goose. Embryonic segments are counted from the anterior end.

During the formation of vertebrae, the most anterior embryonic segment that expresses *Hoxc6* marks the end of the cervical (neck) vertebrae and the beginning of the thoracic (rib) vertebrae. All mammals have seven cervical vertebrae.

Which of the following statements is most likely to be true?

(A) The chick and the goose have the same number of thoracic vertebrae.

(B) The most anterior expression of *Hoxc6* is the eighth vertebra in mammals. ✓

(C) *Hoxc6* is expressed in the same embryonic segments in birds and mammals.

(D) *Hoxc6* is expressed in the same vertebra at the anterior end of all bird embryos.

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66. The insulin receptor is a transmembrane protein that plays a role in the regulation of glucose homeostasis. The receptor's extracellular domain binds specifically to the peptide hormone insulin. The receptor's intracellular domain interacts with cellular factors. The binding of insulin to the receptor stimulates a signal transduction pathway that results in the subcellular translocation of GLUT4, a glucose transport protein that is stored in vesicles inside the cell. A simplified model of the insulin receptor–signaling pathway is shown in Figure 1.

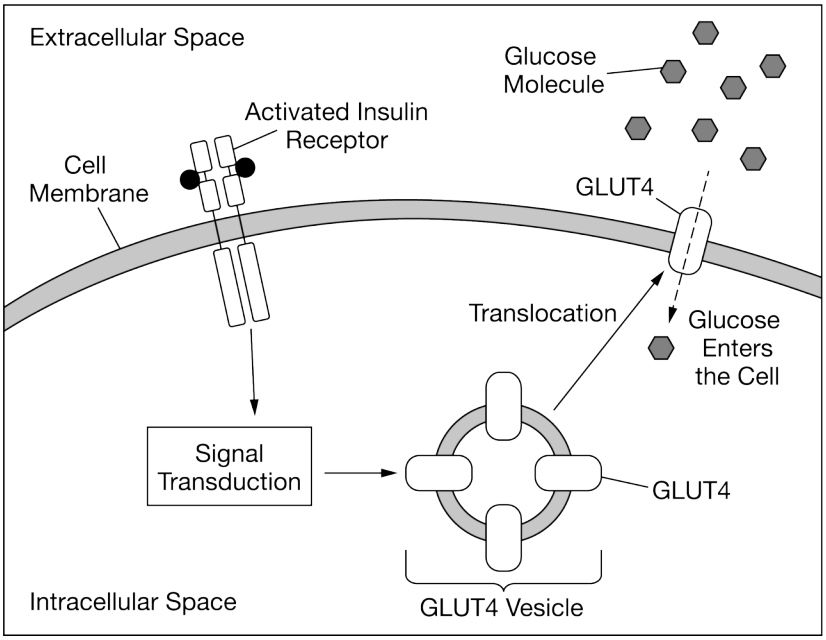

Figure 1. A simplified model of the insulin receptor–signaling pathway

Which of the following statements best predicts the effect of a loss of function of the insulin receptor's intracellular domain?

(A) The stimulation of the signal transduction pathway will increase.

(B) The storage of GLUT4 in vesicles inside the cell will increase. ✓

(C) The number of GLUT4 molecules in the plasma membrane will increase.

(D) The concentration of glucose inside the cell will increase.

Answer B

Correct. The insulin receptor's intracellular domain is required for the stimulation of the signal transduction pathway. If the signal transduction pathway is not stimulated by the insulin receptor, the GLUT4 vesicles will not be translocated to the plasma membrane. Consequently, the storage of GLUT4 in vesicles inside the cell will increase.

4 Cell Communication and Cell Cycle MCQ

67. Signal transduction may result in changes in gene expression and cell function, which may alter phenotype in an embryo. An example is the expression of the SRY gene, which triggers the male sexual development pathway in mammals. This gene is found on the Y chromosome.

Which statement provides the evidence to justify the claim that signal transduction may result in an altered phenotype?

(A) If the SRY gene is absent or nonfunctional, the embryo will exhibit male sexual development.

(B) If the SRY gene is absent or nonfunctional, the embryo will exhibit female sexual development. ✓

(C) An embryo with a male sex chromosome will always exhibit male sexual development.

(D) An embryo with two male sex chromosomes will always exhibit male sexual development.

Answer B

Correct. The SRY gene must be present and functional for normal male sexual development. It is the proper functioning of the SRY gene that is important, and this is the best statement to support the claim.

68. Two types of cholesterol transport proteins, low-density lipoproteins (LDL) and high-density lipoproteins (HDL), bind to cholesterol and carry it through the bloodstream. Familial hypercholesterolemia (FH) is characterized by high cholesterol levels in the blood, which can lead to cardiovascular disease.

FH is associated with a loss-of-function mutation of a gene that encodes LDL receptors in liver cells. Individuals who are heterozygous produce lower-than-normal amounts of the LDL receptors, and individuals who are homozygous for the mutant allele have no LDL receptor function.

Individuals with FH can be treated with drugs that result in increased production of LDL receptors in liver cells. Which of the following best explains the observation that the drugs can effectively control blood cholesterol levels in individuals who are heterozygous but are not effective in individuals homozygous for the mutant allele?

(A) The drugs repair the mutant allele by copying the wild-type allele.

(B) The drugs prevent cholesterol from entering the liver cells in individuals who are heterozygous but not in individuals who are homozygous for the mutant allele.

(C) Cholesterol molecules primarily bind to HDL receptors in individuals with FH.

(D) There must be at least one copy of the wild-type LDL receptor allele to produce functional LDL receptors. ✓

69. The mechanism of action of many common medications involves interfering with the normal pathways that cells use to respond to hormone signals. Which of the following best describes a drug interaction that directly interferes with a signal transduction pathway?

4 Cell Communication and Cell Cycle MCQ

(A) A medication causes the cell to absorb more of a particular mineral, eventually poisoning the cell.

(B) A medication enters the target cell and inhibits an enzyme that normally synthesizes a second messenger. ✓

(C) A medication enters the target cell's nucleus and acts as a mutagen.

(D) A medication interrupts the transcription of ribosomal RNA genes.

Answer B

This option is correct. The student is asked to justify the claim that changes in signal transduction pathways can alter cellular response. Second messengers are involved in signal transduction pathways, and if the second messenger is not present, the pathway is blocked.

70. The relative amounts of DNA present in the nucleus of a cell at four different stages of the life cycle are shown in Figure 1.

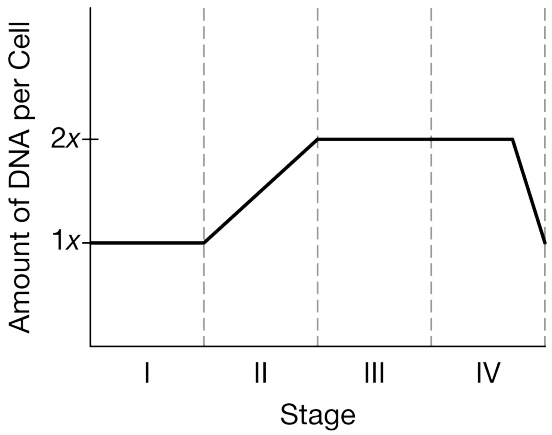

Figure 1. Relative amounts of DNA present in the nucleus of a cell

Based on Figure 1, which of the following statements correctly links a stage of the cell cycle with the event occurring at that stage?

(A) Stage I represents the G_2 phase of the cell cycle.

(B) Synthesis of sufficient DNA for two daughter cells occurs in stage II. ✓

(C) Stage III includes mitosis.

(D) The replication of genetic material occurs in stage IV.

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Answer B

Correct. Stage II represents the S (synthesis) phase of interphase. The genetic information (DNA) is doubling. If this did not occur, the daughter cells formed as a result of the cell cycle would have too little genetic information (chromosomes). This process is visible on the graph, as the copies of genetic information increase from one to two copies.

71. Which of the following best describes the role of mitosis in the cell cycle?

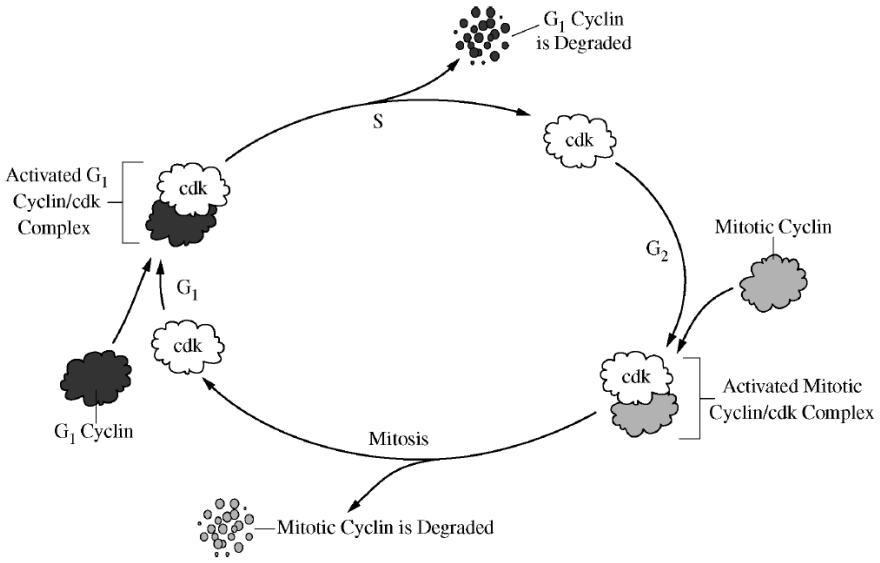
(A) Distributing replicated chromosomes to daughter nuclei ✓

(B) Dividing the cytoplasm to form four gametes

(C) Producing organelles and replicating chromosomes

(D) Exchanging genetic material between homologous chromosomes

72.



Based on the model of eukaryotic cell cycle regulation shown in the figure, which of the following best describes the effect of a drug that blocks the production of the mitotic cyclin?

(A) The cell cycle would proceed uncontrollably, and the cell would become cancerous.

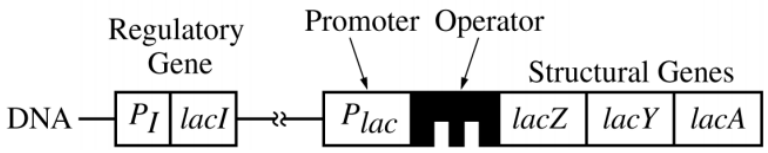
(B) The G1 cyclin would functionally replace mitotic cyclin, and the cell would continue dividing normally.

(C) DNA synthesis would be prevented, and the cell would stop dividing.

(D) The cell would be prevented from entering mitosis, and the cell would stop dividing. ✓

4 Cell Communication and Cell Cycle MCQ

LAC OPERON STRUCTURE



The functions of the loci of the *lac* operon shown in the diagram are described in the table below.

Locus	Function
P_I	Attachment site for RNA polymerase
lacI	Encodes a repressor protein that prevents transcription of the structural genes of the *lac* operon
P_{lac}	Attachment site for RNA polymerase
Operator	Binding site for the repressor protein
lacZ	Encodes beta-galactosidase, the enzyme that digests lactose to glucose and galactose
lacY	Encodes lactose permease, the channel through which lactose moves into the cell
lacA	Encodes galactoside transacetylase

The diagram above represents a segment of the *E. coli* chromosome that contains the *lacI* gene and part of the *lac* operon, a coordinately regulated set of genes that are required for the metabolism of lactose. The presence of lactose, which causes the repressor to be released from the operator, results in increased transcription of the *lac* operon.

73. Which of the following describes the most likely consequence of a negative feedback pathway involving the *lac* operon?

(A) The breakdown products of lactose block cell division.

(B) Diffusion of lactose across the plasma membrane would stop in the presence of lactose.

(C) Expression of the *lac* operon diminishes as lactose is depleted. ✓

(D) Binding of DNA by the repressor would increase in the presence of lactose.

4 Cell Communication and Cell Cycle MCQ

74. Figure 1 shows a model of a signal transduction cascade, initiated by the binding of a ligand to the transmembrane receptor protein A.

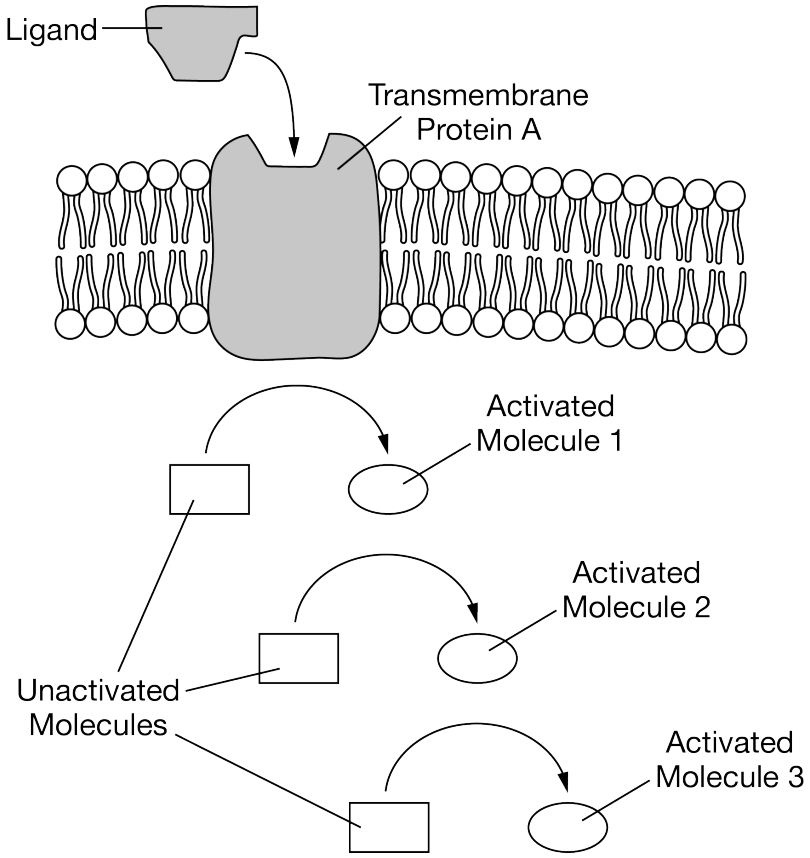


Figure 1. Model of signal transduction cascade incorporating protein A

A DNA mutation changes the shape of the extracellular domain of transmembrane receptor protein A produced by the cell. Which of the following predictions is the most likely consequence of the mutation?

(A) Production of activated molecule 1 will stop, but production of activated molecules 2 and 3 will continue.

(B) The molecule that normally binds to protein A will no longer attach, deactivating the cellular response. ✓

(C) The molecule that normally binds to protein A will not enter the cell, thus no cellular response will occur.

(D) Since protein A is embedded in the membrane, the mutation will be silent and not affect the cellular response.

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Answer B

Correct. In this signal transduction model, the molecule does not enter the cell but binds to the extracellular surface receptor of protein A, thus causing a cascade reaction that produces the cellular response. Ligand binding and activation depends upon the ligand, and the receptor have the appropriate complementary shapes. Since the specified mutation changes the shape of the ligand-binding site, the cascade will not be activated, and the cell will not respond to the presence of the ligand.

75. Cancer cells behave differently than normal body cells. For example, they ignore signals that tell them to stop dividing.

Which of the following conditions will most likely cause a normal body cell to become a cancer cell?

(A) The environment already contains cancer cells.

(B) The environment has an abundance of nutrients.

(C) The environment lacks signals that would otherwise tell the cell to stop dividing.

(D) The environment contains mutagens that induce mutations that affect cell-cycle regulator proteins. ✓

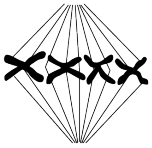
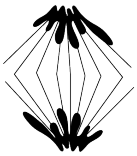
Answer D

Correct. Cancerous cells often produce abnormal forms of cell-cycle regulators, allowing them to divide even when it is not appropriate to do so.

4 Cell Communication and Cell Cycle MCQ

76. A student used microscopy to investigate the relative lengths of the different stages of mitosis. The student prepared slides of cells isolated from a growing onion root tip and viewed the slides under a dissecting microscope. The student then made diagrams of cells that were in different stages of mitosis and counted the number of cells that were in each of those stages. The student's data are presented in Table 1.

Table 1. Number of cells in each of four different stages of mitosis

Stage of Mitosis	Number of Cells
	7
	4
	25
	2

Based on the data, the percent of the mitotic cells that were in metaphase is closest to which of the following?

(A) 5%

(B) 11%

(C) 18% ✓

(D) 66%

Answer C

Correct. The number of cells that were in metaphase equals 7, and the total number of mitotic cells equals 38. The percent of mitotic cells in metaphase is calculated as follows: divide 7 by 38 and multiply

by 100. The correct answer is 18 percent.

77. The tumor suppressor protein p53 binds to DNA and activates target genes, which results in the synthesis of p21, CD95, and other proteins. The p21 protein promotes cell-cycle arrest, whereas the CD95 protein promotes apoptosis.

Which of the following will most likely result from a loss of p53 function?

(A) Rapid cell growth without cell division

(B) Immediate activation of apoptosis pathways

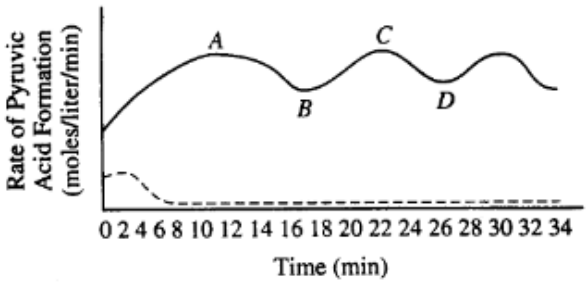
(C) Uncontrolled cell proliferation ✓

(D) Increased expression of p53 target genes

Answer C

Correct. A loss of p53 function will result in unregulated cell proliferation because the p21 protein will not be synthesized. In the absence of the p21 protein, a cell will continue to progress through the cell cycle without stopping.

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.



A tissue culture of vertebrate muscle was provided with a constant excess supply of glucose under anaerobic conditions starting at time zero and the amounts of pyruvic acid and ATP produced were measured. The solid line in the graph above represents the pyruvic acid produced in moles per liter per minute. ATP levels were also found to be highest at points *A* and *C*, lowest at *B* and *D*. A second culture was set up under the same conditions, except that substance **X** was added, and the results are indicated by the dotted line.

78. The rate of pyruvic acid formation fluctuates because

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(A) all glucose has reacted
(B) all enzymes have been used up
(C) the reaction is accelerated by positive feedback
(D) the reaction is affected by negative feedback ✓
(E) coenzymes have begun to function

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79. The binding of erythropoietin to the erythropoietin receptor (EPoR) stimulates intracellular signaling in certain cells, which causes them to differentiate to red blood cells. The EPoR is a transmembrane protein with two cytoplasmic domains, each of which is bound by a kinase protein, JAK 2. The binding of erythropoietin to the EPoR causes the cytoplasmic domains of the EPoR to undergo conformational changes that activate the bound JAK 2 kinases. The JAK 2 kinases then phosphorylate several intracellular proteins, including STAT 5, whose resulting activity causes the cells to differentiate to red blood cells.

Which of the following explains how a mutation that removes a portion of each cytoplasmic domain of EPoR most likely affects differentiation of the cells?

(A) The cells will differentiate more rapidly than they typically do because the JAK 2 kinases will always be active.
(B) The cells will mature to another cell type because the JAK 2 kinases will phosphorylate inappropriate cellular proteins.
(C) STAT 5 will not be activated because erythropoietin cannot bind to the EPoR.
(D) STAT 5 will not be activated because it will not be phosphorylated by the JAK 2 kinases. ✓

Answer D

Correct. If a portion of the EPoR cytoplasmic domain is missing, the JAK 2 kinases may be unable to bind to the EPoR cytoplasmic domains. Even if the kinases can bind, the EPoR cytoplasmic domains are unlikely to undergo the normal conformational changes that are required to activate the JAK 2 kinases. In turn, STAT 5 will not be phosphorylated and activated.

80. A student claims that the Y chromosome contains the sex-determining region gene, known as the *SRY* gene, which causes male fetuses to develop testes.

Which of the following provides correct information about cell signaling that supports the claim?

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(A) The SRY gene produces a protein that binds to specific regions of DNA in certain tissues, which affects the development of these tissues. ✓

(B) The SRY gene produces a protein that deletes portions of the X chromosome in males so that male characteristics can develop.

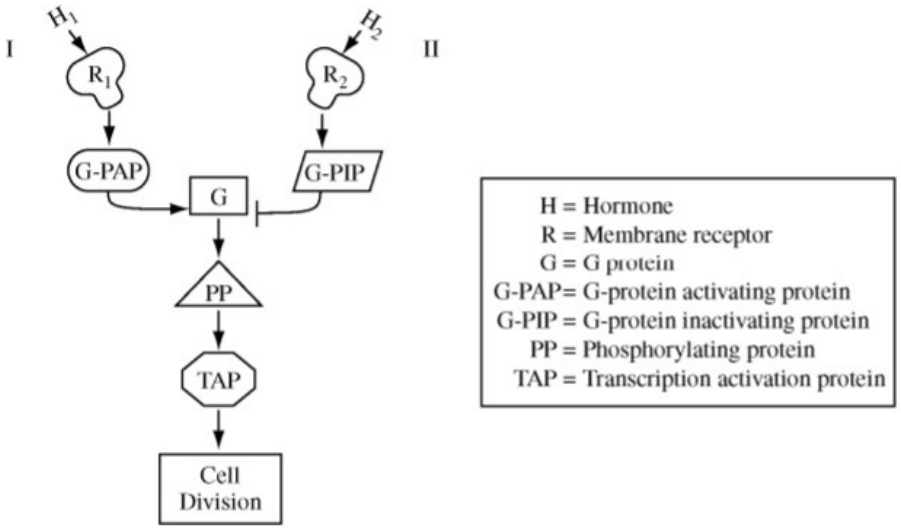
(C) The SRY gene produces an RNA segment that is exported from specific cells and targets the developing gonads.

(D) The SRY gene is found only in tissues of the developing gonads.

Answer A

Correct. SRY activation results in a signal transduction pathway that produces the male phenotype. SRY activation produces transcription factors that bind to specific regions of DNA and results in male sexual development.

81.



The diagram above represents a model of signal transduction pathways (I and II) in a cell that is targeted by two different hormones (H_1 and H_2). The components of the signal transduction pathways are identified in the figure legend. Each cellular molecule in both pathways can exist in an inactive or active form. When the components in pathway I are sequentially activated, the TAP molecules promote cell division. When the components in pathway II are sequentially activated, downstream signaling by the G protein is inhibited. Based on the model, which of the following mutations is most likely to result in a cell that will generate a cancerous tumor?

(A) A mutation in the gene encoding PP that results in a nonfunctional protein

(B) A mutation in the gene encoding G-PIP that results in a nonfunctional protein ✓

(C) A mutation in the gene encoding R_1 so that it is inactive even in the presence of H_1

(D) A mutation in the gene encoding R_2 so that it is active even in the absence of H_2

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82. The vertebrate forelimb initially develops in the embryo as a solid mass of tissue. As development progresses, the solid mass near the end of the forelimb is remodeled into individual digits. Which of the following best explains the role of apoptosis in remodeling of the forelimb?

(A) Apoptosis replaces old cells with new ones that are less likely to contain mutations.

(B) Apoptosis involves the regulated activation of proteins in specific cells of the developing forelimb that leads to the death of those cells. ✓

(C) Apoptosis involves the destruction of extra cells in the developing forelimb, which provides nutrients for phagocytic cells.

(D) Apoptosis in the developing forelimb triggers the differentiation of cells whose fate was not already determined.

83. The coagulation cascade controls blood clot formation in response to blood vessel injury. Thrombin is an enzyme that plays a key role in regulating the coagulation cascade. A simplified model of thrombin's role in regulating the coagulation cascade is represented in Figure 1.

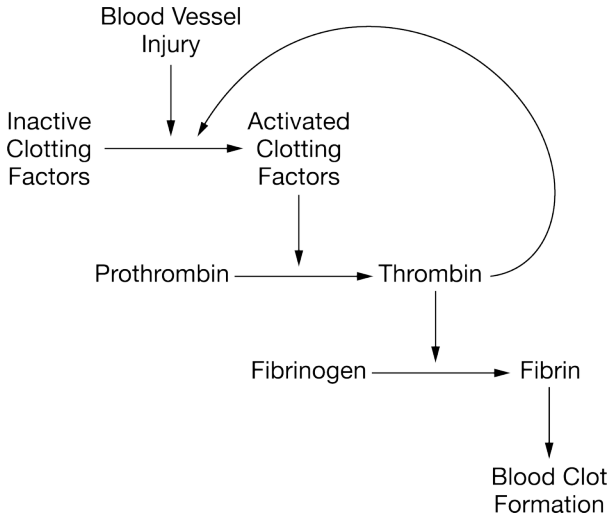

Figure 1. A simplified model of thrombin's role in regulating the coagulation cascade

Argatroban is a competitive inhibitor of thrombin. Which of the following effects on the coagulation cascade is most likely to result from inhibiting thrombin activity with argatroban?

(A) The activation of clotting factors will be blocked.

(B) The rate of fibrin formation will decrease. ✓

(C) Thrombin will be converted to prothrombin.

(D) The rate of blood clot formation will increase.

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Answer B

Correct. Thrombin is involved in a positive feedback loop. When activated, thrombin activates clotting factors that convert prothrombin to more thrombin. Activated thrombin also catalyzes the conversion of fibrinogen to fibrin, which is an important step in blood clot formation. Although activated clotting factors will catalyze the conversion of prothrombin to thrombin, inhibiting thrombin activity with argatroban will interfere with the positive feedback loop that generates more active thrombin. A decrease in the rate of thrombin formation will result in a decrease in the rate of fibrin formation.

84.

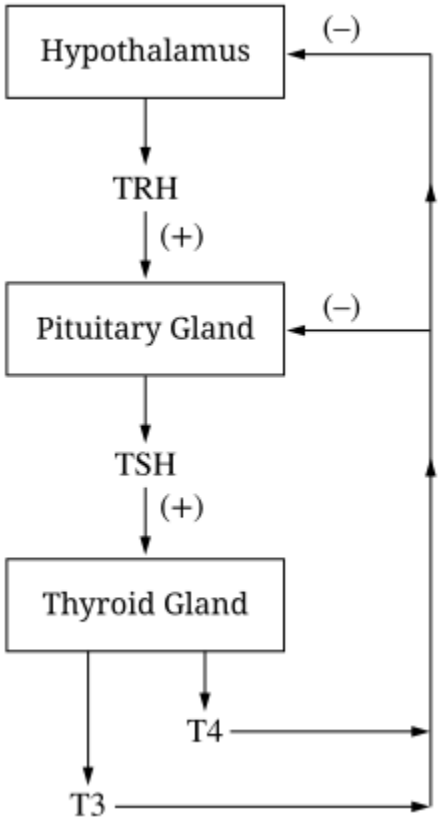


Figure 1. Regulation of hormones secreted by the thyroid gland. The + indicates stimulation and the − indicates inhibition.

Based on Figure 1, which of the following most likely occurs in response to an increase in the secretion of thyroid hormones T3 and T4 ?

4 Cell Communication and Cell Cycle MCQ

(A) The hypothalamus secretes more TRH to maintain the increase in thyroid hormone secretion.
(B) The pituitary gland decreases secretion of TSH, which results in an increase in TRH secretion.
(C) The thyroid hormones directly inhibit the hypothalamus, which results in a decrease in TRH secretion. ✓
(D) The thyroid gland increases secretion of T4 in order to cause a decrease in T3 secretion.

Answer C

Correct. Based on the figure, both thyroid hormones have an inhibitory effect on the hypothalamus; thus it would be predicted that less TRH would be secreted as T3 and T4 secretions increase.

85. Thyroxin is a hormone that increases metabolic activities within various tissue targets. Low levels of circulating thyroxin trigger the secretion of thyroid-stimulating hormone (TSH) from the anterior pituitary. TSH secretion then stimulates thyroxin production and release by the thyroid gland. The increased level of circulating thyroxin inhibits further secretion of TSH from the anterior pituitary.

Based on the information provided, which of the following can most likely be concluded about the TSH-thyroxin loop?

(A) A person taking thyroxin to supplement low thyroxin secretion will produce more TSH.
(B) Increased thyroxin production would cause elevated ribosomal activity in the anterior pituitary.
(C) The structure of the loop would lead to elevated thyroid and tissue activity due to positive feedback.
(D) The feedback mechanism would maintain relatively constant levels of thyroxin throughout tissue targets. ✓

4 Cell Communication and Cell Cycle MCQ

86. Researchers determined the average amount of time that a particular type of eukaryotic cell spends in each phase of the cell cycle. The data collected by the researchers are represented in Figure 1.

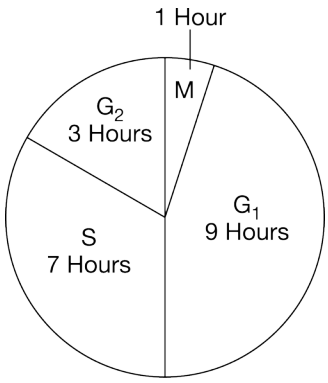


Figure 1. The average amount of time spent by a particular type of eukaryotic cell in each phase of the cell cycle

Based on Figure 1, what percent of the time required to complete a full cycle do the cells typically spend in interphase?

(A) 5%

(B) 35%

(C) 50%

(D) 95% ✓

Answer D

Correct. Interphase is composed of the G_1, S, and G_2 phases. Based on Figure 1, the cells typically spend 19 hours in interphase and require 20 hours total to complete one full cycle. Therefore, $\frac{19}{20} \times 100 = 95\%$.

87. Which of the following is true of mitosis?

(A) It is also known as cytokinesis.

(B) It maintains the same chromosome number in the daughter cells as in the parent cell. ✓

(C) It is the last phase of interphase.

(D) It regulates the transfer of genetic information from one daughter cell to another.

(E) It moves homologous chromosomes to opposite poles.

4 Cell Communication and Cell Cycle MCQ

88. Trypsinogen is split by the enzyme enterokinase to form an activated molecule of the protease trypsin. Which of the following would confirm that the activation of trypsin is an example of how a positive feedback mechanism can amplify a biological process?

(A) The activated trypsin enzyme can use enterokinase as a substrate.

(B) The trypsin produced by the reaction is capable of splitting and activating additional trypsinogen molecules. ✓

(C) If levels of trypsin were to get too high, the trypsin molecules would inhibit the enzyme enterokinase.

(D) Each mRNA molecule that codes for trypsinogen can be translated repeatedly to form many peptide molecules.

89. Type 1 diabetes results from the destruction of insulin-producing cells in the pancreas. Individuals with type 1 diabetes produce insufficient amounts of insulin, a hormone that regulates the concentration of glucose in the blood.

Which of the following best explains how treatment with a drug that stimulates the production of insulin receptors on target cells will affect the insulin signaling pathway in an individual with type 1 diabetes?

(A) The drug will have little or no effect on the signaling pathway because the receptors will not be activated in the absence of insulin. ✓

(B) The drug will have little or no effect on the signaling pathway because insulin receptors will not be allowed to enter the cells.

(C) The drug will restore the function of the signaling pathway because insulin levels will return to normal.

(D) The drug will restore the function of the signaling pathway because nonpancreatic cells will begin to produce insulin receptors.

90. Researchers studying cell cycle regulation in budding yeast have observed that a mutation in the *CDC15* gene causes cell cycle arrest in telophase when the yeast cells are incubated at an elevated temperature. Which of the following statements best predicts the effect of the cell cycle arrest on proliferating yeast cells?

(A) The yeast cells will transition out of G_0 but will fail to complete the G_1 phase.

(B) The yeast cells will initiate mitosis but will fail to complete the G_2 phase.

(C) The yeast cells will replicate their chromosomes but will fail to complete cytokinesis. ✓

(D) The yeast cells will replicate their organelles but will fail to complete the S phase.

Answer C

Correct. Because the cell cycle arrest occurs in telophase, the yeast cells will progress through the cell cycle, including replicating their chromosomes in the S phase, and will enter the M phase but will fail to complete mitosis and cytokinesis.

4 Cell Communication and Cell Cycle MCQ

91. A group of researchers cultured yeast cells in a nutrient-rich environment and a nutrient-poor environment and observed the duration of the stages of their cell cycles. The results of their study are summarized in Table 1.

Table 1. Duration (in minutes) of yeast cell cycle phases in a nutrient-rich environment and a nutrient-poor environment

Cell Cycle Stages	Nutrient-rich environment	Nutrient-poor environment
S and G_2	23	33
Mitosis	48	61
G_1	20	57
Cytokinesis	5	10

The cell cycle of yeast cells grown in the nutrient-poor environment is approximately what percent of the cell cycle of yeast cells grown in the nutrient-rich environment?

(A) 168 ✓

(B) 127

(C) 179

(D) 160

Answer A

Correct. Yeast cells in the nutrient-rich environment spend 96 minutes in the cell cycle compared to 161 minutes in the nutrient-poor environment. The percent can be determined by dividing 161 by 96, which results in 1.68 or 168 percent.

4 Cell Communication and Cell Cycle MCQ

92. A model of the typical life cycle of a cell is shown in Figure 1.

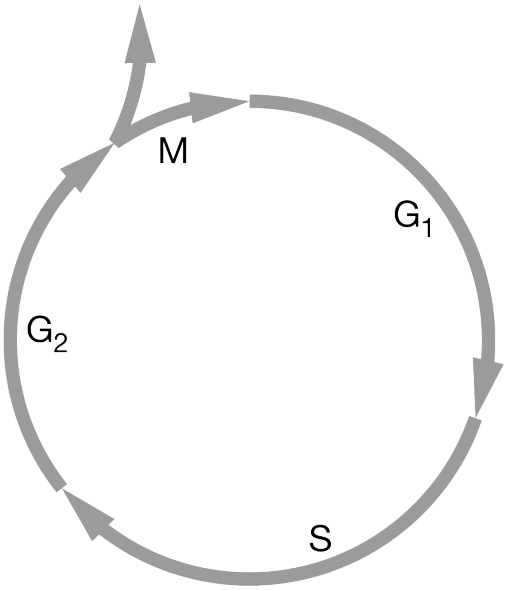


Figure 1. Typical life cycle of a eukaryotic cell

Scientists have estimated that it takes yeast cells approximately 20 hours to complete the entire cycle. Table 1 shows the amount of time in each phase of the life cycle for yeast cells.

Table 1. Amount of time spent in each stage of the cell cycle by yeast cells

Stage	M	G_1	S	G_2
Time (hours)	2	10	5	3

Based on Table 1, what percent of the life cycle of yeast cells is spent in DNA replication?

(A) 5 percent

(B) 10 percent

(C) 25 percent ✓

(D) 50 percent

Answer C

Correct. The yeast cells spend 5 hours out of a total of 20 hours in S phase, which is 25%.

4 Cell Communication and Cell Cycle MCQ

93. In mammals, an increase in the concentration of sodium in the blood triggers the release of antidiuretic hormone (ADH) from the pituitary gland. As the concentration of sodium in the blood returns to previous levels, the release of ADH from the pituitary gland is reduced. Based on the information presented, which of the following describes the most likely role of ADH in maintaining blood osmolarity?

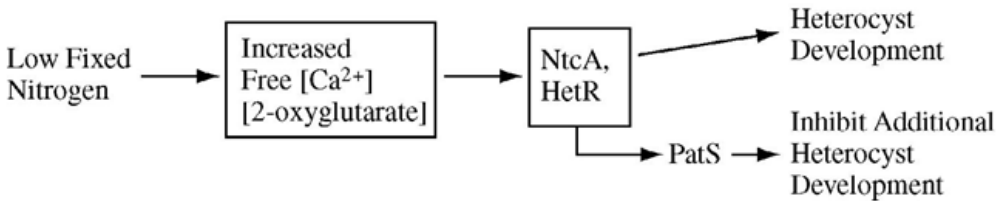
(A) ADH promotes an increase in the movement of sodium into the bloodstream.

(B) ADH promotes an increase in the movement of water into the bloodstream. ✓

(C) ADH promotes an increase in the excretion of water from the body.

(D) ADH promotes an increase in the secretion of additional ADH from the pituitary gland.

94. *Anabaena* is a simple multicellular photosynthetic cyanobacterium. In the absence of fixed nitrogen, certain newly developing cells along a filament express genes that code for nitrogen-fixing enzymes and become nonphotosynthetic heterocysts. The specialization is advantageous because some nitrogen-fixing enzymes function best in the absence of oxygen. Heterocysts do not carry out photosynthesis but instead provide adjacent cells with fixed nitrogen, in exchange receiving fixed carbon and reduced energy carriers.



As shown in the diagram above, when there is low fixed nitrogen in the environment, an increase in the concentration of free calcium ions and 2-oxyglutarate stimulates the expression of genes that produce two transcription factors (NtcA and HetR) that promote the expression of genes responsible for heterocyst development. HetR also causes production of a signal, PatS, that prevents adjacent cells from developing as heterocysts.

Based on your understanding of the ways in which signal transmission mediates cell function, which of the following predictions is most consistent with the information given above?

(A) In an environment with low fixed nitrogen, treating the *Anabaena* cells with a calcium-binding compound should prevent heterocyst differentiation. ✓

(B) A strain that overexpresses the *patS* gene should develop many more heterocysts in a low fixed nitrogen environment.

(C) In an environment with abundant fixed nitrogen, free calcium levels should be high in all cells so that no heterocysts develop.

(D) In environments with abundant fixed nitrogen, loss of the *hetR* gene should induce heterocyst development.

95. Cell communication is critical for the function of both unicellular and multicellular eukaryotes. Which of the following is likely true of cell signaling?

4 Cell Communication and Cell Cycle MCQ

(A) Cell signaling uses the highest molecular weight molecules found in living cells.

(B) Cell signaling has largely been replaced by other cell functions in higher mammals.

(C) Similar cell signaling pathways in diverse eukaryotes are evidence of conserved evolutionary processes. ✓

(D) Cell signaling functions mainly during early developmental stages.

96. An antigen can induce an immune response in a host organism. Antigens are targeted by antibodies that are produced by the organism's immune system in response to contact with the antigen. Antibodies are specific to antigens. Many different cell types are involved in immune responses.

Which of the following best explains the role of cell-to-cell communication during a response to an invasion of antigens?

(A) Chemicals that are secreted from antigen-presenting cells then activate helper T cells. ✓

(B) A macrophage cell engulfs a pathogen in the blood.

(C) Antigens attaching to receptors on memory B cells stimulate the memory B cells to become plasma cells.

(D) Antigen-presenting cells engulf antigens at the first exposure.

Answer A

Correct. Cell signaling pathways are activated by cell-to-cell communication.

97. Which of the following best explains how a group of cells can send chemical signals to cells in another part of the body?

(A) The chemical is released into the extracellular space and immediately binds to specific receptors on the surface of the target cells that released it.

(B) The chemical diffuses through the extracellular space until it reaches target cells with specific receptors for that chemical.

(C) The chemical is released into blood vessels, where it circulates until it reaches target cells that have specific receptors for that chemical. ✓

(D) The chemical is sent from the group of cells through neurons to the target cells, where it binds to specific receptors.

98. A cell signaling pathway begins when a hormone binds to its receptor. After binding, relay molecules within the cell become activated and trigger cellular responses. When drug K is added to cells, the cellular response is heightened and prolonged.

Which of the following mechanisms best explains how drug K increases the cellular response?

4 Cell Communication and Cell Cycle MCQ

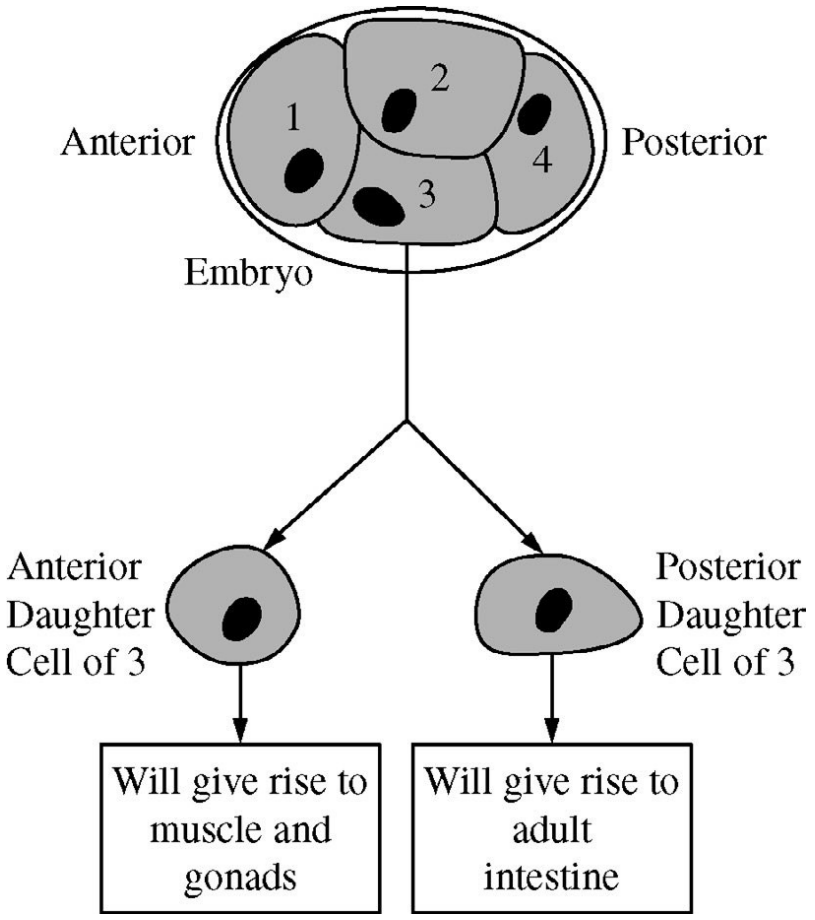
(A) Drug K inhibits the activation of the receptor protein.

(B) Drug K promotes the activation of relay molecules. ✓

(C) Drug K inhibits the deactivation of the receptor protein.

(D) Drug K promotes the circulation of relay molecules.

99.



The diagram above shows a developing worm embryo at the four-cell stage. Experiments have shown that when cell 3 divides, the anterior daughter cell gives rise to muscle and gonads and the posterior daughter cell gives rise to the intestine. However, if the cells of the embryo are separated from one another early during the four-cell stage, no intestine will form. Other experiments have shown that if cell 3 and cell 4 are recombined after the initial separation, the posterior daughter cell of cell 3 will once again give rise to normal intestine. Which of the following is the most plausible explanation for these findings?

(A) A cell surface protein on cell 4 signals cell 3 to induce formation of the worm's intestine. ✓

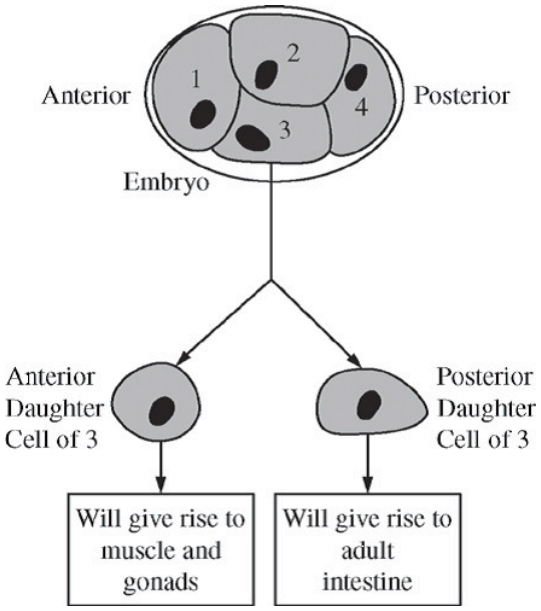
(B) The plasma membrane of cell 4 interacts with the plasma membrane of the posterior portion of cell 3, causing invaginations that become microvilli.

(C) Cell 3 passes an electrical signal to cell 4, which induces differentiation in cell 4.

(D) Cell 4 transfers genetic material to cell 3, which directs the development of intestinal cells.

4 Cell Communication and Cell Cycle MCQ

100.



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(C) Cell 3 passes an electrical signal to cell 4, which induces differentiation in cell 4.

(D) Cell 4 transfers genetic material to cell 3, which directs the development of intestinal cells.

Answer A

This option is correct. The stem of the question indicates that cell 4 and cell 3 must be touching in order for cell 3 to induce formation of the intestine. This is the mechanism by which the correct timing of this developmental event occurs.

101. A particular cell signaling pathway requires a hormone that is released into the bloodstream and travels to target cells. Scientists learned that not all cells respond to the hormone.

Which of the following best explains why some cells do not respond to the hormone?

4 Cell Communication and Cell Cycle MCQ

(A) Some cells lack hormone receptors. ✓

(B) Some cells lack regulators.

(C) Some cells lack cholesterol.

(D) Some cells lack embedded proteins.

102. Ethylene is an organic compound produced by ripening fruits. In a controlled experiment, researchers found that ethylene gas stimulated the ripening process in newly harvested fruits. Which of the following describes the most likely connection between natural ethylene production and fruit ripening?

(A) As a result of metabolic inactivity, newly harvested fruits are unable to absorb ethylene gas from the atmosphere.

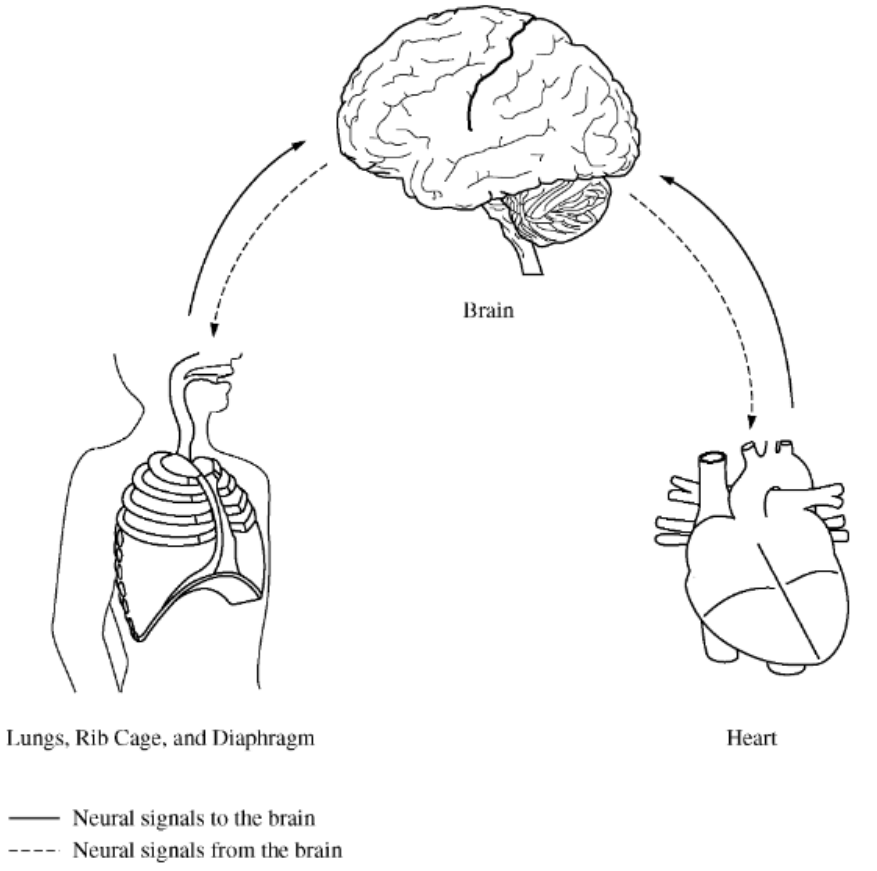
(B) Ethylene gas is a chemical signal through which ripening fruits trigger the ripening process in other fruits. ✓

(C) Because of normal phenotypic variation, only some of the fruits in a given generation are expected to produce ethylene gas.

(D) The rate of ethylene gas production by ripening fruits is an indicator of the relative age of an ecosystem.

4 Cell Communication and Cell Cycle MCQ

103. The brain coordinates the circulatory and respiratory systems of the human body. The control of breathing, for example, involves neural pathways among the structures represented in the figure above. One important stimulus in the control of breathing is an increase in blood CO_2 concentration, which is detected as a decrease in blood pH.



Which of the following best explains the physiological response to an overall increase in cellular respiration in the body?

(A) In response to depleted blood CO_2 levels, the pH sensors send signals directly to the rib muscles, resulting in an increase in the rate of CO_2 uptake by the lungs and a decrease in CO_2 utilization by the brain.

(B) In response to low blood pH, the pH sensors send a signal to the brain, which then sends a signal to the diaphragm, resulting in an increased rate of breathing to help eliminate excess blood CO_2 . ✓

(C) In response to high blood pH, the pH sensors send a signal directly to the lungs, resulting in a slower rate of breathing, and the lungs send a signal back to the heart once CO_2 availability has been restored.

(D) In response to an increased rate of breathing, the rib muscles send a signal to the brain, which then sends a signal to the heart, resulting in a decrease in heart activity and slower flow of blood through the body.

4 Cell Communication and Cell Cycle MCQ

104. Notch is a receptor protein displayed on the surface of certain cells in developing fruit fly embryos. Notch's ligand is a membrane-bound protein called Delta that is displayed on the surface of adjacent cells. When Notch is activated by its ligand, the intracellular tail of the Notch protein becomes separated from the rest of the protein. This allows the intracellular tail to move to the cell's nucleus and alter the expression of specific genes.

Which of the following statements best explains Delta's role in regulating cell communication through the Notch signaling pathway?

(A) Delta transmits a chemical signal to all the cells of a developing embryo.

(B) Delta allows the cells of a developing embryo to communicate without making direct contact.

(C) Delta restricts cell communication to short distances within a developing embryo. ✓

(D) Delta determines which cells in a developing embryo express the gene that encodes the Notch protein.

Answer C

Correct. Delta restricts cell communication to short distances by requiring direct cell-to-cell contact for activation of the Notch signaling pathway.

105. Vertebrate immune responses involve communication over short and long distances. Which of the following statements best helps explain how cell surface proteins, such as MHC proteins and T cell receptors, mediate cell communication over short distances?

(A) The proteins receive electrical signals from nerve cells.

(B) The proteins leave the cell and travel in the bloodstream to other cells.

(C) The proteins interact directly with proteins on the surfaces of other cells. ✓

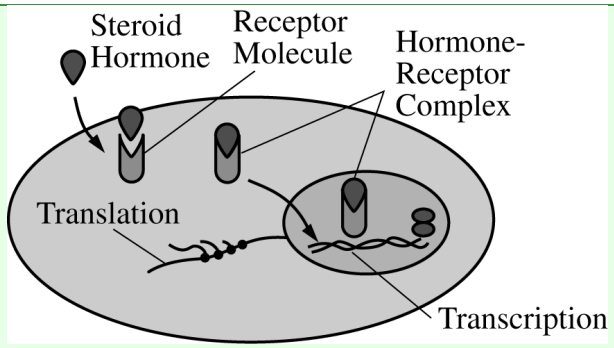
(D) The proteins bind to molecules secreted by cells located in other parts of the body.

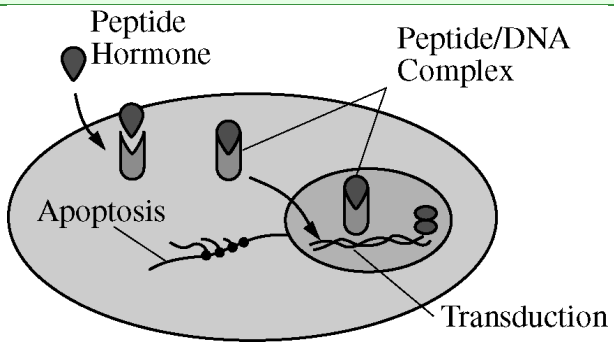
Answer C

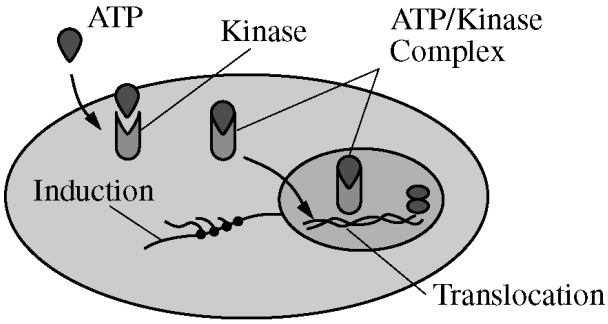
Correct. In vertebrate immune responses, cell surface proteins, including MHC proteins and T cell receptors, mediate cell communication over short distances by interacting directly with proteins on the surfaces of other cells. In many cases, the protein-protein interaction activates signal transduction pathways that control specific cellular responses. The interaction by cell surface proteins is an example of cell communication that is limited to short distances because the communicating cells must be near to each other.

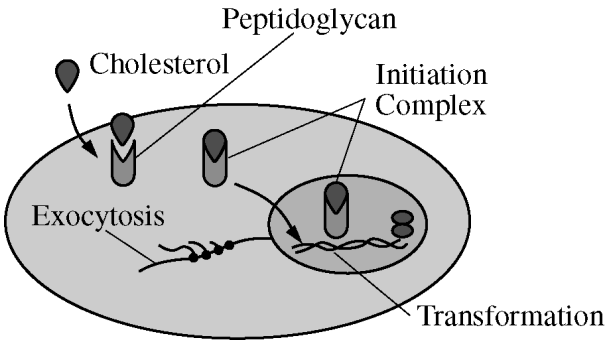
106. Which of the following diagrams best represents hormone-activated gene expression?

4 Cell Communication and Cell Cycle MCQ

(A) 

(B) 

(C) 

(D) 

107. Which of the following best explains how small molecules move between adjacent cells in a plant shoot?

(A) The molecules are actively transported by motor proteins along the cytoskeleton.

(B) The molecules pass freely through plasmodesmata, which are cytoplasmic strands connecting two cells. ✓

(C) The molecules are swept along in the extracellular fluid by cilia projecting from cell membranes.

(D) The molecules bind reversibly to receptors on the cell membranes of xylem.

4 Cell Communication and Cell Cycle MCQ

108. In flowering plants, plasmodesmata are narrow channels through cell walls that connect the cytoplasms of adjacent cells. An explanation of how plant cells communicate across cell walls will most likely refer to the diffusion through plasmodesmata of which of the following?

(A) Membrane-bound organelles

(B) Condensed, duplicated chromosomes

(C) Branched polysaccharides

(D) Small, water-soluble molecules ✓

Answer D

Correct. Because plasmodesmata are narrow channels through plant cell walls, communication between adjacent cells most likely involves the diffusion of small, water-soluble molecules through the plasmodesmata, which connect the cytoplasms of the adjacent cells.

109. Adjacent plant cells have narrow channels called plasmodesmata that pass through the cell walls of the connected cells and allow a cytoplasmic connection between the cells.

Which of the following statements best explains a primary function of plasmodesmata?

(A) They allow the movement of molecules from one cell to another, enabling communication between cells. ✓

(B) They prevent the cell membrane from pulling away from the cell wall during periods of drought.

(C) They eliminate the need to produce signaling molecules and eliminate the need for cells to have receptors for signaling molecules.

(D) They increase the surface area available for attachment of ribosomes and thus increase protein synthesis.

Answer A

Correct. Plasmodesmata are channels through which cytosol exchange can occur, moving select molecules directly between cells. This enables cells enclosed by a cell wall to communicate.

4 Cell Communication and Cell Cycle MCQ

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

The following questions refer to the data in Figures 1 through 4 below, which were collected during a study of the growth of plant seedlings.

4 Cell Communication and Cell Cycle MCQ

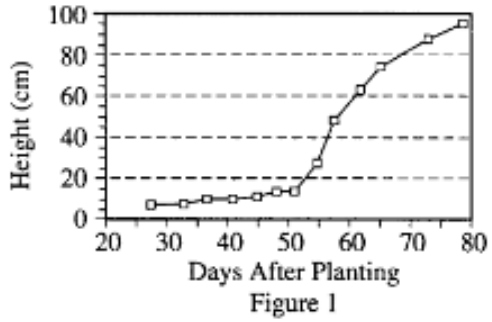


Figure 1

Figure 1 shows the average height of control seedings as a function of time after planting.

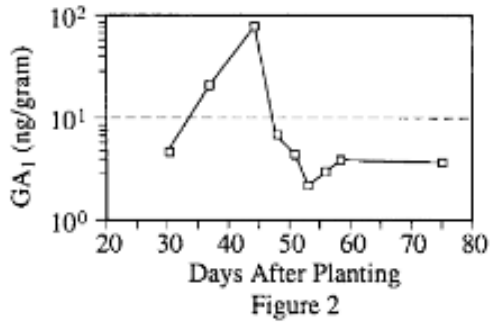


Figure 2

Figure 2 shows the concentration of one specific gibberellin compound, GA_1, that can be extracted from the plant seedling tissues (in units of nanograms of GA_1 per gram of plant tissue).

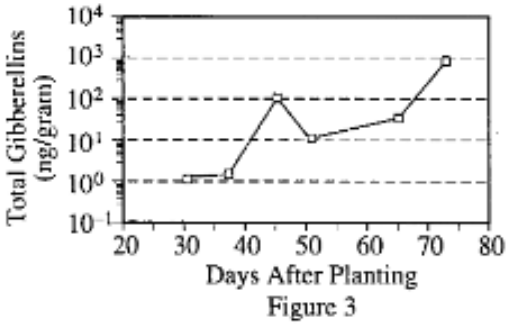


Figure 3

Figure 3 shows the concentration of all gibberellins that have physiological activity, including GA_1.

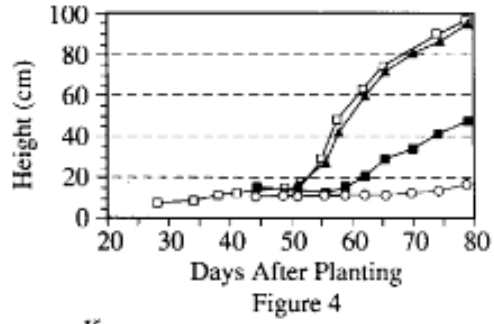


Figure 4

Figure 4 shows the patterns of growth that result when solutions containing different concentrations of an inhibitor of GA_1 synthesis are sprayed on the plant seedlings at regular intervals during the growth period.

Key

- Control seedlings not treated with inhibitor
- Seedlings treated with inhibitor at a concentration of 0.6 mg/ml
- Seedlings treated with inhibitor at a concentration of 6.0 mg/ml
- Seedlings treated with inhibitor at a concentration of 60.0 mg/ml

4 Cell Communication and Cell Cycle MCQ

110. When the concentration of GA_1 is highest, the average height of the control seedlings is approximately

(A) 10 cm ✓

(B) 30 cm

(C) 60 cm

(D) 85 cm

(E) 95 cm

111. The concentration of all gibberellins 65 days after planting is approximately

(A) 5 ng/gram

(B) 10 ng/gram

(C) 20 ng/gram ✓

(D) 120 ng/gram

(E) 150 ng/gram

112. Which of the following is a correct conclusion that can be drawn based only on the data in Figure 4?

(A) Seedling growth rates decrease between 50 and 80 days after planting at all concentrations of the inhibitor.

(B) The greater the inhibition of GA_1 synthesis, the lower the plant height after 80 days. ✓

(C) The spraying of GA_1 on seedlings results in an increase in seedling growth rates.

(D) The inhibitor kills the seedlings when it is applied in very high concentrations.

(E) The growth of the seedlings is directly proportional to the concentrations of inhibitor.

113. Which of the following is a correct conclusion that can be drawn from the data in Figures 1, 2, and 3?

(A) Most of the increase in the concentration of all gibberellins 40 to 45 days after planting is due to an increase in the concentration of GA_1. ✓

(B) The concentration of all gibberellins is three times as great at 75 days as it is at 30 days after planting.

(C) An increase in GA_1 levels occurs a day or two before the seedlings start to grow rapidly.

(D) An increase in the concentration of GA_1 inhibits the synthesis of other gibberellins.

(E) The concentration of GA_1 is a thousand times as great at 45 days as it is at 30 days after planting.

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114. G proteins are a family of receptor proteins that are involved in transmitting signals from outside a cell to inside a cell. When a signaling molecule binds to a G protein, the G protein is activated. The G protein then activates an enzyme that produces a second messenger called cAMP.

Which of the following describes a critical role of cAMP during the transduction stage of a G protein signal transduction pathway?

4 Cell Communication and Cell Cycle MCQ

(A) cAMP carries the signal to the nucleus of the cell and results in new sequences of nucleotides being added to the cell's DNA.

(B) cAMP binds the extracellular signal molecule and carries it to the intracellular target specified by the signal.

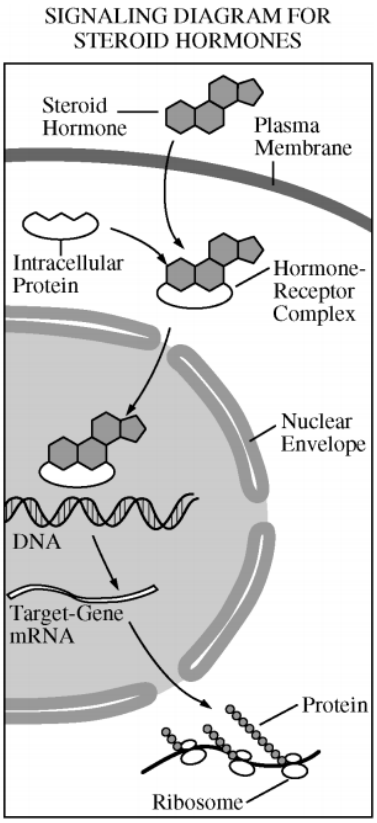
(C) cAMP modifies a specific monomer so that it can be added to an elongating structural macromolecule.

(D) cAMP results in the activation of an enzyme that amplifies the signal by acting on many substrate molecules. ✓

Answer D

Correct. Typically, cAMP immediately activates protein kinase A, which phosphorylates various other proteins. Kinases are a type of enzyme that activates proteins by transferring a phosphate group from ATP. Kinases are involved in many signaling pathways in a wide variety of organisms.

115. Steroid hormones, such as testosterone, pass through the plasma membrane and bind to an intracellular protein, as shown in the diagram below. The hormone-receptor complex then enters the nucleus, where it interacts with DNA to promote transcription of a specific gene.



Based on the information presented, which of the following will also occur in response to steroid signaling?

4 Cell Communication and Cell Cycle MCQ

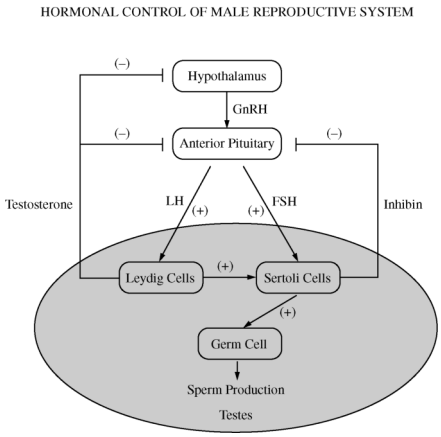
(A) Histone protein synthesis will increase because histones maintain the DNA in an optimal conformation for chromosome assembly.

(B) Ribosome production will increase because ribosomes are specific for the mRNA with which they bind during translation.

(C) DNA replication will increase as a result of the binding of the hormone-receptor complex to the DNA.

(D) Production of a specific mRNA will increase as a result of the binding of the hormone-receptor complex to the DNA. ✓

116.



Precise regulation of specific hormone levels is required for optimal sperm production in mammals, as summarized in the figure above. Anabolic-androgenic steroids (AAS) are synthetic variants of testosterone that are sometimes abused by persons who desire to enhance their athletic performance or alter their physique. Assuming that AAS function in the same way as naturally occurring testosterone, it is most likely that long-term abuse of AAS would

(A) stimulate FSH secretion

(B) stimulate testosterone production

(C) stimulate LH secretion

(D) reduce sperm production ✓

117. Insulin is a protein hormone that is secreted in response to elevated blood glucose levels. When insulin binds to its receptors on liver cells, the activated receptors stimulate phosphorylation cascades that cause the translocation of glucose transporters to the plasma membrane.

Based on the information provided, which of the following best describes the role of insulin in this liver cell signal transduction pathway?

(A) It acts as a ligand. ✓

(B) It acts as a receptor.

(C) It acts as a secondary messenger.

(D) It acts as a protein kinase.

118. Which of the following steps in a signaling pathway typically occurs first once a chemical messenger reaches a target cell?

4 Cell Communication and Cell Cycle MCQ

(A) Specific genes are activated.

(B) A second messenger molecule is produced.

(C) A ligand binds to a receptor. ✓

(D) Specific proteins are synthesized.

Answer C

Correct. The chemical messenger (ligand) first binds to a receptor, which then changes shape to initiate the signaling pathway.

119. A hydrophilic peptide hormone is produced in the anterior pituitary gland located at the base of the brain. The hormone targets specific cells in many parts of the body.

Which of the following best explains a possible mechanism that would enable the hormone to efficiently reach all of the target cells in the body?

(A) The hormone interacts with the nerves at the base of the brain and directs signals to the target cells through the nervous system.

(B) The hormone diffuses into target cells adjacent to the anterior pituitary gland, where the hormone is degraded.

(C) The hormone is released into the bloodstream where it can be transported to all cells with the correct receptors. ✓

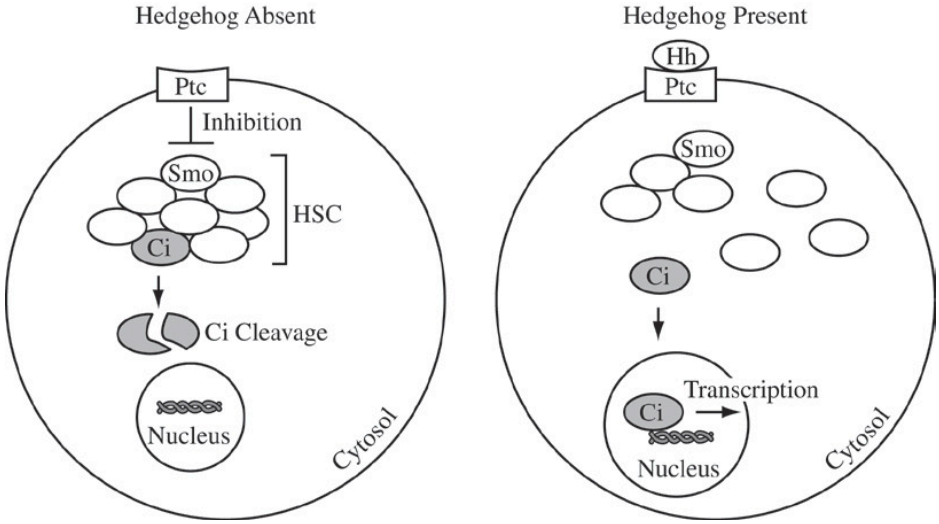
(D) The hormone moves through cytoplasmic connections between cells until it has reached all cells with the correct intracellular binding sites.

Answer C

Correct. The bloodstream is the most effective way to distribute a hormone that has to reach multiple target cells that are relatively long distances from the source of the hormone.

4 Cell Communication and Cell Cycle MCQ

120. The Hedgehog protein (Hh) plays a critical role during a certain period of embryo development, but it normally has no role in adults except for the maintenance of adult stem cells. However, the Hedgehog protein has been detected in 70 percent of pancreatic cancer cell samples. As illustrated in the figures below, the Hedgehog protein binds to an integral membrane protein receptor known as Patched (Ptc), thus initiating a pathway of gene expression. When Hedgehog is absent, Ptc inhibits another protein known as Smoothened (Smo), which, in turn, blocks the activation of a group of proteins collectively known as the Hedgehog signaling complex (HSC). The inactivation is the result of proteolytic cleavage of one component of the HSC complex, a transcription factor known as Cubitus interruptus (Ci). When Hedgehog is present, it binds to Ptc, which prevents the inhibition of Smo by Ptc. The result is that Ci remains intact and can enter the nucleus, where it binds to and activates certain genes.



One approach to treating patients with pancreatic cancer and other cancers in which the Hedgehog protein is detected is to modify the Hedgehog signaling pathway. Which of the following is the most useful approach?

(A) Treating patients with a molecule that is structurally similar to Hedgehog and that will bind to and interact with Ptc in the same fashion as Hedgehog

(B) Injecting patients with embryonic cells so that Hedgehog will bind to those cells instead of the cancer cells

(C) Treating patients with a membrane-soluble compound that can bind to Smo and block its activity ✓

(D) Injecting patients with a preparation of purified membrane-soluble Ci that will enter the nuclei of the cancer cells and induce gene transcription

121. Which of the following statements best describes how a growth factor stimulates cell division from outside a cell?

(A) The growth factor binds to other cells in the same area and holds them together to form a large, multicellular structure.

(B) The growth factor binds to receptors on the cell surface, initiating a signal transduction pathway that activates specific target genes. ✓

(C) The growth factor binds to sugar molecules in the extracellular fluid and provides them to the cell as a source of energy.

(D) The growth factor binds to phospholipids in the plasma membrane, creating a channel through which substances enter the cell.

4 Cell Communication and Cell Cycle MCQ

122. Over 100 different types of G-protein-linked receptors have been observed in eukaryotes, including scent receptors in animals, mating-signal receptors in yeasts, and light-sensing proteins in vertebrates. The proton pumps of bacteria have protein structures similar to G-protein-linked receptors. Based on this information, which of the following scientific questions will best guide research on the evolution of G-protein-linked signal transduction?

(A) Did the G-protein-linked signal transduction pathway in eukaryotes arise from a precursor in prokaryotes? ✓

(B) Are G-protein-linked receptors acquired from the environment as they are needed?

(C) Do eukaryotes require symbiotic prokaryotes in order to use G-protein-linked receptors?

(D) Do single-celled organisms have more efficient cell communication than do multicellular organisms?

Answer A

Correct. Prokaryotes have protein structures similar to eukaryotic G-protein-linked receptors. Since prokaryotes arose earlier in evolution, it is feasible to research the evolution of the G-protein-linked signal transduction pathway in eukaryotes as arising from a prokaryotic precursor.

123. Epinephrine is a protein hormone found in many animals. Epinephrine stimulates a signaling pathway that results in the breakdown of glycogen to glucose in the liver cells. Which of the following describes the initial steps in the process whereby epinephrine stimulates glycogen breakdown?

(A) Epinephrine binds to a cell-surface receptor; the activated receptor stimulates production of the second messenger, cAMP. ✓

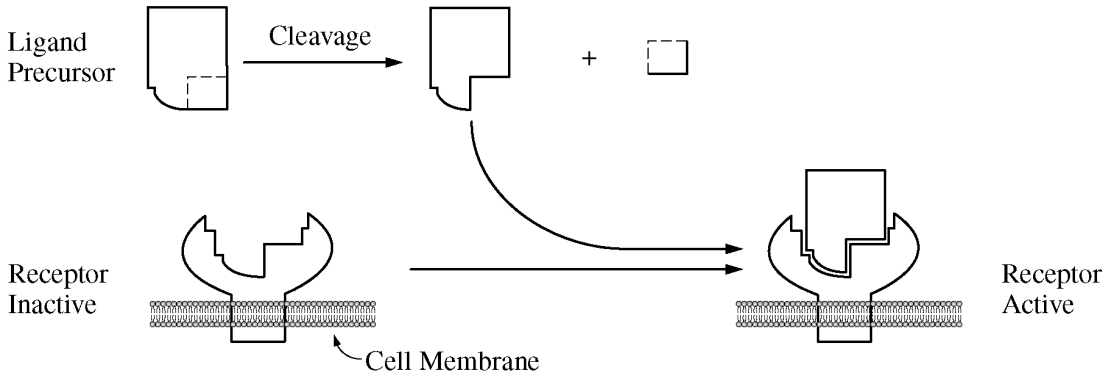
(B) Epinephrine binds to a cell-surface receptor; the activated receptor catalyzes the conversion of glycogen to glucose.

(C) Epinephrine diffuses through the plasma membrane; the hormone dimerizes in the cytosol.

(D) Epinephrine is taken into the cell by endocytosis; glycogen is converted to glucose in the endocytotic vesicle.

4 Cell Communication and Cell Cycle MCQ

124.



The figure above shows a model of a ligand precursor being cleaved to produce an active ligand that binds to a specific receptor. Which of the following is most likely to reduce the binding of the active ligand to its receptor?

(A) A change in the cytoskeletal attachment of transmembrane proteins

(B) The presence of a large amount of the precursor form of the ligand

(C) An increase in the ratio of the number of unsaturated to the number of saturated fatty acid tails of the membrane lipids

(D) A mutation in the receptor gene that causes a substitution of a charged amino acid for a nonpolar amino acid in the ligand binding site of the receptor ✓

125. Which of the following best describes a molecule that relays messages from receptors to other molecules only inside the cell?

(A) The molecule is a ligand.

(B) The molecule is a hormone.

(C) The molecule is a secondary messenger. ✓

(D) The molecule is a transport protein.

126. Metformin is a drug used to treat type 2 diabetes by decreasing glucose production in the liver. AMP-activated protein kinase (AMPK) is a major cellular regulator of glucose metabolism. Metformin activates AMPK in liver cells but cannot cross the plasma membrane. By blocking AMPK with an inhibitor, researchers found that AMPK activation is required for metformin to produce an inhibitory effect on glucose production by liver cells.

Which of the following best explains the component that metformin represents in a signal transduction pathway that regulates glucose production in the liver?

(A) It is a secondary messenger that amplifies a signal through a cascade reaction.

(B) It is a ligand that activates the signal transduction pathway of the activation of AMPK. ✓

(C) It is an allosteric regulator that binds to a crucial section of the DNA that makes the enzymes needed for glucose uptake.

(D) It is a competitive inhibitor that binds to glucose and prevents it from entering the cell.

4 Cell Communication and Cell Cycle MCQ

Answer B

Correct. Since metformin does not cross the membrane, it is likely a ligand that binds to a receptor on the surface of the plasma membrane, initiating the signal transduction pathway.

127. In a certain signal transduction pathway, the binding of an extracellular molecule to a cell-surface protein results in a rapid increase in the concentration of cyclic AMP inside the cell. The cyclic AMP binds to and activates cytosolic enzymes that then activate other enzymes in the cell.

Which of the following statements best describes the role of cyclic AMP in the signal transduction pathway?

(A) It acts as a signaling molecule that passes the signal from the cell to other cells.

(B) It acts as a receptor that carries the signal from outside the cell to inside the cell.

(C) It acts as a second messenger that helps relay and amplify the signal within the cell. ✓

(D) It acts as a channel protein that transmits the signal across the cell's nuclear membrane.

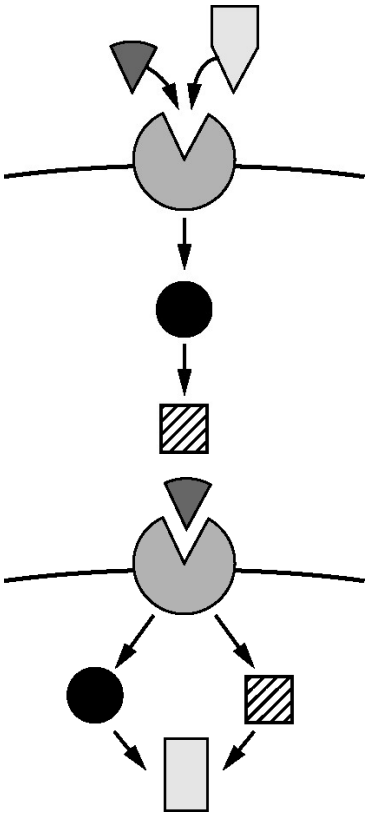
Answer C

Correct. In the signal transduction pathway, cyclic AMP acts as a second messenger by relaying the signal from the plasma membrane to cytosolic enzymes. The signal is amplified inside the cell by the production of many cyclic AMP molecules, which then activate enzyme molecules that are capable of catalyzing a specific reaction repeatedly.

128. Which of the following best represents two different signaling pathways that share a second messenger?

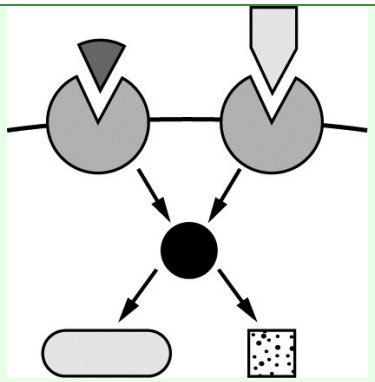
4 Cell Communication and Cell Cycle MCQ

(A)

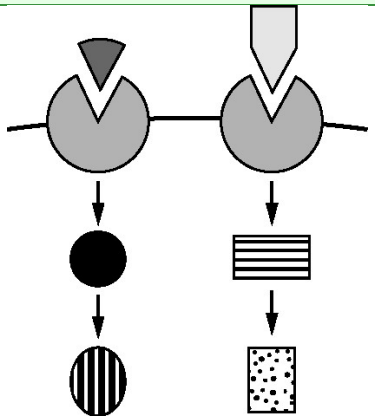


(B)

(C) ✓



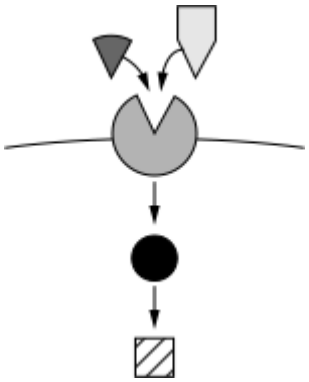
(D)



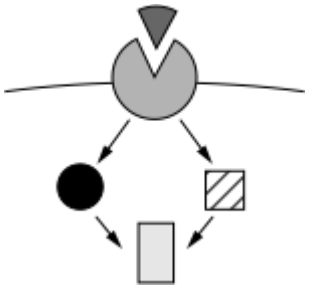
129. Which of the following best represents a signaling pathway that will lead to multiple responses from the same signaling molecule?

4 Cell Communication and Cell Cycle MCQ

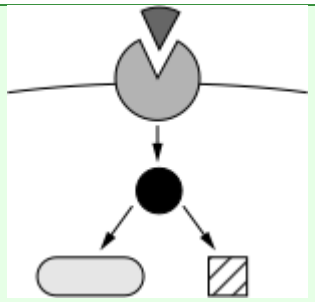
(A)



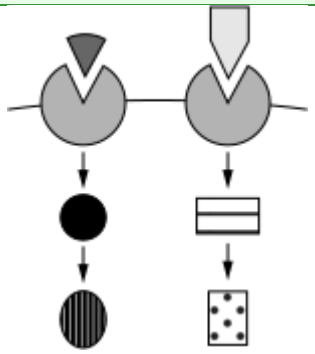
(B)



(C)



(D)



Answer C

Correct. The initial signal activates an intermediary that activates two different pathways and produces two unique responses.

5 Heredity MCQ

1. A blue-flowered African violet of unknown ancestry self-pollinated and produced 50 seeds. These seeds germinate and grow into flowering plants. Of these plants, 36 produce blue flowers and 14 produce pink flowers. What is the best explanation for the pink-flowered offspring?

(A) Blue flowers are incompletely dominant to pink flowers.

(B) Pink flower color is a trait recessive to blue flower color. ✓

(C) Pink flower color is the result of somatic mutations in the flower color gene.

(D) A previous generation of the blue-flowered parent must have included 50 percent pinkflowered plants.

2. Several members of a family are found to involuntarily sneeze when exposed to bright lights, such as sunlight. Following analysis of the condition in the family, a doctor predicts that the symptoms have an underlying genetic basis. A pedigree of the family is shown in Figure 1.

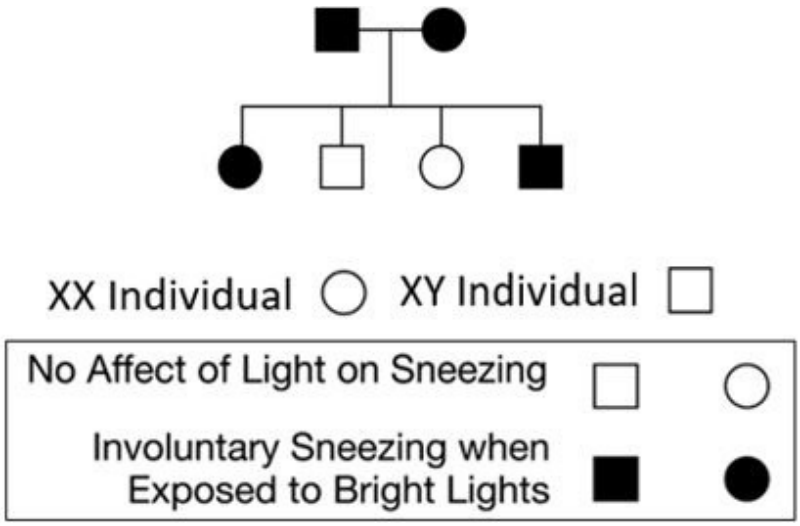


Figure 1. Pedigree showing family members with and without symptoms

For this condition, which of the following modes of inheritance is most consistent with the observations?

(A) Autosomal dominant ✓

(B) Autosomal recessive

(C) X-linked dominant

(D) X-linked recessive

Answer A

Correct. Based on the pedigree, the most likely explanation is autosomal dominant inheritance. Both

5 Heredity MCQ

parents are likely heterozygous for the condition, making it possible to have two children that are not affected. The unaffected children have a homozygous recessive genotype.

5 Heredity MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

The three-spined stickleback (*Gasterosteus aculeatus*) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mainly of individuals with armor-like plates covering most of their body surface (completely plated). Approximately 10,000 years ago, some marine sticklebacks colonized freshwater environments. After many generations in the freshwater environments, the freshwater stickleback populations lacked the armor plating (low plated) typical of marine stickleback populations.

Over the period between 1957 and 2005, one freshwater population, in Lake Washington, a lake in a coastal region of the northwestern United States, changed from having a majority of individuals of the low-plated phenotype to having more individuals of the completely-plated phenotype than of the low-plated phenotype. Figure 1 shows the distribution of plated phenotypes in Lake Washington sticklebacks at four time points between 1957 and 2005.

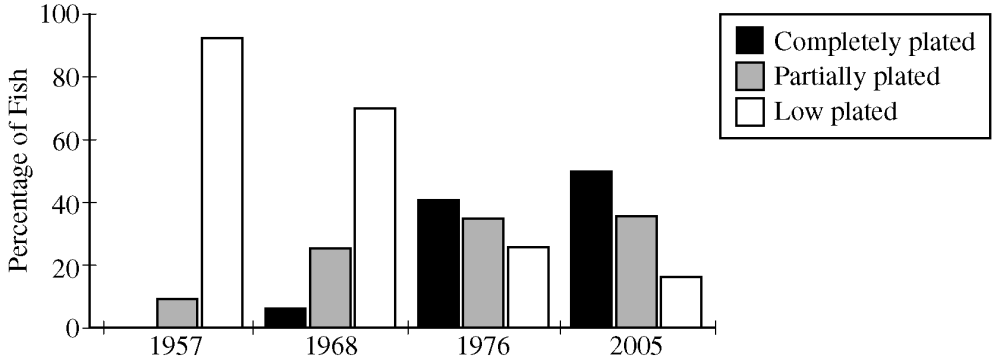


Figure 1. Armor plating phenotypes in Lake Washington stickleback population

A single gene, *ectodysplasin* (*EDA*), is thought to be responsible for the variation in the number of armor plates in sticklebacks. Figure 2 shows a phylogenetic tree constructed by comparing DNA sequences of the *EDA* gene from a number of stickleback populations with low-plated or completely plated phenotypes. Figure 3 shows a phylogenetic tree constructed by comparing the sequences of 25 genes that were randomly selected from the same populations as shown in Figure 2. In both figures, shaded populations display the completely plated phenotype.

5 Heredity MCQ

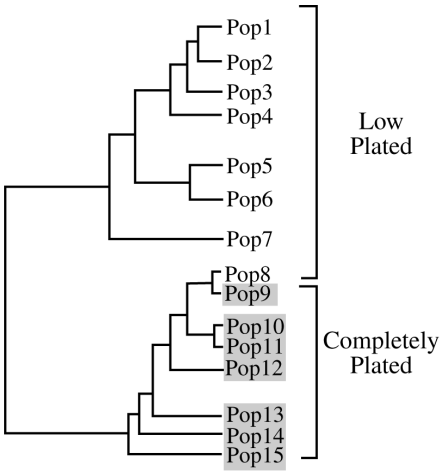


Figure 2. Phylogeny based on *EDA* gene only

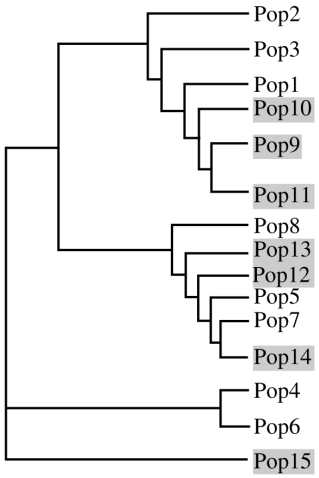


Figure 3. Phylogeny based on 25 random genes

3. A completely-plated stickleback from a marine population was mated to a low-plated stickleback from a freshwater population. The resulting F_1 hybrids all displayed a completely plated phenotype. When the F_1 hybrids were allowed to interbreed, the resulting F_2 generation included completely plated offspring and low-plated offspring in an approximate 3:1 ratio. Which of the following conclusions is best supported by the results of the breeding experiments?

(A) Phenotypic variation in the F_2 generation suggests that armor morphology is controlled by many alleles of a single gene.

(B) The completely-plated phenotype is controlled by a dominant allele of a single gene. ✓

(C) Armor loss is an acquired characteristic that is affected by one or more environmental factors.

(D) Patterns of armor plating in stickleback populations are regulated by sex-specific signals.

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4. A couple has 5 children, all sons. If the woman gives birth to a sixth child, what is the probability that the sixth child will be a son?

(A) 5/6

(B) 1/2 ✓

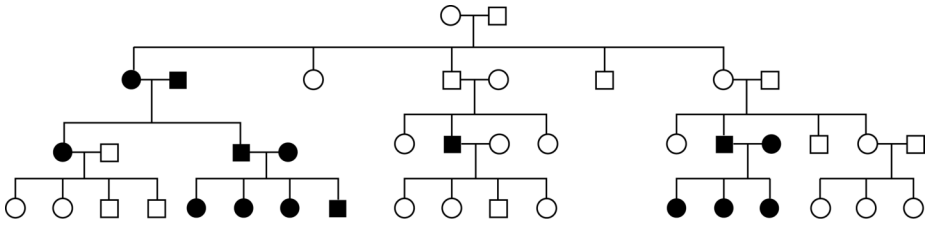
(C) 1/5

(D) 1/6

(E) 1/64

5 Heredity MCQ

5.



In the pedigree above, circles represent females, squares represent males, and shaded figures represent individuals expressing a specific trait. The expression of this trait is most likely due to which of the following?

(A) Sex-linked dominant inheritance

(B) Sex-linked recessive inheritance

(C) Autosomal dominant inheritance

(D) Autosomal recessive inheritance ✓

(E) A codominant relationship of a single pair of alleles

6. An African violet grower observes that genetically identical African violet plants growing near the walls of the greenhouse have white flowers, that plants growing farther away from the walls have pale blue flowers, and that plants growing nearest the center of the greenhouse have dark blue flowers.

Which of the following best explains the differences in flower color of the African violets in the greenhouse?

(A) Warmer temperatures result in genotypic alterations, which result in flower color differences.

(B) The plants along the walls of the greenhouse are homozygous recessive and therefore have white flowers.

(C) An enzyme responsible for flower color does not fold correctly in cooler temperatures, and the greenhouse is warmest in the center. ✓

(D) More light is available along the walls of the greenhouse, so the flowers need less pigment to absorb sunlight for photosynthesis.

Answer C

Correct. If an enzyme responsible for pigment production does not fold correctly in cooler temperatures, then the enzyme's active site will not be in the correct shape and it will not be able to catalyze the required reaction effectively. If temperatures are cooler along the walls of the green house, then the flowers of plants along the walls will have less pigment.

5 Heredity MCQ

7. A genetic counselor is consulted by a young man who is worried about developing Huntington's disease, an inherited disorder caused by a dominant allele of a single gene. The young man explains that his cousin was recently diagnosed with Huntington's disease, and the news has caused him to consider his own risk of developing the disorder. Which of the following questions will best help the genetic counselor to evaluate the risk of the young man developing Huntington's disease and transmitting it to his children?

(A) Were you and your cousin born in the same geographical area?

(B) Were your parents or grandparents ever diagnosed with Huntington's disease? ✓

(C) Were you in physical contact with a person diagnosed with Huntington's disease?

(D) Were you ever exposed to substances that are suspected of being mutagens?

Directions: Each group of questions below concerns an experimental or a laboratory situation. In each case, first study the description of the situation. Then choose the one best answer to each question following it.

In dogs, one pair of alleles determines coat color (dark and albino). Another pair of alleles determines hair length (short and long). Thus, each gamete will contain one of the coat-color alleles, *C* or *c* and one of the hair-length alleles, *B* or *b*. In repeated crosses of a specific dark, short-haired dog with an albino, long-haired dog, all the offspring were dark with short hair, as shown in cross I. However, in subsequent crosses of another dark, short-haired dog with a dark, long-haired dog, the ratios shown in cross II below were obtained.

Cross	*Parents*	*Offspring*
I.	Dark, short hair x Albino, long hair	All dark, short hair
II.	Dark, short hair x Dark, long hair	3 dark, short hair 3 dark, long hair 1 albino, short hair 1 albino, long hair

8. In cross II, the genotype of the dark, short-haired parent is

(A) *CcBb* ✓

(B) *ccbb*

(C) *CCBB*

(D) *CCbb*

(E) *ccBB*

9. Which of the following is probably the genotype of the dark, short-haired parent in cross I?

(A) *CcBb*

(B) *ccbb*

(C) *CCBB* ✓

(D) *CCbb*

(E) *ccBB*

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10. Which of the following correctly describes the relationship of the dark-coat-color allele to the albino condition?

(A) It is dominant. ✓

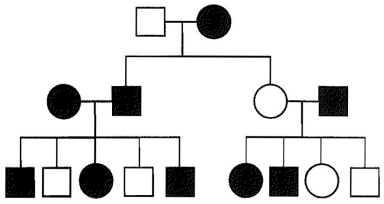
(B) It is recessive.

(C) It is codominant.

(D) It is a polygenic inheritance pattern.

(E) The alleles are linked.

11.



In the pedigree above, squares represent males and circles represent females. Individuals who express a particular trait are represented by shaded figures. Which of the following patterns of inheritance best explains the transmission of the trait?

(A) Sex-linked dominant

(B) Sex-linked recessive

(C) Autosomal recessive

(D) Autosomal dominant ✓

(E) Incompletely dominant

12. A series of crosses is performed with fruit flies (*Drosophila melanogaster*) to examine inheritance of the genes *vestigial* (*vg*) and *cinnabar* (*cn*). The recessive *vg* allele causes small, malformed wings called vestigial wings. The recessive *cn* allele causes bright-red eyes called cinnabar eyes.

In the first cross, a female with wild-type wings and eyes is mated with a male with vestigial wings and cinnabar eyes. All the F_1 individuals have wild-type wings and eyes. In the second cross, female F_1 flies are mated with males with vestigial wings and cinnabar eyes. The phenotypes of 500 F_2 individuals are shown in the table

Phenotype	Number of Individuals
Wild-type wings, wild-type eyes	226
Wild-type wings, cinnabar eyes	25
Vestigial wings, wild-type eyes	26
Vestigial wings, cinnabar eyes	223

Which of the following is the most likely explanation of the results?

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(A) The two genes are located on two different chromosomes.

(B) The two genes are sex-linked.

(C) The two genes are located on mitochondrial DNA.

(D) The two genes are linked on an autosome. ✓

13. Assume that genes *A* and *B* are not linked. If the probability of allele *A* in a gamete is 1/2 and the probability of allele *B* in a gamete is 1/2, then the probability that both *A* and *B* are in the same gamete is

(A) 1/2 x 1/2 ✓

(B) 1/2 + 1/2

(C) (1/2) / (1/2)

(D) $\sqrt{1/2}$

(E) |1/2 — 1/2|

14. A student in a biology class crossed a male *Drosophila melanogaster* having a gray body and long wings with a female *D. melanogaster* having a black body and apterous wings. The following distribution of traits was observed in the offspring.

Phenotype	Number of Offspring
Gray body, long wings	42
Black body, apterous wings	41
Gray body, apterous wings	9
Black body, long wings	8

Which of the following is supported by the data?

(A) The alleles for gray body and long wings are dominant.

(B) The alleles for gray body and long wings are recessive.

(C) Genes for the two traits are located on two different chromosomes, and independent assortment occurred.

(D) Genes for the two traits are located close together on the same chromosome, and crossing over occurred between the two gene loci. ✓

5 Heredity MCQ

Table I shows the results of breeding experiments to examine the inheritance of flower color (purple versus white) and pod shape (inflated versus constricted). For the crosses recorded in Table I, true-breeding parents were crossed to produce F_1 offspring, which were then testcrossed to homozygous recessive individuals. Table II shows the results of computer-simulated crosses to model the inheritance of leaf shape (broad versus narrow) and flower color (purple versus white).

TABLE I: RESULTS FROM CROSSES WITH PEA PLANTS

Parental Cross	Phenotypes of F_1 Offspring	Phenotypes of Testcross Offspring (numbers of individuals)			
Purple × White	Purple	Purple (461)		White (468)	
Inflated × Constricted	Inflated	Inflated (593)		Constricted (588)	
Purple, Inflated × White, Constricted	Purple, Inflated	Purple, Inflated (315)	Purple, Constricted (312)	White, Inflated (320)	White, Constricted (317)

TABLE II: RESULTS OF COMPUTER-SIMULATED CROSSES

Parental Cross	Phenotypes of F_1 Offspring	Phenotypes of Testcross Offspring (numbers of individuals)			
Broad × Narrow	Broad	Broad (4870)		Narrow (4862)	
Purple × White	Purple	Purple (4253)		White (4259)	
Broad, White × Narrow, Purple	Broad, Purple	Broad, White (672)	Broad, Purple (75)	Narrow, White (61)	Narrow, Purple (664)

15. Based on the data in Table I, which of the following best explains why there are no individuals with constricted pods in the F_1 generation?

(A) Inflated pod shape is dominant to constricted pod shape. ✓

(B) The inflated-pod offspring in the F_1 generation are homozygous.

(C) Constricted pod shape typically arises from a new mutation in the F_1 generation.

(D) The constricted-pod offspring are carriers for the inflated pod shape allele.

16. In Table I, the ratio of phenotypes in the offspring from the testcross with F_1 plants that had purple flowers and inflated pods suggests that the genes for flower color and pod shape are located

(A) close together on the same autosome

(B) on the X chromosome

(C) on different chromosomes ✓

(D) on a mitochondrial chromosome

5 Heredity MCQ

17. Which of the following provides the best justification for an assumption that might have been used in the computer simulation (Table II)?

(A) The broad allele is recessive to the narrow allele because broad leaves appear in every generation.

(B) The purple allele is dominant to the white allele because all the offspring from the cross of purple-flowered and white-flowered plants had purple flowers. ✓

(C) The narrow allele is codominant with the purple allele because the purple-flower trait and the narrow-leaf trait segregate together.

(D) The white allele is dominant to both the broad and narrow alleles because plants with either type of leaf shape can have white flowers.

18. In Table II, the F_1 offspring of the cross between broad-leaved, white-flowered plants with narrow-leaved, purple-flowered plants have a phenotype that differs from that of either parent. However, many testcross offspring have the same phenotype as one of the two plants in the parental cross, but relatively few testcross offspring have the same phenotype as the F_1 offspring. Which of the following best explains the observation?

(A) Recombination between the leaf-shape and flower-color genes resulted in chromosomes carrying a dominant allele of both genes. ✓

(B) Recombination between the broad and narrow alleles of the leaf-shape gene resulted in chromosomes carrying three different alleles at the same genetic locus.

(C) Independent assortment of homologous chromosomes resulted in the combinations of alleles present in the parental generation.

(D) The computer model cannot capture the possible assortments of gametes when multiple genes are considered.

19.

	Cross 1		Cross 2		Cross 3	
Parents	Blue Male × Blue Female		Blue Male × White Female		White Male × Blue Female	
Offspring	Males	Females	Males	Females	Males	Females
	60 blue	124 blue	0 blue	115 blue	48 blue	42 blue
	52 white	0 white	108 white	0 white	53 white	49 white

The data above represent the results of three different crosses involving the inheritance of a gene that determines whether a certain organism is blue or white. Which of the following best explains the mechanism of inheritance of the gene?

(A) The allele for white is an autosomal dominant allele because a $1:1$ phenotype ratio of blue to white among both sexes is observed in cross 3.

(B) The allele for blue is an autosomal dominant allele because an approximate $3:1$ phenotype ratio of blue to white is observed in cross 1.

(C) The allele for white is an X-linked dominant allele because no white females are produced in cross 1.

(D) The allele for blue is an X-linked dominant allele because there are no blue male offspring in cross 2. ✓

5 Heredity MCQ

20. Butterflies of the genus Colias live in the Rocky Mountains, where they experience a wide range of temperatures. Different variants of a particular glycolytic enzyme in the flight muscles are optimally active at different temperatures. Within the same population, some individual butterflies fly most effectively at 29°C , while others fly most effectively at 40°C. Still others can be equally active at both temperatures. Which of the following claims is most consistent with the observed butterfly behavior?

(A) Butterflies that express two variants of the enzyme are active over a greater range of temperature. ✓

(B) Butterflies that are active over a wide range of temperatures produce greater amounts of the enzyme.

(C) Temperature has little effect on the activity of butterflies.

(D) Butterflies that are active at warmer temperatures produce more offspring.

21. A scientist studying phenotypic variation in a species of butterfly observed that genetically identical caterpillars grown in similar cages but exposed to different colored lights developed into butterflies with differences in wing color and body size, as shown in Table 1.

Table 1. Effect of Exposing Identical Caterpillars to Specific Colors of Light

Phenotype of Adult Butterfly	Caterpillars Exposed to Red Light	Caterpillars Exposed to Blue Light
Wing color	Darker	Lighter
Body size	Smaller	Larger

Which of the following best explains the cause of the phenotypic variation observed in the butterflies?

(A) Different mutations occurred in the caterpillars that were exposed to different colors of light.

(B) The energy used to grow a larger body results in butterflies with lighter colored wings.

(C) Individual caterpillars evolved adaptations to survive in each of the light conditions they were exposed to.

(D) There was differential gene expression of wing color and body size in response to the colors of light the caterpillars were exposed to. ✓

Answer D

Correct. Phenotype is a product of both genes and environment. Since the butterflies are genetically identical, their phenotype differences must be a product of the environment interacting with the genotype.

5 Heredity MCQ

22. A gene that influences coat color in domestic cats is located on the X chromosome. A female cat that is heterozygous for the gene ($\mathrm{X}^{B}\mathrm{X}^{O}$) has a calico-colored coat. In a genetics experiment, researchers mate a calico-colored female cat ($\mathrm{X}^{B}\mathrm{X}^{O}$) with an orange-colored male cat ($\mathrm{X}^{O}\mathrm{Y}$) to produce an F_1 generation. The researchers record observations for the cats in the F_1 generation and plan to use the data to perform a chi-square goodness-of-fit test for a model of X-linked inheritance. The data for the chi-square goodness-of-fit test are presented in Table 1.

Table 1. Data for the chi-square goodness-of-fit test

Phenotype	Genotype	Observed	Expected
Calico-colored female	$\mathrm{X}^{B}\mathrm{X}^{O}$	15	10
Orange-colored female	$\mathrm{X}^{O}\mathrm{X}^{O}$	6	10
Black-colored male	$\mathrm{X}^{B}\mathrm{Y}$	11	10
Orange-colored male	$\mathrm{X}^{O}\mathrm{Y}$	8	10

The researchers calculate a chi-square value of 4.6 and choose a significance level of $p = 0.05$. Which of the following statements best completes the chi-square goodness-of-fit test?

(A) The null hypothesis can be rejected because the chi-square value is greater than the critical value.

(B) The null hypothesis can be rejected because the chi-square value is less than the critical value.

(C) The null hypothesis cannot be rejected because the chi-square value is greater than the critical value.

(D) The null hypothesis cannot be rejected because the chi-square value is less than the critical value. ✓

Answer D

Correct. There are four potential phenotypic categories, so there are three degrees of freedom. At the $p = 0.05$ significance level, the critical value is 7.82. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected.

5 Heredity MCQ

23.

TABLE 1. NUMBER OF CHICKENS OF EACH PHENOTYPE

Phenotype	Number
Black, solid	47
Black, spotted	12
Gray, solid	16
Gray, spotted	4
Total	79

In a certain type of chicken, the allele for gray feathers is recessive to the allele for black feathers, and the allele for a spotted pattern is recessive to the allele for a solid pattern. Chickens that are heterozygous for both traits were crossed and the phenotypes for the offspring were recorded (Table 1).

To determine whether the data are consistent with the expectations for independent assortment, students performed a chi-square analysis.

Which of the following is the closest to the calculated chi-square value for the experiment?

(A) -0.26

(B) 0.20

(C) 0.95 ✓

(D) 1.25

Answer C

Correct. Chi-square is calculated by first determining the expected phenotype values based on the phenotype ratio of $9:3:3:1$ for a dihybrid cross with unlinked genes. Then the observed values provided in Table 1 and the expected values are plugged into the chi-square formula. The expected number of black, solid chickens is $79 \times (9/16) = 44.44$. The expected number of black, solid chickens and of gray, solid chickens is $79 \times (3/16) = 14.81$ each. The expected number of gray, spotted chickens is $79 \times (1/16) = 4.94$. Using these values in the chi-square formula yields 0.95.

5 Heredity MCQ

24. In fruit flies of the genus *Drosophila*, the allele for vestigial wings is recessive to the allele for round wings, and the allele for brown eye color is recessive to the allele for red eye color. A scientist crossed flies that are heterozygous for both traits and determined the number of offspring with each combination of phenotypes. The scientist performed a chi-square analysis to determine if the data are consistent with the expectations for independent assortment. The chi-square calculated value for the experiment was 6.03.

Based on the chi-square calculated value, which of the following statements is most accurate?

(A) The chi-square calculated value is greater than the chi-square critical value; therefore the null hypothesis should be rejected.

(B) The chi-square calculated value is greater than the chi-square critical value; therefore the null hypothesis should not be rejected.

(C) The chi-square calculated value is less than the chi-square critical value; therefore the null hypothesis should be rejected.

(D) The chi-square calculated value is less than the chi-square critical value; therefore the null hypothesis should not be rejected. ✓

Answer D

Correct. Degrees of freedom (df) equals $n - 1$, where n is the number of different categories, or in this example, phenotypes, into which the data are divided. Degrees of freedom is important because a greater number of categories means a greater likelihood of deviation from the expected value. The chi-square critical value for three degrees of freedom ($\mathrm{df} = 3$) and $\mathrm{p} > 0.05$ is 7.81. Because the calculated value 6.03 is less than 7.81, the observed frequencies do not significantly deviate from the expectations for independent assortment, and the null hypothesis is not rejected.

5 Heredity MCQ

25. Rubber rabbitbrush plants display heritable variation in plant height and insect-induced gall formation. In a field study, researchers investigated the relationship between plant height and gall number for the rubber rabbitbrush plants in a natural population. The researchers used the data they collected to perform a chi-square test of independence. The null hypothesis for the chi-square test was that plant height and gall number are independent. The data for the chi-square test are presented in Table 1.

Table 1. Data for the chi-square test of independence

Height Category (cm)	Number of Plants with Ten or Fewer Galls	Number of Plants with More than Ten Galls
0–30	34	6
31–60	22	16
61–90	14	21
91–120	10	27

The researchers calculated a chi-square value of 29.25. If there are three degrees of freedom and the significance level is $p = 0.05$, which of the following statements best completes the chi-square test?

(A) The critical value is 0.05, and the null hypothesis cannot be rejected because the calculated chi-square value is greater than the critical value.

(B) The critical value is 0.05, and the null hypothesis can be rejected because the calculated chi-square value is greater than the critical value.

(C) The critical value is 7.82, and the null hypothesis cannot be rejected because the calculated chi-square value is greater than the critical value.

(D) The critical value is 7.82, and the null hypothesis can be rejected because the calculated chi-square value is greater than the critical value. ✓

Answer D

Correct. The critical value is 7.82. Because the calculated chi-square value is greater than the critical value, the null hypothesis can be rejected.

5 Heredity MCQ

26. Researchers performed a dihybrid cross with coffee bean plants to investigate whether the inheritance of two traits (height and stem circumference) follows Mendel's law of independent assortment. The data for the F_2 generation are presented in Table 1.

Table 1. Data for the F_2 generation

Phenotype	Observed	Expected
Dwarf with small stem circumference	191	172
Dwarf with large stem circumference	26	57
Tall with small stem circumference	39	57
Tall with large stem circumference	49	19
TOTAL	305	305

Which of the following is closest to the calculated chi-square (χ^2) value for the data presented in Table 1?

(A) 8.35

(B) 72.01 ✓

(C) 98.00

(D) 2,546.00

Answer B

Correct. 72.01 is the correct calculated answer using the formula $\chi^2 = \Sigma \frac{(o-e)^2}{e}$.

5 Heredity MCQ

27. In corn plants, purple kernel color is dominant to yellow kernel color, and smooth kernels are dominant to wrinkled kernels. Researchers carried out a genetics experiment to investigate whether the genes controlling kernel color and kernel texture segregate independently. In their experiment, the researchers crossed two corn plants that were each heterozygous for both kernel color and kernel texture. The results of the experiment are presented in Table 1.

Table 1. Results of a genetic cross between two heterozygous corn plants

Offspring Phenotype	Observed	Expected
Purple, smooth kernels	918	900
Purple, wrinkled kernels	281	300
Yellow, smooth kernels	314	300
Yellow, wrinkled kernels	87	100

Using a significance level of $p = 0.05$, which of the following statements best completes a chi-square goodness-of-fit test for a model of independent assortment?

(A) The calculated chi-square value is 0.66, and the critical value is 0.05. The null hypothesis can be rejected.

(B) The calculated chi-square value is 0.66, and the critical value is 3.84. The null hypothesis cannot be rejected.

(C) The calculated chi-square value is 3.91, and the critical value is 5.99. The null hypothesis can be rejected.

(D) The calculated chi-square value is 3.91, and the critical value is 7.82. The null hypothesis cannot be rejected. ✓

Answer D

Correct. Because the null hypothesis is based on a model of independent assortment, the phenotypic ratios are expected to be $9{:}3{:}3{:}1$. Using the equation $\chi^2 = \Sigma\frac{(o-e)^2}{e}$, the calculated chi-square value is 3.91. With three degrees of freedom (one less than the number of phenotypic categories) and a significance level of $p = 0.05$, the critical value is 7.82. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected. In other words, the data collected in the genetics experiment fit a model of independent assortment.

5 Heredity MCQ

R. C. Punnett conducted experiments on the inheritance of traits in the sweet pea, *Lathyrus odoratus*. In one experiment, he crossed two different true-breeding sweet pea plant strains, one with erect petals and long pollen, and the other with hooded petals and round pollen. All the offspring (F_1 generation) had erect petals and long pollen (Figure 1).

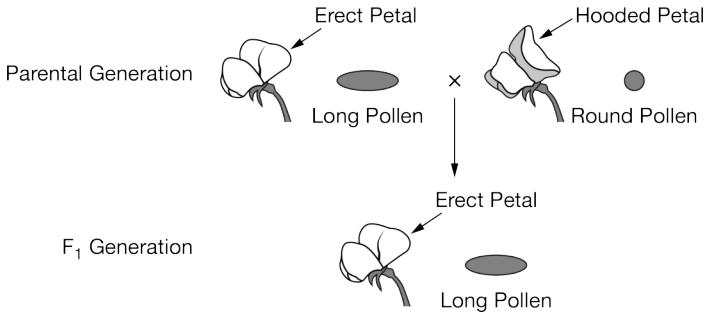

Figure 1. Sweet pea plant cross

Next, Punnett allowed the F_1 generation to self-fertilize and recorded the phenotypes of their offspring. The data are shown in Table 1.

Table 1. F_2 results of F_1 self-fertilization cross

Phenotype of F_2	Number of Offspring
Erect petals, long pollen	765
Erect petals, round pollen	62
Hooded petals, long pollen	71
Hooded petals, round pollen	220
Total	1,118

28. How many degrees of freedom should be used when looking up the critical value for a chi-square analysis of the ratios of phenotypes observed among the F_2 offspring versus the expected phenotypic ratio assuming independent assortment?

(A) 2

(B) 3 ✓

(C) 4

(D) 5

5 Heredity MCQ

Answer B

Correct. The number of degrees of freedom is equal to the number of categories minus 1. In this experiment, there are 4 different phenotype combinations, and $4 - 1 = 3$.

29. Sweet pea plants have a diploid $(2n)$ chromosome number of 14.

Which of the following best explains how the sweet pea plants in the parental generation produce F_1 offspring with 14 chromosomes?

(A) Meiosis I and II lead to the formation of cells with 14 chromosomes. When two cells combine during fertilization, extra chromosomes are randomly broken down, leading to offspring with 14 chromosomes.

(B) Meiosis I and II lead to the formation of cells with 14 chromosomes. When two cells combine during fertilization, extra chromosomes with recessive traits are broken down, leading to offspring with 14 chromosomes.

(C) Meiosis I and II lead to the formation of cells with 7 chromosomes. During meiosis I, homologous chromosomes separate. During meiosis II, sister chromatids separate. Two cells combine during fertilization to produce offspring with 14 chromosomes. ✓

(D) Meiosis I and II lead to the formation of cells with 7 chromosomes. During meiosis I, sister chromatids separate. During meiosis II, homologous chromosomes separate. Two cells combine during fertilization to produce offspring with 14 chromosomes.

Answer C

Correct. Meiosis produces haploid cells with half the number of chromosomes of the parent. When cells combine during fertilization, the diploid chromosome number is restored. Before meiosis, chromosomes replicate, forming duplicated chromosomes, each made of two sister chromatids. During meiosis I, homologous duplicated chromosomes line up in pairs, forming a set of 4 chromatids. The homologous chromosomes separate, forming cells with half the normal number of chromosomes, but the chromosomes are still duplicated. In meiosis II, the sister chromatids separate, forming cells with half the normal number of unduplicated chromosomes.

30. Which of the following questions would be most useful to researchers trying to determine the role of meiosis in the F_2 phenotypic frequencies?

(A) What is the molecular mechanism underlying the dominance of erect petals and long pollen?

(B) Which phenotypes give pea plants the highest level of fitness: erect or hooded petals and long or round pollen?

(C) How do the phases of meiosis differ between sweet pea plants and other organisms?

(D) What is the recombination frequency between the genes for petal shape and pollen shape? ✓

5 Heredity MCQ

Answer D

Correct. The F_2 generation in this experiment showed a greater-than-expected frequency of phenotype combinations from the parental generation. The erect petals, round pollen, and hooded petals, long pollen phenotype combinations are the result of crossing over of homologous chromosomes during meiosis I. The recombination frequency in this experiment was $\frac{(62+71)}{1118} = 12\ \%$. The closer genes are to each other, the lower the chance of them being separated from each other by crossing over and the lower the recombination frequency.

A male fruit fly (*Drosophila melanogaster*) with red eyes and long wings was mated with a female with purple eyes and vestigial wings. All of the offspring in the F_1 generation had red eyes and long wings. These F_1 flies were test crossed with purple-eyed, vestigial-winged flies. Their offspring, the F_2 generation, appeared as indicated below.

	F_2 Generation
125	red eyes, long wings
124	purple eyes, vestigial wings
18	purple eyes, long wings
16	red eyes, vestigial wings
283	Total

31. If in the F_1 and F_2 generations the same characteristics appeared in both males and females, it would be safe to assume that these traits for eye color and wing length

(A) are sex-linked

(B) vary in dominance according to sex

(C) are sex-influenced characteristics

(D) are autosomal characteristics ✓

(E) follow the Mendelian rule of independent assortment

32. In the F_2 generation, the results are best explained by the fact that

(A) the test cross with the F_1 flies resulted in sterile offspring

(B) these genes for eye color and wing shape do not pass through the F_1 generation

(C) these genes for eye color and wing shape are found on the same chromosome ✓

(D) crossing-over decreases variability

(E) the genes are sex-linked

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5 Heredity MCQ

Dystrophin is a protein that is expressed in certain muscle cells. In combination with other cellular proteins, dystrophin strengthens protein fibers in muscle cells to allow muscles to contract without injury.

Nucleotide deletions in the gene that encodes dystrophin are associated with the genetic disorder Duchenne Muscular Dystrophy (DMD). Individuals with DMD do not produce functional dystrophin and, as a result, the protein fibers, and then entire muscle cells, become damaged.

The history of DMD for three generations of a family is shown in Figure 1.

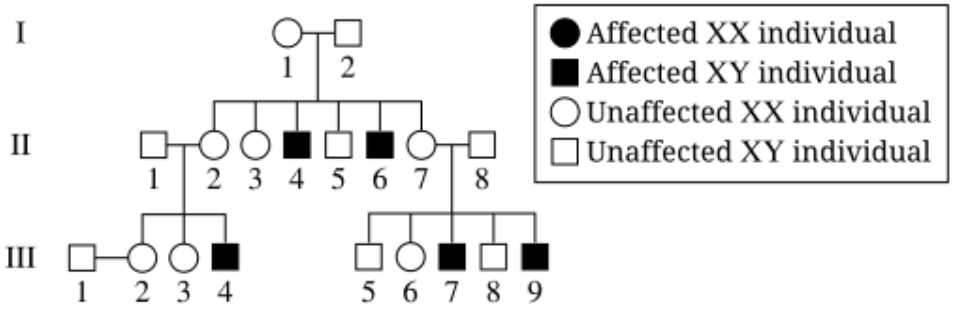


Figure 1. Pedigree showing instances of DMD in a family

Individuals III–1 and III–2 plan to have children and wish to first determine whether individual III–2 is a carrier of DMD. Individual III–2 undergoes genetic testing to determine whether individual III-2 carries a particular allele for the mutated dystrophin that is associated with the disorder in this family. The results of gel electrophoresis analysis of the individual's dystrophin alleles and the alleles of several family members are shown in Figure 2.

5 Heredity MCQ

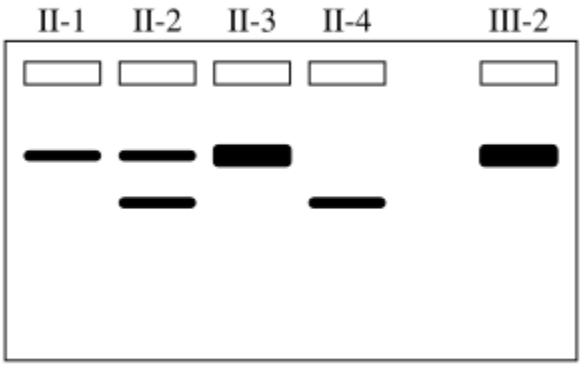


Figure 2. Gel electrophoresis showing the dystrophin alleles present in five members of the family

33. Based on the information provided, which of the following describes the most likely pattern of inheritance for the disorder?

(A) Autosomal recessive

(B) Autosomal dominant

(C) X-linked recessive ✓

(D) Mitochondrial

Answer C

Correct. The trait shows a recessive inheritance pattern because unaffected parents can produce affected offspring. The disorder is most likely X-linked because XY individuals are more likely to be affected. Based on Figure 2, XY individual $\mathrm{II}-4$ appears to have only one copy of the allele, further supporting that the allele is on the X chromosome.

34. Based on the information provided in Figures 1 and 2, which of the following best describes the chances of individuals $\mathrm{III}{-}1$ and $\mathrm{III}{-}2$ passing on the mutant dystrophin allele and/or DMD disorder to their offspring?

(A) There is a 50% chance that an XY child will have DMD.

(B) There is a 25% chance that a child will have DMD.

(C) There is a 0% chance that any of the children will have the disorder or be carriers of the mutant allele. ✓

(D) There is a 0% chance that an XY child will have DMD but a 50% chance that an XX child will be a carrier of the mutant allele.

5 Heredity MCQ

Answer C

Correct. Individual $\mathrm{III}-2$ is homozygous for the nonmutated dystrophin allele, as confirmed by the single upper band in Figure 2, and individual $\mathrm{III}-1$ does not possess the mutant allele either because this individual is XY and unaffected. If both parents do not possess the mutant allele, they will not be able to pass it on to their offspring, which means the offspring have 0% chance of having the disorder or being a carrier.

---

35. Eye color in a particular strain of fly is influenced by one gene with two alleles: a dominant allele that results in red eyes and a recessive allele that results in sepia eyes.

A red-eyed female from a true-breeding population is mated with a sepia-eyed male. The F1 offspring are all red-eyed. The F1 flies are allowed to interbreed, producing the following in the F2 generation.

Females: 40 red eyes; 13 sepia eyes

Males: 39 red eyes; 11 sepia eyes

Which of the following best describes the likely mode of inheritance for the eye-color gene?

(A) The eye-color gene is likely autosomal because males and females have similar phenotype ratios. ✓

(B) The eye-color gene is likely autosomal because more females have sepia eyes than males do.

(C) The eye-color gene is likely sex-linked because the males and females have similar phenotype ratios.

(D) The eye-color gene is likely sex-linked because the males and females display both phenotypes.

5 Heredity MCQ

36. Researchers hypothesized that red eye color in Japanese koi, a type of fish, is due to a mutation. To study the inheritance of red eye color in koi, the researchers conducted breeding experiments over several generations. The results are summarized in Figure 1.

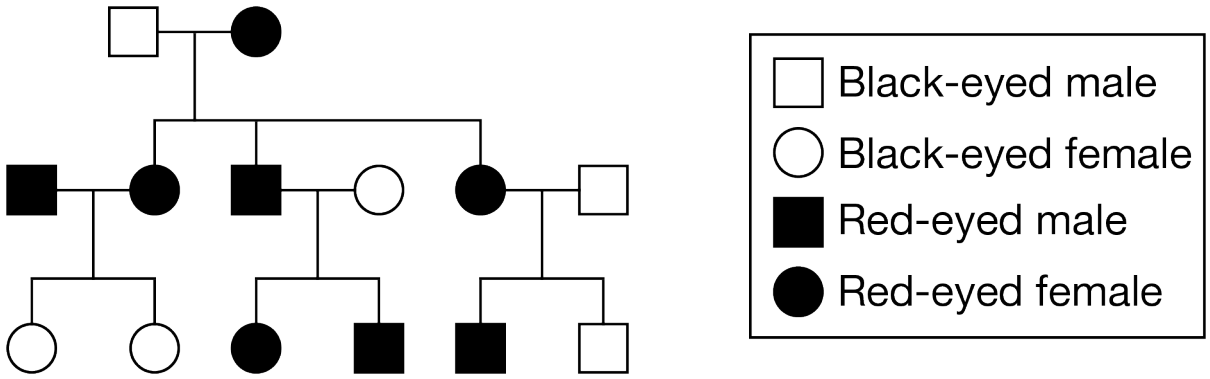


Figure 1. Inheritance of eye color in Japanese koi

Based on the data in Figure 1, which of the following is the best prediction of the mode of inheritance of red eyes in Japanese koi?

(A) The allele for red eyes is inherited in an autosomal dominant pattern. ✓

(B) The allele for red eyes is inherited in an autosomal recessive pattern.

(C) The allele for red eyes is inherited in an X-linked recessive pattern.

(D) The allele for red eyes is inherited in an X-linked dominant pattern.

Answer A

Correct. Based on the pedigree, red eye color in koi is inherited in an autosomal dominant pattern. In the second generation, two koi with red eyes produced female offspring with black eyes. This could only happen if the parents were heterozygous with the dominant allele.

37. In sheep, eye color is controlled by a single gene with two alleles. When a homozygous brown-eyed sheep is crossed with a homozygous green-eyed sheep, blue-eyed offspring are produced. If the blue-eyed sheep are mated with each other, what percent of their offspring will most likely have brown eyes?

5 Heredity MCQ

(A) 0%

(B) 25% ✓

(C) 50%

(D) 75%

(E) 100%

38. Eye pigment in a particular strain of fly is determined by two genes. An autosomal gene that controls the color of the pigments in the eye has two alleles: a dominant allele *(R)* that results in red eyes and a recessive allele *(r)* that results in sepia eyes. A sex-linked gene that controls the expression of the colored pigments also has two alleles: a dominant allele *(T)* that allows for expression of the colored pigments and a recessive allele *(t)* that does not allow for expression of the colored pigments. Individuals without a T allele have white eyes regardless of the alleles of other eye-color genes.

Which of the following represents a cross between a white-eyed female and a red-eyed male?

(A) $TtX^RX^R \times ttX^rY$

(B) $TtX^rX^r \times ttX^RY$

(C) $RRX^TX^T \times RrX^TY$

(D) $RrX^tX^t \times RrX^TY$ ✓

5 Heredity MCQ

In a laboratory population of fruit flies, body color is determined by a single gene with a wild-type allele associated with black body color and a mutant allele associated with yellow body color. A group of researchers hypothesizes that the mutation in fruit flies that causes the yellow body is sex-linked. They also hypothesize that this mutation results in a change in the structure of the sex combs, bristle-like structures on the front legs of male flies that facilitate mating.

To determine the inheritance pattern of the mutant allele, researchers crossed yellow-bodied females with wild-type males. Flies from the F_1 generation were then crossed. The distribution of phenotypes observed in the F_1 and F_2 generations is shown in Table 1.

TABLE 1. DISTRIBUTION OF PHENOTYPES OBSERVED IN THE P, F_1, and F_2 GENERATIONS

	Wild-Type Females*	Yellow-Bodied Females*	Wild-Type Males**	Yellow-Bodied Males**
P Generation	0	3	3	0
F_1 Generation	38	0	0	41
F_2 Generation	117	124	121	130

*Female fruit flies have two X chromosomes.

**Male fruit flies have one X chromosome and one Y chromosome.

39. Based on the researchers' hypothesis that the mutant allele is sex-linked, which of the following is closest to the expected number of wild-type females and the expected number of yellow-bodied males in the F_2 generation?

(A) 123 wild-type females and 123 yellow-bodied males ✓

(B) 61 wild-type females and 184 yellow-bodied males

(C) 123 wild-type females and 369 yellow-bodied males

(D) 184 wild-type females and 123 yellow-bodied males

Answer A

Correct. Based on Mendel's Laws of Segregation and Independent Assortment, and based on the P generation, the phenotypic ratio of the F_2 generation should be 1 : 1 : 1 : 1. Thus, 123 wild-type females and 123 yellow-bodied males would most closely match the expected numbers.

5 Heredity MCQ

40. Using a p-value of 0.05, researchers performed a chi-square analysis of the distribution of phenotypes in the F_2 generation.

Which of the following chi-square calculated values would cause the researchers to reject their null hypothesis?

(A) 0.73
(B) 2.16
(C) 4.35
(D) 7.98 ✓

Answer D

Correct. Because four different phenotypes would be produced in the F_2 generation, there should be 3 degrees of freedom in the chi-square analysis. Based on the chi-square table, a chi-square value that is greater than 7.81 is cause to reject the null hypothesis. The 7.98 value is higher than 7.81, and thus the researchers should reject the null hypothesis.

41. Based on the data in Table 1, which of the following is the most likely inheritance pattern for the mutant allele?

(A) The mutant allele is X-linked dominant.
(B) The mutant allele is X-linked recessive. ✓
(C) The mutant allele is autosomal (non-sex-linked) recessive.
(D) The mutant allele is Y-linked.

Answer B

Correct. To produce the values in the table, the mutant allele must be an X-linked recessive trait. Thus, in the P generation, if X^B represents the allele for black bodies and X^b represents the allele for yellow bodies, the 3 females would be X^bX^b and the 3 males would be X^By. In the F_1 generation, all females would be X^BX^b, and all males would be X^bY. The F_2 generation should produce an approximately $1:1:1:1$ ratio of X^BX^b, X^bX^b, X^By, X^bY flies.

42. Which of the following most likely explains how the sex combs of yellow-bodied flies have a different structure than do the sex combs of wild-type flies?

5 Heredity MCQ

(A) The protein encoded by the mutant body color allele interferes with the development of sex combs. ✓

(B) The protein encoded by the mutant body color allele is only produced if a copy of the wild-type body color allele is present.

(C) The gene associated with body color and the gene associated with sex comb development both exhibit codominance in flies.

(D) The gene associated with body color and the gene associated with sex comb development are located far apart on the same chromosome in flies.

Answer A

Correct. It is likely that the protein encoded by the mutant body color allele is also important to the normal development of sex combs in male flies.

---

43. The relative location of four genes on a chromosome can be mapped from the following data on crossover frequencies

Genes	*Frequency of Crossover*
B and *D*	5%
C and *A*	15%
A and *B*	30%
C and *B*	45%
C and *D*	50%

Which of the following represents the relative positions of these four genes on the chromosome?

(A) *ABCD*

(B) *ADCB*

(C) *CABD* ✓

(D) *CBAD*

(E) *DBCA*

44. Which of the following occurs in all species of living organisms and may lead to an increase in genetic variation?

(A) Mutations in the genome ✓

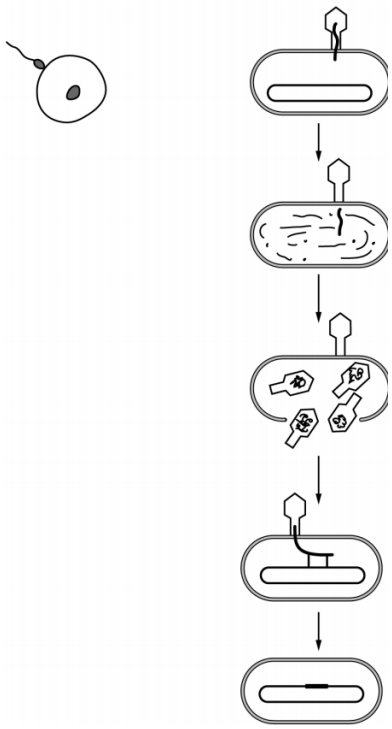
(B) Crossing-over in meiosis

(C) Random assortment of chromosomes

(D) Alternative splicing of mRNA

5 Heredity MCQ

45.



Which of the following correctly compares the two processes shown above?

(A) Both processes increase genetic variation. ✓

(B) Both processes represent aspects of sexual reproduction.

(C) Both processes require partitioning of genetic material and organelles.

(D) The amount of genetic material per cell remains constant in both processes.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

5 Heredity MCQ

46. Which of the following best supports the hypothesis that the difference in leaf color is genetically controlled?

(A) The number of yellow-leaved seedlings in dish A on day 7

(B) The number of germinated seeds in dish A on days 7 and 14

(C) The death of all the yellow-leaved seedlings

(D) The existence of yellow-leaved seedlings as well as green-leaved ones on day 14 in dish B ✓

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48. Gregor Mendel's pioneering genetic experiments with pea plants occurred before the discovery of the structure and function of chromosomes. Which of the following observations about inheritance in pea plants could be explained only after the discovery that genes may be linked on a chromosome?

(A) Pea color and pea shape display independent inheritance patterns.

(B) Offspring of a given cross show all possible combinations of traits in equal proportions.

(C) Most offspring of a given cross have a combination of traits that is identical to that of either one parent or the other. ✓

(D) Recessive phenotypes can skip a generation, showing up only in the parental and F_2 generations.

5 Heredity MCQ

49. Sex chromosomes determine the phenotype of sex in humans. Embryos containing XX chromosomes develop into females, and embryos containing XY chromosomes develop into males. The sex chromosomes separate during meiosis, going to different gamete cells.

A woman is heterozygous for the X-linked recessive trait of hemophilia A. Her sex chromosomes can be designated as X^HX^h. During meiosis the chromosomes separate as shown in Figure 1.

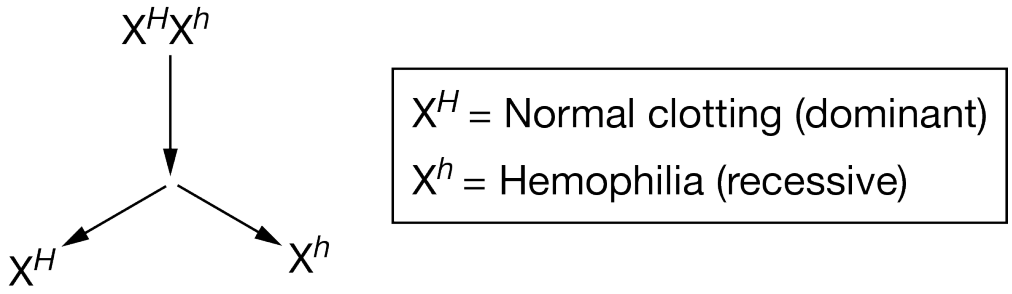


Figure 1. Transmission pattern for sex chromosomes of a woman heterozygous for hemophilia A into gametes

If the woman and a man with normal clotting function have children, what is the probability of their children exhibiting hemophilia A?

(A) 50 percent for daughters, 0 percent for sons

(B) 50 percent for sons, 0 percent for daughters ✓

(C) 50 percent for all children

(D) 0 percent for all children

Answer B

Correct. Half of the sons would inherit the X^H allele from the woman and a Y chromosome from the man. Without a X^H allele, these sons would develop hemophilia A.

50. Hydrangea flowers differ in color based on interactions between a pigment in the flower cells and aluminum ions dissolved in water absorbed from the soil. When aluminum ions are present, the pigment is blue; when aluminum ions are absent, the pigment is pink. The amount of aluminum ions the plant can absorb is dependent on the pH of the soil.

Which of the following questions will best help a researcher design an experiment to learn about the relationship between hydrangea flower color, aluminum, and pH?

5 Heredity MCQ

(A) How do aluminum-pigment interactions change the structure of the pigment and flower color?

(B) Can the pigment be used as an indicator of pH, because flower color depends on pH and aluminum?

(C) Does an increase in hydrogen ion concentration affect the ability of the pigment to dissolve in water?

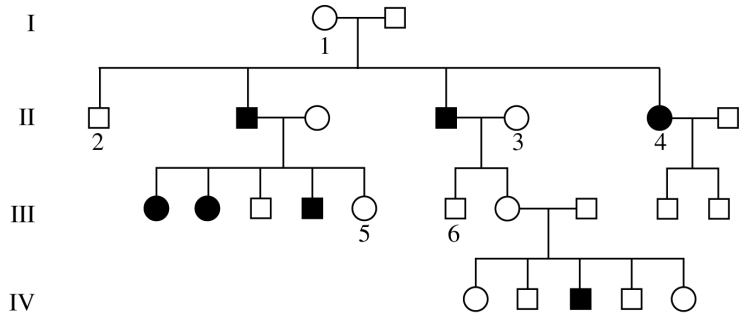
(D) What will be the flower color of hydrangeas grown in soils of varying pH with or without aluminum? ✓

Answer D

Correct. This question addresses how aluminum and pH interact to affect flower color.

For following group of questions first study the description of the situation and diagram and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

The pedigree of a family with a history of a particular genetic disease is shown below. Squares represent males and circles represent females. Shaded symbols represent those who have the disease.



51. If Individual 2 were to marry a woman with no family history of the disease, which of the following would most likely be true of their children?

(A) All of the children would have the disease.

(B) None of the children would have the disease. ✓

(C) Only the sons would have the disease.

(D) All of the sons would be carriers of the disease.

(E) None of the daughters would be carriers of the disease.

52. If Individual 6 marries a woman with the disease, what is the probability that their first child will have the disease?

5 Heredity MCQ

(A) 0
(B) 25%
(C) 50% ✓
(D) 75%
(E) 100%

53. Which of the following is the most likely mode of inheritance for the disease?

(A) Codominance
(B) Autosomal dominant
(C) Autosomal recessive ✓
(D) Sex-linked dominant
(E) Sex-linked recessive

---

54. Individuals with an inherited autosomal recessive disorder called primary ciliary dyskinesia (PCD) often have severe respiratory problems due to defective cilia. Males with PCD are often sterile because they produce sperm with defective flagella. Which of the following most likely explains the effect of the recessive allele?

(A) The mitochondria are defective and do not produce sufficient protein to synthesize microtubules in the cilia and flagella.
(B) The plasma membrane of the alveoli is not permeable to carbon dioxide during respiration because it is too hydrophobic.
(C) The Golgi bodies secrete an enzyme that destroys the proteins in the flagella and cilia.
(D) The cells do not produce functional motor proteins in flagella and cilia. ✓

55. In garden peas, a single gene controls stem length. The recessive allele (*t*) produces short stems when homozygous. The dominant allele (*T*) produces long stems. A short-stemmed plant is crossed with a heterozygous long-stemmed plant. Which of the following represents the expected phenotypes of the offspring and the ratio in which they will occur?

(A) 3 long-stemmed plants: 1 short-stemmed plant
(B) 1 long-stemmed plant: 1 short-stemmed plant ✓
(C) 1 long-stemmed plant: 3 short-stemmed plants
(D) Long-stemmed plants only
(E) Short-stemmed plants only

5 Heredity MCQ

56. The Russian blue is a rare breed of cat that is susceptible to developing cataracts on the eyes. Scientists hypothesize that this condition is inherited as a result of a mutation. Figure 1 shows a pedigree obtained in a study of cats owned by members of the Russian Blue Club in Sweden.

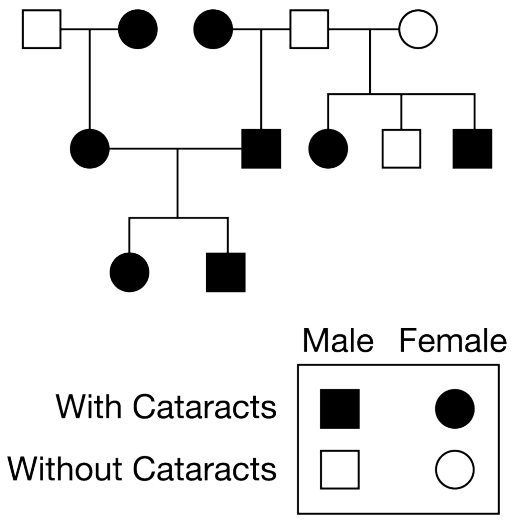


Figure 1. Pedigree of Russian blue cats owned by Russian Blue Club

Based on the inheritance pattern shown in Figure 1, which of the following best predicts the nature of the original mutation?

(A) A recessive mutation on the X chromosome

(B) A recessive mutation on a somatic chromosome ✓

(C) A dominant mutation on the X chromosome

(D) A dominant mutation on a somatic chromosome

Answer B

Correct. The inheritance pattern shows offspring phenotypes that are possible only if the trait is somatic and recessive. Since two cats without cataracts produce cats with cataracts, the trait must be recessive. This trait cannot be X-linked recessive because unaffected males are able to pass on the trait to their offspring.

57. In humans, red-green color blindness is a sex-linked recessive trait. If a man and a woman produce a color-blind son, which of the following must be true?

5 Heredity MCQ

(A) The father is color-blind.

(B) Both parents carry the allele for color blindness.

(C) Neither parent carries the allele for color blindness.

(D) The father carries the allele for color blindness.

(E) The mother carries the allele for color blindness. ✓

58. In peas the trait for tall plants is dominant (*T*) and the trait for short plants is recessive (*t*). The trait for yellow seed color is dominant (*Y*) and the trait for green seed color is recessive (*y*). A cross between two plants results in 296 tall yellow plants and 104 tall green plants. Which of the following are most likely to be the genotypes of the parents?

(A) *TTYY* x *TTYY*

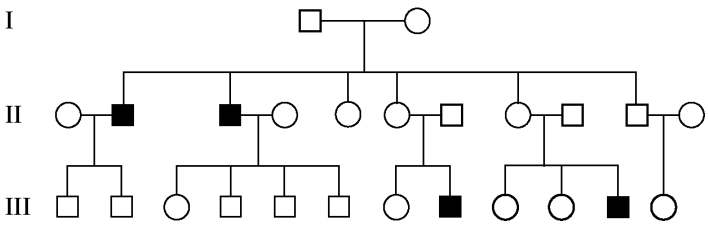
(B) *TTyy* x *TTYy*

(C) *TtYy* x *TtYy*

(D) *TtYy* x *TTYy* ✓

(E) *TtYY* x *Ttyy*

59. In the following human pedigree, squares represent XY individuals, circles represent XX individuals, and shaded symbols indicate individuals affected with a disorder.



One of the affected XY individuals from the third generation has a child with an XX individual who is a carrier. For the pedigree shown above, which of the following best expresses the probability that the couple's first XY offspring will be affected with the disorder?

(A) 25%

(B) 50% ✓

(C) 75%

(D) 100%

5 Heredity MCQ

60.

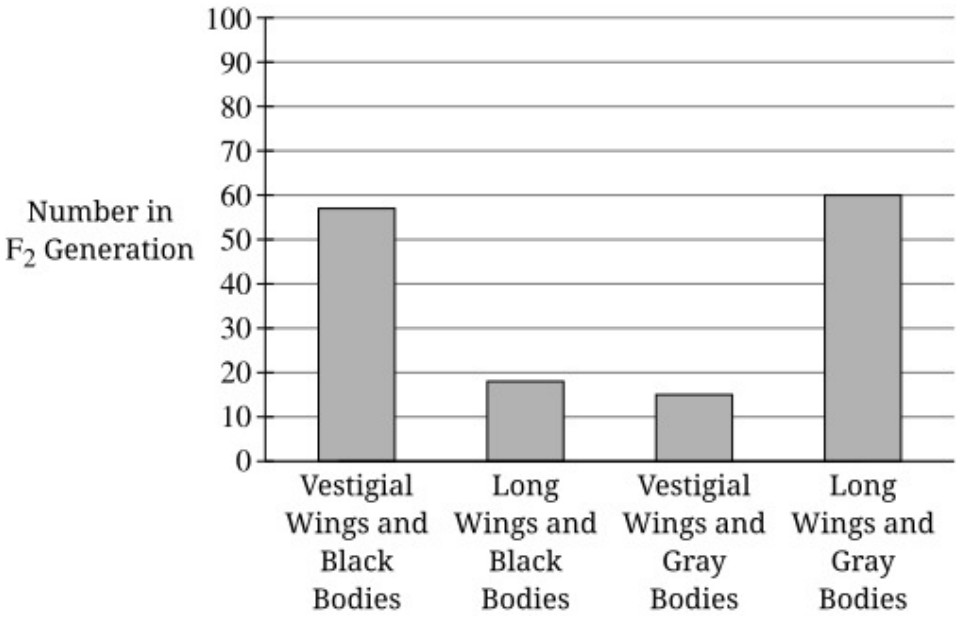


Figure 1. The distribution of phenotypes in the F_2 generation of fruit flies

In fruit flies, genes that control body color and wing length are linked. A researcher crossed a true-breeding fruit fly with long wings and a gray body with a true-breeding fruit fly with vestigial wings and a black body. All of the F_1 flies had long wings and gray bodies. The researcher then crossed the F_1 flies with true-breeding flies that have vestigial wings and black bodies. The number of flies with each phenotype in the resulting F_2 generation is shown in Figure 1.

What percent of the F_2 generation flies carry chromosomes produced by crossing-over between the two linked genes?

(A) 10%

(B) 15%

(C) 22% ✓

(D) 33%

Answer C

Correct. There are 33 flies with crossing-over phenotypes (18 have long wings with black bodies, and 15 have vestigial wings with gray bodies). The total number of 33 is then divided by 150, the total number of F_2 flies: $\frac{33}{150} = 22\%$.

5 Heredity MCQ

61. In sweet pea plants, purple flower color is dominant over red flower color and long pollen grain shape is dominant over round pollen grain shape. Two sweet pea plants that are heterozygous for both flower color and pollen grain shape are crossed with one another. A geneticist is surprised to observe that there are far fewer round, purple offspring and long, red offspring than were predicted by the $9:3:3:1$ expected phenotypic ratio.

Which of the following statements provides the most reasonable prediction to account for the deviation from the expected results?

(A) In sweet pea plants, the gene for flower color and the gene for pollen grain shape are genetically linked. ✓

(B) In sweet pea plants, the genes for flower color and for pollen grain shape both exhibit codominance.

(C) Several mutations occurred spontaneously producing a deviation from the expected phenotypic ratios of the offspring.

(D) The genes for flower color and pollen grain shape are inherited independently because of the law of independent assortment.

Answer A

Correct. There are fewer recombinant phenotypes than expected due to genetic linkage. The gene for flower color and the gene for pollen grain shape have a strong tendency to segregate as a unit, so it is most likely that they are close together on the same chromosome and thus, the offspring exhibit parental phenotypes more than expected.

62. Four trials measuring recombination frequency between gene R and gene L were conducted, and the results are shown in Table 1.

Table 1. Recombination frequency

	Trial 1	Trial 2	Trial 3	Trial 4
Frequency of recombination	24%	33%	26%	29%

The mean map distance between gene R and gene L is closest to which of the following?

(A) 0.28 map units

(B) 28 map units ✓

(C) 0.14 map units

(D) 14 map units

5 Heredity MCQ

Answer B

Correct. The recombination frequency is equal to the map distance between genes on the same chromosome. Thus, the mean recombination frequency should be determined, and this is equal to the mean map distance between the two genes.

63. Two fruit fly mutations are ebony body (e) and sepia eyes (s). Four different students performed dihybrid crosses with flies that were heterozygous with a mutant allele and a wild-type allele for both genes $(EeSs \times EeSs)$. The results are shown in Table 1.

Table 1. Offspring of four separate dihybrid crosses

	Number of Offspring			
Phenotype	**Student 1**	**Student 2**	**Student 3**	**Student 4**
Wild-type eyes, wild-type body	89	92	83	95
Wild-type eyes, ebony body	22	17	15	20
Sepia eyes, wild-type body	21	24	30	16
Sepia eyes, ebony body	28	27	32	29

The mean number of fruit flies per student that are homozygous recessive for both genes is closest to which of the following?

(A) 89.75

(B) 29.0 ✓

(C) 22.75

(D) 18.5

Answer B

Correct. The mean number of flies that express both recessive mutations homozygous recessive is 29.0. The sepia eyes, ebony body phenotype is homozygous recessive $(eess)$.

5 Heredity MCQ

64. A model showing the cells in anaphase I and anaphase II of meiosis during a nondisjunction event is shown in Figure 1.

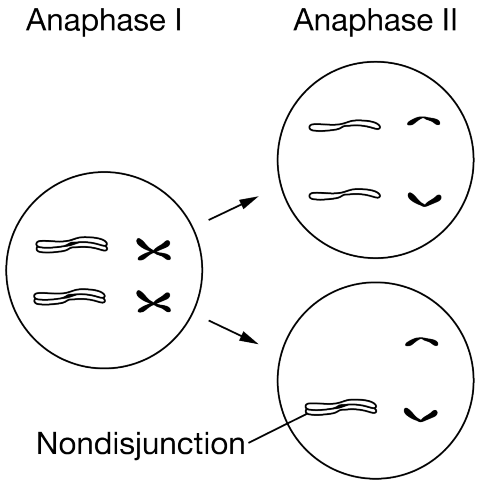


Figure 1. Model of a nondisjunction event

Which of the following best predicts the effect of the chromosomal segregation error shown in Figure 1?

(A) All of the resulting gametes will have an extra chromosome.

(B) All of the resulting gametes will be missing a chromosome.

(C) Half of the resulting gametes will have an extra chromosome and the other half will be missing a chromosome.

(D) Half of the resulting gametes will have the correct number of chromosomes, and the other half will have an incorrect number of chromosomes. ✓

Answer D

Correct. In this instance, nondisjunction of sister chromatids during meiosis II will produce two gametes with the correct number of chromosomes (n). The remaining two gametes that are produced will be $n + 1$ and $n - 1$, respectively.

5 Heredity MCQ

Mice live in territorial social groups, and each group has a single dominant male with the rest of the males in the group being subordinate. Male mice mark the area in which they live with their scent by leaving urine wherever they walk. There are two classes of proteins that are found in mouse urine, major urinary proteins (MUPs) and major histocompatibility complex (MHC) proteins. The proteins provide a unique scent for each mouse.

In an experiment, wild male mice were kept isolated in separate cages and their urination patterns were recorded. They were then combined into a single cage that was divided into two compartments by a mesh screen, and again their urination patterns were recorded (Figure 1).

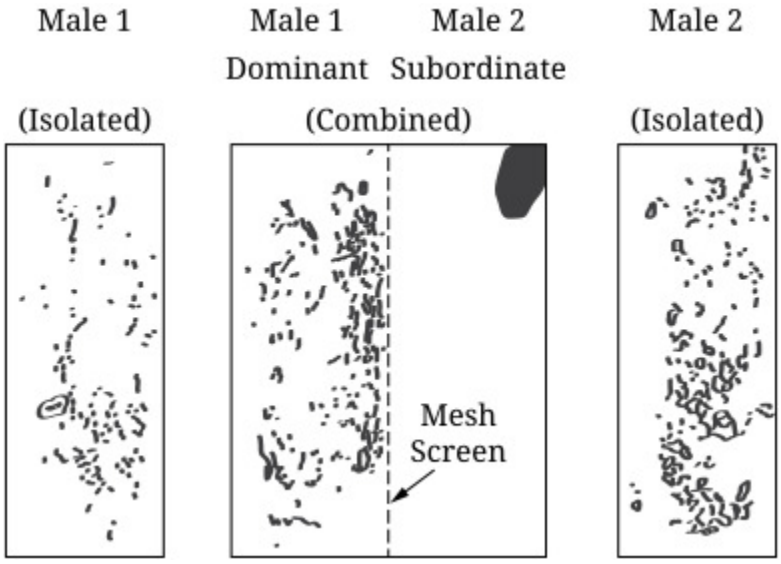


Figure 1. Urination patterns recorded during an experiment

65. Which of the following best explains why particular MUP protein variants are associated with male dominance in individuals in populations of wild mice?

(A) Viral infections introduced new MUP alleles into affected populations.

(B) Environmental stress caused a decreased rate of mutations in the MUP genes of subordinate mice.

(C) Interbreeding between different mouse species introduced new MUP alleles into the populations.

(D) The MUP genes are located on a chromosome near genes associated with phenotypes of dominant males. ✓

Answer D

Correct. Male dominance is an inherited trait encoded by particular alleles of genes. Genes that are located near each other on the same chromosome tend to stay together during meiosis. This explains why certain MUP variants are associated with other traits that produce dominant males.

5 Heredity MCQ

66. In order to determine the effects of age on the accumulation of mitochondrial mutations, mitochondrial DNA samples from young mice (3 months) and old mice (30 months) were observed for mutations.

Table 1. Number of mitochondrial mutations in 10^6 base pairs

Mouse	Young	Old
A	1	
B	1	
C	3	
D	2	
E	4	
F		10
G		15
H		12
I		9
J		11

Which of the following is a correct analysis of this data set?

(A) There is an increase in the mean number of mutations for the two age groups of 9.0 mutations per 10^6 base pairs. This is more critical in male mammals since mitochondria are paternally inherited.

(B) There is an increase in the mean number of mutations for the two age groups of 9.0 mutations per 10^6 base pairs. This is more critical in female mammals since mitochondria are maternally inherited.

(C) There is an increase in the mean number of mutations for the two age groups of 9.2 mutations per 10^6 base pairs. This is more critical in male mammals since mitochondria are paternally inherited.

(D) There is an increase in the mean number of mutations for the two age groups of 9.2 mutations per 10^6 base pairs. This is more critical in female mammals since mitochondria are maternally inherited. ✓

Answer D

Correct. The mean number of mutations for the young mice is $\frac{1+1+3+2+4}{5} = 2.2$ per 10^6 base pairs, and the mean number of mutations for the old mice is $\frac{10+15+12+9+11}{5} = 11.4$ per 10^6 base pairs. This yields an increase in the mean number of mutations of 9.2 per 10^6 base pairs between the young and old mice. Mitochondria are normally passed to offspring only from their mother.

5 Heredity MCQ

67. A true-breeding variety of wheat that produces deep-red-colored grain was crossed with a true-breeding variety that produces a white-colored grain. The resulting F_1 offspring all had medium-red-colored grain. Five crosses of the F_1 plants were all grown under the same conditions and resulted in plants with a variety of grain color, as indicated in Table 1.

Table 1. F_2 phenotypes resulting from five crosses

Grain Color	Cross 1	Cross 2	Cross 3	Cross 4	Cross 5
Deep red	1	2	1	2	0
Dark red	6	7	7	5	6
Red	15	13	16	17	14
Medium red	20	22	21	20	21
Light red	15	16	14	16	15
Very light red	6	4	3	3	6
White	1	0	2	1	2

Which of the follow indicates the mean number per cross of F_2 plants producing medium-red grain and correctly explains the distribution of the phenotypes?

(A) The mean number of medium-red phenotypes per cross is 20.8. The distribution of phenotypes suggests that grain color is under environmental control.

(B) The mean number of medium-red phenotypes per cross is 20.8. The distribution of phenotypes suggests that multiple genes are involved in grain color determination. ✓

(C) The mean number of medium-red phenotypes per cross is 104. The distribution of phenotypes suggests that grain color is under environmental control.

(D) The mean number of medium-red phenotypes per cross is 104. The distribution of phenotypes suggests that multiple genes are involved in grain color determination.

Answer B

Correct. This is the correct mean, $\frac{(20+22+21+20+21)}{5} = 20.8$. The phenotype distribution shows a continuous range of grain color from plants grown under the same conditions, suggesting that grain color is controlled by multiple alleles.

5 Heredity MCQ

68. An experiment was performed to determine the mode of inheritance of two mouse genes, one for fur color and one for fur length. It is known that black fur (B) is dominant over white fur (b) and that long fur (L) is dominant over short fur (l). To determine how the genes are inherited, a cross was performed between two true-breeding mice, one with long black fur and one with short white fur. Their progeny, the F_1 generation, all had long black fur. Five F_1 male-female pairs were then crossed with one another. The F_2 generation phenotypes for each cross are shown in Table 1.

Table 1. Number of F_2 generation phenotypes for five crosses

Phenotype	Cross 1	Cross 2	Cross 3	Cross 4	Cross 5
Long black fur	6	5	5	6	7
Long white fur	1	1	0	2	2
Short black fur	0	2	1	1	1
Short white fur	2	3	3	2	2

Which of the following is the mean number per cross of F_2 generation offspring that are the result of crossing over?

(A) 1

(B) 2.2 ✓

(C) 2.4

(D) 5.8

Answer B

Correct. This is the mean of the six F_2 offspring with long white fur and the five F_2 offspring with short black fur. These two phenotypes are both the result of crossing over. The 11 offspring divided by the 5 crosses results in a mean per cross of 2.2.

5 Heredity MCQ

69. In pea plants, purple flower color is dominant to red flower color and long pollen grains are dominant to round pollen grains. Researchers crossed two pure-breeding lines of the pea plants to investigate whether the genes controlling flower color and pollen shape segregate independently. The procedure for the genetics experiment is summarized in Figure 1.

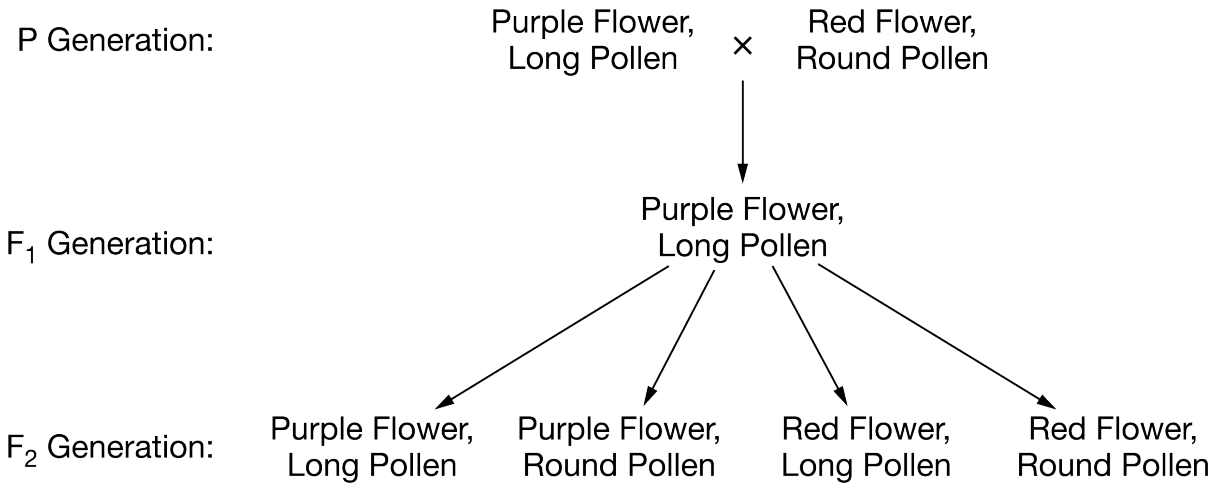


Figure 1. Summary of a genetics experiment using pea plants

Which of the following tables best shows the expected values in the F_2 generation for a chi-square goodness-of-fit test for a model of independent assortment?

5 Heredity MCQ

(A)

Phenotype	Expected
Purple, long	533
Purple, round	533
Red, long	533
Red, round	533

(B) ✓

Phenotype	Expected
Purple, long	1199
Purple, round	400
Red, long	400
Red, round	133

(C)

Phenotype	Expected
Purple, long	1067
Purple, round	355
Red, long	355
Red, round	355

(D)

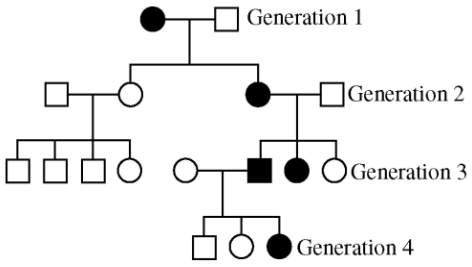
5 Heredity MCQ

Phenotype	Expected
Purple, long	1002
Purple, round	515
Red, long	515
Red, round	1002

Answer B

Correct. Because the F_1 offspring used to produce the F_2 generation were heterozygous for both traits, the expected phenotypic ratio for a model of independent assortment is $9:3:3:1$.

70. The pedigree below shows the inheritance of a dominant allele of a gene in a family over several generations. Circles represent females and squares represent males. Shaded symbols indicate individuals carrying the allele.



The pedigree suggests that the gene is on a nuclear chromosome, and not on mitochondrial DNA, because

(A) mitochondrial genes are not heritable

(B) mitochondrial mutations cannot produce dominant traits

(C) maternal mitochondrial mutations are inherited by all of a mother's offspring ✓

(D) mitochondrial DNA is circular, whereas chromosomal DNAis linear

5 Heredity MCQ

71. Pigeons demonstrate ZW sex determination, such that a ZZ genotype produces a male and a ZW genotype produces a female. The gene for feather color is located on the Z chromosome, and the red allele is dominant over the brown allele. Three crosses between brown male pigeons and red female pigeons were performed, and the results are shown below.

Table 1. Offspring from three separate crosses of a brown male pigeon and a red female pigeon

	Number of Offspring		
Phenotype	Cross 1	Cross 2	Cross 3
Red	11	9	7
Brown	9	11	13

Which of the following is the mean number of male offspring produced by the three crosses?

(A) 27

(B) 20

(C) 11

(D) 9 ✓

Answer D

Correct. For a pigeon to be red, a dominant allele must have been inherited from the female parent. Thus, red pigeons produced by the cross described above will always have two Z chromosomes, which makes them male. The genotype of the offspring would be either Z^RZ^r (phenotype: red male) or Z^rW (phenotype: brown female).

5 Heredity MCQ

72. A student crosses a pure-breeding line of red-flowered poinsettias with a pure-breeding line of white-flowered poinsettias. The student observes that all the plants in the F_1 generation have pink flowers. The student then crosses the F_1 plants with one another and records observations about the plants in the F_2 generation. The student will use the F_2 data to perform a chi-square goodness-of-fit test for a model of incomplete dominance. The setup for the student's chi-square goodness-of-fit test is presented in Table 1.

Table 1. A chi-square goodness-of-fit test for incomplete dominance

Phenotype	Observed	Expected
Red flowers	64	56
Pink flowers	107	112
White flowers	53	56

The critical value for a chi-square test with a significance level of $p = 0.05$ and 2 degrees of freedom is 5.99. Which of the following statements best completes the student's chi-square goodness-of-fit test?

(A) The calculated chi-square value is 1.53, and the null hypothesis cannot be rejected. ✓

(B) The calculated chi-square value is 1.53, and the null hypothesis can be rejected.

(C) The calculated chi-square value is 98, and the null hypothesis cannot be rejected.

(D) The calculated chi-square value is 98, and the null hypothesis can be rejected.

Answer A

Correct. The calculated chi-square value is 1.53. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected. Consequently, the student may conclude that the data fit a model of incomplete dominance.

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the probabilities below.

Assume that the alleles referred to all assort independently.

(A) 0
(B) 1/16
(C) 1/4
(D) 1/2
(E) 3/4

5 Heredity MCQ

73. Probability that the genotype *Aa* will be produced by the parents *Aa* x *Aa*

(A) 0
(B) 1/16
(C) 1/4
(D) 1/2 ✓
(E) 3/4

74. Probability that the genotype *ccdd* will be produced by the parents *CcDd* x *CcDd*

(A) 0
(B) 1/16 ✓
(C) 1/4
(D) 1/2
(E) 3/4

75. Probability that the genotype *Rr* will be produced by the parents *Rr* x *rr*

(A) 0
(B) 1/16
(C) 1/4
(D) 1/2 ✓
(E) 3/4

76. Probability that the genotype *TTSs* will be produced by the parents *TTSs* x *TtSS*

(A) 0
(B) 1/16
(C) 1/4 ✓
(D) 1/2
(E) 3/4

5 Heredity MCQ

77. In fruit flies, purple eyes and ebony body are traits that display autosomal recessive patterns of inheritance. In a genetics experiment, students cross wild-type flies with flies that have purple eyes and ebony bodies. The students observe that all the flies in the F_1 generation have normal eyes and a normal body color. The students then allow the F_1 flies to mate and produce an F_2 generation. The students record observations about the flies in the F_2 generation and use the data to perform a chi-square goodness-of-fit test for a model of independent assortment. The setup for the students' chi-square goodness-of-fit test is presented in Table 1.

Table 1. The students' chi-square goodness-of-fit test for a model of independent assortment

Phenotype	Observed	Expected
Normal eyes, normal body	187	171
Normal eyes, ebony body	49	57
Purple eyes, normal body	41	57
Purple eyes, ebony body	27	19

The students choose a significance level of $p = 0.01$. Which of the following statements best completes the next step of the chi-square goodness-of-fit test?

(A) The calculated chi-square value is 2.11, and the critical value is 7.82.

(B) The calculated chi-square value is 2.11, and the critical value is 11.35.

(C) The calculated chi-square value is 10.48, and the critical value is 7.82.

(D) The calculated chi-square value is 10.48, and the critical value is 11.35. ✓

Answer D

Correct. The calculated chi-square value is 10.48, and the critical value for 3 degrees of freedom and a significance level of $p = 0.01$ is 11.35. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected. Consequently, the students can conclude that the data fit a model of independent assortment.

5 Heredity MCQ

78.

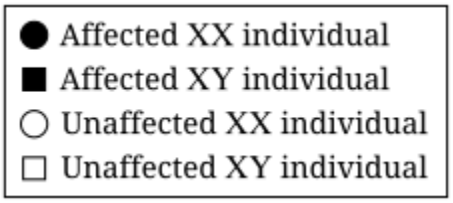
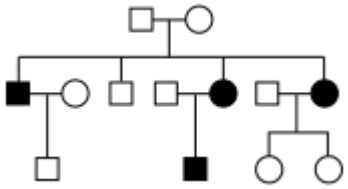
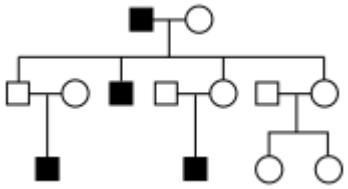
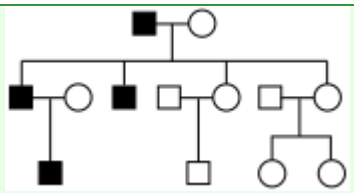


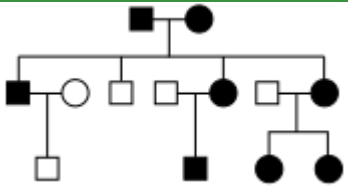
Figure 1. Key for interpreting a pedigree

Which of the following pedigrees represents the inheritance of a trait most likely associated with a gene on the Y chromosome?

(A) 

(B) 

(C)  ✓

(D) 

Answer C

Correct. A Y chromosome is only present in an XY individual and is passed by the XY individual to all XY offspring. In this pedigree, only the F_1 XY individuals inherit the trait; F_1 XX individuals do not. The only F_2 individual to inherit the trait is also an XY offspring of an F_1 individual with the trait.

5 Heredity MCQ

79. In fruit flies, sepia eyes and ebony body are traits that display autosomal recessive patterns of inheritance. To investigate whether the traits are genetically linked, students cross wild-type flies with a line of flies that have sepia eyes and ebony bodies. The students observe that all the flies in the F_1 generation have normal eyes and normal bodies. The students allow the flies in the F_1 generation to mate and produce an F_2 generation. The students then record observations for the flies in the F_2 generation and use the data to perform a chi-square goodness-of-fit test for a model of independent assortment. The setup for the chi-square goodness-of-fit test is presented in Table 1.

Table 1. Setup for the students' chi-square goodness-of-fit test

Phenotype	Observed	Expected
Normal eyes, normal body	231	279
Normal eyes, ebony body	86	93
Sepia eyes, normal body	97	93
Sepia eyes, ebony body	82	31

The students calculate a chi-squared value of 92.86 and compare it with a critical value of 7.82. Which of the following best completes the chi-square goodness-of-fit test?

(A) The null hypothesis cannot be rejected, and the students should conclude that the data fit a model of independent assortment.

(B) The null hypothesis cannot be rejected, and the students should conclude that the data may have resulted from genetic linkage.

(C) The null hypothesis can be rejected, and the students should conclude that the data fit a model of independent assortment.

(D) The null hypothesis can be rejected, and the students should conclude that the data may have resulted from genetic linkage. ✓

Answer D

Correct. Because the chi-square value is greater than the critical value, the null hypothesis can be rejected. The rejection of the null hypothesis indicates that the data do not fit a model of independent assortment. Consequently, the students can infer that the traits may be genetically linked.

80. Australian dragon lizards have a ZW sex-determination system. The male genotype is homogametic (ZZ), and the female genotype is heterogametic (ZW). However, all eggs incubated at temperatures above $32^\circ C$ tend to develop into females.

Which of the following best explains how the development of phenotypic female Australian dragon lizards with a ZZ genotype occurs when incubation temperatures are above $32^\circ C$?

5 Heredity MCQ

(A) Lizard embryos with a ZZ genotype cannot develop at temperatures above $32^{\circ}\mathrm{C}$.

(B) At incubation temperatures above $32^{\circ}\mathrm{C}$, Z chromosomes are mutated into W chromosomes.

(C) At incubation temperatures above $32^{\circ}\mathrm{C}$, crossing over transfers genes from the W chromosome to the Z chromosome, producing females.

(D) Incubation temperatures above $32^{\circ}\mathrm{C}$ inhibit the genes on the Z chromosome that produce proteins necessary for male development. ✓

Answer D

Correct. Similar in function to other SRY genes found on the Y chromosome in mammals, the Z chromosome contains genes that need to actively produce proteins during development of the embryo to trigger production of male characteristics. Above $32^{\circ}\mathrm{C}$, these genes are inhibited, so the default female phenotype develops.

81. A student carries out a genetics experiment with fruit flies to investigate the inheritance pattern of the white eye trait. The student crosses a homozygous white-eyed female with a wild-type male and records observations about the flies in the F_1 generation. The student plans to use the F_1 data to perform a chi-square goodness-of-fit test for a model based on an X-linked recessive pattern of inheritance. The student will use one degree of freedom and a significance level of $p = 0.05$. The setup for the student's chi-square goodness-of-fit test is presented in Table 1.

Table 1. Setup for the student's chi-square goodness-of-fit test

Phenotype	Observed	Expected
Red-eyed female	53	50
White-eyed male	47	50

The student calculates a chi-square value of 0.36. Which of the following statements best completes the student's chi-square goodness-of-fit test?

(A) The critical value is 0.05, and the student cannot reject the null hypothesis.

(B) The critical value is 0.05, and the student can reject the null hypothesis.

(C) The critical value is 3.84, and the student cannot reject the null hypothesis. ✓

(D) The critical value is 3.84, and the student can reject the null hypothesis.

Answer C

Correct. The critical value is 3.84. Because the calculated chi-square value is less than the critical value, the student cannot reject the null hypothesis. Consequently, the student may conclude that the data fit a

model in which the white eye trait has an X-linked recessive pattern of inheritance.

82. When a mustard plant seedling is transferred to an environment with higher levels of carbon dioxide, the new leaves have a lower stomata-to-surface-area ratio than do the seedling's original leaves.

Which of the following best explains how the leaves from the same plant can have different stomatal densities when exposed to an elevated carbon dioxide level?

(A) Increased photosynthesis leads to larger leaves that need more stomata for photosynthesis, leading to an increase in stomatal density.

(B) Leaf growth is promoted through increased photosynthesis, but the genetically regulated rate of stomatal production is not altered, leading to a decrease in stomatal density. ✓

(C) Leaf growth is inhibited by decreased photosynthesis, and the genetically regulated rate of stomatal production remains the same, leading to an increase in stomatal density.

(D) Leaf growth is inhibited by decreased photosynthesis, and the genetically regulated rate of stomatal production remains the same, leading to a decrease in stomatal density.

Answer B

Correct. Elevated carbon dioxide levels promote leaf growth due to increased photosynthetic activity. If the number of stomata produced remains the same as the leaves grow larger, the stomatal density decreases. This is an example of phenotypic plasticity, which is an adaptive trait.

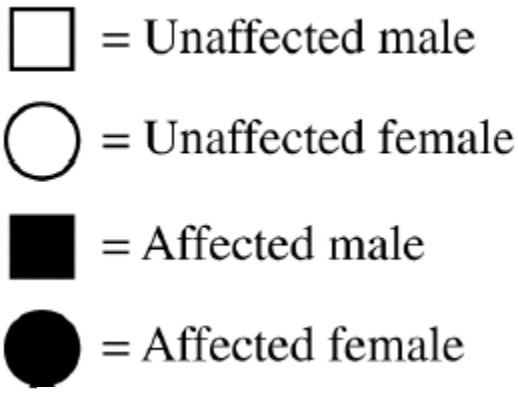
Tay-Sachs disease is a rare inherited disorder caused by an autosomal recessive allele of the *HEXA* gene. Affected individuals exhibit severe neurological symptoms and do not survive to reproductive age. Individuals who inherit one copy of the allele (Tay-Sachs carriers) typically show no symptoms of the disorder. The frequencies of Tay-Sachs carriers in the general population of North America and in three different subpopulations are presented in the table.

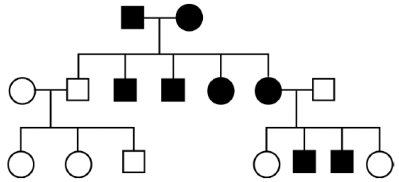
FREQUENCY OF TAY-SACHS CARRIERS IN DIFFERENT POPULATIONS

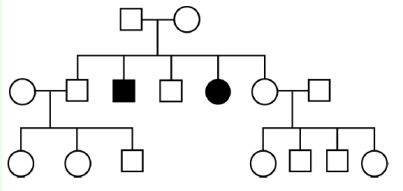
Population	Frequency of Tay-Sachs Carriers
General population	0.004
Subpopulation I	0.037
Subpopulation II	0.035
Subpopulation III	0.020

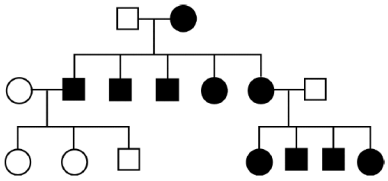
5 Heredity MCQ

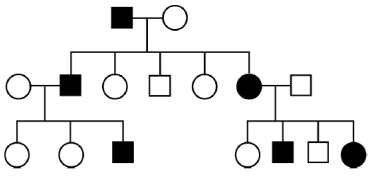
83. Which of the following pedigrees most accurately represents a family with a history of Tay-Sachs disease?



(A) 

(B)  ✓

(C) 

(D) 

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84. Himalayan rabbits are a breed of rabbits with highly variable fur color. If genetically similar rabbits are raised in environments that have different temperature conditions, the rabbits can have different color patterns.

Which of the following statements best explains how the fur color can be different in Himalayan rabbits raised under different temperature conditions?

5 Heredity MCQ

(A) The genotype does not contribute to coat color in Himalayan rabbits.

(B) The phenotype determines the genotype of coat color in Himalayan rabbits.

(C) Different environments cause specific mutations in the genes controlling pigment production.

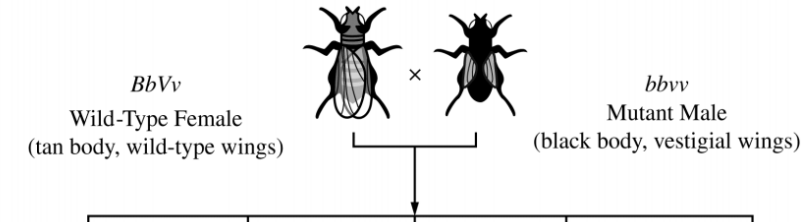
(D) The environment determines how the genotype is expressed. ✓

Answer D

Correct. The environment influences gene activity, which leads to a change in the phenotype of fur color. Temperature is the most significant environmental factor in determining the coat color of these rabbits.

85.

FRUIT FLY EXPERIMENTAL RESULTS



Genotypes	*BbVv* Tan, Wild Type	*bbvv* Black, Vestigial	*Bbvv* Tan, Vestigial	*bbVv* Black, Wild Type
Expected results	862	862	862	862
Observed phenotypes	1,447	1,416	309	277

In *Drosophila melanogaster* the allele for wild-type tan body color (B) is dominant to the recessive allele for black body color (b). Similarly, the allele for wild-type wing shape (V) is dominant to the recessive allele for vestigial wing phenotype (v). In the cross diagrammed above, the expected and observed results are shown. Which of the following best explains the observed results of the cross?

(A) The alleles for body color and wing shape assort independently, as predicted by Mendel's laws.

(B) The genes for body color and wing shape are located close to each other on the same chromosome. ✓

(C) The traits of body color show complete dominance over the traits of wing shape.

(D) The observed variations in body color and wing shape are detectable in males but not in females

5 Heredity MCQ

86. A culture of white-eyed fruit flies (*Drosophila melanogaster*) was maintained for many generations. Females from the stock white-eyed culture were crossed with red-eyed (wild-type) males. The F_1 females were crossed with the white-eyed males from the original culture. The resulting phenotypes of the progeny are summarized below.

Parental Generation Cross	**F_1 Generation (at least 500 flies)**
White-eyed females × red-eyed males	100% of females are red-eyed 100% of males are white-eyed

F_1 Generation Cross	**F_2 Generation (at least 500 flies)**
F_1 red-eyed females × white-eyed males	50% of females are red-eyed and 50% are white-eyed 50% of males are red-eyed and 50% are white-eyed

There are white-eyed females in the F_2 generation because

(A) white is a dominant allele

(B) the white allele is autosomal

(C) a mutation has occurred

(D) these F_2 females have two white alleles ✓

(E) the white allele is located on the *Y* chromosome

87. The tiny blue-eyed Mary flower is often one of the first flowers seen in the spring in some regions of the United States. The flower is normally blue, but sometimes a white or pink flower variation is found.

The following data were obtained after several crosses.

Parents	F_1	F_2
Blue x white	Blue	196 blue, 63 white
Blue x pink	Blue	149 blue, 52 pink
Pink x white	Blue	226 blue, 98 white, 77 pink

Which of the following statements best explains the data?

5 Heredity MCQ

(A) The appearance of blue in the F_1 generation of the pink and white cross demonstrates that flower color is not an inherited trait but is determined by the environment.

(B) Flower color depends on stages of flower development, and young flowers are white, advancing to pink and then blue.

(C) Since the F_1 and F_2 phenotypes of the pink and white cross do not fit the expected genotypic and phenotypic ratios, blue-eyed Mary must reproduce by vegetative propagation.

(D) Flower color is an inherited trait, and the F_1 and F_2 phenotypes of the flowers arising from the pink and white cross can best be explained by another gene product that influences the phenotypic expression. ✓

Answer D

This option is correct. The student is asked to evaluate the data given. The crosses between blue / white and blue / pink show expected 3:1 ratios in the F_1 generation. The appearance of a majority of blue flowers in the F_1 cross between pink and white indicate that there is another gene product affecting the outcome.

88. The tiny blue-eyed Mary flower is often one of the first flowers seen in the spring in some regions of the United States. The flower is normally blue, but sometimes a white or pink flower variation is found.

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(D) Flower color is an inherited trait, and the F_1 and F_2 phenotypes of the flowers arising from the pink and white cross can best be explained by another gene product that influences the phenotypic expression. ✓

5 Heredity MCQ

89. The tadpoles of Mexican spadefoot toads are known to exhibit phenotypic plasticity depending on food availability. Tadpole mouthparts can vary significantly, prompting researchers to categorize them as either omnivore-morph or carnivore-morph. Carnivore-morph tadpoles are larger and have mouthparts that are better suited for predation. Remarkably, carnivore-morph tadpoles can change into omnivore-morph tadpoles when the food supply changes.

Which of the following best describes an advantage of the phenotypic plasticity displayed by the tadpoles?

(A) It allows the tadpoles to change their genome in response to environmental pressures.

(B) It enables the tadpoles to develop into a distinct species of toads.

(C) It gives the tadpoles increased versatility with respect to diet. ✓

(D) It allows the tadpoles to delay metamorphosis until there is maximal food available for the adults.

90. In a strain of tomato plants, short plant height and small fruit size are traits that display autosomal recessive patterns of inheritance. To investigate whether the traits segregate independently, researchers cross a pure-breeding line of tall tomato plants that have large fruits with a pure-breeding line of short tomato plants that have small fruits. The researchers observe that all the plants in the F_1 generation are tall and have large fruits. The researchers cross the F_1 plants with one another to generate an F_2 generation. The researchers record observations for the F_2 generation and will use the data to perform a chi-square goodness-of-fit test for a model of independent assortment. The setup for the chi-square goodness-of-fit test is shown in Table 1.

Table 1. Observed and expected counts of tomato plants in the F_2 generation

Phenotype	Observed	Expected
Tall plant with large fruits	104	90
Tall plant with small fruits	17	30
Short plant with large fruits	26	30
Short plant with small fruits	13	10

The researchers choose a significance level of $p = 0.05$. Which of the following best completes the chi-square goodness-of-fit test?

(A) The calculated chi-square value is 9.24, and the critical value is 7.82. The null hypothesis of independent assortment can be rejected. ✓

(B) The calculated chi-square value is 9.24, and the critical value is 9.49. The null hypothesis of independent assortment cannot be rejected.

(C) The calculated chi-square value is 13.13, and the critical value is 7.82. The null hypothesis of independent assortment can be rejected.

(D) The calculated chi-square value is 13.13, and the critical value is 9.49. The null hypothesis of independent assortment cannot be rejected.

5 Heredity MCQ

Answer A

Correct. The calculated chi-square value is 9.24. The critical value for a chi-square test with 3 degrees of freedom and a significance level of $p = 0.05$ is 7.82. Because the calculated chi-square value is greater than the critical value, the null hypothesis of independent assortment can be rejected. The researchers can conclude that the data do not fit a model of independent assortment and that the traits may be genetically linked.

91. A culture of white-eyed fruit flies (*Drosophila melanogaster*) was maintained for many generations. Females from the stock white-eyed culture were crossed with red-eyed (wild-type) males. The F_1 females were crossed with the white-eyed males from the original culture. The resulting phenotypes of the progeny are summarized below.

Parental Generation Cross	**F_1 Generation (at least 500 flies)**
White-eyed females × red-eyed males	100% of females are red-eyed 100% of males are white-eyed

F_1 Generation Cross	**F_2 Generation (at least 500 flies)**
F1 red-eyed females × white-eyed males	50% of females are red-eyed and 50% are white-eyed 50% of males are red-eyed and 50% are white-eyed

Which of the following best describes the mode of inheritance of eye color in the white culture?

(A) Autosomal

(B) Dominant

(C) Located on the *Y* chromosome

(D) Sex-linked ✓

(E) Lethal

5 Heredity MCQ

92. Students carry out a genetics experiment to investigate the inheritance pattern of the white-eye trait in fruit flies. In the experiment, the students cross a red-eyed female with a white-eyed male to produce an F_1 generation. The students observe that all the flies in the F_1 generation have red eyes. The students then allow the F_1 flies to mate and produce an F_2 generation. The students will use the F_2 data to perform a chi-square goodness-of-fit test based on a null hypothesis of autosomal recessive inheritance. The observed and expected values for the chi-square goodness-of-fit test are shown in Table 1.

Table 1. The observed and expected values for a chi-square goodness-of-fit test

Phenotype	Observed	Expected
Red-eyed female	497	375
Red-eyed male	254	375
White-eyed female	0	125
White-eyed male	249	125

The students plan to use a significance level of $p = 0.01$. Which of the following is the most appropriate critical value for the students to use in their chi-square goodness-of-fit test?

(A) 7.82

(B) 11.34 ✓

(C) 13.28

(D) 326.7

Answer B

Correct. The critical value for a significance level of $p = 0.01$ and 3 degrees of freedom is 11.34. A calculated chi-square value greater than 11.34 will allow the students to reject the null hypothesis of autosomal recessive inheritance.

93. A human cell in prophase of mitosis can be distinguished from a human cell in prophase I of meiosis by the presence of

(A) kinetochores in the mitotic cell

(B) a spindle in the mitotic cell

(C) twice as many chromosomes in the meiotic cell

(D) half as many chromosomes in the meiotic cell

(E) paired homologous chromosomes in the meiotic cell ✓

5 Heredity MCQ

94. For sexually reproducing diploid parent cells, which of the following statements best explains the production of haploid cells that occurs in meiosis but not in mitosis?

(A) Separation of chromatids occurs once, and there is one round of cell division in meiosis.

(B) Separation of chromatids occurs twice, and there are two rounds of cell division in mitosis.

(C) Separation of chromatids occurs once, and there are two rounds of cell division in meiosis. ✓

(D) Separation of chromatids occurs twice, and there is one round of cell division in mitosis.

Answer C

Correct. Separation of chromatids occurs once in meiosis, and there are two rounds of cell division, which ensures that haploid gamete cells are formed in sexually reproducing diploid organisms.

95. Humans have a diploid number ($2n$) of 46. Which of the following statements best predicts the consequence if meiosis did not occur during gametogenesis?

(A) The gametes would get larger from one generation to the next.

(B) The chromosome number would double with each generation. ✓

(C) The chromosome number would be halved with each generation.

(D) The chromosome number would triple with each generation.

96. Which of the following best describes the cells that result from the process of meiosis in mammals?

(A) They are diploid.

(B) They can be used to repair injuries.

(C) They are genetically different from the parent cell. ✓

(D) They are genetically identical to all the other cells in the body.

(E) They are identical to each other.

97. All of the following are true statements about meiosis in mammals EXCEPT:

(A) It serves as a factor in bringing about variation among offspring.

(B) It follows DNA replication.

(C) It occurs only in reproductive structures.

(D) It produces cells with the haploid number of chromosomes.

(E) It produces four genetically identical gametes. ✓

5 Heredity MCQ

98. Table 1 shows the stage and number of cells and chromosomes per cell at the end of the stage in a $2n = 24$ organism.

Table 1. Cell and chromosome count during selected phases of meiosis

Stage	Number of Cells	Number of Chromosomes per Cell
Prophase I	1	24
Metaphase I	1	24
Anaphase I	1	24
Telophase I	1	24
Beginning of Prophase II	2	12

Which of the following statements correctly describes the chromosomes in each daughter cell at the end of meiosis I ?

(A) Each daughter cell contains 12 chromatids. Each chromatid is one of two from a single chromosome with the other one of the pair found in the other daughter cell.

(B) Each daughter cell contains 12 chromosomes, each composed of two chromatids. Since the chromosomes were randomly divided, one daughter cell may contain both of a pair of homologous chromosomes, while the other cell contains both of another pair of homologous chromosomes.

(C) Each daughter cell contains 12 chromosomes, each composed of two chromatids. Each chromosome is one of a pair of homologous chromosomes from the parent cell, with the other homologue found in the other daughter cell. ✓

(D) Each daughter cell contains 24 separate chromatids. Since every two chromatids were originally joined, forming one homologous chromosome, the number of chromatids is divided by two to determine the number of chromosomes.

Answer C

Correct. The described distribution of chromosomes ensures that each daughter cell contains a complete haploid genome. Each chromosome still consists of two chromatids, formed during the S phase of the cell cycle. Chromatids will separate into two daughter cells during meiosis II.

99. Both mitosis and meiosis begin with a parent cell that is diploid. Which of the following best describes how mitosis and meiosis result in daughter cells with different numbers of chromosomes?

5 Heredity MCQ

(A) In mitosis, the chromosomes consist of a single chromatid, which is passed to two haploid daughter cells. In meiosis, the chromosomes consist of two chromatids during the first round of division and one chromatid during the second round of division, resulting in two haploid daughter cells.

(B) In mitosis, synapsis of homologous chromosomes results in four haploid daughter cells after one division. In meiosis, synapsis of homologous chromosomes occurs during the second division and results in four diploid daughter cells.

(C) Mitosis produces one identical daughter cell after one round of division. Meiosis has two rounds of division and doubles the number of chromosomes in the second round of division, producing four diploid cells.

(D) Mitosis produces two identical diploid daughter cells after one round of division. Meiosis produces four haploid daughter cells after two rounds of division. ✓

Answer D

Correct. The function of mitosis is to produce identical daughter cells, while the function of meiosis is to produce gametes. Meiosis I reduces the chromosome number, and meiosis II separates the sister chromatids in the remaining chromosomes.

100. Which of the following best explains a distinction between metaphase I and metaphase II?

(A) The nuclear membrane breaks down during metaphase I but not during metaphase II.

(B) Chromosomes align at the equator of the cell during metaphase II but not during metaphase I.

(C) The meiotic spindle is needed during metaphase I but not during metaphase II.

(D) Homologous pairs of chromosomes are aligned during metaphase I, but individual chromosomes are aligned during metaphase II. ✓

Answer D

Correct. During metaphase I, tetrads (homologous pairs of chromosomes) are aligned along the metaphase plate. During metaphase II, however, single chromosomes (each composed of two chromatids) are aligned along the metaphase plate.

101. Which of the following occurs in both meiosis and mitosis?

(A) Separation of homologous chromosomes

(B) Reduction of chromosome number

(C) Separation of sister chromatids ✓

(D) Crossing-over between homologous chromosomes

(E) Production of four daughter cells

5 Heredity MCQ

102. Which of the following occurs during mitosis but not during meiosis I ?

(A) The chromosomes are pulled to opposite poles of the spindle apparatus.

(B) The chromatids of each chromosome are separated. ✓

(C) The nuclear envelope breaks down.

(D) Both synapsis and crossing-over take place.

(E) The diploid number of chromosomes is reduced to the haploid number.

103. *Saccharomyces cerevisiae* is a diploid yeast species that can reproduce either sexually or asexually. An experiment was performed to induce mitotically dividing *S. cerevisiae* cells in G_2 to undergo meiosis. Which of the following best describes the steps these cells will follow to form gametes?

(A) The first division will result in crossing over between homologous chromosomes, and the second division will reduce the original number of chromosomes by half in the daughter cells.

(B) The first division will reduce the number of chromosomes by half for each daughter cell, and the second division will result in each daughter cell having one-fourth of the original number of chromosomes.

(C) The first division will move single chromatids to each daughter cell, and the second division will double the number of chromosomes in each daughter cell.

(D) The first division will reduce the number of chromosomes by half for each daughter cell, and the second division will move single chromatids to each daughter cell. ✓

Answer D

Correct. The homologous pairs of chromosomes separate in meiosis I, resulting in two haploid daughter cells. The chromosomes had replicated prior to G_2 and consist of two sister chromatids. The chromatids separate during meiosis II, resulting in a total of four haploid cells.

5 Heredity MCQ

104.

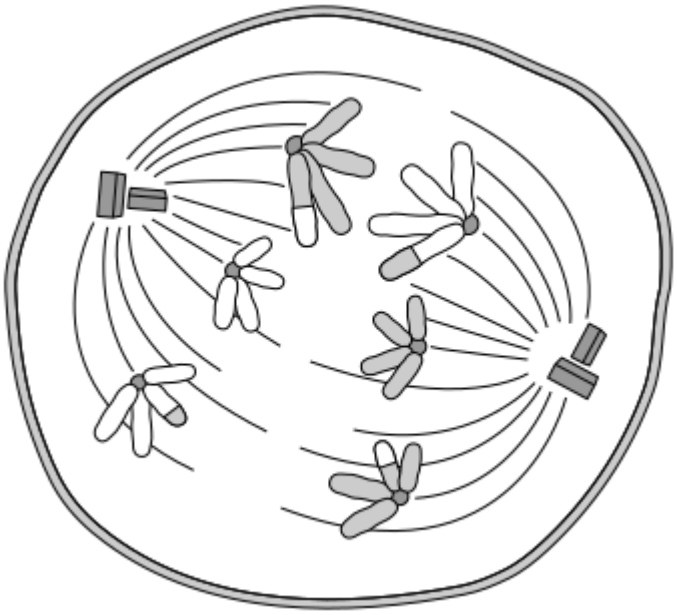


Figure 1. A cellular process

Which of the following best describes the cellular process illustrated in Figure 1 ?

(A) Sister chromatids separating during anaphase of mitosis

(B) Chromosomes lining up along the midline of the cell during mitosis

(C) Reducing the chromosome number during anaphase I of meiosis ✓

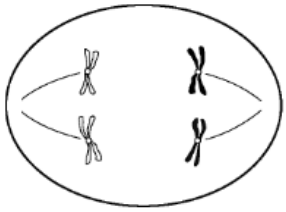
(D) Chromatids failing to separate during meiosis

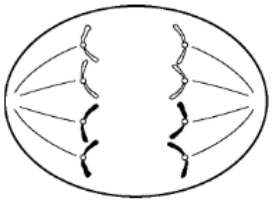
Answer C

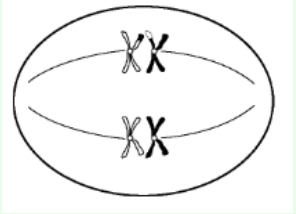
Correct. The fact that each chromosome is composed of a pair of chromatids, some of the chromatids have undergone crossing-over, indicates that these are homologous chromosomes in meiosis. The movement of the chromosomes toward opposite poles supports the claim that this is an illustration of anaphase I.

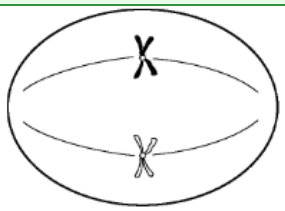
105. Within a forest ecosystem, there is a large amount of diversity among members of a warbler species. Of the following stages of meiosis illustrated for a typical cell, which contributes most to diversity among the warblers?

5 Heredity MCQ

(A) 

(B) 

(C)  ✓

(D) 

106. In most vertebrates, the sperm cell normally contributes which of the following to the new organism?

(A) Many mitochondria

(B) Significant amounts of RNA

(C) A haploid complement of chromosomes ✓

(D) Most of the cytoplasm of the zygote

(E) Two sex chromosomes

5 Heredity MCQ

107.

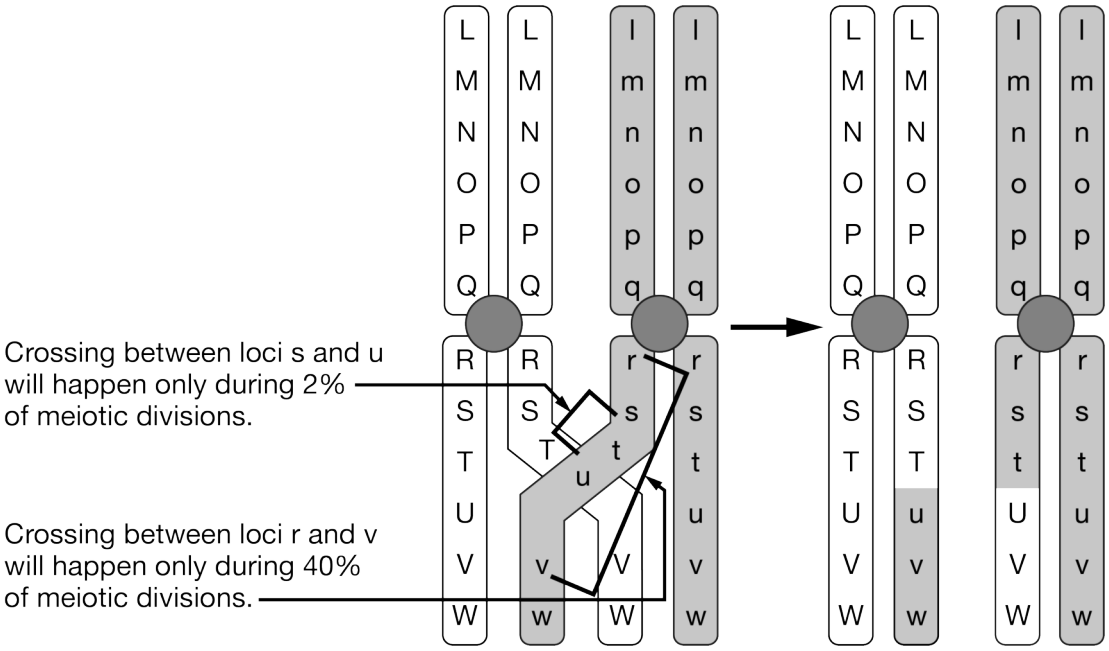


Figure 1. Model of crossing over between homologous chromosomes, indicating crossing over rate of selected loci.

During prophase I replicated homologous chromosomes pair up and undergo synapsis. What testable question is generated regarding synapsis and genetic variability by Figure 1 ?

(A) Is the distance between two gene loci related to crossover rate? ✓

(B) Does crossing over occur more often in some chromosomes than in others?

(C) Is crossing over inhibited by methylation?

(D) Is crossing over promoted by methylation?

Answer A

Correct. During meiosis I, homologous, non-sister chromatids exchange genetic material via a process called crossing over (recombination). The likelihood that a recombination event will separate two genes on the same chromosome increases as the distance between the two gene loci increases.

5 Heredity MCQ

108. A model of crossing over during gamete formation is shown in Figure 1.

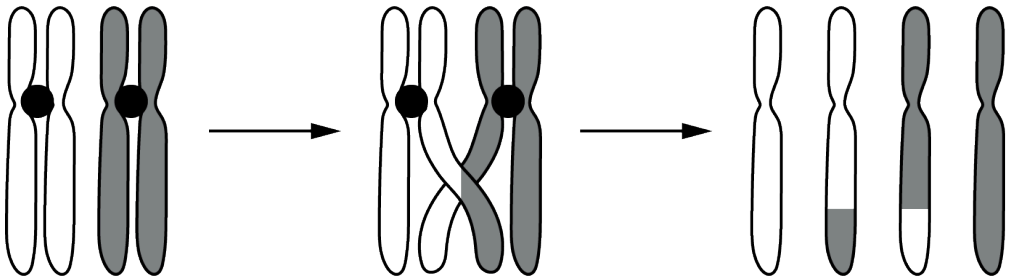


Figure 1. Model of crossing over during meiosis

Based on Figure 1, which of the following questions could best be addressed?

(A) Does synapsis of homologous chromosomes in the parent cell contribute to an increase in genetic diversity in the daughter cells? ✓

(B) Do sister chromatids separate and form diploid daughter cells?

(C) Do chromatids from nonhomologous chromosomes rearrange to produce identical daughter cells?

(D) Does synapsis of nonhomologous chromosomes produce daughter cells that are identical to the parent cell?

Answer A

Correct. Based on the model, synapsis of homologous chromosomes allows the opportunity for crossing over to occur, which can result in chromosomal rearrangement and increased genetic variation in the daughter cells.

109. Scientists have found that DNA methylation suppresses crossing-over in the fungus *Ascobolus immersus*. Which of the following questions is most appropriately raised by this specific observation?

(A) Is the level of genetic variation in the gametes related to the amount of DNA methylation observed? ✓

(B) Without crossing-over, will gametes be viable and be able to produce zygotes?

(C) Does DNA methylation result in shorter chromosomes?

(D) Is this species of fungus a diploid organism?

Answer A

Correct. DNA methylation is shown to suppress crossing-over, which is an important source of genetic

5 Heredity MCQ

variation. Thus, it is reasonable to investigate how the level of genetic variation in gametes being produced is related to how much DNA has been methylated.

110. In anaphase I of meiosis, cohesion between the centromeres of sister chromatids is maintained while homologous chromosomes migrate to opposite poles of the cell along the meiotic spindle as represented in Figure 1.

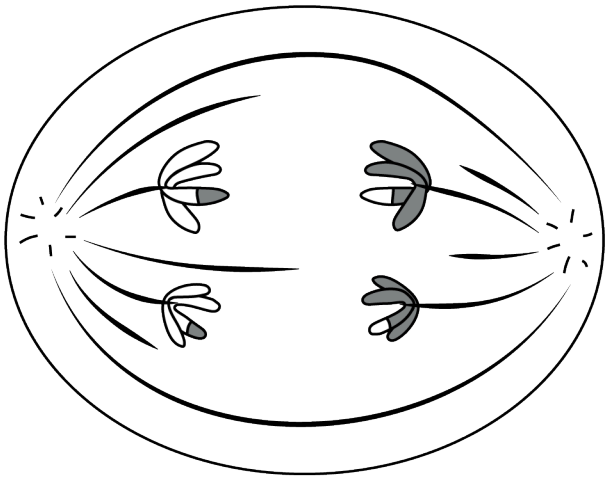


Figure 1. Migration of homologous chromosomes during anaphase I of meiosis

A compound that prevents the separation of the homologous chromosomes in anaphase I is being studied. Which of the following questions can be best answered during this study?

(A) Will the cells produced at the end of meiosis still be genetically identical to each other in the presence of this compound?

(B) Will the long-term development of the individual be affected by this meiotic error?

(C) When do the centrosomes start to move apart during meiosis I as compared to meiosis II?

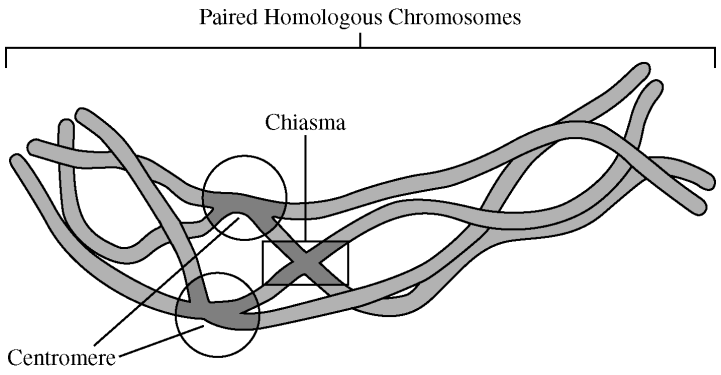
(D) Is there a pattern to the movement of homologous chromosomes in the presence of this compound? ✓

Answer D

Correct. The compound prevents separation of homologous chromosomes, meaning that both the maternal and paternal chromosomes will migrate along the spindle fibers together. Patterns and other aspects of the movement of these homologous chromosomes in the presence of this compound could be recognized effectively during this study.

5 Heredity MCQ

111.



The process depicted in the image above is best summarized by which of the following descriptions?

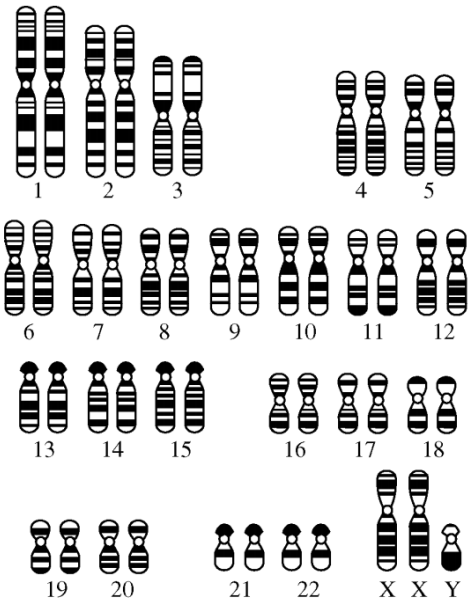
(A) During the synthesis phase of the cell cycle, DNA molecules replicate to generate identical daughter cells.

(B) Centromeres align specific gene sequences of homologous chromosomes during mitotic divisions.

(C) The spindle apparatus attaches at chiasma during metaphase of mitosis.

(D) During meiosis, crossing over leads to recombination of alleles between homologous chromosomes. ✓

112.



The diagram above depicts a karyotype of an individual human. Which of the following statements concerning the karyotype in the diagram is true?

(A) The diagram illustrates a genetic condition found in females.

(B) The diagram indicates a mechanism for increasing genetic diversity in subsequent generations.

(C) The diagram illustrates the results of nondisjunction during gamete formation. ✓

(D) The diagram indicates Down syndrome, a genetic condition.

5 Heredity MCQ

113. A model showing two possible arrangements of chromosomes during meiosis is shown in Figure 1.

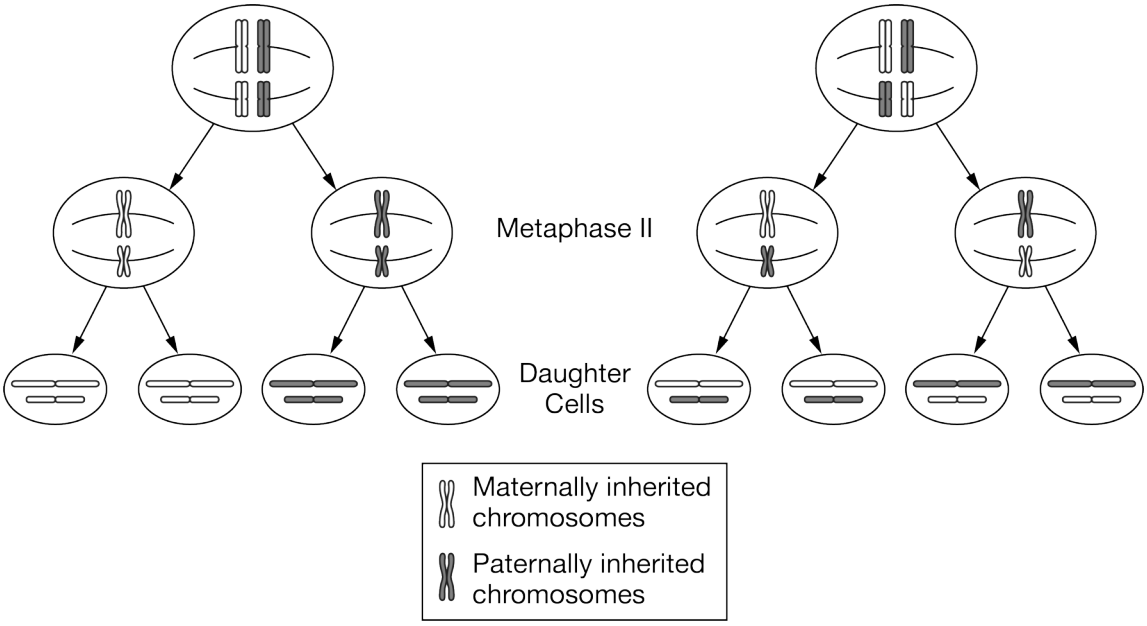


Figure 1. Two possible arrangements of chromosomes during meiosis

Which of the following questions about genetic diversity could most appropriately be answered by analysis of the model in Figure 1 ?

(A) Does crossing-over generate more genetic diversity than the fusion of gametes does?

(B) Does DNA methylation prevent independent assortment during metaphase II?

(C) How does the independent assortment of the two sets of homologous chromosomes increase genetic diversity? ✓

(D) Do daughter cells that are not genetically identical to parent cells produce viable zygotes?

Answer C

Correct. By mixing the maternally and paternally inherited chromosomes in the parent and passing different combinations on to each daughter cell or gamete, the number of combinations of alleles in the gametes is increased. Thus, the gametes have a greater potential to increase genetic diversity in future generations.

114. If 2n = 48 for a particular cell, then the chromosome number in each cell after meiosis would be

5 Heredity MCQ

(A) 96
(B) 48
(C) 24 ✓
(D) 12
(E) 6

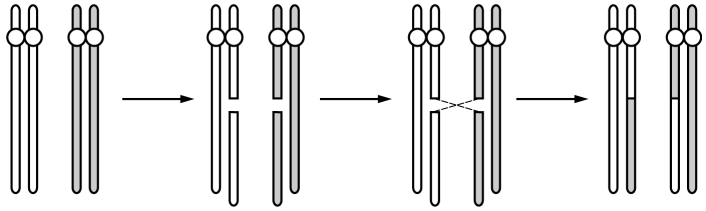
115. The events listed below generally take place during meiosis.

I. Synapsis occurs.
II. Crossing-over is completed.
III. Condensation of chromosomes begins.
IV. Separation of homologous chromosomes begins.

Which of the following is the correct sequence of these events?

(A) I, II, III, IV
(B) II, I, III, IV
(C) III, I, II, IV ✓
(D) III, IV, II, I
(E) IV, II, III, I

116.



The diagram above illustrates which of the following processes?

(A) Crossing-over ✓
(B) Base pair substitution
(C) Duplication
(D) Deletion
(E) Posttranscriptional processing

117. In a breeding pair of eukaryotic organisms, no crossing over occurs between chromatids of homologous chromosome pairs during either egg formation or sperm formation. Which of the following best explains the genetic makeup of the offspring produced when eggs are fertilized by sperm in this breeding pair?

5 Heredity MCQ

(A) All offspring will be genetically identical to the parent that has the greatest number of dominant traits.

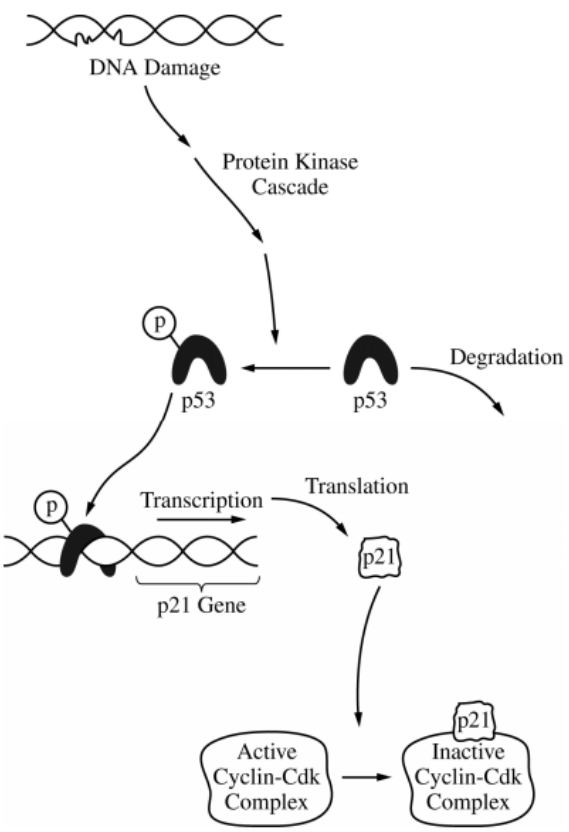
(B) Half of the offspring will be genetically identical to one parent and the other half will be genetically identical to the other parent.

(C) All offspring will be genetically different from each other and from the parents. ✓

(D) The zygotes from fertilization will not develop into any offspring because of nondisjunction.

6 Gene Expression and Regulation MCQ

1.



The p53 protein regulates a cellular response to DNA damage. Based on the diagram above, which of the following best describes the role of p53 in the response to DNA damage?

(A) Phosphorylated p53 binds to DNA and repairs the damage.

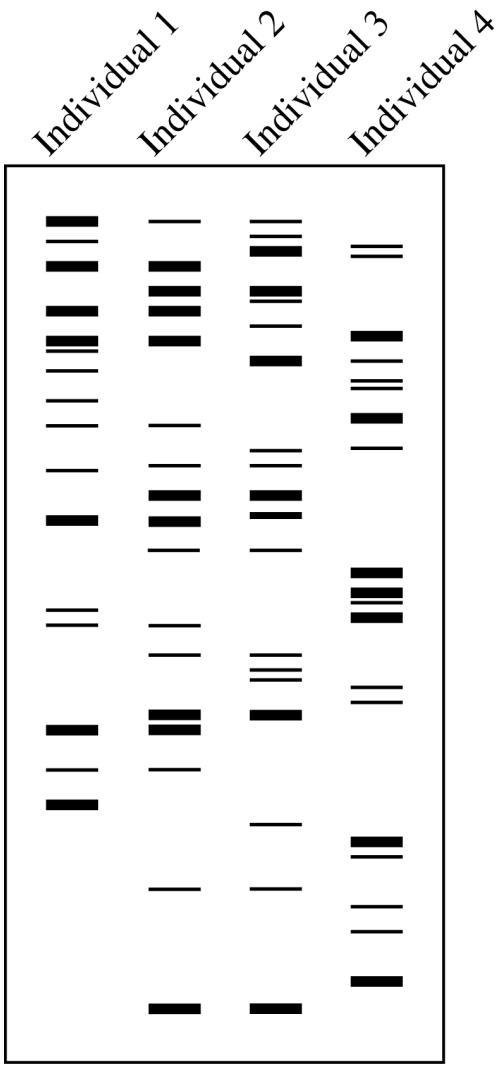
(B) Phosphorylated p53 stimulates transcription of p21, and the resulting p21 protein suppresses cell division until the DNA damage is repaired. ✓

(C) Phosphorylated p53 binds cyclin − Cdk complexes, and the resulting protein complex repairs the DNA damage.

(D) Phosphorylated p53 activates p21 proteins, and the p21 proteins in turn repair the DNA damage.

6 Gene Expression and Regulation MCQ

2.



Samples of DNA were isolated from four different individuals and each sample was digested by the same restriction enzymes. Gel electrophoresis was used to separate the resulting DNA fragments and the results are shown above. These data best support which of the following hypotheses?

(A) Individual 1 is the offspring of 2 and 3.

(B) Individual 1 is the offspring of 3 and 4.

(C) Individual 2 is the offspring of 1 and 3. ✓

(D) Individual 2 is the offspring of 1 and 4.

(E) Individual 3 is the offspring of 1 and 4.

6 Gene Expression and Regulation MCQ

3. A genetic counselor is consulted by a young man who is worried about developing Huntington's disease, an inherited disorder caused by a dominant allele of a single gene. The young man explains that his cousin was recently diagnosed with Huntington's disease, and the news has caused him to consider his own risk of developing the disorder. Which of the following questions will best help the genetic counselor to evaluate the risk of the young man developing Huntington's disease and transmitting it to his children?

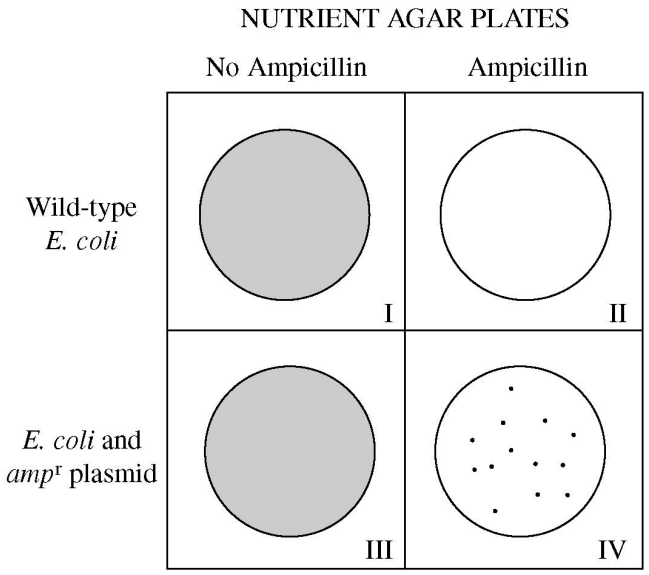
(A) Were you and your cousin born in the same geographical area?

(B) Were your parents or grandparents ever diagnosed with Huntington's disease? ✓

(C) Were you in physical contact with a person diagnosed with Huntington's disease?

(D) Were you ever exposed to substances that are suspected of being mutagens?

In a transformation experiment, a sample of *E. coli* bacteria was mixed with a plasmid containing the gene for resistance to the antibiotic ampicillin (*amp*^{r}). Plasmid was not added to a second sample. Samples were plated on nutrient agar plates, some of which were supplemented with the antibiotic ampicillin. The results of *E. coli* growth are summarized below. The shaded area represents extensive growth of bacteria; dots represent individual colonies of bacteria.



4. Plates that have only ampicillin-resistant bacteria growing include which of the following?

(A) I only

(B) III only

(C) IV only ✓

(D) I and II

5. Which of the following best explains why there is no growth on plate II?

6 Gene Expression and Regulation MCQ

(A) The initial *E. coli* culture was not ampicillin-resistant. ✓

(B) The transformation procedure killed the bacteria.

(C) Nutrient agar inhibits *E. coli* growth.

(D) The bacteria on the plate were transformed.

6. Plates I and III were included in the experimental design in order to

(A) demonstrate that the *E. coli* cultures were viable ✓

(B) demonstrate that the plasmid can lose its amp^r gene

(C) demonstrate that the plasmid is needed for *E. coli* growth

(D) prepare the *E. coli* for transformation

7. Which of the following statements best explains why there are fewer colonies on plate IV than on plate III?

(A) Plate IV is the positive control.

(B) Not all *E. coli* cells are successfully transformed. ✓

(C) The bacteria on plate III did not mutate.

(D) The plasmid inhibits *E. coli* growth.

8. In a second experiment, the plasmid contained the gene for human insulin as well as the amp^r gene. Which of the following plates would have the highest percentage of bacteria that are expected to produce insulin?

(A) I only

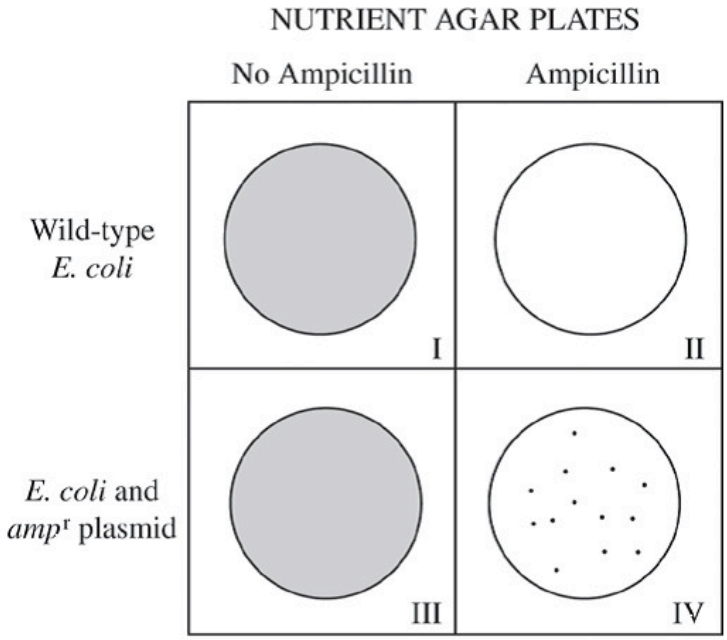
(B) III only

(C) IV only ✓

(D) I and III

6 Gene Expression and Regulation MCQ

In a transformation experiment, a sample of *E. coli* bacteria was mixed with a plasmid containing the gene for resistance to the antibiotic ampicillin (*amp*^{r}). Plasmid was not added to a second sample. Samples were plated on nutrient agar plates, some of which were supplemented with the antibiotic ampicillin. The results of *E. coli* growth are summarized below. The shaded area represents extensive growth of bacteria; dots represent individual colonies of bacteria.



9. Plates that have **only** ampicillin-resistant bacteria growing include which of the following?

(A) I only

(B) III only

(C) IV only ✓

(D) I and II

Answer C

This option is correct. Plate IV shows that only *E. coli* with the plasmid containing the gene for ampicillin-resistant plasmid grew individual colonies with ampicillin in the agar, thus showing the competent cells that transformed successfully.

10. Which of the following best explains why there is no growth on plate II?

6 Gene Expression and Regulation MCQ

(A) The initial *E. coli* culture was not ampicillin-resistant. ✓

(B) The transformation procedure killed the bacteria.

(C) Nutrient agar inhibits *E. coli* growth.

(D) The bacteria on the plate were transformed.

Answer A

This option is correct. The initial wild-type *E. coli* did not contain the plasmid containing the gene for resistance to the ampicillin (*amp*r) and, when exposed to ampicillin, did not grow on Plate II.

11. Plates I and III were included in the experimental design in order to

(A) demonstrate that the *E. coli* cultures were viable ✓

(B) demonstrate that the plasmid can lose its amp^r gene

(C) demonstrate that the plasmid is needed for *E. coli* growth

(D) prepare the *E. coli* for transformation

Answer A

This option is correct. Both Plates I and III had extensive *E. coli* growth, showing that the cells were viable before and after the transformation procedure.

12. Which of the following statements best explains why there are fewer colonies on plate IV than on plate III?

(A) Plate IV is the positive control.

(B) Not all *E. coli* cells are successfully transformed. ✓

(C) The bacteria on plate III did not mutate.

(D) The plasmid inhibits *E. coli* growth.

Answer B

This option is correct. Only the *E. coli* that have been transformed and contain the amp^r gene will grow and produce colonies.

6 Gene Expression and Regulation MCQ

13. In a second experiment, the plasmid contained the gene for human insulin as well as the *amp*^{r} gene. Which of the following plates would have the highest percentage of bacteria that are expected to produce insulin?

(A) I only

(B) III only

(C) IV only ✓

(D) I and III

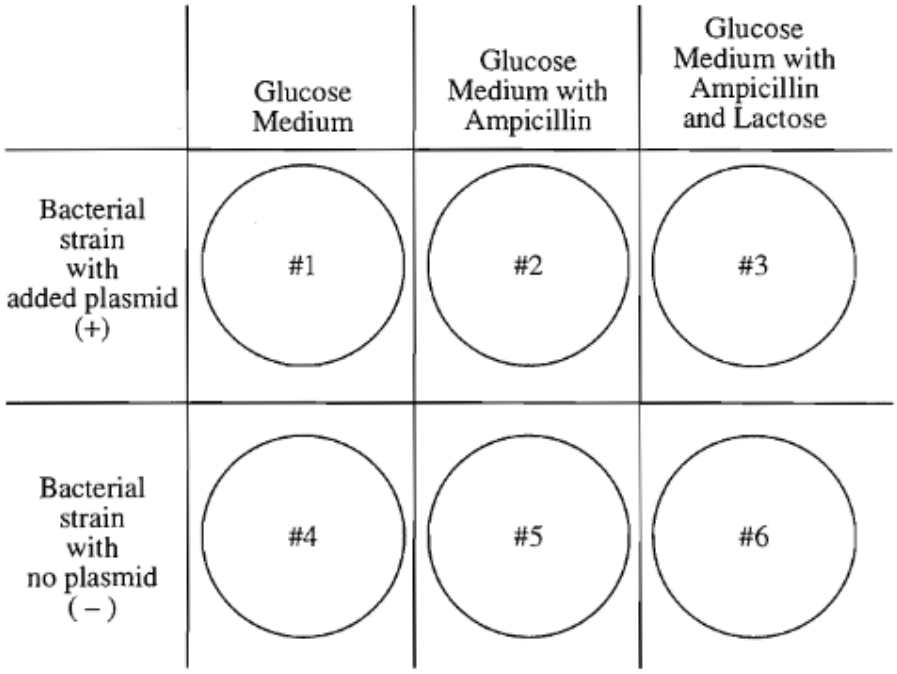
Answer C

This option is correct. In Plate IV the colonies of *E. coli* were successfully transformed and expressed the *amp*^{r} gene; adding another gene to the plasmid would express the human insulin gene the same way.

6 Gene Expression and Regulation MCQ

- **Directions:** Each qroup of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

A scientist is using an ampicillin-sensitive strain of bacteria that cannot use lactose because it has a nonfunctional gene in the *lac* operon. She has two plasmids. One contains a functional copy of the affected gene of the *lac* operon, and the other contains the gene for ampicillin resistance. Using restrictions enzymes and DNA ligase, she forms a recombinant plasmid containing both genes. She then adds a high concentration of the plasmid to a tube of the bacteria in a medium for bacterial growth that contains glucose as the only energy source. This tube (+) and a control tube (-) with similar bacteria but no plasmid are both incubated under the appropriate conditions for growth and plasmid uptake. The scientist then spreads a sample of each bacterial culture (+ and -) on each of the three types of plates indicated below.



14. If no new mutations occur, it would be most reasonable to expect bacterial growth on which of the following plates?

(A) 1 and 2 only
(B) 3 and 4 only
(C) 5 and 6 only
(D) 4, 5, and 6 only
(E) 1, 2, 3, and 4 only ✓

15. If the scientist had forgotten to use DNA ligase during the preparation of the recombinant plasmid, bacterial growth would most likely have occurred on which of the following?

6 Gene Expression and Regulation MCQ

(A) 1 and 2 only

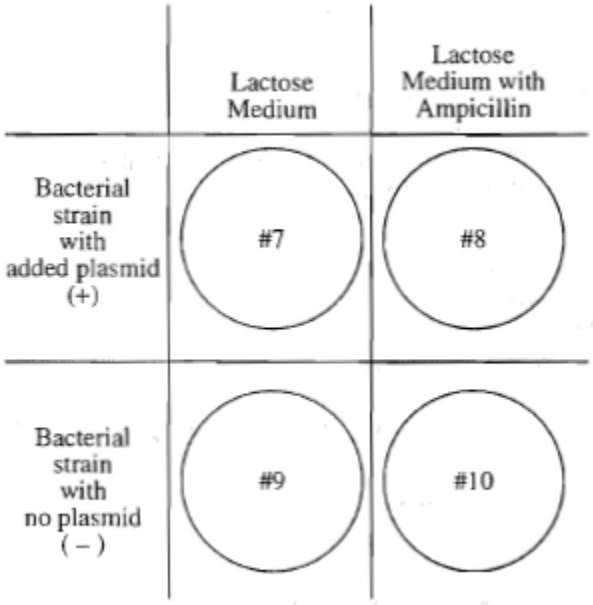
(B) 1 and 4 only ✓

(C) 4 and 5 only

(D) 1, 2, and 3 only

(E) 4, 5, and 6 only

16.



If the scientist used the cultures to perform another experiment as shown above, using medium that contained lactose as the only energy source, growth would most likely occur on which of the following plates?

(A) 10 only

(B) 7 and 8 only ✓

(C) 7 and 9 only

(D) 8 and 10 only

(E) 9 and 10 only

---

17. A new mutation that arose in one copy of gene *X* in a somatic cell resulted in the formation of a tumor. Which of the following pieces of evidence best describes how the new mutation directly caused the tumor?

(A) Protein X normally stimulates cell division, and the mutation created an overactive version of protein X. ✓

(B) Protein X normally activates a growth hormone receptor, and the mutation decreased the stability of protein X.

(C) Protein X normally prevents passage through the cell cycle, and the mutation created an overactive version of protein X.

(D) Protein X normally regulates gene expression, and the mutation created an underactive version of protein X that blocked the cell cycle.

6 Gene Expression and Regulation MCQ

Directions: Each group of questions below concerns an experimental or a laboratory situation. In each case, first study the description of the situation. Then choose the one best answer to each question following it.

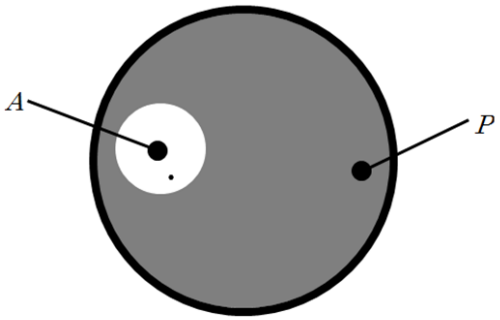


Plate I

A sterile agar plate, I, is streaked with a pure culture of bacteria by means of aseptic techniques. Paper discs treated with the antibiotics Aureomycin (*A*) and penicillin (*P*) are placed at opposite sides of the plate, as shown in the diagram above. The plate is examined after a 24-hour incubation period, and a clear ring is discovered around disc *A*, but not around disc *P*. Within the clear ring around disc *A*, a single bacterial colony with physical characteristics like those of the pure culture is observed. A second sterile agar plate, II, is then streaked with this single colony and also incubated with antibiotics.

18. The single colony found within the clear ring in plate I is most likely made up of the descendants of a bacterial cell that

(A) contaminated the agar plate

(B) contained information conferring resistance to Aureo-mycin ✓

(C) changed its response to Aureomycin as a result of being exposed to the antibiotic

(D) was susceptible to both penicillin and Aureomycin

(E) emigrated from another area of the agar plate

19. Which of the following would most likely be observed in plate II after 24 hours?

(A) A clear ring larger than that around disc *A* in plate I would appear around disc *A* only.

(B) A clear ring larger than that around disc *A* in plate I would appear around disc *P* only.

(C) A clear ring smaller than that around disc *A* in plate I would appear around disc *P* only.

(D) There would be a clear ring around both disc *A* and disc *P*.

(E) There would not be a clear ring around either disc *A* or disc *P*. ✓

6 Gene Expression and Regulation MCQ

20. Antigens are foreign proteins that invade the systems of organisms. Vaccines function by stimulating an organism's immune system to develop antibodies against a particular antigen. Developing a vaccine involves producing an antigen that can be introduced into the organism being vaccinated and which will trigger an immune response without causing the disease associated with the antigen. Certain strains of bacteria can be used to produce antigens used in vaccines.

Which of the following best explains how bacteria can be genetically engineered to produce a desired antigen?

(A) The gene coding for the antigen can be inserted into plasmids that can be used to transform the bacteria. ✓

(B) The bacteria need to be exposed to the antigen so they can produce the antibodies.

(C) The DNA of the antigen has to be transcribed in order for the mRNA produced to be inserted into the bacteria.

(D) The mRNA of the antigen has to be translated in order for the protein to be inserted into the bacteria.

Answer A

Correct. The bacterial plasmid containing the antigen gene would be taken up by bacterial cells through transformation. This gene would be transcribed and translated by the bacteria to produce the desired antigen.

Tay-Sachs disease is a rare inherited disorder caused by an autosomal recessive allele of the *HEXA* gene. Affected individuals exhibit severe neurological symptoms and do not survive to reproductive age. Individuals who inherit one copy of the allele (Tay-Sachs carriers) typically show no symptoms of the disorder. The frequencies of Tay-Sachs carriers in the general population of North America and in three different subpopulations are presented in the table.

FREQUENCY OF TAY-SACHS CARRIERS IN DIFFERENT POPULATIONS

Population	Frequency of Tay-Sachs Carriers
General population	0.004
Subpopulation I	0.037
Subpopulation II	0.035
Subpopulation III	0.020

21. A researcher claims that Tay-Sachs carriers are protected against the infectious disease tuberculosis (TB). Which of the following observations about the annual incidence of tuberculosis in subpopulation II could best be used to support the researcher's claim?

6 Gene Expression and Regulation MCQ

(A) The incidence of TB in subpopulation II is roughly equal to the incidence of TB in the general population.

(B) The incidence of TB in subpopulation II is greater than the incidence of TB in the general population.

(C) The incidence of TB in subpopulation II is lower than the incidence of TB in the general population. ✓

(D) The incidence of TB in subpopulation II is roughly equal to the incidence of Tay-Sachs disease in the general population.

22. Based on the information presented, which of the following best explains the difference in phenotype between Tay-Sachs carriers and homozygous recessive individuals?

(A) Tay-Sachs carriers received a vaccination that homozygous recessive individuals did not receive.

(B) Tay-Sachs carriers inherited an extra chromosome that homozygous recessive individuals did not inherit.

(C) Tay-Sachs carriers have access to a critical nutrient that homozygous recessive individuals did not inherit.

(D) Tay-Sachs carriers synthesize an essential enzyme that homozygous recessive individuals cannot synthesize. ✓

23. Which of the following is an ethical question about Tay-Sachs disease that cannot be answered using scientific methods?

(A) Would a difference in the *HEXA* alleles in subpopulations I and II affect the severity of the neurological symptoms?

(B) Should genetic testing be required before individuals in subpopulation III are advised to not have children? ✓

(C) Could the frequency of Tay-Sachs carriers in subpopulation I be a consequence of a genetic bottleneck?

(D) Should a statistical test be used to evaluate whether the general population is in Hardy-Weinberg equilibrium?

6 Gene Expression and Regulation MCQ

24. Arsenic is a toxic element found in both aquatic and terrestrial environments. Scientists have found genes that allow bacteria to remove arsenic from their cytoplasm. Arsenic enters cells as arsenate that must be converted to arsenite to leave cells. Figure 1 provides a summary of the arsenic resistance genes found in the operons of three different bacteria. *E. coli* R773 is found in environments with low arsenic levels. *Herminiimonas arsenicoxydans* and *Ochrobactrum tritici* are both found in arsenic-rich environments.

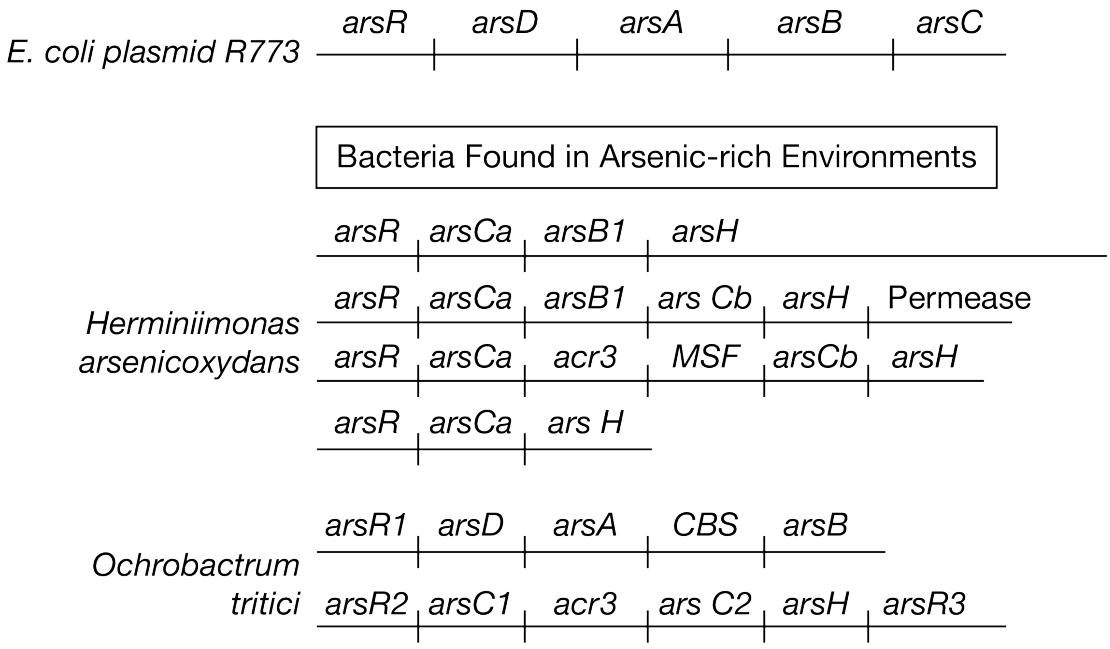


Figure 1. Operons found in three selected bacteria for arsenic removal

Researchers claim that bacteria that live in environments heavily contaminated with arsenic are more efficient at processing arsenic into arsenite and removing this toxin from their cells. Justify this claim based on the evidence shown in Figure 1.

(A) There are multiple operons controlling the production of proteins that process and remove arsenite from cells in both *H. arsenicoxydans* and *O. tritici*. In contrast, *E. coli* has only one operon devoted to arsenic removal. ✓

(B) Both *H. arsenicoxydans* and *O. tritici* contain the $arsR$ gene that codes for a repressor that turns on the operon to eliminate arsenite from the cell.

(C) Both *O. tritici* and *E. coli* contain the $arsD$ gene, which codes for a protein that helps remove arsenite from the cell.

(D) Both *H. arsenicoxydans* and *O. tritici*. have more arsenic resistance genes than has *E. coli*.

Answer A

Correct. Species found in arsenic-rich environments have several operons allowing for several pathways to process and remove arsenite. Also, this allows for a greater capacity of gene regulation.

6 Gene Expression and Regulation MCQ

Ascorbic acid (vitamin C) is an organic molecule necessary for the health of plants and animals. The majority of animals, including most mammals, synthesize ascorbic acid from organic precursors, but some primates are unable to synthesize ascorbic acid and must instead acquire it from dietary sources, such as certain fruits and vegetables.

The *L-gulonolactone oxidase (GULO)* gene encodes an enzyme that catalyzes a required step in the biosynthesis of ascorbic acid. Most mammals carry a functional copy of the *GULO* gene, but some primates carry only a *GULO* pseudogene, which is a nonfunctional variant.

A comparison of *GULO* genes and *GULO* pseudogenes from different animals can provide insight into the evolutionary relatedness of the animals. In Table I, selected members of some mammalian groups are listed, along with an indication of their ability to synthesize ascorbic acid. Table II shows an alignment of amino acid coding sequences from homologous regions of the *GULO* genes and *GULO* pseudogenes of the organisms listed in Table I. Figure 1 represents the universal genetic code.

TABLE I: SELECTED MAMMALIAN GROUPS

Group	Selected Members	Biosynthesis of Ascorbic Acid
Nonprimate mammals	Elephant, mouse	Yes
Primate mammals	Lemur	Yes
	Orangutan, chimpanzee	No
	Human	No

TABLE II: DNA SEQUENCE ALIGNMENT*

	Relative Positions of Nucleotides in Nontemplate (Coding) Sequence																									
	1 (5′)	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27 (3′)
Elephant	G	A	C	A	C	C	C	A	T	C	T	G	A	A	G	A	A	G	T	C	G	G	A	A	T	A	C
Mouse	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	A	G	T	C	T	G	A	G	T	A	C
Lemur	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	G	G	T	C	C	G	A	G	T	A	C
Orangutan	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Chimp	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Human	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C

*For each DNA segment, the alternating shaded and unshaded nucleotides indicate the triplet codons of the open reading frame, shown from left (5′) to right (3′) as the nontemplate (coding) strand. An "–" indicates the absence of a nucleotide.

6 Gene Expression and Regulation MCQ

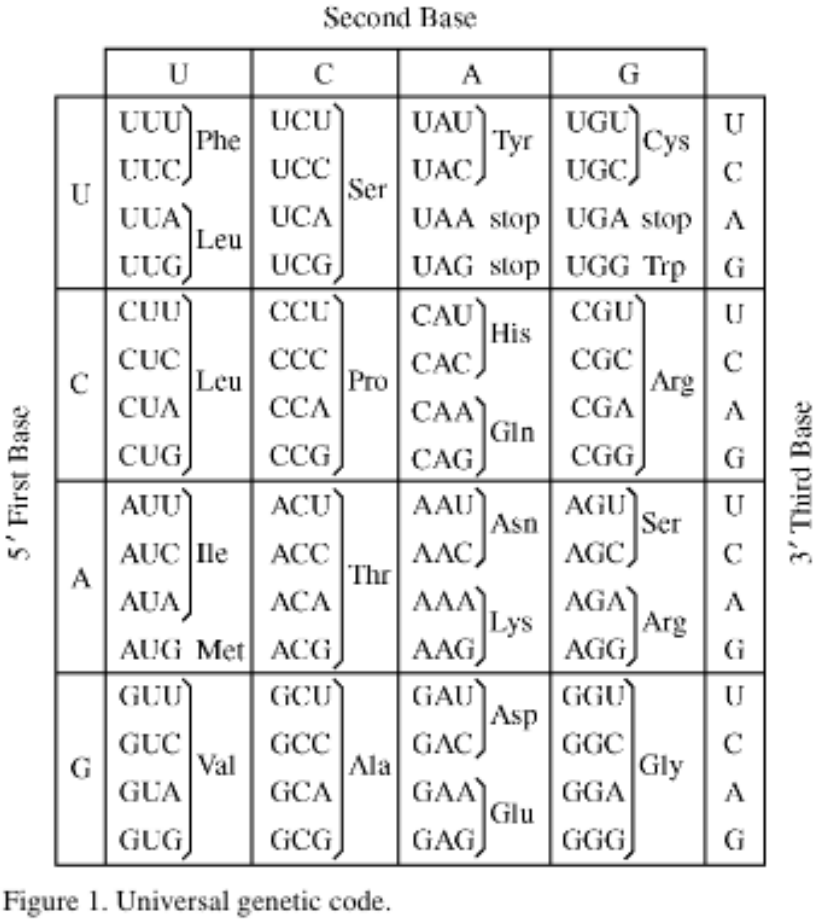


Figure 1. Universal genetic code.

25. Comparison of DNA sequences in Table II suggests that a functional *GULO* gene in lemurs can have a G, C, or T at position 21 but only a G at position 22. Which of the following pairs of predictions is most helpful in explaining the discrepancy?

6 Gene Expression and Regulation MCQ

(A)

A substitution at position 21 would result in	A substitution at position 22 would result in
No change to the protein	A premature stop codon or an amino acid with different biochemical characteristics

✓

(B)

A substitution at position 21 would result in	A substitution at position 22 would result in
A different amino acid	A premature stop codon or an amino acid with different biochemical characteristics

(C)

A substitution at position 21 would result in	A substitution at position 22 would result in
No change to the protein	A frame shift producing an inactive protein

(D)

A substitution at position 21 would result in	A substitution at position 22 would result in
An amino acid with different biochemical characteristics	No transcription of the gene

26. Which of the following is the most likely effect of the mutation at nucleotide position 7 in the *GULO* gene of humans?

6 Gene Expression and Regulation MCQ

(A) The mutation results in the deletion of the *GULO* gene, so no polypeptide can be translated.

(B) The deletion of the single nucleotide causes a frame shift, changing the primary structure downstream of the mutation and resulting in a nonfunctional protein. ✓

(C) The point mutation causes a substitution of the amino acid isoleucine (Ile)for histidine (His) at position 7, resulting in a protein with higher than normal activity.

(D) The substitution of a single nucleotide in the *GULO* coding region results in a stop codon. This results in a smaller nonfunctional protein.

27. Lemurs are primates that live on the island of Madagascar off the coast of Africa. Lemurs have a functional *GULO* gene and are able to produce ascorbic acid. However, primates that live in other places (e.g., humans, chimpanzees, and orangutans) have a *GULO* pseudogene and are unable to produce ascorbic acid. Which of the following best explains the genetic variation among primate species?

(A) Food sources on the island of Madagascar were deficient in ascorbic acid, which caused the primates to migrate to the mainland where ascorbic acid–rich foods are more widely available.

(B) Human activity on Madagascar, including poaching, drove the other primates into extinction on the island.

(C) Food sources where non-Madagascar primates lived provided ample ascorbic acid in the diet, which removed the selective pressure for maintaining a functional *GULO* gene. ✓

(D) Food sources on the island of Madagascar were deficient in ascorbic acid, which caused lemurs to adapt by developing the ability to produce an active GULO protein.

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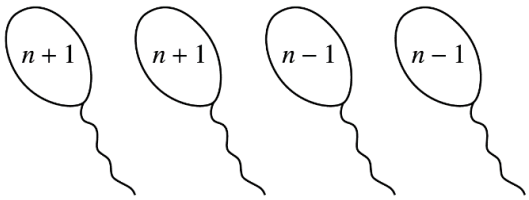
28. A spermatocyte produces the following four sperm cells.



These cells are the result of nondisjunction during which of the following phases?

(A) Interphase (G1 or G2)

(B) Interphase (S)

(C) Mitosis

(D) Meiosis I ✓

(E) Meiosis II

29. Which of the following is a correct statement about mutations?

6 Gene Expression and Regulation MCQ

(A) They are a source of variation for evolution. ✓

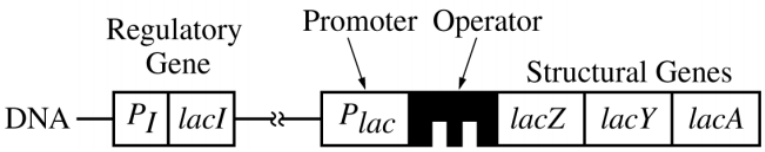
(B) They drive evolution by creating mutation pressures.

(C) They are irreversible.

(D) They occur in germ cells but not in somatic cells.

(E) They are most often beneficial to the organisms in which they occur.

LAC OPERON STRUCTURE



The functions of the loci of the *lac* operon shown in the diagram are described in the table below.

Locus	Function
P_I	Attachment site for RNA polymerase
lacI	Encodes a repressor protein that prevents transcription of the structural genes of the *lac* operon
P_{lac}	Attachment site for RNA polymerase
Operator	Binding site for the repressor protein
lacZ	Encodes beta-galactosidase, the enzyme that digests lactose to glucose and galactose
lacY	Encodes lactose permease, the channel through which lactose moves into the cell
lacA	Encodes galactoside transacetylase

The diagram above represents a segment of the *E. coli* chromosome that contains the *lacI* gene and part of the *lac* operon, a coordinately regulated set of genes that are required for the metabolism of lactose. The presence of lactose, which causes the repressor to be released from the operator, results in increased transcription of the *lac* operon.

30. Bacterial cells that contain green fluorescent protein (GFP) will fluoresce under ultraviolet light. Which of the following is the most likely outcome of replacing the *lacZ* gene in the *E. coli lac* operon with the gene encoding GFP?

(A) Bacteria growing in the presence of lactose will fluoresce under ultraviolet light. ✓

(B) Beta-galactosidase will be made only when bacteria are cultured under ultraviolet light.

(C) Ultraviolet light will cause a bond to form between glucose and galactose monomers.

(D) Ultraviolet light will cause a duplication of the *lac* operon.

31. Which of the following best explains the contribution of the *lac* operon to the metabolic efficiency of a bacterial cell?

6 Gene Expression and Regulation MCQ

(A) Expression of the *lacI* gene requires lactose.

(B) RNA polymerase is rapidly degraded by the product of the *lacP* locus.

(C) The repressor binds to DNA only when the cellular concentration of glucose is low.

(D) The *lacZ* gene is highly expressed only when lactose is available. ✓

32. Which of the following is the most likely consequence of a mutation at the operator locus that prevents binding of the repressor protein?

(A) Expression of the structural genes will be repressed, even in the presence of lactose.

(B) Beta-galactosidase will be produced, even in the absence of lactose. ✓

(C) RNA polymerase will attach at the P_{lac} locus, but transcription will be blocked.

(D) The operator locus will code for a different protein and thereby prevent transcription of the structural gene.

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33. Living cells typically have biosynthetic pathways to synthesize at least some of the amino acids used in making proteins. Some strains of *E. coli*, a prokaryote, can synthesize the amino acid tryptophan, while other *E. coli* strains cannot. Similarly, some strains of the yeast *S. cerevisiae*, a eukaryote, can synthesize tryptophan, while other *S. cerevisiae* strains cannot.

Which of the following describes the most likely source of genetic variation found in the tryptophan synthesis pathways of both species?

(A) Exchange of genetic information occurs through crossing over.

(B) Viral transmission of genetic information required to synthesize tryptophan occurs.

(C) Random assortment of chromosomes leads to genetic variation.

(D) Errors in DNA replication lead to genetic variation. ✓

Answer D

This option is correct. The most likely source of genetic variation found in the tryptophan synthesis pathways of both species would have occurred primarily with the imperfect nature of DNA replication, which could have led to an increase in genetic variation in both species.

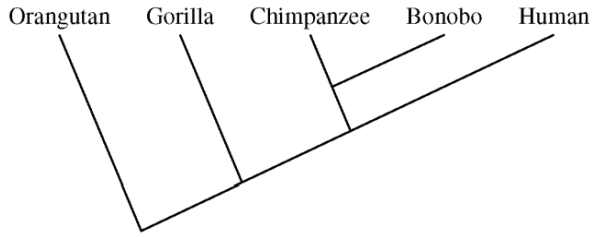
6 Gene Expression and Regulation MCQ

The *TAS2R38* gene encodes a receptor protein that influences the ability to taste bitterness. The gene has two alleles: a dominant, wild-type allele that enables an individual (taster) to taste bitterness and a recessive, mutant allele that interferes with the ability of an individual (nontaster) to taste bitterness. Three single nucleotide mutations in the coding region of the *TAS2R38* gene are associated with the nontaster allele. The nucleotides present at the three positions are shown in the table below.

NUCLEOTIDE VARIATION IN THE *TAS2R38* GENE

	Position in the Nucleotide Sequence		
	145	785	886
Human nontaster	G	T	A
Human taster	C	C	G
Bonobo	C	C	G
Chimpanzee	C	C	G
Gorilla	C	C	G

A cladogram representing the evolutionary relatedness of selected primates is shown below.



34. The TAS2R38 receptor protein has been detected on the surface of cells from individuals who are homozygous for the nontaster allele of the *TAS2R38* gene. Which of the following is the most likely effect of the mutations associated with the nontaster allele on *TAS2R38* gene expression?

(A) The mutations change the primary structure of the encoded receptor protein. ✓

(B) The mutations increase the stability of the *TAS2R38* mRNA.

(C) The mutations prevent transcription of the *TAS2R38* gene.

(D) The mutations prevent translation of the *TAS2R38* mRNA.

6 Gene Expression and Regulation MCQ

The following questions refer to the following DNA strand and table of codons.

	3′ TAG	TTC	AAA	CCG	CGT	AAC	ATT 5′
Triplet Number	1	2	3	4	5	6	7

Second Letter

First Letter (5′ End)	U	C	A	G	Third Letter (3′ End)
U	UUU phe	UCU ser	UAU tyr	UGU cys	U
	UUC phe	UCC ser	UAC tyr	UGC cys	C
	UUA leu	UCA ser	UAA stop	UGA stop	A
	UUG leu	UCG ser	UAG stop	UGG trp	G
C	CUU leu	CCU pro	CAU his	CGU arg	U
	CUC leu	CCC pro	CAC his	CGC arg	C
	CUA leu	CCA pro	CAA gln	CGA arg	A
	CUG leu	CCG pro	CAG gln	CGG arg	G
A	AUU ile	ACU thr	AAU asn	AGU ser	U
	AUC ile	ACC thr	AAC asn	AGC ser	C
	AUA ile	ACA thr	AAA lys	AGA arg	A
	AUG met	ACG thr	AAG lys	AGG arg	G
G	GUU val	GCU ala	GAU asp	GGU gly	U
	GUC val	GCC ala	GAC asp	GGC gly	C
	GUA val	GCA ala	GAA glu	GGA gly	A
	GUG val	GCG ala	GAG glu	GGG gly	G

35. Which of the following modifications of the DNA would produce the greatest change in the primary structure of the polypeptide chain?

(A) Deleting the first T in the second triplet ✓

(B) Changing the second triplet to read 3’ CTC 5’

(C) Changing the third triplet to read 3’ AAC 5’

(D) Changing the fourth triplet to read 3’ CCA 5’

(E) Deleting the sixth triplet

36. In which of the following would there NOT be a change in the amino acid sequence of the peptide coded for by this DNA?

(A) Changing 3’ AAA 5’ to read 3’ AAG 5’ ✓

(B) Changing 3’ TTC 5’ to read 3’ ATC 5’

(C) Changing 3’ CCG 5’ to read 3’ GGC 5’

(D) Deleting the first A from 3’ AAA 5’

(E) Deleting the last triplet

6 Gene Expression and Regulation MCQ

37. The mRNA transcribed from the DNA would read

(A) 5’ TAG TTC AAA CCG CGT AAC AAT 3’

(B) 5’ ATC AAG TTT GGC GCA TTG TAA 3’

(C) 5’ AUC AAG UUU GGC GCA UUG UAA 3’ ✓

(D) 5’ AAU CAA UGC GCC AAA CUU GAU 3’

(E) 5’ AUU GUU ACG CGG UUU GAA CUA 3’

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38. Red-green color blindness in humans is caused by a recessive allele located on the X chromosome. Figure 1 shows the potential offspring of a female who is red-green color-blind and a male with full-color vision. All of the possible male offspring would be color-blind, and all of the possible female offspring would have full-color vision.

If during the production of male gametes an error in meiosis occurred, sperm containing both an X and a Y chromosome could be produced.

Potential Offspring With No Error in Meiosis				Potential Offspring With Error in Meiosis		
	X^B	Y				$\mathrm{X}^B\mathrm{Y}$
X^b	$\mathrm{X}^B\mathrm{X}^b$	$\mathrm{X}^b\mathrm{Y}$	$\mathrm{X}^B = \text{Full color vision}$ $\mathrm{X}^b = \text{Red-green colorblind}$	X^b	X^b	$\mathrm{X}^b\mathrm{X}^B\mathrm{Y}$
X^b	$\mathrm{X}^B\mathrm{X}^b$	$\mathrm{X}^b\mathrm{Y}$		X^b	X^b	$\mathrm{X}^b\mathrm{X}^B\mathrm{Y}$

Figure 1. Possible offspring of a female who is red-green color-blind and a male who has full-color vision.

How would the extra chromosome affect the male offspring produced by the gamete?

(A) None of the potential offspring would be male, because the potentially male zygote would have two X chromosomes, and the Y chromosome would be ignored.

(B) The male offspring would all be red-green color-blind, because of interference from alleles on the Y chromosome.

(C) The male offspring would have full-color vision, because of the presence of the extra X chromosome. ✓

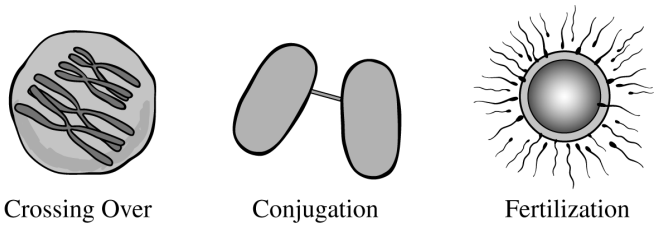
(D) There would be no change to the phenotypes of the possible offspring, because the extra X chromosome would not be active.

Answer C

Correct. The presence of the Y chromosome would trigger the development of a male, and the extra X chromosome would provide an allele for full-color vision, a dominant trait.

6 Gene Expression and Regulation MCQ

39.



Crossing Over Conjugation Fertilization

The processes illustrated in the models depicted above all result in which of the following?

(A) Transcription

(B) An increase in genetic variation ✓

(C) An increase in the chromosome number

(D) Horizontal gene transfer

40. Cystic fibrosis (CF) is a progressive genetic disease that causes persistent lung infections and affects the ability to breathe. CF is inherited in an autosomal recessive manner, caused by the presence of mutations in both copies of the gene for the cystic fibrosis transmembrane conductance regulator (CFTR) protein. Partial nucleotide sequences and the corresponding amino acid sequences for an unaffected individual and an affected individual are modeled in Figure 1.

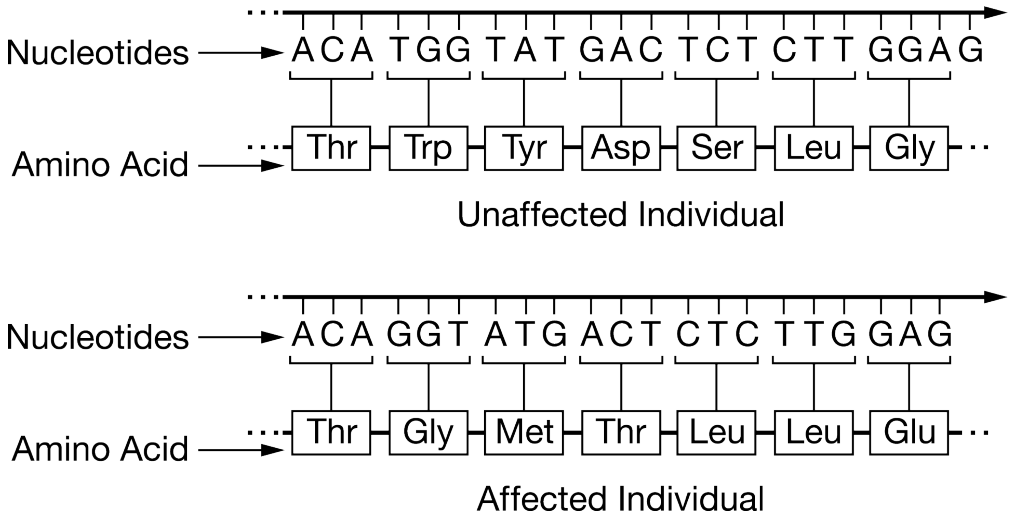


Figure 1. CFTR protein sequences in unaffected and affected individuals

Based on the information in Figure 1, which type of mutation explains the nature of the change in DNA that resulted in cystic fibrosis in the affected individual?

(A) Substitution, because the amino acid tryptophan is replaced with glycine.

(B) Insertion, because an extra guanine is present, which changes the reading frame.

(C) Deletion, because a thymine is missing, which changes the reading frame. ✓

(D) Duplication, because the amino acid leucine occurs twice, which changes the reading frame.

6 Gene Expression and Regulation MCQ

Answer C

Correct. Deletion of the thymine nucleotide results in a frameshift that alters the triplet code sequence and changes the rest of the amino acid sequence.

Figure 1. Domain structure of two proteins

Investigators studied the interactions between two different proteins, protein I and protein II. Each protein has two structural domains (Figure 1) that represent different functional parts of the protein. Different combinations of the protein domains were tested for their ability to bind to a known DNA sequence, interact with each other, and activate transcription. The results are shown in Table 1.

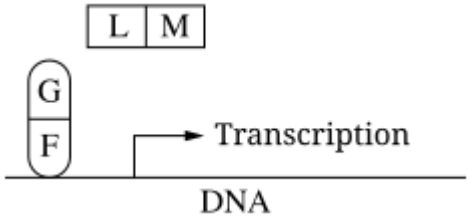
TABLE 1. INTERACTION AND FUNCTION OF COMBINATIONS OF STRUCTURAL VARIANTS OF PROTEINS I AND II

Test	Domain(s) of Protein I Used	Domain(s) of Protein II Used	DNA Binding	Protein-Protein Interaction	Transcription Activation
1	FG	LM	+	+	+
2	FG	None	+	−	−
3	None	LM	−	−	−
4	F	LM	+	−	−
5	G	LM	−	+	−
6	FG	L	+	−	−
7	FG	M	+	+	+

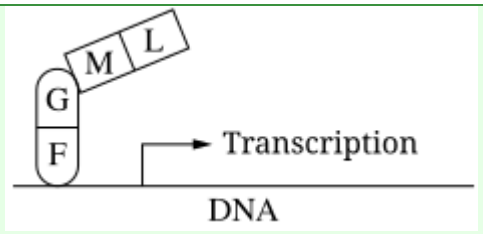
6 Gene Expression and Regulation MCQ

41. Which of the following best models the interaction of protein I and protein II that results in the transcription of target genes?

(A)

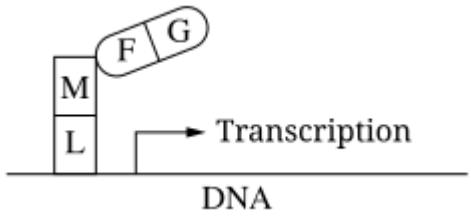


(B)



(C)

(D)



Answer B

Correct. Based on the data for test 4 (in comparison with the data for tests 1, 2, and 5), the F domain of protein I is responsible for DNA binding. The data for tests 5 and 7 (in comparison with the data for tests 4 and 6) demonstrate that the G domain of protein I interacts with the M domain of protein II.

6 Gene Expression and Regulation MCQ

Dystrophin is a protein that is expressed in certain muscle cells. In combination with other cellular proteins, dystrophin strengthens protein fibers in muscle cells to allow muscles to contract without injury.

Nucleotide deletions in the gene that encodes dystrophin are associated with the genetic disorder Duchenne Muscular Dystrophy (DMD). Individuals with DMD do not produce functional dystrophin and, as a result, the protein fibers, and then entire muscle cells, become damaged.

The history of DMD for three generations of a family is shown in Figure 1.

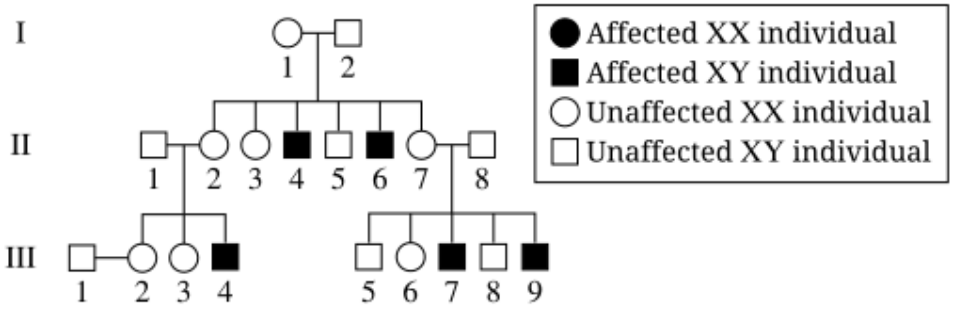


Figure 1. Pedigree showing instances of DMD in a family

Individuals III–1 and III–2 plan to have children and wish to first determine whether individual III–2 is a carrier of DMD. Individual III–2 undergoes genetic testing to determine whether individual III-2 carries a particular allele for the mutated dystrophin that is associated with the disorder in this family. The results of gel electrophoresis analysis of the individual's dystrophin alleles and the alleles of several family members are shown in Figure 2.

6 Gene Expression and Regulation MCQ

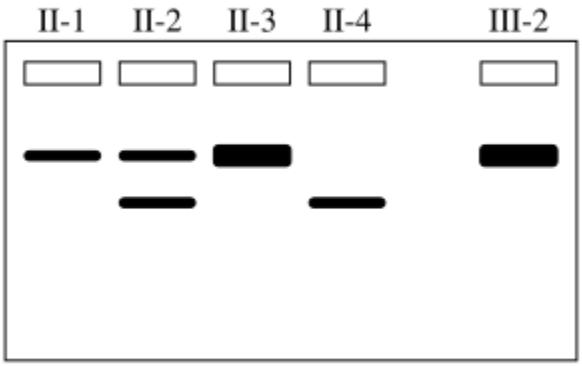


Figure 2. Gel electrophoresis showing the dystrophin alleles present in five members of the family

42. Which of the following best explains how the dystrophin protein is synthesized by certain muscle cells of an individual but not by other cell types?

(A) Each cell has different genes as a result of changes that occur during development and cell specialization.

(B) The presence of different transcription factors in different cell types results in differential gene expression. ✓

(C) A cell can only translate mRNA transcripts that have gene-specific sequences that are recognized by the ribosomes.

(D) The promoter region of the gene is necessary for expression to occur, and this region is only present in certain cell types.

Answer B

Correct. Transcription factors, in general, determine which parts of the genome will be transcribed. Tissue- and cell-specific transcription factors result in differential gene expression that leads to different cell types synthesizing different proteins.

43. Another form of muscular dystrophy, BMD, has a less severe phenotype. Individuals with BMD produce less dystrophin protein or abnormal dystrophin protein with reduced function, whereas individuals with DMD produce no dystrophin protein. Both BMD and DMD are commonly caused by the deletion of a portion of the dystrophin gene.

Which of the following best explains why there is a difference in phenotype between BMD and DMD?

6 Gene Expression and Regulation MCQ

(A) BMD can be caused by the deletion of exons, while DMD is caused by the deletion of introns.

(B) BMD can be caused by a deletion that affects a single amino acid, while DMD is caused by a deletion that alters all the amino acids that follow the mutation. ✓

(C) BMD can be caused by a deletion that removes the transcription start site, while DMD is caused by a deletion that removes the transcription stop signal.

(D) BMD can be caused by the inheritance of two alleles containing deletions, while DMD is caused by the inheritance of one allele containing the deletion and one normal allele.

Answer B

Correct. A single amino acid change may alter the function of the protein but would be expected to be less detrimental than a frameshift mutation that alters every amino acid added after the mutation.

44. Over 2,000 different mutations in the gene encoding dystrophin are associated with different forms of muscular dystrophy. The mutations include deletions of a portion of the gene, duplications of a portion of the gene, and sometimes substitutions of base pairs in certain regions of the gene.

Which of the following correctly predicts the movement of a mutant allele compared with the movement of the normal allele when analyzed by gel electrophoresis?

(A) An allele with base pair substitutions in an exon will travel farther than the normal allele does.

(B) An allele with a duplication in an exon will travel a shorter distance than the normal allele does. ✓

(C) An allele with base pair deletions in an exon will travel the same distance as the normal allele does.

(D) An allele with a duplication in an exon will travel a longer distance than the normal allele does.

Answer B

Correct. An allele with a duplication will contain more nucleotides than in a normal allele. Given the same charge and amount of time, longer DNA sequences with a larger molecular mass will not move as far as shorter DNA sequences will.

6 Gene Expression and Regulation MCQ

45. An evolutionary biologist hypothesizes that two morphologically similar plant species are not closely related. To test the hypothesis, the biologist collects DNA samples from each of the two plant species and then uses restriction enzymes to cut the DNA samples into fragments, which are then subjected to gel electrophoresis. The results are shown in Figure 1.

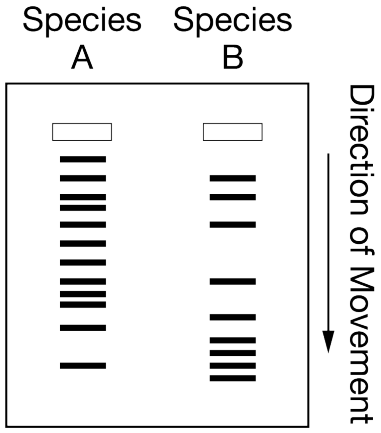


Figure 1. DNA analysis of two species

Given the results shown in Figure 1, which of the following correctly explains a relationship between the two species?

(A) Species B is the ancestor of species A because it has fewer bands.

(B) Species A is more complex than species B because it has more bands.

(C) Species B has more short fragments of DNA than species A does. ✓

(D) Species A has more short fragments of DNA than species B does.

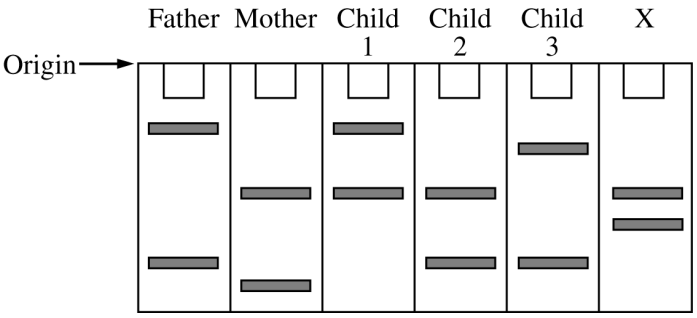
Answer C

Correct. Electrophoresis separates DNA fragments by size and charge. The smaller fragments will move more quickly through the gel. In this example, species B has more fragments that move further through the gel. Thus, species B has more short fragments than species A does.

6 Gene Expression and Regulation MCQ

For following group of questions first study the description of the situation or data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

The diagram below illustrates the results of electrophoresis of DNA sequences obtained from a family of two adults and three children, and amplified using PCR. The bands represent short repeating sequences of variable length. Results for another female (X) are included for comparison.



46. The banding patterns of the DNA fragments reveal that

(A) child 1 and child 2 cannot be biological siblings

(B) child 1 and child 3 probably look like the mother

(C) the mother cannot be the biological parent of all three children ✓

(D) the mother's DNA has the same DNA sequence as the father's DNA

(E) child 2 and child 3 inherited all of their DNA from the father

47. Which of the following is the best explanation for the fragment pattern for individual X ?

(A) She has only one member of this chromosome pair.

(B) She has only one living parent.

(C) She is homozygous for this particular DNA fragment.

(D) She is the mother's child from another marriage. ✓

(E) She is not related to any member of the family being tested.

48. Gel electrophoresis can be used to directly determine which of the following?

(A) The amino acid sequence of a protein

(B) The number of base pairs in a fragment of DNA ✓

(C) The organelle in a cell toward which a protein moves after its initial synthesis

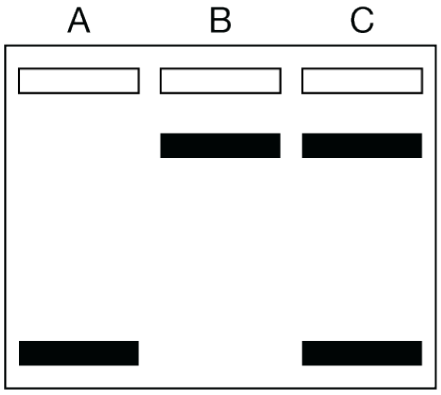
(D) Whether one allele of a gene is more subject to genetic drift than are other alleles of the same gene

6 Gene Expression and Regulation MCQ

Answer B

Correct. By using an electrophoresis gel, the number of base pairs in a fragment of DNA can be estimated by comparing the unknown DNA fragment to a set of standard DNA fragments with a known number of base pairs.

49. Scientists conducted a transformation experiment using *E. coli bacteria* and the pTru plasmid. Samples of the pTru plasmid (lane A) and the chromosomal DNA from two different *E. coli* strains that the scientists attempted to transform (lane B and lane C) were compared using gel electrophoresis. The results are shown in Figure 1.



Lane	Sample
A	pTru
B	*E. coli* strain I
C	*E. coli* strain II

Figure 1. Results of *E. coli* transformation with pTru plasmid

Which of the following statements best explains the experimental results observed in Figure 1 ?

(A) *E. coli* in both lanes B and C have been successfully transformed and contain additional genetic information.

(B) *E. coli* in lane B have been successfully transformed and contain additional genetic information.

(C) *E. coli* in lane C have been successfully transformed and contain additional genetic information. ✓

(D) Which *E. coli* have been transformed cannot be determined from this gel.

Answer C

Correct. A successful transformation would contain a band of DNA from the pTru plasmid and a band of DNA from the *E. coli* chromosome. Lane C has both bands correctly identified and is experimental evidence of a successful bacteria transformation of *E. coli* with the pTru plasmid.

6 Gene Expression and Regulation MCQ

50. Students subjected three samples of five different molecules to gel electrophoresis as shown in Figure 1.

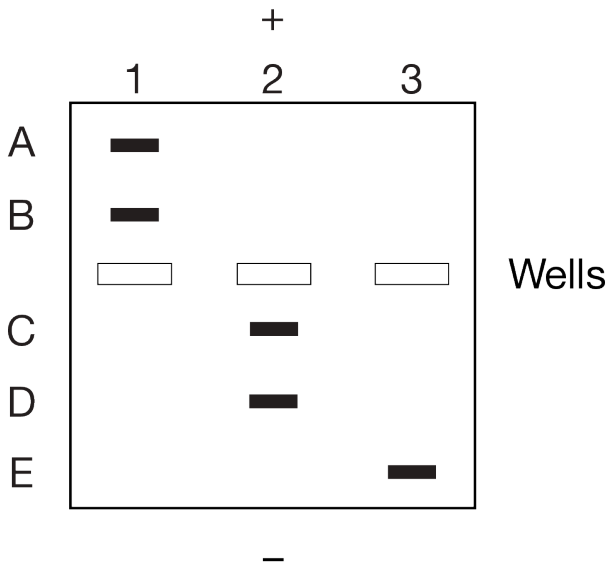


Figure 1. Gel electrophoresis of three prepared samples

Which of the following statements best explains the pattern seen on the gel with regard to the size and charge of molecules A and B?

(A) Molecules A and B are positively charged, and molecule A is smaller than molecule B.

(B) Molecules A and B are positively charged, and molecule A is larger than molecule B.

(C) Molecules A and B are negatively charged, and molecule A is smaller than molecule B. ✓

(D) Molecules A and B are negatively charged, and molecule A is larger than molecule B.

Answer C

Correct. Only negatively charged molecules will move toward the positive electrode. The smaller molecule A will migrate further from the well than the larger molecule B will.

6 Gene Expression and Regulation MCQ

51. Mutations in the *MYO6* and *POU4F3* genes have been associated with a form of hereditary hearing loss in humans. Researchers studying the genes have proposed that *POU4F3* encodes a transcription factor that influences the regulation of *MYO6*.

Which of the following questions will best help guide the researchers toward a direct test of their proposal?

(A) Have mutations in other genes also been associated with hearing loss?

(B) In what types of cells are the mutant forms of the *POU4F3* gene expressed?

(C) Are mutations in the *MYO6* and *POU4F3* genes also found in mice?

(D) Do mutations in the *POU4F3* gene affect *MYO6* mRNA levels in cells? ✓

52. Which of the following is most likely to create genetic variation in a population?

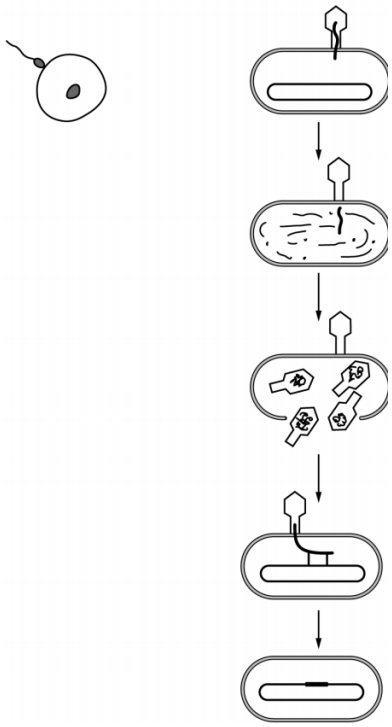
(A) RNA polymerase errors during transcription

(B) Helicase failure to unwind DNA during DNA replication

(C) DNA polymerase errors during replication ✓

(D) Misincorporation of amino acids by tRNA during translation

53.



Which of the following correctly compares the two processes shown above?

(A) Both processes increase genetic variation. ✓

(B) Both processes represent aspects of sexual reproduction.

(C) Both processes require partitioning of genetic material and organelles.

(D) The amount of genetic material per cell remains constant in both processes.

6 Gene Expression and Regulation MCQ

54. Gibberellic acid stimulates the cells of germinating grass seeds to produce mRNA molecules that code for hydrolytic enzymes. In this case the role of gibberellic acid can best be described as that of

(A) a regulator of gene activity ✓

(B) a stimulator of hydrolase secretion

(C) a stimulator of DNA replication

(D) an allosteric activator of hydrolase

(E) an activator of translation

55. Huntington's disease has been traced to the number of CAG repeats in the *HTT* gene, which is located on chromosome 4. The phenotypic influence of individual alleles with different numbers of repeats is shown in Table 1.

Table 1: Effect of CAG repeat number on Huntington's disease expression

CAG Repeats	Expression
< 27	No symptoms
27–35	No symptoms but increased risk for offspring
36–39	May develop symptoms in old age
> 40	Will express symptoms from an early age

Which of the following is most likely the immediate cause of the first appearance of Huntington's disease in a person?

(A) A point mutation occurs in the *HTT* gene.

(B) The first appearance of the CAG repeat occurs in the *HTT* gene.

(C) An allele with more than 39 CAG repeats was inherited by the affected person. ✓

(D) The person inherited two alleles that each contained 20 CAG repeats.

Answer C

Correct. At some point, all alleles for the *HTT* gene contain less than 39 CAG repeats. Only after this threshold is crossed will a person inherit an allele that will cause symptoms of Huntington's disease.

6 Gene Expression and Regulation MCQ

56. Genetic engineering techniques can be used when analyzing and manipulating DNA and RNA. Scientists used gel electrophoresis to study transcription of gene L and discovered that mRNA strands of three different lengths are consistently produced.

Which of the following explanations best accounts for this experimental result?

(A) Gel electrophoresis can only be used with DNA (not mRNA), so experimental results are not interpretable.

(B) RNA polymerase consistently makes the same errors during transcription of gene L.

(C) Gene L is mutated, so RNA polymerase does not always transcribe the correct sequence.

(D) Pre-mRNA of gene L is subject to alternative splicing, so three mRNA sequences are possible. ✓

Answer D

Correct. Excision of different introns and splicing and retention of different exons can generate different versions of the resulting mRNA, which may have different lengths.

57. Insulin is a hormone produced by some pancreatic cells. Scientists have isolated the DNA sequence that codes for human insulin production.

Which of the following best predicts the effect of inserting this gene into the DNA of a bacterial cell?

(A) The recombinant bacterium will produce human insulin using its own transcription and translation machinery. ✓

(B) The recombinant bacterium will not transcribe the human insulin gene because this gene is not normally found in the bacterial genome.

(C) The recombinant bacterium will transcribe the gene but will be unable to translate the mRNA.

(D) The recombinant bacterium will die because it has been exposed to foreign DNA.

Answer A

Correct. DNA is the carrier of genetic information, and its insertion into the bacterium is sufficient to produce human insulin. The bacterium can transcribe the gene with RNA polymerase and translate the mRNA with its ribosomes.

6 Gene Expression and Regulation MCQ

Iron is an essential nutrient that is acquired by organisms from the environment. When intracellular levels of iron are relatively high, living cells synthesize an iron-storage protein called ferritin.

The induction of ferritin synthesis by iron was investigated in rats. Figure 1 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and actinomycin D, a drug that inhibits transcription.

Figure 2 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and cycloheximide, a drug that inhibits translation.

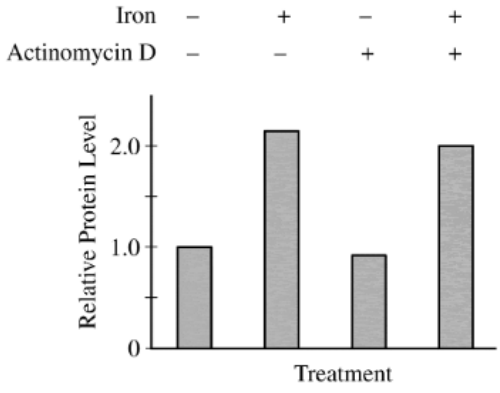


Figure 1. Ferritin Protein Levels

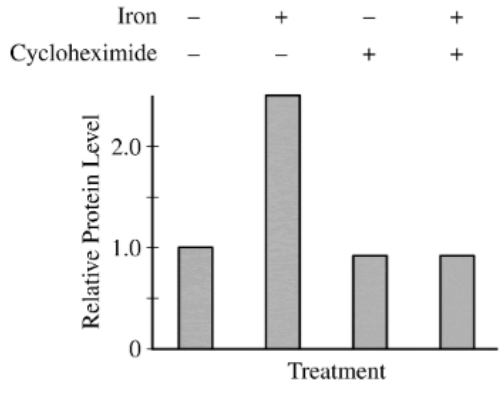


Figure 2. Ferritin Protein Levels

58. Which of the following biotechnology approaches could be used to identify ferritin mRNA in a sample of total cellular RNA?

(A) RNA samples could be directly cloned into a DNA plasmid, grown in bacteria, and tested for the ability to bind iron.

(B) RNA samples could be separated by size using agarose gel electrophoresis and incubated with labeled single-stranded DNA molecules that are complementary to the ferritin mRNA. ✓

(C) RNA samples could be converted to protein and subsequently cut with a restriction endonuclease that recognizes DNA sequences.

(D) RNA samples could be examined under a high-power microscope to visually identify the ferritin mRNA.

59. In individuals with iron-rich diets, intracellular levels of iron are higher than in individuals with iron-poor diets. Which of the following best explains how organisms regulate ferritin protein levels in response to available iron?

6 Gene Expression and Regulation MCQ

(A) Translation of ferritin mRNA is inhibited in the presence of iron.

(B) Translation of ferritin mRNA is stimulated by iron. ✓

(C) Ferritin protein is more stable in the presence of iron.

(D) Ferritin protein is less stable in the presence of iron.

60. The data can best be used to support which of the following claims about the mechanism for regulating ferritin gene expression?

(A) Iron increases ribosome binding to ferritin mRNA. ✓

(B) Iron causes the ferritin mRNA to be degraded.

(C) Iron increases the effect of actinomycin D on RNA polymerase.

(D) Iron decreases the ability of ferritin mRNA to bind to ribosomes.

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61. The enzyme lactase aids in the digestion of lactose, a sugar found in the milk of most mammals. In most mammal species, adults do not produce lactase. Continuing to produce lactase into adulthood in people is called lactase persistence. A number of different alleles have been identified that result in lactase persistence. Figure 1 shows the percentage of people in different geographic areas parts of the Old World that exhibit lactase persistence.

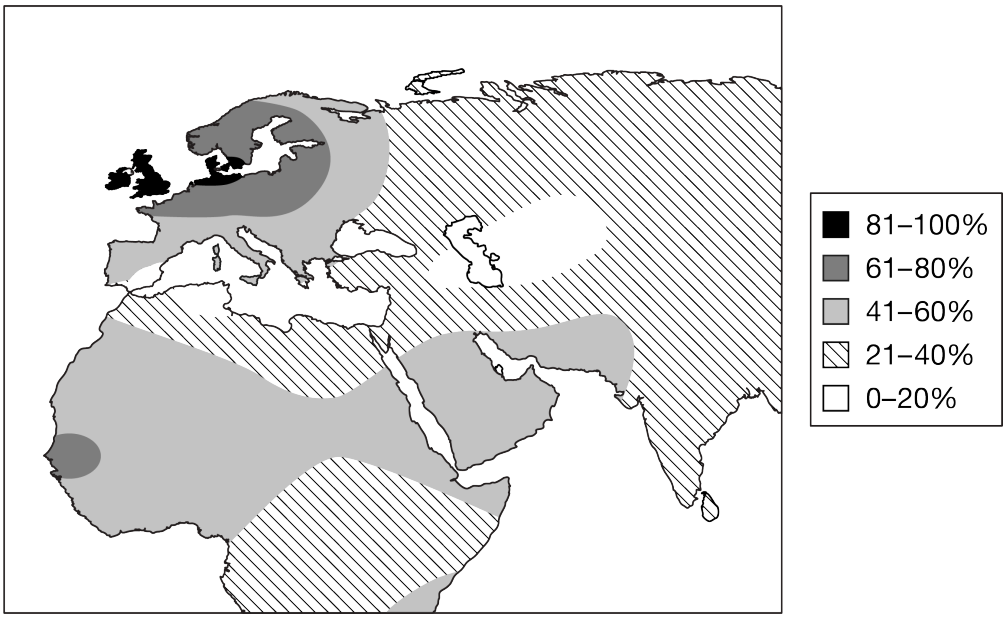

Figure 1. Distribution of lactase persistence in Europe, North Africa, and parts of Asia

Which of the following best explains the distribution of lactase persistence in the areas shown in Figure 1 ?

6 Gene Expression and Regulation MCQ

(A) Lactase persistence developed because people were malnourished in Europe.

(B) Lactase persistence alleles are present in all human populations and are expressed when lactose is consumed.

(C) Mutations conferring lactase persistence likely arose independently in different geographic areas and offered a selective advantage. ✓

(D) The mutations that cause lactase persistence are detrimental to humans and will eventually disappear from the gene pool.

Answer C

Correct. The map shows two widely separated areas in which a high percentage of the population shows lactase persistence. Based on the distribution, it is likely that different mutations arose in these two different areas. In areas where dairy products are a significant portion of the adult diet, individuals with a lactase persistence allele are more likely to survive, reproduce, and pass on the mutation.

62. Both liver cells and lens cells have the genes for making the proteins albumin and crystalline. However, only liver cells express the blood protein albumin and only lens cells express crystalline, the main protein in the lens of the eye. Both of these genes have enhancer sequences associated with them.

The claim that gene regulation results in differential gene expression and influences cellular products (albumin or crystalline) is best supported by evidence in which of the following statements?

(A) Liver cells possess transcriptional activators that are different from those of lens cells. ✓

(B) Liver cells and lens cells use different RNA polymerase enzymes to transcribe DNA.

(C) Liver cells and lens cells possess the same transcriptional activators.

(D) Liver cells and lens cells possess different general transcription factors.

Answer A

Correct. The specific regulatory transcription factors produced in a given cell determines which genes are actually expressed. For example, the transcriptional activators that allow for substantial transcription of the albumin gene exist only in the liver cells.

6 Gene Expression and Regulation MCQ

63. Lynch syndrome is an inherited condition associated with an increased risk for colon cancer, as well as certain other cancers. Mutations in one of several genes involved in DNA repair during DNA replication have been associated with Lynch syndrome. DNA sequencing was performed for an individual. The results indicated that the individual carries one of the dominant alleles that has been associated with Lynch syndrome.

Which of the following best explains how the results should be interpreted?

(A) The individual does not have an increased risk of developing cancer because one dominant allele is insufficient to cause the disease.

(B) The individual has an increased risk of developing colon cancer. ✓

(C) Because the person's DNA has the mutation, other family members must have cancer.

(D) Results cannot be interpreted until testing determines if additional mutated alleles are present.

Answer B

Correct. The individual may have an increased risk of developing colon cancer. Not all people with alleles associated with Lynch syndrome develop cancer.

6 Gene Expression and Regulation MCQ

In mammals, the dark color of skin, hair, and eyes is due to a pigment called melanin. Melanin is produced by specialized skin cells called melanocytes. The melanin is then transferred to other skin cells called keratinocytes. Melanocytes synthesize melanin in a multistep metabolic pathway (Figure 1). The amount of melanin produced is dependent on the amount of the enzymes TYR, $\mathrm{TRP2}$, and $\mathrm{TRP1}$ present inside melanocytes.

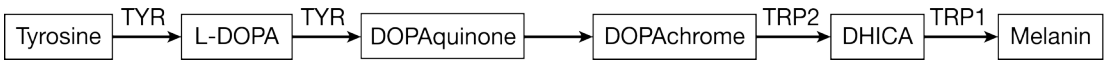

Figure 1. Melanin synthesis pathway

The peptide hormone α–melanocyte stimulating hormone (α–MSH) activates a signal transduction pathway leading to the activation of MITF. MITF is a transcription factor that increases the expression of the TYR, $TRP2$, and $TRP1$ genes (Figure 2).

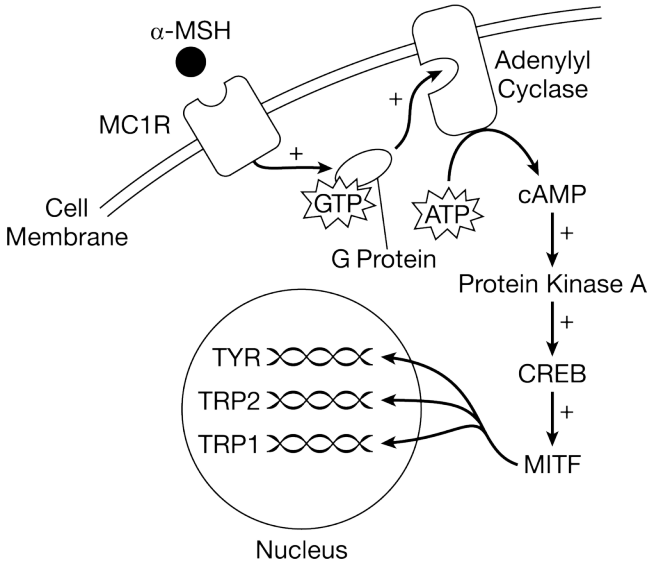

Figure 2. Activation of melanin synthesis genes in melanocytes

Some mammals increase melanin production in response to ultraviolet (UV) radiation. The UV radiation causes damage to DNA in keratinocytes, which activates the $\mathrm{p53}$ protein. $\mathrm{p53}$ increases the expression of the $POMC$ gene. The POMC protein is then cleaved to produce α–MSH. The keratinocytes secrete α–MSH, which signals nearby melanocytes. The increased melanin absorbs UV radiation, reducing further DNA damage.

6 Gene Expression and Regulation MCQ

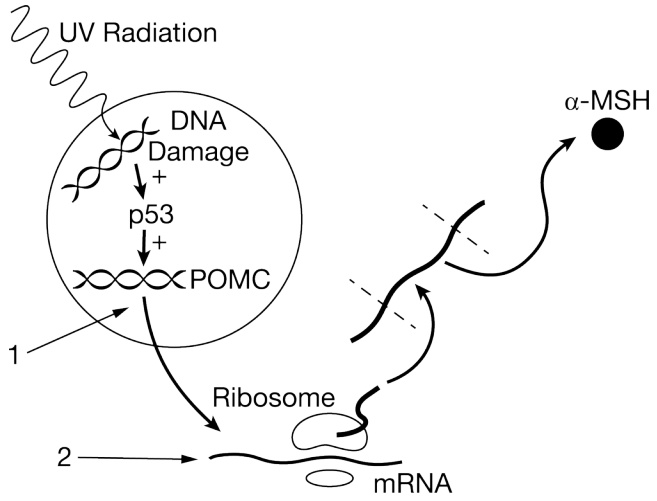

Figure 3. Production of α–MSH in keratinocytes in response to UV radiation

64. Mice have melanocytes in the skin on their ears and show a tanning response to UV radiation. Researchers were studying a mutant population of mice that do not show a tanning response. Genetic testing of these mutant mice showed that the pathway causing the production of α–MSH by keratinocytes in response to UV radiation was fully functional. Thus, the researchers claimed that the lack of tanning response was due a nonfunctional MC1R.

Which of the following pieces of evidence would best support the researchers' claim above?

(A) When researchers applied a drug that activates adenylyl cyclase to the mutant mice's ears, the level of melanin increased. ✓

(B) When researchers viewed sections of mutant mouse ears under the microscope, they found melanocyte numbers comparable to nonmutant mice.

(C) When researchers exposed the mutant mice to UV radiation, the amount of POMC mRNA in keratinocytes did not change.

(D) When researchers exposed the mutant mice to UV radiation, the level of melanin production did not change.

Answer A

Correct. Application of a drug that activates adenylyl cyclase would increase the production of cAMP, which would activate protein kinase A, which would activate CREB, which would activate MITF, which would activate the *TYR*, *TYR2*, and *TRP1* genes for the enzymes required to synthesize melanin. This evidence demonstrates that the parts of the signal transduction pathway from adenylyl cyclase onward are functional and that the enzymes required for melanin production are functional.

6 Gene Expression and Regulation MCQ

65. Based on the information provided in Figure 1 and Figure 2, which of the following best explains the effects of a mutation in the promoter of the TYR gene that prevents it from being transcribed?

(A) DNA damage due to UV radiation will be strongly inhibited, resulting in a positive selection pressure.

(B) DNA damage due to UV radiation will be strongly inhibited, resulting in a negative selection pressure.

(C) Skin pigmentation will not be able to change, resulting in a positive selection pressure.

(D) Skin pigmentation will not be able to change, resulting in a negative selection pressure. ✓

Answer D

Correct. As shown in Figure 1, TYR is required for two steps in the melanin synthesis pathway. If it is not expressed properly, then this will prevent melanocytes from synthesizing melanin regardless of exposure to UV radiation. This would result in a negative selection pressure.

66. Which of the following best explains a process occurring between point 1 and point 2 in Figure 3 ?

(A) α–MSH is produced.

(B) The TYR gene is transcribed.

(C) Polypeptides are removed from a protein.

(D) A poly-A tail is added to RNA. ✓

Answer D

Correct. Eukaryotic mRNA is modified after transcription and a poly-A tail is added before it is translated at the ribosome.

67. Researchers discovered a mutant form of the TYR gene with a deletion of a single guanine nucleotide in the beginning of the coding sequence.

Which of the following best predicts the phenotype of an individual who is homozygous for this TYR mutation?

6 Gene Expression and Regulation MCQ

(A) The mutation will cause a single amino acid change in the TYR protein, which will not be enough to disrupt its function. Therefore, those with this mutation will produce melanin in the hair, skin, and eyes and tan in response to UV radiation.

(B) The mutation will cause a single amino acid change in the TYR protein, leading to a nonfunctional TYR protein. Therefore, those with this mutation will lack melanin in the hair, skin, and eyes and will not tan in response to UV radiation.

(C) The mutation will change all subsequent amino acids in the TYR protein, leading to nonfunctional TYR protein. Since the *TRP1* and *TRP2* genes were not affected, the TRP1 and TRP2 proteins will fill the role of the TYR protein. Therefore, those with this mutation will produce melanin in the hair, skin, and eyes in response to UV radiation.

(D) The mutation will change all subsequent amino acids in the TYR protein, leading to nonfunctional TYR protein. Individuals with this mutation will lack melanin in their hair, skin, and eyes and will not tan in response to UV radiation. ✓

Answer D

Correct. Without a functional TYR enzyme to catalyze the first two steps of the melanin synthesis pathway, melanin will not be produced. Without melanin, individuals will not have a tanning response to UV radiation. Note that UV radiation exposure will still lead to the production of α–MSH in keratinocytes. The α–MSH will then activate a signal transduction pathway in melanocytes, leading to the production of MITF, which will activate the*TYR*, *TYR2*, and *TYR1* genes. However, the TYR enzyme will still be nonfunctional because of the major frameshift mutation.

68. Which of the following claims about the *TYR*, *TRP2*, and *TRP1* mammalian genes is most likely to be accurate?

(A) The *TYR*, *TRP2*, and *TRP1* genes are located next to each other on a single chromosome and are organized into an operon.

(B) The *TYR*, *TRP2*, and *TRP1* genes may be located on different chromosomes but are activated by the same transcription factor. ✓

(C) The *TYR*, *TRP2*, and *TRP1* genes are identical genes since they are activated by the same transcription factor.

(D) The *TYR*, *TRP2*, and *TRP1* genes may be located on different chromosomes but with identical operator sequences.

Answer B

Correct. Related eukaryotic genes are often located on different chromosomes. In humans, *TYR* is on chromosome 11, *TRP2* is on chromosome 13, and *TRP1* is on chromosome 9. The melanocytes control these genes together, as the same transcription factor, MITF, activates the transcription of all

three genes.

69. Which of the following claims best explains why keratinocytes do not produce melanin?

(A) Keratinocytes do not contain the TYR, $TRP2$, and $TRP1$ genes.

(B) Keratinocytes do not contain the $MC1R$ gene.

(C) Keratinocytes do not express the $MITF$ gene. ✓

(D) Keratinocytes do not express the $POMC$ gene.

Answer C

Correct. MITF stands for melanocyte inducing transcription factor, and its expression during embryonic development is critical to the differentiation of melanocytes. The genes activated by MITF include the TYR, $TRP2$, and $TRP1$ genes that are involved in producing melanin. In keratinocytes, MITF is not expressed, so it cannot activate transcription of genes involved in melanin production.

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70. Methicillin-resistant *Staphylococcus aureus* (MRSA) can be a serious threat to human health. There is evidence that *S. aureus* infections are common in hospitals and that MRSA have become resistant to other antibiotics besides methicillin. This suggests that the rapid evolution of resistance in the bacteria poses a serious public-health challenge. Which of the following best explains the ability of MRSA to evade existing drug therapies?

(A) MRSA have very long generation times and very large population sizes.

(B) MRSA develop new alleles by intentionally introducing specific mutations that will give them a selective advantage over other bacteria.

(C) MRSA metabolize many drugs in their lysosomes and therefore evolve resistance at a high rate.

(D) MRSA exchange genetic material with other antibiotic-resistant bacteria, which can spread resistance in the *S. aureus* population. ✓

71. Small single-stranded RNA molecules called microRNAs (miRNAs) are capable of base pairing with specific binding sites in the $3'$ untranslated region of many mRNA transcripts. Transcription of gene *Q* yields an mRNA transcript that contains such an miRNA binding site, which can associate with miRNA-delta, a specific miRNA molecule.

Which of the following best supports the claim that binding of miRNA-delta to the miRNA binding site inhibits translation of gene *Q* mRNA?

6 Gene Expression and Regulation MCQ

(A) When the promoter for gene *Q* is altered, transcription is inhibited.

(B) Translation of *Q* mRNA is inhibited regardless of whether the miRNA binding site sequence is altered.

(C) Translation of *Q* mRNA is inhibited in the absence of miRNA-delta.

(D) When the miRNA binding site sequence is altered, translation of *Q* mRNA occurs in the presence of miRNA-delta. ✓

Answer D

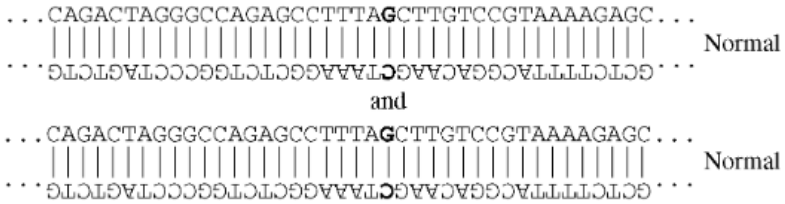
Correct. If the miRNA binding site sequence is altered, it cannot base-pair with miRNA-delta because they will no longer be complementary with one another. If miRNA-delta cannot bind to the miRNA binding site, translation of *Q* mRNA will proceed.

72. Mutations in DNA are usually caused by chemical or radiation damage to DNA molecules, followed by imperfect repair of the damage. Immediately after this kind of imperfect repair, there may be a mismatched base pair in the DNA. The illustration below shows an example of a mismatch, with the relevant pair bases in bold.

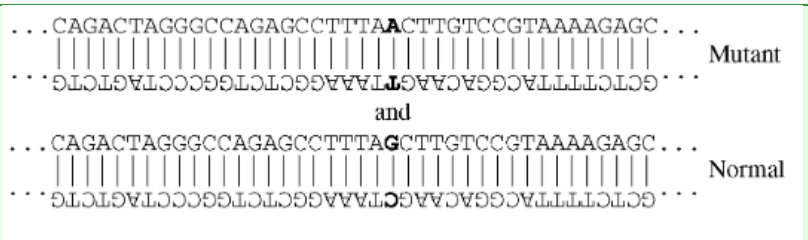
Which of the following best represents the DNA of the two daughter cells produced when a bacterial cell with this particular mismatch replicates its DNA and divides?

6 Gene Expression and Regulation MCQ

(A)



(B) ✓



(C)

(D)

6 Gene Expression and Regulation MCQ

73. Figure 1 illustrates the X and Y chromosomes during meiosis I and meiosis II of normal spermatogenesis in a mammal species.

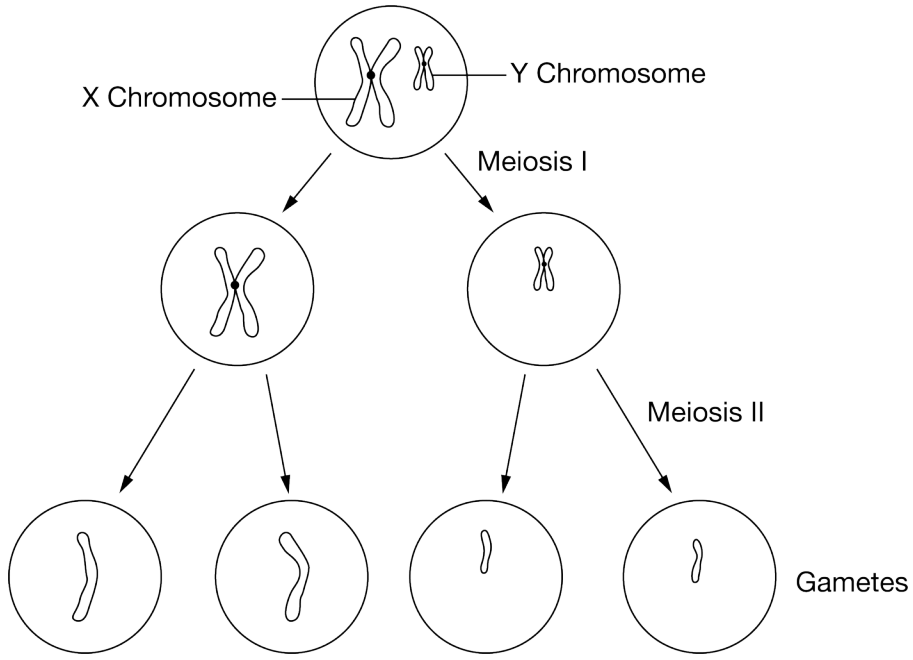


Figure 1. X and Y chromosomes during meiosis I and meiosis II

If the normal spermatogenesis is disrupted, the gametes can have different chromosomes than expected. Which of the following is the most likely cause of one of the four gametes having two X chromosomes and one having neither an X nor a Y chromosome?

(A) Nondisjunction of the chromosomes during meiosis I

(B) Nondisjunction of both the X and Y chromosomes during meiosis II

(C) Nondisjunction of the Y chromosome during meiosis II

(D) Nondisjunction of the X chromosome during meiosis II ✓

Answer D

Correct. A nondisjunction event of the X chromosome in meiosis II would result in both of the X chromatids going to one gamete and no X chromatids going to the other gamete.

6 Gene Expression and Regulation MCQ

74. Nondisjunction during meiosis can negatively affect gamete formation. A model showing a possible nondisjunction event and its impact on gamete formation is shown in Figure 1.

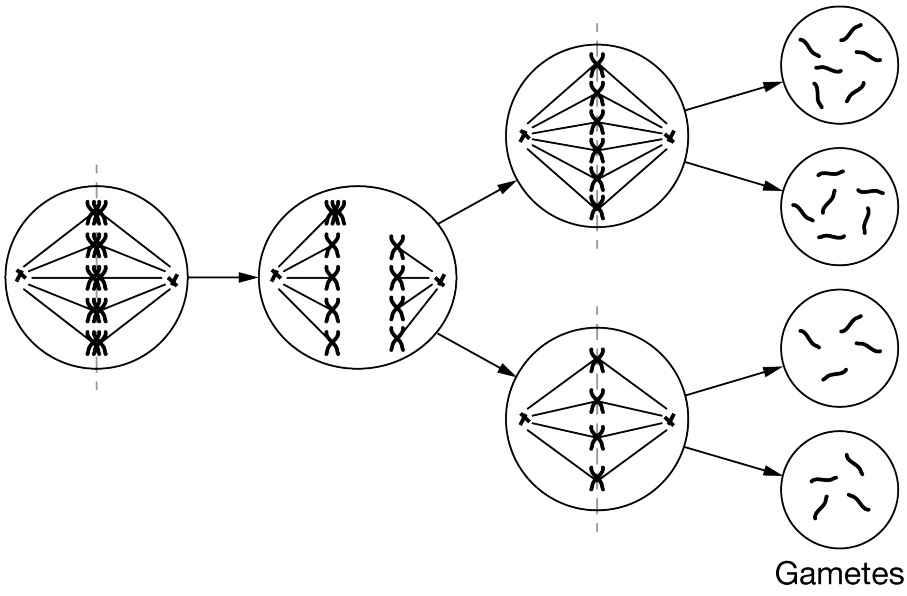


Figure 1. Model of a nondisjunction event

Which of the following best describes the most likely impact on an individual produced from fertilization between one of the daughter cells shown and a normal gamete?

(A) Because nondisjunction occurred in anaphase I, all gametes will be normal and the resulting individual will be phenotypically normal.

(B) Because nondisjunction occurred in anaphase I, all gametes will have an abnormal chromosome number and the individual will likely exhibit phenotypic evidence of the nondisjunction event. ✓

(C) Because nondisjunction occurred in anaphase II, all gametes will be normal and the resulting individual will be phenotypically normal.

(D) Because nondisjunction occurred in anaphase II, all gametes will have an abnormal chromosome number and the individual will likely exhibit phenotypic evidence of the nondisjunction event.

Answer B

Correct. Nondisjunction occurred in anaphase I and will result in an abnormal chromosome number, as the resulting chromosome numbers in the daughter cells are either $n+1$ or $n-1$.

6 Gene Expression and Regulation MCQ

75. Phytochromes are molecules that change light stimuli into chemical signals, and they are thought to target light-activated genes in plants. A study was conducted to determine how certain cell proteins were made in a plant cell using a phytochrome. Figures 1 and 2 represent findings from the study.

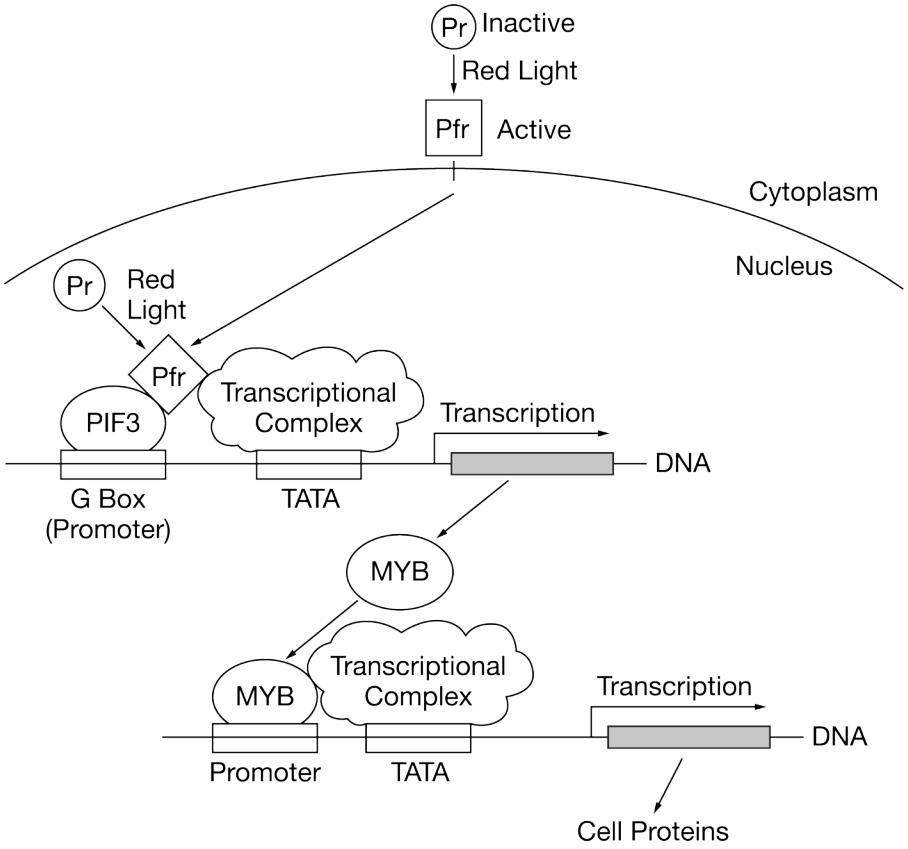


Figure 1. Phytochrome response to exposure to red light

6 Gene Expression and Regulation MCQ

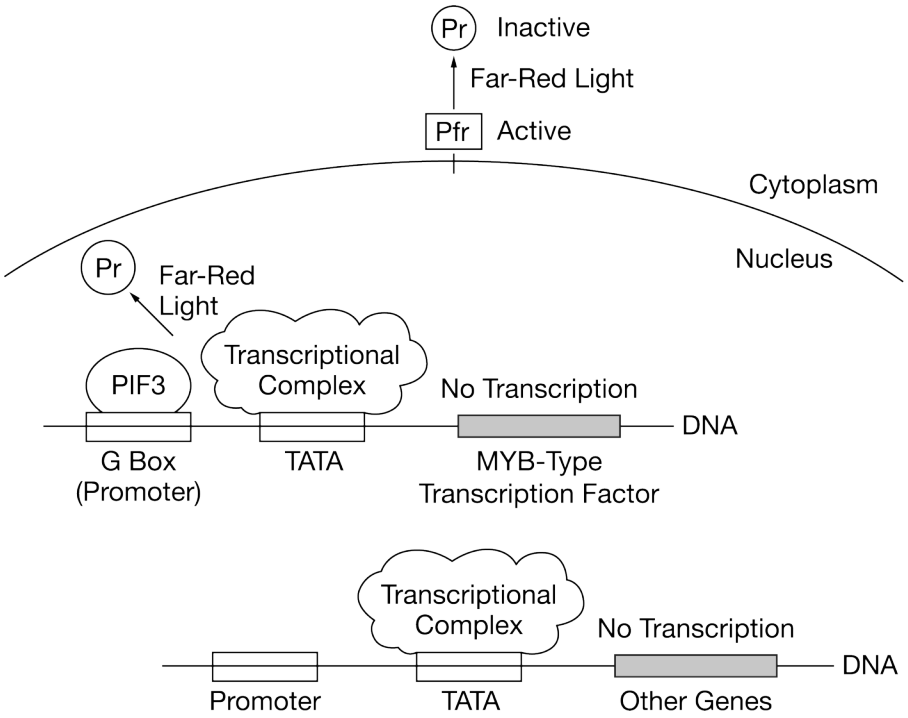


Figure 2. Phytochrome response to exposure to far-red light

Use the response models shown in Figures 1 and 2 to justify the claim that phytochromes regulate the transcription of genes leading to the production of certain cellular proteins.

(A) When inactive phytochrome Pr is activated by red light to become phytochrome Pfr, it is transported into the nucleus where it binds to the transcription factor PIF3 at the promoter. This stimulates transcription, ultimately leading to protein production. Far-red light inactivates the phytochrome, which will turn transcription off by not binding to PIF3. ✓

(B) Far-red light activates phytochrome Pr, causing it to travel to the nucleus where it binds to PIF3 at the promoter. This stimulates transcription, ultimately leading to protein production. Red light inactivates the phytochrome, which will turn transcription off by not binding to PIF3.

(C) MYB, and not Pfr, is activated by red light, causing it to bind to the promoter and stimulate transcription and translation of cellular proteins.

(D) PIF3 binds to the promoter only in the presence of red light and Pfr. Any time PIF3 is bound to the promoter, MYB is transcribed, initiating transcription of various other proteins in the cell.

Answer A

Correct. The diagram indicates the phytochrome is light sensitive, becoming activated by red light. Activated Pfr goes into the nucleus and binds to PIF3, which activates the transcriptional complex, thus turning on the genes responsible for making cellular proteins.

6 Gene Expression and Regulation MCQ

- **Directions:** This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

(A) Transcription
(B) Translation
(C) Transformation
(D) Replication
(E) Reverse transcription

76. Process in which naked DNA is taken up by a bacterial or yeast cell

(A) A
(B) B
(C) C ✓
(D) D
(E) E

77. Process in which RNA is produced by using a DNA template

(A) A ✓
(B) B
(C) C
(D) D
(E) E

78. Process in which a protein is assembled at a ribosome

(A) A
(B) B ✓
(C) C
(D) D
(E) E

79. Which of the following describes a process that increases genetic diversity in prokaryotic cells?

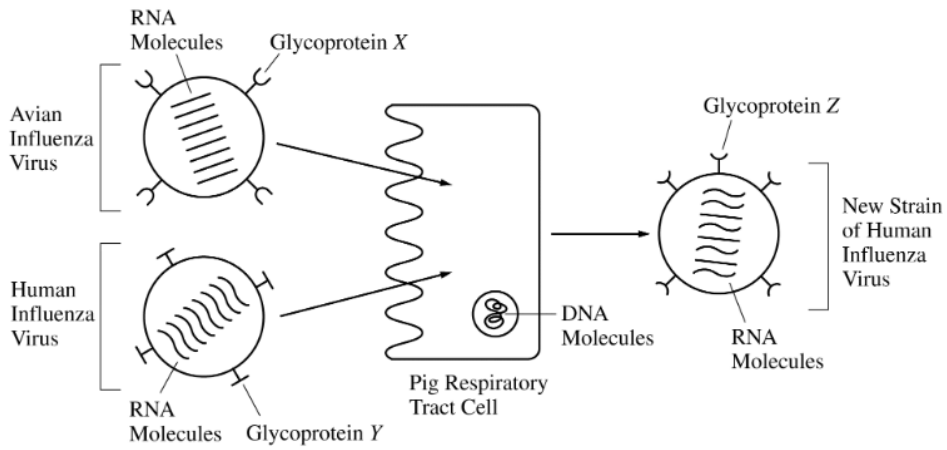
(A) Chromosomes segregate independently and form different combinations in daughter cells.
(B) Homologous chromosomes exchange genetic material during meiosis.
(C) Conjugation transfers DNA from one cell to the other cell. ✓
(D) A cell infected with two genetically different viruses can produce viruses with new allele combinations.

6 Gene Expression and Regulation MCQ

Answer C

Correct. When a cytoplasmic bridge forms between two prokaryotic cells, DNA can be exchanged between the cells, increasing genetic diversity. This process is called conjugation.

80.



The cells lining the respiratory tract of pigs have receptors for both avian and human influenza viruses. Based on the model above, which of the following best explains the origin of the new strain of human influenza virus?

(A) The new viral strain inherited a mixture of genetic material from both avian influenza virus and human influenza virus. ✓

(B) The new viral strain inherited RNA molecules from the avian influenza virus and packaged them inside the human influenza virus membrane.

(C) The new viral strain inherited a mutant DNA molecule from the pig respiratory tract cell.

(D) The new viral strain inherited an RNA molecule that had recombined with a DNA molecule from the pig respiratory tract cell.

81. Sickle-cell anemia is associated with a mutation in the gene encoding the beta subunit of hemoglobin that results in a change from glutamic acid to valine at position 6. All other amino acids are identical to a normal hemoglobin molecule.

Based on the information above, which of the following mutations is the most likely cause of sickle-cell anemia?

(A) A single base-pair substitution in the gene encoding the beta subunit ✓

(B) A single base-pair insertion in the gene encoding the beta subunit

(C) A single base-pair deletion in the gene encoding the beta subunit

(D) A translocation of DNA from one chromosome to another

6 Gene Expression and Regulation MCQ

82. Sickle-cell anemia results from a point mutation in the *HBB* gene. The mutation results in the replacement of an amino acid that has a hydrophilic R-group with an amino acid that has a hydrophobic R-group on the exterior of the hemoglobin protein. Such a mutation would most likely result in altered

(A) properties of the molecule as a result of abnormal interactions between adjacent hemoglobin molecules ✓

(B) DNA structure as a result of abnormal hydrogen bonding between nitrogenous bases

(C) fatty acid structure as a result of changes in ionic interactions between adjacent fatty acid chains

(D) protein secondary structure as a result of abnormal hydrophobic interactions between R-groups in the backbone of the protein

Answer A

This option is correct. The mutation results in the replacement of an amino acid, which alters the properties of the R-group both structurally and functionally; thus, the interactions between adjacent hemoglobin molecules would also be altered. In this case, hemoglobin molecules would tend to stick together as a result of hydrophobic interactions between amino acids on the surface.

83. Sickle-cell anemia results from a point mutation in the *HBB* gene. The mutation results in the replacement of an amino acid that has a hydrophilic R-group with an amino acid that has a hydrophobic R-group on the exterior of the hemoglobin protein. Such a mutation would most likely result in altered

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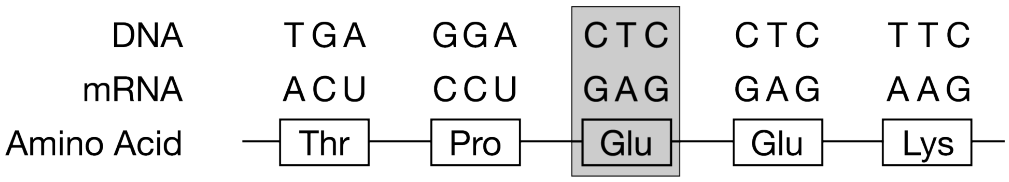
(C) fatty acid structure as a result of changes in ionic interactions between adjacent fatty acid chains

(D) protein secondary structure as a result of abnormal hydrophobic interactions between R-groups in the backbone of the protein

6 Gene Expression and Regulation MCQ

84. Sickle-cell anemia is an inherited blood disorder in which one of the hemoglobin subunits is replaced with a different form of hemoglobin. Partial DNA sequences of the *HBB* gene for normal hemoglobin and for sickle-cell hemoglobin are shown in Figure 1.

Partial sequence for normal hemoglobin:



Partial sequence for sickle-cell hemoglobin:

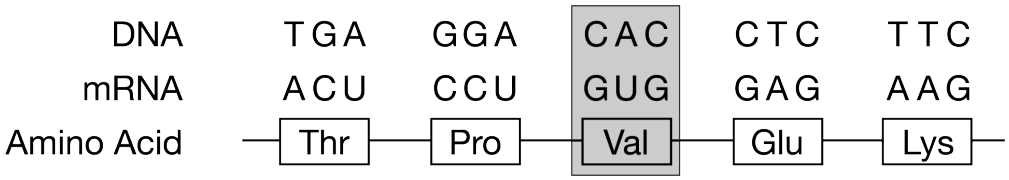


Figure 1. Comparison of partial DNA sequences for normal hemoglobin and hemoglobin with a sickle-cell mutation

Which of the following best describes the type of mutation shown in Figure 1 that leads to sickle-cell anemia?

(A) Insertion

(B) Deletion

(C) Substitution ✓

(D) Frameshift

Answer C

Correct. The change in the DNA sequence from thymine to adenine resulted in a different amino acid in the protein, valine instead of glutamic acid.

6 Gene Expression and Regulation MCQ

Snowshoe hares are small mammals that live in forests throughout Canada and the northern United States. Hares feed primarily on vegetation and are preyed on by lynx and owls.

In the fall, the hares shed their brown coat (molt), and a new white coat grows in over a period of 40 days. In the spring, the hares molt again and their white coat is replaced with a brown coat over a period of approximately 30 to 50 days.

In a recent study, researchers observed that the majority of hares in an isolated population began the molting process during the same week in the fall, even though the ground was not snow covered. The majority of hares also began the molting process during the same week in the spring, even though the ground was still snow covered. The researchers formulated a hypothesis that the timing of the molting process is influenced by seasonal changes in the daylight period.

85. A second isolated hare population has white coats all year, although the area is not covered by snow.

Which of the following is a testable question the researchers can ask to explain the origin of the different coat color of the second population?

(A) Do the genes associated with coat color in both populations have the same sequence of DNA nucleotides? ✓

(B) Are hares that have a white coat all year round better able to find food than those that change coat color?

(C) Is the reproductive fitness greater in hares that have a white coat all year round than in those that change coat colors?

(D) Is the same genetic code used to produce the enzymes involved in the production of coat color pigments in both populations?

Answer A

Correct. A constant all-white coat color would strongly suggest a change in the genetic makeup of the hare population. To determine whether this is the case, sequencing of the coat color genes in both populations would indicate whether the genes are the same or different.

86. Some strains of the bacterium *Streptococcus pyogenes* secrete poisonous substances called exotoxins. The genes encoding the exotoxins are thought to have originated in bacteriophages, which are viruses that infect bacteria.

Which of the following is the most likely mechanism by which the *S. pyogenes* acquired the ability to produce the exotoxins?

(A) Bacteriophages engulfed cellular debris from dead bacteria.

(B) Bacteriophages in the environment activated bacterial cell division.

(C) Bacteriophage DNA became integrated in the bacterial chromosome. ✓

(D) Bacteriophage proteins were absorbed into bacteria cells by endocytosis.

6 Gene Expression and Regulation MCQ

87. Trisomy 21 is a condition in which a child is born with an extra chromosome in pair 21. Researchers assessed the frequency of children born with trisomy 21 by age of the mothers at birth (maternal age) and primary cause of the error leading to trisomy 21. The findings are presented in Figure 1.

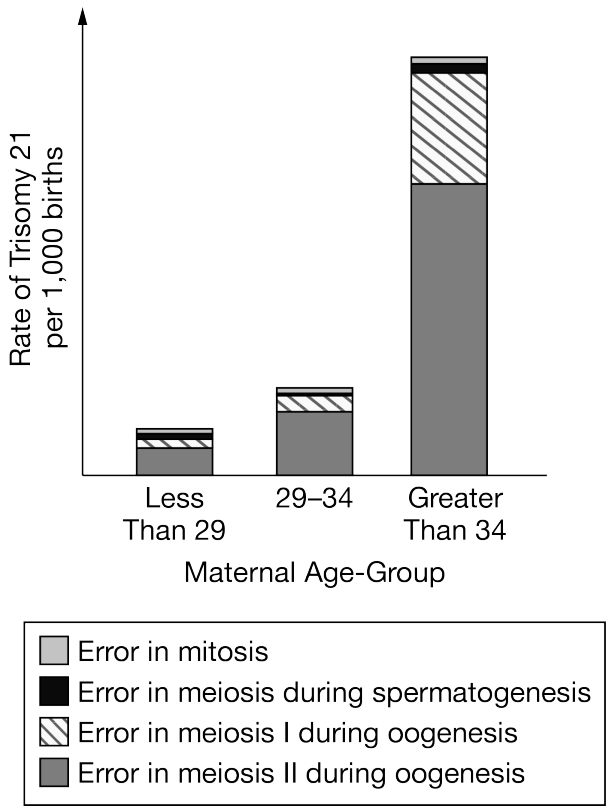


Figure 1. Incidence and primary cause of trisomy 21 by maternal age-group

Based on the data in Figure 1, which of the following is most likely the primary cause of the pattern of frequency of trisomy 21 births in the selected maternal age-groups?

(A) At older maternal ages, there is an increase in the number of errors during mitosis, which leads to an increase in nondisjunction during egg production.

(B) The incidence of nondisjunction errors in meiosis during sperm production is positively correlated with increasing maternal age.

(C) At older maternal ages, the incidence of errors in meiosis during egg production increases, which leads to an increase in nondisjunction. ✓

(D) Errors in meiosis leading to nondisjunction are more likely to occur during meiosis I than during meiosis II.

Answer C

Correct. Maternal age is correlated with a significant increase in meiotic errors during egg maturation.

6 Gene Expression and Regulation MCQ

Figure 1 shows that most of trisomy 21 births are due to these errors, with the number of these errors observed increasing significantly with maternal age.

88. The bacterium *Vibrio cholerae* is harmless unless a lysogenic bacteriophage provides the gene coding for the cholera toxin, which converts the bacterium to the virulent form that causes cholera. Which of the following best explains how the gene encoding cholera toxin becomes part of the bacterial genome?

(A) The bacteriophage inserts the toxin gene into the host cell DNA, and the gene is expressed with the rest of the host cell's genes. ✓

(B) The bacteriophage makes copies of the toxin gene and expresses the copies inside the bacteriophage.

(C) The bacteriophage converts its toxin gene into mRNA, which is then translated by the host cell.

(D) The bacteriophage transforms itself into a self-replicating protein that can survive inside the host cell.

89. The trp operon in *E. coli* is an example of a repressible operon that consists of genes coding for enzymes used to synthesize tryptophan. When tryptophan levels are high, the operon is turned off and these genes are not transcribed. However, it is also known that tryptophan does not bind directly to the operator DNA sequence. A regulatory gene called $trpR$ has also been discovered although it is not part of the trp operon. The proposed model of how tryptophan acts as a corepressor is shown in Figure 1.

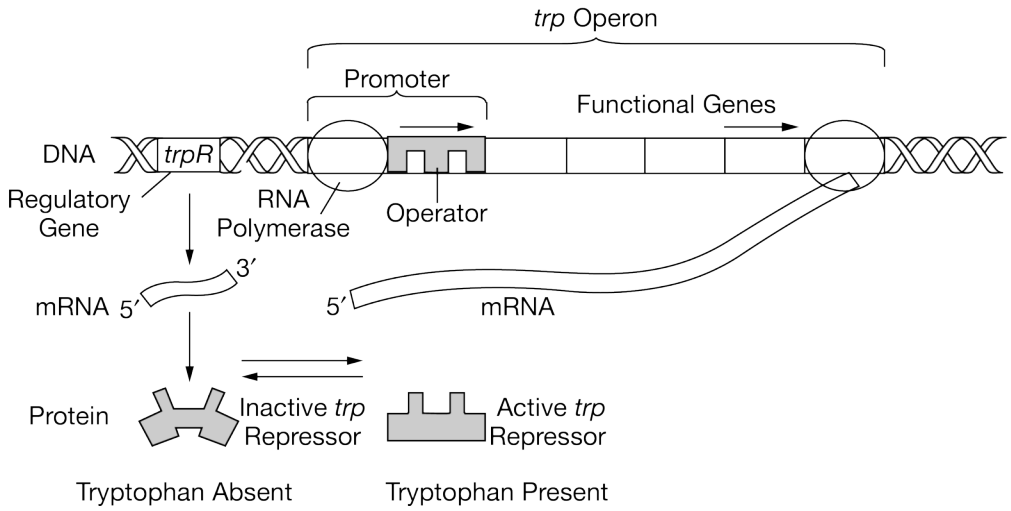


Figure 1. Model of proposed regulation of the trp operon by corepressors trp repressor and tryptophan

Which of the following evidence best supports a claim that tryptophan functions as a corepressor?

(A) Normal expression of $trpR$ causes the trp operon to be transcribed regardless of tryptophan levels.

(B) When the operator sequence is mutated, the trp operon is not transcribed.

(C) The $trpR$ gene codes for a repressor protein that has a DNA binding domain.

(D) When $trpR$ is mutated, the trp operon is transcribed regardless of tryptophan levels. ✓

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Answer D

Correct. As tryptophan accumulates in the cell, it inhibits its own production by activating the trp repressor protein (coded by the $trpR$ gene), which then binds to the operator and blocks transcription of the trp operon. If the $trpR$ gene is mutated, then tryptophan cannot bind to the allosteric binding site to activate trp repressor protein. Thus, no matter how high tryptophan levels get, the trp operon will not be repressed.

6 Gene Expression and Regulation MCQ

In prokaryotes, the genes of the trp operon encode enzymes used in the synthesis of tryptophan. The genes of the trp operon are actively expressed (Figure 1A) unless abundant tryptophan is present in the environment. When tryptophan is abundant, some tryptophan binds to the trp repressor protein, which in turn causes the repressor to bind to the operator and block transcription of the genes (Figure 1B). The repressor protein is encoded by the $trpR$ gene which is not a part of the trp operon.

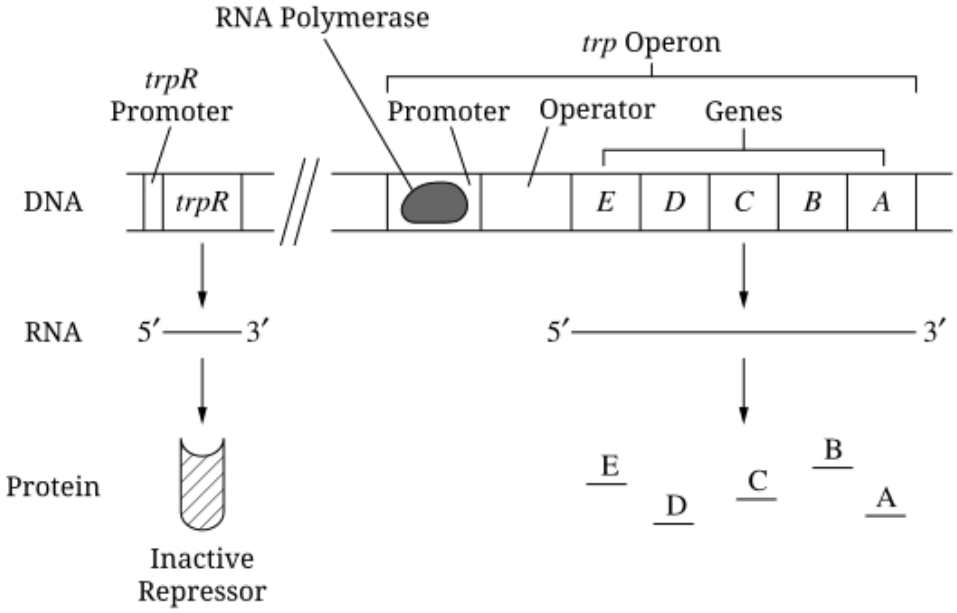

Figure 1A. When tryptophan is absent, the trp operon is active.

6 Gene Expression and Regulation MCQ

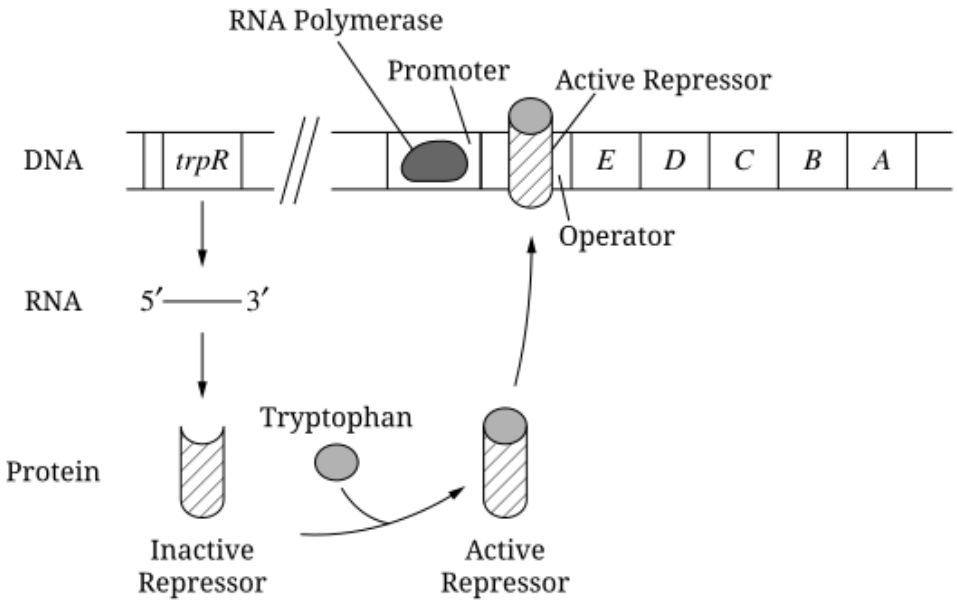

Figure 1B. When tryptophan is abundant, the *trp* operon is inactivated by the binding of the repressor protein to the operator.

90. A scientist claimed that an *E. coli* strain had either a mutated *trpR* gene or a mutated operator. Which of the following observations most likely supports the claim?

(A) Transcription from the operon occurred only in the presence of abundant tryptophan.

(B) The strain of *E. coli* required more tryptophan for its metabolic processes than does a strain of *E. coli* with typical tryptophan regulatory controls.

(C) Enzymes required for the synthesis of tryptophan were continuously produced whether tryptophan was absent or present in large quantities. ✓

(D) The cells died when they were grown in nutrient medium that lacked tryptophan.

Answer C

Correct. Under high tryptophan levels, the wild-type repressor protein binds to tryptophan and the complex binds to the operator, blocking transcription and ultimately the production of the enzymes that produce more tryptophan. In this mutant, high tryptophan levels do not reduce the production of more tryptophan. This suggests either that the repressor protein is unable to bind to tryptophan or the operator or, alternatively, that the operator cannot interact with the repressor-tryptophan complex to repress transcription.

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91. Which of the following questions will most likely allow the scientist to design an experiment to determine whether unregulated tryptophan synthesis in a strain of *E. coli* is caused by a mutation in the $trpR$ gene or in the operator of the trp operon?

(A) Is normal regulation of tryptophan synthesis restored when the operator sequence of this strain of *E. coli* is duplicated?

(B) Is normal regulation of tryptophan synthesis restored when a gene encoding a normal repressor protein is introduced into and expressed by this strain of *E. coli*? ✓

(C) Is normal regulation of tryptophan synthesis restored when a plasmid that contains a normal operator sequence is introduced into this strain of *E. coli*?

(D) Is normal regulation of tryptophan synthesis restored when the repressor protein gene is deleted in this strain of *E. coli*?

Answer B

Correct. Providing a functional copy of the $trpR$ gene that encodes the repressor protein would allow the scientist to observe whether regulatory functionality is restored. Restored functionality would indicate that the unregulated tryptophan synthesis was due to a mutation in the $trpR$ gene. If functionality is not restored, that would suggest a mutant operator sequence.

92. Attenuation is another mechanism by which tryptophan synthesis is regulated when tryptophan is abundant in the environment. During attenuation, ribosomes quickly translate the 5′ end of the mRNA transcript, causing the transcript to fold into a loop that forces RNA polymerase to detach from the DNA.

Which of the following best explains why attenuation is a mechanism that is unique to prokaryotes?

(A) Prokaryotic genes do not contain introns.

(B) Only prokaryotes can translate an mRNA while the mRNA is being transcribed. ✓

(C) Eukaryotes require more amino acids than prokaryotes do and therefore do not shut down production of tryptophan.

(D) Eukaryotes do not have groups of genes that are regulated together.

Answer B

Correct. Attenuation requires simultaneous transcription and translation. In eukaryotes, mRNA is transcribed and processed in the nucleus before being transported to the cytosol for translation. Prokaryotes have no nuclear envelope to separate these processes.

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93. Which of the following best supports a claim that an *E. coli* strain has a mutation in gene E that results in the production of a nonfunctional enzyme?

(A) Cells of this strain synthesize very little of the repressor protein and produce a large amount of tryptophan.

(B) RNA polymerase does not bind to the promoter of the operon in cells of this strain unless tryptophan is added to the nutrient medium.

(C) The operator sequence of the operon always has repressor proteins bound to it, independent of the amount of tryptophan in the nutrient medium.

(D) mRNA is continuously transcribed from the operon, but the amount of tryptophan produced from the cells of this strain remains low. ✓

Answer D

Correct. Continuous transcription of the operon during low levels of tryptophan is expected, but the lack of resultant tryptophan production suggests that one of the enzymes in the tryptophan synthesis pathway is nonfunctional.

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94. Exposure to ultraviolet (UV) radiation is the leading cause of skin cancer in humans. Figure 1 shows a model of how UV exposure damages DNA.

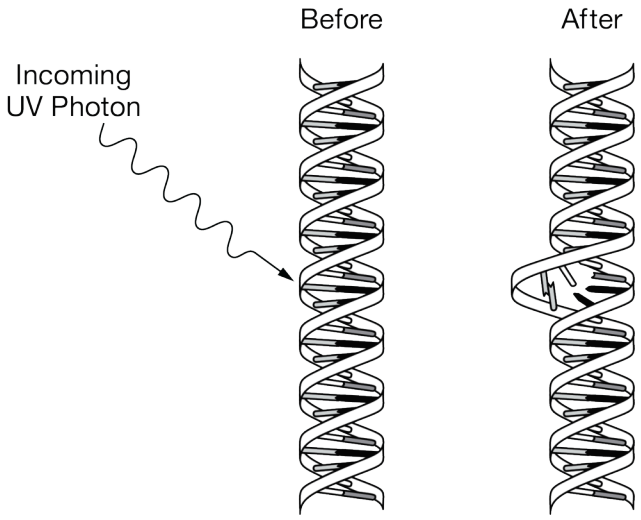

Figure 1. Model of damage to DNA caused by UV exposure

Which of the following statements best explains what is shown in Figure 1 ?

6 Gene Expression and Regulation MCQ

(A) UV exposure triggers DNA replication, which results in rapid cell proliferation.

(B) Naturally occurring dimers are removed by the UV photons, causing misshapen DNA, which results in replication errors.

(C) The hydrogen bonds between base pairs absorb the UV photons, separating the two DNA strands, which results in rapid DNA replication.

(D) UV photons cause dimers to form, leading to misshapen DNA, which results in replication and transcription errors. ✓

Answer D

Correct. The model shows the partial separation of a segment of DNA caused by the structural change resulting from dimer formation. The presence of dimers can lead to errors in both transcription and replication.

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question.

In the 1940's, Avery MacCleod, and McCarty transformed nonencapsulated bacteria into encapsulated forms by growing the nonencapsulated cells in a culture containing an extract made from dead encapsulated cells. The transformed cells produced colonies of encapsulated bacteria. Three different procedures and their results are outlined below.

Procedure I:

Extract made from dead encapsulated cells added to culture medium.
Nonencapsulated bacteria added to culture medium.

Results: Both nonencapsulated and encapsulated bacteria grow.

Procedure II:

Extract made from dead encapsulated cells treated with protein-degrading enzymes before adding extract to culture medium.
Nonencapsulated bacteria added to culture medium.

Results: Both nonencapsulated and encapsulated bacteria grow.

Procedure III:

Extract made from dead encapsulated cells treated with DNAse (an enzyme that selectively destroys DNA) before adding extract to culture medium.
Nonencapsulated bacteria added to culture medium.

Results: Only nonencapsulated bacteria grow.

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95. A resonable conclusion to draw from the results of the experiment is that

(A) DNA is the genetic material ✓

(B) DNA replication is semiconservative

(C) DNA is a double helix

(D) DNA is translated into protein

(E) mutation is a change in the genetic material

96. What was the purpose of treating the extract with DNAse in Procedure III ?

(A) To remove the encapsulated bacteria from the extract

(B) To serve as a positive control by demonstrating that a protein in the extract is the transforming factor

(C) To serve as a negative control by demonstrating that transformation does not occur without DNA ✓

(D) To destroy enzymes in the extract

(E) To destroy any capsules that might be in the extract

97. What was the purpose of treating the extract with protein-degrading enzymes in Procedure II ?

(A) To demonstrate that the transforming factor is an enzyme

(B) To demonstrate that the transforming factor is not a protein ✓

(C) To destroy nuclei acids in the extract

(D) To destroy any capsules in the extract

(E) To prevent the extract from being contaminated by nonencapsulated bacteria

---

98. Erwin Chargaff investigated the nucleotide composition of DNA. He analyzed DNA from various organisms and measured the relative amounts of adenine (A), guanine (G), cytosine (C), and thymine (T) present in the DNA of each organism. Table 1 contains a selected data set of his results.

Table 1. Nucleotide composition of sample DNA from selected organisms

Organism	Relative Amounts of Bases					
	%A	%G	%C	%T	%(G+C)	%(A + T)
Octopus	33.2	17.6	17.6	31.6	35.2	64.8
Chicken	28.0	22.0	21.6	28.4	43.7	56.4
Rat	28.6	21.4	20.5	28.4	42.9	56.0
Grasshopper	29.3	20.5	20.7	29.3	41.2	58.6
Wheat	27.3	22.7	22.8	27.1	45.5	54.4

Which of the following statements best explains the data set?

6 Gene Expression and Regulation MCQ

(A) Since the $\%\mathrm{A}$ and the $\%\mathrm{G}$ add up to approximately 50 percent in each sample, adenine and guanine molecules must pair up in a double-stranded DNA molecule.

(B) Since the $\%\mathrm{A}$ and the $\%\mathrm{T}$ are approximately the same in each sample, adenine and thymine molecules must pair up in a double-stranded DNA molecule. ✓

(C) Since the $\%(\mathrm{A}+\mathrm{T})$ is greater than the $\%(\mathrm{G}+\mathrm{C})$ in each sample, DNA molecules must have a poly-A tail at one end.

(D) Since the $\%\mathrm{C}$ and the $\%\mathrm{T}$ add up to approximately 50 percent in each sample, cytosine and thymine molecules must both contain a single ring.

Answer B

Correct. The percentages of adenine and thymine are approximately the same because adenine aligns with thymine in double-stranded DNA and they are therefore present in each organism in the same amounts.

99. Cells that contain only circular chromosomes are most probably which of the following?

(A) Protist cells
(B) Fungal cells
(C) Bacterial cells ✓
(D) Plant cells
(E) Animal cells

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the following groups of biological compounds.

(A) Proteins
(B) Carbohydrates
(C) Nucleic acids
(D) Lipids
(E) Steroids

100. Used to carry the genetic code

6 Gene Expression and Regulation MCQ

(A) Proteins
(B) Carbohydrates
(C) Nucleic acids ✓
(D) Lipids
(E) Steroids

---

101. A scientist separated a sample of cells of the same type into two dishes, dish X and dish Y. Each dish had the same number of cells. After a period of time, the scientist measured the quantity of histone proteins in each dish (Table 1).

Table 1. Histone Protein Quantities

Dish	Histone Proteins (\(\text{ng}\))
X	6.62
Y	13.57

Which of the following statements best explains the greater quantity of histone proteins in dish Y shown in Table 1?

(A) Prokaryotic cells are preparing for cell division, because prokaryotes must double the quantity of histone proteins each time they replicate their DNA.
(B) Eukaryotic cells are preparing for cell division, because eukaryotes must double the quantity of histone proteins each time they replicate their DNA. ✓
(C) Prokaryotic cells are preparing for protein synthesis, because prokaryotes must double the quantity of histone proteins when they manufacture proteins.
(D) Eukaryotic cells are preparing for protein synthesis, because eukaryotes must double the quantity of histone proteins when they manufacture proteins.

102. Huntington's disease, an autosomal dominant disorder, is caused by a mutation in the HTT gene. The HTT gene contains multiple repeats of the nucleotide sequence CAG. A person with fewer than 35 CAG repeats in the HTT gene is unlikely to show the neurological symptoms of Huntington's disease. A person with 40 or more CAG repeats almost always becomes symptomatic. Due to errors in meiosis, an individual without symptoms of Huntington's disease can produce gametes with a larger number of CAG repeats than there are in their somatic cells.

A woman develops Huntington's disease. Her father had the disorder. Her mother did not, and there is no history of the disorder in the mother's family. Which of the following best explains how the woman inherited Huntington's disease?

6 Gene Expression and Regulation MCQ

(A) She inherited an allele with fewer than 40 CAG repeats in the *HTT* gene because her mother did not have Huntington's disease.

(B) She inherited an allele with more than 40 CAG repeats in the *HTT* gene from her father. ✓

(C) Her mother produced eggs that all have more than 40 repeats in the *HTT* gene.

(D) Her mother produced eggs that all have fewer than 40 CAG repeats in the *HTT* gene.

Answer B

Correct. If her father was heterozygous for Huntington's disease, half of his sperm would carry an *HTT* allele with more than 40 CAG repeats. His children would have a fifty-fifty chance of inheriting the allele responsible for Huntington's disease.

103. Antibiotics can be used to kill the specific pathogenic bacterium, *Mycobacterium tuberculosis*, that causes tuberculosis. The appearance of antibiotic-resistant strains has made it more difficult to cure *M. tuberculosis* infections. These antibiotic-resistant bacteria survive and pass on the genes to their offspring, making the resistant phenotype more common in the population.

DNA analysis indicates that the genes for antibiotic resistance are not normally present in bacterial chromosomal DNA.

Which of the following statements best explains how the genes for antibiotic resistance can be transmitted between bacteria without the exchange of bacterial chromosomal DNA?

(A) The antibiotic-resistant bacteria release a hormone that signals neighboring bacteria to become resistant.

(B) The genes for antibiotic resistance are located on a plasmid that can be passed to neighboring bacteria. ✓

(C) The antibiotic-resistant bacteria are the result of bacteria that specifically modify their own chromosomal DNA to neutralize the antibiotics.

(D) The antibiotic alters the bacterial genome of each bacterium, which results in an antibiotic-resistant population.

Answer B

Correct. Bacteria can exchange and express the genes found on plasmids, which are foreign, extrachromosomal loops of DNA that they pick up. Genes for antibiotic resistance are located on these plasmids.

104. Which of the following statements best explains the structure and importance of plasmids to prokaryotes?

6 Gene Expression and Regulation MCQ

(A) Plasmids are circular, single-stranded RNA molecules that transfer information from the prokaryotic chromosome to the ribosomes during protein synthesis.

(B) Plasmids are circular, double-stranded DNA molecules that provide genes that may aid in survival of the prokaryotic cell. ✓

(C) Plasmids are single-stranded DNA molecules, which are replicated from the prokaryotic chromosome, that prevent viral reproduction within the prokaryotic cell.

(D) Plasmids are double-stranded RNA molecules that are transmitted by conjugation that enable other prokaryotic cells to acquire useful genes.

Answer B

Correct. Plasmids are small, circular, double-stranded DNA molecules. Plasmids can be acquired and can be replicated by prokaryotes. Plasmids may carry genes, such as antibiotic resistance genes, that may help the prokaryote survive.

105. Which of the following best explains how the pattern of DNA arrangement in chromosomes could be used, in most cases, to determine if an organism was a prokaryote or a eukaryote?

(A)

Prokaryotic DNA	**Eukaryotic** DNA
Single circular chromosome	Multiple circular chromosomes

(B)

Prokaryotic DNA	**Eukaryotic** DNA
Multiple chromosomes	Single chromosome

(C)

Prokaryotic DNA	**Eukaryotic** DNA
Single linear chromosome	Multiple linear chromosomes

(D) ✓

Prokaryotic DNA	**Eukaryotic** DNA
Single circular chromosome	Multiple linear chromosomes

6 Gene Expression and Regulation MCQ

Answer D

Correct. This describes the typical arrangement of chromosomes: prokaryotes have DNA arranged in a single circular chromosome and eukaryotes have DNA arranged in multiple linear chromosomes.

106. Nucleotide base pairing in DNA is universal across organisms. Each pair (T − A; C − G) consists of a purine and a pyrimidine. Which of the following best explains how the base pairs form?

(A) Ionic bonds join two double-ringed structures in each pair.

(B) Hydrogen bonds join two single-ringed structures in each pair.

(C) Hydrogen bonds join a double-ringed structure to a single-ringed structure in each pair. ✓

(D) Covalent bonds join a double-ringed structure to a single-ringed structure in each pair.

Answer C

Correct. Each pair consists of one single-ringed structure (pyrimidine) joined to a double-ringed structure (purine). The hydrogen bonds allow the strands to separate for DNA replication and for transcription from DNA to RNA.

107. A tobacco plant can be made to express a gene from fireflies, resulting in the emission of light. Which of the following is the basis for this phenomenon?

(A) Chloroplasts can be made to produce light if firefly proteins are injected into plant cells.

(B) Fireflies and tobacco plants share a recent common ancestor.

(C) Fireflies and tobacco plants are infected by the same kinds of bacteria.

(D) Transcription and translation are fundamen-tally similar in both fireflies and tobacco plants. ✓

(E) Most enzymes in fireflies have the same amino acid sequence as the enzymes in tobacco plants.

108. A student analyzed a viral genome and found that the genome had the following nucleotide composition.

- 28% adenine
- 20% thymine
- 35% cytosine
- 17% guanine

Which of the following best describes the structure of the viral genome?

6 Gene Expression and Regulation MCQ

(A) Double-stranded DNA

(B) Single-stranded DNA ✓

(C) Double-stranded RNA

(D) Single-stranded RNA

109. A model of a process involving nucleic acids is shown in Figure 1.

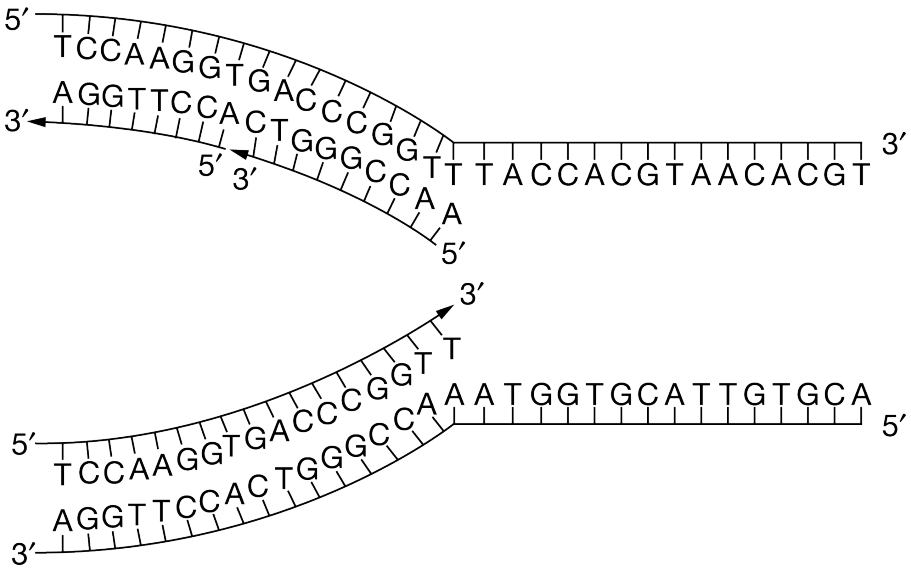


Figure 1. Model of a process involving nucleic acids

Which of the following best explains what process is represented in Figure 1 ?

(A) New DNA strands are being synthesized in the 3′ to 5′ direction from their DNA templates.

(B) New DNA strands are being synthesized in the 5′ to 3′ direction from their DNA templates. ✓

(C) A new RNA strand is being synthesized in the 3′ to 5′ end from its DNA template.

(D) Two new RNA strands are being synthesized in both directions from their DNA templates.

Answer B

Correct. New DNA strands are synthesized in the 5′ to 3′ direction from the DNA template, and this is depicted in the diagram. The leading strand is being synthesized on the bottom, while the lagging strand is being synthesized in fragments on top.

6 Gene Expression and Regulation MCQ

110. Figure 1 shows some relevant details of a model of how a deoxynucleotide, in this case dTMP, is added to a growing strand of DNA.

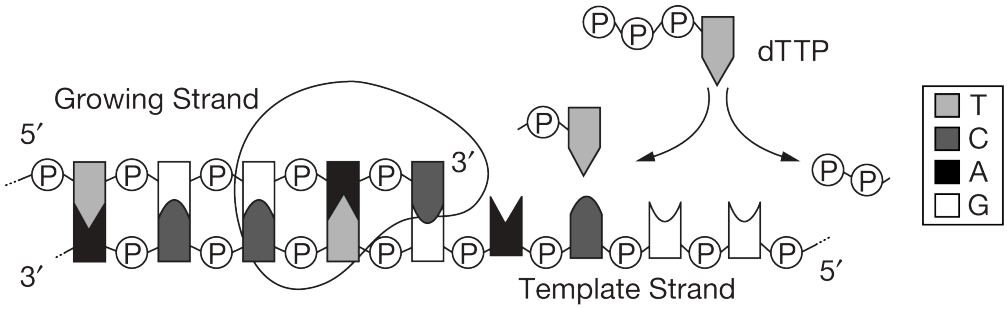


Figure 1. Model showing details of adding a deoxythymidine monophosphate (dTMP) nucleotide to a growing strand of DNA

The features of this model provide evidence for which explanation of why all growing strands are synthesized in a 5′ to 3′ direction?

(A) The two strands need to be antiparallel to bond properly.

(B) Thymine and adenine would not bond properly if the strand grew from 3′ to 5′.

(C) The translation of mRNA occurs in the 5′ to 3′ direction; therefore, the growing DNA strand must also grow in the 5′ to 3′ direction.

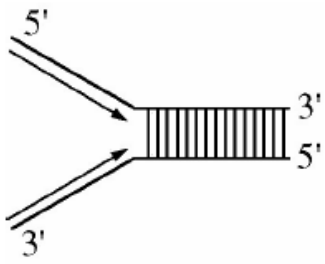
(D) The phosphate group, attached to the 5′ carbon of the dTMP, forms a covalent bond with the oxygen atom attached to the 3′ carbon of the growing strand. ✓

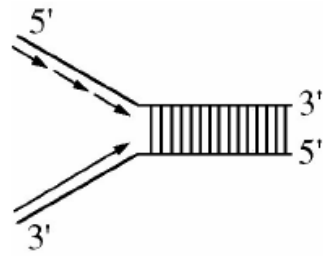
Answer D

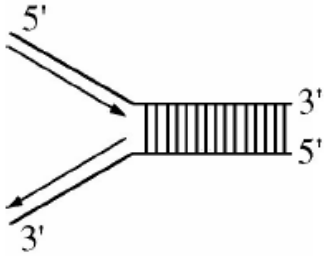
Correct. Synthesizing a growing strand requires the interaction of the phosphate groups found at the 5′ location on the new deoxynucleotide with the OH group attached to the 3′ carbon of the existing end of the growing strand. These specific interactions and the release of diphosphate explain the directionality requirement for a growing strand of DNA.

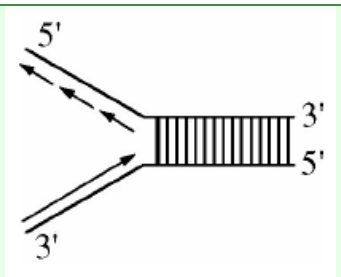
111. When DNA replicates, each strand of the original DNA molecule is used as a template for the synthesis of a second, complementary strand. Which of the following figures most accurately illustrates enzyme-mediated synthesis of new DNA at a replication fork?

6 Gene Expression and Regulation MCQ

(A) 

(B) 

(C) 

(D) 

6 Gene Expression and Regulation MCQ

112. Figure 1 depicts a simplified model of a replication bubble.

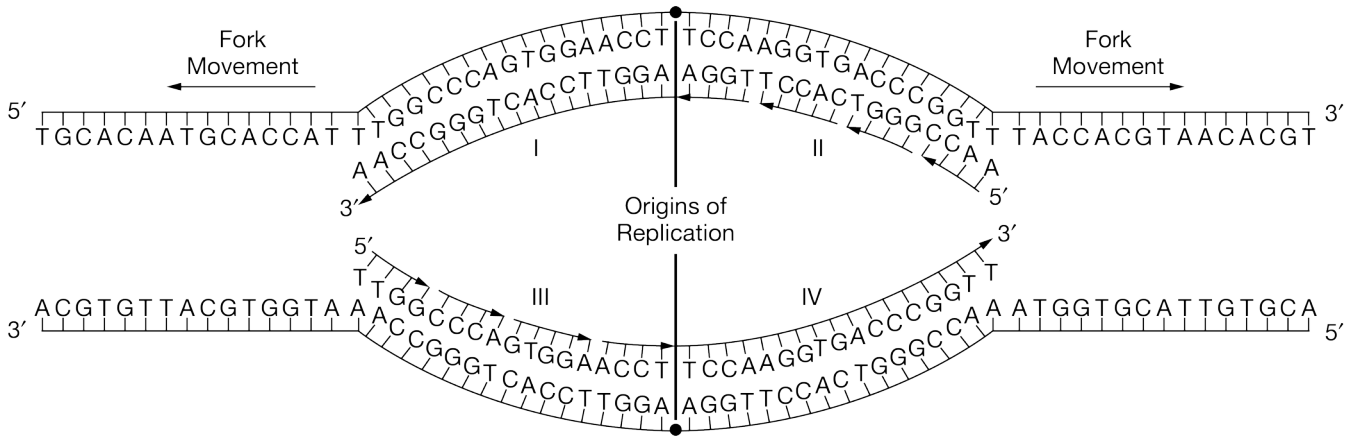


Figure 1. DNA replication bubble.

Which of the following best explains how this model illustrates DNA replication of both strands as a replication fork moves?

(A) I and IV are synthesized continuously in the 5′ to 3′ direction.

(B) II and III are synthesized in segments in the 3′ to 5′ direction.

(C) I is synthesized continuously in the 5′ to 3′ direction, and III is synthesized in segments in the 5′ to 3′ direction. ✓

(D) II is synthesized in segments after DNA polymerase is released from synthesizing strand IV.

Answer C

Correct. As a replication fork moves, DNA polymerase can continuously replicate one strand in the 5′ to 3′ direction as the double stranded DNA is separated. The other strand (lagging strand) must be replicated in segments as the fork moves, since DNA polymerase only replicates DNA in the 5′ to 3′ direction, and DNA ligase later joins these fragments together.

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Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the following enzymes.

(A) DNA ligase
(B) DNA polymerase
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase

113. Enzyme used during replication to attach Okazaki fragments to each other

(A) DNA ligase ✓
(B) DNA polymerase
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase

114. Enzyme used to position nucleotides during DNA replication

(A) DNA ligase
(B) DNA polymerase ✓
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase

115. Enzyme used in the synthesis of mRNA

(A) DNA ligase
(B) DNA polymerase
(C) RNA polymerase ✓
(D) Restriction enzyme
(E) Reverse transcriptase

116. Enzyme found in retroviruses that produce DNA from an RNA template

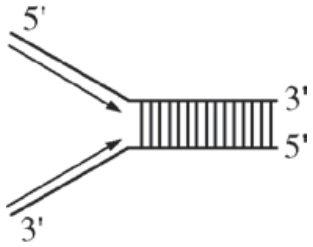
6 Gene Expression and Regulation MCQ

(A) DNA ligase
(B) DNA polymerase
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase ✓

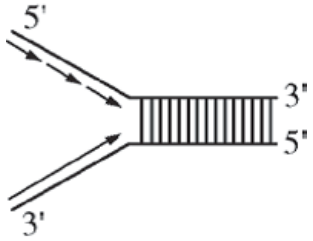
117. When DNA replicates, each strand of the original DNA molecule is used as a template for the synthesis of a second, complementary strand. Which of the following figures most accurately illustrates enzyme-mediated synthesis of new DNA at a replication fork?

6 Gene Expression and Regulation MCQ

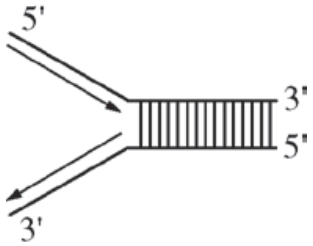
(A)



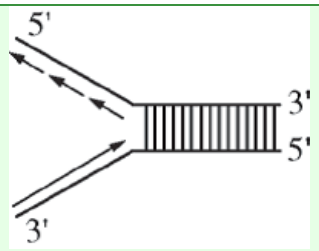
(B)



(C)



(D)



Answer D

This option is correct. It demonstrates an understanding of the structure of DNA and the process of replication. The two strands of the DNA molecule run antiparallel to each other; the 5' end of one strand pairs with the 3' end of the other strand. Replication is semi-conservative, with each strand serving as the template for the creation of new complementary strands. Helicase unzips the DNA molecules between

6 Gene Expression and Regulation MCQ

the hydrogen bonds connecting the two strands. DNA polymerase “reads” each template strand in the 3'-to-5' direction and assembles the growing DNA chain in a 5'-to-3' direction. The “leading” strand is produced continuously, but on the “lagging” strand, Okazaki fragments are produced and are connected by ligase to produce the daughter molecule.

118. Figure 1 illustrates a model of the molecules involved in DNA replication and their placement relative to each other.

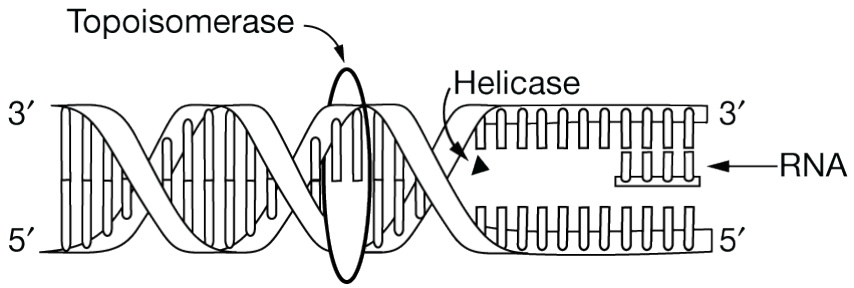


Figure 1. Model including molecules involved in DNA replication

Which of the following correctly explains where DNA replication will begin on the strand oriented 5′ → 3′, reading from left to right?

(A) DNA replication will be randomly initiated along the unwound portion of the DNA strand since base pairing will occur.

(B) DNA replication cannot occur since there is already RNA base pairing with the template strand.

(C) DNA replication will be initiated immediately to the left of the RNA, since DNA polymerase requires an RNA primer. ✓

(D) DNA replication will be initiated at the site of the topoisomerase since that is where DNA begins to uncoil.

Answer C

Correct. The RNA primer marks the beginning of DNA synthesis in the 5′ to 3′ direction. There is only one RNA primer present on the top strand in the diagram so DNA replication will begin there.

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119.

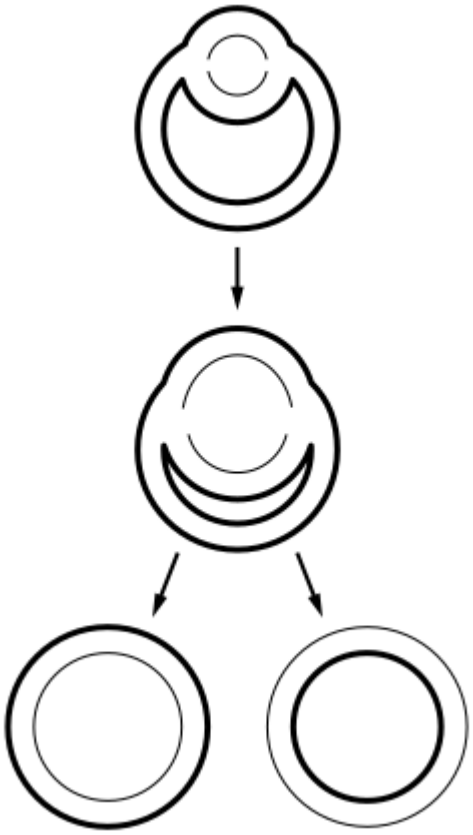

Figure 1. A cellular process

Which of the following processes is best represented by Figure 1 ?

(A) Cell signaling to increase gene expression

(B) Translation of an mRNA to produce a protein

(C) Regulation of gene expression by microRNA

(D) Replication of a plasmid ✓

Answer D

Correct. Plasmids are small circles of bacterial DNA. DNA replication starts by separating the two complementary, original strands of the plasmid, synthesizing two new strands that are complementary to each of the original strands, and finishing with two new circles of DNA, each of which contains one of the original plasmid strands and one new complementary strand.

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120. A simplified model of a DNA replication fork is represented in Figure 1. The protein labeled Enzyme 1 carries out a specific role in the DNA replication process.

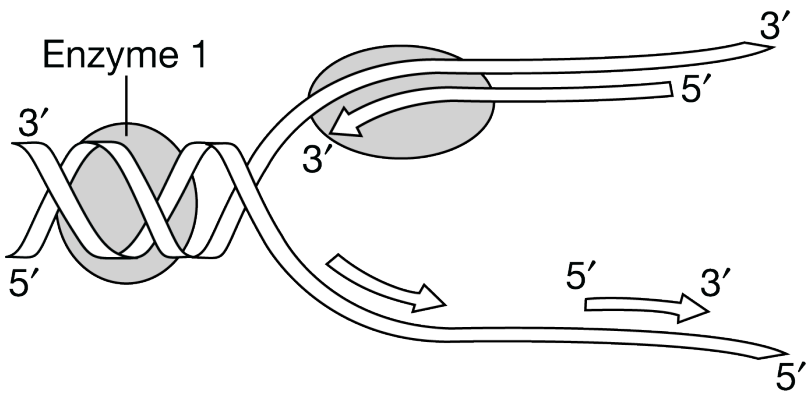

Figure 1. A simplified model of a DNA replication fork

Which of the following statements best explains the role of Enzyme 1 in the DNA replication process?

(A) Enzyme 1 is a DNA ligase that joins together the DNA fragments at a replication fork to form continuous strands.

(B) Enzyme 1 is a DNA primase that catalyzes the synthesis of RNA primers on the lagging strand of a replication fork.

(C) Enzyme 1 is a DNA polymerase that synthesizes new DNA by using the leading and lagging strands of a replication fork as templates.

(D) Enzyme 1 is a topoisomerase that relieves tension in the overwound DNA in front of a replication fork. ✓

Answer D

Correct. Because Enzyme 1 carries out its role in the DNA replication process ahead of the DNA replication fork, it is most likely a topoisomerase. Topoisomerases relieve tension in the overwound DNA in front of replication forks.

6 Gene Expression and Regulation MCQ

121. Labeled nucleotides were supplied to a cell culture before the cells began DNA replication. A simplified representation of the process for a short segment of DNA is shown in Figure 1. Labeled DNA bases are indicated with an asterisk (*).

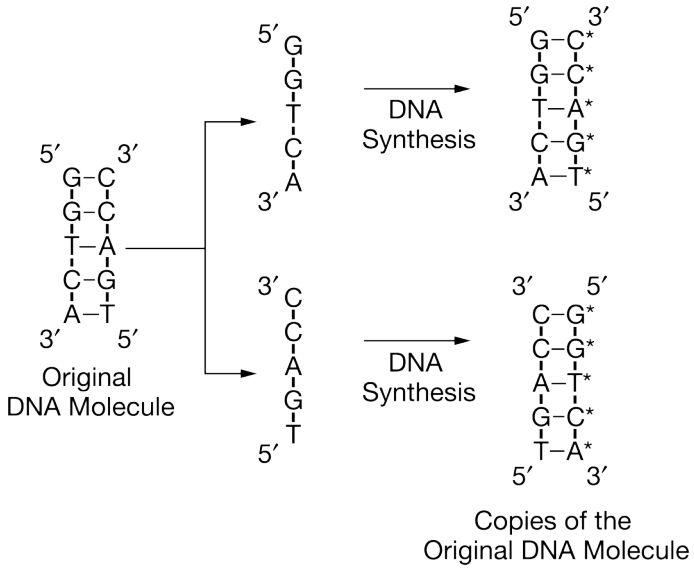

Figure 1. A simplified representation of the DNA replication process

Which of the following best helps explain how the process represented in Figure 1 produces DNA molecules that are hybrids of the original and the newly synthesized strands?

(A) Each template strand is broken down into nucleotides, which are then used to synthesize both strands of a new DNA molecule.

(B) Each template strand is broken into multiple fragments, which are randomly assembled into two different DNA molecules.

(C) Each newly synthesized strand is associated with another newly synthesized strand to form a new DNA molecule.

(D) Each newly synthesized strand remains associated with its template strand to form two copies of the original DNA molecule. ✓

Answer D

Correct. Figure 1 shows a simplified representation of semiconservative DNA replication. The process of semiconservative replication produces DNA molecules that consist of one template strand and one newly synthesized strand.

6 Gene Expression and Regulation MCQ

122. Figure 1 represents part of a process essential to gene expression.

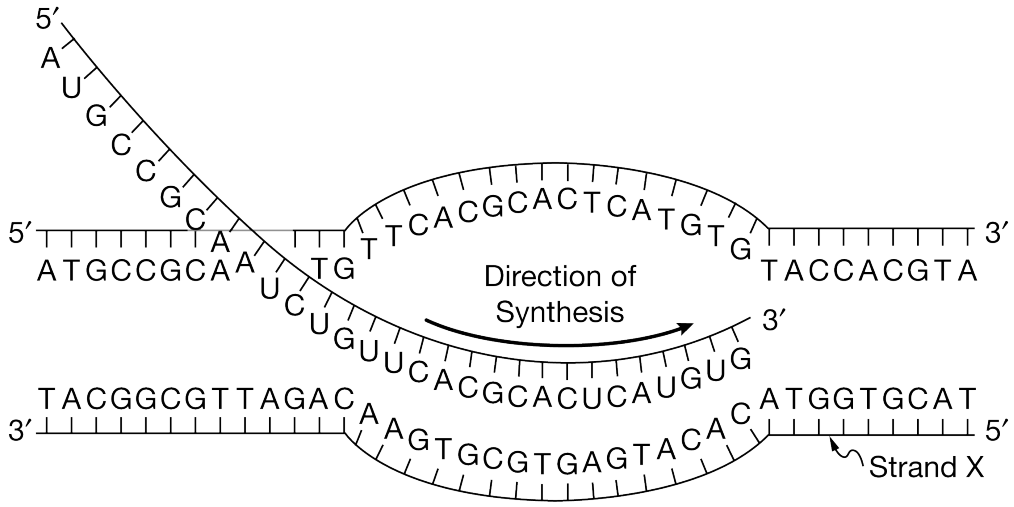

Figure 1. Model of process involved in gene expression.

Which of the following best explains what strand X represents?

(A) A complementary RNA sequence, because it contains thymine

(B) The coding strand in this process, because it is being read $3'$ to $5'$

(C) The antisense strand, because it is serving as a template ✓

(D) The pre-mRNA, because it does not yet have a GTP cap

Answer C

Correct. Strand X is serving as the template strand for the synthesis of a complementary strand of RNA. The template strand is also called the antisense strand.

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123. A model that represents a process occurring in a cell of a particular organism is shown in Figure 1.

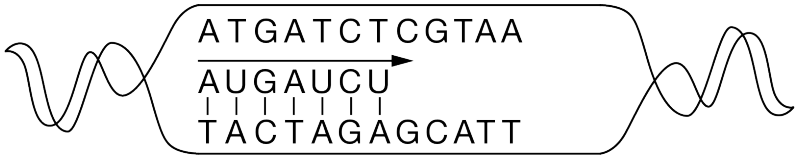

Figure 1. Process occurring in a cell

Which of the following correctly explains the process shown in Figure 1 ?

(A) DNA replication is occurring because replication is semi-conservative and the new strand is a copy of the template strand.

(B) Initiation of transcription is occurring because a strand of RNA is being produced from a DNA template strand. ✓

(C) Translation is occurring because the two strands have separated and a new strand is being produced.

(D) Alternative splicing of mRNA is occurring because the mRNA strand is being synthesized from only one strand of DNA.

Answer B

Correct. Transcription is occurring. Since uracil (U) is present in the new strand being formed, RNA is being formed from a DNA template.

124. The human *TPM1* gene encodes members of the tropomyosin family of cytoskeletal proteins. Which of the following best explains how different proteins can be made in different cell types from the one *TPM1* gene?

(A) Different introns are selectively converted to exons.

(B) Different exons are retained or spliced out of the primary transcript. ✓

(C) The GTP cap is selectively added to and activates different exons.

(D) Different portions of the primary transcript remain bound to the template DNA.

125.

Step 1 Step 2 Step 3

DNA ⟶ Pre-mRNA ⟶ mRNA ⟶ Protein

Which of the following best describes an event during step 2 in the simplified model above?

6 Gene Expression and Regulation MCQ

(A) A new RNA molecule is synthesized using a DNA template.

(B) A new polypeptide is synthesized using an RNA template.

(C) Thymine nucleotides in an RNA molecule are replaced with uracil nucleotides.

(D) Noncoding sequences are removed from a newly synthesized RNA molecule. ✓

126. If a segment of DNA is 5'-TAC GAT TAG-3', the RNA that results from the transcription of this segment will be

(A) 3'-TAC GAT TAU-5'

(B) 3'-ATG CTA ATA-5'

(C) 3'-UAC GAU UAG-5'

(D) 3'-AAC GAU UAA-5'

(E) 3'-AUG CUA AUC-5' ✓

127. Eukaryotes transcribe RNA from DNA that contains introns and exons. Alternative splicing is one posttranscriptional modification that can create distinct mature mRNA molecules that lead to the production of different proteins from the same gene. Figure 1 shows a gene and the RNA produced after transcription and after alternative splicing.

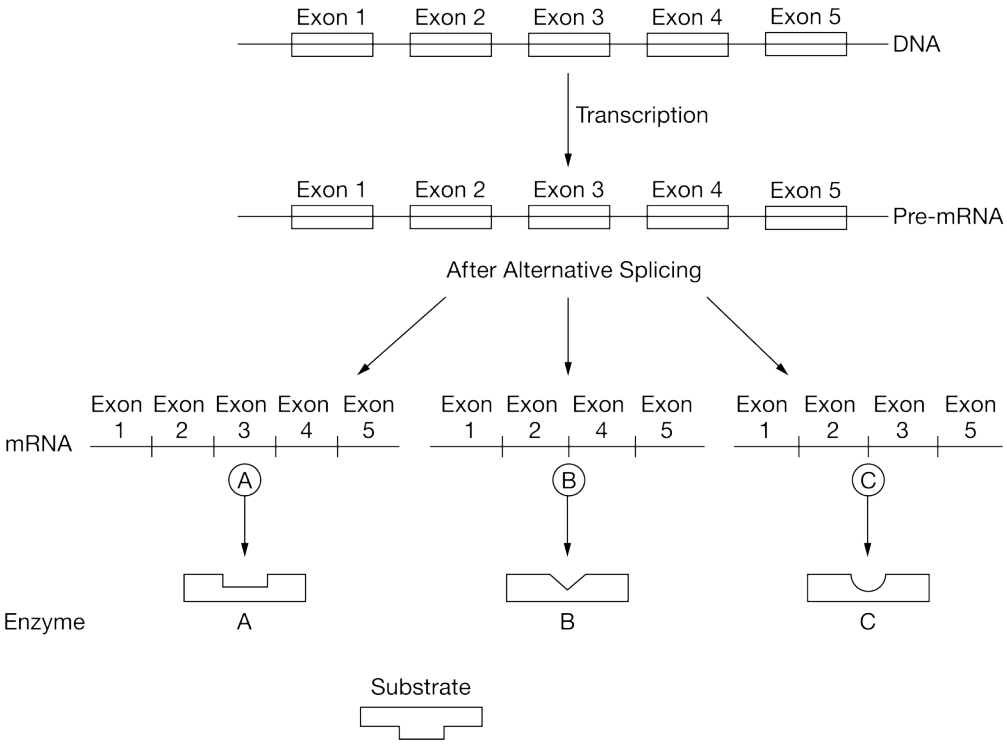

Figure 1. Model of posttranscriptional alternate splicing of mRNA

A cell needs to metabolize the substrate illustrated in Figure 1 for a vital cellular function. Which of the following best explains the long-term effect on the cell of splicing that yields only enzyme C mRNA?

6 Gene Expression and Regulation MCQ

(A) The cell will die because it is unable to metabolize the substrate without enzyme A, which is structurally specific for the substrate shown. ✓

(B) The cell will remain healthy because all three of the above enzymes can metabolize the substrate, as they are from the same gene.

(C) The cell will remain healthy because the enzyme C mRNA will undergo alternative splicing again until it transformed into enzyme A mRNA.

(D) The cell will remain healthy because enzyme-substrate interactions are nonspecific and enzyme C will eventually metabolize the substrate.

Answer A

Correct. Alternative splicing allows all three enzymes to be produced. If only enzyme C mRNA is available for translation, then only enzyme C (which is not specific for the substrate) will be produced. Thus, the cell will not be able to perform a vital cellular function and it will die.

128.

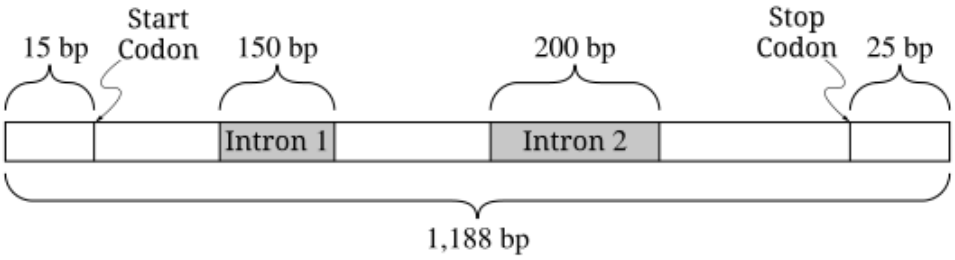

Figure 1. A model of the pre-mRNA transcribed from a gene

Which of the following is closest to the number of amino acids encoded by the mRNA produced from the pre-mRNA shown in Figure 1 ?

(A) 265 ✓

(B) 396

(C) 838

(D) 1,188

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Answer A

Correct. The total length of noncoding sequences [$150\ \text{bp} + 200\ \text{bp}$ for introns, $15\ \text{bp}$ for the 5' untranslated region (UTR), and $25\ \text{bp}$ for the 3' UTR $= 390$] as well as $3\ \text{bp}$ for the stop codon (which does not code for an amino acid) is subtracted from the total length of the $\text{pre} - \text{mRNA}$ ($1{,}188\ \text{bp}$). That total is then divided by 3 ($3\ \text{bp}$ per codon) to determine the total number of amino acids. $\frac{1{,}188-390-3}{3} = 265$

129. Which of the following normally leads to the production of functional messenger RNA in eukaryotic cells?

(A) A decrease in the rate of ribosome synthesis

(B) The removal of portions of RNA known as intervening sequences (introns) ✓

(C) A decrease in RNA polymerase activity

(D) The replication of new messenger RNA molecules from existing messenger RNA molecules

(E) The formation of peptide bonds between adjacent nucleotides

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- **Directions**: Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

mRNA Codons	Amino Acids
AGA	arginine
GGA	glycine
AGC	serine
GCA	alanine
CAG	glutamine

130. . . . . glutamine-glutamine-glutamine . . .

. . . serine-serine-serine . . .

Which of the following messenger RNA sequences could code for both of the two amino acid sequences above, simply by a shift in the reading frame?

(A) . . . AGCAGCAGCAGC . . . ✓

(B) . . . AGUAGUAGUAGU . . .

(C) . . . CAACAACAACAA . . .

(D) . . .GCUGCUGCUGCU . . .

(E) . . . GCAAGCGCAAGC . . .

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131. . . . . glycine-serine-glycine . . .

Which of the following DNA strands will code for the amino acid sequence shown above?

(A) . . . ACTCCTTCT . . .

(B) . . . TCTCCGTCG . . .

(C) . . . CCGTCGACT . . .

(D) . . . CCTCCGTCG . . .

(E) . . . CCTTCGCCT . . . ✓

132.

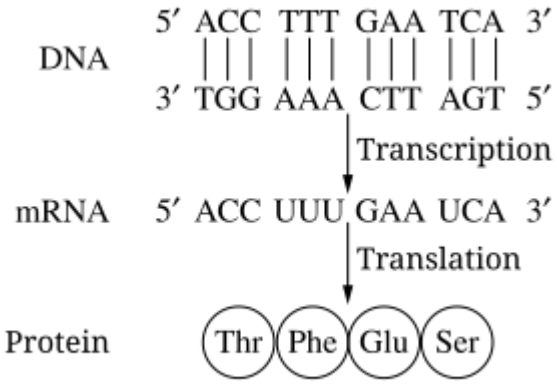


Figure 1. Model of a cellular process

Which of the following best describes the model shown in Figure 1 ?

(A) The model shows how genetic information directs the synthesis of polypeptides. ✓

(B) The model shows how a mutation in the DNA results in a nonfunctional protein.

(C) The model shows how information in the mRNA can be used to make both protein and DNA.

(D) The model shows how proteins are needed for the synthesis of both DNA and RNA.

Answer A

Correct. The model shows the central dogma of molecular biology in which DNA is transcribed to RNA, which then directs the formation of a polypeptide chain.

6 Gene Expression and Regulation MCQ

133. Figure 1 represents part of a process that occurs in eukaryotic cells. There are untranslated regions (UTR) in this sequence.

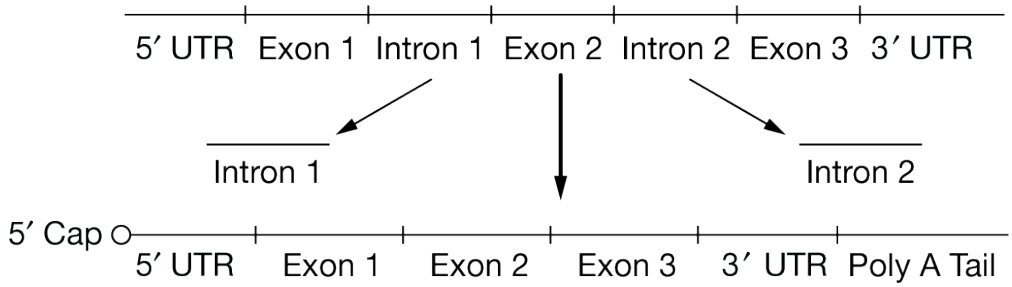


Figure 1. Cellular process involving nucleic acids

Which of the following best explains the process represented by Figure 1 ?

(A) The synthesis of mRNA in the 5′ to 3′ direction from DNA

(B) The modification of a protein to produce a functional form of that protein

(C) The translation of an mRNA molecule into a polypeptide

(D) The enzyme-regulated processing of pre-mRNA into mature mRNA ✓

Answer D

Correct. The removal of introns and the addition of a 3′ poly-A tail and a 5′GTP cap are associated with mRNA processing in eukaryotic cells, and the diagram reflects these changes.

6 Gene Expression and Regulation MCQ

134. All cells must transcribe rRNA in order to construct a functioning ribosome. Scientists have isolated and identified rRNA genes that contribute to ribosomal structure for both prokaryotes and eukaryotes. Figure 1 compares the transcription and processing of prokaryotic and eukaryotic rRNA.

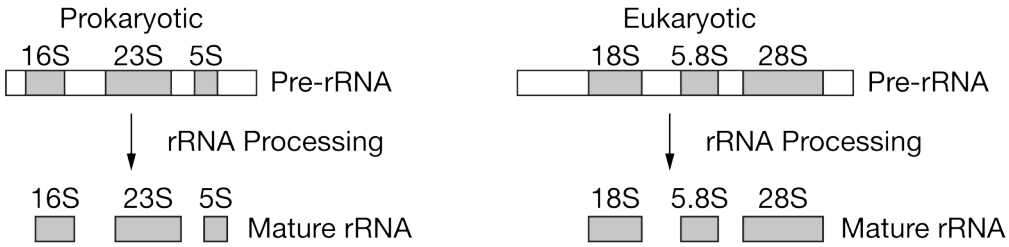


Figure 1. Comparison of rRNA processing in prokaryotes and eukaryotes

Which of the following statements provides the best explanation of the processes illustrated in Figure 1 ?

(A) Introns are removed from the pre-rRNA, and the mature rRNA molecules are joined and then translated to produce the protein portion of the ribosome.

(B) Introns are removed from the pre-rRNA, and each mature rRNA molecule is translated to produce the proteins that make up the ribosomal subunits.

(C) Sections of the pre-rRNA are removed, and the mature rRNA molecules are available to combine with proteins to form the ribosomal subunits. ✓

(D) Sections of the pre-rRNA are removed, and the mature rRNA molecules are available to bring different amino acids to the ribosome.

Answer C

Correct. In both prokaryotes and eukaryotes, the mature rRNA molecules are removed from a longer transcript. These mature rRNA molecules, together with proteins, form the large and small ribosomal subunits.

135. Which of the following statements concerning a gene is correct?

(A) A gene can code for a specific protein. ✓

(B) A gene can exist in alternate forms called introns.

(C) A gene undergoes crossing-over during DNA replication.

(D) A gene that is very similar in sequence in a human and in a bacterium is probably a recent mutation.

(E) A gene that is expressed in every offspring of every generation is recessive.

6 Gene Expression and Regulation MCQ

136. The following the DNA sequence is a small part of the coding (nontemplate) strand from the open reading frame of *β-hemoglobin* gene. Given the codon chart listed below, what would be the effect of a mutation that deletes the G at the beginning of the DNA sequence?

5'- GTT TGT CTG TGG TAC CAC GTG GAC TGA - 3'

Second Base in Codon

First Base in Codon	U	C	A	G	Third Base in Codon
U	UUU Phe UUC UUA Leu UUG	UCU UCC UCA Ser UCG	UAU Tyr UAC UAA Stop UAG Stop	UGU Cys UGC UGA Stop UGG Trp	U C A G
C	CUU CUC CUA Leu CUG	CCU CCC CCA Pro CCG	CAU His CAC CAA Gln CAG	CGU CGC CGA Arg CGG	U C A G
A	AUU AUC Ile AUA AUG Met or Start	ACU ACC ACA Thr ACG	AAU Asn AAC AAA Lys AAG	AGU Ser AGC AGA Arg AGG	U C A G
G	GUU GUC GUA Val GUG	GCU GCC GCA Ala GCG	GAU Asp GAC GAA Glu GAG	GGU GGC GGA Gly GGG	U C A G

(A) The mutation precedes the gene, so no changes would occur.

(B) Lysine (lys) would replace glutamine (gln), but there would be no other changes.

(C) The first amino acid would be missing, but there would be no other change to the protein.

(D) The reading frame of the sequence would shift, causing a change in the amino acid sequence after that point. ✓

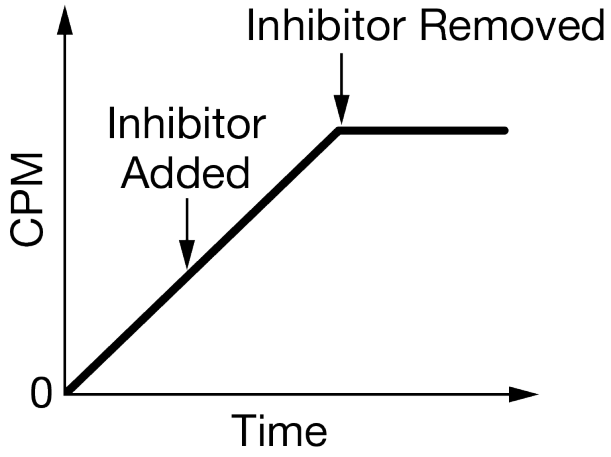
137. Cycloheximide (CHX) is a eukaryote protein synthesis inhibitor. It is used in biomedical research to inhibit protein synthesis in eukaryotic cells studied *in vitro*. Its effects are rapidly reversed by simply removing it from the culture medium.

In a translation experiment using a fungus culture, radiolabeled amino acids were added to the culture, allowing the researchers to measure the growth of a single polypeptide chain by measuring counts per minute (CPM). As the chain grew, the CPM increased. After a certain amount of time, CHX was added to the mixture, and the experiment continued. After an additional amount of time, the CHX was removed from the culture medium.

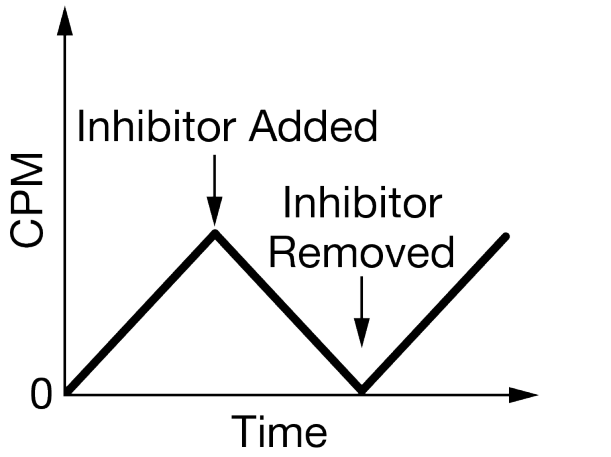
Which of the following graphs best predicts the data collected during the experiment?

6 Gene Expression and Regulation MCQ

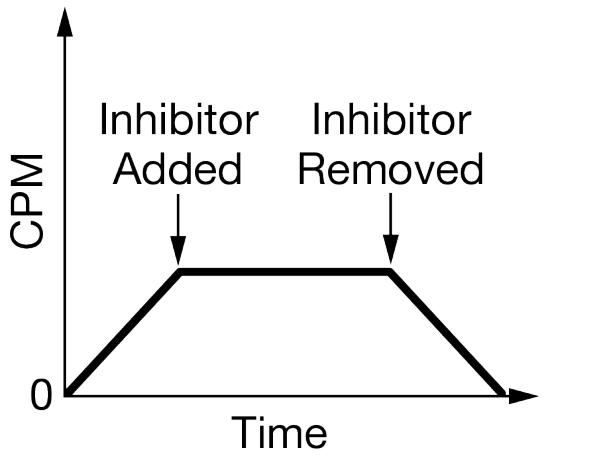
(A)



(B)

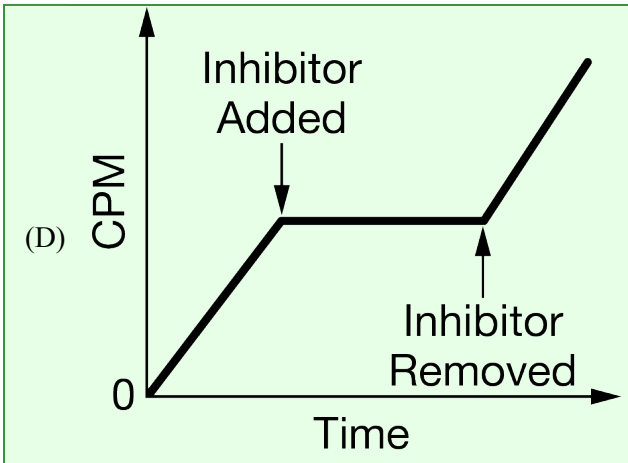


(C)



6 Gene Expression and Regulation MCQ

(D)



Answer D

Correct. In the presence of the drug, protein synthesis halted and the polypeptide chain stopped growing. When the drug was removed, however, elongation resumed and the polypeptide chain started growing again.

138.

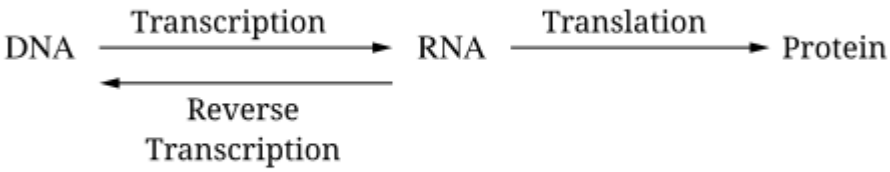


Figure 1. A model of a biological process

Which of the following processes requires all of the steps represented in Figure 1 ?

6 Gene Expression and Regulation MCQ

(A) Cell signaling to increase gene expression in an animal cell

(B) Replication of a retrovirus after it enters a host cell ✓

(C) Regulation of gene expression in a cell by microRNA

(D) Production of plasmids in a bacterial cell

Answer B

Correct. Retroviruses insert viral RNA and the enzyme reverse transcriptase into a host cell. The reverse transcriptase copies the RNA into DNA, and then the cell's own transcription and translation machinery synthesizes new viral proteins in the process of producing new viruses.

139. Analysis of DNA sequences from two individuals of the same species results in a greater estimate of genetic variability than does analysis of amino acid sequences from the same individuals because

(A) different DNA sequences can code for the same amino acid ✓

(B) some amino acid variations cannot be detected by protein electrophoresis

(C) DNA sequencing is a more reliable technique than protein electrophoresis

(D) proteins are more easily damaged than is DNA

(E) DNA is more heat-sensitive and therefore varies more

140. During the infection cycle for a typical retrovirus, which uses RNA as genetic material, the genetic variation in the resulting population of new virus particles is very high because of

(A) damage to the virus particle from envelope loss during infection

(B) errors introduced in the DNA molecule through reverse transcription ✓

(C) errors in the protein molecules produced in translation

(D) recombination of the genomes of free virus particles

Answer B

This option is correct. The process of reverse transcription does not involve proofreading.

6 Gene Expression and Regulation MCQ

141.

TABLE 1. RATE OF POLYPEPTIDE ELONGATION

Species	Number of Amino Acids Added	Duration of Measurement (seconds)
E. coli	1,200	60
S. cerevisiae	800	80

During translation of an mRNA, amino acids are added one-by-one to the growing end of a polypeptide. Researchers wanted to determine whether there is a difference in the rate of amino acid addition in *E. coli*, a prokaryote, and *S. cerevisiae*, a single-celled eukaryote (Table 1).

Based on the data in Table 1, how does the rate of polypeptide elongation in *E. coli* compare with the rate of polypeptide elongation in *S. cerevisiae*?

(A) The rate in *E. coli* is one-tenth the rate in *S. cerevisiae*.

(B) The rate in *E. coli* is half the rate in *S. cerevisiae.*

(C) The rate in *E. coli* is one-and-one-half times the rate in *S. cerevisiae.*

(D) The rate in *E. coli* is twice the rate in *S. cerevisiae*. ✓

Answer D

Correct. The rate of elongation in *E. coli* is $20 \text{ amino acids/second}$, while the rate of elongation in *S. cerevisiae* is $10 \text{ amino acids/second}$. Thus, *E. coli* has a rate that is twice that of *S. cerevisiae*.

6 Gene Expression and Regulation MCQ

142. Molecular biologists are studying the processes of transcription and translation and have found that they are very similar in prokaryotes and eukaryotes, as summarized in Table 1.

Table 1. Comparison of Selected Features of Transcription and Translation

Feature of Protein Synthesis	Prokaryotes	Eukaryotes
Transcription		
RNA polymerase	Present	Present
mRNA	Synthesized	Synthesized
Location	Cytoplasm	Nucleus
Translation		
Large Ribosomal Subunit	50S	60S
Small Ribosomal Subunit	30S	40S
tRNA	Present	Present
Location	Cytoplasm	Cytoplasm

Based on the information in Table 1, which of the following best predicts a key difference in prokaryotes and eukaryotes with regard to transcription and translation?

(A) The two processes will occur simultaneously in prokaryotes but not eukaryotes. ✓

(B) Prokaryotic mRNA is shorter than eukaryotic mRNA.

(C) Eukaryotic mRNA contains more coding regions than prokaryotic DNA.

(D) The processing of mRNA by eukaryotes is required for the mRNA to leave the nucleus.

Answer A

Correct. Since transcription and translation both occur in the cytoplasm, they can occur simultaneously in prokaryotes. Note that the spatial separation of these two processes in eukaryotes increases the capacity for gene regulation.

143. Retroviruses have an RNA genome. HTLV-1 is a lysogenic retrovirus that establishes a latent infection in human cells. By which of the following mechanisms does infection by a retrovirus such as HTLV-1 most likely cause long-lasting genetic changes to host cells?

(A) The host-cell ribosomes translate the viral RNA genome that enters the cell upon initial viral infection.

(B) The viral RNA polymerase that transcribes host genes has a high error rate.

(C) The RNA viral genome is reverse transcribed into DNA that integrates into the host genome. ✓

(D) The RNA viral genome integrates into the host genome.

6 Gene Expression and Regulation MCQ

144. Retroviruses such as HIV and hepatitis B virus use RNA as their genetic material rather than DNA. In addition, they contain molecules of reverse transcriptase, an enzyme that uses an RNA template to synthesize complementary DNA.

Which of the following best predicts what will happen when a normal cell is exposed to a retrovirus?

(A) The reverse transcriptase will cut the host DNA into fragments, destroying the host cell.

(B) The reverse transcriptase will insert the viral RNA into the host's genome so it can be transcribed and translated.

(C) The reverse transcriptase will produce DNA from the viral RNA, which can be incorporated into the host's genome and then transcribed and translated. ✓

(D) The reverse transcriptase will force the host ribosomes to translate the viral RNA prior to polypeptide assembly.

Answer C

Correct. Reverse transcriptase will transcribe DNA from RNA. The DNA integrates into the host genome and is transcribed and translated for the assembly of new viral progeny.

145. Antibiotics interfere with prokaryotic cell functions. Streptomycin is an antibiotic that affects the small ribosomal subunit in prokaryotes. Specifically, streptomycin interferes with the proper binding of tRNA to mRNA in prokaryotic ribosomes.

Which of the following best predicts the most direct effect of exposing prokaryotic cells to streptomycin?

(A) Amino acid synthesis will be inhibited.

(B) No mRNA will be transcribed from DNA.

(C) Posttranslational modifications will be prevented.

(D) Synthesis of polypeptides will be inhibited. ✓

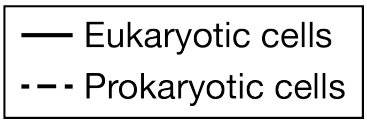
Answer D

Correct. Translation of mRNA involves assembling polypeptides from amino acids attached to tRNA specific for each amino acid. Streptomycin prevents the ribosome from interacting properly with tRNA, preventing the synthesis of needed proteins and leading to the death of the prokaryote.

6 Gene Expression and Regulation MCQ

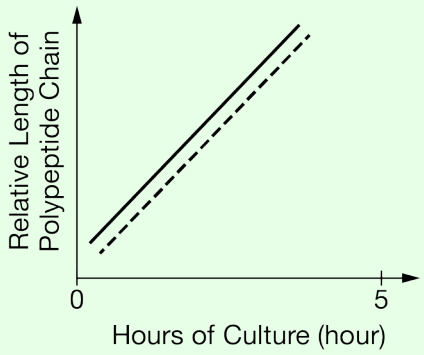
146. Nuclear pores regulate the passage of substances into and out of the nucleus. Antibodies such as mAb414 have been used to inhibit the movement of substances through the nuclear pores of rat liver cells. Scientists cultured rat liver cells (eukaryotic) and bacteria cells (prokaryotic) in separate dishes with radioactively labeled amino acids. A specific gene in both cell types was engineered to synthesize identical polypeptide chains, and translation of this gene was measured. The procedure was repeated with mAb414 (the inhibitor) added to each of the two cell cultures, and translation was monitored again.

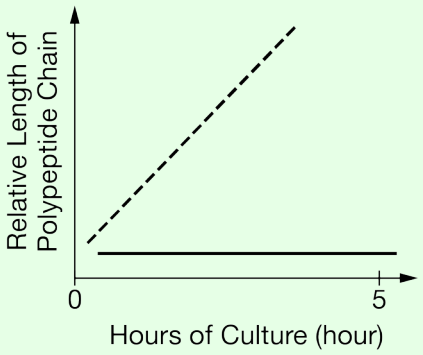
Which of the following sets of graphs best summarizes the results of the experiments?



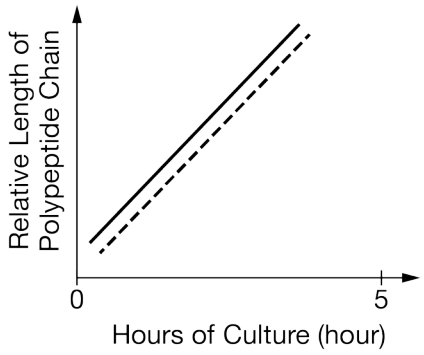
6 Gene Expression and Regulation MCQ

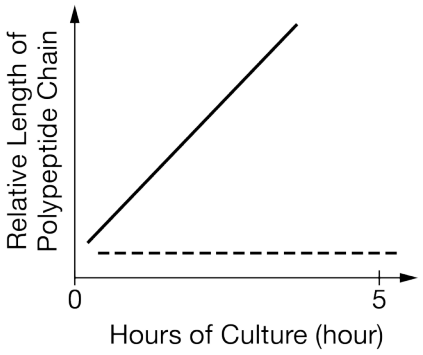
(A)



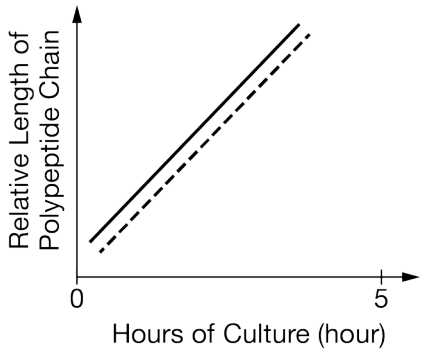


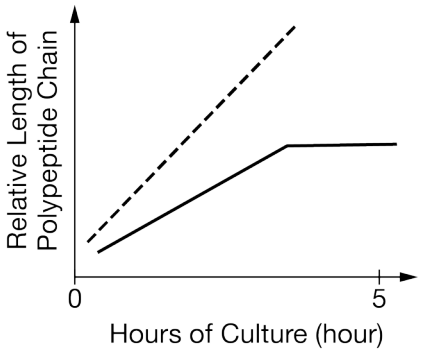
(B)



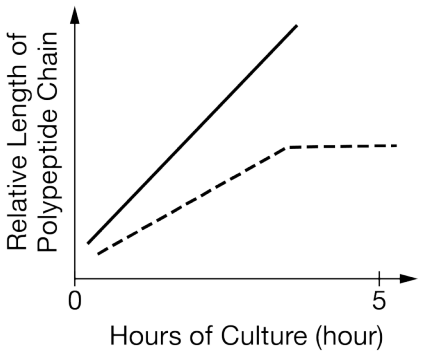


(C)





(D)



6 Gene Expression and Regulation MCQ

Answer A

Correct. In eukaryotes, transcription and translation occur at different areas in the cell, whereas in prokaryotes, they occur simultaneously. In the experimental dish with the eukaryotes, the drug inhibited the translocation of mRNA to the cytoplasm, thereby preventing the translation, so no polypeptide chain was formed. However, in the prokaryotes, with no nucleus and both processes occurring simultaneously, translation occurred and the polypeptide chain was formed.

147.

Second Base in Codon

First Base in Codon	U	C	A	G	Third Base in Codon
U	UUU, UUC: Phe; UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr; UAA Stop; UAG Stop	UGU, UGC: Cys; UGA Stop; UGG Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAC: His; CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile; AUG: Met or Start	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn; AAA, AAG: Lys	AGU, AGC: Ser; AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp; GAA, GAG: Glu	GGU, GGC, GGA, GGG: Gly	U C A G

5′-GACCGCAUGGUGACGAAAUUUGGCCAUUAA-3′

Based on the universal genetic code, which of the following represents the correct polypeptide that will result from translation of the mRNA molecule shown, beginning with the first available start codon?

(A) Asp – Arg – Met – Val – Thr – Lys – Phe – Gly – His

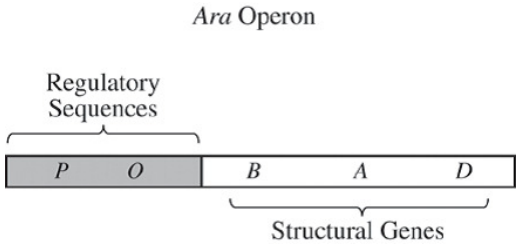
(B) Met – Arg – Asp – Stop – His – Gly – Phe – Lys – Thr – Val

(C) Met – Val – Thr – Lys – Phe – Gly – His ✓

(D) Val – Thr – Lys – Phe – Gly – His

6 Gene Expression and Regulation MCQ

148.



The regulatory sequences of the operon controlling arabinose metabolism (*ara* operon) were studied to determine whether bacteria can respond to changes in nutrient availability. It is predicted that if those regulatory sequences are functioning properly, the bacteria will produce the enzymes involved in arabinose metabolism (structural genes *B*, *A,* and *D*) in the presence of arabinose.

If a gene that encodes a green fluorescent protein (GFP) is substituted for the structural genes of the operon, activation of the regulatory sequences can be assayed by GFP expression. A culture of *E. coli* cells underwent a transformation procedure with a plasmid containing the regulatory sequences of the *ara* operon directly upstream of the gene encoding the GFP. The plasmid also confers ampicillin resistance to bacteria. Samples were then plated on different types of culture media. (Note: The GFP fluoresces only under UV light, not under white light.) The table below shows the results.

Transformation Results

Type of Culture Media			Color of Colonies Under White Light	Color of Colonies Under UV Light
Agar	Ampicillin	Arabinose		
+	-	-	All white	All white
+	+	-	All white	All white
+	-	+	All white	Mostly white, some green
+	+	+	All white	All green

+ Indicates the presence of the indicated substance in the culture media
- Indicates the absence of the indicated substance in the culture media

Which of the following can best be used to justify why the GFP is expressed by *E. coli* cells after transformation with the plasmid?

(A) The presence of arabinose in the nutrient agar activated the expression of the genes located downstream of the *ara* operon regulatory sequences. ✓

(B) The combination of ampicillin and arabinose in the nutrient agar inhibited the expression of certain gene products, resulting in the increased expression of the GFP.

(C) The nutrient agar without arabinose but with ampicillin activated the expression of the genes located downstream of the *ara* operon regulatory sequences.

(D) Both arabinose and ampicillin were required in the nutrient agar to activate the expression of genes located downstream of the *ara* operon regulatory sequences.

6 Gene Expression and Regulation MCQ

Answer A

This option is correct. Arabinose will bind with the repressor protein allowing expression of the genes controlled by the *ara* regulatory sequences. Because the genes for ampicillin resistance and GFP were inserted downstream of the regulatory sequences, these genes will be expressed and the products will cause the resultant colonies to be green under UV light. This answer indicates that the student is able to describe the connection between the regulation of gene expression and observed differences between different kinds of organisms.

149. Arctic foxes typically have a white coat in the winter. In summer, when there is no snow on the ground, the foxes typically have a darker coat.

Which of the following is most likely responsible for the seasonal change in coat color?

(A) The decrease in the amount of daylight in winter causes a change in gene expression, which results in the foxes growing a lighter-appearing coat. ✓

(B) The diet of the foxes in summer lacks a particular nutrient, which causes the foxes to lose their white coat and grow a darker-colored coat.

(C) Competition for mates in the spring causes each fox to increase its camouflage with the environment by producing a darker-appearing coat.

(D) The lower temperatures in winter denature the pigment molecules in the arctic fox coat, causing the coat to become lighter in color.

Answer A

This option is correct. It demonstrates the ability to explain the influence of environmental factors on the phenotype of an organism. When the genes of the arctic fox that produce the dark coat are blocked by cold temperatures, a white coat results. In the summer these genes are not blocked, and a darker coat results.

6 Gene Expression and Regulation MCQ

150. The table below describes the action of two genes involved in the regulation of nervous system development in the nematode *C. elegans*.

	Gene *A*	Gene *B*	Observation
Pattern 1	Inactive	Inactive	No neurons develop.
Pattern 2	Inactive	Active	No neurons develop.
Pattern 3	Active	Inactive	Greater-than-normal number of neurons develop.
Pattern 4	Active	Active	Normal number of neurons develop.

Which of the following claims is best supported by the data?

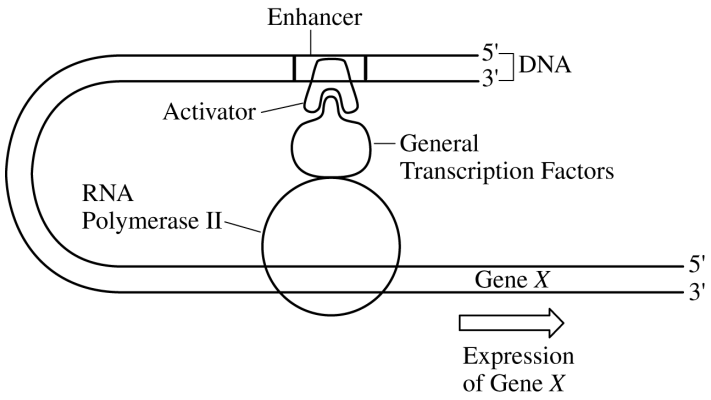
(A) Gene *A* promotes neuron development; gene *B* promotes programmed cell death in neuronal precursors. ✓

(B) Gene *A* promotes programmed cell death in neuronal precursors; gene *B* promotes neuron development.

(C) Gene *B* must be active before gene *A* can function.

(D) Gene *B* must be inactive before gene *A* can function.

151.



The figure above depicts the DNA-protein complex that is assembled at the transcriptional start site of gene *X* when the expression of gene *X* is activated in liver cells. Previous studies have shown that gene *X* is never expressed in nerve cells. Based on the diagram, which of the following most likely contributes to the specific expression pattern of gene *X*?

(A) Expression of gene *X* produces large amounts of tRNA but undetectable amounts of mRNA.

(B) The general transcription factors inhibit the activation of gene *X* in liver cells by blocking the activator from binding to RNA polymerase II.

(C) The activator is a sequence-specific DNA-binding protein that is present in some tissues but not in other tissues. ✓

(D) The enhancer is a unique DNA segment that is added to the nuclear DNA of some cells of an organism during the process of mitotic cell division but not other cells.

6 Gene Expression and Regulation MCQ

152. Histone methyltransferases are a class of enzymes that methylate certain amino acid sequences in histone proteins. A research team found that transcription of gene R decreases when histone methyltransferase activity is inhibited.

Which scientific claim is most consistent with these findings?

(A) DNA methylation inhibits transcription of gene R.

(B) Histone modifications of genes are usually not reversible.

(C) Histone methylation condenses the chromatin at gene R so transcription factors cannot bind to DNA.

(D) Histone methylation opens up chromatin at gene R so transcription factors can bind to DNA more easily. ✓

Answer D

Correct. This claim is consistent with the findings presented. Histone methylation in this case increases transcription of gene R, so blocking histone methyltransferase activity would inhibit transcription.

153. Pit vipers are snakes that have deep sensory openings (pits) between their nostrils and eyes. These pits enable the snakes to sense heat emitted by endotherms, such as mammals, allowing the snakes to find their prey. In an experiment, researchers discovered that nerve cells in the pits transcribe the gene $TRPA1$ 400 times more often than do other nerve cells in the snakes. The researchers claim that the $TRPA1$ gene is involved in the snakes' ability to sense heat.

Which of the following would be the best control to add to the researchers' experiment?

(A) Measuring translation of $TRPA1$ mRNA in pit nerve cells

(B) Measuring translation of $TRPA1$ mRNA in nerve cells of mammals

(C) Measuring $TRPA1$ transcription in nerve cells of snakes lacking pits ✓

(D) Measuring $TRPA1$ transcription in nerve cells of mammals

Answer C

Correct. Comparing the transcription rate of the $TRPA1$ gene in the pit nerve cells of snakes with pits relative to nerve cells in snakes lacking pits would serve as a control to indicate if the high level of $TRPA1$ gene transcription is exclusive to pit nerve cells.

154. Maximal transcription of the *lac* operon requires

6 Gene Expression and Regulation MCQ

(A) the presence of high levels of glucose

(B) the removal of the operator region of the operon

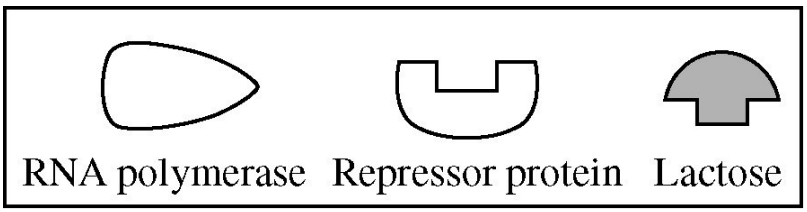
(C) the presence of cyclic AMP and lactose ✓

(D) the activation of the repressor protein

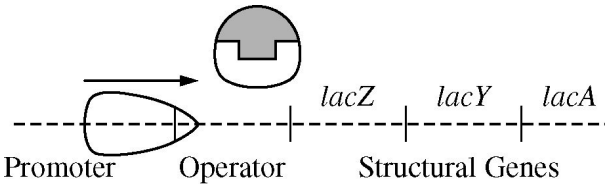
(E) the presence of cyclin and cyclin-dependent kinase

155. Lactose digestion in *E. coli* begins with its hydrolysis by the enzyme *β*-galactosidase. The gene encoding *β*-galactosidase, *lacZ*, is part of a coordinately regulated operon containing other genes required for lactose utilization.

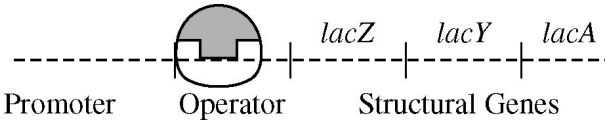
Which of the following figures correctly depicts the interactions at the *lac* operon when lactose is NOT being utilized? (The legend below defines the shapes of the molecules illustrated in the options.)



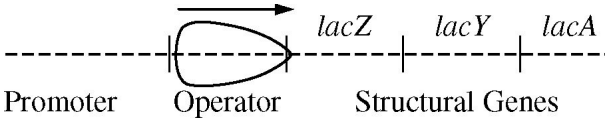
(A)



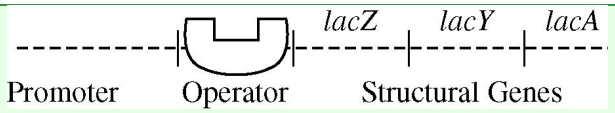
(B)



(C)



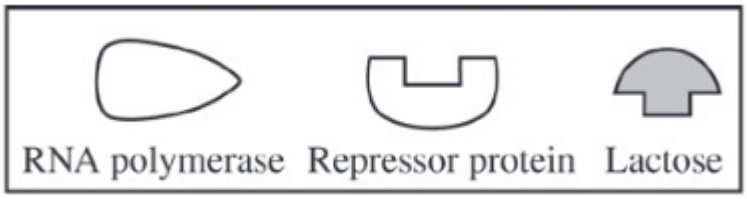
(D) ✓



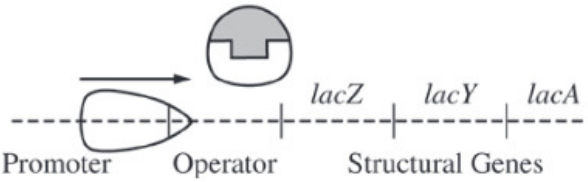
6 Gene Expression and Regulation MCQ

156. Lactose digestion in *E. coli* begins with its hydrolysis by the enzyme *β*-galactosidase. The gene encoding *β*-galactosidase, *lacZ*, is part of a coordinately regulated operon containing other genes required for lactose utilization.

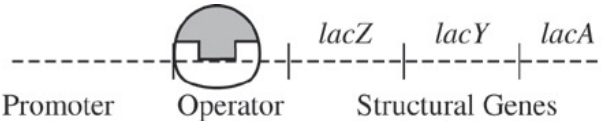
Which of the following figures correctly depicts the interactions at the *lac* operon when lactose is NOT being utilized? (The legend below defines the shapes of the molecules illustrated in the options.)



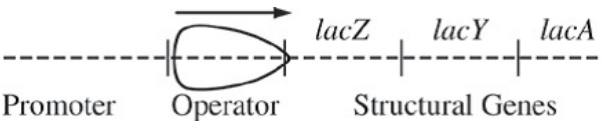
(A)



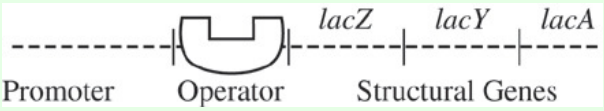
(B)



(C)



(D) ✓



6 Gene Expression and Regulation MCQ

157. Figure 1 represents a metabolic process involving the regulation of lactose metabolism by *E. coli* bacteria. Lactose is utilized for energy by *E. coli* when glucose is not present. Allolactose is an isomer of lactose that is in the environment of these bacteria when lactose is present. The CAP site prevents the binding of RNA polymerase when glucose is present in the environment. The $lacZ$, $lacY$, and $lacA$ genes code for proteins needed for lactose metabolism.

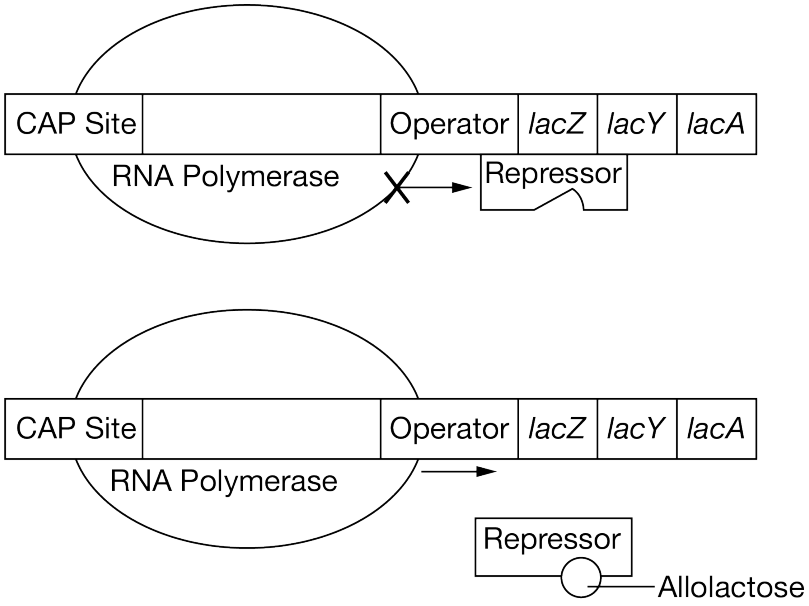


Figure 1. Model of *lac* operon, comparing repressed and active states

Which is a scientific claim that is consistent with the information provided and Figure 1 ?

(A) The presence of excess lactose blocks the functioning of RNA polymerase in this operon.

(B) When bound to the operator, the repressor protein prevents lactose metabolism in *E. coli*. ✓

(C) The binding of the repressor protein to the operator enables *E. coli* to metabolize lactose.

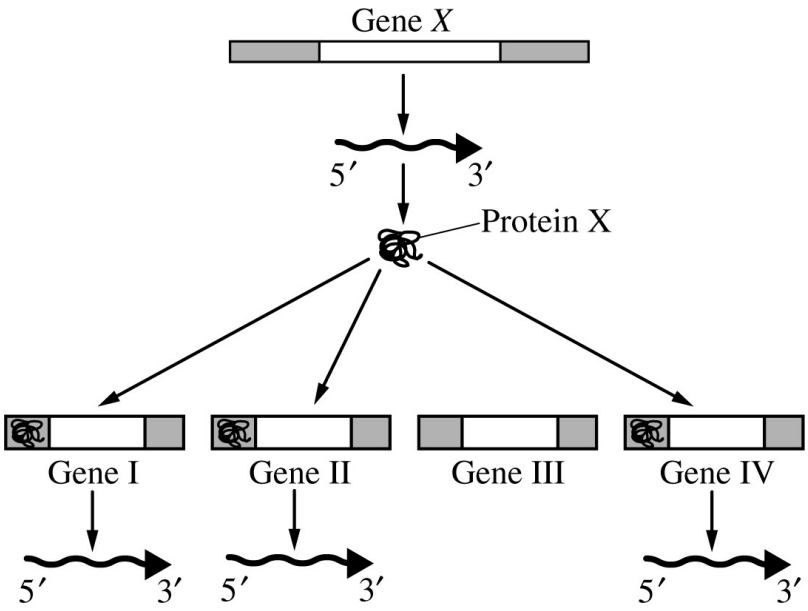
(D) Allolactose acts as an inducer that binds to the operator, allowing *E. coli* to metabolize lactose.

Answer B

Correct. When the repressor protein is bound to the operator, the *lac* operon is turned off and the genes for lactose metabolism cannot be transcribed.

6 Gene Expression and Regulation MCQ

158. Phenotype is determined, in part, by which genes are expressed. The diagram below illustrates how the product of gene *X* regulates the expression of other genes.



Which of the following statements best explains how protein X regulates gene expression?

(A) Protein X is responsible for processing pre-mRNA.

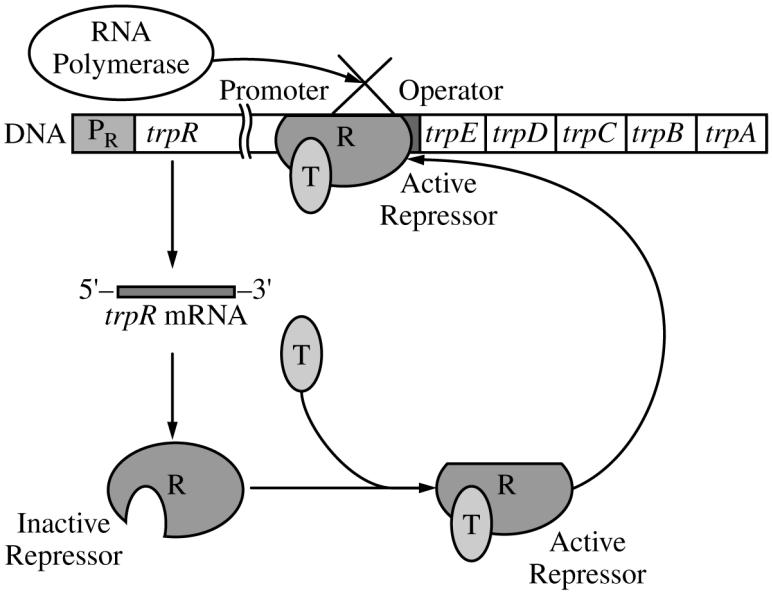
(B) Protein X is responsible for activating transcription of some genes but not others. ✓

(C) Protein X is a member of some cytoplasmic protein complexes but not others.

(D) Protein X causes specific base-pair changes to produce new alleles.

6 Gene Expression and Regulation MCQ

159.



R	= Repressor molecule
T	= Tryptophan molecule
P_R	= Promoter for the *trpR* gene
trpR	= Repressor gene
Operator	= Binding site for repressor molecule

The *Trp* operon is a coordinately regulated group of genes (*trpA–trpE*) that are required for tryptophan biosynthesis in *E. coli*. Based on the figure above, which of the following correctly explains the regulation of the *Trp* operon?

(A) In the absence of tryptophan, the repressor is active and binds to the *Trp* operator, preventing RNA polymerase from transcribing the operon.

(B) In the presence of tryptophan, the repressor is active and binds to the *Trp* operator, preventing RNA polymerase from transcribing the operon. ✓

(C) In the absence of tryptophan, the *trpR* gene is inactive, preventing the production of the repressor that blocks expression of the operon.

(D) In the presence of tryptophan, the *trpR* gene is inactive, preventing the production of the repressor that blocks expression of the operon.

6 Gene Expression and Regulation MCQ

160. Figure 1 illustrates processes related to control of transcription and translation in a cell.

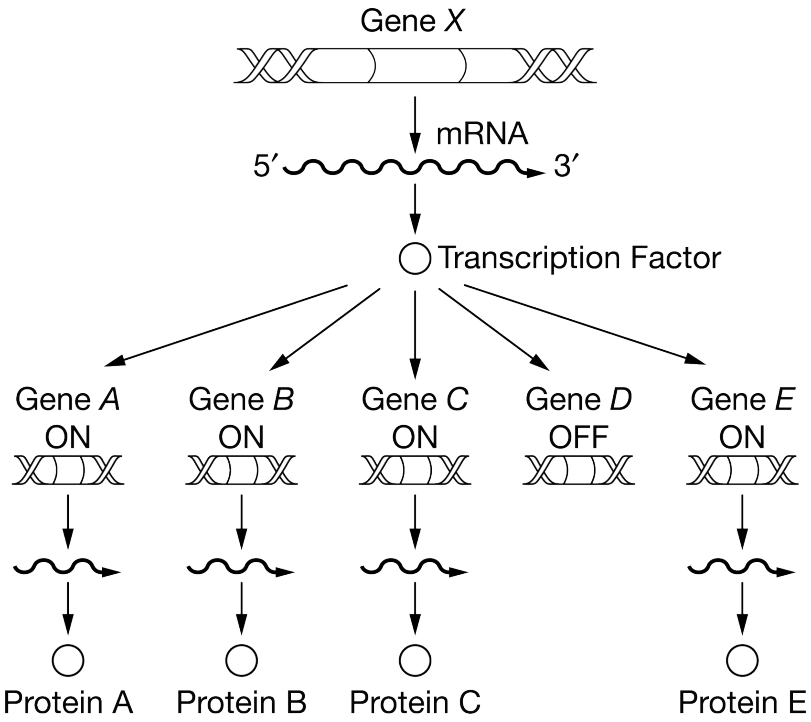


Figure 1. Model of a relationship between a transcription factor and selected genes

Which of the following scientific claims is most consistent with the information provided in Figure 1 ?

(A) Gene X codes for a transcription factor required for transcription of gene D.

(B) A single transcription factor regulates transcription similarly, regardless of the specific gene.

(C) Transcription of genes A, B, and C is necessary to transcribe gene E.

(D) Different genes may be regulated by the same transcription factor. ✓

Answer D

Correct. The same transcription factor may regulate the transcription of multiple genes and exhibit differential regulatory effects on different genes.

6 Gene Expression and Regulation MCQ

161. Figure 1 represents a portion of a process that occurs during protein synthesis.

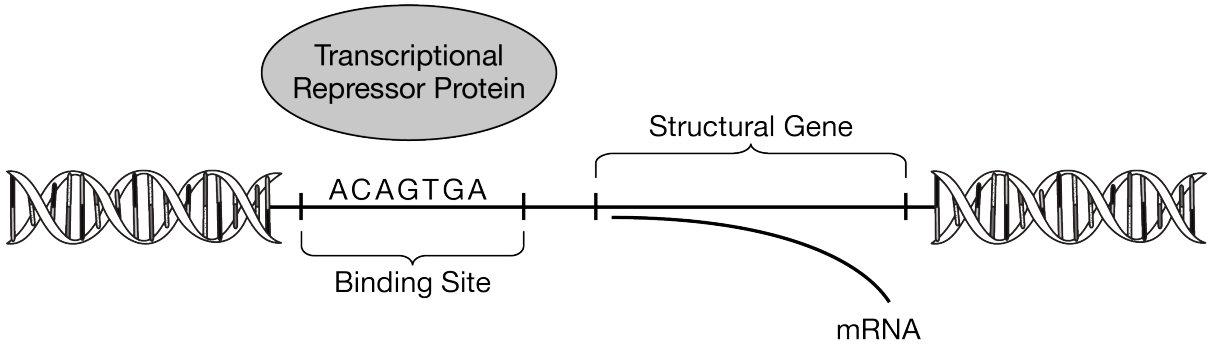


Figure 1. Model of selected features of DNA transcription

Which claim is most consistent with the information provided by the diagram and current scientific understanding of gene regulation and expression?

(A) Reversible changes in the DNA sequence may influence how a gene is expressed in a cell.

(B) Some sequences of DNA can interact with regulatory proteins that control transcription. ✓

(C) This is an inducible operon controlled by several regulatory factors.

(D) The transcription factor may produce mutations in the binding site at the promoter sequence inhibiting the synthesis of the protein.

Answer B

Correct. The regulatory base sequence (called a silencer) interacts with regulatory proteins such as this transcriptional repressor, which inhibits transcription.

6 Gene Expression and Regulation MCQ

The three-spined stickleback (Gasterosteus aculeatus) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mostly of individuals with pronounced pelvic spines, as shown in Figure 1. Individuals in freshwater stickleback populations, on the other hand, typically have reduced pelvic spines, as shown in Figure 2.

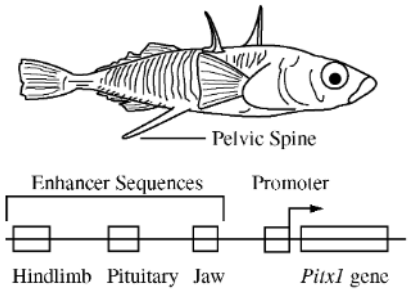


Figure 1. Marine stickleback

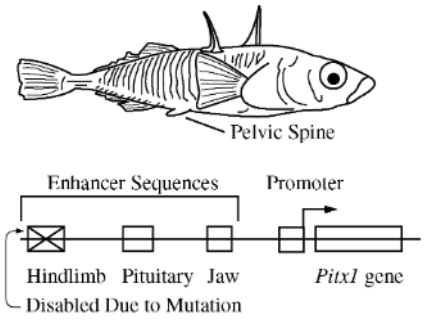


Figure 2. Freshwater stickleback

As represented in Figure 1 and Figure 2, the phenotypic difference between marine and freshwater sticklebacks involves Pitx1, a gene that influences the formation of the jaw, pituitary gland, and pelvic spine. Enhancer sequences upstream of the Pitx1 genetic locus regulate expression of the Pitx1 gene at the appropriate times and in the appropriate tissues during development. Previous studies have found that a mutation in the hindlimb enhancer interferes with the formation of a pronounced pelvic spine.

162. Which of the following best explains how sticklebacks in the same population with identical copies of the Pitx1 gene can still show phenotypic variation in the pelvic spine character?

(A) The Pitx1 gene is carried on different chromosomes in different individuals.

(B) Expression of the Pitx1 gene is affected by mutations at other genetic loci. ✓

(C) The genetic code of the Pitx1 gene is translated differently in males and females.

(D) The subcellular location of the Pitx1 gene changes when individuals move to a new environment.

163. A mutation that affects Pitx1 gene function in all tissue types is most likely to be at which of the following genetic loci?

(A) Hindlimb enhancer

(B) Pituitary enhancer

(C) Jaw enhancer

(D) Promoter ✓

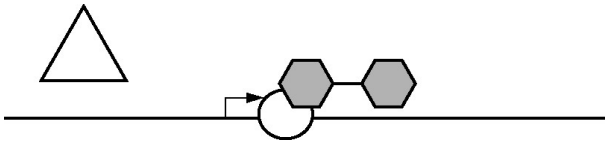
6 Gene Expression and Regulation MCQ

164. 26. The *lac* operon in *E. coli* consists of genes that code for enzymes necessary for the breakdown of lactose. When lactose is absent, the operon is inactive because a repressor protein binds to a specific site in the *lac* operon. When lactose is present, lactose molecules bind to the repressor protein, causing the repressor protein to dissociate from the binding site. In the absence of glucose (a preferred energy source for bacteria), the protein CAP binds to a regulatory site near the *lac* promoter to activate transcription of the *lac* operon.

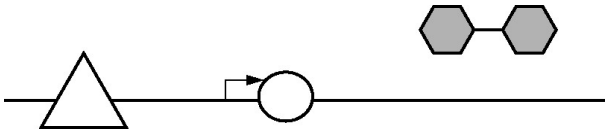
The following symbols represent the macromolecules involved in regulation of the *lac* operon.

In the diagrams below, the horizontal line represents the *lac* operon and the bent arrow represents the transcription start site of the *lac* operon. Which of the following diagrams best represents the scenario in which lactose is available to the cell and glucose is absent?

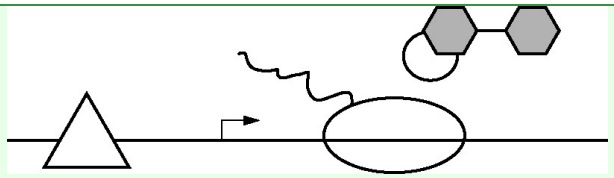
(A)



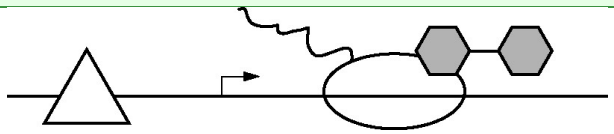
(B)



(C) ✓



(D)



7 Natural Selection MCQ

1. All of the following are examples of prezygotic genetic isolating mechanisms EXCEPT:

(A) Male fireflies of different species have differing flash patterns.

(B) Three closely related orchid species flower on different days.

(C) The progeny of a cross between two different lizard species fail to develop properly. ✓

(D) One species of snake lives in water and another is terrestrial.

(E) Peeper frogs mate in April, and tree frogs mate in June.

2. All of the following were likely present on the primitive Earth during the evolution of self-replicating molecules EXCEPT

(A) amino acids and nucleotides

(B) nitrogen

(C) simple carbohydrates

(D) freestanding liquid water

(E) an O_2-rich atmosphere ✓

3.

TABLE 1. PERCENT HOMOLOGY OF $MEF2A$ NUCLEOTIDE SEQUENCES

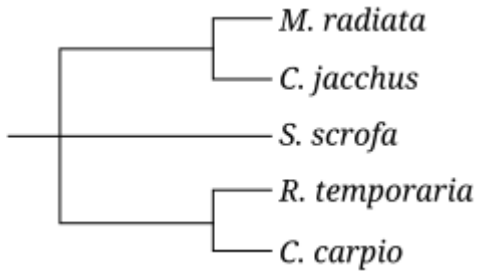
Species	*M. radiata*	*C. jacchus*	*S. scrofa*	*C. carpio*	*R. temporaria*
M. radiata		90.3	71.5	42.7	50.9
C. jacchus			70.6	42.1	50.5
S. scrofa				40.8	48.9
C. carpio					44.5
R. temporaria					

The $MEF2A$ gene is critical for muscle development in many vertebrates. A researcher sequenced the $MEF2A$ gene from five vertebrate species and calculated the percent homology between the nucleotide sequences from each pair of species (Table 1).

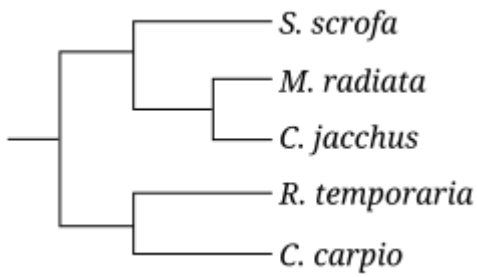
Which of the following cladograms is the most accurate representation of the data in Table 1 ?

7 Natural Selection MCQ

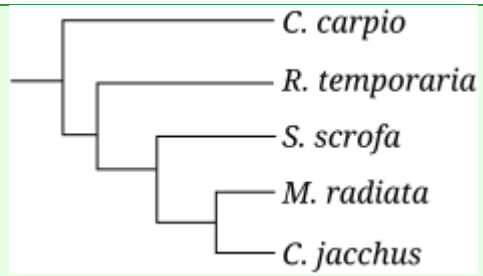
(A)



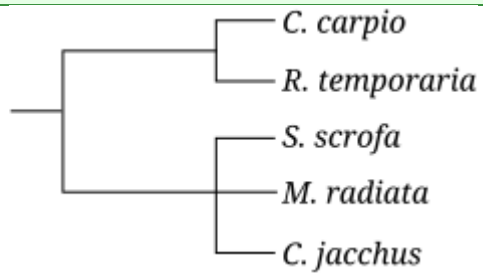
(B)



(C) ✓



(D)



Answer C

Correct. This cladogram assumes that *M. radiata* is the most recently evolved species and has thus undergone the most genetic changes. *M. radiata* also shares the most sequence homologies with *C. jacchus*, and this is indicated in the cladogram. As other species are compared with *M. radiata* and the level of homology decreases, it is assumed that these species are more closely related to the most recent common ancestor.

7 Natural Selection MCQ

4. A small number of lizards from a mainland population have been deposited on four isolated islands because of the effects of a rare strong storm.

Which of the following best predicts the outcome of these lizards reproducing for many generations on the islands?

(A) Courtship rituals specific to each island lizard species prevent the lizards from interbreeding.

(B) Speciation results from bottleneck events that happened before the ancestral species reached the islands.

(C) The different species that currently exist are the result of hybridization between lizards from different islands.

(D) The isolation prevents gene flow; thus, the lizards on different islands experience prezygotic isolation. ✓

Answer D

Correct. The primary cause of speciation is the prevention of gene flow between different populations. In this case, the original lizard populations were geographically isolated because they were on different islands, which prevented migration. Each island's *Anolis* species radiated from a similar-appearing common ancestor that underwent speciation in isolation.

5. A number of different phylogenies (evolutionary trees) have been proposed by scientists. These phylogenies are useful because they can be used to

(A) determine when two similar populations of a species evolved into two separate species

(B) evaluate which groups of organisms may be most closely related ✓

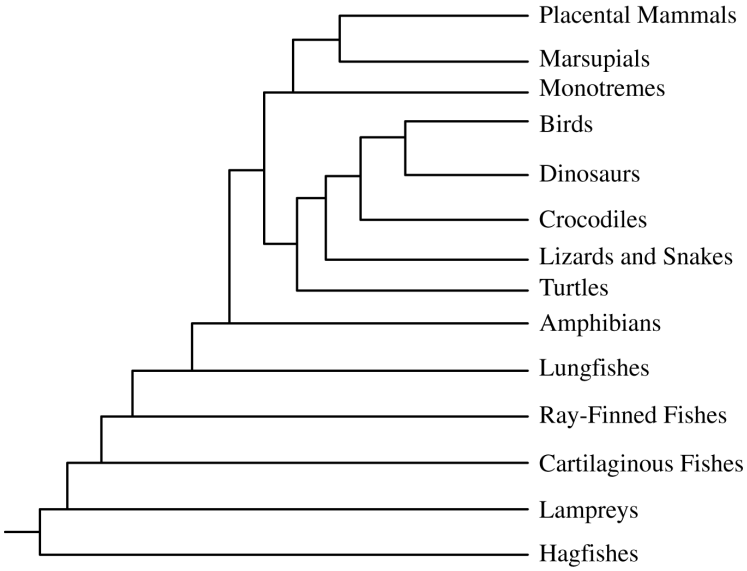
(C) demonstrate that all photosynthetic organ-isms are members of the Kingdom Plantae

(D) demonstrate that natural selection is the prevailing force in evolution

(E) demonstrate which taxa (groups of organisms) contain the most highly evolved species

7 Natural Selection MCQ

6.



To determine the evolutionary history and relationships among organisms, scientists gather evidence from a wide variety of sources including paleontology, embryology, morphology, behavior, and molecular biology. A phylogenetic tree of vertebrates is shown.

Which of the following statements is most consistent with the phylogenetic tree shown?

(A) Birds and turtles evolved their own means of gas exchange independently of the other vertebrates.

(B) Mammals are most closely related to birds because they share a direct common ancestor.

(C) The common ancestor of reptiles, birds, and mammals produced amniotic eggs. ✓

(D) Crocodiles are direct descendents of ray-finned fishes since they live in the same environment.

7. A researcher hypothesizes that RNA molecules were present in the most recent common ancestor of all living organisms. Which of the following scientific questions would best test the hypothesis?

(A) Is it possible to produce an RNA polymer in a laboratory setting?

(B) How many distinct functions can a particular RNA molecule perform in a cell?

(C) How many different monomers of RNA are found in a eukaryotic cell?

(D) Do any known organisms function entirely without RNA? ✓

7 Natural Selection MCQ

Ascorbic acid (vitamin C) is an organic molecule necessary for the health of plants and animals. The majority of animals, including most mammals, synthesize ascorbic acid from organic precursors, but some primates are unable to synthesize ascorbic acid and must instead acquire it from dietary sources, such as certain fruits and vegetables.

The *L-gulonolactone oxidase (GULO)* gene encodes an enzyme that catalyzes a required step in the biosynthesis of ascorbic acid. Most mammals carry a functional copy of the *GULO* gene, but some primates carry only a *GULO* pseudogene, which is a nonfunctional variant.

A comparison of *GULO* genes and *GULO* pseudogenes from different animals can provide insight into the evolutionary relatedness of the animals. In Table I, selected members of some mammalian groups are listed, along with an indication of their ability to synthesize ascorbic acid. Table II shows an alignment of amino acid coding sequences from homologous regions of the *GULO* genes and *GULO* pseudogenes of the organisms listed in Table I. Figure 1 represents the universal genetic code.

TABLE I: SELECTED MAMMALIAN GROUPS

Group	Selected Members	Biosynthesis of Ascorbic Acid
Nonprimate mammals	Elephant, mouse	Yes
Primate mammals	Lemur	Yes
	Orangutan, chimpanzee	No
	Human	No

TABLE II: DNA SEQUENCE ALIGNMENT*

	Relative Positions of Nucleotides in Nontemplate (Coding) Sequence																										
	1 (5′)	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27 (3′)
Elephant	G	A	C	A	C	C	C	A	T	C	T	G	A	A	G	A	A	G	T	C	G	G	A	A	T	A	C
Mouse	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	A	G	T	C	T	G	A	G	T	A	C
Lemur	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	G	G	T	C	C	G	A	G	T	A	C
Orangutan	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Chimp	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Human	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C

*For each DNA segment, the alternating shaded and unshaded nucleotides indicate the triplet codons of the open reading frame, shown from left (5′) to right (3′) as the nontemplate (coding) strand. An "–" indicates the absence of a nucleotide.

7 Natural Selection MCQ

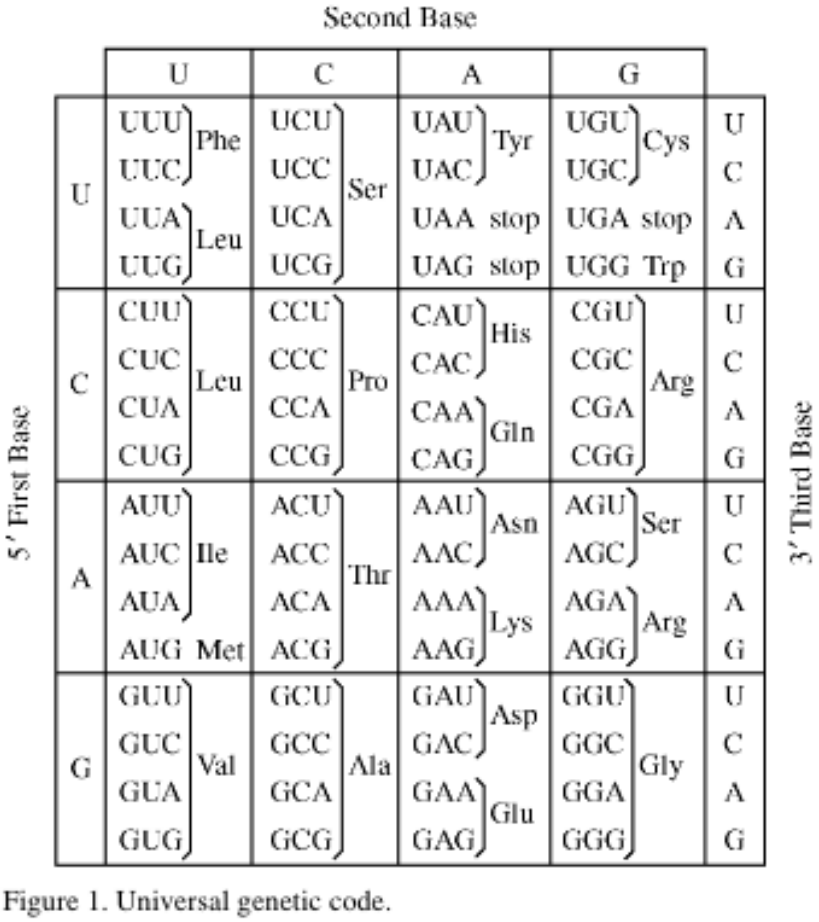
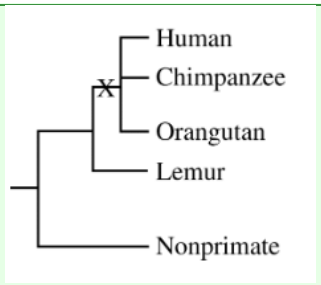
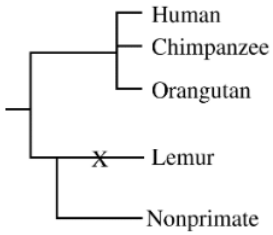


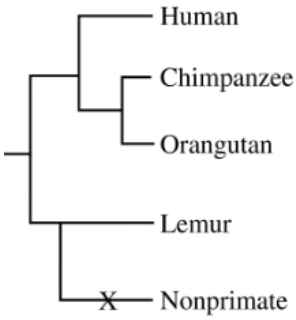
Figure 1. Universal genetic code.

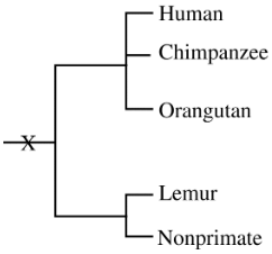
8. Which of the following phylogenetic trees best illustrates (with the X) the point at which the mutation in the *GULO* gene most likely occurred during the evolutionary history of these organisms?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

A biologist spent many years researching the rate of evolutionary change in the finch populations of a group of islands. It was determined that the average beak size (both length and mass) of finches in a certain population increased dramatically during an intense drought between 1981 and 1987. During the drought, there was a reduction in the number of plants producing thin-walled seeds.

9. Which of the following procedures was most likely followed to determine the change in beak size?

7 Natural Selection MCQ

(A) A few finches were trapped in 1981 and again in 1987, and their beak sizes were compared.

(B) The beak size in fifteen finches was measured in 1987, and the beak size in the original finches was determined by estimation.

(C) The beak size in a large number of finches was measured every year from 1981 to 1987. ✓

(D) Finches were captured and bred in 1981, and the beak size of the offspring was measured.

Answer C

This option is correct. It demonstrates ability to apply the scientific method to evaluate data that illustrate evolution as an ongoing process. Based on observation, the beak sizes of a large number of individuals were measured between the time interval (1981 to 1987) correlating with the period of drought.

10. The biologist discovered that from 1988 to 1993, the average beak size declined to pre-1981 levels. The reversal in beak size from 1988 to 1993 was most likely related to which of the following events?

(A) A loss of food supply for the finches

(B) The end of the drought ✓

(C) An increase in drought conditions

(D) An increase in predators consuming finches

Answer B

This option is correct. It demonstrates an ability to evaluate data providing evidence for evolution as an ongoing process. At the end of the drought period, there was a natural return to a variety of seeds available — including a decrease in thick-walled seeds. Thus, allele frequencies shifted to pre-drought numbers because no longer was there selective pressure favoring larger beak sizes.

11. Which of the following statements might best explain the increase in average beak size in the finch population during the drought?

(A) Finches with bigger beaks are better able to crack thick-walled seeds and produce more surviving offspring. ✓

(B) Finches with bigger beaks can attack and kill finches with smaller beaks.

(C) Finches with bigger beaks possess more powerful flight muscles and are able to find more food.

(D) Finches that crack large seeds develop larger beaks over time.

7 Natural Selection MCQ

Answer A

This option is correct. It demonstrates understanding of evolution through natural selection and explains the relationship between changes in genotype, phenotype, and environment. During the drought period, plants with thick- walled seeds were favored; in turn, finches with larger beaks that could crack the seeds to obtain nourishment were selected for, thus increasing chances for survival.

12. Which of the following best explains the mechanism behind the change in beak size in the finch population?

(A) The formation of two new finch species from a single parent species

(B) A change in gene frequencies in the finch population due to selective pressure from the environmental change ✓

(C) A new allele appearing in the finch population as a result of mutation

(D) The achievement of dynamic equilibrium in the finch population as a result of homeostasis

Answer B

This option is correct. It demonstrates an understanding of the role of natural selection in evolution. Natural selection acts on phenotypic variations in populations. During the drought period the environment favored plants with thick-walled seeds, and finches with larger beaks had a selective advantage over finches with smaller beaks in using the thick-walled seeds as food. More food led to greater reproductive success for the large-beaked finches. Thus, allele frequencies within the finch population changed, with an increase in alleles conferring the more advantageous phenotype — larger beaks. Changes in allele frequencies in a population provide evidence for evolution.

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13. A widely accepted hypothesis about the origin of life on Earth is that life arose approximately 3.5 billion years ago as the result of a complex sequence of chemical reactions that took place spontaneously in Earth's atmosphere. Another hypothesis about the origin of life suggests that life began somewhere else in the universe and arrived on Earth by chance.

Which of the following questions might scientists ask to most reliably determine if there has ever been life on Mars?

(A) How far from Earth was Mars 3.5 billion years ago?

(B) Is there scientific proof that Earth ever had an oxygen-free atmosphere?

(C) Did Mars have the elements present in its atmosphere to support life? ✓

(D) Was there oxygen gas on Mars 3.5 billion years ago?

7 Natural Selection MCQ

14. Commercial bananas are grown as a monoculture, with all banana plants cloned from one original banana plant. The commercial strains of bananas are seedless, so each new banana plant has to be manually planted from a cutting of an existing banana root. In the 1950s, the Gros Michel banana strain, the dominant export banana at that time, was destroyed by the fungus *Fusarium oxysporum*. A new *Fusarium* resistant variety, the Cavendish banana, was developed and is currently the banana strain grown for export. Recently, a *Fusarium* strain that successfully attacks the Cavendish strain has been documented.

Which of the following best provides reasoning supporting a method that would help protect commercial banana crops from infection by pathogenic organisms such as *Fusarium* fungi?

(A) The commercial banana strains should be exposed to X-rays to encourage random mutations that will then be passed to offspring, producing resistance to pathogenic organisms.

(B) The Cavendish banana plants should be exposed to pathogenic organisms under controlled conditions, so the plants can be encouraged to mutate and develop resistance to the pathogens.

(C) The commercial banana strains should not be grown in monocultures, since many pathogenic organisms are able to evolve rapidly in response to a single selective pressure. ✓

(D) Growing the Cavendish strain under different conditions will allow natural selection to produce the variation needed to resist infection by pathogenic organisms.

Answer C

Correct. The commercial banana strains are clones that lack both genetic diversity and the ability to generate diversity through sexual reproduction. Any single clone grown in a monoculture will not prevent selection for a pathogen able to successfully infect the plants.

15. The condition in which there are barriers to successful interbreeding between individuals of different species in the same community is referred to as

(A) latent variations

(B) sterility

(C) structural differences

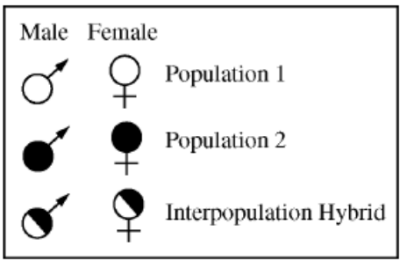
(D) geographic isolation

(E) reproductive isolation ✓

7 Natural Selection MCQ

Over many generations, two populations of the same species can diverge into separate species through reproductive isolation. The figures below represent a model of speciation and show the results of matings between individuals from two diverging populations at four different stages of speciation. The males represented in the model are heterogametic, which means they have two different sex-determining chromosomes, (e.g., XY). The females are homogametic, which means they have two similar sex-determining chromosomes, (e.g., XX). The offspring from each mating are labeled interpopulation hybrids.

Figure 1 shows the results of a mating between a male and a female from the two populations. In each subsequent figure, the males are from one of the diverging populations and the females are from the other population. The fertility and viability of the offspring from each mating are indicated in the figures.



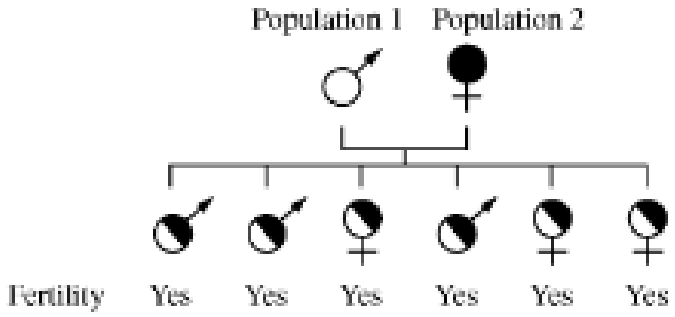


Figure 1. Mating between individuals from the initial populations

7 Natural Selection MCQ

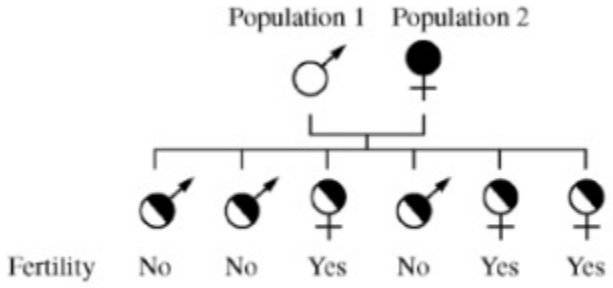


Figure 2. Mating between individuals from diverging populations at an intermediate stage of speciation

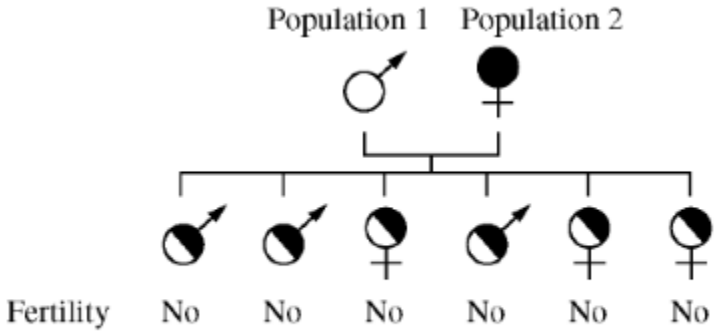


Figure 3. Mating between individuals from diverging populations at a late stage of speciation

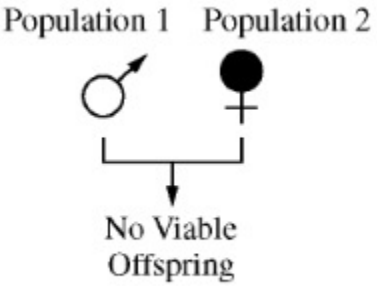


Figure 4. Mating between individuals from divergent populations after speciation is complete

16. Based on the model of speciation presented, which of the following describes the most likely consequence to the populations over time?

7 Natural Selection MCQ

(A) Hybrid individuals are less likely to pass their genetic information on to subsequent generations. ✓

(B) Hybrid individuals living together are physically unable to mate with each other.

(C) Geographically isolated populations gradually acquire similar heritable traits.

(D) Differences between hybrid individuals of a species disappear over time.

17. In a separate investigation, individual mice from two populations that in nature are geographically isolated from each other are mated in the laboratory. The hybrid offspring were then mated with individuals from either of the original populations. Only the female hybrid offspring were fertile. The experimental results are most consistent with which of the stages that are depicted in the model?

(A) Initial population (Figure 1)

(B) Intermediate stage (Figure 2) ✓

(C) Late stage (Figure 3)

(D) Terminal stage (Figure 4)

18. Which of the following best describes the reason for excluding hybrid males when calculating the allele frequencies of two interbreeding populations at the intermediate stage of speciation (Figure 2) ?

(A) The process of mate selection in large groups often favors females over males.

(B) The frequency of sex-determining chromosomes is usually equal to 0.5.

(C) Sterile individuals make no genetic contribution to the next generation. ✓

(D) The chance of inheriting a recessive allele from a male is too small to calculate.

19. Using the model of speciation and applying it to a different population, which of the following outcomes is most consistent for a different species in which the males are homogametic and the females are heterogametic?

(A) Sterility would appear in females before appearing in males. ✓

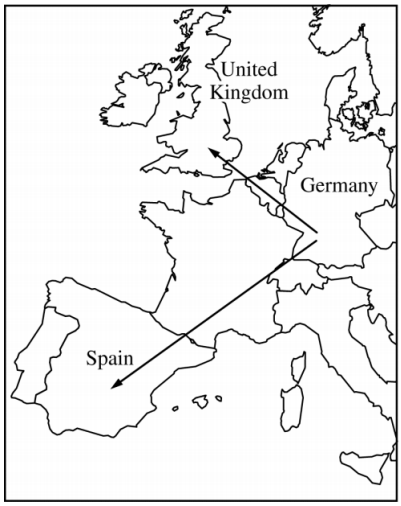
(B) Speciation would occur more rapidly because females would produce more offspring.

(C) Behavioral isolation would occur sooner in species exhibiting nonrandom mating.

(D) The population would reach Hardy-Weinberg equilibrium at an accelerated rate.

7 Natural Selection MCQ

20.



Blackcap birds (*Sylvia atricapilla*) migrate out of Germany before wintertime. Prior to the 1960s, all members of a particular blackcap population flew to Spain, which had an abundant natural food source. Now, some members of the same blackcap population fly to the United Kingdom, where food placed in feeders by humans is abundant. The blackcaps return to the same forests in Germany to nest during the breeding season.

Some blackcaps that migrate to the United Kingdom have become distinguishable by certain physical and behavioral traits from blackcaps that migrate to Spain. Which of the following best predicts the effect on the blackcap population if humans in the United Kingdom continue to place food in feeders during the winter?

(A) The blackcaps that migrate to Spain will selectively mate with the blackcaps that migrate to the United Kingdom, resulting in increased genetic variation in the blackcap population.

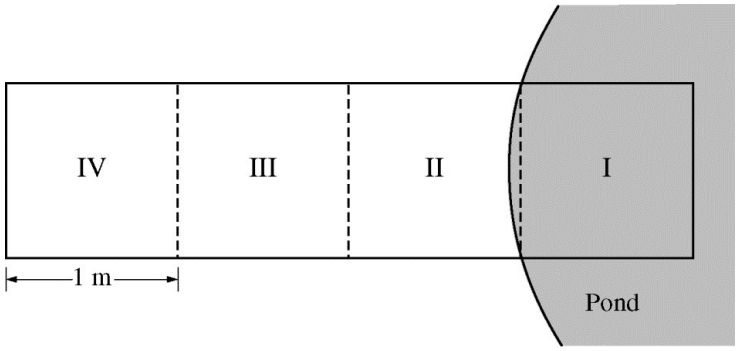
(B) The blackcaps that migrate to the United Kingdom will become reproductively isolated from the blackcaps that migrate to Spain, resulting in speciation in the blackcap population. ✓

(C) The blackcaps in Spain and the United Kingdom will migrate to other geographical locations in search of potential mating partners, resulting in a smaller blackcap population in Germany.

(D) The blackcaps will begin nesting at their wintering sites in Spain or the United Kingdom, resulting in a larger blackcap population migrating back to Germany after the breeding season has ended.

7 Natural Selection MCQ

Researchers investigated the habitat preferences of two species of garter snakes, *Thamnophis sirtalis* and *Thamnophis atratus*. To create a choice chamber, the researchers built a meshed enclosure and positioned one end of the enclosure at the edge of a small pond. Zone I of the enclosure was located in the water, whereas zone IV of the enclosure was located 2–3 meters away from the water, as represented in the figure below. Snakes inside the enclosure were able to move freely between zones.



In a series of experiments, the researchers introduced a single snake into zone IV of the enclosure at 7:00 A.M. The researchers recorded the location of the snake at six time points throughout the day. In a related experiment, the researchers introduced two snakes, one of each species, into the enclosure at the same time and observed the location of each of the two snakes at the same six time points as before. The researchers repeated both the one-snake and two-snake experiments using different individual snakes of each species. The results are presented in the table.

ZONES MOST FREQUENTLY OCCUPIED BY GARTER SNAKES IN A MESHED ENCLOSURE

	Species Introduced Inside the Meshed Enclosure			
	T. atratus Only	*T. sirtalis* Only	*T. atratus* and *T. sirtalis*	
Time of Day	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*
8:00 A.M.	I	II	III	II
10:00 A.M.	II	I	II	II
12:00 P.M.	I	I	III	I
2:00 P.M.	I	I	IV	I
4:00 P.M.	II	I	IV	I
6:00 P.M.	II	II	IV	I

21. Both species of garter snakes prey on the California newt, *Taricha torosa*, a small amphibian that produces a potent neurotoxin (TTX) in its skin. However, neither species of garter snake is affected by TTX. The resistance to TTX is associated with mutations in the *SCN4A* gene. Which of the following best supports a claim that TTX resistance arose independently in *T. atratus* and *T. sirtalis* ?

7 Natural Selection MCQ

(A) The stomach contents in fossils of both species include the remains of food items that contained TTX.

(B) Both species of snakes possess the *SCN4A* gene.

(C) The two species of snakes have different genetic mutations in the *SCN4A* gene. ✓

(D) *T. atratus* and *T. sirtalis* are sister species that share many of the same morphological features.

---

22. By discharging electric sparks into a laboratory chamber atmosphere that consisted of water vapor, hydrogen gas, methane, and ammonia, Stanley Miller obtained data that showed that a number of organic molecules, including many amino acids, could be synthesized. Miller was attempting to model early Earth conditions as understood in the 1950s. The results of Miller's experiments best support which of the following hypotheses?

(A) The molecules essential to life today did not exist at the time Earth was first formed.

(B) The molecules essential to life today could not have been carried to the primordial Earth by a comet or meteorite.

(C) The molecules essential to life today could have formed under early Earth conditions. ✓

(D) The molecules essential to life today were initially self-replicating proteins that were synthesized approximately four billion years ago.

Answer C

This option is correct. It demonstrates the ability to evaluate scientific questions about the origin of life on Earth by recognizing that the Miller experiment modeled the presumed early atmospheric conditions and, under laboratory conditions, produced biological molecules, such as amino acids, that were required for early life.

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7 Natural Selection MCQ

24. A group of researchers noticed that using a particular drug to treat cancer in mice would reduce the size of the tumor, but some of the tumor cells appeared to be resistant to the drug since some of these cells would increase in number over time. The researchers wish to increase the reliability of their experimental observations before investigating new claims that chemotherapy-resistant tumors are an example of continuing evolution.

Of the following, which is the best next step the researchers might take to increase the reliability of their evidence?

(A) Increasing the dose of chemotherapy being administered to the mice

(B) Increasing the number of trials and the number of mice in each treatment group ✓

(C) Plotting the tumor volume data as a graph rather than as a table

(D) Isolating resistant tumor cells and looking for novel mutations in proto-oncogenes

Answer B

Correct. The statistical power and reliability of the experimental results will increase if more replicates with increased sample sizes are used. Experimental findings must be consistently reproducible for them to be reliable.

25. Researchers observed selected internal structures of four different microscopic organisms as part of a larger study on the divergence between eukaryotes and prokaryotes. Their observations are recorded in Figure 1.

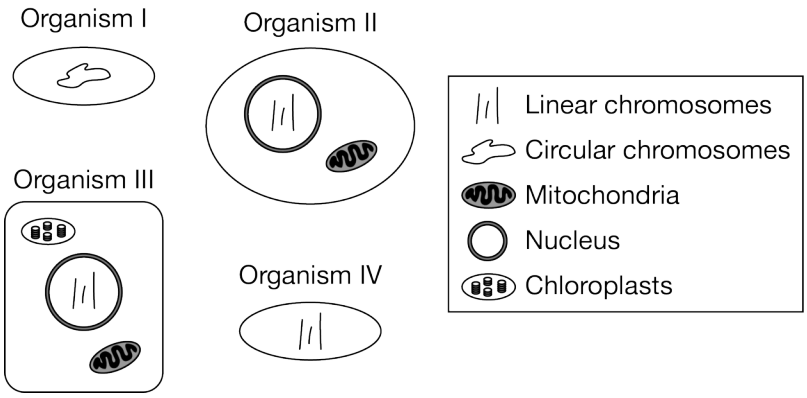


Figure 1. Selected cellular structures in four organisms

Which organism would the researchers most likely predict to be the most distantly related to eukaryotes?

7 Natural Selection MCQ

(A) Organism I ✓

(B) Organism II

(C) Organism III

(D) Organism IV

Answer A

Correct. The circular chromosome in Organism I identifies it as a prokaryote.

26. Because cheetahs, *Acinonyx jubatus*, experienced a bottleneck event about ten thousand years ago, living cheetahs exhibit very little genetic variation compared to other living species of large cats.

Which of the following principles best justifies a claim that cheetahs, compared to other large cats, have a relatively low resilience to environmental perturbations?

(A) A captive population of cheetahs kept in a confined area will have a greater risk of contracting and passing on a virus.

(B) When a population is reduced, new niches become available in the environment, which can trigger adaptive radiation.

(C) Populations with little genetic diversity are less likely to contain individuals that can withstand different selective pressures. ✓

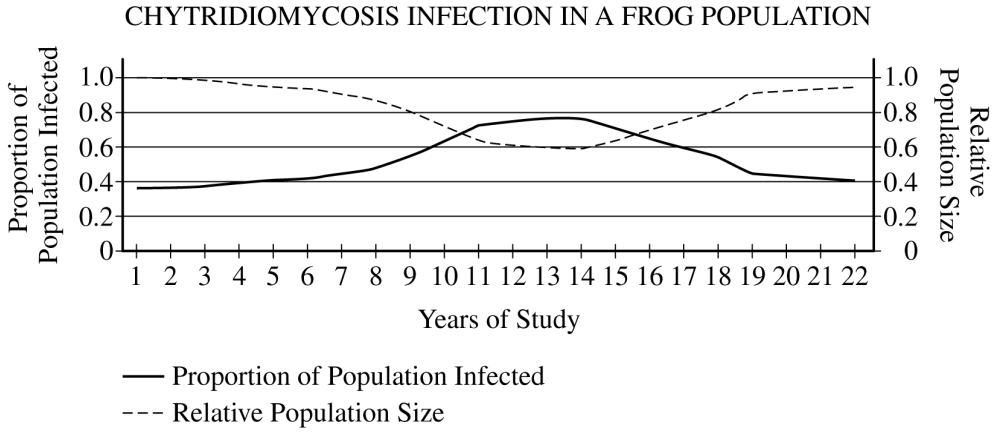
(D) Populations with low genetic diversity are more likely to contain individuals that can withstand different selective pressures.

Answer C

Correct. Cheetahs exhibit less genetic diversity than other cats, which is why they are at an increased risk of decline and extinction. The ability of a population to respond to changes in the environment is influenced by the level of genetic diversity among the individuals.

7 Natural Selection MCQ

27.



Chytridiomycosis is a potentially lethal fungal infection that adversely affects some frog populations. The incidence and severity of the fungal infection can vary over time. Which of the following statements best describes the changes in the frog population that are depicted in the model above?

(A) The fungus that infected the frog population was initially virulent but it later became harmless. Based on the trend over the past few years of the study, the fungus has become extinct.

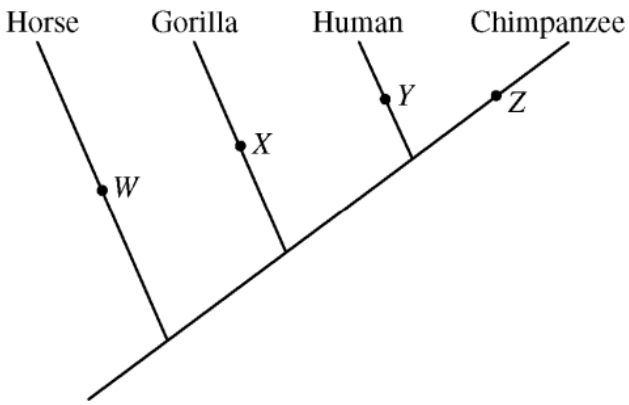
(B) Infected individuals gradually died out, and genetically resistant individuals became more common. The frog population recovered because of the increased frequency of resistant individuals. ✓

(C) The patterns of the curves on the graph indicate a sudden increase in the severity of the fungal infection. A large proportion of the frog population died because the individuals could not evolve fast enough.

(D) The size of the frog population decreased sharply upon initial contact with the fungus. The fungus eventually became symbiotic with individual frogs, and the frog population began to recover.

7 Natural Selection MCQ

28. The cladogram shown below depicts an accepted model of the evolutionary relationships among selected species.



The amino acid at position 104 in the beta-hemoglobin protein for each of these four organisms is listed below.

Species	Amino Acid 104
Horse	Arginine
Gorilla	Leucine
Human	Arginine
Chimpanzee	Arginine

The validity of the cladogram is best supported by molecular evidence for which of the following changes in the amino acid composition of the beta-hemoglobin protein during the evolution of these species?

(A) Arginine to leucine at position *X* on the cladogram ✓

(B) Arginine to leucine at position *Y* on the cladogram

(C) Leucine to arginine at position *W* on the cladogram

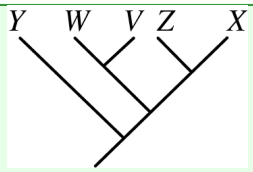
(D) Leucine to arginine at position *Z* on the cladogram

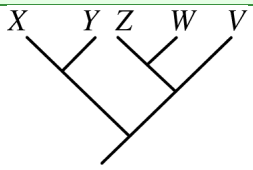
7 Natural Selection MCQ

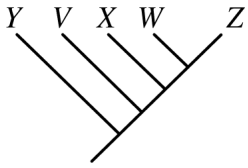
29. Data regarding the presence (+) or absence (-) of five derived traits in several different species are shown in the table below.

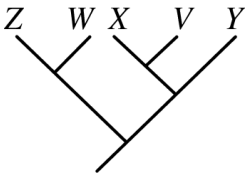
	Trait				
Species	1	2	3	4	5
V	+	+	+	–	–
W	+	+	–	–	–
X	+	–	–	+	+
Y	–	–	–	–	–
Z	+	–	–	–	+

Which of the following cladograms provides the simplest and most accurate representation of the data in the table?

(A) 

(B) 

(C) 

(D) 

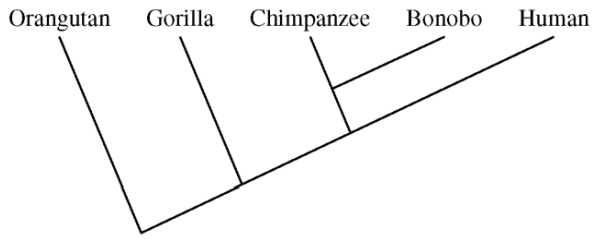
7 Natural Selection MCQ

The *TAS2R38* gene encodes a receptor protein that influences the ability to taste bitterness. The gene has two alleles: a dominant, wild-type allele that enables an individual (taster) to taste bitterness and a recessive, mutant allele that interferes with the ability of an individual (nontaster) to taste bitterness. Three single nucleotide mutations in the coding region of the *TAS2R38* gene are associated with the nontaster allele. The nucleotides present at the three positions are shown in the table below.

NUCLEOTIDE VARIATION IN THE *TAS2R38* GENE

	Position in the Nucleotide Sequence		
	145	785	886
Human nontaster	G	T	A
Human taster	C	C	G
Bonobo	C	C	G
Chimpanzee	C	C	G
Gorilla	C	C	G

A cladogram representing the evolutionary relatedness of selected primates is shown below.



30. Which of the following conclusions is supported by the data?

(A) Humans and bonobos belong to the same genus because the nucleotide sequence of the *TAS2R38* gene is the same

(B) Chimpanzees are equally related to humans and bonobos because all three groups possess variants of the *TAS2R38* gene.

(C) Humans are more closely related to bonobos than to chimpanzees because the branch of the tree with humans is closest to the branch of the tree with bonobos.

(D) Bonobos are more closely related to humans than to gorillas because bonobos and humans share a more recent common ancestor than bonobos and gorillas do. ✓

31. Toxic substances often have a bitter taste that causes animals who try to eat such substances to spit them out rather than swallow them. Additional data suggest that gorilla populations have a very low frequency of nontasters. Which of the following best predicts the likely evolution of the *TAS2R38* locus in the gorilla population?

7 Natural Selection MCQ

(A) Gorillas who could taste bitter toxins were more likely to survive and reproduce than nontasters. ✓

(B) Gorillas who could taste bitter toxins became resistant to the toxins, so the taster gene was no longer under selective pressure.

(C) Gorillas acquired the taster allele by horizontal gene transfer from chimpanzees.

(D) Gorillas who could taste bitter toxins could utilize food resources that nontasters could not.

32. Which of the following scientific questions will best help researchers determine when the nontaster allele arose in the evolutionary history of the selected primates?

(A) How does the nontaster allele contribute to individual fitness?

(B) Does the nontaster allele exist in any nonhuman primate populations? ✓

(C) What caused the mutations in the *TAS2R38* gene?

(D) How do the mutations in the nontaster allele affect the structure of the resulting protein?

33. In a sample of 2,400 people, 1,482 were found to have the dominant (taster) phenotype. Assuming that the population is in Hardy-Weinberg equilibrium, approximately how many individuals in the sample are expected to be heterozygous for *TAS2R38*?

(A) 741

(B) 918

(C) 1,133 ✓

(D) 1,482

34.

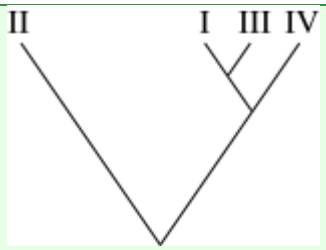
TABLE 1. PERCENT DNA SEQUENCE DIFFERENCES IN THE *MT-CO1* GENE

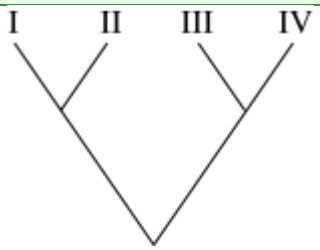
Species	I	II	III	IV
I		13.5	1.5	3.0
II			12.7	12.2
III				3.8
IV				

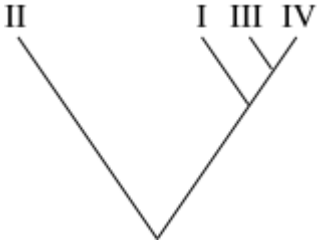
The gene encoding the catalytic subunit of cytochrome c oxidase (*MT-CO1*) is found in the mitochondria of all animals. The gene evolves rapidly and is often used to determine evolutionary relationships among species. Table 1 shows DNA sequence differences among four animal species (I, II, III, and IV).

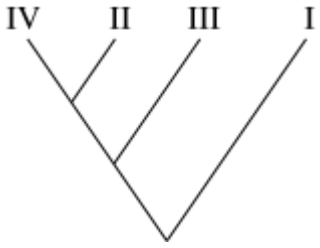
Which of the following cladograms best represents the evolutionary relationships among the four species?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

Answer A

Correct. The cladogram indicates that species II split off early in the evolutionary lineage and that species IV split off prior to the separation of species I from species III. The cladogram is supported by the data in Table 1 because the greatest percent difference in the sequence of the $MT-CO1$ gene is between species II and each of the other species. Additionally, the lowest percent difference is between species I and III.

7 Natural Selection MCQ

35. In a particular region, a species of lizard lives in the lowlands. Even though there are numerous mountains in the region, the climate on the mountains is too cold for the lizards to survive. Scientists are concerned that the lowlands will become much warmer by the end of the twenty-first century and will become unsuitable for this lowland lizard species.

Which of the following best predicts the impact on the lizard species if there is a gradual increase in the average temperatures found in both the lowlands and mountains in this region?

(A) Some lizards will migrate to new habitats in the mountains and become isolated, leading to allopatric speciation. ✓

(B) The species will undergo artificial selection, as the genes will adapt to warmer temperatures.

(C) The species will experience a genetic bottleneck, and sympatric speciation will result in the development of several new species.

(D) The lizards will hybridize with a different lizard species that is adapted to a warmer environment, leading to increased genetic variation in the species.

Answer A

Correct. In allopatric speciation, a new species forms in geographic isolation from its parent population. In this case, the lizards would be migrating to the mountainsides in small groups, each group isolated from the others by the now inhospitable lowlands.

36. Two species of fish that live in extremely cold environments produce near-identical antifreeze glycoproteins, called AFGPs. Scientists have determined the origins of the AFGP gene in both species. Selected characteristics of the two fish species are provided in Table 1.

Table 1. Characteristics of two fish species producing nearly identical antifreeze glycoproteins

Fish Species	Order	Location	Evolutionary origin of the *AFGP* Gene
Boreogadus saida	Gadiformes	Arctic	No sequence identity with trypsinogen
Dissostichus mawsoni	Perciformes	Antarctic	Pancreatic trypsinogen

Based on the information in Table 1, which of the following statements best describes the production of nearly identical AFGPs in these two species of fish?

(A) The fish eat the same type of food, which resulted in the evolution of similar digestive proteins.

(B) The fish live in environments with similar selective pressures, and those that produce AFGPs are better able to survive. ✓

(C) The Antarctic fish species evolved into a separate species after being geographically isolated from the Arctic population because of commercial fishing.

(D) The production of similar AFGPs was due to random splicing of exons in both species.

7 Natural Selection MCQ

Answer B

Correct. This is an example of convergent evolution. This occurs when similar selective pressures lead to similar phenotypic adaptations in different populations or species.

37. Whales, bony fishes, and cartilaginous fishes all have similar adaptations that enable them to live in the ocean. Figure 1 shows the phylogenetic relationships of selected vertebrates.

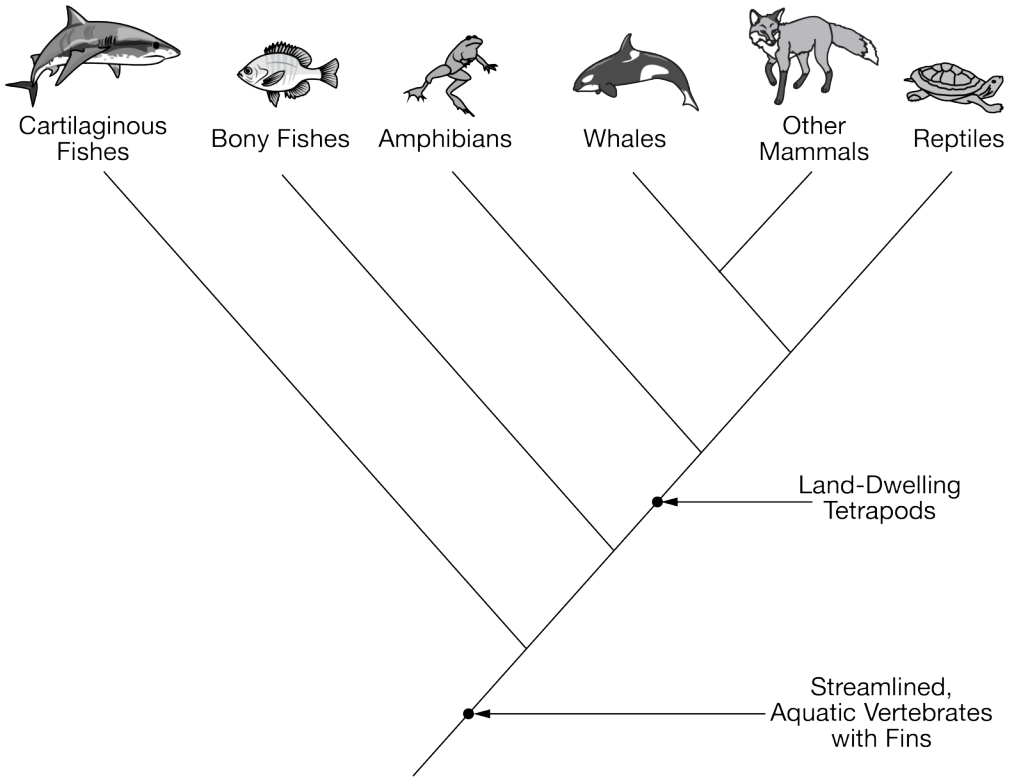


Figure 1. Cladogram of selected vertebrate groups

Based on the phylogenetic relationships shown in Figure 1, which of the following best explains the similarities exhibited by whales, bony fishes, and cartilaginous fishes?

(A) Convergent evolution, because these organisms do not share a recent common ancestor. ✓

(B) Adaptive radiation, because all three groups evolved from a unique, shared common ancestor from the cartilaginous fishes group.

(C) Mutation, because the shared traits can be traced back to a shared recent common ancestor.

(D) Founder effect, because all three groups formerly existed together as one population that then was disrupted, causing speciation.

7 Natural Selection MCQ

Answer A

Correct. Convergent evolution states that similar physical traits in unrelated groups are the result of the same selective pressures experienced by each group.

38. Female European corn borer moths (*Ostrinia nubilalis*) produce the sex pheromone $11-$tetradecenyl acetate ($11-\mathrm{TDA}$), which attracts males of their species. $11-\mathrm{TDA}$ can be produced in two forms, E and Z, each with a different three-dimensional shape. Researchers have discovered two different strains of the European corn borer moth living in the same area. Females of the E strain produce primarily the E form of $11-\mathrm{TDA}$ and females of the Z strain produce primarily the Z form of $11-\mathrm{TDA}$. Males of both strains are more attracted to the form of $11-\mathrm{TDA}$ produced by the females of the same strain.

Which of the following best predicts a long-term effect of the differences between the E and Z strains?

(A) The difference between the forms of the sex pheromone $11-\mathrm{TDA}$ produced by E and Z strains will act as a prezygotic reproductive barrier leading to sympatric speciation. ✓

(B) The difference between the forms of the sex pheromone $11-\mathrm{TDA}$ produced by E and Z strains will act as a postzygotic reproductive barrier, leading to allopatric speciation.

(C) Since the E and Z strains are both part of the same species, they will produce fertile offspring, leading to a single strain that produces both forms of $11-\mathrm{TDA}$.

(D) Since the E and Z strains are found in the same geographic area, they will hybridize, leading to a single strain that produces the dominant form of $11-\mathrm{TDA}$.

Answer A

Correct. There are many different mechanisms that can provide reproductive isolation within a population leading to speciation without a geographic barrier. Barriers that prevent fertilization are called prezygotic barriers and those that act after fertilization are called postzygotic barriers. Sex pheromone preferences affecting mate choice would reduce the likelihood of mating between members of the E and Z European corn borer strains, acting as a prezygotic barrier by preventing mating that could lead to fertilization. Speciation that occurs without geographic isolation is called sympatric speciation.

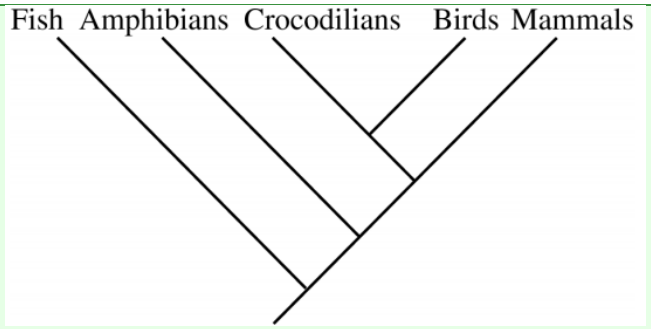
7 Natural Selection MCQ

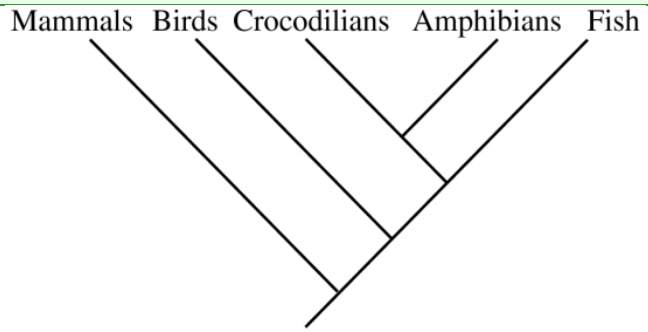
39.

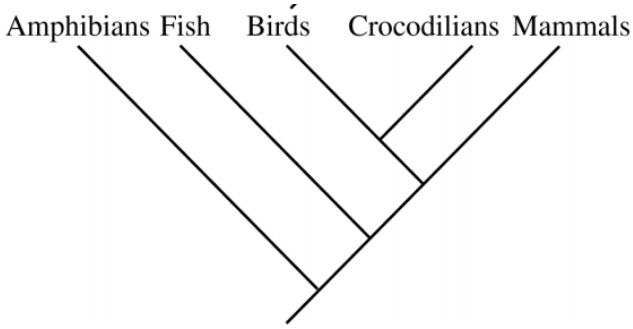
Organisms	Derived Characters (+ indicates the character is present)					
	Hair	Jaws	Feathers	Amniotic Eggs	Two Pairs of Limbs	β-Keratin Scales
Mammals	+	+		+	+	
Crocodilians		+		+	+	+
Fish		+				
Amphibians		+			+	
Birds		+	+	+	+	+

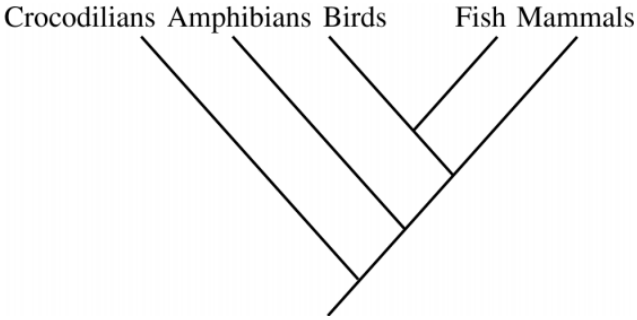
The table above shows derived characters for selected organisms. Based on the information in the table, which of the following cladograms best represents the evolutionary relationships among the organisms?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

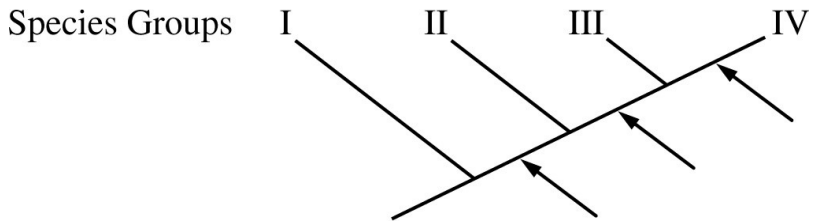
(D) 

7 Natural Selection MCQ

40.

	Derived Characters		
Plant Species	**Vascular Tissue**	**Seeds**	**Flowers**
P	+	+	+
Q	–	–	–
R	+	+	–
S	+	+	+
T	–	–	–
U	+	–	–
W	+	–	–

The table above shows the presence (+) or absence (-) of three different derived characters (vascular tissue, seeds, and flowers) for several selected plant species. The cladogram below was created based on the information in the table to represent groups of related species. The presence of the derived characters is indicated with arrows.



Group IV of the cladogram most likely includes which of the following species?

(A) P and S only ✓

(B) Q and T only

(C) R and W only

(D) U and W only

41. The apple maggot fly, *Rhagoletis pomonella*, is native to North America and originally fed on fruit of the wild hawthorn. Since the mid-1800s, a population of flies has emerged that instead feed on domesticated apples. Apple maggot flies typically mate on or near the fruit of their host plants. Many varieties of apples ripen three to four weeks before the hawthorn fruits do.

The different fruit preferences of the two fly populations will most likely have which of the following effects?

7 Natural Selection MCQ

(A) The flies that eat hawthorn fruit will increase in number, while the flies that eat apples will decrease in number because of the use of insecticides on apple trees.

(B) The single fly species will evolve into two distinct species because of the lack of gene flow between the two populations. ✓

(C) The ability to survive on a diet of two different fruits will help the flies learn to eat many more types of fruit.

(D) The flies that eat hawthorn fruit will lay some of their eggs on the earlier-ripening apples to minimize competition among the larvae.

42. One of the oldest known mammalian fossils belongs to a small furry species, *Hadrocodium wui,* that lived 195 million years ago. For the following 100 million years, only a small number of groups of mammalian fossils are found in the fossil record. By 65 million years ago, the nonavian dinosaurs were extinct. By 55 million years ago, there were 130 mammal genera, including 4,000 different species that occupied a wide variety of habitats.

Which of the following best describes the cause of the rapid increase in the number of mammalian species between 65 and 55 million years ago?

(A) It took over 100 million years for enough random mutations to occur in the genomes of early mammals to provide the variation needed for the observed speciation.

(B) It took over 100 million years for the early mammals to disperse over a wide enough geographic area to allow allopatric speciation to occur.

(C) After the dinosaur extinction, many ecological niches became available, leading to the adaptive radiation of mammals. ✓

(D) After the dinosaur extinction, mammal population sizes increased, allowing species to become more easily detectable in the fossil record.

Answer C

Correct. From 245 to 65 million years ago, dinosaurs were the dominant land animals and occupied a wide range of ecological roles. Early mammals were nocturnal, allowing them to avoid predation and competition with dinosaurs. With the dinosaur extinction 65 million years ago, many ecological niches were suddenly left empty. Speciation can occur rapidly when new niches become available, and a small number of ancestral species rapidly diversify into many new species, each adapted to a different role in the environment. This process is called adaptive radiation and has occurred many times on many scales throughout evolutionary history.

7 Natural Selection MCQ

43. A group of birds is found in an island chain that have all descended from a single ancestral species. Figure 1 shows the phylogeny of the birds.

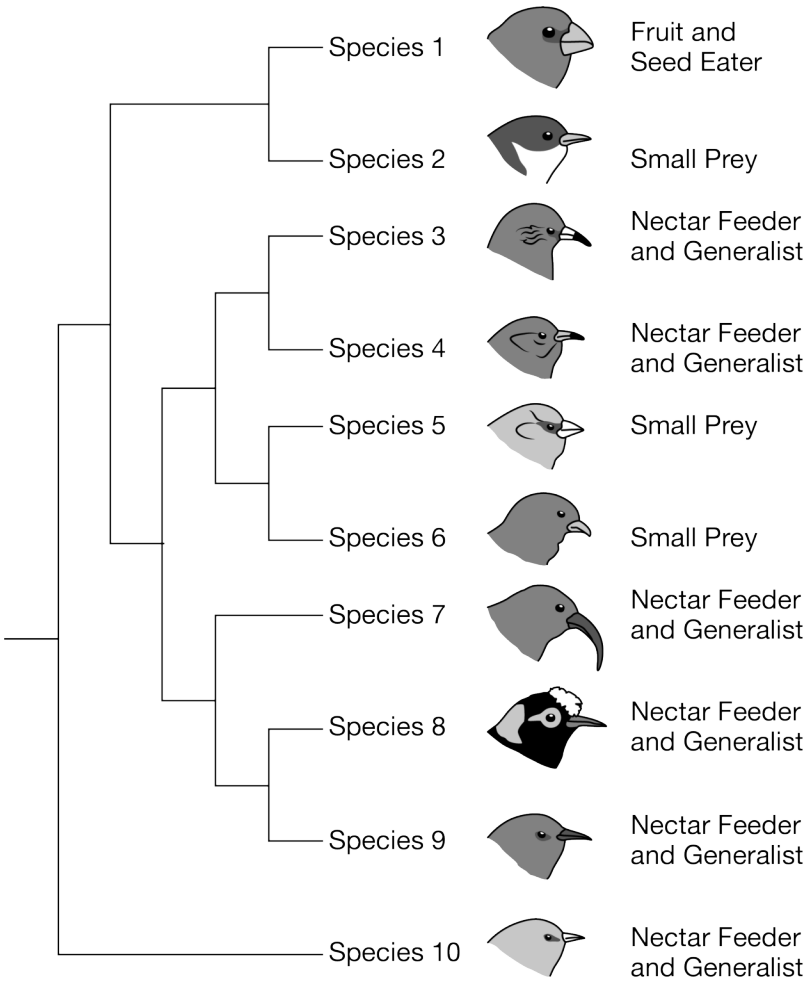


Figure 1. Phylogeny of island birds

Which of the following best explains the divergence in morphology in the birds?

(A) The original habitat contained mutagens, leading to the rise of new species as a result of accumulated mutations.

(B) The original habitat had many available unoccupied ecological niches that were colonized by populations of the original species. ✓

(C) The original habitat contained many predators, and the species diversified in order to avoid extinction.

(D) The original habitat was tropical, which increased the rate of mutation and led to rapid diversification.

7 Natural Selection MCQ

Answer B

Correct. This is an example of adaptive radiation and divergent evolution.

44. In plants, members of the euphorbia family and the cactus family have separate origins, with the euphorbia evolving in the desert regions of Asia and Africa and the cacti in the desert regions of the Americas. Both have fleshy stems adapted for water storage, protective spines, and leaves that are greatly reduced or absent. The similarities between these two families of plants represent an example of

(A) coevolution

(B) convergent evolution ✓

(C) genetic drift

(D) sympatric speciation

(E) balanced polymorphism

45. A scientist is attempting to provide support for the hypothesis that RNA was the first genetic material. Which of the following would be a workable alternate hypothesis?

(A) RNA, like DNA, is a carrier of genetic information and contains the nitrogen bases adenine, guanine, and cytosine.

(B) RNA can be observed to self-replicate without the assistance of proteins, while DNA always requires protein-based enzymes to replicate. ✓

(C) RNA exists in both single-stranded and double-stranded configurations; however, it is less stable compared with DNA.

(D) Transfer RNA and ribosomal RNA are products of transcription, but they are not translated as is mRNA.

Answer B

Correct. Both DNA and RNA can serve as genetic storage, but only RNA can also catalyze enzymatic reactions needed for replication and metabolic activities, so it may have been the earliest genetic material. Stating that there is a difference between the two types of nucleic acids is a characteristic of an alternate hypothesis.

7 Natural Selection MCQ

46. A group of researchers claim that chance events play an important role in the evolution of populations. To test their claim, they monitored 12 populations of *E. coli* for 10,000 generations under the same environmental conditions. They found that although all 12 populations were exposed to the same environmental pressures, each population had differentiated from one another genetically after 10,000 generations.

Which of the following statements includes a reasonable refinement that researchers could use in repeating the experiment?

(A) Start each population with genetically identical bacterial cells from a single culture to make sure that the initial genetic variation in all of the populations was exactly the same. ✓

(B) Start each population with a different species of bacteria to determine whether the same environment has an effect on genetic differentiation.

(C) Start each population with bacterial cells that have different genotypes to make sure there is enough genetic variation for selection to occur.

(D) Start each population with bacteria that differ from one another by a single mutation to ensure that the initial genetic variation in all populations is known.

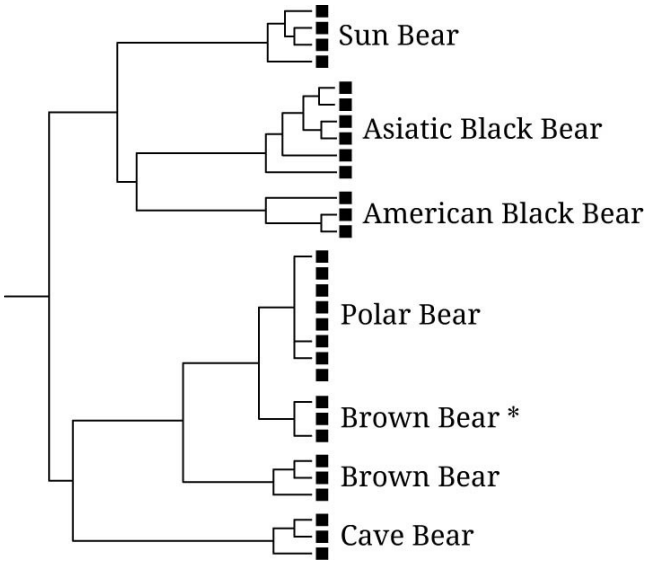
Answer A

Correct. The only way to know whether chance events were influencing the evolution of the bacteria would be if all of the bacteria had the exact genetic makeup at the beginning of the experiment.

7 Natural Selection MCQ

47. Scientists studying the evolution of bears analyzed the phylogenetic relationship among several extant bear species using 27 mitochondrial DNA (mtDNA) genomes. In addition to the five extant bear species, three genomes from an extinct species of cave bear were examined. The scientists constructed the phylogenetic tree (Figure 1).

Figure 1. Phylogenetic tree of bears based on 27 mtDNA genomes. Each square represents one mtDNA genome. The asterisk symbol (*) denotes brown bears from Admiralty, Baranof, and Chichagof Islands.



Which of the following statements best explains the characteristics of two species that are on the same branch of the phylogenetic tree in Figure 1?

(A) Their mtDNA genomes are identical in sequence.

(B) Their mtDNA genomes share more similarities with each other than with the mtDNA genomes of other species on the tree. ✓

(C) Their mtDNA genomes have the same number of genes and are the same length.

(D) Their mtDNA genomes must have evolved at exactly the same rate since their divergence from a common ancestor.

48. Geologic events such as mountain building can directly affect biodiversity. The Sino-Himalayan fern, *Lepisorus clathratus* is widely distributed on the Qinghai-Tibetan plateau.

If geologic events lead to a series of new mountain ranges that divide the plateau into several discrete valleys, which of the following best describes how the *L. clathratus* population would be affected?

7 Natural Selection MCQ

(A) Geographic isolation will create separate gene pools, leading to speciation over time. ✓

(B) Gene flow will be reduced by sexual selection and polyploidy.

(C) Chance events will cause allele frequencies to fluctuate unpredictably from one generation to the next.

(D) Ferns reproduce asexually, so there will be no impact on the gene pool of the species.

Answer A

Correct. A geographic barrier will likely lead to allopatric speciation because the gene pools of the species will be separated.

49. The appearance of a fertile, polyploid individual within a population of diploid organisms is a possible source of a new species. If this individual is capable of reproducing to form a new population, scientists would consider this to be an example of

(A) allopatric speciation

(B) sympatric speciation ✓

(C) polygenic inheritance

(D) genetic drift

(E) Hardy-Weinberg equilibrium

50. Researchers examined the ability of cultures of the bacterium *Pseudomonas aeruginosa* to adapt to the antibiotics ceftazidime and avibactam when they are administered together. The researchers grew four replicate cultures in growth medium with no antibiotics added (control) and four additional replicate cultures in the same growth medium with added ceftazidime and avibactam. At the conclusion of the experiment, samples of the replicates exposed to the antibiotics were removed and their genomes were sequenced. All of the antibiotic-resistant mutants were missing three genes ($hmgA$, $galU$, $mexXY$). It is known that one of these three genes makes wild-type *P. aeruginosa* susceptible to the two antibiotics. However, the researchers do not know which gene it is.

Which of the following experiments would specifically determine which gene is responsible for ceftazidime-avibactam sensitivity in *P. aeruginosa* ?

(A) Clone the three deleted genes from a wild-type *P. aeruginosa* strain, and insert these sequences into the mutant bacterial strains to restore sensitivity to ceftazidime-avibactam.

(B) Delete genes that are homologous to the three deleted genes in other bacterial species, and determine if doing so also confers ceftazidime-avibactam resistance in those bacteria.

(C) Perform gene knockout (targeted mutation) of either the $hmgA$, $galU$, or $mexXY$ gene on separate cultures of the wild-type *P. aeruginosa*. For each individual mutant, determine whether or not ceftazidime-avibactam resistance has occurred. ✓

(D) Perform gene knockout (targeted mutation) of both the $hmgA$ and $galU$ genes of the wild-type *P. aeruginosa* in a single culture, and then determine if ceftazidime-avibactam resistance has occurred in the mutant strain.

7 Natural Selection MCQ

Answer C

Correct. After specifically mutating the three genes separately, a check for ceftazidime-avibactam resistance in each different mutant strain would indicate which of the three genes is required for ceftazidime-avibactam sensitivity.

51.

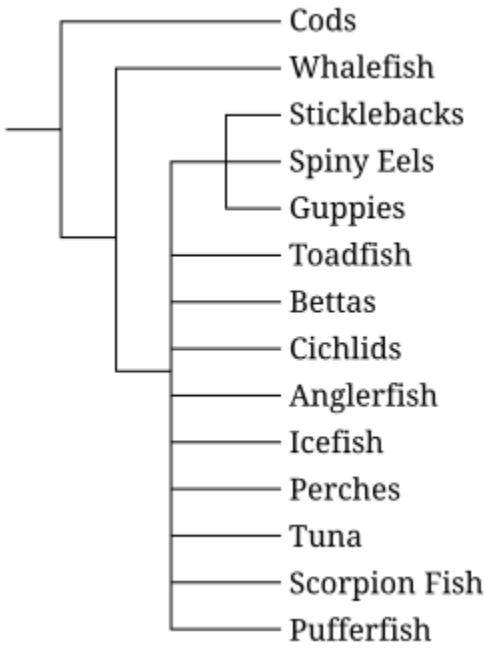


Figure 1. Simplified cladogram of several orders of fish

Cods and icefish are the only orders of fish that include species that produce antifreeze proteins in their blood. Based on the cladogram in Figure 1, which of the following scenarios describes the most likely evolutionary history of antifreeze protein synthesis in fish?

(A) Antifreeze proteins arose twice, once in the cod lineage and once in the icefish lineage. ✓

(B) Antifreeze proteins arose once in a common ancestor of cods and icefish.

(C) Antifreeze proteins arose once in the most recent common ancestor of whalefish and icefish.

(D) Antifreeze proteins were lost repeatedly in fish lineages.

Answer A

Correct. This is an example of convergent evolution. Cods and icefish do not share a recent common ancestor. Additionally, none of the other fish to which icefish are more closely related produce antifreeze

7 Natural Selection MCQ

proteins. This suggests that the presence of antifreeze proteins evolved separately in the cods and icefish lineages, and similar environmental conditions selected for the proteins to continue to be produced in future generations.

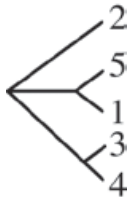
52. Five new species of bacteria were discovered in Antarctic ice core samples. The nucleotide (base) sequences of rRNA subunits were determined for the new species. The table below shows the number of nucleotide differences between the species.

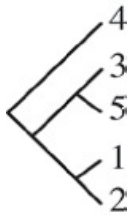
NUCLEOTIDE DIFFERENCES

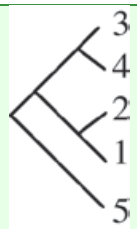
Species	1	2	3	4	5
1	-	3	19	18	27
2		-	19	18	26
3			-	1	27
4				-	27
5					-

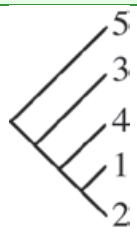
Which of the following phylogenetic trees is most consistent with the data?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

Answer C

This option is correct. It demonstrates the ability to select the phylogenetic tree that correctly represents evolutionary history and speciation from the data set. The data indicate that species 5 is not closely related to the others and that 3 and 4 are very closely related. This cladogram incorporates those differences as well as the close relationship between 1 and 2.

7 Natural Selection MCQ

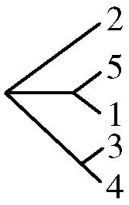
53. Five new species of bacteria were discovered in Antarctic ice core samples. The nucleotide (base) sequences of rRNA subunits were determined for the new species. The table below shows the number of nucleotide differences between the species.

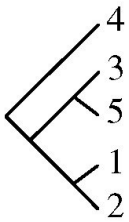
NUCLEOTIDE DIFFERENCES

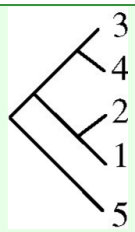
Species	1	2	3	4	5
1	--	3	19	18	27
2		--	19	18	26
3			--	1	27
4				--	27
5					--

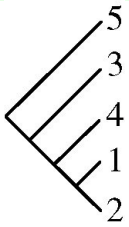
Which of the following phylogenetic trees is most consistent with the data?

7 Natural Selection MCQ

(A) 

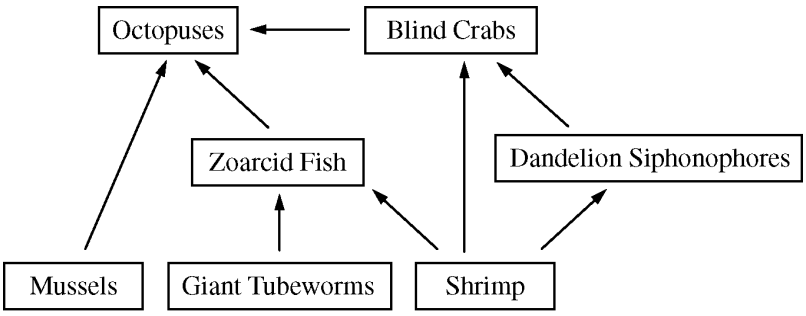
(B) 

(C) 

(D) 

7 Natural Selection MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



The food web above represents feeding relationships in a biological community near a deep-sea hydrothermal vent. Hydrothermal vents are geysers on the seafloor that gush super-heated, mineral-rich water. The seawater surrounding hydrothermal vents typically contains carbon dioxide (CO_2), molecular hydrogen (H_2), hydrogen sulfide (H_2S), and methane (CH_4). Sunlight, however, fails to reach the seafloor where deep-sea hydrothermal vents are located.

As part of an investigation, researchers collected living specimens from an area near a deep-sea hydrothermal vent. Mussels in the collection were found to be dependent on molecular hydrogen in seawater. Also, the researchers discovered multiple species of bacteria living in the gills of the mussels. Mussels use gills for filter-feeding and gas exchange with the surrounding seawater. On the basis of their experimental results, the researchers hypothesized that some bacteria living in the gills of the mussels are capable of chemosynthesis.

54. Researchers are investigating the evolutionary relationships among organisms found near deep-sea hydrothermal vents and similar organisms found closer to the ocean surface. Which of the following scientific questions is most relevant to the investigation?

(A) What are the nucleotide sequences of ribosomal RNA genes that are found in the genomes of the different species? ✓

(B) What large-scale geological events have occurred recently in the Mid-Atlantic Ocean?

(C) Does water temperature at different ocean depths affect the relative levels of dissolved oxygen?

(D) Do species found near deep-sea hydrothermal vents all have the same haploid number of chromosomes?

55. The fossil record indicates that in some cases reasonably well-defined species appear suddenly and remain unchanged for a long time before they become extinct. This phenomenon is referred to as

(A) allopatric speciation

(B) sympatric speciation

(C) phyletic gradualism

(D) punctuated equilibrium ✓

(E) parallel evolution

7 Natural Selection MCQ

56. Finch beak depth was measured before and after a severe drought in 1977. The averaged results are shown in Figure 1. After the drought, there were fewer seeds available for the finches to eat and the seeds that did remain were larger and harder to crack open. Researchers claim that when large, hard seeds are the predominate food source, the average beak depth of these finches tends to increase in subsequent generations.

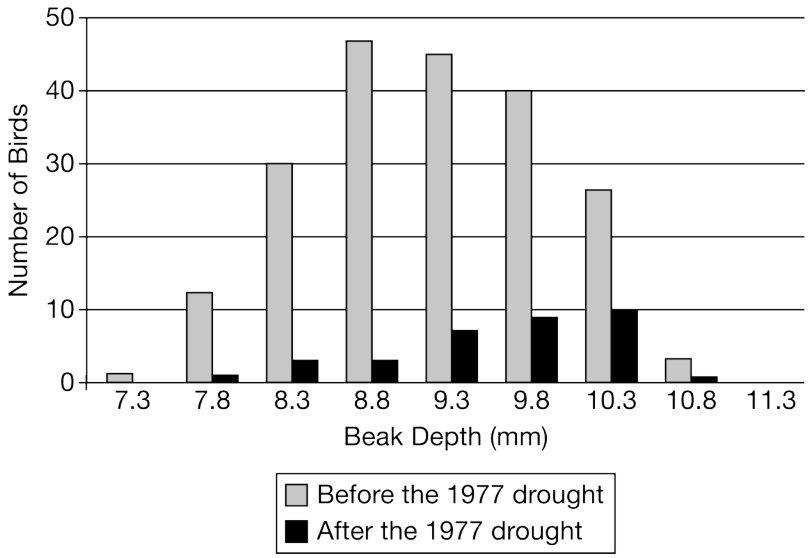


Figure 1. Finch beak depth before and after the 1977 drought

Using the evidence provided, which of the following best justifies the researchers' claim?

(A) Finch beak depth increased after the drought because the overall numbers of finches decreased, reducing competition for resources.

(B) Finch beak depth increased as a consequence of selective pressure after the drought because the remaining seeds were larger and harder. ✓

(C) Finch beak depth increased after the drought due to intraspecific competition since finches with deeper beaks attract more mates.

(D) Finch beak depth increased after the drought due to increased predation since finches with deeper beaks can fight off predators more effectively.

Answer B

Correct. Increased beak depth is associated with the ability to crack open harder and larger seeds. Seeds were harder and larger after the drought, so finches with increased beak depth could get more food and had a survival advantage after the drought.

7 Natural Selection MCQ

57. A researcher uses the protocol shown in Figure 1 to isolate a eukaryotic gene for the purpose of genetically engineering bacteria to synthesize the corresponding protein.

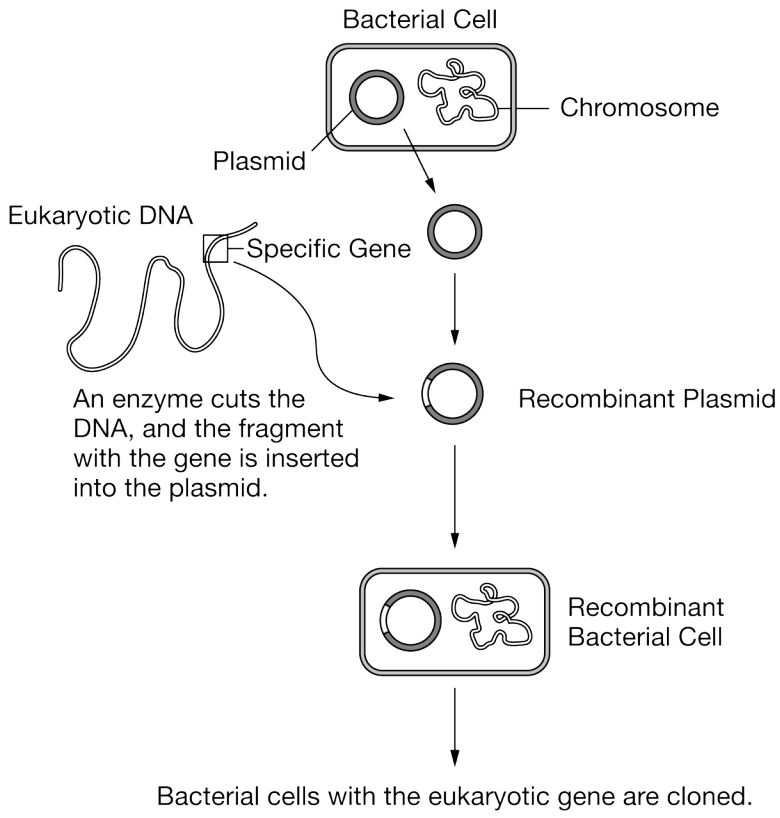


Figure 1. Creating recombinant bacteria that contain a specific gene

Which of the following best predicts why the recombinant bacteria will fail to produce the eukaryotic protein?

(A) Introns must be removed from eukaryotic DNA before the gene is inserted into the plasmid. ✓

(B) Bacterial ribosomes cannot recognize eukaryotic mRNA and will not translate the codon sequence.

(C) Restriction enzymes in the cytoplasm of the bacteria degrade linear DNA.

(D) Bacteria and eukaryotes use different mRNA codons for the same amino acids during translation.

Answer A

Correct. Because eukaryotic genes contain introns (noncoding segments), inserting eukaryotic genes directly into bacteria cells will not yield usable instructions for protein synthesis. Prokaryotic genes do not have introns, and bacteria are prokaryotic. The presence of introns is evidence at the molecular level that indicates common ancestry of all eukaryotes.

7 Natural Selection MCQ

58. Two populations of a species of squirrel are geographically isolated from each other. Although they have the same population density, one population is significantly larger in number than the other. A new bacterial disease, which is easily spread and extremely virulent, affects both populations at the same time. Which of the following is the best prediction of how the new disease will affect the two populations?

(A) The two populations will be equally affected, because the ability to trigger an immune response is randomly distributed among all squirrels of that species.

(B) The larger population will be less affected by the disease than will the smaller population, because the mutation rate of the larger population is higher than the mutation rate of the smaller population.

(C) The smaller population will be more affected than will the larger population, because the smaller population has less genetic variation than the larger population has. ✓

(D) The smaller population will be less affected than will the larger population, because the smaller population exhibits more genetic drift than the larger population exhibits.

59. In a certain small mammal, fur color is controlled by a pair of alleles, with B being dominant over b. The genotypes BB and Bb result in dark-colored fur, the genotype bb results in light-colored fur. A population of the species has the following allele frequencies: $B = 0.85$ and $b = 0.15$. If a fire caused the surface of the ground in the habitat of the species to become darker in color, which of the following claims about the impact on the allele frequencies in this population is most reasonable?

(A) The frequency of homozygous recessive individuals will increase because light-colored fur absorbs less heat.

(B) The frequency of the B allele will increase because light-colored individuals will be more visible to predators. ✓

(C) The frequency of the b allele will increase because there will be more heterozygous individuals.

(D) The frequency of both alleles will increase because the population will enter a period of rapid growth.

Answer B

Correct. A darker-colored habitat will give darker-colored individuals a selective advantage as they will be less visible to both predators and prey. More dark-colored individuals will survive and reproduce. This will result in an increase in the frequency of the B allele.

60. Dr. Robert Hazen has worked to develop a hypothesis that explains the origin of life on Earth. His work has focused on hydrothermal vents, which are cracks in the ocean floor. Water heated by molten rock beneath the crust escapes from these vents at very high temperatures, producing hot, high-pressure environments at the vents on the ocean floor.

In order to provide evidence that these vents may be areas where life originated, which of the following states a null hypothesis Hazen might have used to begin his research?

7 Natural Selection MCQ

(A) Organic molecules were transported to the hydrothermal vents via a meteorite, thereby seeding them with precursors for more complex molecules.

(B) The temperatures and pressures found in hydrothermal vents resemble conditions described in currently accepted origins of life hypotheses. ✓

(C) Organic waste sinking to the ocean floor would have used the unique hot water environment as an energy source to begin manufacturing food.

(D) Chemosynthetic organisms can best develop in the hydrothermal environment, manufacturing nutrients for other living things to feed on.

Answer B

Correct. This states the null hypothesis that the similarity between the conditions at the surface of the early Earth and the conditions of these deep-sea vents made them similarly eligible as the source of the first life.

61.

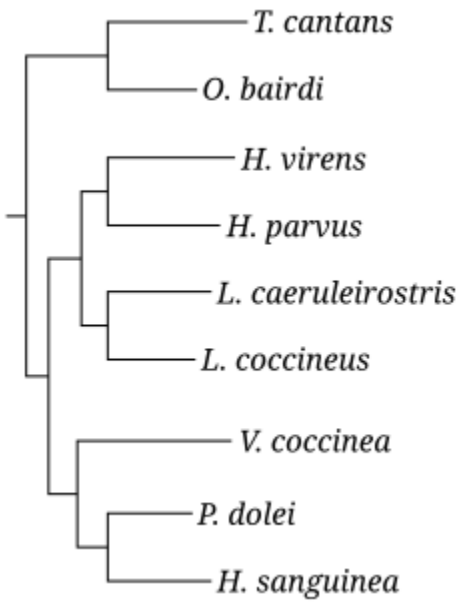


Figure 1. A hypothesized phylogenetic relationship among some members of the Hawaiian honeycreeper bird family

One hypothesized phylogenetic relationship among selected members of the Hawaiian honeycreeper bird family is shown in Figure 1. Based on the phylogenetic tree, birds in which of the following pairs share the most recent common ancestor?

7 Natural Selection MCQ

(A) *H. parvus* and *O. bairdi*

(B) *L. coccineus* and *V. coccinea*

(C) *L. caeruleirostris* and *H. virens* ✓

(D) *T. cantans* and *H. sanguinea*

Answer C

Correct. There are only two nodes separating *L. caeruleirostris* and *H. virens*. They share a more recent common ancestor compared with the other pairs listed.

Rhagoletis pomonella is a parasitic fly native to North America that infests fruit trees. The female fly lays her eggs in the fruit. The larvae hatch and burrow through the developing fruit. The next year, the adult flies emerge.

Prior to the European colonization of North America, the major host of *Rhagoletis* was a native species of hawthorn, *Crataegus marshallii*. The domestic apple tree, *Malus domestica*, is not native to North America, but was imported by European settlers in the late 1700s and early 1800s.

When apple trees were first imported into North America, there was no evidence that *Rhagoletis* could use them as hosts. Apples set fruit earlier in the season and develop faster, where hawthorns set later and develop more slowly.

Recent analysis of *Rhagoletis* populations has shown that two distinct populations of flies have evolved from the original ancestral population of flies that were parasitic on hawthorns. One population infests only apple trees, and the other infests only hawthorns. The life cycles of both fly populations are coordinated with those of their host trees. The flies of each population apparently can distinguish and select mates with similar host preferences and reject mates from the population specific to the other host tree. There is very little hybridization (only about 5 percent) between the two groups.

62. Initially, which of the following isolating mechanisms is likely to have been the most important in preventing gene flow between the two populations of *Rhagoletis*?

(A) Gamete incompatibility

(B) Temporal isolation ✓

(C) Mechanical isolation

(D) Reduced hybrid viability

63. Matings between individuals from the two populations of *Rhagoletis* produce hybrid flies that appear to be healthy and have normal life spans. The eggs laid by these hybrid flies, however, hatch less often than those of flies from either of the two populations. What isolating mechanism seems to be important in this hybrid population?

7 Natural Selection MCQ

(A) Prezygotic isolation
(B) Mechanical isolation
(C) Reduced hybrid fertility ✓
(D) Habitat isolation

64. The divergence between the two populations of *Rhagoletis* must have occurred very rapidly because

(A) the apple tree was imported into North America with European settlement approximately 200 years ago ✓
(B) flies were imported into North America with European settlement approximately 200 years ago
(C) long-distance rail transport of fruit increased only after the American Civil War (1861–1865)
(D) heavy use of gunpowder during the American Civil War (1861–1865) led to increased mutation rates in many natural populations of plants and animals

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65. Many scientists claim that the synthesis of the first organic molecules from inorganic precursors was possible because of the highly reducing atmosphere found on primitive Earth.

Which of the following is an appropriate null hypothesis that could be used when investigating the claim?

(A) The level of atmospheric oxygen on modern Earth is significantly higher than on primitive Earth.
(B) Organic molecules were transported to primitive Earth by a meteorite or other celestial event.
(C) The synthesis of organic molecules from inorganic molecules is possible under current atmospheric oxygen levels as well as those found on primitive Earth. ✓
(D) The absence of a significant quantity of atmospheric oxygen is required for the synthesis of organic molecules from inorganic precursors.

Answer C

Correct. A null hypothesis is a hypothesis of no difference. In this case, it should stipulate that the highly reducing atmosphere found on primitive Earth is not needed for the synthesis of organic molecules from inorganic precursors.

7 Natural Selection MCQ

66.

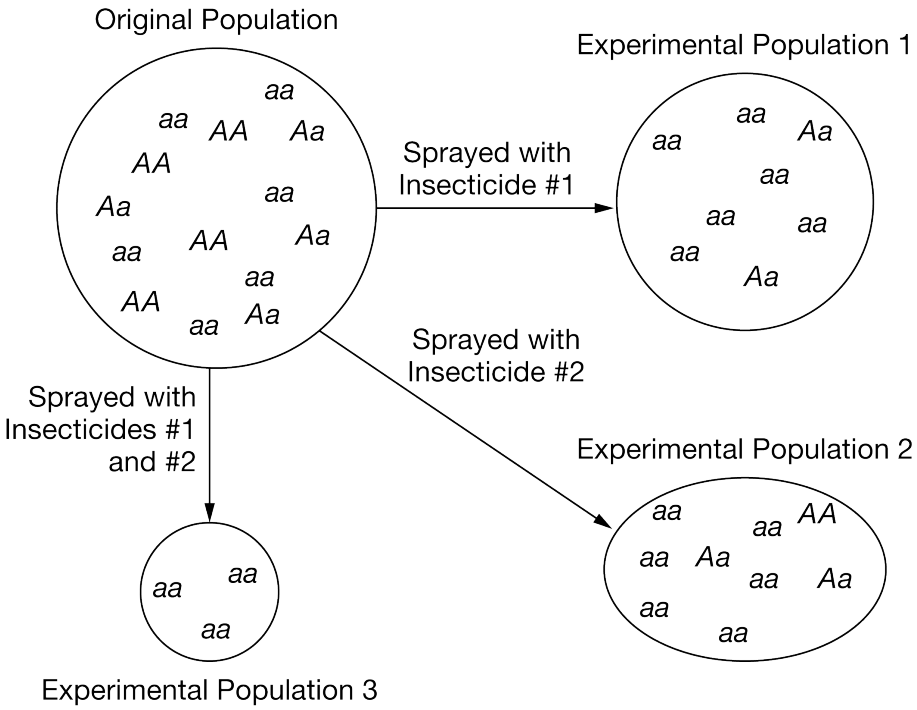


Figure 1. Effectiveness of Insecticides #1 and #2

Of the following, which is the best next step for the researchers to best determine the long-term effectiveness of these insecticides with regards to the development of resistance?

(A) Continue treating the three insect populations with the same insecticides and determine whether their efficiencies decline significantly after several generations. ✓

(B) Add an additional treatment group to the experiment to measure the effectiveness of another insecticide (Insecticide #3) over several generations.

(C) Set up a new experiment in which these insects are sprayed with two different insecticides to see if they are more effective than Insecticides #1 and #2.

(D) Spray these insecticides on a different species of insect (whose genotypes are also known) to determine its effectiveness on insects that are not pests.

Answer A

Correct. If the researchers continued the experiment for several generations and found that one of the insect population sizes started to increase, this would indicate that they have adapted and become resistant. Thus, this insecticide would probably not be useful in controlling the populations long-term.

7 Natural Selection MCQ

67. In the "RNA world" model for the origin of life, the first protocells (probionts) relied on RNA, not DNA, for information storage and transmission. Which of the following could best be considered evidence in support of the RNA world model?

(A) DNA is more stable than is RNA.

(B) Some RNA molecules in contemporary cells have catalytic function. ✓

(C) RNA is so complex in both structure and function that it must have preceded DNA.

(D) RNA has secondary structure whereas DNA does not.

68. Which of the following observations best supports the hypothesis that a large object collided with Earth in a period of time associated with a mass extinction?

(A) A species found only in Australia resembles a species found only in North America.

(B) An analysis of radiation reaching Earth from outer space suggests that water might have existed on distant planets.

(C) A fossil of an extinct species of mammal is found to have morphological characteristics that are similar to those of a living nocturnal mammal.

(D) Iridium, which is common in meteorites but rare on Earth, is found in sedimentary rock that contains the last appearance of many species in the fossil record. ✓

69. On a large volcanic island, researchers are studying a population of annual herbaceous plants. Which of the following observations best supports the prediction that speciation will occur within the existing plant population?

(A) Individuals of the species sometimes reproduce asexually by producing runners.

(B) Lava has separated the population into two areas: an upland forest and a lowland marsh. ✓

(C) Multiple groups of birds depend on the fruit produced by the plants as a source of food.

(D) The plants produce more seeds during warm summers than they do during cool summers.

7 Natural Selection MCQ

70. Figure 1 shows some major structural changes between a typical prokaryotic cell, the first eukaryotic common ancestor, and the last eukaryotic common ancestor.

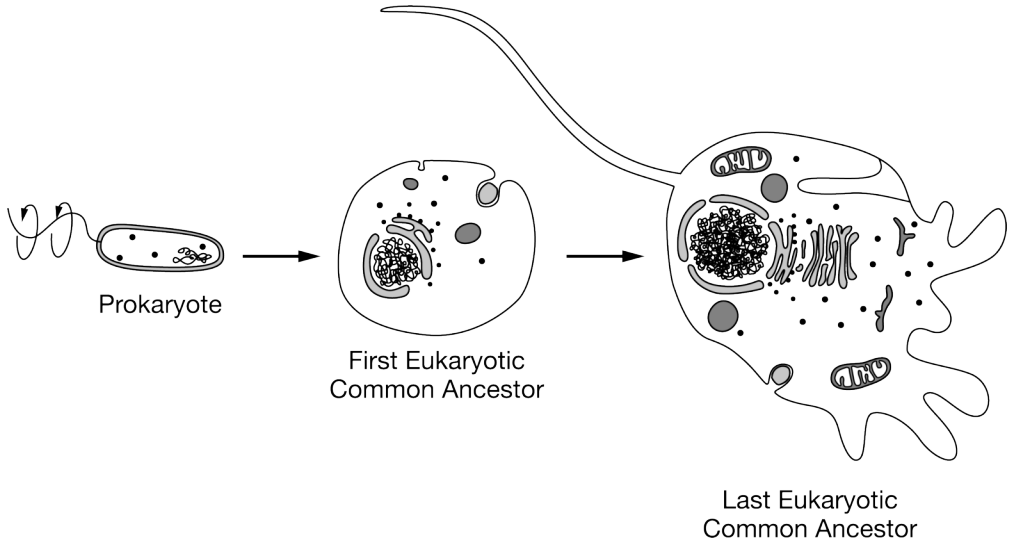


Figure 1. Models of a prokaryote, first eukaryotic common ancestor, and last eukaryotic common ancestor.

Based on the information provided in the figure, which of the following best predicts how membrane-bound organelles provided unique evolutionary advantages to the last eukaryotic common ancestor?

(A) Cells with nuclear membranes were better able to protect their genetic material against damage.

(B) Cells with flagella were better able to find food and escape predators.

(C) Cells with ribosomes were better able to produce proteins needed for a variety of metabolic functions.

(D) Cells with mitochondria were able to utilize metabolic energy more efficiently. ✓

Answer D

Correct. Eukaryotic cells acquired mitochondria via endosymbiosis. Most eukaryotic cells today have mitochondria because the last eukaryotic common ancestor had a selective advantage because it used mitochondria to perform aerobic cellular respiration. There are a few eukaryotic cell types (e.g., mature mammalian red blood cells) and a few parasitic single-celled eukaryotic organisms that have lost their mitochondria.

7 Natural Selection MCQ

71. Low doses of antibiotics are often added to livestock feed to increase production. Studies have shown that bacterial populations constantly exposed to the sublethal doses can evolve resistance to the antibiotics. A research group claimed that when a population of bacteria are constantly exposed to sublethal doses of streptomycin, their fitness declines due to the increased energy requirements for survival in the presence of the antibiotic. For subsequent studies, the researchers wish to determine whether adding a low, sublethal dose of an additional antibiotic causes further decline in the fitness of the bacteria.

Which of the following best represents the next step the researchers should take with respect to experimental design?

(A) Group 1: high dose of streptomycin + low dose of additional antibiotic
Group 2: low dose of streptomycin + high dose of additional antibiotic

(B) Group 1: low dose of streptomycin + no dose of additional antibiotic
Group 2: low dose of streptomycin + low dose of additional antibiotic ✓

(C) Group 1: no dose of streptomycin + no dose of additional antibiotic
Group 2: low dose of streptomycin + low dose of additional antibiotic

(D) Group 1: low dose of streptomycin + low dose of additional antibiotic
Group 2: high dose of streptomycin + high dose of additional antibiotic

Answer B

Correct. Since both groups have low doses of streptomycin, any deviation in the fitness of the bacteria in Group 2 may be attributed to the sublethal dose of the additional antibiotic. Group 1 bacteria serve as an appropriate control group for the experiment.

72. Malaria is caused by several different species of *Plasmodium*, a protozoan parasite. *Plasmodium* resistance to the common drugs used to treat malaria has increased in recent years.

In a scientific study, *Plasmodium* samples were analyzed in blood drawn from a large number of infected patients before drug treatment and subsequently from the subset of infected patients with drug-resistant *Plasmodium*. DNA sequences of four different *Plasmodium* genes thought to be involved in resistance were compared between samples from patients with drug-sensitive *Plasmodium* and patients with drug-resistant *Plasmodium*.

Which of the following best supports the hypothesis that preexisting mutations confer drug resistance?

7 Natural Selection MCQ

(A) Drug resistance can sometimes be reversed by simultaneous treatment with a drug that inhibits the drug-pumping mechanism of *Plasmodium*.

(B) Some of the mutations that occur most frequently in samples from patients treated with antimalarial drugs are detectable in *Plasmodium* samples collected prior to use of antimalarial drugs. ✓

(C) Continued use of the same antimalarial drugs leads to increased *Plasmodium* resistance.

(D) The same mechanism of drug resistance in *Plasmodia* isolated from separate populations of patients resulted from an accumulation of new genetic mutations.

73. Scientists claim that a population of mammals with high genetic diversity is better suited to survive and reproduce in a given geographic region. Which of the following reasonings best justifies the scientists' claim?

(A) High genetic diversity attracts more mates from other areas and increases the population size.

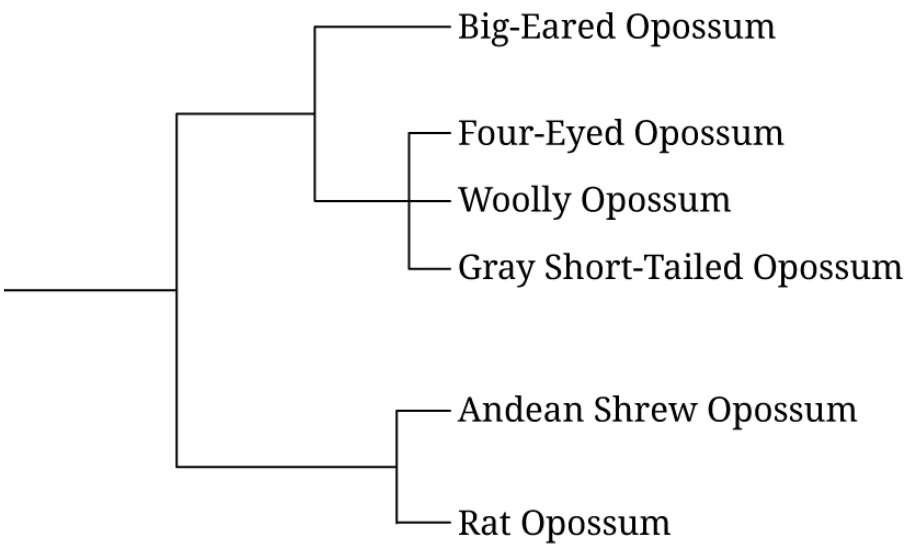
(B) High genetic diversity improves the ability of individuals in the population to hide from predators and survive.

(C) High genetic diversity increases the likelihood that individuals in the population possess traits that can withstand environmental changes. ✓

(D) High genetic diversity ensures that most individuals in the population have identical genetic makeup so that they can withstand environmental pressures.

Scientists studied the marsupial population in South America to determine species relatedness. The hypothetical relationships are presented in Figure 1.

Figure 1. Simplified cladogram of marsupial species in South America



74. Based on the cladogram in Figure 1, which of the following marsupials is most distantly related to the woolly opossum?

7 Natural Selection MCQ

(A) Big-eared opossum
(B) Four-eyed opossum
(C) Gray short-tailed opossum
(D) Rat opossum ✓

Answer D

Correct. The rat opossum branches off in the cladogram, indicating it shares a distant common ancestor with the woolly opossum compared to other species shown. This makes it the most distantly related species.

75. Which of the following best explains the relationships illustrated in Figure 1?

(A) The gray short-tailed opossum is more closely related to the Andean shrew opossum than to the other opossum species.
(B) The rat opossum and the gray short-tailed opossum share a more recent common ancestor than the rat opossum and the Andean shrew opossum do.
(C) The Andean shrew opossum and the rat opossum form a separate branch from the other opossum species. ✓
(D) The four-eyed opossum and the woolly opossum are more distantly related to each other than to the big-eared opossum.

Answer C

Correct. The Andean shrew opossum and the rat opossum form a distinct evolutionary branch separate from the other opossum species, indicating a different evolutionary lineage.

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76. Iridium is an element that is rare on Earth but commonly found in meteorites. A scientist believes that the first organic molecules may have come to Earth on meteorites 3.6 billion years ago. Which of these would be an appropriate null hypothesis to test related to whether meteorites hit the Earth 3.6 billion years ago?

(A) If sediment layers that are 3.6 billion years old are tested, they will contain relatively high iridium levels, because that is when many meteorites crashed into the Earth.
(B) There is no difference in iridium levels in sediments that were deposited on Earth 3.6 billion years old, compared to recent sediments. ✓
(C) If a sediment layer that is 3.6 billion years old is tested, it will contain organic material.
(D) If recently fallen meteorites and meteoroids collected in space are tested, they will not contain organic molecules.

7 Natural Selection MCQ

Answer B

Correct. This hypothesis states that there is no difference in iridium levels 3.6 billion years ago compared to other geological time periods.

77. MRSA is the acronym for methicillin-resistant *Staphylococcus aureus*. Many of the strains of the common bacterium are also resistant to other antibiotics in use today. The resistance is linked to a collection of genes carried on plasmids that are passed from one bacterium to another by conjugation. Suppose a newly discovered, chemically different antibiotic is used in place of methicillin. Which of the following would be the most likely effect on *Staphylococcus aureus* antibiotic resistance?

(A) The gene for methicillin resistance, no longer needed, would disappear entirely from *Staphylococcus aureus* populations within a few generations.

(B) Transmission of the methicillin-resistance plasmid by conjugation would increase among the *Staphylococcus aureus* population as the genes would confer resistance to the new antibiotic.

(C) Transmission of the methicillin-resistance plasmid would gradually decrease but the plasmid would not entirely disappear from the *Staphylococcus aureus* population. ✓

(D) Transmission of the methicillin-resistance plasmid by conjugation would increase among the *Staphylococcus aureus* population due to destruction of bacteria without the plasmid through use of the new antibiotic.

Answer C

This option is correct. It demonstrates an understanding that the plasmid would remain in the population even though it is not favored by selection and because it is not being selected against.

78. Mollusk fossils were excavated and collected from a coastal cliff. Scientists studying the fossils observed patterns of changing shell shape in the mollusks. The scientists identified a period of stability or stasis, followed by rapid change, and then stasis again with regard to shell shape.

Which of the following statements best predicts the result of a comparison between the fossils found before and after the period of rapid change?

(A) The same species will be found in both static periods, but they will have different shell shapes.

(B) The same species will be found in both static periods, but the species in the second static period will have smaller shells due to the stress of undergoing rapid change.

(C) Most of the species found in the first static period will be absent in the second static period. ✓

(D) Most of the species found in the second static period will resemble each other more than the species found in the first static period.

7 Natural Selection MCQ

Answer C

Correct. The relatively rapid changes occurring during the period of change indicate that both extinction and speciation were occurring. Most of the species observed in the second static period only date back to the period of rapid change. A majority of those species present before the rapid change will have become extinct or will have changed sufficiently to be considered a new species in the second static period. The scenario described exemplifies punctuated equilibrium.

Researchers were modeling the effects of repeated cycles of isolation with occasional interbreeding among five hypothetical fish species (species A, B, C, D, and E) found in two separate lakes (lakes I and II) that are occasionally joined by flooding, as shown in Figure 1. The research team indicated that species A would be considered to be the single common ancestor of species B, C, D, and E.

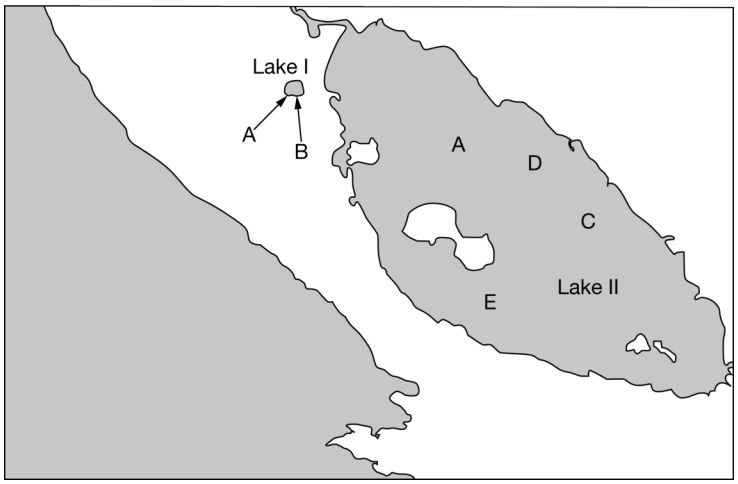

Figure 1. Current locations of populations of fish species A, B, C, D, and E

79. Which of the following mechanisms could most likely account for the origin of species B in Lake I?

(A) Allopatric speciation

(B) Directional selection

(C) Stabilizing selection

(D) Sympatric speciation ✓

7 Natural Selection MCQ

Answer D

Correct. Since species A and species B coexist, there must have been some isolating mechanism other than a physical barrier. The two species likely occupy separate niches, accounting for the lack of gene flow between the species.

80. Which of the following data could best support the claim that species B arose from a speciation event in Lake I?

(A) All of the fish species occupy the same niches in their respective lakes.

(B) DNA analysis shows fewer differences between species A and species B in Lake I than between species B and the populations in Lake II. ✓

(C) Species A and species B have similar body shapes due to convergent evolution in their habitats.

(D) Individuals of species A from Lake I can mate and produce viable offspring with individuals of species A from Lake II.

Answer B

Correct. Speciation results from accumulated genetic changes. DNA differences between the two lake populations would support the claim that a speciation event occurred.

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81. Old World vultures are birds that have digestive systems that enable them to feed mainly on carrion (dead animals). They have large wings used to soar as they search for dead animals with their vision. New World vultures have similar digestive systems and wingspans; however, they search for food using their sense of smell. Old World vultures and New World vultures are not closely related.

Which of the following best explains how Old World vultures and New World vultures both evolved to feed mainly on carrion?

(A) Old World vultures and New World vultures had a common ancestor that fed mainly on carrion, and this trait was selected for in populations of New World and Old World vultures.

(B) Old World vultures and New World vultures were exposed to similar selective pressures, which contributed to the independent evolution of carrion feeding in two separate lineages. ✓

(C) Old World vultures were introduced into areas with New World vultures, and the trait for carrion feeding was selected for in the population of New World vultures.

(D) Old World vultures and New World vultures have similar digestive systems, which drove them to consume a food source that was not available to other bird species.

7 Natural Selection MCQ

Answer B

Correct. This is an example of convergent evolution, where similar environmental pressures select for similar characteristics, including the ability to feed on carrion, in unrelated species.

82. Which of the following questions about the origin of life on Earth is most scientifically testable?

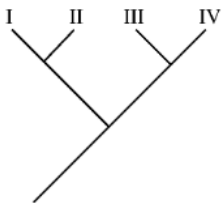
(A) Why were the earliest life-forms created using only twenty amino acids?

(B) Did life originate to make the universe a better place?

(C) Was catalytic RNA used by ancient organisms as a stepping stone to acquire protein enzymes?

(D) Can simple organic molecules form spontaneously in an oxygen-free atmosphere? ✓

83.



Based on morphological characteristics, the phylogenetic tree above has been developed for species I, II, III, and IV. DNA sequencing has recently been completed for a particular gene found in all four species. The sequencing will provide additional information for answering which of the following questions?

(A) Is species I the common ancestor of species II and III?

(B) Does species I contain the greatest number of vestigial structures?

(C) Are species III and IV the most closely related? ✓

(D) Should the positions of species III and IV be reversed?

7 Natural Selection MCQ

84. The degrees of relatedness in several native populations of prairie chickens in Marion and Jasper Counties in Illinois were determined. The degree of relatedness is inversely proportional to genetic diversity; that is, populations with a high degree of relatedness have less genetic diversity. Table 1 shows the collected data. Populations J1 through J5 are located in Jasper County. Populations M1 through M5 are located in Marion County. Numbers above 0.125 indicate a high degree of relatedness.

Table 1. RELATEDNESS OF PRAIRIE CHICKEN POPULATIONS

Population	Degree of Relatedness
J1	0.215
J2	0.215
J3	0.339
J4	0.280
J5	0.362
M1	0.305
M2	0.365
M3	0.199
M4	0.432
M5	0.405

Scientists claim that some of the populations in Marion County are at an increased risk of decline or extinction. Based on the data in table 1, which of the following best provides evidence to support this claim?

(A) The Marion County populations have a higher rate of mutation than the Jasper County populations, as evidenced by the higher degrees of relatedness.

(B) The Marion County populations are less stable because they are smaller than the Jasper County populations.

(C) The M2 population has a higher degree of relatedness than the M1 population, meaning that it has more genetic diversity.

(D) The M4 and M5 populations have the lowest genetic diversity of all populations. ✓

Answer D

Correct. Genetic diversity is inversely proportional to the degree of relatedness. The M4 and M5 populations have degrees of relatedness above 0.4; therefore, they have very low genetic diversity. Thus, these populations are at an increased risk of decline or extinction.

7 Natural Selection MCQ

85. In an exercise to understand how genetically controlled physical traits could affect speciation, a theoretical population was imagined that contained organisms with three shapes and three levels of pigmentation, as illustrated in Figure 1.

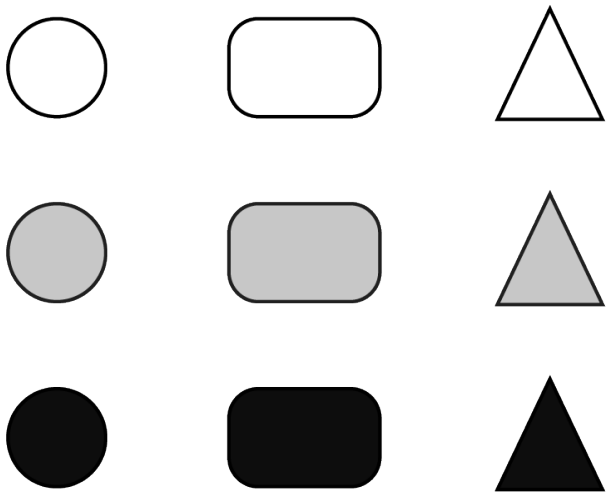
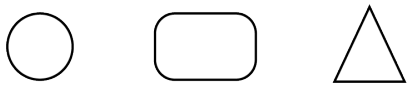
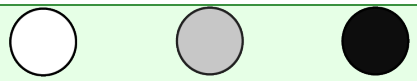


Figure 1. A single population with individuals of different shapes and pigmentation

Which of the following would most likely represent the types of individuals found in one of the species resulting from shape becoming a prezygotic isolating mechanism in the initial population represented in Figure 1 ?

(A) 

(B)  ✓

(C)

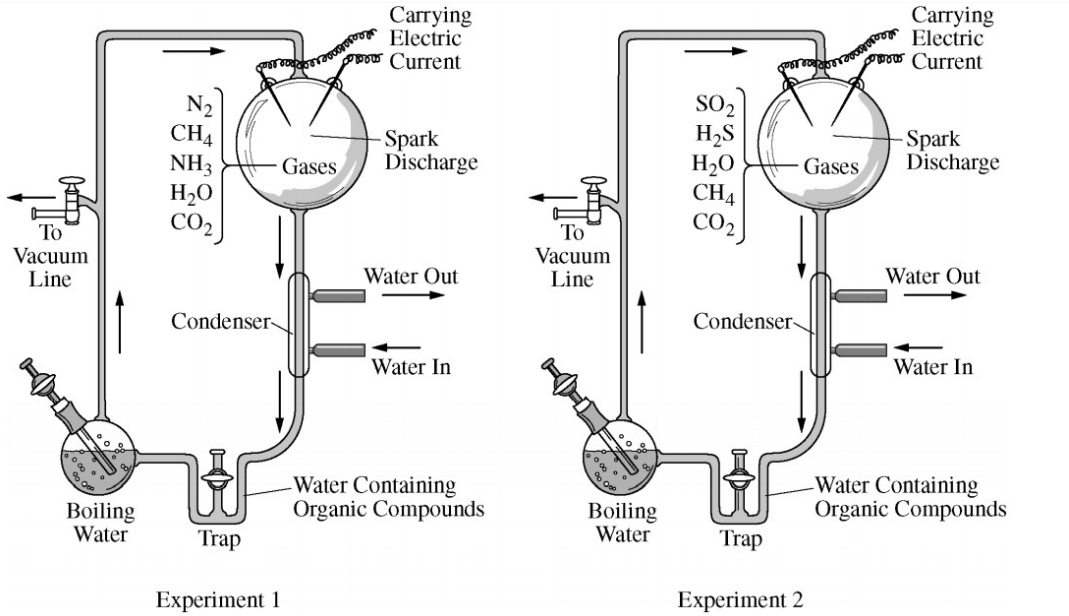
(D)

Answer B

Correct. Since shape was the prezygotic isolating mechanism, individuals with different shapes would not be able to mate and produce viable offspring. This would lead to different species based on shape.

7 Natural Selection MCQ

86. Two groups of students attempted to re-create the primitive atmospheric conditions of early Earth using the apparatus represented below. Each group ran the experiment with different gas mixtures in the apparatus.



Which of the following statements best justifies the claim that the conditions in at least one of the experiments could generate the molecular building blocks essential for life?

(A) The carbon dioxide (CO_2) gas and water (H_2O) vapor in experiment 1 could react spontaneously to produce the phospholipids required by the first life-forms.

(B) The nitrogen (N_2) gas and ammonia (NH_3) gas in experiment 1 could provide the elemental nitrogen required for the formation of amino acids. ✓

(C) The sulfur dioxide (SO_2) gas in experiment 2 could donate the excited electrons required to drive the process of photosynthesis.

(D) The methane (CH_4) gas in experiment 2 could act as the electron acceptor required to complete the process of cellular respiration

7 Natural Selection MCQ

87. Figure 1 compares two models of speciation, A and B.

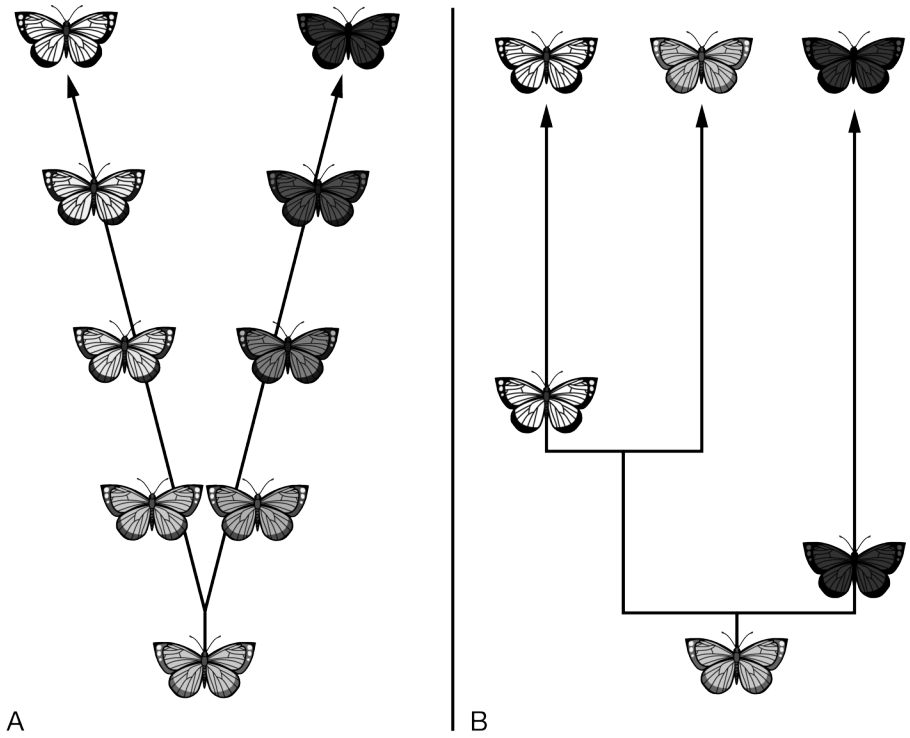


Figure 1. A comparison of two speciation models

Which of the following best explains how the ecological conditions are likely to be different in the two models?

(A) In model A the ecological conditions remain unchanged for long periods of time and then change drastically; in model B the ecological conditions change drastically from generation to generation.

(B) In model A the ecological conditions remain unchanged; in model B the ecological conditions change gradually over long periods of time.

(C) In model A the ecological conditions are changing drastically from generation to generation; in model B the ecological conditions remain unchanged.

(D) In model A the ecological conditions change gradually over a long period of time; in model B the ecological conditions remain unchanged for long periods of time and then change drastically. ✓

Answer D

Correct. Model A reflects gradualism while model B reflects punctuated equilibrium.

88. Which of the following statements best expresses the concept of punctuated equilibrium?

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(A) Small variations gradually accumulate in evolving lineages over periods of millions of years.

(B) Random mating ensures that the proportions of genotypes in a population remain unchanged from generation to generation.

(C) Stability is achieved when selection favors the heterozygote, while both types of homozygotes are at a relative disadvantage.

(D) Evolutionary changes consist of rapid bursts of speciation alternating with long periods in which species remain essentially unmodified. ✓

(E) Under competition for identical resources, one of the two competing species will be eliminated or excluded.

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89. The distribution of four populations of giant kangaroo rats, *Dipodomys ingens*, living on the same island when it was first surveyed in 1789 are shown in Figure 1.

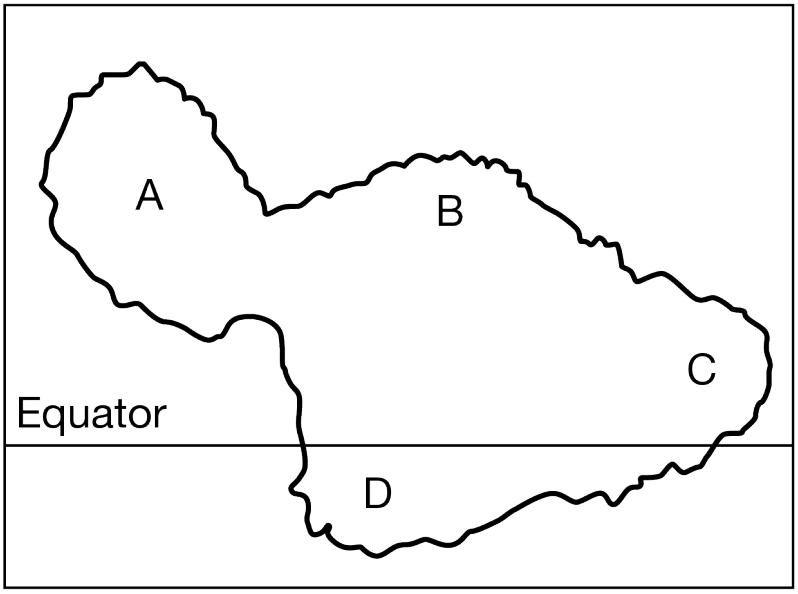

Figure 1. Four populations of *D. ingens* in 1789

In 1852, there was a large earthquake which altered the island's geography, as shown in Figure 2.

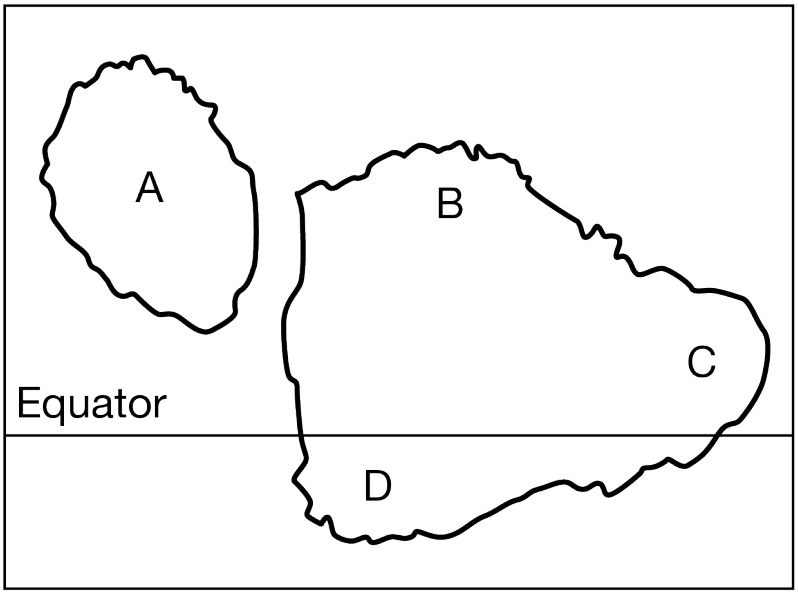

Figure 2. Map of the island following the 1852 earthquake

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The island was surveyed again in 2015, and scientists determined that the giant kangaroo rats originally in population A are now to be considered as a new species.

Which of the following best explains the mechanism that caused this speciation event in the giant kangaroo rats of population A?

(A) The animals in population A were exposed to different predators and had to modify their fur color to be better camouflaged.

(B) The animals in population A were reproductively isolated from those in populations B, C, and D. ✓

(C) The animals in population A accumulated fewer genetic mutations, adapting to their microclimate.

(D) The animals in population A mate at different times of year, causing their gametes to be incompatible with those from animals in populations B, C, and D.

Answer B

Correct. This is an example of reproductive isolation; one species diverges into two or more when some members of the species cannot reproduce with other members of the same species due to a physical barrier.

90. Rock pocket mice live in the deserts of the American southwest. Ancestral pocket mice all had light-colored coats that blended in with the region's rocks and sandy soil, keeping the mice hidden from predatory owls. About 1.7 million years ago, a series of volcanic eruptions spewed out wide trails of black lava into the middle of rock pocket mouse territory. Currently there are two color morphs of rock pocket mice: light-colored mice that are typically found in habitats with sandy soil, and dark-colored mice that are typically found in habitats with dark-colored lava rocks.

Which of the following best justifies the claim that alleles that may be adaptive in one type of environment can be deleterious in another because of different selective pressures?

(A) Light- and dark-colored mice survive in many environments because they are able to avoid predation by moving into tunnels.

(B) Light-colored mice are more likely to survive than dark-colored mice in habitats with lava rocks and sandy soil.

(C) Light-colored mice are significantly more susceptible to predation by owls in habitats with lava rocks than in habitats with sandy soil. ✓

(D) Dark-colored mice are more likely to survive than light-colored mice in habitats with lava rocks and sandy soil.

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Answer C

Correct. On the darker-colored lava rocks, alleles that confer light coat colors in mice are now deleterious because these mice are more easily identified by predators, such as owls. Thus, the dark lava rock environment places different selective pressures on the light-colored mice than the sandy soil environment does.

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Researchers have undertaken a long-term study of populations of rock pocket mice, *Chaetodipus intermedius*, in Arizona and New Mexico. Rock pocket mouse fur is either dark or light in color. Most of the populations studied live in areas with light-colored granite rocks; however, some areas are dominated by dark-colored rocks from ancient lava flows. Researchers collected mice from six sites in Arizona and recorded their coat colors. Figure 1 shows the study sites, the rock colors, and the distribution of coat colors.

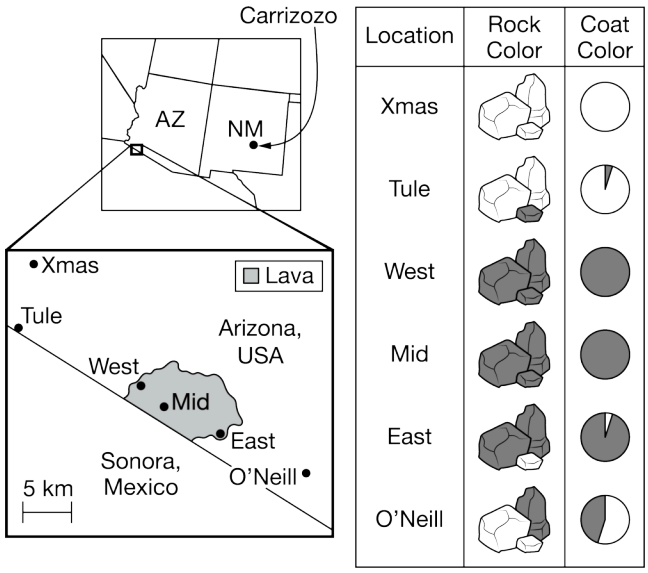

Figure 1. Rock color and coat color at three sites on a lava flow (West, Mid, and East), and three nearby sites: O'Neill Hills (O'Neill), the Tule Mountains (Tule), and Christmas Pass (Xmas)

In the Arizona populations, the melanocortin-1 receptor $(MC1R)$ gene is responsible for coat color phenotype (light or dark). The $MC1R$ D allele, which results in a dark coat, is dominant to the d allele. Researchers collected samples of mice from all six study sites and sequenced the $MC1R$ gene. They also collected mice with dark coats from a location in New Mexico, the Carrizozo lava flow, to determine if the $MC1R$gene was also responsible for the dark coat colors observed in that population. The rock pocket mouse population in Carrizozo is isolated from the populations in Arizona. Table 1 presents the genotypic data from sampled mice from all seven locations.

Table 1. $MC1R$ Genotypes of Rock Pocket Mice at Seven Collection Sites

State	Collection Site	$MC1R$ Genotype			Total Collected
		DD	Dd	dd	
Arizona	Christmas Pass	0	0	6	6
Arizona	Tule Mountains	0	5	80	85
Arizona	West	6	1	0	7
Arizona	Mid	3	2	0	5
Arizona	East	35	7	3	45
Arizona	O'Neill Hills	9	34	34	77
New Mexico	Carrizozo	0	0	9	9

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91. Based on the data in Table 1, which of the following best describes the relationship between the $MC1R$ gene and coat color in the Carrizozo, New Mexico, rock pocket mouse population?

(A) The $MC1R$ D allele is responsible for dark coat color in the Carrizozo population and likely spread from the Arizona population through migration.

(B) The $MC1R$ D allele is not responsible for dark coat color in the Carrizozo population, so the dark coat color is likely a result of convergent evolution. ✓

(C) The $MC1R$ D allele is absent from the Carrizozo population, so all the mice in this population must have coats that are light in color.

(D) The $MC1R$ D allele is absent from the Carrizozo population because of different environmental pressures in the Carrizozo population.

Answer B

Correct. In the Arizona rock pocket mouse populations, mice with dark coats have a homozygous dominant or heterozygous genotype for the $MC1R$ gene. However, according to the data table, the dark mice from the Carrizozo population are all homozygous recessive for the $MC1R$ gene. The Carrizozo population is also found on a dark lava flow. Convergent evolution occurs when different species or populations show a similar adaptation to a similar environment, but the adaptation is due to different genes. In this case, the Carrizozo mice population have a different mutation that results in a dark coat color.

92. Which of the following best describes the processes occurring between the mouse populations and their environments?

(A) Natural selection favors the mice with the dark coat color, causing the mice in all of the populations to be better adapted.

(B) Natural selection favors the mice with the light coat color, causing the mice in all of the populations to be better adapted.

(C) Dark color is selected for in mice from the West, Mid, and East populations. ✓

(D) Light color is selected for in mice from the West, Mid, and East populations.

Answer C

Correct. On the dark lava flow rocks (West, Mid, and East locations), most of the mice have dark coats. These dark-colored mice are better camouflaged, thus they are more likely to survive and reproduce. Therefore, on the dark lava flow rocks, natural selection favors the mice with dark coats.

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93. Owls (primary predators of rock pocket mice) are nocturnal visual predators that are able to distinguish effectively between dark and light colors in low-light conditions.

Which of the following best explains the relationship between coat color and predation rate in the rock pocket mouse population on the lava flow rocks with respect to the different coat color phenotypes?

(A) A dark coat color provides camouflage to mice on the lava flow rocks; therefore, mice with dark coats have a lower predation rate in that habitat. ✓

(B) At night, predators have reduced vision, so there is no relationship between coat color and predation rate on rock pocket mice.

(C) A dark coat color reflects less light and therefore mice with dark coats have a lower predation rate.

(D) A light coat color reflects more light and therefore mice with light coats have a lower predation rate.

Answer A

Correct. Many prey species have adaptations that allow them to blend in with their environment, thus escaping predators. In this case, the dark coat color of rock pocket mice blends in with the dark rocks on the lava flows but is more obvious on the light granite rocks. Therefore, mice with dark coats would have a lower predation rate on the lava flow rocks and a higher predation rate on the granite rocks.

94. The researchers built an enclosure to prevent migration into and out of the O'Neill rock pocket mouse population for a period of two years.

Which of the following best predicts the effect of the enclosure on the O'Neill population?

(A) The O'Neill population will rapidly evolve into a separate species.

(B) The O'Neill population will become less diverse due to the bottleneck effect.

(C) The frequency of the $MC1R\ D$ allele will decline due to natural selection. ✓

(D) One of the alleles for the $MC1R$ gene will become fixed due to genetic drift.

Answer C

Correct. The $MC1R\ D$ allele occurs at a higher frequency in the O'Neill population than in the other populations of rock pocket mice on the light granite rocks. This higher-than-expected frequency is likely due to immigration from the East rock pocket mouse population on the lava flow rocks. An enclosure that would prevent this gene flow would reinforce the selective pressures on the O'Neill population due to the light-colored granite rocks. Since mice with dark coats are more visible to predators on light

7 Natural Selection MCQ

granite rock, the mice with dark coats would be less likely to survive and reproduce, so the $MC1R\ D$ allele frequency would decrease over time.

95. The researchers determined that the East lava flow population is not in Hardy-Weinberg equilibrium.

Which of the following best explains why the population is not in Hardy-Weinberg equilibrium?

(A) The presence of heterozygotes for the $MC1R$ gene in the East lava flow population indicates that mating is occurring randomly between mice with different coat colors.

(B) The selective pressures on the East lava flow population are not strong enough to maintain Hardy-Weinberg equilibrium.

(C) The East lava flow population does not experience migration, which is a required condition for Hardy-Weinberg equilibrium.

(D) The presence of $MC1R\ d$ alleles in the East lava flow population is an indication of gene flow from mice populations living on light granite rocks. ✓

Answer D

Correct. One of the conditions of Hardy-Weinberg equilibrium is the absence of migration. The higher frequency of the $MC1R\ d$ allele in the East lava flow population compared to other lava flow populations suggests there is a high migration rate into the East lava flow population from nearby populations of rock pocket mice living on the light-colored granite rocks.

96. Which of the following is closest to the frequency of the $MC1R\ D$ allele in the Tule Mountain population?

(A) 0.03 ✓

(B) 0.06

(C) 0.94

(D) 0.97

Answer A

Correct. Since the frequencies of each genotype are provided, and it is not certain that the population is in equilibrium, it is neither necessary nor valid to resort to the Hardy-Weinberg equation for calculations. The frequency of the $MC1R\ D$ allele is equal to the number of $MC1R\ D$ alleles divided by the total number of alleles. The total number of alleles is equal to the number of mice times two; $85 \times 2 = 170$. The number of the $MC1R\ D$ alleles is equal to the number of homozygous dominant mice (DD)

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times two plus the number of heterozygous mice; $(Dd) : 0 \times 2 + 5 = 5$. Therefore, the frequency of the $MC1R\ D$ allele is $5 \div 170 = 0.03$.

Staphylococcus aureus is a pathogenic bacterium that can infect a wide range of host species, including humans. *S. aureus* has a particular protein that binds with hemoglobin from the host organism. Hemoglobin is the iron-containing protein used to transport oxygen in the blood. Since iron is important for growth, *S. aureus* have evolved the ability to absorb the iron from the host's hemoglobin.

Different *S. aureus* strains preferentially infect different hosts and have different amino acid sequences at their hemoglobin-binding domains (Table 1; letters indicate different amino acids). In an experiment, different *S. aureus* strains were mixed with hemoglobin from macaque monkeys and their binding ability was measured (Figure 1). The differences in amino acid sequences contributed to the differential binding abilities observed.

Table 1. Selected amino acid sequences and preferred host for four strains of *S. aureus*

S. aureus Strain	Amino Acid Sequence	Host Species
1	Q Q F Y H Y A R S	Species A
2	R Q A Y H Y A R T	Species B
3	Q Q A Y H Y A R T	Macaque
4	R Q A A H Y Q L T	Species C

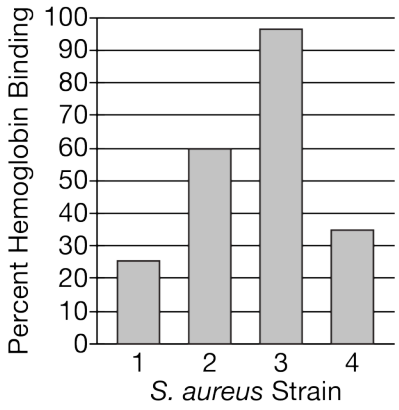

Figure 1. Macaque hemoglobin binding ability of different strains of *S. aureus*

97. Which of the following experiments would be most appropriate to determine whether populations of *S. aureus* are continuously adapting in order to obtain iron from hosts more effectively?

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(A) Grow *S. aureus* in media with macaque hemoglobin and abundant free iron. Record bacterial growth over time and compare it to growth of the original population.

(B) Culture *S. aureus* bacteria with hemoglobin from a mixture of host species. Then transfer the bacteria to media with a known effective antibiotic and record bacterial growth.

(C) Sequence the genes of different *S. aureus* strains and construct a cladogram representing their relatedness.

(D) Culture *S. aureus* bacteria with hemoglobin from a novel host species as the only source of iron for many generations, then compare protein structure from bacteria in this culture with bacteria from the original culture. ✓

Answer D

Correct. If the sequences change over time, this suggests that the bacteria adapted to better obtain iron from the hemoglobin of the novel host.

98. Which of the following processes is most consistent with the differences in the amino acid sequences listed in Table 1 ?

(A) Each host transcribes and translates this gene differently.

(B) *S. aureus* develops a mutualistic relationship with each host species.

(C) Changes in amino acid sequence are unrelated to protein function.

(D) Each strain is best adapted to a specific host species. ✓

Answer D

Correct. Since different strains of *S. aureus* preferentially infect different hosts, they are adapted to use the hemoglobin of their specific host. A strain that infects humans has a protein that binds well to hemoglobin found in humans but not as well to the hemoglobin found in other species.

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99. The heterotroph hypothesis of the origin of life states that basic organic building blocks formed from inorganic molecules under the primitive conditions present early in Earth's development. These building blocks combined to form polymers, such as crude proteins. Dr. Sidney Fox wished to add to the heterotroph hypothesis of the origin of life by synthesizing simple proteins from a mixture of amino acids.

Which statement below best states an alternative hypothesis Fox could have used to frame his research?

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(A) Formation of crude proteins occurs by chance and occurs regardless of atmospheric O_2 levels or available free energy.

(B) Given aerobic conditions and large amounts of lightning, amino acids will spontaneously form crude proteins.

(C) Formation of simple proteins on primitive Earth only occurs spontaneously if the required building blocks were transported to Earth by a celestial event or a meteorite.

(D) In the presence of available energy and the absence of a significant quantity of atmospheric O_2 simple proteins will form spontaneously from a mixture of amino acids. ✓

Answer D

Correct. The alternative hypothesis should stipulate that the unique conditions of primitive Earth are what enabled the formation of crude protein molecules from amino acids.

100. Figure 1 illustrates a process that occurs in a certain group of organisms.

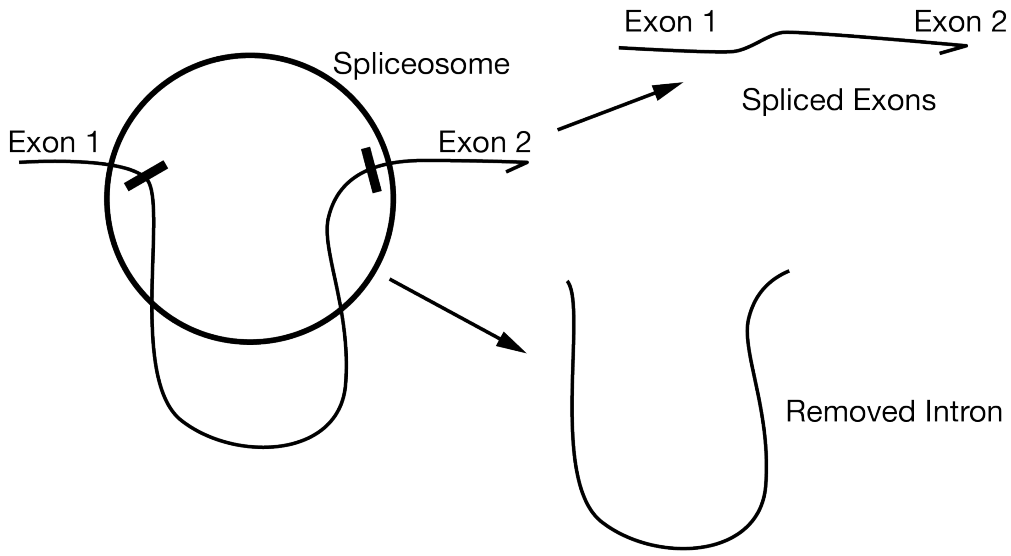

Figure 1. Model of spliceosome activity

Which of the following best predicts the importance of this process?

(A) This process is common to all members of Eukarya and allows noncoding regions of RNA to be removed so that proteins with the correct amino acid sequence are produced. ✓

(B) This process is common to all members of Archaea and allows these organisms to survive harsh environments.

(C) This process is common to all organisms and is a conserved DNA repair mechanism.

(D) This process is common to vertebrates and allows the production of large and complicated proteins.

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Answer A

Correct. All eukaryotes have introns that must be removed, and exons must be joined together in order to translate eukaryotic mRNA and produce a functional protein.

101. The Stanley Miller apparatus demonstrated that organic molecules could assemble spontaneously in an environment lacking free oxygen and containing water, methane, and ammonia in the presence of an abundant energy source, such as an electric discharge. The research was considered supportive of the organic soup hypothesis, which states that the primitive atmosphere provided inorganic precursors from which organic molecules could have been synthesized in the presence of an energy source. Based on subsequent research, the primordial atmosphere was determined to contain less methane and more carbon dioxide. The new data about the composition of the early atmosphere had which of the following effects on origin-of-life hypotheses?

(A) Miller's work was discarded as not relevant to an origin-of-life hypothesis.

(B) Miller's work was used to refute the new hypothesis about the composition of the atmosphere.

(C) The organic soup hypothesis was abandoned as no longer being supportable.

(D) A new organic soup hypothesis was proposed to account for the new data about the atmosphere. ✓

102. To prevent the loss of genes as chromosome ends wear down during DNA replication, the tips of eukaryotic chromosomes have specialized DNA "caps" called telomeres. Telomeres consist of hundreds or thousands of repeats of the same short DNA sequence, which varies between organisms. The telomeric repeat sequences for selected organisms are given in Table 1.

Table 1. Telomeric Repeat Sequences for Selected Organisms

Organism	Telomeric Repeat Sequence (5′ to 3′)
Human	TTAGGG
Mouse	TTAGGG
Paramecium	TTGGGT
Green alga	TTTTAGGG
Onion	CTCGGTTATGGG

Which of the following best predicts why humans and mice have identical telomeric repeat sequences?

(A) Humans and mice have the same number of genes.

(B) Humans and mice have the same number of chromosomes.

(C) Humans and mice share a recent common ancestor that had the 5′–TTAGGG–3′ repeat sequence in its telomeres. ✓

(D) Mice utilized the 5′-TTAGGG-3′ telomeric repeat sequence first, but the sequence independently evolved in humans as well.

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Answer C

Correct. The most likely cause of homology in telomeric repeat sequences is common ancestry between humans and mice. This is an example of structural evidence on the molecular level that indicates common ancestry.

103. One facet of the study of the origin of life focuses on the nature of the first genetic material. Figure 1 describes a possible scenario, known as the RNA World Hypothesis, that could have led to the evolution of our modern-day genetic material. This hypothesis states that the first genetic material was RNA, not DNA, and a transitional period led to the emergence of DNA as the storage molecule for genetic material.

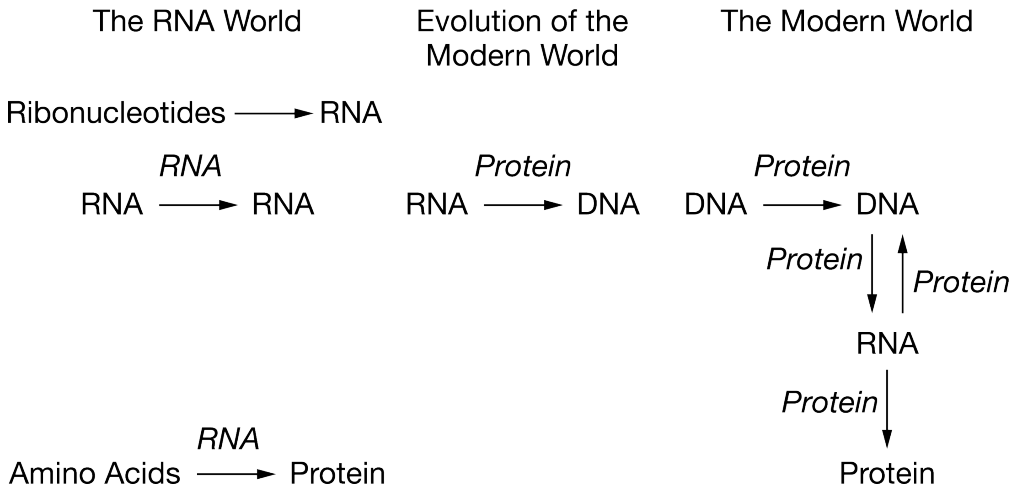

Figure 1. A model of the RNA World Hypothesis

A scientist wishes to provide experimental evidence to support the model shown in Figure 1 by demonstrating the ability to synthesize an RNA molecule. Which of the following is an alternative hypothesis that can be tested to support the RNA World Hypothesis?

(A) A protein template that matches proteins found on the primitive Earth added to a mixture of deoxyribonucleotides will produce double-stranded DNA molecules.

(B) Double strands of DNA placed in a harsh environment will unwind and form RNA molecules for long term information storage.

(C) A pool of amino acids can spontaneously form into single-stranded RNA, which would then code for proteins.

(D) A mixture of ribonucleotides such as adenine and uracil will spontaneously form single-stranded chains of RNA. ✓

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Answer D

Correct. Producing this result in a laboratory would support the claim that RNA molecules could have formed spontaneously in the primitive Earth environment.

104. Transposable elements are significant features in the genomes of almost all eukaryotic organisms. The accumulation of these elements in various species can help determine their relatedness. Table 1 indicates the presence (+) or absence (−) of 5 segments of DNA that have been moved by transposition (transposons) in 6 primate species.

Table 1. Presence or absence of 5 transposons in primate species

Species	1	2	3	4	5
Human	−	+	+	+	+
Bonobo	+	+	+	+	+
Chimpanzee	+	+	+	+	+
Gorilla	−	−	+	+	+
Orangutan	−	−	−	+	+
Gibbon	−	−	−	−	−

Which of the following best predicts how phylogenetic relationships might be revised if transposon 1 was not found in chimpanzees?

(A) Bonobos would be more closely related to humans than to chimpanzees.

(B) Chimpanzees would be more closely related to humans than to bonobos. ✓

(C) Orangutans would be more closely related to humans than to chimpanzees.

(D) Chimpanzees would be more closely related to gibbons than to gorillas.

Answer B

Correct. If chimpanzees lost transposon 1, their transposon pattern would be identical to that of humans.

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105.

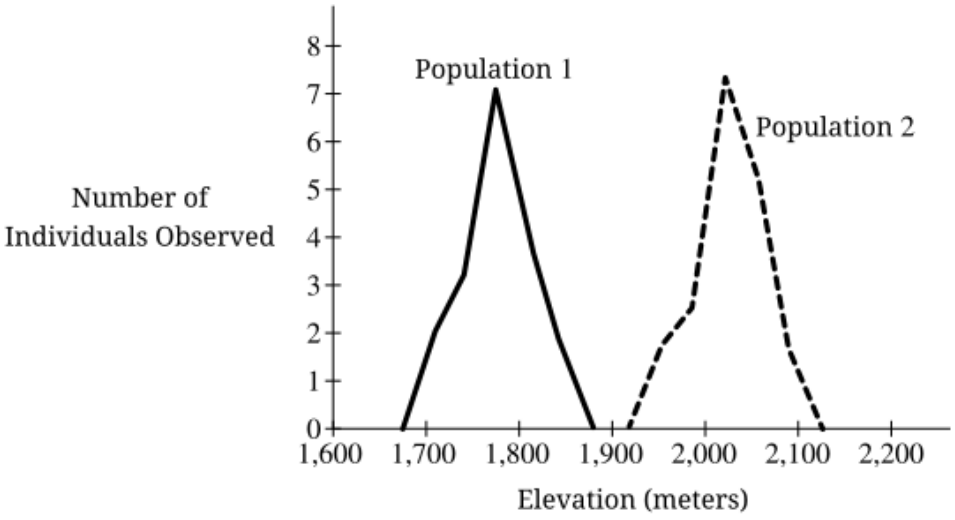

Figure 1. The distribution of individuals from two populations of the same species at different elevations

Researchers investigating two populations of the same species recorded the elevation at which individuals from each population were found (Figure 1).

Which of the following claims is best supported by the data in Figure 1 ?

(A) Gene flow is likely to occur between the two populations because they share the same habitat.

(B) Gene flow is likely to occur between the two populations because they are the same species.

(C) Gene flow is unlikely to occur between the two populations because they compete for food.

(D) Gene flow is unlikely to occur between the two populations because they are geographically isolated. ✓

Answer D

Correct. Based on the data in Figure 1, the individuals of the two populations are found in non-overlapping habitats at different elevations. Because individuals from the two populations do not meet, there will be no reproduction between members of the populations and thus no gene flow between the populations.

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106. A 100 acre woodlot was observed for the last forty years. Five plant species (1 through 5) were surveyed, with the results shown in Table 1.

Table 1. Data from Forty-Year Survey of Five Plant Species in a Woodlot

Year of Survey	Species 1 Population	Species 2 Population	Species 3 Population	Species 4 Population	Species 5 Population
1	35	30	23	32	25
20	35	42	20	2	1
40	50	50	2	0	0

Some researchers claim that the decrease in plant species richness of this woodlot is unrelated to human activities over the last forty years.

Which of the following would be an appropriate alternative hypothesis?

(A) Interspecific competition caused the extinction of species 4 and 5.

(B) The woodlot environment is arid and species 1 and 2 are well-adapted to conserve water.

(C) The woodlot is shaded, with little direct sunlight, and species 3, 4, and 5 are shade-intolerant canopy trees.

(D) During the last few decades, tall buildings have been built around the woodlot, and species 3, 4, and 5 are shade-intolerant plants ✓

Answer D

Correct. The construction of tall buildings over the last few decades that surround the woodlot has made the woodlot heavily shaded. This means that shade-intolerant species will be eliminated from the woodlot over time. Therefore, human activities have changed the environment of the woodlot and caused extinction of species 3, 4, and 5.

107. A species of snail lives in the intertidal zone along the coast of New England. The dark-colored variety of the species is more common in northern New England, the light-colored variety is more common two hundred miles away in southern New England, and both varieties are commonly found together in central New England. Which of the following best explains the observed distribution pattern of the snails?

7 Natural Selection MCQ

(A) The founder effect suggests that dark-colored snails migrated from the southern regions to the north and established the populations found there.

(B) Genetic drift at the shell-color locus caused the northern population to become homozygous for the dark-color allele.

(C) The mutation rate is higher in the south, as the longer days expose the snails to more ultraviolet radiation than in the north.

(D) Dark-colored snails absorb more solar energy and so survive more readily in the colder northern waters. ✓

108. Hunting of African elephants for their tusks resulted in a 90% reduction in the size of one particular elephant population by 1992. Since then, protections of elephant populations have improved, and the population size has begun to increase. Before 1992, tuskless elephants were rare, but now a significant proportion of the female elephants in this population are tuskless. An allele of the gene $AMELX$ appears to be responsible for the tuskless trait.

Which of the following best explains the change in the relative proportion of tuskless elephants since 1992 ?

(A) With human protections in place, tusks are less important to the protection of the herds and in particular to the females, so tusks are selected against.

(B) Female tuskless elephants were more likely to have survived than were tusked females, resulting in a greater proportion of the offspring having the genes of the tuskless elephants. ✓

(C) Female elephants that survived removal of their tusks were more likely to pass the tuskless allele than the tusked allele of the $AMELX$ gene to their female offspring.

(D) Overhunting of the elephants caused the emergence of the tuskless mutation in the $AMELX$ gene, leading to an increase in the frequency of the tuskless phenotype.

Answer B

Correct. Since tusked elephants were hunted, tuskless elephants were more likely to survive before protective measures were enacted. Traits of the tuskless elephants, such as that encoded by the $AMELX$ mutation, would have passed on to offspring, allowing the population to evolve over time to include more tuskless females.

109. Which of the following is probably the best explanation for the fact that Antarctic penguins cannot fly, although there is evidence that millions of years ago their ancestors could do so?

(A) Penguins live on land and feed in the water; therefore they have no need to fly.

(B) The Antarctic home of penguins is flat and barren; therefore there is no place to fly.

(C) Ancestral penguins without large wings were better able to swim and feed in the water; therefore they passed their genes for shorter wing structure on to their offspring. ✓

(D) Ancestral penguins did not use their wings for long periods of time; therefore today's penguins have only tiny, nonfunctional wings.

(E) The cold and wind of Antarctica make flight impossible; therefore penguins that live there have lost the ability to fly.

7 Natural Selection MCQ

110. Competition for food would probably be most severe between two

(A) closely related species in different niches

(B) closely related species in similar niches ✓

(C) unrelated species in different communities

(D) unrelated species in the same community occupying different niches

(E) ecological equivalents in different niches

111. In Darwinian terms, the fittest individuals of a species are those that

(A) attract the most mates

(B) grow to the largest size

(C) are best adapted to the temperature extremes to which they are exposed

(D) are best equipped to cope with the predators to which they are exposed

(E) leave the greatest number of reproducing descendants ✓

7 Natural Selection MCQ

112. African elephants, *Loxodonta africana*, are often hunted illegally for their tusks. Both male and female elephants have tusks, although the tusks are much larger in the males. Researchers have followed the elephant population in Gorongosa National Park in Mozambique for many years. Figure 1 shows the percent of tuskless female elephants expected in wild populations throughout the species' range, as well as the percent of tuskless females that survived the hunting between 1977 and 1992 (those over 25 years old) and the percent of female elephants born after 1992 without tusks in the park. The data are based on 200 known female elephants in the park.

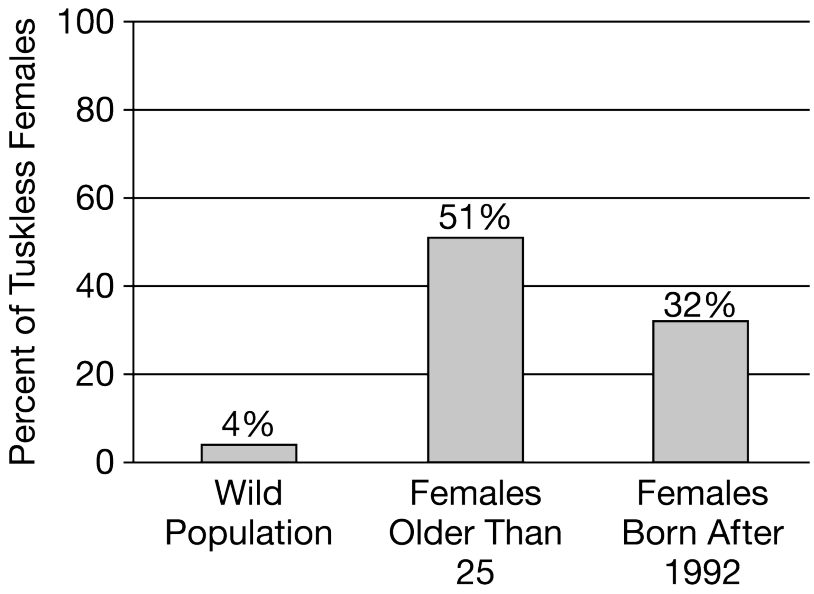


Figure 1. Comparison of the frequency of tuskless females by age cohort

Which of the following best describes the process responsible for the change in the percent of tuskless female elephants in the Gorongosa National Park population shown in Figure 1 ?

(A) Human hunting of elephants with tusks between 1977 and 1992. ✓

(B) Sexual selection for females without tusks between 1977 and 1992.

(C) Tusklessness in female elephants is a dominant trait that is only expressed in females.

(D) After 1992, female elephants from neighboring populations merged, with the population in the park adding more tuskless individuals to the population.

Answer A

Correct. The increased survival of elephants without tusks has resulted in tuskless female elephants leaving more offspring to reproduce than the female elephants that were hunted for their tusks.

113. Evolutionary fitness is measured by

7 Natural Selection MCQ

(A) physical strength

(B) reproductive success ✓

(C) length of life

(D) resistance to disease

(E) competitiveness

114. Scientists investigated the role that beak depth plays in the ability of one species of seed-eating finch to reproduce. The scientists calculated the average beak depth of finches in mating pairs and then observed whether or not the pairs produced at least one offspring that survived to the next season. The data are represented in Figure 1.

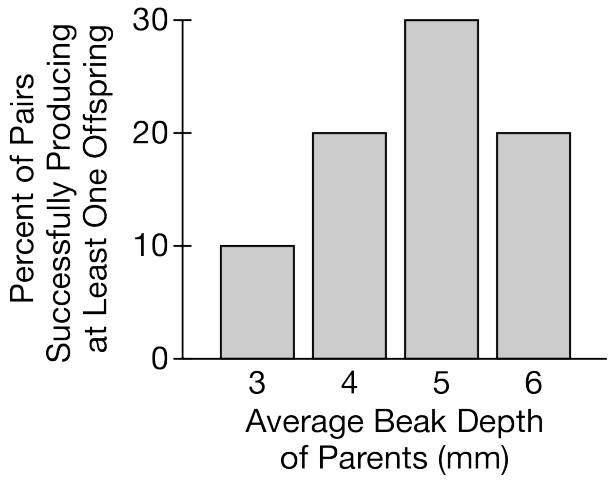


Figure 1. Beak depth and offspring survival in a species of finch

Based on the data in Figure 1, which of the following best describes the concept illustrated?

(A) Parental pairs with a specific beak depth had the highest reproductive fitness. ✓

(B) Parental pairs with a specific beak depth ate the most nutritious seeds.

(C) Finches with a certain beak depth rarely find mates.

(D) Increasing average beak depth results in increasing finch fitness.

Answer A

Correct. Darwinian fitness is the ability to successfully reproduce, meaning the finches leave surviving offspring. The parental pairs with an average beak depth of 5 mm had the greatest reproductive success.

7 Natural Selection MCQ

115. Tobacco hawk moths, *Manduca sexta*, feed exclusively on sugar-rich nectar produced in the flowers of tobacco plants. To feed, the moth inserts its proboscis (a long, tubular mouthpart) into the flower to reach the nectary, which is usually at the end of a long tube in the flower. Moths will feed on the nectar of the tobacco plant species whose nectaries they can access. A study determined the proboscis sizes in one tobacco hawk moth population. The data are shown in Figure 1. The most abundant tobacco species in the population's habitat has a nectary tube that is $5\ \text{cm}$ long.

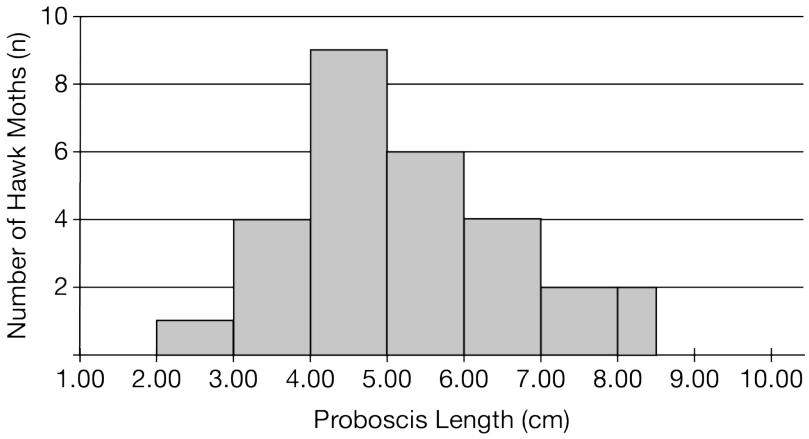


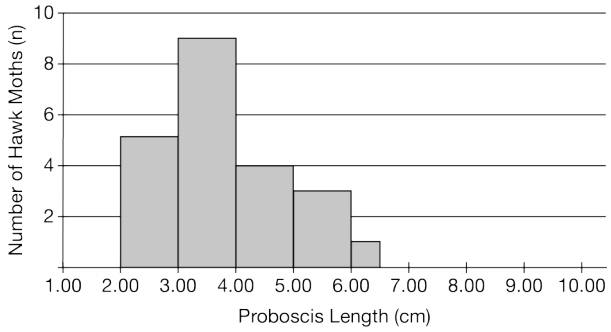
Figure 1. Proboscis length distribution in a population of tobacco hawk moths

Over the next year, a virus attacks and rapidly kills almost all plants of the tobacco species. A closely related species of tobacco that is resistant to the virus becomes established during that same year. The nectary tube in the resistant species averages $7.5\ \text{cm}$. Scientists monitored the area for the next five years to determine how this change affected the moth population.

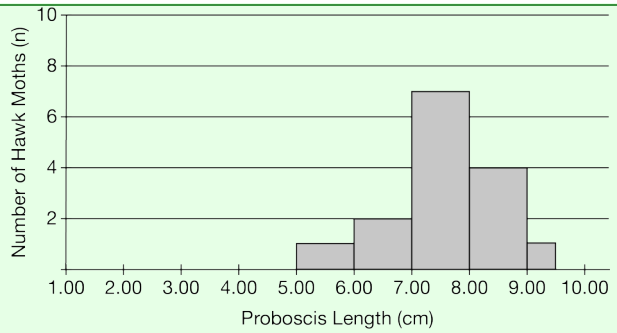
Which of the following best describes the expected distribution of proboscis size of the moths at the end of the five-year study?

7 Natural Selection MCQ

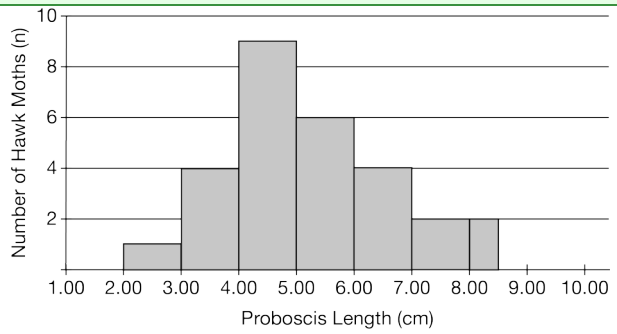
(A)



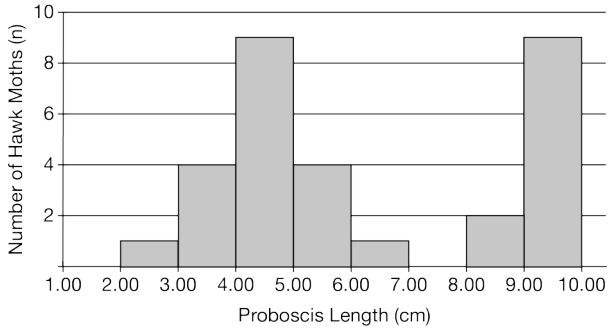
(B) ✓



(C)



(D)



Answer B

Correct. Since the moths will be relying more heavily on tobacco plants with necataries that are at the ends of longer tubes, moths with longer proboscises should have easier access to food, allowing them to survive and reproduce and pass on the genes that result in longer proboscises to their offspring.

7 Natural Selection MCQ

116. Students created a model showing one possible mechanism for speciation in which a geographic barrier separated one large population into two smaller populations, A and B. Over time, populations A and B developed into two distinct species. The model is shown in Figure 1.

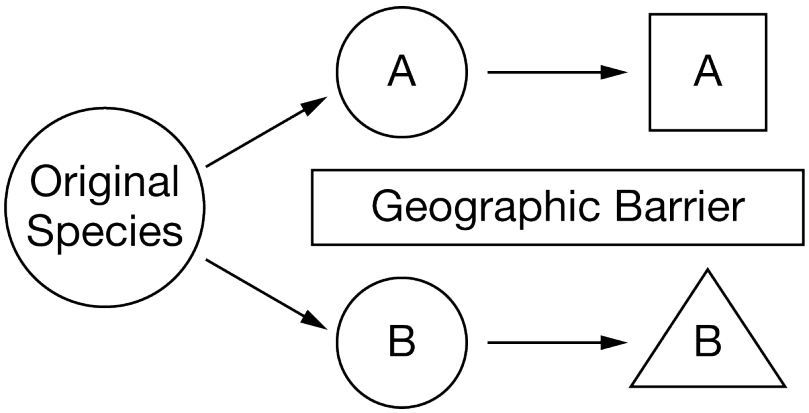


Figure 1. Model of speciation

Which of the following best describes what occurred during the periods indicated by the arrows?

(A) Natural selection acted on variation that was present in both populations A and B, and the lack of gene flow eventually led to the formation of two new species. ✓

(B) Population B was separated for a longer time from the original population than population A was, and population B evolved into a more distinct new species than population A did.

(C) Gene flow between populations A and B maintained sufficient variation in both gene pools to facilitate the development of two distinct species.

(D) In response to different environments, populations A and B experienced different mutation rates, leading to the emergence of two different gene pools and, over time, new species A and B.

Answer A

Correct. Allopatric speciation is a result of differential natural selection acting on two populations that have been separated geographically. Populations A and B became sufficiently different due to accumulated changes to their gene pools.

117. Which of the following principles is NOT part of Darwin's theory of evolution by natural selection?

7 Natural Selection MCQ

(A) Evolution is a gradual process that occurs over long periods of time.
(B) Variation occurs among individuals in a population.
(C) Mutations are the ultimate source of genetic variation. ✓
(D) More individuals are born than will survive.
(E) Individuals that possess the most favorable variations have the best chance of repro-ducing.

118. Which of the following statements best summarizes organic evolution as it is viewed by modern evolutionists?

(A) It is goal directed.
(B) It represents the results of selection for acquired characteristics.
(C) It is synonymous with the process of gene flow.
(D) It is the descent of humans from the present-day great apes.
(E) It is the differential survival and reproduction of certain phenotypes. ✓

Stickleback fish are found in both marine and freshwater habitats. The marine fish have no scales but have hardened, armorlike plates along their sides. The plates are thought to protect sticklebacks from certain predators.

In the late 1980s, sticklebacks from a marine population colonized Loberg Lake, a freshwater lake in Alaska. Starting in 1990, researchers sampled fish from the lake every four years and recorded the armor-plate phenotypes of the male sticklebacks in each sample. The armor-plate phenotypes were categorized as either complete (plates extending from head to tail), partial (plates extending from head to abdomen), or low (a few plates near the head only). The results are shown in the table below.

ARMOR-PLATE VARIATION IN THE STICKLEBACK POPULATION OF LOBERG LAKE

Percent of Males in the Sample with Each Armor-Plate Phenotype

Year	Low	Partial	Complete
1990	1%	2%	97%
1994	45%	14%	41%
1998	58%	16%	26%
2002	76%	15%	9%
2006	90%	6%	4%

119. Which of the following is an independent variable in the investigation?

7 Natural Selection MCQ

(A) Year in which the sample was collected ✓

(B) Salinity of the water in Loberg Lake

(C) Size of the stickleback population

(D) Percent of males with each armor-plate phenotype

120. To evaluate the reliability of the results, it would be best to know which of the following?

(A) The surface area of Loberg Lake

(B) The number of males in each sample ✓

(C) The average daily temperature of Loberg Lake

(D) The age of the sticklebacks in each sample

121. Which of the following best explains the changes in the phenotype frequencies of the stickleback population in Loberg Lake?

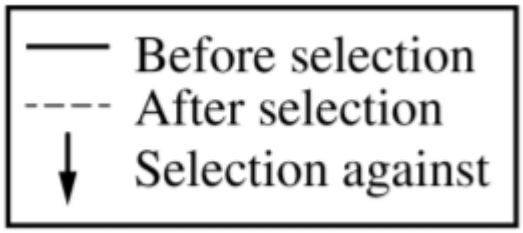
(A) Predation in the marine environment is different from predation in Loberg Lake. ✓

(B) Marine predators and sticklebacks both colonized Loberg Lake.

(C) Sticklebacks with the partial armor-plate phenotype have the highest rate of reproduction and survival.

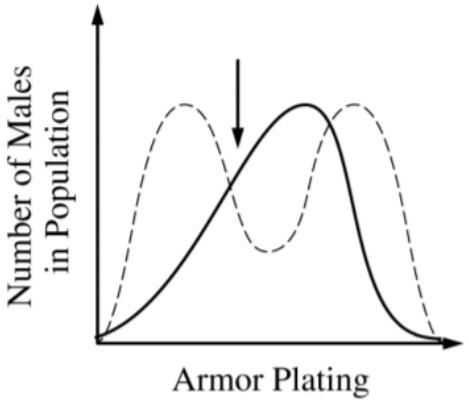
(D) The population of sticklebacks in Loberg Lake is subject to genetic drift.

122. Which of the following graphs best represents the type of selection most likely operating in the stickleback population of Loberg Lake?

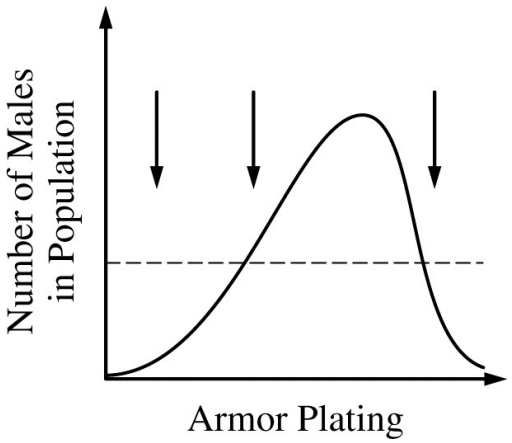


7 Natural Selection MCQ

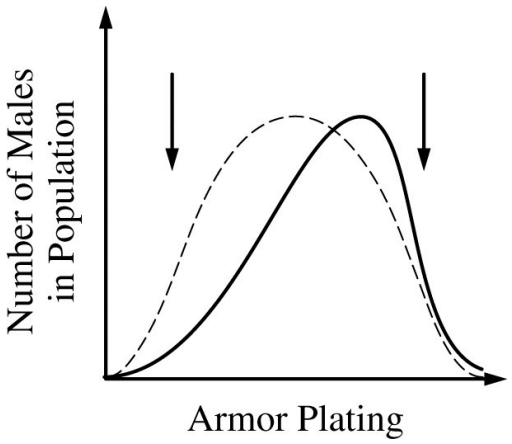
(A)



(B)

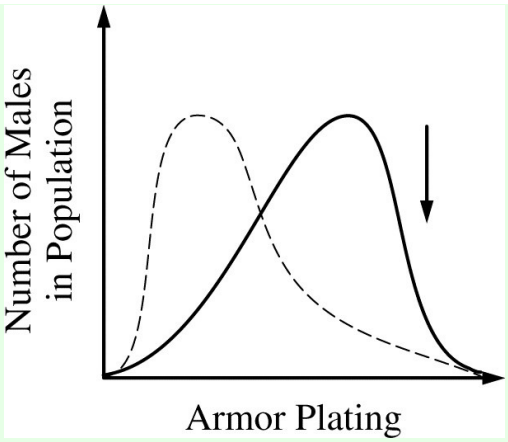


(C)



(D) ✓

7 Natural Selection MCQ



123. In a population of coyotes, there is variation in the size of individual animals. Larger animals are able to bring down larger prey but also require more food to survive. In a particular area, the primary food source for many years had been rabbits, but disease led to a sudden crash in the rabbit population. However, the deer population in the area increased at about the same time. Over time, naturalists noted that the average size of coyotes in the population had increased.

Which of the following is the best explanation for the increase in average coyote size following the crash in the rabbit population?

(A) Deer run faster than rabbits, so coyotes needed to get larger to survive.

(B) Smaller coyotes were at a hunting disadvantage, so fewer survived long enough to reproduce. ✓

(C) Since the coyotes needed to be larger to bring down the deer, they grew larger.

(D) Since smaller coyotes struggled to obtain enough food, they produced larger offspring to improve their fitness.

Answer B

Correct. The information provided indicates that larger coyotes can catch larger prey, so they would have a hunting advantage over smaller coyotes when forced to catch larger prey after the population size of the preferred smaller food source, rabbits, had decreased.

7 Natural Selection MCQ

124. In an experiment, 100 mice were released into a field to which no other mice had access. Immediately after their release, a representative sample of mice was captured, their fur color was recorded, and they were returned to the field. After twenty years, a representative sample of mice was captured and the distribution of fur color was again recorded.

Fur Color	Initial Distribution (%)	Distribution after Twenty Years (%)
Brown	80	20
Gray	10	75
Black	10	5

Which of the following could best explain the change in fur color distribution, as shown in the table above?

(A) The allele for black fur color is unstable, and over twenty years most of the black fur alleles mutated to become alleles for gray fur.

(B) The field was primarily composed of light-colored soil and little vegetation, affording gray mice protection from predators. ✓

(C) Sexual selection led to increased mating frequency of black and brown versus gray and brown.

(D) The gray mice were harder to catch, and so were underrepresented in the twenty-year sample.

125. A researcher claims that genetic variation provides organisms with the ability to survive and reproduce in different environments. To support the claim, the researcher makes the following observation: bacteria that contain plasmids (small DNA molecules) are resistant to a wider range of antibiotics than are bacteria that contain no plasmids.

Which of the following statements best establishes a connection between the observation and the researcher's claim?

(A) Some antibiotics inhibit the synthesis of bacterial cell walls.

(B) Some antibiotics inhibit protein synthesis in bacteria cells.

(C) Some plasmids cannot exist in a bacterial cell with certain other plasmids.

(D) Some plasmids contain antibiotic resistance genes. ✓

Answer D

Correct. Because some plasmids contain antibiotic resistance genes, bacteria that contain plasmids are more likely to be resistant to antibiotics than are bacteria that do not contain plasmids. The observation is evidence that genetic variation in the form of plasmids provides bacteria with the ability to survive and reproduce in different environments.

7 Natural Selection MCQ

126. A researcher claims that different metabolic pathways allow bacteria to use different molecules as sources of matter and energy.

Which of the following statements best helps justify the researcher's claim by providing a relevant example?

(A) Rhizobia bacteria form close associations with the roots of bean plants.

(B) *E. coli* bacteria reproduce in liquid media containing either glucose or galactose. ✓

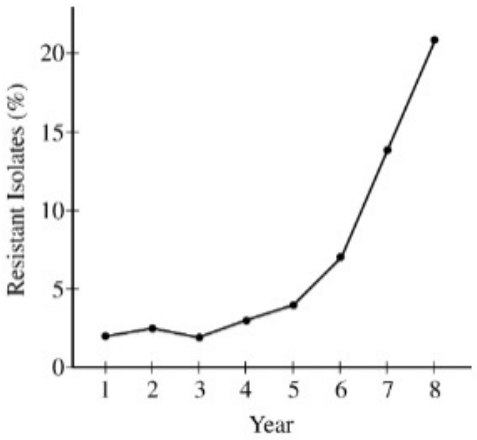
(C) The antibiotic rifampicin inhibits the growth of some bacterial strains but not of others.

(D) Some viruses that infect bacteria reproduce by either the lysogenic cycle or the lytic cycle.

Answer B

Correct. Glucose and galactose are two different monosaccharides that can be used by organisms as sources of matter and energy. The ability of *E. coli* bacteria to reproduce in liquid media containing either glucose or galactose provides a relevant example that will help justify the researcher's claim.

127.



Over several years, bacteria were isolated from members of a human population and tested for antibiotic resistance. The percent of bacterial isolates that were found to be antibiotic resistant is presented in the graph above for each year of the study. Which of the following conclusions is best supported by the information presented in the graph?

(A) The greatest increase in antibiotic resistance occurred from the first year to the third year of the study.

(B) Gene flow in the population occurred at the same rate for the entire duration of the eight-year study.

(C) Because evolution in bacterial populations occurs over millions of years, the results are due to random variation.

(D) Natural selection over multiple generations is favoring antibiotic resistance in the population of bacteria. ✓

7 Natural Selection MCQ

128. A current challenge for doctors involves the bacterial strain *Clostridioides difficile,* which no longer responds to traditional antibiotic treatments.

Which of the following best explains why this particular strain of bacteria is resistant to antibiotic treatment?

(A) Mutations creating new alleles occur more frequently in this species of bacteria than in other species.

(B) The bacteria have high genetic variability and high reproductive rates. ✓

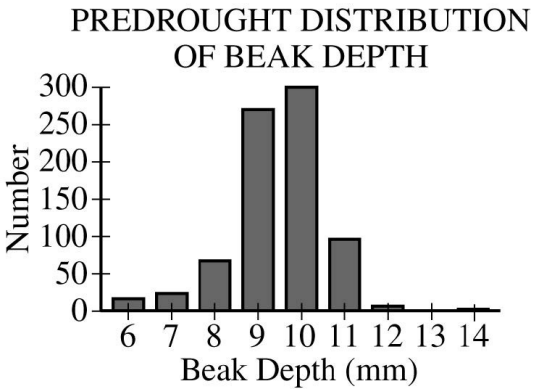
(C) Crossing over during meiosis increased genetic variability in the bacteria.

(D) The bacteria are able to recognize and destroy the antibiotics by breaking them down extracellularly.

Answer B

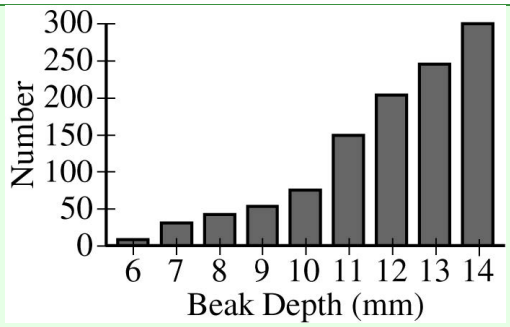
Correct. These processes promote rapid evolution of a species. Bacteria can produce several generations in a single day; each generation produced under the severe selective pressures of antibiotics is more likely to be antibiotic resistant.

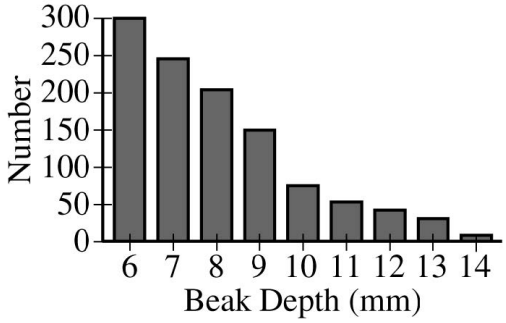
129.

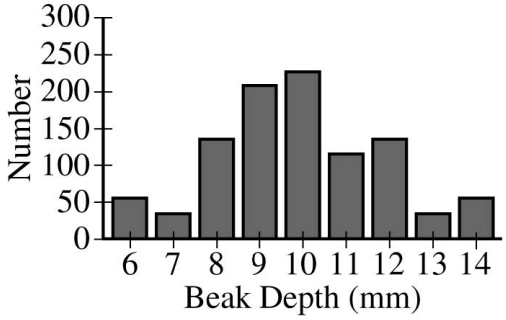


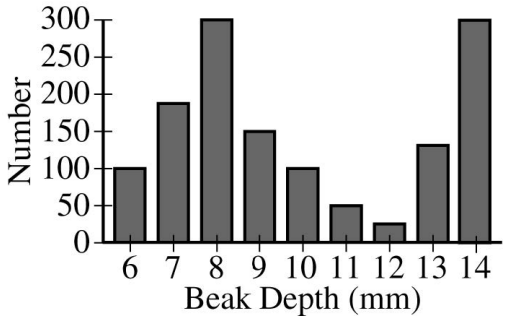
The graph above shows the distribution of beak depth in a finch population that had been living on an island under conditions of normal rainfall. During a subsequent drought, the small seeds normally eaten by the finches were less available. Most of the available seeds were large seeds that could be eaten most easily by finches with deep beaks. Which of the following graphs best predicts the distribution of beak depth in the finch population after several years of drought?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

130. The bird, *Parus major*, is commonly found in man-made habitats. During the summer these birds primarily feed on insects, and in the winter, when insects are scarce, they forage in gardens and bird feeders for seeds, nuts, and berries. Recently, climate change has led to a decrease in food available from gardens, and scientists have observed that the average beak length of the birds has increased.

Which of the following best explains the directional shift in beak length in these birds?

7 Natural Selection MCQ

(A) Longer beaks allow the birds to better fend off predators.

(B) Longer beaks increase the fitness of the birds in wild habitats.

(C) Longer beaks allow the birds to better access seeds in bird feeders. ✓

(D) Longer beaks decrease the fitness of the birds in wild habitats.

Answer C

Correct. Since there are fewer resources available in gardens, it is advantageous for the birds to be able to obtain food from bird feeders more effectively. The observed phenotypic variation is likely the result of birds with longer beaks being able to better extract seeds form bird feeders, and therefore be better able to survive and pass on the traits for longer beaks to their offspring.

131.

TABLE 1. PHOTOPIGMENTS IN CYANOBACTERIA

Pigment	Color of Reflected Light
Chlorophyll *a*	Green
Phycocyanin	Blue
Allophycocyanin	Blue
Phycoerythrin	Red

Cyanobacteria contain a variety of pigment molecules, as shown in Table 1. As a result, the color of cyanobacteria cultures can vary significantly based on the relative amount of each pigment produced.

A researcher placed a culture of cyanobacteria under green lights. Within a few weeks, the appearance of the cyanobacteria changed from green to red. The researcher claimed the color change in the culture was the result of an adaptation allowing greater photosynthesis.

Which of the following provides the best reasoning to justify the researcher's claim?

(A) In green light, more chlorophyll *a* molecules are produced, reflecting more light to other cyanobacteria to be used for photosynthesis.

(B) In green light, more phycoerythrin molecules are produced, allowing more green light to be absorbed, thus increasing photosynthesis. ✓

(C) In green light, cyanobacteria that have more phycocyanin molecules are less likely to survive and reproduce.

(D) In green light, cyanobacteria that have more allophycocyanin molecules are more likely to survive and reproduce.

7 Natural Selection MCQ

Answer B

Correct. Cyanobacteria exposed to green light would increase production of pigment molecules that can absorb green light and reflect red light, such as phycoerythrin. As a result, the color of the culture would change from green to red. In this way, the organisms are better able to survive under different conditions.

132. Ciprofloxacin is given as an antibiotic to healthy livestock to promote efficient weight gain. Strains of the bacterium *Campylobacter jejuni* naturally colonize the digestive tracts of livestock, and the *C. jejuni* can be transferred to people through the handling and consumption of contaminated meat. People infected with *C. jejuni* also are treated with ciprofloxacin.

Which of the following is the most likely consequence of adding ciprofloxacin to animal feed?

(A) Naturally occurring strains of *C. jejuni* will reproduce more rapidly in the digestive tracts of livestock, reducing the risk of food-borne illnesses in people.

(B) Ciprofloxacin-resistant strains of *C. jejuni* will have a selective advantage in the digestive tracts of livestock, increasing the risk of serious infections in people. ✓

(C) Ciprofloxacin-sensitive strains of *C. jejuni* will be consumed in meat that contains a chemical that inhibits bacterial growth, reducing the risk of food poisoning in people.

(D) Actively growing strains of *C. jejuni* will metabolize the nutrients in meat produced by livestock operations, increasing the risk of protein deficiencies in people.

133. Phycobiliproteins are a complex of accessory pigments and proteins found in cyanobacteria but not in green algae. A researcher claims that the phycobiliprotein pigments in cyanobacteria allow the cyanobacteria to survive in certain aquatic niches better than green algae can.

Which of the following statements best justifies the researcher's claim?

(A) The additional pigments allow the cyanobacteria to store light energy so that it can be used at night to continue photosynthesis.

(B) The additional pigments block light and prevent it from reaching photosynthetic organisms at greater depths, so no photosynthetic organisms can live below the surface waters containing cyanobacteria.

(C) The additional pigments require energy and cellular resources to produce, so they can be used as an energy source during times of insufficient light.

(D) The additional pigments absorb light at wavelengths that green algae cannot absorb; this may allow cyanobacteria to capture more light energy for photosynthesis than green algae can in certain areas. ✓

Answer D

Correct. Since the phycobiliprotein pigments absorb light wavelengths not absorbed by green algae, cyanobacteria are able to better perform photosynthesis and survive in areas where there is insufficient light at the needed frequencies for green algae to survive.

7 Natural Selection MCQ

Diapause is the interruption of an organism's life cycle in response to environmental cues. The soil nematode *Caenorhabditis elegans* is capable of entering adult reproductive diapause (ARD) when food is scarce. In *C. elegans*, individuals normally become reproductively mature 2 days after hatching and remain fertile for 18 days. They reproduce either by self-fertilization or by mating with another individual.

In an investigation, researchers examined the survival and reproductive success of *C. elegans* following different times in ARD. In the first experiment, groups of *C. elegans* were held in ARD without food for 0-30 days. Upon reintroduction of food, average brood sizes (average numbers of offspring per adult) were determined following either self-fertilization or mating with a well-fed male. The results are shown in Figure 1.

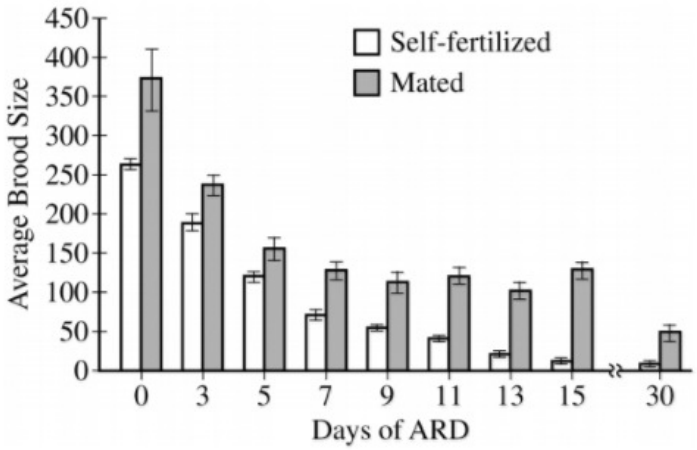


Figure 1. Mean brood sizes $\pm 2SE_{\overline{X}}$ after different times in ARD.

Individual *C. elegans* were held in ARD and subsequently

allowed to reproduce either via self-fertilization (unshaded bars)

or by mating with well-fed males (shaded bars).

In a second experiment, individuals were held in ARD without food for $0-30$ days and monitored for average survival times following reintroduction of food (Figure 2).

7 Natural Selection MCQ

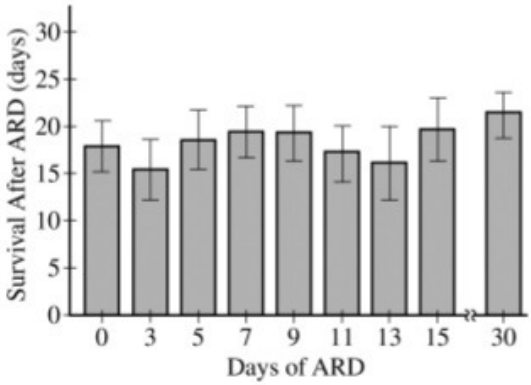


Figure 2. Mean survival $\pm 2SE_{\overline{x}}$ following different times in ARD. Individual *C. elegans* were held in ARD and subsequently given access to food, whereupon their survival times were determined.

134. Based on the experimental results, which of the following is the best evolutionary explanation for the occurrence of ARD in *C. elegans* ?

(A) The ability to enter ARD provides a strong selective advantage because reproduction can occur despite periods of food scarcity. ✓

(B) Acquiring the genes for ARD gives individuals a selective advantage because they produce more offspring than do individuals who cannot enter ARD.

(C) Individuals who can enter ARD are selected for in the population because they live longer than do individuals who cannot enter ARD.

(D) Individuals who can enter ARD have high fitness because they can reproduce even when food is scarce.

7 Natural Selection MCQ

Excess intracellular iron is toxic to cells (iron-induced toxicity). Ferritin is an intracellular iron storage protein that binds excess iron. The presence of ferritin can protect cells from iron-induced toxicity.

In an experiment to investigate the effects of dietary iron intake on ferritin synthesis, rats were given food containing different amounts of iron. Subsequently, the levels of ferritin protein in the liver were measured. The results are shown in Figure 1.

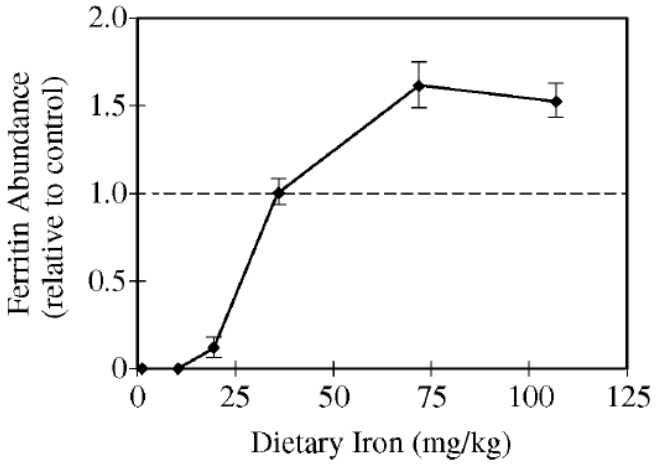
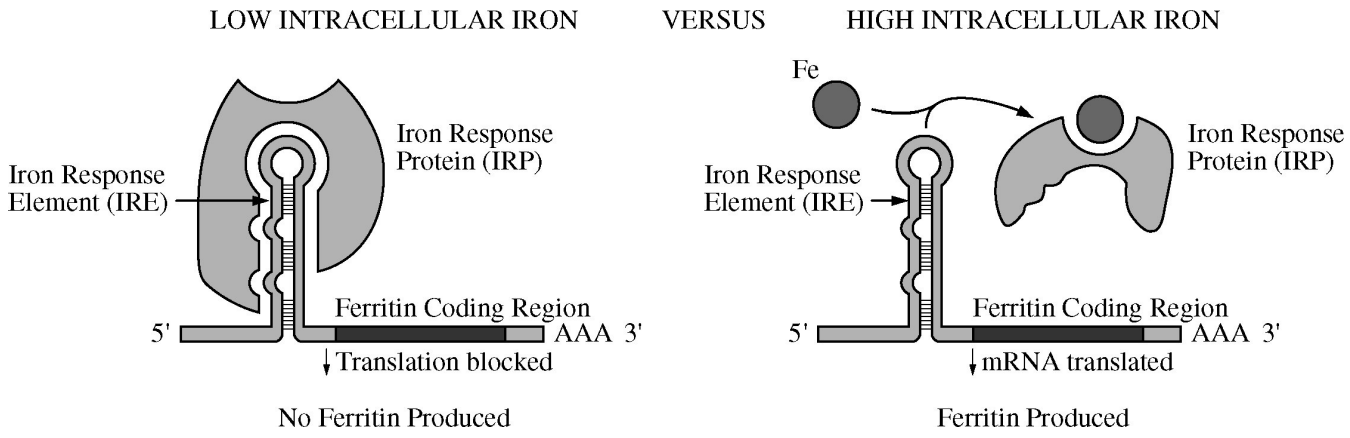


Figure 1. Effects of dietary iron on ferritin levels in rat liver

Based on these and other data, researchers have developed the following model demonstrating how ferritin synthesis is regulated by iron. When iron levels are low, a repressor of translation, iron response protein (IRP), binds to an iron response element (IRE), which is a stem-loop structure near the 5¢ end of ferritin mRNA. When iron levels are high, intracellular iron binds to the IRP, and the iron-IRP complex dissociates from the IRE, permitting ribosomes to proceed with the translation of ferritin mRNA. Figure 2 represents the model of the regulation of ferritin mRNA translation by iron.



7 Natural Selection MCQ

Figure 2. Model of regulation of ferritin synthesis by iron

135. Based on the data in Figure 1, the concentration of iron in the control diet used in the investigation is most likely within which of the following ranges?

(A) 20–25 mg/kg

(B) 35–40 mg/kg ✓

(C) 70–75 mg/kg

(D) 100–125 mg/kg

136. Which of the following conclusions about dietary iron and ferritin synthesis is best supported by the data in Figure 1?

(A) A dietary iron concentration of 25 mg/kg has no effect on ferritin gene expression compared with the control diet.

(B) At a dietary iron concentration of 50 mg/kg, rats make twice as much ferritin as rats fed the control diet do.

(C) Maximum activation of ferritin synthesis occurs at dietary iron concentrations of 75 mg/kg or greater. ✓

(D) The maximum rate of ferritin absorption from the gut occurs at dietary iron concentrations greater than 75 mg/kg.

137. After a search of nucleotide sequence databases, researchers identified an IRE in the 5¢ untranslated region of a gene encoding aconitase, an enzyme involved in the Krebs cycle. Which of the following pieces of experimental evidence best supports the claim that the synthesis of aconitase is controlled by a mechanism similar to ferritin regulation?

(A) IRP binds to aconitase mRNA in the presence of iron.

(B) The relative amount of aconitase protein increases in the presence of high levels of iron. ✓

(C) Oxygen consumption by cells increases in the presence of high levels of iron.

(D) The levels of reduced electron carriers, NADH and $FADH_2$, increase in the presence of high levels of iron.

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138. Freshwater bony fish secrete large volumes of very dilute urine. Many marine bony fish secrete small volumes of concentrated urine. Which of the following best explains the differences in nitrogenous-waste excretion between freshwater and marine bony fish?

7 Natural Selection MCQ

(A) The differences arose during divergence, as fish from a common ancestor faced different selective pressures in environments of different osmolarity. ✓

(B) The differences resulted from the evolution of kidneys in independent evolutionary lineages that were adapted to different environmental conditions.

(C) The differences are accidents of evolutionary divergence and are unrelated to the osmotic differences in freshwater and marine environments.

(D) The differences arise in each generation because of the physiological adjustment of individual fish to fluctuating salt concentrations in the environments in which they live.

139. While there is only one species of Galápagos Island tortoise, there are several subspecies. Larger islands with more wet highlands have lush vegetation near the ground. Tortoises there tend to have high-domed shells and shorter necks, which restrict upward head movement. They also have shorter limbs. They are the heaviest and largest of the subspecies.

Smaller, drier islands are inhabited by tortoises with longer necks and limbs and with shells that are elevated above the neck, which allow them to browse taller vegetation.

Based on the information given, which of the following is a plausible explanation for the ancestry of the tortoise subspecies?

(A) The subspecies share a recent common ancestor whose neck length, shell shape, and leg length were intermediate between the two subspecies.

(B) The tortoises with shorter legs and necks were most easily preyed on as young animals by the rats that were introduced, so they survived only on a few islands.

(C) Random mutations coupled with the inheritance of acquired characteristics resulted in distinct subspecies.

(D) Individuals with different adaptations in shell shape and leg length best exploited the food resources and left more surviving offspring on each island. ✓

7 Natural Selection MCQ

Individuals of a particular species of ground beetle are either light tan or dark brown. Light-tan beetles are predominant in habitats with light-colored sandy soils, and dark-brown beetles are predominant in habitats with dark-colored loam soils. In an experiment designed to determine the survival rates of light-tan beetles and dark-brown beetles in different habitats, 500 light-tan beetles and 500 dark-brown beetles were released in each of four habitats. Each beetle had been marked with a small spot of red paint on the underside of its abdomen before it was released. One week after the beetles had been released, any marked beetles that could be found were recaptured. The results are presented in the table below. It is assumed that differences in the numbers of beetles recaptured are directly related to differences in survival rates.

	Habitat 1 Sandy soil, no insectivorous birds present		Habitat 2 Sandy soil, insectivorous birds present		Habitat 3 Loam soil, no insectivorous birds present		Habitat 4 Loam soil, insectivorous birds present	
Color of Beetle	Light tan	Dark brown	Light tan	Dark brown	Light tan	Dark brown	Light tan	Dark brown
Number Released	500	500	500	500	500	500	500	500
Number Recaptured	130	114	123	22	65	74	13	87

140. Which of the following processes best explains why fewer light-tan beetles than dark-brown beetles were recaptured in habitat 4?

(A) Immigration

(B) Genetic drift

(C) Mutation

(D) Adaptive radiation

(E) Predation ✓

141. If all insectivorous birds and remaining beetles were removed from habitat 2 and 500 additional dark-brown beetles and 500 additional light-tan beetles were then released into habitat 2, which of the following is the best estimate of the number of additional dark-brown beetles that would be expected to be recaptured in habitat 2 after one more week?

(A) 0

(B) 22

(C) 120 ✓

(D) 220

(E) 500

142. Which of the following can be inferred from the data in the table?

7 Natural Selection MCQ

(A) Ground beetles do not emigrate from the habitat in which they live.

(B) Insectivorous birds prefer to eat light-tan beetles rather than dark-brown beetles.

(C) Ground beetles have higher rates of survival in habitats with loam soil.

(D) Insectivorous birds are predators of this species of ground beetle. ✓

(E) The reproductive success of beetles in habitats with sandy soils is greater than that of beetles in habitats with loam soils.

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143. High blood cholesterol (hypercholesterolemia, HC) can lead to cardiovascular problems such as atherosclerosis and heart attack. Exercise and monitoring of diet can often control cholesterol levels; however, in certain cases HC is inherited as an autosomal dominant disease caused by a mutation in a single gene.

A man with high cholesterol levels is about to marry a woman whose total cholesterol levels are also higher than average. A physician has suggested they get tested for the *HC allele*. Which of the following is a valid ethical question concerning the test?

(A) How important are the results of the test to the couple's health, since they already know they have HC?

(B) How can they eliminate the allele for HC from their cells?

(C) If both have the allele for HC, should an insurance company raise their rates because of the results of the test? ✓

(D) What cures are available for HC other than dietary changes and exercise?

144. Evolutionary biologists have observed variation in the average age and size of mature individuals in a population of small freshwater guppies. Guppies found in pools with pike cichlids— a predator fish that preys primarily on larger guppies—reproduce at a young age and are small at maturity. Guppies found in pools with killifish —a different predator fish that preys primarily on smaller guppies—reproduce at an older age and are larger at maturity.

Which of the following predicts the most likely outcome of moving a population of guppies from a pool with pike cichlids to a pool with killifish?

(A) The introduced guppy population will remain phenotypically similar to the original population because the killifish will not recognize them as prey.

(B) Because the introduced population was in Hardy-Weinberg equilibrium, successive generations in the new population will continue to display the same allele frequencies as the original population.

(C) After many generations, the offspring of the introduced guppies will mature at an older age and larger size because of selective pressure from the new predator. ✓

(D) Within two or three generations, the introduced guppies will become extinct because the killifish prefer the juveniles.

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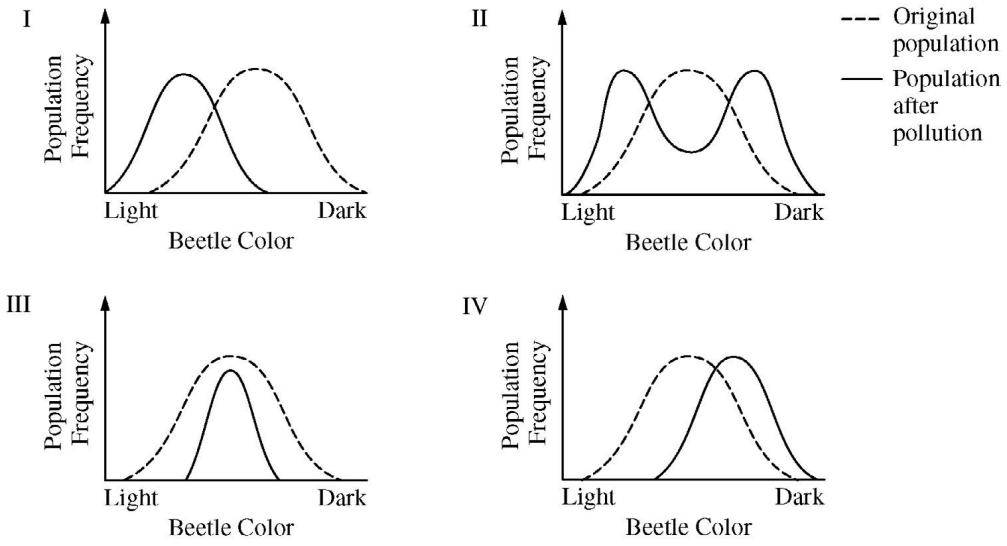
145. In a hypothetical population of beetles, there is a wide variety of color, matching the range of coloration of the tree trunks on which the beetles hide from predators. The graphs below illustrate four possible changes to the beetle population as a result of a change in the environment due to pollution that darkened the tree trunks.



Which of the following includes the most likely change in the coloration of the beetle population after pollution and a correct rationale for the change?

(A) The coloration range shifted toward more light-colored beetles, as in diagram I. The pollution helped the predators find the darkened tree trunks.

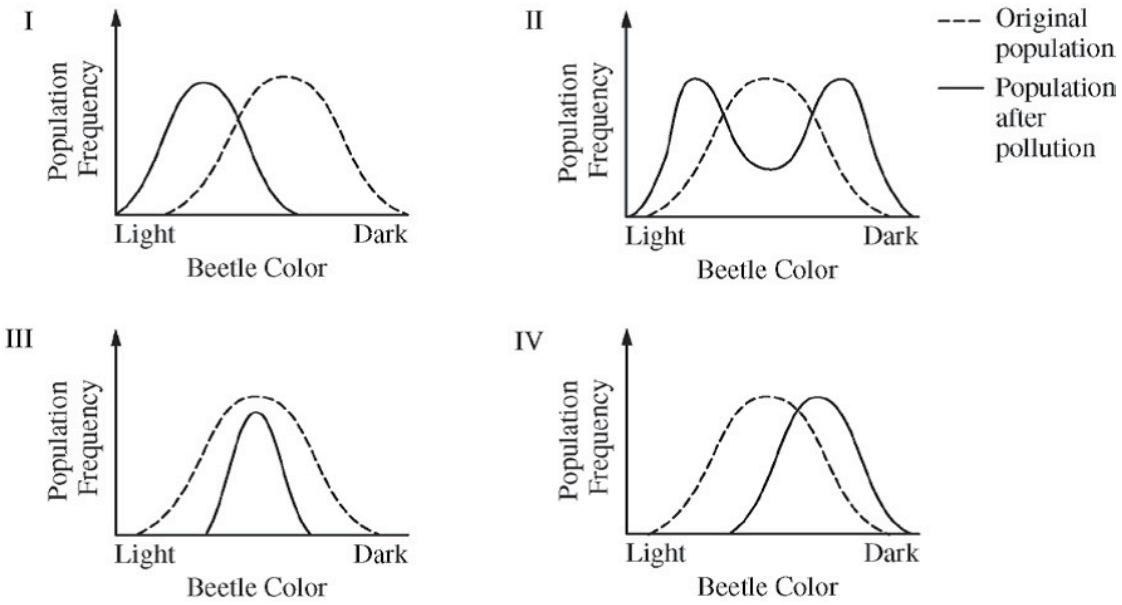
(B) The coloration in the population split into two extremes, as in diagram II. Both the lighter-colored and the darker-colored beetles were able to hide on the darker tree trunks.

(C) The coloration range became narrower, as in diagram III. The predators selected beetles at the color extremes.

(D) The coloration in the population shifted toward more darker-colored beetles, as in diagram IV. The lighter-colored beetles were found more easily by the predators than were the darker-colored beetles. ✓

7 Natural Selection MCQ

146. In a hypothetical population of beetles, there is a wide variety of color, matching the range of coloration of the tree trunks on which the beetles hide from predators. The graphs below illustrate four possible changes to the beetle population as a result of a change in the environment due to pollution that darkened the tree trunks.



Which of the following includes the most likely change in the coloration of the beetle population after pollution and a correct rationale for the change?

(A) The coloration range shifted toward more light-colored beetles, as in diagram I. The pollution helped the predators find the darkened tree trunks.

(B) The coloration in the population split into two extremes, as in diagram II. Both the lighter-colored and the darker-colored beetles were able to hide on the darker tree trunks.

(C) The coloration range became narrower, as in diagram III. The predators selected beetles at the color extremes.

(D) The coloration in the population shifted toward more darker-colored beetles, as in diagram IV. The lighter colored beetles were found more easily by the predators than were the darker-colored beetles. ✓

147. If an individual with diabetes consumes food that is high in simple carbohydrates, blood-sugar levels can rise above normal levels. Which of the following questions would provide the best direction for a researcher who wanted to study the impact of abnormally high blood-sugar levels on cellular homeostasis in diabetics?

(A) Are cells in diabetics larger in size than those in nondiabetics?

(B) Do the cells in diabetics have more potassium ion channels in the cell membrane than the cells in nondiabetics do?

(C) Does water move from cells into blood vessels more rapidly in diabetics than in nondiabetics when blood-sugar levels are higher than normal? ✓

(D) Do the cells of diabetics use simple sugars as an energy source?

7 Natural Selection MCQ

Iron is an essential nutrient that is acquired by organisms from the environment. When intracellular levels of iron are relatively high, living cells synthesize an iron-storage protein called ferritin.

The induction of ferritin synthesis by iron was investigated in rats. Figure 1 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and actinomycin D, a drug that inhibits transcription.

Figure 2 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and cycloheximide, a drug that inhibits translation.

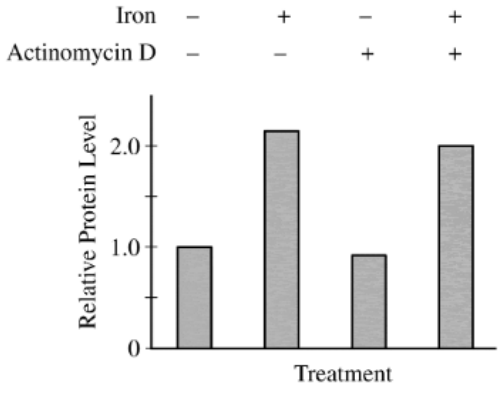


Figure 1. Ferritin Protein Levels

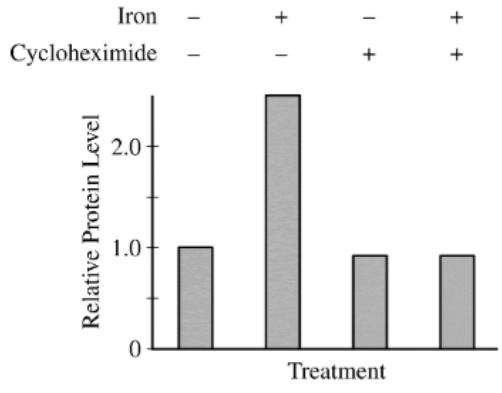


Figure 2. Ferritin Protein Levels

148. Hereditary hemochromatosis (HHC) is an iron storage disease that results in reduced uptake of iron into cells. Based on the data, which of the following is the most likely effect of HHC on ferritin protein levels?

(A) Elevated levels of ferritin protein because there is more iron in the blood of affected individuals

(B) Reduced levels of ferritin protein because there is less iron in the cytosol of affected individuals ✓

(C) Elevated levels of ferritin protein because ferritin replaces the mutant protein in affected individuals

(D) Reduced levels of ferritin protein because iron stimulates ferritin breakdown

149. The gene sequences responsible for the iron mediated changes in ferritin protein levels are highly conserved and are called iron response elements (IREs). IREs have been observed in a number of genes involved in iron transport and metabolism. The IRE sequences found in the ferritin gene are found in all multicellular organisms, whereas other nonferritin IRE sequences are found only in certain vertebrate organisms. Which of the following is the best explanation for the observations?

7 Natural Selection MCQ

(A) Ferritin IREs arose in the common ancestor of multicellular organisms, whereas nonferritin IREs arose later in specific evolutionary lineages. ✓

(B) Nonferritin IREs arose at different times in the common ancestor of multicellular organisms.

(C) Ferritin IREs arose in diverse organisms by convergent evolution, while the nonferritin IREs arose by recombination events between closely related species.

(D) Ferritin IREs have evolved independently in different evolutionary lineages, whereas nonferritin IREs arose in the common ancestor of multicellular organisms.

Mice live in territorial social groups, and each group has a single dominant male with the rest of the males in the group being subordinate. Male mice mark the area in which they live with their scent by leaving urine wherever they walk. There are two classes of proteins that are found in mouse urine, major urinary proteins (MUPs) and major histocompatibility complex (MHC) proteins. The proteins provide a unique scent for each mouse.

In an experiment, wild male mice were kept isolated in separate cages and their urination patterns were recorded. They were then combined into a single cage that was divided into two compartments by a mesh screen, and again their urination patterns were recorded (Figure 1).

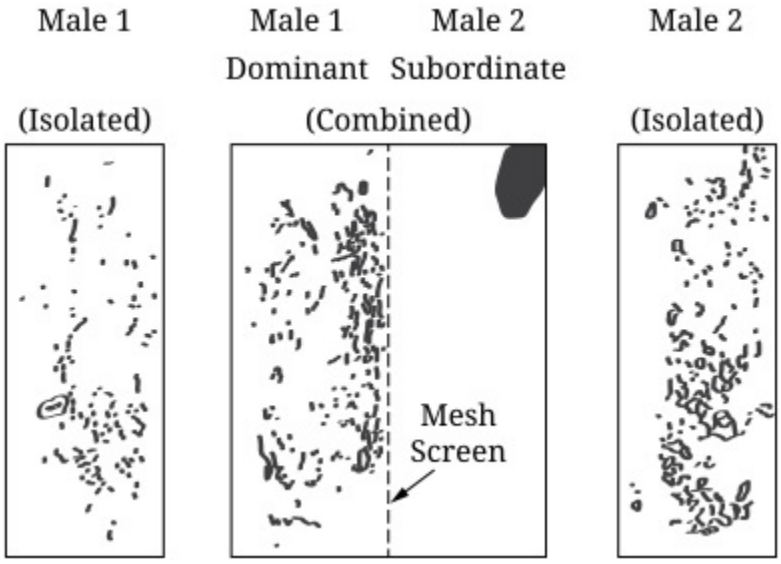


Figure 1. Urination patterns recorded during an experiment

150. A researcher claims that the life spans of dominant males are longer than the life spans of subordinate males. Based on the results of the experiment, which of the following best supports the researcher's claim?

(A) Dominant males have proteins in their urine that protect them from harmful genetic mutations.

(B) Dominant males have proteins in their urine that enable them to communicate with subordinate males in their group.

(C) Dominant males are able to move more freely throughout their territory, which gives them greater access to resources. ✓

(D) Dominant males are able to move more freely throughout their territory, which reduces the number of pathogens they are exposed to.

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Answer C

Correct. The scent patterns produced in the combined experiment indicate that the dominant male occupies his entire enclosure, whereas the subordinate male occupies only one small area of his enclosure. This would limit a subordinate male's access to resources and could limit his life span.

151.

TABLE 1. POLLINATOR VISITATION RATES TO TWO SPECIES OF *MIMULUS*

***Mimulus* Species and Flower Color**	**Bumblebees** (10^{-3} visits / flower / hour)	**Hummingbirds** (10^{-3} visits / flower / hour)
M. lewisii (wild-type, pink)	15.4	0.021
M. lewisii (mutant, orange)	2.63	1.44
M. cardinalis (wild-type, red)	0.148	189
M. cardinalis (mutant, pink)	10.9	168

Plant species in the genus *Mimulus* have evolved specialized flower structures and colors that attract pollinators. Researchers identified mutant forms of two of the species, *Mimulus lewisii* and *Mimulus cardinalis*, that have different flower colors from the wild-type forms. They measured the number of visits by two pollinators, bumblebees and hummingbirds, to the flowers of the wild-type and mutant species (Table 1).

Which of the following statements best describes the data shown in Table 1 ?

(A) Bumblebees visited wild-type *M. lewisii* approximately 13 times more frequently than they did mutant *M. lewisii*.

(B) Hummingbirds preferred wild-type *M. lewisii* over wild-type *M. cardinalis*.

(C) Hummingbirds preferred the pink flower type in both *M. lewisii* and *M. cardinalis*.

(D) Bumblebees visited mutant *M. cardinalis* approximately 70 times more frequently than they did wild-type *M. cardinalis*. ✓

7 Natural Selection MCQ

Answer D

Correct. Dividing 10.9 (the number of bumblebee visits to mutant *M. cardinalis*) by 0.148 (the number of bumblebee visits to wild-type *M. cardinalis*) produces a value of 73.6, which is close to 70.

152.

PERCENT OF MOTHS RECAPTURED IN TWO DIFFERENT ENVIRONMENTS		
Trial	Moth Color and Environment	Percent of Released Moths Recaptured
I	Light-colored moths were released in an unpolluted environment.	13%
II	Light-colored moths were released in a polluted environment.	12%
III	Dark-colored moths were released in an unpolluted environment.	7%
IV	Dark-colored moths were released in a polluted environment.	28%

A researcher released large numbers of moths into different environments in an attempt to better understand a mechanism of evolution. The moths were released in four trials as described in the table above. Each of the released moths had a small mark on the underside of a wing for identification. After an appropriate amount of time, the researcher recaptured as many of the released moths as possible. Data from the experiment are included in the table above. Which of the following claims is best supported by the data?

(A) Light-colored moths were more likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success.

(B) Dark-colored moths were more likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success. ✓

(C) Light-colored moths were less likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success.

(D) Dark-colored moths were less likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success.

153. In addition to the pigments commonly associated with photosynthesis, a certain photosynthetic species contains two additional pigment types.

Which of the following best supports the claim that this species is better adapted to environmental changes than other photosynthetic species are?

7 Natural Selection MCQ

(A) The increased pigment concentration better facilitates energy production within the cells of the species.

(B) The pigment combination allows the organism to absorb heat as well as light, making better use of available energy.

(C) The additional pigments allow the species to outcompete other species for the wavelengths of light commonly used in photosynthesis.

(D) The additional pigments allow the species containing them to harvest energy from wavelengths of light that the other photosynthetic species cannot use. ✓

Answer D

Correct. The ability of the pigment combination to absorb a wider range of sunlight allows the organism to use photosynthesis to produce energy in order to survive in environments with different ranges of available wavelengths of sunlight.

154. Which of the following factors is the most effective in changing the allele frequency in a natural population?

(A) Large population size

(B) Low rate of mutation

(C) Negligible migration

(D) Random mating

(E) Selection ✓

155. Some plant species flower in response to increasing daily temperatures in the spring. Many of these species rely on pollinators that migrate based on changes in day length and the position of the Sun. The current global warming trend is placing new selective pressures on the species involved in these relationships.

Which of the following best explains the impact of these new selective pressures on the organisms involved?

(A) If the environment for the plant species becomes too warm, the pollinators will no longer migrate to that area in the spring, continuing on to a more northern environment instead.

(B) The warmer temperatures will lead to a drier environment, so the plants will no longer produce enough nectar to attract the pollinators.

(C) The plant species will flower earlier in the spring in response to rising temperatures before the arrival of the pollinators, so seeds will not be produced. ✓

(D) Migrating pollinators will start migrating later in the year, switching from spring-flowering plants to summer-flowering plants.

Answer C

Correct. The migrating pollinators will still arrive at the same time, as their environmental cues to

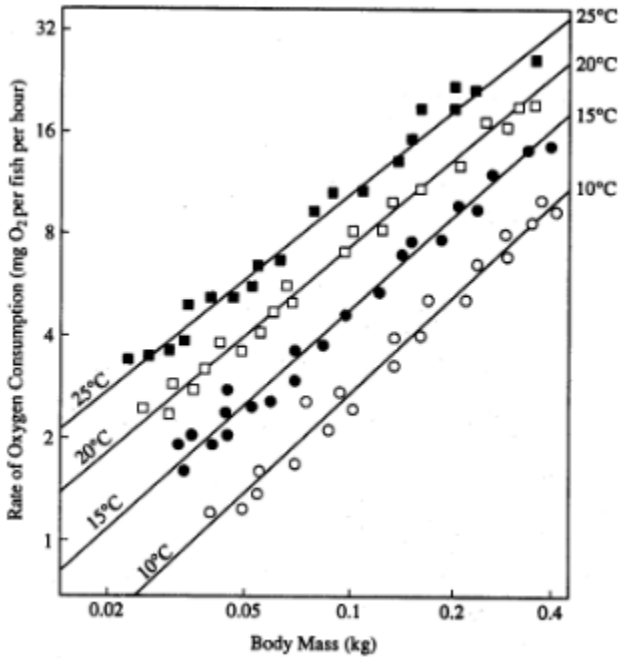
7 Natural Selection MCQ

migrate are not affected by warming. However, the plants will flower earlier, as the proper temperature cues they use will occur earlier in the spring. If the plants contain sufficient variability, they may be able to attract different pollinators.

- Directions: Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

The following questions refer to the following information and graph.

The data presented in the figure below are measurements of the rate of oxygen consumption at differing body masses in a species of fish. Each point represents measurements from a different fish. Measurements were taken at different temperatures. (○= 10°C,●= 15°C, □= 20°C, ■= 25°C.)



156. The fact that each line on the graph rises from left to right means that

7 Natural Selection MCQ

(A) higher temperatures produce higher rates of metabolism
(B) there were more large fish in the samples taken at high temperatures
(C) larger fish consume more oxygen than smaller fish at all four temperatures ✓
(D) when measurements are taken for larger fish late in the day, observed values are higher
(E) larger fish prefer to live at higher temperatures than do smaller fish

157. The best explanation for the fact that not all points lie on any given line is that

(A) the thermometer was incorrectly calibrated
(B) the scale used to weigh the fish registered 0.001 kg too little
(C) the fish grew during the course of the experiment
(D) errors were made in plotting the data
(E) organisms within populations show variability ✓

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158. The peppered moth, *Biston betularia* was generally white in color with black speckles. Over time a more solid black moth became the more common variety. Which of the following best explains the relationship between wing pigmentation and the coloration of a moth's surroundings?

(A) The frequencies of the different phenotypes in the moth population changed randomly due to genetic drift.
(B) If the surroundings of the peppered moth became darker, those backgrounds conferred a selective advantage to a black variety, as described by natural selection. ✓
(C) There was a gene flow of the black variety, spreading it throughout the moth's normal range.
(D) The black variety was a nineteenth century mutation, dominant to the peppered variety. The allele for the peppered trait, being recessive, decreased in frequency over time

Answer B

Correct. Natural selection occurred as surfaces became darker from burning coal, giving an advantage to the black variation, as they were able to be better camouflaged. The switch to cleaner fuel sources led to a return to lighter surfaces, restoring the adaptive advantage of the peppered variation.

7 Natural Selection MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

The three-spined stickleback (*Gasterosteus aculeatus*) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mainly of individuals with armor-like plates covering most of their body surface (completely plated). Approximately 10,000 years ago, some marine sticklebacks colonized freshwater environments. After many generations in the freshwater environments, the freshwater stickleback populations lacked the armor plating (low plated) typical of marine stickleback populations.

Over the period between 1957 and 2005, one freshwater population, in Lake Washington, a lake in a coastal region of the northwestern United States, changed from having a majority of individuals of the low-plated phenotype to having more individuals of the completely-plated phenotype than of the low-plated phenotype. Figure 1 shows the distribution of plated phenotypes in Lake Washington sticklebacks at four time points between 1957 and 2005.

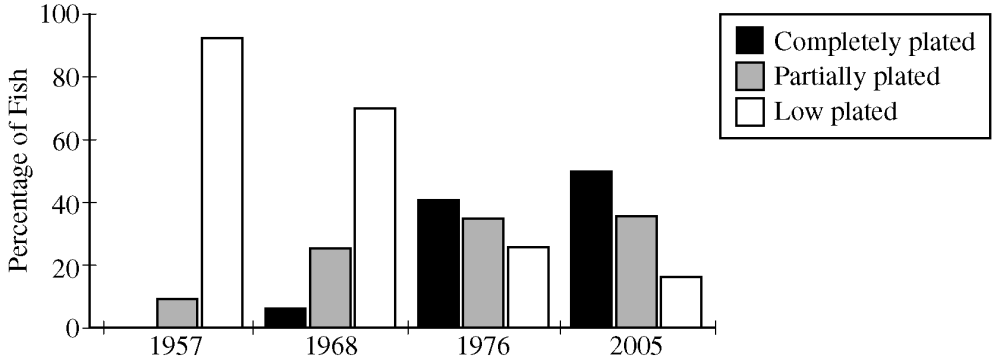


Figure 1. Armor plating phenotypes in Lake Washington stickleback population

A single gene, *ectodysplasin* (*EDA*), is thought to be responsible for the variation in the number of armor plates in sticklebacks. Figure 2 shows a phylogenetic tree constructed by comparing DNA sequences of the *EDA* gene from a number of stickleback populations with low-plated or completely plated phenotypes. Figure 3 shows a phylogenetic tree constructed by comparing the sequences of 25 genes that were randomly selected from the same populations as shown in Figure 2. In both figures, shaded populations display the completely plated phenotype.

7 Natural Selection MCQ

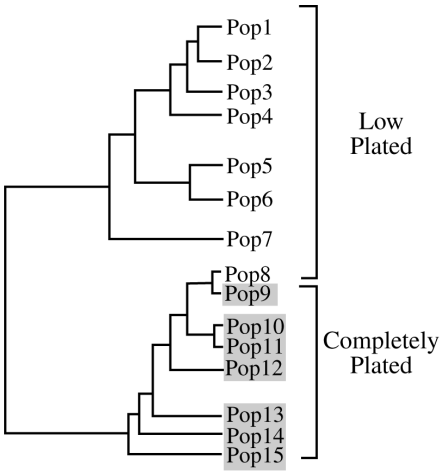


Figure 2. Phylogeny based on *EDA* gene only

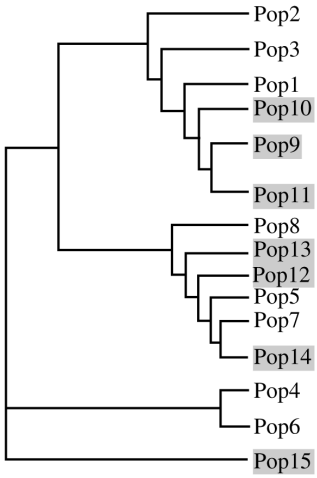


Figure 3. Phylogeny based on 25 random genes

159. Which of the following best explains the differences in the armor of the Lake Washington stickleback population summarized in Figure 1 ?

(A) Analysis of somatic cells using chromosomal staining and light microscopy indicates that stickleback fish have a diploid number of 42.

(B) Stickleback males from natural freshwater populations are typically more aggressive when competing for mates than are stickleback males from laboratory-bred populations.

(C) Fish exhibiting the low-plated phenotype were selected against in the Lake Washington stickleback population over the last 50 years. ✓

(D) Migration of individuals from other freshwater environments to Lake Washington led to gene flow between populations that were once geographically isolated.

160. Evolution of a new trait typically takes many generations. Yet a dramatic shift in the extent of armor plating in the Lake Washington stickleback population occurred in the 50 years following the cleanup of the lake. Which of the following best describes the mechanism of the rapid evolution of the armor phenotype in the Lake Washington sticklebacks?

(A) Pollutants in Lake Washington forced the sticklebacks to increase their mutation rate.

(B) The rapid change in the Lake Washington ecosystem required individual sticklebacks to evolve complete armor quickly.

(C) The increase in visibility in Lake Washington allowed sticklebacks to visually select mates with low armor.

(D) New selective pressures favored individuals with the plated phenotype, causing the plated allele frequency in the population to quickly increase. ✓

161. The phylogenetic trees in Figures 2 and 3 depict two different phylogenies of the same populations of sticklebacks. Which of the following questions will best help determine which tree represents the most accurate phylogeny?

7 Natural Selection MCQ

(A) Is the *EDA* gene as representative of the differences between the populations as the 25 random genes that were examined for Figure 3? ✓

(B) Are the low-plated populations found only in freshwater and the high-plated populations found only in saltwater environments?

(C) Is the common ancestor of the organisms represented in Figure 2 different from the common ancestor of the organisms represented in Figure 3?

(D) Is the expression level of the *EDA* gene analyzed in Figure 2 significantly greater than the expression levels of the 25 genes analyzed in Figure 3?

162.

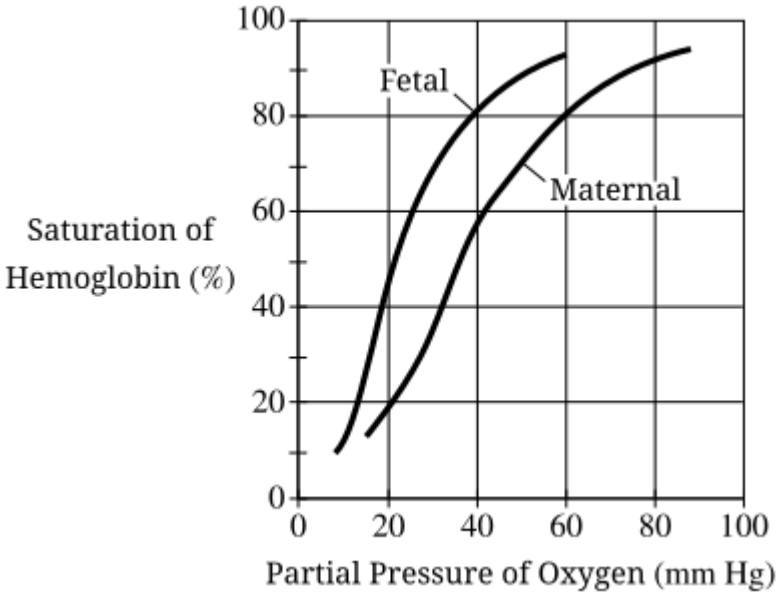


Figure 1. Percent saturation of human hemoglobins as a function of the partial pressure of oxygen

The graph in Figure 1 shows the percent saturation of fetal hemoglobin and maternal hemoglobin in humans as a function of the partial pressure of oxygen. Based on the data, which of the following values is closest to the predicted difference in the percent saturation of fetal hemoglobin and maternal hemoglobin at $40\ \text{mm Hg}$?

(A) 10

(B) 12

(C) 20 ✓

(D) 40

Answer C

Correct. Based on the data in Figure 1, at $40\ \text{mm Hg}$, the percent saturation of fetal hemoglobin is

approximately 81% and the percent saturation of maternal hemoglobin is approximately 58%. The difference in saturation is 23%, which is closest to the choice of 20%.

163. In a region of Africa where malaria is common, 9% of the population is born with a severe form of sickle-cell anemia (ss). Carriers of the sickle-cell allele (heterozygotes) are resistant to malaria and display only mild symptoms of malaria if exposed to the malaria parasite. If an effective antimalarial program is instituted in the region, which of the following is most likely to occur over many generations?

(A) The frequency of the sickle-cell allele will increase, as there will be more heterozygous individuals.

(B) The incidence of malaria will decrease, but the incidence of sickle-cell anemia will increase.

(C) The frequency of the sickle-cell allele s will decrease as the incidence of malaria decreases. ✓

(D) The frequency of the sickle-cell allele will increase because antimalarial treatments cause mutations.

Answer C

Correct. If malaria is eradicated, there is no longer a benefit to individuals to be heterozygous, particularly because these individuals may still display some of the symptoms of sickle-cell anemia. Over many generations, the frequency of the harmful s allele will be eliminated through natural selection.

Snowshoe hares are small mammals that live in forests throughout Canada and the northern United States. Hares feed primarily on vegetation and are preyed on by lynx and owls.

In the fall, the hares shed their brown coat (molt), and a new white coat grows in over a period of 40 days. In the spring, the hares molt again and their white coat is replaced with a brown coat over a period of approximately 30 to 50 days.

In a recent study, researchers observed that the majority of hares in an isolated population began the molting process during the same week in the fall, even though the ground was not snow covered. The majority of hares also began the molting process during the same week in the spring, even though the ground was still snow covered. The researchers formulated a hypothesis that the timing of the molting process is influenced by seasonal changes in the daylight period.

164. Several climate models for the original study area predict that the length of time the ground is covered in snow will decrease over time while the daylight periods will remain unchanged.

Assuming that the researchers' hypothesis about the timing of molting and the climate models are correct, which of the following will be the most likely immediate effect of the decrease in time the ground is snow covered on the hare population?

(A) The hare population will stop molting and maintain a single coat all year.

(B) The hare population will migrate to an area with an even shorter time during which the ground is snow covered.

(C) Hares will experience longer times with coat colors that do not blend with the ground color. ✓

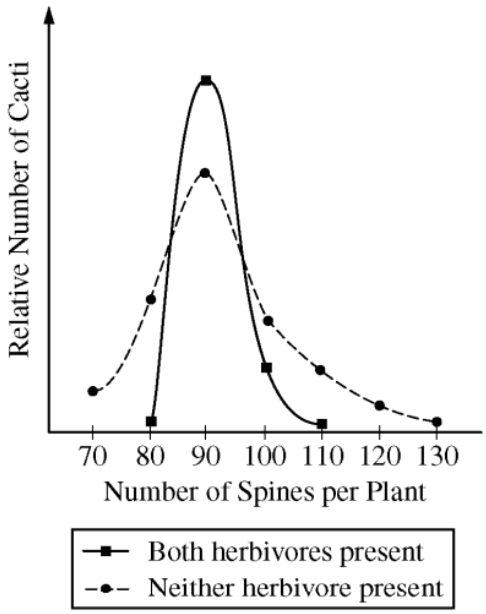
(D) The process of molting from a white to a brown coat in hares will take longer.

7 Natural Selection MCQ

Answer C

Correct. Hares will retain white coats even though the ground will not be covered by snow. This is because molting is determined by daylight length and the hares' genetic makeup rather than by the presence or absence of snow.

165.



In a species of cactus, the number of spines on a plant is genetically determined. The graph above shows frequency distributions for populations of the cactus species growing in the presence or absence of two herbivores: peccaries (a New World pig) and wasp larvae. Which of the following best accounts for the different frequency distributions in the graph?

(A) Peccaries eat cacti with the greatest number of spines, and wasp larvae show no preference with respect to spine number.

(B) Peccaries eat cacti with a smaller number of spines, and wasp larvae eat cacti with a greater number of spines. ✓

(C) Wasps have increasing difficulty depositing eggs on cacti with more spines, and peccaries show no preference with respect to spine number.

(D) Both peccaries and wasp larvae will eat the most abundant types of cacti, which are the plants with the greatest number of spines.

7 Natural Selection MCQ

166. Climate change is disrupting the length of winter in various ecosystems, resulting in less snowfall and milder winters. Scientists in Finland have been observing the tawny owl, a fairly large owl species that is commonly found in woodlands across much of Europe and Asia. Typically, the feathers of the upper body are either brown (BB) or gray (bb). Scientists studying tawny owls in Finland have found that the percentage of tawny owls that have brown feathers has increased over the past 28 years.

Which of the following best explains why the brown-feathered phenotype in tawny owls has increased over the past 28 years?

(A) Since the brown genotype is dominant, it will eventually eliminate the recessive gray genotype from the population.

(B) Darker feather color in tawny owls makes them less visible to predators as they sit in the trees.

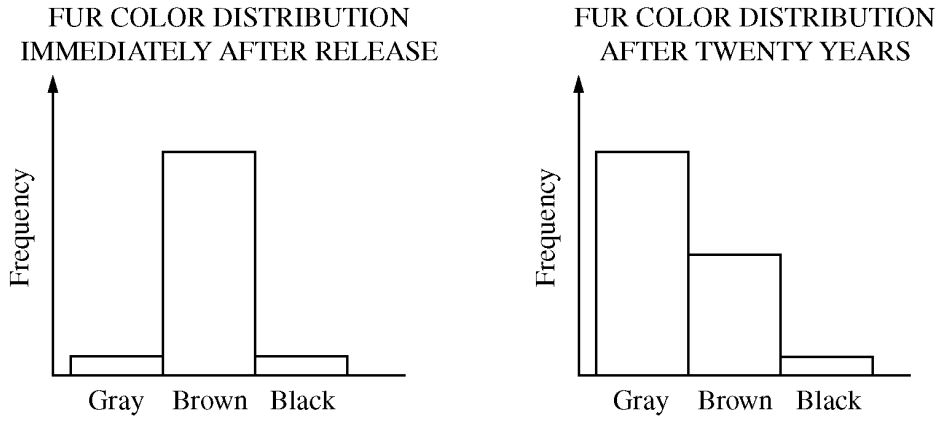
(C) With milder winters, the advantage of gray feathers that blend into a snowy environment has decreased. ✓

(D) The directional selection toward a darker phenotype is random and likely due to a bottleneck event.

Answer C

Correct. With less snowfall and milder winters, the advantages of gray feather color as camouflage has decreased.

167.



A group of mice was released into a large field to which no other mice had access. Immediately after the release, a representative sample of the mice was captured, and the fur color of each individual in the sample was observed and recorded. The mice were then returned to the field. After twenty years, another representative sample of the mice was captured, and the fur color of each individual in the sample was again recorded. Which of the following best explains the change in the frequency distribution of fur color phenotypes in the mouse population, as shown in the figures above?

7 Natural Selection MCQ

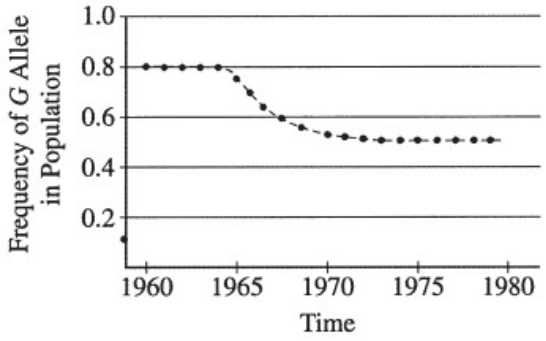
(A) The allele for gray fur color is unstable, and over twenty years most of those alleles mutated to become alleles for black fur.

(B) The field was composed primarily of light-colored soil and little vegetation, affording gray mice protection from predators. ✓

(C) Sexual selection led to increased mating frequency of black and brown versus gray and brown.

(D) The gray mice were hardest to capture and so were underrepresented in the twenty-year sample.

168. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Which of the following is the most likely reason for the observed differences in the frequency of the *G* allele between 1965 and 1972?

(A) Emigration of white moths from the population

(B) Chance

(C) Selection against gray phenotypes ✓

(D) Speciation

(E) Mutation

169. A researcher claims that budding yeast are able to survive in different environments because they produce enzymes that allow them to use different molecules as sources of matter and energy.

Which of the following statements best helps justify the researcher's claim by providing a relevant example?

(A) Yeast cells produce protein kinases, which are enzymes that catalyze the transfer of phosphate groups from ATP to protein substrates.

(B) Yeast cells produce DNA polymerases, which are enzymes that catalyze the conversion of free nucleotides into strands of DNA.

(C) Yeast cells produce invertase, which is an enzyme that catalyzes the conversion of the disaccharide sucrose into glucose and fructose. ✓

(D) Yeast cells produce catalase, which is an enzyme that catalyzes the conversion of hydrogen peroxide into water and oxygen gas.

7 Natural Selection MCQ

Answer C

Correct. The disaccharide sucrose is not transported efficiently into yeast cells. By producing the enzyme invertase, budding yeast can use the sucrose, glucose, and fructose in the surrounding environment as sources of matter and energy. As such, the production of invertase by budding yeast is a relevant example that will help justify the researcher's claim.

170. A team of scientists grew small pieces of coral in controlled tanks. Once these pieces of coral reached sufficient size, they were transferred to natural reefs to replace lost corals in the ocean. A scientist from the team suggested revising the experiment so that the water temperature in all the tanks would be increased to $0.5^{\circ}C$ above the normal ocean temperature. The corals pieces that survived the increased temperature in the tank would be separated and allowed to grow into different colonies. When these new colonies grown from those coral pieces grown in warmer water reached sufficient size, they would be transferred to natural reefs.

Which of the following statements predicts the most likely outcome of the revised experiment?

(A) The increased temperature will induce corals to create new genes, which will allow the corals to withstand future increases in ocean temperature.

(B) Stress from the increased temperature will cause corals to evolve into a new species that can tolerate future increases in ocean temperature.

(C) After being transferred to the natural reef, the new colonies of corals will return to their previous heat tolerance level and will not survive future increases in ocean temperature.

(D) The new colonies of corals are artificially selected for tolerating an increased temperature, so they will be better adapted to moderate future increases in ocean temperature. ✓

171. A researcher created an experimental chamber that allowed a population of fruit flies to mate randomly, but where female flies had to choose one of three side-chambers in which to lay eggs. All three side-chambers were the same except for the added scent of apple in side-chamber A, banana in side-chamber B, and cucumber in side-chamber C. Only eggs deposited in side-chamber C were allowed to hatch and produce offspring. Adult flies did not know which side-chamber was producing offspring. Every day for 100 days, the researcher counted the number of flies in each side-chamber. By day 100, the number of flies in side-chamber C greatly increased, while the number of flies in side-chambers A and B had greatly decreased.

Which of the following statements best explains why there were more flies in side-chamber C, than in side-chambers A or B, on day 100?

(A) More flies were entering side-chamber C because flies that prefer the scent of cucumber were artificially selected for. ✓

(B) More flies were entering side-chamber C because female flies learned which side-chamber allowed for successful reproduction.

(C) More flies were entering side-chamber C because female fruit fly offspring produced in side-chambers A and B were less fit for the environment.

(D) More flies were entering side-chamber C because male fruit fly offspring produced in side-chambers A and B were less fit for the environment.

7 Natural Selection MCQ

172. Bt crops have been genetically modified to express a gene from the bacterium *Bacillus thuringiensis*. Bt crops produce a protein that is harmful to insects that consume it but is harmless to humans and other mammals. Scientists have observed that some insects appear to be resistant to the protein produced by Bt crops. They collected data to compare the amount of Bt crops being grown with the number of cases of resistance to the Bt crop protein in insects (Table 1).

Table 1. Comparison of the Amount of Bt Crops Grown with the Number of Cases of Insect Resistance to Bt Crops

Year	Amount of Bt Crops Grown (million hectares*)	Total Number of Cases of Insect Resistance to Bt Crops
1996	0	0
2000	11	0
2004	22	2
2008	46	6
2012	70	11
2016	98	16

*1 hectare = $10,000\ m^2$

Based on the data in Table 1, which of the following best describes how the amount of Bt crops grown is related to incidence of insect resistance to Bt crops?

(A) As the amount of Bt crops grown decreases, the total number of cases of insect resistance remains stable.

(B) As the amount of Bt crops grown decreases, the total number of cases of insect resistance increases.

(C) As the amount of Bt crops grown increases, the total number of cases of insect resistance increases. ✓

(D) As the amount of Bt crops grown increases, the total number of cases of insect resistance decreases.

7 Natural Selection MCQ

173. Over many years of a breeding program, a zoo has an established population of foxes that is well adapted for living in captivity. A representative sample of wild foxes from the neighboring forest was used to start the zoo population. A study was conducted to compare the behavior of the zoo fox population with the wild fox population in the neighboring forest. The behaviors of equal numbers of foxes from each population were assessed. Each fox's behavior was scored on a continuum from docile to aggressive based on its interaction with a trained behaviorist. The data is shown in Figure 1.

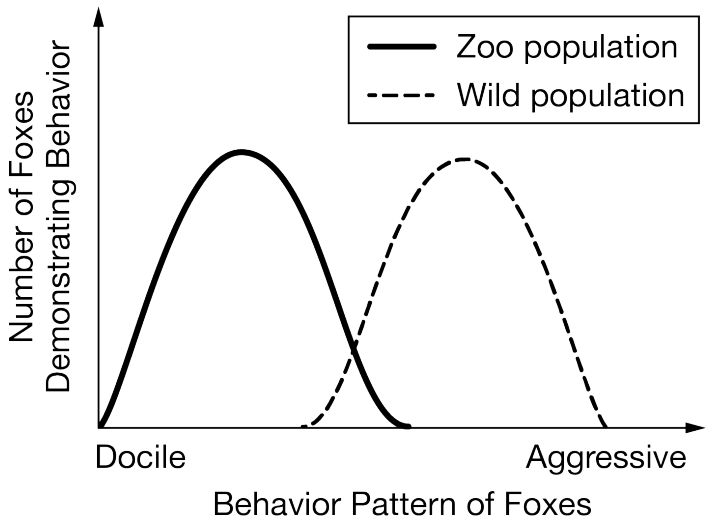


Figure 1. Aggressive behavior in zoo and wild fox populations

The phenotypic variation in behavior between the two populations can best be described as resulting from

(A) sexual selection

(B) natural selection

(C) artificial selection ✓

(D) genetic drift

Answer C

Correct. The zoo staff specifically selected the most docile foxes to produce offspring, an example of artificial selection.

7 Natural Selection MCQ

174. Table 1 shows a trend of increasing milk yield for cows. Part of the increase in milk production can be attributed to improved nutrition and health protocols for dairy cows.

Table 1. Average milk production per cow (in kilograms per cow per year)

Year	Average Annual Milk Production($\frac{\text{Kg}}{\text{Cow}}$)
1924	2,300
1937	2,300
1950	2,600
1963	3,400
1976	4,800
1989	6,400
2002	8,400
2015	10,000

Based on the information provided, which of the following best describes the source of the rapid increase in milk production from 1976 to 2015 ?

(A) From the 1970s to the early 2000s, dairy cows were not used for beef production.

(B) Starting in the 1970s, farmers increased the selective breeding of cows and bulls for offspring that produced more milk. ✓

(C) Reproduction rates in cows and bulls increased significantly in the 1970s.

(D) Cows were healthier after the 1970s, so they lived longer and produced more milk during their lifetime.

Answer B

Correct. Artificial selection by humans led to the production of offspring that produced more milk.

7 Natural Selection MCQ

175. Thoroughbred race horses are subject to intense artificial selection to produce faster-running horses. The results of breeding programs can be seen in the change in winning times of the Kentucky Derby from 1880 to the present, as shown in Figure 1.

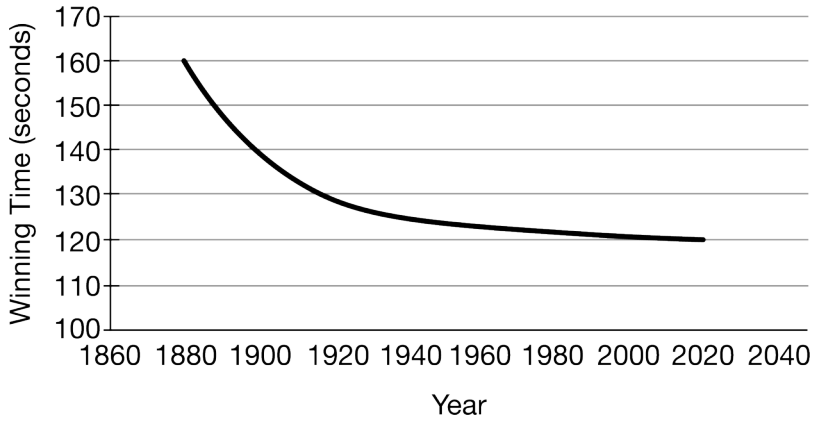


Figure 1. Best-fit curve for winning times in the Kentucky Derby since 1880

Which of the following best explains the trend shown in Figure 1?

(A) The distance the horses ran became shorter between 1880 and 1940.

(B) All of the selective breeding programs ended in 1940.

(C) After 1940, natural selection prevented any additional speed increases.

(D) By 1940, most genetic variation for speed had been lost from the gene pool. ✓

Answer D

Correct. Intense selection likely removed most of the alleles that produce slower speed by 1940, and few new combinations of the remaining alleles were possible, resulting in limited genetic variation relating to speed.

176. In eastern gray squirrels, *Sciurus carolinensis*, the allele for black fur (B) is dominant to the allele for gray fur (b). In a particular population of gray squirrels, 64% have black fur and 36% have gray fur. A researcher calculated the allelic frequencies to be $B = 0.4$ and $b = 0.6$. Five years later, the researcher returned to the location and determined the allelic frequencies within the squirrel population to be $B = 0.6$ and $b = 0.4$.

Which of the following could best explain the increase in the frequency of the B allele in the population after five years?

7 Natural Selection MCQ

(A) The frequency of the B allele increased because it is the dominant allele.

(B) The frequency of the B allele increased because squirrels with gray fur exhibited greater fitness.

(C) The frequency of the B allele increased due to the selective pressures of the environment. ✓

(D) The frequency of the B allele increased because black squirrels randomly mate with other eastern gray squirrels.

Answer C

Correct. It is most likely that the squirrels with black fur had a selective advantage, as the frequency of the allele for black fur increased.

177. In a small group of people living in a remote area, there is a high incidence of "blue skin", a condition that results from a variation in the structure of hemoglobin. All of the "blue-skinned" residents can trace their ancestry to one couple, who were among the original settlers of this region. The unusually high frequency of "blue skin" in the area is an example of

(A) mutation

(B) genetic drift ✓

(C) natural selection

(D) sexual selection

(E) heterozygote advantage

178. In humans, one allele of the $APOE$ gene, called $APOE$–$\varepsilon 2$, can result in a high tolerance of cholesterol. Cholesterol is a vital substance for humans but may lead to heart disease in an older adult with a history of high cholesterol diets. High cholesterol diets are becoming more prevalent in the United States. Currently only about 2% of humans carry the $APOE$–$\varepsilon 2$ allele.

Which of the following states a valid null hypothesis about the future distribution of $APOE$ alleles in future generations in the United States?

(A) The $APOE$–$\varepsilon 2$ allele was the result of chance mutations so it is not possible to predict how its frequency will change in the future.

(B) As high cholesterol diets become more common in the United States, individuals with the $APOE$–$\varepsilon 2$ allele will have a better survival rate from heart disease than those without the allele, so the allele will increase in frequency.

(C) The low frequency of the $APOE$–$\varepsilon 2$ allele indicates it is probably a recessive allele, so it will become less frequent as the dominant allele becomes more frequent.

(D) The variant protects an individual from a condition that is only common among humans beyond reproductive age, so the frequency of the allele will likely not change much in the future because it is not influenced by natural selection. ✓

7 Natural Selection MCQ

Answer D

Correct. Heart disease is a progressive condition that does not affect most humans during their child-bearing years. The survival rate of individuals with the $APOE$–$\varepsilon 2$ allele in later life will not affect its frequency in the next generation. The statement that there is no cause-and-effect relationship between the $APOE$–$\varepsilon 2$ allele and fitness is a null hypothesis.

179. A scientist maintains a large population of the fruit fly *Drosophila affinis* in the laboratory. After 20 generations of allowing this population of about 1000 flies to mate randomly, 10 females and 10 males were isolated from the rest of the population. Both the larger population and the isolated population were maintained under identical conditions and allowed to mate randomly for another 20 generations. After the 20 generations, DNA sequencing was used to determine the genetic composition of each population.

Which of the following predicts the expected outcome of the experiment?

(A) The two populations will show significant genetic differences because the founder effect will result in significant losses in diversity in the smaller population. ✓

(B) The two populations will show significant genetic differences because there will be much higher rates of mutation in the smaller population.

(C) The two populations will not show any differences because they were reared under identical conditions.

(D) The two populations will not show any differences because they originated from the same gene pool.

Answer A

Correct. Genetic drift, in the form of the founder effect, is likely to have much greater consequences for the smaller population than for the larger, resulting in differences between the two, including, but not limited to, lower diversity in the smaller population.

180. The different species of finches on the Galapagos Islands are believed to have arisen as a result of natural selection acting on populations of finches that had experienced

(A) convergent evolution

(B) gene flow

(C) the bottleneck effect

(D) geographic isolation ✓

(E) hybrid sterility

7 Natural Selection MCQ

181. In a large, isolated population of an insect species, a specific gene locus has one dominant allele (A) and one recessive allele (a). The genotype frequencies of the gene were collected for ten generations, as shown in Table 1.

Table 1. Genotypic frequencies over ten generations in an insect species

	AA	Aa	aa
Generation 1	0.47	0.43	0.10
Generation 5	0.32	0.60	0.08
Generation 10	0.20	0.75	0.05

Which of the following could best account for the change in genotypic frequencies over the ten generations?

(A) Because the A allele is unstable, over the course of the study most of the A alleles mutated to become a alleles.

(B) The population is in Hardy-Weinberg equilibrium, which allows for changes in genotypic frequencies every generation.

(C) The population is not exhibiting random mating between individuals. ✓

(D) The population is becoming extinct because the a allele is disappearing.

Answer C

Correct. Nonrandom mating that preferentially led to mating between AA and aa individuals could result in the observed increase in Aa genotype frequencies.

182. A group of people whose ancestors immigrated to North America 200 years ago have certain allele frequencies that differ significantly from those in surrounding populations in the United States. Which of the following is the most likely explanation for the difference in allele frequencies?

(A) Mutation

(B) Independent assortment

(C) Reproductive isolation ✓

(D) Sex linkage

(E) Gametic isolation

7 Natural Selection MCQ

183. A scientist is studying the impact of mutagens on genetic variation in mice. In an experiment, the scientist creates two groups of twenty highly inbred mice. Because the mice are highly inbred, they are essentially identical to one another. One group is given food with an added substance that is known to be a mutagen. The second group is fed the same food without the mutagen. After several generations, both groups of mice are given a new type of food that is harder to digest than the previous food. After rearing five additional generations of the mice on this new food, the two groups are assessed for how efficiently this new food is digested.

Which of the following is the most reasonable prediction about the results of the experiment?

(A) Neither group of mice will show any change in the ability to digest the food because thousands of years are required to develop such a change.

(B) Both groups of mice will show an increase in the ability to digest the new food, because they experienced the same selective pressures.

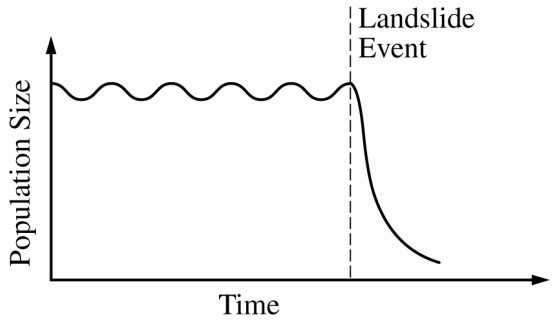
(C) The group of mice fed the mutagen will show a greater ability to digest the food, because the higher genetic variation within the group makes it more likely that some individuals will have a combination of genes better adapted to the new food. ✓

(D) The group fed the mutagen will show a decrease in the ability to digest the new food, because the mutagen prevented natural selection from acting on them.

Answer C

Correct. Inducing mutations provided the mouse population the variation necessary for a population to adapt to the new food.

184.



Undersea landslides can disrupt marine habitats by burying organisms that live on the ocean floor. The graph above shows the size of a population of a certain organism that lives on the ocean floor. The population was affected by a recent landslide at the time indicated on the graph. Which of the following best predicts how the population will be affected by the landslide?

(A) The surviving organisms will evolve into a new species.

(B) The reduced population will likely have allelic frequencies that are different from the initial population. ✓

(C) The population will adapt to deeper waters to avoid future landslides.

(D) The reduced population will have a greater number of different genes than the initial population.

7 Natural Selection MCQ

185. In addition to the ABO blood groups, humans have an MN blood type system in which the M and N alleles are codominant to one another. These alleles code for glycoproteins that act as antigens on the red blood cells. Scientists determined the frequencies of the M and N alleles in three localized and stable human populations in the Philippines. In Isabela, M and N alleles had a frequency of .8 and .21, respectively. Butuan had M and N allele frequencies of .82 and .18, respectively. Metro Manila had M and N allele frequencies of .55 and .51, respectively.

Which of the following would most affect the allelic frequencies of the populations on Isabela and Butuan?

(A) Migration of individuals between the Isabela and Butuan populations

(B) Immigration of individuals from Metro Manila into the two populations ✓

(C) Random mating within each population

(D) Increased population sizes over time in the Isabela population, but no change in the Butuan population

Answer B

Correct. Immigration of individuals from different populations can impact allelic frequencies. The Metro Manila population has a significantly different distribution of the M and Nalleles, so adding new individuals to the existing populations is likely to change the allelic frequencies in the populations.

186.

	Population 1		Population 2	
Year	Allele *R*	Allele *r*	Allele *R*	Allele *r*
1980	0.3	0.7	0.37	0.63
2010	0.0	1.0	0.75	0.25

The table shows the changes in allele frequencies of a specific gene in two populations of randomly mating small mammals after 30 years. The populations inhabit adjacent equatorial islands that have similar topography and climate. Which of the following is the most reliable conclusion that can be drawn from analysis of the data above?

(A) Genetic drift has occurred in population 1. ✓

(B) Population 2 is in Hardy-Weinberg equilibrium.

(C) Selection for allele r is occurring in both populations.

(D) The reduced frequency of allele R in population 1 will eventually lead to the extinction of population 1.

187. Retinitis pigmentosa is a rare genetic disorder that causes the breakdown of cells in the eye, resulting in vision loss. Worldwide, the occurrence of retinitis pigmentosa in the population is approximately 1 in $4,000$. However, on a small group of islands in the Atlantic Ocean, the occurrence is approximately 1 in 58.

Which of the following best explains how the occurrence of a rare, genetic disorder can be so much more prevalent on a small group of islands as compared with the rest of the world?

7 Natural Selection MCQ

(A) An unusually high number of individuals with the disease initially colonized the islands, and there has been little immigration to the islands since that time. ✓

(B) Individuals on the islands evolved adaptations to the vision loss, and because the disorder does not affect an individual's fitness, the disorder remains in the population.

(C) The population of individuals on the islands has greater genetic diversity compared with the worldwide population, resulting in a larger number of individuals with genetic disorders.

(D) The mutation rate is greater in the population of individuals on the islands than in the worldwide population, resulting in a greater incidence of rare disorders.

Answer A

Correct. The best explanation for the difference in prevalence is genetic drift, specifically a founder effect in which a population is founded by an initially small group of individuals with a large percentage of a particular trait. With very little immigration to the population, the initial percentage of the trait will remain the same over the generations.

188. Students observed the distribution of different color phenotypes in northern ravine salamanders (*Plethodon electromorphus*) before and after a spring flood. The data are shown in Table 1.

Table 1. Observed number of salamanders by phenotype before and after a spring flood

	Salamander Distribution before the Flood (n)	Salamander Distribution after the Flood (n)
Black phenotype	13	7
Dark-brown phenotype	25	14
Light-brown phenotype	6	3

Which of the following is an appropriate null hypothesis regarding the phenotypic frequencies of this population of salamanders before and after the flood?

(A) More black salamanders survived because they could better hide from predators on the dark muddy soil, so the frequency of the black phenotype significantly increased in the population after the flood.

(B) Fewer light-brown salamanders survived because they had a reduced chance of finding mates due to the high water levels, so the frequency of the light-brown phenotype in the population decreased after the flood.

(C) More dark-brown salamanders migrated to a new, drier area because of the flood and fewer of them remained in the original habitat, so the frequency of dark-brown salamander significantly decreased in the population after the flood.

(D) The proportions of black, dark-brown, and light-brown salamanders before and after the flood are not significantly different, and any observed differences are due to random chance. ✓

7 Natural Selection MCQ

Answer D

Correct. This is a null hypothesis that attributes any deviation in the phenotypic frequencies before and after flood to chance.

The three-spined stickleback (Gasterosteus aculeatus) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mostly of individuals with pronounced pelvic spines, as shown in Figure 1. Individuals in freshwater stickleback populations, on the other hand, typically have reduced pelvic spines, as shown in Figure 2.

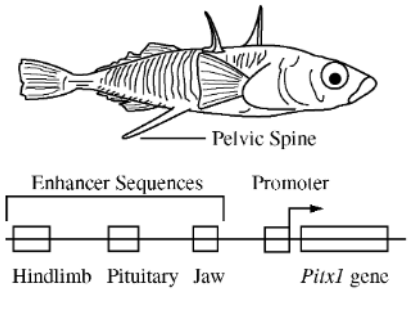


Figure 1. Marine stickleback

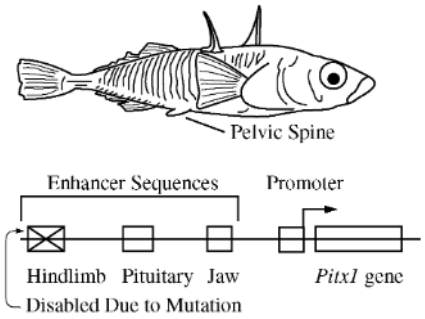


Figure 2. Freshwater stickleback

As represented in Figure 1 and Figure 2, the phenotypic difference between marine and freshwater sticklebacks involves Pitx1, a gene that influences the formation of the jaw, pituitary gland, and pelvic spine. Enhancer sequences upstream of the Pitx1 genetic locus regulate expression of the Pitx1 gene at the appropriate times and in the appropriate tissues during development. Previous studies have found that a mutation in the hindlimb enhancer interferes with the formation of a pronounced pelvic spine.

189. Which of the following explains a possible selective mechanism causing freshwater sticklebacks to typically have reduced pelvic spines?

(A) Reduced pelvic spines increase the likelihood of escaping predators in freshwater environments. ✓

(B) Reduced pelvic spines expand the range of foods that can be used as sources of energy in freshwater environments.

(C) Reduced pelvic spines result in males becoming effectively invisible to potential mating partners in freshwater environments.

(D) Reduced pelvic spines allow individual fish to pass through narrow waterways on their way to marine environments.

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7 Natural Selection MCQ

190. The California newt, *Taricha torosa,* lives in the coastal areas around Los Angeles. Which of the following is a valid null hypothesis relating fitness to survival of a bottleneck event in a coastal area where a small, isolated population of California newts resides?

(A) Only those salamanders with the lowest evolutionary fitness will be eliminated by the bottleneck effect, lowering the allelic frequencies of their traits.

(B) Only those salamanders with the highest evolutionary fitness will survive the bottleneck event, raising the allelic frequencies of their traits.

(C) Surviving the bottleneck event will be random, so any change in the allelic frequencies of the salamander population is not attributed to fitness. ✓

(D) Surviving the bottleneck event will be random, so there will be no changes to the allelic frequencies due to the bottleneck event.

Answer C

Correct. A bottleneck event is an example of genetic drift that randomly changes gene frequencies within a small population. Survival is a matter of chance, not fitness.

191. Members of one population of the striped stem borer insect engage in mating behavior in the early hours of the night, whereas members of a second population mate much later during the night. Crossbreeding between individuals from the two populations produces offspring that mate at an intermediate time.

Which of the following is the most likely cause of the different timing of mating behavior between the two populations?

(A) Genetic variation between the populations ✓

(B) The seasonal difference in day length

(C) The difference in learned behavior between the populations

(D) Gene flow between the populations

Answer A

Correct. Certain behavioral characteristics are encoded in the genome. The evidence that this is true for the striped stem borer insect is found in the crossbreeding experiment, which produced first-generation hybrid insects with a breeding time between the times of the two original populations.

7 Natural Selection MCQ

192. Cystic fibrosis is an autosomal (non-sex-linked) recessive genetic disorder in humans. In a certain geographical area, the incidence of cystic fibrosis is currently estimated at 1 in 377 live births. Which of the following is most likely closest to the frequency of the recessive allele in the human population in the area after 10 generations, assuming that the population remains in Hardy-Weinberg equilibrium?

(A) 0.003

(B) 0.052 ✓

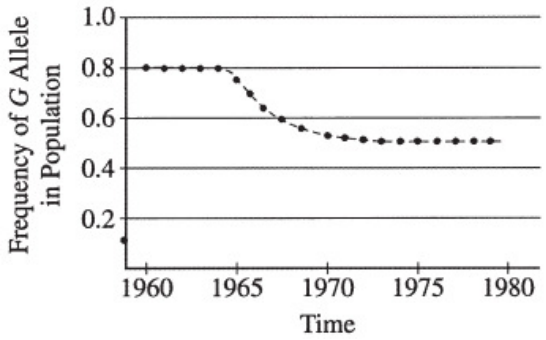
(C) 0.098

(D) 0.948

Answer B

Correct. The frequency of the recessive allele (q) in the population is determined by taking the square root of q^2 and is calculated as follows: $\frac{1}{377} = 0.00265 = q^2$, so $q = 0.051$, rounded to 0.052.

193. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Assuming that the population was in Hardy-Weinberg equilibrium for the *G* locus, what was the frequency of allele *G* in the gray moths that emerged in 1980?

(A) 0.33

(B) 0.50 ✓

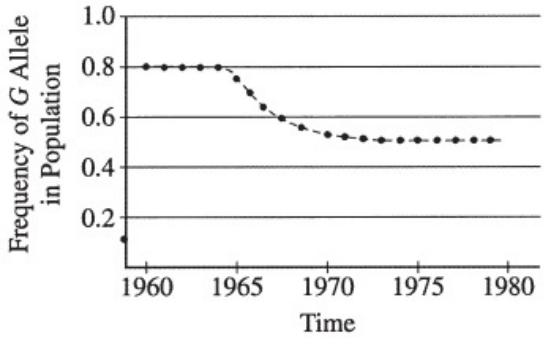
(C) 0.67

(D) 0.75

(E) 1.00

7 Natural Selection MCQ

194. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Assuming that the population was in Hardy-Weinberg equilibrium for the *G* locus, what percentage of the gray moths that emerged in 1980 was heterozygous?

(A) 0%

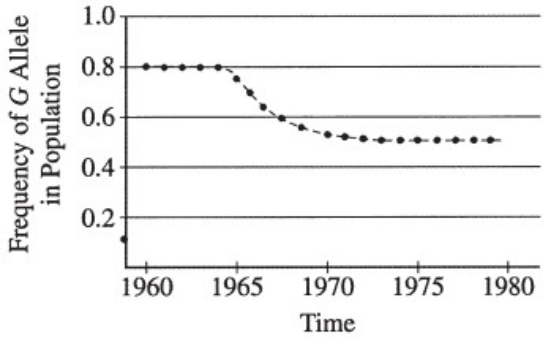
(B) 25%

(C) 33%

(D) 67% ✓

(E) 100%

195. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Assuming that the population was in Hardy-Weinberg equilibrium for the *G* locus, what percentage of moths in the natural population was white in 1962 ?

7 Natural Selection MCQ

(A) 2%

(B) 4% ✓

(C) 8%

(D) 20%

(E) 64%

For following group of questions first study the description of the situation and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

A survey reveals that 25 percent of a population of 1,000 individuals have attached earlobes (are homozygous recessivefor the trait). For the following questions, assume that the population fits the parameters of the Hardy-Weinberg law.

196. What is the frequency of the recessive allele?

(A) $\sqrt{0.75}$

(B) 0.75

(C) $\sqrt{0.50}$

(D) $\sqrt{0.25}$ ✓

(E) 0.25

197. Unlike most natural populations, this population is best characterized in which of the following ways?

(A) There is genetic equilibrium. ✓

(B) There is gene flow.

(C) There is genetic drift.

(D) Mutations occur.

(E) Mating is nonrandom.

198. If p equals the frequency of the dominant allele and q equals the frequency of the recessive allele, which of the following terms represents the frequency of the individuals who show the dominant phenotype?

(A) p^2

(B) q^2

(C) $2pq$

(D) $p^2 + 2pq$ ✓

(E) $p^2 + q^2$

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7 Natural Selection MCQ

199. A population geneticist is conducting an investigation on a population of mice. The geneticist determines the genotypes for a particular trait of all of the mice in the population during generation 1 and then removes all of the mice with the genotype *aa*. The remaining members of the population are allowed to randomly mate with one another to produce generation 2 and future generations. A model of the investigation is shown in Figure 1. Each of the fifteen genotypes shown in generation 1 and generation 2 represents 100 mice.

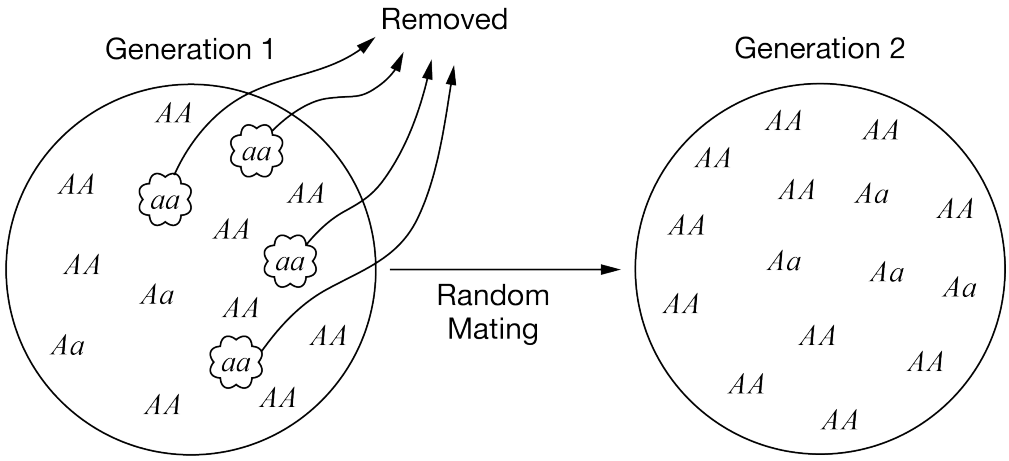


Figure 1. Model of effect of selective mouse removal on a population

Which of the following best illustrates the change in the frequencies of the *A* and *a* alleles from generation 1 to generation 2 and to future generations, assuming Hardy-Weinberg equilibrium is present after generation 1 ?

7 Natural Selection MCQ

(A) Future generations will have the same frequencies of the A and a alleles as generation 2. Individuals with the aa genotype could be produced. ✓

	Generation 1	Generation 2	Generation n
A	.67	.87	.87
a	.33	.13	.13

(B) Future generations will eventually completely lose the a allele from the gene pool. There will not be any new individuals with the aa genotype.

	Generation 1	Generation 2	Generation n
A	.67	.87	1.0
a	.33	.13	0

(C) Future generations will completely lose the a allele from the gene pool. There will no longer be individuals with the aa genotype.

	Generation 1	Generation 2	Generation n
A	.73	1.0	1.0
a	.27	0	0

(D) Future generations will have fewer a alleles in the gene pool but will not completely lose the allele. There will be a few individuals with the aa genotype.

	Generation 1	Generation 2	Generation n
A	.73	.87	.95
a	.27	.13	.05

7 Natural Selection MCQ

Answer A

Correct. This answer is correct because in generation 1 there are 2,000 alleles and 1,000 a alleles. $\frac{2,000}{3,000} = .67 = A$ and $\frac{1,000}{3,000} = .33 = a$. In generation 2 and future generations the frequency of A is $\frac{2,600}{3,000} = .87$ and the frequency of a is .13 . The population will not completely lose the a allele because there are still heterozygotes in the population that could mate with one another. Random mating would produce a stable gene frequency similar to that in generation 2.

200. There are two species of elephant seals: northern elephant seals, *Mirounga angustirostris*, found off the western coast of North America from Baja California to the Gulf of Alaska, and southern elephant seals, *M. leonina*, found in the southern oceans. Hunting in the nineteenth century has been blamed for a drastic reduction in the northern elephant seal population. Even though the population has rebounded following restrictions on seal hunting, genetic variation is greatly reduced in the northern species compared to that in the southern species.

Which of the following conditions of Hardy-Weinberg equilibrium was most likely not met in the northern elephant seal population following overhunting?

(A) Absence of selection

(B) No new mutations

(C) Random mating

(D) Large population size ✓

Answer D

Correct. A small population typically shows decreased genetic variation, and if the population is reduced sufficiently, it will undergo a bottleneck, where some previously existing genetic variants will be lost, thus disturbing Hardy-Weinberg equilibrium.

201. Ellis-van Creveld syndrome is a recessive genetic disorder that includes the characteristics of short stature and extra fingers or toes. In the general population, this syndrome occurs in approximately 1 in 150,000 live births. In a particular isolated population, however, the incidence of this syndrome among live births is 1 in 500.

Assume that both the isolated population and the general population are in Hardy-Weinberg equilibrium with respect to this syndrome. Which of the following best describes the difference between the frequency of the allele that causes the syndrome in the general population and the frequency of the allele in the isolated population?

7 Natural Selection MCQ

(A) The frequency of the Ellis-van Creveld allele is 0.002 in the isolated population and 0.0000066 in the general population, which suggests that selection for this trait is occurring in both populations.

(B) The frequency of the Ellis-van Creveld allele is 0.0447 in the isolated population and 0.0026 in the general population, showing that the rate of genetic mutation is highest among individuals in the isolated population.

(C) The frequency of the Ellis-van Creveld allele is 0.002 in the isolated population and 0.0000066 in the general population, which demonstrates gametic incompatibility between the populations.

(D) The frequency of the Ellis-van Creveld allele is 0.0447 in the isolated population and 0.0026 in the general population, which suggests that genetic drift has occurred in the isolated population. ✓

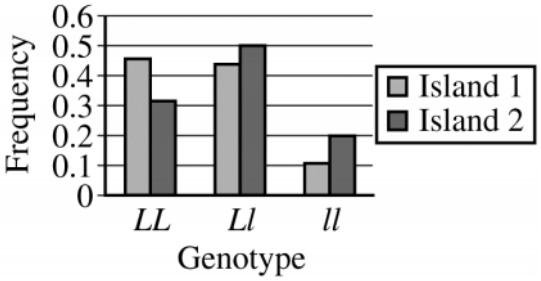
202. Scientists are studying several populations of finches on neighboring islands in the South Pacific. Previous genetic analysis has shown that a single gene controls tail-feather length in the finch populations and that the allele for long tail feathers (L) is dominant to the allele for short tail feathers (l). On two separate islands, the scientists recorded the number of finches with long tail feathers and the number of finches with short tail feathers. The results are shown in the table below.

Location	Phenotype	Number of Individuals
Island 1	Long tail feathers	1,582
	Short tail feathers	598
	Total count	2,180
Island 2	Long tail feathers	2,432
	Short tail feathers	1,110
	Total count	3,542

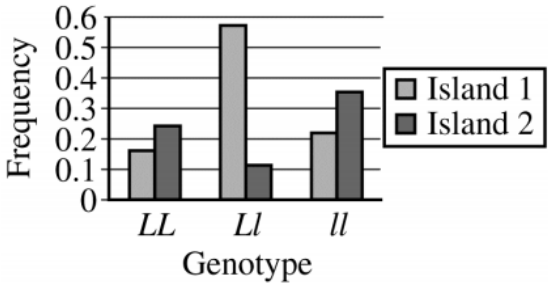
If the two finch populations are each in Hardy-Weinberg equilibrium and are isolated from each other, then which of the following graphs correctly displays the relative genotype frequencies?

7 Natural Selection MCQ

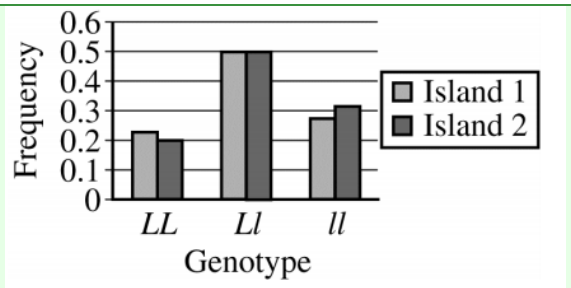
(A)



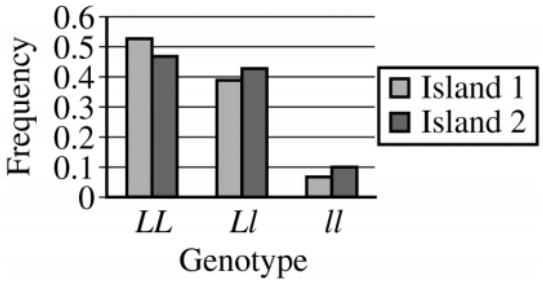
(B)



(C) ✓



(D)



203. In a certain flock of sheep, 4 percent of the population has black wool and 96 percent has white wool. Assume that the population is in Hardy-Weinberg equilibrium.

What percentage of the population is homozygous for white wool?

(A) 20%

(B) 40%

(C) 64% ✓

(D) 80%

(E) 96%

7 Natural Selection MCQ

204. Gaucher disease type 1 (GD1) is a recessive genetic disease that affects 1 in 900 individuals in a particular population. GD1 is caused by a mutation in the enzyme glucocerebrosidase.

Assuming the population is in Hardy-Weinberg equilibrium, calculate the frequency of the wild-type (nonmutant) allele for the enzyme glucocerebrosidase.

(A) 0.001

(B) 0.033

(C) 0.967 ✓

(D) 0.999

Answer C

Correct. Assuming the population is in Hardy-Weinberg equilibrium, the frequency of the dominant wild-type allele can be determined from the Hardy-Weinberg equation $p^2 + 2pq + q^2 = 1$, where p represents is the frequency of the dominant allele, q is the frequency of the recessive allele and $p + q = 1$. Since GD1 is a recessive condition, and it affects 1 in 900 individuals in the population, $q^2 = \frac{1}{900} = 0.001$. Therefore, $q = \sqrt{q^2} = \sqrt{\frac{1}{900}} = \frac{1}{30} = 0.033$. Since $p + q = 1$, $p = 1 - q$, the frequency of the dominant wild-type allele $p = 1 - 0.033 = 0.967$.

205. A large population of laboratory animals has been allowed to breed randomly for a number of generations. After several generations, 49 percent of the animals display a recessive trait (*bb*), the same percentage as at the beginning of the breeding program. The rest of the animals show the dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.

What proportion of the population is heterozygous (*Bb*) for this trait?

(A) 0.51

(B) 0.42 ✓

(C) 0.21

(D) 0.09

(E) 0.07

206. A large population of laboratory animals has been allowed to breed randomly for a number of generations. After several generations, 49 percent of the animals display a recessive trait (*bb*), the same percentage as at the beginning of the breeding program. The rest of the animals show the dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.

What is the frequency of allele *b* in the gene pool?

7 Natural Selection MCQ

(A) 0.70 ✓

(B) 0.51

(C) 0.49

(D) 0.30

(E) 0.07

207. Humans vary in their ability to taste the bitter chemical compound phenylthiocarbamide (PTC). The taster phenotype is dominant to the nontaster phenotype. Researchers tested the PTC taster phenotype of individuals from an isolated population. In the sample, 780 individuals were able to taste PTC and 70 individuals were not able to taste PTC.

Assuming the population is in Hardy-Weinberg equilibrium, what is the frequency of the non-taster allele?

(A) 0.08

(B) 0.29 ✓

(C) 0.41

(D) 0.71

Answer B

Correct. Since the population is in equilibrium, the frequency of the nontaster allele can be determined by using the Hardy-Weinberg equation, $p^2 + 2pq + q^2 = 1$, where p is the frequency of the dominant allele, q is the frequency of the recessive allele, and $p + q = 1$. In this equation, q^2 is the frequency of homozygous recessive nontasters. In this population the frequency of nontasters (q^2) is $\frac{70}{780+70} = 0.082$; therefore, the frequency of the recessive nontaster allele (q) is $\sqrt{0.082} = 0.29$.

208. In certain Native American groups, albinism due to a homozygous recessive condition in the biochemical pathway for melanin is sometimes seen. If the frequency of the allele for this condition is 0.06, which of the following is closest to the frequency of the dominant allele in this population? (Assume that the population is in Hardy-Weinberg equilibrium.)

(A) 0.04

(B) 0.06

(C) 0.16

(D) 0.36

(E) 0.94 ✓

7 Natural Selection MCQ

209. In a long-term study of a population of gray squirrels, researchers observed that most of the squirrels had gray fur. However, there were some individuals in the population with white fur (albino). Genetic analysis revealed that the albino condition is expressed by individuals who are homozygous recessive (gg). Over a ten-year period, the average frequency of albino squirrels in the population was 0.18. If the population is assumed to be in Hardy-Weinberg equilibrium, the average frequency of the dominant allele is closest to which of the following?

(A) 0.42

(B) 0.50

(C) 0.58 ✓

(D) 0.75

Answer C

Correct. If the frequency of *gg* individuals (q^2) is 0.18, then the frequency of the recessive allele (g) is equal to 0.42 $\left(q = \sqrt{.18}\right)$. Subtraction provides the frequency of the dominant allele (*p*): $1 - 0.42 = 0.58$.

210. In addition to the ABO blood groups, humans have an MN blood type system in which the M and N alleles are codominant to one another. A group of scientists found that in one population the frequency of the M allele was 0.82, and the frequency of the N allele was 0.18.

Assuming that the population is in Hardy-Weinberg equilibrium, which of the following is the expected frequency of individuals with the genotype MN?

(A) 0.148

(B) 0.295 ✓

(C) 0.640

(D) 0.672

Answer B

Correct. If the population is in Hardy-Weinberg equilibrium, the frequency of heterozygotes is determined by the expression $2pq$, where p is the frequency of one allele and q is the frequency of the other. The calculation would be $(2)(0.82)(0.18) = 0.295$.

211. Nine percent of a population is homozygous recessive (*aa*) at a certain locus. Assuming that the population is in Hardy-Weinberg equilibrium, which of the following is closest to the frequency of the recessive allele (*a*) ?

7 Natural Selection MCQ

(A) 0.09
(B) 0.27
(C) 0.30 ✓
(D) 0.90
(E) 0.97

212. Which of the following statements best explains how a condition of Hardy-Weinberg equilibrium results in a population that exhibits stable allele frequencies (i.e., a nonevolving population) ?

(A) Large populations are not subject to natural selection.
(B) Random mating prevents gene flow from changing allele frequencies.
(C) Without migration or mutation, new alleles cannot be introduced to the population. ✓
(D) In the absence of selection, allele frequencies in a population will not change.

Answer C

Correct. One of the conditions of Hardy-Weinberg equilibrium is that there is no gene flow and no net mutation, which prevent the introduction of new alleles and thus new phenotypes.

213.

Bacteria	P M V V G H E Y V G E V V G I G Q
Yeast	P L V G G H E G A G V V V G M G E
Corn	P R I L G H E A G G I V E S V G E
Horses	P V I A G H E A A G I V E S I G E
Sheep	P M V L G H E A S G T V V K V G S

Figure 1. Amino acid sequences of certain dehydrogenase enzymes in different species

Portions of the amino acid sequences of some dehydrogenase enzymes found in bacteria, yeast, corn, horses, and sheep are shown in Figure 1. Each letter represents a different amino acid in the sequence.

Based on Figure 1, which of the following conclusions about the enzymes is most likely correct?

7 Natural Selection MCQ

(A) Corn and horse dehydrogenase enzymes arose separately during each organism's evolution and now exhibit strong sequence similarity as a result of their similar catalytic roles.

(B) These enzymes all evolved from a common prokaryotic dehydrogenase enzyme, so they exhibit regions of sequence similarity. ✓

(C) Any evolutionary relationship is unlikely, because there is little sequence similarity between these enzymes.

(D) No conclusion can be drawn based on amino acid sequence, because there are twenty amino acids and only four nucleotides.

Answer B

Correct. The data in Figure 1 indicate only random amino acid differences in the dehydrogenase enzymes of the five diverse species, including a bacterium. This is expected if all enzymes evolved from the enzyme of a prokaryotic ancestor.

214.

Representative Vertebrate	Primary Nitrogenous Waste Product		
	Ammonia	Urea	Uric Acid
Fish	X		
Tadpole	X		
Frog		X	
Reptile			X
Bird			X
Mammal		X	
Property of Nitrogenous Waste Product			
Solubility in Water	High	High	Low
Toxicity	High	Medium	Low

The table above shows the types and properties of nitrogencontaining wastes produced by several vertebrates. Which of the following is the best evolutionary explanation of the data?

(A) The data support convergent evolution for nitrogen excretion in organisms occupying dramatically different environmental niches.

(B) The nitrogenous waste excreted by each vertebrate is the result of evolutionary adaptations that have decreased vulnerability to predators.

(C) Nitrogen homeostasis in terrestrial vertebrates reflects an adaptation for more frequent and higher-volume urination than that occurring in aquatic vertebrates.

(D) Ammonia secretion requires a large volume of water and was therefore selected against in terrestrial vertebrates. ✓

7 Natural Selection MCQ

215. Which of the following pieces of evidence best supports the hypothesis that birds and crocodilians (crocodiles and alligators) are more closely related to each other than they are to other organisms?

(A) Modern birds and crocodilians utilize different niches within the same geographical region.

(B) Like turtles and snakes, modern birds and crocodilians build nests in which they lay shelled eggs.

(C) The skulls of birds and crocodilians have an opening that is absent in the skulls of other reptiles. ✓

(D) Fossil evidence indicates that both modern birds and crocodilians originated during the Jurassic Period.

216. Students analyzed several photosynthetic eukaryotic organisms in the laboratory to determine their phylogenetic relationships. The organisms were *Spirogyra* (a green alga), a moss, a spruce tree, and an apple tree. Table 1 compares several characteristics in the organisms (+ indicates the trait is present, − indicates the trait is absent).

Table 1. Characteristics present in four photosynthetic species

	Green Alga	Moss	Spruce Tree	Apple Tree
Cell wall composed of cellulose	+	+	+	+
Vascular tissue	−	−	+	+
Roots	−	−	+	+
Seeds enclosed in fruit	−	−	−	+
Flagellated sperm cells	−	+	−	−

Which of the following rows of data listed in table 1 best supports the possibility of a common ancestor for the organisms listed there?

(A) Vascular tissue

(B) Seeds enclosed in fruit

(C) Cell wall composed of cellulose ✓

(D) Flagellated sperm cells

Answer C

Correct. The presence of a cell wall composed of cellulose is a characteristic of all four organisms and suggests a common ancestor.

7 Natural Selection MCQ

217. The amino acid sequence of cytochrome *c* is exactly the same in humans and chimpanzees. There is a difference of 13 amino acids between the cytochrome *c* of humans and dogs, and a difference of 20 amino acids between the cytochrome *c* of humans and rattlesnakes.

Which of the following statements is best supported by these data?

(A) Rattlesnakes apparently gave rise evolutionarily to the dog, chimpanzee, and human.

(B) Cytochrome *c* apparently has an entirely different function in rattlesnakes than in mammals, which explains the difference in the number of amino acids.

(C) Cytochrome *c* is not found universally in animals.

(D) Cytochrome *c* from a rattlesnake could function in a dog, but not in a chimpanzee.

(E) The human is apparently more closely related to the chimpanzee than to the dog or rattlesnake. ✓

218. Scientists want to determine the age of four fossil specimens. They have analyzed the fossils and gathered the data shown in Table 1.

Table 1. Data collected from four fossil specimens

	Specimen 1	Specimen 2	Specimen 3	Specimen 4
Amino acid differences*	5	21	18	11
Percent of original carbon-14 remaining	38	73	25	11
Composition	Sandstone	Sandstone	Limestone	Shale
DNA analysis	35% Guanine	28% Guanine	12% Guanine	29% Guanine

*amino acid differences of a conserved protein compared with a fifth fossil specimen

Which of the specimens indicated in the table is the oldest?

(A) Specimen 1

(B) Specimen 2

(C) Specimen 3

(D) Specimen 4 ✓

Answer D

Correct. Carbon−14 is incorporated into living organisms based on the percent of carbon−14 in the

7 Natural Selection MCQ

atmosphere. After the organism dies, carbon−14 breaks down by radioactive decay. Therefore the specimen with the lowest percent of the original carbon−14 remaining would be the oldest.

219. Which of the following explains the most accurate way to determine common ancestry?

(A) Observations of behavior provide evidence of common ancestry for organisms.
(B) Observations related to morphological homologies among organisms provide evidence of relatedness.
(C) Examination of fossil records provides evidence of organisms' evolution over time.
(D) Comparisons of DNA nucleotide sequences provide evidence of the origin of the organisms' lineages. ✓

A scientist used the amino acid sequence of cytochrome *c* in different species to consider evolutionary relationships.

The data below summarize the number of differences in the amino acid sequences of cytochrome *c* found in selected species.

Species Compared	Number of Differences
Humans–Chimpanzees	0
Humans–Rhesus monkeys	1
Humans–Horses or donkeys	7
Humans–Cows or pigs or sheep	7
Humans–Rabbits	7
Mammals–Birds and reptiles	10–15
Mammals–Fish	18–20
Animals–Plants	45–48

220. Interpretation of the data supports which of the following statements?

(A) Rhesus monkeys are more closely related to rabbits than they are to horses.
(B) Horses and cows have identical amino acid sequences in their cytochrome *c*.
(C) Humans are more closely related to rabbits than they are to rhesus monkeys.
(D) Plants and animals have no similarities at all.
(E) Mammals are more closely related to reptiles than they are to fish. ✓

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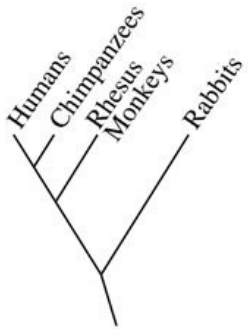
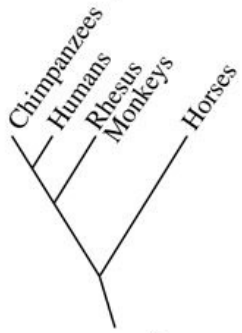
221. Which of the following phylogenetic trees can be supported by the data?

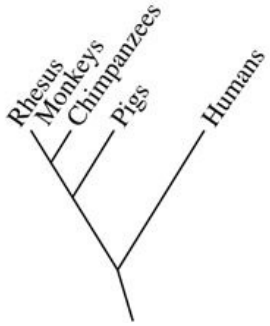
I.



II.



III.



(A) I only
(B) II only
(C) III only
(D) I and II only ✓
(E) I, II, and III

7 Natural Selection MCQ

222. Table 1 summarizes selected characteristics of organisms from each of the three domains of life.

Table 1. A Summary of Selected Characters across Domains of Life

Domain	Bacteria	Archaea	Eukarya			
Major divisions	Bacteria	Archaea	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Genetic code bases	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine
Genetic material	DNA	DNA	DNA	DNA	DNA	DNA
Metabolic pathway(s)	Glycolysis	Glycolysis	Glycolysis, citric acid cycle, oxidative phosphorylation	Glycolysis, citric acid cycle, oxidative phosphorylation	Glycolysis, citric acid cycle, oxidative phosphorylation	Glycolysis, citric acid cycle, oxidative phosphorylation
Mode of nutrition	Autotrophic, heterotrophic	Autotrophic, heterotrophic	Autotrophic, heterotrophic	Heterotrophic	Autotrophic	Heterotrophic
Number of cells	Unicellular	Unicellular	Unicellular; some colonial	Multicellular	Multicellular	Multicellular

Which of the following sets of data provides evidence that best supports common ancestry for organisms in all three domains?

(A) There are autotrophic and heterotrophic organisms found in Bacteria, Archaea, and Eukarya.

(B) There is a progression of pathways in organisms found in Bacteria, Archaea, and Eukarya.

(C) Organisms in Bacteria, Archaea, and Eukarya each have adenine, thymine, guanine, and cytosine bases forming their DNA. ✓

(D) There is a progression of cellular organization in organisms found in Bacteria, Archaea, and Eukarya.

Answer C

Correct. Genetic information is contained in DNA in all three domains, and the same nucleotides are found in the DNA in all three domains.

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223. Researchers studying the evolutionary relationships of three species of snails collected the data shown in Table 1.

Table 1. Comparison of selected characteristics of three snail species

	Species A	Species B	Species C
Location of Fossils	North America	North America	Europe
Shell Orientation	Counterclockwise	Counterclockwise	Clockwise
Shell Composition	Calcium Carbonate	Calcium Carbonate	Calcium Carbonate
Trophic Level	Carnivore	Carnivore	Detritivore
DNA Analysis	$\%\ \mathrm{A+T} = 43.2$	$\%\ \mathrm{A+T} = 43.4$	$\%\ \mathrm{A+T} = 43.0$
	$\%\ \mathrm{G+C} = 56.8$	$\%\ \mathrm{G+C} = 56.6$	$\%\ \mathrm{G+C} = 57.0$

Which of the following data sets is most likely to provide reliable information about the evolutionary relationships among these three snail species?

(A) Shell composition and shell orientation

(B) Trophic level and DNA analysis

(C) Location of fossils and shell orientation ✓

(D) Shell composition and location of fossils

Answer C

Correct. Of the characteristics given, fossil location and shell orientation are the most informative. Both species A and B are found on the same continent and have the same shell orientation, suggesting a closer relationship than either has with species C.

224. The wing of a bat, the flipper of a whale, and the forelimb of a horse appear very different, yet detailed studies reveal the presence of the same basic bone pattern. These structures are examples of

(A) analogous structures

(B) homologous structures ✓

(C) vestigial structures

(D) balanced polymorphism

(E) convergent ‘evolution

7 Natural Selection MCQ

225. Catalase is a protein enzyme that catalyzes the breakdown of hydrogen peroxide (H_2O_2) into water and oxygen gas in the cells of many living organisms. Students used an online database, which used the structure of catalase from different species, to generate the phylogenetic tree shown in Figure 1.

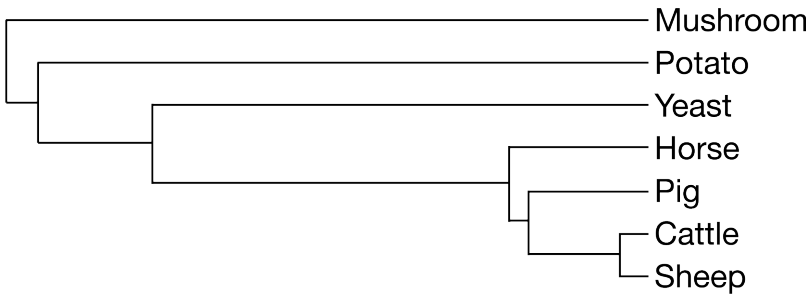

Figure 1. Relationships of selected organisms based on catalase structure

Which of the following student claims is best supported by the data provided?

(A) Mushrooms and potatoes are most closely related, since the node they share gave rise to all of the other organisms.

(B) Cattle are equally related to pigs and sheep, because the phylogeny places cattle between pigs and sheep.

(C) Yeast and mushrooms are the most closely related, because they are both fungi.

(D) Cattle and sheep are separated by only one node, and they are separated from the other organisms by more than one node. ✓

Answer D

Correct. In a phylogeny the number of nodes separating organisms indicates how closely related they are.

226. Lobe-finned fishes were present in the oceans of the world approximately 400 million years ago. The first tetrapods (vertebrates that had limbs and could move on land) date to about 365 million years ago. One hypothesis states that early tetrapods evolved from lobe-finned fishes. Which of the following is the best plan for testing the hypothesis?

(A) Determining the number of fins found on lobe-finned fishes from different geographical locations

(B) Measuring the proportion of radioisotopes found in fossils of early tetrapods

(C) Comparing the arrangements of bones in the fins of lobe-finned fishes and limbs of the earliest tetrapods ✓

(D) Sequencing DNA isolated from fossils of lobe-finned fishes to determine the size of the genome

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227. Hemoglobin is a highly conserved protein used by all mammals to transport oxygen. Table 1 was constructed by comparing the amino acid sequence in the hemoglobin molecules from five different mammal species.

Table 1. Hemoglobin amino acid differences among five different mammal species

	Species 1	Species 2	Species 3	Species 4	Species 5
Species 1	–	21	11	13	5
Species 2		–	18	17	20
Species 3			–	3	10
Species 4				–	12
Species 5					–

Which of the following best describes the importance of the number of amino acid differences indicated in Table 1 ?

(A) They reflect the different environments where the organisms were captured.

(B) They reflect the number of base pair changes in the DNA. ✓

(C) They reflect random changes due to translation errors.

(D) They reflect the number of oxygen molecules each hemoglobin molecule carries.

Answer B

Correct. Differences in the amino acid sequences are due to differences in the sequences of the DNA nucleotides of the hemoglobin genes and can be used as evidence of relatedness.

228. The placement and development of two bones of the reptilian jaw are very similar to that of two bones of the mammalian middle ear, which probably evolved from the reptile bones. The similarity of these bones is an example of

(A) structural homology ✓

(B) speciation

(C) vestigial structures

(D) convergent evolution

(E) adaptive radiation

8 Ecology MCQ

1. A scientist is evaluating a proposal for raising large numbers of fish in ocean pens for human consumption. As part of the evaluation, the scientist is designing a plan for investigating how the fish in the ocean pens might affect nearby ecosystems.

Which of the following is the most appropriate factor to use as the dependent variable in the experimental investigation?

(A) The amount of metabolic wastes in the water where the fish are being raised ✓

(B) The water temperature in the natural habitat of the fish being studied

(C) The types of fish that are preferentially consumed by humans

(D) The amount of time it takes for fish to digest a meal

2. A biologist is studying territoriality in three related species of birds found in the same section of a forest. For each observed pair of nesting adults, the biologist measured the distance to the next closest nesting pair of the same species (nearest-neighbor distance). An analysis of the results is shown in the table below.

Species	Mean Nearest-Neighbor Distance (km)	Median Nearest-Neighbor Distance (km)	Range of Nearest-Neighbor Distances (km)
I	0.35	0.47	0.25–0.50
II	0.31	0.32	0.30–0.32
III	0.80	1.10	0.10–1.50

Which of the following would be most appropriate for identifying the species with the most even distribution of nests?

(A) Area of the forest in which the nearest-neighbor measurements were made

(B) Total number of nearest-neighbor observations collected during the study period

(C) Standard deviation of the nearest-neighbor distances for each species ✓

(D) Proportion of nearest-neighbor distances that were less than the median

3. A colony of termites was exposed to an atmosphere of 100 percent oxygen for three days. The insects were not immediately harmed by the treatment, but the protozoa that lived in the termites' guts were eliminated. The treated termites continued to behave normally and to eat wood, but they began to starve after a short time.

When the treated termites were instead fed wood contaminated with the feces of untreated termites, the treated termites regained the ability to digest wood and no longer starved. The best analysis of the results of the experiment is that

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(A) infection with protozoa causes the termites to switch to fermentation

(B) cooperative interactions between the termites and the protozoa allow termites to extract energy from wood to survive ✓

(C) the termite immune system eliminates protozoa from the gut

(D) termites digest protozoa as their main source of energy

4. A group of students summarized information on five great extinction events.

Mass Extinction	Time of Extinction	Organisms Greatly Reduced or Made Extinct
End of the Ordovician period	443 million years ago	Trilobites, brachiopods, echinoderms, and corals
End of the Devonian period	354 million years ago	Marine families on tropical reefs, corals, brachiopods, and bivalves
End of the Permian period	248 million years ago	Trilobites, mollusks, brachiopods, and many vertebrates
End of the Triassic period	206 million years ago	Mollusks, sponges, marine vertebrates, and large amphibians
End of the Cretaceous period	65 million years ago	Ammonites, dinosaurs, brachiopods, bivalves, and echinoderms

The students are sampling a site in search of fossils from the Devonian period. Based on the chart, which of the following would be the most reasonable plan for the students to follow?

(A) Searching horizontal rock layers in any class of rock and try to find those that contain the greatest number of fossils

(B) Collecting fossils from rock layers deposited prior to the Permian period that contain some early vertebrate bones ✓

(C) Looking in sedimentary layers next to bodies of water in order to find marine fossils of bivalves and trilobites

(D) Using relative dating techniques to determine the geological ages of the fossils found so they can calculate the rate of speciation of early organisms

8 Ecology MCQ

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(D) Using relative dating techniques to determine the geological ages of the fossils found so they can calculate the rate of speciation of early organisms

6. All of the following statements concerning characteristics of predator-prey relationships are correct EXCEPT:

(A) A rise in the population of prey is often followed by a rise in the population of predators.

(B) A rise in the population of predators is followed by a decrease in the population of prey.

(C) Camouflage is an adaptation that protects prey.

(D) The production of large numbers of offspring within very short periods of time ensures the survival of some prey populations.

(E) The population of predators most often eliminates the population of prey. ✓

7. Some scientists claim that amphibians are currently experiencing a period of mass extinction.

Which of the following should be included in an alternative hypothesis that would best support this claim?

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(A) The current extinction rate of amphibians compared with the background rate of extinction as determined by the fossil record ✓

(B) The extinction rate of amphibians in the 21st century compared with the extinction rate of amphibians during the 20th century

(C) The number of species of amphibians currently alive compared with the number of amphibian species known to be extinct

(D) The number of new amphibian species being discovered per year compared with the number of amphibian species becoming extinct during the same time period

Answer A

Correct. A mass extinction requires that the extinction rate of a certain group of organisms, in this case amphibians, be greater than the background extinction rate. Claiming that this difference exists would be an alternative hypothesis.

8. Scientists have found that the existing populations of a certain species of amphibian are small in number, lacking in genetic diversity, and separated from each other by wide areas of dry land. Which of the following human actions is most likely to improve the long-term survival of the amphibians?

(A) Cloning the largest individuals to counteract the effects of aggressive predation

(B) Reducing the population size by one-fifth to decrease competition for limited resources

(C) Constructing a dam and irrigation system to control flooding

(D) Building ponds in the areas of dry land to promote interbreeding between the separated populations ✓

9. An overlap in the niches of two species will most frequently result in

(A) interspecific cooperation

(B) a hybridization of species

(C) a mutualistic symbiotic relationship

(D) an increase in the biomass

(E) interspecific competition ✓

10. A parasite may be described as an organism that derives nourishment from another living organism and

(A) becomes necessary to the life of the host

(B) is neutral in its effect on the host

(C) benefits the host nutritionally

(D) provides a supportive structure for the host

(E) harms the host in the process ✓

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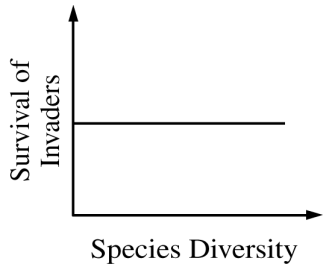
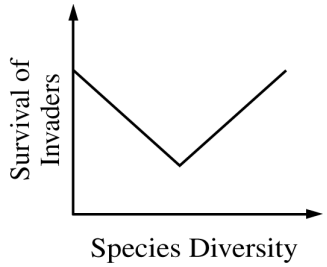
11. A researcher is investigating the relationship between the existing species diversity in a community and the ability of an introduced nonnative species to destabilize the community.

Which of the following graphs is most consistent with the claim that communities with high diversity are more resistant to change than are communities with low diversity?

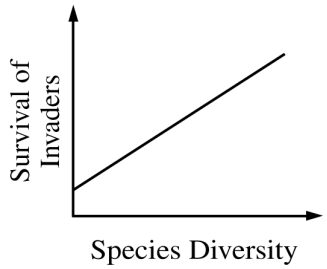
(A)



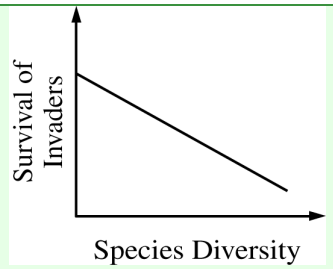
(B)



(C)

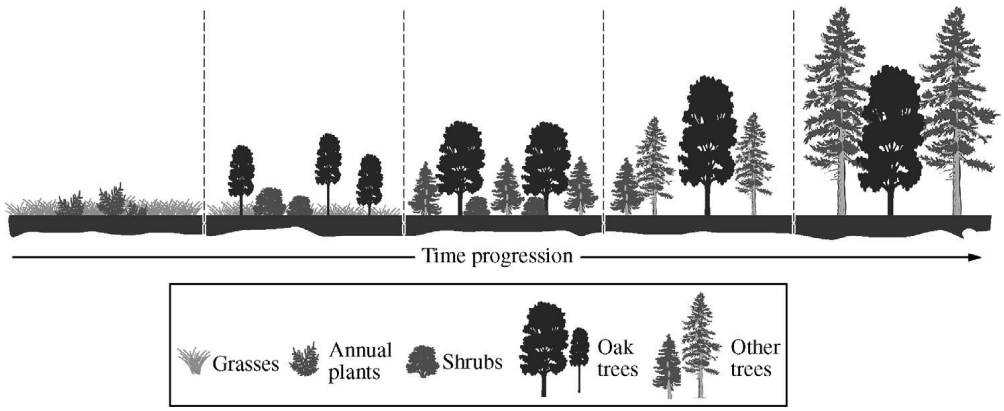


(D) ✓



8 Ecology MCQ

12.



The diagram above shows the progression of ecological events after a fire in a particular ecosystem. Based on the diagram, which of the following best explains why the oak trees are later replaced by other trees?

(A) Eventually the other trees grow taller than the oak trees and form a dense canopy that shades the understory. ✓

(B) Oak trees alter the pH of the soil, making the forest better suited for shrubs and other trees.

(C) Roots of shrubs proliferate in the soil of the forest and prevent the oak trees from obtaining water.

(D) Oak trees succumb to environmental pollutants more readily than do either the shrubs or the other trees.

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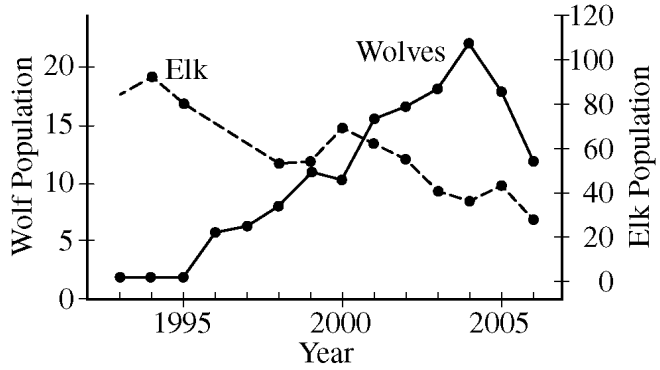

Figure 1. Wolf and Elk Population Sizes in Yellowstone National Park

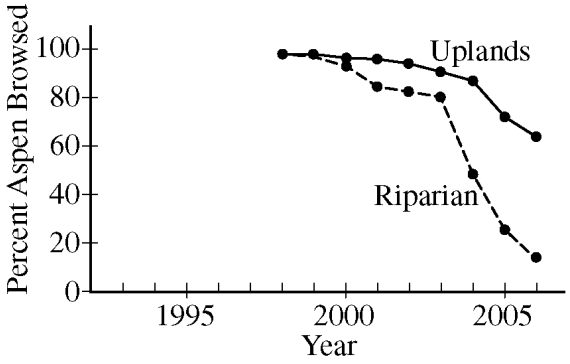

Figure 2. Browsing of Aspen in Yellowstone National Park

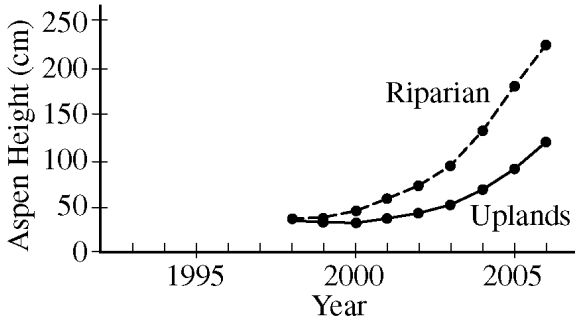

Figure 3. Growth of Aspen in Yellowstone National Park

Wolves, a top predator, were reintroduced to Yellowstone National Park in 1995 after a 50-year absence. In a multiyear study, the numbers of wolves and elk were monitored. The data are shown in Figure 1. In two different environments scientists monitored the percent of aspen trees browsed by herbivores (Figure 2) as well as the growth of the trees (Figure 3). The upland environments consist mostly of flat forested areas. The riparian environments are areas along streams with steep, wooded banks.

13. Based on the data, which of the following behaviors in elk could account for the differences between the percent of aspens browsed by herbivores and the height of aspen trees in riparian and upland environments?

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(A) Elk tend to avoid riparian areas where the steep, wooded riverbanks make it difficult to escape predators. ✓

(B) Elk tend to prefer riparian areas where there is easy access to water.

(C) Elk tend to avoid upland areas where trees are too tall to be easily eaten.

(D) Elk tend to prefer upland areas where there are richer sources of mineral nutrients, such as potassium and iodine.

14. Based on the data, which of the following is the best explanation for the changes in the elk population size between 2000 and 2005 ?

(A) The heights of aspen trees increased during that time period.

(B) Predation by wolves was higher than before 1995. ✓

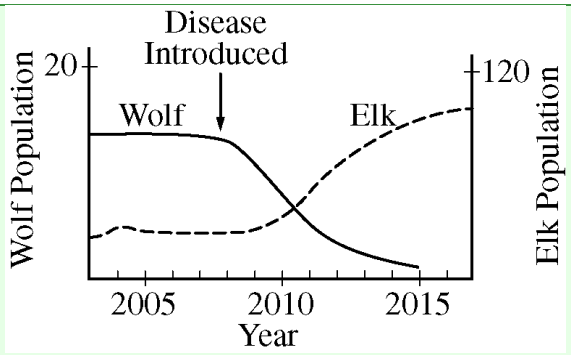
(C) The numbers of aspen trees increased during that time period.

(D) Wolf populations increased more rapidly in the upland areas.

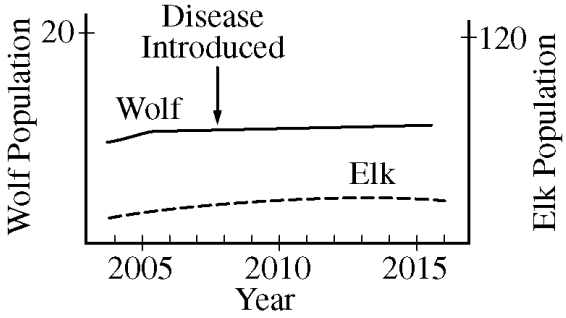
15. Which of the following models best predicts the effect on the ecosystem if a deadly communicable disease is introduced into the wolf population?

8 Ecology MCQ

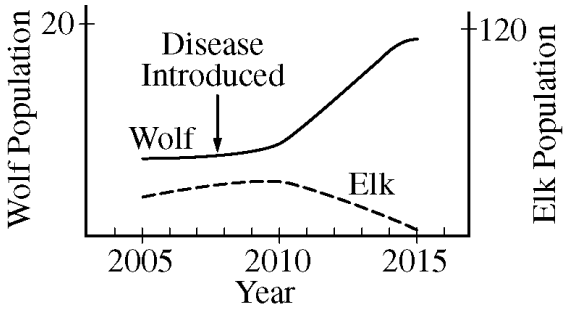
(A)



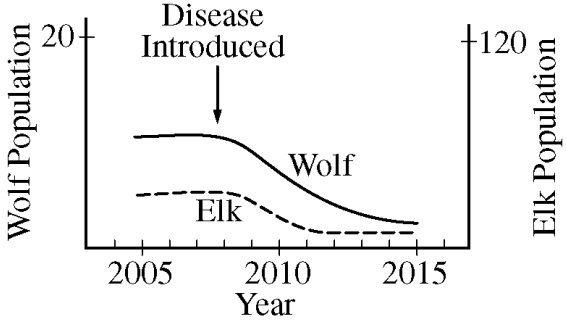
(B)



(C)



(D)



16. Which of the following predictions about the community is most likely true?

(A) A decrease in the elk population will cause wolves to feed on aspen trees.

(B) An increase in the wolf population will lead to a decrease in aspen growth.

(C) An increase in the growth of aspens will lead to a decrease in the wolf population.

(D) A decrease in the wolf population will lead to a decrease in the mean aspen height.

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In an investigation of interspecies competition, researchers grew the unicellular protozoan *Paramecium aurelia* in a 5 mL culture and *Paramecium caudatum* in a separate 5 mL culture. *P. aurelia and P. caudatum* were grown together in a third 5 mL culture. Each day a small sample of each culture was removed so the total number of individuals could be estimated, and the remainder of the population was transferred to fresh growth medium. The experimental results are represented in the graphs below.

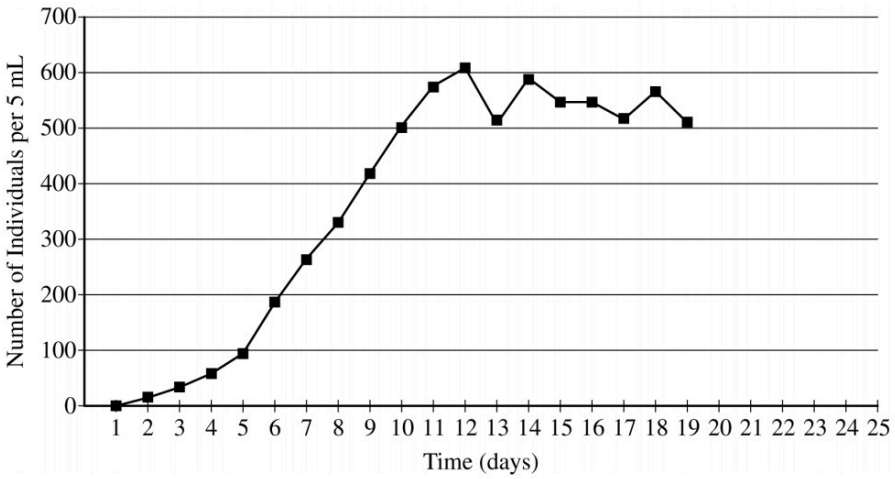


Figure 1. Population of *P. aurelia* cultured alone

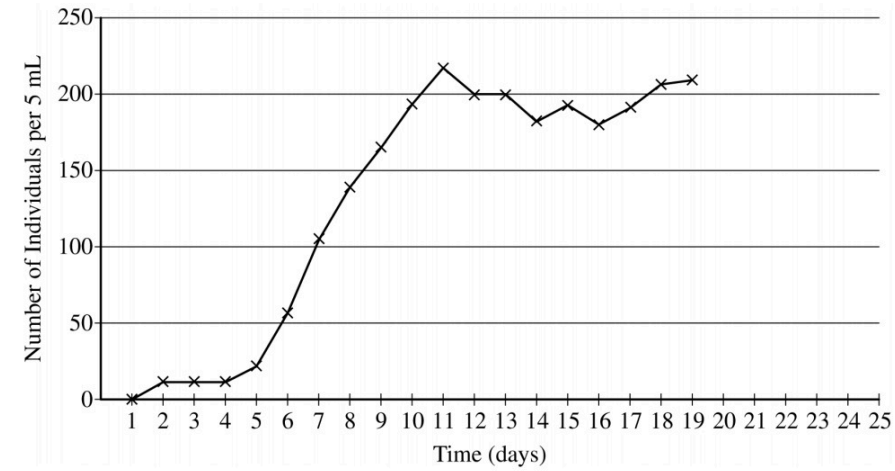


Figure 2. Population of *P. caudatum* cultured alone

8 Ecology MCQ

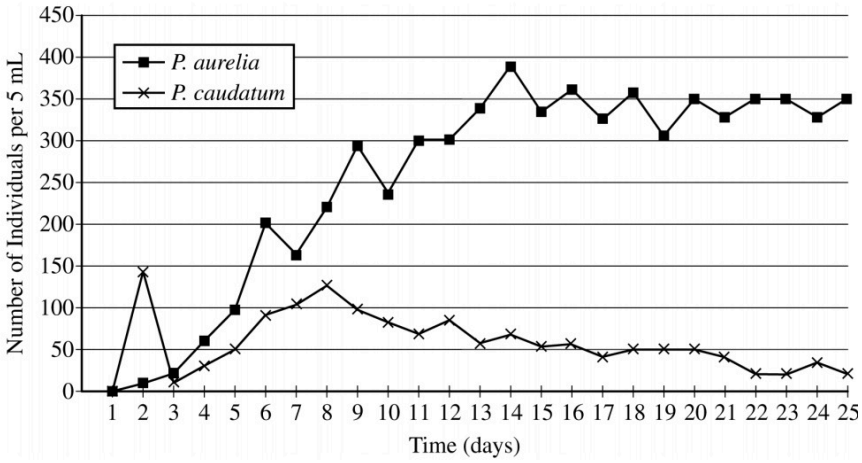


Figure 3. Populations of *P. aurelia* and *P. caudatum* cultured together

17. Based on the experimental results, which of the following statements best describes the relationship of the two populations that were studied in the investigation?

(A) The populations have a parasitic relationship in which one species requires the presence of the other to reproduce.

(B) The populations have a symbiotic relationship in which both species require the presence of the other to reproduce.

(C) The populations have a competitive relationship in which each species limits the ability of the other to reproduce. ✓

(D) The populations have a predator/prey relationship in which one species positively influences the ability of the other to reproduce.

18. If the mixed populations were cultured for an additional 25 days, which of the following predictions would be consistent with the data?

(A) Both populations would stabilize at a small size.

(B) Both populations would stabilize at a large size.

(C) Both populations would be eliminated from the culture.

(D) One population would stabilize at a large size, and the other population would be eliminated from the culture. ✓

19. Which of the following conclusions is best supported by the results of the experiment?

(A) Neither population grows as large when cultured together as each does when cultured separately. ✓

(B) *P. caudatum* prefers different food when cultured with *P. aurelia* than when cultured separately.

(C) Competition for mates increases when the species are cultured together compared with when they are cultured separately.

(D) Individuals of the species *P. caudatum* are smaller when cultured together with *P. aurelia* than when cultured separately.

8 Ecology MCQ

20. Which of the following statements best justifies the use of the experimental results in an investigation of interspecies competition?

(A) When two populations use different sources of energy, one of the populations will thrive in a shared environment and the other will become extinct.

(B) When two populations use the same sources of energy, resource availability in a shared environment limits the growth of both populations. ✓

(C) Because the change in entropy of a system containing two different species is always positive, populations typically grow faster in shared environments.

(D) Because the free energy of a system containing two different species is always negative, populations that utilize similar resources cannot coexist in nature.

21. The difference in carrying capacity between the two species when cultured separately is closest to which of the following values?

(A) 0 individuals per 5.0 mL

(B) 60 individuals per 5.0 mL

(C) 150 individuals per 5.0 mL

(D) 350 individuals per 5.0 mL ✓

---

22. Bt corn varieties are genetically modified to produce a toxin that specifically kills corn borers, a type of insect that eats and damages corn crops. Insects that are not closely related to corn borers are not affected by the toxin. Figure 1 shows the increasing production of Bt corn varieties and the change in the use of broad-spectrum pesticides that were the primary means of killing corn borers before the invention of Bt corn.

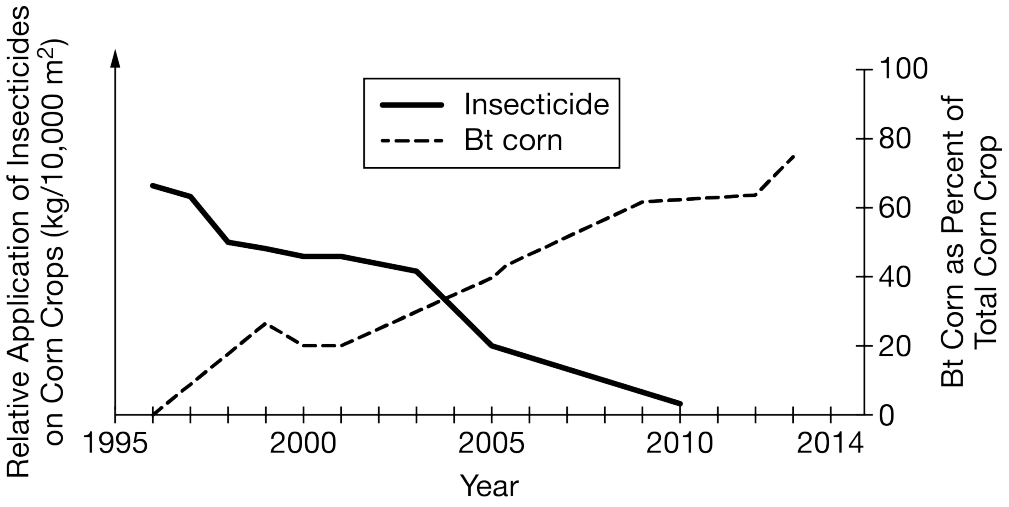


Figure 1. Relative use of pesticides on corn fields and percent of Bt corn in crops

How do the data in Figure 1 support the alternative hypothesis that increased use of Bt corn reduces the impact of corn farming on the natural environment?

8 Ecology MCQ

(A) The increased use of Bt corn will result in rapid mutation in corn borers.

(B) The toxin in Bt corn kills only the corn pests, leaving other insects unharmed. ✓

(C) Other insect species will replace corn borers and require additional applications of insecticides.

(D) Since Bt corn is not natural like non-Bt corn, it will not interact with the rest of the environment.

Answer B

Correct. Bt corn is modified to be toxic to certain insects; thus, it doesn't require as much insecticide. The graph shows this trend.

23. Paleoclimatic analysis has generated a claim that there is a possible correlation between an extinction event around 29 million years ago that allowed grasslands to become abundant and the adaptive radiation of several groups of butterflies.

Which of the following proposes the best alternative hypothesis for the claim?

(A) Climate analysis conducted by sampling pollen preserved in amber shows an increase in grass pollen 29 million years ago, but there is no significant difference between speciation rate, as well as species richness with regard to butterflies, before and after 29 million years ago.

(B) Climate analysis conducted by sampling pollen preserved in amber shows an increase in grass pollen 29 million years ago, which indicates there was an increase in grasslands that correlates to an increase in butterfly speciation during the same time period. ✓

(C) There is a correlation between the rapid and wide-ranging decrease in species richness of insects in general 29 million years ago and the increase in species richness and diversity among the butterfly population during the same time period.

(D) Evidence of rapid speciation among butterfly species 29 million years ago is evenly distributed among all environments of that period.

Answer B

Correct. Of the two conditions being studied, grassland distribution seems to correlate to butterfly species radiation. Saying the change in the butterfly speciation rate is due to the change in grassland is an alternative hypothesis.

24. In the year 2000, specimens of *Caulerpa taxifolia*,a green alga used in tropical aquariums, were found off the coast of California. Native to the Indian Ocean, *C. taxifolia* is known for aggressive growth and an ability to compete with sea grasses.It is currently on an international list of invasive species. Which of the following best predicts the consequences of the introduction of *C. taxifolia* to the California coast?

8 Ecology MCQ

(A) Without natural herbivores or competitors,*C. taxifolia* will grow rapidly and crowd out native species of producers. ✓

(B) *C. taxifolia* will have a hard time establishing itself because it will have to compete against native species of sea grasses, which are better adapted to the environment.

(C) *C. taxifolia* will grow rapidly, leading to an increase in the diversity of producers.

(D) Because it is not in the Indian Ocean, its natural environment, *C. taxifolia* will not be able to grow efficiently.

25. There is strong evidence that two species of cliff-dwelling birds have nested in the same environment and fed in the same water for thousands of years, with each having stable population sizes. Based on this information, which of the following statements is the best conclusion?

(A) One species is a better diver than the other species is.

(B) The two birds compete for food.

(C) One species will displace the other.

(D) The two birds have different ecological niches that allow coexistence. ✓

(E) One species eats a more diverse diet than the other does.

26. Which of the following communities is likely to be most stable?

(A) One where a keystone predator has been removed

(B) One with uniformly spaced vegetation

(C) One with high species diversity ✓

(D) One that lacks decomposers

8 Ecology MCQ

27. Researchers studied the diversity of macroscopic invertebrates in different areas of watershed ecosystems. They collected samples from 58 sites along 11 streams in the same geographical area. Their data were classified by stream type and can be found in Figure 1. The stream-type classifications range from the perennial stream type, which always has water, to the intermittent stream type, which is dry for more than nine months during the year.

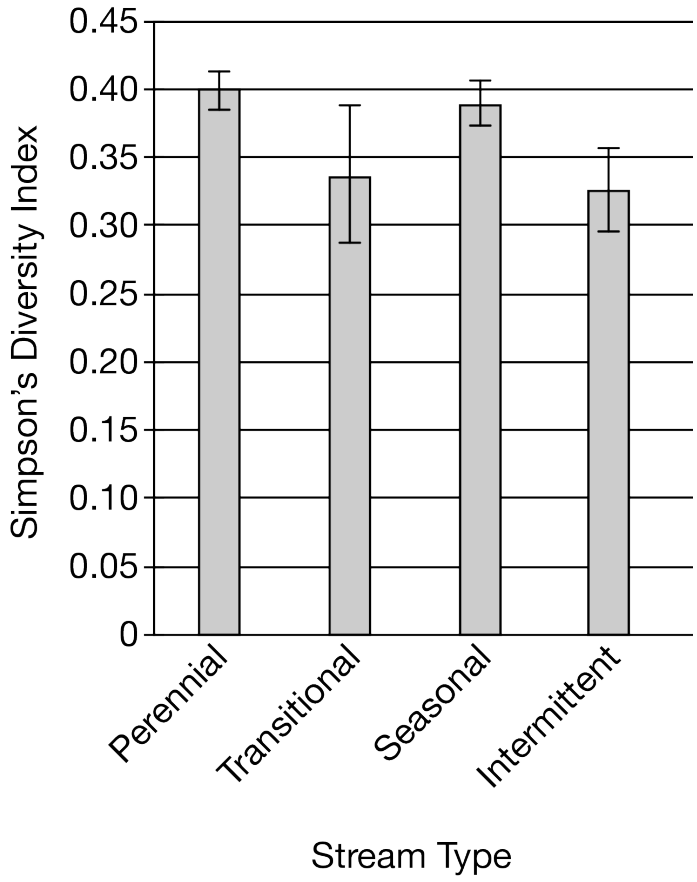


Figure 1. Diversity of macroscopic invertebrates in different stream types. Error bars represent a 95% confidence interval (± two Standard Errors from the mean).

Which of the stream types had a statistically lower macroscopic invertebrate species diversity than the seasonal streams?

(A) Transitional streams only

(B) Intermittent streams only ✓

(C) Perennial and intermittent streams only

(D) Transitional and intermittent streams only

8 Ecology MCQ

Answer B

Correct. In order to determine whether two sample means are statistically significant, the standard errors of the means $(SE_{\overline{x}})$ must be considered. The 95% confidence interval for intermittent streams does not overlap with the 95% confidence interval for seasonal streams.

8 Ecology MCQ

The coral species *Acropora millepora* is a marine animal that has a symbiotic relationship with algae species in the genus *Symbiodinium*. An increase in the concentration of dissolved CO_2 in oceans has a negative impact on this relationship and the health of the coral species. Scientists investigated the effect of CO_2 concentrations on the number of algae cells found in the coral (Figure 1) and on both the gross (total) and net rates of photosynthesis of those algae (Figure 2). The net rate of photosynthesis is equal to the gross rate of photosynthesis minus the rate of respiration by the algae.

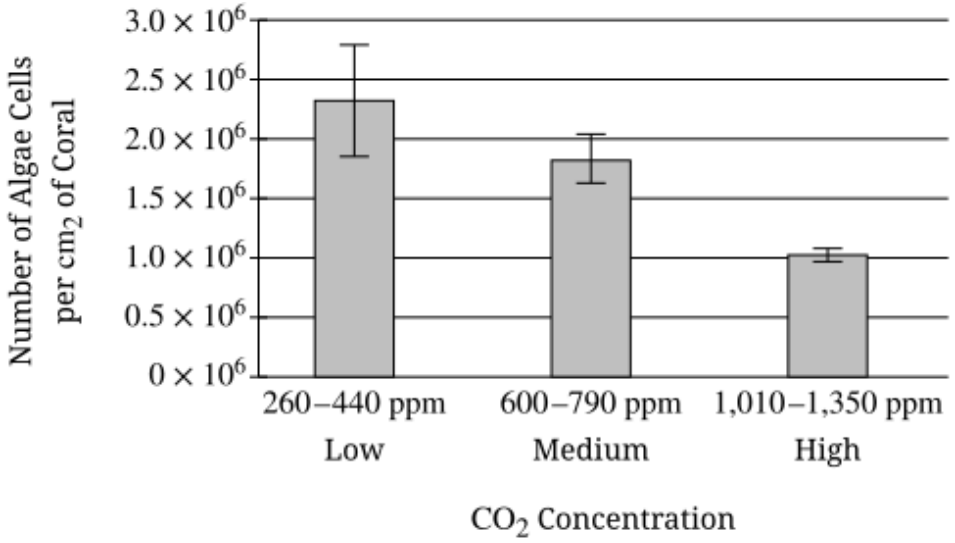


Figure 1. Effect of CO_2 concentration on the number of algae per cm^2 of coral. Error bars represent $\pm 2SE_{\bar{x}}$.

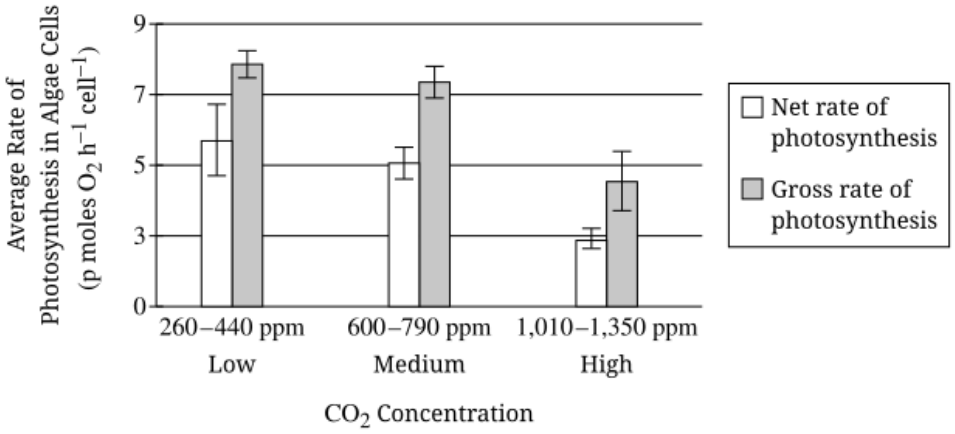


Figure 2. Average gross and net rates of photosynthesis in algae under various concentrations of CO_2. Error bars represent $\pm 2SE_{\bar{x}}$.

28. Which of the following best explains the symbiotic relationship between the coral and the algae?

8 Ecology MCQ

(A) The relationship is an example of commensalism because the algae add CO_2 to the oceans and the coral use CO_2.

(B) The relationship is an example of commensalism because the coral provide shelter for the algae and the algae provide chlorophyll to the coral.

(C) The relationship is an example of mutualism because the coral use sugars made by the algae and provide shelter for the algae. ✓

(D) The relationship is an example of mutualism because the algae perform photosynthesis for the coral and the coral perform respiration for the algae.

Answer C

Correct. Through photosynthesis, the algae produce sugars that the coral can use, and the coral provide living space for the algae. Because both species benefit from the interaction, this symbiosis is mutualism.

---

29. Based on Figure 1, which of the following values best represents the change in the number of algal cells at the high concentration of CO_2 as compared with the number at the low concentration of CO_2 ?

(A) 17.4% increase

(B) 43.5% increase

(C) 56.5% decrease ✓

(D) 65.2% decrease

Answer C

Correct. The percent change is calculated by $[(1.0 \times 10^6 - 2.3 \times 10^6)/2.3 \times 10^6] \times 100 = -56.5\%$. This indicates a 56.5% decrease.

8 Ecology MCQ

30. Interactions among populations determine how they access resources within a community. In Costa Rica, two different species of mites, *Rhinoseius colwelli* and *Rhinoseius richardsoni*, exhibit a high degree of interspecies competition. Both mite species live within the flowers of plants, and adult male mites tend to kill mites if they belong to a different species. An experiment was carried out to determine whether these aggressive interactions have resulted in niche partitioning by *R. colwelli* and *R. richardsoni*, of different flower species, with each mite species living in different plant species. The results are summarized in Figure 1.

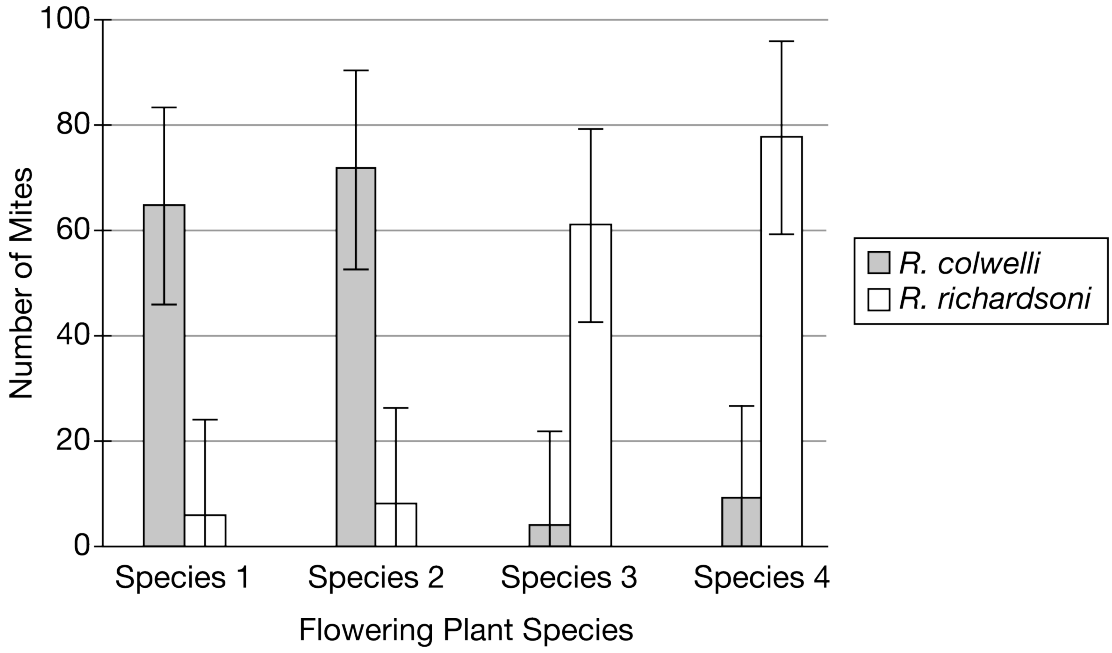


Figure 1. Mite Habitat Preferences

Which of the following is an accurate interpretation of the data in Figure 1 ?

(A) *R. richardsoni* is found significantly more frequently on Species 4 compared to Species 3.

(B) *R. colwelli* is found significantly more frequently on Species 1 compared to Species 2.

(C) *R. richardsoni* is found significantly more frequently on Species 3 and Species 4 compared to Species 1 and Species 2. ✓

(D) *R. colwelli* is found significantly more frequently on Species 3 and Species 4 compared to Species 1 and Species 2.

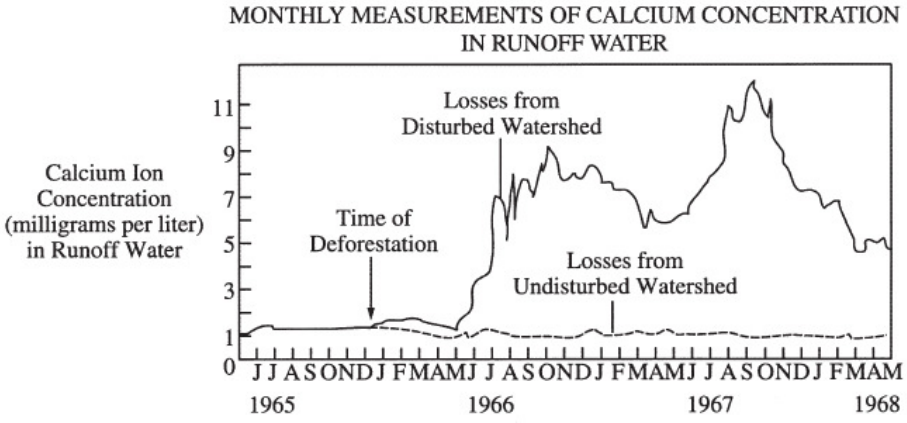
Answer C

Correct. The high degree of interspecific competition between these two interacting populations of mites has influenced community structure. Specifically, the aggressive adult male mites have promoted species-specific separation of mites on to flowers of different plant species. In general within this community, *R. richardsoni* is restricted to Species 3 and Species 4 and *R. colwelli* is restricted to Species

8 Ecology MCQ

1 and Species 2.

31. In an experiment on the effects of deforestation of a portion of a northern temperate deciduous forest, all the trees were removed from a small mountainside watershed area and herbicides were applied for three years to prevent regrowth. Measurements of the concentration of calcium (an important nutrient) in runoff water were taken both before and after the deforestation. Monthly measurements also were made in a similar watershed in another part of the same forest that had been left undisturbed. The findings are displayed in the graph below.



Which of the following is best supported by the data?

(A) Deforestation does not change the amount of water flowing through a watershed.

(B) The time of year during which deforestation occurs has no impact on the negative effects related to it.

(C) The loss of calcium affects several other variables in the forest ecosystem.

(D) Calcium loss can be rated as the most damag-ing aspect of mountainside deforestation.

(E) The presence or absence of plants greatly affects the loss of calcium from the soil. ✓

8 Ecology MCQ

32. Climate change can affect oceanic ecosystems and their food webs. In the East China Sea (ECS), three major climactic events were considered:

the Pacific Decadal Oscillation (PDO),
the East Asia winter monsoon (EAWM), and
the El Niño Southern Oscillation.

Figure 1 shows the relationships among these climactic events, surface sea temperature (SST), amount of phytoplankton, and copepod abundance. Copepods are important primary consumers in this ecosystem, and many species of fish depend on copepods as a food source. Solid arrows on the diagram indicate a statistical effect. The thicker the arrow, the greater the effect. Dashed arrows indicate statistically insignificant relationships.

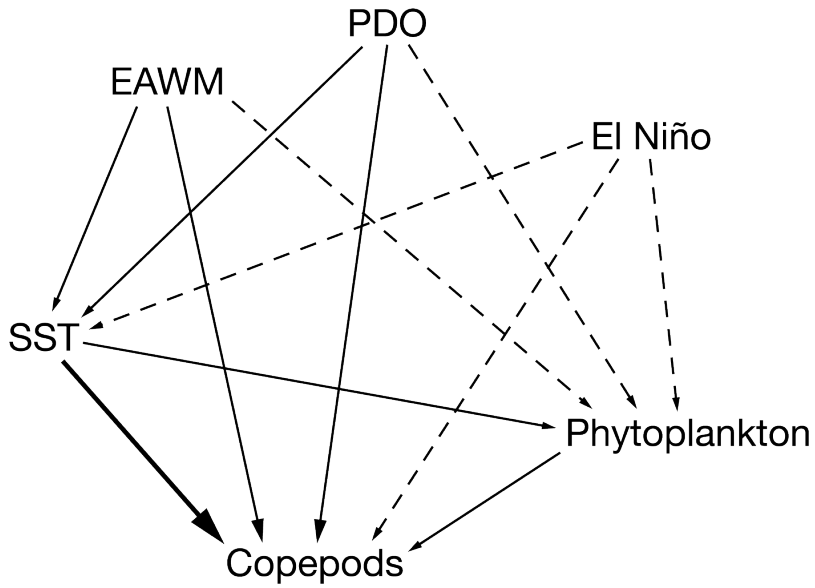


Figure 1. Structural equation model deciphering interactions among climate, phytoplankton, and copepods in the southern East China Sea. Solid arrows indicate statistical effects, with the thickness of the arrow reflecting the relative effect. Dashed arrows indicate inconsequential effects.

What do the data in Figure 1 suggest about a null hypothesis that the East China Sea ecosystem is unaffected by climactic events?

(A) The null hypothesis is rejected since there are statistical effects of EAWM and PDO on surface sea temperature and copepod abundance. ✓

(B) The null hypothesis is rejected because the El Niño Southern Oscillation has a statistical impact on the abundance of copepods in the ECS.

(C) The null hypothesis is rejected because the EAWM and PDO do not have statistical effects on the abundance of copepods and phytoplankton in the ECS.

(D) The null hypothesis is rejected because the El Niño Southern Oscillation does not have a statistical impact on the abundance of copepods in the ECS.

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Answer A

Correct. The null hypothesis should be rejected because the data indicate that the EAWM and the PDO have significant effects on surface sea temperature as well as copepod abundance.

33. Figure 1 shows a food web that describes the relationships within a community containing nine species. Each circle represents a separate species. The size of the circle represents the population size of the species relative to other species in the same trophic level.

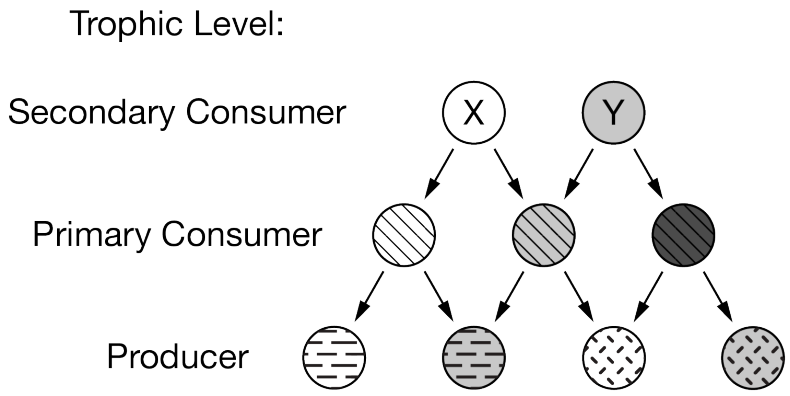
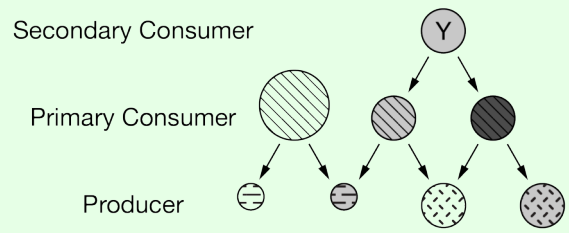


Figure 1. Trophic relationships among organisms in a community

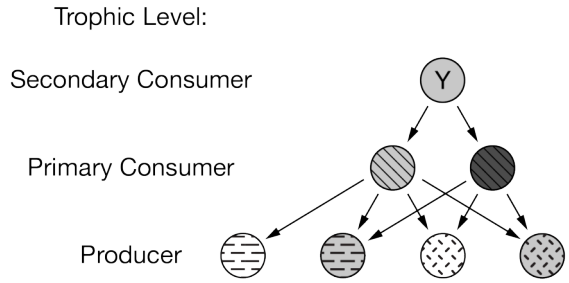
Based on the information in Figure 1, which of the following best predicts a short-term effect of removing secondary consumer X from the community?

8 Ecology MCQ

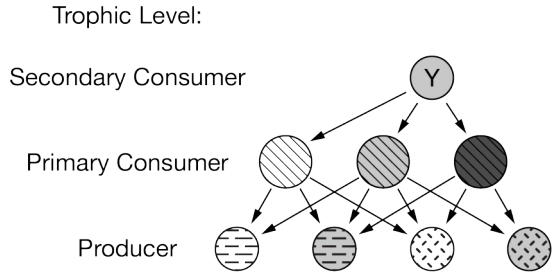
(A)



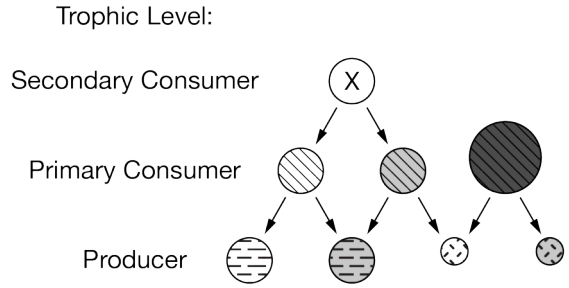
(B)



(C)



(D)



Answer A

Correct. If secondary consumer X is removed, then the leftmost primary consumer will increase in number. Consequently, the two producer species that this primary consumer eats will decrease in number. Note that disturbance events that remove a secondary consumer species can cause large shifts in community composition when the trophic web consists of only a few strong links instead of many weak links.

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34. Phytoplankton (dashed line) are important because they occupy the bottom of the food web absorbing CO_2, functioning as a carbon sink. A scientist suggests that an El Niño event in early 1998 caused a significant decrease in the abundance of phytoplankton that year in the equatorial Pacific Ocean.

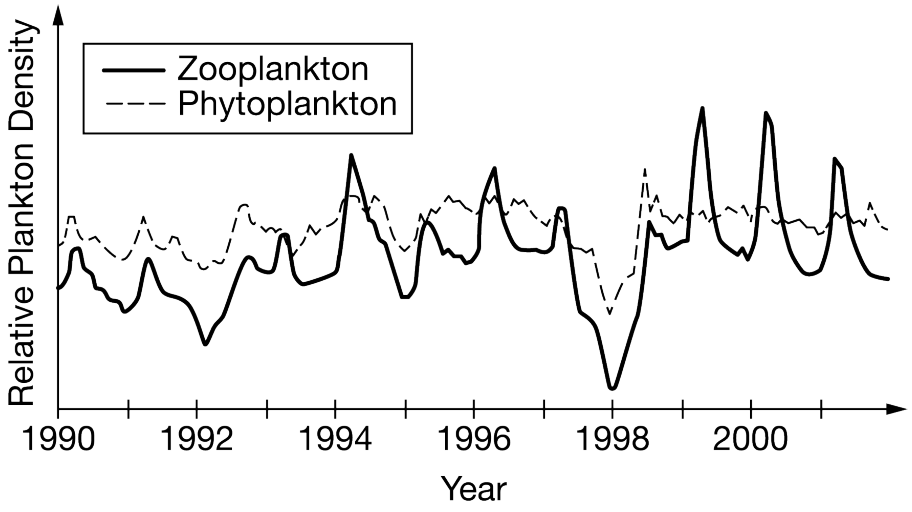


Figure 1. Relative abundance of plankton over time in the eastern equatorial Pacific Ocean

Which of the following observations would best support this alternative hypothesis?

(A) Previous dips in plankton abundance coincided with other El Niño events. ✓

(B) Primary consumers also experienced a dip in abundance in early 1998.

(C) After mid-1998, there is no correlation between phytoplankton and zooplankton abundance.

(D) CO_2 levels in surface waters increased significantly in this region during 1998.

Answer A

Correct. Observing that each dip in phytoplankton abundance occurred during an El Niño event is strong evidence of a cause-and-effect relationship between the two events.

35. The marine iguana, *Amblyrhynchus cristatus*, is endemic to the Galápagos Islands and is especially vulnerable to El Niño events. These iguanas feed on red and green algae. During an El Niño climate event, surface waters and their currents in the central and eastern Pacific Ocean become significantly warmer than usual. Also during El Niño years, land masses experience increased rainfall.

A researcher suggests that El Niño conditions cause a significant decrease in the size of the Galápagos marine iguana population. Which of the following would best support this alternative hypothesis?

8 Ecology MCQ

(A) Increased rainfall on land leads to increased nutrient runoff into the ocean and increased food resources for marina iguanas.

(B) Warm surface water contains fewer nutrients needed by red and green algae than cooler surface waters does. ✓

(C) Warm surface water shortens the incubation time of marine iguana embryos.

(D) Cool surface water decreases the metabolic rate of marine iguanas, so they need more food to compensate for the lowered metabolic rate.

Answer B

Correct. Upwelling brings cold, nutrient-rich waters to the surface. During El Niño years this is inhibited, so algae populations decrease because nutrients are limiting, resulting in decreased marine iguana populations.

In a laboratory population of fruit flies, body color is determined by a single gene with a wild-type allele associated with black body color and a mutant allele associated with yellow body color. A group of researchers hypothesizes that the mutation in fruit flies that causes the yellow body is sex-linked. They also hypothesize that this mutation results in a change in the structure of the sex combs, bristle-like structures on the front legs of male flies that facilitate mating.

To determine the inheritance pattern of the mutant allele, researchers crossed yellow-bodied females with wild-type males. Flies from the F_1 generation were then crossed. The distribution of phenotypes observed in the F_1 and F_2 generations is shown in Table 1.

TABLE 1. DISTRIBUTION OF PHENOTYPES OBSERVED IN THE P, F_1, and F_2 GENERATIONS

	Wild-Type Females*	Yellow-Bodied Females*	Wild-Type Males**	Yellow-Bodied Males**
P Generation	0	3	3	0
F_1 Generation	38	0	0	41
F_2 Generation	117	124	121	130

*Female fruit flies have two X chromosomes.

**Male fruit flies have one X chromosome and one Y chromosome.

8 Ecology MCQ

36. Which of the following best supports the claim that the fitness of yellow-bodied flies is lower than the fitness of wild-type flies?

(A) Males with the yellow-body phenotype are selected by females for mating more often than are wild-type males.

(B) Males with the yellow-body phenotype blend in with their environment and avoid detection by predators better than do wild-type males.

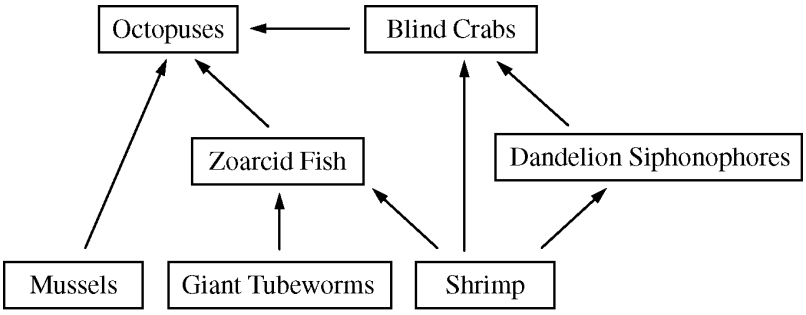
(C) Males with the yellow-body phenotype reproduce less but produce a greater number of viable offspring than do wild-type flies.

(D) Males with the yellow-body phenotype have weaker sex-comb bristles and decreased mating success compared with the bristles and mating success of wild-type males. ✓

Answer D

Correct. Evolutionary fitness is measured by an individual's or species' ability to successfully produce viable offspring. If yellow-bodied males have altered sex-comb bristles that reduce their reproductive success, they have lower fitness.

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



The food web above represents feeding relationships in a biological community near a deep-sea hydrothermal vent. Hydrothermal vents are geysers on the seafloor that gush super-heated, mineral-rich water. The seawater surrounding hydrothermal vents typically contains carbon dioxide (CO_2), molecular hydrogen (H_2), hydrogen sulfide (H_2S), and methane (CH_4). Sunlight, however, fails to reach the seafloor where deep-sea hydrothermal vents are located.

As part of an investigation, researchers collected living specimens from an area near a deep-sea hydrothermal vent. Mussels in the collection were found to be dependent on molecular hydrogen in seawater. Also, the researchers discovered multiple species of bacteria living in the gills of the mussels. Mussels use gills for filter-feeding and gas exchange with the surrounding seawater. On the basis of their experimental results, the researchers hypothesized that some bacteria living in the gills of the mussels are capable of chemosynthesis.

8 Ecology MCQ

37. On the basis of the food web, which of the following members of a deep-sea biological community is most likely to also have a symbiotic relationship with chemosynthetic organisms?

(A) Octopuses

(B) Blind crabs

(C) Zoarcid fish

(D) Shrimp ✓

38. To refine their model of deep-sea biological communities, the researchers investigated areas of the seafloor that are distant from any active hydrothermal vents. Which of the following is the best interpretation of the observation that some octopus species are only found near active hydrothermal vents?

(A) Octopuses depend indirectly on inorganic compounds in the seawater surrounding hydrothermal vents. ✓

(B) Organisms capable of moving across the seafloor are more difficult to observe than are organisms that are rooted in place.

(C) Genetic bottleneck events in isolated areas of the deep ocean adversely affect some octopus populations but not others.

(D) Biomass in a typical marine food web is expected to be greater in higher trophic levels than in lower trophic levels.

39. Based on an analysis of the food web, an observation that deep-sea bacteria consume molecular hydrogen (H_2) is most relevant to resolving which of the following apparent contradictions?

(A) Water gushing from deep-sea hydrothermal vents can be as hot as 400°C, which is a lethal temperature for most organisms.

(B) Some deep-sea organisms appear to be primary consumers, but no plants live near the hydrothermal vents. ✓

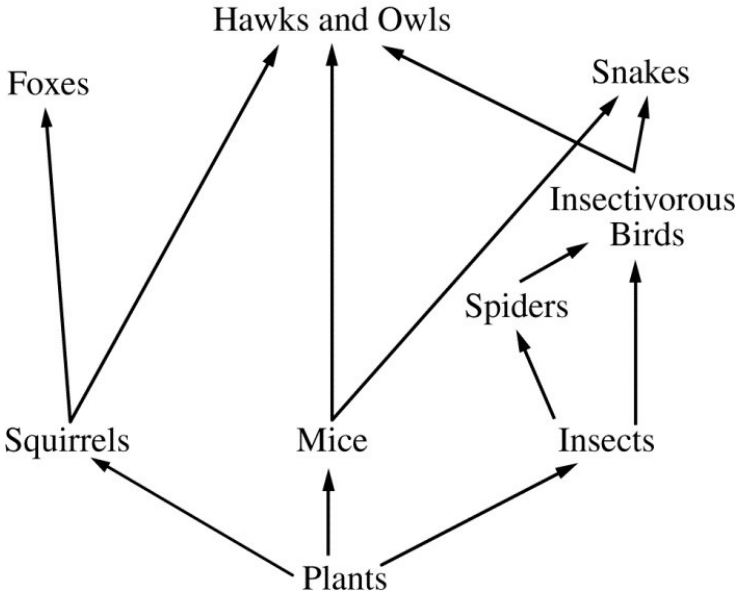
(C) Zoarcid fish are thought to be aggressive predators, but they are frequently described as being lazy swimmers.

(D) Some tissues of tubeworms contain hemoglobin, which is an oxygen-carrying molecule, but there is little free oxygen at this depth.

8 Ecology MCQ

40.

FOOD WEB IN A PARTICULAR MEADOW COMMUNITY



Which of the following changes to the environment will most likely lead to more energy entering the meadow community represented above?

(A) Increasing the number of nesting sites for hawks and owls

(B) Removing squirrels from the area

(C) Increasing the light available to the plants ✓

(D) Applying a chemical pesticide that is specific for spiders

41. The fossils in a layer of rock are significantly different from those in the rock layer directly above it. One hypothesis to explain the difference is that a major extinction event occurred at the point in time represented by the transition between the two rock layers.

Which of the following is the best plan for collecting data to use in a test of the hypothesis?

(A) Examining the upper part of the lower rock layer for evidence of a catastrophic event, such as the presence of shocked quartz, iridium, or soot particles ✓

(B) Digging farther into the hillside to look for a section of sedimentary rock that has a third layer below the two layers that were already identified

(C) Comparing the genomes of organisms from different evolutionary lineages to estimate the relative geologic age of the upper rock layer

(D) Searching fossil collections in several museums for examples of the types of features once shared among organisms that are now extinct

8 Ecology MCQ

42. Fur color is a genetically controlled trait in mice. A population of a species of mouse that typically lives in habitats with light-colored soil was studied for several generations. The background color of the habitat of this population became darker after the second generation due to human activity. A researcher sampled approximately 120 mice in each generation and recorded the fur color phenotypes as shown in Figure 1.

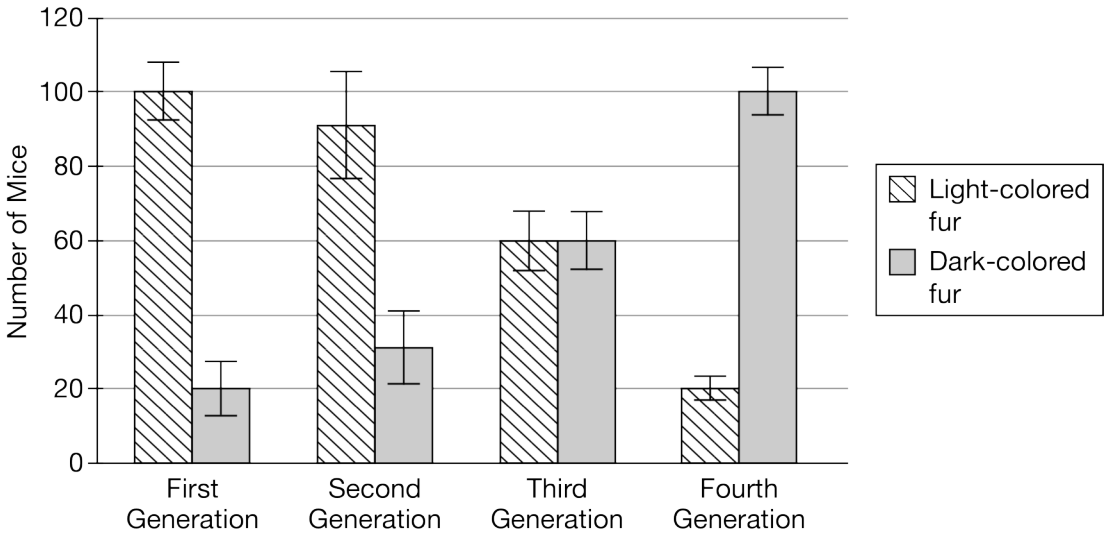


Figure 1. Fur color phenotype distribution of mice over four generations

The researcher claims that there is no selective advantage to fur color, so light and dark fur color phenotypes are present in similar frequencies in the fourth generation of this mouse population. Based on the data in Figure 1, which of the following best evaluates this null hypothesis?

(A) The null hypothesis is rejected, because the change to a darker background color led to an increase in the dark-colored fur phenotype. ✓

(B) The null hypothesis is rejected, because there is no significant difference between the frequencies of the two phenotypes right after the human-caused change to the habitat in the third generation.

(C) The null hypothesis cannot be rejected, because there is no significant difference between the frequencies of the two phenotypes right after the human-caused change to the habitat in the third generation.

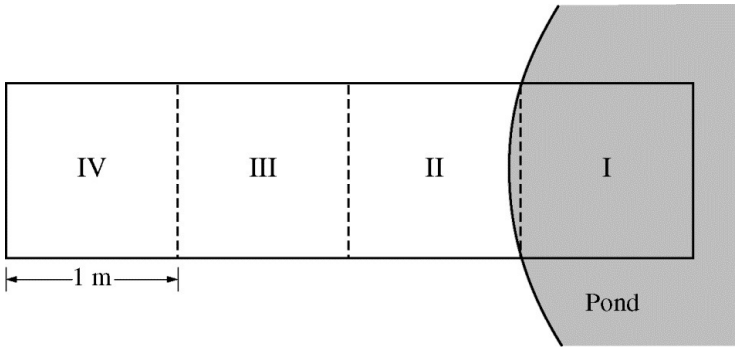
(D) The null hypothesis cannot be rejected, because the darker background color caused by the human-caused change to the habitat led to a decrease in the light fur color phenotype.

Answer A

Correct. The null hypothesis is rejected because there are significantly more dark-colored mice than light-colored mice in the fourth, and most recent, generation. This occurs because the mice with dark fur are more effectively camouflaged on the dark background created by the change in habitat background color.

8 Ecology MCQ

Researchers investigated the habitat preferences of two species of garter snakes, *Thamnophis sirtalis* and *Thamnophis atratus*. To create a choice chamber, the researchers built a meshed enclosure and positioned one end of the enclosure at the edge of a small pond. Zone I of the enclosure was located in the water, whereas zone IV of the enclosure was located 2–3 meters away from the water, as represented in the figure below. Snakes inside the enclosure were able to move freely between zones.



In a series of experiments, the researchers introduced a single snake into zone IV of the enclosure at 7:00 A.M. The researchers recorded the location of the snake at six time points throughout the day. In a related experiment, the researchers introduced two snakes, one of each species, into the enclosure at the same time and observed the location of each of the two snakes at the same six time points as before. The researchers repeated both the one-snake and two-snake experiments using different individual snakes of each species. The results are presented in the table.

ZONES MOST FREQUENTLY OCCUPIED BY GARTER SNAKES IN A MESHED ENCLOSURE

	Species Introduced Inside the Meshed Enclosure			
	T. atratus Only	*T. sirtalis* Only	*T. atratus* and *T. sirtalis*	
Time of Day	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*
8:00 A.M.	I	II	III	II
10:00 A.M.	II	I	II	II
12:00 P.M.	I	I	III	I
2:00 P.M.	I	I	IV	I
4:00 P.M.	II	I	IV	I
6:00 P.M.	II	II	IV	I

43. Which of the following additions to the experimental design will best help test whether the observed habitat preferences were the result of competition between species?

8 Ecology MCQ

(A) Placing two individuals from the same population together inside the enclosure ✓

(B) Introducing different types of plants into the enclosure together with the snakes

(C) Doubling the number of repetitions for the experimental treatment with *T. atratus* alone

(D) Repeating the investigation with an enclosure that is twice as long as the original

44. The researchers discovered that both species of garter snakes feed almost exclusively on amphibians found in or near the pond. Based on the results of the investigation, which of the following is most likely to occur if the amphibians become a limited source of food?

(A) Individuals of *T. atratus* will no longer prefer to be near the water.

(B) *T. sirtalis* will shift its food preference to herbivory.

(C) The reproduction rate of *T. sirtalis* will increase.

(D) The population size of *T. atratus* will decrease. ✓

45. Based on the data in the table, which of the following best predicts the results of a study in which natural populations of *T. atratus* and *T. sirtalis* are observed together in an environment that includes a freshwater pond?

(A) Neither *T. atratus* nor *T. sirtalis* will be observed near the water.

(B) *T. atratus* will be observed near the water whether or not *T. sirtalis* is present.

(C) *T. sirtalis* will be observed near the water whether or not *T. atratus* is present. ✓

(D) *T. atratus* and *T. sirtalis* will be observed together near the water.

46. Based on the data in the table, which of the following best describes the habitat preference of *T. atratus* when introduced alone inside the meshed enclosure?

(A) *T. atratus* exhibited equal preferences for all four zones.

(B) *T. atratus* exhibited a preference for the zones in or near water. ✓

(C) *T. atratus* exhibited a preference for the zones most distant from the water.

(D) *T. atratus* exhibited a preference to remain within the zone in which it was initially introduced.

8 Ecology MCQ

47. A study recorded changes in the distribution of some commercial marine species off the East Coast of the United States over a period of several decades. The results of the study are summarized in Figure 1.

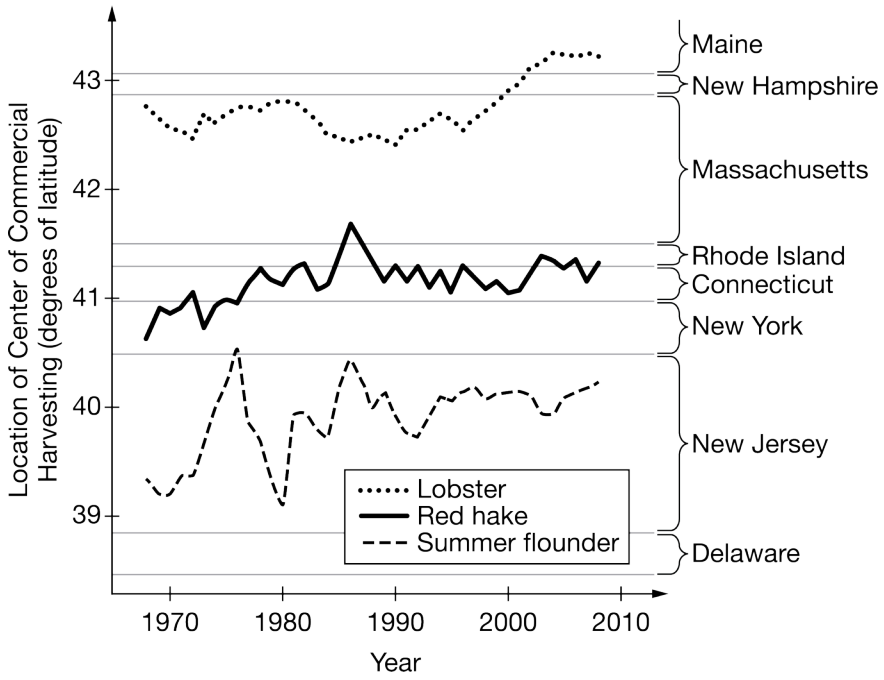


Figure 1. Latitude that is the center of commercial harvesting of specified marine species

Some researchers link changes in the distribution of these species to warming of the oceans due to climate change. Which of the following pieces of data best supports this alternative hypothesis?

(A) Decreased acidity of ocean water in the more southern latitudes caused a migration of these species northward.

(B) These species evolved new adaptations over time, allowing them to survive in warmer ocean environments.

(C) The prey of these species require colder waters to survive, and owing to warming in the oceans, they are moving farther north, to higher latitudes, where the water is colder. ✓

(D) Diseases devastated the prey of these species, forcing them to move northward as insufficient food was available in their original range.

48. The area covered by tropical rain forest is reduced by millions of hectares per year due to agriculture and logging. Which of the following best describes a likely result of tropical rain forest deforestation?

(A) Populations of plants and animals will decrease as more rain forest disappears, leading to a decrease in biodiversity. ✓

(B) An increase of soil moisture will lead to a rapid increase in new vegetation coverage.

(C) An increase in atmospheric carbon dioxide will lead to higher levels of ultraviolet radiation reaching the surface of Earth.

(D) More oxygen will be available to other organisms as plant numbers decrease.

8 Ecology MCQ

Answer A

This option is correct. It demonstrates an understanding of the human impact on ecosystems and correctly predicts that the loss of habitat from deforestation likely will lead to a loss of species diversity due to a decrease of available niches.

49. Henslow's sparrow is a songbird found in North American grasslands. The logging industry created vast meadow habitats for the sparrow in the late 1800s. Henslow's sparrow populations came to a peak in the mid-1900s. A recent survey of ten Midwest counties are shown in Figure 1.

Table 1. Year of Last Documented Observations of Henslow's Sparrow in Ten Selected Midwest Counties

County	Number of Confirmed Observations	Last Documented Observation
1	8	2007
2	1	2007
3	2	2007
4	2	2015
5	6	2015
6	3	2006
8	1	2004
9	4	2007
10	4	2007

Human activities can drive changes in ecosystems that affect the risk of extinction. An experiment is designed to help prevent the further decline and extinction of the Henslow's sparrow in which controlled burns will be set in the each of the ten Midwest counties in order to prevent the regrowth of the forest.

Which of the following represents an alternate hypothesis tested by this experiment with regard to the population sizes of Henslow's sparrows?

(A) There will be no difference in the populations observed in the controlled fire sites and the rest of counties 5 and 6.

(B) Population numbers will continue to decline even in the controlled fire sites because human activities always lead to extinction.

(C) Controlled fires in counties 6 and 8 will have no effect on the sparrow populations.

(D) Controlled fires in counties 4 and 5 will lead to an increase in the sparrow population. ✓

8 Ecology MCQ

Answer D

Correct. Counties 4 and 5 show evidence of existing sparrow populations, so the claim that controlled burns will affect these populations in an appropriate alternate hypothesis for this experiment.

50. Zebra mussels are an invasive species that has become widely established throughout the United States. Figure 1 shows the percent change in the population sizes of selected groups of organisms in the Hudson River since the introduction of zebra mussels. In this study chlorophyll-containing bacteria are considered phytoplankton and all other bacteria are considered bacterioplankton.

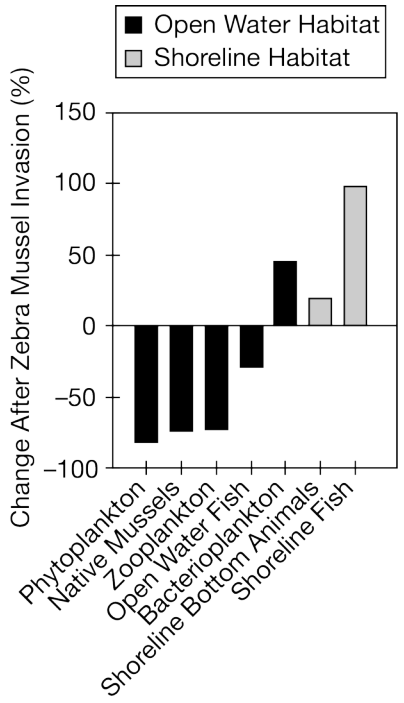


Figure 1. Change in biomass of selected Hudson River organisms after the introduction of zebra mussels

Which of the following hypotheses about the effect of zebra mussels on the Hudson River community is best supported by the data in Figure 1 ?

(A) Shoreline fish have replaced open water fish as top predators in a stable community.

(B) Zebra mussels have integrated into the existing community with little effect on the community stability.

(C) The reduction in primary productivity has resulted in an unstable community that will eventually collapse. ✓

(D) Bacterioplankton have replaced phytoplankton as the energy harvesters in the community.

8 Ecology MCQ

Answer C

Correct. Phytoplankton are primary producers that have experienced a large decline in number and have not been replaced by another primary producer. Thus, energy is not being harvested or distributed in the community, which contributes to instability and eventually the inability of the consumers to be supported.

Pacific salmon and black bears have often been cited as examples of keystone species. Pacific salmon spawn in freshwater streams but spend most of their lives at sea. When mature salmon return to the freshwater streams to spawn, they are preyed upon by bears and other predators. When salmon migrate from their marine habitat to the freshwater streams, they bring nitrogen and other marine-derived nutrients that subsequently remain in the areas surrounding the streams—a process called nitrogen influx.

In an investigation, the relationship between black bears, salmon, and influx of marine nitrogen into the area around a southwestern Alaskan stream was studied. The investigators established several test plots of the same size along the stream with the following species composition: no salmon or black bears (N), bears but not salmon (B), salmon but not bears (S), and a plot where salmon and bears interact (SB). Nitrogen influx in the different sampling areas was measured as a means of assessing the impact of the different species on the health of the ecosystem. The data are plotted in Figure 1.

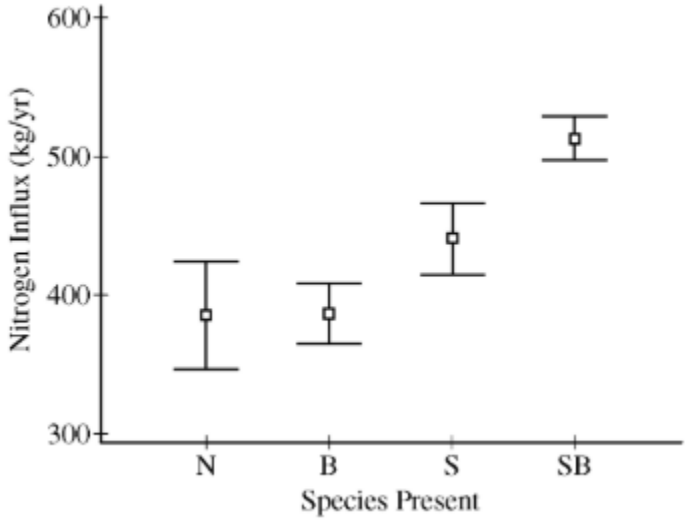


Figure 1. Mean nitrogen influx $\pm SE_{\bar{x}}$ as a function of species present. N = neither salmon nor bears present, B = only bears present, S = only salmon present and SB = both salmon and bears present.

51. If a dam is built downstream and prevents salmon migration to the test sites, which of the following most accurately predicts the impact on nitrogen influx?

8 Ecology MCQ

(A) Nitrogen influx will increase because the bears will no longer store nitrogen from the salmon.

(B) Nitrogen influx will decrease because there will be less bear-salmon interaction. ✓

(C) Nitrogen influx will remain stable because organisms other than bears and salmon will mobilize nitrogen in the environment.

(D) Nitrogen influx will increase because bears will no longer lose nitrogen to the salmon.

52. Which of the following statements is best supported by the data?

(A) Black bears are a keystone species.

(B) Salmon are a keystone species.

(C) Black bear urine is a significant source of nitrogen in the environment.

(D) The presence of black bears and salmon correlates with a significant increase in nitrogen influx. ✓

53. Which of the following most likely describes how the interaction between bears and salmon influences nitrogen dynamics in the environment?

(A) When bears consume salmon, they leave parts of the carcasses on the ground, which decompose, releasing nitrogen into the environment. ✓

(B) When salmon swim upstream to spawn, bears migrate to the rivers and deposit nitrogen from the forest with their excrement.

(C) Bears are a reservoir for nitrogen because they are the top consumer in the environment.

(D) Bears urinate in rivers and streams, which provides nitrogen for the salmon.

54. Which of the following pieces of additional data would help further investigate the relationship between bears, salmon, and influx of nitrogen into the local environment?

(A) The experimenters should remove the remains of salmon carcasses immediately after the salmon are eaten by the bears and determine the nitrogen content of the carcasses. ✓

(B) The experimenters should increase the number of bears in the area and measure the amount of nitrogen available for uptake by plants.

(C) The experimenters should set up a net to catch salmon before they enter the area and then measure nitrogen influx.

(D) The experimenters should clear-cut the trees from the area and determine how much nitrogen remains in the streams.

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55. Which of the following statements correctly describes a typical trend during the initial stages of succession in a terrestrial ecosystem?

8 Ecology MCQ

(A) Food webs become less complex.

(B) Diversity of plant species increases. ✓

(C) The amount of biomass and nonliving organic matter decreases.

(D) The litter layer becomes thinner because of increased assimilation of nutrients by shrubs and trees.

(E) The storage of inorganic nutrients within individual plants decreases.

The three-spined stickleback (Gasterosteus aculeatus) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mostly of individuals with pronounced pelvic spines, as shown in Figure 1. Individuals in freshwater stickleback populations, on the other hand, typically have reduced pelvic spines, as shown in Figure 2.

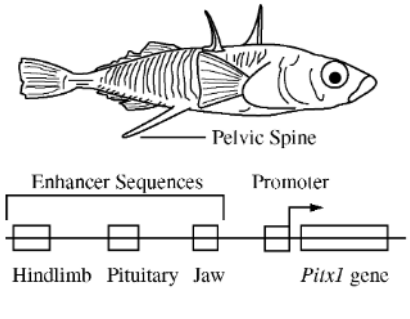

Figure 1. Marine stickleback

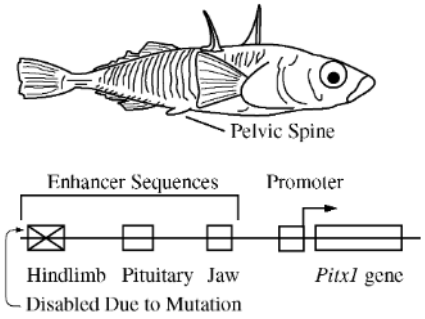

Figure 2. Freshwater stickleback

As represented in Figure 1 and Figure 2, the phenotypic difference between marine and freshwater sticklebacks involves Pitx1, a gene that influences the formation of the jaw, pituitary gland, and pelvic spine. Enhancer sequences upstream of the Pitx1 genetic locus regulate expression of the Pitx1 gene at the appropriate times and in the appropriate tissues during development. Previous studies have found that a mutation in the hindlimb enhancer interferes with the formation of a pronounced pelvic spine.

56. In sticklebacks, which of the following is most likely to occur if the jaw enhancer is disabled instead of the hindlimb enhancer?

(A) The jaw and a pronounced pelvic spine develop normally because the Pitx1 gene is expressed in both developing tissues.

(B) Neither the jaw nor a pronounced pelvic spine develop normally because there is no Pitx1 gene expression in either developing tissue.

(C) The jaw develops normally, but a pronounced pelvic spine does not develop because the Pitx1 gene is expressed in the developing jaw but not in the developing pelvis.

(D) The jaw does not develop normally, but a pronounced pelvic spine does develop because the Pitx1 gene is expressed in the developing pelvis but not in the developing jaw. ✓

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8 Ecology MCQ

57. Researchers were studying species diversity in a savanna rangeland community in southwest Queensland, Australia. Researchers counted the numbers of plants species in four different categories at two nearby sites within the savanna rangeland; one lightly grazed by herbivores and one heavily grazed by herbivores (Figure 1).

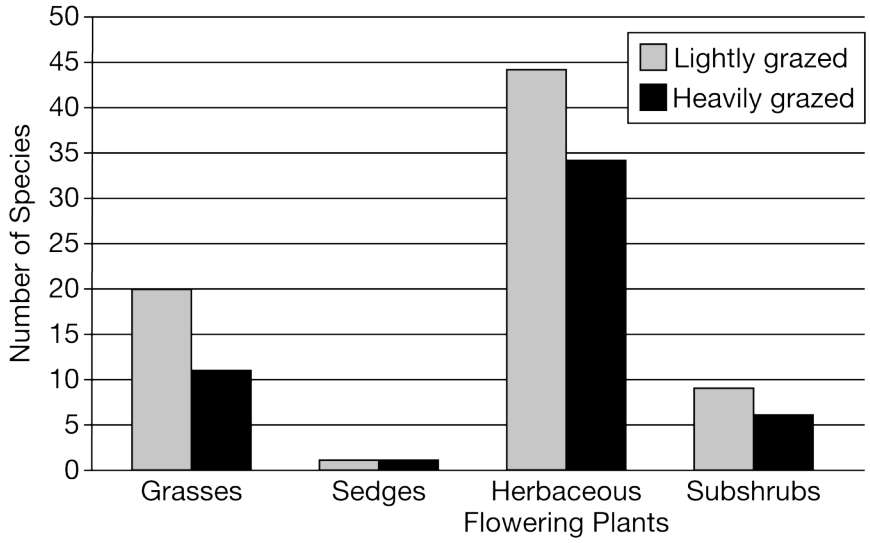


Figure 1. Plant species richness in lightly and heavily grazed savanna communities.

Which of the following best predicts the results of a drought in the savanna research area with regards to community resilience to this environmental change?

(A) The heavily grazed area is more likely to recover more quickly from the drought due to reduced competition between species.

(B) The lightly grazed area is likely to recover more quickly from the drought due to greater biodiversity. ✓

(C) The lightly and heavily grazed areas are likely to recover at equal rates due to the overlapping categories of plant species.

(D) The lightly and heavily grazed areas are likely to recover at equal rates due to the similar climates and locations.

Answer B

Correct. As shown in Figure 1, the lightly grazed area has a greater number of species of grasses, herbaceous flowering plants, and subshrubs. Higher levels of biodiversity in an ecosystem are associated with the greater ability of an ecosystem to respond to environmental changes. After a drought, some species will recover more quickly than others and the greater the biodiversity, the faster the ecosystem is likely to recover.

8 Ecology MCQ

58. The lionfish is a venomous fish found primarily in the Red Sea and the Indian Ocean. In the 1990s, lionfish were accidentally released into the Atlantic Ocean, where they found abundant resources and favorable environmental conditions. Which of the following scenarios is most likely to result in the lionfish having a major impact on the communities into which they were introduced?

(A) With no natural predators, the lionfish population will become very large. ✓

(B) Some native species of invertebrates will develop a resistance to lionfish venom.

(C) Random mating will allow the lionfish population to reach Hardy-Weinberg equilibrium.

(D) A virus that specifically infects lionfish will become more prevalent.

59. Scientific evidence has revealed five mass extinctions in the history of life on Earth.

Table 1. Five Mass Extinctions

Time period (million years ago)	Percent of species lost	Probable cause
65 mya	75%	Asteroid impact and volcanism
200 mya	75%	Asteroid impact and volcanism
250 mya	95%	Cold climate, formation of Pangea, Siberian volcanism
355 mya	70-80%	Global cooling
435 mya	22%	Global cooling

Some scientists propose that due to human activity and climate change, a sixth mass extinction has already begun. Select the best alternate hypothesis to support this claim.

(A) Current temperature changes will correlate with temperature changes during previous extinctions.

(B) Observed modern extinction rates will resemble the rates found early in previous extinctions. ✓

(C) The number of species present on Earth will equal the number of species on Earth before the last extinction.

(D) The number of species present on Earth will equal the number of species on Earth after the last extinction.

Answer B

Correct. By testing this hypothesis, we could establish a similarity to known extinctions. The null hypothesis would state that there is no correlation between the current extinction rate and those of previous extinctions.

8 Ecology MCQ

60. Scientists want to determine whether the rapid economic growth in China between 1950 and 2000 caused a mass extinction of animal species during that time period. A mass extinction event is considered to occur when the rate of species extinction far exceeds the background extinction rate. The scientists collected data on the number of extinctions of selected vertebrate specie in China between 1950 and 2000, as shown in Table 1. Extinction rates were ascertained by measuring the average percent of species lost (PSL).

Table 1. The average percent of extinctions of 252 protected vertebrate species in 2,365 counties in China from 1950 t 2000.

	Mammals	Birds	Amphibians and Reptiles	Background Rate
Average PSL ± 1 standard deviation	47.4 ± 33.59	19.75 ± 17.70	28.80 ± 28.86	27.21 ± 16.26

Which of the following is the null hypothesis most relevant for evaluating the data in Table 1 ?

(A) There is no significant difference between the extinction rate of animals between 1950 and 2000 in China and the background extinction rate. ✓

(B) There is no significant difference between the extinction rates of mammals, birds, amphibians, and reptiles in China.

(C) There is a significant difference between the extinction rate for mammals in China and the background extinction rate for mammals.

(D) Because of climate change, habitat loss, pollution, and excessive exploitation of resources, China experienced a mass extinction event between 1950 and 2000.

Answer A

Correct. The null hypothesis is that there is no significant difference between two groups of data. In this experiment, the null hypothesis would therefore be that there is no difference between the rates of extinction between 1950 and 2000 in China and the background rate of extinction.

8 Ecology MCQ

61. Potatoes as a food crop originated in South America. There are over 4,000 variations of potato grown there. Potatoes were brought to Europe by Spanish colonists in the late 1500s. One variety became the chief crop grown in much of Ireland by the early 1800s. Table 1 summarizes the spread of potatoes as a food crop.

Table 1. Spread of potatoes as food crop from origin in South America

Time Period	Location	Number of Potato Varieties Grown (approximate)
Pre–1567	South America	4,000
1567–1800	Europe	12
1840–1850	Ireland	1

The Irish Lumper variety of potato was introduced into Ireland in the early 1800s. The Lumper soon became the only variety of potato grown in much of Ireland. The potato blight destroyed most of the Lumper crop between 1845 and 1852 resulting in the death of about 1 million Irish from starvation.

A researcher suggests that the Irish monoculture (growing a single crop) of the Lumper potato did not significantly change the overall resistance of the potato crop in Ireland to diseases. Which of the following observations best refutes this null hypothesis?

(A) Before the arrival of the potato blight, there were already two other potato plant diseases in Ireland.

(B) The number of fatalities from the Irish potato famine was greater than the rest of the European countries combined. ✓

(C) During the early nineteenth century, the vast majority of people in Ireland depended on the Lumper potato for food.

(D) Growing more varieties of potatoes increases competition for available resources, which may yield fewer potatoes.

Answer B

Correct. Table 1 shows that twelve different varieties of potatoes were grown across Europe, as compared to the Lumper variety in Ireland. This suggests that the European potato crop, with more varieties, was less affected by the potato blight. This observation would support the alternative hypothesis that the variability of potato crops did affect the impact of the potato famine in Europe compared to Ireland.

62. Which of the following observations best represents a mutualistic relationship?

8 Ecology MCQ

(A) Some bacteria and fungi obtain their nutrients by enzymatically digesting larger organisms.

(B) Chloroplasts in green algae provide sugars for use in cellular metabolism.

(C) Hermit crabs utilize empty sea snail shells to protect themselves from predators.

(D) Flowers produce nectar that bees gather to make honey, and in the process the bees pollinate the flowers. ✓

63. Sea otters living along the Pacific coast were hunted to near extinction in the nineteenth century. After being protected from hunting in the early 1900s, a remnant population of otters near Adak Island, Alaska, recovered rapidly. Otters did not return to the environmentally similar nearby island, Alaid Island. Sea otters eat sea urchins, which eat kelp, a brown alga. Researchers surveyed both islands in 1988, to measure sea urchin biomass and kelp density. The data are presented in Table 1.

Table 1. Kelp and sea urchin data from two Alaskan islands in 1988

	Alaid Island	Adak Island
Understory Kelp Density (Number $\cdot 0.25\mathrm{m}^{-2}$)	0.23	6.56
Sea Urchin Biomass ($\mathrm{g} \cdot 0.25\mathrm{m}^{-2}$)	336.4	38.4

In 1991, researchers at Adak Island observed the first attack by a killer whale on a sea otter in historical times. The researchers hypothesized that the population sizes of the larger marine mammals that the killer whales normally prey on declined, so the killer whales were starting to prey on different prey, including the smaller sea otters.

Which of the following best predicts the effects of an increase in killer whale predation on sea otters on the Adak Island ecosystem?

(A) Both the sea urchin population and the kelp population will increase, leading to greater total biomass and diversity in the Adak Island ecosystem.

(B) Both the sea urchin population and the kelp population will decrease, leading to reduced total biomass and diversity in the Adak Island ecosystem.

(C) The sea urchin population will increase and the kelp population will decrease, leading to greater total biomass and diversity in the Adak Island ecosystem.

(D) The sea urchin population will increase and the kelp population will decrease, leading to reduced total biomass and diversity in the Adak Island ecosystem. ✓

Answer D

Correct. Increased predation by killer whales will lead to a decrease in the sea otter population at Adak Island. A decline in the sea otter population will allow the sea urchin population to increase without predation as a limiting factor. An increasing sea urchin population will lead to overgrazing and reduced kelp densities. Kelp forests provide important habitat and food for many other organisms, and their decline would cause the entire ecosystem to decline. Due to this cascade effect, sea otters are described as a keystone species because when they are removed from an ecosystem, the ecosystem declines

rapidly.

64.

TABLE 1. OZONE DAMAGE IN SIX PLANT SPECIES

Species	Number of Plants Observed	Number of Damaged Plants	Percent of Plants with Damage
Ash	544	41	7.5
Cherry	30	2	6.7
Dogwood	41	2	4.9
Sassafras	132	3	2.3
Sweet gum	170	2	1.2
Wild grape	124	20	16.1

To investigate the effects of ground-level ozone pollution on different species of plants, researchers recorded the number of plants in a certain area that had ozone-induced leaf damage. Based on the results of the survey shown in Table 1, which of the following plant species was most affected by ozone pollution?

(A) Ash

(B) Cherry

(C) Sweet gum

(D) Wild grape ✓

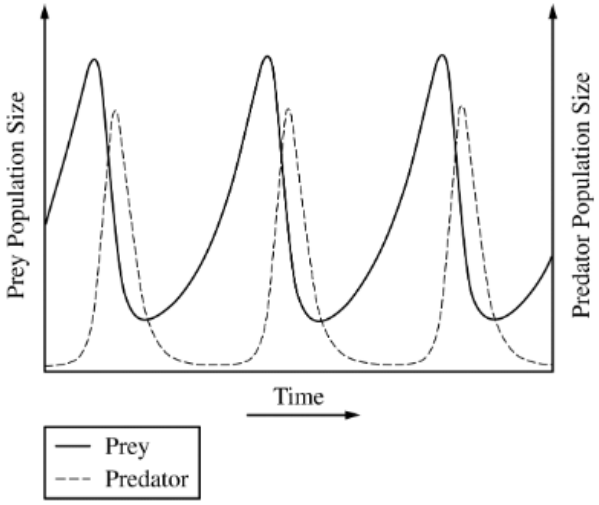
Answer D

Correct. Wild grape plants had the second-highest number of damaged plants observed. However, when calculated as a percentage of total plants observed, wild grape plants had the highest percentage of plants

8 Ecology MCQ

damaged.

65. The graph below shows a model of population changes over time in a predator and its prey.



Which of the following data would be most helpful in predicting possible disruptions to the relationship between the predator and its prey?

(A) Time of day, because as days become shorter, nighttime hunting increases

(B) Average birth weight of predators, because heavier offspring will begin hunting at younger ages

(C) Numbers of additional predator species, because additional predators will increase competition for prey ✓

(D) Prey metabolic rates, because predators are less successful against prey with higher metabolic rates

8 Ecology MCQ

66.

	Diet	
	Before Change	After Change
Principal diet	Hay	Grain
Main type of carbohydrate supplied by diet	Cellulose	Starch
Abundance of *Streptococcus bovis* in the rumen	Low	High
Abundance of *Ruminococcus albus* in the rumen	High	Low
pH of rumen fluid	6 – 7	< 5.6
Clinical warning signs in cattle	None	Weakness

Researchers conducted a controlled experiment to investigate the effect of diet on the health of beef cattle. The initial hypothesis was that an abrupt change in diet will benefit beef cattle by reducing the sizes of bacterial populations living in the digestive systems of the cattle.

In the experiment, the researchers determined the relative abundance of two bacterial species found in the rumen of cattle. The rumen is a part of the stomach that acts as a fermentation chamber in cattle and other ruminants. Shown in the table are the results from before and after an abrupt change in the cattle's diet. Based on the results, which of the following best explains why the initial hypothesis should be revised?

(A) The diets were too similar, since cellulose and starch are both carbohydrates obtained from eating plants.

(B) The abundance of one of the bacterial species increased in response to the change in diet. ✓

(C) The change in the diversity of microorganisms had no observable effect on cattle physiology.

(D) The ability of rumen fluid to resist changes in pH counteracted the changes in microorganism diversity.

8 Ecology MCQ

In the Florida Everglades, Burmese pythons are an invasive species. They were introduced into southern Florida in 1992. These pythons feed on many of the native Florida species, establishing the pythons as the top predator in the environment. By the year 2000, their population had increased dramatically.

Figures 1 and 2 display data collected by ecologists studying the results of the Burmese python invasion. Figure 1 shows counts of animals collected from nighttime road surveys in southern Florida, which are used to estimate population size. Figure 2 shows data collected from mosquitoes captured from the wild. DNA sequencing was used to identify the species of blood that the mosquitoes had in their stomachs, identifying various hosts used by the mosquitoes.

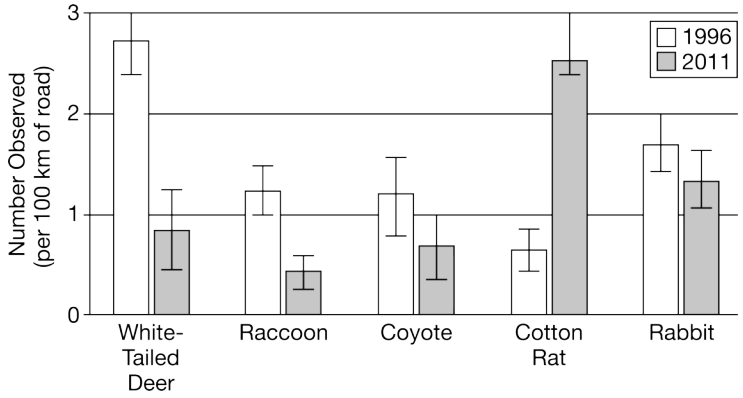


Figure 1. Comparison of observations of selected mammals in 1996 and 2011

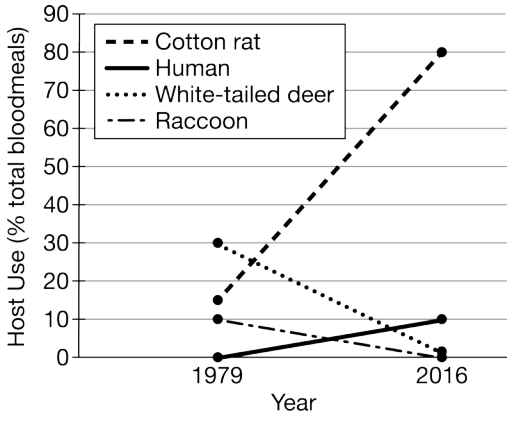


Figure 2. Change in host preference by *Culex cedecei* between 1979 and 2016. Numbers do not add up to one hundred percent because these represent a subset of all the host species.

In 1996, the native *Culex cedecei* mosquitoes in southern Florida preferentially took blood meals from white-tailed deer and raccoons. It was predicted that changes in host population size would alter these host preferences. Additionally, it is known that cotton rats are often infected by the Everglades virus, which normally exists in animals, but is capable of infecting humans. Ecologists predict that increased feeding on cotton rats by *C. cedecei* may significantly increase the tendency of this virus to infect humans.

8 Ecology MCQ

67. Which of the following most accurately explains an impact of Burmese pythons on the Everglades community in southern Florida using the data provided?

(A) Python parasitism has reduced the population of white-tailed deer and raccoons.

(B) Python predation has caused a trophic cascade that changed the mosquitoes' niche. ✓

(C) Pythons have a mutual symbiosis with cotton rats, allowing their population to increase.

(D) Intraspecific competition for food between white-tailed deer and raccoons has increased due to pythons.

Answer B

Correct. Since predation by invasive Burmese pythons has decreased the number of white-tailed deer and raccoons, the mosquitoes are forced to use alternative hosts, including cotton rats.

68. Which of the following populations have significantly decreased in size between 1996 and 2011 ?

(A) White-tailed deer only

(B) Cotton rats and rabbits only

(C) White-tailed deer and raccoons only ✓

(D) White-tailed deer, raccoons, and cotton rats

Answer C

Correct. The error bars do not overlap for the 1996 and 2011 population estimates for white-tailed deer and for raccoons. This indicates that there is a significant decrease between the 1996 and 2011 population sizes.

69. Based on the data, which of the following most accurately matches a potential change with its predicted effects on the southern Florida ecosystem?

(A) Introduction of a new top predator species that preys solely on Burmese pythons would cause an increase in the raccoon population. ✓

(B) If the population of Burmese pythons were to decrease, then the cotton rat population would increase.

(C) If more Burmese pythons are introduced in the Everglades, then mosquitoes will begin feeding on white-tailed deer frequently again.

(D) Introduction of a new snake species that competes with the Burmese python for food would lead to an increase in the white-tailed deer population.

8 Ecology MCQ

Answer A

Correct. If the Burmese python population were reduced, then there would be less predation on raccoons, and the raccoon population would be able to grow larger. The same would be true of the white-tailed deer population.

70. Scientists hypothesize that the invasive Burmese pythons are responsible for recent increases in human Everglades virus infections. Evaluate this alternate hypothesis using the data provided.

(A) The alternate hypothesis is supported because invasive Burmese pythons caused an increase in the mosquito population. This enabled them to transmit the Everglades virus at a higher frequency if they were infected.

(B) The alternate hypothesis is supported because the invasive Burmese pythons caused mosquitoes to switch to feeding on cotton rats. This means that mosquitoes are more likely to carry the Everglades virus when they bite humans. ✓

(C) The alternate hypothesis is refuted because humans are used as hosts for only a small percentage of mosquito feedings. Humans therefore will not have an increase in Everglades virus infection.

(D) The alternate hypothesis is refuted by the data because the population of cotton rats increased. Since there are more cotton rats, they will contain the Everglades virus and prevent it from infecting humans.

Answer B

Correct. As mosquitoes change hosts from white-tailed deer and raccoons to cotton rats, they will more frequently be carriers of the Everglades virus. Thus, interactions with humans are more likely to result in transmission of the Everglades virus.

8 Ecology MCQ

71. Soluble reactive phosphate (SRP) is a term used to describe the amount of phosphate available for uptake by living things. A study of river plants species and water phosphate levels was completed in Great Britain. Recently, in a section of one of the rivers studied, a population of fennel-leaved pondweed was replaced by spiked water-milfoil, even though these two plants are associated with different environmental conditions as indicated in the table below.

Table 1. Concentrations of phosphate and river plant correlations in Britain

SRP **Concentration** (mg/L)	**Plant Species Observed**
< 0.1	Bog pondweed *(Potamogeton polygonifolius)* and river water-crowfoot *(Ranunculus fluitans)*
$0.1 - 0.4$	Fennel-leaved pondweed *(Potamogeton pectinatus)*
0.4–1.0	Yellow water-lily *(Nuphar lutea)* and arrowhead *(Sagittarias agittifolia)*
> 1.0	Spiked water-milfoil *(Myriophyllum spicatum)*

A group of researchers suggest that the concentration of available phosphate in this particular section of the British river does not have a significant impact on which plant species grow. Alternatively, another group of researchers hypothesize that levels of available phosphate do affect which plant species grow at this location. Which of the following observations would best support this alternative hypothesis?

(A) Reforestation efforts in Britain near this section of the river have caused changes in the plant community structure.

(B) Increased fertilizer use near this section of the river has led to increased runoff of phosphates into the river. ✓

(C) Spiked water-milfoil adapted to lower phosphorous conditions has been observed to outcompete fennel-leaved pondweed in this section of the river.

(D) Measured levels of SRP in this section of the river have remained constant over a period of time.

Answer B

Correct. Human activities such as the addition of organic or chemical fertilizers to lawns, golf courses, and agricultural lands and the observed increase in runoff of the phosphorus in these fertilizers into the rivers is the most likely contributor to the increased SRP level in the water that resulted in the change in plant species.

8 Ecology MCQ

Researchers investigated the effects of nonnative trout on the behavior of native salmon inhabiting a freshwater stream. Juvenile trout compete with juvenile salmon for resources, and adult trout prey on juvenile salmon.

Researchers built enclosed channels in a freshwater stream. A transparent plastic mesh screen divided each channel into two sections, an upstream predator section and a downstream competitor section. Similar screens placed at the ends of each channel permitted the flow of water and small invertebrate organisms, eaten by juvenile members of both species of fish, through the channel. Juvenile salmon were placed in only the competitor section of each channel, in the absence or presence of juvenile trout, and in the absence or presence of adult trout placed only in the predator section (Figure 1). The researchers then recorded the percent of juvenile salmon that were active during the day and night, under the different conditions (Figure 2).

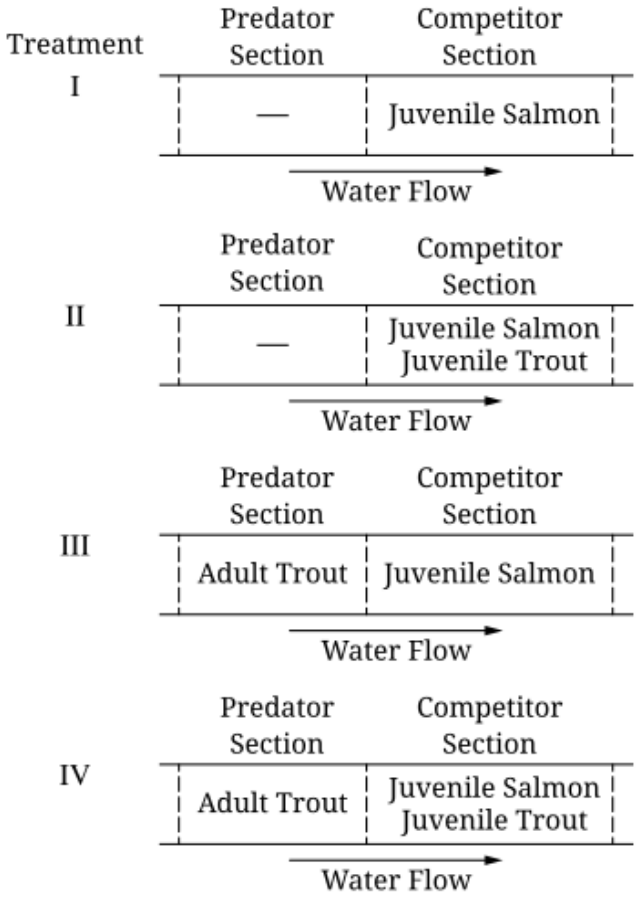


Figure 1. Experimental design of four freshwater stream channels used to monitor juvenile salmon behavior in the presence or absence of predator and/or competitor fish

8 Ecology MCQ

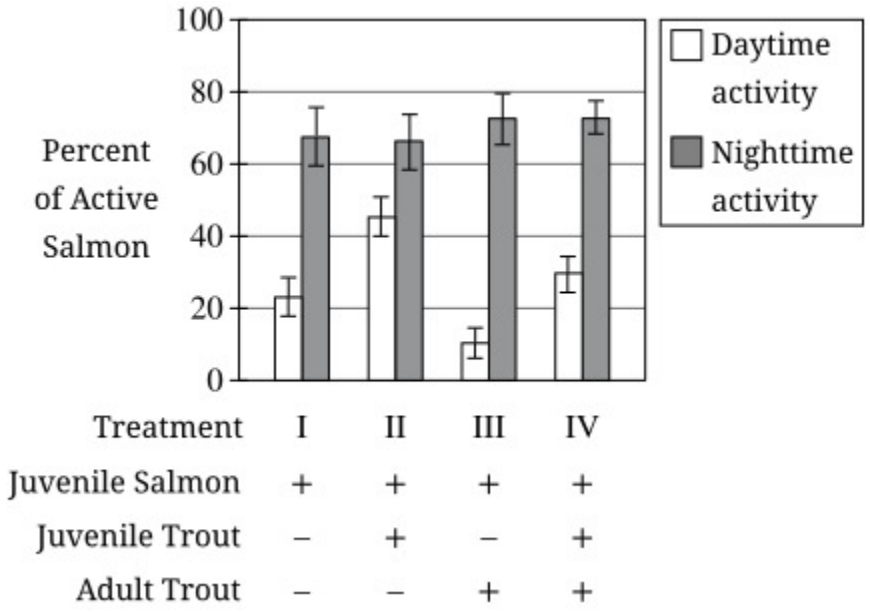


Figure 2. Activity of juvenile salmon in freshwater stream channels in the presence (+) or absence (−) of juvenile and/or adult trout. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

72. Which of the following descriptions of juvenile salmon activity is best supported by the data in Figure 2 ?

(A) Daytime activity increases when only a predator is present.

(B) Nighttime activity decreases when only a competitor is present.

(C) Daytime activity increases when only a competitor is present. ✓

(D) Nighttime activity increases in the presence of competitors and predators.

Answer C

Correct. Juvenile salmon daytime activity increased when only a competitor was present (Treatment II) in comparison with daytime activity in the absence of a competitor (Treatment I) and was additionally the highest daytime activity observed in the experiment.

73. The researchers claim that the risk to juvenile salmon from predation is about the same as the risk due to competition. Which of the following best supports the researchers' claim?

8 Ecology MCQ

(A) The daytime activity in treatment IV is about the same as the daytime activity in treatment I and intermediate between the daytime activity in treatments II and III. ✓

(B) In treatment III, the daytime activity is the lowest of all of the treatments and the nighttime activity is the highest of all of the treatments.

(C) In treatment IV, the nighttime activity is twice as great as the daytime activity.

(D) The nighttime activity is about the same in all four treatments.

Answer A

Correct. The presence of both the competitor and the predator in treatment IV results in the juvenile salmon exhibiting a daytime activity level that is intermediate between the daytime activity in the presence of the competitor only (treatment II) or the predator only (treatment III), thus resembling the daytime activity in treatment I. In treatment IV, the juvenile trout must increase their activity because their food competitor is present (as in treatment II), but they are also cautious because the predator is present (as in treatment III).

74. A researcher claims that the predator avoidance behavior of juvenile salmon observed in the study has evolved because of natural selection. Which of the following data could best be used to support the claim?

(A) The salmon that avoid predators have longer life spans as compared with salmon that do not avoid predators.

(B) The salmon that avoid predators can grow to a larger size as compared with salmon that do not avoid predators.

(C) The salmon that avoid predators are able to access more resources as compared with salmon that do not avoid predators.

(D) The salmon that avoid predators have a greater number of viable offspring as compared with salmon that do not avoid predators. ✓

Answer D

Correct. Natural selection is the result of increased survival and reproduction of individuals with beneficial phenotypes. Traits of such individuals are then passed on to offspring, allowing the population to evolve over time.

75. The researchers hypothesize that juvenile salmon use chemical cues to detect predators that are nearby. Which of the following modifications to the experiment will help the researchers test their hypothesis?

8 Ecology MCQ

(A) Remove the mesh screen from the channel so that the predators have access to the juvenile salmon.

(B) Change the light schedule so that there are twice as many nighttime hours as daytime hours.

(C) Add rocks to the channel that the juvenile salmon can use to hide from predators.

(D) Replace the mesh screen with a solid, transparent barrier that allows light to pass through. ✓

Answer D

Correct. A solid barrier will prevent the movement of chemical signals in the water and therefore allow researchers to determine the influence of chemical cues on juvenile salmon behavior.

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76. Students in a class were divided into five groups to analyze community structure. Each group of students assessed the species of plants in four environments and determined the Simpson's Diversity Index for each environment. The Simpson's Diversity Index takes into account the number of species present as well as the abundance of each species. The index values are shown in Figure 1.

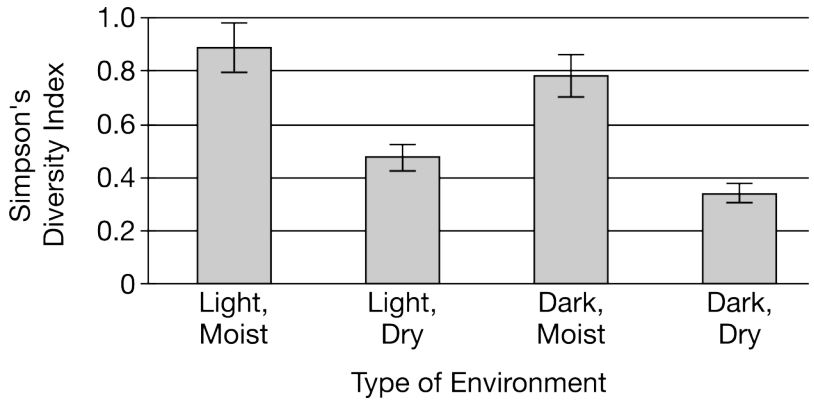


Figure 1. Simpson's Diversity Index for four environments

Based on the data in Figure 1, which of the following statements is most accurate about the species diversity of the four different environments that the students analyzed?

(A) The light, moist environment is the most ideal environment for plants.

(B) The most diverse environment is the dark, dry environment because plants are best able to protect themselves from drying out during the summer months.

(C) Based on the data, there is no statistical difference between the species diversity of plants in the light, moist environment and in the dark, moist environment. ✓

(D) Based on the data, there is no statistical difference between the species diversity of plants in light, dry and dark, dry environments.

8 Ecology MCQ

Answer C

Correct. Because the error bars overlap, the light, moist and dark, moist environments show no significant statistical difference in diversity.

77.

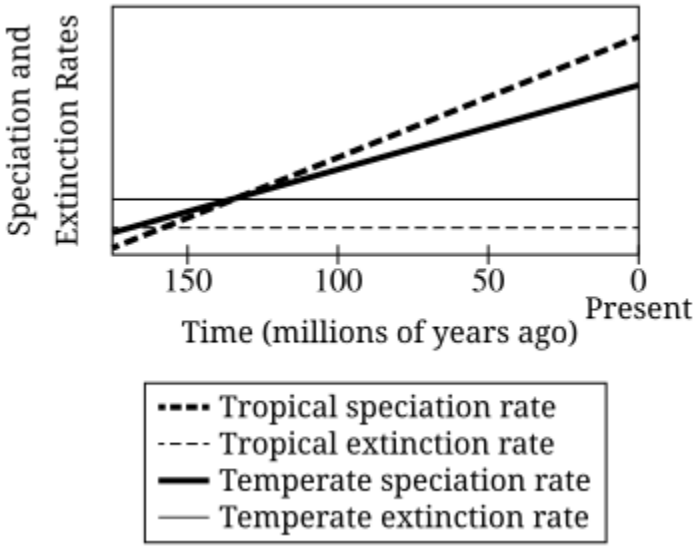


Figure 1. Estimate of rates of speciation and rates of species extinction in temperate and tropical biomes over the last 175 million years

Assuming both temperate and tropical biomes had equivalent biodiversity 175 million years ago, which of the following claims is best supported by the data in Figure 1 ?

(A) Tropical biomes in the present have less biodiversity than do temperate biomes.

(B) Tropical biomes in the present have greater biodiversity than do temperate biomes. ✓

(C) The rate of extinction is lower in temperate biomes than in tropical biomes.

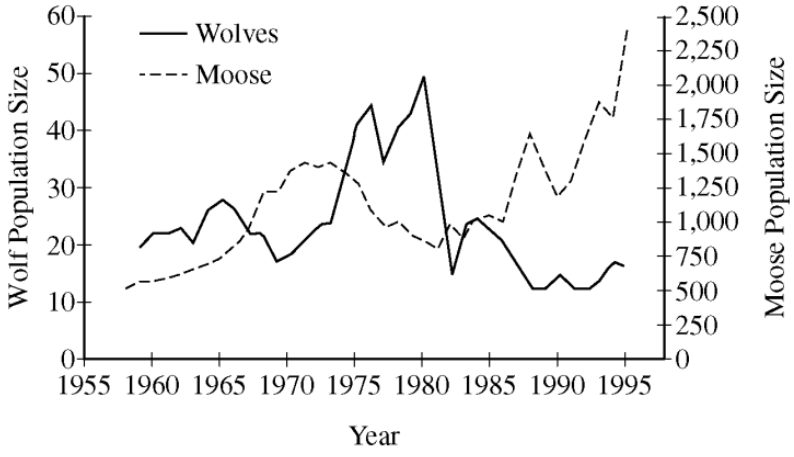
(D) The rate of speciation in both temperate and tropical biomes has not changed over time.

Answer B

Correct. Based on the data, tropical biome speciation has increased more rapidly than has temperate biome speciation and has a lower constant extinction rate than that of temperate biomes. Given this difference in overall net speciation, tropical biomes are expected to have more biodiversity than temperate biomes.

8 Ecology MCQ

78.



The graph above represents the number of individuals in a population of wolves and in a population of moose observed in the same isolated geographic area over a 40-year period, from 1955 through 1995. Which of the following statements about the two populations is best supported by the information presented in the graph?

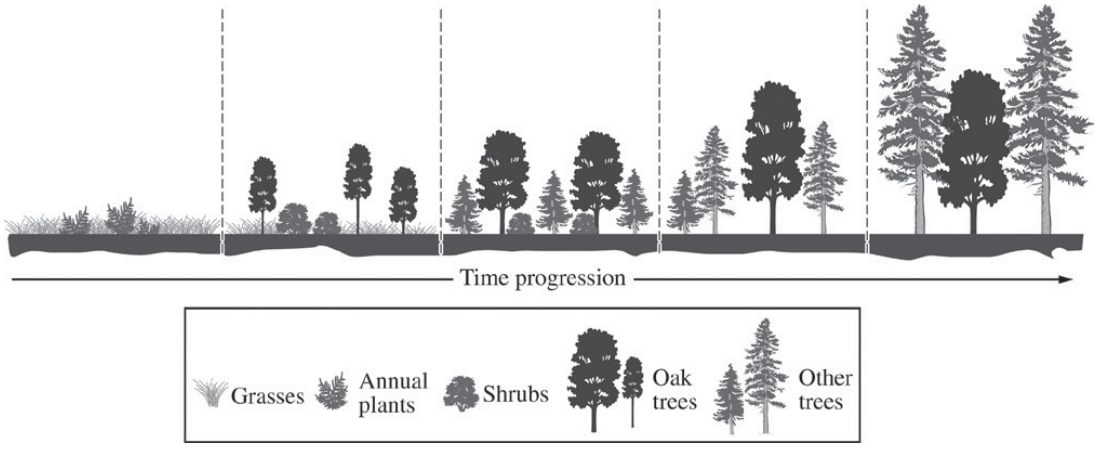
(A) The reproductive rate of the wolves was greater than the reproductive rate of the moose.

(B) Mutualism allowed the two populations to reproduce while occupying the same ecological niche.

(C) Speciation occurred when the two populations became reproductively isolated from each other.

(D) The wolves were predators of the moose, which were otherwise reproductively successful. ✓

79.



The diagram above shows the progression of ecological events after a fire in a particular ecosystem. Based on the diagram, which of the following best explains why the oak trees are later replaced by other trees?

(A) Eventually the other trees grow taller than the oak trees and form a dense canopy that shades the understory. ✓

(B) Oak trees alter the pH of the soil, making the forest better suited for shrubs and other trees.

(C) Roots of shrubs proliferate in the soil of the forest and prevent the oak trees from obtaining water.

(D) Oak trees succumb to environmental pollutants more readily than do either the shrubs or the other trees.

8 Ecology MCQ

Answer A

This option is correct. The other trees shade out the oak trees. Oak seedlings are relatively intolerant to shade. Oak trees need sunlight for photosynthesis, and although they do survive in poor sunlight, they do not proliferate enough to be the dominant plant in the forest. Some oak species need sunlight for full development.

80. Thrips are insects that feed on rose pollen. Scientists noted that the thrips population increased in the spring and decreased dramatically during the summer. The researchers hypothesized that food abundance was the limiting factor for the population. Which of the following types of data would be most useful for the scientists to collect at regular intervals on a designated test plot of rose plants?

(A) Amount of sunlight (hours/day)

(B) Mean temperature (^{o}C)

(C) Density of rose pollen produced (g/m^2) ✓

(D) Amount of pollen produced by each flower (g/flower)

8 Ecology MCQ

81. Figure 1 shows the changing pattern of human population sizes in rural and urban areas from 1960 to 2016 with regard to the global population.

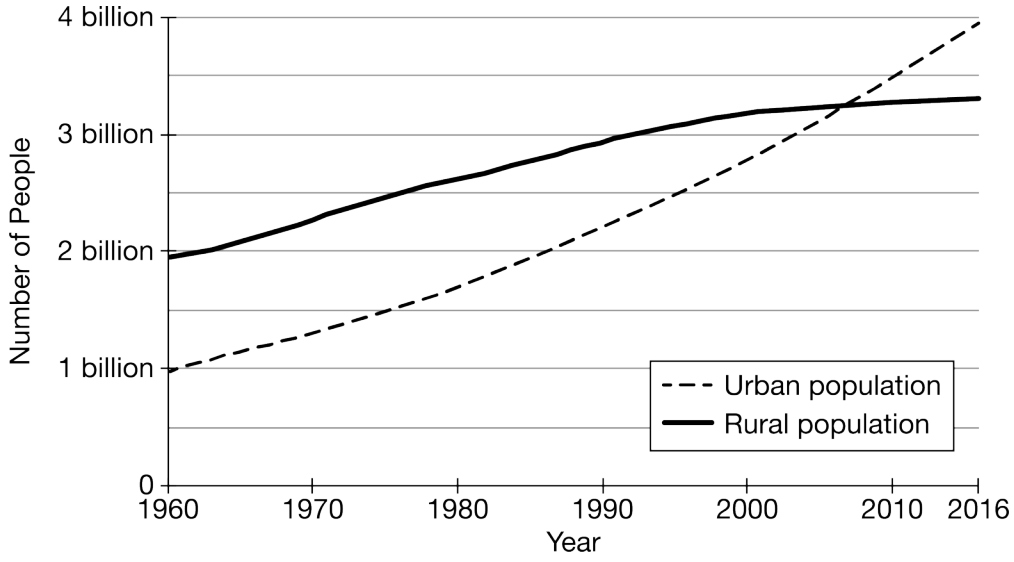


Figure 1. Comparison of population size in urban and rural areas from 1960 to 2016

A scientist interested in investigating how human population trends might affect local animal species' richness stipulates a null hypothesis that animal biodiversity in rural areas will be unaffected in the future by the human population trends shown in Figure 1. Which of the following would best refute this null hypothesis?

(A) More rural areas will become available for animal habitats over time.

(B) Urbanization causes habitat fragmentation. ✓

(C) Urbanization decreases the per person fossil fuel use in urban areas.

(D) Water in urban areas is treated before it is released into the rivers that flow into rural areas.

Answer B

Correct. As humans infringe on species' habitats, causing fragmentation, the organisms must adapt to new, smaller habitats, and biodiversity typically decreases.

8 Ecology MCQ

82. Water in a pond contaminated with the weed killer atrazine is suspected of inhibiting metamorphosis in northern leopard frogs. A team of scientists collected fertilized northern leopard frog eggs from a different pond that is not contaminated. Which of the following is the best experimental design to determine whether atrazine is responsible for inhibiting metamorphosis in northern leopard frogs?

(A) Place half of the fertilized eggs in a pool of water with the same concentration of atrazine as the contaminated pond and place the other half of the fertilized eggs in a pool of water that has no atrazine. Monitor the development of the embryos through metamorphosis into adulthood. ✓

(B) Place all of the fertilized eggs in a pool of pond water with the same concentration of atrazine as the contaminated pond and compare the number of frogs that reach metamorphosis to those that reach adulthood in the contaminated pond.

(C) Allow all fertilized eggs to develop into adults. Expose one-third of the frogs to one-half of the concentration of atrazine in the contaminated pond and expose another one-third of the frogs to the same concentration of atrazine as the contaminated pond. Leave the last one-third of the frogs in water with no atrazine and note any adverse changes in the physical condition of the atrazine-treated frogs in three months.

(D) Divide the fertilized eggs into three groups and expose each group to a different concentration of atrazine. Release the eggs back into the contaminated pond and check for metamorphosis after three months.

8 Ecology MCQ

In the mid-1920s, gray wolves, *Canis lupus*, were eliminated from Yellowstone National Park in the northwestern United States. Over the course of 1995 and 1996, 31 wolves from Canada were relocated to the park. Researchers have tracked the growth of the wolf population from 21 in 1995 to 108 at the end of the study in 2016 (Figure 1).

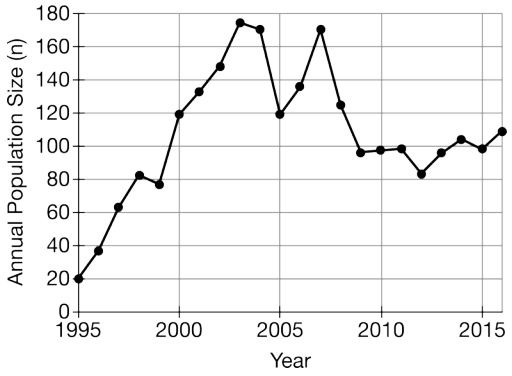


Figure 1. Yellowstone National Park annual wolf population from 1995 to 2016

The wolves prey on large herbivores, primarily elk. The carcasses of wolf kills are a valuable food source for coyotes, *Canis latrans*. When coyotes are feeding on carcasses, they may be chased off the carcass and/or killed by wolves. Coyotes also hunt small mammals for food.

Researchers were studying the effect of wolves' presence on the behavior of coyotes. Researchers observed coyotes in a specific area of Yellowstone National Park and recorded the proportion of the coyotes' time spent on five different activities when wolves were present in the study area and when wolves were absent (Figure 2).

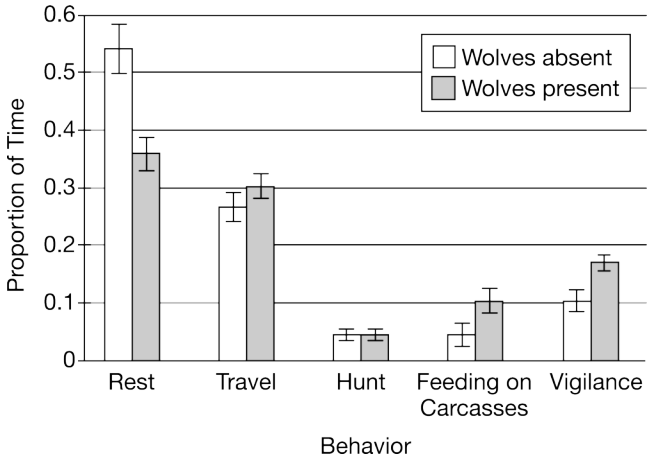


Figure 2. Coyote behavior in the presence and absence of wolves. Error bars represent a confidence interval of 95% (± 2 standard errors of the mean).

83. Which of the following best describes the change in coyote behavior in the presence of wolves?

8 Ecology MCQ

(A)

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Increases	No effect	No effect	Decreases	Decreases

(B)

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Increases	Decreases	No effect	Decreases	Decreases

(C) ✓

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Decreases	No effect	No effect	Increases	Increases

(D)

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Decreases	Increases	No effect	Increases	Increases

Answer C

Correct. According to Figure 2, in the presence of wolves (white bars), the proportion of time spent on rest is less, the proportion of time spent feeding on carcasses is more, and the proportion of time spent on vigilance is more than in the absence of wolves (gray bars). Although the mean proportion of time spent on travel increases in the presence of wolves, the error bars overlap, showing that the difference is not statistically significant.

8 Ecology MCQ

84. Following the reintroduction of wolves to Yellowstone National Park, researchers measured the mean height of young aspen trees in upland and streamside areas.

Table 1. Mean Height of New Aspen Growth in Early Spring

	Year	
	1998	2010
Upland	31 cm	176 cm
Streamside	33 cm	237 cm

Which of the following is the most likely cause of the change in mean aspen height from 1998 to 2010 ?

(A) The reintroduction of wolves led to a decrease in the population of elk, allowing aspen trees to grow taller. ✓

(B) The reintroduction of wolves led to a decrease in the population of small mammals, allowing aspen trees to grow taller.

(C) The reintroduction of wolves led to an increase in the number of carcasses that provided fertilizer for young aspen trees.

(D) The reintroduction of wolves led to an increase in the number of carcasses, so coyotes no longer ate young aspen trees.

Answer A

Correct. Top carnivores are keystone species, meaning they have a larger role in maintaining an ecosystem than would be expected by their population size. After wolves were eliminated from Yellowstone National Park, the population of elk rose dramatically. The high elk population led to the overgrazing of plant species, including aspen. With the reintroduction of wolves, the elk population has declined, allowing plant species to recover.

85. Identify the independent variable for the experimental results shown in Figure 2.

(A) The presence of wolves ✓

(B) The proportion of time

(C) Coyote behavior

(D) The study area

8 Ecology MCQ

Answer A

Correct. The independent variable is selected by the researchers, and the researchers test its effect on the dependent variable. In this experiment, the researchers were looking at the effect of wolves' presence on the behavior of coyotes. Therefore, the presence of wolves is the independent variable, and the behavior of coyotes is the dependent variable.

86. Based on information in Figure 1, which of the following is closest to the annual growth rate of the wolf population in Yellowstone National Park from 1996 to 2004 ?

(A) 4 wolves per year

(B) 17 wolves per year ✓

(C) 87 wolves per year

(D) 135 wolves per year

Answer B

Correct. The equation for calculating a rate is $\frac{dY}{dt}$, which is the difference in the variable on the y-axis divided by the difference in the variable on the x-axis. In this case, that equation becomes annual population growth rate = (final population size - initial population size)/(number of years). According to Figure 1, in 1996 the population was 36 wolves, and 8 years later, in 2004, the population was 171 wolves. Therefore, $\frac{171-36}{8} = 16.875$, which can be rounded to 17 wolves per year.

87. Based on the data in the figure, a student claimed that since 2007, the carrying capacity of wolves in Yellowstone National Park has been 100 wolves, and the maximum annual per capita growth rate of gray wolves is 0.6 wolves per wolf per year.

Which of the following is closest to the calculated wolf population size in 2017 based on the student's claim?

(A) 100

(B) 103 ✓

(C) 113

(D) 173

8 Ecology MCQ

Answer B

Correct. According to the logistic growth model, the change in population size $\left(\frac{dN}{dt}\right)$ can be calculated using the equation $\frac{dN}{dt} = r_{\max}N\left(\frac{K-N}{K}\right)$. According to the student's claim, $r_{\max} = 0.6$ wolves/year and $K = 100$ wolves. From Figure 1, $N = 108$ wolves in 2016. Therefore, $\frac{dN}{dt} = 0.6 \times 108 \times \left(\frac{100-108}{100}\right) = -5.184$. The –5.184 needs to be added to the 2016 population size to get the population size for 2017: $108 + (-5.184) = 102.816$, which is rounded to 103 to get the final answer.

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88. Researchers were studying the effects of microbes on the growth of the yellow bedstraw plant, *Galium verum*. They grew yellow bedstraw plants under four treatments: in the presence of soil bacteria only, arbuscular mycorrhizal (AM) fungi only, both bacteria and AM fungi, or neither. After 6 months, they measured the biomass of the plants (Figure 1).

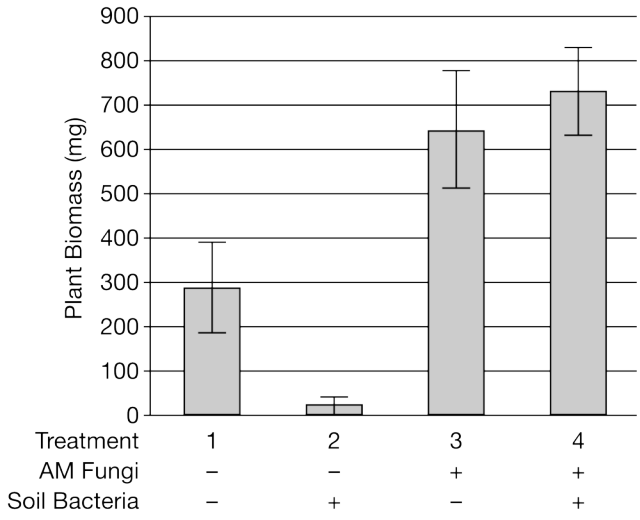


Figure 1. The Effect of Soil Microbes on the Biomass of *Galium verum* $(\text{Mean} \pm 2SE_{\overline{x}})$

Which of the following best describes a statistical relationship indicated in Figure 1 ?

(A) Yellow bedstraw plants grow significantly more in the presence of both soil bacteria and AM fungi (treatment 4) than in the presence of soil bacteria alone (treatment 2). ✓

(B) Yellow bedstraw plants, in the presence of AM fungi (treatment 3), grow significantly more if soil bacteria (treatment 4) are also present.

(C) Yellow bedstraw plants grow significantly more when soil bacteria (treatment 2) are present compared to when there are no soil bacteria (treatment 1).

(D) Yellow bedstraw plants grown in soil without bacteria and AM fungi (treatment 1) grow significantly more than when only AM fungi is present in the soil (treatment 3).

8 Ecology MCQ

Answer A

Correct. The error bars in treatments 2 and 4 do not overlap, which means that the differences between the two data sets are statistically significant. Thus, AM fungi significantly increases yellow bedstraw growth when these plants are exposed to soil bacteria.

89. Zebra mussels, *Dreissena polymorpha*, were introduced into the Great Lakes on shipping vessels, first appearing in Lake St. Clair in the 1980s. Since that time, they have rapidly expanded in range, as shown in Figure 1.

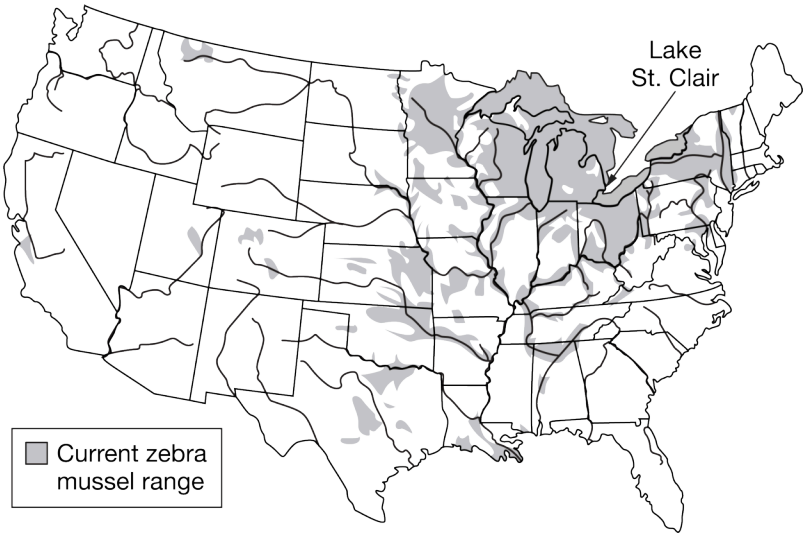


Figure 1. Map showing the current range of zebra mussels in the United States

Which of the following best supports the alternate hypothesis that zebra mussels display characteristics of an invasive species?

(A) Lacking natural predators and able to outcompete native species of mussels and other freshwater species for food, zebra mussels have experienced exponential population growth and rapid expansion into new habitats. ✓

(B) Since their arrival, zebra mussels have undergone rapid adaptive radiation, producing new species in many different habitats.

(C) Throughout their current range, zebra mussels coexist with native mussel species, whose populations have been unaffected by the arrival of zebra mussels.

(D) Zebra mussels can crossbreed with native mussels, producing hybrids that are better adapted in the range shown in Figure 1.

8 Ecology MCQ

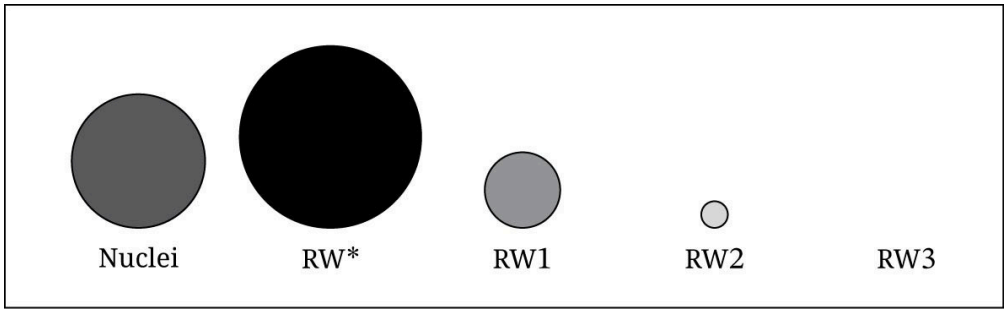
Answer A

Correct. Because invasive species typically do not have natural predators in the habitats into which they are introduced, they can experience rapid exponential population growth. In addition, because these mussels are able to outcompete native species, zebra mussel range has expanded rapidly.

90. A researcher conducted an experiment in a brightly lit aquarium containing a large population of single-celled green algae. Water containing radioactive phosphorus was added to the aquarium. After 7 days, the researcher took a water sample containing the algae and separated the algae from the radioactive water (RW*).

The researcher added deionized water to rinse the algae and saved the rinse water and then repeated that process twice. The rinse water from each of the three rinses was saved as RW1, RW2, and RW3, respectively. The researcher also removed and kept the nuclei of the algal cells. The nuclei were mixed with deionized water, and a drop of the mixture was placed on a piece of filter paper. A drop of the radioactive water and a drop of each of the rinse water samples were also placed on the same paper. These samples were exposed to X-ray to determine their radioactivity. The results after 24 hours are shown in Figure 1.

Figure 1. Radioactivity of saved algal water samples. Darker color and larger circle size indicate a higher level of radioactivity.



Based on Figure 1, which of the following statements best explains why the nuclei sample had a higher level of radioactivity than any of the rinse water samples?

(A) Phosphorus is used in the building of nucleic acids in the nucleus. ✓

(B) The nuclei sample was contaminated with radioactive water.

(C) Radioactive phosphorus is dissolved from rocks and becomes part of the sugar molecules created during photosynthesis.

(D) Radioactive phosphorus is dissolved from rocks and is used to create cholesterol stored in the nucleus.

8 Ecology MCQ

91.

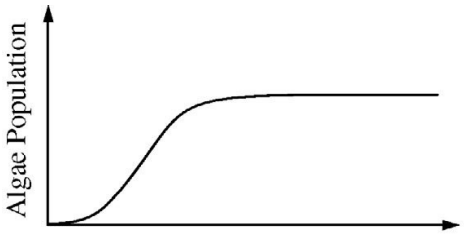


Figure I

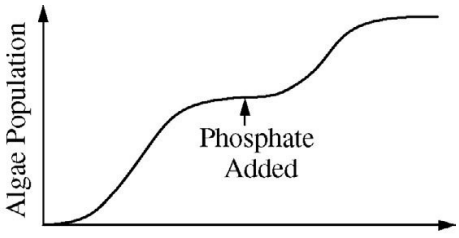
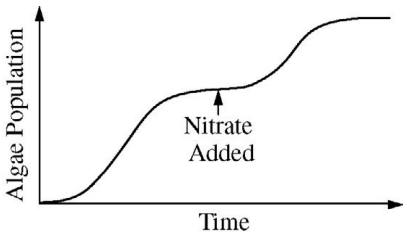


Figure II

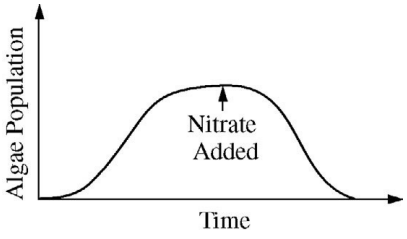
Figure I shows the growth of an algal species in a flask of sterilized pond water. If phosphate is added as indicated, the growth curve changes as shown in Figure II.

Which of the following is the best prediction of the algal growth if nitrate is added instead of phosphate?

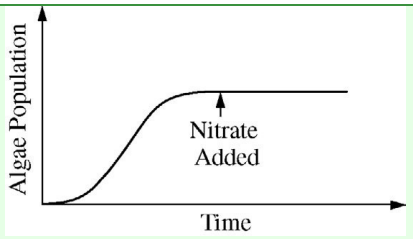
(A)



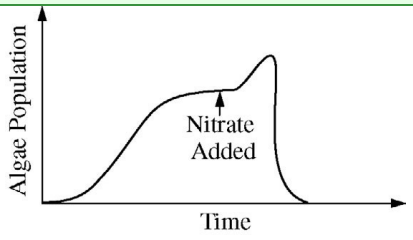
(B)



(C)



(D)



8 Ecology MCQ

92.

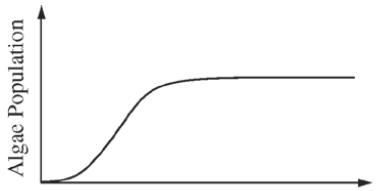
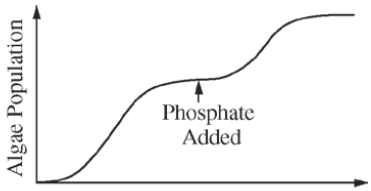


Figure I



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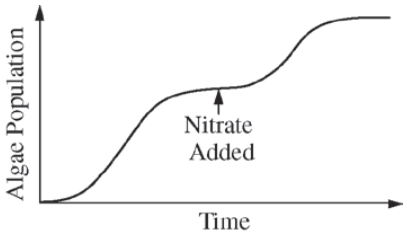
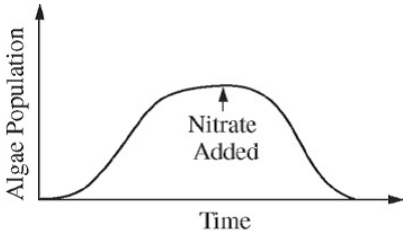
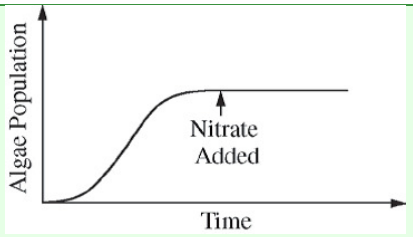
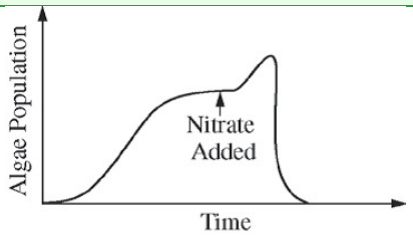


(B)


(C)


(D)


8 Ecology MCQ

In the American Southwest, annually emerging cicadas are dependent upon the cottonwood trees in the floodplain. Adult male cicadas perch in the cottonwood trees and chirp to attract females. Females lay their eggs in the branches of the cottonwoods, and, upon hatching, the cicada nymphs drop to the ground, burrow into the soil, feed on the tree roots, and later emerge as adults. Other organisms in the floodplain community prey on the cicadas, as shown in the food web.

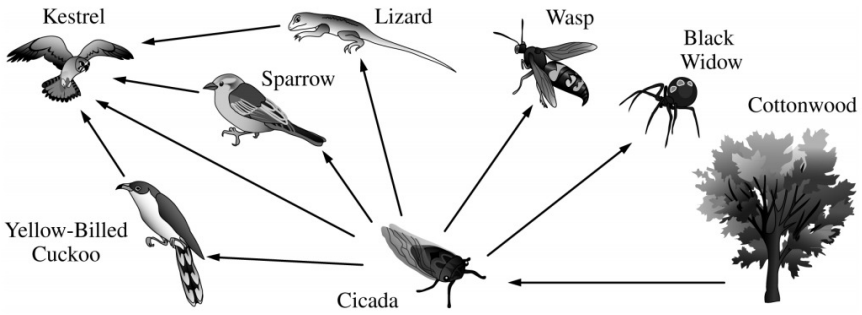


Figure 1. Food web in floodplain community

Human activity has caused a decrease in the amount of flooding and an increase in the incidence of wildfires in the floodplain. In an investigation into the recent changes in the floodplain ecosystem, researchers monitored the soil temperature, amount of cottonwood ground coverage (i.e., area of the ground that is shaded by leaves), and cicada emergence for a period from mid-June until late July. The results of the investigation are represented in Figure 2 and Figure 3.

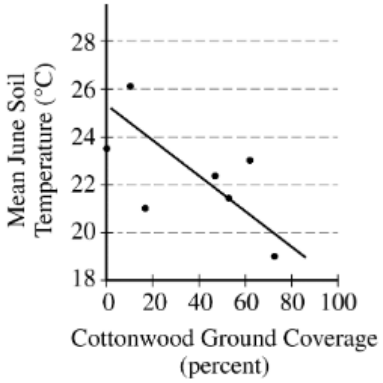


Figure 2. Effect of cottonwood ground coverage on soil temperature

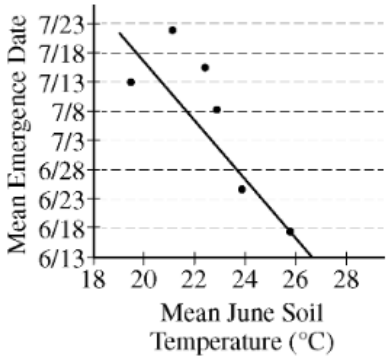


Figure 3. Relationship between soil temperature and cicada emergence date

To assess the impact of wildfires on soil temperature and cicada emergence, the researchers compared mean emergence dates for two natural sites where portions had been affected by wildfire. In addition, cicada emergence was monitored at an experimental site where the soil temperature was experimentally maintained. The data are shown in the table.

8 Ecology MCQ

MEAN CICADA EMERGENCE DATES AT SITES AFFECTED BY WILDFIRES OR WITH EXPERIMENTALLY CONTROLLED SOIL TEMPERATURE

	Natural Site 1		Natural Site 2		Experimental Site	
	Unburned	Burned	Unburned	Burned	24°C	27°C
Mean Emergence Date	July 8	June 18	July 15	July 3	July 14	July 5

93. Based upon the data, which of the following best describes the effect on the timing of cicada emergence if the cottonwood ground cover decreases from 50 percent to 25 percent?

(A) Cicadas will emerge approximately 10 days earlier. ✓

(B) Cicadas will emerge approximately 2 days earlier.

(C) Cicadas will emerge approximately 10 days later.

(D) Cicadas will emerge approximately 2 days later.

94. Based on the data, which of the following describes the most likely relationship between wildfires, soil temperature, and the timing of cicada emergence in the same year?

(A) Wildfires caused a reduction in ground cover, which in turn caused increased soil temperatures leading to earlier emergence of cicadas. ✓

(B) Wildfires caused an increase in ground cover, which in turn caused increased soil temperatures delaying the emergence of cicadas.

(C) Wildfires caused an increase in growth of new plants, which in turn caused a decrease in soil temperatures which led to earlier emergence of cicadas.

(D) Wildfires caused a decrease in ground cover, which in turn caused a decrease in soil temperatures leading to later emergence of cicadas.

95. The data from the temperature-controlled experimental plots can best be used to support which of the following conclusions about cicada development?

(A) Adult cicadas lay more eggs when the soil is warmer than when the soil is cooler.

(B) Adult cicadas mate more frequently when the soil is warmer than when the soil is cooler.

(C) Cicada nymphs grow larger in warmer soil than in cooler soil.

(D) Cicada nymphs mature to adults faster in warmer soil than in cooler soil. ✓

96. Yellow-billed cuckoos typically hatch in midJuly. Emerging cicadas are a primary food source for nesting cuckoos. Which of the following best predicts the effect of wildfires on yellow-billed cuckoo populations?

8 Ecology MCQ

(A) The yellow-billed cuckoo population will decline because the decreased ground cover will allow lizards to prey on cuckoo nests.

(B) The yellow-billed cuckoo population will decline because the cicadas will emerge before the hatching season begins. ✓

(C) The yellow-billed cuckoo population will grow because the adults will more easily see and eat the cicada nymphs.

(D) The yellow-billed cuckoo population will remain unchanged because cuckoos do not nest in areas affected by wildfires.

97. Which of the following statements best predicts the effect of increased cottonwood growth on the food web?

(A) Increased cottonwood growth will lead to an increase in the number of primary consumers, resulting in more competition between kestrels and sparrows for food.

(B) Increased cottonwood growth will decrease the habitat available for lizards, resulting in a decline in the number of kestrels.

(C) Increased cottonwood growth will provide more resources for cicadas, resulting in an increase in the kestrel population. ✓

(D) Increased cottonwood growth will lead to a large accumulation of dead wood that serves as fuel for wildfires, resulting in the collapse of the food web.

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98. In some species of bats, adult females regurgitate (bring up food from the stomach) to feed their young. In addition, adult bats regurgitate to feed other adults that have not recently eaten. The adults receiving the food may be unrelated bats who have previously participated in food sharing and are members of the same social group. Scientists claim that sharing among unrelated adult bats increases the fitness of the species.

Which of the following best supports the scientists' claim?

(A) Bats that do not find food over a 24-hour period may die, preventing their alleles from being passed on within the social group. ✓

(B) Bats that do not find food may adapt to eating other types of food, adding new alleles to the social group.

(C) Food sharing reduces the number of unrelated animals that cluster in one area, thus increasing the genetic diversity in the social group.

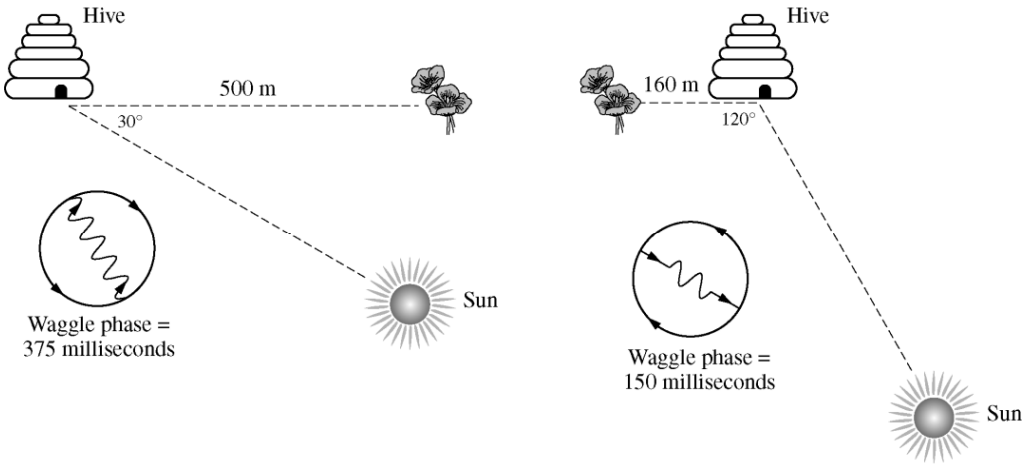
(D) Food sharing increases the number of unrelated animals that cluster in one area, thus decreasing competition for food among members of the social group.

Answer A

Correct. The death of individuals within a social group could potentially reduce the overall genetic diversity of the entire group. Loss of genetic diversity is directly proportional to loss of overall fitness for a species.

8 Ecology MCQ

99.



As depicted in the diagram, honeybees communicate the location of flower patches to members of their hives with waggle dances that give information about the direction and distance to the flowers. Which of the following statements about how honeybees communicate the position of flower patches is most consistent with the model?

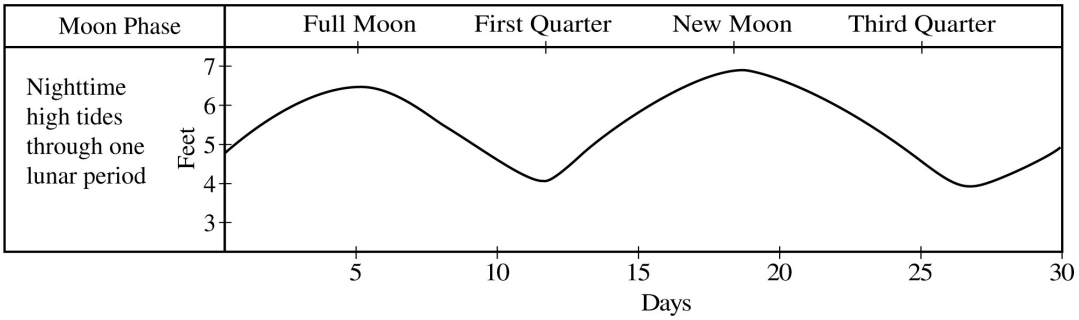
(A) The number of repetitions of the waggle dance indicates the quality of the pollen source.

(B) The total area covered by any waggle dance leads bees to the target flowers.

(C) The angle of the waggle phase relative to the vertical plane indicates the position of the target flowers relative to another beehive.

(D) The farther the target flowers from the hive, the longer the waggle phase. ✓

100.



The California grunion (*Leuresthes tenuis*) is a small marine fish that lives in shallow waters near the ocean shore. Grunions swim as far onto the beach as possible to mate and lay their eggs (spawn).

A researcher proposes that the spawning behavior takes place when the nighttime tides are highest during the month. Which of the following pieces of evidence would best support the researcher's claim?

(A) Grunion spawning occurs every two weeks during the spawning season. ✓

(B) Grunion spawning occurs when nighttime illumination levels are the lowest.

(C) High tide occurs one time per lunar cycle.

(D) The light intensity of the moon varies within the lunar cycle.

8 Ecology MCQ

101. A group of students designed an experiment to determine the effect of compost on the germination and growth of plants. The students set up experimental plots by mixing different ratios of soil and compost. They planted 20 pea and 20 melon seeds in each plot and watered each plot regularly. The students recorded the number of seeds that germinated, and as the plants grew, they recorded plant length, number and size of leaves, and general health observations.

The following observations were recorded.

- In all treatments, more pea seeds germinated than melon seeds.
- There was a week of rainy weather at three weeks.
- Melon plants developed a fungal growth at four weeks.
- Melon plants grew longer than pea plants, but many melon leaves showed signs of yellowing as the weeks passed.

The students concluded that pea plants grew better in compost than did melon plants.

Which of the following best addresses the validity of the conclusion made by the students?

(A) The conclusion is valid because the independent and dependent variables in the experiment were controlled.

(B) The conclusion is valid because the experimental design included a large sample size.

(C) The conclusion is invalid because other variables in the experiment (both biotic and abiotic) affected the results. ✓

(D) The conclusion is invalid because the units of measurement were not given in the experimental design.

102. The pesticide DDT was widely used in the 1940s as a method of insect control. In the late 1950s the first DDT-resistant mosquitoes were discovered, and eventually DDT-resistant mosquitoes were found globally. When DDT is used now, the development of DDT resistance in mosquito populations occurs in months rather than years.

Which of the following best explains the observations concerning DDT resistance in mosquitoes?

(A) Competition for limited resources causes mosquitoes to migrate to geographical areas that have richer supplies of DDT.

(B) The proportion of DDT-resistant mosquitoes in a population remains constant due to the metabolic costs of DDT utilization

(C) Natural selection favors DDT-resistant mosquitoes that are already present in a population when DDT exposure occurs. ✓

(D) DDT is a chemical signal that delays normal reproductive cycles in many mosquito populations.

8 Ecology MCQ

Diapause is the interruption of an organism's life cycle in response to environmental cues. The soil nematode *Caenorhabditis elegans* is capable of entering adult reproductive diapause (ARD) when food is scarce. In *C. elegans*, individuals normally become reproductively mature 2 days after hatching and remain fertile for 18 days. They reproduce either by self-fertilization or by mating with another individual.

In an investigation, researchers examined the survival and reproductive success of *C. elegans* following different times in ARD. In the first experiment, groups of *C. elegans* were held in ARD without food for 0-30 days. Upon reintroduction of food, average brood sizes (average numbers of offspring per adult) were determined following either self-fertilization or mating with a well-fed male. The results are shown in Figure 1.

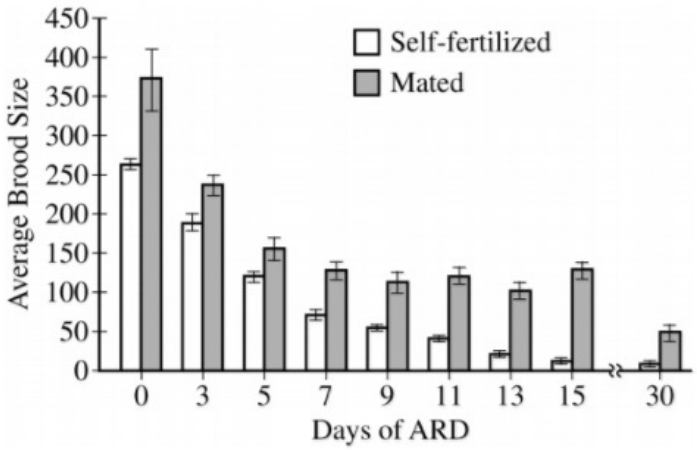


Figure 1. Mean brood sizes $\pm 2SE_{\overline{X}}$ after different times in ARD.

Individual *C. elegans* were held in ARD and subsequently

allowed to reproduce either via self-fertilization (unshaded bars)

or by mating with well-fed males (shaded bars).

In a second experiment, individuals were held in ARD without food for $0-30$ days and monitored for average survival times following reintroduction of food (Figure 2).

8 Ecology MCQ

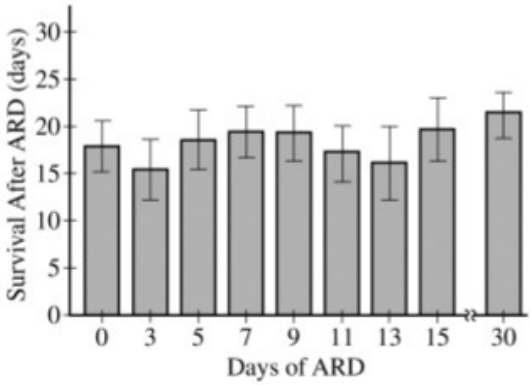


Figure 2. Mean survival $\pm 2SE_{\overline{X}}$ following different times in ARD. Individual *C. elegans* were held in ARD and subsequently given access to food, whereupon their survival times were determined.

103. Which of the following best describes the reproductive ability of *C. elegans* following the ARD induced in the first experiment?

(A) Mating with a well-fed male consistently produced more offspring than did reproduction via self-fertilization. ✓

(B) The numbers of progeny produced by self-fertilization and by mating with well-fed males were not statistically different.

(C) *C. elegans* stopped reproducing after 20 days without food.

(D) There was no relationship between days without food and average brood size.

104. The average brood size per mated individual upon reintroduction of food following 30 days of ARD is closest to which of the following?

(A) 10

(B) 50 ✓

(C) 250

(D) 400

105. Which of the following conclusions is most consistent with the data shown in Figure 2?

(A) Animals that spend $3\text{–}5$ days in ARD are more likely to survive periods of food scarcity than are animals that spend $13\text{–}15$ days in ARD.

(B) Animals that spend 30 days in ARD live significantly longer after reintroduction of food than animals that spend only 3 days in ARD.

(C) The number of days an animal spends in ARD does not significantly affect its time of survival after reintroduction of food. ✓

(D) The large standard errors of the means make conclusions from the data impossible.

8 Ecology MCQ

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

106. The most probable cause for the difference in mean stem length between plants in dish A and plants in dish B is which of the following?

(A) Shortening of cells in the stem in response to the lack of light

(B) Elongation of seedlings in response to the lack of light ✓

(C) Enhancement of stem elongation by light

(D) Genetic differences between the seeds

Answer B

This option is correct. In response to the lack of light, the plant elongates cells in an attempt to reach light necessary for photosynthesis at the expense of leaf development.

8 Ecology MCQ

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(D) Genetic differences between the seeds

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108. Meerkats live together in groups to help raise and protect their offspring. Meerkat pups are most vulnerable when they are away from their burrow foraging for food. Adult meerkats exhibit sentinel behavior, standing upright and making alarm calls to warn other members in the group that predators are nearby.

In a study of meerkat behaviors, scientists observed meerkat foraging groups and recorded the presence or absence of pups along with the frequency of adult sentinel behavior. The scientists determined that sentinel behavior increased when pups were present in the foraging group.

Which of the following correctly identifies the dependent and independent variables in the described study?

(A) The frequency of sentinel behavior is the independent variable, and the presence of pups is the dependent variable.

(B) The presence of pups is the independent variable, and the frequency of sentinel behavior is the dependent variable. ✓

(C) The number of predators is the independent variable, and the presence of pups is the dependent variable.

(D) The distance from the burrow is the independent variable, and the number of predators is the dependent variable.

8 Ecology MCQ

Answer B

Correct. The scientists are studying the relationship between the presence of pups (independent variable) and the frequency of sentinel behavior (dependent variable).

109. Hydrangea macrophylla is a species of plant with blue or pink flowers. Flower color in Hydrangea macrophylla plants is thought to be determined primarily by soil conditions rather than by inherited differences. Which of the following will provide the most direct evidence that flower color in Hydrangea macrophylla is due primarily to soil conditions?

(A) Comparing DNA sequence information for Hydrangea macrophylla to DNA sequence information for a different species of flowering plant

(B) Performing a genetic cross between a pink-flowered plant and a blue-flowered plant and then determining the ratio of phenotypes in the offspring

(C) Growing cuttings from the same Hydrangea macrophylla plant under controlled conditions that vary only with regard to soil pH ✓

(D) Analyzing fossilized remains of an ancestral species that grew in geographic areas similar to the habitats of modern Hydrangea macrophylla

8 Ecology MCQ

110. Scientists are studying the two primary vocalizations made by Belding's ground squirrels (*Urocitellus beldingi*). The trill call consists of a series of medium-pitched notes given rapidly, while the whistle call consists of a single high-pitched note. The objective of the experiment is to determine the function of each vocalization. Recordings of the calls were played to four different groups of squirrels, and the response of each individual squirrel was documented. The responses are summarized in Table 1.

Table 1. Response of ground squirrels to recorded vocalizations

	Percent of Squirrels Responding							
Alarm Call	Trill				Whistle			
Response*	Cr	Po	R/S	O	Cr	Po	R/S	O
Group A	12	65	22	1	42	2	51	5
Group B	9	58	31	2	38	0	60	2
Group C	15	64	18	3	48	4	44	4
Group D	11	66	21	2	1	1	57	1

*Responses

Cr = Crouching
Po = Posting (standing on hind legs)
R/S = Run to shelter
O = Other or no response

Which of the following correctly identifies the dependent variable and the independent variable for the experiment?

(A) The independent variable is the percent of squirrels responding, and the dependent variable is the type of alarm call.

(B) The independent variable is the percent of squirrels responding, and the dependent variable is the type of response.

(C) The independent variable is the group of squirrels, and the dependent variable is the percent of squirrels responding.

(D) The independent variable is the type of alarm, and the dependent variable is the percent of squirrels responding. ✓

8 Ecology MCQ

Answer D

Correct. The type of alarm is the independent variable because it is the factor manipulated by the scientists. The response of the squirrels is being studied to determine how it is affected by the nature of the alarm and is therefore the dependent variable.

Mice live in territorial social groups, and each group has a single dominant male with the rest of the males in the group being subordinate. Male mice mark the area in which they live with their scent by leaving urine wherever they walk. There are two classes of proteins that are found in mouse urine, major urinary proteins (MUPs) and major histocompatibility complex (MHC) proteins. The proteins provide a unique scent for each mouse.

In an experiment, wild male mice were kept isolated in separate cages and their urination patterns were recorded. They were then combined into a single cage that was divided into two compartments by a mesh screen, and again their urination patterns were recorded (Figure 1).

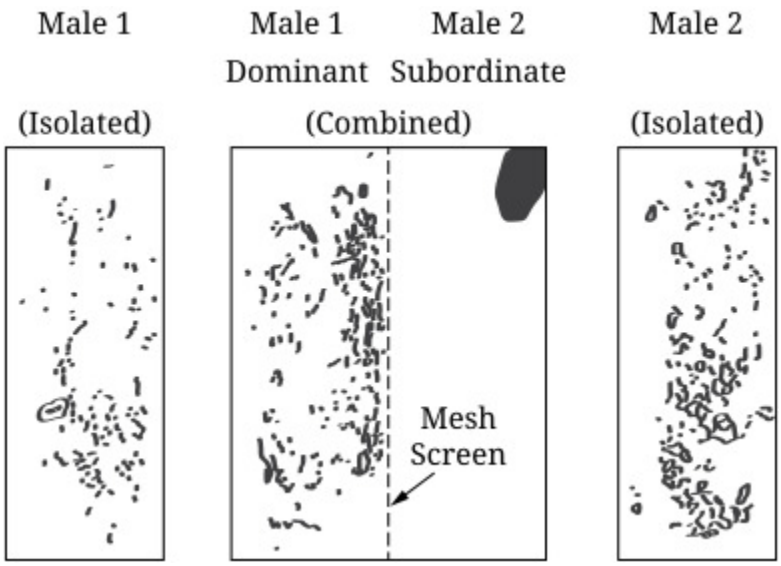

Figure 1. Urination patterns recorded during an experiment

111. Which of the following claims about communication and behavior in mice is best supported by the data from the experiment?

(A) Subordinate males use scent to avoid territory occupied by a dominant male. ✓

(B) Dominant males use urine to communicate sex differences.

(C) Subordinate males occupy more space when in the presence of a dominant male.

(D) Dominant males occupy less space when in the presence of a subordinate male.

8 Ecology MCQ

Answer A

Correct. The scent patterns produced in the combined experiment indicate that the subordinate male is occupying only one small area of his enclosure, which is located as far as possible from the dominant male's territory.

112. In addition to their role in urine scent marking, MHC proteins are used by body cells in both sexes of mice to distinguish self cells that originate in an individual's body from nonself cells that do not originate in an individual's body.

Which of the following best explains how MHC-related odor-producing molecules might be used to maintain genetic diversity in mouse populations?

(A) They could be used by female mice for mate selection to avoid mating with related individuals. ✓

(B) They could be used by subordinate males to attract dominant males to the area to enrich the gene pool.

(C) They could be used by subordinate males to decrease competition with other males resulting in increased genetic diversity.

(D) They could be used by females to decrease bacterial infections that occur during pregnancy.

Answer A

Correct. Close relatives are more likely to share the same MHC proteins. It is possible that female mice could detect and avoid the scent of MHC proteins coming from close relatives and instead choose to mate with males producing different MHC proteins. This would increase the genetic diversity in the mice population.

8 Ecology MCQ

113. A biologist was investigating the effects of herbivory on the amount of nicotine produced by the leaves of tobacco plants (*Nicotiana rustica*). Six groups of plants were grown from seed in the same greenhouse with identical environmental conditions. Three groups were then each exposed for a different interval of time to tobacco hornworms (*Maduca sexta*), which feed on the foliage of tobacco plants. The amount of nicotine present in the leaves was measured at the beginning and end of each time interval. As a control, the other three groups were not exposed to hornworms. The results of the experiment are summarized in Table 1.

Table 1. Response of tobacco plants to hornworm herbivory

Test Group	Time Interval (Weeks)	Hornworms	Initial Concentration of Nicotine$(\mathrm{g/mL})$	Final Concentration of Nicotine$(\mathrm{g/mL})$	% Change in Nicotine Concentration
1	1	Present	0.2	0.8	300
2	1	Absent	0.3	0.2	-33
3	2	Present	0.2	1.4	600
4	2	Absent	0.2	0.3	50
5	3	Present	0.3	2.5	733
6	3	Absent	0.4	0.5	25

Which of the following correctly identifies the dependent variable and the independent variable for the experiment?

(A) The dependent variable is the time interval the plants were exposed to the hornworms and the independent variable is the percent change in the amount of nicotine.

(B) The dependent variable is the absence or presence of hornworms, and the independent variable is the initial amount of nicotine.

(C) The dependent variable is the initial amount of nicotine, and the independent variable is the final amount of nicotine.

(D) The dependent variable is the percent change in the amount of nicotine, and the independent variable is the time interval the plants were exposed to the hornworms. ✓

Answer D

Correct. The change in the concentration of nicotine is the factor being studied and is therefore the dependent variable. The time interval the plants were exposed to the hornworms is changed by the biologist and therefore the independent variable.

8 Ecology MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

The three-spined stickleback (*Gasterosteus aculeatus*) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mainly of individuals with armor-like plates covering most of their body surface (completely plated). Approximately 10,000 years ago, some marine sticklebacks colonized freshwater environments. After many generations in the freshwater environments, the freshwater stickleback populations lacked the armor plating (low plated) typical of marine stickleback populations.

Over the period between 1957 and 2005, one freshwater population, in Lake Washington, a lake in a coastal region of the northwestern United States, changed from having a majority of individuals of the low-plated phenotype to having more individuals of the completely-plated phenotype than of the low-plated phenotype. Figure 1 shows the distribution of plated phenotypes in Lake Washington sticklebacks at four time points between 1957 and 2005.

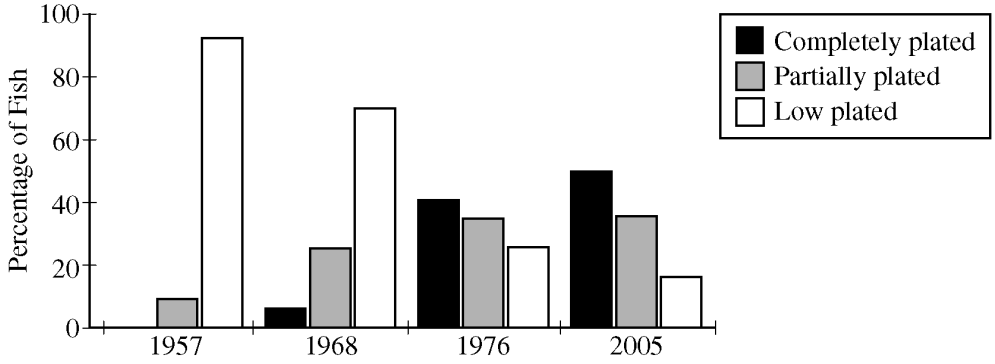

Figure 1. Armor plating phenotypes in Lake Washington stickleback population

A single gene, *ectodysplasin* (*EDA*), is thought to be responsible for the variation in the number of armor plates in sticklebacks. Figure 2 shows a phylogenetic tree constructed by comparing DNA sequences of the *EDA* gene from a number of stickleback populations with low-plated or completely plated phenotypes. Figure 3 shows a phylogenetic tree constructed by comparing the sequences of 25 genes that were randomly selected from the same populations as shown in Figure 2. In both figures, shaded populations display the completely plated phenotype.

8 Ecology MCQ

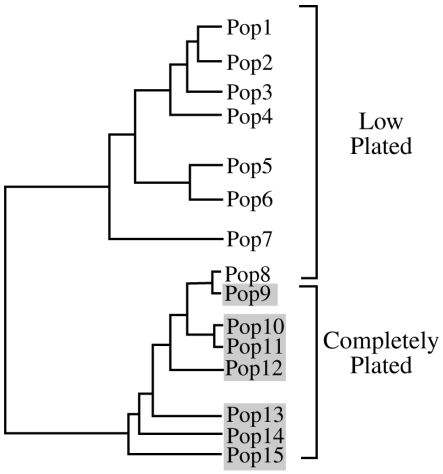

Figure 2. Phylogeny based on *EDA* gene only

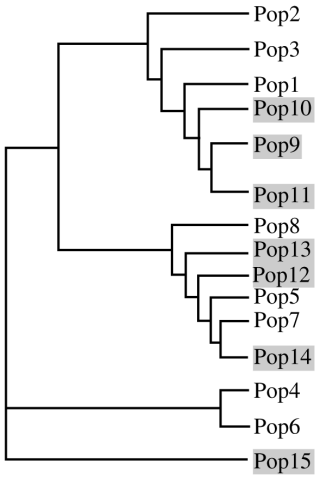

Figure 3. Phylogeny based on 25 random genes

114. Prior to 1960, Lake Washington was highly polluted and underwater visibility was limited to one or two meters. In the late 1960s, a large cleanup effort reduced pollution, resulting in visibility that increased to six to seven meters by 1976. Which of the following best explains how the change in underwater visibility affected armor plating in Lake Washington sticklebacks between 1957 and 1976?

(A) Higher visibility allowed sticklebacks to increase their food consumption to make armor production easier.

(B) Higher visibility made the sticklebacks more susceptible to large-toothed predators in the lakes, giving complete armor a selective advantage. ✓

(C) Clearer water allowed sunlight to penetrate to deeper depths, so the sticklebacks had to increase the amount of armor to protect themselves from the resulting increase in water temperature.

(D) Clearer water contained fewer of the molecular building blocks needed for armor production, so sticklebacks showed a reduction in the amount of armor.

8 Ecology MCQ

115. A group of biologists were investigating the effects of light color on phototropism (growing toward the light source) in tomato plants. Four groups of tomato plants were grown in the same greenhouse under identical conditions. Each group was then placed in a different room, each room having a different color of light, exposed to that single color of light from different directions for one week at a time. After each week, the biologists counted the number of tomato plants in the groups that were showing evidence of phototropism. The results are summarized in Table 1.

Table 1. Phototropic responses of plants to different colors of light

Test Group	Direction of Light	Percent of Plants Showing Phototropism
Group 1 (red light)	North	98
	South	99
	East	98
	West	96
Group 2 (green light)	North	4
	South	8
	East	5
	West	5
Group 3 (blue light)	North	65
	South	58
	East	65
	West	62
Group 4 (white light)	North	100
	South	100
	East	100
	West	99

Which of the following correctly identifies the dependent and independent variables in this experiment?

(A) The color of the light is the dependent variable, and the percentage of plants showing phototropism is the independent variable.

(B) The percentage of plants showing phototropism is the dependent variable, and the color of the light is the independent variable. ✓

(C) The direction of the light is the dependent variable, and the percentage of plants showing phototropism is the independent variable.

(D) The color of the light is the dependent variable, and the direction of the light is the independent variable.

8 Ecology MCQ

Answer B

Correct. The dependent variable is a variable whose value depends on that of another. For this experiment, this would be the percentage of plants showing phototropism. The independent variable represents the potential reason or cause for a given outcome. In this case, it is the color of the light.

116. Prairie dogs are small mammals that live in large colonies in burrows in the ground. Prairie dogs that are near their own relatives when a predator approaches are much more likely to issue a warning bark than those that are near unrelated prairie dogs. The prairie dogs that hear a warning bark are more likely to hide in their burrows than to remain above ground. However, the prairie dog that gives the warning bark is putting itself at increased risk of being identified and killed by the predator.

Which of the following presents the most likely evolutionary explanation for the behaviors described?

(A) The warning bark changes the behavior of the related prairie dogs nearby, allowing the prairie dog's family to have increased survival and reproductive success. ✓

(B) The barking prairie dog chooses to warn other prairie dogs, leading to more prairie dogs living above ground.

(C) The barking prairie dog is alerting unrelated prairie dogs to the predator, so it is not giving any advantage to its own relatives.

(D) The failure of the individual to bark when surrounded by unrelated prairie dogs ensures survival of the individual.

In a study on seed germination of radish seeds, five beakers were set up containing equal numbers of seeds. The beakers were subjected to the conditions indicated in the table below.

	Beaker 1	Beaker 2	Beaker 3	Beaker 4	Beaker 5
Incubation temperature	22°C	22°C	22°C	22°C	5°C
Treatment of radish seeds	None	Boiled	None	None	None
Addition of deoxygenated water	Yes	Yes	No	Yes	Yes
Addition of mineral oil to block oxygen diffusion	No	No	Yes	Yes	No

117. The experimental design allows testing of which of the following?

(A) Only the effect of temperature on germination

(B) Only the need for oxygen for germination

(C) Only the effect of light on germination

(D) The effects of both temperature and oxygen on germination ✓

(E) The effects of both temperature and light on germination

8 Ecology MCQ

118. Germination would be expected to occur first in which beaker?

(A) 1 ✓

(B) 2

(C) 3

(D) 4

(E) 5

119. Comparing results from which of the following pairs of beakers would help determine if enzymes are necessary for germination?

(A) 1 and 2 ✓

(B) 1 and 3

(C) 1 and 4

(D) 2 and 4

(E) 3 and 4

8 Ecology MCQ

To investigate the influence of predation risk on ray behavior, a student observed and counted the large marine animals swimming in a shallow, nearshore section of a coral reef ecosystem. The time of each observation was recorded relative to the time of high tide. The student noted that at low tide, when the water level is low, many of the large animals are forced out of the study area and into the deeper waters of the outer reef. During high tides, when the water level is high, the large animals are able to reenter the study area.

Over a three-day period, the student observed a total of 604 individual rays belonging to three species: cowtail rays, giant shovelnose rays, and black stingrays. For each ray that was sighted, its body length was estimated and its status as either alone (ungrouped) or found with other rays (grouped) was noted. Occasionally, rays were observed sifting through the sandy substrate of the study area to capture food items such as molluscs and crustaceans. In one instance, an injured ray with bite marks that were likely sustained in a shark attack was sighted. In addition to the rays, the student observed lemon sharks (n = 46) and blacktip reef sharks (n = 39). The results of the study are presented in the figures below.

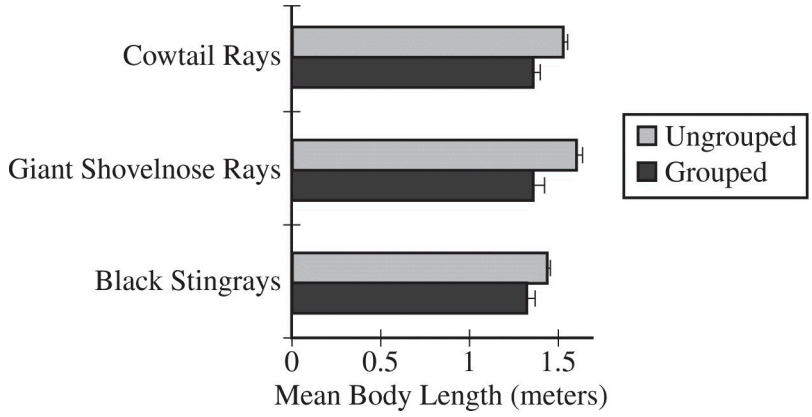

Figure 1. Comparison of mean body lengths of the grouped and ungrouped rays that were observed in a nearshore section of a coral reef ecosystem. Error bars represent $2SE_{\bar{x}}$

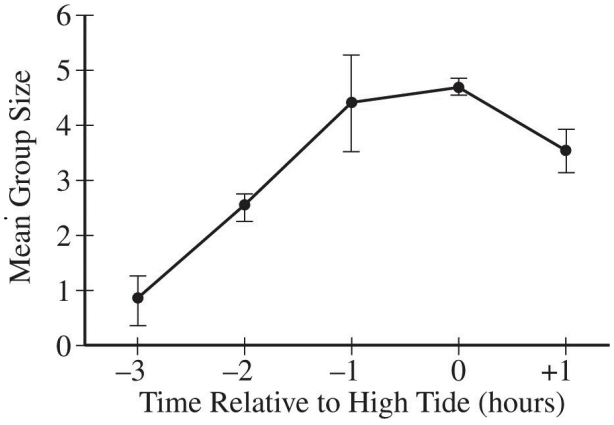

Figure 2. Mean numbers of rays per group in the study area at different stages of the tide cycle. High tide occurs at T = 0 hours.

8 Ecology MCQ

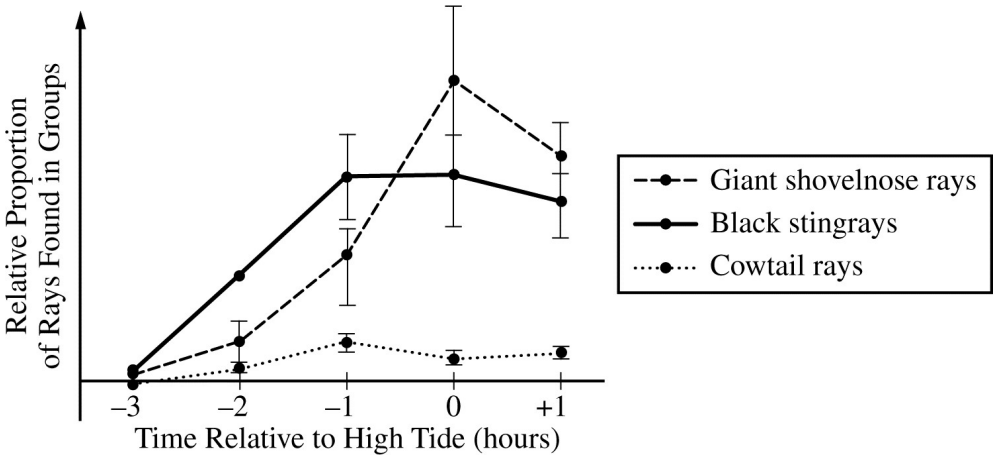

Figure 3. Relative proportions of rays in groups at different stages of the tide cycle for each of the three different populations. High tide occurs at T = 0 hours.

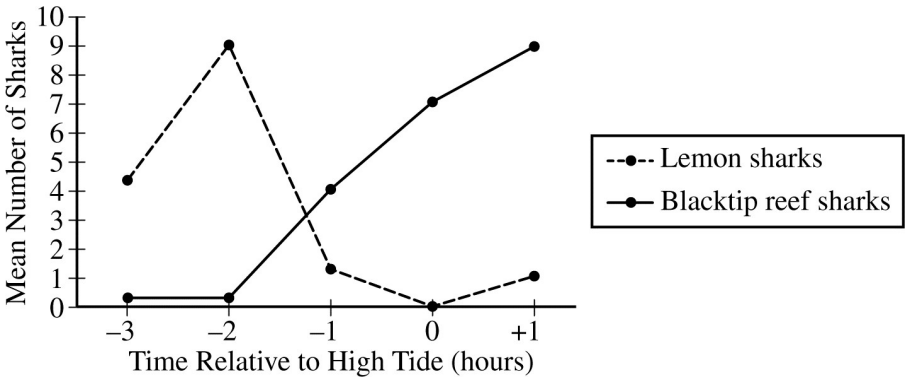

Figure 4. Mean numbers of lemon sharks and blacktip reef sharks in the study area at different stages of the tide cycle. High tide occurs at T = 0 hours.

120. Which of the following scientific claims about the survival strategies used by organisms in a coral reef ecosystem is best supported by the data presented in Figure 1?

(A) Smaller members of a vulnerable population are more likely to act collectively. ✓

(B) Smaller members of a vulnerable population are more likely to rely on defensive coloration.

(C) Larger members of a vulnerable population are more likely to exhibit aggression.

(D) Larger members of a vulnerable population are more likely to behave altruistically.

121. Based on the data presented in Figure 2, which of the following behavioral patterns do rays in a coral reef ecosystem typically exhibit three hours before high tide?

8 Ecology MCQ

(A) Group learning

(B) Mate selection

(C) Solitary foraging ✓

(D) Social interaction

122. Which of the following scientific claims about interacting populations of giant shovelnose rays and blacktip reef sharks is best supported by the results shown in Figures 3 and 4?

(A) Some parasitic species cause significant changes in the behavior of their hosts.

(B) Some invasive species outcompete native species for limited resources.

(C) Some migratory species transmit infectious diseases to resident populations.

(D) Some prey species form protective groups in the presence of predators. ✓

123. Which of the following best justifies the use of the study area to investigate how one species influences the behavior of another?

(A) Black stingrays were present in the study area in the presence and absence of blacktip reef sharks. ✓

(B) Giant shovelnose rays and cowtail rays spent long periods of time at rest while inside the study area.

(C) Natural barriers in the study area prevented lemon sharks from competing with blacktip reef sharks for limited resources.

(D) Cowtail rays in the study area formed groups at some stages of the tide cycle but remained alone at other stages.

124. Based on the results of the study, which of the following is the most likely connection between behavior and evolutionary fitness in a nearshore coral reef environment?

(A) Rays that search for food alone at low tide typically grow to larger sizes than do rays that search for food in groups.

(B) Rays that join groups during rising tides are reproductively more successful than are rays that do not join groups. ✓

(C) Rays that swim far from shore at high tide often encounter a greater variety of species than do rays that remain near the shore.

(D) Rays that roam across large distances during falling tides become stronger swimmers than do rays that spend more time at rest.

8 Ecology MCQ

125. Quorum sensing is a form of cell-to-cell communication used by some bacterial species to regulate their behavior and population size. *Staphylococcus aureus* uses quorum sensing to activate its virulence genes once its population reaches a minimum size. Virulence refers to the ability of a pathogen to cause disease.

S. aureus secretes autoinducing peptide (AIP), which binds to the AgrC receptor on the bacterial cell membrane. When the *S. aureus* population is sufficiently large and produces enough AIP, this concentration activates AgrC and stimulates a signal cascade resulting in the production of hemolysin, a toxin used to attack the host organism. In order to lessen the impact of the toxin produced by an *S. aureus* infection on patients, scientists created AIP mimics intended to act as competitive inhibitors of AIP − AgrC binding at the cell surface.

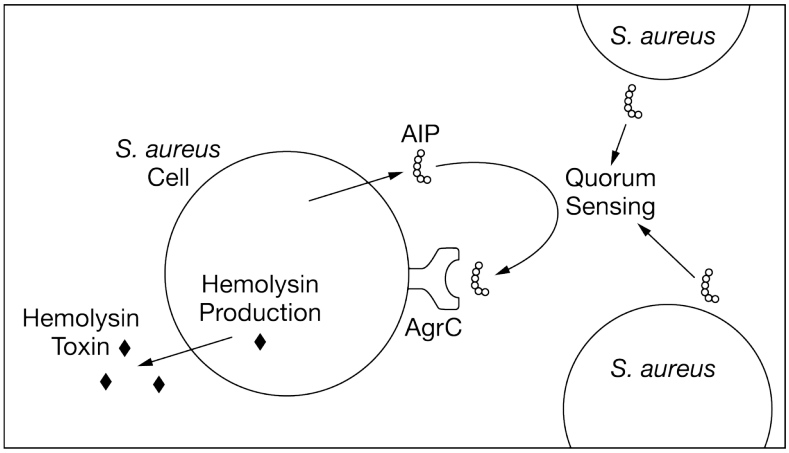

Figure 1. AIP cascade producing hemolysin toxin

Identify the experimental design that would determine which AIP mimic would most effectively inhibit the production of hemolysin toxin.

(A) Grow *S. aureus* in separate culture media containing one AIP mimic each and measure the amount of AIP mimic transported into the cells. The most effective AIP mimic will be absorbed the fastest.

(B) Grow *S. aureus* in separate culture media containing one AIP mimic each and measure the production of hemolysin toxin. The most effective AIP mimic will be in the culture producing the lowest amount of hemolysin. ✓

(C) Grow *S. aureus* in culture media containing hemolysin and measure the production of AIP mimic in response to hemolysin intake. The most effective AIP mimic will be produced in the highest amount.

(D) Grow *S. aureus* in separate culture media containing one AIP mimic each and measure the amount of live cells in each culture. The most effective AIP mimic will be the one that killed the most *S. aureus*.

Answer B

Correct. The most effective mimic will compete with AIP the best, prevent activation of the AgrC receptor, and decrease hemolysin production.

8 Ecology MCQ

126. In short-day (long-night) plants, the critical environmental factor that induces flowering is the

(A) duration of uninterrupted dark only ✓

(B) duration of uninterrupted light only

(C) total combined duration of the dark and light periods

(D) proportion of light to dark

(E) intensity of light during the light period

Snowshoe hares are small mammals that live in forests throughout Canada and the northern United States. Hares feed primarily on vegetation and are preyed on by lynx and owls.

In the fall, the hares shed their brown coat (molt), and a new white coat grows in over a period of 40 days. In the spring, the hares molt again and their white coat is replaced with a brown coat over a period of approximately 30 to 50 days.

In a recent study, researchers observed that the majority of hares in an isolated population began the molting process during the same week in the fall, even though the ground was not snow covered. The majority of hares also began the molting process during the same week in the spring, even though the ground was still snow covered. The researchers formulated a hypothesis that the timing of the molting process is influenced by seasonal changes in the daylight period.

127. Which of the following would best support the claim that the timing of the molting process in certain species is influenced by seasonal changes in the daylight period?

(A) More species that molt seasonally do so when the daylight period is the longest, than when the daylight period is the shortest.

(B) More species that molt seasonally do so when the daylight period is the shortest, than when the daylight period is the longest.

(C) More species that molt seasonally are found in the arctic, where the daylight period varies, than near the equator where the daylight period is relatively constant. ✓

(D) More species that molt seasonally are found near the equator, where the daylight period is relatively constant, than in the arctic, where the daylight period varies.

Answer C

Correct. Molt timing is strongly correlated with seasonal changes in the daylight period. Thus, these results would best support the claim.

128. To test their hypothesis, the researchers conducted a laboratory experiment in which groups of hares were exposed to gradually decreasing periods of daylight. The lengths of the daylight period when molting began were recorded.

Which of the following experimental controls should have been included to ensure that the results could be attributed to changes in the daylight period and not to some other variable?

8 Ecology MCQ

(A) Adjusting the temperature in the lab to simulate seasonal temperature changes

(B) Exposing one group of hares to a fixed daylight period for the duration of the experiment ✓

(C) Feeding the hares at various times of the day to simulate the availability of food in nature

(D) Using lights of different intensities to simulate different weather conditions

Answer B

Correct. The second group of hares in a fixed daylight period serves as a control. If all other environmental factors between the two groups of hares are the same and the first group shows a change in the start of molting, then this would suggest that the length of daylight is the cause.

129. If a viral infection significantly reduces the size of the snowshoe hare population, which of the following predicts the most likely effect on the community during that year?

(A) The biomass of producers will decrease.

(B) The biomass of the lynx will increase.

(C) New predators will feed on the owls.

(D) The lynx and owl populations will decrease. ✓

Answer D

Correct. Because hares are a primary food source for both lynx and owls, a decrease in the hare population size will result in fewer surviving lynx and owls; thus, these populations will also decrease.

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130. Which of the following statements most directly supports the claim that different species of organisms use different metabolic strategies to meet their energy requirements for growth, reproduction, and homeostasis?

(A) During cold periods pond-dwelling animals can increase the number of unsaturated fatty acids in their cell membranes while some plants make antifreeze proteins to prevent ice crystal formation in tissues.

(B) Bacteria lack introns while many eukaryotic genes contain many of these intervening sequences.

(C) Carnivores have more teeth that are specialized for ripping food while herbivores have more teeth that are specialized for grinding food.

(D) Plants generally use starch molecules for storage while animals use glycogen and fats for storage. ✓

8 Ecology MCQ

Answer D

This option is correct. Starch, glycogen, and fat are all molecules used for energy storage, yet plants have evolved to use starch while animals have evolved to use fat for long-term energy storage and glycogen for short-term energy storage.

131. Which of the following statements most directly supports the claim that different species of organisms use different metabolic strategies to meet their energy requirements for growth, reproduction, and homeostasis?

(A) During cold periods pond-dwelling animals can increase the number of unsaturated fatty acids in their cell membranes while some plants make antifreeze proteins to prevent ice crystal formation in tissues.

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132.

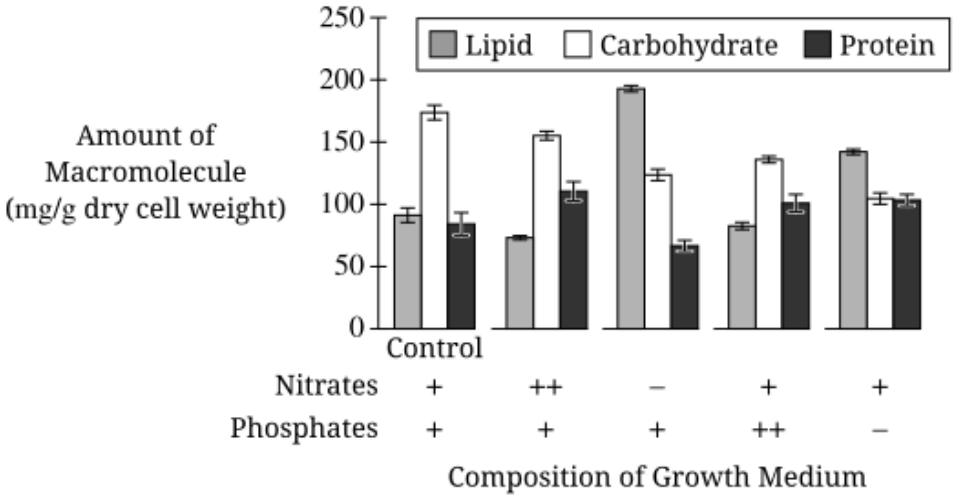


Figure 1. Effect of different abiotic factors on the macromolecule content of bacterial cells. Cells were grown in media with a standard amount (+), twice the standard amount (++), or a reduced amount (−) of each substance. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

To test the effect of different abiotic factors on the macromolecule content of bacterial cells, researchers grew identical samples of bacterial cells on growth media in which the amount of nitrates and phosphates differed. The amounts of carbohydrates, lipids, and proteins in each sample of cells were then determined (Figure 1).

Based on Figure 1, use of which of the following growth media resulted in a decrease in the amount of protein but an increase in the amount of lipids compared with the control?

8 Ecology MCQ

(A) Medium with a standard amount of nitrates and a reduced amount of phosphates

(B) Medium with a reduced amount of nitrates and a standard amount of phosphates ✓

(C) Medium with twice the standard amount of nitrates and a standard amount of phosphates

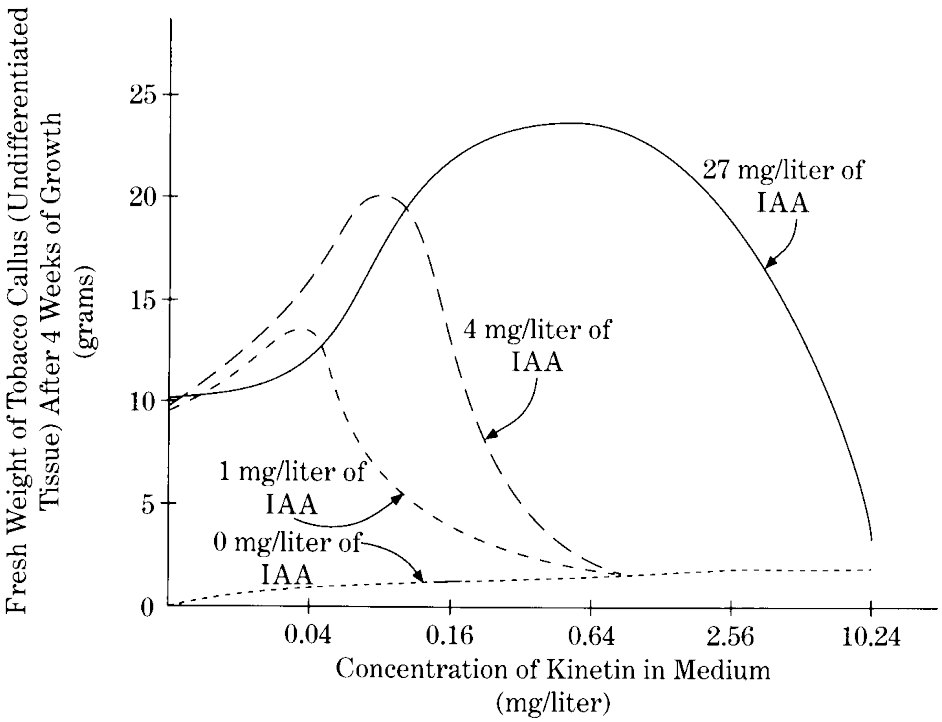
(D) Medium with a standard amount of nitrates and twice the standard amount of phosphates

Answer B

Correct. Based on the graph, this medium produced bacteria with an approximately 100% increase in lipid but an approximately 30% reduction in protein compared with the control.

Directions: Each group of questions below concerns an experimental or a laboratory situation. In each case, first study the description of the situation. Then choose the one best answer to each question following it.

The following questions refer to the data presented in the graph below of tobacco cells grown in tissue culture. The numbers on the curves indicate the concentrations of indoleacetic acid (IAA) in milligrams per liter.



8 Ecology MCQ

133. The optimum concentrations of hormones for promoting maximum tobacco cell growth are

(A) 27 mg/liter of IAA alone

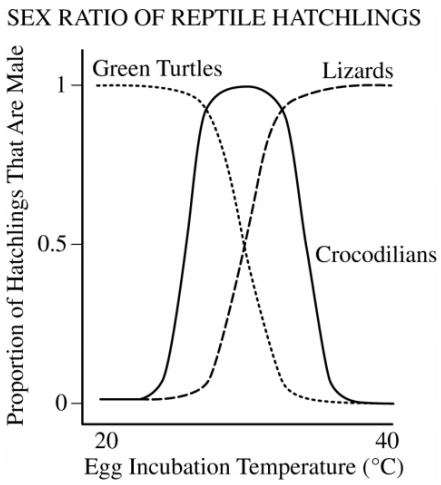
(B) 27 mg/liter of IAA and 2.56 mg/liter of kinetin

(C) 27 mg/liter of IAA and 0.64 mg/liter of kinetin ✓

(D) 4 mg/liter of IAA and 0.64 mg/liter of kinetin

(E) 1 mg/liter of IAA and 0.64 mg/liter of kinetin

134.



The graph above represents the proportions of hatchlings of certain types of reptiles that are male at different egg incubation temperatures. Which of the following claims is best supported by the data?

(A) The expression of sex-determination genes in certain reptiles is influenced by temperature. ✓

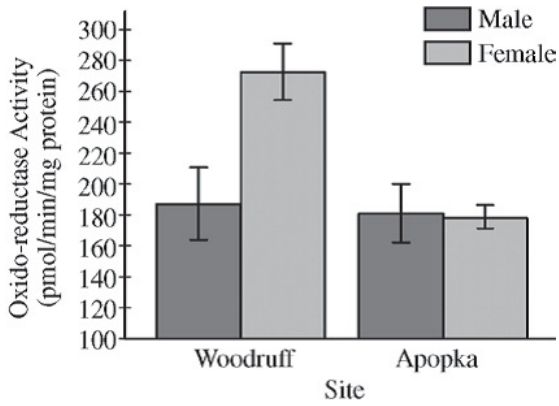
(B) Green turtle eggs do not hatch at temperatures above 35°C.

(C) In reptiles, sex determination is influenced by the sex ratio of the existing population.

(D) Incubating lizard eggs at constant low temperatures produces a large proportion of males.

8 Ecology MCQ

135.



Testosterone oxido-reductase is a liver enzyme that regulates testosterone levels in alligators. One study compared testosterone oxido-reductase activity between male and female alligators from Lake Woodruff, a relatively pristine environment, and from Lake Apopka, an area that has suffered severe contamination. The graph above depicts the findings of that study.

The data in the graph best support which of the following claims?

(A) Environmental contamination elevates total testosterone oxido-reductase activity in females.

(B) Environmental contamination reduces total testosterone oxido-reductase activity in females. ✓

(C) Environmental contamination elevates total testosterone oxido-reductase activity in males.

(D) Environmental contamination reduces total testosterone oxido-reductase activity in males.

Answer B

This option is correct. In Lake Woodruff, the lake that is relatively pristine, females had high levels of oxido-reductase enzyme activity, but in Lake Apopka, where environmental contamination conditions were severe, the enzyme activity was low, indicating that environmental factors had an influence on the organism.

8 Ecology MCQ

Students investigated the effect of light on the carbon cycle in aquatic ecosystems by performing the controlled experiment summarized below. The students placed equal amounts of water (pH 7.0) from a large aquarium in glass beakers. The students transferred aquatic plants from the aquarium to several of the beakers, and then they placed equal numbers of the beakers in the light or the dark (Figure 1: groups I and II). Similarly, the students transferred goldfish from the same aquarium to other beakers, and then they placed equal numbers of those beakers in the light or dark (Figure 1: groups III and IV). Finally, the students placed an equal number of beakers containing water only in the light or dark (Figure 1: groups V and VI).

After exposing the samples to light or dark for one hour, the students recorded the pH of the water in each beaker. Carbon dioxide dissolved in water will lower the pH of an aqueous solution. In the experiment, the students used changes in pH to monitor changes in the amount of carbon dioxide in the water. For each treatment group, the students calculated the mean pH and standard error, as documented in the table below.

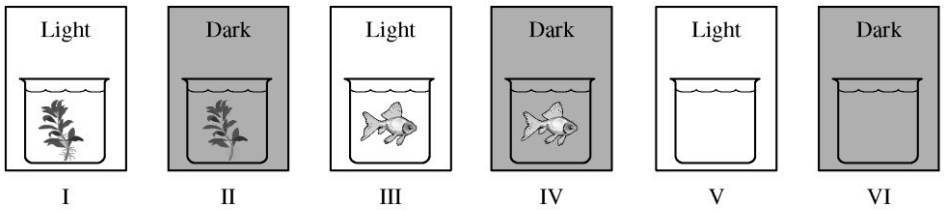


Figure 1. Treatment groups

MEAN pH OF TREATMENT GROUPS AFTER 1 HOUR

Treatment group (n = 10)	I	II	III	IV	V	VI
Mean pH	8.2	6.4	5.9	5.6	7.3	6.8
Standard error of the mean	0.3	0.3	0.2	0.3	0.5	0.4

136. Which of the following observations provides the best evidence that photosynthesis occurred in treatment group I?

(A) The specimens in the beakers were aquatic plants from a large aquarium.

(B) The beakers were placed in the light.

(C) The mean pH of the samples increased after one hour. ✓

(D) The standard error of the mean was smaller than that for treatment group V.

137. Which of the following modifications to the experimental design will best help reduce the standard errors of the means?

(A) Using pond water instead of aquarium water

(B) Exposing samples to light for a greater amount of time

(C) Increasing the sample size of each treatment group ✓

(D) Collecting organisms from a natural water source instead of an aquarium

8 Ecology MCQ

138. The results for treatment groups V and VI could suggest which of the following questions about the design of the experiment?

(A) Do the glass beakers prevent light from reaching the test samples?

(B) Is the method used for measuring pH harmful to aquatic organisms?

(C) Is the availability of carbon dioxide a limiting factor in aquatic ecosystems?

(D) Does the aquarium water contain living microorganisms? ✓

139. To investigate whether an organism in the study is capable of both photosynthesis and respiration, a comparison of which treatment groups is most appropriate?

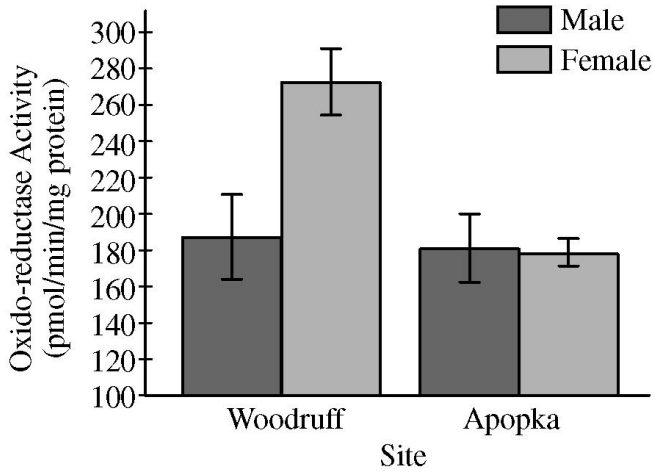
(A) I and II ✓

(B) II and IV

(C) III and V

(D) IV and VI

140.



Testosterone oxido-reductase is a liver enzyme that regulates testosterone levels in alligators. One study compared testosterone oxido-reductase activity between male and female alligators from Lake Woodruff, a relatively pristine environment, and from Lake Apopka, an area that has suffered severe contamination. The graph above depicts the findings of that study.

The data in the graph best support which of the following claims?

(A) Environmental contamination elevates total testosterone oxido-reductase activity in females.

(B) Environmental contamination reduces total testosterone oxido-reductase activity in females. ✓

(C) Environmental contamination elevates total testosterone oxido-reductase activity in males.

(D) Environmental contamination reduces total testosterone oxido-reductase activity in males.

141. Which of the following describes the mechanism by which a plant stem grows toward light?

8 Ecology MCQ

(A) The plant seeks light in order to maximize photosynthesis.

(B) Nervelike impulses stimulate contractile cells on the lighted side of the stem.

(C) Cells on the dark side of the stem elongate more than those on the lighted side. ✓

(D) The plant grows into an open area where its leaves will not be shaded by competing plants.

(E) The greater energy supply on the lighted side of the stem stimulates metabolism and growth on that side.

142.

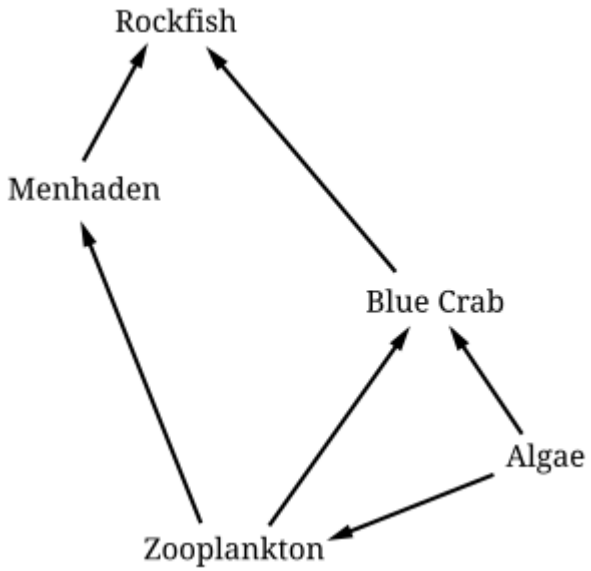


Figure 1. Feeding relationships between selected organisms in the Chesapeake Bay

Figure 1 represents the feeding relationships between selected organisms in the Chesapeake Bay. Decades of overharvesting have reduced the populations of both blue crabs and rockfish. However, recent efforts have increased the number of rockfish in the bay. Based on Figure 1, what is the most likely result of the increase in rockfish?

(A) The population of menhaden will increase because the energy available from zooplankton will increase.

(B) The population of menhaden will decrease because the energy resources available to them will decrease.

(C) The population of zooplankton will decrease because the energy available from algae will decrease.

(D) The population of blue crabs will decrease because predation by rockfish will increase. ✓

Answer D

Correct. Based on the figure, rockfish prey on blue crabs. An increase in the rockfish population will result in an increase in predation, which will reduce the blue crab population.

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143. The deep ocean floor communities contain very little life. In constant darkness, the water at these depths hovers around $2^{\circ}C$. Sparsely distributed in this environment are hydrothermal vent ecosystems. Superheated water streams out of the vents at temperatures approaching $400^{\circ}C$. These waters are rich in inorganic materials, including carbon dioxide and hydrogen sulfide. Hydrothermal vent zones have a density of organisms 10,000 to 100,000 times greater than found on the surrounding seafloor.

Scientists have studied these vents to determine the energy source for these communities and the identity of the primary producers in the perpetual darkness of the ocean floor. As expected, they found a total absence of photosynthetic organisms at all vents, identifying mats of bacteria as the producers of the carbon compounds that are used by the rest of the community for growth, repair, and as an energy source.

Which of the following describes how the vent bacteria harvest energy to produce organic compounds for use by the vent community?

(A) The bacteria use the heat from the hot water to assemble carbon-based materials from inorganic molecules.

(B) The bacteria function as chemoautotrophs, extracting energy from hydrogen-rich inorganic molecules. ✓

(C) Organic matter drifting down from the ocean's surface is used to make molecules with enough energy to sustain the vent community.

(D) The bacteria can recycle the energy from the waste products and dead material produced by the heterotrophs of the vent community into energy-rich carbon compounds that can be recycled.

Answer B

Correct. The bacteria use the energy stored in hydrogen sulfide to produce carbon compounds that are used by the heterotrophs in the community. This process is known as chemosynthesis.

144. Beaked whales feed at various depths, but they defecate at the ocean's surface. Nitrogen-rich whale feces deposited in surface waters supply nutrients for algae that are eaten by surface dwelling fish. Which of the following best predicts what would happen if the whale population decreased?

(A) There would be a reduction in surface nitrogen concentration, which would cause an algal bloom.

(B) The surface fish populations would decline due to reduced populations of algae. ✓

(C) The remaining whales would accumulate mutations at a faster rate.

(D) The remaining whales would be forced to forage in the deepest parts of the ocean.

145. ATP serves as a common energy source for organisms because

8 Ecology MCQ

(A) it is the smallest energy molecule

(B) it stores the least energy of any energy source

(C) its energy can be easily transferred to do cellular work ✓

(D) it is extremely stable and can be stored in the cell for long periods of time

(E) traces of it have been found in fossils of ancient organisms dating back to the beginning of life on Earth

146. A student created a terrarium by placing soil in the bottom of a large glass container. He added several species of mosses and tiny grasses to the soil. The student ensured that the soil and plants were damp by adding water from a nearby creek. He added a few dead, decaying leaves and twigs, some small insects, and a few small garden slugs to the terrarium. He placed an airtight seal on the container and placed the container near a window where it would receive some sunlight but would not overheat. After three years, the terrarium still contains living organisms.

Which of the following statements best explains how this terrarium could sustain living organisms for three years?

(A) After a short time, the seal began to leak, allowing gaseous forms of carbon, phosphorus, and nitrogen to enter the terrarium.

(B) Small amounts of carbon, phosphorus, and nitrogen atoms have been leaking through the terrarium's glass.

(C) Carbon, phosphorus, and nitrogen atoms have been cycling through the terrarium's organisms and environment. ✓

(D) New carbon, phosphorus, and nitrogen atoms have been created through natural processes inside the terrarium.

8 Ecology MCQ

147. In an experiment, three different animals were exposed to a series of environmental temperatures and their body temperatures were assessed at multiple temperature points between 10 and 40 degrees Celsius. Ten different individuals of each type of animal were included in the study to produce an average body temperature profile for each type of animal. Figure 1 shows the data collected.

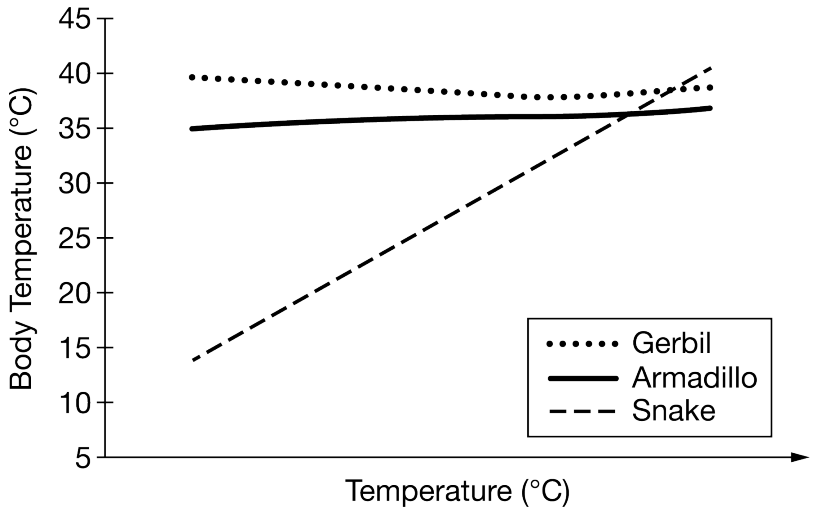


Figure 1. Effect of environmental temperature on body temperature of selected animals

Which of the following statements is most consistent with the data in Figure 1 ?

(A) Armadillos tend to be more metabolically active than either gerbils or snakes are.

(B) Armadillos cannot survive at environmental temperatures over $30^{\circ}C$.

(C) Snakes are found in terrestrial rather than aquatic environments.

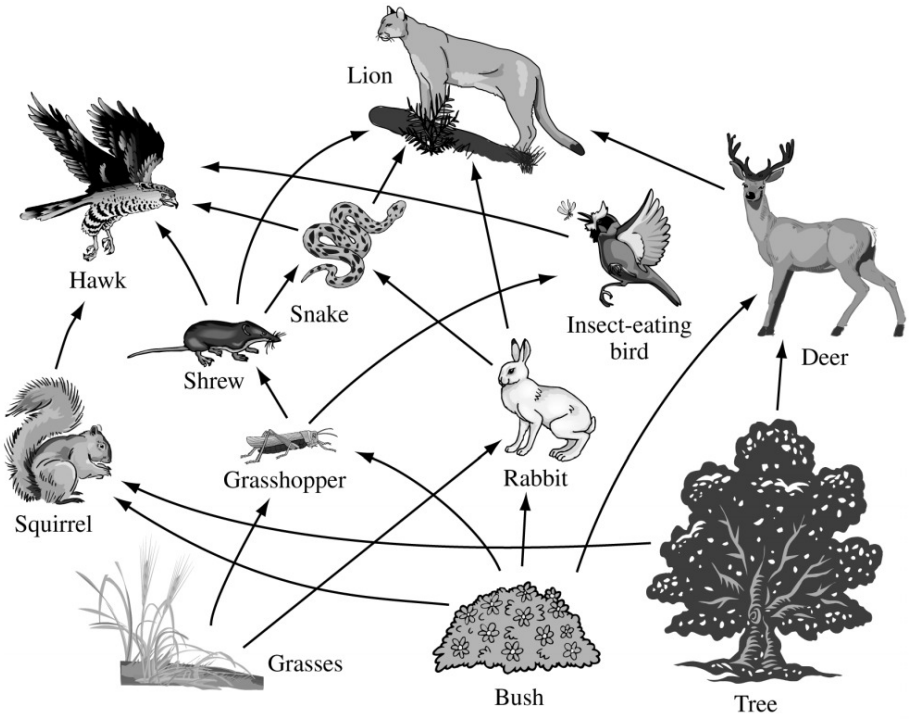
(D) Snakes display behaviors that allow them to absorb or radiate heat as required. ✓

Answer D

Correct. Figure 1 shows that snakes cannot control their internal temperatures through metabolic processes. In low temperatures, snake will bask on rocks and flatten their profiles to absorb heat to raise their body temperatures above ambient temperature.

8 Ecology MCQ

148.



The food web represented above does not include bacteria and fungi. Which of the following best describes a consequence of having no bacteria and fungi in the food web?

(A) Grasses will have no direct access to chemical components recycled from dead organisms. ✓

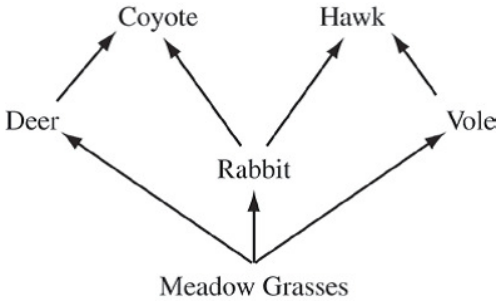
(B) Rabbits will have no direct access to energy absorbed from secondary consumers.

(C) Shrews will have no direct access to matter transferred from tertiary consumers.

(D) Hawks will have no direct access to nitrogen acquired from the atmosphere.

8 Ecology MCQ

149. The following is a food web for a meadow habitat that occupies 25.6 km^2. The primary producers' biomass is uniformly distributed throughout the habitat and totals 1,500 kg/km^2.



Developers have approved a project that will permanently reduce the primary producers' biomass by 50 percent and remove all rabbits and deer.

Which of the following is the most likely result at the completion of the project?

(A) The biomass of coyotes will be 6 kg, and the biomass of hawks will be 0.5 kg.

(B) The biomass of coyotes will be dramatically reduced. ✓

(C) The coyotes will switch prey preferences and outcompete the hawks.

(D) There will be 50 percent fewer voles and 90 percent fewer hawks.

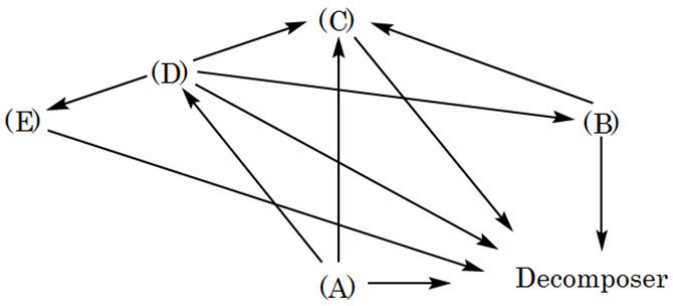
Answer B

This option is correct. It demonstrates an understanding of the components of a food web and interactions between all of the participants. Coyotes prey on deer and rabbits, and if developers remove them, coyotes will lose their primary source of nutrition/energy. With this loss of nutrition/energy they will experience a decrease in reproductive success and therefore a significant decline in their population.

8 Ecology MCQ

Directions: Each set of lettered choices below refers to the numbered words or statements immediately following it. Select the one lettered choice that best fits each word or statement. A choice may be used once, more than once, or not at all in each set.

The figure below represents a food web in a particular ecosystem. Each letter represents a species. The arrows indicate the direction of energy flow.



150. Which species would most likely represent humans if they were part of this ecosystem?

(A) A
(B) B
(C) C ✓
(D) D
(E) E

151. A photosynthetic organism would be represented by which species?

(A) A ✓
(B) B
(C) C
(D) D
(E) E

152. Members of which species are herbivores?

(A) A
(B) B
(C) C
(D) D ✓
(E) E

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153. Members of which species are most likely to be omnivorous?

(A) A

(B) B

(C) C ✓

(D) D

(E) E

8 Ecology MCQ

154. In an experiment to determine the effect of light availability on species richness in an environment, a grassland was divided into 26 plots that were assigned to one of two treatment groups:

1. Clipped – plots were mowed to simulate the grazing of herbivores that maintain the low height of grasses.

2. Enclosure – plots were enclosed in fences, and grasses were allowed to grow undisturbed to their maximum height.

After eight years, data was collected on the amount of light penetration to the soil (bottom light density) and the number of species in the two treatment groups. Figure 1 shows the averaged light penetration measurements, and Figure 2 shows the averaged number of species per plot type.

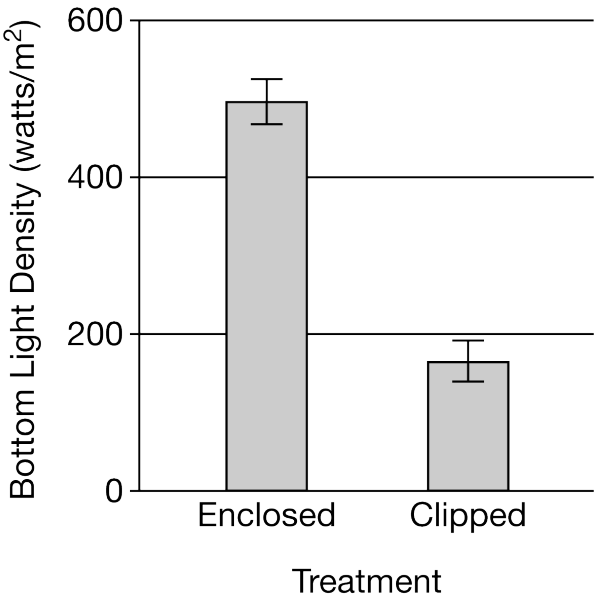

Figure 1. Average bottom light intensity in clipped plots versus enclosed plots

8 Ecology MCQ

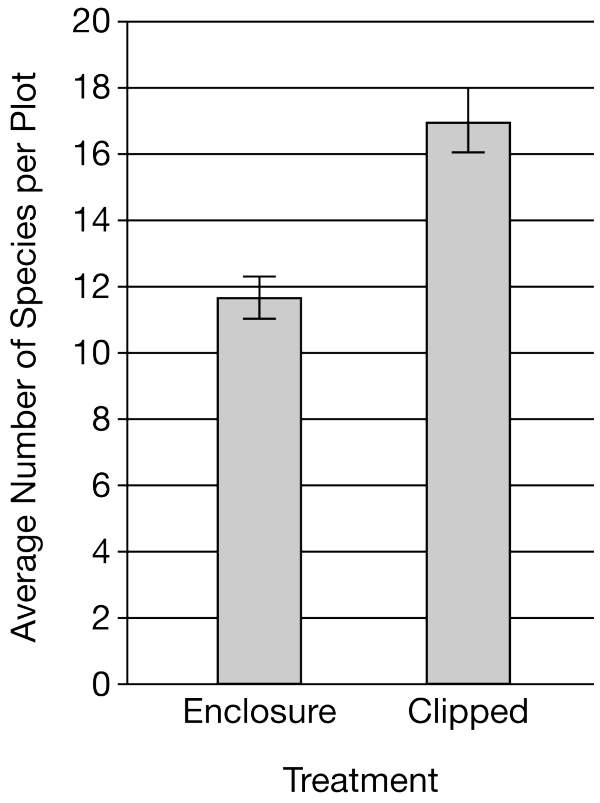

Figure 2. Richness comparison of two treatments

Which of the following statements is consistent with the data shown in the figures?

(A) Grazing activity in a field decreases the species richness of the field.

(B) Different patterns of energy availability in ecosystems will affect species richness. ✓

(C) Forests that develop a closed canopy with a shaded floor will exhibit increased species richness.

(D) The enclosed plots are more efficient at capturing and distributing energy than the clipped plots.

Answer B

Correct. When light energy is allowed to reach the soil surface of an ecosystem, species richness is greater than when most of the light is absorbed before it can reach the soil surface.

8 Ecology MCQ

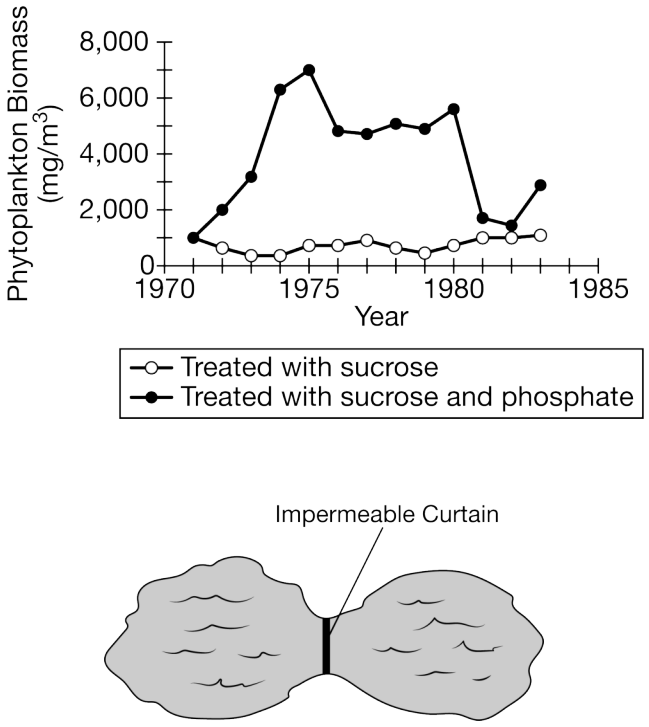

Figure 1. Phytoplankton biomass in two sides of a small lake that is divided by an impermeable curtain

In the early 1970s, researchers hypothesized that carbon was the limiting nutrient in many aquatic ecosystems. To test this hypothesis, the researchers divided a small lake in two roughly equal halves with an impermeable curtain that was fastened and sealed to the bedrock of the lake. Beginning in 1971 the researchers treated one side of the lake with sucrose and the other side with both sucrose and phosphate. From 1971 to 1983 the researchers monitored the phytoplankton biomass in both parts of the lake. The results are shown in Figure 1.

155. Which of the following was most likely a direct consequence of the addition of phosphate to the lake?

(A) The amount of biomass in the first trophic level decreased.

(B) The amount of biomass in the second trophic level decreased.

(C) The amount of energy available to producers in the lake increased.

(D) The amount of energy available to consumers in the lake increased. ✓

156. Which of the following treatments would have been the best control treatment for the experiment?

(A) An untreated section of the lake ✓

(B) A section of the lake that was treated with phosphate but not sucrose

(C) A different lake that was treated with sucrose and phosphate

(D) A small pool of the lake water maintained in a controlled laboratory environment

8 Ecology MCQ

157. The average growth rate of the phytoplankton population from 1971 to 1975 in the side of the lake treated with sucrose and phosphate is closest to which of the following?

(A) $125\ \frac{\text{mg}}{\text{m}^3}$ per year

(B) $1,000\ \frac{\text{mg}}{\text{m}^3}$ per year

(C) $1,500\ \frac{\text{mg}}{\text{m}^3}$ per year ✓

(D) $6,000\ \frac{\text{mg}}{\text{m}^3}$ per year

158. Which of the following claims is best supported by the data?

(A) Carbon was a limiting factor for phytoplankton in the lake.

(B) Phosphate was a limiting factor for phytoplankton in the lake. ✓

(C) Both carbon and phosphate were limiting factors for phytoplankton in the lake.

(D) Neither carbon nor phosphate was a limiting factor for phytoplankton in the lake.

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159. An ecologist collected a large sample of forest soil and distributed it evenly into two samples, sample A and sample B. Sample A was placed in a temperature control chamber that was maintained at the forest's average high temperature. Sample B was placed in a separate temperature control chamber that was maintained at $5^{\circ}C$ above the forest's average high temperature. Both samples were kept in their chambers for 60 days and were watered with only enough sterile water to keep the soil moist. After 60 days, soil from both samples was removed and tested for ammonia levels. Results are shown in Table 1.

Table 1. Ammonia Levels $(\text{mg/kg},\ \pm 2SE_{\overline{x}})$

Sample A	Sample B
11.4 \(±\) 0.7	3.3 \(±\) 0.4

Which of the following is the best explanation of the ammonia level in sample B shown in Table 1?

(A) The ammonia level was lower in sample B because the elevated temperature caused the ammonia to evaporate into the atmosphere.

(B) The ammonia level was lower in sample B because the added water washed ammonia out of the soil.

(C) The ammonia level was lower in sample B because the increased temperature reduced the number of nitrogen-fixing bacteria in the soil. ✓

(D) The ammonia level was lower in sample B because the elevated temperature caused the ammonia molecules to break down into simple atoms.

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Answer C

Correct. Nitrogen gas in the atmosphere is fixed into ammonia and ammonium in the soil by nitrogen-fixing bacteria.

160. Which of the following best explains why there are seldom more than five trophic levels in a food chain?

(A) Most carnivores function at more than one trophic level.
(B) Trophic levels above this number contain too many individuals.
(C) Top carnivores are too few in number to prey effectively.
(D) The ecosystem contains too much biomass.
(E) Energy is lost from each trophic level. ✓

161.

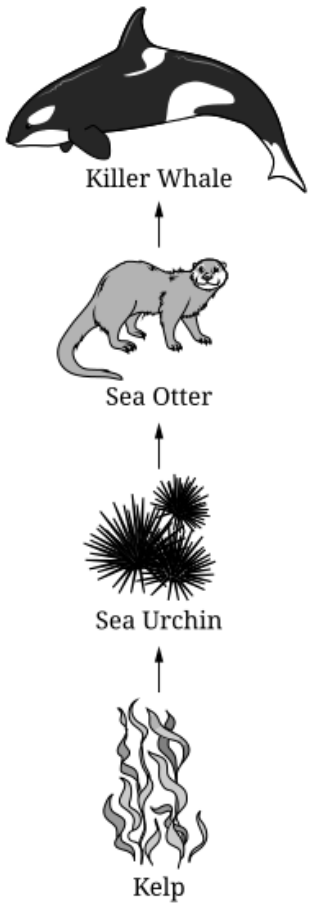


Figure 1. A food chain in the North Pacific Ocean

Based on Figure 1, which of the following is the most likely short-term effect of a decrease in the killer whale population?

8 Ecology MCQ

(A) The sea otter and kelp populations will increase, while the sea urchin population will decrease. ✓

(B) The sea otter and sea urchin populations will increase, while the kelp population will decrease.

(C) The sea otter population will stay the same, while the sea urchin and kelp populations will increase.

(D) There will be no change to the sea otter, sea urchin, and kelp populations because another predator species will fill the niche previously occupied by the killer whales.

Answer A

Correct. A decrease in killer whales will reduce predation on sea otters, allowing their population to increase. An increase in sea otters will decrease the population of sea urchins, which eat kelp. Therefore, the population size of kelp will also increase.

162. In a study of energy flow, the following data were collected:

- Each human requires 55 grams of protein per day.
- An acre of soybeans produces 200,000 grams of protein per year.
- A cow grazing on one acre of land produces 19,000 grams of protein per year.

Based on these data, an acre of land would support:

- 10 humans per year on a diet of soy, or
- 0.9 human per year on a diet of beef.

Which of the following statements best explains these relationships in terms of energy transfer?

(A) Humans are better adapted to make efficient use of soy protein than beef protein, so they can extract more energy from a gram of soy protein than from a gram of beef protein.

(B) A cow can harvest energy only at a certain rate, so adding another cow to the acre of land would double the number of humans supported by the acre.

(C) Preparing meat for human consumption takes more energy per gram of protein than preparing soy for consumption.

(D) Only 10% of the energy on each level of the trophic pyramid is available for use by the next level. The rest is lost as heat or used for growth and repair. ✓

Answer D

Correct. The 10% rule states that approximately 90% of the energy at any trophic level is lost to heat, growth, and repair of the organisms, and only 10% is available for harvesting by the next trophic level.

8 Ecology MCQ

163. Producers within very deep regions of the ocean most likely introduce energy into ecosystems through which of the following processes?

(A) Absorbing photons of light energy

(B) Capturing electrons from inorganic molecules ✓

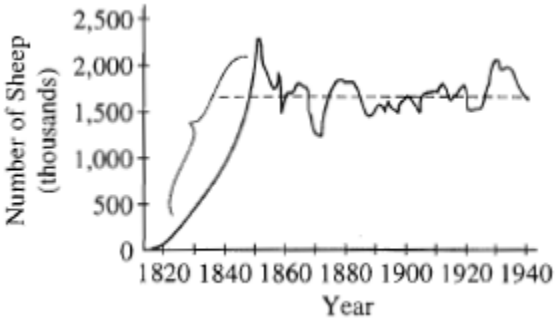
(C) Absorbing heat energy from hydrothermal vents on the ocean floor

(D) Living in other organisms

Answer B

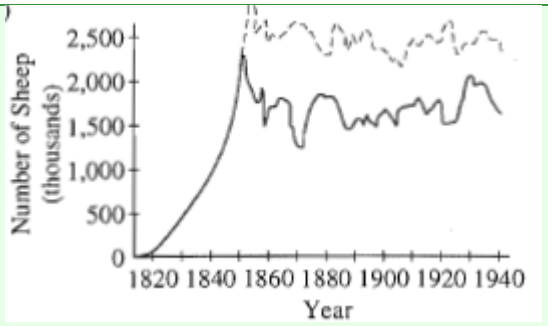
Correct. Breaking chemical bonds in inorganic molecules provides energy for the transfer of electrons in the bonds. Organisms that utilize electrons captured from the bonds of such molecules are classified as producers and are referred to as chemosynthetic organisms.

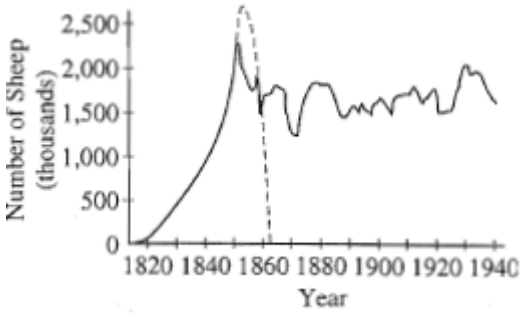
The graph below shows changes in a population of wild sheep that were introduced to the island of Tasmania in the early 1800's.

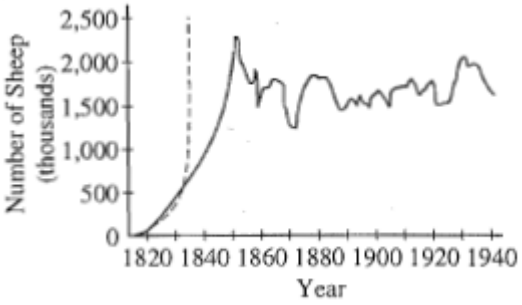


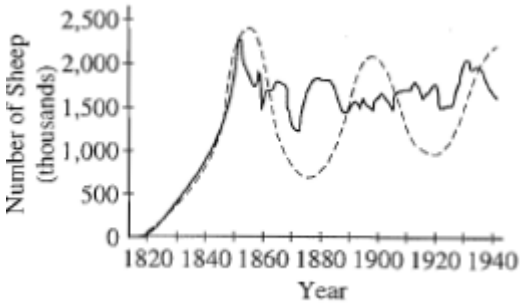
164. In the graphs below, the solid line represents the original population. The dotted line on which graph best represents the sheep population that would have resulted from a sustained increase in the primary productivity of the environment?

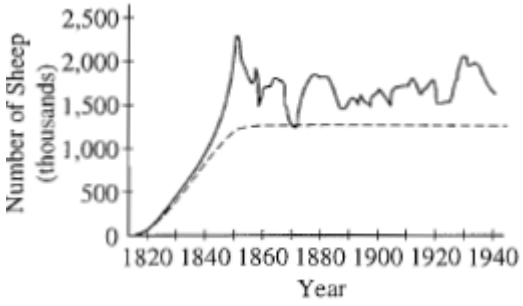
8 Ecology MCQ

(A) 

(B) 

(C) 

(D) 

(E) 

8 Ecology MCQ

165. The type of population growth represented by that portion of the graph line enclosed in the bracket is most accurately termed

(A) stable

(B) exponential ✓

(C) density-dependent

(D) arithmetic

(E) decelerating

166. The graph indicates that the sheep population most likely is

(A) growing in excess of its carrying capacity, since fluctuations in population size occurred after 1850

(B) headed for extinction because of the population explosion about 1930

(C) regulated by density-independent factors, because there appears to be about a 10-year cycle of sharp declines in size

(D) shifting from a *K*-selected strategy to an *r*-selected strategy

(E) stable after 1850 under the effects of density-dependent regulating factors ✓

167. The dashed line on the graph represents the

(A) maximum population size

(B) average birth rate

(C) biotic potential of the population

(D) carrying capacity of the environment ✓

(E) point of maximum effect for density-independent factors

8 Ecology MCQ

168. Figure 1 shows the population fluctuations of phytoplankton and zooplankton in the eastern Pacific Ocean over the course of six years. The major dip in phytoplankton was attributed to a lack of nutrients for phytoplankton.

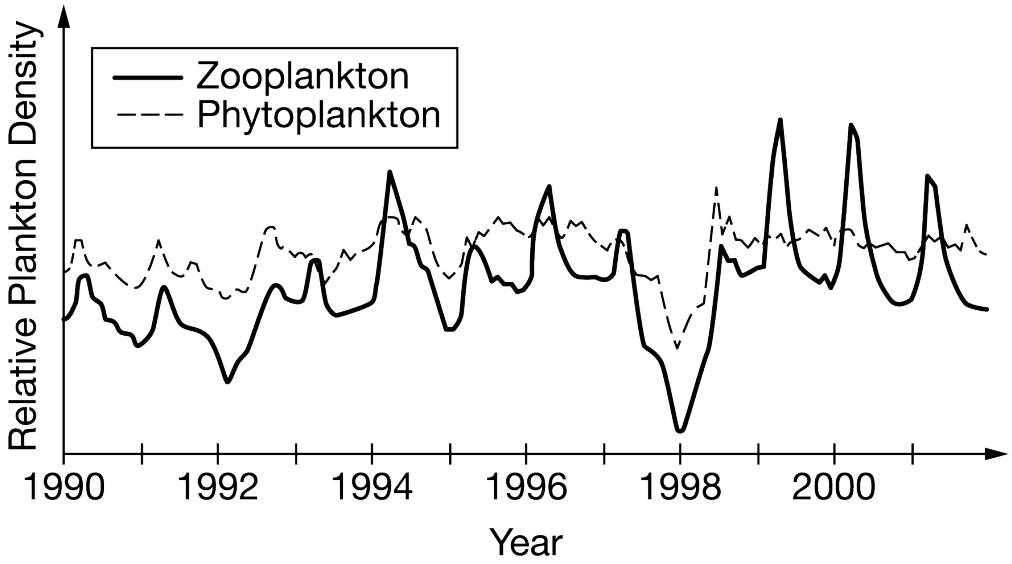


Figure 1. Relative abundance of plankton over time in the eastern equatorial Pacific Ocean

Of the following, which explains why the zooplankton also dipped during the same year and then seemed to cycle over the next three years?

(A) The zooplankton rely on the same nutrients for energy harvesting as the phytoplankton, so their population was also unable to acquire the necessary energy to maintain their population level. After that major dip there were enough nutrients for both categories of plankton.

(B) Zooplankton are more temperature-sensitive than the phytoplankton, so the annual seasonal changes will affect their population sizes in a cyclical fashion. There was a sudden, extreme temperature change in the first dip that also affected the phytoplankton.

(C) A migration of whales was blown off course by a storm one year and ate most of the phytoplankton and zooplankton. In the following years the whales returned to their normal migration route.

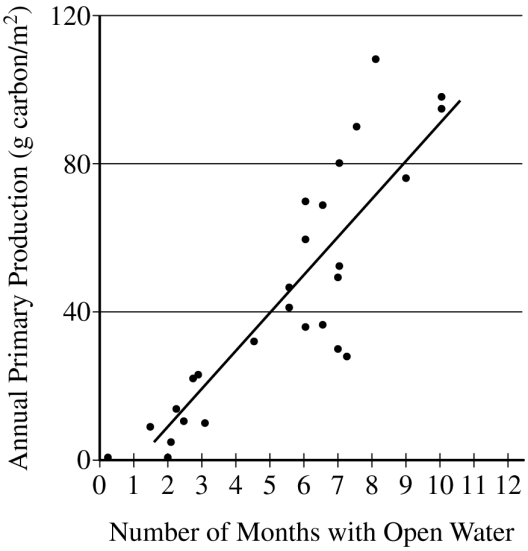
(D) The reduction in the phytoplankton population meant that the zooplankton's energy source was drastically reduced, leading to the decrease in the zooplankton population size. After the dip, the phytoplankton provided a steady energy source, and the zooplankton population fluctuated around its carrying capacity. ✓

Answer D

Correct. The energy source of zooplankton is the carbon compounds fixed by phytoplankton. Without the proper inorganic nutrients, the phytoplankton population is limited in its photosynthetic activity, leading to an energy deficit in the zooplankton populations. It is also correct that the annual fluctuations in zooplankton represent a stable population cycling around its carrying capacity.

8 Ecology MCQ

169.



In the Arctic Ocean, the predominant primary producers are phytoplankton. Phytoplankton are consumed by zooplankton, which in turn are eaten by codfish. In years when there is more open water (less ice coverage), there are more zooplankton and fish than in years with less open water (more ice coverage). Based on the graph above, the difference is most likely because

(A) when there is less open water, light is blocked from the zooplankton, so they cannot produce as much food for the fish

(B) when there is more open water, the temperature is warmer, so the zooplankton and fish populations increase in size

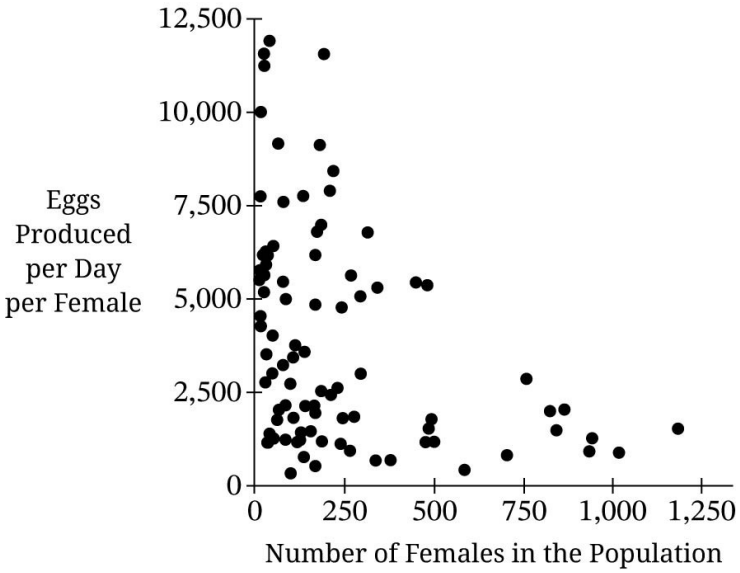
(C) the ice blocks the light, so in years with more ice coverage, there is less photosynthesis by the phytoplankton ✓

(D) the ice increases the light available for photosynthesis, so primary production increases and zooplankton populations increase in size

8 Ecology MCQ

170. Researchers studying parasitic hookworms investigated the effect of the size of the female hookworm population on the number of eggs produced per worm (Figure 1).

Figure 1. Eggs produced per hookworm in female populations of various sizes



Based on the data in Figure 1, which of the following statements best explains the relationship between numbers of females and eggs produced per day per female?

(A) As the female population increases, each individual expends more energy on competition, leaving less energy available for egg production. ✓

(B) Females in larger populations reduce their energy output for egg-laying to conserve resources for maintaining the overall population size.

(C) The decrease in eggs produced per female reflects a shift in energy allocation from reproduction to improved egg quality in denser populations.

(D) As the number of females grows, the total energy output of the population for egg production remains stable, resulting in fewer eggs produced per female.

Answer A

Correct. As the female population increases, competition for resources increases, leaving each individual with less energy to invest in egg production, resulting in fewer eggs per female.

8 Ecology MCQ

171. Baker's yeast, *Saccharomycetes cerevisiae*, is the species of yeast that is commonly used in baking, brewing, and making wine. It is a single-celled organism; each cell is capable of undergoing asexual reproduction by mitosis and sexual reproduction by meiosis, forming spores that combine with other spores.

To find out if energy and nutritional resource availability determines which form of reproduction is used, researchers grew twenty yeast colonies on an agar medium plate providing ideal nutrient environments. Each colony was composed of a population of yeast that originated from a single yeast cell that had reproduced asexually to form a population of millions of yeast clones. That plate was used to produce replicate plates of yeast colonies. Each replicate plate was a mirror image of the original cultures, so each colony could be subjected to each treatment group. The replicate plates contained different media as follows.

Robust media: Contains all nutrients required by yeast in excess amounts
Complete media: Contains all nutrients at a level promoting exponential growth at $r_{\max}$
Minimal media: Nutrients available at a low level, allowing growth, but stressing the yeast colonies
Deficient media: Nutrient content less than what is needed to sustain growth, extreme stress on yeast colonies

Eight replica plates were made, two plates of each media, and all plates were incubated at 30°C for 24 hours. Yeast from each colony were observed through a microscope and their means of reproduction determined. Table 1 summarizes these results.

Table 1: Reproductive Strategy of *S. cerevisiae* under different environmental stress levels (20 different colonies per medium $\times 2$)

Medium	Nutritional Conditions	Colonies Reproducing Sexually	Colonies Reproducing Asexually	Colonies Showing No Reproduction
Robust	Ideal	0	38	2
Complete	Adequate	2	35	3
Minimal	Mild Stress	24	13	3
Deficient	Extreme Stress	28	1	11

Which of the following best explains the data in Table 1 ?

(A) All forty colonies are genetically identical. Differences in stress responses are due to experimental error.

(B) Asexual reproduction allows stressed colonies to divide rapidly and compete for limited resources with other colonies.

(C) Mild nutritional stress does not trigger a change in reproductive strategy among yeast cultures.

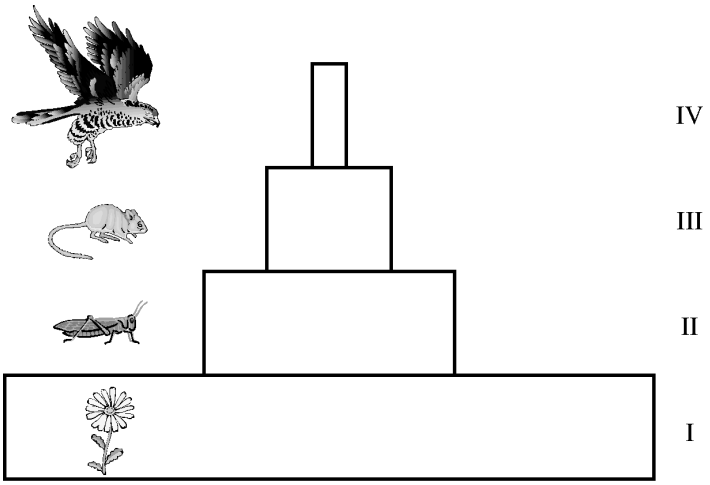
(D) Yeast respond to nutritional stress with sexual reproduction, producing new genetic combinations, or dormancy. ✓

8 Ecology MCQ

Answer D

Correct. Increasing stress leads to an increased number of colonies using sexual reproduction. Sexual reproduction could lead to new combinations of alleles that might allow the yeast to survive using a different energy source or produce a dormant form that could survive until the yeast cells were in a more favorable environment.

172.



Assume there are 50,000 joules (J) of energy available in trophic level II in the figure. According to the conventional model of energy flow in ecosystems, which of the following statements correctly describes the flow of energy in the system?

(A) Trophic level I generates a maximum of 50,000 J of energy.

(B) Trophic level I has approximately 5,000 J of available energy.

(C) Trophic level III has approximately 50 J of available energy.

(D) Trophic level IV has approximately 500 J of available energy. ✓

173. In the process of succession, which of the following is true of *K*-selected plant species?

(A) They keep their carrying capacity at a minimum level.

(B) They usually reproduce early in life.

(C) They are usually found in climax stages of succession. ✓

(D) They are the dominant species in early stages of succession.

(E) They have a short lifespan.

8 Ecology MCQ

174.

Level	Population
4	4
3	360
2	780
1	5,782

In a simple ecosystem, a census of the populations in four successive trophic levels was taken as shown above. If level 1 is composed of photosynthetic autotrophs, then the trophic level with 780 individuals will most likely represent

(A) primary consumers ✓

(B) secondary consumers

(C) tertiary consumers

(D) decomposers

(E) producers

An experiment to measure the rate of respiration in crickets and mice at 10^{o} C and 25^{o} C was performed using a respirometer, an apparatus that measures changes in gas volume. Respiration was measured in mL of O_2 consumed per gram of organism over several five-minute trials, and the following data were obtained.

Organism	Temperature (°C)	Average respiration (mL O_2/g/min)
Mouse	10	0.0518
Mouse	25	0.0321
Cricket	10	0.0013
Cricket	25	0.0038

175. According to the data, the crickets at 25^{o} C have greater oxygen consumption per gram of tissue than do the crickets at 10^{o} C. This trend in oxygen consumption is the opposite of that in the mice. The difference in trends in oxygen consumption among crickets and mice is due to their

(A) relative size

(B) mode of nutrition

(C) mode of internal temperature regulation ✓

(D) mode of ATP production

---

8 Ecology MCQ

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(A) relative size

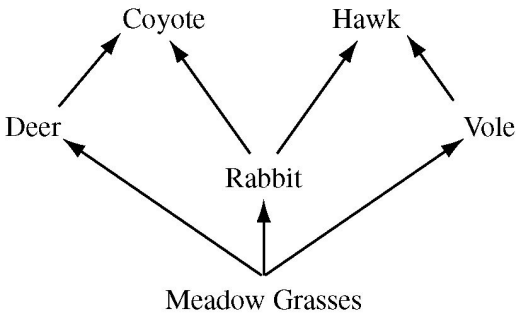
(B) mode of nutrition

(C) mode of internal temperature regulation ✓

(D) mode of ATP production

8 Ecology MCQ

177. The following is a food web for a meadow habitat that occupies 25.6km^2. The primary producers' biomass is uniformly distributed throughout the habitat and totals 1,500 kg/km^2



Developers have approved a project that will permanently reduce the primary producers' biomass by 50 percent and remove all rabbits and deer.

Which of the following is the most likely result at the completion of the project?

(A) The biomass of coyotes will be 6 kg, and the biomass of hawks will be 0.5 kg.

(B) The biomass of coyotes will be dramatically reduced. ✓

(C) The coyotes will switch prey preferences and outcompete the hawks.

(D) There will be 50 percent fewer voles and 90 percent fewer hawks.

178. All of the following statements about the pyramid of biomass are correct EXCEPT:

(A) Biomass is the total dry mass of the organisms present.

(B) The base of the pyramid generally represents primary consumers. ✓

(C) The amount of biomass at a particular level of the pyramid depends on the amount of energy available.

(D) Certain toxins tend to become concentrated at upper levels of the pyramid.

(E) Biomass pyramids tend to vary for different ecosystems.

179. Decades of seal hunting eliminated many fur seal breeding colonies. Since that time, fur seals have recolonized several islands in the Pacific Ocean. Scientists studying the populations of fur seals on three recently colonized islands observed different population growth rates among the three populations.

Which of the following best explains the different population growth rates on the recently colonized islands?

8 Ecology MCQ

(A) The islands differed in the number of predators of seals in the surrounding waters, resulting in lower growth rates on islands with fewer predators nearby.

(B) The islands differed in the amount of available prey, resulting in a higher population growth rate on the islands with fewer prey.

(C) The islands differed in the amount of competition for resources, resulting in higher population growth rates on islands with greater competition.

(D) The islands differed in the amount of available space for breeding, resulting in higher growth rates on islands with more available breeding space. ✓

Answer D

Correct. The islands are likely to differ in the amount of available space for breeding. More breeding space can support more breeding seals and offspring, so if other resources are also plentiful, it is expected that the seal population growth rate will be higher on those islands with more available breeding space.

180. An ecologist is studying a population of 70 penguins. The ecologist estimates that each year, 7 to 10 penguin chicks hatch and 4 to 5 of those chicks grow into viable adults.

Assuming that both the birth and death rates remain at their current maxima, what will the penguin population be after 20 years?

(A) 110

(B) 150

(C) 170 ✓

(D) 210

181. Which of the following best describes opportunistic, or *r*-strategy, organisms?

(A) They reach sexual maturity rapidly. ✓

(B) They reach their adult size slowly.

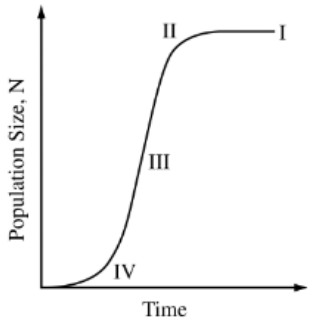
(C) They attain a large body size.

(D) They live for a long time.

(E) The size of their population remains fairly constant.

8 Ecology MCQ

182.



One model of a sustainable fisheries practice is for individual fish to be removed from a natural population at a rate equal to the highest possible growth rate of an ideal population. The graph above represents a population of bluefin tuna living along the Atlantic coast. At which labeled point in the graph is the population growth rate the highest?

(A) I

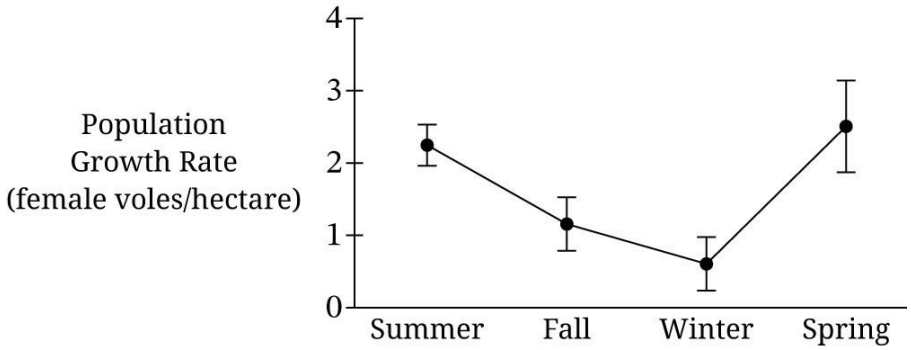
(B) II

(C) III ✓

(D) IV

183. Scientists studying small mammals called voles investigated how the growth rate of the female vole population changed during different seasons. Once a month, researchers captured voles, counted the females, and then released the voles back into their habitat. The data were used to estimate the total female population size and calculate the growth rate of the female population through each of the seasons (Figure 1).

Figure 1. Growth rate of female vole population across four seasons. The error bars represent $\pm SE_{\overline{x}}$



Which of the following statements about the growth rate of the female vole population is best supported by the data in Figure 1?

8 Ecology MCQ

(A) It is higher in the spring than it is in the summer.

(B) It decreases from summer to fall. ✓

(C) It decreases from fall to winter.

(D) It is three times greater in the spring than it is in the winter.

Answer B

Correct. The population growth rate declines from summer to fall, reflecting a downward trend during this seasonal transition.

184. A population of 250 birds inhabit the canopy of a tropical rain forest that has a carrying capacity of 400 birds.

What is the maximum population growth rate $(r_{\max})$ if the population grows to 283 in one year?

(A) 0.12

(B) 0.13

(C) 0.35 ✓

(D) 0.39

Answer C

Correct. This calculation uses the correct population size of $N = 250$ and includes consideration of the carrying capacity $(K = 400)$ as part of the logistical growth pattern, as described by $\frac{dN}{dt} = (283 - 250) = r_{\max} N\left(\frac{K-N}{K}\right) : 33 = r_{\max} \times 250 \times \left(\frac{400-250}{400}\right)$ and $r_{\max} = 0.35$.

185. Bison in Yellowstone National Park have a food-limited carrying capacity that determines how their population grows. If the population grows too large, some bison will either starve or migrate to search for more food. The park is able to support a maximum population of 4,500 bison. Park conservation officials must watch the population and prevent the bison from overpopulating, since they would migrate to nearby farms, causing agricultural damage and creating hazards to drivers. The bison begin to migrate when the population reaches 4,000 because of competition for food.

If the current population size of bison is 3,652 and the maximum growth rate of the population is 0.28, calculate the population size after one year and determine whether the park will need to take measures to control the population.

8 Ecology MCQ

(A) The population will be 3,845 after a year. The bison will not pose a problem until the following year, and no conservation action will need to take place. ✓

(B) The population will be 4,675 after a year. Some bison will need to be removed to prevent migration, since they will surpass the resource availability.

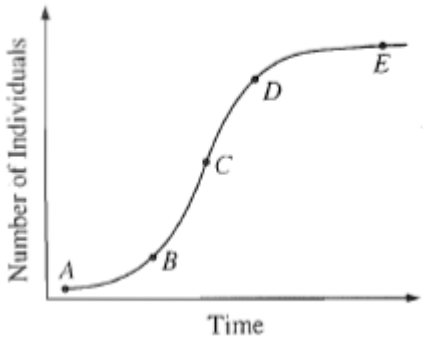
(C) The population will be 1,023 after a year, since the growth rate is too low. Conservation efforts will need to focus on increasing the bison population.

(D) The population will be 4,500 after a year. The bison cannot exceed the carrying capacity, so they will stop reproducing. Some bison will need to be removed.

Answer A

Correct. $\frac{dN}{dt} = r_{\max} N\left(\frac{K-N}{K}\right)$ should be used to determine the one-year change in population size: $0.28 \times 3,652\left(\frac{4,500-3,652}{4,500}\right) = 193$. Adding 193 to 3,652 projects a population of 3,845, so the population will not reach the point at which bison migration occurs. However, it will reach a level to initiate migration the following year.

186.



Which point on the curve in the diagram above best represents the carrying capacity of the environment for the population shown?

(A) *A*

(B) *B*

(C) *C*

(D) *D*

(E) *E* ✓

187. All of the following are density-dependent factors that limit animal populations EXCEPT

8 Ecology MCQ

(A) weather ✓
(B) predation
(C) birthrate
(D) food competition
(E) mortality

188. A fish population of 250 in a pond has a maximum annual rate of increase of 0.8.

If the carrying capacity of the pond is 1,500 for this species, what is the expected population size after one year?

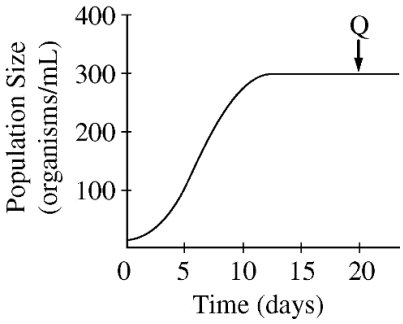
(A) 167
(B) 417 ✓
(C) 450
(D) 500

Answer B

Correct. Using the formula for logistic growth, $\frac{dN}{dt} = r_{\max}N\left(\frac{K-N}{K}\right)$, the increase in population is calculated: $8 \times 250\left(\frac{1{,}500-250}{1{,}500}\right) = 166.7$. Adding 166.7 to 250 (the current population) provides a predicted population of 417 after one year.

For following group of question first study the description of the diagram and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

The graph below illustrates a logistic growth curve for a population of *Euglena*.



189. The population would be most likely to remain at the level shown at Q under which of the following conditions?

8 Ecology MCQ

(A) All predators are eliminated.
(B) The population size is well below its carrying capacity.
(C) *Euglena* is a primary producer.
(D) *Euglena* is a primary consumer.
(E) Environmental conditions do not fluctuate. ✓

190. The population density at point Q indicates that
(A) exponential population growth is occurring
(B) predation is beginning to occur
(C) the reproductive rate equals the death rate ✓
(D) the average age of individuals is decreasing
(E) the reproductive capacity for each individual is increasing

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191. The population of Japanese sika deer in central Japan was determined each year from 2005 to 2014. The sika deer population underwent logistic growth starting at 26,000 deer in 2005 and stabilized at 58,000 deer between 2012 and 2014. Based on these data, the $\mathrm{r}_{\max}$ for this population was determined to be 0.57. Central Japan contains a variety of habitats, including forests and grasslands. Sika deer benefit from the resources in grasslands more than forests; if deforestation occurred, producing more grasslands in the region, the carrying capacity for sika deer population would increase.

What would the population size of sika deer be one year after the carrying capacity increased to 70,000 as a result of deforestation? Assume $\mathrm{r}_{\max}$ does not change.

(A) The population would grow to 70,000 deer in a year, because they would occupy the entire environmental niche.
(B) The population would grow to 91,060 deer in a year, but then they would die off from a lack of resources.
(C) The population would grow to 63,667 deer in a year, and the growth rate would decrease the following year. ✓
(D) The population would remain at 58,000 deer, since it had already stabilized.

Answer C

Correct. $\frac{dN}{dt} = r_{\max} N\left(\frac{K-N}{K}\right)$ should be used to calculate the increase in the population size after a year: $0.57 \times 58,000 \times \left(\frac{70,000-58,000}{70,000}\right) = 5,667$. The new population size would be $5,667 + 58,000 = 63,667$. The growth rate would decrease the following year because the population would be closer to the new carrying capacity and $\left(\frac{K-N}{K}\right)$ would get smaller.

8 Ecology MCQ

192. Which of the following is an example of a density-independent limitation on a population?

(A) In larger population densities of rabbits, the percentage of rabbits infected with bacteria is greater.

(B) At greater concentrations of bacteria, nutrient supplies diminish and population growth ceases.

(C) At higher temperatures, more tomato plants in a population become infested with nematodes than at lower temperatures. ✓

(D) In denser stands of oak trees, greater percentages of trees are attacked by gypsy moths than in less dense stands.

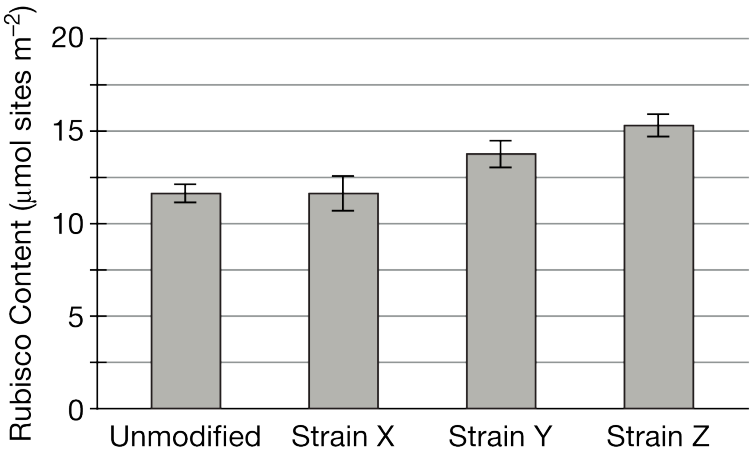
(E) Once the population of grouse exceeds a certain number, the predation rate becomes very high.

FRQ

1. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Increasing the efficiency of photosynthesis is one way to increase crop yield and help to feed human populations. One enzyme vital to the Calvin-Benson cycle of photosynthesis is Rubisco. Rubisco catalyzes the fixation of atmospheric CO_2 into organic molecules so that the carbon atoms can be used to produce carbohydrates. Rubisco is composed of two different types of polypeptide subunits: large subunits and small subunits. It is hypothesized that another protein called Rubisco assembly factor (RAF) is needed to help the polypeptide subunits fold together to form a functional enzyme.

Researchers engineered three genetically modified strains of maize (corn). Strain X was modified to produce additional Rubisco polypeptides of both types in the cells. Strain Y was modified to produce additional RAF protein. Strain Z was modified to produce both additional Rubisco polypeptides and additional RAF protein. The Rubisco content of each of the maize strains studied is shown in Figure 1.

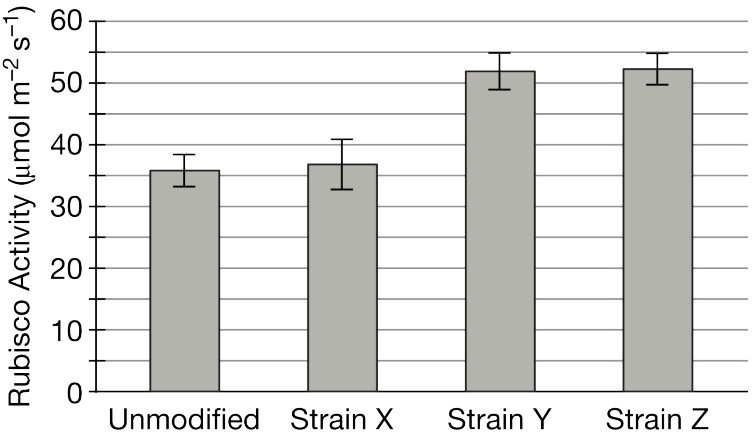


Additional Rubisco Polypeptides	−	+	−	+
Additional Rubisco Assembly Factor (RAF)	−	−	+	+

Figure 1. Rubisco content in genetically modified and unmodified strains of maize. (−) indicates unmodified levels of Rubisco polypeptides and/or RAF protein. (+) indicates additional Rubisco polypeptides and/or RAF protein.

The researchers then grew plants of each strain at a light intensity of $500\ \mu\text{mol of photons} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$ and a temperature of $25\,^{\circ}\text{C}$. After 40 days, the amount of Rubisco activity (as determined by the rate of carbon fixation) in each modified strain was determined (Figure 2).

FRQ



Additional Rubisco Polypeptides	−	+	−	+
Additional Rubisco Assembly Factor (RAF)	−	−	+	+

Figure 2. Rubisco activity (in $\mu mol\ of\ photons \cdot m^{-2} \cdot s^{-1}$) in genetically modified and unmodified strains of maize. $(-)$ indicates unmodified levels of Rubisco polypeptides and/or RAF protein. $(+)$ indicates additional Rubisco polypeptides and/or RAF protein.

Researchers also evaluated several characteristics of the plants of each strain: the height, the fresh mass of the plants, and the mass of plants dehydrated in an oven (dry mass). The data are presented in Table 1.

TABLE 1. AVERAGE PLANT HEIGHT, FRESH MASS, AND DRY MASS OF FOUR STRAINS OF MAIZE

Strain	Rubisco Production	RAF Production	Average Plant Height (cm) $\pm 2\mathrm{SE}_{\bar{x}}$	Average Fresh Mass (g) $\pm 2\mathrm{SE}_{\bar{x}}$	Average Dry Mass (g) $\pm 2\mathrm{SE}_{\bar{x}}$
Unmodified	$-$	$-$	73.0 ± 4.0	230.0 ± 25.0	29.0 ± 2.0
Strain X	$+$	$-$	83.0 ± 1.0	292.0 ± 13.0	32.0 ± 2.0
Strain Y	$-$	$+$	81.0 ± 3.0	281.0 ± 37.0	34.0 ± 2.0
Strain Z	$+$	$+$	86.0 ± 6.0	294.0 ± 32.0	37.0 ± 1.0

(a) The small subunit of Rubisco is made up of many monomers. **Describe** the general structure of one of these

FRQ

monomers, including the characteristic that would allow it to interact with a negatively charged monomer in a second Rubisco polypeptide.

(b) **Identify** the dependent variable in the experiment whose data are shown in Figure 1. Based on the data in Table 1, **identify** the variable measured that best represents the net amount of carbon fixation that occurs in the experimental plants. Based on the data in Table 1, **describe** the effect of additional Rubisco polypeptide production and normal RAF production on the dry mass of plants, indicating the strains you compare. **Calculate** the percent mass of the fresh plants in strain Z that was due to water.

(c) Researchers engineer a strain of maize with a mutation that results in a decrease in the expression of RAF. **Predict** the most likely effect of this RAF mutation on $\mathrm{CO_2}$ consumption in plants grown under the same conditions as those of the experimental strains. **Justify** your prediction. In addition to binding $\mathrm{CO_2}$, Rubisco can bind $\mathrm{O_2}$ when $\mathrm{CO_2}$ is not readily available. Once Rubisco binds to $\mathrm{O_2}$, an energetically wasteful cycle called photorespiration must be completed before Rubisco can bind $\mathrm{CO_2}$ again. **Predict** the initial effect of a higher than normal $\mathrm{O_2 : CO_2}$ ratio in the atmosphere on the rate of carbon fixation in most plants.

(d) A mutation was found to be naturally occurring in switchgrass, a grass species related to maize. This mutation leads to overexpression of RAF. Using the experimental results shown in Figure 2, **explain** how the mutant switchgrass could have increased fitness compared with nonmutant switchgrass. The small subunits of Rubisco are encoded by a nuclear gene, while the large subunits of Rubisco are encoded by a chloroplast gene. A particular maize plant with a mutation in the allele encoding the large subunit of Rubisco is crossed with a plant that contains the normal allele. **Explain** how examination of the offspring will enable a researcher to determine whether the mutant plant contributed the ovule or the pollen to the cross.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that the monomers (amino acids) have an amino group, a carboxyl group, and a variable (R) group with a positive/partially positive charge.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- [ ] An identification that the dependent variable is Rubisco content/level/amount
- [ ] An identification that the variable measured that best represents the net amount of carbon fixation that occurs in the experimental plants is average dry mass

FRQ

- [ ] A description that there is no (statistical) difference between the dry mass of plants with additional Rubisco production (strain X) and control plants (Unmodified)
- [ ] The percent mass of the fresh plants in strain Z that was due to water is calculated to be 87.4%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] A prediction that $\mathrm{CO_2}$ consumption will be reduced
- [ ] A justification that Rubisco activity will be reduced resulting in less carbon fixation/$\mathrm{CO_2}$ entering the Calvin-Benson cycle
- [ ] A prediction that the rate of carbon fixation will decrease

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] An explanation that because additional RAF would allow increased Rubisco activity in the mutant strain, the mutant switchgrass could fix more carbon/produce more carbohydrates/increase biomass more quickly
- [ ] An acceptable explanation. Acceptable explanations include the following.
  · If all of the offspring have the mutation, the mutant plant contributed the ovules, because chloroplast genes are only passed on by the ovule.
  · If none of the offspring have the mutation, the mutant plant contributed the pollen, because the pollen does not pass down chloroplast genes.

FRQ

2. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

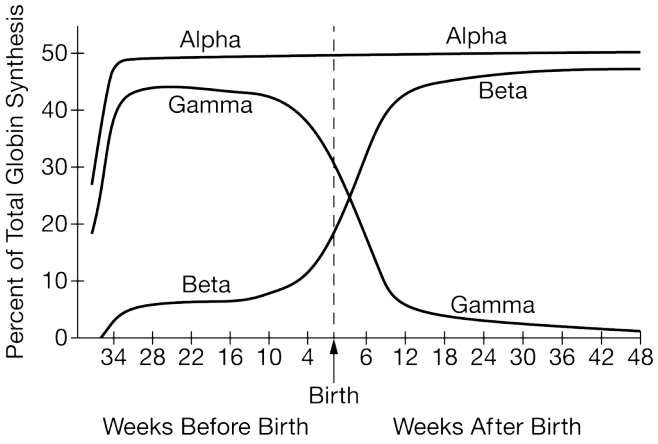
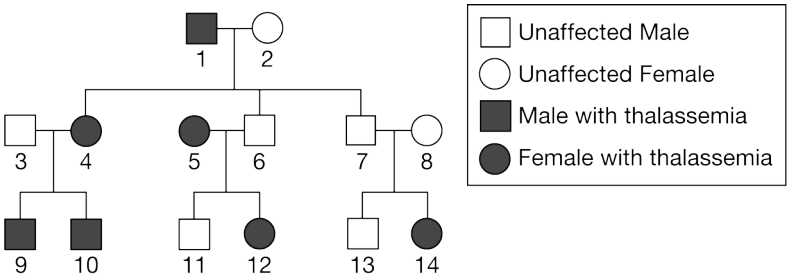


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.



FRQ

Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

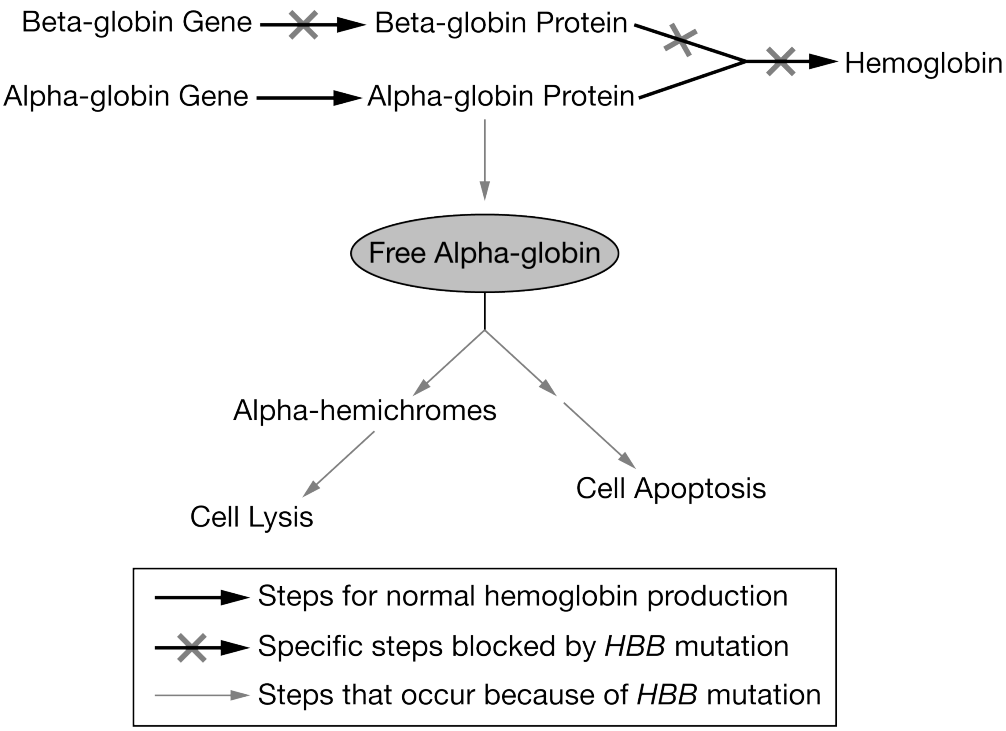


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Individual 13 in Figure 2 undergoes genetic testing and it is determined that there is a 0% chance that he will pass along the mutant allele. **Describe** the HBB alleles inherited by individual 13, including the parent from which each allele came.

(b) The binding of one oxygen molecule to one subunit of the hemoglobin protein complex facilitates the binding of oxygen to the remaining three subunits. **Explain** how this could happen. Both adult and fetal hemoglobins bind to CO more tightly than to O_2. **Explain** why the hemoglobin of a one-week-old infant is able to bind CO more tightly than the hemoglobin of an adult is able to bind CO.

(c) Researchers investigating the effects of mutant beta-globin found that many individuals with beta-thalassemia have a higher concentration of lactic acid in the blood than do individuals without the disease. **Predict** a possible cause for the increase in lactic acid. **Justify** your prediction.

(d) When abnormal human hemoglobin genes are introduced into mice, the mice develop symptoms similar to

FRQ

those of humans with the abnormal hemoglobin genes. Provide reasoning to **support** the claim that mutant human genes can be expressed in mice to model human genetic conditions.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 13 inherited a normal allele from both his parents.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that the binding of the first oxygen molecule could alter the shape/charge of other subunits, increasing the compatibility between the binding sites and the oxygen molecule
- ☐ An explanation that a one-week-old infant still has gamma globin in addition to alpha globin, and its hemoglobin will bind more tightly to oxygen than adult hemoglobin does, so it is expected that fetal hemoglobin will also bind CO more tightly

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that individuals with beta-thalassemia (have less oxygen available and) perform more fermentation than do individuals without the disease
- ☐ A justification that if individuals with beta-thalassemia cannot properly deliver oxygen they will likely rely more heavily on glycolysis to produce ATP than a healthy individual does. Fermentation, which produces lactic acid, will need to occur to keep glycolysis going in the absence of oxygen.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes supportive reasoning that because the genetic code is shared by mice and humans, human genes can be expressed in mouse cells.

FRQ

3. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Brook trout (*Salvelinus fontinalis*) is a species of fish that lives in freshwater streams. Brook trout are adapted to tolerate wide ranges in several environmental factors, including temperature and water flow. Brook trout are well adapted to water temperatures from near freezing ($3^\circ C$) to about $18^\circ C$.

A researcher investigated whether the phenotypes of the brook trout are affected by the rate of water flow. The researcher collected a random sample of $5{,}000$ brook trout eggs and incubated the eggs until the embryos matured. Four weeks after hatching, the juvenile fish were randomly divided into two groups of equal numbers. One group was placed in tanks that simulated a freshwater stream with slowly moving water. The other group was placed in tanks that simulated a freshwater stream with rapidly moving water. After 16 weeks, the researcher collected the fish from the tanks and measured several phenotypic traits (Figure 1). Some of the researcher's measurements are shown in Table 1.

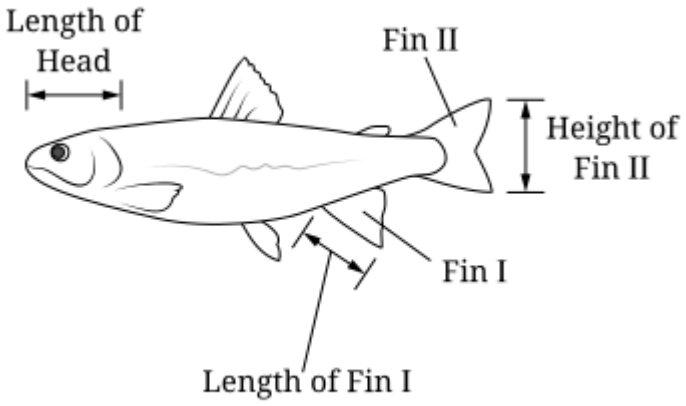

Figure 1. Phenotypic traits of the brook trout measured under different water flow rates

TABLE 1. MEASUREMENTS OF PHENOTYPIC TRAITS OF THE JUVENILE FISH

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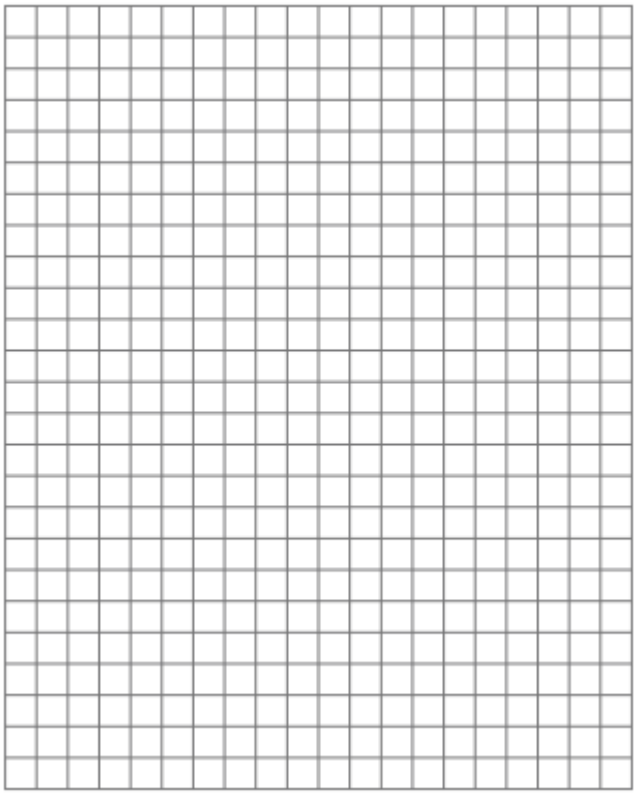
Phenotypic Trait	Juvenile Fish in Slowly Moving Water $(\text{mm}; \text{mean} \pm 2\text{SE}_{\bar{x}})$	Juvenile Fish in Rapidly Moving Water $(\text{mm}; \text{mean} \pm 2\text{SE}_{\bar{x}})$
Length of head	20.95 ± 0.20	20.96 ± 0.20
Length of fin I	15.83 ± 0.70	14.98 ± 0.78
Height of fin II	18.08 ± 0.64	19.62 ± 0.34

(a) Brook trout that live in bodies of freshwater with surface temperatures that reach freezing (0°C) can survive in the water under sheets of ice. **Describe** a property of water that allows the ice to float.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data shown in Table 1. Based on the data, **determine** whether there is a statistical difference in the length of fin I between the groups of fish in slowly moving water and rapidly moving water.



(c) Based on the data, **describe** the effect of the rate of water flow on head length and the height of fin II.

(d) In a follow-up experiment, the researcher collects fertilized eggs produced by fish raised in a tank with rapidly moving water and places the eggs into a tank with slowly moving water. The researcher measures the phenotypic

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traits when the offspring reach the same age as the fish measured in the original experiment. Assuming no natural selection occurred over the course of the experiments, **predict** the phenotypes of the offspring in comparison with the phenotypes of their parents. **Explain** how the results of the original experiment provide evidence of phenotypic plasticity.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response does not include an acceptable description. Acceptable descriptions include the following.

· Frozen water is less dense than liquid water.

· More space is present between the water molecules at $0^{\circ}C$ than at higher temperatures.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ Appropriate labeling
- ☐ Data are plotted in a bar graph or modified bar graph
- ☐ Accurately plotted error bars
- ☐ A determination that there is no (statistical) difference in the length of fin I between the groups of fish in slowly moving water and rapidly moving water

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that head length was not different for the fish in slowly or rapidly moving water and

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that the height of fin II was greater for the fish in rapidly moving water.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- An acceptable prediction. Acceptable predictions include the following.
  · The fish will appear more like the fish in the tank with slowly moving water (and less like the parent fish).
  · The head and fin I of the offspring fish will be similar to those of their parents, but fin II will be shorter (than that of the parents).
- An acceptable explanation that the fish in each treatment group were genetically similar, but they showed measurable phenotypic differences in different environments, suggesting that phenotypes are influenced by water flow/environmental factors

FRQ

4. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In order to investigate the possible toxic effects of chemicals on organisms living in rivers, scientists collected water samples from locations downstream of four factories that released wastewater into a river.

The scientists established six treatment groups of bulbs of the common onion, *Allium cepa*, to which they added clean freshwater, an aqueous solution of 300 mM mutagen X, or a water sample from one of the four river locations (A through D). Mutagen X is a compound that causes chromosomal mutations. After incubating the bulbs for 24 hours, the scientists removed root tips emerging from the bulbs in each treatment group and examined individual root tip cells under a microscope.

The scientists examined 6,000 root tip cells from each slide and recorded both the total number of cells that were in some stage of mitosis and the total number of mitotic cells in which either the chromosomes or the mitotic arrangement of the chromosomes appeared to be abnormal (Table 1). They calculated the mitotic index, the percent of all cells examined that were in mitosis. The mitotic index of treatment groups 2 through 6 was then determined relative to the mitotic index of cells in freshwater, which was set at 100 (Figure 1).

TABLE 1. MITOTIC APPEARANCE OF ROOT TIP CELLS AFTER TREATMENT

Treatment Group	Solution Added to Bulbs	Number of Cells in Mitosis	Number of Cells with Chromosomal or Mitotic Abnormalities
1	Freshwater	301	6
2	Mutagen X	144	117
3	Water from location A	150	41
4	Water from location B	210	35
5	Water from location C	186	29
6	Water from location D	270	5

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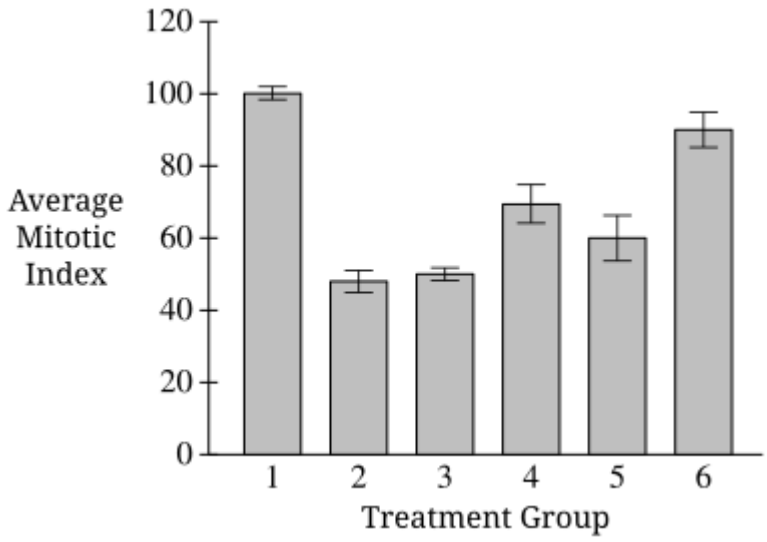

Figure 1. Average mitotic index of root tip cells in each treatment group after 24 hours of incubation. Error bars represent $\pm 2\text{SE}_{\bar{x}}$.

(a) **Describe** the structural features of the two ends of a single nucleotide chain. **Explain** how the addition of a chemical that disrupts hydrogen bonds will alter the structure of DNA isolated from onion root tip cells.

(b) Based on Table 1, **identify** an independent variable in the scientists' investigation. **Justify** the use of mutagen X in one of the treatment groups. **Justify** the scientists' examining the same number of cells in each treatment group.

(c) Based on the data in Table 1 and Figure 1, **describe** the general relationship between the mitotic index and the percent of cells with chromosomal or mitotic abnormalities.

(d) The scientists then analyzed the DNA from root tip cells of *A. cepa* bulbs that were maintained in water from location A for one week and then transferred to freshwater for one week. Assuming mutations do not result in cell death, use the data in Table 1 and Figure 1 to **predict** how the amount of cells with mutations in these bulbs will compare with the amount of cells expected to have mutations in bulbs maintained in freshwater and in bulbs maintained in water from location A for the full two-week period. Based on the data, provide reasoning to **justify** your prediction. **Explain** why the mitotic index of cells in a root tip might be altered by a mutation that affects the mitotic spindle.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

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- [ ] A description that a phosphate group/-PO_4 is attached to a $5'$ carbon of the sugar/one end, and a hydroxyl group/-OH is attached to a $3'$ carbon of the sugar/the other end
- [ ] An explanation that the hydrogen bonds (that hold the complementary base pairs of the two nucleotide chains together) will be broken, causing the two chains/backbones to separate

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] An identification that the independent variable is the incubation/treatment solution
- [ ] A justification that the use of mutagen X was a (positive) control for the effects of a chemical that is known to cause mutations
- [ ] A justification that the scientists' examining the same number of cells was necessary to (accurately) compare the fraction of mitotic cells in each sample

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- As the mitotic index decreases, the percent/number of cells (with chromosomal or mitotic abnormalities) increases.
- As the mitotic index increases, the percent/number of cells (with chromosomal or mitotic abnormalities) decreases.
- There is an inverse/negative relationship.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

FRQ

The response includes all of the following criteria.

- A prediction that the amount (of cells with mutations in the bulbs that were moved from location A water to freshwater) should be intermediate between the number of cells with mutations in the bulbs in freshwater and in the bulbs in water from location A
- A justification that water from location A induces more mutations than does freshwater, so the longer the bulbs are in location A water, the more mutations they should have
- An explanation that (if the mutation is in a gene encoding a spindle protein, then) chromosomes will not align/separate correctly (and the cells may not complete cell division)

FRQ

5. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

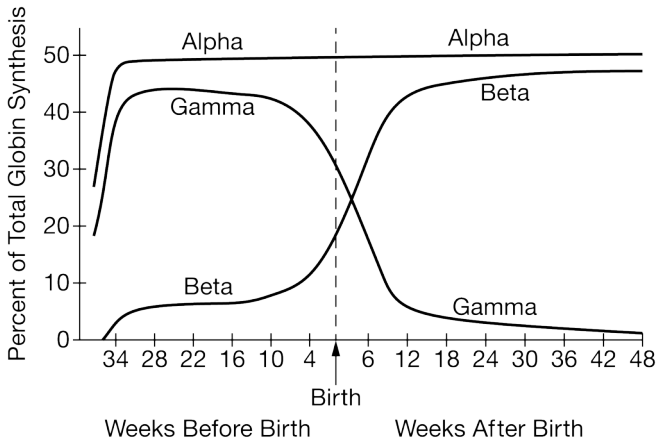

Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.

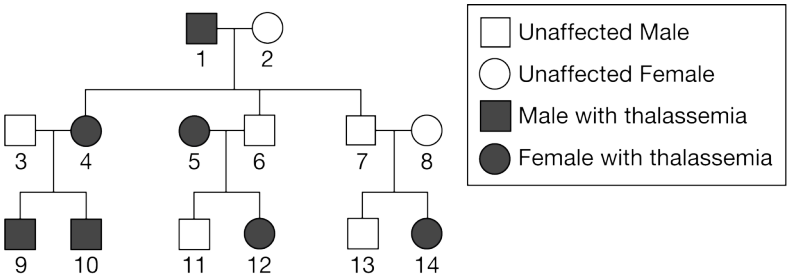

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Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

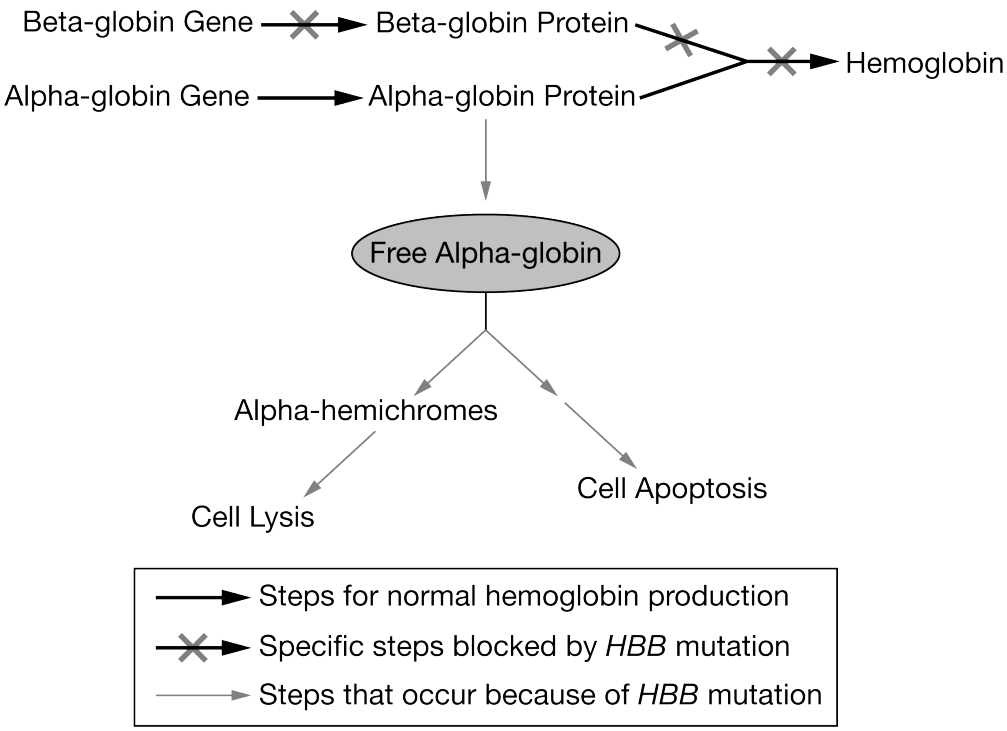


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Based on the pedigree shown in Figure 2, **describe** the HBB alleles inherited by individual 14, including the parent from which each allele came.

(b) **Explain** how a change in the primary sequence of the amino acid chain in the beta-globin subunit could interfere with the assembly of the entire hemoglobin complex. Carbon monoxide (CO) gas binds to the oxygen-binding site of hemoglobin more tightly than O_2 binds. **Explain** why exposure to CO affects ATP production.

(c) Alpha-globin subunits can interact with either beta-globin or gamma-globin subunits. **Predict** the effect of a mutation that enables the continued expression of gamma-globin on the number of cells undergoing lysis in adults who do not produce beta-globin. **Justify** your prediction.

(d) A researcher claims that the pattern of inheritance for beta-thalassemia is autosomal recessive. Use the pedigree in Figure 2 to **support** the researcher's claim.

FRQ

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 14 inherited a mutant allele from both of her parents.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] An explanation that a change in the primary sequence of the amino acid chain could result in an altered secondary and tertiary structure, which could prevent the beta-globin subunits from properly interacting with the alpha-globin subunits to form the hemoglobin complex
- [ ] An acceptable explanation. Acceptable explanations include the following.
  · CO interferes with the delivery of oxygen to cells, which is necessary for oxidative phosphorylation.
  · If CO is bound to hemoglobin, oxygen cannot be carried to the cells. Without oxygen to accept electrons from the electron transport chain, the cell cannot make as much ATP.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that the number of cells undergoing lysis would decrease
- [ ] A justification that these adults would be able to make functional hemoglobin with alpha- and gamma-subunits, resulting in less/no free alpha-globin, the accumulation of which is responsible for cell lysis

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes acceptable supporting information. Acceptable supporting information is as follows.

· Individuals 7 and 8 in the pedigree are unaffected but have an affected child, and individual 7 must be a carrier/ heterozygous.

· Individuals 7 and 8 in the pedigree are unaffected but have an affected child, and the trait must be autosomal because individual 5 did not pass it to individual 11.

FRQ

6. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

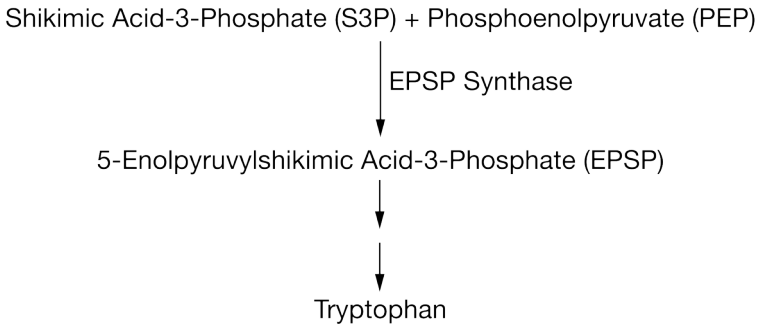


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

Scientists investigated the effect that Compound X has on EPSP synthase activity by performing an experiment to examine the relationship between EPSP synthase activity, PEP concentration, and Compound X concentration (Figure 2).

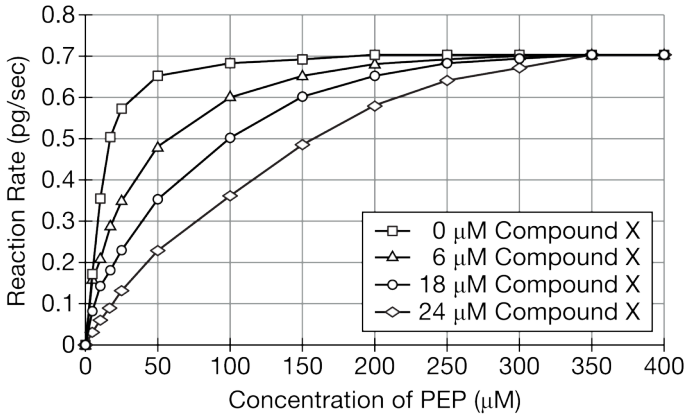


Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

Some populations of the plant *Lolium rigidum* have evolved resistance to Compound X. To investigate the

inheritance of this resistance, researchers crossed true-breeding plants from a Compound X-susceptible (S) population with true-breeding plants from a Compound X-resistant (R) population. In half of these crosses, pollen from S plants was used to fertilize ovules in R plants. In the other half, pollen from R plants was used to fertilize ovules in S plants. Survival of parental and F_1 seedlings, differentiated by the maternal phenotype, in response to varying concentrations of Compound X is shown in Figure 3.

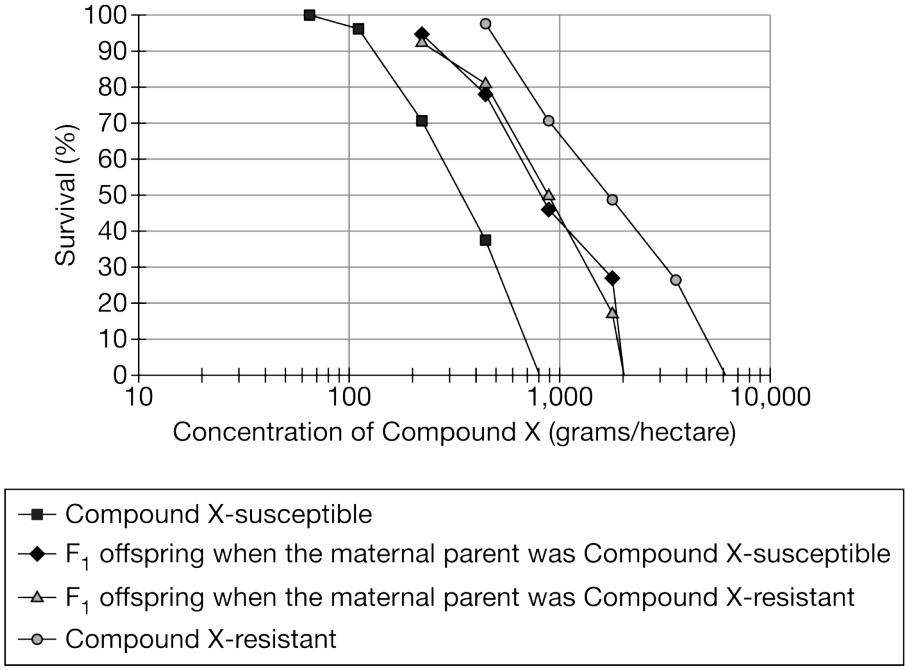


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots. $1\ \text{Hectare} = 10{,}000\ \text{m}^2$

(a) **Explain** how an increased concentration of EPSP synthase would change the rate of the reaction between S3P and PEP in the presence of a low concentration of Compound X.

(b) **Identify** one independent variable in the experiment shown in Figure 2. **Identify** the experimental group(s) in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in grams/hectare, required to kill 30% of the Compound X-resistant (R) population. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by the reaction in one minute at a PEP concentration of $100\ \mu\text{M}$ and a Compound X concentration of $6\ \mu\text{M}$.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 female with a male from the original resistant (R) population. The resulting offspring were treated with $500\ \text{grams/hectare}$ of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. Based on the evidence in Figure 3, researchers claim that the allele for Compound X-resistance is carried on a chromosome in the nucleus. Provide reasoning to **justify** this claim. Researchers discovered a variant of the EPSP synthase gene in *L. rigidum* in which there was a mutation in the promoter region of this gene. This mutation increased the likelihood that RNA polymerase binds to the EPSP synthase gene. **Predict** the most likely effect of this mutation on the ability of *L. rigidum* plants with this

FRQ

variant of the EPSP synthase gene to produce tryptophan.

(d) Researchers have engineered Compound X-resistant corn by inserting a Compound X-resistance gene from a prokaryote into the corn genome. **Explain** why the upstream regulatory region, including the promoter, of the prokaryotic DNA would need to be altered in order to express the Compound X-resistance gene in corn. In addition to a conserved genetic code, **explain** how the prokaryotic EPSP synthase protein can be translated from identical mRNAs by both a prokaryote and a corn plant.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the reaction rate will increase because there will be more enzymes available to bind to substrates.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of an independent variable. Acceptable identifications include the following.
  - Concentration of PEP
  - Concentration of Compound X
- ☐ An acceptable identification of the experimental group. Acceptable identifications include the following.
  - Both F_1 generations
  - F_1 offspring when the maternal parent is Compound X-susceptible and F_1 offspring when the maternal parent is Compound X-resistant
- ☐ An identification that the concentration of Compound X, in $\text{grams} / \text{hectare}$, required to kill 30% of the Compound X-resistant (R) population is 900 ($\text{grams} / \text{hectare}$); NOTE: acceptable student responses can be between 850 and 950.
- ☐ The mass, in pg, of EPSP that would be produced by the reaction in one minute at a PEP concentration of $100\mu\text{M}$ and a Compound X concentration of $6\mu\text{M}$ is calculated to be 36 pg.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable prediction of the percent survival of one genotype of the resulting offspring. Acceptable predications include the following.
  · 75% survival RS; NOTE: acceptable student responses can be between 72% and 78%.
  · 94% survival RR; NOTE: acceptable student responses can be between 92% and 96%.
- ☐ An acceptable justification. Acceptable justifications include the following.
  · When susceptible and resistant strains are crossed, the offspring have an intermediate phenotype. Therefore, each offspring received an allele from each parent.
  · If the allele were carried in the chloroplast or mitochondrion, then the offspring would have the same phenotype as the maternal parent.
- ☐ A prediction that the likely effect of this mutation is an increased production of tryptophan

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  · Because eukaryotic transcription factors differ from prokaryotic transcription factors, eukaryotic transcription factors will not recognize the prokaryotic promoter sequence.
  · Because eukaryotic RNA polymerase differs from prokaryotic RNA polymerase, eukaryotic RNA polymerase will not recognize the prokaryotic promoter sequence.
- ☐ An explanation that the mRNAs can be translated because both have ribosomes/transfer RNA to read the mRNA code

FRQ

7. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

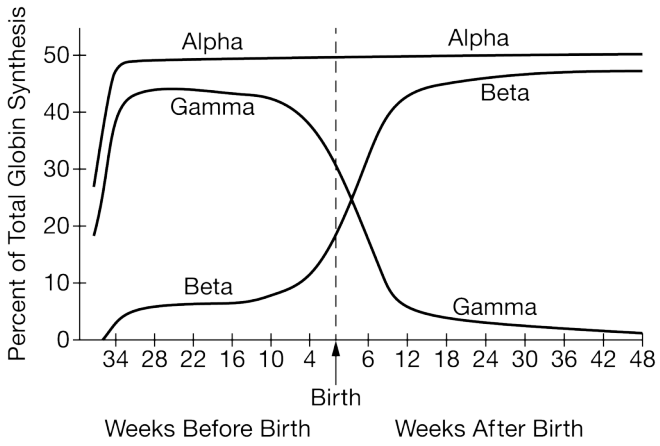
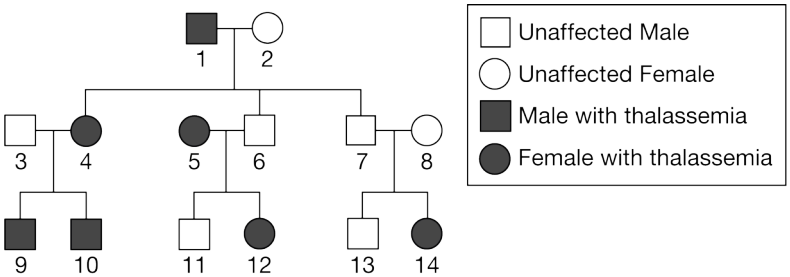


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.



FRQ

Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

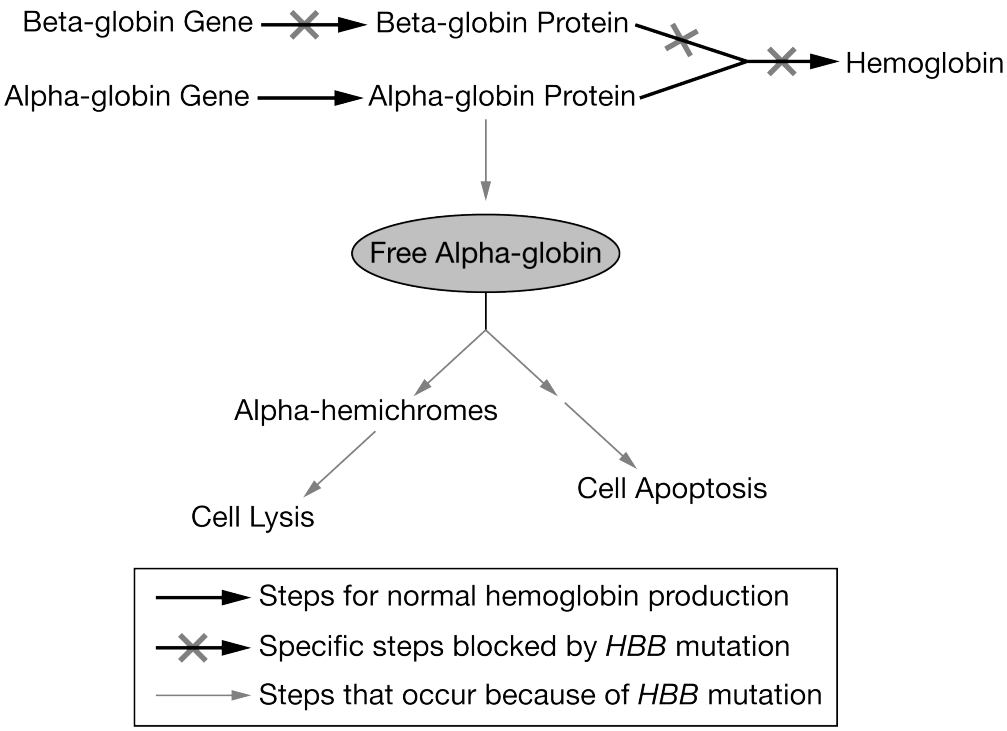


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Based on the pedigree shown in Figure 2, **describe** the HBB alleles inherited by individual 11, including the parent from which each allele came.

(b) **Explain** how mutant hemoglobin interferes with a cell's ability to synthesize ATP. Red blood cells damaged by the accumulation of alpha-hemichromes undergo apoptosis and are taken into other cells by a process similar to endocytosis. **Explain** how these cells will digest and recycle the components of the apoptotic red blood cells.

(c) While studying the family shown in the pedigree shown in Figure 2, researchers notice that the HBB mutant allele and an allele for another trait are passed on together at a frequency greater than that which would be predicted if the genes assorted independently. **Predict** the cause of this increased frequency of inheritance. **Justify** your prediction.

(d) A researcher claims that HBB exhibits a maternal mitochondrial inheritance pattern. Use the pedigree to **refute** the researcher's claim.

FRQ

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 11 inherited a normal allele from his father and a mutant allele from his mother.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- An explanation that oxygen will not be delivered to the cells to act as the terminal electron acceptor of the electron transport chain, and the ETC is the main source of ATP in cells/for cellular respiration
- An explanation that a vesicle containing the apoptotic red blood cell will fuse with a lysosome containing digestive enzymes (which will allow the components to be recycled)

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- A prediction that the genes are close together on a chromosome
- An acceptable justification. Acceptable justifications include the following.
  · Genes that are located close together on the same chromosome are less likely to be separated by crossing over than are genes that are far apart from each other on the same chromosome.
  · Genes carried on separate chromosomes assort independently, but genes carried on the same chromosome tend to be inherited together unless crossing over separates them.

Part D

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a refutation stating that mitochondrial genes are passed only from the maternal parent to the offspring. Therefore, all offspring should have the same phenotype as their mother, but #4/#11/#14 has a phenotype that differs from that of her/his mother.

FRQ

8. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, photosynthetic guard cells on the surface of leaves form pores called stomata (a single pore is called a stoma). Stomata help prevent water loss and control gas exchange with the environment to maintain plant homeostasis. The turgor pressure of the guard cells on either side of the stomata determines whether the stomata are open or closed based on the transport of certain ions into and out of the guard cells (Figure 1). A proton pump in the membrane of the guard cells pumps protons into the extracellular space. This makes the interior of the cell more negatively charged than the extracellular space.

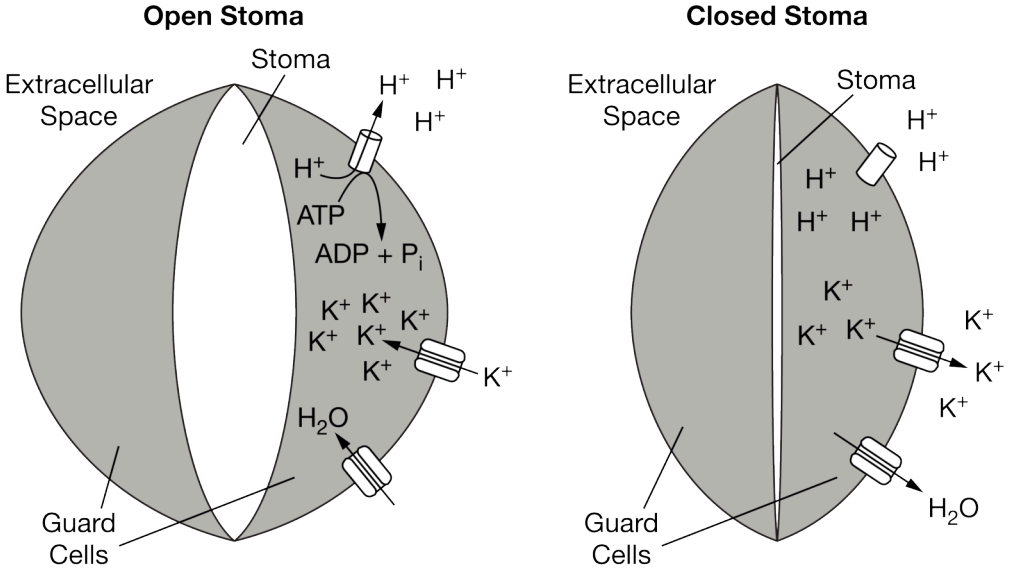


Figure 1. Models of open and closed stomata and the transport proteins in the guard cell membranes

Scientists interested in understanding drought tolerance in plants discovered a mutant variety of a plant with increased cell division, increased water content, and reduced density of leaf stomata compared with the wild type. Analysis of the mutant plant showed a mutation in the gene encoding a cyclin-dependent kinase (CDK). This mutation prevents CDK-cyclin interaction that normally results in the phosphorylation of RNA polymerase (Figure 2).

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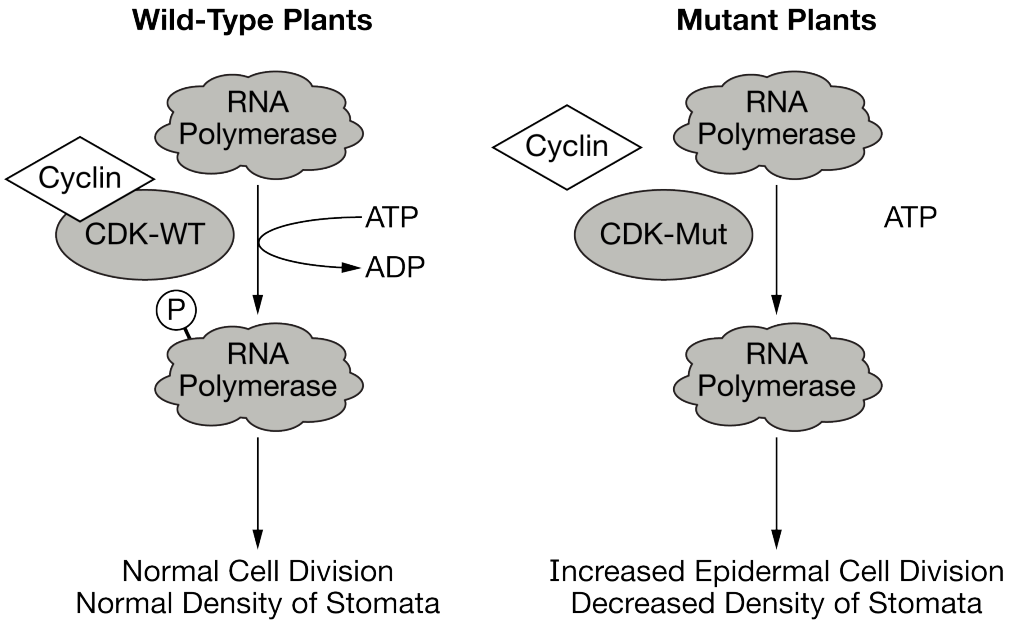

Figure 2. In wild-type plants, the wild-type CDK protein ($\mathrm{CDK\text{-}WT}$) allows interaction with a cyclin and leads to the phosphorylation of RNA polymerase. In mutant plants, the mutant CDK protein ($\mathrm{CDK\text{-}Mut}$) prevents interaction with cyclin and the phosphorylation of RNA polymerase.

(a) **Describe** the role of ATP in the model shown in Figure 2. Scientists crossed plants that were homozygous for the cyclin dependent kinase-wild type ($CDK\text{-}WT$) allele with plants that were homozygous for the cyclin dependent kinase-mutant ($CDK\text{-}Mut$) allele. Crossing offspring from the F_1 generation produced F_2 offspring with a phenotypic ratio of $3\ \text{wild-type} : 1\ \text{mutant}$. **Explain** how the F_2 generation includes offspring with the wild-type phenotype.

(b) Based on Figure 2, **explain** how the expression of the $CDK\text{-}Mut$ allele most likely causes a decrease in the density of stomata.

(c) A mutation in some guard cells inactivates the proton pump in these cells. **Predict** how the membrane potential across the membrane of these guard cells compares with the membrane potential across the membrane of guard cells with functioning proton pumps. Provide reasoning to **justify** your prediction.

(d) Drought conditions result in a decrease in turgor pressure in plant cells. Researchers claim that this reduction in turgor pressure negatively affects gas exchange in the wild-type plants but not in the mutant plants. **Refute** the researchers' claim using the information provided.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

FRQ

The response includes both of the following criteria.

- [ ] A description that ATP provides the energy and the phosphate (for the phosphorylation of RNA polymerase)
- [ ] An explanation that the F_2 generation was produced from two heterozygous parents, thus the wild-type offspring inherited at least one copy of a dominant wild-type allele

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· The mutation leads to the alteration/prevents the phosphorylation of RNA polymerase which induces the synthesis of proteins/expression of genes that encode proteins that block formation of stomata.

· The mutation leads to the alteration/prevents the phosphorylation of RNA polymerase which prevents the synthesis of proteins/expression of genes that encode proteins that promote the formation of stomata.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that the membrane potential will be less in the mutated guard cells than in the normal guard cells
- [ ] A justification that the mutated proton pump will not be able to pump out positively charged protons, so the inside of the guard cell be less negative with respect to the outside of the cell when compared with guard cells without the mutation

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

FRQ

The response includes a refutation for the claim by stating that, using the information provided, the decrease in turgor pressure in the guard cells causes the stomata to close and prevents gas exchange in both types of plants.

9. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Aquaporin channel proteins are found in the plasma membrane of various cell types. Scientists studied the production of aquaporins in the keratinocytes (skin cells) of healthy individuals and found that these proteins play a role in hydration of the skin. Some individuals have a skin disorder that causes the absorption of excess water by the skin of their hands and feet when they are soaked in water. Scientists determined that the keratinocytes of these individuals are characterized by the presence of abnormal aquaporins. Scientists studied a family in which some individuals have this skin disorder and constructed a pedigree (Figure 1).

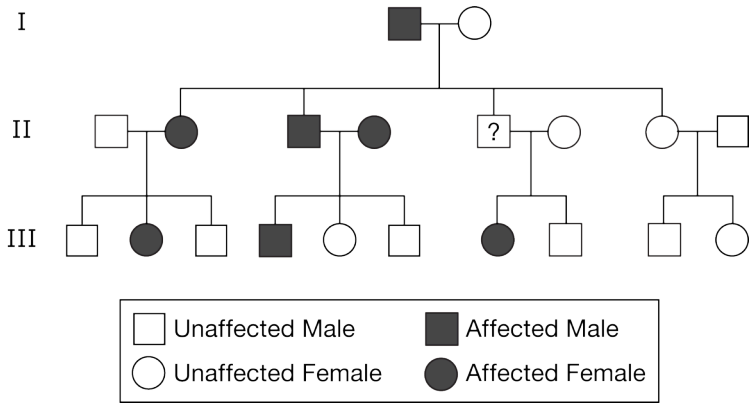

Figure 1. Pedigree of a human family in which some members have a skin disorder. An individual with an unknown phenotype is marked by a question mark.

(a) Based on the pedigree in Figure 1, **predict** the most likely genotype AND phenotype of the individual marked by "?". **Justify** your prediction.

(b) **Describe** the characteristics of the keratinocyte membrane that restrict water movement across the membrane. If isolated keratinocytes are placed in a hypotonic solution for 24 hours, **explain** why there are likely to be far fewer cells at the end of this time than there were at the start.

(c) **Explain** how the aquaporin polypeptides are transported from the Golgi complex and inserted into the plasma membrane.

(d) A student claims that individuals who have a lower than normal number of aquaporins in their skin cells will easily dehydrate. **Refute** this claim with evidence from biological concepts.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that the individual marked by "?" is heterozygous (e.g., Rr) and has the disorder
- ☐ A justification that the inheritance pattern is autosomal dominant (because two affected individuals had unaffected children), and the individual marked by "?" and an unaffected female produced both affected and unaffected offspring

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable description. Acceptable descriptions include the following.
  · The membrane has a non-polar/hydrophobic interior.
  · The membrane includes phospholipids with fatty acid tails.
- ☐ An explanation that because of the greater solute concentration inside the keratinocytes, water will move into the keratinocytes, causing many of them to lyse by the end of the 24 hours

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· A portion of the Golgi complex containing the aquaporin polypeptide buds off as a vesicle that fuses with the plasma membrane (allowing the aquaporin to embed in the membrane).

· The aquaporin polypeptide is carried by a vesicle that fuses with the plasma membrane.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes a refutation of the claim by stating that fewer aquaporin channels will result in less/slower water loss compared with water loss from cells with a normal number of aquaporins.

FRQ

10. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Increased expression and activity of the ligand-gated ion channel TRPV1 has been shown to block cell division and induce apoptosis in human cells. The structure of the TRPV1 protein consists of several transmembrane domains that are embedded in the membrane as well as a carboxy-terminus and an amino-terminus that are located inside the cell (Figure 1).

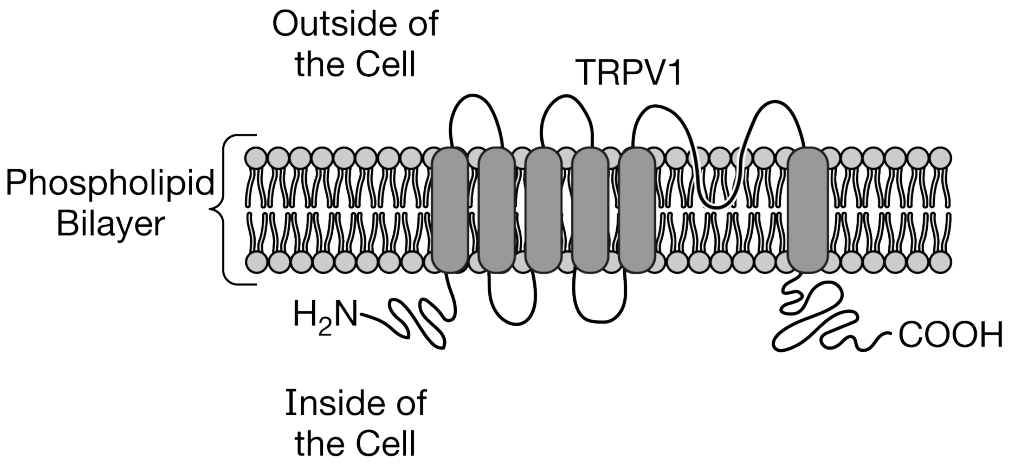

Figure 1. Structure of the TRPV1 protein channel in the membrane

TRPV1 is activated by several stimuli, including the binding of capsaicin, a chemical found in chili peppers. Capsaicin enters the cell by simple diffusion and then binds to one of the transmembrane domains of TRPV1, which opens the ion channel.

The opening of the TRPV1 channel allows Ca^{2+} ions to enter the cell, leading to the activation of the enzyme calcineurin. Calcineurin removes phosphate groups from the phosphorylated form of the transcription factor NFAT2, which is typically found in the cytoplasm. Once dephosphorylated, NFAT2 moves into the nucleus, where it blocks the transcription of another protein, ATF3. ATF3, when active, prevents the cell cycle regulatory protein pGoG from inducing apoptosis and blocking cell division (Figure 2).

FRQ

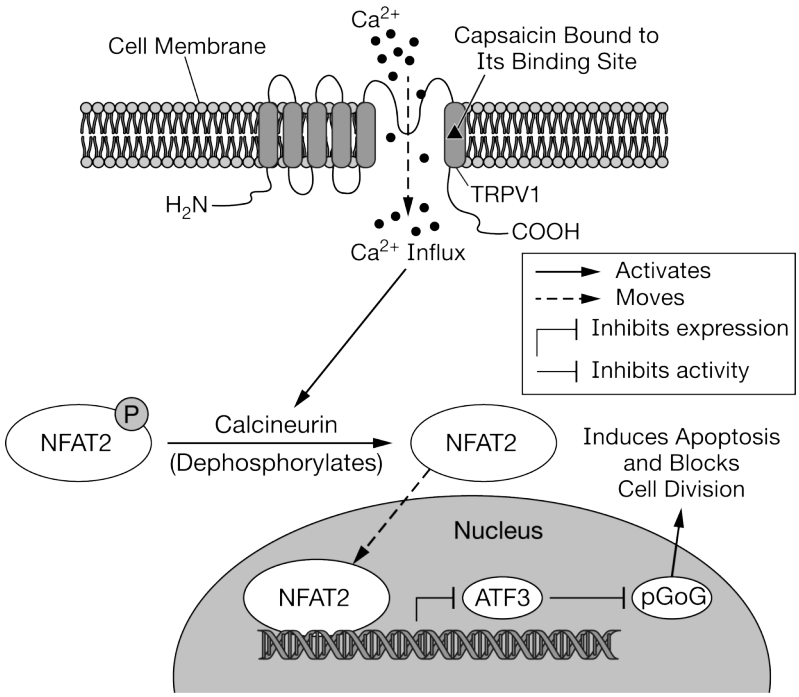

Figure 2. Capsaicin-activated TRPV1 signaling pathway in human cells

(a) **Describe** the differences between the R groups of the amino acids located in the transmembrane domains and those located in the amino- or carboxy-terminus of TRPV1. **Describe** the specific location in the cell where TRPV1 protein is synthesized.

(b) The pGoG protein leads to the inhibition of cyclin activity. **Explain** how this interferes with cell cycle progression.

(c) Researchers discover a mutation that prevents NFAT2 from being phosphorylated. **Predict** the effect of this mutation on apoptosis by the pathway shown in Figure 2. Provide reasoning to **justify** your prediction.

(d) The drug cyclosporin A is known to inhibit calcineurin. Researchers claim that cyclosporin A should not be given to patients with cancer. **Support** the researchers' claim based on the pathway shown in Figure 2.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A description that R groups in the transmembrane domain should be hydrophobic/nonpolar, while R groups in either terminus should be hydrophilic/polar/charged
- [ ] A description that TRPV1 protein is synthesized on the ribosomes attached to the endoplasmic reticulum/ ER /rough ER

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that cyclins cannot activate cyclin-dependent kinases which normally (phosphorylate other proteins to) move the cell through a cell cycle checkpoint.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that apoptosis will be induced
- [ ] A justification that if NFAT2 cannot be phosphorylated, it will be able to (move to the nucleus and) inhibit ATF3 expression, which enables activity of pGoG which will induce apoptosis

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes support for the claim by stating that if calcineurin is inhibited, then the steps in this pathway that follow from calcineurin activation will be unable to block cell division/induce apoptosis. Since cancer is characterized by excessive cell division, this drug would be an inappropriate treatment for cancer.

FRQ

11. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Gitelman syndrome (GS) is a heritable disorder caused by a mutation in the *SLC* gene that encodes a Na^+/Cl^- cotransporter protein (A in Figure 1) that is expressed in kidney cells. When molecules or ions are moved by cotransporters, the molecules or ions must move across a membrane together, either in the same direction or in opposite directions. The Na^+/Cl^- cotransporter simultaneously transports Na^+ and Cl^- from one region of the kidney (Region I) into the cell. Other transporters in kidney cell membranes include the Na^+/K^+ ATPase (B in Figure 1) and a Na^+/Ca^{2+} cotransporter (D in Figure 1) that allow ions to move across the membrane between the kidney cell and a second region of the kidney (Region II). Kidney cells also have channel proteins like the Ca^{2+} channel (C in Figure 1) that transports Ca^{2+} from Region I into the kidney cell. A model representing the exchange of various ions between the kidney cell and the two extracellular regions is shown in Figure 1. The size of an ion's symbol indicates the relative concentration of that ion in a given area. The larger symbol indicates the area of higher concentration, and the smaller symbol indicates the area of lower concentration.

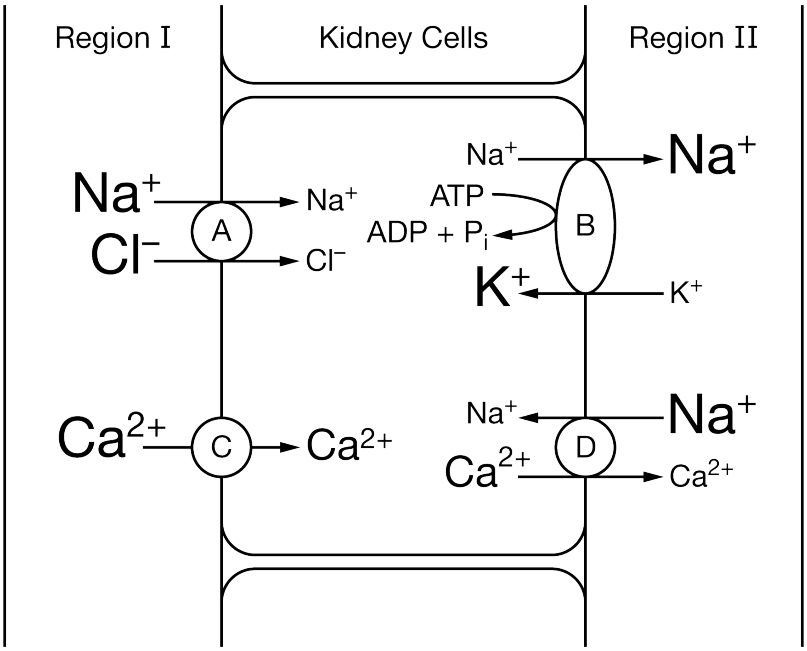

Figure 1. Model representing the exchange of ions between a kidney cell and two isolated extracellular regions. The larger symbols indicate the areas of higher ion concentration, and the smaller symbols indicate the areas of lower ion concentration.

Researchers have determined that there are a number of mutations to the SLC wild-type allele (*SLC*) that result in GS. One mutant allele, *SLC-gs1*, results from a single nucleotide deletion that creates a stop codon midway through the coding region of the gene. A second mutant allele, *SLC-gs2*, results from a mutation that disrupts the sequence of a splicing site between an intron and an exon, preventing the removal of an intron. To better understand how GS is inherited, researchers studied a number of families with a history of GS. The pedigree of

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one of these families is shown in Figure 2.

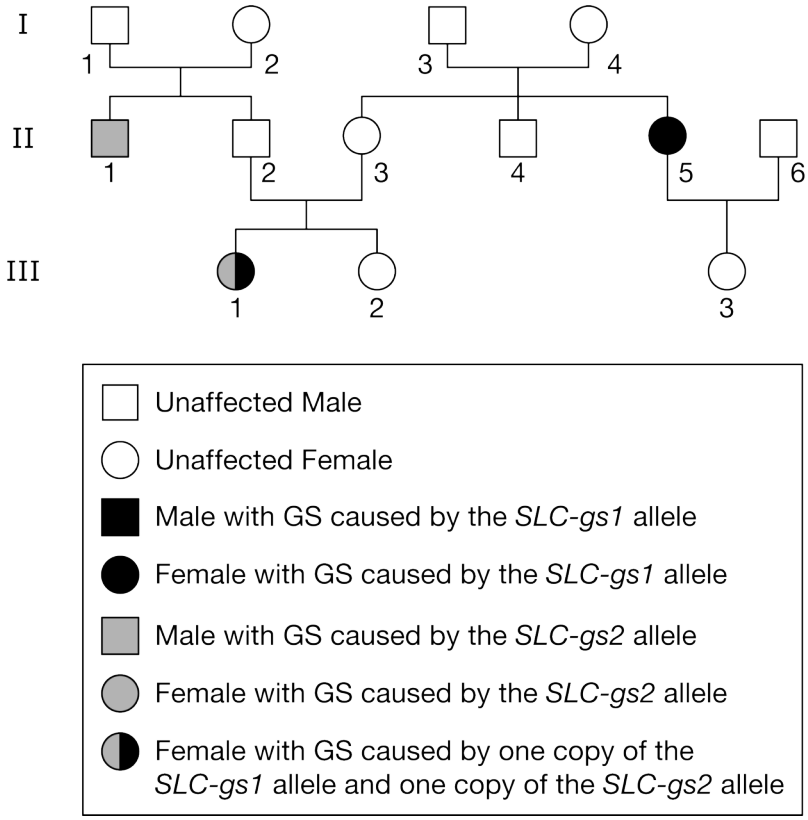

Figure 2. Pedigree of a family with a history of GS caused by two different mutant alleles

(a) **Describe** the characteristics of the Ca^{2+} ions and the plasma membrane of the kidney cell that prevent the Ca^{2+} ions from moving from Region I into the kidney cell without the channel labeled C.

(b) **Explain** the effect of the mutation in the *SLC-gs2* allele on the $\mathrm{Na}^{+}/\mathrm{Cl}^{-}$ cotransporter protein produced by the cell. **Explain** why the transporter labeled B in Figure 1 requires ATP to move ions between Region II and the kidney cells.

(c) **Predict** the immediate effect on the movement of Na^{+} AND Ca^{2+} by the cotransporter labeled D in Figure 1 if the $\mathrm{Na}^{+}/\mathrm{Cl}^{-}$ cotransporter (A in Figure 1) is nonfunctional. Based on Figure 1, provide reasoning to **justify** your prediction.

(d) Based on the information provided in Figure 2, **predict** the two alleles of the SLC gene that make up the genotype of individual $\mathrm{II\text{-}2}$ AND the two alleles of the SLC gene that make up the genotype of individual $\mathrm{II\text{-}3}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an explanation that Ca^{2+} ions are charged/hydrophilic and the interior of the membrane is nonpolar/hydrophobic.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  · The sequence of the intron will be translated and (as long as a stop codon is not introduced) this will result in a longer protein.
  · The sequence of the intron will be translated, which could introduce a stop codon before the existing stop codon, resulting in a shorter protein.
  · The sequence of the intron will be translated resulting in a change in the structure/function of the protein.
- ☐ An explanation that energy (from ATP) is required because the transporter is moving both sodium and potassium ions against their concentration gradient/from low concentration to high concentration

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  · Na^{+} will continue to enter the cell and Ca^{2+} will continue to leave the cell.
  · The movement of Na^{+} AND Ca^{2+} by D will not change.
- ☐ A justification that (with A not functioning) there is still a concentration gradient because transporter B/C still functions, so Na^{+} and Ca^{2+} continue to be transported through transporter D

Part D

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a prediction that that the two alleles that make up the genotype of individual II-2 are SLC and $SLC-gs2$, AND the two alleles that make up the genotype of individual II-3 are SLC, $SLC-gs1$.

FRQ

12. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Increased expression and activity of the ligand-gated ion channel TRPV1 has been shown to block cell division and induce apoptosis in human cells. The structure of the TRPV1 protein consists of several transmembrane domains that are embedded in the membrane as well as a carboxy-terminus and an amino-terminus that are located inside the cell (Figure 1).

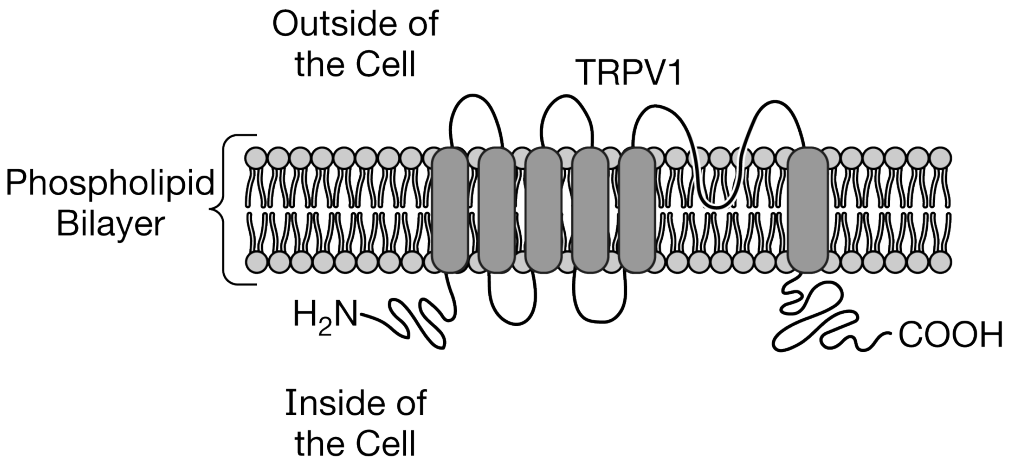

Figure 1. Structure of the TRPV1 protein channel in the membrane

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The opening of the TRPV1 channel allows Ca^{2+} ions to enter the cell, leading to the activation of the enzyme calcineurin. Calcineurin removes phosphate groups from the phosphorylated form of the transcription factor NFAT2, which is typically found in the cytoplasm. Once dephosphorylated, NFAT2 moves into the nucleus, where it blocks the transcription of another protein, ATF3. ATF3, when active, prevents the cell cycle regulatory protein pGoG from inducing apoptosis and blocking cell division (Figure 2).

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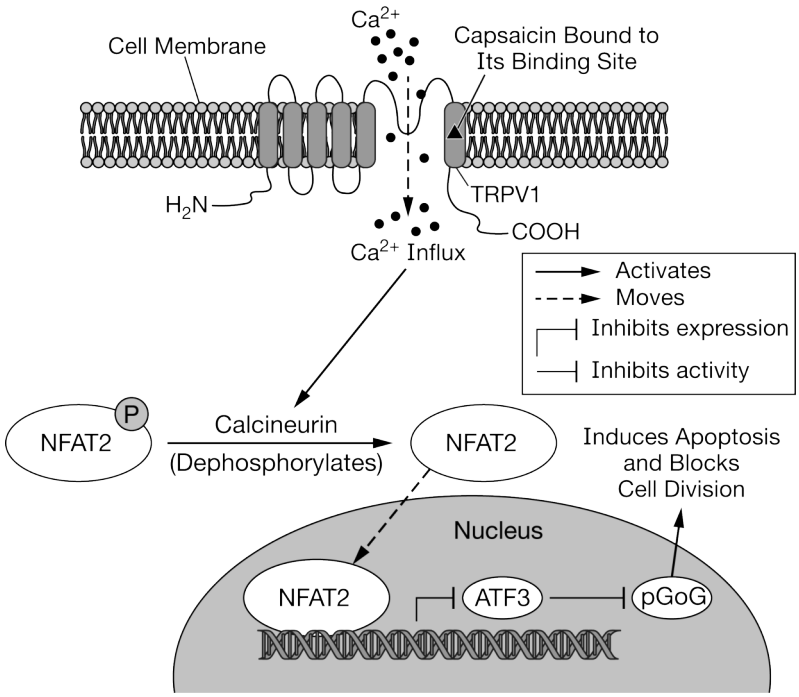


Figure 2. Capsaicin-activated TRPV1 signaling pathway in human cells

(a) **Describe** the characteristics of capsaicin that allow it to enter the cell before binding to a transmembrane domain of TRPV1 . **Describe** what prevents NFAT2 from inhibiting ATF3 expression in the absence of calcineurin.

(b) The pGoG protein is known to indirectly interfere with the interaction between cyclins and cyclin-dependent kinases. **Explain** why this prevents DNA replication from happening in a cell.

(c) Researchers discover a mutation that results in the constant expression of ATF3 regardless of the presence of NFAT2. **Predict** the effect of this mutation on the cell cycle of cells treated with capsaicin. Provide reasoning to **justify** your prediction.

(d) Researchers claim that capsaicin could be used as a potential treatment for melanoma, a type of skin cancer. **Support** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A description that capsaicin is relatively small and nonpolar
- [ ] A description that NFAT2 stays phosphorylated and remains in the cytoplasm/does not enter the nucleus

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the indirect interference by pGoG would prevent entrance into S phase because the cyclins are not able to activate the CDKs to move the cell cycle forward.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that the cell cycle will continue
- [ ] A justification that (since ATF3 will always be expressed) ATF3 will always block activity of pGoG, and pGoG will be unable to block the cell cycle and/or induce apoptosis

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes support for the claim by stating that since cancer is characterized by excessive cell division, capsaicin will be an effective treatment because it leads to the activation of pGoG, which decreases cell division and/or induces apoptosis.

FRQ

13. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, photosynthetic guard cells on the surface of leaves form pores called stomata (a single pore is called a stoma). Stomata help prevent water loss and control gas exchange with the environment to maintain plant homeostasis. The turgor pressure of the guard cells on either side of the stomata determines whether the stomata are open or closed based on the transport of certain ions into and out of the guard cells (Figure 1). A proton pump in the membrane of the guard cells pumps protons into the extracellular space. This makes the interior of the cell more negatively charged than the extracellular space.

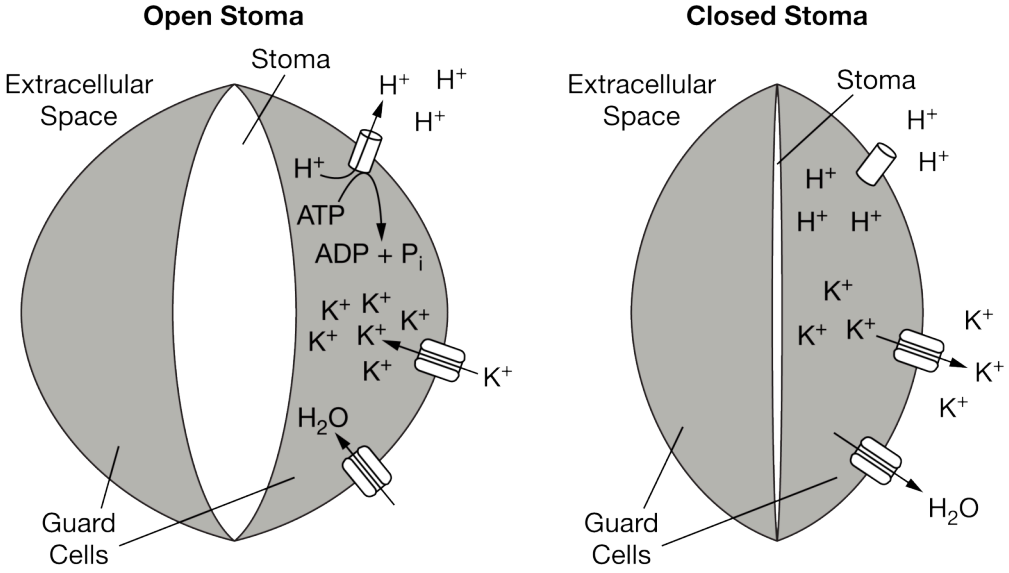


Figure 1. Models of open and closed stomata and the transport proteins in the guard cell membranes

Scientists interested in understanding drought tolerance in plants discovered a mutant variety of a plant with increased cell division, increased water content, and reduced density of leaf stomata compared with the wild type. Analysis of the mutant plant showed a mutation in the gene encoding a cyclin-dependent kinase (CDK). This mutation prevents CDK-cyclin interaction that normally results in the phosphorylation of RNA polymerase (Figure 2).

FRQ

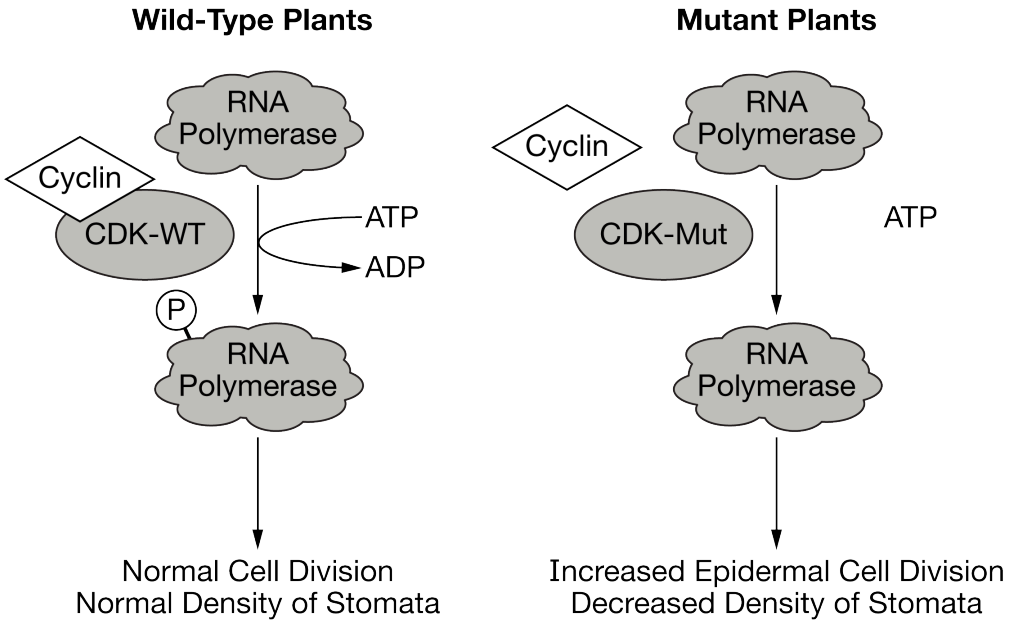


Figure 2. In wild-type plants, the wild-type CDK protein ($\mathrm{CDK\text{-}WT}$) allows interaction with a cyclin and leads to the phosphorylation of RNA polymerase. In mutant plants, the mutant CDK protein ($\mathrm{CDK\text{-}Mut}$) prevents interaction with cyclin and the phosphorylation of RNA polymerase.

(a) **Describe** the movement of K^{+} ions into the guard cell based on the model in Figure 1. Scientists crossed plants that were homozygous for the cyclin dependent kinase-wild type ($CDK\text{-}WT$) allele with plants that were homozygous for the cyclin dependent kinase-mutant ($CDK\text{-}Mut$) allele. The F_1 offspring of this cross were then crossed, producing F_2 offspring. **Explain** how the ratio of phenotypes in the F_2 generation would demonstrate that the $CDK\text{-}Mut$ allele is recessive.

(b) **Explain** the relationship between the turgor pressure of the guard cell and the pH of the extracellular space.

(c) In another strain of the same species of plant, a deletion has removed a portion of the promoter region of the gene encoding a polypeptide of the proton pump shown in Figure 1. **Predict** the effect of this mutation on the expression of the proton pump. Provide reasoning to **justify** your prediction.

(d) Researchers claim that the $CDK\text{-}Mut$ allele has a positive effect on plant survival in drought conditions. Use the information provided to **support** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] An acceptable description. Acceptable descriptions include the following.
  · K^+ ions move against their concentration gradient through a transmembrane channel.
  · K^+ ions move as a result of the membrane potential established by the pumping of the protons.
- [ ] An explanation that there will be a phenotypic ratio of 3 wild-type : 1 mutant/25% mutant in the F_2 generation, because the F_1 parents are heterozygous

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· As the pH of the extracellular space decreases, turgor pressure increases because protons are being pumped into the extracellular space.

· As protons are pumped into the extracellular space, the guard cells fill up with water, resulting in a lower extracellular pH and a higher turgor pressure.

· There is an inverse relationship because as pH decreases in the extracellular space, turgor pressure increases in guard cells.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that there will be a decrease/no expression (of the proton pump)
- [ ] A justification that transcription factors and RNA polymerase must bind to the promoter to initiate transcription

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

FRQ

The response includes support for the claim by stating that, based on the information provided, the reduction in density of stomata in the mutant plants reduces water loss from the plants.

14. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Aquaporin channel proteins are found in the plasma membrane of various cell types. Scientists studied the production of aquaporins in the keratinocytes (skin cells) of healthy individuals and found that these proteins play a role in hydration of the skin. Some individuals have a skin disorder that causes the absorption of excess water by the skin of their hands and feet when they are soaked in water. Scientists determined that the keratinocytes of these individuals are characterized by the presence of abnormal aquaporins. Scientists studied a family in which some individuals have this skin disorder and constructed a pedigree (Figure 1).

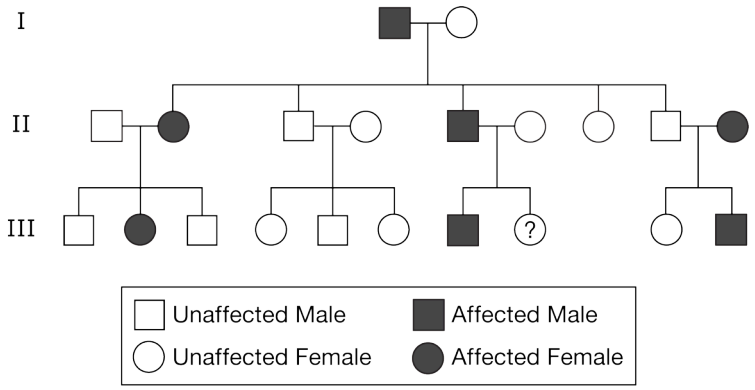


Figure 1. Pedigree of a human family in which some members have a skin disorder. An individual with an unknown phenotype is marked by a question mark.

(a) Based on the pedigree in Figure 1, **predict** the probability that the individual marked by the "?" will have the skin disorder. **Justify** your prediction.

(b) **Describe** the characteristics of the interior of the aquaporin channel protein that allow water to flow through AND exclude glucose. If isolated keratinocytes are placed in a hypotonic solution for 24 hours, **explain** the net direction of water movement AND why the rate of water movement slows down over time.

(c) **Explain** how specific membrane-bound organelles of the keratinocytes are used to produce, transport, and modify aquaporin polypeptides from mRNA.

(d) A scientist claims that the presence of aquaporins in chloroplast membranes supports the endosymbiotic theory. **Support** the claim with evidence from biological concepts.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that the probability that the individual marked by the "?" will have the skin disorder is 0.5 OR $1/2$ OR 50%
- ☐ An acceptable justification. Acceptable justifications include the following.
  · Based on the autosomal dominant inheritance pattern, the mother of "?" must be homozygous recessive, and the father must be heterozygous because his mother is unaffected.
  · Based on the autosomal recessive inheritance pattern, the father of "?" must be homozygous recessive, and the mother must be heterozygous because the sibling is affected.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A description that the interior of the aquaporin channel is hydrophilic/polar and is large enough for water to pass through but too small for glucose to do so
- ☐ An explanation that because of the greater solute concentration inside the keratinocyte, the water will move quickly into the keratinocyte, and then the rate will decrease as equilibrium is reached (if the cells do not lyse)

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the aquaporin polypeptide is synthesized by the rough endoplasmic reticulum, transported by the (rough) endoplasmic reticulum/a vesicle, and then modified by the rough endoplasmic reticulum/Golgi complex.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes supporting evidence that chloroplasts evolved when an ancestral cell engulfed a (free-living prokaryotic) cell. Aquaporins in chloroplasts are likely derived from aquaporins in (the plasma membrane of) the engulfed cell.

FRQ

15. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Once thought of as only a toxic by-product of cellular metabolism, the chemical hydrogen peroxide (H_2O_2) is now known to play an integral role in maintaining cell homeostasis. Recent research suggests that H_2O_2 is one of a small group of molecules that can be transported across the plasma membrane through aquaporins, where it can affect cell signaling. To investigate the transport of H_2O_2 through aquaporins, researchers made four different groups of liposomes (spherical vesicles surrounded by a phospholipid bilayer). One group of liposomes was constructed as a simple vesicle of phospholipids (group A). Into the phospholipid bilayer of the three other groups of liposomes, researchers embedded either a type of human aquaporin (group B) or one of two types of plant aquaporin (groups C and D). The rate at which H_2O_2 was transported into the liposomes of each group was then determined (Table 1).

TABLE 1. RATE OF HYDROGEN PEROXIDE TRANSPORT INTO LIPOSOMES

Group	Type of Aquaporin Embedded in Liposome Phospholipid Bilayer	Average Rate of H_2O_2 Transport ($\mu L/s \pm 2SE_{\bar{x}}$)
A	None	0.9 ± 0.1
B	Human	1.4 ± 0.1
C	Plant I	1.7 ± 0.1
D	Plant II	1.8 ± 0.1

Figure 1 shows a simplified model of the relationship between H_2O_2 concentration and an intracellular signaling pathway. When the concentration of H_2O_2 is low, the enzyme $Tpx\ 1$ is active, leading to the activation of the transcription factor $Pap\ 1$ in a reaction that consumes H_2O_2. Once activated, $Pap\ 1$ moves to the nucleus, where it can activate transcription of a group of $Pap\ 1$-regulated genes.

As the concentration of H_2O_2 starts to increase, $Tpx\ 1$ becomes inactivated, allowing for even more H_2O_2 to accumulate, because it is no longer consumed in the activation of $Pap\ 1$. The accumulation of H_2O_2 leads to the activation of the enzyme $Sty\ 1$, which then activates transcription factor $Atf\ 1$. $Atf\ 1$ moves to the nucleus and activates transcription of a group of $Atf\ 1$-regulated genes.

FRQ

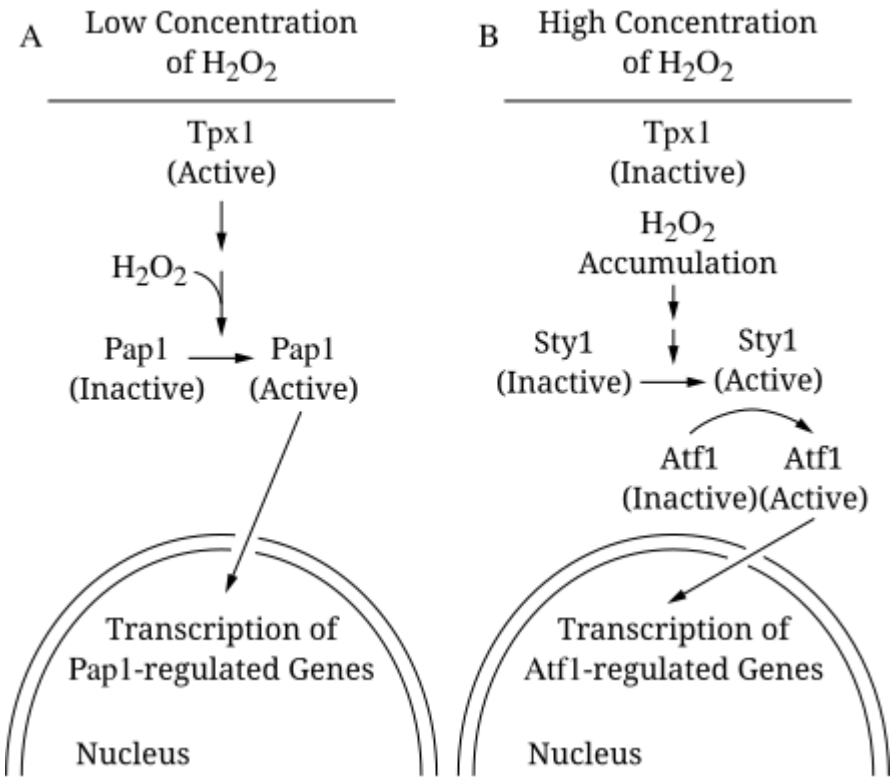
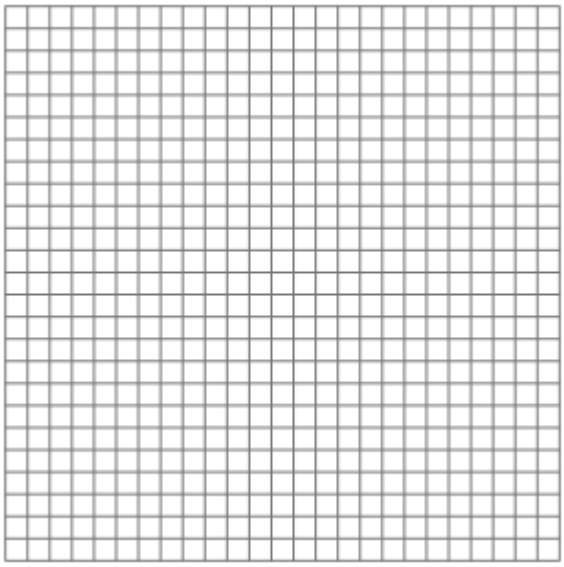


Figure 1. Simplified model of a signaling pathway involving H_2O_2 at both low (A) and high (B) intracellular concentrations of H_2O_2

(a) **Describe** the advantage to a cell of having aquaporins when the substances that pass directly through aquaporins can also pass through the phospholipid bilayer.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. **Determine** the type(s) of aquaporins that transport(s) H_2O_2 at a rate that is different from the rate at which the plant type II aquaporins transport H_2O_2.

FRQ



(c) Based on the data in Table 1, **identify** the rate in $\mu L/s$ at which H_2O_2 can be transported by simple diffusion through the phospholipid bilayer. **Calculate** the difference between the average microliters of H_2O_2 transported into group A liposomes in five minutes and the average microliters of H_2O_2 transported into group C liposomes in five minutes.

(d) An enzyme called $Srx\ 1$ is required for the activation of $Tpx\ 1$. Researchers identified a mutation that results in a nonfunctional form of $Srx\ 1$. Based on the model in Figure 1, **predict** the effect of this nonfunctional $Srx\ 1$ on the expression of $Pap\ 1$-regulated genes AND $Atf\ 1$-regulated genes. Provide reasoning to **justify** your prediction.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that aquaporins are able to transport these substances at a faster rate/in large quantities.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

FRQ

The response includes three of the following criteria.

- [ ] Appropriate labeling
- [ ] Data graphed on a bar/modified bar graph
- [ ] Accurately plotted data points and error bars
- [ ] An acceptable determination. Acceptable determinations include the following.
  - The human aquaporin/Group B
  - None and the human aquaporin/Groups A and B

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes all of the following criteria.

- [ ] An acceptable identification of the rate. Acceptable identifications include the following.
  - $0.9\ (\pm\ 0.1)$
    - $0.8 - 0.1$
- [ ] A calculation that the difference in microliters transported equals 240

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes all of the following criteria.

- [ ] An acceptable prediction. Acceptable predictions include the following.
  - Expression of $\mathrm{Pap1}$-regulated genes will not occur/will decrease, while expression of $\mathrm{Atf1}$ -regulated genes will stay the same/increase.
  - The expression levels of $\mathrm{Pap1}$ - and $\mathrm{Atf1}$-regulated genes will be similar to expression levels

under a high concentration of H_2O_2.

- A justification that (if Tpx1 cannot be activated) H_2O_2 will accumulate/Sty1 will be activated, resulting in the activation of Atf1 but not Pap1

FRQ

16. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Chemical Z blocks the cell cycle of T cells, cells that divide in order to respond to pathogens to prevent infectious diseases. Scientists activated T cells that were in cell cycle phase G_0 to make them reenter the cell cycle. The scientists treated the activated T cells with chemicals A, B, C, or Z. Chemicals A, B, and C are known to block cell cycle progression at certain phases of interphase (Figure 1).

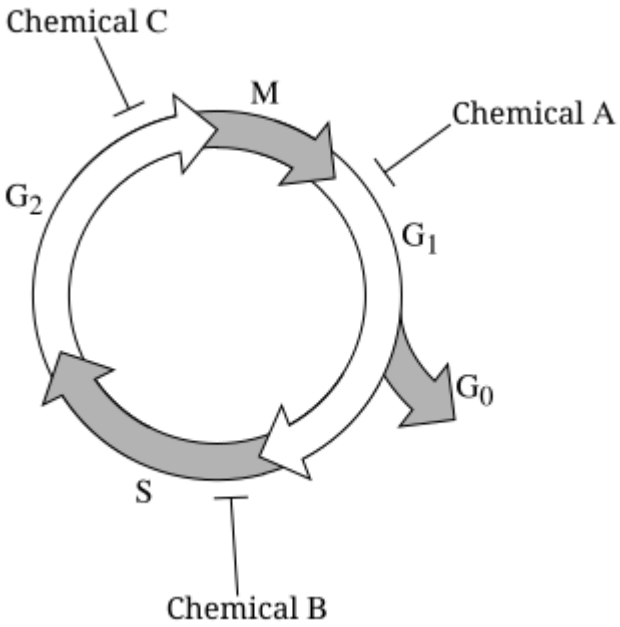


Figure 1. A model of the eukaryotic cell cycle showing the points (marked by bars, ⊥) at which chemicals A, B, and C block progression through interphase

By measuring the amount of DNA in the T cells in each sample, the scientists were able to determine the percent of cells in each phase of interphase (Figure 2).

FRQ

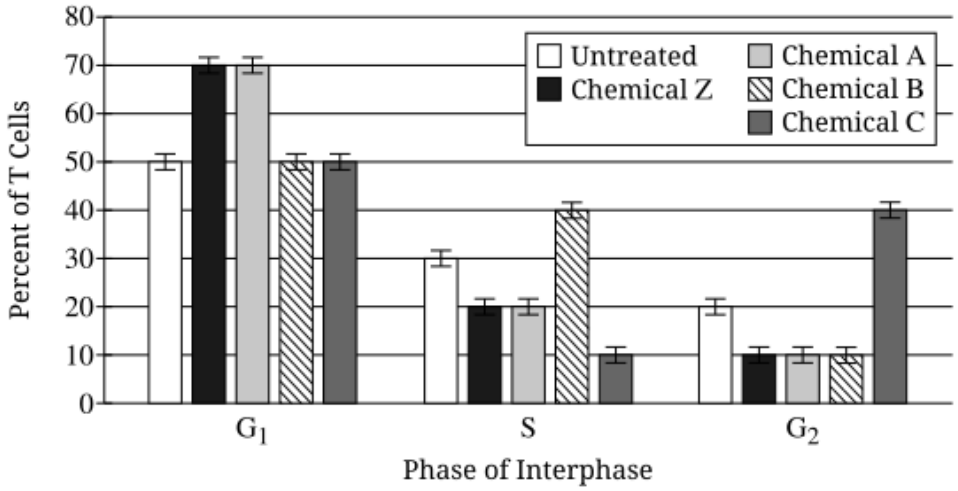


Figure 2. Percent of activated T cells in each phase of interphase in untreated cells and cells treated with chemicals A, B, C, and Z. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

(a) Chemical Z is a large, polar molecule. **Describe** the process by which chemical Z moves across the plasma membrane without an input of cellular energy. **Explain** why the amount of DNA in a cell indicates whether the cell is in G_1, S, or G_2.

(b) **Identify** a dependent variable in the experiment. **Justify** the use of untreated cells in the experiment. **Justify** the use of both chemicals B and C in the experiment.

(c) Based on the data in Figure 2, **identify** the phase in interphase at which chemical Z treatment blocks the cell cycle. **Describe** the trends in the data that support your identification. Based on the mean values shown in Figure 2, **calculate** the difference in the number of cells in G_1 between the sample treated with chemical Z and the untreated sample if a total of 600 cells are counted for each sample.

(d) Cancerous T cells result from genetic mutations and lack normal cell cycle controls. Scientists tested whether chemical Z blocks the cell cycle of cancerous T cells. **State** the null hypothesis for their experiment. Individuals with cancerous T cells also have normal T cells. Scientists find that chemical Z is an effective treatment against cancerous T cells. However, they claim that using chemical Z can have a negative effect on a patient's ability to fight off an infection. Provide reasoning to **justify** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- An acceptable description. Acceptable descriptions include the following.
  · Chemical Z requires a transport protein and moves from a high to a low concentration.
  · Chemical Z moves by facilitated diffusion.
- An explanation that there will be twice as much DNA per cell in cells that are (at the end of S and) in G_2 than in cells that are in G_1 because DNA replication occurs during S phase

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- An acceptable identification of a dependent variable in the experiment. Acceptable identifications include the following.
  · Percent of T cells (in each phase of the cell cycle/interphase)
  · Amount of DNA in the T cells (in each phase of the cell cycle/interphase)
- A justification that the use of untreated cells in the experiment provides a control to see the percent of cells in each phase of the cell cycle in the/effects of an absence of chemicals/chemical Z
- A justification that the use of both Chemicals B and C in the experiment shows the percent of cells in/effect on each phase when blocked by chemical B/in S and by chemical C /in G_2 in order to help interpret the data from cells treated with chemical Z

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- An identification that the phase in interphase at which chemical treatment blocks the cell cycle is G_1
- An acceptable description. Acceptable descriptions include the following.
  · In comparison with untreated T cells, a greater percent of the chemical Z-treated cells is in G_1 .
  · Chemical A blocks the cell cycle in G_1, and chemical Z-treated cells have the same pattern as chemical A-treated cells.
- A calculation that the difference in the number of cells in G_1 between the sample treated with chemical and the untreated sample if a total of cells are counted for each sample is 120 (Treated: $600 \times 0.7 = 420$; untreated: $600 \times 0.5 = 300$; $420 - 300 = 120$)

Part D

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable null hypothesis. Acceptable null hypotheses include the following.
  - · There will be no difference between the cell cycle of untreated cancerous T cells and the cell cycle of (cancerous) T cells treated with chemical Z.
  - · Chemical Z has no effect on the cell cycle of cancerous T cells.
- ☐ An acceptable justification that chemical Z can have a negative effect because chemical Z will prevent cell division/block the cell cycle of normal T cells (in response to an infection)

FRQ

17. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The red king crab, *Paralithodes camtschaticus*, inhabits shallow coastal waters in the northern Pacific Ocean that can vary in temperature from $1^{\circ}C$ to $13^{\circ}C$. The body temperature of the crab is typically similar to that of the surrounding water. In order to study the effect of temperature on crab metabolism, scientists purified the enzyme phosphofructokinase (PFK) from red king crabs. PFK catalyzes a reaction early in the glycolysis pathway (Figure 1).

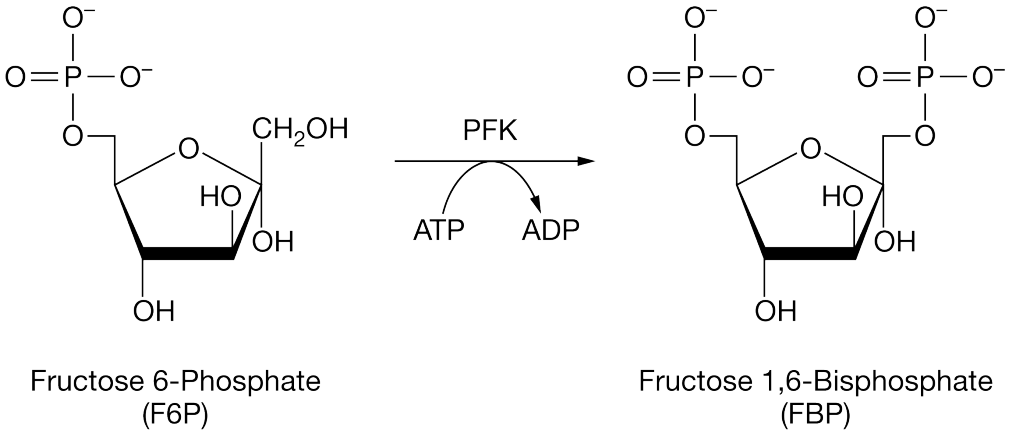


Figure 1. The conversion of fructose 6-phosphate (F6P) to fructose 1,6-bisphosphate (FBP) by PFK

The scientists determined the rate of the PFK-catalyzed reaction using a constant amount of PFK and increasing concentrations of the substrate F6P at $15^{\circ}C$ and $25^{\circ}C$. To determine whether the reaction rate is affected by the nucleotide AMP, the analyses were additionally performed in the presence of 0.5 mM AMP (Figure 2).

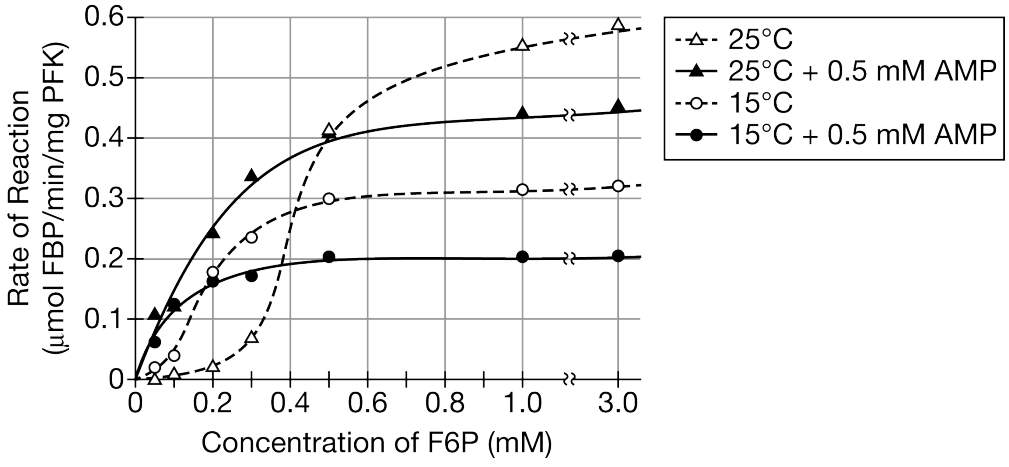


Figure 2. Effect of temperature and AMP on the rate of the reaction catalyzed by PFK

FRQ

In a second analysis of PFK activity at different temperatures, the scientists determined how the concentration of citric acid in the reaction mixture affects the rate of the reaction at $5^{\circ}C$, $15^{\circ}C$, and $25^{\circ}C$ (Figure 3) in the presence of a constant amount of PFK. Citric acid is the first metabolic intermediate produced by the Krebs cycle. All reactions were performed in the presence of 1 mM F6P and 0.5 mM ATP.

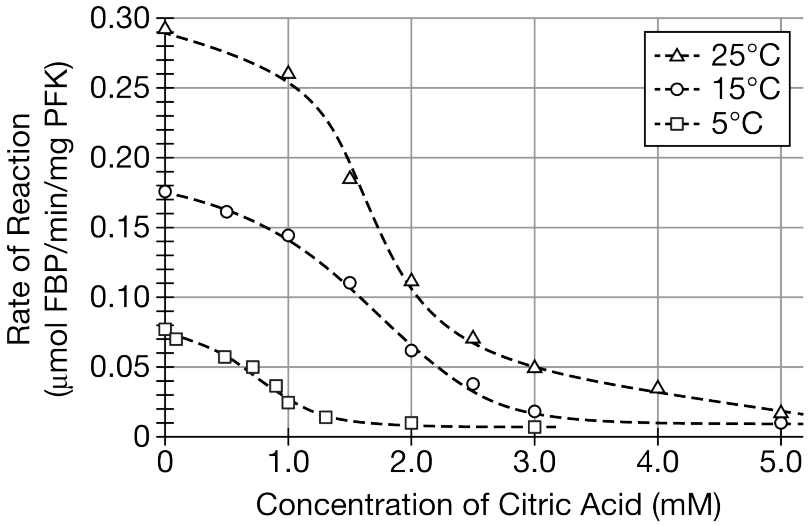


Figure 3. Effect of citric acid on the rate of the reaction catalyzed by PFK. All reactions were carried out in the presence of 1 mM F6P and 0.5 mM ATP.

(a) **Describe** the role of ATP in the reaction shown in Figure 1.

(b) **Identify** an independent variable in the experiment shown in Figure 2. **Justify** the use of increasing concentrations of the substrate F6P when the scientists performed the experiments shown in Figure 2. Based on Figure 2, **describe** the effect of adding AMP to the reactions carried out at $15^{\circ}C$. Based on Figure 3, **calculate** the fold increase in the reaction rate (how many times greater the reaction rate is) between the rate at $5^{\circ}C$ and at $25^{\circ}C$ in the presence of 2 mM citric acid.

(c) **State** the alternative hypothesis for the experiments shown in Figure 3. The scientists claim that, in the absence of AMP, PFK of red king crabs functions efficiently at the crabs' typical low environmental temperatures because the active site of the enzyme can bind substrate more tightly at the low temperatures than it does at higher temperatures. **Support** the scientists' claim based on the data provided. **Predict** the effect on the rate of the reaction catalyzed by PFK in the experiment shown in Figure 2 if increasing concentrations of FBP are added to the reaction.

(d) When crustaceans such as crabs are kept in seawater with a lower-than-normal pH, ATP-requiring pumps that maintain ion gradients in the animals' bodies significantly increase their activity in comparison with normal levels. **Explain** how the increased use of ATP by these pumps will most likely affect the ability of the animals to carry out other metabolic reactions such as that catalyzed by PFK. Hemolymph is the fluid in the red king crab's body that surrounds its cells and is typically slightly hypertonic to the seawater in which the crabs live. When adult crabs were removed from their native environment and placed into seawater solutions with gradually decreasing salt concentrations, their body weight increased. **Explain** how the crabs gained weight during this procedure.

Part A

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that ATP provides both the phosphate group and the energy required to convert $\mathrm{F6P}$ to FBP.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of an independent variable. Acceptable identifications include the following.
  - · Temperature at which reaction was performed
  - · Concentration of $\mathrm{F6P}$ in the reaction mixture
  - · Presence or absence of AMP in the reaction mixture
- ☐ A justification that the scientists increased substrate concentration to determine the substrate concentration at which the reaction rate would reach a maximum/level off
- ☐ A description that at any given $\mathrm{F6P}$ concentration below $0.15\ \mathrm{mM}$ (accept $0.1\ \mathrm{mM}\ -\ 0.2\ \mathrm{mM}$), the reaction rate is more rapid in the presence of AMP, whereas at any given $\mathrm{F6P}$ concentration above $0.15\ \mathrm{mM}$, the reaction rate is slower in the presence of AMP
- ☐ The fold increase in the reaction rate is calculated to be 11.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable stated alternative hypothesis. Acceptable statements include the following.
  - · Increasing concentration of citric acid causes a change (increase/decrease) in reaction rate.
  - · The reaction rate changes (increases/decreases) at different temperatures.
- ☐ Support for the claim based on data that at low concentrations of substrate, the reaction rate is greater at $15^{\circ}\mathrm{C}$ (than at $25^{\circ}\mathrm{C}$ in the absence of AMP)

FRQ

- [ ] A prediction that the reaction rate will decrease

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] An explanation that the rate of other metabolic reactions is likely to decrease because the cells have only so much ATP to use
- [ ] An explanation that (more) water moved from the dilute seawater into the hypertonic hemolymph of the crabs (than normally moves into the hemolymph)

FRQ

18. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

ATCase is a complex enzyme involved in the production of the nucleotide CTP. ATCase is composed of many subunits that are grouped into two functional domains. The catalytic domain of ATCase (C in Figure 1) contains an active site which binds two substrates (S1 and S2) and catalyzes a chemical reaction, resulting in the formation of two products (P1 and P2). P2 then acts as a substrate in the pathway leading to CTP production. The regulatory domain of ATCase (R in Figure 1) contains an allosteric site capable of binding with the nucleotides CTP and ATP.

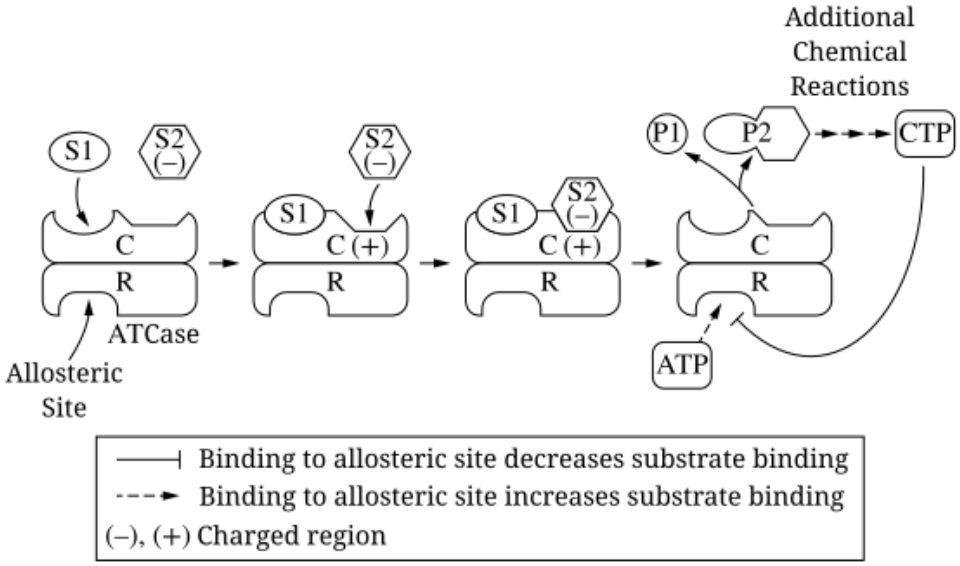


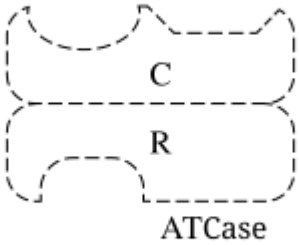
Figure 1. Simplified model of the regulation of an ATCase-catalyzed reaction

(a) **Describe** the effect that the binding of a noncompetitive inhibitor has on the structure of an enzyme.

(b) Based on Figure 1, **explain** how the binding of S1 increases the likelihood that S2 will bind.

(c) Researchers studying ATCase observed that, when ATCase is heated at 60°C for 4 minutes, it is still able to form P1 and P2, but the presence of regulatory molecules has no effect on substrate binding. Using the template in the space provided for your response, **represent** the effect of the increased temperature on ATCase structure by modifying the model of ATCase provided.

FRQ



(d) **Explain** how the concentration of CTP in a cell contributes to the regulation of CTP production.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the description that the binding of a noncompetitive inhibitor changes the conformation/shape/structure of the enzyme/active site.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- It changes the conformation/shape of the active site to better match the shape of S2.
- The active site becomes more positively charged, which attracts negatively charged S2.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes a representation that shows an unchanged active site and a modified allosteric site that is not compatible with the regulators.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- The concentration of CTP determines whether ATP or CTP is more likely to bind to the allosteric site, which affects ATCase activity.
- Competition between ATP and CTP for the allosteric site of ATCase affects ATCase activity.
- When CTP levels are elevated, CTP binds to the allosteric site, decreasing the production of additional CTP (a negative feedback loop).
- When CTP levels are low, ATP binds to the allosteric site, increasing the production of CTP.

FRQ

19. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Glycogen is a complex polymer of glucose molecules involved in the storage of energy in many organisms. Depending on the signals received by a cell, different enzymes are activated, which will lead to either the synthesis or breakdown of glycogen. The product of glycogen breakdown, glucose 1-phosphate, can be modified into glucose 6-phosphate, an intermediate of glycolysis (Figure 1).

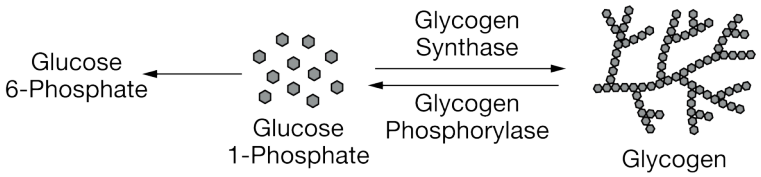
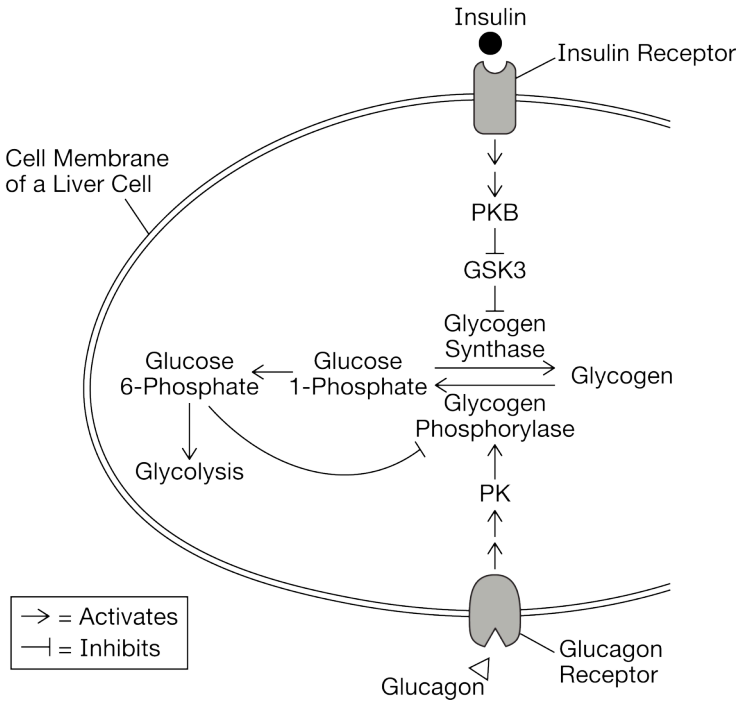


Figure 1. Glycogen synthesis and breakdown

In the liver, glycogen synthase and glycogen phosphorylase are regulated by kinases. These kinases transfer phosphate groups from ATP to amino acids in glycogen synthase and glycogen phosphorylase. Phosphorylase kinase B (PKB) is a kinase that regulates glycogen synthase kinase 3 (GSK3) activity. GSK3 regulates the activity of glycogen synthase. Phosphorylase kinase (PK) regulates the activity of glycogen phosphorylase. The activity of these kinases depends on the activation of signaling pathways in response to the binding of specific cell signals, including insulin and glucagon, to their receptors (Figure 2).



FRQ

Figure 2. A simplified model of signaling pathways in a liver cell that are activated by insulin and glucagon

(a) A cell has a mutation that results in the permanent activation of the glucagon receptor, even in the absence of glucagon. Based on the information in Figure 2, **predict** the effect of this mutation on the activity of glycogen phosphorylase. **Justify** your prediction.

(b) **Describe** the role of ATP in the regulation of glycogen synthase activity. **Explain** how the addition of a phosphate group to certain amino acids of glycogen synthase can result in the inhibition of glycogen synthase.

(c) **Explain** why an increase in glycogen phosphorylase activity might result in an increase in pyruvate production in the cell.

(d) Diabetes often results in the production of reduced levels of insulin. Researchers claim that a drug that binds to and blocks the active site of GSK3 might mimic the role of insulin in the pathway shown in Figure 2. Provide reasoning to **justify** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  - · Glycogen phosphorylase will be active.
  - · Glycogen phosphorylase will have the same activity as normal.
- ☐ An acceptable justification. Acceptable justifications include the following.
  - · The activation of the glucagon receptor leads to the activation of glycogen phosphorylase, provided there is not enough glucose 6-phosphate to inhibit glycogen phosphorylase.
  - · Glucose 6-phosphate will regulate the activity of glycogen phosphorylase even if the glucagon receptor is permanently active.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A description that ATP provides the energy to form a (covalent) bond between a phosphate group and glycogen synthase
- [ ] An explanation that the addition of a phosphate group to glycogen synthase could alter the structure and/or charge of glycogen synthase such that the substrate can no longer interact with the active site/enzyme can no longer catalyze the reaction

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the explanation that an increase in glycogen phosphorylase activity can lead to an increase in glycolysis which produces pyruvate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable justification. Acceptable justifications include the following.

· Both insulin and this drug will prevent the binding of substrate to GSK3, effectively inhibiting the enzyme.

· The drug mimics the activity of PKB, which is activated by insulin.

FRQ

20. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

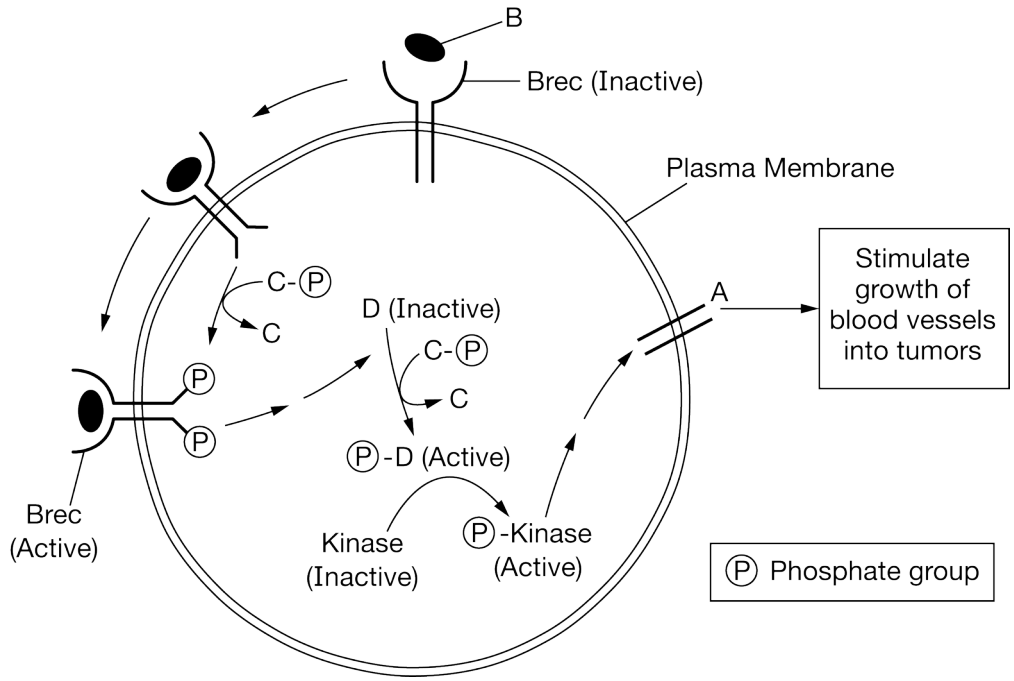


Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of molecule B.

(b) **Explain** how the addition of a phosphate group to certain amino acids of Brec likely affects the tertiary structure and function of Brec. **Explain** how the addition of a small RNA could prevent the activation of kinase D.

(c) Based on the proposed signaling pathway, **predict** the relative amount of phosphorylated to unphosphorylated

FRQ

kinase D in the Brec-WT cells when the cells are grown in nutrient broth lacking B. **Justify** your prediction.

(d) Cells are treated with a compound that prevents the inactivation of kinase D. Researchers claim that Brec-WT cells treated with this compound will stimulate blood vessel-forming cell division to the same degree as Brec-MUT cells do in the absence of B. Based on the information provided, provide reasoning to **support** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that molecule B is the ligand/binds to the receptor and activates the receptor/the signaling pathway.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that the addition of a phosphate group could alter the structure and/or charge of Brec so that Brec can now interact with other signaling molecules in the cell
- ☐ An acceptable explanation. Acceptable explanations include the following.
  · The small RNA could inhibit translation of an mRNA encoding one of the proteins in the pathway leading to kinase D activation.
  · The small RNA could lead to the degradation of an mRNA encoding a protein in the pathway leading to kinase D activation.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A prediction that there will be more unphosphorylated than phosphorylated kinase D
- [ ] A justification that because the wild-type Brec is inactive in the absence of B, the receptor will not stimulate intracellular signaling, and kinase D will remain unphosphorylated

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes support for the claim by stating that, based on the information provided, both cell types will now have continuously active kinase D, which leads to the expression of A, and the continuous expression of A will stimulate blood vessel-forming cell division by both cell types.

FRQ

1. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In response to an increase in the frequency of thunderstorms that produce damaging hail, some apple growers have started to cover their apple orchards with large nets. While these nets protect the apple trees from hail damage, the nets also decrease the amount of light that reaches the trees. In an attempt to reduce the impact of shading by the nets, researchers investigated the ability of different ground covers placed around the base of the trees to reflect light back to the apple trees. An apple orchard with genetically identical apple trees was divided in half. A reflective ground cover was applied to one half of the orchard, while grass was left as the ground cover for the other half. A sample of apples was taken from every tree. The percent of light reflected by the two ground covers was measured at three distances above the ground (Figure 1).

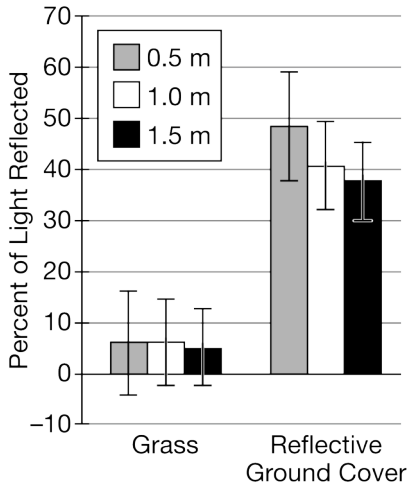


Figure 1. Percent of light reflected by ground covers, measured at three distances above the ground. Error bars equal $\pm 2\mathrm{SE}_{\bar{x}}$.

The amount of light that apple trees receive changes the expression of a certain transcription factor. This transcription factor affects the transcription rate of the gene encoding the enzyme phenylalanine ammonium-lyase (PAL). The PAL enzyme is required for the production of two types of pigments, anthocyanins and flavonoids, which contribute to the coloration of apples. Light also affects the activity of the PAL enzyme by a signal transduction pathway (Figure 2).

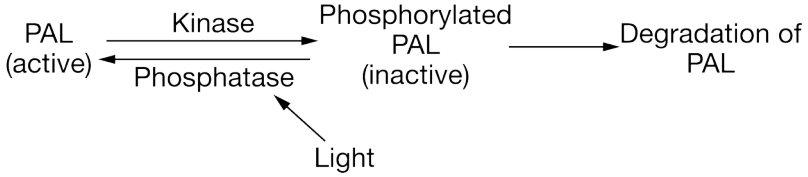


Figure 2. Simplified model of the signal transduction pathway leading to PAL activation and breakdown

FRQ

The researchers measured the average concentrations of anthocyanins and flavonoids in apples from trees grown with both types of ground cover (Table 1).

TABLE 1. CONCENTRATION OF ANTHOCYANINS AND FLAVONOIDS IN APPLES

Type of Pigment	Concentration of Pigment (mean $\pm 2\mathrm{SE}_{\bar{x}}$ in $\mathrm{nmol\ cm^{-2}}$) in Apples Grown with Grass Ground Cover	Concentration of Pigment (mean $\pm 2\mathrm{SE}_{\bar{x}}$ in $\mathrm{nmol\ cm^{-2}}$) in Apples Grown with Reflective Ground Cover
Anthocyanins	14.51 ± 1.9	24.10 ± 3.0
Flavonoids	29.24 ± 3.6	44.50 ± 5.7

(a) Using the information provided in Figure 2, **explain** the role of light in the activation of the PAL enzyme.

(b) **Identify** the variable in the researchers' experiment that is dependent on the percent of reflected light the apple trees receive. **Justify** the use of genetically identical apple trees as a controlled factor in the experiment. Using the data provided, **describe** the effect of the different ground covers on the percent of light reflected. **Calculate** the percent change in the average concentration of flavonoids in apples grown with the reflective ground cover compared with the apples grown with the grass ground cover.

(c) Researchers also study apple trees that have been genetically modified so that the DNA binding domain of the transcription factor that stimulates expression of the PAL enzyme has been deleted. The researchers repeat the ground cover experiment with the genetically modified apple trees. **Predict** the effect of the two ground covers on pigment concentrations in the genetically modified apples. Provide reasoning to **justify** your prediction.

(d) Based on the information provided in Figure 2, **predict** the effect of a constantly active kinase on flavonoid pigment production. In leaves, the PAL enzyme is also required for the production of other pigment molecules that are capable of absorbing UV and other radiation from the sun. **Explain** how the relationship between light and pigment production might serve as a protective mechanism for the cells of the apple tree. **Explain** how the PAL enzyme interacts with its substrate in the absence of light.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

FRQ

The response includes an explanation that the role of light in the activation of the PAL enzyme is to activate phosphatase which removes the phosphate group from the PAL enzyme, thereby activating the enzyme.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An identification of the concentration of pigments (anthocyanins and flavonoids) in the apples as the variable dependent on the percent of reflected light the apple trees receive
- ☐ A justification that, because the apple trees are genetically identical, any differences in the concentration of pigments can be attributed to the differences in the experimental treatment (type of ground cover/ amount of reflected light)
- ☐ A description that there is (statistically) more light reflected from the reflective ground cover
- ☐ The percent change in the average concentration of flavonoids calculated to be 52.19%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  · These genetically modified trees will have low levels of/no pigments regardless of the type of ground cover.
  · The type of ground cover will have no effect on pigment concentrations in these genetically modified trees.
- ☐ An acceptable justification. Acceptable justifications include the following.
  · If the transcription factor cannot bind to (the regulatory region of) the DNA encoding the PAL enzyme, the gene cannot be transcribed (to mRNA), and no PAL enzyme will be produced.
  · If PAL cannot be produced, the amount of light (reflected by the ground cover) will not affect the production or activation of PAL.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] A prediction that flavonoid production will decrease/there will be no pigment production
- [ ] An explanation that, by increasing the amount of pigment molecules in the leaves in response to an increase in sunlight, plants reduce radiation damage to DNA
- [ ] An explanation that, in the absence of light, the PAL enzyme and substrate would not interact because the phosphate group would still be attached, which would lead to the degradation of PAL/an inactive shape of PAL

FRQ

2. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

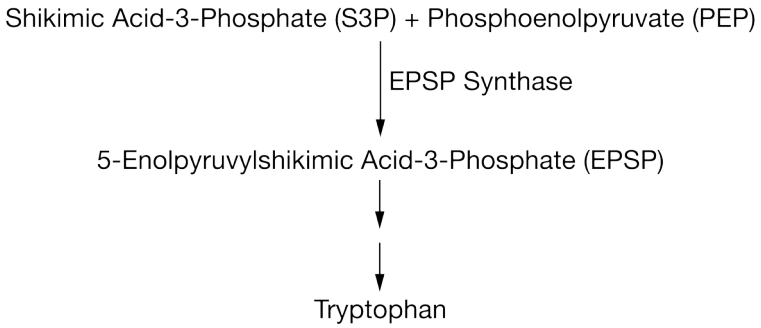


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

Scientists investigated the effect that Compound X has on EPSP synthase activity by performing an experiment to examine the relationship between EPSP synthase activity, PEP concentration, and Compound X concentration (Figure 2).

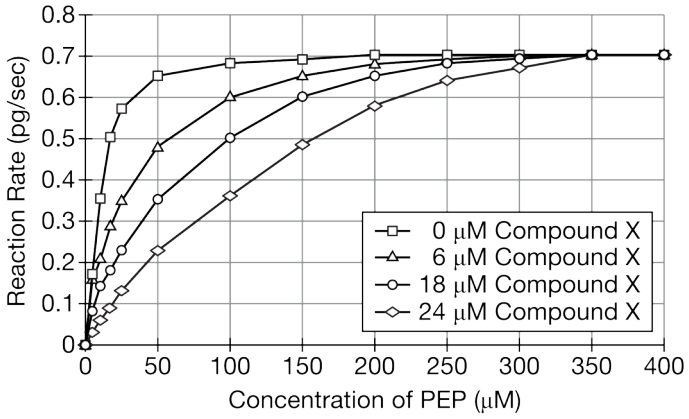


Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

Some populations of the plant *Lolium rigidum* have evolved resistance to Compound X. To investigate the

inheritance of this resistance, researchers crossed true-breeding plants from a Compound X-susceptible (S) population with true-breeding plants from a Compound X-resistant (R) population. In half of these crosses, pollen from S plants was used to fertilize ovules in R plants. In the other half, pollen from R plants was used to fertilize ovules in S plants. Survival of parental and F_1 seedlings, differentiated by the maternal phenotype, in response to varying concentrations of Compound X is shown in Figure 3.

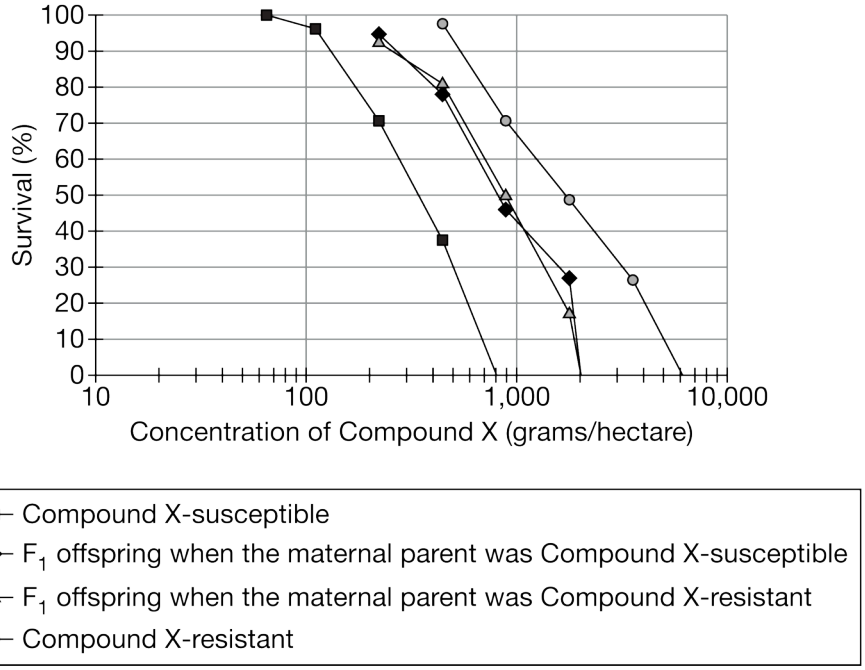


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots. $1\ \text{Hectare} = 10{,}000\ \text{m}^2$

(a) Compound X competes with PEP for binding to the active site of EPSP synthase. **Explain** how increasing the concentration of PEP can reverse the inhibitory effect of Compound X on the reaction catalyzed by EPSP synthase.

(b) **Identify** a control group in the experiment shown in Figure 2. **Identify** the dependent variable in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in grams/hectare required to kill 60% of F_1 offspring when the maternal parent was Compound X-susceptible. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of $200\ \mu\text{M}$ and a Compound X concentration of $0\ \mu\text{M}$.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 female with a male from the original susceptible (S) population. The resulting offspring were treated with $300\ \text{grams/hectare}$ of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. A student claims that the allele for the Compound X -resistance gene exhibits incomplete dominance over the Compound X-susceptible allele. Based on data provided in Figure 3, provide reasoning to **support** this claim. Researchers discovered a variant of the Compound X -resistance gene in *L. rigidum* in which there was a two base pair deletion early in the coding region of both copies

FRQ

of the gene. **Predict** the most likely effect on the phenotype of the plant.

(d) Researchers have engineered Compound X-resistant corn by inserting a Compound X-resistance gene from a prokaryote into the corn genome. **Explain** how scientists could ensure that the gene would be reliably transmitted to the offspring by either the pollen or the ovule. **Explain** why the *EPSP* synthase gene and the pathway it regulates can exist in both prokaryotes and corn plants.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that as the concentration of PEP/ratio of PEP to Compound X increases, the chance of PEP instead of Compound X binding to the active site of EPSP synthase increases.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of a control group. Acceptable identifications include the following.
  - 0 µM Compound X
  - 0 µM PEP
- ☐ An identification that the dependent variable is percent survival
- ☐ An identification that the concentration of Compound X in grams / hectare, required to kill 60 % of F_1 offspring when the maternal parent was Compound X-susceptible is 1100 (grams / hectare); NOTE: acceptable student responses can be between 1050 and 1200.
- ☐ The mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of 200μM and a Compound X concentration of 0 µM is calculated to be 42 pg.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable prediction of the percent survival of one genotype of the resulting offspring. Acceptable predictions include the following.
  - 87% survival RS; NOTE: acceptable student responses can be between 85% and 90%.
  - 55% survival SS (susceptible); NOTE: acceptable student responses can be between 52% and 58%.
- ☐ The reasoning that heterozygous F_1 offspring are more resistant than the original susceptible population is but not as resistant as is the original resistant population
- ☐ An acceptable prediction of the most likely effect on the genotype of the plant. Acceptable predictions include the following.
  - No Compound X resistance
  - Compound X-susceptibility
  - Death following exposure to Compound X

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that the gene would have to be placed in a nuclear chromosome because nuclear chromosomes are transmitted in both ovules and pollen
- ☐ An acceptable explanation. Acceptable explanations include the following.
  - Corn and prokaryotes share a common ancestor.
  - Core metabolic pathways are conserved in multiple domains of life.

FRQ

3. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Saccharomyces cerevisiae, also known as baker's yeast, is a single-celled eukaryote that can make ATP both aerobically and anaerobically. To obtain a readily accessible form of chemical energy for cellular activities, many strains of yeast convert glucose into ethyl alcohol and carbon dioxide.

Researchers conducted an experiment to determine the effect of temperature on the rate of glucose conversion to ethyl alcohol and carbon dioxide. They added an equal mass of yeast cells to equal volumes of a glucose solution in a series of sealed test tubes. Groups of tubes were incubated at temperatures ranging from $28^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$, the optimal range for yeast growth. The researchers measured carbon dioxide production in the tubes.

(a) **Describe** the importance of fermentation to eukaryotic cells such as yeast.

(b) **Identify** the independent variable in the experiment.

(c) **Predict** the relative rates of glucose breakdown at $4^{\circ}\mathrm{C}$ AND at $56^{\circ}\mathrm{C}$ compared with the rate at $37^{\circ}\mathrm{C}$.

(d) **Provide reasoning** to support your prediction in part (c).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that the importance of fermentation to cells such as yeast is that it (fermentation) enables the yeast cells to regenerate NAD^{+}/make ATP/obtain energy without oxygen.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies the independent variable as temperature.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response predicts that the rates [of glucose breakdown] will be lower at both $4^{\circ}C$ and $56^{\circ}C$ (than at $37^{\circ}C$).

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response supports the prediction in part (c) because it provides reasoning that at $4^{\circ}C$, molecules move more slowly, so there are fewer collisions between enzymes and substrates. At $56^{\circ}C$, enzymes denature/change their structure and function less efficiently.

FRQ

4. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

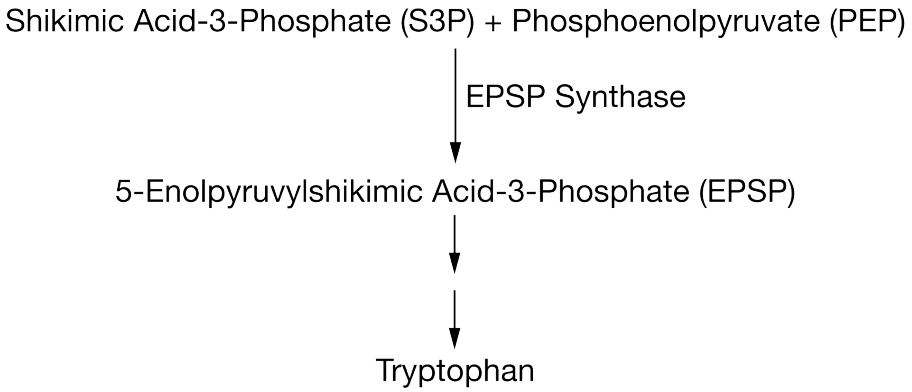


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

Scientists investigated the effect that Compound X has on EPSP synthase activity by performing an experiment to examine the relationship between EPSP synthase activity, PEP concentration, and Compound X concentration (Figure 2).

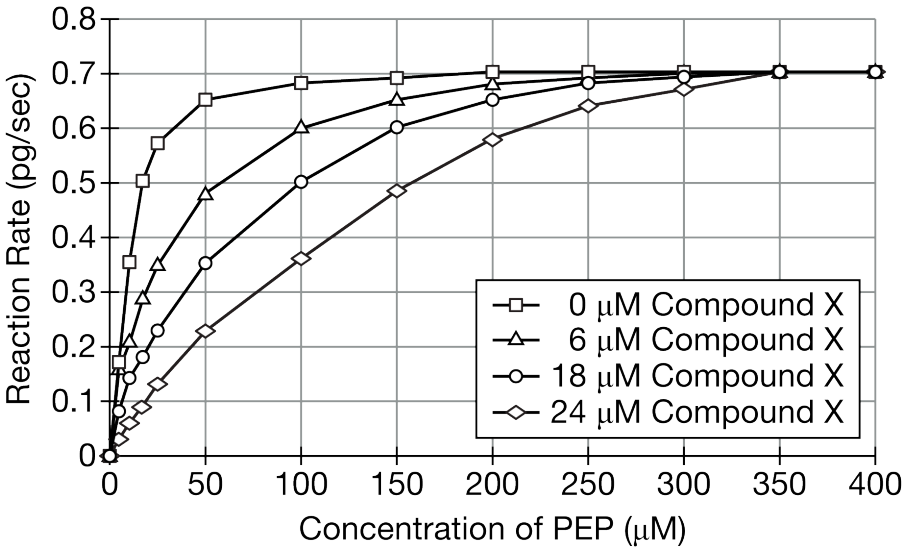


Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

Some populations of the plant *Lolium rigidum* have evolved resistance to Compound X. To investigate the inheritance of this resistance, researchers crossed true-breeding plants from a Compound X-susceptible (S) population with true-breeding plants from a Compound X-resistant (R) population. In half of these crosses, pollen from S plants was used to fertilize ovules in R plants. In the other half, pollen from R plants was used to fertilize ovules in S plants. Survival of parental and F_1 seedlings, differentiated by the maternal phenotype, in response to varying concentrations of Compound X is shown in Figure 3.

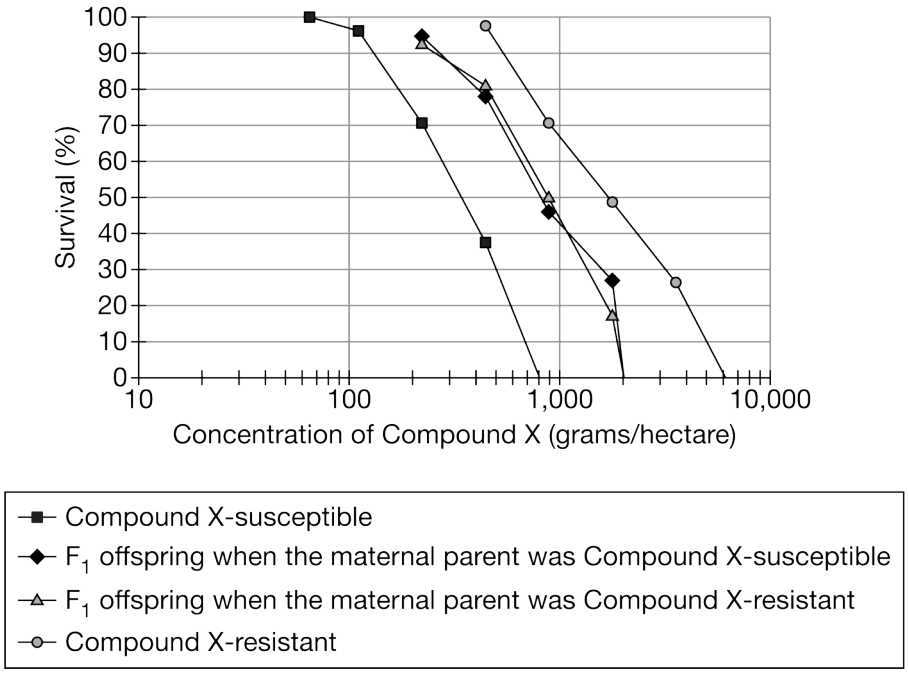


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots. $1\ \text{Hectare} = 10{,}000\ \text{m}^2$

(a) **Explain** how Compound X can affect binding of S3P and PEP to EPSP synthase.

(b) **Identify** one dependent variable in the experiment shown in Figure 2. **Identify** a control group in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in grams/hectare, required to kill 50% of the Compound X-susceptible (S) population. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of 50 µM and a Compound X concentration of 18 µM.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 male with a female from the original resistant (R) population. The resulting offspring were treated with 900 grams/hectare of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. Researchers claim that if the Compound X-resistance gene

FRQ

were carried on a chloroplast chromosome, the F_1 offspring would not have a phenotype intermediate to that of the two parents. Provide reasoning to **justify** the claim. Researchers discovered a variant of the Compound X -resistance gene in *L. rigidum* in which there was a single base pair insertion upstream of the promoter region of one copy of the gene. **Predict** the most likely effect on the phenotype of the plant.

(d) Researchers have engineered Compound X-resistant corn using prokaryotic genes. Other genetic engineering experiments have placed eukaryotic genes into prokaryotic cells for mass production of their gene products. **Explain** how the eukaryotic DNA would need to be altered in order for prokaryotic cells to express the eukaryotic gene. **Explain** why the *EPSP* synthase gene from a prokaryote can be properly expressed in a corn plant.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· The binding of Compound X to EPSP synthase alters the shape and/or charge of the active site such that S3P and PEP are no longer compatible.

· Compound X can bind to the active site and block the binding of S3P and PEP.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An identification that the dependent variable is the reaction rate
- ☐ An acceptable identification of a control group. Acceptable identifications include the following.
  · Compound X-susceptible
  · Compound X-resistant
- ☐ An identification that the concentration of Compound X, in grams / hectare, required to kill 50% of the Compound X-susceptible (S) population is 350 (grams / hectare); NOTE: acceptable student responses can be between 300 and 400
- ☐ The mass, in pg, of EPSP that would be produced by the reaction in one minute at a PEP concentration of $50\mu M$ and a Compound X concentration of $18\mu M$ is calculated to be 21 pg.

FRQ

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable prediction of the percent survival of one genotype of the resulting offspring. Acceptable predictions include the following.
  - 48% survival RS; NOTE: acceptable student responses can be between 45% and 50%.
  - 71% survival RR (resistant); NOTE: acceptable student responses can be between 70% and 72%.
- ☐ A justification that the offspring would inherit the Compound X resistance gene only from the maternal parent, and the phenotype would either be fully susceptible or fully resistant
- ☐ An acceptable prediction of the mostly likely effect on the phenotype of the plant. Acceptable predictions include the following.
  - No change (to the phenotype)
  - Compound X resistance

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  - Introns would need to be removed because prokaryotes do not perform RNA splicing.
  - The gene would need to be under the control of a prokaryotic promoter.
- ☐ An explanation that the genetic code is shared in all living organisms (and therefore the protein would have the same amino acids)

FRQ

5. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

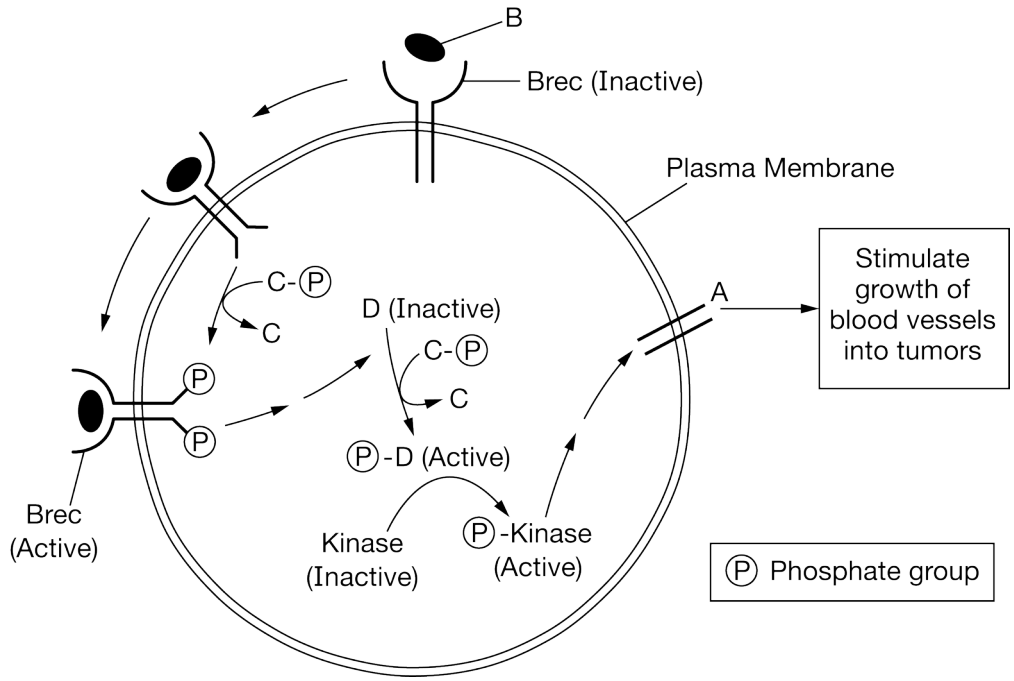


Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of molecule C-P in the activation of D.

(b) **Explain** how a molecule such as kinase D can amplify an extracellular signal. **Explain** how the mutant Brec protein is able to activate kinase D even in the absence of its ligand.

(c) Based on the proposed signaling pathway, **predict** the relative activity of kinase D in the Brec-WT cells compared with kinase D in the Brec-MUT cells when both cell types are grown in nutrient broth lacking B.

FRQ

Justify your prediction.

(d) The researchers claim that treatment with a compound that inhibits dephosphorylation of the intracellular domain of the Brec protein will inhibit expression of A in Brec-WT cells but not in Brec-MUT cells. Based on the information provided, provide reasoning to **refute** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that molecule C-P provides the energy and the phosphate (to activate D).

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that kinase D can act on multiple intracellular substrates/phosphorylate multiple kinases
- ☐ An explanation that the mutant Brec protein already has the appropriate shape to be phosphorylated/ activate kinase D/cause the release of a second messenger

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that there will be less kinase D activity in the Brec-WT cells than in the Brec-MUT cells
- ☐ A justification that the Brec-WT cells require B to stimulate intracellular signaling including activation of kinase D, but the Brec-MUT cells do not

FRQ

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a refutation of the claim by stating that if dephosphorylation of Brec (with WT or mutant) is blocked, then downstream signaling will continue to occur, and A will be expressed in both cell types.

FRQ

6. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Glycogen is a complex polymer of glucose molecules involved in the storage of energy in many organisms. Depending on the signals received by a cell, different enzymes are activated, which will lead to either the synthesis or breakdown of glycogen. The product of glycogen breakdown, glucose 1-phosphate, can be modified into glucose 6-phosphate, an intermediate of glycolysis (Figure 1).

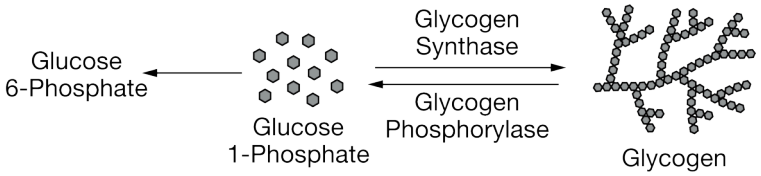


Figure 1. Glycogen synthesis and breakdown

In the liver, glycogen synthase and glycogen phosphorylase are regulated by kinases. These kinases transfer phosphate groups from ATP to amino acids in glycogen synthase and glycogen phosphorylase. Phosphorylase kinase B (PKB) is a kinase that regulates glycogen synthase kinase 3 (GSK3) activity. GSK3 regulates the activity of glycogen synthase. Phosphorylase kinase (PK) regulates the activity of glycogen phosphorylase. The activity of these kinases depends on the activation of signaling pathways in response to the binding of specific cell signals, including insulin and glucagon, to their receptors (Figure 2).

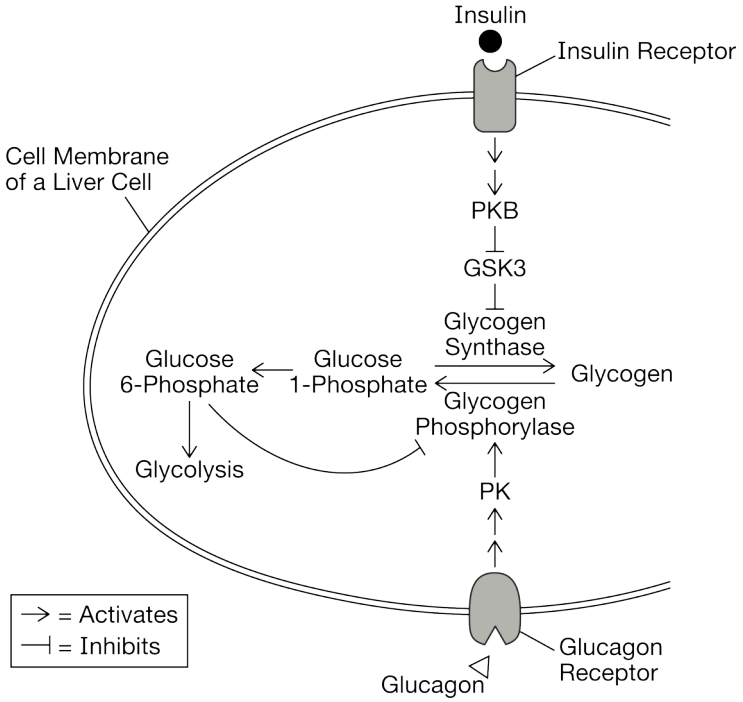


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Figure 2. A simplified model of signaling pathways in a liver cell that are activated by insulin and glucagon

(a) A cell where PKB is always active has a mutation that results in the permanent activation of GSK3. Based on the information in Figure 2, **predict** the effect of this mutation on the activity of glycogen synthase. **Justify** your prediction.

(b) **Describe** the process that results in the activation of multiple copies of PKB in response to the binding of a single molecule of insulin to its receptor. **Explain** why insulin can stimulate the activation PKB of but not the activation of PK.

(c) **Explain** why an increase in glycogen phosphorylase activity might result in an increase in O_2 consumption in the cell.

(d) A certain type of tumor results in the overproduction of glucagon. Researchers claim that treatment with insulin can counteract the effects of the excess glucagon on the pathway shown in Figure 2. Provide reasoning to **justify** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that glycogen synthase will be inactive
- ☐ A justification that when active, the role of GSK3 is to phosphorylate glycogen synthase, resulting in its inhibition, even in the presence of active PKB, an upstream signaling molecule

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable description. Acceptable descriptions include the following.
  · In the signal transduction pathway activated by insulin binding, each enzyme in the pathway can catalyze multiple phosphorylation reactions.
  The insulin receptor activates many second messengers.
- ☐ An explanation that insulin can bind to the insulin receptor but not the glucagon receptor, because the

FRQ

shapes are not compatible

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that an increase in glycogen phosphorylase activity can lead to an increase in glycolysis (and the Krebs/citric acid cycle) and the electron transport chain, which consumes O_2.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a justification that glucagon stimulates glycogen breakdown, but insulin stimulates the synthesis of glycogen.

FRQ

7. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The SIRT3 protein has been reported to increase the growth of certain types of cancer cells. Researchers studying the metabolism and development of stomach cancer cells investigated the potential role of the SIRT3 protein in these cells. In the first investigation, the researchers quantified the relative amount of SIRT3 in normal stomach cells and in each of four different lines of stomach cancer cells (Figure 1). Using data from a group of 100 patients with stomach cancer, the researchers also analyzed the relative amount of SIRT3 protein detected in samples of the cancer cells and in normal cells immediately adjacent to the cancer cells (Figure 2).

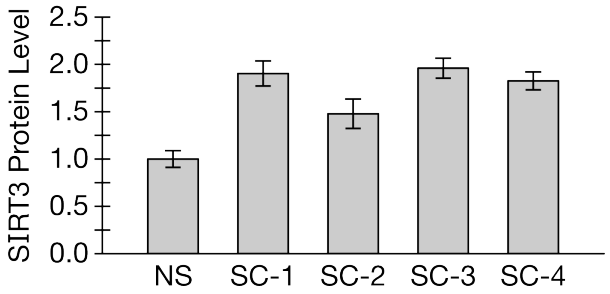


Figure 1. SIRT3 protein levels in normal stomach cells (NS) and stomach cancer cell lines from four different patients (SC-1, SC-2, SC-3, SC-4), shown relative to the level in normal stomach cells (NS). Error bars represent $\pm SE_{\bar{x}}$.

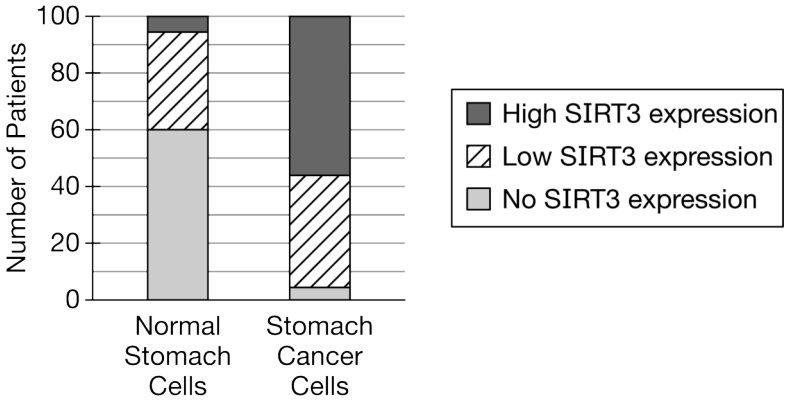
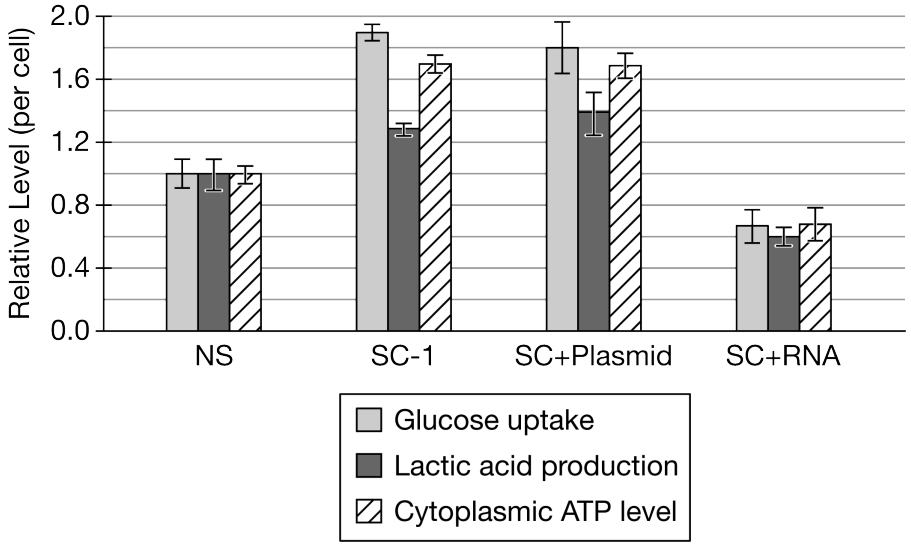


Figure 2. The number of patients, from 100 tested, whose stomach cancer cells or normal stomach cells adjacent to the cancer cells express high, low, or no SIRT3 protein

- In the second investigation, the researchers grew four groups of stomach cells in petri dishes containing nutrient broth:• NS, normal stomach cells• SC-1, a line of stomach cancer cells that the researchers had examined for SIRT3 expression• SC + RNA, the SC-1 line of stomach cancer cells to which a plasmid was added that encodes a small RNA complementary to a portion of the *SIRT3* mRNA. When these small RNAs bind to the complementary portion of *SIRT3* mRNA,

the *SIRT3* mRNA is broken down rather than translated.• SC + plasmid, the SC-1 line of stomach cancer cells to which a plasmid was added that does NOT encode the complementary small RNA

- After 24 hours of growth, the researchers measured several indicators of metabolism by an equal number of all four cell groups: glucose uptake, lactic acid production, and the level of cytoplasmic ATP production in the presence of a compound that blocks the electron transport chain (Figure 3). The values were calculated relative to the values obtained for normal stomach cells.



- Figure 3. Glucose uptake, lactic acid production, and cytoplasmic ATP levels in stomach cells. NS, normal stomach cells; SC-1, stomach cancer cells; SC+ plasmid, SC-1 cells with added plasmid only; SC + RNA, SC-1 cells with added plasmid encoding small RNA complementary to a portion of *SIRT3*. Error bars represent $\pm \mathrm{SE}_{\bar{x}}$.

- (a) A stomach cell line growing in nutrient broth has a mutation in the gene encoding a particular type of tRNA. Assuming the mutation is not lethal to the cells, **explain** why the amount of *SIRT3* mRNA in this cell line is identical to the amount produced by a normal stomach (NS) cell line but the amount of SIRT3 protein is less than in the NS cells.

(b) **Identify** a dependent variable in the researchers' experiment shown in Figure 3. **Justify** the analysis of normal stomach cells (NS) in the experiments shown in Figure 3. Based on the data in Figures 1 and 3, **describe** the relationship between SIRT3 expression and glucose uptake. **Calculate** the percent change in lactic acid production by SC-1 cells compared with SC+RNA cells.

(c) Researchers analyzing regulation of *SIRT3* gene expression engineer a *SIRT3* gene that is identical to the normal gene except that the normal *SIRT3* gene promoter is replaced by the promoter for a gene encoding a protein typically found only in human nerve cells. The researchers introduce this gene into normal human stomach cells to determine how much *SIRT3* mRNA AND protein will be expressed from the engineered *SIRT3* gene in the stomach cells. **Predict** the results of this experiment. The researchers claim that metabolic activity increases in cancerous cells. Provide reasoning based on the data provided to **justify** the claim. Based on the information provided, **predict** the cause of a decrease in CO_2 production by the three groups of cancer cells (SC-1, SC + plasmid, and SC + RNA) treated with the compound that blocks the electron transport chain compared

FRQ

with the same groups of cells not treated with the compound.

(d) Based on the data provided, **explain** how testing for $SIRT3$ gene expression in samples of stomach cells might be used by physicians to predict which individuals have an increased risk of developing stomach cancer. Human stomach cells have plasma membrane receptors for the binding of certain extracellular signaling molecules such as growth factors. **Explain** why the irreversible binding of chemicals that mimic the growth factors will most likely change the normal cells into cancer cells.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that tRNAs are involved in translation not transcription, (so amount of $SIRT3$ mRNA should be the same in both cell lines). If a tRNA is mutated, it will be unable to carry the correct amino acid or unable to bind to the $SIRT3$ mRNA, so the cell line with the mutation will produce less $\mathrm{SIRT3}$ protein.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An identification that the dependent variable is (relative level of) glucose uptake/lactic acid production/ cytoplasmic ATP
- ☐ A justification that normal stomach cells are used (as a control) for comparison to determine whether the metabolic indicators of the other groups of cells have changed
- ☐ An acceptable description. Acceptable descriptions include the following.
  · The greater the expression of $SIRT3$, the greater the amount of glucose uptake by the cells
  · There is a positive correlation.
- ☐ The percent change in lactic acid production by $\mathrm{SC\text{-}1}$ cells compared with $\mathrm{SC+RNA}$ cells is calculated to be 117%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ A prediction that no mRNA or protein will be produced from the introduced gene
- ☐ A justification that the cancer cells/ $\mathrm{SC\text{-}1}$ cells/ $\mathrm{SC+plasmid}$ cells studied in Figure 3 demonstrate increased glucose uptake/lactic acid production/ ATP production, indicating that there has been an increase in glycolysis/fermentation/cellular respiration
- ☐ An acceptable prediction. Acceptable predictions include the following.
  · If the ETC is not active, the rate of NADH accumulation will increase, so the rate of the Krebs/citric acid cycle will decrease, and therefore less $\mathrm{CO_2}$ will be produced.
  · Cells switched to fermentation because Krebs/citric acid cycle stopped, and in the absence of the Krebs cycle, $\mathrm{CO_2}$ is not produced.
  · Pyruvate will be unable to enter the mitochondria because the hydrogen ion gradient is not being maintained and therefore pyruvate cannot enter the Krebs/citric acid cycle, which produces $\mathrm{CO_2}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ ð An explanation that because $SIRT3$ is expressed at higher levels in most stomach cancer cells than in normal stomach cells, determining the level of expression might help indicate if an individual is likely to develop stomach cancer
- ☐ An explanation that because irreversible attachment can continually stimulate a signaling cascade, this can lead to continued stimulation of the cell cycle/continued cell division

FRQ

8. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Friedreich's ataxia (FA) is an autosomal recessive genetic disorder that causes a lack of muscle control in patients. Heart defects are common in FA patients. FA is associated with a mutated *FXN* gene (*f* allele) that results in decreased levels of the mitochondrial protein frataxin, the *FXN* gene product. Muscle cells in individuals with FA produce decreased amounts of ATP . The normal *FXN* gene contains several copies of a triplet of nucleotides in an intron. Mutant alleles of the gene have many more copies of the nucleotide triplet, which causes the *FXN* gene to be abnormally condensed.

To investigate the effects of the *FXN* mutation, researchers studied heart cells from homozygous dominant individuals (*FF*) without FA (genotype A), heart cells from patients with FA (genotype B), and heart cells from individuals heterozygous for the *FXN* gene (genotype C). The researchers isolated two different proteins from an equal number of each of the three cell genotypes. For each cell genotype, they compared the amount of frataxin protein with the amount of glyceraldehyde 3-phosphate dehydrogenase (GAPDH), an enzyme that functions in glycolysis (Figure 1). They then quantified the relative amount of frataxin protein in each cell genotype (Figure 2).

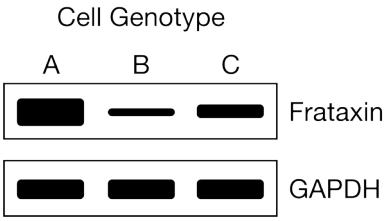


Figure 1. Amounts of frataxin and GAPDH proteins in the three different cell genotypes A, B, and C, as revealed by band width following protein gel electrophoresis

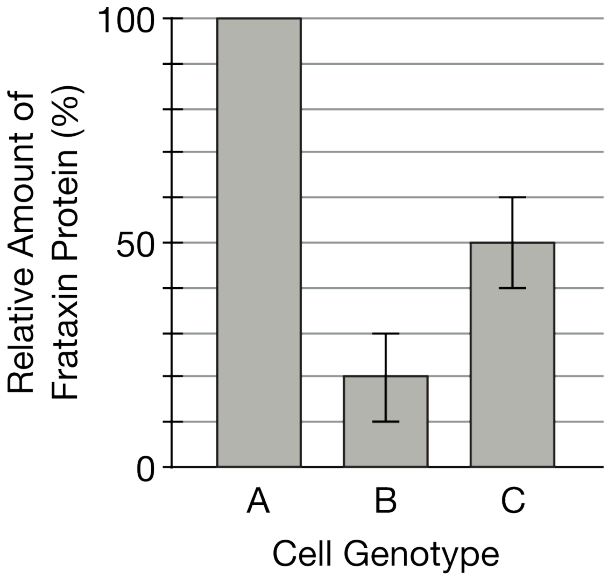


Figure 2. Quantified relative amount of frataxin protein in cell genotypes A, B, and C. Data represent the average from six trials, as shown in Figure 1. Error bars represent $\pm 2SE_{\bar{x}}$.

FRQ

To investigate a potential role for frataxin in cellular respiration, another group of researchers introduced the human FXN gene into a mouse cell line and overexpressed the frataxin protein. Identical control cells were similarly treated but did not receive the human FXN gene. The researchers measured the activity of a Krebs cycle enzyme in an equal number of control cells and cells overexpressing the human FXN gene (Figure 3). In a related experiment, an equal number of both cell types were grown in medium with ^{14}C-labelled glucose, and the researchers measured the amount of ^{14}C-labelled CO_2 produced by the cells (Figure 3). They also determined the concentration of lactic acid in both cell types (Figure 3).

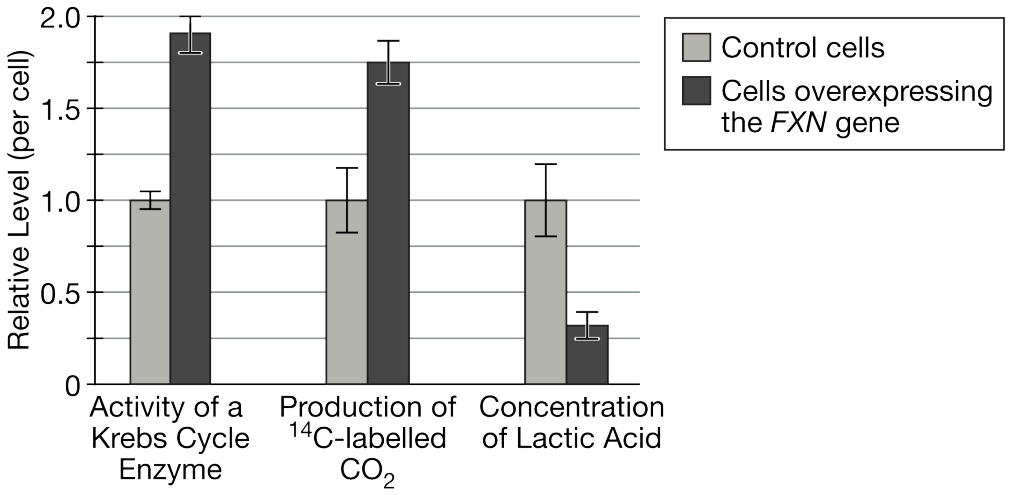


Figure 3. The activity of a Krebs cycle enzyme, the production of ^{14}C-labelled CO_2 after growth in a medium with ^{14}C-labelled glucose, and the concentration of lactic acid in control cells and in cells overexpressing the FXN gene. Values are shown relative to the values in the control cells. Error bars represent $\pm 2\text{SE}_{\bar{x}}$.

(a) **Describe** the possible genotypes of the heart cells of an individual if meiosis in this individual results in two sperm cells carrying the f allele and two sperm cells carrying the F allele.

(b) **Identify** a dependent variable for the experiment with data shown in Figure 2. **Justify** the measurement of ^{14}C-labelled CO_2 in the experiment with data shown in Figure 3. **Describe** the difference between the mouse cells overexpressing the FXN gene and the control mouse cells with respect to the ^{14}C-labelled CO_2 production shown in Figure 3. Based on the data in Figure 3, **calculate** how many times more lactic acid is produced on average in the control cells as compared with cells overexpressing FXN.

(c) Researchers produced mice that contained three copies of the FXN gene in each cell, producing genotype fff, rather than the typical two copies. Based on the data provided, **predict** the amount of frataxin protein most likely to be produced by the mice compared with the amount of frataxin in cells with genotypes A, B, and C. **Justify** your prediction.

(d) **Explain** why the phenotypes associated with the human FXN gene can be studied in a mouse cell. Unlike skeletal muscle cells, heart muscle cells cannot rely on fermentation for long. **Explain** the inability of heart muscle cells to function using only fermentation for metabolism. ATP acts as a negative feedback regulator of phosphofructokinase, one of the enzymes in glycolysis. **Predict** the eventual effect of ATP on pyruvate production in heart cells that express excess amounts of the normal frataxin.

Part A

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· All the heart cells are heterozygous/Ff.

· The only possible genotype is Ff.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- An identification that the dependent variable is the relative amount of frataxin protein
- A justification that the measurement of ^{14}C-labelled CO_2 allows researchers to analyze the rate at which the cells are performing respiration/the Krebs/citric acid cycle
- An acceptable description. Acceptable descriptions include the following.
  · The cells overexpressing the FXN gene produced about 1.75 times as much CO_2as the control cells.
  · Cells overexpressing FXN produced more (^{14}C -labelled) CO_2.
- The number of times more lactic acid produced on average in the control cells as compared with cells overexpressing FXN is calculated to be 3.33.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- An acceptable prediction. Acceptable predictions include the following.
  · The amount of frataxin protein should be about 30% of the amount produced by genotype A.
  · The amount of frataxin protein should be similar to/the same as the amount of frataxin in cells with genotype B.

FRQ

· The amount of frataxin protein should be slightly greater than the amount of frataxin in cells with genotype B but less than the amount of frataxin in cells with genotype C.

An acceptable justification. Acceptable justifications include the following.

· Because the three mutant alleles are expressed at a low level, the total amount of frataxin will be similar to (or slightly more than) the amount in a cell that is homozygous recessive.

· Two f alleles produce about 20%, so three f alleles could be expected to produce 30% of the amount produced by an individual with FF.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

An acceptable explanation. Acceptable explanations include the following.

· Metabolic pathways are conserved across all domains/mammals, so similar genes should produce similar phenotypes in different mammals.

· The genetic code is shared by mice and humans as are the mechanisms by which genes are expressed.

An acceptable explanation. Acceptable explanations include the following.

· Fermentation (and glycolysis) cannot generate sufficient ATP for the high energy demands of heart muscle cells.

· Fermentation cannot regenerate NAD^{+} fast enough to support glycolysis at a rate that would produce sufficient ATP.

An acceptable prediction. Acceptable predictions include the following.

· Pyruvate production will decrease.

· Pyruvate production will stay the same.

FRQ

9. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The SIRT3 protein has been reported to increase the growth of certain types of cancer cells. Researchers studying the metabolism and development of stomach cancer cells investigated the potential role of the SIRT3 protein in these cells. In the first investigation, the researchers quantified the relative amount of SIRT3 in normal stomach cells and in each of four different lines of stomach cancer cells (Figure 1). Using data from a group of 100 patients with stomach cancer, the researchers also analyzed the relative amount of SIRT3 protein detected in samples of the cancer cells and in normal cells immediately adjacent to the cancer cells (Figure 2).

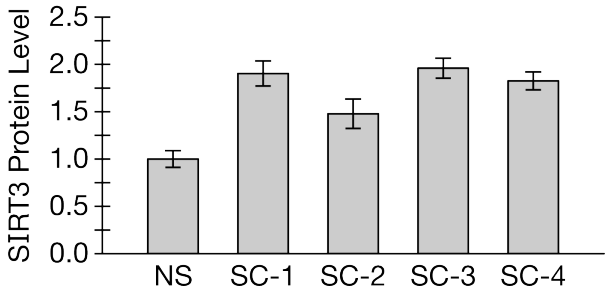


Figure 1. SIRT3 protein levels in normal stomach cells (NS) and stomach cancer cell lines from four different patients (SC-1, SC-2, SC-3, SC-4), shown relative to the level in normal stomach cells (NS). Error bars represent $\pm SE_{\bar{x}}$.

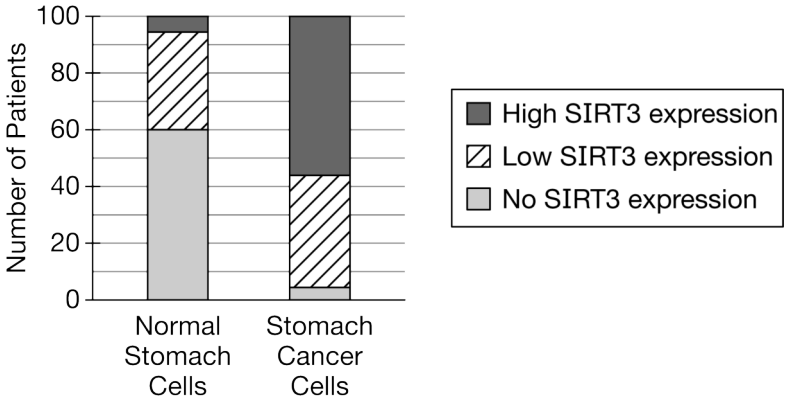
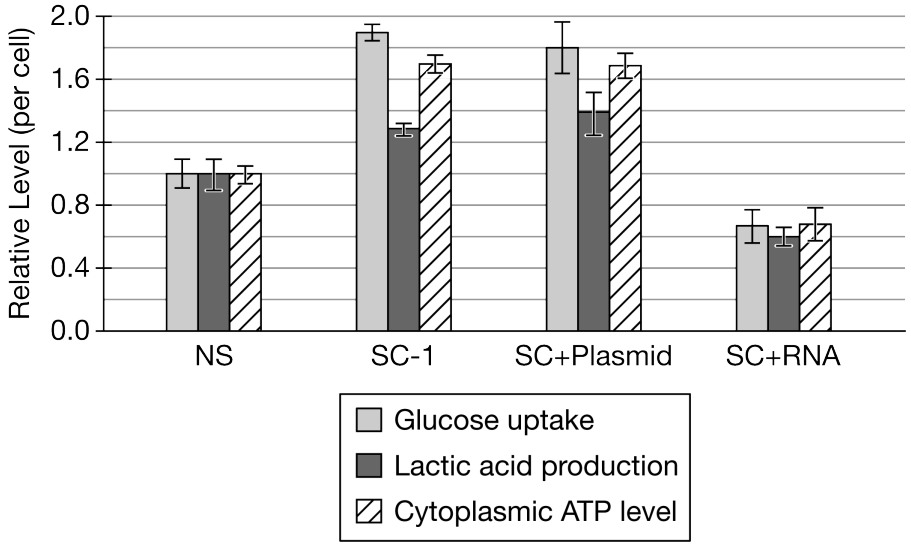


Figure 2. The number of patients, from 100 tested, whose stomach cancer cells or normal stomach cells adjacent to the cancer cells express high, low, or no SIRT3 protein

- In the second investigation, the researchers grew four groups of stomach cells in petri dishes containing nutrient broth:• NS, normal stomach cells• SC-1, a line of stomach cancer cells that the researchers had examined for SIRT3 expression• SC + RNA, the SC-1 line of stomach cancer cells to which a plasmid was added that encodes a small RNA complementary to a portion of the *SIRT3* mRNA. When these small RNAs bind to the complementary portion of *SIRT3* mRNA,

the *SIRT3* mRNA is broken down rather than translated.• SC + plasmid, the SC-1 line of stomach cancer cells to which a plasmid was added that does NOT encode the complementary small RNA

- After 24 hours of growth, the researchers measured several indicators of metabolism by an equal number of all four cell groups: glucose uptake, lactic acid production, and the level of cytoplasmic ATP production in the presence of a compound that blocks the electron transport chain (Figure 3). The values were calculated relative to the values obtained for normal stomach cells.



- Figure 3. Glucose uptake, lactic acid production, and cytoplasmic ATP levels in stomach cells. NS, normal stomach cells; SC-1, stomach cancer cells; SC+ plasmid, SC-1 cells with added plasmid only; SC + RNA, SC-1 cells with added plasmid encoding small RNA complementary to a portion of *SIRT3*. Error bars represent $\pm \mathrm{SE}_{\bar{x}}$.

- (a) **Explain** why an mRNA transcribed from the template strand of the *SIRT3* gene in normal stomach (NS) cells is translated to produce a SIRT3 protein of correct sequence, but if an mRNA were transcribed from the nontemplate strand of the *SIRT3* gene in these cells, it would not produce a SIRT3 protein of correct sequence.

(b) **Identify** a control group for the analysis shown in Figure 3. **Justify** analyzing SIRT3 protein level in four different cancer cell lines, as shown in Figure 1. Based on Figures 1 and 3, **describe** the relationship between SIRT3 expression and cytoplasmic ATP levels. **Calculate** the percent change in cytoplasmic ATP levels by SC + RNA cells compared with SC + plasmid cells.

(c) Researchers analyzing the *SIRT3* gene sequence engineer a *SIRT3* gene that is identical to the normal gene except that a single nucleotide has been deleted from the $5'$ end of the second exon of the gene; the deletion does not affect splicing. The researchers introduce this gene into normal stomach cells to determine how the primary AND tertiary structures of the protein expressed from the engineered *SIRT3* gene will compare with the protein expressed from the normal *SIRT3* gene in these cells. **Predict** the results of this experiment. The researchers claim that the data in Figure 3 provide evidence that the stomach cancer cells perform fermentation to a greater extent than do normal stomach cells. Provide reasoning to **justify** the claim. Based on the information provided, **predict** the effect on the ratio of $\mathrm{NADH}/\mathrm{NAD}^+$ in the cells treated with the compound that blocks the electron

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transport chain compared with similar cells not treated with the compound.

(d) Based on the data provided, **explain** how cellular processes must be altered in stomach cancer cells compared with normal stomach cells to result in the different levels of $\mathrm{SIRT3}$ expression observed. A mutation to the gene encoding a cyclin-dependent kinase (CDK) in human stomach cells results in a CDK that is continually active in the cells. **Explain** why the continually active CDK will most likely change the normal cells into cancer cells.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the nontemplate strand has a sequence that is complementary to the template strand, so transcription of this strand followed by translation will produce a protein of a different sequence from the normal $\mathrm{SIRT3}$ protein.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification. Acceptable identifications include the following.
  - · Normal stomach cells (NS)
  - · $\mathrm{SC+plasmid}$ cells
- ☐ A justification that analyzing several different cell lines that all show an increase in $SIRT3$ expression allows researchers to attribute the increases in $\mathrm{SIRT3}$ protein to cancer rather than to variations among individuals
- ☐ An acceptable description. Acceptable descriptions include the following.
  - · The greater the expression of $SIRT3$, the greater the cytoplasmic ATP level.
  - · There is a positive correlation.
- ☐ The percent change in cytoplasmic ATP levels by $\mathrm{SC+RNA}$ cells compared with $\mathrm{SC+plasmid}$ cells is calculated to be -59%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] An acceptable prediction. Acceptable predictions include the following.
  · The protein will be a different length if the nucleotide deletion changes the position of the stop codon. The polypeptide produced will fold differently/have a different 3D structure from the normal protein.
  · Every amino acid encoded following the nucleotide deletion is different from that encoded by the normal gene. The polypeptide produced will fold differently from the normal protein.
- [ ] A justification that the increase in lactic acid production in the cancer cells/ SC-1 and SC+plasmid means that there is an increase in fermentation by these cells
- [ ] A prediction that the ratio will increase/be comparatively greater in the treated cells

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] An acceptable explanation. Acceptable explanations include the following.
  · An increase in cell signaling that leads to increased rates of transcription/translation results in higher levels of SIRT3 in cancer cells.
  · Increased rates of transcription and translation result in higher levels of SIRT3 in cancer cells.
- [ ] An explanation that CDK will push the cell cycle forward/stimulate cell division by moving the cell through checkpoints regardless of the levels of cyclin, because the CDK is active

FRQ

10. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

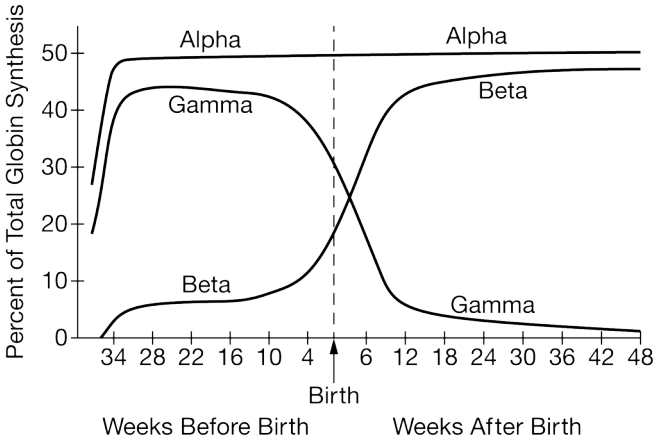

Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.

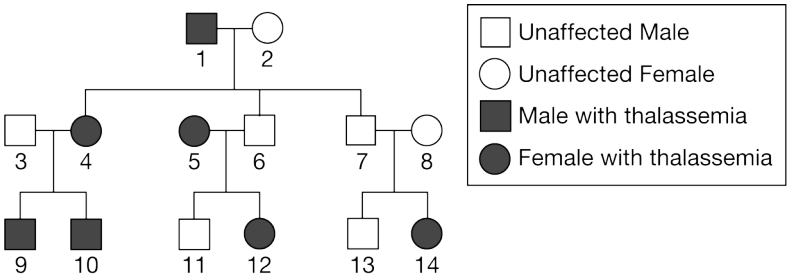

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Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

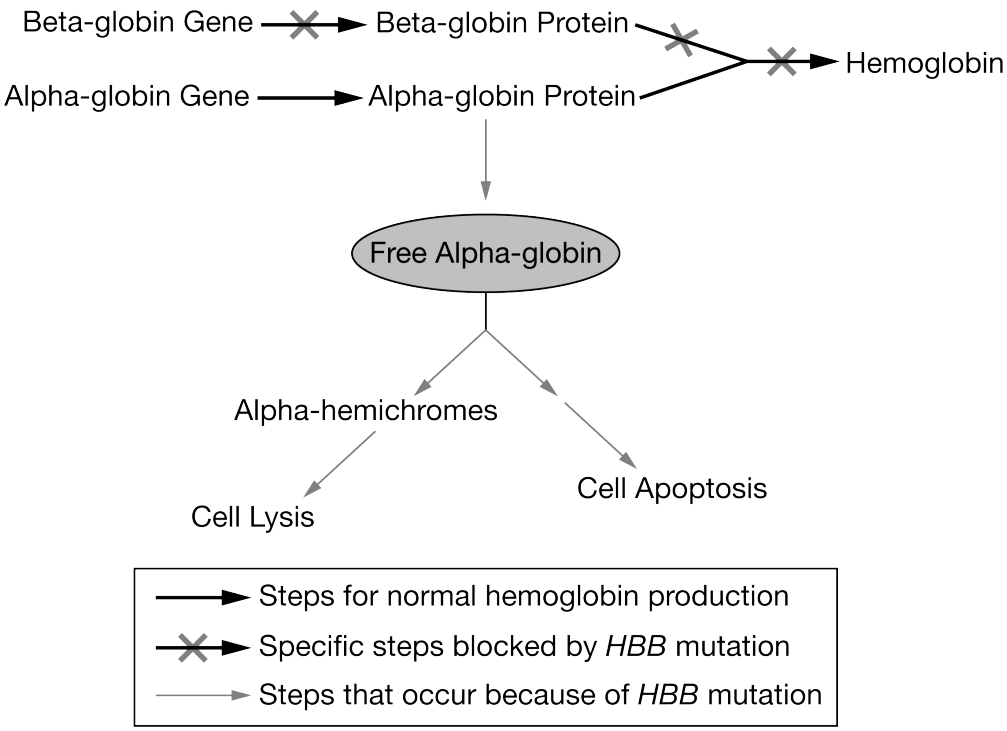

Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Based on the pedigree shown in Figure 2, **describe** the HBB alleles inherited by individual 6, including the parent from which each allele came.

(b) **Explain** why the symptoms associated with beta-thalassemia are often undetectable at time of birth. A study of the gene encoding the gamma-globin subunit showed that, shortly after birth, the gene becomes methylated (CH_3 groups are added to the DNA). **Explain** how DNA methylation could cause the change in the amount of gamma-globin protein that is represented in Figure 1.

(c) **Predict** the effect of a treatment that reduces the expression of $HBA1$ on the symptoms of individuals with beta-thalassemia. **Justify** your prediction.

(d) Researchers claim that the ability of fetal hemoglobin to bind oxygen more tightly than the hemoglobin of the mother increases the fitness of the fetus. Provide reasoning to **support** the claim.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 6 inherited a mutant allele from his father and a normal allele from his mother.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  · The symptoms are the result of nonfunctional beta-globin subunits, which are not part of hemoglobin until a few months after birth.
  · At birth, the baby would have enough gamma-globin subunits to carry oxygen, so the lack of beta globin would not be noticed.
- ☐ An explanation that methylation of DNA can lead to repression of gene expression, which could result in the decrease in the amount of gamma-globin present after birth

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction of the effect of a treatment. Acceptable predictions include the following.
  · Their symptoms would be less severe.
  · There would be less cell lysis/apoptosis.
- ☐ A justification that the individuals will produce less alpha-globin, so there will be less free alpha-globin to form hemichromes and result in the cells lysing or undergoing apoptosis

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the reasoning that the ability of fetal hemoglobin to bind oxygen more tightly allows the fetus to obtain the oxygen needed to perform cellular respiration/make the ATP/transform the energy, so they can survive.

FRQ

11. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

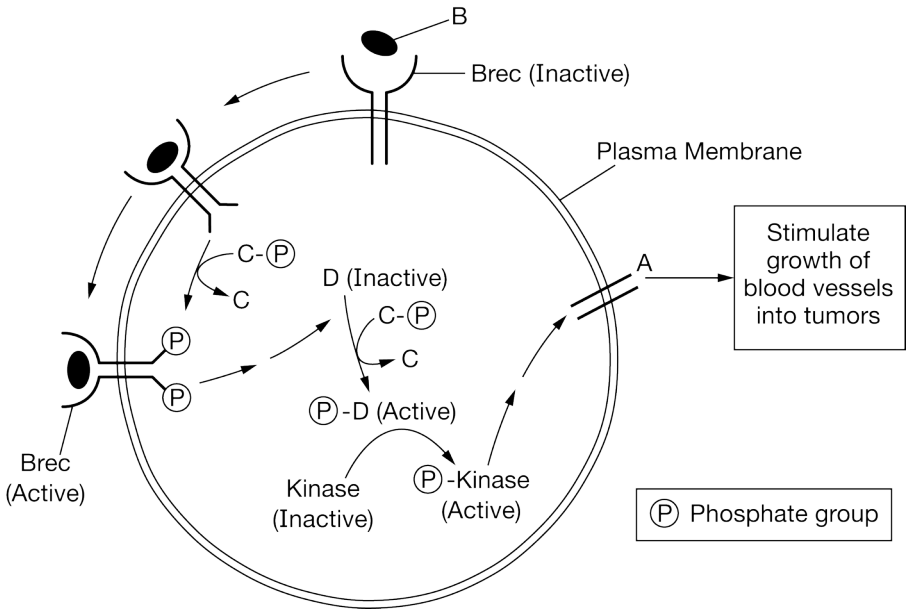

Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of kinase D.

(b) A certain group of Brec-MUT cells expresses low levels of protein A, unlike the high expression levels of most Brec-MUT cells. Researchers determine that the cells have an excess of ATP and that there are no mutations in the gene encoding protein A. **Describe** a change in the Brec signaling pathway that would lead to the decreased expression of protein A even in the presence of Brec-MUT. **Explain** why B is able to bind to Brec but not to A.

(c) Based on the proposed signaling pathway, **predict** the relative amount of phosphorylated to unphosphorylated

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mutant Brec in the Brec-MUT cells when the cells are grown in nutrient broth lacking B. **Justify** your prediction.

(d) The researchers claim that treatment with a compound that inhibits phosphorylation of kinase D will affect expression of A in the Brec-WT cells but not in the Brec-MUT cells. Based on the information provided, provide reasoning to **refute** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· Kinase D phosphorylates and activates downstream kinases in the pathway.

· Kinase D activates downstream kinases, stimulating the production of protein A.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable description. Acceptable descriptions include the following.
  · There is a mutation in the gene encoding any protein in the pathway between Brec and protein A.
  · Any protein in the pathway between Brec and protein A is inhibited.
- ☐ An explanation that B has a shape/charge that is complementary to the binding site of Brec but not to any domain of A

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

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The response includes both of the following criteria.

- [ ] A prediction that there will be more of the phosphorylated Brec protein than of the unphosphorylated
- [ ] A justification that the mutant Brec protein does not require B for activation/phosphorylation

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a refutation for the claim by stating that because the signaling pathway is blocked after activation of the receptor protein/Brec, there will be no difference in expression of A between the two cell types.

FRQ

12. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Human hair grows in a cycle that includes a growth phase and a rest phase. A small fraction of the hairs on the scalp are in the rest phase, while the majority of the hairs on the scalp are in the growth phase, which allows the hair on the scalp to grow long. Conversely, the majority of hairs in areas such as the eyelashes are in the rest phase, which keeps those hairs short.

Familial trichomegaly (FT) is a genetic condition that results in excessively long eyelashes in affected individuals. In an investigation into the causes of FT, a team of researchers identified mutations in the gene that codes for fibroblast growth factor-5 (FGF5) that are associated with FT. FGF5 is a protein that is secreted by certain cells and functions as a signaling molecule that influences the ratio of growth phase to rest phase in hair. The structure of the *FGF5* gene includes 3 exons and 2 introns. Alternate splicing of the pre-mRNA transcribed from the gene creates a long version of the mRNA, *FGF5-L*, and a short version of the mRNA, *FGF5-S* (Figure 1).

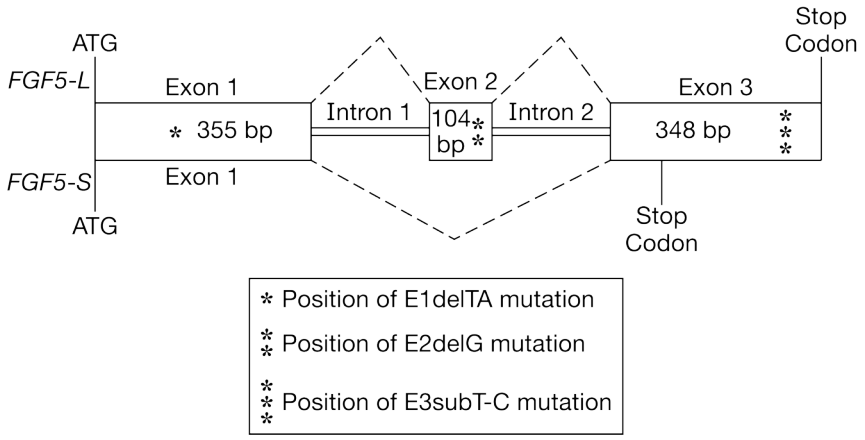

Figure 1. A map of the splicing patterns that lead to the production of both the FGF5-L protein and the FGF5-S protein. Labels below the diagram refer to gene components for *FGF5-S*, and labels above the diagram refer to gene components for *FGF5-L*. Dotted lines indicate the portion(s) of the sequence that will be removed from the transcribed pre-mRNA to form either *FGF5-L* or *FGF5-S* mRNA .

- Researchers identified three different mutations in the *FGF5* gene that have been associated with FT. These mutations, the positions of which are shown in Figure 1, include: • E1delTA: a mutation in which a thymine and an adenine are deleted from Exon 1 of the *FGF5* gene• E2delG: a mutation in which a guanine is deleted from Exon 2 of the *FGF5* gene• E3subT-C: a mutation in which a thymine is substituted for a cytosine in Exon 3 of the *FGF5* gene

The researchers collected data on a family with a history of FT and created a pedigree representing the instances of FT in the family as shown in Figure 2.

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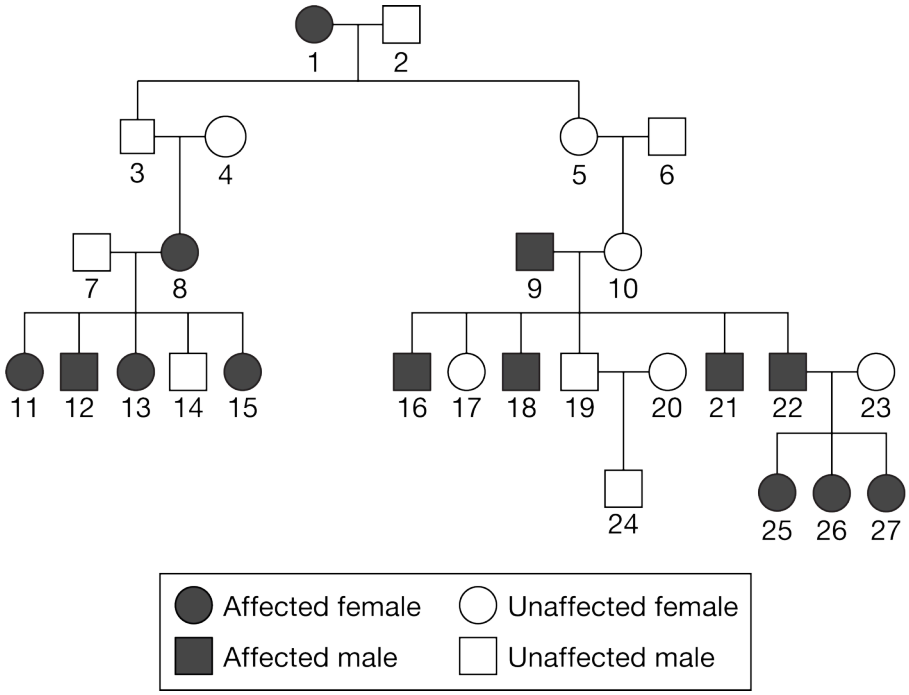

Figure 2. A pedigree representing five generations of a family with a history of FT

The researchers analyzed the length of hairs from individuals with FT and from unaffected individuals and plotted the length of each hair as shown in Figure 3A. They also determined the ratio of hairs in growth phase to hairs in rest phase as shown in Figure 3B.

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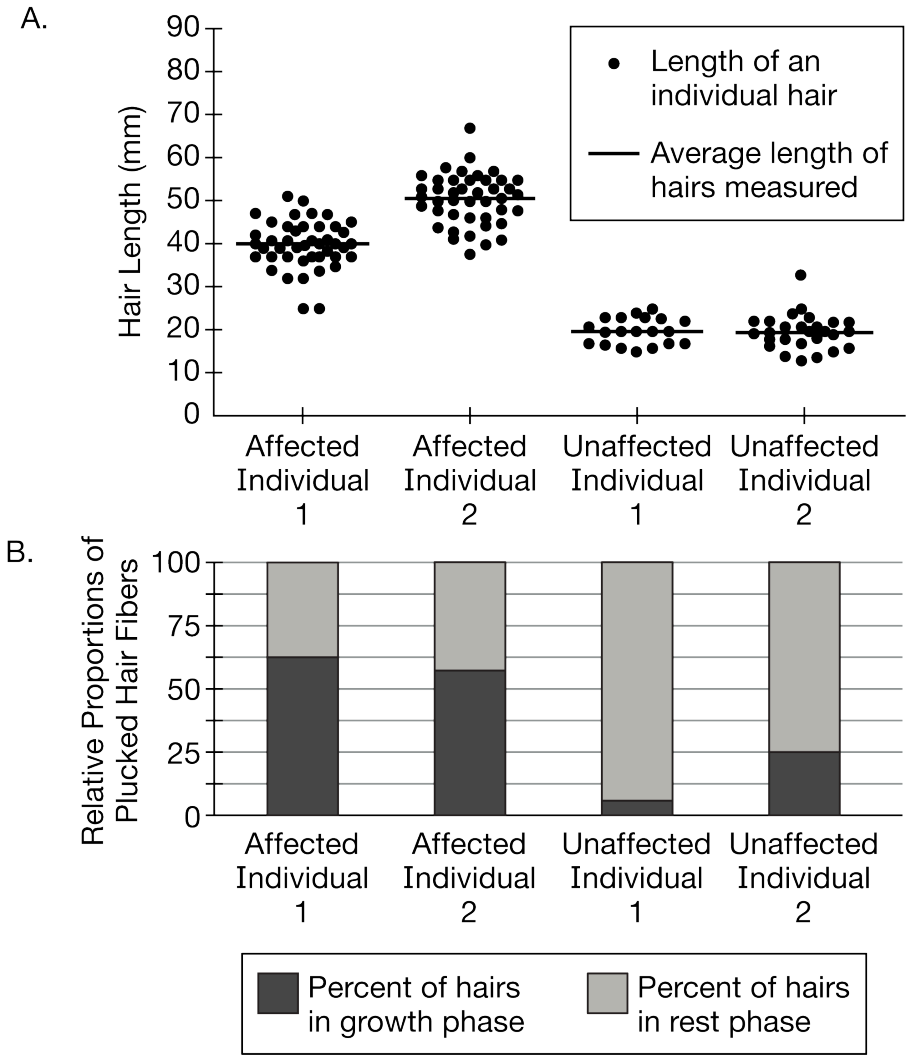

Figure 3A. Length of hairs taken from individuals with and without FT
Figure 3B. Proportion of hairs examined that were in growth phase compared with the proportion of examined hairs in rest phase

In a final experiment, the researchers grew human hair in a petri dish with and without adding FGF5 to the dish. The growth of the hair over an 8-day period was measured. The data from this experiment are shown in Figure 4.

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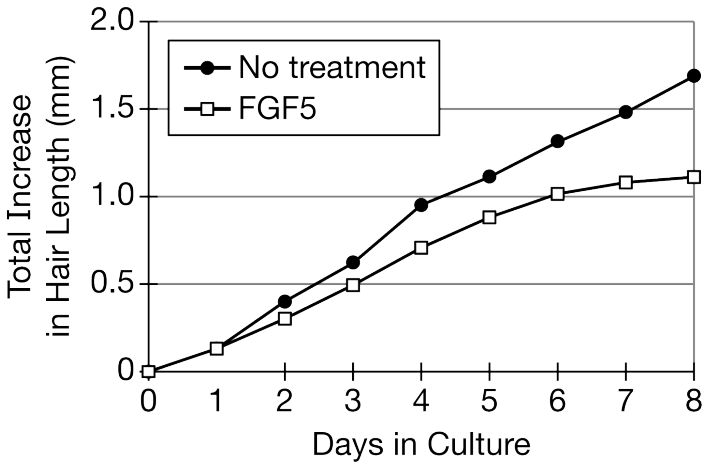

Figure 4. Growth of human hair in culture with and without FGF5

(a) **Describe** what happens to an *FGF5* mRNA to which a poly(A) tail has not been added.

(b) **Identify** a dependent variable in the experiment represented in Figure 4. **Justify** using the culture without FGF5 as the control in the experiment represented in Figure 4. Based on an analysis of the pedigree provided in Figure 2, **describe** the most likely mode of inheritance of FT in this family. Based on Figure 2, if individuals 3 and 4 had another child, **calculate** the probability that the child would have been an affected female.

(c) **Predict** the effect of the E3subT-C mutation on the function of FGF5-S. **Justify** your prediction. Researchers identify a new mutation that results in an FGF5 receptor that can bind the FGF5 protein but cannot stimulate production of second messengers in the cell. Based on the data provided in Figure 4, **predict** the effect of the new mutation on hair growth, regardless of hair type.

(d) Researchers introduced a single copy of the normal mouse *FGF5* allele into human cells homozygous for the *FGF5* mutation and found that, while the human cells could express the mouse *FGF5* gene, the mouse FGF5 protein was not able to restore normal FGF5 signaling in the human cells. **Explain** how the mouse FGF5 protein could be expressed but not activate the FGF5 pathway in the human cells. Bacterial cells can be used to produce large quantities of eukaryotic proteins. **Explain** why a functional FGF5 protein is not produced when bacteria are transformed with the full-length eukaryotic *FGF5* gene, even when the eukaryotic promoter is replaced with a bacterial promoter.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· The mRNA is degraded more rapidly (in the cytoplasm).

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· The mRNA will be unable to move from the nucleus to the cytoplasm.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of a dependent variable. Acceptable identifications include the following.
  · Hair growth (in the culture system)
  · Increase in hair length
- ☐ An acceptable justification. Acceptable justifications include the following.
  · Comparing the hair grown with $\mathrm{FGF5}$ to the hair without $\mathrm{FGF5}$ verifies whether $\mathrm{FGF5}$ has an effect on hair growth.
  · Growing hair in culture without $\mathrm{FGF5}$ verifies that hair will grow in the culture medium.
- ☐ A description that FT is most likely autosomal recessive
- ☐ The probability that the child would have been an affected female is calculated to be 0.125 OR $1/8$ OR 12.5% .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ A prediction that $\mathrm{FGF5\text{-}S}$ will be unaffected
- ☐ A justification that the mutation occurs after the stop codon in the $FGF5-S$ transcript and therefore does not affect the translation of the protein
- ☐ A prediction that hair growth should increase

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	✓ 2

The response includes both of the following criteria.

- [ ] An explanation that the genetic code is conserved in all species, so human cells can transcribe and translate the mouse allele, but the shape/charge of the mouse protein must be different enough from the human $\mathrm{FGF5}$ such that it is unable to bind to and/or activate the receptor.
- [ ] An explanation that the full length eukaryotic $\mathrm{FGF5}$ gene contains introns, which are not removed in bacterial cells, therefore these introns will be translated, and the resulting protein will be different from the eukaryotic protein

FRQ

13. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Crossing-over between homologous chromosomes during meiosis is necessary for the proper segregation of the chromosomes. In the fly species *Drosophila melanogaster*, two conserved genes, $mei\text{-}9$ and $pch2$, are thought to play a role in crossing-over. The protein encoded by $mei\text{-}9$ is part of a large protein complex required for the pairing of homologous chromosomes, while the protein encoded by $pch2$ is associated with a delay in prophase I that is believed to serve as a cell cycle checkpoint.

To investigate the role of $mei\text{-}9$ and $pch2$ in crossing-over, researchers determined the frequency of recombination for each of four regions of chromosome 2 in wild-type flies (WT), flies expressing a mutant form of $mei\text{-}9$ ($\mathrm{mei\text{-}9\text{-}mut}$ flies), and flies expressing mutant forms of both $mei\text{-}9$ and $pch2$ ($\mathrm{mei\text{-}9\text{-}mut/pch2\text{-}mut}$ flies). The frequency of recombination of each region in both types of mutant flies is shown in Figure 1 as a percentage of the frequency of recombination observed in WT flies.

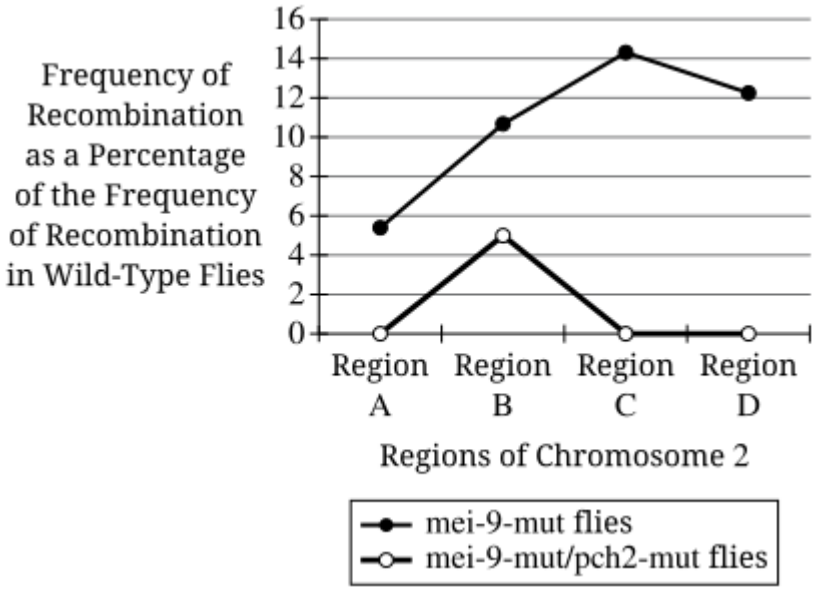

Figure 1. Frequency of recombination of four regions of chromosome 2 in $\mathrm{mei\text{-}9\text{-}mut}$ flies and $\mathrm{mei\text{-}9\text{-}mut/pch2\text{-}mut}$ flies as a percentage of the frequency of recombination in the same region in wild-type flies

(a) Based on the data in Figure 1, **identify** the region of chromosome 2 with the greatest frequency of recombination in $\mathrm{mei\text{-}9\text{-}mut/pch2\text{-}mut}$ flies.

(b) Based on the data in Figure 1, **describe** the frequency of recombination in $\mathrm{mei\text{-}9\text{-}mut}$ flies compared with the frequency of recombination in WT flies.

(c) Scientists hypothesize that the function of the protein encoded by wild-type $pch2$ allows some of the defects caused by the $mei\text{-}9$ mutation to be overcome, increasing the likelihood that crossing-over will occur. Use the

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data in Figure 1 to **evaluate** their hypothesis.

(d) Based on the information provided, **explain** why the fertility of mei-9-mut flies might be reduced compared with that of WT flies.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an identification that region of chromosome 2 with the greatest frequency of recombination in mei-9-mut/pch2-mut flies is Region B.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- The frequency of recombination (in each region) is lower (in mei-9-mut flies than it is in WT flies).
- The frequency of recombination for mei-9-mut does not go above 15% of the WT.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides an acceptable evaluation of the scientists' hypothesis. Acceptable evaluations include the following.

- The scientists' hypothesis is supported because the frequency of recombination is higher in mei-9-mut flies (where the protein encoded by *pch2* is functional than it is in the mei-9-mut/pch2-mut flies where it is not).
- The scientists' hypothesis is supported because the frequency of recombination is lower in the mei-9-mut/pch2-mut flies (where the protein encoded by *pch2* is nonfunctional than it is in mei-9-mut

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flies where it is functional).

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- The reduced incidence of crossing-over/frequency of recombination observed in *mei-9-mut* flies could lead to (an increase in) nondisjunction/the formation of gametes with the incorrect numbers of chromosomes.
- Gametes with incompatible chromosome numbers are less likely to result in viable/fertile offspring.

FRQ

14. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Tuberculosis is an infectious disease caused by the bacterium *Mycobacterium tuberculosis.* Many of the mycobacteria are resistant to the drugs typically used for treatment. As a part of the process of developing new drugs, scientists identified two mycobacterial proteins, L10 and L12, that are part of the ribosome and that must bind to each other to allow ribosomal function. The scientists aimed to identify a compound that blocks the interaction between the two proteins, thereby preventing the survival and reproduction of *Mycobacterium tuberculosis*.

As a first step, the scientists performed an *in vitro* (in a test tube) experiment to test the ability of different compounds to block the binding between the L10 and L12 proteins. The scientists measured the amount of binding in the presence of increasing concentrations of three compounds: I, II, and III. Results are shown in Figure 1.

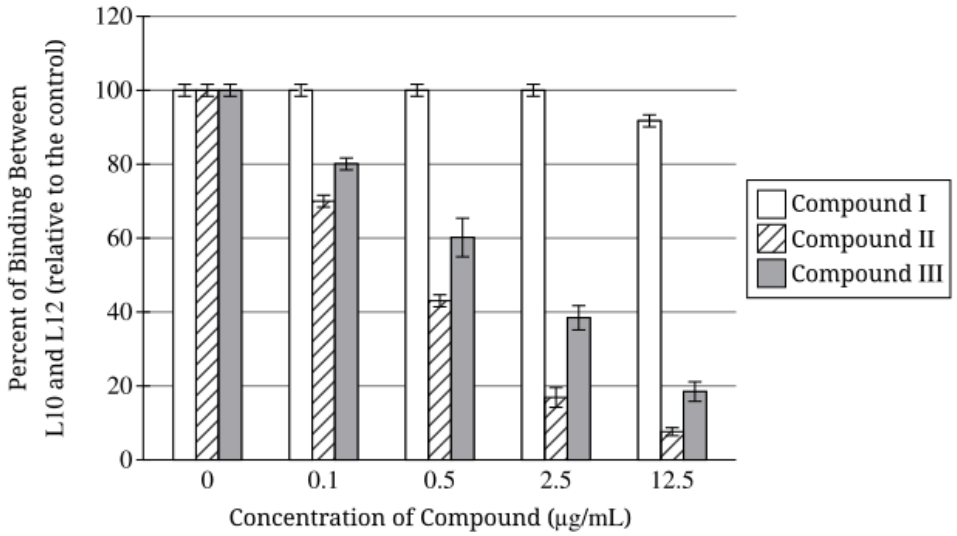


Figure 1. Average percent of binding between L10 and L12 in the presence of increasing concentrations of compounds I, II, and III, relative to the amount of binding in the absence of any compound ($0\ \mu g/mL$, the control). Error bars represent $\pm 2SE_{\bar{x}}$.

In a second experiment, the scientists performed an *in vitro* experiment to test the amount of inhibition of translation in the presence of compounds I, II, and III. They added the three compounds to test tubes containing molecules needed for translation, including a specific mRNA, and ribosomes from one of three organisms: *Mycobacterium tuberculosis*, the bacterium *Escherichia coli*, or the eukaryote *Oryctolagus cuniculus* (rabbit). They measured the concentration of each compound required to reduce translation by 50% (Table 1).

TABLE 1. CONCENTRATION ($\mu g/mL$) REQUIRED TO REDUCE TRANSLATION BY 50% FOR COMPOUNDS I, II, AND III

FRQ

Compound	*M. tuberculosis* ribosomes	*E. coli* ribosomes	*O. cuniculus* ribosomes
I	162.7	143.1	30.2
II	16.9	117.7	118.1
III	25.9	34.2	60.1

(a) Based on the data in Figure 1, **identify** the concentration in $\mu g/mL$ of compound III that reduces the amount of binding between L10 and L12 to 60% of the control.

(b) Based on the data in Figure 1, **describe** the specific decrease in L10 and L12 binding in relation to each specific increase in compound III concentration from $0.1\,\mu g/mL$ to $12.5\,\mu g/mL$.

(c) Based on the data in Table 1, the scientists predict that compound II is the best of the three compounds to treat tuberculosis infections. **Evaluate** the data to support or refute their prediction.

(d) One of the compounds becomes routinely used as a drug to treat tuberculosis. Based on the experiments, **explain** how a mutation in *Mycobacterium tuberculosis* would most likely cause resistance to the drug.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an identification that the concentration in $\mu g/mL$ of compound III that reduces the amount of binding between L10 and L12 to 60% of the control is 0.5.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that each five-fold (five times) increase in concentration of compound III should decrease binding by an additional 20% (relative to the control).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an acceptable evaluation of the data to support or refute the scientists' prediction. Acceptable evaluations include the following.

· The prediction is supported. Almost 7 times more/a much greater amount of/a much higher concentration of compound II is required to block translation by 50% in rabbits than to block translation in *M. tuberculosis*, and rabbits are a mammal model for humans.

· The prediction is supported. Compound II works best against *M. tuberculosis* and is least effective against eukaryotes (the rabbit), and therefore is unlikely to be toxic to humans.

· The prediction is supported. Compound II works best against *M. tuberculosis* and is much less effective against *E. coli*, and therefore is unlikely to be toxic to those *E. coli* that are beneficial to humans.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that a mutation to the DNA encoding ribosomal proteins/ L10 / L12 can change the structure of the proteins/ L10 / L12 in such a way that the compound can no longer block the binding between the two proteins (so the proteins still interact and allow the ribosomes to function).

FRQ

15. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Several studies indicate that a particular forest biome in the Northern Hemisphere will become significantly warmer over the next century. Researchers studying a population of trees in the forest hypothesize that warming will negatively impact the evolutionary fitness of many of the trees. The researchers also hypothesize that warming will shift the range of the population further north to cooler habitats.

(a) **Describe** the way the evolutionary fitness of an individual tree is measured.

(b) The range shift of the tree population may result in the fragmentation of the population into several smaller, isolated populations. **Explain** how the fragmentation may result in speciation.

(c) Recent research suggests that the range in which trees can grow is due in large part to their symbiotic relationships with soil microbes that are sensitive to temperature. **Predict** the effects on a population of microbes if the average temperature in an area increases above the temperature at which most of the microbes are able to successfully reproduce.

(d) **Justify** your prediction in part (c) using evidence from biological principles.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description of the way the evolutionary fitness of an individual tree is measured. Acceptable descriptions include the following.

· It is measured by counting the number of offspring/seedlings/alleles in the next generation that came from this tree.

· It is measured by determining the ability of a tree to produce offspring.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the explanation that the separate populations will be reproductively isolated/have no gene flow

FRQ

between them, and genetic differences between the populations may accumulate over time due to natural selection/genetic drift/mutations.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable prediction. Acceptable predictions include the following.

· Option 1: Future generations of the microbes would be composed of microbes able to reproduce at higher temperatures (shifting the gene frequency toward those alleles that enable reproduction at higher temperatures).

· Option 2: The population would go extinct.

· Option 3: The population would decrease in size.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable justification for the prediction provided in part (c). Acceptable justifications include the following.

· Justification for option 1: The alleles that allow the microbes to survive would have been passed down to their offspring, and the alleles would become more prevalent in the population because of natural selection.

· Justification for option 1: Microbes with different heat tolerances have different degrees of reproductive success.

· Justification for option 1: More-heat-tolerant individuals reproduce more than do less-heat-tolerant individuals.

· Justification for option 2: The (effective) population size will be too small/will have insufficient genetic variation to allow continued survival of the population.

· Justification for option 3: Most of the microbes cannot reproduce (therefore there will be fewer offspring).

FRQ

16. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

During the Late Devonian Period, there was a massive loss of marine biodiversity. An analysis of fossil marine invertebrates that existed during the Late Devonian has revealed that the rate of extinction was relatively constant, as compared with the rates of extinction before and after the Late Devonian. The fossil evidence also suggests that frequent increases in sea level during the Late Devonian effectively removed geographical barriers and allowed generalist species (species that can inhabit a wide range of habitats) to migrate into new areas.

(a) **Describe** how the geographical separation of two populations can lead to speciation.

(b) **Explain** how a decrease in biodiversity during the Late Devonian could occur even though the extinction rate was relatively constant.

(c) Scientists claim that the modern rate of biodiversity loss may be even greater than that observed during the Late Devonian. In recent decades, the number of invasive species has increased. **Predict** the most likely effect of the current increase in invasive species on speciation rates of native species.

(d) Provide reasoning to **justify** your prediction in part (c).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- The populations (on either side of the geographical barrier) are no longer able to exchange genetic material.
- There is no gene flow between the populations (on either side of the geographical barrier).
- The populations (on either side of the geographical barrier) are no longer able to reproduce with each other.
- The populations on either side of the geographical barrier have different selective pressures/have different mutations/experience genetic drift in different directions.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- The speciation rate could have been lower than the extinction rate.
- A decrease in the speciation rate could reduce biodiversity (even if extinction rates do not change).
- Genetic biodiversity could be reduced (if species undergo selection), without an increase in extinction rate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the prediction that there will be/has been a decrease (in speciation rates, compared with previous rates).

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable justification. Acceptable justifications include the following.

- Invasive species can outcompete native species/specialists for resources, leading to the extinction of many native species/limiting the ability of native species to spread to new habitats.
- An invasive species can prey on and drive (native prey) species to extinction (which would prevent these species from diverging).

FRQ

17. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

A population of invasive feral hogs in a wilderness area is approaching carrying capacity. The feral hogs in each age-group of this population are counted once a year. Figure 1 shows the age pyramid of this population in one particular year. Feral hogs reproduce between 1 and 5 years of age. Mothers care for their young until the young are three months old.

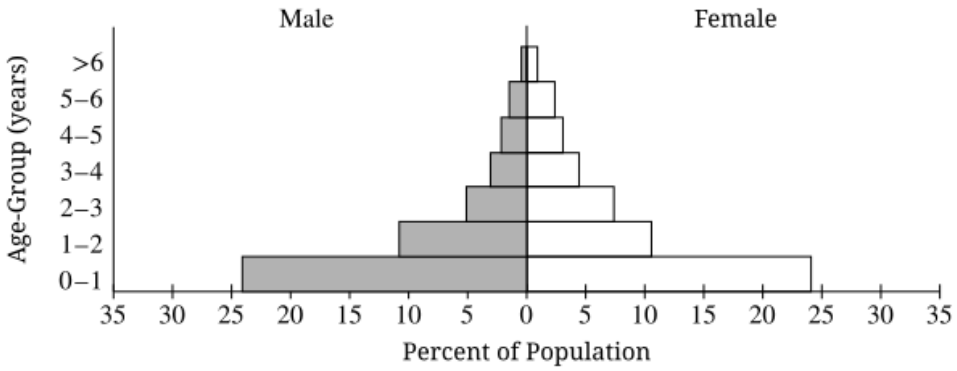


Figure 1. Age pyramid of a feral hog population in a wilderness area

Ten years later a researcher again counts the number of individual feral hogs in each age-group in the same population and produces the age pyramid shown in Figure 2.

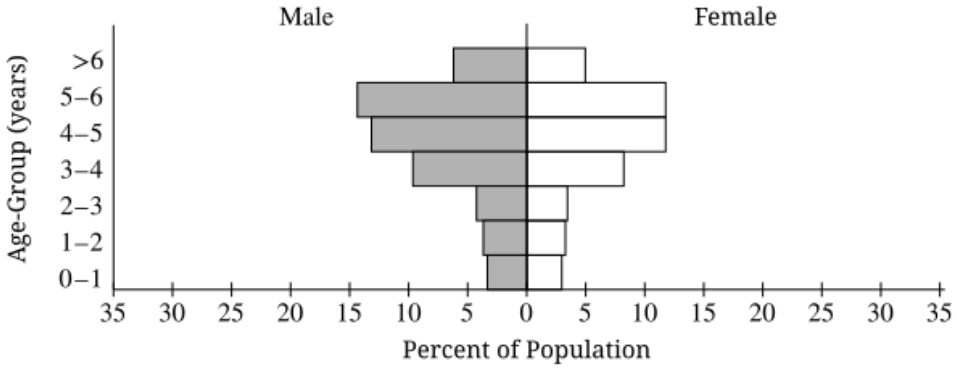


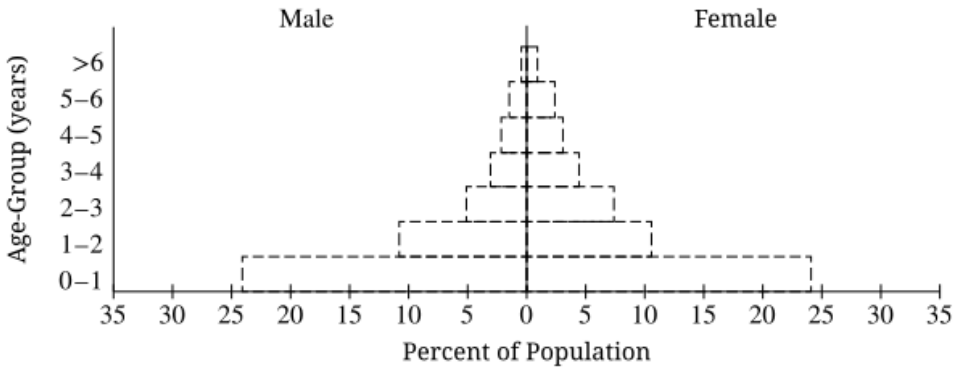
Figure 2. Age pyramid of the feral hog population ten years later

(a) **Describe** a characteristic of an ecosystem that allows an introduced species to become an invasive species.

FRQ

(b) A population age structure, as shown in Figure 1, is the result of multiple factors affecting birth and death rates of animals of various ages. **Explain** how ecological factors lead to the difference between the numbers of hogs in the $0-1$ year age-group and the number of hogs in the $1-2$ year age-group shown in Figure 1.

(c) An outbreak of a viral infection causes approximately half of the male hogs in each age-group to die. Using the template in the space provided for your response, **draw** new bars for each age-group to represent the approximate distribution of the entire hog population immediately after the death of the males. The dashed lines of the template represent the original age pyramid (Figure 1) before the viral outbreak.



(d) Using biological principles, **explain** what is likely to have caused the shift from the age structure in Figure 1 to the age structure shown in Figure 2.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· There is an available niche.

· There are sufficient resources.

· There are few/no competitors.

· There are few/no predators.

· There is a lack of parasites.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an explanation that the 0-1 year age-group is more susceptible to predation/diseases/competition for food/abiotic fluctuations, and this is likely to contribute to higher death rates of individuals in this age group.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a representation showing a decrease (approx 50%) in males on the left side and an increase in females on the right side.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explantion. Acceptable explanations include the following.

· The population exceeded the carrying capacity of the environment, and younger hogs were unable to compete for resources.

· The carrying capacity of the environment was reduced, and younger hogs became unable to compete for resources.

· There was a decrease in the reproductive rate, so there are fewer young hogs.

· There is a (new) predator/disease that causes lower survival rates among young hogs (compared with older hogs).

FRQ

18. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Vernal ponds, ponds that fill with water each spring and then evaporate in late summer, pose particular challenges to the organisms that live in them. Copepods, a type of crustacean, produce eggs that survive in dry sediment (the soil and other solid material at the bottom of a pond) for long periods of time. Once the water returns, the copepod larvae emerge from the eggs and begin to eat, consuming primarily algae. However, algae often emerge from the sediment later than the copepod larvae do, and the algal populations grow slowly in cool spring temperatures.

To investigate whether the copepods that live in vernal ponds have other adaptations to allow them to supplement their diet of algae, scientists examined related species that they categorized as large-bodied or small-bodied copepods. After examining the mouthparts of each species, the scientists determined a mouthpart carnivory score for each. A high score indicates a tendency toward a more carnivorous diet. The scientists also recorded the gut contents of each species in both early spring and summer. They hypothesized that large-bodied copepods have mouth adaptations that enable them to eat small animals that emerge early from sediments, while the small-bodied copepods do not have such adaptations.

(a) **Describe** the effect of periodic changes in resource availability on population density.

(b) **Justify** the researchers' performing their measurements and analyzing gut contents in both early spring and summer.

(c) **State** a null hypothesis for this investigation.

(d) Climate change is causing water in vernal ponds to evaporate earlier in the summer. Scientists claim that under these conditions, copepod species with a lower mouthpart carnivory score are more likely to go extinct in these vernal ponds than are species with a greater mouthpart carnivory score. Provide reasoning to **support** the scientists' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- The population density will fluctuate.
- The population will fluctuate around the carrying capacity.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response justifies that the researchers' performing their measurements in both early spring and summer allowed the scientists to see whether/how the diets of the copepods changed (with changing conditions).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response states an acceptable null hypothesis. Acceptable null hypotheses include the following.

- There is no (statistical) difference in diet among copepods of all sizes.
- There is no difference in (mouthpart) adaptations between large-bodied and small-bodied copepods.
- The (mouth) adaptations of a copepod do not correlate with the copepod's diet.
- The mouthpart carnivory score of a copepod does not correlate with the copepod's diet.
- The time of year does not correlate with diet.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the support that (if the ponds dry up) the algae will die, so the copepods (that have a lower mouthpart carnivory score) will die before they have a chance to lay eggs.

2023

AP® Biology

Scoring Guidelines

Question 1: Interpreting and Evaluating Experimental Results with Experimental Design — 9 points

In eukaryotic microorganisms, the PHO signaling pathway regulates the expression of certain genes. These genes, *Pho* target genes, encode proteins involved in regulating phosphate homeostasis. When the level of extracellular inorganic phosphate (Pi) is high, a transcriptional activator Pho4 is phosphorylated by a complex of two proteins, Pho80–Pho85. As a result, the *Pho* target genes are not expressed. When the level of extracellular Pi is low, the activity of the Pho80–Pho85 complex is inhibited by another protein, Pho81, enabling Pho4 to induce the expression of these target genes. A simplified model of this pathway is shown in Figure 1.

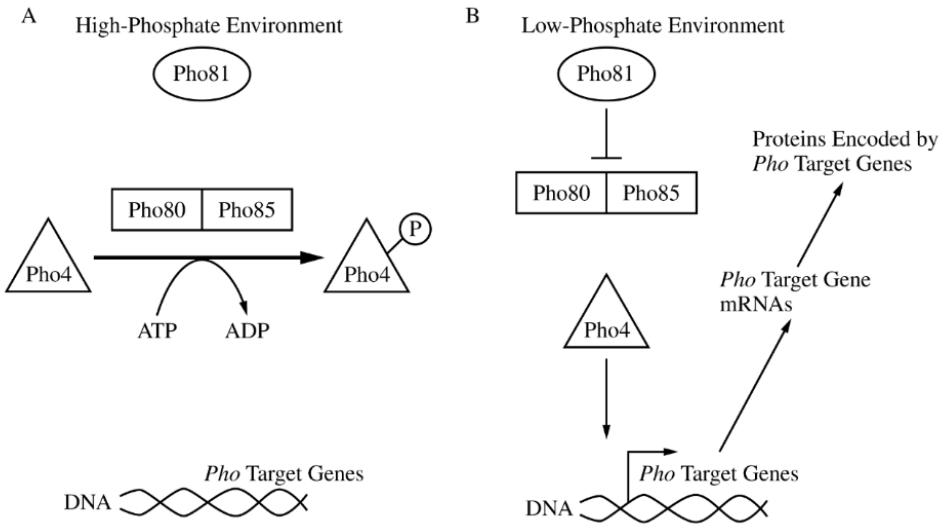


Figure 1. A simplified model of the regulation of expression of *Pho* target genes in (A) a high-phosphate (high-Pi) environment and (B) a low-phosphate (low-Pi) environment

To study the role of the different proteins in the PHO pathway, researchers used a wild-type strain of yeast to create a strain with a mutant form of Pho81 (*pho81mt*) and a strain with a mutated form of Pho4 (*pho4mt*). In each of these mutant strains, researchers measured the activity of a particular enzyme, APase, which removes phosphates from its substrates and is encoded by *PHO1*, a *Pho* target gene (Table 1). They then determined the level of *PHO1* mRNA relative to that of the wild-type yeast strain, which was set to 10.

TABLE 1. APase ACTIVITY AND RELATIVE AMOUNTS OF *PHO1* mRNA IN WILD-TYPE AND MUTANT STRAINS OF YEAST IN HIGH- AND LOW-PHOSPHATE ENVIRONMENTS

Yeast Strain	Mutation	APase Activity in High-P_i Environment (mU/mL/OD_{600}) $\pm 2SE_{\bar{x}}$	APase Activity in Low-P_i Environment (mU/mL/OD_{600}) $\pm 2SE_{\bar{x}}$	Relative Amounts of *PHO1* mRNA in High-P_i Environment $\pm 2SE_{\bar{x}}$	Relative Amounts of *PHO1* mRNA in Low-P_i Environment $\pm 2SE_{\bar{x}}$
Wild-type	None	0.5 ± 0.1	17.3 ± 0.9	0.1 ± 0.0	10 ± 2.0
pho81mt	Nonfunctional Pho81	0.4 ± 0.1	0.6 ± 0.1	0.7 ± 0.2	0.9 ± 0.8
pho4mt	Nonfunctional Pho4	0.5 ± 0.0	0.8 ± 0.2	0.6 ± 0.4	0.3 ± 0.1

(a) **Describe** the effect the addition of a charged phosphate group can have on a protein that would cause the protein to become inactive. **1 point**

- It changes the structure/shape of the protein.

Explain how a signal can be amplified during signal transduction in a pathway such as the PHO signaling pathway. **1 point**

- Each enzyme (in a signal transduction pathway) can act on many copies of a protein.

Total for part (a) **2 points**

(b) Based on Table 1, **identify** a dependent variable in the researchers' experiment. **1 point**

Accept one of the following:

- APase activity
- (Relative) amount of *PHO1* (mRNA)

Justify the researchers' using the wild-type strain for the creation of the mutant strains. **1 point**

Accept one of the following:

- It ensures that any observed differences (in experimental results) between the strains are due to the introduced mutations (and not to other genetic differences between the yeast strains).
- It ensures that the strains are genetically identical except for the introduced mutations.

Justify the researchers' using mutant strains in which only a single component of the pathway was mutated in each strain. **1 point**

Accept one of the following:

- It allows them to test the effect of each mutation separately.
- It allows them to (better) determine which component is responsible for any observed differences.

Total for part (b) **3 points**

(c)	Based on the data in Table 1, **identify** the yeast strain and growth conditions that lead to the highest relative amount of *PHO1* mRNA. • Wild-type yeast in a low-Pi environment	**1 point**
	Calculate the percent change in APase activity in wild-type yeast cells in a high-Pi environment compared with that of wild-type cells in a low-Pi environment. Accept one of the following: • 3,360% [(17.3–0.5)/0.5 × 100%] • -97% [(0.5–17.3)/17.3 × 100%]	**1 point**
	Total for part (c)	**2 points**
(d)	In a follow-up experiment, researchers created a strain of yeast with a mutation that resulted in a nonfunctional Pho85 protein. Based on Figure 1, **predict** the effects of this mutation on *PHO1* expression in the mutant strain in a high-Pi environment. • It/*PHO1*/Target genes will be expressed.	**1 point**
	Provide reasoning to **justify** your prediction. • (In a high-Pi environment) a nonfunctional Pho85 will be unable to phosphorylate/inhibit Pho4.	**1 point**
	Total for part (d)	**2 points**
	Total for question 1	**9 points**

Question 2: Interpreting and Evaluating Experimental Results with Graphing — 9 points

Elevated levels of CO_2 increase the rate of photosynthesis and growth in plants. Scientists studying the mechanisms involved in these increases examined a variety of species and found that when plants are exposed to elevated levels of CO_2, there is an increase in the number of chloroplasts per cell. To investigate whether the elevated levels of CO_2 have a similar effect on the number of mitochondria in plant cells, the scientists then selected six of these species to quantify the number of mitochondria per cell when the plants were exposed to both normal and elevated levels of CO_2 (Table 1).

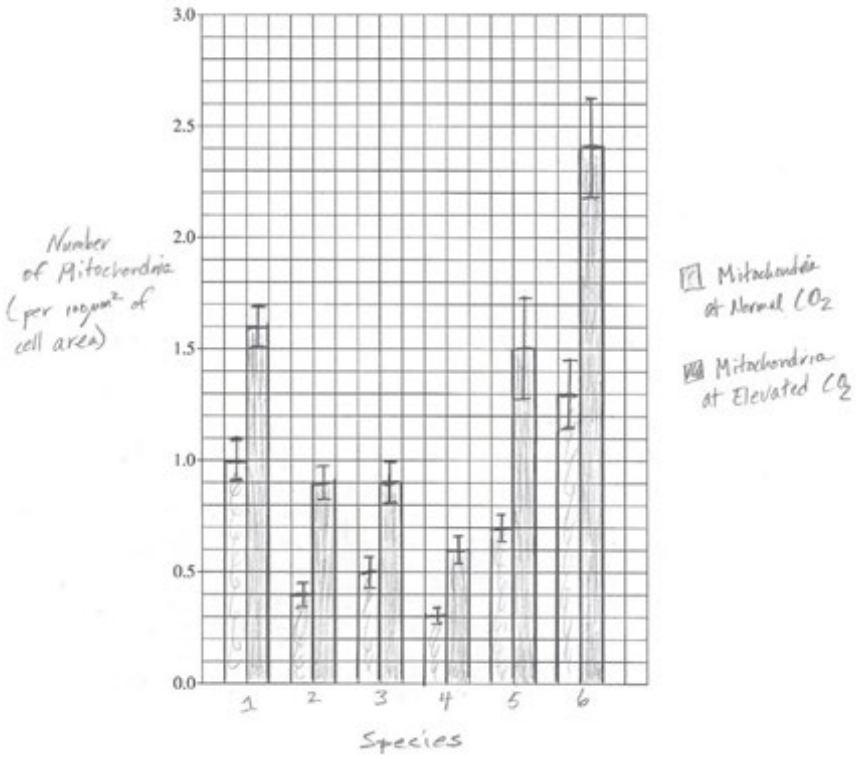
TABLE 1. AVERAGE NUMBER OF MITOCHONDRIA IN PLANTS EXPOSED TO NORMAL AND ELEVATED LEVELS OF CO_2

Species	Mitochondria at Normal CO_2 (per 100 µm^2 of cell area) $\pm 2SE_{\bar{x}}$	Mitochondria at Elevated CO_2 (per 100 µm^2 of cell area) $\pm 2SE_{\bar{x}}$
1	1.0 ± 0.10	1.6 ± 0.10
2	0.4 ± 0.05	0.9 ± 0.08
3	0.5 ± 0.07	0.9 ± 0.10
4	0.3 ± 0.03	0.6 ± 0.06
5	0.7 ± 0.06	1.5 ± 0.22
6	1.3 ± 0.15	2.4 ± 0.22

(a) **Describe** the role of the inner mitochondrial membrane in cellular respiration. **1 point**

Accept one of the following:

- It provides the location for the components of the electron transport chain/ATP synthase/oxidative phosphorylation.
- It separates (reactions in) the intermembrane space from (reactions in) the matrix.
- It allows the establishment of a proton gradient.

(b)	Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. Sample response:  • Appropriate labelling	**1 point**
	Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. • Data are represented in a bar/modified bar graph.	**1 point**
	Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. • Data points and error bars are correctly plotted.	**1 point**
	Determine which species show(s) a difference in the number of mitochondria between normal and elevated levels of CO_2. • All of the species	**1 point**
		Total for part (b) 4 points
(c)	Based on the data in Table 1, **describe** the relationship between the level of CO_2 and the average number of mitochondria per unit area of a cell. Accept one of the following: • The number of mitochondria is greater under conditions of elevated CO_2 (than under normal CO_2). • It is a positive relationship/correlation.	**1 point**

(d)	The leaves of a particular plant species are typically green, but scientists notice a plant in which the leaves have white stripes. They determine that the stripes result from a mutation in mitochondrial DNA that interferes with the development of chloroplasts. The scientists crossed plants using pollen from the plant with white-striped leaves and ovules from a plant with green leaves. **Predict** the phenotype(s) of the leaves of offspring produced from this cross. • The leaves will be (all) green/not have white stripes.	**1 point**
	Provide reasoning to **justify** your prediction. • (All offspring will have the same leaf phenotype as the ovule-producing plant because) mitochondria are maternally inherited/transferred by the ovule.	**1 point**
	Explain why plants with the same genotype are able to differ in the structure and/or number of certain organelles in response to changes in atmospheric levels of CO_2. • (Plants have different phenotypes because) changes in CO_2 levels/the environment affect the expression of certain genes.	**1 point**
	Total for part (d)	**3 points**
	Total for question 2	**9 points**

Question 3: Scientific Investigation — 4 points

Sand lances of the genus *Ammodytes* are small fish that function as keystone organisms in several coastal ecosystems. These sand lances are prey fish that support organisms at higher trophic levels. Scientists performed experiments to examine how sand lance populations are likely to be affected by the rising temperatures and CO_2 levels associated with climate change.

Sand lance embryos typically develop and mature into adult fish at low temperatures (approximately 5°C) and stable, low CO_2 levels (approximately 400 µatm). Over the course of two years, the scientists measured the survival rate of sand lance embryos allowed to develop and mature in a laboratory at three different temperatures, 5°C, 7°C, and 10°C, with the level of CO_2 maintained at 400 µatm, 1,000 µatm, and 2,100 µatm for each temperature.

(a) **Describe** the effect of increased biodiversity on the resilience of an ecosystem in a changing environment. — **1 point**

- (Ecosystem) resilience/it will be greater (with increased biodiversity).

(b) **Justify** the scientists' selecting 5°C as the lowest temperature and 400 µatm as the lowest CO_2 level in their study of sand lance embryo survival. — **1 point**

Accept one of the following:

- These are the normal/current conditions at which the embryos develop and were used as a basis for comparison.
- These (current) conditions were used as a basis to compare the effects of changes in environmental conditions/increases in temperature and CO_2.

(c) **State** a null hypothesis for the experiment. — **1 point**

Accept one of the following:

- Climate change will have no effect on sand lance (embryo) survival/sand lance development/the size of sand lance populations.
- (Increases in) temperature/ CO_2 levels will have no effect on sand lance (embryo) survival/sand lance development /the size of sand lance populations.
- There will be no difference in the sand lance (embryo) survival rates/sand lance development/the size of sand lance populations measured at all/different temperatures and CO_2 levels.

(d) The scientists claim that a reduction in the population size of the *Ammodytes* sand lances will affect the stability of the entire coastal ecosystem. Provide reasoning to **support** the scientists' claim. — **1 point**

Accept one of the following:

- There will be a negative effect on other trophic levels because the sand lance provides food for many other species.
- There will be a negative effect on other trophic levels because there will be reduced energy to transfer (to higher trophic levels).

Total for question 3 — 4 points

Question 4: Conceptual Analysis — 4 points

Noncyclic electron flow and cyclic electron flow are two major pathways of the light-dependent reactions of photosynthesis. In noncyclic electron flow, electrons pass through photosystem II, then components of a chloroplast electron transport chain, and then photosystem I before finally reducing $NADP^+$ to NADPH. In cyclic electron flow, electrons cycle through photosystem I and some components of the electron transport chain (Figure 1).

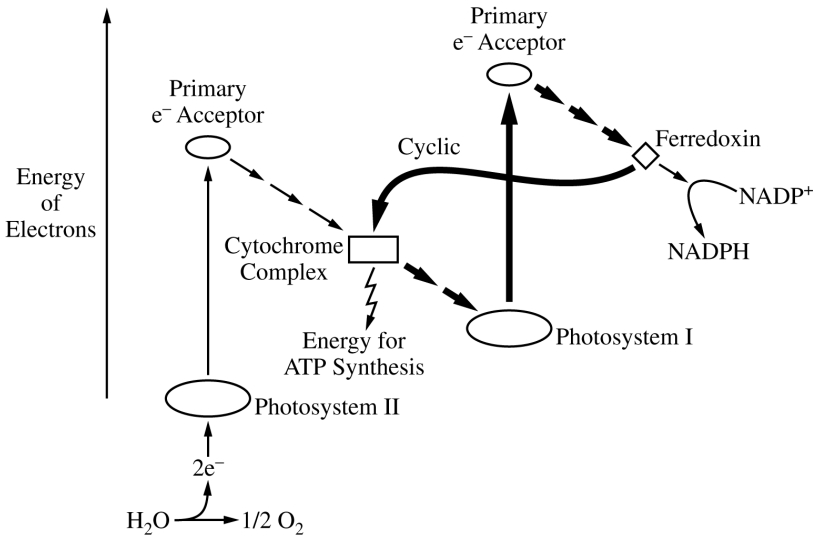


Figure 1. The pathways of noncyclic and cyclic (heavy arrows) electron flow. The cytochrome complex is a component of the electron transport chain between the two photosystems.

(a)	**Describe** the role of chlorophyll in the photosystems of plant cells. Accept one of the following: • Chlorophyll captures/absorbs light (energy). • Chlorophyll receives electrons (from water)/receives electrons (from an electron transport chain)/transfers electrons (to an electron transport chain).	**1 point**
(b)	Based on Figure 1, **explain** why an increase in the ratio of NADPH to $NADP^+$ will cause an increase in the flow of electrons through the cyclic pathway. • There is less/no $NADP^+$ to accept the electrons, so the electrons pass (instead) to the cyclic pathway/from ferredoxin to the cytochrome complex.	**1 point**
(c)	Using rice plants, scientists examined the effect of a mutation that results in the loss of the protein CRR6. CRR6 is a part of the photosystem I complex, and its absence reduces the activity of photosystem I. **Predict** the effect of the mutation on the rate of biomass (dry weight) accumulation. • The rate (of biomass accumulation)/Biomass/It will be lower (in comparison with plants without the mutation).	**1 point**
(d)	**Justify** your prediction in part (c). • There will be insufficient ATP/NADPH produced for the synthesis of carbohydrates/the Calvin cycle.	**1 point**
		Total for question 4 — 4 points

Question 5: Analyze Model or Visual Representation of a Biological Concept or Process — 4 points

Ruminants are hoofed animals, including cattle and sheep, that have a unique four-chambered stomach specialized to digest tough, fiber-filled grasses. Researchers studying ruminants are investigating the morphological and molecular characteristics of different ruminant families in order to determine the evolutionary relationships among the families. Cladograms of several ruminant families were constructed based on morphological data (Figure 1A) and molecular data (Figure 1B). Table 1 shows a sample of the morphological characteristics present in each family used to construct the cladogram in Figure 1A.

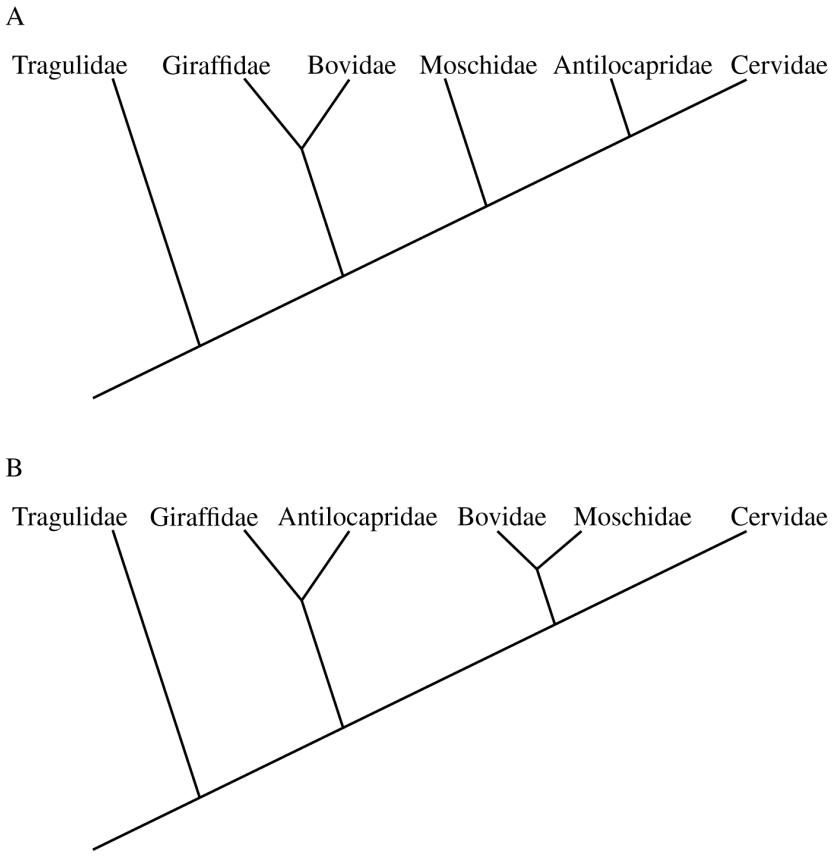
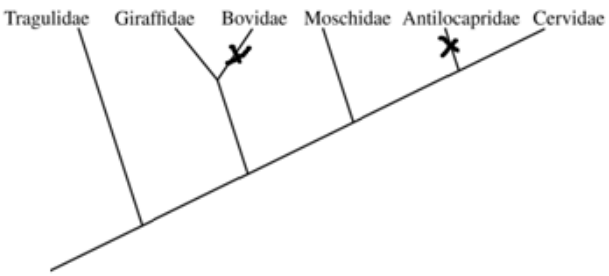


Figure 1. Cladogram of six ruminant families based on (A) morphological data and (B) molecular data

TABLE 1. MORPHOLOGICAL CHARACTERISTICS FOUND IN EACH RUMINANT FAMILY

Characteristic Number	Morphological Characteristic	Tragulidae	Giraffidae	Bovidae	Moschidae	Antilocapridae	Cervidae
1	Extra tooth material			X		X	
2	Third stomach		X	X	X	X	X
3	Double opening for tear ducts					X	X

(a)	**Describe** how a scientist would use a comparison of the DNA sequences of different organisms to suggest the most likely evolutionary relationship among the organisms. • The (DNA) sequences of organisms that are more closely related are more similar (than those of organisms that are less closely related).	**1 point**
(b)	Based on Figure 1, **explain** why Bovidae is likely to be more closely related to Moschidae than it is to Giraffidae. Accept one of the following: • (The molecular data/Figure 1B support(s) this relationship, and) molecular data are more reliable (than are morphological data). • (The molecular data/Figure 1B support(s) this relationship.) Morphological similarities may not reliably indicate evolutionary relatedness.	**1 point**
(c)	Using the template in the space provided for your response, **represent** the point(s) at which characteristic 1, listed in Table 1, evolved by marking "X" on the line(s) of the cladogram in the correct location(s). Sample Response:  • Response must show an X placed on the line leading to Bovidae and an X placed on the line leading to Antilocapridae.	**1 point**
(d)	Based on Figure 1A, **explain** why a characteristic found only in the Cervidae and Bovidae families is more likely evidence of convergent evolution than it is of common ancestry. Accept one of the following: • There are other families that have the same common ancestor as the Bovidae and Cervidae families but do not have the characteristic. • It is more likely that the characteristic arose independently in Cervidae and Bovidae than it arose in their common ancestor and was lost in Giraffidae, Moschidae, and Antilocapridae.	**1 point**
		Total for question 5 **4 points**

Question 6: Analyze Data — 4 points

Housekeeping genes encode proteins involved in universally important processes such as transcription, translation, and glycolysis. Because these genes appear to be expressed in all cells at constant levels, the expression of housekeeping genes is often used as a control when comparing how the expression of other genes varies under different conditions.

Researchers studying the effect of pesticides on declining bee populations wanted to determine whether the expression of four housekeeping genes (*GAPDH*, *RPL32*, *RPS5*, and *TBP-AF*) was in fact constant in bees across different variables. The researchers collected samples of mRNA for each of the four genes and compared how their expression varied across the developmental stage of the bee, the sex of the bee, and the cell type from which the sample was taken. The mRNA from the samples was reverse transcribed to produce DNA copies of each gene. PCR was then used to amplify the DNA, and the Cq value was determined. The Cq value is the number of PCR cycles needed to produce a specified number of DNA copies (Figure 1). A high Cq value for a sample indicates the gene was expressed at a low level.

To analyze whether any of the examined variables affected expression of the housekeeping genes, researchers examined the range of Cq values for each gene in response to each variable. Genes with a wide range of Cq values were determined to be affected by the variable, while genes with a narrow range of Cq values were determined to be unaffected by the variable.

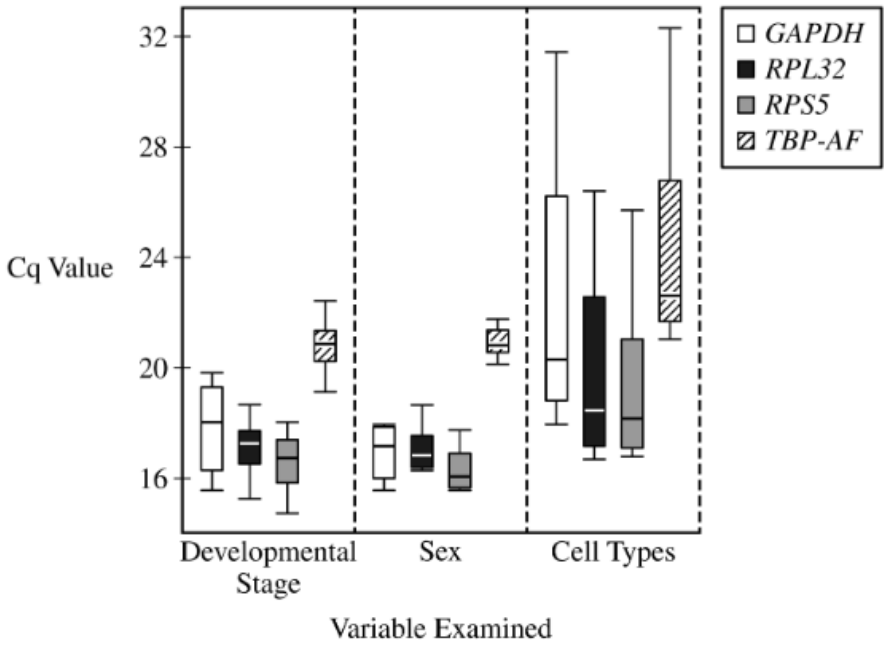


Figure 1. The effect of developmental stage, sex, and cell type on the Cq value of four housekeeping genes

(a)	Based on the data in Figure 1, **identify** the gene that had the lowest median Cq value when bees of different developmental stages were compared. • *RPS5*	**1 point**
(b)	The Cq value is inversely proportional to the amount of mRNA from that gene in the starting sample. Based on the data in Figure 1, **identify** the gene that has the lowest level of gene expression regardless of variable. • *TBP-AF*	**1 point**
(c)	The scientists investigated the effect of pesticides on the expression of other genes in one cell type of a group of bees containing males and females of the same developmental stage. They hypothesized that *TBP-AF* would serve as the best control gene for this experiment. Use the data to **evaluate** their hypothesis. • Their hypothesis is supported because *TBP-AF* has the smallest Cq range/most constant expression (when comparing sexes).	**1 point**
(d)	**Explain** how expression of a gene such as *GAPDH* can vary from one cell type to another within the same bee. • Different cell types contain different levels of/different transcription factors, and therefore regulate the expression of genes in different ways.	**1 point**
	Total for question 6	**4 points**

AP® Biology

Scoring Guidelines

Question 1: Interpreting and Evaluating Experimental Results with Experimental Design — 9 points

Crossing over in meiosis I is required for homologous chromosomes to properly align during metaphase and segregate during the first cell division.

Some regions of a chromosome called hotspots display a higher frequency of crossing over than other regions do. Crossing over is suppressed in chromosomal regions near the centromeres. The centromere region of a duplicated chromosome includes a collection of proteins that form a structure called the kinetochore. Scientists hypothesized that one or more of these kinetochore proteins are responsible for suppressing crossing over around the centromere.

To investigate their hypothesis, scientists modified chromosome 8 in yeast such that, in each cell, one chromosome from the pair of homologous chromosome 8s contained the gene encoding red fluorescent protein (RFP), while the other chromosome from the pair contained the gene encoding green fluorescent protein (GFP). Cells expressing RFP emit (give off) red light, and cells expressing GFP emit green light. Models of the modified chromosome 8 both before and after crossing over are shown in Figure 1.

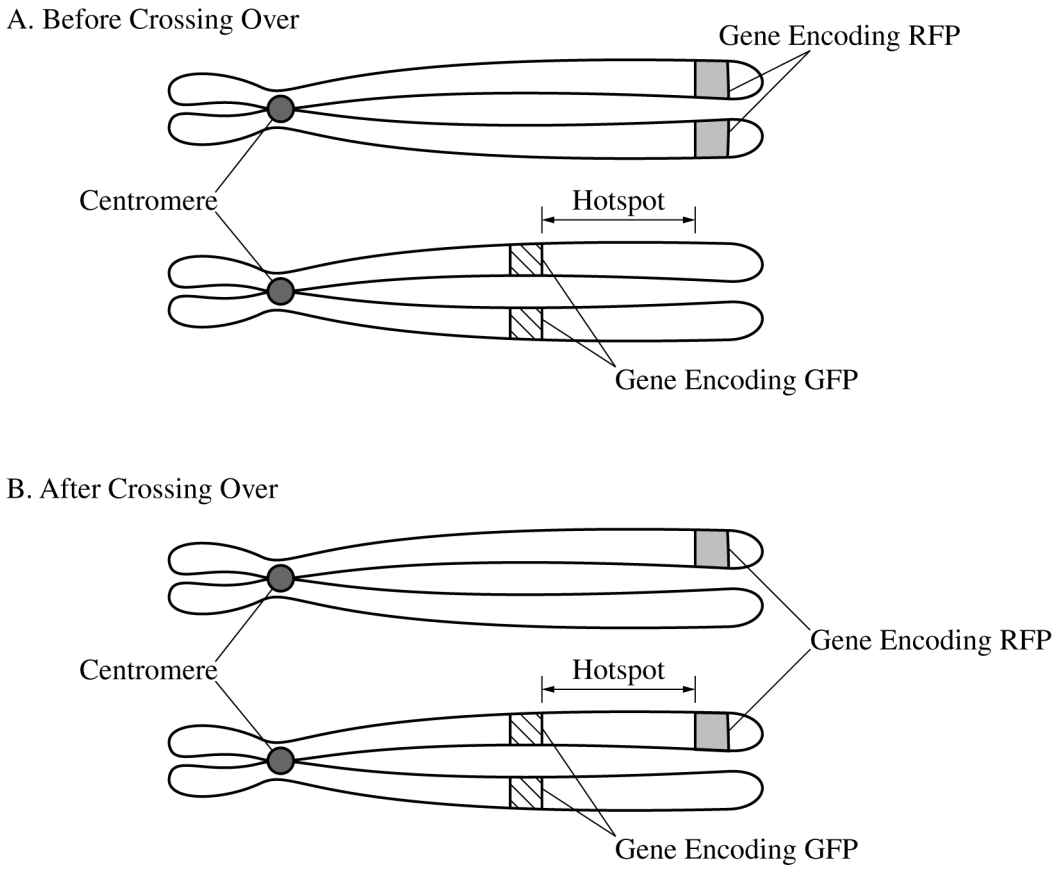


Figure 1. Models of modified chromosome 8 used in the experiment (A) before and (B) after crossing over occurs at the hotspot

The scientists then investigated whether attaching individual kinetochore proteins to a specific DNA sequence present in a known crossing-over hotspot on chromosome 8 affected the frequency of crossing

over at this location. In their first experiment, they examined three groups of yeast cells containing the modified chromosome 8. Group 1 contained no kinetochore proteins attached to the hotspot, group 2 contained the kinetochore protein CTF attached to the hotspot, and group 3 contained the kinetochore protein IML attached to the hotspot. For each group, the scientists determined the frequency of crossing over between the RFP and GFP genes. To determine the frequency, the scientists added the number of cells emitting both red and green light to the number of cells that emitted no light and divided by the total number of cells (Figure 2).

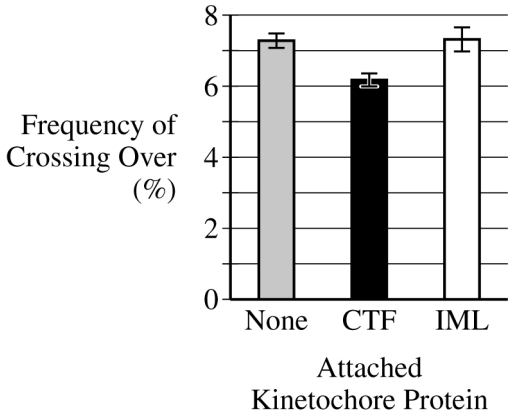


Figure 2. The frequency of crossing over in a hotspot on yeast chromosome 8 for cell groups treated with different kinetochore proteins. Error bars represent $\pm 2\mathrm{SE}_{\bar{X}}$.

(a)	**Describe** the function of S phase of interphase. Accept one of the following: • (The function of S phase is to) replicate/duplicate/synthesize the DNA/chromosomes. • (The function of S phase is to) double the amount of DNA.	**1 point**
	Explain why some haploid cells formed after meiosis in this experiment will have only one fluorescent marker. Accept one of the following: • Some cells will receive a chromosome/(sister) chromatid that did not undergo crossing over/recombination (in the hotspot). • Two (or a multiple of two) crossing-over events occurred (in the hotspot). • Each daughter cell will receive one of the four sister chromatids. Two of the sister chromatids shown in Figure 1 have only one fluorescent protein gene. • The two chromosomes that were not involved in crossing over will have only the RFP gene or the GFP gene. After meiosis II, a gamete with one of these chromosomes will have only the RFP or the GFP gene.	**1 point**
		Total for part (a) 2 points
(b)	**Identify** the control group for the scientists' first experiment, shown in Figure 2. Accept one of the following: • Group 1 • (The group called) None • Cells with no (kinetochore) protein (attached to the chromosome)	**1 point**

	In a follow-up experiment, the scientists created a modified version of CTF in which the DNA-binding portion had been removed. They compared the frequency of crossing over in yeast cells in the presence and absence of unmodified CTF with that in yeast cells in the presence and absence of the modified CTF protein (data not shown). In the follow-up experiment, **justify** why the scientists used a modified CTF protein that is unable to bind to DNA as a control. Accept one of the following: • (Using a modified CTF enabled the scientists) to determine whether DNA binding of the CTF/kinetochore protein affects/inhibits crossing over/recombination. • (Using a modified CTF enabled the scientists) to determine whether just the presence of the CTF/kinetochore protein is enough to affect/inhibit the crossing over/recombination frequency.	**1 point**
	Identify the independent variable in the follow-up experiment. Accept one of the following: • The type of CTF used • The presence or absence of (modified/unmodified) CTF	**1 point**
		Total for part (b) 3 points
(c)	Based on Figure 2, **describe** the effect on the frequency of crossing over when CTF is attached to the chromosome 8 hotspot compared with the effect when IML is attached to the hotspot. Accept one of the following: • CTF attachment results in a decreased/lower frequency of crossing over/recombination, (whereas IML had no effect). • IML attachment results in an increased/a higher frequency of crossing over/recombination (compared with CTF attachment).	**1 point**
(d)	**Predict** the effect on the number of copies of chromosome 8 likely to be present in the resulting daughter cells when CTF is attached to the hotspot. Accept one of the following: • There will be zero/two (copies). • There will be one less/one extra (copy).	**1 point**
	Provide reasoning to **justify** your prediction. • Cells (with attached CTF molecules) undergo crossing over/recombination at a lower frequency, so it is more likely that nondisjunction would occur/chromosomes would not separate properly.	**1 point**
	Explain how the presence of hotspots (Figure 1) could increase the likelihood that a population will survive in the presence of selective pressures. • Hotspots would increase genetic diversity; therefore, it would be more likely that some individuals would survive and reproduce.	**1 point**
		Total for part (d) 3 points
		Total for question 1 9 points

Question 2: Interpreting and Evaluating Experimental Results with Graphing — 9 points

To investigate how increases in environmental temperatures affect the metabolism of certain organisms, researchers incubated liver cells from toads at different temperatures and measured two markers of metabolic activity (Table 1): the rate of oxygen consumption and the rate of ATP synthesis.

TABLE 1. RATE OF OXYGEN CONSUMPTION AND ATP SYNTHESIS AT DIFFERENT TEMPERATURES

Metabolic Marker	20°C	25°C	30°C
Rate of Oxygen Consumption (nmol/min/mg of mitochondrial protein $\pm 2\mathrm{SE}_{\bar{X}}$)	12.8 ± 2.2	16.5 ± 2.0	22.1 ± 0.7
Rate of ATP Synthesis (nmol/min/mg of mitochondrial protein $\pm 2\mathrm{SE}_{\bar{X}}$)	12.6 ± 1.6	16.8 ± 2.0	21.07 ± 0.8

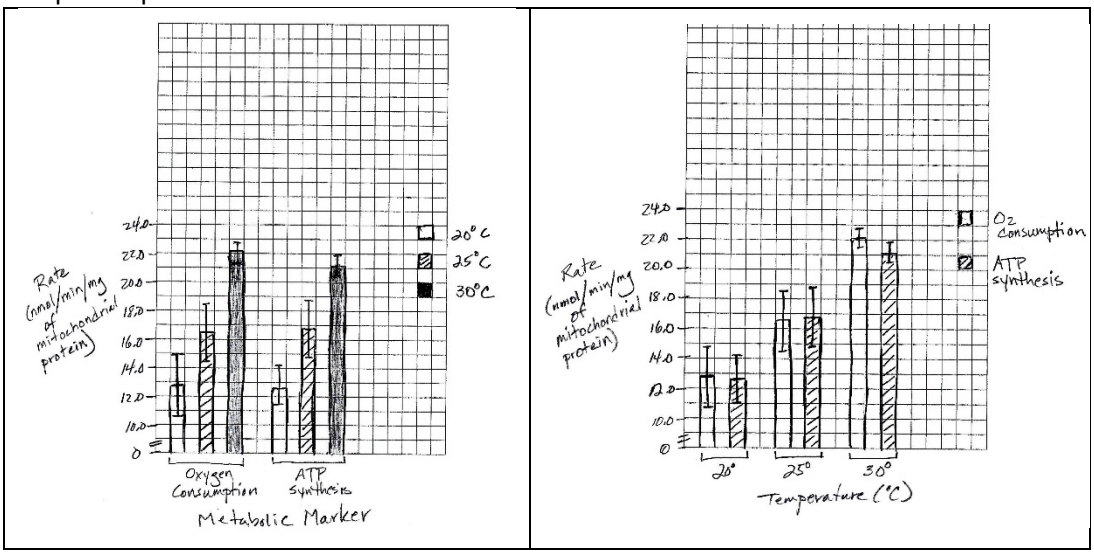
(a) **Describe** the role of water in the hydrolysis of ATP. **1 point**

Accept one of the following:

- Water is added in the process of cleaving/splitting (a phosphate from) ATP.
- Water breaks down/splits ATP.

(b) Using the template in the space provided for your response, **construct** a bar graph that represents the data shown in Table 1. Your graph should be appropriately plotted and labeled. **1 point**

Sample Responses:



- Data are represented in a bar graph.

Using the template in the space provided for your response, **construct** a bar graph that represents the data shown in Table 1. Your graph should be appropriately plotted and labeled. **1 point**

- Graph is appropriately labeled.

	Using the template in the space provided for your response, **construct** a bar graph that represents the data shown in Table 1. Your graph should be appropriately plotted and labeled. • Data points and error bars are correctly plotted.	**1 point**
	Based on the data provided, **determine** the temperature in $°C$ at which the rate of oxygen consumption is different from the rate of oxygen consumption at $25°C$. • 30	**1 point**
		Total for part (b) 4 points
(c)	Based on the data in Table 1, **describe** the effect of temperature on the rate of ATP synthesis in liver cells from toads. Accept one of the following: • As the temperature increases, the rate of ATP synthesis also increases. • There is a positive relationship (between temperature and ATP synthesis). • Temperature and ATP synthesis are directly correlated.	**1 point**
	Based on the data in Table 1**, calculate** the average amount of oxygen consumed, in $nmol$, for 10 mg of mitochondrial protein after 10 minutes at $25°C$. 1,650 [$16.5\ nmol/min/mg \times 10\ mg \times 10\ min$]	**1 point**
		Total for part (c) 2 points
(d)	Oligomycin is a compound that can block the channel protein function of ATP synthase. **Predict** the effects of using oligomycin on the proton gradient across the inner mitochondrial membrane. Accept one of the following: • (The proton gradient) will increase/become steeper (and may eventually plateau). • The difference in the concentration of protons/pH (across the inner mitochondrial membrane) will increase. • There will be an increase in the concentration of protons/a decrease in pH in the intermembrane space relative to that found within the mitochondrial matrix.	**1 point**
	Justify your prediction. Accept one of the following: • (Without protons being able to flow back into the matrix through ATP synthase), more protons will accumulate in the intermembrane space/between the two mitochondrial membranes. • (Without protons being able to flow back into the matrix through ATP synthase), there will be a lower pH in the intermembrane space/between the two mitochondrial membranes. • Protons will not be able to flow across the membrane (through ATP synthase), but the electron transport chain will still pump protons into the intermembrane space.	**1 point**
		Total for part (d) 2 points
		Total for question 2 9 points

Question 3: Scientific Investigation — 4 points

To investigate whether red blood cells of animals lose the ability to take in glucose from their environment as they age, scientists collected red blood cells from guinea pigs that ranged in age from one day old to seven months old. Scientists incubated an equal number of red blood cells in separate culture dishes that contained a $300\ \text{nM}$ solution of radioactively labeled glucose. The amount of radioactively labeled glucose present inside the red blood cells of each group was measured over time.

(a) **Describe** a difference between passive transport and active transport. **1 point**

Accept one of the following:

- Active transport requires energy/ATP (while passive transport does not).
- Passive transport does not require energy/ATP (while active transport does).
- In passive transport, substances move from a high concentration to a low concentration, (while in active transport substances move from a low concentration to a high concentration).
- In active transport, substances move from a low concentration to a high concentration (while in passive transport substances move from a high concentration to a low concentration).

(b) **Justify** why the scientists used an equal number of red blood cells in each culture dish as a control. **1 point**

Accept one of the following:

- (Scientists used an equal number of cells) to attribute differences in results/glucose transport to guinea pig age, (rather than to the number of cells used in the experiment).
- (Scientists used an equal number of cells) to compare results from the different dishes (containing cells from guinea pigs of different ages).
- (Scientists used an equal number of cells) to eliminate the number of cells as a variable (that might affect the amount of glucose in each group).
- (Scientists used an equal number of cells) because the number of cells used might affect the results/amount of glucose (present inside the red blood cells).

(c) Glucose transporters are required for the facilitated diffusion of glucose into red blood cells. The scientists claim that the expression of the gene encoding these transporters decreases as guinea pigs age. If the scientists' claim is supported by experimental data, **predict** the effect of increased age on the amount of radioactively labeled glucose present inside the cells of each group. **1 point**

- (As guinea pig age increases) the amount of glucose (inside the cells) decreases.

(d) **Justify** your prediction in part (c). **1 point**

- With fewer transporters, fewer glucose molecules will be moved into the cells.

Total for question 3 **4 points**

Question 4: Conceptual Analysis — 4 points

The common wild oat is native to regions of Europe and Asia but is an invasive species in central California grasslands. In California, the common wild oat has almost completely replaced some species of native bunchgrass. Researchers found that aphids, a type of small insect that often carries plant viruses, have a much higher reproductive rate in grasslands that include the common wild oat than in grasslands composed of only native bunchgrass species. Additionally, the viruses carried by the aphids appear to affect only the native bunchgrasses and not the common wild oat. Native bunchgrasses infected by the virus have much higher death rates than do native bunchgrasses that are not infected.

(a)	**Describe** the change in the resilience of an ecosystem when there is a decrease in the number of species. • (The resilience of the ecosystem) will decrease.	**1 point**
(b)	**Explain** how the addition of the common wild oat affects the number of native bunchgrass plants that can be supported by the California grasslands ecosystem. Accept one of the following: • (The addition of the wild oat) limits the resources available (to the native plants), resulting in a decrease (in the population size). • (The addition of the wild oat) <u>enables the aphid population to increase</u>/<u>increases the exposure to viruses</u>, resulting in a decrease (in the population size of the native bunchgrass).	**1 point**
(c)	Researchers suggest adding ladybugs, predators of aphids, to the California grasslands. **Predict** the effect of adding ladybugs on the abundance of the native bunchgrass population. • (The native bunchgrass population) will increase (in abundance).	**1 point**
(d)	**Justify** your prediction in part (c). • (Adding ladybugs) will decrease the number of aphids, which will cause a decrease in the <u>transmission of</u>/<u>infection by</u> plant viruses.	**1 point**
	Total for question 4	**4 points**

Question 5: Analyze Model or Visual Representation of a Biological Concept or Process — 4 points

Researchers study mechanisms that enable or prevent speciation.

New genes can evolve from noncoding regions of DNA. It is not until certain regulatory elements are present in the DNA that a noncoding region becomes a new, functional gene that encodes a protein. These regulatory elements include a promoter, a 5′ untranslated region (UTR) followed by a start codon, and a 3′ UTR following a stop codon (Figure 1).

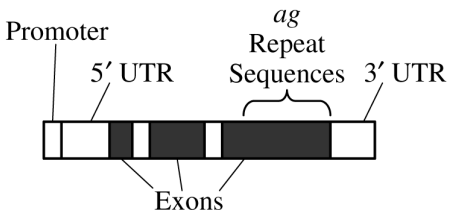
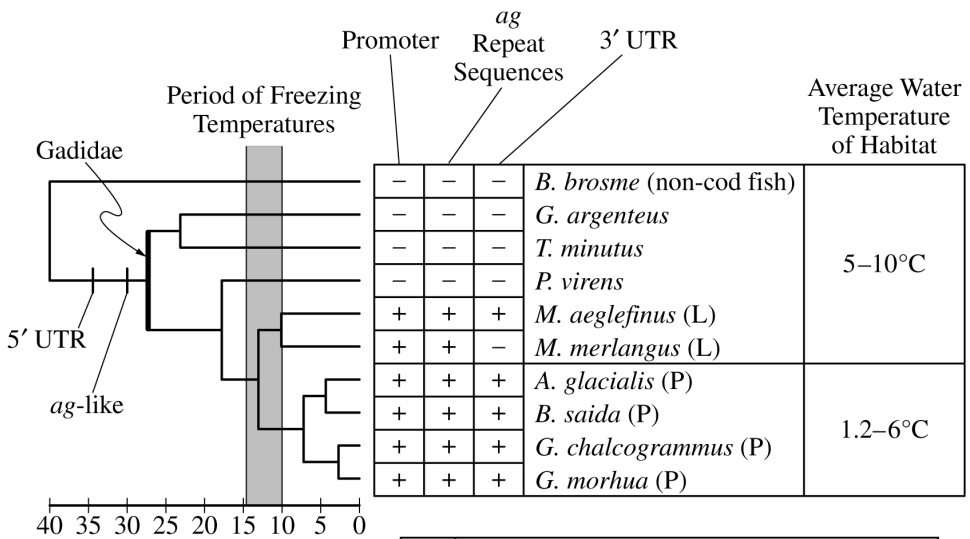


Figure 1. Basic structure of a functional *ag* gene

Researchers studied the evolution of the family of antifreeze-glycoprotein (AG) encoding genes in Gadidae, a family of marine fish known as cods. When present in the fish, these glycoproteins reduce the freezing temperature of the fish. The researchers compared genomic sequences in nine cod species and one non-cod fish species, *B. brosme*. They recorded the presence or absence of the elements of functional *ag* genes as well as *ag*-like sequences that are similar to a functional gene but have undergone mutation and do not contain all the elements required to enable protein production (Figure 2).



				Average Water Temperature of Habitat
–	–	–	*B. brosme* (non-cod fish)	5–10°C
–	–	–	*G. argenteus*	
–	–	–	*T. minutus*	
–	–	–	*P. virens*	
+	+	+	*M. aeglefinus* (L)	
+	+	–	*M. merlangus* (L)	
+	+	+	*A. glacialis* (P)	1.2–6°C
+	+	+	*B. saida* (P)	
+	+	+	*G. chalcogrammus* (P)	
+	+	+	*G. morhua* (P)	

–	Element is absent.
+	Element is present.
L	Functional *ag* genes are lost.
P	Functional *ag*-encoded proteins are present.

Figure 2. Phylogenetic tree showing the evolution of *ag* genes

(a) **Describe** a post-zygotic mechanism that prevents gene flow and thus enables speciation. **1 point**

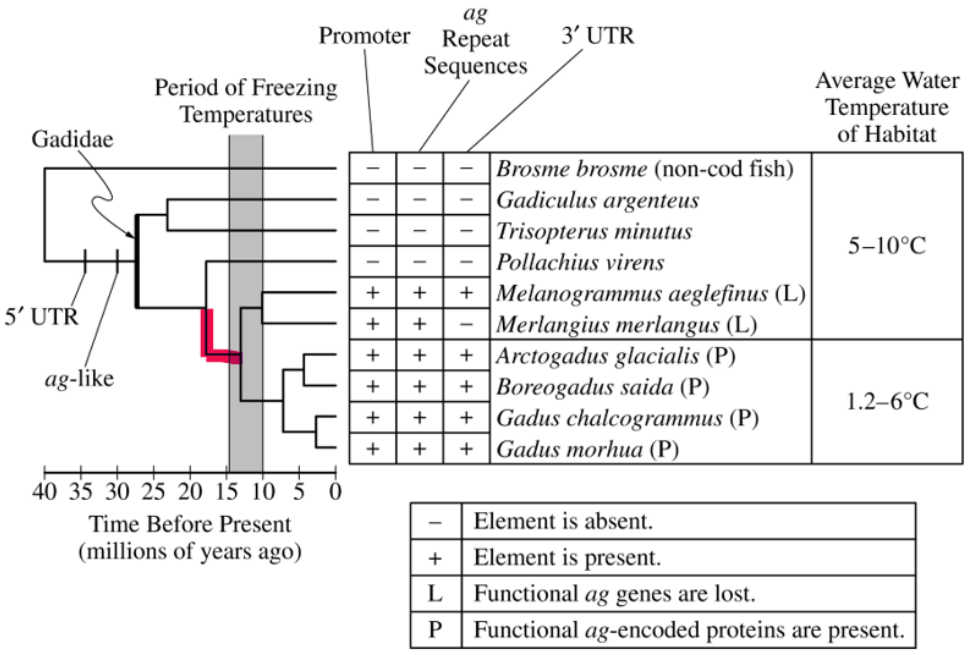
- The offspring do not survive (long enough to reproduce)/cannot (successfully) reproduce.

(b) Based on the data in Figure 2, **explain** how changes to the genome enabled cods to survive and reproduce after a period of freezing temperatures between 10 and 15 million years ago. **1 point**

- (Over time) the addition of the promoter/*ag* repeat sequences/3′ UTR/regulatory elements led to emergence of new genes/*ag* genes/functional gene products (that prevent freezing).

(c) Using the template in the space provided for your response, place an "X" on the phylogenetic tree to **represent** the origin of the functional *ag* gene. **1 point**

- An X is placed anywhere along the colored L-shaped line in the figure.



–	–	–	*Brosme brosme* (non-cod fish)	5–10°C
–	–	–	*Gadiculus argenteus*	
–	–	–	*Trisopterus minutus*	
–	–	–	*Pollachius virens*	
+	+	+	*Melanogrammus aeglefinus* (L)	
+	+	–	*Merlangius merlangus* (L)	
+	+	+	*Arctogadus glacialis* (P)	1.2–6°C
+	+	+	*Boreogadus saida* (P)	
+	+	+	*Gadus chalcogrammus* (P)	
+	+	+	*Gadus morhua* (P)	

–	Element is absent.
+	Element is present.
L	Functional *ag* genes are lost.
P	Functional *ag*-encoded proteins are present.

(d) Based on Figure 2, **explain** how genetic differences among the species in the Gadidae family determine the habitats in which they can survive. **1 point**

- Species with the functional *ag* gene/antifreeze glycoprotein are able to live in colder water/lower temperatures (than are species without the functional gene).

Total for question 5 **4 points**

Question 6: Analyze Data — 4 points

Scientists can quantify the rate of translation as ribosomes move along an mRNA from one codon to the next. Using a procedure called ribosome profiling, the scientists measured how long a ribosome remains stationary at each codon of each mRNA. They determined the average translation rate across all codons is 5.2 amino acids per second but that the average translation rate for specific codons in different mRNA sequences can vary widely. These variations in translation rates are thought to facilitate correct folding of the protein being produced. The rate at which three different codons were translated was measured in 100 different mRNAs. The scientists determined the distribution of rate (number of times each rate was recorded) for each of the three codons: GAC (Figure 1A), AUU (Figure 1B), and UGG (Figure 1C).

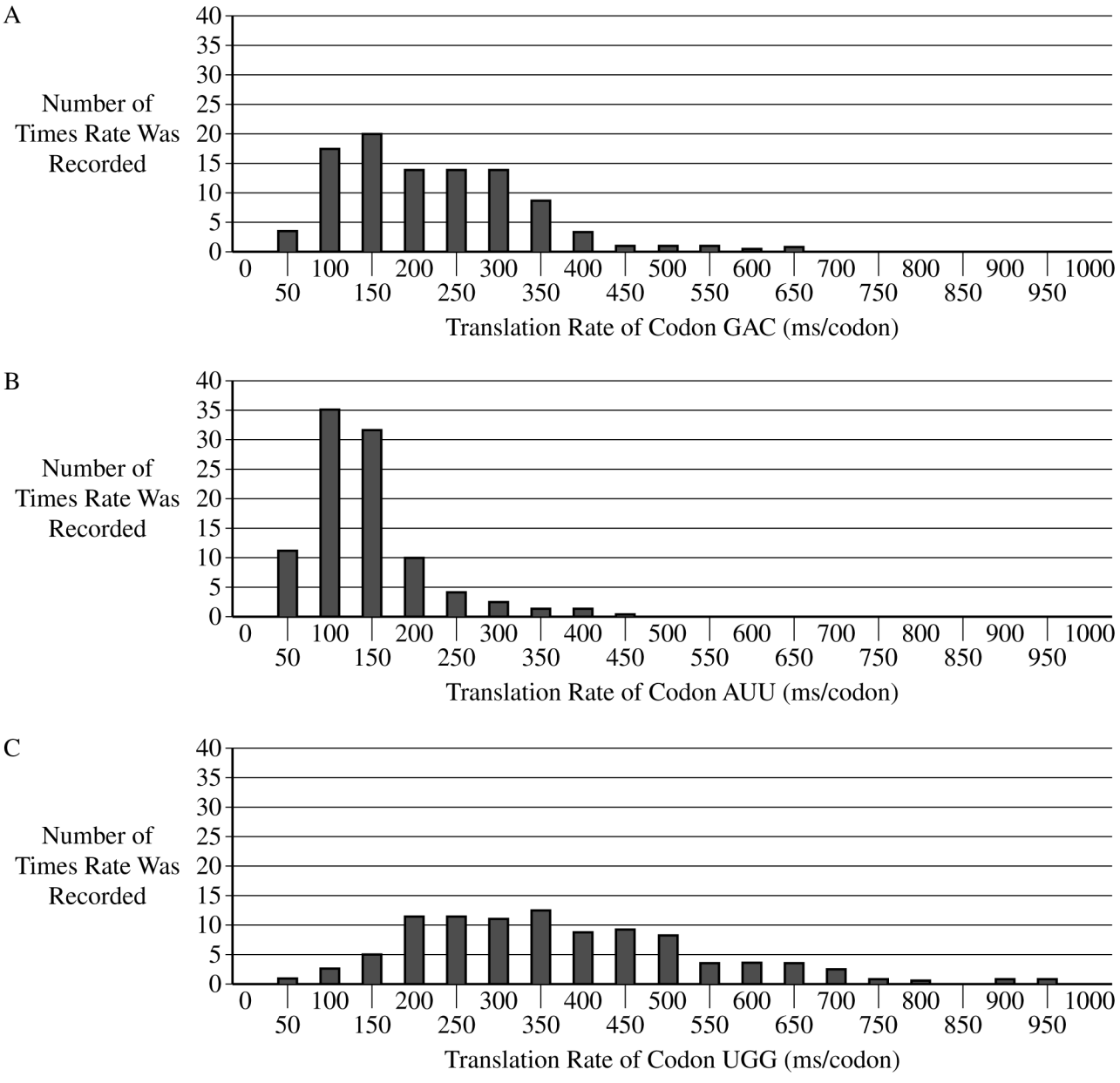


Figure 1. The distribution of translation rates for three different codons (A) GAC, (B) AUU, and (C) UGG

(a)	Using the data in Figure 1, graph A, **identify** the rate (in ms/codon) that was recorded the greatest number of times for the GAC codon. • 150	**1 point**
(b)	Using the data in Figure 1, graphs B and C, **describe** the variation in translation rate of the AUU codon compared with that of the UGG codon. Accept one of the following: • There is greater variation in (the translation rate of) UGG codons (than in the translation rate of AUU codons). • (The translation rate of) UGG ranges from 50 (ms/codon) to 950 (ms/codon), while (the translation rate of) AUU ranges from 50 (ms/codon) to 450 (ms/codon).	**1 point**
(c)	Scientists hypothesize that tRNA molecules that bind to UGG codons are available in lower abundance than are tRNAs that bind to AUU codons. **Support** the scientists' hypothesis using the data in Figure 1. Accept one of the following: • The (average) translation rate of UGG is slower (than that of AUU). • Translation of UGG takes longer (per codon than does translation of AUU). • More of the UGG codons were translated at slower rates (than AUU codons were).	**1 point**
(d)	Amino acids can be encoded by multiple codons. In many organisms, certain codons for the same amino acid occur more frequently in an mRNA than do other codons. Based on the data provided, **explain** why the use of one codon over another for the same amino acid might result in increased levels of protein production from a particular mRNA. • Certain codons are translated at faster rates than are others (and result in increased levels of protein production from a particular mRNA).	**1 point**
	Total for question 6	**4 points**

AP® Biology

Scoring Guidelines

Question 1: Interpreting and Evaluating Experimental Results 9 points

Most proteins that are secreted from a cell must be transported to the endoplasmic reticulum (ER) either during translation or after translation.

For proteins transported during translation, this process begins in the cytosol and pauses when a specific sequence of amino acids is translated. The translation complex is then transported to the surface of the ER where translation continues. Proteins that are transported after translation are translated entirely in the cytosol and then transported to the ER. In both instances, the translated proteins enter the ER through a protein channel in the membrane of the ER.

Researchers studying the two types of protein transport identified that the ER membrane protein SR is necessary for transport during translation, while the ER membrane protein Sec62 is necessary for transport after translation. To investigate which transport mechanism is used for different proteins, researchers first created small interfering RNAs (siRNAs) that reduce expression of either SR or Sec 62. They then treated groups of cells with either the SR siRNA or the Sec62 siRNA and determined the relative amount of SR and Sec 62 protein in each group of cells compared with cells treated with a control siRNA. (Figure 1).

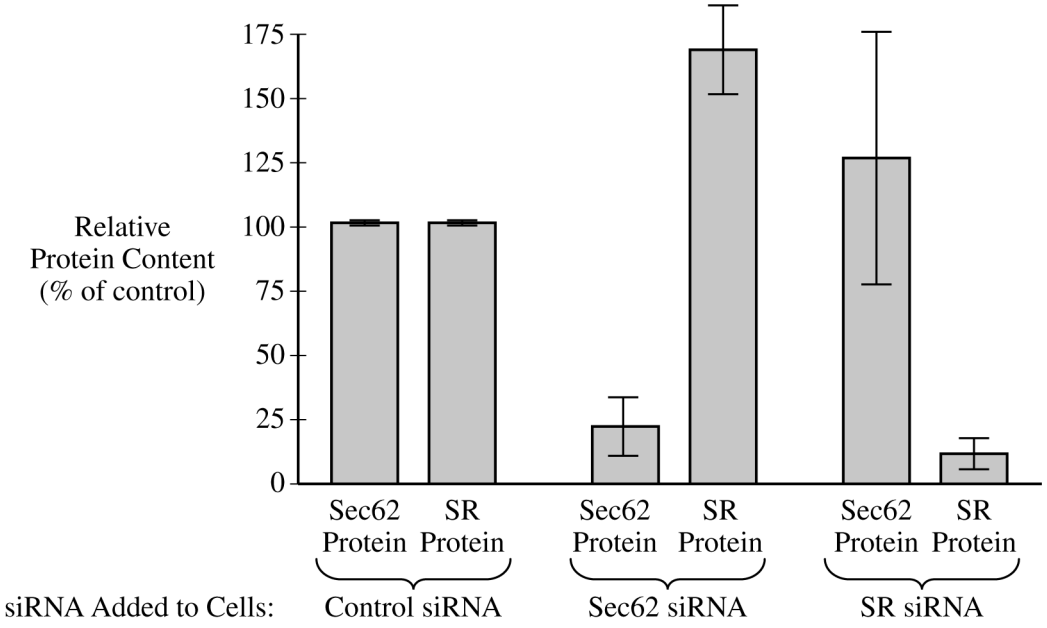


Figure 1. Average relative amounts of Sec62 and SR proteins in cells treated with control siRNA, Sec62 siRNA or SR siRNA. Error bars represent $\pm SE_{\bar{X}}$.

The researchers then measured the amount of each of three different proteins that was transported to the ER in cells treated with Sec62 siRNA or SR siRNA. The researchers calculated the percent transported relative to the cells treated with control siRNA (Figure 2).

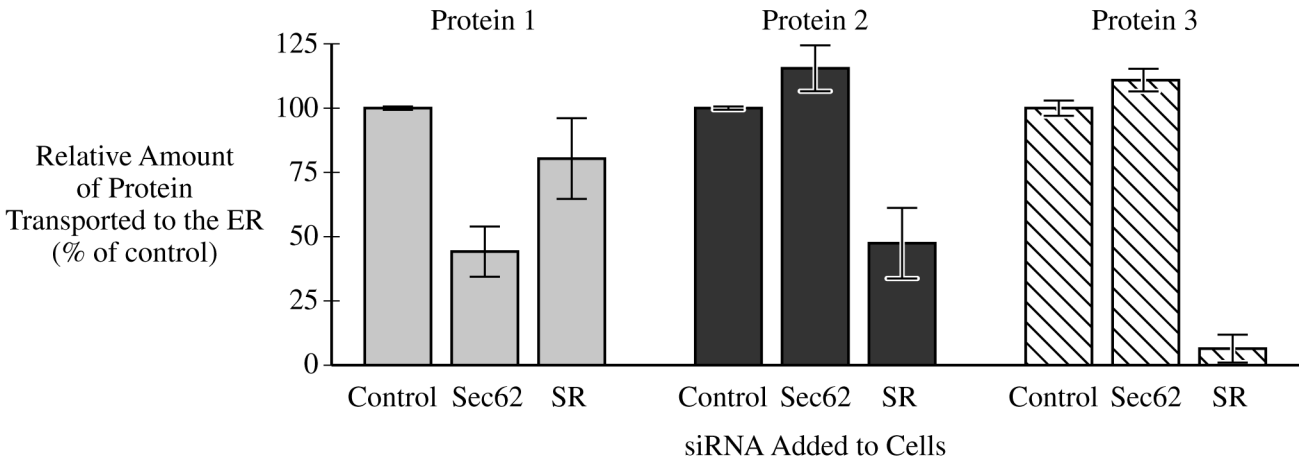


Figure 2. Average relative amounts of three proteins that were transported to the ER when treated with control siRNA, Sec62 siRNA, or SR siRNA. Error bars represent $\pm SE_{\bar{X}}$.

A	**Describe** the function of ribosomes. Examples of acceptable responses may include the following: • Ribosomes synthesize polypeptides/proteins. • Ribosomes perform translation. • Ribosomes are sites of polypeptide/protein synthesis.	**Point A1**
B	(i) **Identify** the dependent variable in the experiments shown in Figure 1. • The (relative) amount of protein/protein content	**Point B1**
	(ii) **Justify** why the researchers included the control of measuring the relative amounts of both Sec62 and SR proteins in cells that were treated with Sec62 siRNA only (data shown in Figure 1). Examples of acceptable responses may include the following: • (The control allowed the researchers) to determine whether the (Sec62) siRNA reduced/affected the content of both proteins (relative to protein content in the presence of the control siRNA). • (The control allowed the researchers) to determine whether the (Sec62) siRNA reduced/affected Sec62 (protein content) only.	**Point B2**
	(iii) Based on Figure 1, **describe** the effect on the production of SR protein when cells are treated with Sec62 siRNA. Examples of acceptable responses may include the following: • (SR protein production) increased. • (SR protein production) increased by 65% (accept 50–80%).	**Point B3**

C	(i) **Identify** the independent variable in the researchers' second experiment (data shown in Figure 2). Examples of acceptable responses may include the following: • (The type of) siRNA (added to the cells) • The type of protein whose amount was measured	**Point C1**
	(ii) Based on Figure 2, **identify** the protein(s) that when treated with Sec62 siRNA showed an increase in percent transport to the ER compared with the control. Examples of acceptable responses may include the following: • Proteins 2 and 3 • Protein 2 • Protein 3	**Point C2**
	(iii) Protein 1 is encoded by 234 nucleotides, while protein 2 is encoded by 495 nucleotides. Assuming all nucleotides for both proteins encode amino acids, **calculate** the difference in the number of amino acids between the two proteins. • 87 (amino acids) [(495/3)−(234/3)]	**Point C3**
D	(i) Researchers claim that protein 1 is the only tested protein that is transported to the ER following its complete translation in the cytosol. Using data from Figure 2, **support** the researchers' claim. • Only protein 1 showed a reduced amount/percentage of transport when cells were treated with Sec62 siRNA.	**Point D1**
	(ii) For any protein that enters the ER, researchers claim that amino acids close to the protein's amino terminus determine how likely the protein is to pass through the protein channel within the ER membrane. **Justify** the researchers' claim based on your understanding of factors that affect the transport of proteins across membranes. • Amino acids (at the amino terminus) that have similar polarity to/opposite charge to (the R groups of amino acids lining) the protein channel are more likely to pass through the channel (than are proteins where the amino terminus contains amino acids with dissimilar polarity or similar charge).	**Point D2**

Question 2: Interpreting and Evaluating Experimental Results with Graphing — 9 points

Many insects rely on pheromones (chemical signals) that are released by the females to find mating partners. Scientists hypothesize that, in a certain type of moth, the behavior of male moths in response to pheromones is regulated by the extracellular signaling molecule 20E.

To investigate whether the binding of 20E to its receptor, DopEcR, affects behavior in moths, scientists injected male moths with saline (control solution) or with small interfering RNA molecules (siRNAs) that inhibit the expression of the gene encoding DopEcR. The scientists then exposed the moths to the pheromone and determined the percent of total time observed that the moths engaged in general activity, defined as movement in any direction. The scientists also determined the percent of the general activity time that the moths spent in oriented activity, defined as movement toward an area of high pheromone concentration (Table 1).

Table 1. Average General and Oriented Activity in Male Moths Injected With Saline or siRNA Molecules

Treatment	General Activity (percent of total time observed, average $\pm 2SE_{\bar{x}}$)	Oriented Activity (percent of general activity, average $\pm 2SE_{\bar{x}}$)
Male moths injected with saline (control solution)	95 ± 5	60 ± 4
Male moths injected with siRNAs that inhibit expression of the gene encoding DopEcR	90 ± 8	25 ± 6

DopEcR is a G protein-coupled receptor. When 20E binds to DopEcR, GTP displaces the GDP bound to the G protein, and a signaling pathway is activated. The scientists hypothesize that this leads to the transcription of genes associated with the oriented activity observed in the male moths (Figure 1).

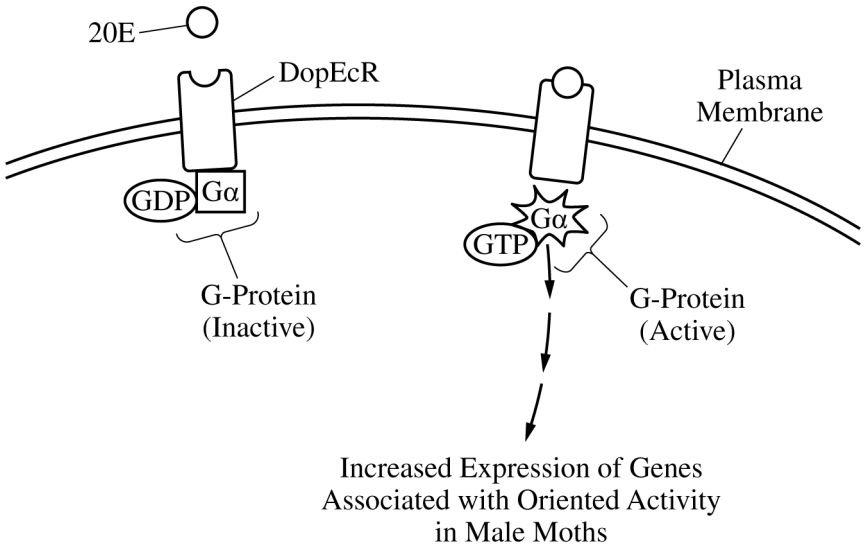
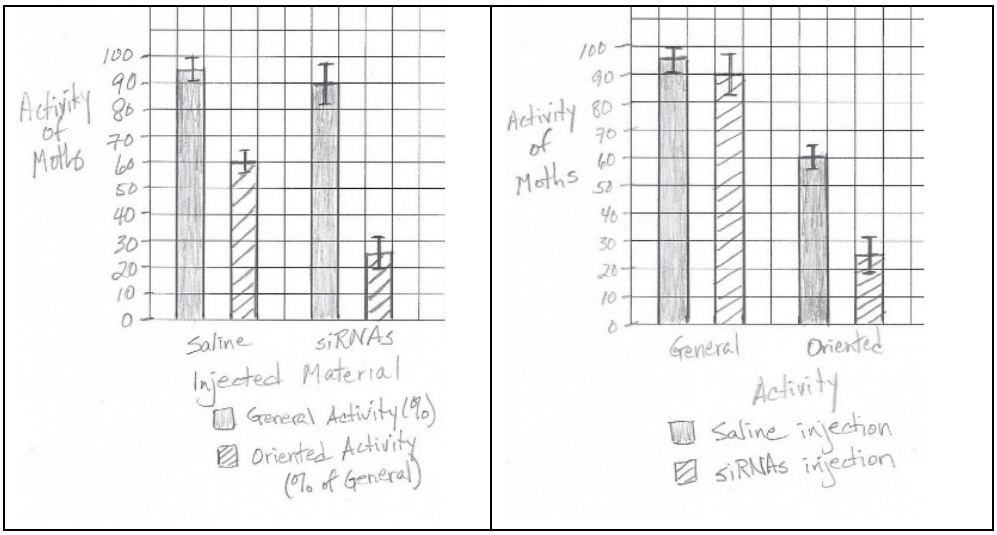


Figure 1. A simplified model of a signaling pathway activated by the binding of 20E to its receptor, DopEcR

A	Many receptors are embedded in the plasma membrane. **Describe** the polarity of the portion of the receptor that is inside the membrane. Examples of acceptable responses may include the following: • (The portion of the receptor inside the membrane) is nonpolar. • (The portion of the receptor inside the membrane) is hydrophobic.	**Point A1**
B	(i) Using the template in the space provided for your response, **construct** an appropriate type of graph that represents the data in Table 1. Your graph should be appropriately plotted and labeled. • Data are represented in a bar/modified bar graph. • Examples of acceptable responses may include the following: 	**Point B1**
	(i) Using the template in the space provided for your response, **construct** an appropriate type of graph that represents the data in Table 1. Your graph should be appropriately plotted and labeled. • Data and error bars are accurately plotted.	**Point B2**
	(i) Using the template in the space provided for your response, **construct** an appropriate type of graph that represents the data in Table 1. Your graph should be appropriately plotted and labeled. • Graph is appropriately labeled.	**Point B3**
	(ii) Based on the data in Table 1, **determine** the type of activity that was affected by inhibiting the expression of the DopEcR receptor. • Oriented activity (was affected).	**Point B4**

C	(i) Based on Table 1, identify the treatment group in which the oriented activity was greater than 50% of the general activity. Examples of acceptable responses may include the following: • The control/saline-treated group • Male moths injected with saline/control solution	**Point C1**
	(ii) The scientists studied some moths with a mutation in the gene encoding the G protein. The mutation prevents GTP from displacing the GDP bound to the G protein. Based on Figure 1, **predict** the effect of this mutation on the oriented activity in male moths exposed to the pheromone. • (The moths will show) decreased (oriented) activity.	**Point C2**
D	(i) Expression of the gene encoding DopEcR is low in the male moths during their first few days as adults, when they are sexually immature. Gene expression rapidly increases as the moths reach sexual maturity. The scientists claim that this increase in gene expression increases the likelihood of males finding females with whom to mate. Use evidence from the information provided to **support** the scientists' claim. Examples of acceptable responses may include the following: • (Increased expression of DopEcR) will increase oriented activity (and the moths' ability to find a mate). • (An increase in DopEcR expression) will enable the binding of 20E to more receptors. • (An increase in DopEcR expression) will make the males more sensitive to pheromones from the females. • (An increase in DopEcR expression) will enable the binding of 20E to more receptors and will make the males more sensitive to pheromones from the females.	**Point D1**
	(ii) Based on Figure 1, **explain** how an inhibitor of the DopEcR pathway might serve as an effective chemical to protect crops from moth damage. • (An inhibitor of the pathway) will reduce oriented activity/expression of genes associated with oriented activity and therefore decrease mating/population growth.	**Point D2**

Question 3: Scientific Investigation — 4 points

Buffelgrass, an invasive grass species in southwestern desert ecosystems, is threatening the saguaro cactus, a keystone species in these ecosystems. Buffelgrass is drought-tolerant and can survive wildfires. However, the dry buffelgrass also acts as fuel for wildfires, causing the fires to be more severe. Older saguaro cacti can survive wildfires; however, many of the young cacti cannot.

Scientists conducted an experiment to determine whether they could control the abundance of the buffelgrass population. The scientists identified several native grass species that, when grown with buffelgrass, might reduce the abundance of buffelgrass. They grew buffelgrass in the presence of several different native grass species in greenhouses, in either nondrought (watered every 3 days) or drought (watered every 9 days) conditions. After twelve weeks, they measured the height and dry weight of the buffelgrass in each treatment group.

A — **Describe** the effect that removing a keystone species will have on an ecosystem. **1 point**

Examples of acceptable responses may include the following:

- (Removal of a keystone species) reduces biodiversity/diversity/resilience (of the ecosystem).
- (Removal of a keystone species) will cause the ecosystem to collapse.

B — **Identify** a control group the scientists should include in their experiment. **1 point**

Examples of acceptable responses may include the following:

- (A treatment group with) only buffelgrass planted
- (A treatment group with) no native grass species

C — **State** the null hypothesis of the experiment in which buffelgrass is grown in the presence of native grass species. **1 point**

Examples of acceptable responses may include the following:

- There will be no difference in/effect on the abundance/dry weight/height/size of buffelgrass (when grown alone as compared with) when grown with native plants.
- There will be no difference in/effect on the abundance/dry weight/height/size of buffelgrass grown in drought and nondrought conditions.

D — Scientists have found that the population growth rates of native grasses are much slower than the population growth rate of buffelgrass following a wildfire. The scientists claim that wildfires will therefore increase the abundance of buffelgrass plants in the ecosystem. Based on the information given, **justify** the scientists' claim. **1 point**

Examples of acceptable responses may include the following:

- (With fewer native plant species after a wildfire) buffelgrass will have less competition/no competition (for resources).
- (With fewer native plant species after a wildfire) buffelgrass will have more resources (for growth, development, and reproduction).

Question 4: Conceptual Analysis — 4 points

Twenty million years ago the Caribbean Sea and Pacific Ocean were connected, and water flowed freely between the two bodies of water. Many of the same marine species were found in both areas. Over millions of years, the land referred to as the Isthmus of Panama formed, eventually closing off the connection between the Caribbean Sea and Pacific Ocean and creating two separate bodies of water. The ecology of these two marine habitats was dramatically altered by this land formation. The warmer Caribbean water could no longer flow west, so the Pacific water cooled and became more nutrient-rich, while the Caribbean water became warmer.

A	**Describe** the genetic evidence that evolution is occurring in a population. Examples of acceptable responses may include the following: • There are changes in allele/gene frequencies. • There are heritable changes in phenotypes.	**1 point**
B	**Explain** how the isolation of marine species by the formation of a land barrier can lead to divergent evolution of those species. Examples of acceptable responses may include the following: • Selective pressures could result in different allele/gene frequencies. • (The land barrier) results in reproductive isolation/lack of gene flow/allopatric speciation. • Different environmental conditions/pressures select for different alleles/genotypes/phenotypes.	**1 point**
C	The formation of the Isthmus of Panama connected two continents, North America and South America. Many North American land animal species migrated to South America after the formation of the isthmus and occupied similar niches as South American species. **Predict** the effect the formation of the isthmus had on resource availability for South American species. • (Resource availability) would have decreased.	**1 point**
D	**Justify** your prediction in part C. • More species would now be competing for the same resources, (resulting in fewer resources for each individual).	**1 point**

Question 5: Analyze Model or Visual Representation of a Biological Concept or Process — 4 points

Figure 1 shows the reactions of the metabolic pathway used to synthesize amino acid B from amino acid A in cells.

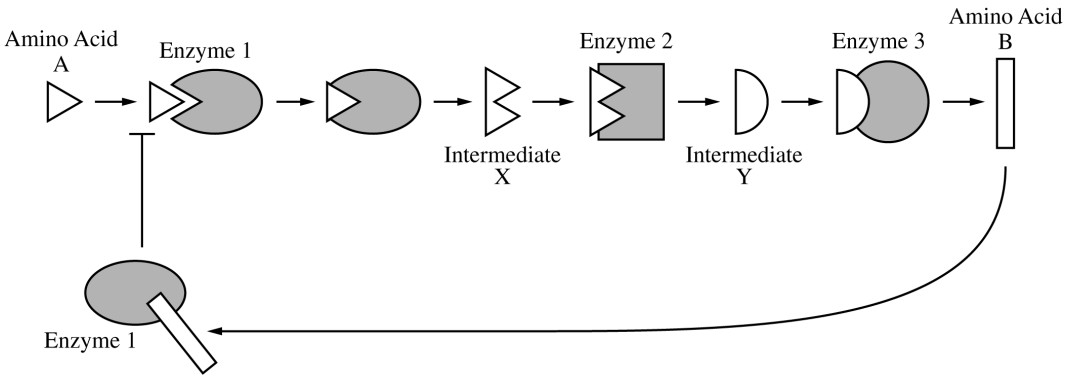


Figure 1. Synthesis of amino acid B from amino acid A

A	**Describe** a characteristic of an enzyme's active site that allows it to catalyze a specific chemical reaction. Examples of acceptable responses may include the following: • (The active site) is able to bind to a (specific) substrate/amino acid. • The shape/charge of the active site of an enzyme is compatible with a (specific) substrate/amino acid. • (The active site) positions the substrate/amino acid in a way that makes the reaction more likely. • Induced fit stresses bonds so that a reaction is more likely to occur.	**1 point**
B	Based on Figure 1, **explain** how the binding of amino acid A to enzyme 1 is regulated by amino acid B. Examples of acceptable responses may include the following: • Amino acid B inhibits the binding of amino acid A to enzyme 1. • Amino acid B binds to an allosteric site of enzyme 1 and prevents amino acid A from binding to enzyme 1. • The binding of amino acid B causes the enzyme to change shape and prevents amino acid A from binding to enzyme 1.	**1 point**
C	Using the information in Figure 1, **identify** the product of the reaction catalyzed by enzyme 2: intermediate X, intermediate Y, or amino acid B. • (Intermediate) Y	**1 point**
D	Based on Figure 1, **explain** how a change in pH could affect enzyme 3 in such a way that amino acid B cannot be produced. Examples of acceptable responses may include the following: • (A change in pH) could change the structure/active site/enzyme so that intermediate Y/the substrate cannot bind. • (A change in pH) could denature (enzyme 3) so that intermediate Y/the substrate cannot bind.	**1 point**

Question 6: Analyze Data — 4 points

The *ald* gene of fruit flies encodes the ALD protein, which is associated with both the centromeres of chromosomes and protein filaments produced during meiosis. In the absence of functional ALD proteins, gamete-producing cells enter anaphase I before homologous chromosomes are correctly aligned. As a result, the gametes produced do not contain the correct numbers of chromosomes.

Scientists generated four mutations in the *ald* gene: *ald1*, *ald3*, *ald23*, and *del*, which was a deletion of the gene. To study the role of the ALD protein in meiosis, scientists used gamete-forming metaphase cells from groups of flies with different *ald* genotypes. Some of the flies were homozygous for the wild-type allele of *ald: WT*/*WT*. Other flies were heterozygous for different *ald* alleles: *WT*/*del*; *ald1*/*del*; *ald3*/*ald23*; *ald23*/*del*. The scientists measured the percent of metaphase cells that contained ALD-associated filaments (Figure 1A) and the amount of ALD protein produced by each of the cell types (Figure 1B).

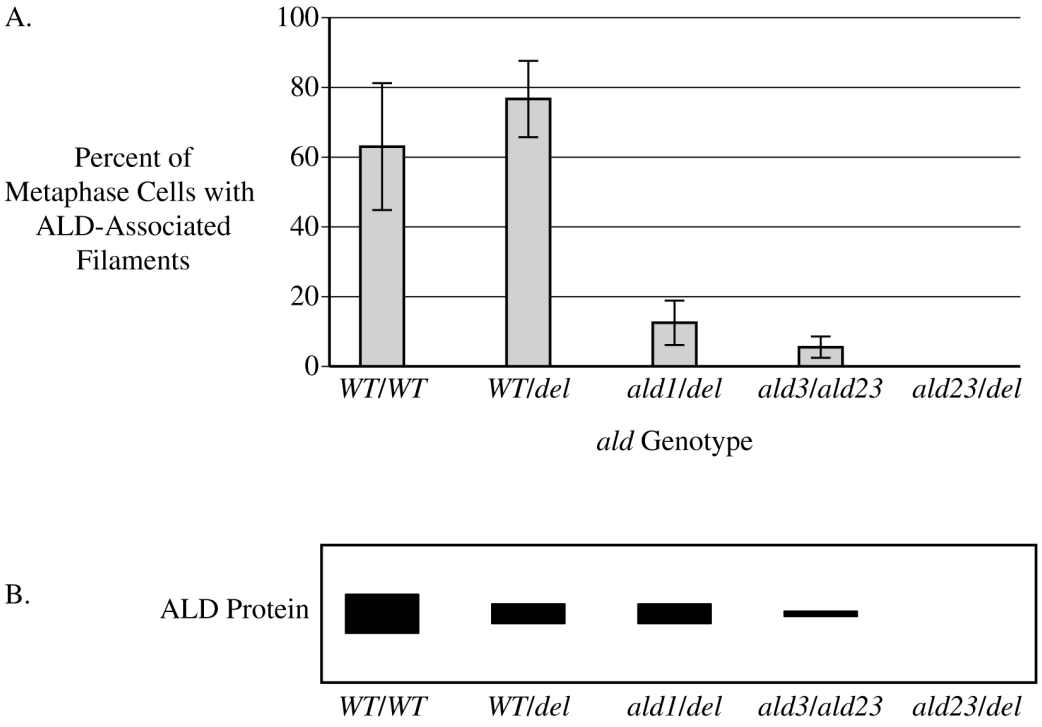


Figure 1. (A) The average percent of gamete-forming metaphase cells that contained filaments associated with ALD and (B) the amount of ALD protein produced by each cell type. A thicker band indicates a greater amount of ALD protein.

A	Based on Figure 1A, **identify** the fly genotype in which the average percent of metaphase cells with ALD-associated filaments is close to 12%. • *ald1/del*	**1 point**
B	Based on Figure 1B, **describe** the difference in ALD protein production between gamete-forming metaphase cells of flies with the genotype *ald3*/*ald23* and flies with the genotype *ald23*/*del*. Examples of acceptable responses may include the following: • More ALD protein is produced by *ald3*/*ald23* cells (than by *ald23*/*del* cells). • Less ALD protein is produced by *ald23*/*del* cells (than by *ald3*/*ald23* cells). • No ALD protein is produced by *ald23*/*del* cells (whereas it is produced by *ald3*/*ald23* cells).	**1 point**
C	Scientists hypothesize that gamete-forming metaphase cells can produce a normal amount of ALD-associated filaments even when they produce about half as much ALD protein as the wild-type cells produce. Use the data in Figures 1A and 1B to **support** the scientists' hypothesis. Examples of acceptable responses may include the following: • (With half as much protein) the *WT/del* cells show no difference in percent of (gamete forming metaphase cells with) ALD-associated filaments. • The data for the *WT/del* relative to/in comparison with the *WT/WT* cells support the scientists' hypothesis.	**1 point**
D	For gamete-forming metaphase cells of the *WT*/*del* and *ald1*/*del* flies, **explain** why the phenotypes observed in Figure 1A differ even though the amount of ALD protein produced (Figure 1B) does not. Examples of acceptable responses may include the following: • (The phenotypes) differ because only the *WT*/*del* flies produce enough functional (ALD) protein to generate a wild-type phenotype. • When one allele encodes functional ALD protein (in *WT/del* flies), the flies can generate a wild-type phenotype/produce ALD-associated filaments in a similar amount as *WT/WT* flies. • Both genotypes produced ALD protein, but the *ald1* mutation resulted in a protein with reduced function (compared with *WT*, resulting in a different phenotype).	**1 point**